

NURTURE COURSE
ATOMIC STRUCTURE

ATOMIC STRUCTURE

1. INTRODUCTION :

The existence of atoms has been proposed since the time of early Indian and Greek philosophers (400 B.C.) who were of the view that atoms are the fundamental building blocks of matter. According to them, the continued subdivisions of matter would ultimately yield atoms which would not be further divisible. The word 'atom' has been derived from the Greek word 'a-tomio' which means 'uncuttable' or 'non-divisible'. These earlier ideas were mere speculations and there was no way to test them experimentally. These ideas remained dormant for a very long time and were revived again by scientists in the nineteenth century. The atomic theory of matter was first proposed on a firm scientific basis by **John Dalton**, a British school teacher in 1808. His theory, called **Dalton's atomic theory**, regarded the atom as the ultimate particle of matter.

2. DALTON ATOMIC THEORY :

- (i) Atom is the smallest particle of any substance which cannot be divided further.
- (ii) Atoms can never be created nor be destroyed. Only the rearrangement of atoms occurs in chemical reactions.
- (iii) All the atoms are hard and dense.
- (iv) All the atoms of an element are identical but the atoms of different elements will be different.
- (v) A compound is formed by the combinations of atoms of different elements in a fixed ratio by mass.

We start with the experimental observations made by scientists towards the end of nineteenth and beginning of twentieth century. These established that atoms can be further divided into subatomic particles, i.e., electrons, protons and neutrons. a concept very different from that of Dalton. The major problems before the scientists at that time were:

- to account for the stability of atom after the discovery of sub-atomic particles,
- to compare the behaviour of one element from other in terms of both physical and chemical properties,
- to explain the formation of different kinds of molecules by the combination of different atoms and,
- to understand the origin and nature of the characteristics of electromagnetic radiation absorbed or emitted by atoms.

3. SUB-ATOMIC PARTICLES

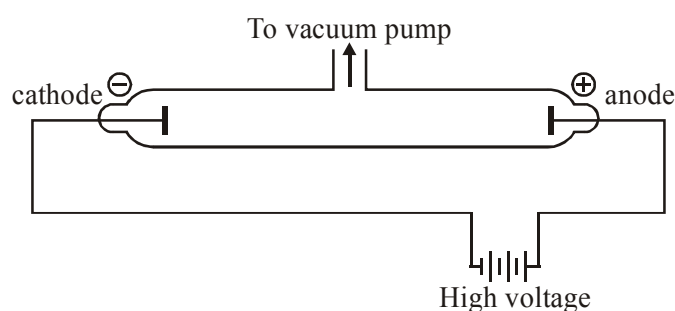
Dalton's atomic theory was able to explain the law of conservation of mass, law of constant composition and law of multiple proportion very successfully. However, it failed to explain the results of many experiments, for example, it was known that substances like glass or ebonite when rubbed with silk or fur generate electricity. Many different kinds of sub-atomic particles were discovered in the twentieth century.

3.1 Discovery of Electron :

In 1830, Michael Faraday showed that if electricity is passed through a solution of an electrolyte, chemical reactions occurred at the electrodes, which resulted in the liberation and deposition of matter at the electrodes. These results suggested the particulate nature of electricity.

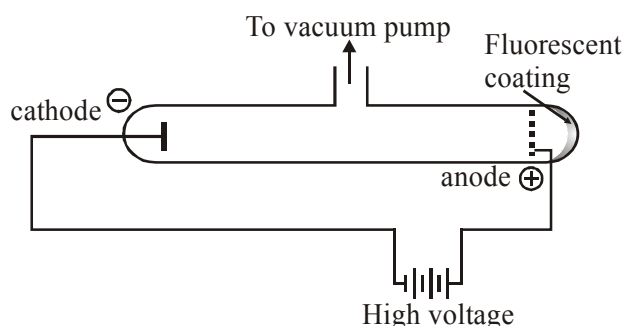
An insight into the structure of atom was obtained from the experiments on electrical discharge through gases. Before we discuss these results we need to keep in mind a basic rule regarding the behaviour of charged particles: "Like charges repel each other and unlike charges attract each other".

In mid 1850s many scientists mainly Faraday began to study electrical discharge in partially evacuated tubes, known as **cathode ray discharge tubes**.



A cathode ray discharge tube

A cathode ray tube is made of glass containing two thin pieces of metal, called electrodes, sealed in it. The electrical discharge through the gases could be observed only at very low pressures and at very high voltages. The pressure of different gases could be adjusted by evacuation. When sufficiently high voltage is applied across the electrodes, current starts flowing through a stream of particles moving in the tube from the negative electrode (cathode) to the positive electrode (anode). These were called **cathode rays or cathode ray particles**. The flow of current from cathode to anode was further checked by making a hole in the anode and coating the tube behind anode with phosphorescent material zinc sulphide. When these rays, after passing through anode, strike the zinc sulphide coating, a bright spot on the coating is developed (same thing happens in a television set)



A cathode ray discharge tube with perforated anode

The results of these experiments are summarised below.

- (i) The cathode rays start from cathode and move towards the anode.
- (ii) These rays themselves are not visible but their behaviour can be observed with the help of certain kind of materials (fluorescent or phosphorescent) which glow when hit by them. Television picture tubes are cathode ray tubes and television pictures result due to fluorescence on the television screen coated with certain fluorescent or phosphorescent materials.

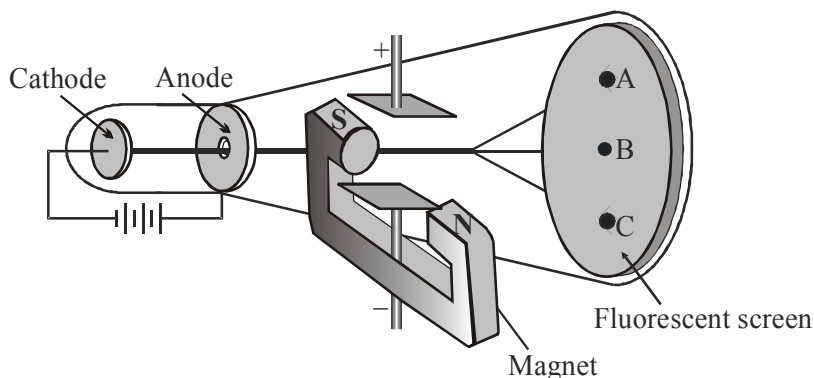
- (iii) In the absence of electrical or magnetic field, these rays travel in straight lines.
- (iv) In the presence of electrical or magnetic field, the behaviour of cathode rays are similar to that expected from negatively charged particles, suggesting that the cathode rays consist of negatively charged particles, called **electrons**.
- (v) The characteristics of cathode rays (electrons) do not depend upon the material of electrodes and the nature of the gas present in the cathode ray tube.

Thus, we can conclude that electrons are basic constituent of all the atoms.

3.1.1 Charge to Mass Ratio of Electron :

In 1897, British physicist J.J. Thomson measured the ratio of electrical charge (e) to the mass of electron (m_e) by using cathode ray tube and applying electrical and magnetic field perpendicular to each other as well as to the path of electrons. Thomson argued that the amount of deviation of the particles from their path in the presence of electrical or magnetic field depends upon :

- (i) the magnitude of the negative charge on the particle, greater the magnitude of the charge on the particle, greater is the interaction with the electric or magnetic field and thus greater is the deflection.
- (ii) the mass of the particle lighter the particle, greater the deflection.
- (iii) the strength of the electrical or magnetic field. the deflection of electrons from its original path increases with the increase in the voltage across the electrodes, or the strength of the magnetic field.



The apparatus to determine the charge to the mass ratio of electron

When only electric field is applied, the electrons deviate from their path and hit the cathode ray tube at point A. Similarly when only magnetic field is applied, electron strikes the cathode ray tube at point C. By carefully balancing the electrical and magnetic field strength, it is possible to bring back the electron to the path followed as in the absence of electric or magnetic field and they hit the screen at point B. By carrying out accurate measurements on the amount of deflections observed by the electrons on the electric field strength or magnetic field strength, Thomson was able to determine the value of e/m_e as:

$$\frac{e}{m_e} = 1.758820 \times 10^{11} \text{ C kg}^{-1}$$

where m_e is the mass of the electron in kg and e is the magnitude of the charge on the electron in coulomb (C). Since electrons are negatively charged, the charge on electron is $-e$.

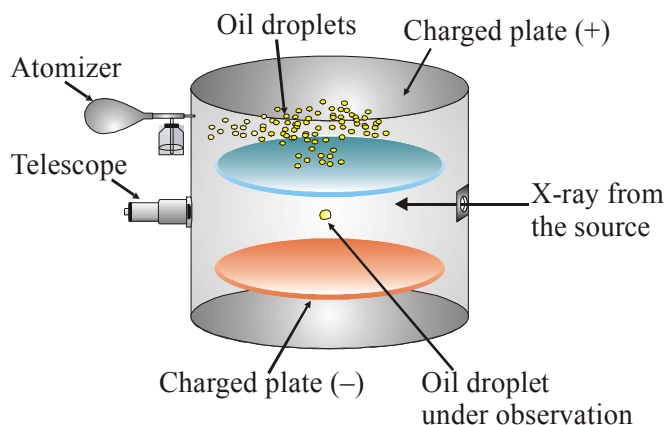
3.1.2 Charge on the Electron :

R.A. Millikan (1868-1953) devised a method known as oil drop experiment (1906-14), to determine the charge on the electrons. He found that the charge on the electron to be -1.6×10^{-19} C. The present accepted value of electrical charge is -1.6022×10^{-19} C. The mass of the electron (m_e) was determined by combining these results with Thomson's value of e/m ratio.

$$m_e = \frac{e}{e/m_e} = \frac{1.6022 \times 10^{-19} \text{ C}}{1.758820 \times 10^{11} \text{ C kg}^{-1}} = 9.1094 \times 10^{-31} \text{ kg}$$

Millikan's Oil Drop Method

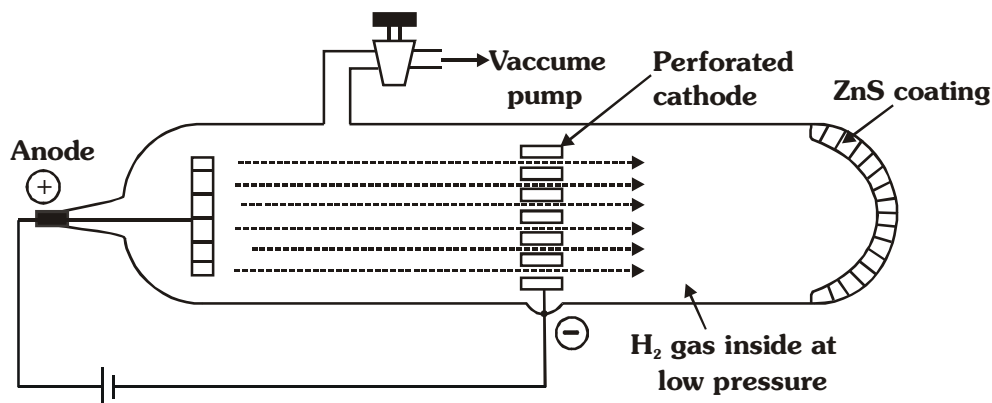
In this method, oil droplets in the form of mist, produced by the atomiser, were allowed to enter through a tiny hole in the upper plate of electrical condenser. The downward motion of these droplets was viewed through the telescope, equipped with a micrometer eye piece. By measuring the rate of fall of these droplets, Millikan was able to measure the mass of oil droplets. The air inside the chamber was ionized by passing a beam of X-rays through it. The electrical charge on these oil droplets was acquired by collisions with gaseous ions. The fall of these charged oil droplets can be retarded, accelerated or made stationary depending upon the charge on the droplets and the polarity and strength of the voltage applied to the plate. By carefully measuring the effects of electrical field strength on the motion of oil droplets, Millikan concluded that the magnitude of electrical charge, q , on the droplets is always an integral multiple of the electrical charge, e , that is, $q = n e$, where $n = 1, 2, 3 \dots$. The highest common factor (HCF) of charges on oil droplet was taken as charge on electron.



The Millikan oil drop apparatus for measuring charge 'e'. In chamber, the forces acting on oil drop are: gravitational, electrostatic due to electrical field and a viscous drag force when the oil drop is moving.

3.2 CANAL RAYS (OR ANODE RAYS) – DISCOVERY OF PROTON :

Atoms are electrically neutral. Hence after the discovery of the negatively charged constituent (electron) of an atom, attempts were made to discover the positively charged counterpart of electrons. By using a discharge tube containing a perforated cathode, **Goldstein (1886)** found that some rays passed through these holes in a direction opposite to that of the cathode rays.



Positive Rays or Canal Rays

Electrical discharge carried out in the modified cathode ray tube led to the discovery of particles carrying positive charge, also known as **canal rays**. The characteristics of these positively charged particles are listed below.

- (i) unlike cathode rays, the positively charged particles depend upon the nature of gas present in the cathode ray tube. These are simply the positively charged gaseous ions.
- (ii) The charge to mass ratio of the particles is found to depend on the gas from which these originate.
- (iii) Some of the positively charged particles carry a multiple of the fundamental unit of electrical charge.
- (iv) The behaviour of these particles in the magnetic or electrical field is opposite to that observed for electron or cathode rays.

The smallest and lightest positive ion was obtained from hydrogen and was called proton. This positively charged particle was characterised in 1919.

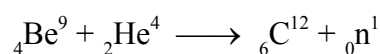
The lightest positively charged particle is called a proton (P or P⁺). Positive rays are atomic or molecular cations from which some electrons have been removed. The removed electrons constitute the cathode rays and the positive cations form the positive or canal rays.

3.3 DISCOVERY OF NEUTRON :

After the discovery of electron and proton **Rutherford (1920)** predicted the existence of a neutral fundamental particle. In **1932, Chadwick** bombarded the element Beryllium with α -particles and noticed the emission of a radiation having the following characteristics.

- (i) The radiation was highly penetrating.
- (ii) The radiation was unaffected by magnetic and electric fields which shows that it is electrically neutral.
- (iii) It was found to have approximately the same mass as the protons.

The name 'neutron' was given to this sub-atomic particle. It is denoted by n or ${}_0n^1$. Bombardment of beryllium by α -particles results in the formation of carbon and neutrons are emitted.



Mass of a neutron is 1.00867 amu (1.67493×10^{-24} g or 1.67493×10^{-27} kg)

3.4 PROPERTIES OF FUNDAMENTAL PARTICLES :

Atoms are made up-essentially, of three fundamental particles, which differ in mass and electric charge as follows :

	Electron	Proton	Neutron
Symbol	e or e ⁻	p	n
Approximate relative mass	1/1836	1	1
Approximate relative charge	-1	+1	0
Mass in kg	9.10939×10^{-31}	1.67262×10^{-27}	1.67493×10^{-27}
Mass in amu	0.00054	1.00727	1.00867
Actual charge/C	-1.6022×10^{-19}	$+1.6022 \times 10^{-19}$	0

Ex.1. Arrange the particle in their increasing order of specific charge ratio.

(a) e⁻, P, n, α-particle

(b) Na⁺, Li⁺, F⁻, Mg²⁺, Al³⁺

Sol. (a) $\left(\frac{e}{m}\right)_e = \frac{1e}{\left(\frac{1}{1836}\right)\text{amu}}$

$$\left(\frac{e}{m}\right)_p = \frac{1}{1} = 1 = \frac{1e}{1\text{amu}}$$

$$\left(\frac{e}{m}\right)_n = \frac{0}{1} = 0 = \frac{0e}{1\text{amu}}$$

$$\left(\frac{e}{m}\right)_\alpha = \frac{2}{4} = \frac{1}{2} = \frac{2e}{4\text{amu}}$$

Ans. $n < \alpha < p < e^-$

(b) $Na^+ = \frac{1}{23}$; $Li^+ = \frac{1}{7}$; $F^- = \frac{1}{19}$; $Mg^{2+} = \frac{2}{24} = \frac{1}{12}$; $Al^{3+} = \frac{3}{27} = \frac{1}{9}$
 $Na^+ < F^- < Mg^{2+} < Al^{3+} < Li^+$

Ex.2. Which of the following pairs have same specific charge $\left(\frac{e}{m}\right)$?

(a) electron & proton

(b) electron & positron

(c) proton & positron

(d) proton & deuteron

(e) α-particle & deuteron

Answer. b, e

Ex.3. Through what potential difference an α -particle should be accelerated to have speed 5×10^6 m/s.

Sol. : $qV = \frac{1}{2}mv^2$

$$(2 \times 1.6 \times 10^{-19} \times V) = \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times (5 \times 10^6)^2$$

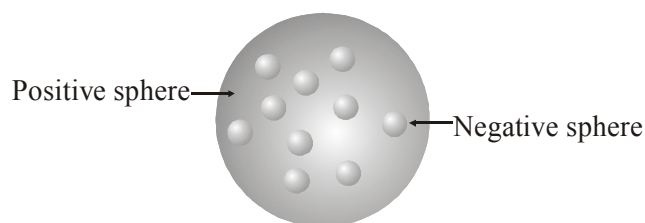
4. ATOMIC MODELS :

Observations obtained from the experiments mentioned in the previous sections have suggested that Dalton's indivisible atom is composed of sub-atomic particles carrying positive and negative charges. Different atomic models were proposed to explain the distributions of these charged particles in an atom. Although some of these models were not able to explain the stability of atoms, two of these models, proposed by J. J. Thomson and Ernest Rutherford are discussed below.

4.1 Thomson Model of Atom :

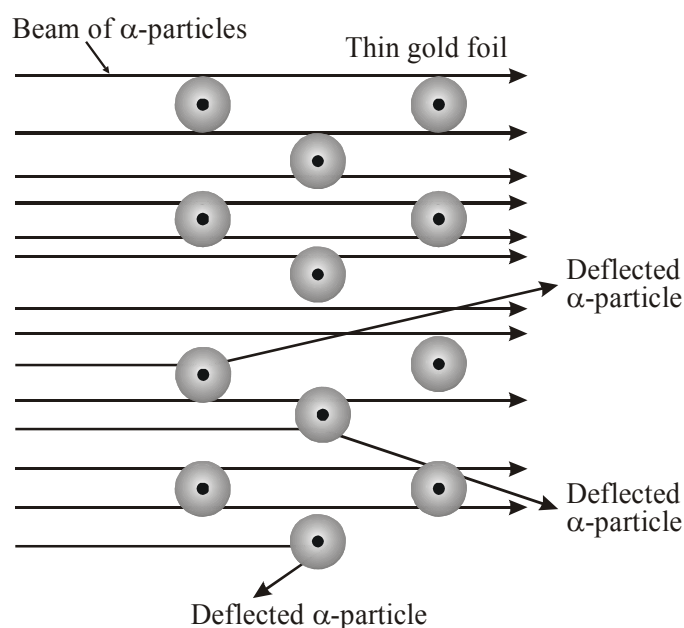
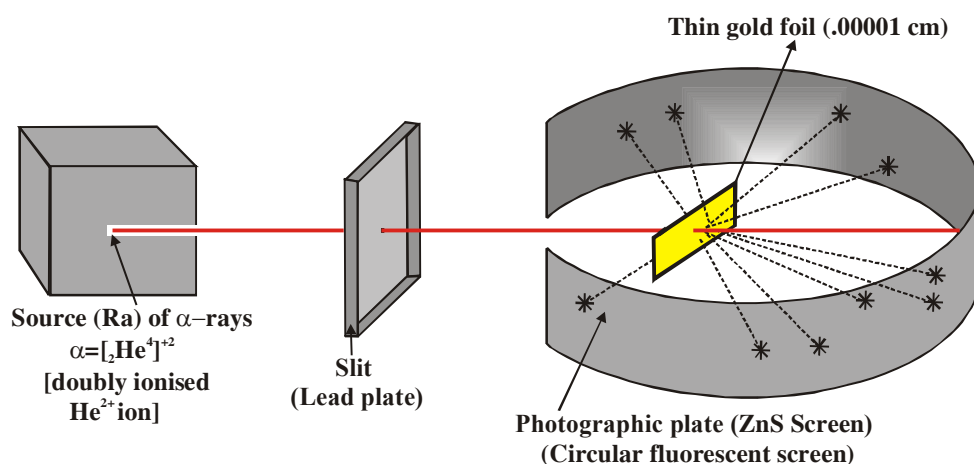
J. J. Thomson, in 1898, proposed that an atom possesses a spherical shape (radius approximately 10^{-10} m) in which the positive charge is uniformly distributed. The electrons are embedded into it in such a manner as to give the most stable electrostatic arrangement. Many different names are given to this model, **for example, plum pudding, raisin pudding or watermelon**. This model can be visualised as a pudding or watermelon of positive charge with plums or seeds (electrons) embedded into it. An important feature of this model is that the mass of the atom is assumed to be uniformly distributed over the atom. Although this model was able to explain the overall neutrality of the atom, but was not consistent with the results of later experiments.

Thomson was awarded Nobel Prize for physics in 1906, for his theoretical and experimental investigations on the conduction of electricity by gases.



Thomson model of atom

4.2 Rutherford's model :

Rutherford's α -scattering experiment

B. Schematic molecular view of the gold foil

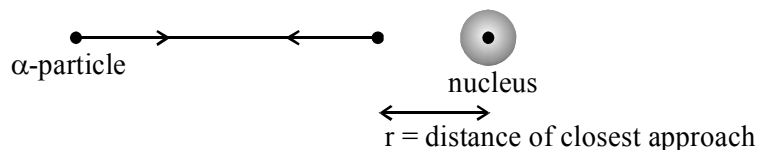
❖ Observations and conclusions :

- (i) Most of the α -particles passed through the gold foil undeflected. Hence, most of the space in the atom is empty.
- (ii) A small fraction of α -particles was deflected by small angles. A very few α -particles (~1 in 20,000) bounced back, that is, were deflected by nearly 180° .

The deflection must be due to enormous repulsive force showing that the positive charge of the atom is not spread throughout the atom as Thomson had presumed. The positive charge has to be concentrated in a very small volume that repelled and deflected the positively charged α -particles. Rutherford given the name **nucleus** to this positively charged center of atom.

- (iii) The volume occupied by the nucleus is negligibly small as compared to the total volume of atom. The radius of the atom is about 10^{-10} m, while that of nucleus is 10^{-15} m.

Rutherford estimated the size of nucleus by calculating the distance of closest approach.



The initial kinetic energy of α -particle must be equal to potential energy at distance of closest approach.

$$\text{K.E.} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 \cdot q_2}{r} = \frac{1}{4\pi\epsilon_0} \cdot \frac{(2e) \cdot (Ze)}{r}$$

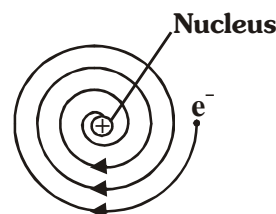
$$r = \frac{2Ze^2}{(4\pi\epsilon_0)(\text{K.E.})}$$

where Z = atomic number of element used in foil.

- (iv) Almost all mass of the atom is densely concentrated in extremely small region (nucleus).
 (v) The nucleus is surrounded by electrons that move around the nucleus with a very high speed in circular paths called orbits. Electrons and the nucleus are held together by electrostatic forces of attraction.

4.2.1 Drawbacks of Rutherford model :

- (i) According to the electromagnetic theory of Maxwell, charged particles when accelerated should emit electromagnetic radiation. Therefore, an electron in an orbit will emit radiation, the energy carried by radiation comes from electronic motion. The orbit will thus continue to shrink.



Calculations show that it should take an electron only 10^{-8} s to spiral into the nucleus. But this does not happen. Thus, the Rutherford model cannot explain the stability of an atom.

- (ii) It does not explain the arrangement of electrons revolving round nucleus.
 (iii) It does not explain the stability of nucleus against strong repulsive forces.
 (iv) It does not explain atomic spectrum.

Ex.4. An α -particles of kinetic energy of 5.4 MeV is projected towards gold nucleus. Calculate the distance of closet approach. (Atomic number of gold = 79, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$)

Sol :
$$\text{K.E.} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r}$$

$$\text{or } 5.4 \times 10^6 \times 1.6 \times 10^{-19} = 9 \times 10^9 \times \frac{(2 \times 1.6 \times 10^{-19}) \times (79 \times 1.6 \times 10^{-19})}{r}$$

$$\therefore r = 4.21 \times 10^{-14} \text{ m}$$

Ex.5 An α -particles is projected towards the following nucleus with same kinetic energy in different experiement the distance of closet approach is maximum for

(A) Na ($Z = 11$)

(B) Ca ($Z = 20$)

(C) Ag ($Z = 47$)

(D) Au ($Z = 79$)

Answer. (A)

Ex.6. An α -particle, a proton, a deuteron and a neutron are projected towards the same nucleus with the same kinetic energy in different experiement. The distance of closet approach is minimum for

(A) α

(B) P

(C) d

(D) n

Answer. (D)

Ex.7 An α -particle having K.E. = 7.7 MeV is scattered by gold ($Z = 79$) nucleus through 180° . Find distance of closest approach.

$$\text{K.E.} = 7.7 \text{ MeV}$$

$$= 7.7 \times 10^6 \times 1.6 \times 10^{-19} \text{ J}$$

$$= 1.23 \times 10^{-12} \text{ J}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$$

$$\text{Using we get : } \frac{9 \times 10^9 \times 2 \times 79 \times (1.6 \times 10^{-19})^2}{1.23 \times 10^{-12}}$$

$$r_0 = 3 \times 10^{-14} \text{ m}$$

From the above example it is clear that nuclear dimension cannot be greater than $3 \times 10^{-14} \text{ m}$.

5. WAVE THEORY :

A wave is defined as a periodic disturbance in space or in a medium that involves elastic displacement of material particles or a periodic change in some physical quantities such as T, P, V etc. Thus, wave motion represents propagation of a periodic disturbance carrying energy. The wave travels at right angles to the vibratory motion of the object.

When an object moves up and down or vibrates continuously, energy in the form of waves is transmitted by a vibrating object to a distant place. For example, when a stone is thrown in a still water of a pond, a disturbance is produced at a place where the stone strikes the water. This disturbance advances outwards in the same form and reaches the edges of the pond. Such a disturbance is called a **mechanical wave**.

The mechanical waves transmit only in a material medium. Besides the mechanical waves, there are waves which do not require any medium for their transmission. These waves are called *electromagnetic waves* or *electromagnetic radiations*.

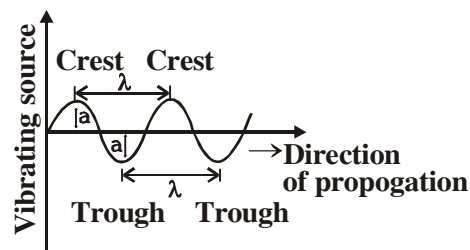
5.1 WAVE CHARACTERISTICS :

I. Wavelength (λ) :

It is defined as the distance between two nearest crests or nearest troughs.

It is measured in terms of a $\text{\AA}$ (Angstrom), pm (Picometre), nm (nanometer), cm (centimetre), m (metre)

$$1 \text{ \AA} = 10^{-10} \text{ m}, \quad 1 \text{ pm} = 10^{-12} \text{ m}, \quad 1 \text{ nm} = 10^{-9} \text{ m}, \quad 1 \text{ cm} = 10^{-2} \text{ m}$$



II. Frequency (ν) :

Frequency of a wave is defined as the number of waves which pass through a point in 1 sec.

- It is measured in terms of Hertz (Hz), sec^{-1} , or cycle per second (cps)

$$1 \text{ Hertz} = 1 \text{ sec}^{-1}$$

III. Time period (T) : Time taken by a wave to pass through one point.

$$T = \frac{1}{\nu} \text{ sec.}$$

IV. Velocity (c) :

Velocity of a wave is defined as distance covered by a wave in 1 sec.

$$c = \lambda / T = \lambda \nu \quad \nu = c / \lambda \quad c = \nu (\text{sec}^{-1}) \times \lambda (\text{m}) \quad c = \nu \lambda (\text{m sec}^{-1})$$

Since c is constant for em-waves.

i.e. frequency is inversely proportional to λ

V. Wave number ($\bar{\nu}$) :

It is defined as number of waves per unit length. It is denoted by $\bar{\nu}$ & is expressed in cm^{-1} .

$$\lambda \text{ m} \rightarrow 1 \text{ wave}$$

$$1 \text{ m} \rightarrow 1/\lambda \text{ waves}$$

$$\bar{\nu} = \frac{1}{\lambda} \quad (1 \text{ cm}^{-1} = 100 \text{ m}^{-1})$$

- It is measured in terms of cm^{-1} , m^{-1} etc.

VI. Amplitude (a) :

It is the height of the crest or depth of the trough of a wave and is denoted by 'a'. It is half the vertical distance from the top of the wave to the bottom of the wave. It determines the intensity or brightness of the beam of light.

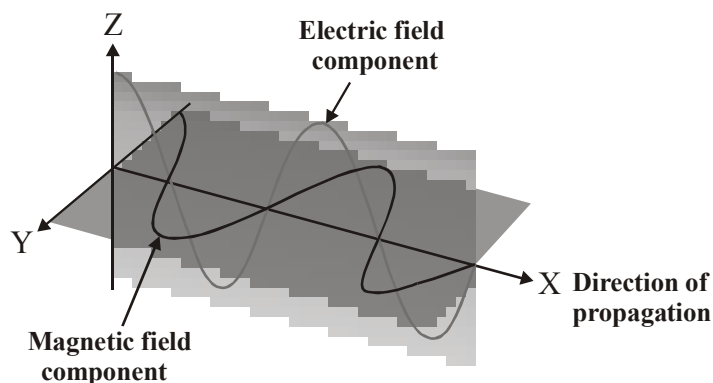
The amplitude of a wave in stretched string is the maximum displacement of the string from its normal position that of water waves is the maximum height of the water surface relative to its normal level, That of a sound wave is the maximum change in pressure relative to the normal pressure.

5.2 ELECTRO MAGNETIC (EM) WAVES :

James Maxwell (1870) suggested that when electrically charged particle moves under acceleration, alternating electrical and magnetic fields are produced and transmitted by it. These fields are transmitted in the forms of waves called electromagnetic waves or electromagnetic radiations.

❖ Main assumptions of Maxwell EM theory :

- (i) The oscillating electric and magnetic fields produced by oscillating charged particles are perpendicular to each other and both are perpendicular to the direction of propagation of the wave.



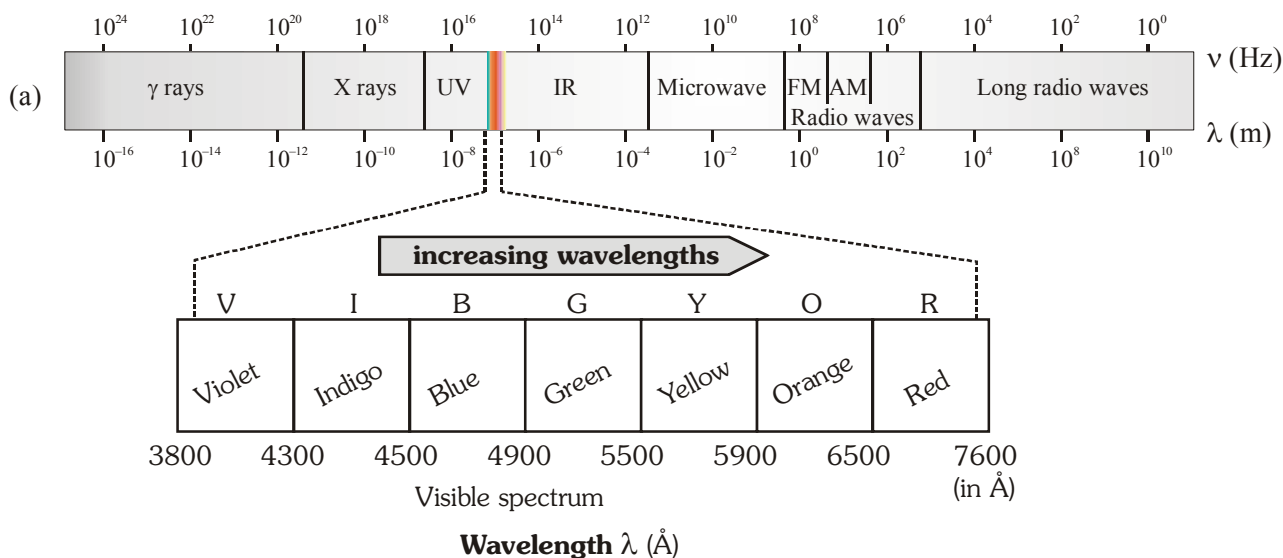
(The electric and magnetic field components of an electromagnetic wave. These components have the same wavelength, frequency, speed and amplitude, but they vibrate in two mutually perpendicular planes.)

- (ii) Unlike sound waves or water waves, electromagnetic waves do not require medium and can move in vacuum.
(However speed changes in different mediums)
- (iii) The EM waves travel with the velocity of light (3×10^8 m/sec) in vacuum.
- (iv) Energy of EM waves depends on its amplitude not on its frequency or wavelength.
(This assumptions was later proved to be wrong in some situations).

5.3 ELECTRO MAGNETIC SPECTRUM :

The arrangement of various types of electromagnetic radiations in the order of their increasing or decreasing wavelength or frequencies is known as electromagnetic spectrum. The wavelength decreases in the following order.

Radio waves > microwaves > Infrared > Television waves > Visible > Ultraviolet > X-rays > γ -rays > cosmic rays



(a) The spectrum of electromagnetic radiation. (b) Visible spectrum.
The visible region is only a small part of the entire spectrum.

Ex.8. Calculate the frequency of EMR (Electromagnetic radiation) of wave number 10^4 cm^{-1} .

Sol : $\nu = \frac{c}{\lambda} = c\bar{\nu} = (3 \times 10^8 \text{ m/s})(10^4 \text{ cm}^{-1}) = (3 \times 10^{10} \text{ cm/s})(10^4 \text{ cm}^{-1}) = 3 \times 10^{14} \text{ Hz}.$

Ex.9. A radio station radiate the radiowaves of frequency 20kHz. What is meter band of that radio station?

Sol : $20\text{kHz} = \frac{3 \times 10^8}{\lambda} \Rightarrow \lambda = 1.5 \times 10^4 \text{ m}$

6. PARTICLE NATURE OF ELECTROMAGNETIC RADIATIONS

(PLANCK'S QUANTUM THEORY OF RADIATION)

Max Planck in 1901, put forward a theory known as "**Planck's Quantum Theory**". It regards electromagnetic radiations made up of particles. This was further extended by *Einstein* in 1905. The main points of the theory are :

- The radiant energy is emitted or absorbed by atoms or molecules discontinuously in the form of small energy packets called quanta. In case of light, these energy packets are known as photons.
- The energy of each quantum is directly proportional to the frequency of the radiation i.e.

$$E \propto \nu \quad \text{or} \quad E = h\nu = \frac{hc}{\lambda}$$

Here h is a constant known as **Planck's constant**. Its value is $6.626 \times 10^{-34} \text{ Joules sec}.$

- (iii) The total amount of energy emitted or absorbed by a body is some whole number multiple of quantum i.e.,

$$E = nh\nu \quad (\text{Here } n \text{ is a positive integer : } 1, 2, 3, 4 \dots \text{etc.})$$

Thus, Planck for the first time has given a relationship between the frequency (or wavelength) of the radiations and the energy associated with them. Cosmic rays, gamma rays and X-rays are high energy radiations since they have very high frequency. At the same time, microwaves and radiowaves with small frequency are regarded as low energy radiations.

❖ **Important features of Photon :**

- Source of energy (light) emits radiation in the form photons, which travel with speed of light.
- Energy of single photon, $E = h\nu = \frac{hc}{\lambda}$.
- All the photons in vacuum travel with speed 3×10^8 m/s but their speed is changed in different medium, however frequency remains same. Speed does not depend on energy.
- Photons travel as waves but are absorbed or emitted as particles.

Ex.10. Calculate the energy per quanta of an EMR of frequency 400 MHz.

Sol : $E = 6.626 \times 10^{-34} \times 4 \times 10^6 \text{ J/quanta}$

Ex.11. Calculate the energy per quanta of an EMR of wavelength 662.6 nm.

Sol : $E = n \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{662.6 \times 10^{-9} \text{ m}} = 3 \times 10^{-19} \text{ J/quanta}$

Ex.12. Calculate the wavelength (in Å) of an EMR of energy 3.1 eV/quanta.

Sol: $E = h\nu = h \frac{c}{\lambda} \Rightarrow E(\text{eV}) \times 1.602 \times 10^{-19} = \frac{1 \times 6.62 \times 10^{-34} \times 3 \times 10^8}{\lambda(\text{Å})}$

$$E(\text{eV}) \approx \frac{12400}{\lambda(\text{Å})} = \frac{1240}{\lambda(\text{nm})}$$

Ex.13. In order to see an object, 10^{-19} J must be received by our eyes. How many photons of green light must be received by our eyes for its visibility. ($\lambda = 550 \text{ nm}$, $h = 6.6 \times 10^{-34} \text{ J}$)

Sol : $E = n \frac{hc}{\lambda}$

$$10^{-19} = n \times \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{5500 \times 10^{-9}}$$

$$n = \frac{5}{18} \approx 1$$

Ex.14. A bulb is rated as 110 watt. If it emits 25% of absorbed energy as red light ($\lambda = 6626\text{\AA}$), how many photons are emitted out by the bulb per second.

Sol: $110 \times \frac{25}{100} = n \times \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{6626 \times 10^{-10}}$

Ex.15. The wavelength of microwave radiation is 0.08 m. How many moles of photons is needed to increase the temperature of 400 gm water from 25° to 45°C, assuming 25% efficiency.

Specific heat capacity = 4.2 J/K-gm

Sol: $\left(\frac{nhc}{\lambda} \right) \times \frac{25}{100} = ms\Delta t$

$$n \times \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{0.08} \times \frac{25}{100} = 400 \times 4.2 \times 20$$

$$\therefore \text{Number of moles of photon} = \frac{n}{N_A}$$

Ex.16. A dye absorbs the radiation of 4000 Å and fluoresces the radiation of 5000 Å. If only 40% of the absorbed energy is emitted out, calculate the ratio of number of quanta emitted out and the number of quanta absorbed.

Sol: $E_a \times \frac{40}{100} = E_e$

$$n_a \times \frac{hc}{4000\text{\AA}} \times \frac{40}{100} = n_e \times \frac{hc}{5000\text{\AA}}$$

$$\frac{n_e}{n_a} = \frac{40}{100} \times \frac{5000}{4000} = \frac{1}{2}$$

Ex.17. The bond dissociation energy of Cl–Cl bond in chlorine gas is 240 kJ/mol. Calculate the longest wavelength of EMR needed to dissociate bond. Assume one photon may dissociate only one bond.

Sol: $\frac{240 \times 10^3}{6 \times 10^{23}} = \frac{1 \times 6.626 \times 10^{-34} \times 3 \times 10^8}{\lambda}$

Ex.18 A near ultra violet photon of wavelength 300 nm is absorbed by a gas and then emitted as two photons. One photon is of red light with wavelength 760 nm. What would be the wave length of the second photon ?

Sol. It may noted that energy of photon which adsorbed is emitted as sum of the energy of two photons.

$$\text{Energy absorbed } h\nu = \frac{hc}{\lambda}$$

According to available information,

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}; \frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}; \frac{1}{\lambda_2} = \left[\frac{1}{\lambda} - \frac{1}{\lambda_1} \right]$$

Now, $\lambda = 300 \text{ nm}$; $\lambda_1 = 760 \text{ nm}$; λ_2 can be calculated as :

$$\frac{1}{\lambda_2} = \left[\frac{1}{300} - \frac{1}{760} \right] = \frac{760 - 300}{300 \times 760} (\text{nm}^{-1})$$

$$\frac{1}{\lambda_2} = \frac{460}{760 \times 300} (\text{nm}^{-1})$$

or $\lambda_2 = 496 \text{ nm}$

7. PHOTOELECTRIC EFFECT :

Hertz in **1887** observed that when a light of certain frequency strikes the surface of a metal, electrons are ejected from the metal. This phenomenon is known as **photoelectric effect** and the ejected **electrons** are called **photoelectrons**.

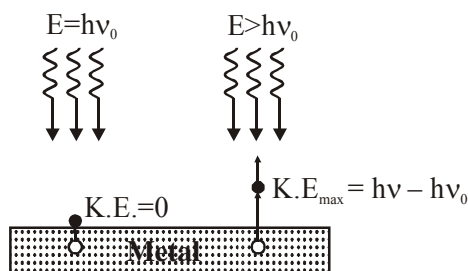
A few metals, which are having low ionisation energy like Cesium, show this effect under the action of visible light but many more show it under the action of more energetic ultraviolet light.

7.1 The experimental findings are summarized as below :

- Electrons come out as soon as the light (of sufficient energy) strikes the metal surface. There is no time lag between the two events.
- The light of any frequency will not be able to cause ejection of electrons from a metal surface. There is a minimum frequency, called the **threshold (or critical) frequency**, which can just cause the ejection. This frequency varies with the nature of the metal. The higher the frequency of the light, the more energy the photoelectrons have. Blue light results in faster electrons than red light.
- Photoelectric current is increased with increase in intensity of light of same frequency, if emission is permitted, i.e. a bright light yields more photoelectrons than a dim one of the same frequency, but the electron energies remain the same.

7.2 Einstein's explanation :

Light must have stream of energy particles or quanta of energy ($h\nu$). Suppose, the threshold frequency of light required for ejecting electrons from a metal is ν_0 , when a photon of light of this frequency strikes a metal it imparts its entire energy ($h\nu_0$) to the electron.



This energy enables the electron to break away from the surface by overcoming the attractive influence of the nucleus. Thus each photon can eject one electron. If the frequency of light is less than ν_0 , there is no ejection of electron. If the frequency of light is higher than ν_0 (let it be ν), the photon of this light having higher energy ($h\nu$), will impart some energy to the electron that

is needed to remove it away from the atom. Einstein proposed that light consisted of quanta, which we call photons with a frequency over a certain threshold would have sufficient energy to eject a single electron, producing the photoelectric effect.

7.3 Einstein's Equation for the Photoelectric Effect :

Einstein's interpretation of the photoelectric effect results in equation :

Energy of photon = Energy needed to remove an electron + Max. Kinetic energy of the emitted electron

The excess energy would give a certain velocity (i.e. kinetic energy) to the electron.

$$h\nu = h\nu_0 + K.E._{max.}$$

$$h\nu = h\nu_0 + \frac{1}{2}mv^2$$

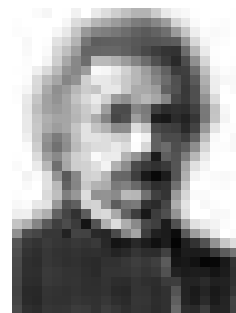
$$\frac{1}{2}mv^2 = h\nu - h\nu_0$$

where ν = frequency of the incident light, ν_0 = threshold frequency

v = max. speed of photoelectron.

$h\nu_0$ is the **threshold energy** (or) the **work function** denoted by $\phi = h\nu_0$ (minimum energy of the photon to liberate electron). It is constant for particular metal.

The maximum kinetic energy of the photoelectrons increases linearly with the frequency of incident light. This, if the energy of the ejected electrons is plotted as a function of frequency, it results in a straight line whose slope is equal to Planck's constant 'h' and whose intercept is $h\nu_0$.



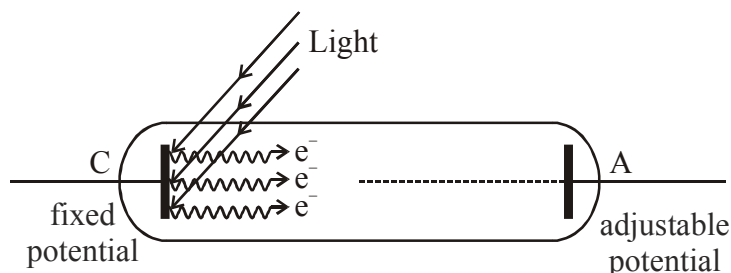
Albert Einstein (1879 - 1955)

Albert Einstein, a German born American physicist, is regarded by many as one of the two great physicists the world has known (the other is Isaac Newton). His three research papers (on special relativity, Brownian motion and the photoelectric effect) which he published in 1905, **Albert Einstein (1879 - 1955)** while he was employed as a technical assistant in a Swiss patent office in Berne have profoundly influenced the development of physics. He received the Nobel Prize in Physics in 1921 for his explanation of the photoelectric effect.

7.4 Important conclusions from photoelectric effect :

- Photoelectric effect demonstrates particle nature of radiation.
- A photon is quanta of energy. Its rest mass is zero. This is why photon can give up its all energy to the particle it strikes.
- There is no effect of frequency of incident light on the number of the emitted photoelectrons.
- There is no effect of intensity of incident light on the K.E. of the emitted photoelectrons.

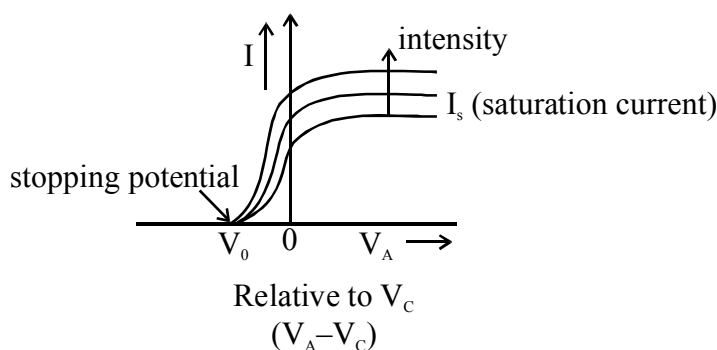
7.5 SATURATION CURRENT & STOPPING POTENTIAL



Case-I : $V_C = V_A$
Some of the ejected electrons reach at electrode A resulting photocurrent.

Case-II : $V_C < V_A$
As electrode (A) is at high potential it attracts the electron & even a slower electron will reach at electrode A. It will result in increase in photocurrent. Further increase in the potential difference, a situation may result when the slowest photocurrent electron reach at electrode. It results maximum called **saturation current**. Further increase in potential will not increase photocurrent.

Case-III : $V_C > V_A$
As electrode (A) is at low potential. It will repel electron resulting decrease in photocurrent. Further decrease in potential at electrode (A) may result a situation when the fastest electron just fails to reach at (A) and the photocurrent drops to zero. The potential of (A) relative to C to just stop photocurrent is called **stopping potential**.



On increasing the intensity of light the stopping potential does not change because the maximum K.E. of photoelectron does not change. But the photocurrent increases, because the number of photon falling on the surface increase.

If the frequency of light is changed, the stopping potential will change.

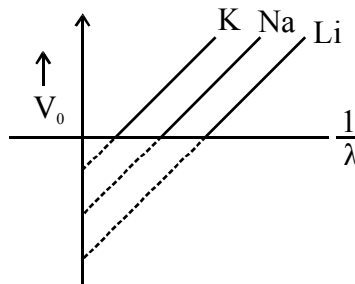
$$(KE)_{\max} = eV_0$$

$$(KE)_{\max} = h\nu - h\nu_0$$

$$eV_0 = \frac{hc}{\lambda} - \phi$$

$$V_0 = \frac{hc}{e} \frac{1}{\lambda} - \frac{\phi}{e}$$

y x



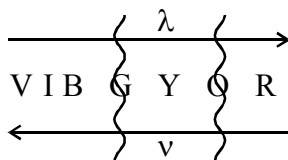
Ex.19. From a metal surface, photoelectron never comes out by orange light but comes from green light. Predict about the injection of photoelectron from the same metal by

(i) Red (ii) Blue (iii) Yellow light

Sol: (i) Red light = No

(ii) Blue light = Yes

(iii) Yellow light = Can't say



Ex.20. The work function of a metal is 3 eV. If EMR of 200 nm fall on the metal surface, calculate the maximum speed of photoelectron ejected.

Sol: $E = \frac{1240}{200} = 6.2 \text{ eV}$

$$(KE)_{\max} = h\nu - \phi$$

$$\frac{1}{2}mv^2 = (6.2 - 3) \text{ eV}$$

$$\frac{1}{2} \times 9.1 \times 10^{-31} v_{\max}^2 = 3.2 \times 1.6 \times 10^{-19}$$

Ex.21. When EMR of frequency $5 \times 10^{15} \text{ Hz}$ fall on a metal surface, the maximum kinetic energy of photoelectron is double than the photoelectron which emits when EMR of frequency $3 \times 10^{15} \text{ Hz}$ fall on the same metal. The threshold frequency for the metal is

Sol: $(K.E.)_1 = h \times 5 \times 10^{15} - h\nu_0$

$$(K.E.)_2 = h \times 3 \times 10^{15} - h\nu_0$$

$$2E_1 = E_2$$

$$\nu_0 = 1 \times 10^{15} \text{ Hz.}$$

Ex.22 A photon of wavelength $3000 \text{ \AA}$ strikes a metal surface, the work function of the metal being 2.20 eV . Calculate (i) the energy of the photon in eV (ii) the kinetic energy of the emitted photo electron and (iii) the velocity of the photo electron.

Sol. (i) Energy of the photon

$$E = h\nu = \frac{hc}{\lambda} = \frac{(6.6 \times 10^{-34} \text{ Js})(3 \times 10^8 \text{ ms}^{-1})}{3 \times 10^{-7} \text{ m}} = 6.6 \times 10^{-19} \text{ J}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$\text{Therefore } E = \frac{6.6 \times 10^{-19} \text{ J}}{1.6 \times 10^{-19} \text{ J/eV}} = 4.125 \text{ eV}$$

(ii) Kinetic energy of the emitted photo electron

$$\text{Work function} = 2.20 \text{ eV}$$

$$\text{Therefore, KE} = 4.125 - 2.20$$

$$= 1.925 \text{ eV} = 3.08 \times 10^{-19} \text{ J}$$

(iii) Velocity of the photo electron

$$\text{KE} = \frac{1}{2}mv^2 = 3.08 \times 10^{-19} \text{ J}$$

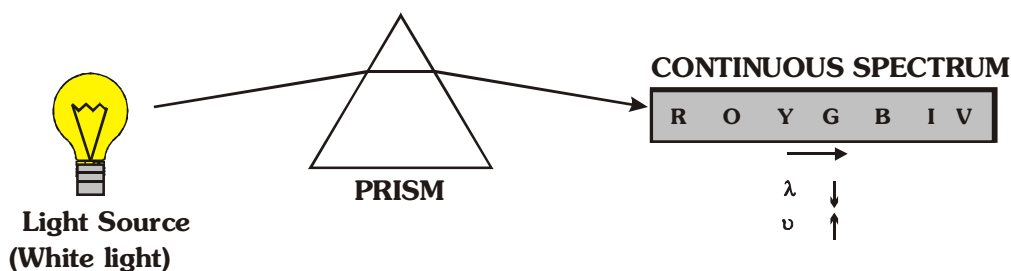
$$\text{Therefore, velocity (v)} = \sqrt{\frac{2 \times 3.08 \times 10^{-19}}{9.1 \times 10^{-31}}} = 8.22 \times 10^5 \text{ ms}^{-1}$$

8. SPECTRUM

It is the impressions produced on any screen when a light falls on it after passing through prism or prism like material.

Classification of spectrum :

- (i) **Continuous and discontinuous spectrum :** In continuous spectrum, the impression produced on screen overlap each other, but in discontinuous spectrum, same gap exist between the impression. The spectrum of white light that we can see ranges from violet at $7.50 \times 10^{14} \text{ Hz}$ to red at $4 \times 10^{14} \text{ Hz}$. This spectrum is called *continuous spectrum* because violet merges into blue, blue into green, and so on. A similar spectrum is produced when a rainbow forms in the sky. In a continuous spectrum, radiations corresponding to all the wavelengths are present.



- (ii) **Emission and Absorption Spectra :** The spectrum of radiation emitted by a substance that has absorbed energy is called an **emission spectrum**. Atoms, molecules or ions that have absorbed radiation are said to be '**excited**'. To produce an emission spectrum, energy is supplied to a sample by heating it or irradiating it and the wavelength (or frequency) of the radiation emitted, as the sample gives up the absorbed energy, is recorded.

An absorption spectrum is like the photographic negative of an emission spectrum. A continuum of radiation is passed through a sample which absorbs radiation of certain wavelengths. The missing wavelength which corresponds to the radiation absorbed by the matter, leave dark spaces in the bright continuous spectrum.

The study of emission or absorption spectra is referred to as **spectroscopy**. The spectrum of the visible light, as discussed above, was continuous as all wavelengths (red to violet) of the visible light are represented in the spectra. The emission spectra of atoms in the gas phase, on the other hand, do not show a continuous spread of wavelength from red to violet, rather they emit light only at specific wavelengths with dark spaces between them. Such spectra are called **line spectra** or **atomic spectra** because the emitted radiation is identified by the appearance of bright lines in the spectra (Fig)

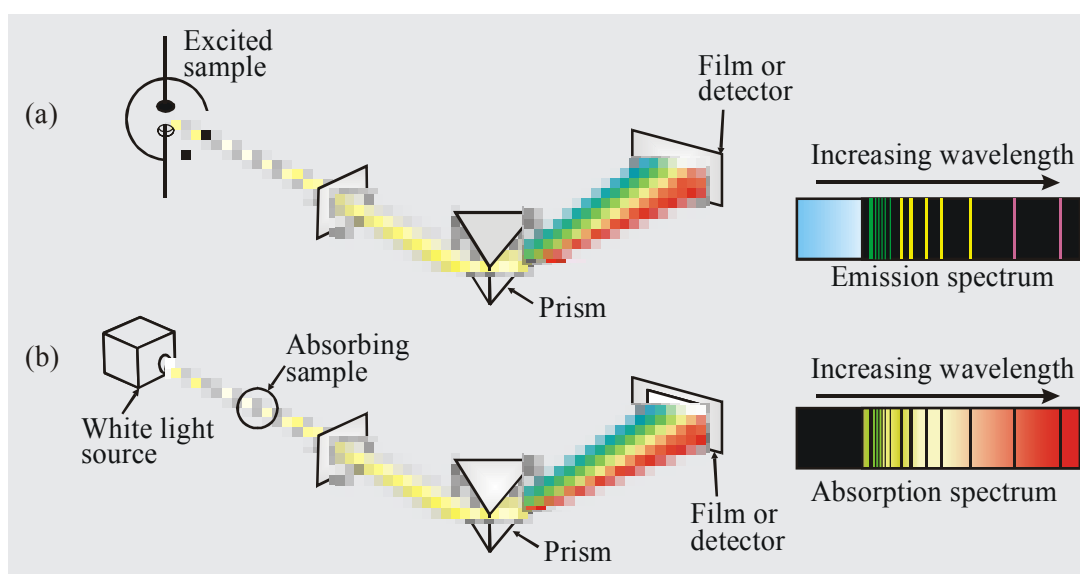


Fig. (a) Atomic emission. The light emitted by a sample of excited hydrogen atoms (or any other element) can be passed through a prism and separated into certain discrete wavelengths. Thus an emission spectrum, which is a photographic recording of the separated wavelengths is called as line spectrum. Any sample of reasonable size contains an enormous number of atoms. Although a single atom can be in only one excited state at a time, the collection of atoms contains all possible excited states. The light emitted as these atoms fall to lower energy states is responsible for the spectrum. **(b) Atomic absorption.** When white light is passed through unexcited atomic hydrogen and then through a slit and prism, the transmitted light is lacking in intensity at the same wavelengths as are emitted in (a) The recorded absorption spectrum is also a line spectrum and the photographic negative of the emission spectrum.

Line emission spectra are of great interest in the study of electronic structure. Each element has a unique line emission spectrum. The characteristic lines in atomic spectra can be used in chemical analysis to identify unknown atoms in the same way as finger prints are used to identify people. The exact matching of lines of the emission spectrum of the atoms of a known element with the lines from an unknown sample quickly establishes the identity of the latter, German chemist, Robert Bunsen (1811-1899) was one of the first investigators to use line spectra to identify elements. Elements like rubidium (Rb), caesium (Cs), thallium (Tl), indium (In), gallium (Ga) and scandium (Sc) were discovered when their minerals were analysed by spectroscopic methods. The element helium (He) was discovered in the sun by spectroscopic method.

9. BOHR'S ATOMIC MODEL

Bohr's model for hydrogen atom is based on the following postulates:

- (i) The electron in the hydrogen atom can move around the nucleus in a circular path of fixed radius and energy. These paths are called **orbits**, stationary states or allowed energy states. These orbits are arranged concentrically around the nucleus.
- (ii) The energy of an electron in the orbit does not change with time. However, the electron will move from a lower stationary state to a higher stationary state when required amount of energy is absorbed by the electron or energy is emitted when electron moves from higher stationary state to lower stationary state. The energy change does not take place in a continuous manner.
- (iii) The frequency of radiation absorbed or emitted when transition occurs between two stationary states that differ in energy by ΔE , is given by :

$$\nu = \frac{\Delta E}{h} = \frac{E_2 - E_1}{h}$$

Where E_1 and E_2 are the energies of the lower and higher allowed energy states respectively. This expression is commonly known as Bohr's frequency rule.

- (iv) The angular momentum of an electron in a given stationary state can be expressed as in equation

$$m_e v r = n \cdot \frac{h}{2\pi} \quad n = 1, 2, 3$$

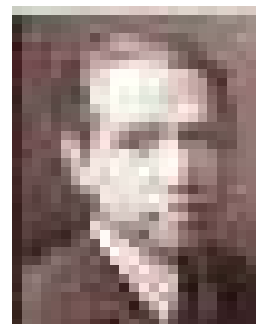
Thus an electron can move only in those orbits for which its angular momentum is integral multiple of $h/2\pi$ that is why only certain fixed orbits are allowed.

9.1 APPLICATION OF BOHR'S MODEL

When electron revolves in fixed circular orbit then electrostatic force of attraction and centrifugal force are equal.

$$\text{Electrostatic force} = \frac{K q_1 q_2}{r^2} = \frac{K \cdot Ze \cdot e}{r^2} = \frac{KZe^2}{r^2}$$

Where, constant $K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$ (MKS) = 1 (CGS)



Niels Bohr
(1885-1962)

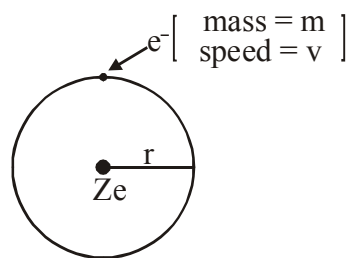
Niels Bohr, was a Danish physicist. After first world war, Bohr worked for peaceful uses of atomic energy. He was awarded the Nobel Prize in physics in 1922.

$$\text{Centrifugal force} = \frac{mv^2}{r}$$

In balanced condition

Electrostatic force = Centrifugal force

$$\frac{KZe^2}{r^2} = \frac{mv^2}{r} \text{ or } \frac{KZe^2}{r} = mv^2 \text{ or } \frac{Ze^2}{r} = mv^2 \text{ (CGS)(i)}$$



9.1.1 Radius of various orbits (shell) :

According to Bohr model, $mvr = \frac{nh}{2\pi}$

$$v = \frac{nh}{2\pi mr} \text{(ii)}$$

Now putting the value of v from eq.(ii) into eq.(i)

$$\frac{KZe^2}{r} = m \left(\frac{nh}{2\pi mr} \right)^2$$

$$\frac{KZe^2}{r} = \frac{mn^2h^2}{4\pi^2m^2r^2}$$

$$r = \frac{n^2h^2}{4\pi^2mKZe^2} \text{ or } r = \frac{n^2h^2}{4\pi^2mZe^2} \text{ (CGS } \because K=1) \text{(iii)}$$

Putting the value of π , h , m , K , & e (Constants) in the above eq. (iii)

$$r = 0.529 \times 10^{-10} \times \frac{n^2}{Z} \text{ m} \quad \{ \text{\AA} = 10^{-10} \text{ m} = 10^{-8} \text{ cm} \}$$

$$\text{or } \boxed{r_n = 0.529 \times \frac{n^2}{Z} \text{\AA}}$$

This formula is only applicable for hydrogen and hydrogen like species i.e. species containing single electron.

9.1.2. Velocity of electron in Bohr orbit :

According to Bohr postulate

$$mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi mr} = \frac{nh}{\frac{2\pi m \times n^2 h^2}{4\pi^2 m K Z e^2}}$$

$$v = \frac{2\pi K Z e^2}{nh} \text{ (MKS)(iv)}$$

$$v = \frac{2\pi Z e^2}{nh} \text{ (CGS)}$$

Putting the value of π , h , K , & e (Constants) in the above eq (iv)

$$v = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}$$

9.1.3. Total energy of electron in Bohr orbit :

Total energy of an electron is the sum of kinetic and potential energy.

i.e. $T.E. = K.E. + P.E.$

(i) Potential energy : $P.E. = -\frac{Kq_1q_2}{r} = -\frac{KZe^2}{r} = -\frac{KZe^2}{r}$

(ii) Kinetic energy : $K.E. = \frac{1}{2}mv^2$

But $\frac{KZe^2}{r} = mv^2$ (By eq. i)

$K.E. = \frac{KZe^2}{2r}$

(iii) Total energy : $T.E. = K.E. + P.E.$

$T.E. = \frac{KZe^2}{2r} - \frac{KZe^2}{r} = -\frac{KZe^2}{2r}$

Now putting the value of r from eq. (iii)

$T.E. = -\frac{KZe^2 \times 4\pi^2 m KZe^2}{2n^2 h^2} \Rightarrow -\frac{2\pi^2 m \times K^2 Z^2 e^4}{n^2 h^2}$

Now putting the value of π , K , e , m , h , we get :

$T.E. = -2.18 \times 10^{-18} \times \frac{Z^2}{n^2} \text{ J / atom} = -1312 \times \frac{Z^2}{n^2} \text{ kJ/mol}$

$= -2.18 \times 10^{-11} \times \frac{Z^2}{n^2} \text{ erg/atom} = -313.6 \times \frac{Z^2}{n^2} \text{ Kcal/mol}$

$= -13.6 \times \frac{Z^2}{n^2} \text{ eV/atom} \Rightarrow E_n = -\frac{13.6Z^2}{n^2} \text{ eV/atom}$

9.1.4 Some extra points :

(i) $K.E. = \frac{KZe^2}{2r}$ i.e. $K.E. \propto \frac{1}{r}$ On increasing radius, K.E. decreases.

(ii) $P.E. = -\frac{KZe^2}{r}$ i.e. $P.E. \propto -\frac{1}{r}$ On increasing radius, P.E. increases.

(iii) $T.E. = -\frac{KZe^2}{2r}$ i.e. $E \propto -\frac{1}{r}$ On increasing radius, total energy increases.

(iv) Relation between T.E., P.E. and K.E.

$P.E. = -2 KE$

$KE = -T.E.$

$P.E. = 2 T.E.$

Ex.26 Calculate the energy of Li^{+2} atom for 2nd excited state.

Sol. $E = -13.6 \times \frac{Z^2}{n^2}$

$\therefore Z = 3$ and e^- exist in 2nd excited state, means e^- present in 3rd shell i.e. $n = 3$

$$\therefore E = -13.6 \times \frac{(3)^2}{(3)^2} = -13.6 \text{ eV/atom}$$

Ex.27 If the P.E. of an electron is -6.8 eV in hydrogen atom then find out K.E., E of orbit where electron exist & radius of orbit.

Sol. (i). P.E. = -2K.E.

$$-6.8 = -2\text{K.E.}$$

$$\frac{6.8}{2} = \text{K.E.} \quad \text{K.E.} = 3.4 \text{ eV}$$

(ii). $E = -\text{K.E.}$
 $= -3.4 \text{ eV}$

(iii). Orbit = 2nd

$$\therefore E = -13.6 \times \frac{Z^2}{n^2}$$

$$\therefore 3.4 = -13.6 \times \frac{1^2}{n^2}$$

$$\Rightarrow n^2 = \frac{-13.6}{-3.4} = 4$$

$$\text{i.e. } n = 2$$

(iv). $r = 0.529 \times \frac{n^2}{Z} \text{ \AA}$

$$r = 0.529 \times \frac{(2)^2}{1} \text{ \AA}$$

$$= 0.529 \times 4 \text{ \AA} = 2.16 \text{ \AA}$$

Ex.28 The ionization energy for the hydrogen atom is 13.6 eV then calculate the required energy in eV to excite it from the ground state to 1st excited state.

Sol. Ionization energy = 13.6 eV

i.e. 1st energy state = -13.6 eV

Energy of 1st excited state

i.e. 2nd orbit = -3.4 eV

so, $E_2 - E_1 = -3.4 + 13.6 = 10.2 \text{ eV}$

Sol. $n_1 \text{ orbit} \rightleftharpoons n_2 \text{ orbit}$

$$\Delta E = E_{n_2} - E_{n_1} = \left(-13.6 \frac{Z_1^2}{n_2^2} \right) - \left(-13.6 \frac{Z_2^2}{n_1^2} \right)$$

$$\Delta E = 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{eV}$$

$$\Delta E = 13.6 \times 3^2 \left(\frac{1}{1^2} - \frac{1}{3^2} \right) = 108.8 \text{ eV}$$

Ex.30 Calculate the excitation energy of Be^{3+} ion in ground state.

Sol. $\Delta E = 13.6 \times 16 \left(\frac{1}{1} - \frac{1}{4} \right) = 163.2 \text{ eV}$

Ex.31 The ionisation energy of He^+ ion is x kJ/mole. Calculate ionisation energy of Li^{2+} ion.

Sol. For I.E. $\Rightarrow n = 1 \longrightarrow n = \infty$

$$IE = 13.6Z^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right) = 13.6 Z^2 eV$$

$$\frac{(\text{I.E.})_{\text{Li}^{2+}}}{(\text{I.E.})_{\text{He}^+}} = \frac{3^2}{2^2} \Rightarrow (\text{I.E.})_{\text{Li}^{2+}} = \frac{9}{4} \times kJ/mol$$

Ex.32 The ionisation energy for a single electron system is 14.4 eV. Calculate the amount of energy released when electron jumps from 3rd orbits to 2nd orbit.

Sol. $\Delta E = (IE) \left(\frac{1}{n_1^2} - \frac{1}{n^2} \right) = 14.4 \times \left(\frac{1}{4} - \frac{1}{9} \right) = 2 \text{ eV}$

Ex.33 Calculate the speed of an electron in the 3rd orbit of the Li^{2+} ion. Also calculate the number of revolutions per second that it makes around the nucleus.

Sol. Radius of 2nd orbit = $r_I x \frac{(n)^2}{Z} = 0.529 \times \frac{(3)^2}{3} = 1.587 \text{\AA}$

Velocity of electron in 2nd orbit, $v = 2.18 \times 10^6 \frac{\text{Z}}{\text{n}} \text{ m/sec} = 2.18 \times 10^6 \text{ m/sec}$

$$\begin{aligned} \text{No. of revolutions/sec} &= \frac{1}{2\pi r/v} = \frac{v}{2\pi r} = \frac{2.18 \times 10^6 \text{ m/sec}}{2 \times 3.14 \times 1.587 \times 10^{-10} \text{ m}} \\ &= 2.187 \times 10^{15} \text{ rev/sec} \end{aligned}$$

9.2 RYDBERG FORMULA

It an electron shows transition from n_2 to n_1 energy level then energy change ΔE will be.

$$\Delta E = E_{n_2} - E_{n_1}$$

$$\Delta E = \frac{-2\pi^2 m K^2 Z^2 e^4}{n_1^2 h^2} - \left[\frac{-2\pi^2 m K^2 Z^2 e^4}{n_2^2 h^2} \right] = \frac{2\pi^2 m K^2 Z^2 e^4}{n_1^2 h^2} - \frac{2\pi^2 m K^2 Z^2 e^4}{n_2^2 h^2}$$

$$\text{But } \Delta E = h\nu = \frac{hc}{\lambda}$$

$$\therefore \frac{hc}{\lambda} = \frac{2\pi^2 m K^2 Z^2 e^4}{h^2} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\frac{1}{\lambda} = \frac{2\pi^2 m K^2 e^4 Z^2}{ch^3} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

where $\frac{2\pi^2 m K^2 e^4}{ch^3}$ is a constant called Rydberg constant (R) (Assume nucleus is stationary)

$$\text{So, } \bar{\nu} = \frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\begin{aligned} \text{value of } R &= 109677 \text{ cm}^{-1} = 10967700 \text{ m}^{-1} \\ &\approx 109700 \text{ cm}^{-1} \approx 10970000 \text{ m}^{-1} \end{aligned}$$

$$\frac{1}{R} = 912 \text{ \AA}$$

Ex.34 What is the wavelength of light emitted when the electron in a hydrogen atom undergoes transition from the energy level with $n = 4$ to the energy level with $n = 1$?

Sol. According to Rydberg's formula, $\bar{\nu}(\text{cm}^{-1}) = 109,677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

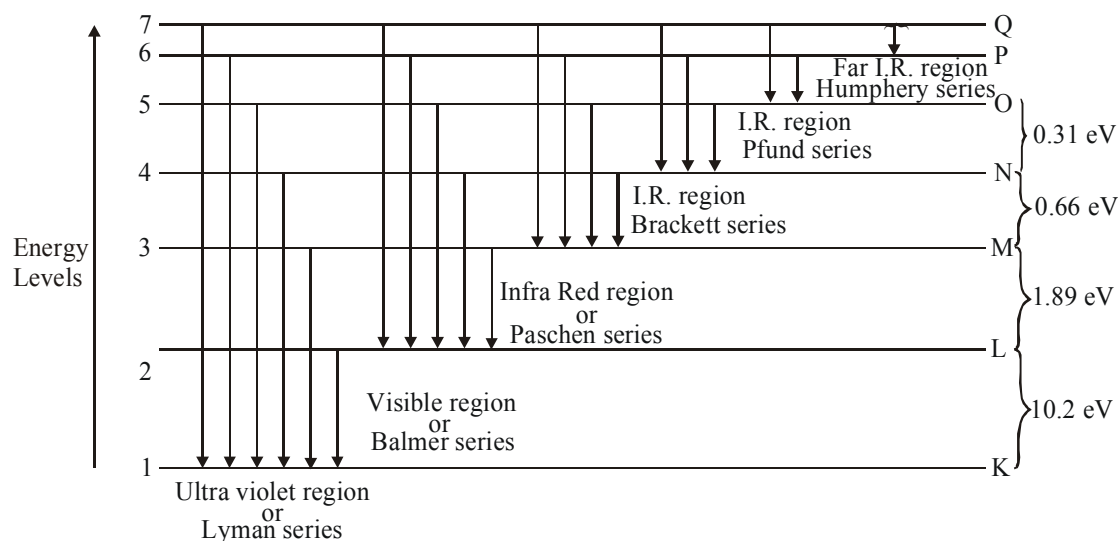
In the present case $n_2 = 4$ and $n_1 = 1$

$$\bar{\nu} = 109,677 \left(\frac{1}{(1)^2} - \frac{1}{(4)^2} \right) = 109,677 \times \frac{15}{16} = 102822 \text{ cm}^{-1}$$

$$\lambda = \frac{1}{\bar{\nu}} = \frac{1}{102822} \text{ cm} = 9.7 \times 10^{-6} \text{ cm} = 9.7 \times 10^{-6} \times 10^7 \text{ nm} = \mathbf{97 \text{ nm.}}$$

9.3 HYDROGEN LINE SPECTRUM :

When an electric excitation is applied on atomic hydrogen gas at low pressure, a bluish light is emitted. When a ray of this light is passed through a prism, a spectrum of several isolated sharp lines is obtained. The wavelength of various lines show that spectrum lines lie in visible, Ultraviolet and Infra red region. These lines are grouped into different series.



Series	Discovered by	Regions	n_2	n_1
Lyman	Lyman	U.V. region	$n_2 = 2, 3, 4 \dots$	$n_1 = 1$
Balmer	Balmer	Visible region	$n_2 = 3, 4, 5 \dots$	$n_1 = 2$
Paschen	Paschen	Infra red (I.R.)	$n_2 = 4, 5, 6 \dots$	$n_1 = 3$
Brackett	Brackett	I.R. region	$n_2 = 5, 6, 7 \dots$	$n_1 = 4$
Pfund	Pfund	I.R. region	$n_2 = 6, 7, 8 \dots$	$n_1 = 5$
Humphery	Humphery	Far I.R. region	$n_2 = 7, 8, 9 \dots$	$n_1 = 6$

□ **KEY POINTS :**

- First line / Starting line / Initial line ($\lambda_{\max.}$ and $\nu_{\min.}$)
- Last line / limiting line / Series limit ($\lambda_{\min.}$ and $\nu_{\max.}$)
- First line of any series = α line
Second line of any series = β line
Third line of any series = γ line
- Total no. of emission lines between n_2 & $n_1 = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$, ($n_2 > n_1$)
- For transition from any orbit 'n' to $n = 1$, total no. of emission lines = $\frac{n(n-1)}{2}$

Ex.35 In a hydrogen spectrum if electron moves from 6^{th} to 2^{nd} by transition in multi steps then find out the number of lines in spectrum

Sol. Total number of line = $4 + 3 + 2 + 1 + 0 = 10$

$$\text{Total number of lines} = \frac{(n_2 - n_1)[(n_2 - n_1) + 1]}{2} = \frac{(6 - 2)(4 + 1)}{2} = 10$$

Ex.36 In the spectrum of He^+ ion the wavelength of α line of Balmer series is $x \text{ \AA}$. What is the wavelength of α line of Paschen series.

Sol. $\frac{1}{\lambda_1} = RZ^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$

$$\frac{1}{\lambda_2} = RZ^2 \left(\frac{1}{3^2} - \frac{1}{5^2} \right)$$

$$\frac{\lambda_2}{\lambda_1} = \frac{\frac{1}{2^2} - \frac{1}{3^2}}{\frac{1}{3^2} - \frac{1}{5^2}} \Rightarrow \frac{\lambda_2}{x \text{ \AA}} = \frac{5}{16} \times \frac{25}{4}$$

Ex.37 A sample of He^+ ions in ground state absorbs the radiation of $x \text{ \AA}$. subsequently, the sample emit radiation of 6 different wavelength. Calculate the value of x .

Sol.
$$\frac{1}{x} = R \left(\frac{1}{1^2} - \frac{1}{4^2} \right) \times 2^2$$
$$x = \frac{16}{15} \times \frac{912}{4} \text{ \AA}$$

Ex.38 In a hydrogen spectrum if electron moves from 6^{th} to 2^{nd} by transition in multi steps then find out the number of lines in spectrum

Sol. Total number of line = $4 + 3 + 2 + 1 + 0$
= 10

$$\text{Total number of lines} = \frac{(n_2 - n_1)[(n_2 - n_1) + 1]}{2} = \frac{(6 - 2)(4 + 1)}{2} \Rightarrow \frac{4 \times 5}{2} = 10$$

9.4 Limitation of the Bohr's model

- Bohr's theory does not explain the spectrum of multi electron atom.
- Why the Angular momentum of the revolving electron is equal to $\frac{nh}{2\pi}$, has not been explained by Bohr's theory.
- Bohr inter-related quantum theory of radiation and classical law of physics without any theoretical explanation. This was the biggest drawback of this model.
- Bohr's theory does not explain the fine structure of spectral lines. Fine structure of the spectral line is obtained when spectrum is viewed by a spectroscopy of high resolving power.
- Bohr's theory does not explain the splitting of spectral lines in the presence of magnetic field (**Zeeman effect**) or electric field (**Stark effect**)

10. DUAL BEHAVIOUR OF MATTER & DE BROGLIE WAVELENGTH :

In 1923, a French physicist, **Louis de Broglie** suggested

that, like light, matter also has dual character. It exhibits wave as well as particle nature. According to de Broglie, the wavelength λ of an electron is inversely proportional to its momentum p .

$$\lambda \propto \frac{1}{p} \quad \text{or} \quad \lambda \propto \frac{1}{mv}$$

$$\lambda = \frac{h}{p} \quad \text{Here } h = \text{Planck's constant}$$

$p = \text{momentum of electron}$

$$\therefore \text{Momentum } (p) = \text{Mass } (m) \times \text{Velocity } (v)$$

The above relation can be derived for a photon as follows by using Einstein's equation, Planck's quantum theory and wave theory of light.



de Broglie'
(1892-1987)

A French physicist, studied history as an undergraduate in the early 1910s. His interest turned to science as a result of his assignment to radio communications in world war 1. He was awarded the Nobel Prize in physics in 1929.

$$E = mc^2 \text{ (Einstein's equation)} \quad \text{.....(i)}$$

Where E is energy, m is mass of a body and c is its velocity.

$$E = h\nu = h \times \frac{c}{\lambda} \text{ (Planck's equation)} \quad \left(\nu = \frac{c}{\lambda} \right) \quad \text{.....(ii)}$$

Combining (i) and (ii)

$$E = mc^2 = h \times \frac{c}{\lambda} \quad \text{or} \quad mc = \frac{h}{\lambda} \quad \text{or} \quad \lambda = \frac{h}{mc}$$

$$\boxed{\lambda = \frac{h}{mv} \quad \text{or} \quad \lambda = \frac{h}{p}}$$

It is clear from the above equation that the value of λ decreases on increasing either m or v or both. The wavelength of many fast-moving objects like an aeroplane or a cricket ball, is very low because of their high mass. Thus wave nature of macroscopic objects can be neglected but for microscopic particles like electrons, protons, atoms etc. wave nature is significant & cannot be neglected.

10.1 DERIVATION OF BOHR'S ANGULAR MOMENTUM QUANTIZATION RULE :

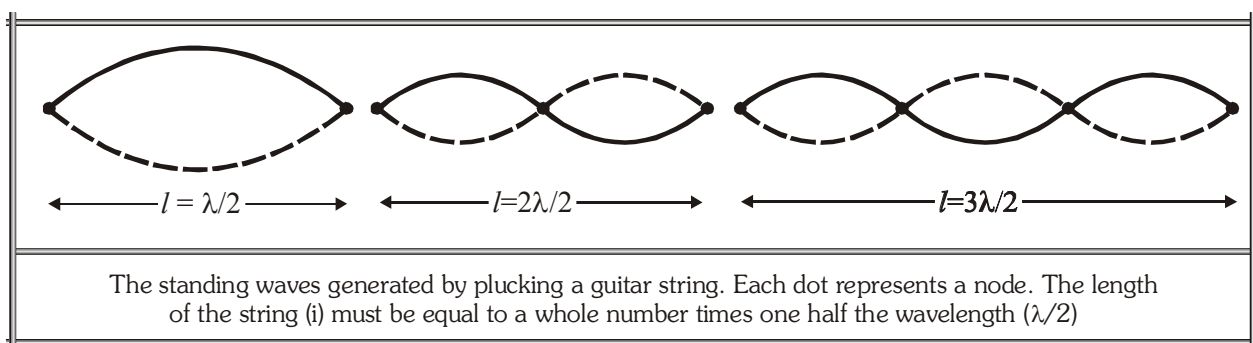
We know that according to Bohr theory, $mvr = \frac{nh}{2\pi}$

$$\text{or} \quad 2\pi r = \frac{nh}{mv} \quad (\because mv = p \text{ momentum})$$

$$\text{or} \quad 2\pi r = \frac{nh}{p} \quad \left(\because \frac{h}{p} = \lambda \text{ de-Broglie equation} \right)$$

According to de Broglie, an electron bound to the nucleus behaves like a standing wave.

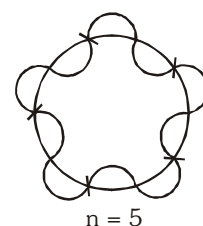
A standing wave – also known as a stationary wave – is a wave that remains in a constant position. Two opposing waves combine to form a standing wave. This phenomenon can occur because the medium is moving in the opposite direction to the wave, or it can arise in a stationary medium as a result of interference between two waves travelling in opposite directions.



For a circular standing wave to persist, a whole number of wavelength must fit into the circumference of the circle (2π).

And if n number of waves of λ wavelength are present in this circle total circumference will be $n\lambda$.

$$2\pi r = n\lambda$$



$\therefore$ Waves made = 5

n = Number of wave made by electron in one complete revolution.

According to de Broglie

$$\lambda = \frac{h}{mv}$$

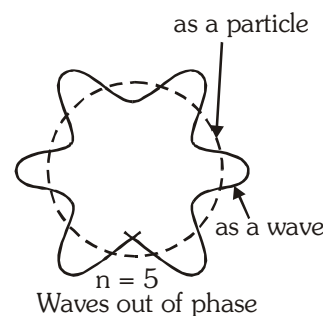
$$2\pi r = n \frac{h}{mv}$$

$$mvr = \frac{nh}{2\pi}$$

Therefore in $2\pi r = n\lambda$ n = Number of shell

i.e., 2nd shell $2\pi r = 2\lambda$

3rd shell $2\pi r = 3\lambda$



Electron in n^{th} orbit of any unielectron system can create 'n' waves in one complete revolution.

Ex.39 Calculate the de Broglie wavelength of a ball of mass 0.1 kg moving with a speed of 30 ms^{-1} .

Sol. $\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34}}{0.1 \times 30} = 2.2 \times 10^{-34} \text{ m}$

This is apparent that this wavelength is too small for ordinary observation.

Although the de Broglie equation is applicable to all material objects but it has significance only in case of microscopic particles.

Ex.40 What should be the mass of the sodium photon if its wavelength is $5894 \text{ \AA}$, the velocity of light is $3 \times 10^8 \text{ metre/second}$ and the value of h is $6.652 \times 10^{-34} \text{ kg m}^2/\text{sec}.$

(A) 3.746×10^{-26} (B) 3.746×10^{-30} (C) 3.746×10^{-34} (D) 3.746×10^{-36}

Sol. $\lambda = \frac{h}{m \times c} \Rightarrow m = \frac{h}{c\lambda}$

($\because \lambda = 5894 \text{ \AA} = 5894 \times 10^{-10} \text{ m}$)

$m = \frac{6.652 \times 10^{-34}}{3 \times 10^8 \times 5894 \times 10^{-10}} \quad \text{or} \quad \frac{6.652}{17682} \times 10^{-32}$

$= 0.0003746 \times 10^{-32} = 3.746 \times 10^{-36} \text{ kg}$

Ex.41 Calculate the de-Broglie wavelength when e^- is accelerated by the following voltage.

(i) 750 V (ii) 300 volt

Sol. (i) $\lambda = \sqrt{\frac{150}{V}} \text{ \AA} = \sqrt{\frac{150}{750}} \text{ \AA} = \frac{1}{\sqrt{5}} \text{ \AA}$ (ii) $\lambda = \sqrt{\frac{150}{300}} \text{ \AA} = \frac{1}{\sqrt{2}} \text{ \AA}$

Ex.42 Find de-Broglie wavelength of electron with $KE = 9.6 \times 10^{-19} \text{ J}$.

Sol. $KE = \frac{9.6 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV} = 6 \text{ eV}$

$\lambda = \sqrt{\frac{150}{6}} \text{ \AA} = 5 \text{ \AA}$

KE of 6 eV means e^- is accelerated by 6 volt.

Ex.43 Calculate the ratio of de-Broglie wavelength of electron and α -particle.

- (i) Moving at same speed
(ii) Moving at same momentum
(iii) Having same K.E.
(iv) Accelerated from rest through the same P.D.

Sol. (i) $\lambda = \frac{h}{mv}$

$$\lambda \propto \frac{1}{m}$$

$$\frac{\lambda_{\text{electron}}}{\lambda_{\alpha}} = \frac{m_{\alpha}}{m_e} = \frac{4 \times 1836}{1}$$

(ii) $\frac{\lambda_{\text{electron}}}{\lambda_{\alpha}} = \frac{1}{1}$

(iii) $\frac{h}{\sqrt{2mE}}$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{\lambda_e}{\lambda_{\alpha}} = \sqrt{\frac{m_{\alpha}}{m_e}} = \sqrt{\frac{1836 \times 4}{1}}$$

(iv) $\lambda = \frac{h}{\sqrt{2mqV}} = \frac{\lambda_e}{\lambda_{\alpha}} = \sqrt{\frac{(mq)_{\alpha}}{(mq)_e}} = \sqrt{\frac{4 \times 2}{1/1836 \times 1}}$

Ex.44 In Li^{2+} ion electron jumps from 2^{nd} to 1^{st} orbit. If the emitted radiation is absorbed by H atom. Calculate the de-Broglie wavelength of the ejected electron.

Sol. $\Delta E = 13.6 \times 9 \left(1 - \frac{1}{4}\right) = 91.8 \text{ eV}$

Excess energy = $91.8 - 13.6 = 78.2 \text{ eV}$

$$\lambda = \sqrt{\frac{150}{78.2}} \text{ \AA} = 1.38 \text{ \AA}$$

Ex.45 Photoelectrons are liberated by ultra violet light of wavelength $2000 \text{ \AA}$ from a metallic surface for which the photoelectric threshold is $4000 \text{ \AA}$. Calculate the de-Broglie wavelength of electrons emitted with maximum kinetic energy.

Solution : K.E. = Quantum Energy – Threshold energy

$$\begin{aligned} &= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{2000 \times 10^{-10}} - \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{4000 \times 10^{-10}} \\ &= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{10^{-10}} \left(\frac{1}{2000} - \frac{1}{4000} \right) = 4.969 \times 10^{-19} \text{ Joule.} \end{aligned}$$

$$\frac{1}{2}mv^2 = 4.969 \times 10^{-19} \Rightarrow m^2v^2 = 2 \times 4.969 \times 10^{-19} \times 9.1 \times 10^{-31}$$

$$mv = 9.51 \times 10^{-25} \Rightarrow \lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}}{9.51 \times 10^{-25}} = 0.696 \times 10^{-9} \text{ m}$$

Ex.46 Calculate the de-Broglie wavelength when proton is accelerated by the 750 V.

Sol. (i) $\lambda = \sqrt{\frac{150}{V \times 1836}} \text{ \AA} = \sqrt{\frac{150}{750 \times 1836}} \text{ \AA}$

10.2 Justification of dual nature of electrons :

I. Particle character :

- If an e^- strikes a screen coated with ZnS, it produces a spot of light called scintillation. On e^- produces only one scintillation point which means e^- are localised not spread out like wave : Photoelectric effect also proves its particle nature.
- Electron possess definite mass, momentum & KE proving their particle nature.

II. Wave character :

- It was confirmed by phenomenon of diffraction, interference, reflection.
- Davisson and Germer showed that when high speed e^- strike Ni crystal a diffraction pattern (having number of rings) is obtained like X-rays of electromagnetic spectrum.

11 HEISENBERG UNCERTAINTY PRINCIPLE :

Bohr's theory considers an electron as a material particle. Its position and momentum can be determined with accuracy. But, when an electron is considered in the form of wave as suggested by de-Broglie, it is not possible to ascertain simultaneously the exact position and velocity of the electron more precisely at a given instant since the wave is extending throughout a region of space.

In 1927, **Werner Heisenberg** presented a principle known as Heisenberg uncertainty principle which states as : **"It is impossible to measure simultaneously the exact position and exact momentum of a body as small as an electron"**.

The uncertainty of measurement of position, Δx , and the uncertainty of momentum Δp or $m\Delta v$, are related by Heisenberg's relationship as : ($p = mv$, $\Delta p = m\Delta v$)

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \quad \text{or} \quad \Delta x \cdot m\Delta v \geq \frac{h}{4\pi}$$

or $\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$

where h is Planck's constant.

$\Delta x \Delta v = \text{uncertainty product}$

For an electron of mass m (9.10×10^{-28} g), the product of uncertainty is quite large.

$$\Delta x \cdot \Delta v \geq \frac{6.624 \times 10^{-27}}{4\pi m} \geq \frac{6.624 \times 10^{-27}}{4 \times 3.14 \times 9.10 \times 10^{-28}}$$

$$= 0.57 \text{ erg sec per gram approximately}$$

When, $\Delta x = 0$, $\Delta v = \infty$ and vice-versa.

In the case of bigger particles (having considerable mass), the value of uncertainty product is negligible. If the position is known quite accurately, i.e., Δx is very small, Δv becomes large and vice-versa.

Sol. Mass of the ball = 40 g = 40×10^{-3} kg

The uncertainty in the speed,

$$\Delta v = 45 \times \frac{2}{100} = 0.9 \text{ ms}^{-1}$$

$$\Delta x = \frac{h}{4\pi m \Delta v} = \frac{6.626 \times 10^{-34} \text{ Js}}{4 \times 3.14 \times (40 \times 10^{-3} \text{ kg})(0.9 \text{ ms}^{-1})} = 1.46 \times 10^{-33} \text{ m.}$$

Ex.48 Calculate the uncertainty in the velocity of a cricket ball of mass 150 g, if the uncertainty in its position is of the order of $1 \text{ \AA}$.

$$(h = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1})$$

Sol. Mass of ball ,

$$m = 150 \text{ g} = 150 \times 10^{-3} \text{ kg} = 0.150 \text{ kg}$$

Uncertainty in position, $\Delta x = 1 \text{ \AA} = 10^{-10} \text{ m}$

$$\Delta x \times m \Delta v = \frac{h}{4\pi}$$

$$\Delta v = \frac{h}{4\pi \times \Delta x \times m}$$

$$= \frac{6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}}{4 \times 3.14 \times 10^{-10} \text{ m} \times 0.150 \text{ kg}}$$

$$= 3.52 \times 10^{-24} \text{ m s}^{-1}.$$

12 QUANTUM MECHANICAL MODEL OF ATOM

Classical mechanics, based on Newton's laws of motion, successfully describes the motion of all macroscopic objects such as a falling stone, orbiting planets etc., which have essentially a particle-like behaviour as shown in the previous section. However it fails when applied to microscopic objects like electrons, atoms, molecules etc. This is mainly because of the fact that classical mechanics ignores the concept of dual behaviour of matter especially for sub-atomic particles and the uncertainty principle. The branch of science that takes into account this dual behaviour of matter is called **quantum mechanics**.

Quantum mechanics is a theoretical science that deals with the study of the motions of the microscopic objects that have both observable wave like and particle like properties. It specifies the laws of motion that these objects obey. When quantum mechanics is applied to macroscopic objects (for which wave like properties are insignificant) the results are the same as those from the classical mechanics.

Quantum mechanics was developed independently in 1926 by Werner Heisenberg and Erwin Schrodinger. Here, however, we shall be discussing the quantum mechanics which is based on the ideas of wave motion. The fundamental equation of quantum mechanics was developed by Schrodinger and it won him the Nobel Prize in Physics in 1933. This equation which incorporates wave-particle duality of matter as proposed by de Broglie is quite complex and knowledge of higher mathematics is needed to solve it. You will learn its solutions for different systems in higher classes.

For a system (such as an atom or a molecule whose energy does not change with time) the Schrodinger equation is written as

$$\hat{H}\psi = E\psi$$

$\hat{H}$ is a mathematical operator called Hamiltonian. Schrodinger gave a recipe of constructing this operator from the expression for the total energy of the system. The total energy of the system takes into account the kinetic energies of all the sub-atomic particles (electrons, nuclei), attractive potential between the electrons and nuclei and repulsive potential among the electrons and nuclei individually. Solution of this equation gives E and ψ .

12.1 Hydrogen Atom and the Schrodinger Equation

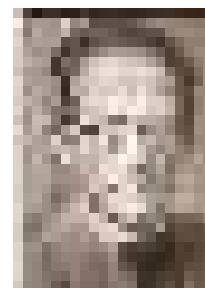
When Schrodinger equation is solved for hydrogen atom, the solution gives the possible energy levels the electron can occupy and the corresponding wave function(s) (ψ) of the electron associated with each energy level. These quantized energy states and corresponding wave functions which are characterized by a set of three quantum numbers (**principal quantum number n , azimuthal quantum number l and magnetic quantum number m_l**) arise as a natural consequence in the solution of the Schrodinger equation. When an electron is in any energy state, the wave function corresponding to that energy state contains all information about the electron.

The wave function is a mathematical function whose value depends upon the coordinates of the electron in the atom and does not carry any physical meaning. Such wave functions of hydrogen or hydrogen like species with one electron are called **atomic orbitals**.

Such wave functions pertaining to one-electron species are called one-electron systems. The probability of finding an electron at a point within an atom is proportional to the $|\psi|^2$ at that point. The quantum mechanical results of the hydrogen atom successfully predict all aspects of the hydrogen atom spectrum including some phenomena that could not be explained by the Bohr model.

Application of Schrodinger equation to multi-electron atoms presents a difficulty: the Schrodinger equation cannot be solved exactly for a multi-electron atom. This difficulty can be overcome by using approximate methods. Such calculations with the aid of modern computers show that orbitals in atoms other than hydrogen do not differ in any radical way from the hydrogen orbitals discussed above. The principal difference lies in the consequence of increased nuclear charge. Because of this all the orbitals are somewhat contracted. Further, as you shall see later, unlike orbitals of hydrogen or hydrogen like species, whose energies depend only on the quantum number n , the energies of the orbitals in multi-electron atoms depend on quantum numbers n and l .

Erwin Schrodinger, an Austrian physicist received his Ph.D. in theoretical physics from the University of Vienna in 1910. In 1927 Schrodinger succeeded Max Planck at the University of Berlin at Planck's request. In 1933, Schrodinger left Berlin because of his opposition to Hitler and Nazi policies and returned to Austria in 1936. After the invasion of Austria by Germany, Schrodinger was forcibly removed from his professorship. He then moved to Dublin, Ireland where he remained for seventeen years. Schrodinger shared the Nobel Prize for Physics with P.A.M. Dirac in 1933.



Erwin Schrodinger (1887-1961)

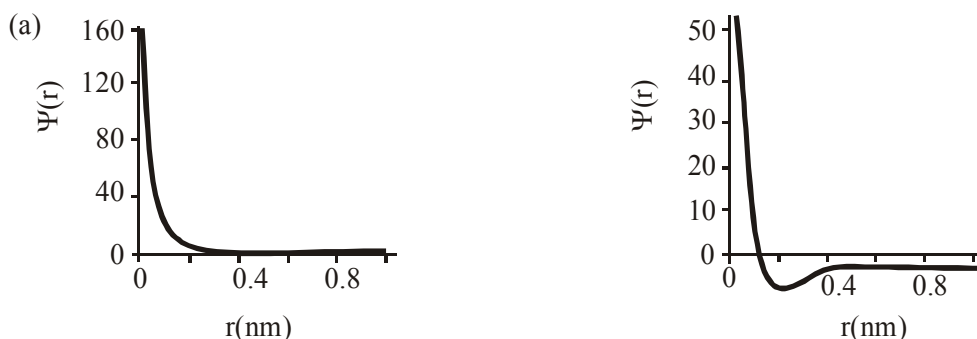
12.2 Important Features of the Quantum Mechanical Model of Atom–

Quantum mechanical model of atom is the picture of the structure of the atom, which emerges from the application of the Schrodinger equation to atoms. The following are the important features of the quantum mechanical model of atom:

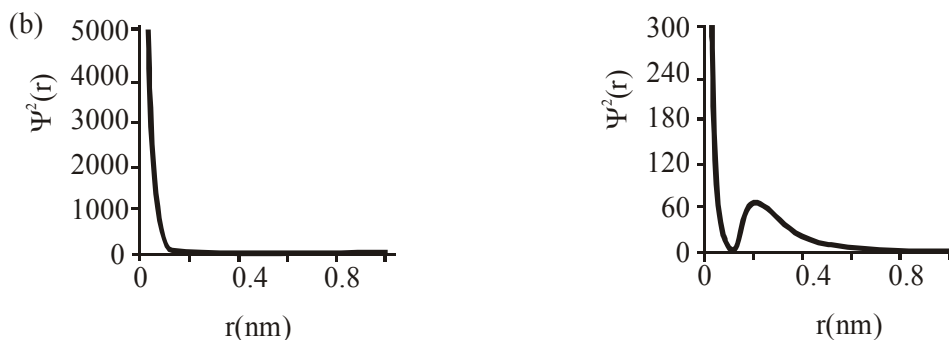
1. The energy of electrons in atoms is quantized (i.e., can only have certain specific values), for example when electrons are bound to the nucleus in atoms.
2. The existence of quantized electronic energy levels is a direct result of the wave like properties of electrons and are allowed solutions of Schrodinger wave equation.
3. Both the exact position and exact velocity of an electron in an atom cannot be determined simultaneously (Heisenberg uncertainty principle). The path of an electron in an atom therefore, can never be determined or known accurately. That is why, as you shall see later on, one talks of only probability of finding the electron at different points in an atom.
4. **An atomic orbital is the wave function ψ for an electron in an atom.** Whenever an electron is described by a wave function, we say that the electron occupies that orbital. Since many such wave functions are possible for an electron, there are many atomic orbitals in an atom. These “one electron orbital wave functions” or orbitals form the basis of the electronic structure of atoms. In each orbital, the electron has a definite energy. An orbital cannot contain more than two electrons. In a multi-electron atom, the electrons are filled in various orbitals in the order of increasing energy. For each electron of a multi-electron atom, there shall, therefore, be an orbital wave function characteristic of the orbital it occupies. All the information about the electron in an atom is stored in its orbital wave function ψ and quantum mechanics makes it possible to extract this information out of ψ .
5. The probability of finding an electron at a point within an atom is proportional to the square of the orbital wave function i.e., $|\psi|^2$ at that point. $|\psi|^2$ is known as **probability density** and is always positive. **From the value of $|\psi|^2$ at different points within an atom, it is possible to predict the region around the nucleus where electron will most probably be found.**

12.3 Shapes of Atomic Orbitals

The orbital wave function or ψ for an electron in an atom has no physical meaning. It is simply a mathematical function of the coordinates of the electron. However, for different orbitals the plots of corresponding wave functions as a function of r (the distance from the nucleus) are different. Such plots for 1s ($n = 1, l = 0$) and 2s ($n = 2, l = 0$) orbitals are

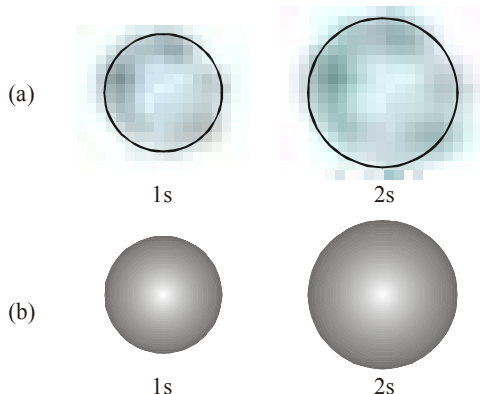


According to the German physicist, Max Born, the square of the wave function (i.e., ψ^2) at a point gives the probability density of the electron at that point. The variation of ψ^2 as a function of r for 1s and 2s orbitals is given in fig. Here again, you may note that the curves for 1s and 2s orbitals are different.



It may be noted that for 1s orbital the probability density is maximum at the nucleus and it decreases sharply as we move away from it. On the other hand, for 2s orbital the probability density first decreases sharply to zero and again starts increasing. After reaching a small maxima it decreases again and approaches zero as the value of r increases further. The region where this probability density function reduces to zero is called **nodal surfaces** or simply **nodes**. In general, it has been found that ns -orbital has $(n - 1)$ nodes, that is, number of nodes increases with increase of principal quantum number n . In other words, number of nodes for 2s orbital is one, two for 3s and so on.

These probability density variation can be visualised in terms of charge cloud diagrams. In these diagrams, the density of the dots in a region represents electron probability density in that region.

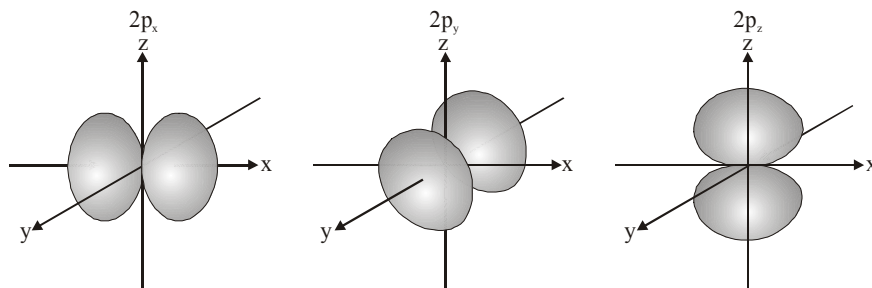


(a) Probability density plots of 1s and 2s atomic orbitals. The density of the dots represents the probability density of finding the electron in that region. (b) Boundary surface diagram for 1s and 2s orbitals.

Boundary surface diagrams of constant probability density for different orbitals give a fairly good representation of the shapes of the orbitals. In this representation, a boundary surface or contour surface is drawn in space for an orbital on which the value of probability density $|\psi|^2$ is constant. In principle many such boundary surfaces may be possible. However, for a given orbital, only that boundary surface diagram of constant probability density is taken to be good representation of the shape of the orbital which encloses a region or volume in which the probability of finding the electron is very high, say, 90%. The boundary surface diagram for 1s and 2s orbitals are given in fig. One may ask a question : Why do we not draw a boundary surface diagram, which bounds a region in which the probability of finding the electron is, 100 %? The answer to this question is that the probability density $|\psi|^2$ has always some value, howsoever small it may be, at any finite distance from the nucleus. It is therefore, not possible to draw a boundary surface diagram of a rigid size in which the probability of finding the electron is 100%. Boundary surface diagram for a s orbital is actually a sphere centred on the nucleus. In two dimensions, this sphere looks like a circle. It encloses a region in which probability of finding the electron is about 90%.

Thus we see that 1s and 2s orbitals are spherical in shape. In reality all the s-orbitals are spherically symmetric, that is, the probability of finding the electron at a given distance is equal in all the directions. It is also observed that the size of the s orbital increases with increase in n , that is, $4s > 3s > 2s > 1s$ and the electron is located further away from the nucleus as the principal quantum number increases.

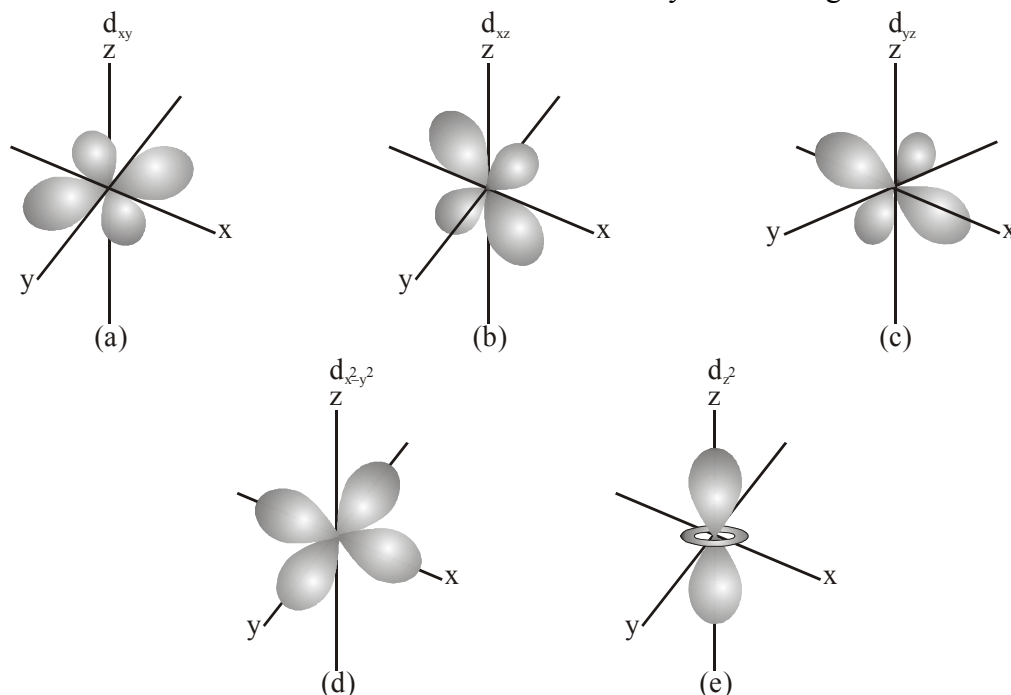
Boundary surface diagrams for three 2p orbitals ($l = 1$) are



Boundary surface diagrams of the three 2p orbitals.

In these diagrams, the nucleus is at the origin. Here, unlike s-orbitals, the boundary surface diagrams are not spherical. Instead each p orbital consists of two sections called lobes that are on either side of the plane that passes through the nucleus. The probability density function is zero on the plane where the two lobes touch each other. The size, shape and energy of the three orbitals are identical. They differ however, in the way the lobes are oriented. Since the lobes may be considered to lie along the x, y or z axis, they are given the designations $2p_x$, $2p_y$, and $2p_z$. It should be understood, however, that there is no simple relation between the values of m_l ($-1, 0$ and $+1$) and the x, y and z directions. For our purpose, it is sufficient to remember that, because there are three possible values of m_l , there are, therefore, three p orbitals whose axes are mutually perpendicular. Like s orbitals, p orbitals increase in size and energy with increase in the principal quantum number and hence the order of the energy and size of various p orbitals is $4p > 3p > 2p$. Further, like s orbitals, the probability density functions for p-orbital also pass through value zero, besides at zero and infinite distance, as the distance from the nucleus increases. The number of nodes are given by the $n - 2$, that is number of radial node is 1 for 3p orbital, two for 4p orbital and so on.

For $l = 2$, the orbital is known as d-orbital and the minimum value of principal quantum number (n) has to be 3, as the value of l cannot be greater than $n - 1$. There are five m_l values $-2, -1, 0, +1$ and $+2$ for $l = 2$ and thus there are five d orbitals. The boundary surface diagram of d orbitals are



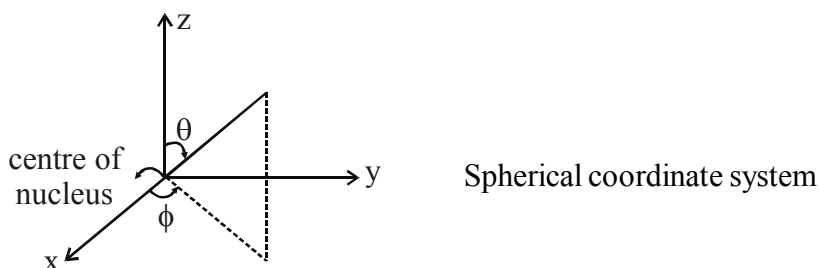
Boundary surface diagrams of the five 3d orbitals.

The five d-orbitals are designated as d_{xy} , d_{yz} , d_{xz} , $d_{x^2-y^2}$ and d_{z^2} . The shapes of the first four d-orbitals are similar to each other, whereas that of the fifth one, d_{z^2} , is different from others, but all five 3d orbitals are equivalent in energy. The d orbitals for which n is greater than 3 (4d, 5d...) also have shapes similar to 3d orbital, but differ in energy and size.

Besides the radial nodes (i.e., probability density function is zero), the probability density functions for the np and nd orbitals are zero at the plane(s), passing through the nucleus (origin). For example, in case of p_z orbital, xy-plane is a nodal plane, in case of d_{xy} orbital, there are two nodal planes passing through the origin and bisecting the xy plane containing z-axis. These are called **angular nodes** and number of angular nodes are given by ' l ', i.e., one angular node for p orbitals, two angular nodes for 'd' orbitals and so on.

The total number of nodes are given by $(n-1)$, i.e., sum of l angular nodes and $(n - l - 1)$ radial nodes.

12.4 SOLUTION OF SCHRODINGER EQUATION :



The solution in spherical coordinates may be represented as :

$$\psi = R(r) \cdot \Theta(\theta) \cdot \Phi(\phi)$$

$R(r)$: Radial function depends on n and l

$\Theta(\theta) \cdot \Phi(\phi)$: Angular function depends on l and m .

12.4.1 Radical part of solution :

$$1s \quad (n = 1, \ell = 0) : \quad R_{1s}(r) = 2 \cdot \left(\frac{Z}{a_0} \right)^{3/2} \cdot e^{-\sigma/2}$$

$$\text{where } \sigma = \frac{2Zr}{na_0} \quad a_0 = 1^{\text{st}} \text{ Bohr's radius} = 0.529 \text{ \AA}$$

$$2s \quad (n = 2, \ell = 0) : \quad R_{2s}(r) = \frac{1}{2\sqrt{2}} \cdot \left(\frac{Z}{a_0} \right)^{3/2} \cdot (2 - \sigma) e^{-\sigma/2}$$

$$2p \quad (n = 2, \ell = 1) : \quad R_{2p}(r) = \frac{1}{2\sqrt{6}} \cdot \left(\frac{Z}{a_0} \right)^{3/2} \cdot \sigma \cdot e^{-\sigma/2}$$

$$3s \quad (n=3, \ell=0): \quad R_{3s}(r) = \frac{1}{9\sqrt{3}} \cdot \left(\frac{Z}{a_0}\right)^{3/2} \cdot (6 - 6\sigma + \sigma^2) e^{-\sigma/2}$$

$$3p \quad (n=3, \ell=1): \quad R_{3p}(r) = \frac{1}{9\sqrt{6}} \cdot \left(\frac{Z}{a_0}\right)^{3/2} \cdot \sigma(4 - \sigma) e^{-\sigma/2}$$

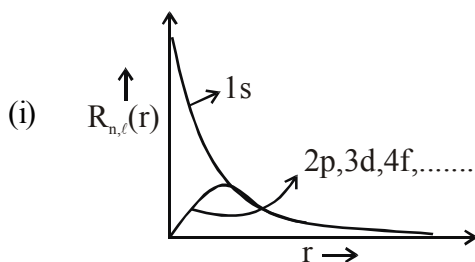
$$3d \quad (n=3, \ell=2): \quad R_{3d}(r) = \frac{1}{9\sqrt{30}} \cdot \left(\frac{Z}{a_0}\right)^{3/2} \cdot \sigma^2 \cdot e^{-\sigma/2}$$

General form :

$$R_{n\ell}(r) = K \cdot e^{-\sigma/2} \cdot \sigma^\ell \text{ (Polynomial of order } n - \ell - 1 \text{)}$$

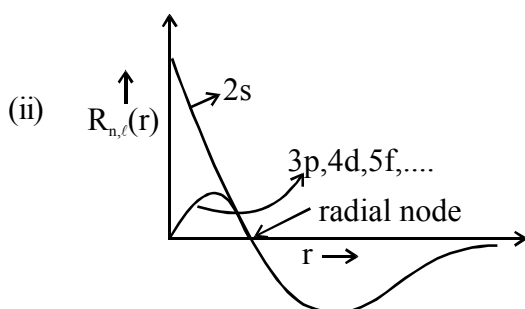
12.4.2 Graph of radial function $[R(r) \text{ or } \Psi(r)] :$

Only the graph of s-orbital does not start from origin.



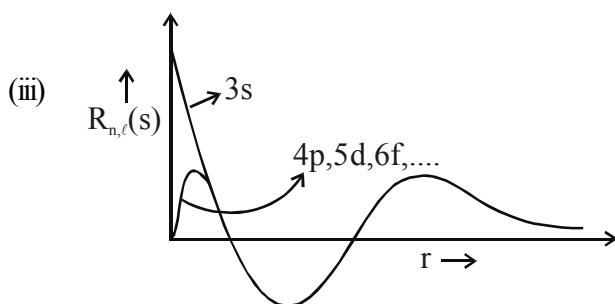
$$n - \ell - 1 = 0$$

$1s, 2p, 3d, 4f \dots$ orbitals



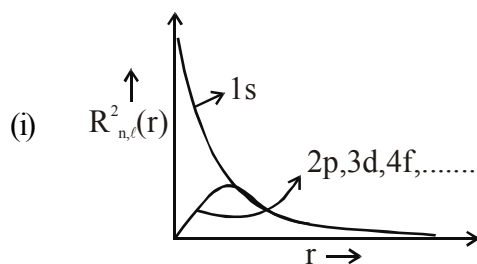
$$n - \ell - 1 = 1$$

$2s, 3p, 4d, 5f \dots$ orbitals

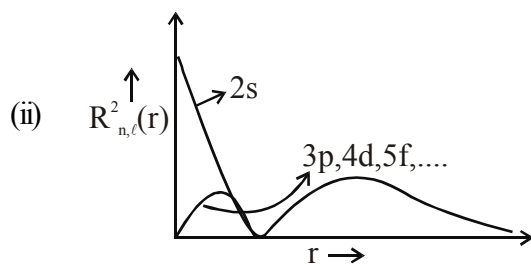


$$n - \ell - 1 = 2$$

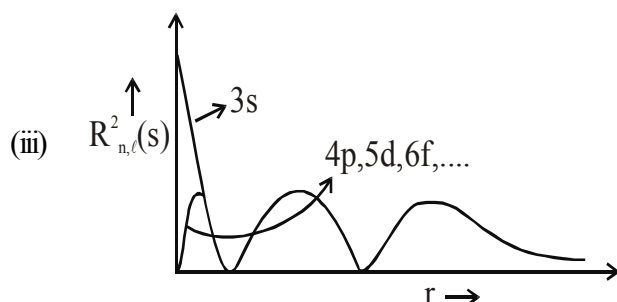
$3s, 5p, 5d, 6f \dots$ orbitals

12.4.3 GRAPH OF RADIAL PROBABILITY DENSITY FUNCTION [$R^2(r)$ or $\psi^2(r)$] :

$$n - \ell - 1 = 0, \quad (1s, 2p, 3d, 4f \dots \text{orbitals})$$



$$n - \ell - 1 = 1 \quad (2s, 3p, 4d, 5f \dots \text{orbitals})$$



$$n - \ell - 1 = 2 \quad (3s, 5p, 5d, 6f \dots \text{orbitals})$$

12.4.3 RADIAL PROBABILITY DISTRIBUTION FUNCTION (RPDF), $4\pi r^2 \psi^2(r)$

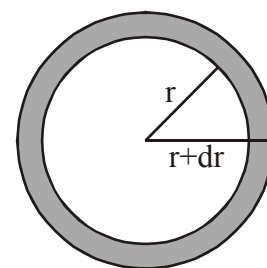
It is often useful to know the likelihood of finding the electron in an orbital at any given distance away from the nucleus. This enables us to say at what distance from the nucleus the electron is most likely to be found, and also how tightly or loosely the electron is bound in a particular atom. This is expressed by the radial probability distribution function, $4\pi r^2 \psi^2(r)$.

Radial distribution function is the measure of the probability of finding the electron in a spherical shell between thickness r and $(r + dr)$ from the nucleus, irrespective of the direction.

❖ Volume of radial shell :

$$\begin{aligned} dV &= \left[\text{Volume of sphere with radius } (r + dr) \right] - \left[\text{Volume of sphere with radius } r \right] \\ &= \frac{4}{3} \pi (r + dr)^3 - \frac{4}{3} \pi r^3 \\ &= \frac{4}{3} \pi (r^3 + 3r^2 dr + 3r dr^2 + dr^3) - \frac{4}{3} \pi r^3 = \frac{4}{3} \pi [r^3 + 3r^2 dr - r^3] \end{aligned}$$

(As dr represents an extremely small thickness, the higher powers of dr such as dr^2 and dr^3 may be neglected.)



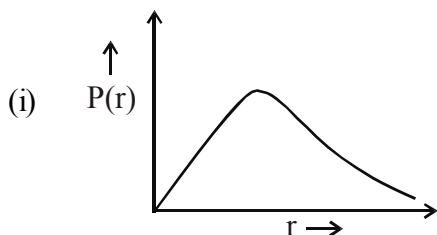
$$\therefore \text{Volume of shell, } dV = \frac{4}{3}(\pi \times 3r^2 dr) = 4\pi r^2 dr$$

$$\text{Now, radial probability density, } R^2(r) = \frac{P}{dV}$$

$\therefore$ Probability of finding electron in the volume element,

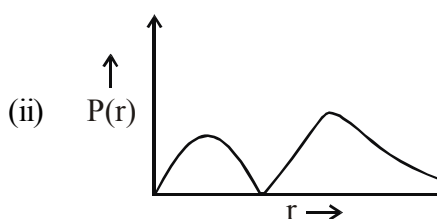
$$P = R^2(r) \cdot dV = R^2(r) \cdot 4\pi r^2 \cdot dr$$

$$\text{Now radial probability distribution function, } P(r) = \frac{P}{dr} = 4\pi r^2 \cdot R^2(r)$$



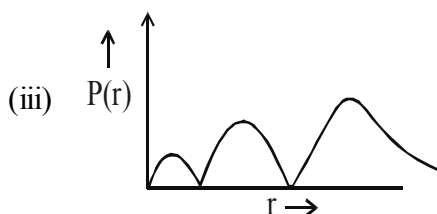
$$n - \ell - 1 = 0$$

1s, 2p, 3d, 4f orbitals



$$n - \ell - 1 = 1$$

2s, 3p, 4d, 5f orbitals



$$n - \ell - 1 = 2$$

3s, 5p, 5d, 6f orbitals

❖ Characteristics of radial distribution function :

- (i) The number of maxima in radial distribution function plot are $(n - \ell)$.
- (ii) The maximum probability of finding the electron, for the ground state hydrogen atom (1s) is found to be at a_0 (first Bohr radius).
- (iii) For 2s, 3s, 3p orbitals, the number of maxima is more than one, indicating that there is maximum probability of finding the electron at the distance corresponding to the highest value of peak.

However, there is lesser probability of finding the electron at the other peaks. It shows that in a certain state, the electron spends some portion of its time very close to the nucleus.

12.5 ANGULAR PART OF SOLUTION :

(1) s-orbital :

$$\ell = 0, m = 0 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{1}{4\pi}}$$

As the probability of finding electron is not depending on angle (direction) then it must be same. In all direction and hence the shape of s-orbital is sphere.

(2) p-orbital :

$$p_x\text{-orbital: } \ell = 1, m = +1 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{3}{4\pi}} \cdot \sin \theta \cos \phi$$

$$p_y\text{-orbital: } \ell = 1, m = -1 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{3}{4\pi}} \cdot \sin \theta \cos \phi$$

$$p_z\text{-orbital: } \ell = 1, m = 0 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{3}{4\pi}} \cdot \cos \theta$$

(3) d-orbital :

$$d_{z^2}\text{-orbital: } \ell = 2, m = 0 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$$

$$d_{x^2-y^2}\text{-orbital: } \ell = 2, m = -2 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin^2 \theta \cos 2\phi$$

$$d_{xy}\text{-orbital: } \ell = 2, m = +2 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin^2 \theta \sin 2\phi$$

$$d_{xz}\text{-orbital: } \ell = 2, m = +1 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin \theta \cos \theta \sin 2\phi$$

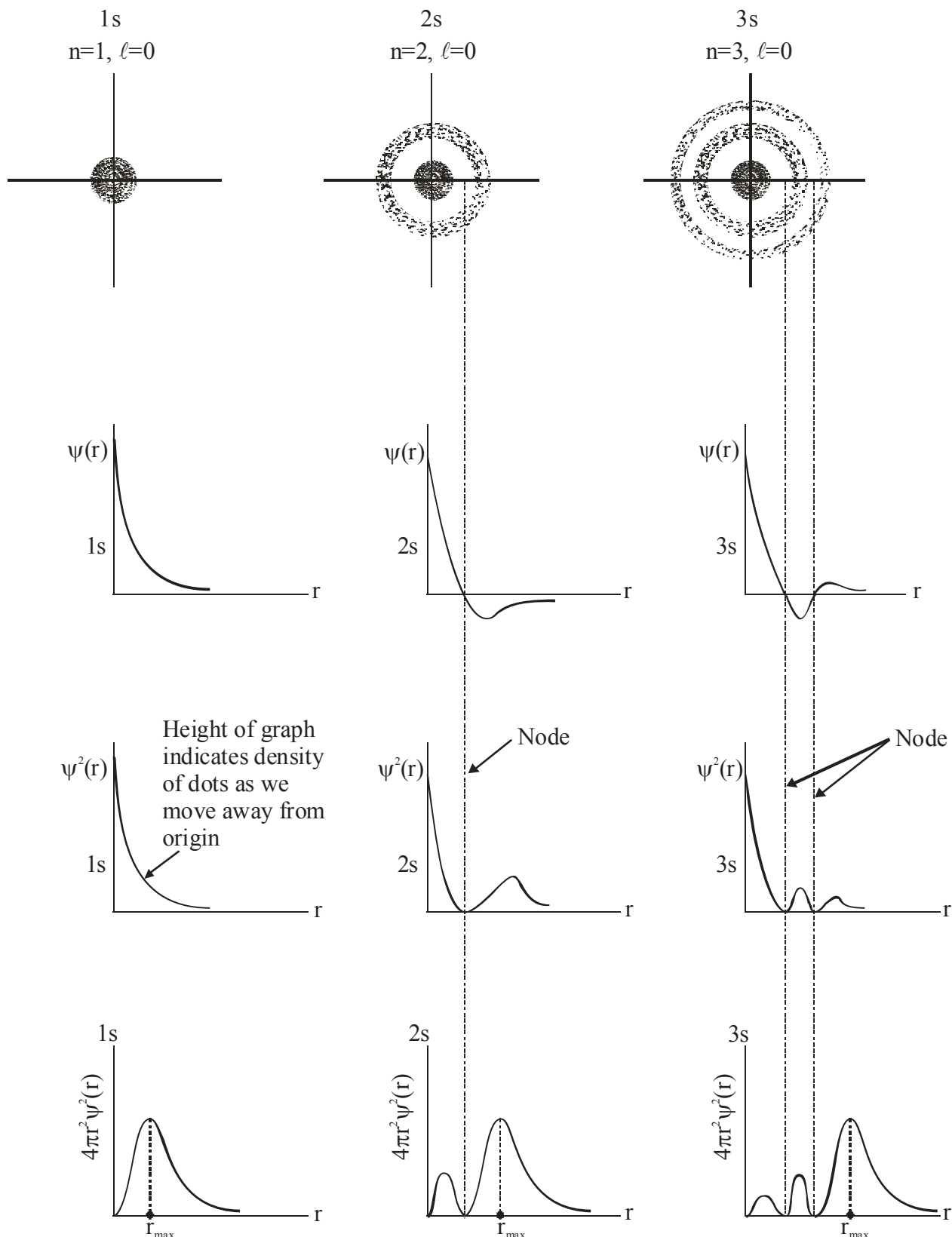
$$d_{yz}\text{-orbital: } \ell = 2, m = +1 \quad \Theta(\theta) \cdot \Phi(\phi) = \sqrt{\frac{15}{4\pi}} \cdot \sin \theta \cos \theta \cdot \sin \phi$$

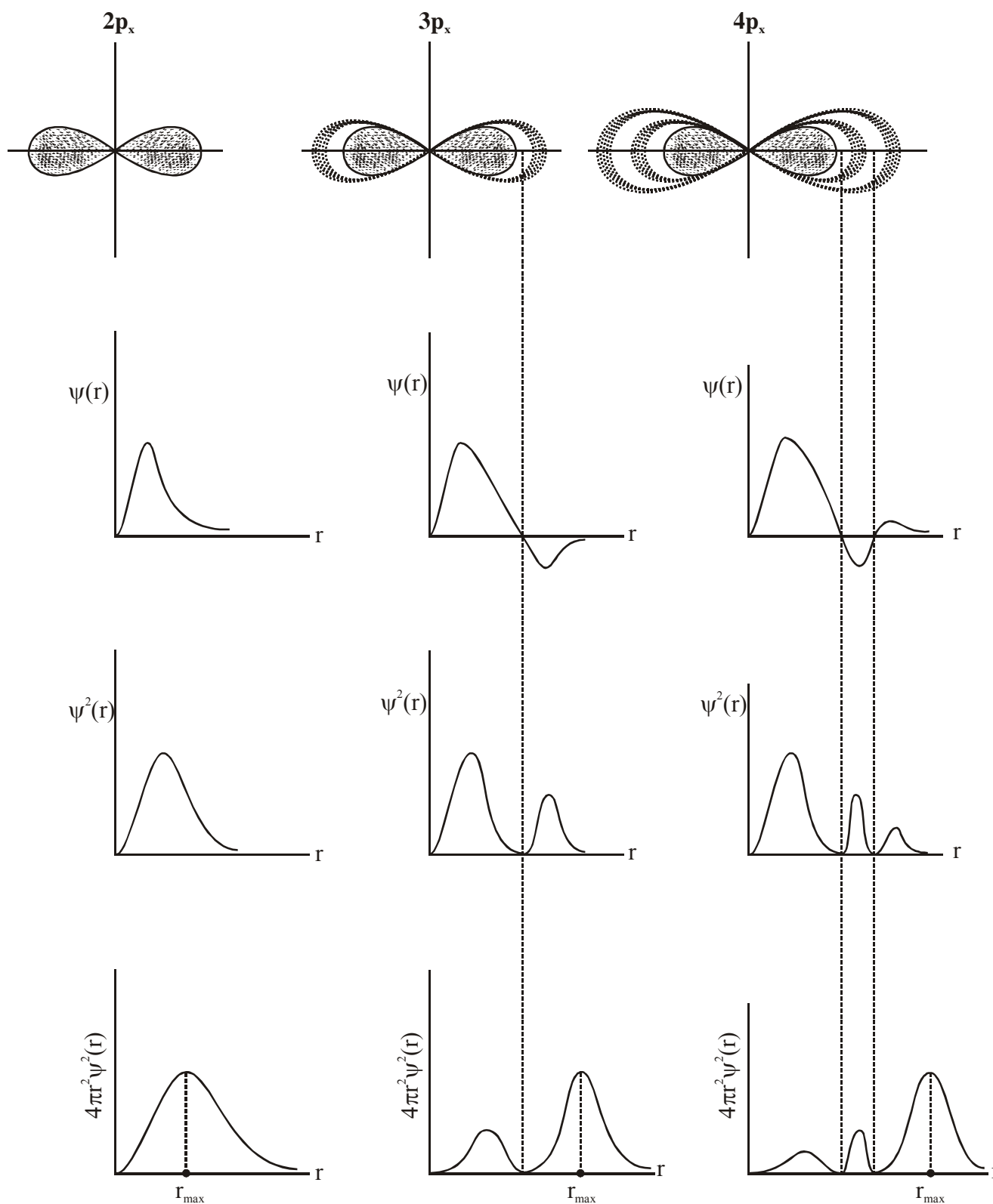
Note : Number of radial nodes = $n - \ell - 1$

Number of angular nodes = ℓ

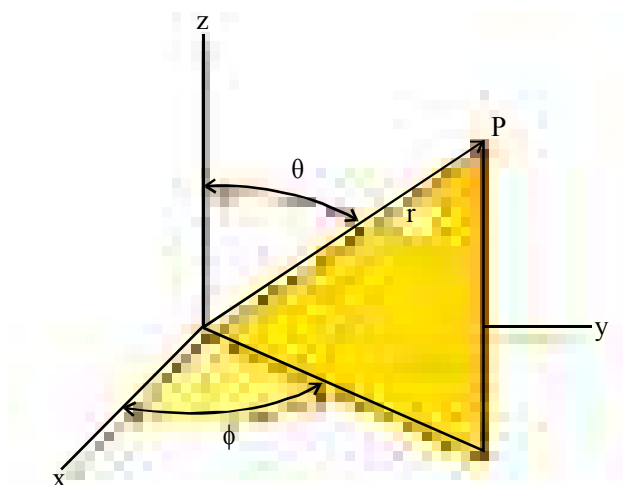
Total of number nodes = $n - 1$

12.6 Electron-density distribution in 1s, 2s and 3s-orbitals. The lower part of the fig. shows how the electron density, represented by ψ^2 varies as a function of distance from the nucleus. In the 2s and 3s - orbitals, the electron-density function drops to zero at certain distances from the nucleus. The spherical surfaces around the nucleus at which ψ^2 is zero are called nodes.

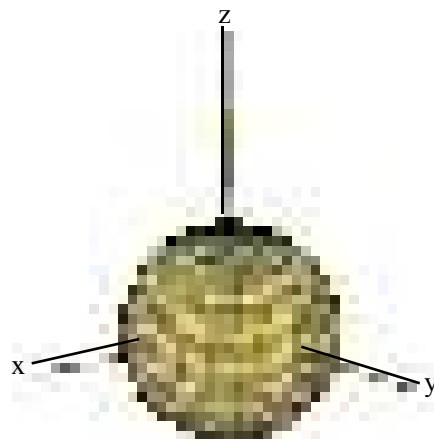


12.7 Electron-density distribution in $2p_x$, $3p_x$ and $4p_x$ -orbitals :

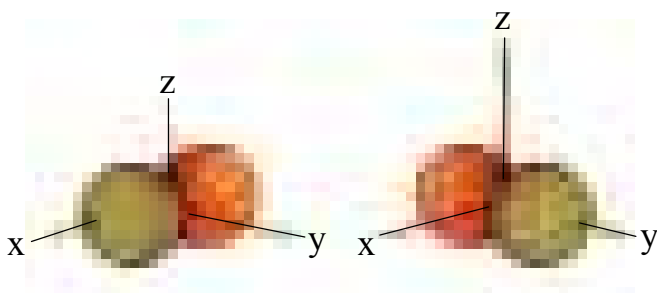
12.8 SHAPES OF ATOMIC ORBITALS



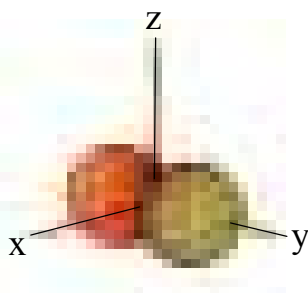
The spherical Polar Coordinates



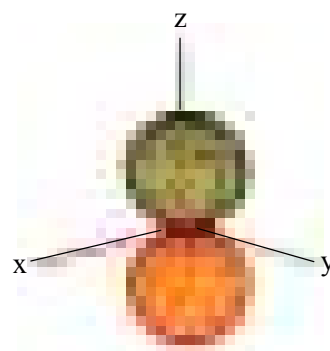
S



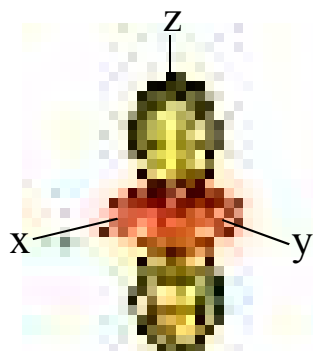
p_x



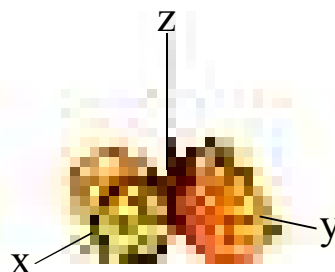
p_y



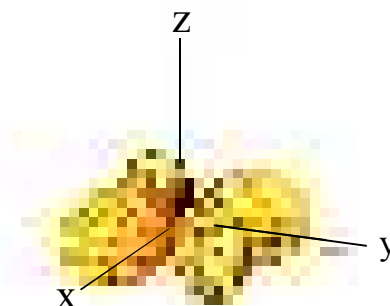
p_z



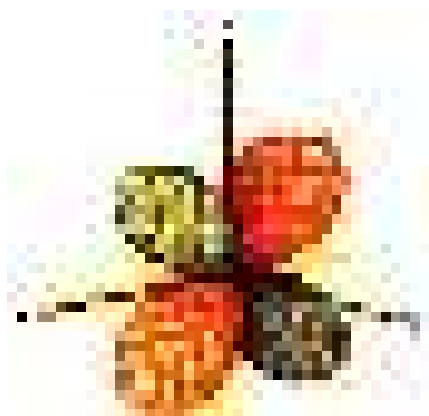
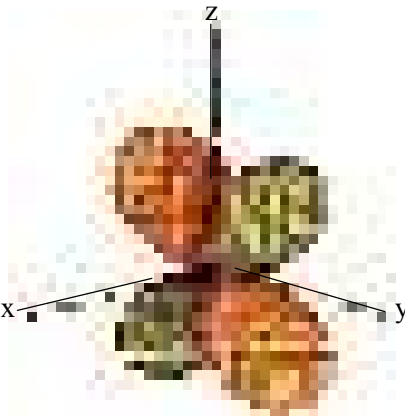
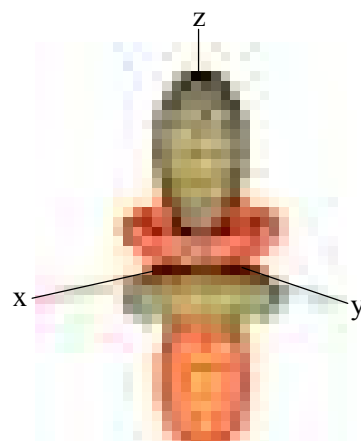
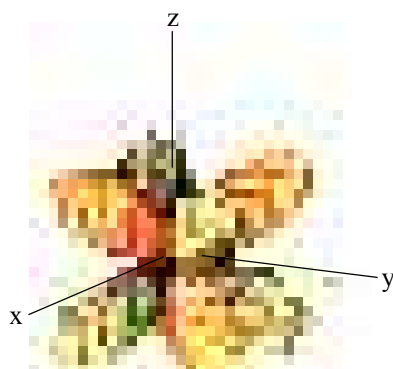
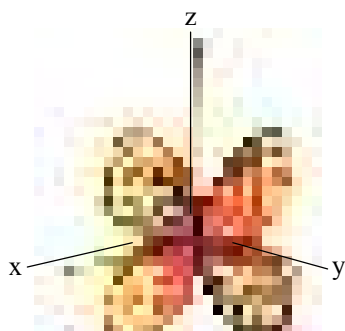
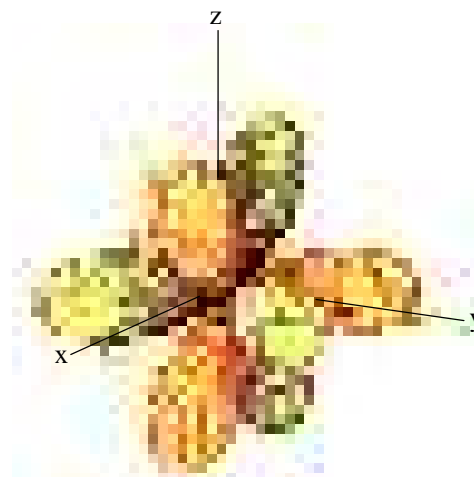
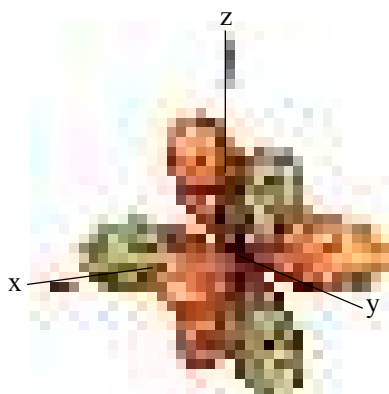
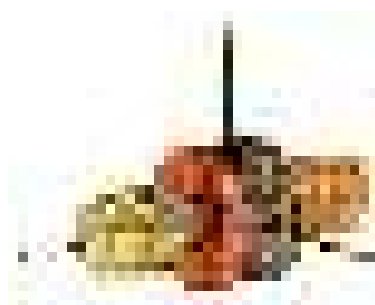
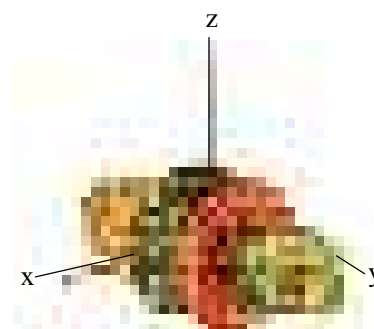
d_{z^2}



$d_{x^2-y^2}$



d_{xy}

 d_{xz}  d_{yz}  f_{z^3}  f_{xyz}  $f_{z(x^2-y^2)}$  $f_{x(y^2-z^2)}$  $f_{y(z^2-x^2)}$  f_{x^3}  f_{y^3}

Ex.49 Find the distance at which probability of finding electron is maximum for 1s orbital in a He atom.
The wave function of orbital is given as.

$$\psi_{1s} = \frac{4}{a_0^{3/2}} \cdot e^{-\frac{2r}{a_0}}$$

Sol. Probability distribution function is $P(r) = \psi^2 \cdot 4\pi r^2 = \left(\frac{16}{a_0^3}\right) e^{-\left(\frac{4r}{a_0}\right)} \cdot 4\pi r^2$

$$\Rightarrow P(r) = k \cdot r^2 \cdot e^{-\frac{4r}{a_0}}$$

$$\text{differentiating } \frac{dP(r)}{dr} = 2r \cdot e^{-\frac{4r}{a_0}} - \left(\frac{4}{a_0}\right) r^2 \cdot e^{-\frac{4r}{a_0}} = 0$$

$$\Rightarrow 1 = \frac{2r}{a_0} \Rightarrow r = \frac{a_0}{2}$$

$\Rightarrow$ probability of finding electron is maximum at distance $\frac{a_0}{2}$ from nucleus.

Ex.50 Consider ψ (wave function) of 2s atomic orbital of H-atom is -

$$\psi_{2s} = \frac{1}{4\sqrt{2\pi} a_0^{3/2}} \left[2 - \frac{r}{a_0} \right] e^{-\frac{r}{2a_0}}$$

Find distance of radial node from nucleus in terms of a_0 .

Sol. $R(r) = 0$

$$\left[2 - \frac{r}{a_0} \right] e^{-\frac{r}{2a_0}} = 0 \Rightarrow 2 - \frac{r}{a_0} = 0 \Rightarrow r = 2 a_0$$

EXERCISE # S-I

FUNDAMENTAL PARTICLES

- Find the ratio of specific charges $\left(\frac{q}{m}\right)$ of an α -particle and a deuteron
- Specific charges of two particles A and B are in ratio 2 : 3. If their mass ratio $m_A : m_B$ is 2 : 3, then find ratio of their charges $\left(\frac{q_A}{q_B}\right)$?
- Arrange electron (e), proton (p), neutron (n) and α -particles (α), in the increasing order of their e/m ratio.
- In oil drop experiment, the following charges were experimentally determined for five oil droplets 1.2×10^{-18} unit, 3.0×10^{-18} unit, 6.0×10^{-18} unit, 2.4×10^{-18} unit and 3.6×10^{-18} unit (unit is arbitrary). What should be the charge on electron in this arbitrary unit?
- The kinetic energy of a charged particle (charge = q) accelerated by a potential difference of 10000 volt is 1.6×10^{-15} Joule. The value of 'q' is.

RUTHERFORD'S ATOMIC MODEL

- If the radius of an atom is $2\text{\AA}$ and the radius of its nucleus is 4×10^{-15} m. The ratio of volume of nucleus to that of atom is.
- Radius of nucleus may be given as $R_N = R_0 A^{1/3}$, where A = mass number and R_0 = constant. Calculate the density of nucleus of an atom if $R_0 = 1.2 \times 10^{-15}$.
[$N_A = 6 \times 10^{23}$, $23 \times 8\pi \times (1.2)^3 = 1000$]
- The ratio of distances of closest approach of a proton and an α -particle projected towards the same nucleus with the same initial kinetic energy is.
- An α -particle of K.E. 5.4 MeV is projected towards Cr-nucleus ($Z = 24$). What is its distance of closest approach? ($e = 1.6 \times 10^{-19}\text{C}$)
- With what speed an α -particle should be projected towards stationary krypton nucleus ($Z = 36$) such that it approaches upto $2.4 \times 10^{-14}\text{m}$ from the nucleus? ($N_A = 6 \times 10^{23}$, $e = 1.6 \times 10^{-19}\text{C}$).

WAVE CHARACTERISTICS

- Calculate the wave length of an electromagnetic radiation of frequency 2×10^{15} Hz.
- A radiostation emits the radiations of wavelength 2000 m (meter band of station). What is the frequency band of that station (Frequency of emitted radio waves)?
- The wave numbers of two electromagnetic radiations are $4 \times 10^6 \text{ m}^{-1}$ and $2 \times 10^5 \text{ cm}^{-1}$. The ratio of their frequencies is.
- An electromagnetic radiation makes 5000 waves in 20 cm. The frequency of radiation is.

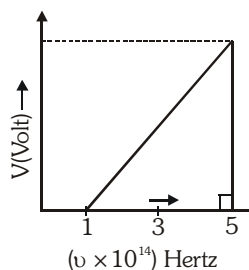
PLANCK'S QUANTUM THEORY

- Calculate the energy in Joule of a photon of wave length 4000 $\text{\AA}$.

16. The eyes of certain member of the reptile family pass a single visual signal to the brain when the visual receptors are struck by photons of wavelength 662 nm. If a total energy of 3.0×10^{-14} J is required to trip the signal, what is the minimum number of photons that must strike the receptor.
($h = 6.62 \times 10^{-34}$ J $\times$ s)
17. Find the number of photons of radiation of frequency 5×10^{13} s⁻¹ that must be absorbed in order to melt one gm ice when the latent heat of fusion of ice is 330 J/g. ($h = 6.6 \times 10^{-34}$ J $\times$ sec.)
18. A certain dye absorbs 4000 Å and fluoresces at 5000 Å these being wavelengths of maximum absorption that under given conditions 40% of the absorbed energy is emitted. Calculate the ratio of the number of quanta emitted to the number absorbed.
19. The reaction between H₂ and Br₂ to form HBr in presence of light is initiated by the photo decomposition of Br₂ into free Br atoms (free radicals) by absorption of light. The bond dissociation energy of Br₂ is 192 KJ/mole. What is the longest wavelength (in Å) of the photon that would initiate the reaction.
 $\left(1 \frac{\text{eV}}{\text{atom}} = 96 \text{ kJ/mol} ; hc = 1240 \text{ eV} \times \text{nm}\right)$
20. The quantum yield for decomposition of HI is 0.2. In an experiment 0.01 moles of HI are decomposed. Find the number of photons absorbed. ($N_A = 6 \times 10^{23}$)

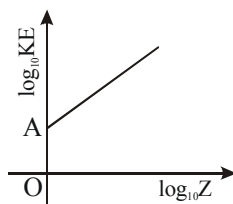
PHOTOELECTRIC EFFECT

21. The minimum energy required for the emission of a metal electron is 13.24×10^{-19} J. Calculate the critical frequency and the corresponding wavelength of the photon (threshold wavelength) required to eject the electron ?
22. A light source of wavelength λ illuminates a metal and ejects photo electron with $(KE)_{\text{max}} = 1$ eV. Another light source of wave length $\frac{\lambda}{3}$, ejects photoelectrons from same metal with $(KE)_{\text{max}} = 5$ eV. Find the value of work function (eV) of metal.
23. The dissociation energy of H₂ is 482.5 KJ/mol. If H₂ is exposed to radiant energy of wavelength 124 nm, what % of radiant energy will be converted into K.E. ($1 \frac{\text{eV}}{\text{atom}} = 96.5 \text{ kJ/mol.}$)
24. The K.E. of an electron emitted from tungsten surface is 3.06 eV. What voltage would be required to bring the electron to rest.
25. Stopping potential [V volts] is plotted against frequency of light used [ν]. Find work function (eV). ($h = 6.62 \times 10^{-34}$ J $\times$ s)



BOHR'S MODEL

26. If the mass of electron is doubled, then find the new radius (in Å) of 1st orbit of H-atom?
27. If an electron in H-atom jumps from one orbit to other its angular momentum doubles. If the new radius is 'x' times the old radius, then find x ?
28. The radius of the an orbit of hydrogen atom is 0.85 nm. Calculate the velocity (in m/sec) of electron in this orbit.
29. The velocity of e^- in a certain Bohr orbit of the hydrogen atom bears the ratio 1:275 to the velocity of light. What is the quantum no. "n" of the orbit and the wave no. of the radiation emitted for the transition from the quatum state (n+1) to the ground state.
30. If the average life time of an excited state of H atom is of order 10^{-8} sec, estimate how many orbits an e^- makes when it is in the state $n = 2$ and before it suffers a transition to $n = 1$ state.
31. Calculate the frequency of e^- in the first Bohr orbit in a H-atom.
32. Calculate energy (in J) of electron which is moving in the orbit of a hydrogen atom that has its radius, sixteen times the radius of first Bohr orbit for H-atom.
33. The energy of an excited H-atom is -3.4 eV. Calculate angular momentum of e^- in the given orbit.
34. In a H-like atom for an electron revolving in n^{th} orbit, the variation of $\log_{10} \text{KE}$ versus $\log_{10} Z$ is plotted as following -



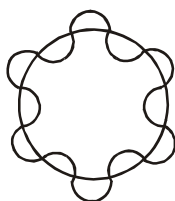
where Z is atomic number & KE is kinetic energy of electron (in eV). If $OA = \log_{10} 3.4$ then find the value of principle quantum number of the electron.

35. The energy of the second stationary state in Li^{+2} ion is -11.025×10^{-18} J. Find ionization energy (in J/mole) for He^+ ions in its ground state :
36. The energy of an electron in the first orbit of He^+ is -871.6×10^{-20} J. The energy of the electron in the first orbit of hydrogen would be :
37. A single electron orbits around a stationary nucleus of charge $+Ze$ where Z is atomic number and ' e ' is the magnitude of the electric charge. The hydrogen like species required 47.2 eV to excite the electron from the second Bohr orbit to the third Bohr orbit. Find
 - (i) the value of Z and give the hydrogen like species formed.
 - (ii) the kinetic energy (in eV) and potential energy (in eV) of the electron in the first Bohr orbit.
38. A hydrogen like atom with atomic number ' Z ' is in higher excited state of quantum number ' n '. This xcited state atom can make a transition to the first excited state by successively emitting two photons of energies 10 eV and 68.2 eV respectivley. Alternatively, the atom from the same excited state can make a transition to the 2nd excited state by emitting two photons of energies 4.25 eV and 5.95 eV respectively. Calculate the value of ' Z '.

39. H-atom is exposed to electromagnetic radiation of $1026 \text{ \AA}$ and gives out induced radiations (radiations emitted when e^- returns to ground state). Calculate λ (in $\text{\AA}$) of induced radiations.
40. Estimate the difference in energy (in eV) between I and II Bohr Orbit for a hydrogen atom. At what minimum atomic number a transition from $n=2$ to $n=1$ energy level would result in the emission of X-rays with $\lambda = 3.0 \times 10^{-8} \text{ m}$? Which hydrogen like species does this atomic number correspond to.
41. A doubly ionised lithium atom is hydrogen like with atomic number $z = 3$. Find the wavelength (in $\text{\AA}$) of the radiation required to excite the electron in Li^{2+} from the first to the third Bohr orbit.
42. If shortest wavelength of H-atom in Balmer series is X then.
 - (i) What is the shortest wave length in Lyman series.
 - (ii) What is the longest wave length in Paschen series.
43. Ionization energy of H-atom is 13.6 eV . If H-atoms in the ground state are excited by monochromatic light of energy 12.1 eV then the maximum number of spectral lines emitted according to Bohr's theory will be -
44. Find the number of spectral lines in Paschen series emitted by atomic H, when electron is excited from ground state to 7^{th} energy level returns back.
45. A stationary He^+ ion emitted a photon corresponding to a first line of the Lyman series. The photon liberated a photoelectron from a stationary H atom in ground state. What is the velocity (in cm/sec.) of photoelectron.

DE-BROGLIE AND HEISENBERG EQUATION

46. The figure shows a sample of H-atoms having electron revolving in higher orbit 'n'.



If this electron makes transition from this orbit 'n' to ground state, No. of paschen lines emtted are.

47. deBroglie wavelength associated with an electron in 4^{th} orbit of hydrogen atom is $a \times (\pi r_0)$ where r_0 is radius of 1^{st} orbit of hydrogen atom, find value of 'a'.
48. The minimum number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number 3 will be -
49. Calculate the de-broglie wavelength (in m) associated with motion of earth (mass $6 \times 10^{24} \text{ Kg}$) orbiting around the sun at a speed of $3 \times 10^6 \text{ m/s}$.
50. What is de Broglie wavelength (in pm) associated with an e^- accelerated through potential difference = 15 KV .
51. To what effective potential (in volt) a proton beam be subjected to give its protons a wavelength of $1 \times 10^{-10} \text{ m}$.
52. A proton is accelerated to one- tenth of the velocity of light. If its velocity can be measured with a precision ± 1 . What must be its uncertainty in position (in m). (Use : $\frac{h}{4\pi} = 5.2 \times 10^{-35}$)

53. The uncertainty in the position of an electron (mass 9.1×10^{-28} gm) moving with a velocity of 3×10^4 cm sec $^{-1}$, uncertainty in velocity is 0.011% will be : ($h = 6.6 \times 10^{-34}$ J $\times$ s)
54. If uncertainties in measurement of position and momentum of an electron are equal, then the uncertainty in measurement of its velocity is $x \times 10^{12}$. The value of x (the closest whole number value) is
 $m_e = 9.1 \times 10^{-31}$ kg, and $h = 6.625 \times 10^{-34}$ Js.
55. With the help of Heisenberg's uncertainty principle, show that electron can never exist in the nucleus.

SCHRODINGER EQUATION

56. Calculate the distance of spherical nodes for '3s' orbital from nucleus?

$$R_{3s} = \frac{1}{9\sqrt{3}a_0^{3/2}} (6 - 6\sigma + \sigma^2) e^{-\frac{\sigma}{2}} \quad \text{where } \sigma = \frac{2r}{na_0}$$

57. How many non-spherical subshell are possible that have atleast two maxima if a curve is plotted between radial probability distribution function versus radial distance for which principal quantum number: $n \leq 4$?
58. The ratio of angular nodes of 3d orbital and radial nodes of 3p orbital is $x : 1$ what is the value of x .
59. For an orbital in B^{+4} radial function is :

$$R(r) = \frac{1}{9\sqrt{6}} \left(\frac{Z}{a_0} \right)^{\frac{3}{4}} (4 - \sigma) \sigma e^{-\sigma/2}$$

where $\sigma = \frac{Zr}{a_0}$ and $a_0 = 0.529 \text{ \AA}$; Z = atomic number, r = radial distance from nucleus.

The radial node of orbital is at distance from nucleus.

60. Total nodes in 5f-orbital is.

EXERCISE # S-II

- ' α particle' of 3.6 MeV are fired towards nucleus A_ZX , at point of closest separation distance between ' α particle' and 'X' is 1.6×10^{-14} m. Calculate atomic number of 'X'
[Given : $1/4\pi\epsilon_0 = 9 \times 10^9$ in S.I. units]
- Suppose the potential energy between electron and proton at a distance r is given by $-\frac{ke^2}{3r^3}$. Use Bohr's theory to obtain energy of such a hypothetical atom.
- In the Bohr's model, for unielectronic species following symbols are used
 $r_{n,z}$ $\longrightarrow$ Radius of n^{th} orbit with atomic number " z "
 $U_{n,z}$ $\longrightarrow$ Potential energy of electron in n^{th} orbit with atomic number " z "
 $K_{n,z}$ $\longrightarrow$ Kinetic energy of electron in n^{th} orbit with atomic number " z "
 $v_{n,z}$ $\longrightarrow$ Velocity of electron in n^{th} orbit with atomic number " z "
 $T_{n,z}$ $\longrightarrow$ Time period of revolution of electron in n^{th} orbit with atomic number " z "
 Calculate z in all in cases.
 (i) $U_{1,2} : K_{1,z} = -8 : 1$ (ii) $r_{1,z} : r_{2,1} = 1 : 8$
 (iii) $v_{1,z} : v_{3,1} = 9 : 1$ (iv) $T_{1,2} : T_{2,z} = 9 : 32$
- In a hypothetical H-atom the mass of electron & its charge is double of what we consider then calculate the total energy (in eV) of electron in the Ist orbit of such a hypothetical H-atom ?
[Assuming all others concepts and parameter to be same as we considered in Bohr's model.]
- In H-spectrum, longest wavelength of Lyman is 120 nm and shortest wavelength of Balmer is 360 nm. From this data, find longest wavelength of photon, that can ionize this H-atom
- A proton and an electron, both at rest initially, combine to form a H-atom in ground state. A single photon is emitted in this process. Find the wavelength (in nm) of this photon
(use : $hc = 1237.6 \text{ eV} \times \text{nm}$)
- A H-like ion is observed to emit six different wavelengths originating from all possible transitions between a group of levels. These levels have energies between -0.85 eV (Min) and -0.544 eV (Max).
 (a) Find atomic number (z) of element
 (b) Find quantum numbers of levels between which transitions occur.
 (c) Calculate largest wavelength emitted in transits between the levels.
- The angular momentum of an electron in a Bohr's orbit of H-atom is $3.1652 \times 10^{-34} \text{ kg-m}^2/\text{sec}$. Calculate the wavenumber in terms of Rydberg constant (R) of the spectral line emitted when an electron falls from this level to the ground state. [Use $h = 6.626 \times 10^{-34} \text{ Js}$]
- The ionisation energy of the hydrogen atom is given to be 13.6 eV. A photon falls on a hydrogen atom which is initially in the ground state and excites it to the ($n = 4$)state.
 (a) show this transition in the energy-level diagram &
 (b) calculate the wavelength (in Å) of the photon.

10. The ionisation energy of a H-like Bohr atom is 4 Rydbergs
- (i) What is the wavelength (in Å) of radiation emitted when the e^- jumps from the first excited state to the ground state.
- (ii) What is the radius (in cm) of first Bohr orbit for this atom. [1 Rydberg = 2.18×10^{-18} J]
11. Photon having wavelength 12.4 nm was allowed to strike a metal plate having work function 25 eV. Calculate the
- (a) Maximum kinetic energy (in eV) of photoelectrons emitted in eV.
- (b) Wavelength (in Å) of electron with maximum kinetic energy in Å.
- (c) Calculate the uncertainty in wavelength (in m) of emitted electron if the uncertainty in the momentum is 6.62×10^{-28} Kg m/sec.
-
12. The diameter of a dust particle of mass 10^{-3} gm is 2Å . If uncertainty in speed of this particle is $\frac{3.31}{\pi} \times 10^{-3}$ m/s, then find minimum uncertainty in its position
13. The vapours of Hg absorb some electrons accelerated by a potential diff. of 4.5 volt as a result of which light is emitted. If the full energy of single incident e^- is supposed to be converted into light emitted by single Hg atom, find the wave no. (in m^{-1}) of the light.
14. A photon having $\lambda = 960\text{Å}$ causes the ionization of a nitrogen atom. Give the I.E. per mole of nitrogen in KJ.
- (Use : $hc = 1240\text{ eV} \times \text{nm}$ and $1 \frac{\text{eV}}{\text{atom}} = 96.0\text{ kJ/mol}$.)
15. Mr. Santa has to decode a number "ABCDEF" where each alphabet is represented by a single digit. Suppose an orbital whose radial wave function is represented as

$$\Psi_{(r)} = k_1 \cdot e^{-r/k_2} (r^2 - 5k_3r + 6k_3^2)$$

From the following information given about each alphabet then write down the answers in the form of "ABCDEF", for above orbital.

Info A = Value of n where "n" is principal quantum number

Info B = No. of angular nodes

Info C = Azimuthal quantum number of subshell to orbital belongs

Info D = No. of subshells having energy between $(n+5)s$ to $(n+5)p$ where n is principal quantum number

Info E = Orbital angular momentum of given orbital.

Info F = Radial distance of the spherical node which is farthest from the nucleus

(Assuming $k_3 = 1$)

EXERCISE # O-I

1. Anode rays are made up of :
(A) only protons (B) only nucleus of atom
(C) Positive residue of atoms (D) only electrons
2. When Lithium (${}^7_3\text{Li}$) vapours were filled in discharge tube for anode ray experiment, the anode rays were found to contain only Li^+ ions. Thus the anode ray particle contains :
(A) 1 proton only (B) 3 proton and 4 neutron only
(C) 3 proton, 4 neutron and 2 electrons (D) 3 proton, 3 neutrons and 3 electrons
3. The specific charge is maximum for
(A) Na^+ (B) Al^{+3} (C) H^+ (D) Mg^{+2}
4. Which of the following particle is not deflected in the magnetic field
(A) Electron (B) proton (C) Neutron (D) Deuteron
5. E/m ratio of a particle of charge 2 unit and mass 4 amu is
(A) $4.8 \times 10^7 \text{ C/kg}$ (B) 0.5 C/kg (C) $4.8 \times 10^4 \text{ C/kg}$ (D) $8 \times 10^{-20} \text{ C/kg}$
6. α -particles are projected towards the nucleus of following metals, with the same kinetic energy. The distance of closest approach will be minimum for which metal?
(A) Cu (B) Ag (C) Au (D) Ca
7. Which of the following is not a conclusion of Rutherford's atomic model -
(A) Most of the part inside an atom is empty
(B) Almost all mass of an atom is concentrated in the nucleus.
(C) The size of nucleus is very small in comparison to the size of atom
(D) Electron revolves around the nucleus in definite orbits.
8. In the different experiments, α -particles, proton, deuteron and neutron are projected towards gold nucleus with the same kinetic energy. The distance of closest approach will be minimum for
(A) α -particle (B) proton (C) dueteron (D) neutron
9. The ratio of the energy of a photon of $2000 \text{ \AA}$ wavelength radiation to that of $4000 \text{ \AA}$ radiation is
(A) $1/4$ (B) 4 (C) $1/2$ (D) 2
10. Small packets of light is called
(A) proton (B) quanta (C) photon (D) spectrum
11. Which of the following electromagnetic radiation have greater frequency?
(A) X - rays (B) Ultraviolet rays (C) Radio waves (D) Visible rays
12. Two electromagnetic radiations have wave numbers in the ratio $2 : 3$. Their energies per quanta will be in the ratio
(A) $3 : 2$ (B) $9 : 4$ (C) $4 : 9$ (D) $2 : 3$

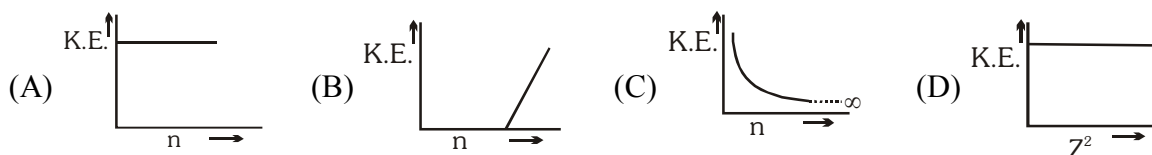
13. A radio station is emitting the radiations of frequency 2×10^4 Hz. If its frequency is doubled,
(A) wavelength will be doubled (B) energy per quanta will be doubled
(C) wave number will be halved (D) all of these
14. A photon of 400 nm is absorbed by a gas molecule and then the molecule re-emits two photons. One re-emitted photon has wavelength 500 nm. Assuming that there is no change in the energy of molecule, the wavelength of second re-emitted photon is
(A) 100 nm (B) 2000 nm (C) -100 nm (D) 900 nm
15. A green bulb and a red bulb are emitting the radiations with equal power. The correct relation between numbers of photons emitted by the bulbs per second is
(A) $n_g = n_r$ (B) $n_g < n_r$ (C) $n_g > n_r$ (D) unpredictable
16. The threshold wavelength for ejection of electrons from a metal is 330 nm. The work function for the photoelectric emission from the metal is ($h = 6.6 \times 10^{-34}$ J-s)
(A) 1.2×10^{-18} J (B) 6.0×10^{-19} J (C) 1.2×10^{-20} J (D) 6.0×10^{-12} J
17. In the emission of photoelectrons, the number of photoelectrons emitted per unit time depends upon
(A) energy of the incident radiation (B) intensity of the incident radiation
(C) frequency of the incident radiation (D) wavelength of the incident radiation
18. Radiations of frequency, ν , are incident on a photosensitive metal. The maximum kinetic energy of photoelectrons is 'E'. When the frequency of the incident radiations is doubled, what is the maximum kinetic energy of the photoelectrons?
(A) 2E (B) E/2 (C) $E + h\nu$ (D) $E - h\nu$
19. Radiation of $\lambda = 155$ nm was irradiated on Li (work function = 5 eV) plate. The stopping potential (in V) is _____.
(A) 3 V (B) 8 V (C) 9 V (D) 5 V
20. Electromagnetic radiations having $\lambda = 310$ Å are subjected to a metal sheet having work function = 12.8 eV. What will be the velocity of photoelectrons with maximum Kinetic Energy..
(A) 0, no emission will occur (B) 2.18×10^6 m/s
(C) $2.18\sqrt{2} \times 10^6$ m/s (D) 8.72×10^6 m/s
21. Bohr's model may be applied to
(A) Na^{10+} ion (B) He atom (C) Be^{2+} ion (D) C^{6+} ion
22. If the radius of 3rd Bohr's orbit of H is x, then radius of 4th orbit of Li^{2+} ion would be :-
(A) $\frac{27}{16}x$ (B) $\frac{16}{27}x$ (C) $\frac{9}{16}$ (D) None of these
23. What would be the approximate quantum number, n, for a circular orbit of hydrogen, 1×10^{-5} cm in diameter?
(A) 31 (B) 43 (C) 40 (D) 39

24. If the mass of electron is doubled, the radius of first orbit of H-atom become about
(A) $0.529 \text{ \AA}$ (B) $0.265 \text{ \AA}$ (C) $1.058 \text{ \AA}$ (D) $0.32 \text{ \AA}$
25. The speed of electron revolving in the 4th orbit of a hydrogen like atom or ion is 1094 km/s . The atom or ion is
(A) H (B) He^+ (C) Li^{2+} (D) Be^{3+}
26. How much distance an electron revolving in 3rd orbit of He^+ ion will travel in one second
(A) $1.458 \times 10^6 \text{ m}$ (B) $3.28 \times 10^6 \text{ m}$ (C) $4.862 \times 10^5 \text{ m}$ (D) $2.917 \times 10^6 \text{ m}$
27. The ratio of time taken by electron in revolutions round the H-nucleus in 2nd and 3rd orbits, are
(A) $2 : 3$ (B) $4 : 8$ (C) $8 : 27$ (D) $27 : 8$
28. Which of the following is not a permissible value of angular momentum of electron in H-atom?
(A) $1.5 \frac{h}{\pi}$ (B) $0.5 \frac{h}{\pi}$ (C) $1.25 \frac{h}{\pi}$ (D) all of these
29. Angular momentum for P-shell electron :-
(A) $\frac{3h}{\pi}$ (B) Zero (C) $\frac{\sqrt{2}h}{2\pi}$ (D) None
30. Angular momentum in 2nd Bohr orbit of H-atom is x . Then find out angular momentum in 1st excited state of Li^{+2} :-
(A) $3x$ (B) $9x$ (C) $x/2$ (D) x
31. The orbit from which when electron will jump in other orbit, energy may be absorbed but not emitted out, will be
(A) 1st orbit (B) 2nd orbit (C) 7th orbit (D) infinite orbit
32. The potential energy of electron revolving in the ground state of H-atom is
(A) -13.6 eV (B) -6.8 eV (C) -27.2 eV (D) Zero
33. If the potential energy (PE) of hydrogen electron is -3.02 eV then in which of the following excited level is electron present :-
(A) 1st (B) 2nd (C) 3rd (D) 4th
34. A single electron is revolving in orbits around a stationary nucleus ($z = 5$). The energy required to excite the electron from third to fourth Bohr orbit will be :-
(A) 4.5 eV (B) 8.53 eV (C) 25 eV (D) 16.53 eV
35. A photon of energy 12.75 eV is completely absorbed by a hydrogen atom initially in ground state. The principle quantum number of the excited state is
(A) 1 (B) 3 (C) 4 (D) ∞

36. An hydrogen atom (ionisation energy 13.6 eV) jumps from third excited state to first excited state. The energy of photon emitted in the process is

(A) 1.89 eV (B) 2.55 eV (C) 12.09 eV (D) 12.75 eV

37. Which is correct graph :-



38. The energy required to remove an electron from the surface of sodium metal is 3.3 eV. What is the longest wavelength of radiation with which it can show photoelectric effect?

Use $h = 6.6 \times 10^{-34} \text{ J.s}$

(A) $1.75 \times 10^{-7} \text{ m}$ (B) $3.75 \times 10^{-5} \text{ m}$ (C) $1.75 \times 10^{-5} \text{ m}$ (D) $3.75 \times 10^{-7} \text{ m}$

39. Which is correct for any H like species :-

(A) $(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3)$ (B) $(E_2 - E_1) < (E_3 - E_2) < (E_4 - E_3)$
 (C) $(E_2 - E_1) = (E_3 - E_2) = (E_4 - E_3)$ (D) $(E_2 - E_1) = 1/4 (E_3 - E_2) = 1/9 (E_4 - E_3)$

40. A single electron orbits a stationary nucleus of charge $+Ze$, where Z is a constant. It requires 47.2 eV to excite electron from second Bohr orbit to third Bohr orbit, find the value of Z :-

(A) 1 (B) 3 (C) 5 (D) 4

41. The energy of H-atom in n^{th} orbit is E_n then energy in n^{th} orbit of singly ionised helium atom will be:

(A) $4E_n$ (B) $E_n/4$ (C) $2E_n$ (D) $E_n/2$

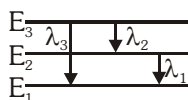
42. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon

(A) 3s (B) 2p (C) 2s (D) 1s

43. The third line in Balmer series corresponds to an electronic transition between which Bohr's orbits in hydrogen

(A) $5 \rightarrow 3$ (B) $5 \rightarrow 2$ (C) $4 \rightarrow 3$ (D) $4 \rightarrow 2$

44. In the following transition which statement is correct



(A) $E_{3-1} = E_{3-2} - E_{2-1}$ (B) $\lambda_3 = \lambda_1 + \lambda_2$
 (C) $\nu_3 = \nu_2 + \nu_1$ (D) All of these

45. The first Lyman transition in the hydrogen spectrum has $\Delta E = 10.2 \text{ eV}$. The same energy change is observed in the second Balmer transition of :-

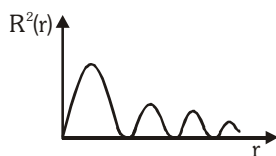
(A) Li^{2+} (B) Li^+ (C) He^+ (D) Be^{3+}

46. In a sample of H-atoms, electron transits from 6th orbit to 2nd orbit in multi step. Then total spectral lines (without Balmer series) will be :-

(A) 6 (B) 10 (C) 4 (D) 0

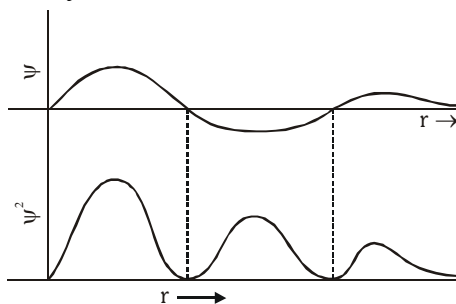
47. What is the shortest wavelength in the Pfund series of He^+ ion :-
 (A) $\frac{25}{R}$ (B) $\frac{4}{R}$ (C) $\frac{4R}{25}$ (D) $\frac{25}{4R}$
48. The shortest wavelength of He^+ ion in Balmer series is x , then longest wavelength in the Paschene series of Li^{+2} is
 (A) $\frac{36x}{5}$ (B) $\frac{16x}{7}$ (C) $\frac{9x}{5}$ (D) $\frac{5x}{9}$
49. The ratio of wave length of photon corresponding to the α -line of Lyman series in H-atom and β -line of Balmer series in He^+ is
 (A) 1 : 1 (B) 1 : 2 (C) 1 : 4 (D) 3 : 16
50. Three energy levels P, Q, R of a certain atom are such that $E_P < E_Q < E_R$. If λ_1 , λ_2 and λ_3 are the wave length of radiation corresponding to transition $R \rightarrow Q$; $Q \rightarrow P$ and $R \rightarrow P$ respectively. The correct relationship between λ_1 , λ_2 and λ_3 is
 (A) $\lambda_1 + \lambda_2 = \lambda_3$ (B) $\frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$ (C) $\lambda_3 = \sqrt{\lambda_1 \lambda_2}$ (D) $\frac{2}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$
51. Number of possible spectral lines which may be emitted in bracket series in H atom, if electrons present in 9th excited level returns to ground level, are
 (A) 21 (B) 6 (C) 45 (D) 5
52. The value of $(n_2 + n_1)$ and $(n_2^2 - n_1^2)$ for He^+ ion in atomic spectrum are 4 and 8 respectively. The wavelength of emitted photon when electron jump from n_2 to n_1 is
 (A) $\frac{32}{9} R_H$ (B) $\frac{9}{32} R_H$ (C) $\frac{9}{32 R_H}$ (D) $\frac{32}{9 R_H}$
53. An electron, a proton and an alpha particle have kinetic energies of 16E, 4E and E respectively. What is the qualitative order of their de Broglie wavelengths?
 (A) $\lambda_e > \lambda_p = \lambda_\alpha$ (B) $\lambda_p = \lambda_\alpha > \lambda_e$ (C) $\lambda_p > \lambda_e > \lambda_\alpha$ (D) $\lambda_\alpha < \lambda_e \gg \lambda_p$
54. The wavelength associated with a golf weighing 200g and moving at a speed of 5m/h is of the order
 (A) 10^{-10}m (B) 10^{-20}m (C) 10^{-30}m (D) 10^{-40}m
55. An electron has kinetic energy $2.8 \times 10^{-23} \text{ J}$. de-Broglie wavelength will be nearly :-
 ($m_e = 9.1 \times 10^{-31} \text{ kg}$)
 (A) $9.28 \times 10^{-24} \text{ m}$ (B) $9.28 \times 10^{-7} \text{ m}$ (C) $9.28 \times 10^{-8} \text{ m}$ (D) $9.28 \times 10^{-10} \text{ m}$
56. What will be de-Broglie wavelength of an electron moving with a velocity of $1.2 \times 10^5 \text{ ms}^{-1}$:-
 (A) $6.068 \times 10^{-9} \text{ m}$ (B) $3.133 \times 10^{-37} \text{ m}$ (C) $6.626 \times 10^{-9} \text{ m}$ (D) $6.018 \times 10^{-7} \text{ m}$
57. For a valid Bohr orbit, its circumference should be:
 (A) $= n \lambda$ (B) $= (n - 1) \lambda$ (C) $> n \lambda$ (D) $< n \lambda$
58. The number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number + 2 :
 (A) 3 (B) 4 (C) 2 (D) 1
59. How fast is an electron moving if it has a wavelength equal to distance travelled in one second -
 (A) $\sqrt{\frac{m}{h}}$ (B) $\sqrt{\frac{h}{m}}$ (C) $\sqrt{\frac{h}{p}}$ (D) $\sqrt{\frac{h}{2 \times \text{K.E.}}}$

60. De-Broglie wavelength of two particles A & B are plotted against $\left(\frac{1}{\sqrt{V}}\right)$; where V is potential on the particles. Which of the following relation is correct about mass of particle (charge is same)
- (A) $M_A = M_B$
 (B) $M_A > M_B$
 (C) $M_A < M_B$
 (D) $M_A \leq M_B$
61. An α -particle is accelerated through a potential difference of V volts from rest. The de-Broglie's wavelength associated with it is -
- (A) $\sqrt{\frac{150}{V}} \text{Å}$ (B) $\frac{0.286}{\sqrt{V}} \text{Å}$ (C) $\frac{0.101}{\sqrt{V}} \text{Å}$ (D) $\frac{0.983}{\sqrt{V}} \text{Å}$
62. Which quantum number is not related with Schrodinger equation
- (A) Principal (B) Azimuthal (C) Magnetic (D) Spin
63. Which is true about ψ :-
- (A) ψ represents the probability of finding an electron around the nucleus
 (B) ψ represent the amplitude of the electron wave
 (C) Both A and B
 (D) None of these
64. According to Schrodinger model nature of electron in an atom is as :-
- (A) Particles only (B) Wave only
 (C) Both simultaneously (D) Sometimes waves and sometimes particle
65. The orbital angular momentum of an electron in 2s orbital is:
- (A) $+\frac{1}{2} \cdot \frac{h}{2\pi}$ (B) Zero (C) $\frac{h}{2\pi}$ (D) $\sqrt{2} \cdot \frac{h}{2\pi}$
66. In an excited state, a calcium atom has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 4d^1$. What is the orbital angular momentum for d electron :- $\left(\hbar = \frac{h}{2\pi}\right)$
- (A) $\sqrt{4}\hbar$ (B) $\sqrt{16}\hbar$ (C) $\sqrt{6}\hbar$ (D) $\sqrt{10}\hbar$
67. Which orbital has two angular nodal planes :-
- (A) s (B) p (C) d (D) f
68. From the following observations predict the type of orbital :
- Observation 1 : x y plane acts as nodal plane
 Observation 2 : The angular function of the orbital intersect the three axis at origin only.
 Observation 3 : $R^2(r)$ v/s r curve is obtained for the orbital is



- (A) $5p_z$ (B) $6d_{xy}$ (C) $6d_{x^2-y^2}$ (D) $6d_{yz}$

69. The wave function ψ vs radial distance (r) and square of wave function $(\psi)^2$ vs radial distance (r) graph of certain orbital is given. Identify the orbital.



- (A) 3s (B) 4d (C) 3p (D) 4p
70. The distance of spherical nodes from nucleus for the given orbital are

$$\psi_{\text{radial}} = \frac{1}{9\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} [\sigma^2 - 4\sigma + 3] \exp(-\sigma/2)$$

where a_0 & Z are the constants and $\sigma = \frac{2Zr}{a_0}$

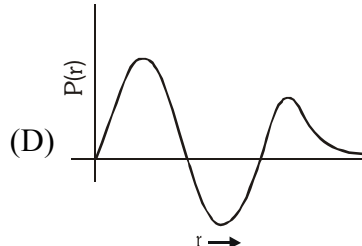
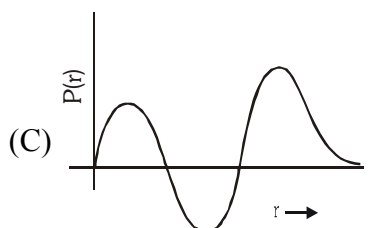
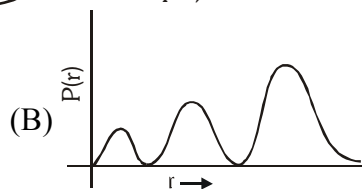
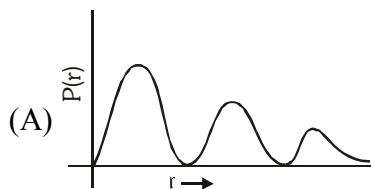
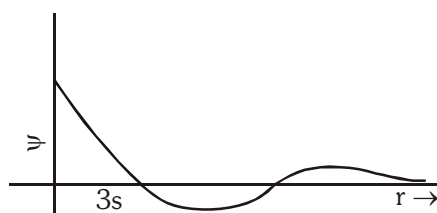
- (A) Zero, infinity (B) $\frac{a_0}{Z}, \frac{1}{2} \frac{a_0}{Z}$ (C) $\frac{3}{2} \frac{a_0}{Z}, \frac{1}{2} \frac{a_0}{Z}$ (D) $\frac{a_0}{Z}, \frac{3}{2} \frac{a_0}{Z}$

71. The radial function for an orbitals is :

$$R(r) = \frac{1}{9\sqrt{6}} \left(\frac{1}{a_0} \right) \cdot (4 - \sigma) \sigma e^{-\frac{\sigma}{3}}; \sigma = \frac{r}{a_0}$$

Pick out the incorrect statement :

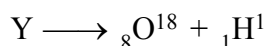
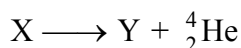
- (A) orbital must be 3p (B) orbital must be $3p_x$
 (C) the orbital has one radial node (D) the orbital must have one angular node
72. The wave function (ψ) verses radial distance (r) curve for certain orbital is given. Predict the shape of $\psi^2 \cdot 4\pi r^2$ (radial probability distribution function) verses r graph.



EXERCISE # O-II

SINGLE CORRECT :

- In a hydrogen like sample electron is in 2nd excited state, the energy of 4th state of this sample is -13.6 eV, then incorrect statement is :
 (A) Atomic number of element is 4.
 (B) 3 different types of spectral line will be observed if electrons make transition upto ground state from the 2nd excited state.
 (C) A 25 eV photon can set free the electron from the 2nd excited state of this sample
 (D) 2nd line of Balmer series of this sample has same energy value as 1st excitation energy of H-atoms.
- An electron in a hydrogen atom in its ground state absorbs energy equal to the ionisation energy of Li²⁺. The wavelength of the emitted electron is:
 (A) 3.32×10^{-10} m (B) 1.17 Å (C) 2.32×10^{-9} nm (D) 3.33 pm
- Given ΔH for the process $\text{Li(g)} \longrightarrow \text{Li}^{3+}(\text{g}) + 3\text{e}^-$ is 19800 kJ/mole & IE_1 for Li is 520 then IE_2 & IE_3 of Li are respectively (approx, value)
 (A) 7505, 11775 (B) 520, 19280 (C) 11775, 19280 (D) Data insufficient
- Consider the following nuclear reactions involving X & Y.



If both neutrons as well as protons in both the sides are conserved in nuclear reaction then moles of neutrons in 4.6 gm of X

- (A) $2.4 N_A$ (B) 2.4 (C) 4.6 (D) $0.2 N_A$

Assertion and Reason :

- Statement-1:** Energy emitted when an electron jump from 5 $\rightarrow$ 2 (energy level) is less than when an electron jump from 2 $\rightarrow$ 1 in all 'H' like atom.
Statement-2: The |total energy difference| between 1st & 2nd energy level is greater than that of any two energy level provided level '1' is not part of those two energy levels.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- Statement-1:** Emitted radiations will fall in visible range when an electron jump from higher level to $n=2$ in Li²⁺ ion.
Statement-2: Balmer series radiations belong to visible range in all H-atoms.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

More than one may be correct :

7. Which of the following could be derived from Rutherford's α -particle scattering experiment-
 - (A) Most of the space in the atom is empty
 - (B) The radius of the atom is about 10^{-10} m while that of nucleus is 10^{-15} m
 - (C) Electrons move in a circular path of fixed energy called orbits
 - (D) Electrons and the nucleus are held together by electrostatic forces of attraction.
8. In a H-like sample electrons make transition from 5th excited state to 2nd excited state
 - (A) 10 different spectral lines will be emitted
 - (B) 6 different spectral lines will be emitted
 - (C) Number of lines belonging to Balmer series will be 4
 - (D) Number of lines belonging to paschen series will be 3
9. The kinetic energy of photoelectron emitted on irradiating a metal surface with frequency ν is related by $KE = h\nu - \phi$. The plots of KE vs. incident frequency ν shows :
 - (A) A straight line with slope equal to Planck's constant.
 - (B) A straight line with intercept on x-axis equal to the product of threshold frequency and Plank's constant.
 - (C) A straight line with extrapolated intercept on y-axis equal to threshold energy.
 - (D) A straight line with intercept on x-axis equal to threshold frequency.
10. Select the correct statement(s):
 - (A) All electromagnetic radiation travel with speed of light in vacuum.
 - (B) Energy of photon of UV light is lower than that of yellow light.
 - (C) He^+ and H have similar spectrum.
 - (D) The total energy of an electron in unielectronic specie is greater than zero
11. Choose the incorrect statement(s):
 - (A) Increasing order of wavelength is
Micro waves > Radio waves > IR waves > visible waves > UV waves
 - (B) The order of Bohr radius is (r_n : where n is orbit number for a given atom) $r_1 < r_2 < r_3 < r_4$
 - (C) The order of total energy is (E_n : where n is orbit number for a given atom) $E_1 > E_2 > E_3 > E_4$
 - (D) The order of velocity of electron in H, He^+ , Li^+ , Be^{3+} species in second Bohr orbit is
 $Be^{3+} > Li^{+2} > He^+ > H$

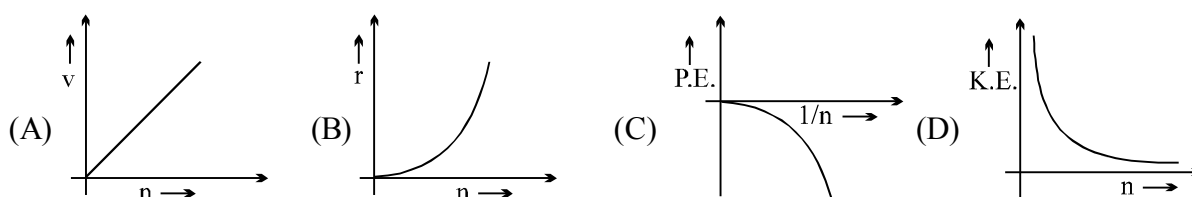
12. Select the correct curve(s):

If v = velocity of electron in Bohr's orbit

r = Radius of electron in Bohr's orbit

P.E. = Potential energy of electron in Bohr's orbit

K.E. = Kinetic energy of electron in Bohr's orbit.



13. Which is / are correct statement.

(A) The difference in angular momentum associated with the electron present in consecutive orbits of

$$\text{H-atom is } (n-1) \frac{h}{2\pi}$$

(B) Energy difference between energy levels will be changed if, P.E. at infinity assigned value other than zero.

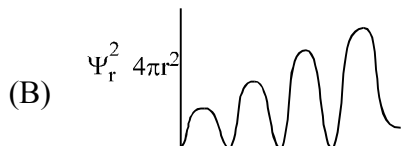
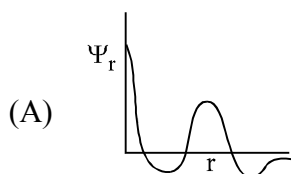
(C) Frequency of spectral line in a H-atom is in the order of $(2 \rightarrow 1) < (3 \rightarrow 1) < (4 \rightarrow 1)$

(D) On moving away from the nucleus, kinetic energy of electron decreases.

MATCH THE COLUMN :

14. Column I & column II contain data on Schrodinger Wave-Mechanical model, where symbols have their usual meanings. Match the columns.

Column I



(C) $\Psi(\theta, \phi) = K$ (independent of θ & ϕ)

(D) at least one angular node is present

Column II (Type of orbital)

(P) 4s

(Q) 5p_x

(R) 3s

(S) 6d_{xy}

15. **Column-I**

(A) Electron moving in 2nd orbit in He⁺ ion electron is

(B) Electron moving in 3rd orbit in H-atom

(C) Electron moving in 1st orbit in Li⁺² ion

(D) Electron moving in 2nd orbit is Be⁺³ ion

Column-II

(P) Radius of orbit in which moving is 0.529 Å

(Q) Total energy of electron is $(-13.6 \times 9 \text{ eV})$

(R) Velocity of electron is $\frac{2.188 \times 10^6}{3} \text{ m/sec}$

(S) De-broglie wavelength of

$$\text{electron is } \sqrt{\frac{150}{13.6}} \text{ Å}$$

COMPREHENSION :

Question No. 16 to 18 (3 questions)

The French physicist Louis de Broglie in 1924 postulated that matter, like radiation, should exhibit a dual behaviour. He proposed the following relationship between the wavelength λ of a material particle, its linear momentum p and planck constant h .

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

The de Broglie relation implies that the wavelength of a particle should decreases as its velocity increases. It also implies that for a given velocity heavier particles should have shorter wavelength than lighter particles. The waves associated with particles in motion are called matter waves or de Broglie waves. These waves differ from the electromagnetic waves as they

- (i) have lower velocities
- (ii) have no electrical and magnetic fields and
- (iii) are not emitted by the particle under consideration.

The experimental confirmation of the de Broglie relation was obtained when Davisson and Germer, in 1927, observed that a beam of electrons is diffracted by a nickel crystal. As diffraction is a characteristic property of waves, hence the beam of electron behaves as a wave, as proposed by de Broglie.

Werner Heisenberg considered the limits of how precisely we can measure properties of an electron or other microscopic particle like electron. He determined that there is a fundamental limit of how closely we can measure both position and momentum. The more accurately we measure the momentum of a particle, the less accurately we can determine its position. The converse is also true. This is summed up in what we now call the "Heisenberg uncertainty principle: It is impossible to determine simultaneously and precisely both the momentum and position of a particle. The product of uncertainty in the position,

Δx and the uncertainty in the momentum $\Delta(mv)$ must be greater than or equal to $\frac{h}{4\pi}$. i.e.

$$\Delta x \Delta(mv) \geq \frac{h}{4\pi}$$

16. The correct order of wavelength of Hydrogen (${}_1H^1$), Deuterium (${}_1H^2$) and Tritium (${}_1H^3$) moving with same kinetic energy is
 (A) $\lambda_H > \lambda_D > \lambda_T$ (B) $\lambda_H = \lambda_D = \lambda_T$ (C) $\lambda_H < \lambda_D < \lambda_T$ (D) $\lambda_H < \lambda_D > \lambda_T$
17. The transition, so that the de-Broglie wavelength of electron becomes 3 times of its initial value in He^+ ion will be
 (A) $2 \rightarrow 5$ (B) $3 \rightarrow 2$ (C) $2 \rightarrow 6$ (D) $1 \rightarrow 2$
18. If the uncertainty in velocity & position is same, then the uncertainty in momentum will be
 (A) $\sqrt{\frac{hm}{4\pi}}$ (B) $m\sqrt{\frac{h}{4\pi}}$ (C) $\sqrt{\frac{h}{4\pi m}}$ (D) $\frac{1}{m}\sqrt{\frac{h}{4\pi}}$

Question No. 19 to 22 (4 questions)

The only electron in the hydrogen atom resides under ordinary conditions on the first orbit. When energy is supplied, the electron moves to higher energy orbit depending on the amount of energy absorbed. When this electron returns to any of the lower orbits, it emits energy. Lyman series is formed when the electron returns to the lowest orbit while Balmer series is formed when the electron returns to second orbit. Similarly, Paschen, Brackett and Pfund series are formed when electron returns to the third, fourth and fifth orbits from higher energy orbits respectively.

Maximum number of lines produced when an electron jumps from n th level to ground level is equal to $\frac{n(n-1)}{2}$. For example, in the case of $n = 4$, number of lines produced is 6. ($4 \rightarrow 3$, $4 \rightarrow 2$, $4 \rightarrow 1$, $3 \rightarrow 2$, $3 \rightarrow 1$, $2 \rightarrow 1$). When an electron returns from n_2 to n_1 state, the number of lines in the spectrum will be equal to

$$\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

If the electron comes back from energy level having energy E_2 to energy level having energy E_1 , then the difference may be expressed in terms of energy of photon as:

$$E_2 - E_1 = \Delta E, \lambda = \frac{hc}{\Delta E}$$

Since h and c are constants, ΔE corresponds to definite energy; thus each transition from one energy level to another will produce a light of definite wavelength. This is actually observed as a line in the spectrum of hydrogen atom.

Wave number of line is given by the formula $\bar{\nu} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$.

where R is a Rydberg's constant ($R = 1.1 \times 10^7 \text{ m}^{-1}$)

19. The energy photon emitted corresponding to transition $n = 3$ to $n = 1$ is [**$h = 6 \times 10^{-34} \text{ J-sec.}$**]
 (A) $1.76 \times 10^{-18} \text{ J}$ (B) $1.98 \times 10^{-18} \text{ J}$ (C) $1.76 \times 10^{-17} \text{ J}$ (D) None of these
20. In a collection of H-atom, electrons make transition from 5th excited state to 2nd excited state then maximum number of different types of photons observed are
 (A) 3 (B) 4 (C) 6 (D) 15
21. The difference in the wavelength of the 1st line of Lyman series and 2nd line of Balmer series in a hydrogen atom is
 (A) $\frac{9}{2R}$ (B) $\frac{4}{R}$ (C) $\frac{88}{15R}$ (D) None
22. The wave number of electromagnetic radiation emitted during the transition of electron in between two levels of Li^{2+} ion whose principal quantum numbers sum is 4 and difference is 2 is
 (A) $3.5 R$ (B) $4 R$ (C) $8 R$ (D) $\frac{8}{9} R$

1 TABLE (3Q)

Column-I	Column-II	Column-III
(A) Energy of electron in H-like atomic orbital	(P) Principal quantum number(n)	(I) Radial function [R(r)]
(B) Orientation and shape of H-like atomic orbital	(Q) Azimuthal quantum number (l)	(II) Angular function
(C) Magnitude & direction of spin angular momentum	(R) Magnetic quantum number (m)	(III) Wave function [Ψ]
(D) Probability density of electron in s-orbital of H-like atom/ion	(S) Spin quantum number(s)	(IV) Cannot be determined from Schrodinger equation

Assume no external electric or magnetic fields in Q.25 to Q.26

23. Which of the following option is incorrectly matched -
 (A) B – Q – II (B) B – R – II (C) B – P – I (D) B – Q – III
24. Which option is correctly matched ?
 (A) C – S – III (B) D – R – III (C) A – P – IV (D) C – S – IV
25. If option (A) is considered in presence of external electric or magnetic field then which option is incorrect -
 (A) A – P – III (B) A – Q – III (C) A – R – III (D) A – S – I

EXERCISE # J-MAINS

- An atom has a mass of 0.02 kg and uncertainty in its velocity is 9.218×10^{-6} m/s then uncertainty in position is ($h = 6.626 \times 10^{-34}$ Js) [AIEEE 2002]
 (1) 2.86×10^{-28} m (2) 2.86×10^{-32} cm (3) 1.5×10^{-27} m (4) 3.9×10^{-10} m
- Energy of H- atom in the ground state is -13.6 eV, Hence energy in the second excited state is- [AIEEE 2002]
 (1) -6.8 eV (2) -3.4 eV (3) -1.51 eV (4) -4.3 eV
- Uncertainty in position of a particle of 25 g in space is 10^{-5} m. Hence uncertainty in velocity (ms^{-1}) is (Planck's constant $h = 6.6 \times 10^{-34}$ Js) [AIEEE-2002]
 (1) 2.1×10^{-28} (2) 2.1×10^{-34} (3) 0.5×10^{-34} (4) 5.0×10^{-24}
- The orbital angular momentum for an electron revolving in an orbit is given by $\sqrt{\ell(\ell+1)} \cdot \frac{h}{2\pi}$. This momentum for an s-electron will be given by [AIEEE-2003]
 (1) $\sqrt{2} \cdot \frac{h}{2\pi}$ (2) $+\frac{1}{2} \cdot \frac{h}{2\pi}$ (3) zero (4) $\frac{h}{2\pi}$
- The number of d-electrons retained in Fe^{2+} (At. no. of Fe = 26) ion is : [AIEEE-2003]
 (1) 6 (2) 3 (3) 4 (4) 5
- The de Broglie wavelength of a tennis ball of mass 60 g moving with a velocity of 10 metres per second is approximately : [AIEEE 2003]
 (1) 10^{-25} metres (2) 10^{-33} metres (3) 10^{-31} metres (4) 10^{-16} metres
- In Balmer series of lines of hydrogen spectrum, the third line from the red end corresponds to which one of the following inter-orbit jumps of the electron for Bohr orbits in an atom of hydrogen ? [AIEEE-2003]
 (1) $2 \rightarrow 5$ (2) $3 \rightarrow 2$ (3) $5 \rightarrow 2$ (4) $4 \rightarrow 1$
- Which of the following sets of quantum number is correct for an electron in 4f orbital ? [AIEEE-2004]
 (1) $n = 3, l = 2, m = -2, s = +\frac{1}{2}$ (2) $n = 4, l = 4, m = -4, s = -\frac{1}{2}$
 (3) $n = 4, l = 3, m = +1, s = +\frac{1}{2}$ (4) $n = 4, l = 3, m = +4, s = +\frac{1}{2}$
- The wavelength of the radiation emitted, when in a hydrogen atom electron falls from infinity to stationary state 1, would be [AIEEE-2004]
 (Rydberg constant = $1.097 \times 10^7 \text{ m}^{-1}$) :
 (1) 9.1×10^{-8} nm (2) 192 nm (3) 406 nm (4) 91 nm
- Which of the following statements in relation to the hydrogen atom is correct ? [AIEEE-2005]
 (1) 3s, 3p and 3d orbitals all have the same energy
 (2) 3s and 3p orbitals are of lower energy than 3d orbitals
 (3) 3p orbital is lower in energy than 3d orbital
 (4) 3s orbitals is lower in energy than 3p orbital

11. According to Bohr's theory angular momentum of electron in 5th shell is :- [AIEEE-2006]
 (1) $1.0 \text{ h}/\pi$ (2) $10 \text{ h}/\pi$ (3) $2.5 \text{ h}/\pi$ (4) $25 \text{ h}/\pi$
12. Uncertainty in the position of an electron (mass = $9.1 \times 10^{-31} \text{ Kg}$) moving with a velocity 300 ms^{-1} , accurate upto 0.001%, will be :- ($h = 6.63 \times 10^{-34} \text{ Js}$) [AIEEE-2006]
 (1) $5.76 \times 10^{-2} \text{ m}$ (2) $1.92 \times 10^{-2} \text{ m}$ (3) $3.84 \times 10^{-2} \text{ m}$ (4) $19.2 \times 10^{-2} \text{ m}$
13. The ionization enthalpy of hydrogen atom is $1.312 \times 10^6 \text{ J mol}^{-1}$. The energy required to excite the electron in the atom from $n=1$ to $n=2$ is [AIEEE-2008]
 (1) $8.51 \times 10^5 \text{ J mol}^{-1}$ (2) $6.56 \times 10^5 \text{ J mol}^{-1}$ (3) $7.56 \times 10^5 \text{ J mol}^{-1}$ (4) $9.84 \times 10^5 \text{ J mol}^{-1}$
14. In an atom, an electron is moving with a speed of 600 m/s with an accuracy of 0.005%. Certainty with which the position of the electron can be located is ($h = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$, mass of electron, $m_e = 9.1 \times 10^{-31} \text{ kg}$):- [AIEEE-2009]
 (1) $1.92 \times 10^{-3} \text{ m}$ (2) $3.84 \times 10^{-3} \text{ m}$ (3) $1.52 \times 10^{-4} \text{ m}$ (4) $5.10 \times 10^{-3} \text{ m}$
15. Calculate the wavelength (in nanometer) associated with a proton moving at $1.0 \times 10^3 \text{ ms}^{-1}$ (Mass of proton = $1.67 \times 10^{-27} \text{ kg}$ and $h = 6.63 \times 10^{-34} \text{ Js}$) :- [AIEEE-2009]
 (1) 2.5 nm (2) 14.0 nm (3) 0.032 nm (4) 0.40 nm
16. The energy required to break one mole of Cl-Cl bonds in Cl_2 is 242 kJ mol^{-1} . The longest wavelength of light capable of breaking a single Cl-Cl bond is ($C = 3 \times 10^8 \text{ ms}^{-1}$ and $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$) [AIEEE-2010]
 (1) 494 nm (2) 594 nm (3) 640 nm (4) 700 nm
17. Ionisation energy of He^+ is $19.6 \times 10^{-18} \text{ J atom}^{-1}$. The energy of the first stationary state ($n=1$) of Li^{2+} is:- [AIEEE-2010]
 (1) $8.82 \times 10^{-17} \text{ J atom}^{-1}$ (2) $4.41 \times 10^{-16} \text{ J atom}^{-1}$
 (3) $-4.41 \times 10^{-17} \text{ J atom}^{-1}$ (4) $-2.2 \times 10^{-15} \text{ J atom}^{-1}$
18. A gas absorbs a photon of 355 nm and emits at two wavelengths. If one of the emissions is at 680 nm, the other is at :- [AIEEE-2011]
 (1) 743 nm (2) 518 nm (3) 1035 nm (4) 325 nm
19. The frequency of light emitted for the transition $n=4$ to $n=2$ of He^+ is equal to the transition in H atom corresponding to which of the following [AIEEE-2011]
 (1) $n=3$ to $n=1$ (2) $n=2$ to $n=1$ (3) $n=3$ to $n=2$ (4) $n=4$ to $n=3$
20. The electrons identified by quantum numbers n and ℓ :- [AIEEE-2012]
 (a) $n=4, \ell=1$ (b) $n=4, \ell=0$ (c) $n=3, \ell=2$ (d) $n=3, \ell=1$
 Can be placed in order of increasing energy as
 (1) (a) < (c) < (b) < (d) (2) (c) < (d) < (b) < (a)
 (3) (d) < (b) < (c) < (a) (4) (b) < (d) < (a) < (c)
21. If the kinetic energy of an electron is increased four times, the wavelength of the de-Broglie wave associated with it would become :- [JEE-Main(online) 2012]
 (1) Two times (2) Half (3) One fourth (4) Four times
22. If the radius of first orbit of H atom is a_0 , the de-Broglie wavelength of an electron in the third orbit is :- [JEE-Main(online) 2012]
 (1) $6 \pi a_0$ (2) $8 \pi a_0$ (3) $2 \pi a_0$ (4) $4 \pi a_0$

23. The wave number of the first emission line in the Balmer series of H-Spectrum is :
(R = Rydberg constant) : **[JEE-Main(online) 2013]**
- (1) $\frac{3}{4}R$ (2) $\frac{9}{400}R$ (3) $\frac{5}{36}R$ (4) $\frac{7}{6}R$
24. The de Broglie wavelength of a car of mass 1000 kg and velocity 36 km/hr is :
($h = 6.63 \times 10^{-34}$ Js) **[JEE-Main(online) 2013]**
- (1) 6.626×10^{-31} m (2) 6.626×10^{-34} m
(3) 6.626×10^{-38} m (4) 6.626×10^{-30} m
25. For which of the following particles will it be most difficult to experimentally verify the de-Broglie relationship? **[JEE-Main(online) 2014]**
- (1) a dust particle (2) an electron (3) a proton (4) an α -particle.
26. If the binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of Li^{++} is : **[JEE-Main(online) 2014]**
- (1) 13.6 eV (2) 30.6 eV (3) 122.4 eV (4) 3.4 eV
27. Based on the equation **[JEE-Main(online) 2014]**
- $$\Delta E = -2.0 \times 10^{-18} J \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$
- the wavelength of the light that must be absorbed to excite hydrogen electron from level $n=1$ to level $n=2$ will be ($h = 6.625 \times 10^{-34}$ Js, $C = 3 \times 10^8$ ms $^{-1}$) **[JEE-Main(online) 2014]**
- (1) 2.650×10^{-7} m (2) 1.325×10^{-7} m
(3) 1.325×10^{-10} m (4) 5.300×10^{-10} m
28. If λ_0 and λ be the threshold wavelength and wavelength of incident light, the velocity of photoelectron ejected from the metal surface is **[JEE-Main(online) 2014]**
- (1) $\sqrt{\frac{2hc}{m} \left(\frac{\lambda_0 - \lambda}{\lambda \lambda_0} \right)}$ (2) $\sqrt{\frac{2h}{m} \left(\frac{1}{\lambda_0} - \frac{1}{\lambda} \right)}$ (3) $\sqrt{\frac{2h}{m} (\lambda_0 - \lambda)}$ (4) $\sqrt{\frac{2hc}{m} (\lambda_0 - \lambda)}$
29. Ionization energy of gaseous Na atoms is 495.5 kJmol $^{-1}$. The lowest possible frequency of light that ionizes a sodium atom is **[JEE-Main(online) 2014]**
- ($h = 6.626 \times 10^{-34}$ Js, $N_A = 6.022 \times 10^{23}$ mol $^{-1}$)
- (1) 3.15×10^{15} s $^{-1}$ (2) 4.76×10^{14} s $^{-1}$ (3) 1.24×10^{15} s $^{-1}$ (4) 7.50×10^4 s $^{-1}$
30. Which of the following is the energy of a possible excited state of hydrogen? **[JEE-Main(offline) 2015]**
- (1) -3.4 eV (2) +6.8 eV (3) +13.6 eV (4) -6.8 eV
31. At temperature T, the average kinetic energy of any particle is $\frac{3}{2}$ kT- The de Broglie wavelength follows the order : **[JEE-Main(online) 2015]**
- (1) Visible photon > Thermal electron > Thermal neutron
(2) Thermal proton > Thermal electron > Visible photon
(3) Visible photon > Thermal neutron > Thermal electron
(4) Thermal proton > Visible photon > Thermal electron

32. A stream of electrons from a heated filament was passed between two charged plates kept at a potential difference V volts. If e and m are charge and mass of an electron respectively, then the value of h/λ (where λ is wavelength associated with electron wave) is given by : [JEE-Main(online) 2016]
- (1) $\sqrt{2meV}$ (2) meV (3) $2meV$ (4) $\sqrt{meV}$
33. The radius of the second Bohr orbit for hydrogen atom is : [JEE-Main(offline) 2017]
- (Plank's const. $h = 6.6262 \times 10^{-34}$ Js ; mass of electron = 9.1091×10^{-31} kg ; charge of electron $e = 1.60210 \times 10^{-19}$ C ; permittivity of vacuum $\epsilon_0 = 8.854185 \times 10^{-12}$ kg $^{-1}$ m 3 A 2)
- (1) $1.65\text{\AA}$ (2) $4.76\text{\AA}$ (3) $0.529\text{\AA}$ (4) $2.12\text{\AA}$
34. If the shortest wavelength in Lyman series of hydrogen atom is A , then the longest wavelength in Paschen series of He^+ is : [JEE-Main(online) 2017]
- (1) $\frac{36A}{5}$ (2) $\frac{9A}{5}$ (3) $\frac{36A}{7}$ (4) $\frac{5A}{9}$
35. The electron in the hydrogen atom undergoes transition from higher orbitals to orbital of radius 211.6 pm. This transition is associated with:- [JEE-Main(online) 2017]
- (1) Brackett series (2) Balmer series (3) Lyman series (4) Paschen series
36. The de-Broglie's wavelength of electron present in first Bohr orbit of 'H' atom is :- [JEE-Main(online) 2018]
- (1) $\frac{0.529}{2\pi}\text{\AA}$ (2) $2\pi \times 0.529\text{\AA}$ (3) $0.529\text{\AA}$ (4) $4 \times 0.529\text{\AA}$
37. Ejection of the photoelectron from metal in the photoelectric effect experiment can be stopped by applying 0.5 V when the radiation of 250 nm is used. The work function of the metal is : [JEE-Main(online) 2018]
- (1) 5 eV (2) 4 eV (3) 5.5 eV (4) 4.5 eV
38. Which of the following statements is false ? [JEE-Main(online) 2018]
- (1) Photon has momentum as well as wavelength.
 (2) Splitting of spectral lines in electrical field is called Stark effect.
 (3) Frequency of emitted radiation from a black body goes from a lower wavelength to higher wavelength as the temperature increases.
 (4) Rydberg constant has unit of energy.

EXERCISE # J-ADVANCE

- With what velocity should an α -particle travel towards the nucleus of a Cu atom so as to arrive at a distance 10^{-13} m. [JEE 1997]
- A compound of Vanadium has magnetic moment of 1.73 BM work out electronic configuration of Vanadium Ion in the compound. [JEE 1997]
- The energy of an electron in the first Bohr orbit of H atom is -13.6 eV. The possible energy value(s) of the excited state(s) for electrons in Bohr orbits of hydrogen is/are :
(A) -3.4 eV (B) -4.2 eV (C) -6.8 eV (D) $+6.8$ eV [JEE 1998]
- The number of nodal planes in a p_x orbital is: [JEE 2000]
(A) one (B) two (C) three (D) zero
- Calculate the energy required to excite one litre of hydrogen gas at 1 atm and 298K to the first excited state of atomic hydrogen. The energy for the dissociation of $H-H$ is 436 kJ mol $^{-1}$.
- The quantum numbers $+1/2$ and $-1/2$ for the electron spin represent: [JEE 2001]
(A) rotation of the electron in clockwise and anticlockwise direction respectively.
(B) rotation of the electron in anticlockwise and clockwise direction respectively.
(C) magnetic moment of the electron pointing up and down respectively.
(D) two quantum mechanical spin states which have no classical analogue.
- Rutherford's experiment, which established the nuclear model of atom, used a beam of :—
(A) β - particles, which impinged on a metal foil and get absorbed. [JEE 2002]
(B) γ - rays, which impinged on a metal foil and ejected electron.
(C) Helium atoms, which impinged on a metal foil and got scattered.
(D) Helium nuclei, which impinged on a metal foil and got scattered.
- The magnetic moment of cobalt of the compound $Hg[Co(SCN)_4]$ is [Given : Co^{+2}]
(A) $\sqrt{3}$ (B) $\sqrt{8}$ (C) $\sqrt{15}$ (D) $\sqrt{24}$ [JEE 2004]
- The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom?
(A) He^+ ($n=2$) (B) Li^{2+} ($n=2$) (C) Li^{2+} ($n=3$) (D) Be^{3+} ($n=2$) [JEE 2004]
- (a) The Schrodinger wave equation for hydrogen atom is [IIT-2004]

$$\psi_{2s} = \frac{1}{4(2\pi)^{1/2}} \left(\frac{1}{a_0} \right)^{3/2} \left(2 - \frac{r_0}{a_0} \right) e^{-r/a}$$

Where a_0 is Bohr's radius. Let the radial node in 2s be at r_0 . Then find r_0 in terms of a_0 .
 (b) A baseball having mass 100 g moves with velocity 100 m/s. Find out the value of wavelength of baseball.
- (a) Calculate velocity of electron in first Bohr orbit of hydrogen atom (Given $r = a_0$) [IIT-2005]
 (b) Find de-Broglie wavelength of the electron in first Bohr orbit.
 (c) Find the orbital angular momentum of 2p-orbital in terms of $h/2\pi$ units.

12. Given in hydrogenic atom r_n , V_n , E , K_n stand for radius, potential energy, total energy and kinetic energy in n^{th} orbit. Find the value of U, v, x, y . [JEE 2006]

(A) $U = \frac{V_n}{K_n}$ (P) 1

(B) $\frac{1}{r_n} \propto E^x$ (Q) -2

(C) $r_n \propto Z^y$ (R) -1
(Z = Atomic number)

(D) v = (Orbital angular momentum of electron in its lowest energy) (S) 0

13. Match the entries in **Column I** with the correctly related quantum number(s) in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

[JEE 2008]

Column I

- (A) Orbital angular momentum of the electron in a hydrogen-like atomic orbital
(B) A hydrogen-like one-electron wave function obeying Pauli principle
(C) Shape, size and orientation of hydrogen like atomic orbitals
(D) Probability density of electron at the nucleus in hydrogen-like atom

Column II

- (P) Principal quantum number
(Q) Azimuthal quantum number
(R) Magnetic quantum number
(S) Electron spin quantum number

Paragraph for questions 14 to 16

[JEE 2010]

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2 . The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

14. The state S_1 is :-

- (A) 1s (B) 2s (C) 2p (D) 3s

15. Energy of the state S_1 in units of the hydrogen atom ground state energy is :-

- (A) 0.75 (B) 1.50 (C) 2.25 (D) 4.50

16. The orbital angular momentum quantum number of the state S_2 is :-

- (A) 0 (B) 1 (C) 2 (D) 3

17. The maximum number of electrons that can have principal quantum number, $n=3$, and spin quantum number, $m_s = -1/2$, is [JEE 2011]

18. The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is :- [JEE 2011]

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
ϕ (eV)	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

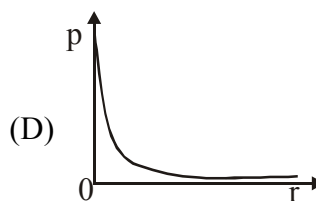
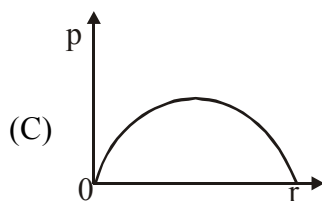
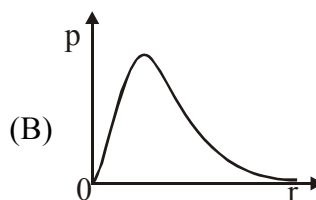
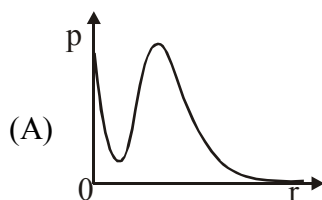
19. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius] [JEE 2012]

(A) $\frac{h^2}{4\pi^2ma_0^2}$ (B) $\frac{h^2}{16\pi^2ma_0^2}$ (C) $\frac{h^2}{32\pi^2ma_0^2}$ (D) $\frac{h^2}{32\pi^2ma_0^2}$

20. The atomic masses of He and Ne are 4 and 20 a.m.u. respectively. The value of the de Broglie wavelength of He gas at -73°C is "M" times that of the de Broglie wavelength of Ne at 727°C . M is. [JEE 2013]

21. In an atom, the total number of electrons having quantum numbers $n = 4$, $|m_l| = 1$ and $m_s = -\frac{1}{2}$ is [JEE 2014]

22. P is the probability of finding the 1s electron of hydrogen atom in a spherical shell of infinitesimal thickness, dr , at a distance r from the nucleus. The volume of this shell is $4\pi r^2 dr$. The qualitative sketch of the dependence of P on r is - [JEE 2016]



Answer Q.23, Q.24 and Q.25 by appropriately matching the information given in the three columns of the following table. [JEE 2017]

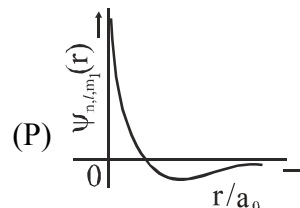
Column-1

Column-2

Column-3

(I) 1s orbital

$$(i) \psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_0}\right)}$$



(II) 2s orbital

(ii) One radial node

(Q) Probability density at

$$\text{nucleus} \propto \frac{1}{a_0^3}$$

(III) 2p_z orbital

$$(iii) \psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{5}{2}} re^{-\left(\frac{Zr}{2a_0}\right)} \cos \theta$$

(R) Probability density is

maximum at nucleus

(IV) 3d_{z²} orbital

(iv) xy - plane is a nodal plane

(S) Energy needed to excite electron from n = 2 state to n = 4 state is $\frac{27}{32}$ times the energy needed to excite electron from n = 2 state to n = 6 state

23. For the given orbital in column 1, the only **CORRECT** combination for any hydrogen - like species is :

(A) (IV) (iv) (R) (B) (II) (ii) (P) (C) (III) (iii) (P) (D) (I) (ii) (S)

24. For He⁺ ion, the only **INCORRECT** combination is

(A) (II) (ii) (Q) (B) (I) (i) (S) (C) (I) (i) (R) (D) (I) (iii) (R)

25. For hydrogen atom, the only **CORRECT** combination is

(A) (I) (iv) (R) (B) (I) (i) (P) (C) (II) (i) (Q) (D) (I) (i) (S)

ANSWER KEY

EXERCISE # S-I

1. (1 : 1)
2. (4 : 9)
3. $n < \alpha < p < e$
4. 6.0×10^{-19} unit.
5. 1.6×10^{-19} Coulomb.
6. $8 \times 10^{-15} : 1$
7. $2.3 \times 10^{17} \text{ kg/m}^3$
8. 1 : 2
9. $1.28 \times 10^{-14} \text{ m}$.
10. $1.44 \times 10^7 \text{ m/s}$
11. 1500 Å
12. 150 KHz
13. 1 : 5
14. $7.5 \times 10^{12} \text{ Hz}$
15. $4.96 \times 10^{-19} \text{ Joule}$
16. 1.0×10^5
17. 10^{22}
18. 0.5
19. 6200 Å
20. 3×10^{22}
21. ($2 \times 10^{15} \text{ Hz}$, $1.5 \times 10^{-7} \text{ m}$)
22. (1)
23. 50 %
24. 3.06 V
25. Ans. 0.41375 eV
26. Ans.(0.2645) Å
27. Ans.(x = 4)
28. $5.44 \times 10^5 \text{ m/s}$
29. 2
30. 8×10^6
31. $6530 \times 10^{12} \text{ Hz}$
32. $-1.36 \times 10^{-19} \text{ Joules}$
33. h/π
34. (2)
35. 1.18×10^7
36. ($-217.9 \times 10^{-20} \text{ J}$)
37. 340 eV, - 680 eV
38. (6)
39. 6563 Å ; 1216 Å ; 1026 Å
40. 10.2 eV, $z = 2$
41. 113.74 Å
42. (i) $\frac{x}{4}$; (ii) $\frac{36x}{7}$
43. (3)
44. (4)
45. $3.09 \times 10^8 \text{ cm/sec}$
46. (3)
47. (8)
48. (4)
49. $3.68 \times 10^{-65} \text{ m}$
50. 10 pm
51. 0.0826 volts
52. $1.05 \times 10^{-13} \text{ m}$
53. (0.175 cm)
54. Ans (8)
56. $\frac{9+3\sqrt{3}}{2} a_0$, $\frac{9-3\sqrt{3}}{2} a_0$
57. (3)
58. (2)
59. (0.423 Å)
60. (4)

EXERCISE # S-II

1. (20)
2. $E = \frac{n^6 h^6}{384 m^3 K^2 e^4 \pi^6}$
3. (i) 1, (ii) 2, (iii) 3, (iv) 3
4. - 435.2 eV
5. (90 nm)
6. 91 nm
7. (a) $Z = 3$; (b) $n_1 = 12$, $n_2 = 15$; (c) $\lambda_{\max} = 15406 \text{ nm}$
8. $R \left(\frac{8}{9} \right)$
9. 973.5 Å
10. 303.89 Å , $2.645 \times 10^{-9} \text{ cm}$
11. (a) 75 eV ; (b) 1.414 Å ; (c) $2 \times 10^{-14} \text{ m}$
12. Ans. ($5 \times 10^{-26} \text{ m}$)
13. $3.63 \times 10^6 \text{ m}^{-1}$
14. Ans. (1240 kJ/mol.)
15. 300303

EXERCISE # O-I

- | | | |
|--------------|-------------|-------------|
| 1. Ans.(C) | 2. Ans.(C) | 3. Ans.(C) |
| 4. Ans.(C) | 5. Ans.(A) | 6. Ans.(D) |
| 7. Ans.(D) | 8. Ans.(D) | 9. Ans.(D) |
| 10. Ans.(C) | 11. Ans.(A) | 12. Ans.(D) |
| 13. Ans.(B) | 14. Ans.(B) | 15. Ans.(B) |
| 16. Ans.(B) | 17. Ans.(B) | 18. Ans.(C) |
| 19. Ans. (A) | 20. Ans.(C) | 21. Ans.(A) |
| 22. Ans.(B) | 23. Ans.(A) | 24. Ans.(B) |
| 25. Ans.(B) | 26. Ans.(A) | 27. Ans.(C) |
| 28. Ans.(C) | 29. Ans.(A) | 30. Ans.(D) |
| 31. Ans.(A) | 32. Ans.(C) | 33. Ans.(B) |
| 34. Ans.(D) | 35. Ans.(C) | 36. Ans.(B) |
| 37. Ans.(C) | 38. Ans.(D) | 39. Ans.(A) |
| 40. Ans.(C) | 41. Ans.(A) | 42. Ans.(D) |
| 43. Ans.(B) | 44. Ans.(C) | 45. Ans.(C) |
| 46. Ans.(A) | 47. Ans.(D) | 48. Ans.(B) |
| 49. Ans.(A) | 50. Ans.(B) | 51. Ans.(B) |
| 52. Ans.(C) | 53. Ans.(A) | 54. Ans.(C) |
| 55. Ans.(C) | 56. Ans.(A) | 57. Ans.(A) |
| 58. Ans.(A) | 59. Ans.(B) | 60. Ans.(B) |
| 61. Ans.(C) | 62. Ans.(D) | 63. Ans.(B) |
| 64. Ans.(C) | 65. Ans.(B) | 66. Ans.(C) |
| 67. Ans.(C) | 68. Ans.(D) | 69. Ans.(D) |
| 70. Ans.(C). | 71. Ans.(B) | 72. Ans.(B) |

EXERCISE # O-II

- | | | |
|---|---------------|------------------|
| 1. Ans.(D) | 2. Ans.(B) | 3. Ans.(A) |
| 4. Ans.(B) | 5. Ans.(A) | 6. Ans.(D) |
| 7. Ans.(A,B,D) | 8. Ans.(B,D) | 9. Ans. (A,D) |
| 10. Ans.(A,C) | 11. Ans.(A,C) | 12. Ans. (B,C,D) |
| 13. Ans.(C,D) | | |
| 14. Ans.(A)-P ; (B)-P,Q,S ; (C)-P,R ; (D)-Q,S | | |
| 15. Ans.(A)-S ; (B)-R ; (C)-Q ; (D)-P | | |
| 16. Ans.(A) | 17. Ans.(C) | 18. Ans.(A) |
| 19. Ans.(A) | 20. Ans.(C) | 21. Ans.(B) |
| 22. Ans.(C) | 23. Ans.(C) | 24. Ans.(D) |
| 25. Ans.(D) | | |

EXERCISE # J-MAINS

- | | | |
|-------------|-------------|-------------|
| 1. Ans.(1) | 2. Ans.(3) | 3. Ans.(1) |
| 4. Ans.(3) | 5. Ans.(1) | 6. Ans.(2) |
| 7. Ans.(3) | 8. Ans.(3) | 9. Ans.(4) |
| 10. Ans.(1) | 11. Ans.(3) | 12. Ans.(2) |
| 13. Ans.(4) | 14. Ans.(1) | 15. Ans.(4) |
| 16. Ans.(1) | 17. Ans.(3) | 18. Ans.(1) |
| 19. Ans.(2) | 20. Ans.(3) | 21. Ans.(2) |
| 22. Ans.(1) | 23. Ans.(3) | 24. Ans.(3) |
| 25. Ans.(1) | 26. Ans.(2) | 27. Ans.(2) |
| 28. Ans.(1) | 29. Ans.(3) | 30. Ans.(1) |
| 31. Ans.(1) | 32. Ans.(1) | 33. Ans.(4) |
| 34. Ans.(3) | 35. Ans.(2) | 36. Ans.(2) |
| 37. Ans.(4) | 38. Ans.(3) | |

EXERCISE #J-ADVANCE

- | | | |
|---|-------------------------------------|--------------|
| 1. Ans. 6.3×10^6 m/s | 2. Ans. [Ar] $3d^1$ | 3. Ans. (A) |
| 4. Ans. (A) | 5. Ans. 97.819 KJ | 6. Ans. (D) |
| 7. Ans. (D) | 8. Ans. (C) | 9. Ans. (D) |
| 10. Ans. (a) $r_0 = 2a_0$, (b) 6.626×10^{-25} Å | | |
| 11. Ans. (a) 2.197×10^6 m/s (b) 3.31 Å | (c) $\sqrt{2} \cdot \frac{h}{2\pi}$ | |
| 12. Ans. (A) Q, (B) P, (C) R, (D) S | | |
| 13. Ans. (A) Q,R (B) P, Q, R,S (C) P, Q, R (D) P, Q | | |
| 14. Ans. (B) | 15. Ans. (C) | 16. Ans. (B) |
| 17. Ans. 9 | 18. Ans. 4 | 19. Ans. (C) |
| 20. Ans. (5) | 21. Ans. (6) | 22. Ans. (B) |
| 23. Ans.(B) | 24. Ans. (D) | 25. Ans. (D) |

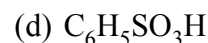
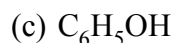
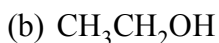
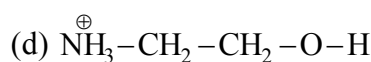
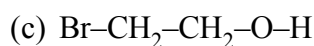
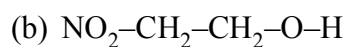
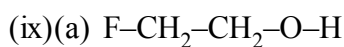
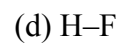
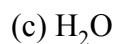
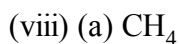
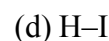
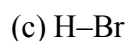
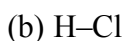
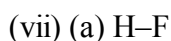
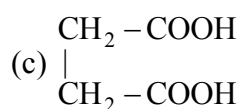
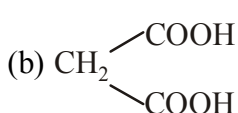
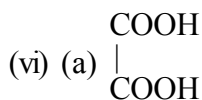
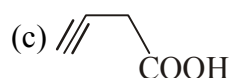
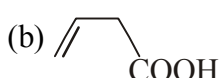
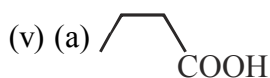
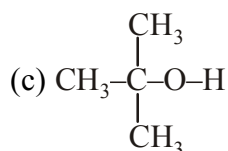
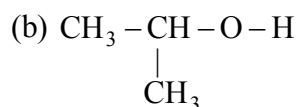
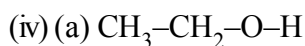
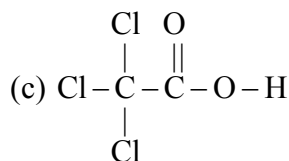
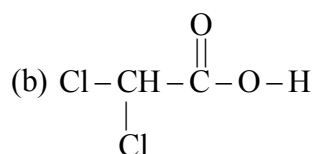
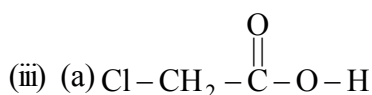
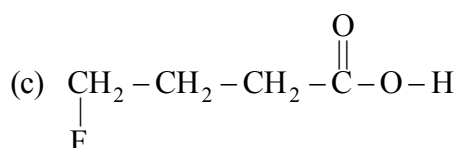
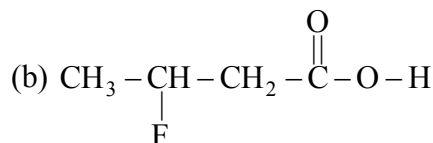
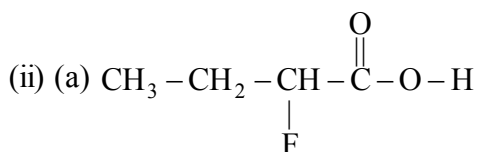
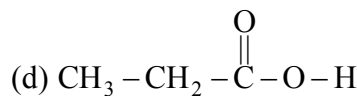
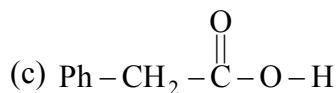
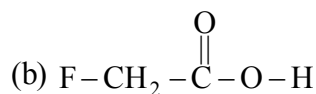
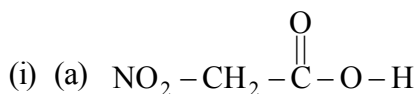
NURTURE COURSE

ACIDIC STRENGTH & BASIC STRENGTH

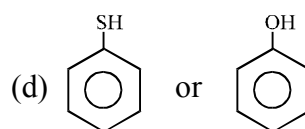
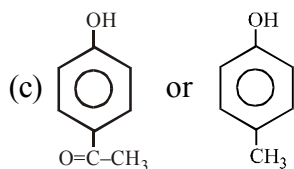
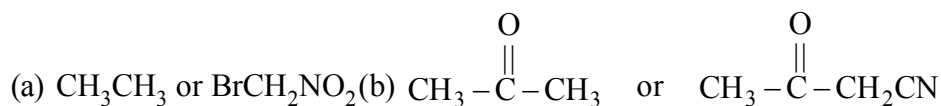
ACIDIC STRENGTH & BASIC STRENGTH

EXERCISE # I

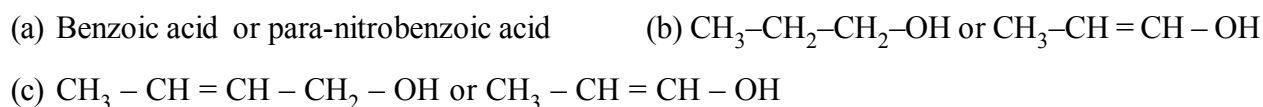
1. Write correct order of acidic strength of following compounds :



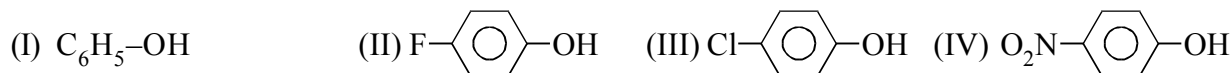
2. Explain which is a stronger acid.



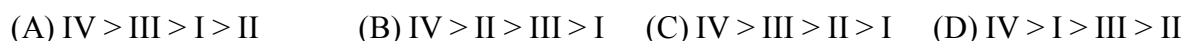
3. Which of the following would you predict to be the stronger acid ?



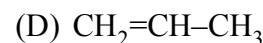
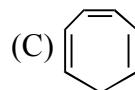
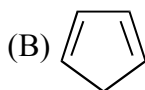
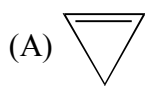
4. Arrange the given phenol & its derivative in their decreasing order of acidity :



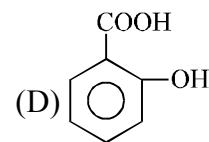
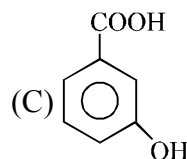
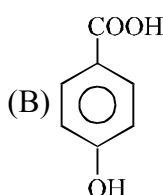
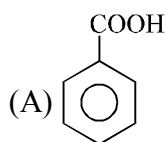
Select the correct answer from the given code:



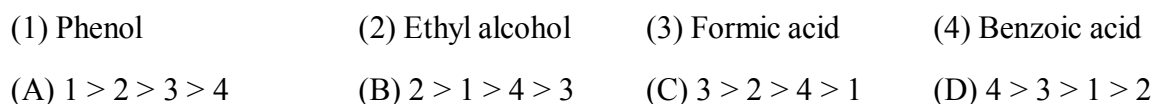
5. Which one of the following is the most acidic?



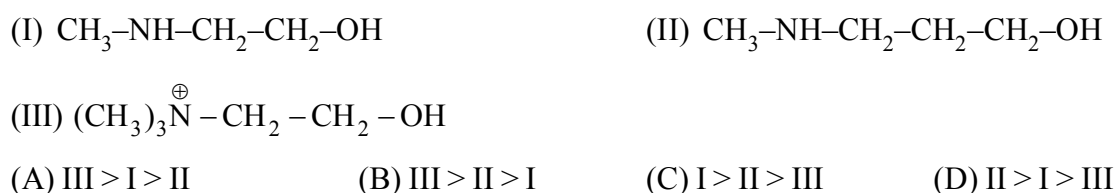
6. Which of the following is weakest acid?



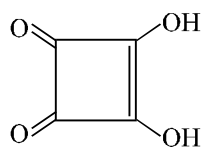
7. Arrange pH of the given compounds in decreasing order:



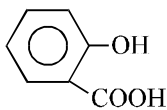
8. Arrange acidity of given compounds in decreasing order:



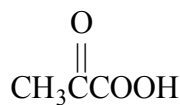
9. Consider the following compound



I



II

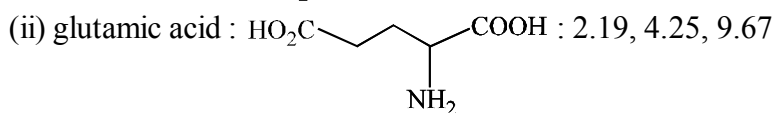
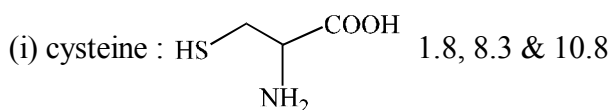


III

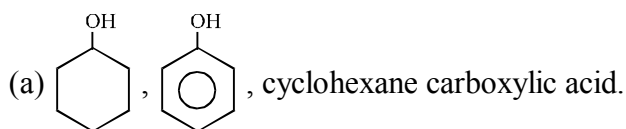
Which of the above compounds reacts with NaHCO_3 giving CO_2

- (A) I, II and III (B) I and III (C) II and III (D) I and II

10. Say which pK_a belong to which functional group in case of following amino acids :



11. Record the following sets of compounds according to increasing pK_a ($= -\log K_a$)



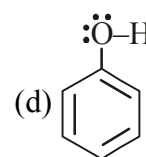
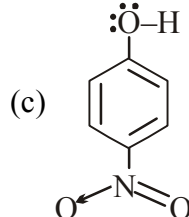
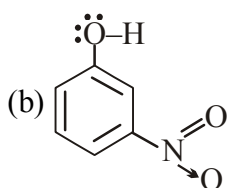
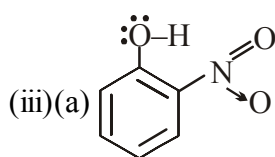
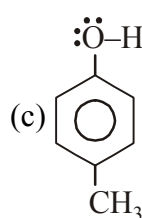
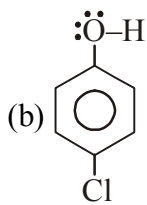
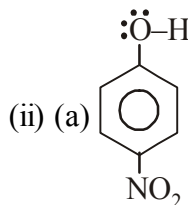
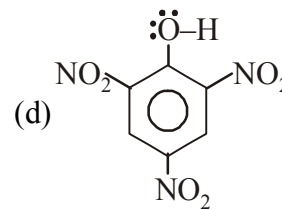
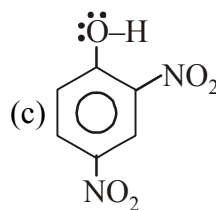
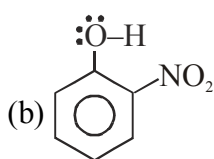
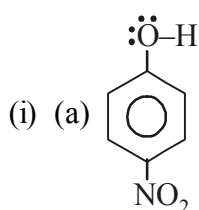
(b) 1-butyne, 1-butene, butane

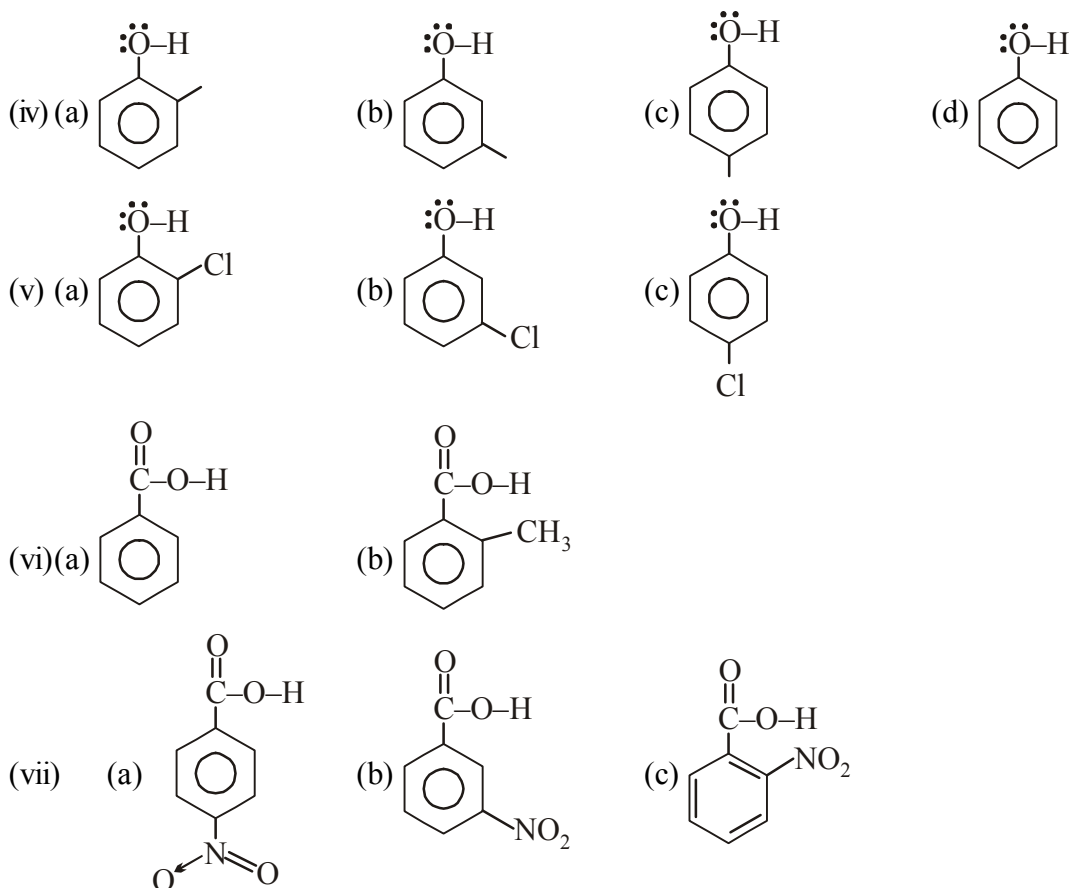
(c) Propanoic acid, 3-bromopropanoic acid, 2-nitropropanoic acid

(d) Phenol, o-nitrophenol, o-cresol

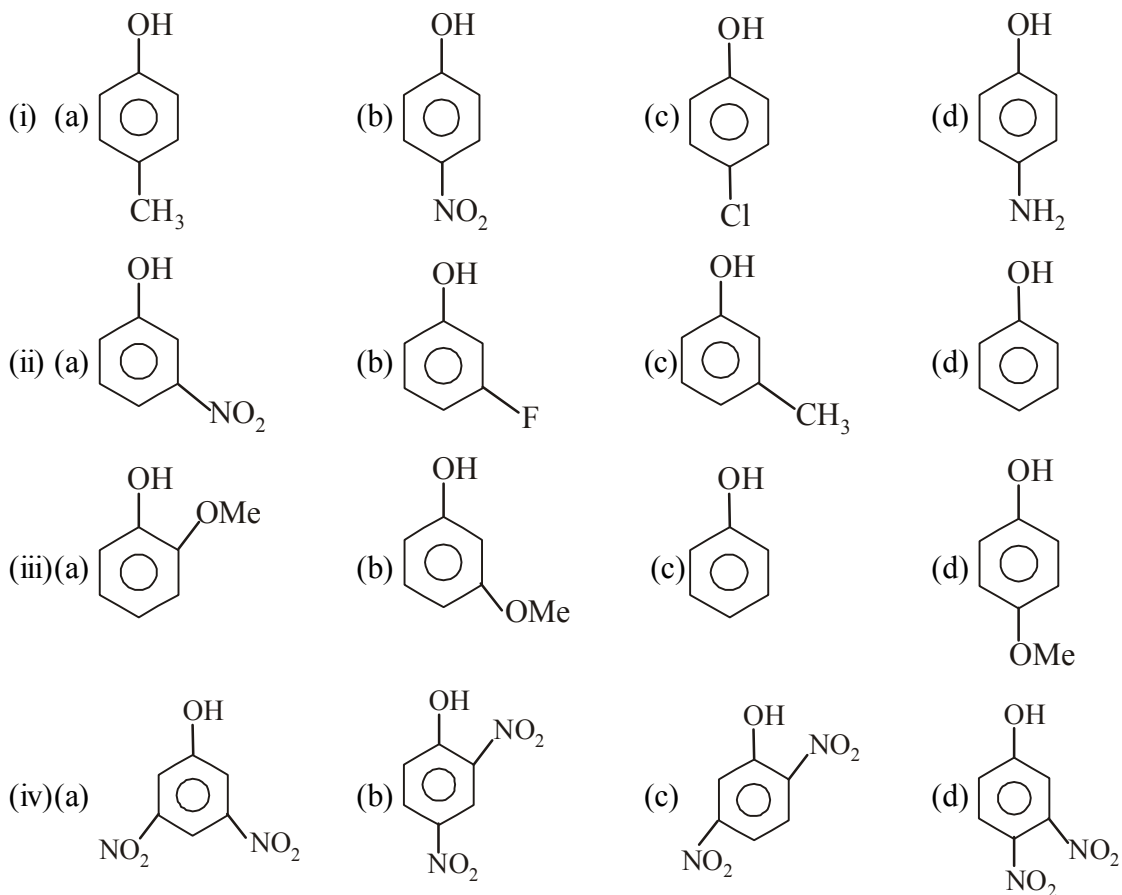
(e) Hexylamine, aniline, methylamine

12. Write correct order of acidic strength of following compounds:

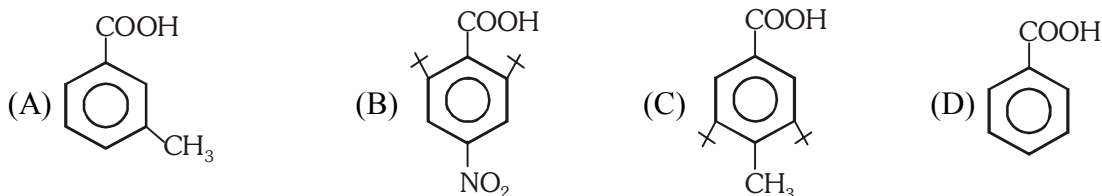




13. Select the strongest acid in each of the following sets :



14. The strongest acid is :
 (A) HF (B) $\text{CH}_3\text{CO}_2\text{H}$ (C) $\text{HF} + \text{SbF}_5$ (D) H_2S
15. The weakest acid (does not show acidic character) is :
 (A) $\text{HC} \equiv \text{CH}$ (B) $\text{CH}_2 = \text{CH}_2$ (C) Me_3CH (D) Ph_3CH
16. Which of the following is most acidic :

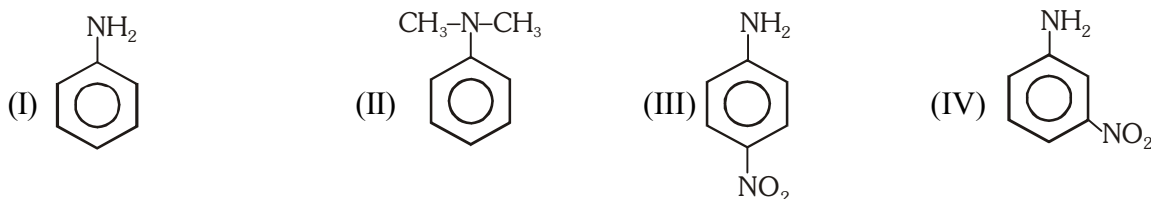


Paragraph for Question 17 to 18

The most important condition for resonance to occur is that the involved atoms in resonating structure must be coplanar or nearly coplanar for maximum delocalisation. If this condition does not fulfil, involved orbitals cannot be parallel to each other and as a consequence delocalisation cannot occur. Bulky groups present on adjacent atoms inhibit the planarity of atoms involved in resonance. This phenomenon is known as steric inhibition of resonance. Steric inhibition of resonance has profound effect on

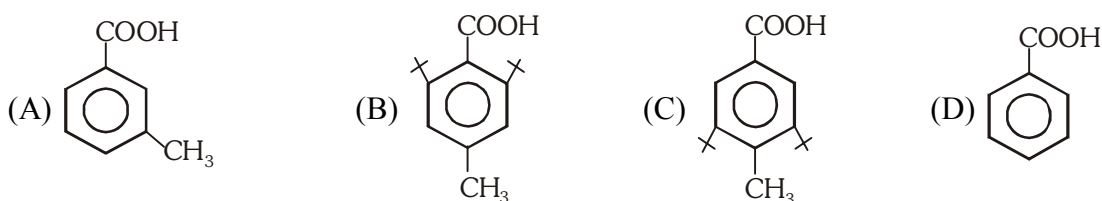
(1) Physical properties (2) Acidity and basicity (3) Reactivity of organic compounds

17. Arrange the following in the decreasing order of basicity :

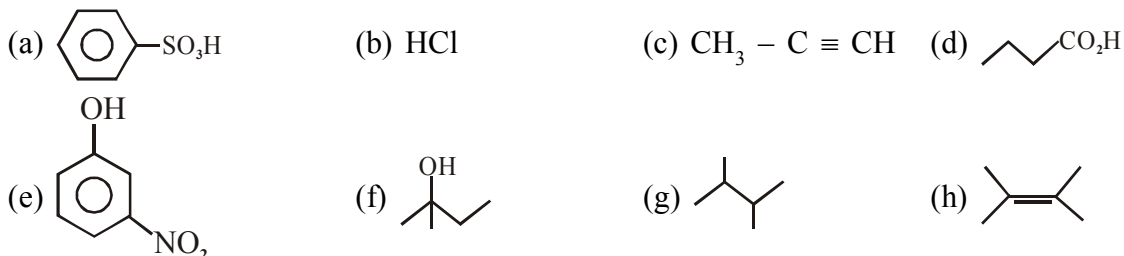


(A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{IV} > \text{III} > \text{II} > \text{I}$ (C) $\text{II} > \text{I} > \text{IV} > \text{III}$ (D) $\text{I} > \text{IV} > \text{III} > \text{II}$

18. Which of the following is most acidic :



19. How many following compounds are more acidic than water ?



(i) NaOH

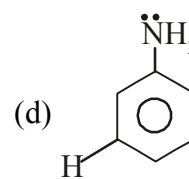
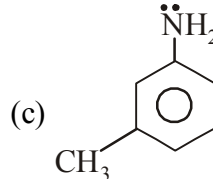
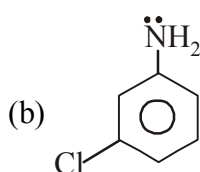
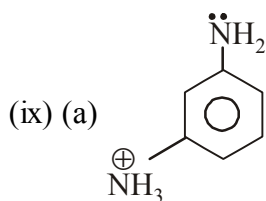
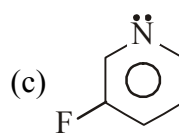
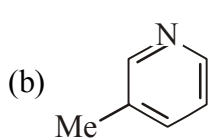
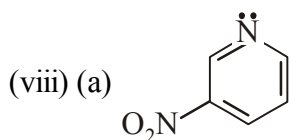
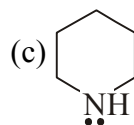
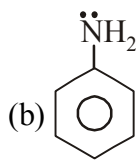
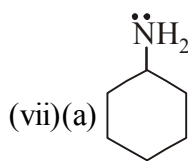
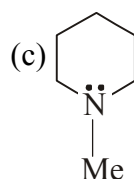
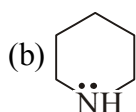
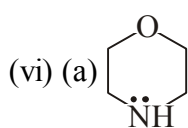
20. Select correct order regarding acidic strength of given compounds :

(1) o-methylbenzoic acid (2) m-methylbenzoic acid
 (3) p-methylbenzoic acid (4) benzoic acid
 (A) $1 > 2 > 3 > 4$ (B) $4 > 3 > 2 > 1$ (C) $1 > 4 > 2 > 3$ (D) $3 > 2 > 4 > 1$

EXERCISE # II

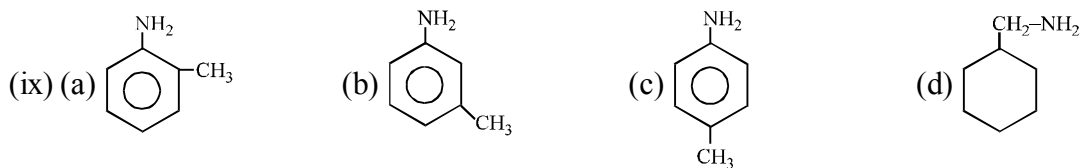
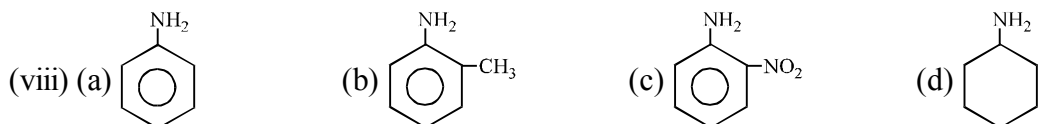
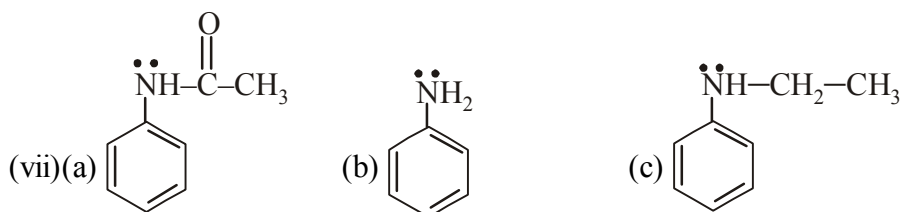
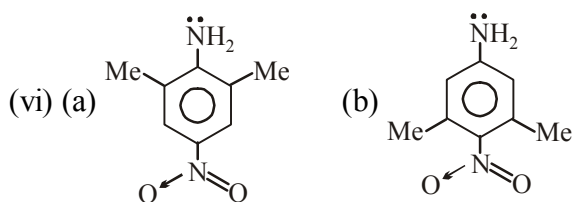
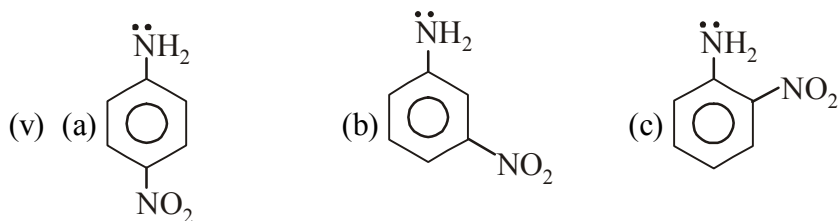
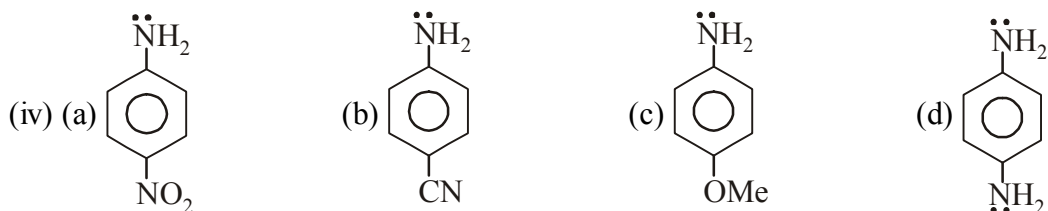
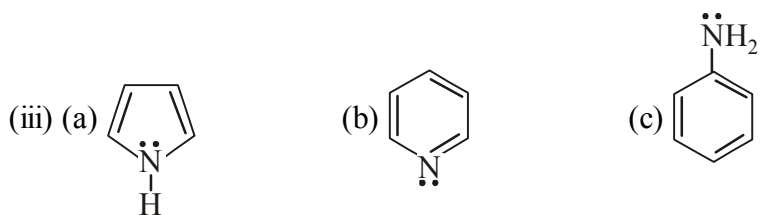
1. Write decreasing order of basic strength of following :

- (i) (a) CH_3^- (b) NH_2^- (c) OH^- (d) F^-
 (ii) (a) F^- (b) Cl^- (c) Br^- (d) I^-
 (iii) (a) NH_3 (b) MeNH_2 (c) Me_2NH (d) Me_3N (in H_2O)
 (iv) (a) NH_3 (b) MeNH_2 (c) Me_2NH (d) Me_3N (Gas phase)
 (v) (a) $\text{R}-\text{NH}_2$ (b) $\text{Ph}-\text{NH}_2$ (c) $\text{R}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{NH}_2$

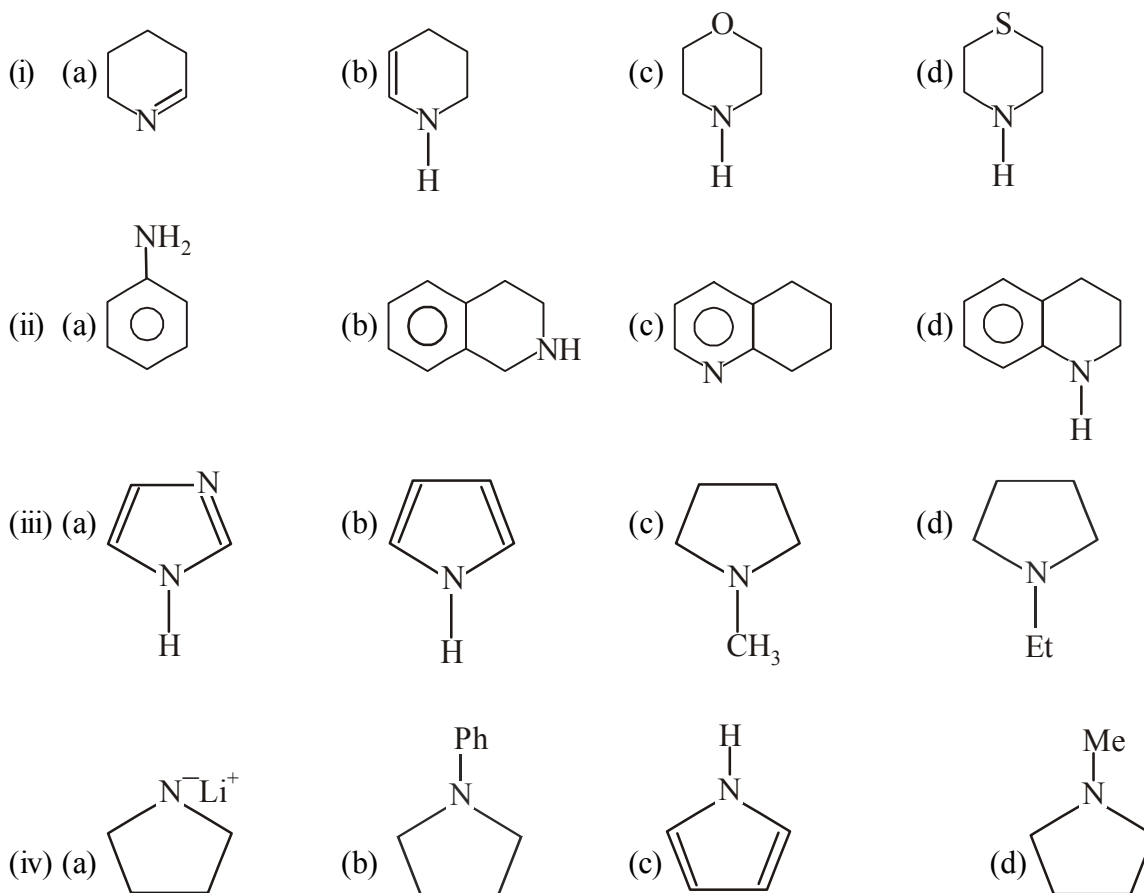


2. Write decreasing order of basic strength of following :

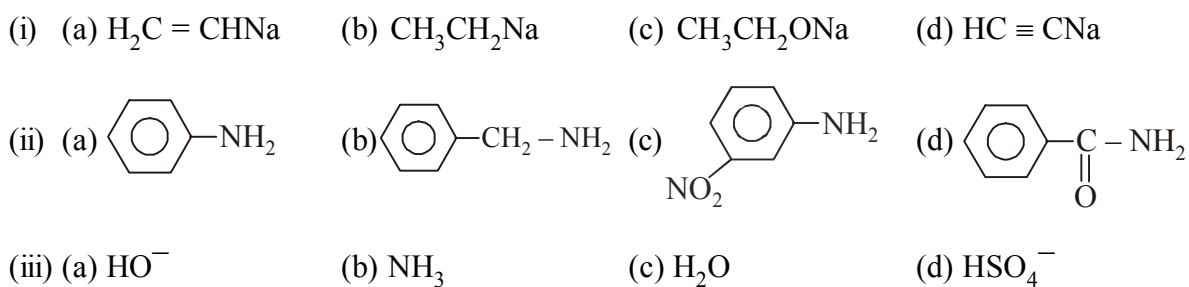
- (i) (a) $\text{CH}_3-\text{CH}_2-\ddot{\text{N}}\text{H}_2$ (b) $\text{CH}_3-\text{CH}=\ddot{\text{N}}\text{H}$ (c) $\text{CH}_3-\text{C}\equiv\ddot{\text{N}}$
 (ii) (a) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\ddot{\text{N}}\text{H}_2$ (b) $\text{CH}_3-\text{CH}_2-\ddot{\text{N}}\text{H}_2$ (c) $\text{CH}_3-\overset{\text{NH}}{\underset{\text{NH}}{\text{C}}}-\ddot{\text{N}}\text{H}_2$ (d) $\ddot{\text{N}}\text{H}_2-\overset{\text{NH}}{\underset{\text{NH}}{\text{C}}}-\ddot{\text{N}}\text{H}_2$



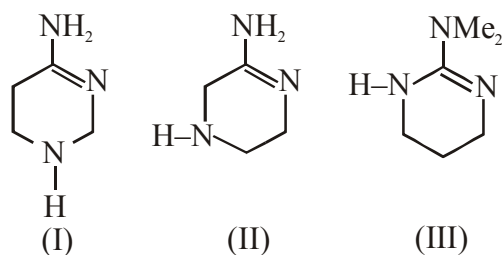
3. Select the strongest base in following compound :



4. Arrange the following compound in decreasing order of their basicity.



5. Correct decreasing order of basic strength -



Of following compound -

- (A) $\text{III} > \text{II} > \text{I}$ (B) $\text{II} > \text{I} > \text{III}$ (C) $\text{I} > \text{II} > \text{III}$ (D) $\text{III} > \text{I} > \text{II}$

6. Consider the following bases:

- (I) o-nitroaniline (II) m-nitroaniline (III) p-nitroaniline

The decreasing order of basicity is:

- (A) $II > III > I$ (B) $II > I > III$ (C) $I > II > III$ (D) $I > III > II$

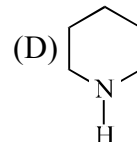
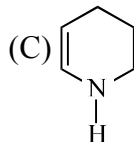
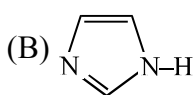
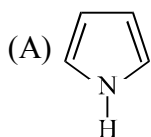
7. Consider the basicity of the following aromatic amines:

- (I) aniline (II) p-nitroaniline (III) p-methoxyaniline (IV) p-methylaniline

The correct order of decreasing basicity is:

- (A) $\text{III} > \text{IV} > \text{I} > \text{II}$ (B) $\text{III} > \text{IV} > \text{II} > \text{I}$ (C) $\text{I} > \text{II} > \text{III} > \text{IV}$ (D) $\text{IV} > \text{III} > \text{II} > \text{I}$

8. Which one of the following is least basic in character?



9. In each of the following pair of compounds, which is more basic in aqueous solution?

Give an explanation for your choice:

- (a) CH_3NH_2 or CF_3NH_2

- (b) CH_3CONH_2 or $\text{H}_2\text{N}-\text{C}(=\text{NH})-\text{NH}_2$

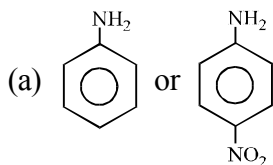
- (c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ or CH_3CN

- (d) $\text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2$ or 2,6-dimethyl-N-N-dimethylaniline

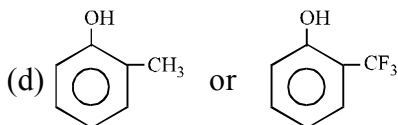
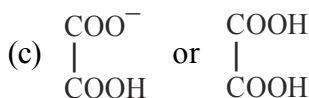
10. Choose the member of each of the following pairs of compounds that is likely to be the weaker base.

- (a) H_2O or H_3O^+ (b) Cl^- , SH^- (c) F^- , OH^- , NH_2^- , CH_3^- (d) HF , H_2O , NH_3
(e) OH^- , SH^- , SeH^-

11. Explain which compound is the weaker base.

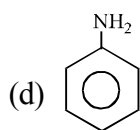


- (b) $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH} - \text{CH}_2^-$ or $\text{CH}_2 = \text{CH} - \text{CH}_2^-$

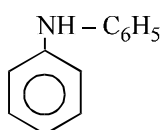


12. Arrange the basic strength of the following compounds.

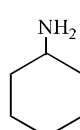
- | | | |
|--|--|--|
| (a) OH^- | CH_3COO^- | Cl^- |
| (i) | (ii) | (iii) |
| (b) $\text{CH} \equiv \text{C}^-$ | $\text{CH}_2 = \text{CH}^-$ | CH_3CH_2^- |
| (i) | (ii) | (iii) |
| (c) $\text{CH}_2 = \text{CHCH}_2\text{NH}_2$ | $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ | $\text{CH} \equiv \text{C} - \text{CH}_2\text{NH}_2$ |
| (i) | (ii) | (iii) |



(i)



(ii)

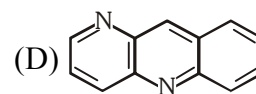
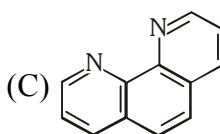
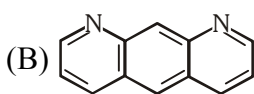
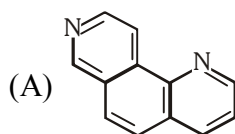


(iii)

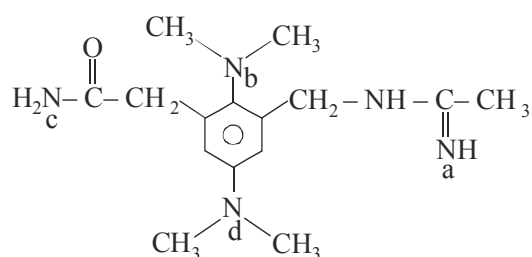
13. Arrange the following compounds in order of increasing basicity.

- (a) CH_3NH_2 , CH_3NH_3^+ , CH_3NH^- (b) CH_3O^- , CH_3NH^- , CH_3CH_2^-

14. Which of the following is most basic :

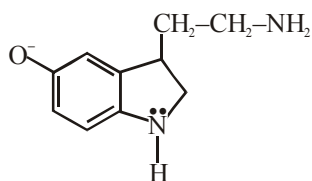


15. Basicity order of N in following compound is :



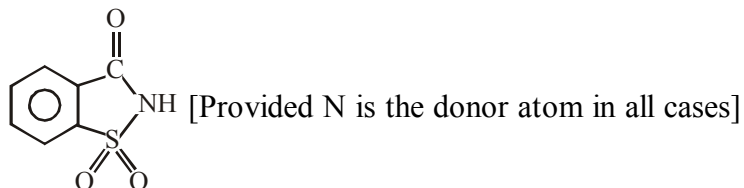
- (A) $b > d > a > c$ (B) $a > b > d > c$ (C) $a > b > c > d$ (D) $a > c > b > d$

16. The conjugate base of serotonin (used as tranquilisers) is given as follows :

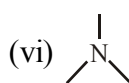
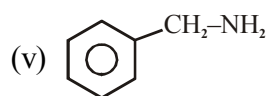
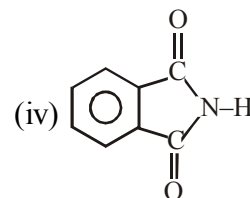
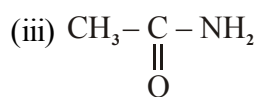
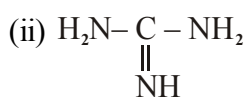
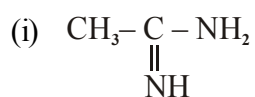


How many basic groups present in following compound ?

17. The structure of saccharin is given as follows :

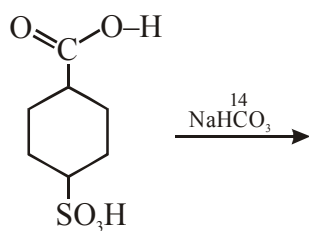


How many following compounds are more basic than saccharin ?

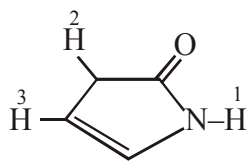


EXERCISE # III

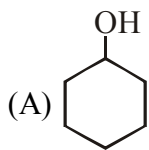
1. In given reaction Gas liberated is/are



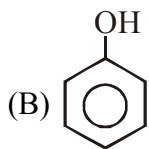
- (A) CO_2 & SO_3 (B) SO_3 & $^{14}\text{CO}_2$ (C) $^{14}\text{CO}_2$ only (D) SO_2 only
2. Arrange marked atom in decreasing order of acidic strength



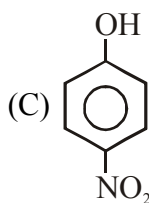
- (A) $1 > 2 > 3$ (B) $3 > 2 > 1$ (C) $2 > 1 > 3$ (D) $2 > 3 > 1$
3. **Column - I** **Column-I**



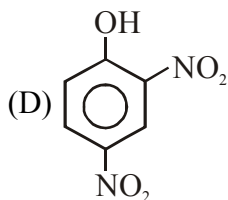
(P) React with NaOH



(Q) React with NaHCO_3



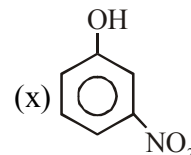
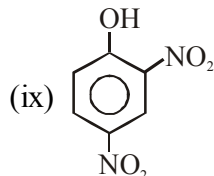
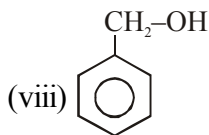
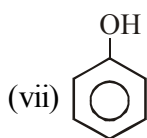
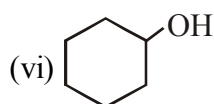
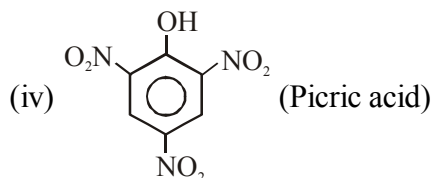
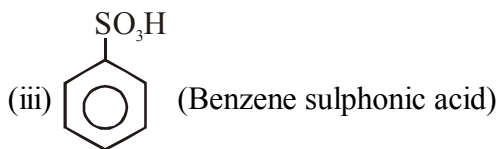
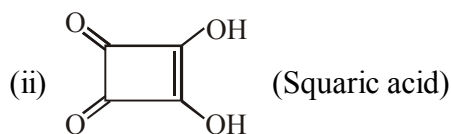
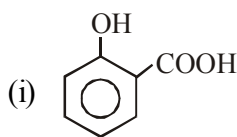
(R) React with NaH



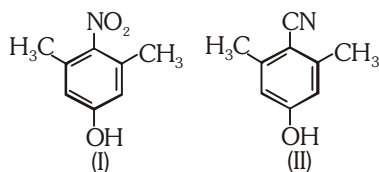
(S) React with Na

(T) React with NaNH_2

4. Compound which can give effevescences with NaHCO_3



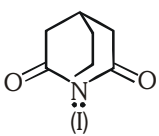
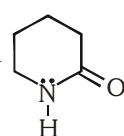
5. **Statement-1** : For the given two compounds-I is more acidic than compounds-II.



and

Statement-2 : Due to presence of $-\text{CH}_3$ group at ortho positions to $-\text{NO}_2$; the plane of $-\text{NO}_2$ deviates, w.r.t plane of ring.

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

6. **Statement 1** :  (I) is more basic than  (II)

and

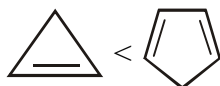
Statement 2 : Lone pair electrons on nitrogen in compound (I) does not participate in resonance.

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

7. Match Column-I with Column-II.

Column - I (Facts)

- (A) Guanidine $\text{H}_2\text{N}-\text{C}(\text{NH})=\text{NH}_2$ is example of strong base
- (B) Carbanion stability $\text{CCl}_3^- > \text{CF}_3^-$
- (C) Alkyne is more acidic than alkene
- (D) Acidity :



Column - II (Reasons)

- (P) Resonance stabilisation of conjugate acid of strong base.
- (Q) Due to s-character of central atoms
- (R) Due to d-orbital resonance
- (S) Due to formation of aromatic anion
- (T) Stability of conjugate base due to more number of identical resonating structure

8. Match Column-I with Column-II.

Column - I (Compounds)

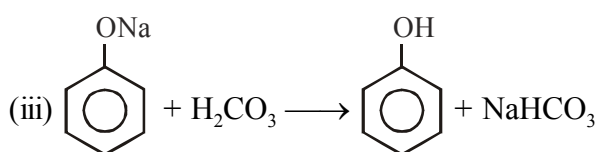
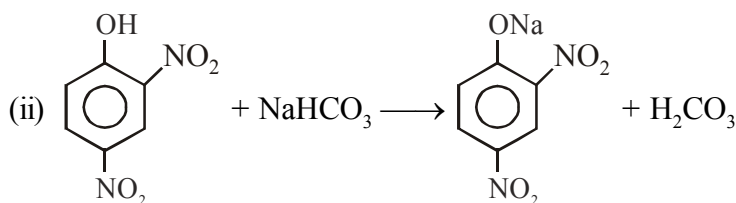
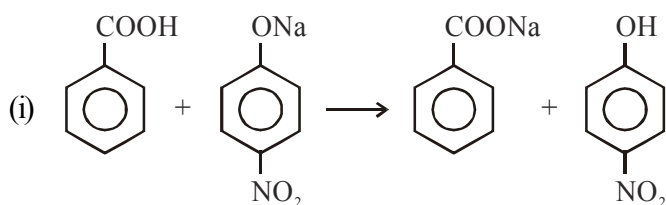
- (A)
- (B)
- (C)
- (D)

Column - II (pKa)

- (P) 7.15
- (Q) 10.14
- (R) 9.98
- (S) 9.38
- (T) pKa is more than phenol

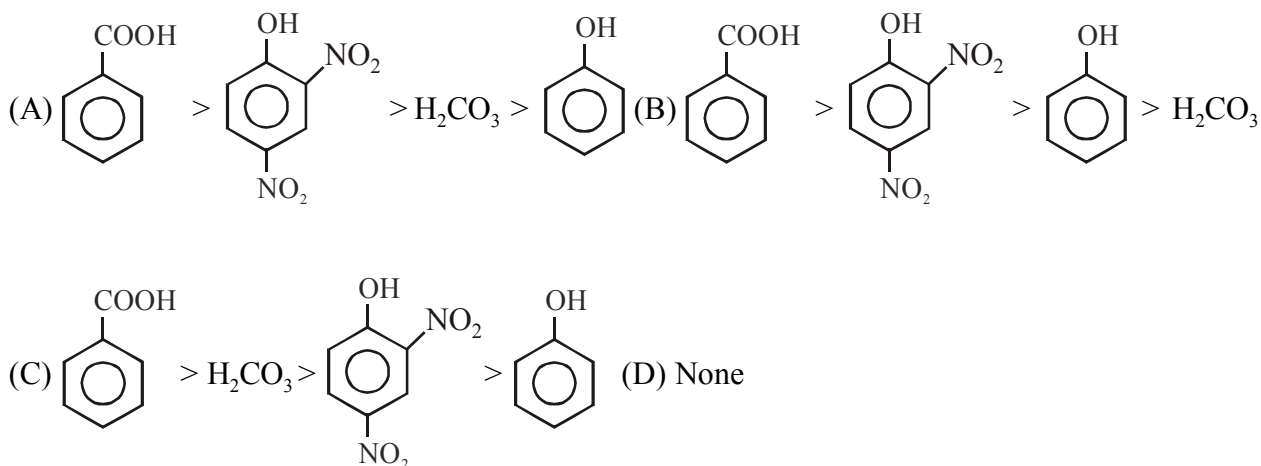
(Comprehension) (Q.9 to Q.11)

Observe the following feasible reactions :

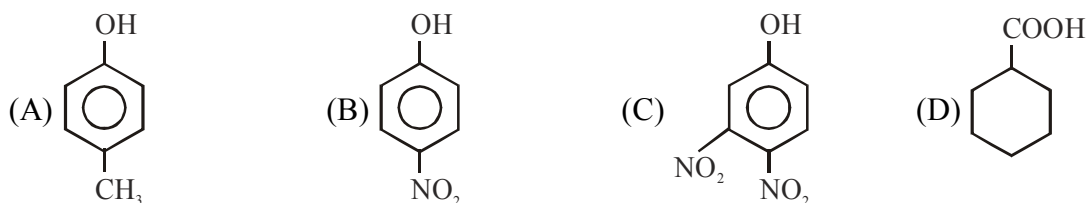


Answer the following question :

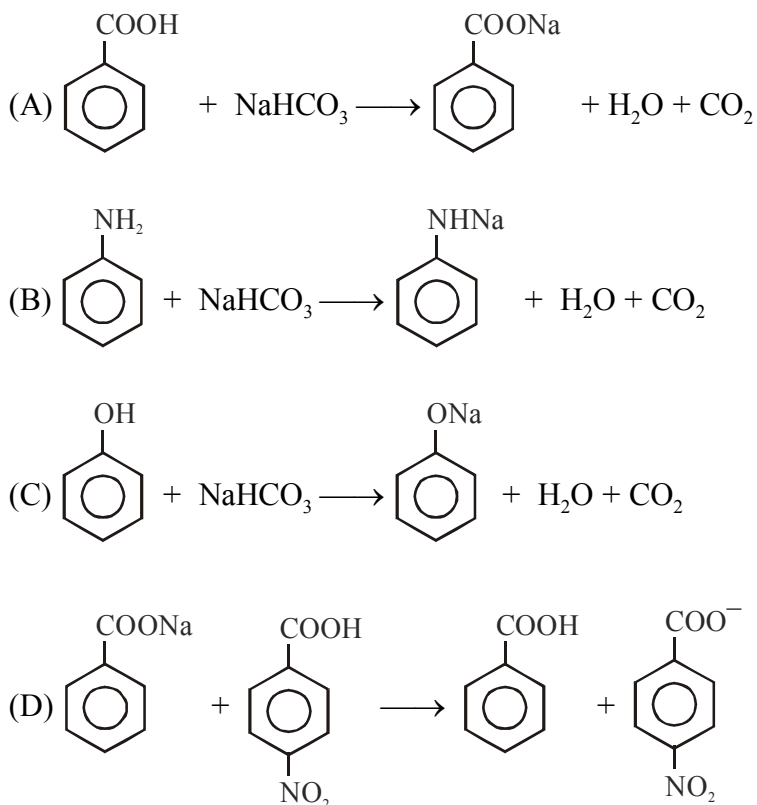
9. Which of the following is the correct order of acidic strength.



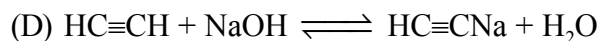
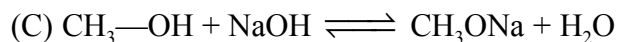
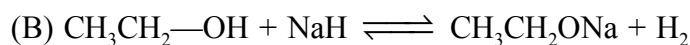
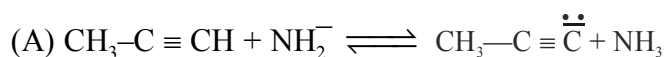
10. Which of the following compound does not react with NaHCO_3



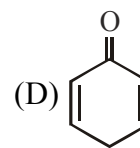
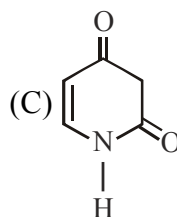
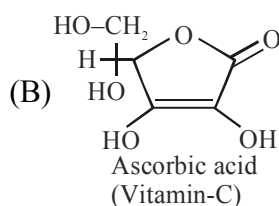
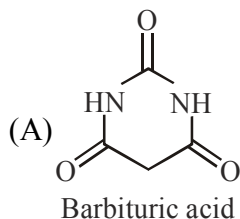
11. Identify the feasible reactions



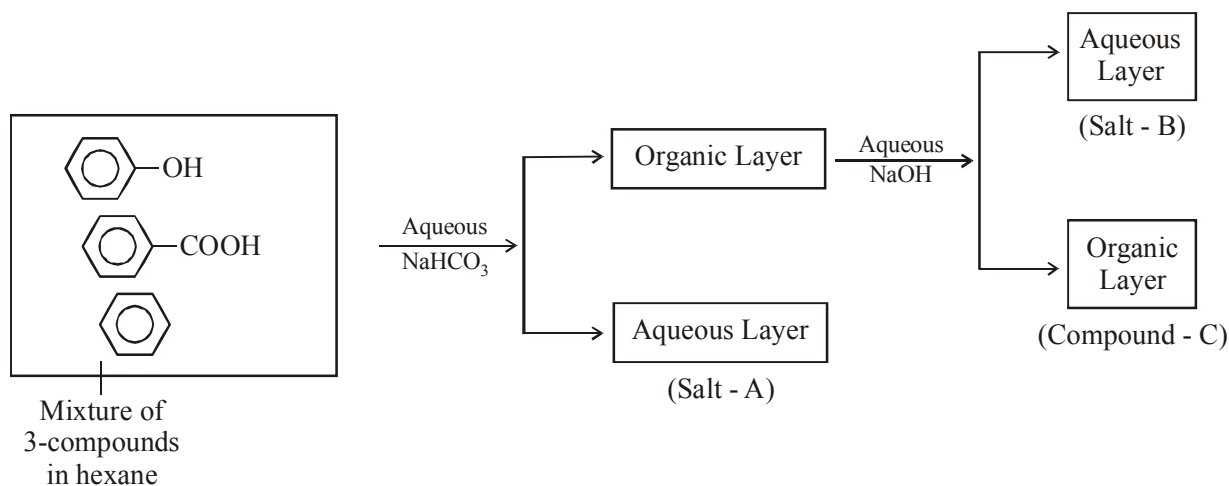
12. Identify the non-feasible reaction



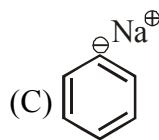
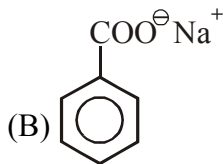
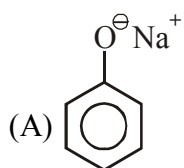
13. Select the number of compounds in which deprotonation gives aromatic anion :



Paragraph for Questions 14 and 15

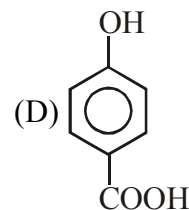
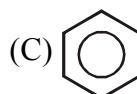
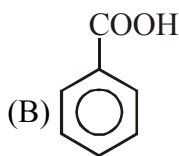
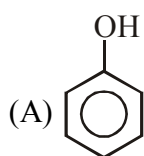


14. Identify salt 'A' ?



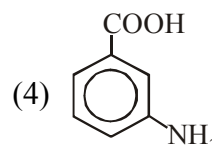
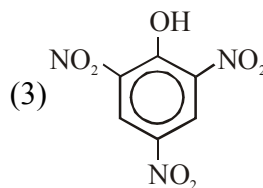
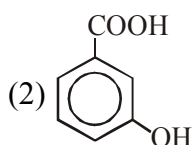
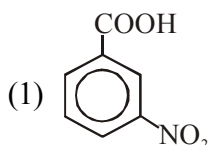
(D) All of these

15. Identify compound 'C' ?



EXERCISE # IV (JEE-MAIN)

1. Picric acid is - [AIEEE-2002]



2. Which of the following species acts both as bronsted acid & base - [AIEEE-2002]

- (1) NH_3 (2) OH^- (3) HSO_4^- (4) 1 and 3 both

3. The correct order of increasing basic nature for the bases NH_3 , CH_3NH_2 and $(\text{CH}_3)_2\text{NH}$ is- [AIEEE-2003]

- (1) $\text{CH}_3\text{NH}_2 < \text{NH}_3 < (\text{CH}_3)_2\text{NH}$ (2) $(\text{CH}_3)_2\text{NH} < \text{NH}_3 < \text{CH}_3\text{NH}_2$
 (3) $\text{NH}_3 < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH}$ (4) $\text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH} < \text{NH}_3$

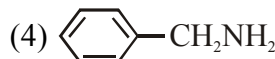
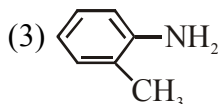
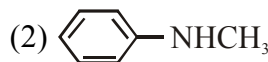
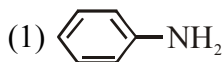
4. Consider the acidity of the carboxylic acids- [AIEEE-2004]

- (i) PhCOOH (ii) $\text{o-NO}_2\text{C}_6\text{H}_4\text{COOH}$
 (iii) $\text{p-NO}_2\text{C}_6\text{H}_4\text{COOH}$ (iv) $\text{m-NO}_2\text{C}_6\text{H}_4\text{COOH}$

which of the following is the correct order of acidity-

- (1) $\text{i} > \text{ii} > \text{iii} > \text{iv}$ (2) $\text{ii} > \text{iv} > \text{iii} > \text{i}$
 (3) $\text{ii} > \text{iv} > \text{i} > \text{iii}$ (4) $\text{ii} > \text{iii} > \text{iv} > \text{i}$

5. Which of the following is the strongest base - [AIEEE-2004]



6. Among the following acids which has the lowest pK_a value- [AIEEE-2005]

- (1) $\text{CH}_3\text{CH}_2\text{COOH}$ (2) $(\text{CH}_3)_2\text{CHCOOH}$
 (3) HCOOH (4) CH_3COOH

7. Amongst the following the most basic compound is- [AIEEE-2005]

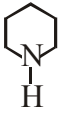
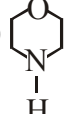
- (1) p-nitro aniline (2) Acetanilide
 (3) Aniline (4) Benzylamine

8. What is the conjugate base of OH^- ? [AIEEE-2005]

- (1) H_2O (2) O_2 (3) O^{2-} (4) O^-

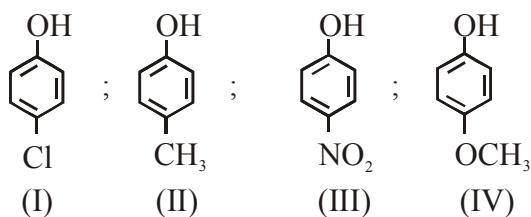
9. Among the following acids which has the lowest pK_a value? [AIEEE-2005]

- (1) HCOOH (2) CH_3COOH
 (3) $\text{CH}_3\text{CH}_2\text{COOH}$ (4) $(\text{CH}_3)_2\text{CH-COOH}$

10. The correct order of increasing acid strength of the compounds is [AIEEE-2006]
- (a) $\text{CH}_3\text{CO}_2\text{H}$ (b) $\text{MeOCH}_2\text{CO}_2\text{H}$
- (c) $\text{CF}_3\text{CO}_2\text{H}$ (d) $\begin{array}{c} \text{Me} \\ \diagup \\ \text{C} \\ \diagdown \\ \text{Me} \end{array} \text{CO}_2\text{H}$
- (1) $d < a < c < b$ (2) $d < a < b < c$
- (3) $a < d < c < b$ (4) $b < d < a < c$
11. Which one of the following is the strongest base in aqueous solution ? [AIEEE-2007]
- (1) Trimethylamine (2) Aniline
- (3) Dimethylamine (4) Methylamine
12. The correct order of increasing basicity of the given conjugate base ($\text{R}=\text{CH}_3$) is :- [AIEEE-2010]
- (1) $\text{RCO}\bar{\text{O}} < \text{HC} \equiv \bar{\text{C}} < \bar{\text{N}}\text{H}_2 < \bar{\text{R}}$ (2) $\text{RCO}\bar{\text{O}} < \text{HC} \equiv \bar{\text{C}} < \bar{\text{R}} < \bar{\text{N}}\text{H}_2$
- (3) $\bar{\text{R}} < \text{HC} \equiv \bar{\text{C}} < \text{RCO}\bar{\text{O}} < \bar{\text{N}}\text{H}_2$ (4) $\text{RCO}\bar{\text{O}} < \bar{\text{N}}\text{H}_2 < \text{HC} \equiv \bar{\text{C}} < \bar{\text{R}}$
13. The strongest acid amongst the following compounds is ? [AIEEE-2011]
- (1) $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{CO}_2\text{H}$ (2) $\text{ClCH}_2\text{CH}_2\text{CH}_2\text{COOH}$
- (3) CH_3COOH (4) HCOOH
14. The correct order of acid strength of the following compounds :-
- A. Phenol B. p-Cresol
- C. m-Nitrophenol D. p-Nitrophenol
- is :- [AIEEE-2011]
- (1) $\text{C} > \text{B} > \text{A} > \text{D}$ (2) $\text{D} > \text{C} > \text{A} > \text{B}$
- (3) $\text{B} > \text{D} > \text{A} > \text{C}$ (4) $\text{A} > \text{B} > \text{D} > \text{C}$
15. In the following compounds : [JEE(Main)-2012]
- (I)  (II)  (III)  (IV) 
- the order of basicity is as follows :
- (1) $\text{IV} > \text{III} > \text{II} > \text{I}$ (2) $\text{II} > \text{III} > \text{I} > \text{IV}$
- (3) $\text{I} > \text{III} > \text{II} > \text{IV}$ (4) $\text{III} > \text{I} > \text{II} > \text{IV}$
16. The most basic compound among the following is :- [JEE(Main)-2012]
- (1) Acetanilide (2) Benzylamine
- (3) p-Nitro aniline (4) Aniline
17. The order of basicity of amines in gaseous state is :- [JEE(Main)-2013]
- (1) $3^\circ > 2^\circ > \text{NH}_3 > 1^\circ$ (2) $1^\circ > 2^\circ > 3^\circ > \text{NH}_3$
- (3) $\text{NH}_3 > 1^\circ > 2^\circ > 3^\circ$ (4) $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$

18. Arrange the following compounds in order of decreasing acidity :

[JEE(Main)-2013]



(1) II > IV > I > III

(2) I > II > III > IV

(3) III > I > II > IV

(4) IV > III > I > II

19. The conjugate base of hydrazoic acid is :-

[JEE(Main)-2014]

(1) HN_3^-

(2) N_3^-

(3) N_2^-

(4) N^{3-}

20. Which one of the following compounds will not be soluble in sodium bicarbonate ?

[JEE(Main)-2014]

(1) Benzene sulphonic acid

(2) Benzoic acid

(3) o-Nitrophenol

(4) 2, 4, 6 - Trinitrophenol

21. Considering the basic strength of amines in aqueous solution, which one has the smallest pK_b value ?

[JEE(Main)-2014]

(1) $(\text{CH}_3)_3\text{N}$

(2) $\text{C}_6\text{H}_5\text{NH}_2$

(3) $(\text{CH}_3)_2\text{NH}$

(4) CH_3NH_2

22. Among the following oxoacids, the correct decreasing order of acid strength is : [JEE(Main)-2014]

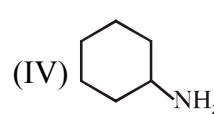
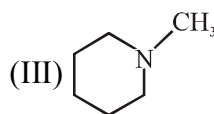
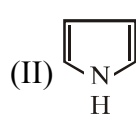
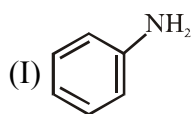
(1) $\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HOCl}$

(2) $\text{HClO}_2 > \text{HClO}_4 > \text{HClO}_3 > \text{HOCl}$

(3) $\text{HOCl} > \text{HClO}_2 > \text{HClO}_3 > \text{HClO}_4$

(4) $\text{HClO}_4 > \text{HOCl} > \text{HClO}_2 > \text{HClO}_3$

23. Among the following compounds, the increasing order of their basic strength is:-



(1) (II) < (I) < (III) < (IV)

(2) (I) < (II) < (IV) < (III)

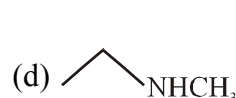
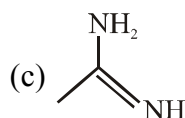
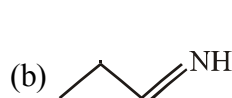
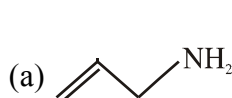
[JEE(Main)-On-Line 2017]

(3) (II) < (I) < (IV) < (III)

(4) (I) < (II) < (III) < (IV)

24. The increasing order of basicity of the following compounds is :

[JEE(Main)-2018]



(1) (b) < (a) < (c) < (d)

(2) (b) < (a) < (d) < (c)

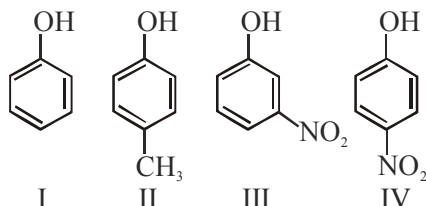
(3) (d) < (b) < (a) < (c)

(4) (a) < (b) < (c) < (d)

EXERCISE # V (J-ADVANCED)

1. In the following compounds

[IIT-JEE-1996]



The order of acidity is -

- (A) III>IV>I>II (B) I>IV>III>II (C) II>I>III>IV (D) IV>III>I>II
2. Although phenoxide ion has more number of resonating structures than benzoate ion, benzoic acid is a stronger acid than phenol. Why? [IIT-JEE-1997]
3. Amongst the following, the most basic compound is - [IIT-JEE-2000]
- (A) $C_6H_5NH_2$ (B) $p\text{-NO}_2\text{-C}_6\text{H}_4\text{NH}_2$ (C) $m\text{-NO}_2\text{-C}_6\text{H}_4\text{NH}_2$ (D) $C_6H_5CH_2NH_2$
4. The correct order of basicities of the following compounds is : [IIT-JEE-2001]



- (A) 2 > 1 > 3 > 4 (B) 1 > 3 > 2 > 4 (C) 3 > 1 > 2 > 4 (D) 1 > 2 > 3 > 4

5. **Statement-I** : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.**Because****Statement-II** : o-Hydroxybenzoic acid has intramolecular hydrogen bonding. [IIT-JEE-2003]

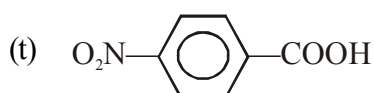
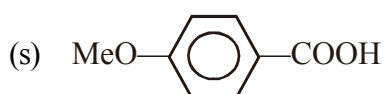
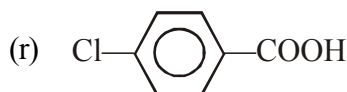
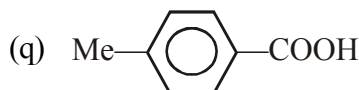
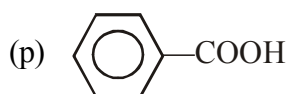
- (A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
- (B) Statement-I is True, Statement-II is True; Statement-II is NOT a correct explanation for Statement-I
- (C) Statement-I is True, Statement-II is False.
- (D) Statement-I is False, Statement-II is True.

6. Match K_a values with suitable acid :

[IIT-JEE-2003]

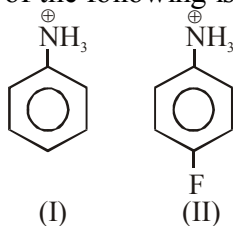
 K_a (A) 3.3×10^{-5} (B) 4.2×10^{-5} (C) 6.3×10^{-5} (D) 6.4×10^{-5} (E) 30.6×10^{-5}

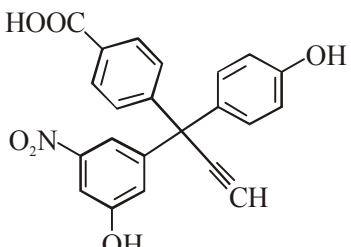
Acid



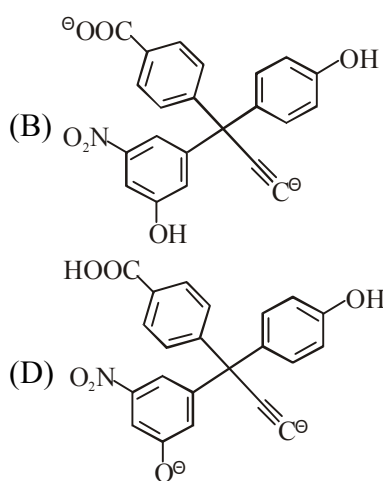
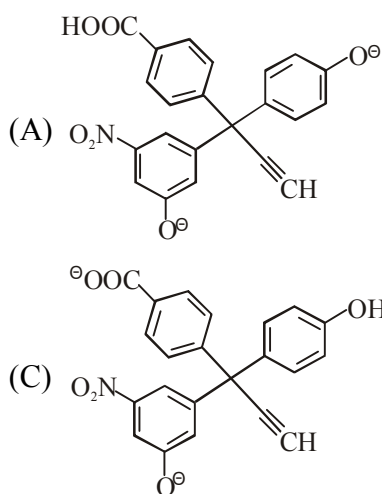
7. (a) Which of the following is more acidic and why ?

[IIT-JEE-2004]



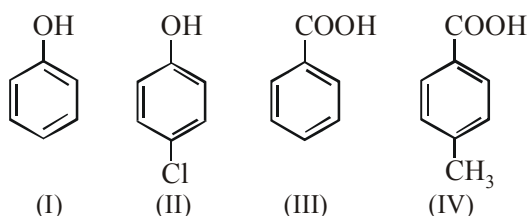
8.  2Moles NaNH₂ → A. The product (A) will be :

[IIT-JEE-2007]



9. The correct acidity order of the following is :

[IIT-JEE-2009]



(A) (III) > (IV) > (II) > (I)

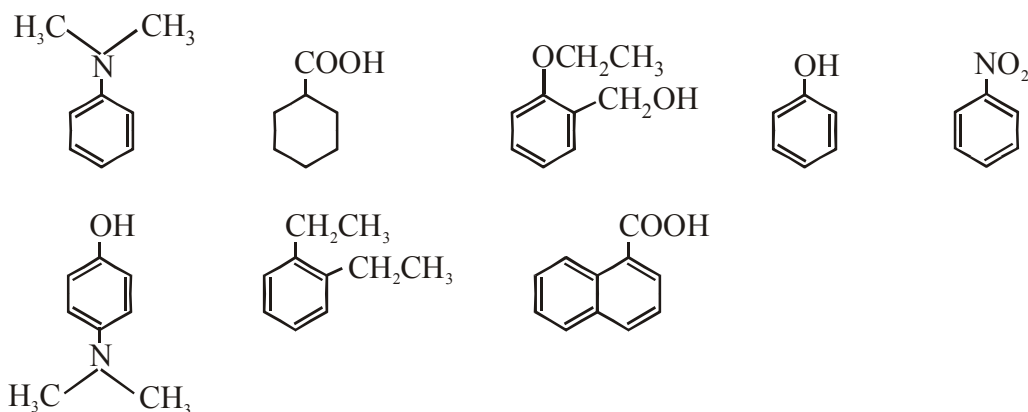
(B) (IV) > (III) > (I) > (II)

(C) (III) > (II) > (I) > (IV)

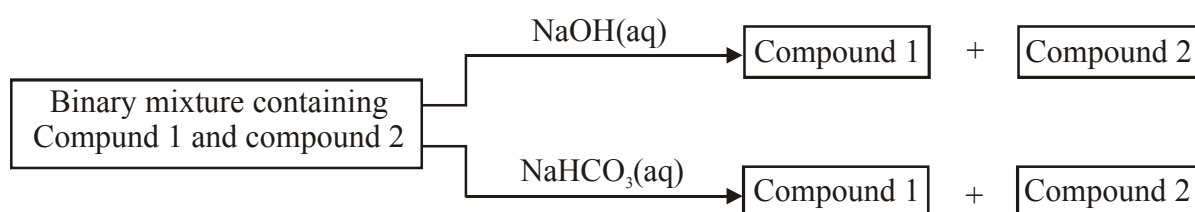
(D) (II) > (III) > (IV) > (I)

10. Amongst the following, the total number of compounds soluble in aqueous NaOH is:

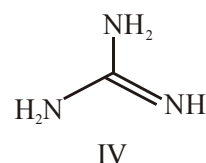
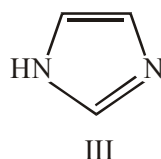
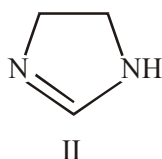
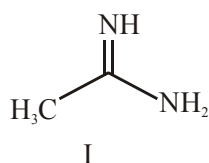
[IIT-JEE-2010]



11. Among the following compounds, the most acidic is [IIT-JEE-2011]
 (A) p-nitrophenol (B) p-hydroxybenzoic acid
 (C) o-hydroxybenzoic acid (D) p-toluic acid
12. The carboxyl functional group ($-\text{COOH}$) is present in - [IIT-JEE-2012]
 (A) picric acid (B) barbituric acid (C) ascorbic acid (D) aspirin
13. Identify the binary mixtures (s) that can be separated into the individual compounds, by differential extraction, as shown in the given scheme - [IIT-JEE-2012]



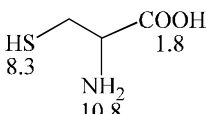
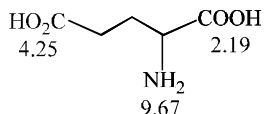
- (A) $\text{C}_6\text{H}_5\text{OH}$ and $\text{C}_6\text{H}_5\text{COOH}$ (B) $\text{C}_6\text{H}_5\text{COOH}$ and $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
 (C) $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ and $\text{C}_6\text{H}_5\text{OH}$ (D) $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ and $\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$
14. The compound that does NOT liberate CO_2 , on treatment with aqueous sodium bicarbonate solution, is - [JEE-ADVANCED-2013]
 (A) Benzoic acid (B) Benzenesulphonic acid (C) Salicylic acid (D) Carboic acid (phenol)
15. Hydrogen bonding plays a central role in the following phenomena [JEE-ADVANCED-2014]
 (A) Ice floats in water
 (B) Higher Lewis basicity of primary amines than tertiary amines in aqueous solutions
 (C) Formic acid is more acidic than acetic acid
 (D) Dimerisation of acetic acid in benzene
16. The order of basicity among the following compounds is [JEE-ADVANCED-2017]



- (A) $\text{II} > \text{I} > \text{IV} > \text{III}$ (B) $\text{IV} > \text{II} > \text{III} > \text{I}$
 (C) $\text{I} > \text{IV} > \text{III} > \text{II}$ (D) $\text{IV} > \text{I} > \text{II} > \text{III}$

ANSWER KEY

EXERCISE # I

1. (i) $a > b > c > d$, (ii) $a > b > c$, (iii) $c > b > a$, (iv) $a > b > c$,
 (v) $c > b > a$, (vi) $a > b > c$, (vii) $d > c > b > a$, (viii) $d > c > b > a$,
 (ix) $d > b > a > c$, (x) $d > a > c > b$
2. (a) 2; (b) 2; (c) 1; (d) 1 3. (a) 2; (b) 2; (c) 2 4. (C) 5. (B)
6. (B) 7. (B) 8. (A) 9. (A)
10. (i) cysteine :  (ii) glutamic acid : 
11. (a) $3 < 2 < 1$; (b) $1 < 2 < 3$; (c) $3 < 2 < 1$; (d) $2 < 1 < 3$; (e) $2 < 3 < 1$
12. (i) $d > c > a > b$, (ii) $a > b > c$, (iii) $c > a > b > d$, (iv) $d > b > c > a$,
 (v) $a > b > c$, (vi) $b > a$, (vii) $c > a > b$
13. (i) b, (ii) a, (iii) b, (iv) b 14. (C) 15. (C) 16. (B)
17. (C) 18. (B) 19. (4) 20. (C)

EXERCISE # II

1. (i) $a > b > c > d$, (ii) $a > b > c > d$, (iii) $c > b > d > a$, (iv) $d > c > b > a$
 (v) $a > b > c$, (vi) $c > b > a$, (vii) $c > a > b$, (viii) $b > c > a$, (ix) $c > d > b > a$
2. (i) $a > b > c$, (ii) $d > c > b > a$, (iii) $b > c > a$, (iv) $d > c > b > a$,
 (v) $b > a > c$, (vi) $b > a$, (vii) $c > b > a$, (viii) $d > a > b > c$
 (ix) $d > c > b > a$
3. (i) d, (ii) b, (iii) a, (iv) a
4. (i) $b > a > d > c$, (ii) $b > a > c > d$, (iii) $a > b > c > d$
5. (A) 6. (A) 7. (A) 8. (A)
9. (a) i, (b) ii, (c) i, (d) ii 10. (a) 2; (b) 1; (c) 1; (d) 1; (e) 3
11. (a) 2; (b) 1; (c) 2; (d) 2 12. (a) $1 > 2 > 3$; (b) $1 < 2 < 3$; (c) $3 < 1 < 2$; (d) $2 < 1 < 3$
13. (a) $2 < 1 < 3$; (b) $1 < 2 < 3$ 14. (C) 15. (B)
16. 3, 3 basic groups are $\ddot{\text{N}}\text{H}_2$; $-\text{NH}-$; O^- 17. (6)

EXERCISE # III

1. (C)
2. (C)
3. (A) – R, S, T ; (B) – P, R, S, T ; (C) – P, Q, R, S, T ; (D) – P, Q, R, S, T
4. (i), (ii) (iii) (iv), (v) (ix) 5. (D) 6. (A)
7. (A) – P, T ; (B) – R ; (C) – Q ; (D) – S, T 8. (A) – R ; (B) – S ; (C) – P ; (D) – Q, T
9. (A) 10. (A) 11. (A, D) 12. (D)
13. (A, B, C, D) 14. (B) 15. (C)

EXERCISE # IV (JEE-MAIN)

1. (3) 2. (4) 3. (3) 4. (4) 5. (4)
6. (3) 7. (4) 8. (3) 9. (1) 10. (2)
11. (3) 12. (1) 13. (1) 14. (2) 15. (3)
16. (2) 17. (4) 18. (3) 19. (2) 20. (3)
21. (3) 22. (1) 23. (3) 24. (2)

EXERCISE # V (J-ADVANCED)

1. (D) 2. Benzoate has equivalent resonating structures
3. (D) 4. (B) 5. (D)
6. A-(s) ; B-(q) ; C-(p) ; D-(r) ; E-(t)
7. (II is most acidic) 8. (C) 9. (A) 10. (4)
11. (C) 12. (D) 13. (B, D) 14. (D)
15. (A, B, D) 16. (D)

NURTURE COURSE

**QUANTUM NUMBER & ELECTRONIC CONFIGURATION
PERIODIC TABLE & PERIODIC PROPERTIES
CHEMICAL BONDING**

CONTENTS

QUANTUM NUMBER & ELECTRONIC CONFIGURATION

• Theory	01
• EXERCISE-(O-1)	14
• EXERCISE-(O-2)	16
• EXERCISE-(S-1)	18
• EXERCISE-(S-2)	19
• EXERCISE-JEE(Main)	22
• EXERCISE-JEE(Advanced)	24
• ANSWER KEY	25

PERIODIC TABLE & PERIODIC PROPERTIES

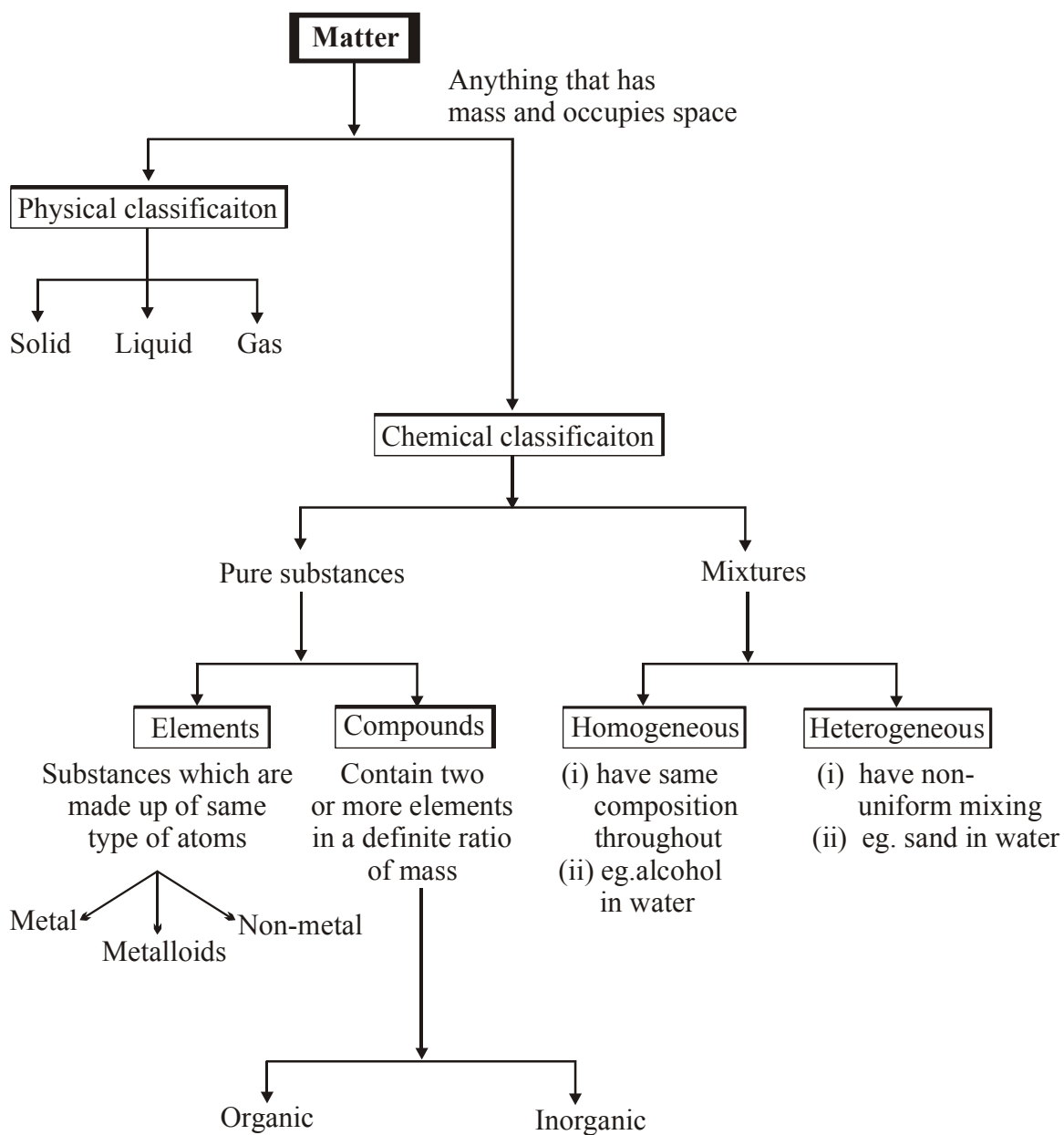
• Theory	26
• EXERCISE-(O-1)	48
• EXERCISE-(O-2)	52
• EXERCISE-(S-1)	54
• EXERCISE-(S-2)	55
• EXERCISE-JEE(Main)	58
• EXERCISE-JEE(Advanced)	62
• ANSWER KEY	63

CHEMICAL BONDING

• Theory	64
• EXERCISE-(O-1)	86
• EXERCISE-(O-2)	89
• EXERCISE-(S-1)	91
• EXERCISE-(S-2)	92
• EXERCISE-JEE(Main)	95
• EXERCISE-JEE(Advanced)	99
• ANSWER KEY	103

QUANTUM NUMBER & ELECTRONIC CONFIGURATION

MATTER & ITS CLASSIFICATION



Example.1 Which of the following is homogeneous mixture :

- (A) Oil + Water (B) Milk
(C) Salt dissolved in water (D) All of these

Example.2 Which of the following molecule is tetra-atomic :

- (A) CH_2Cl_2 (B) NH_3 (C) H_2O (D) Both (B) and (C)

ATOM : An atom is the smallest particle of an element (made up of still smaller particle like electrons, protons, neutrons etc.) which can take part in a chemical reaction. It may or may not exist free in nature.

Name of particle	Mass	Nature of charge	Amount of charge	Presence in the atom
(i) Electron symbol = (e) Notation = ${}_{-1}\text{e}^{\circ}$ Discoverer J.J. Thomson (1897)	$9.11 \times 10^{-28} \text{ g}$ $\frac{1}{1837}$ th of H-atom	Negatively charged	-1.602×10^{-19} Coulomb or $-4.8 \times 10^{-10} \text{ e.s.u}$	Outside the nucleus
(ii) Proton symbol = (p) Notation = $({}_1\text{H}^1)$ Discoverer Rutherford (1911)	$1.6725 \times 10^{-24} \text{ g}$	Positively charged	$+ 1.602 \times 10^{-19}$ coulomb $+ 4.8 \times 10^{-10} \text{ e.s.u.}$	Inside the nucleus of an atom
(ii) Neutron symbol = (n) Notation = $({}_0\text{n}^1)$ Discoverer J. Chadwick (1932)	$1.675 \times 10^{-24} \text{ g}$	Neutral	0	Inside the nucleus of an atom

Representation of atom : ${}_Z^AX^A$

A → Mass number : (total number of protons + total number of neutrons present in an atom.)

Z → Atomic number : (total number of protons present in an atom.)

⇒ **Isotope** : Atoms of given element which have same atomic number but different mass number are called isotope : e.g. ${}_1\text{H}^1$, ${}_1\text{H}^2$, ${}_1\text{H}^3$ etc.

⇒ **Isobar** : Atoms of different elements with the same mass number but different atomic number .

e.g. ${}_{18}\text{Ar}^{40}$, ${}_{19}\text{K}^{40}$ and ${}_{20}\text{Ca}^{40}$

⇒ **Iso-electronic species** : Species (atom, molecules or ions) having same number of electrons are called iso-electronic e.g H^- , He, Li^+ and Be^{2+} have 2 valence electrons each.

Note : Now a days this concept is extended to consider the same valence shell electron also.

⇒ **Iso-sters** : Species having same number of electrons & same number of atoms. eg. N_2O , CO_2

⇒ **Iso-diaphers** : Species having same difference in number of neutrons and protons. eg. $^{12}_6\text{C}$, $^{16}_8\text{O}$ and $^{14}_7\text{N}$.

⇒ **Orbital** : An orbital is defined as that zone in space where electron is most likely to be found. The orbitals are characterized by a set of 3 quantum numbers (n, l, m).

QUANTUM NUMBERS : Quantum numbers give complete information about an electron or orbital in an atom.

1. Principal Quantum number (n) :

- Permissible value of **n** → 1 to ∞
- It represents shell number/energy level
- The energy states corresponding to different principal quantum numbers are denoted by letters K, L, M, N etc.

n	:	1	2	3	4	5	6
Designation of shell	:	K	L	M	N	O	P

- It indicates the distance of an electron from the nucleus.
- It also determines the energy of the electron. In general higher the value of 'n', higher is the energy of a electron.
- It give an idea of total number of orbitals & electron (which may) present in a shell & that equal to **n^2 & $2n^2$** respectively.

2. Azimuthal Quantum number (l) :

- The values of **l** depends upon the value of 'n' and possible values are '0' to (n - 1).
- It gives the name of subshells associated with the energy level and number of subshells within an energy level.
- The different value of 'l' indicates the shape of orbitals and designated as follows :

Value	Notation	Name	Shape
$l = 0$	s	Sharp	Spherical
$l = 1$	p	Principal	Dumbell
$l = 2$	d	Diffused	Double Dumbell
$l = 3$	f	Fundamental	Complex

- It also determines the energy of orbital along with n.

For a particular energy level/shell energy of subshell is in the following order → **s < p < d < f**

- (v) It gives the total number of orbitals in a subshell & that equals to $(2l + 1)$ and number of electron in a subshell = $2(2l + 1)$

3. Magnetic Quantum number (m or m_l) :

- (i) The value of m depends upon the value of l and it may have integral value $-l$ to $+l$ including zero.
- (ii) It gives the number of orbitals in a given subshell and orientation of different orbitals in space.
e.g. for $n = 4$, $l = 0$ to 3 .

l	0	1	2	3
m	0	+1, 0, -1	+2, 1, 0, -1, -2	+3, +2, +1, 0, -1, -2, -3
Possible Orientation	1	3	5	7
Orbitals	s	p_x, p_y, p_z	$d_{z^2}, d_{x^2-y^2}$ d_{xy}, d_{yz}, d_{xz}	Not in syllabus

- (iii) The orbitals having same value of n and l but different value of m , have same energy in absence of external electric & magnetic field. These orbitals having same energy of a particular subshell is known as Degenerate orbitals.

4. Spin Quantum number (s) & magnetic spin quantum number (m_s) :

- (i) While moving around the nucleus, the electron always spins about its own axis either clockwise or anticlockwise. The magnetic spin quantum number represents the direction of electron spin (rotation) around its own axis (clockwise or anticlockwise).
- (ii) There are two possible values of m_s are $+\frac{1}{2}$ & $-\frac{1}{2}$ and represented by the two arrows $\uparrow$ (spin up) and $\downarrow$ (spin down).

RULES FOR FILLING ELECTRONS :

1. Pauli's exclusion principle

'No two electrons in an atom can have same values of all the four quantum numbers.

An orbital accommodates two electron with opposite spin. These two electrons have same values of principal, azimuthal and magnetic quantum number but the fourth, i.e. magnetic spin quantum number will be different. i.e.

For K, shell ($n = 1$)

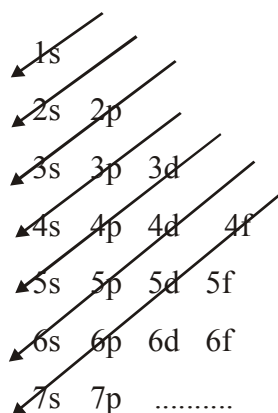
$$l = 0, m = 0$$

For 1st Electron $n = 1, l = 0, m = 0, m_s = +\frac{1}{2}$

For 2nd Electron $n = 1, l = 0, m = 0, m_s = -\frac{1}{2}$

2. Aufbau Principle (Means Building up) :

The electrons are added progressively to the various orbitals in the order of increasing energies starting with the orbital of the lowest energy



$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s < 5f < 6d < 7p$$

Alternatively, the order of increase of energy of orbitals can be calculated from $(n + l)$ rule.

- (i) Lower the value of $(n + l)$ for an orbital, the lower will be its energy.
- (ii) If two orbitals have the same $(n + l)$ value, then orbital with lower value of n has the lower energy.

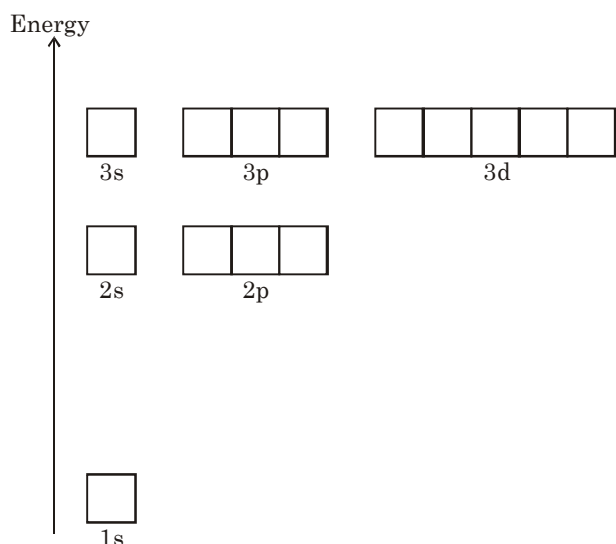
e.g. $2p$ & $3s$

For $2p$, $n = 2$, $l = 1$, $(n + l) = 2 + 1 = 3$

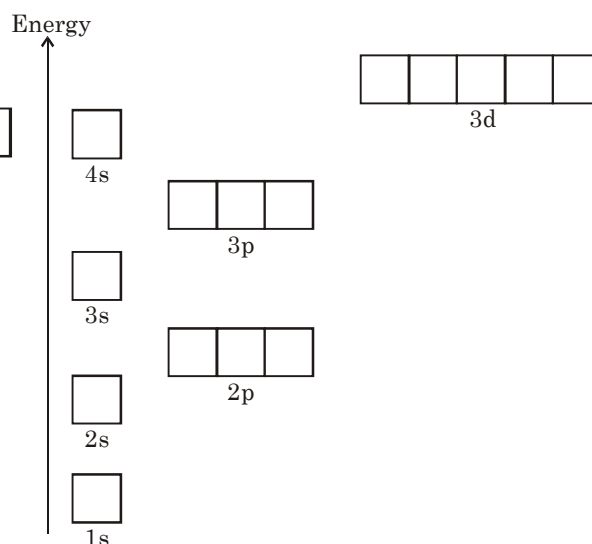
For $3s$, $n = 3$, $l = 0$, $(n + l) = 3 + 0 = 3$

Then for $2p$, n is lesser than for $3s$, so $2p$ has lower energy than $3s$.

- (iii) $1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f \dots$ energy order of different orbitals for single electron system like H, He^+ , Li^{+2} etc.



(A) For single electron or hydrogenic atom



(B) Multi electronic atoms

Energy level diagram for few electronic shells :

Example.4 Write the increasing order of energies of 4s, 3p, 4p and 3d.

Ans. For 4s, $n = 4, l = 0, (n + l) = 4$
 For 3p, $n = 3, l = 1, (n + l) = 4$
 For 4p, $n = 4, l = 1, (n + l) = 5$
 For 3d, $n = 3, l = 2, (n + l) = 5$
 $\Rightarrow 3p < 4s < 3d < 4p$ increasing order

3. **Hund's rule of maximum multiplicity :**

This rule deals with the filling of electrons into the orbitals belonging to the same subshell i.e. orbitals of equal energy, called degenerate orbitals.

“Electrons are distributed among the orbitals of a subshell in such a way as to give the maximum number of unpaired electron with parallel spins.”

“Pairing of electrons in the orbitals belonging to the same subshell (p, d, f) does not take place until each orbital belonging to that subshell has got one electron each i.e. singly occupied. Moreover, the singly occupied orbitals must have the electrons with the parallel spin multiplicity”

Multiplicity = $2|S| + 1$, where S = Total spin.



Find total spin & multiplicity

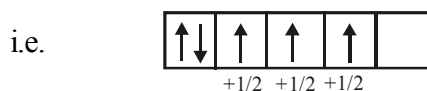
$$\text{Total spin } S = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

$$\text{Multiplicity} = 2 \times \frac{3}{2} + 1 = 4$$



$$\text{Total spin } S = \frac{1}{2} - \frac{1}{2} + \frac{1}{2} = \frac{1}{2}$$

$$\text{Multiplicity} = 2 \times \frac{1}{2} + 1 = 2$$



$$\text{Total spin } S = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

$$\text{Multiplicity} = 2 \times \frac{3}{2} + 1 = 4$$

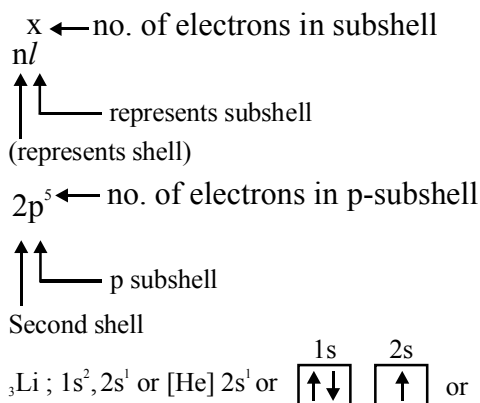


$$\text{Total spin} = 5 \times \frac{1}{2} = \frac{5}{2}$$

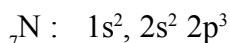
$$\text{Multiplicity} = 2 \times \frac{5}{2} + 1 = 6$$

ELECTRONIC CONFIGURATION OF ATOMS :

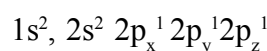
The distribution of electrons in various shells, subshells and orbitals, in an atom of an element, is called its electronic configuration.

**Electronic configuration :****Example.5**

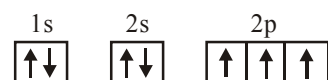
Nitrogen

Ans.

[Orbital notation method]

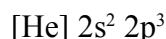


or



[Orbital diagram method]

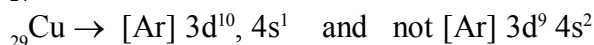
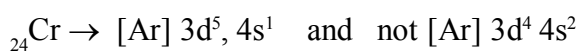
or



[Condensed form]

Extra stability of Half-filled and fully-filled orbitals.

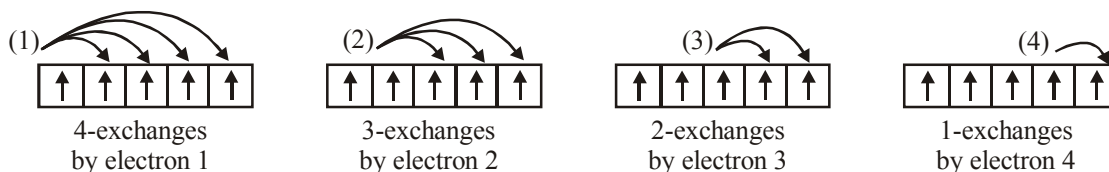
The electronic configuration of most of the atoms follows the Aufbau's rule. However, in certain elements such as Cr, Cu etc. Where the two subshells (4s and 3d) differ slightly in their energies ($4s < 3d$), an electron shifts from a subshell of lower energy (4s) to a subshell of higher energy (3d), provided such a shift results in all orbitals of the subshell of higher energy getting either completely filled or half-filled.



It has been found that there is extra stability associated with these electronic configuration. This stabilization is due to the following two factors.

- (i) **Symmetrical distribution of electron :** It is well known that symmetry leads to stability. The completely filled or half-filled subshell have symmetrical distribution of electron in them and are therefore more stable. This effect is more dominant in d and f-orbitals. This means three or six electrons in p-subshell, 5 or 10 electrons in d-subshell and 7 or 14 in f-subshell forms a stable arrangement.

- (ii) **Exchange energy** : This stabilizing effect arises whenever two or more electrons with the same spin are present in the degenerate orbitals of a subshell. These electrons tend to exchange their positions and the energy released due to this exchange is called exchange energy. The number of exchanges that can take place is maximum when the subshell is either half filled or fully filled. As a result the exchange energy is maximum and so is the stability.



Total exchange pairs = 10

$$\frac{n(n-1)}{2} \rightarrow \text{Number of exchange pairs}$$

$n \rightarrow$ Number of electron with parallel spins.

e.g. $\uparrow\downarrow \uparrow \uparrow \uparrow \uparrow$
Only 6 total exchange possible

Exceptional electronic configuration

S.No.	Element	Z	Configuration
1	Cr	24	$[\text{Ar}]4s^1 3d^5$
2.	Cu	29	$[\text{Ar}]4s^1 3d^{10}$
3.	Nb	41	$[\text{Kr}]5s^1 4d^4$
4.	Mo	42	$[\text{Kr}]5s^1 4d^5$
5.	Ru	44	$[\text{Kr}]5s^1 4d^7$
6.	Rh	45	$[\text{Kr}]5s^1 4d^7$
7.	Pd	46	$[\text{Kr}]4d^{10}$
8.	Ag	47	$[\text{Kr}]5s^1 4d^{10}$
9.	La	57	$[\text{Xe}]6s^2 5d^1$
10.	Ce	58	$[\text{Xe}]6s^2 4f^1 5d^1$
11.	Gd	64	$[\text{Xe}]6s^2 4f^7 5d^1$
12.	Pt	78	$[\text{Xe}]6s^1 4f^{14} 5d^9$
13.	Au	79	$[\text{Xe}]6s^1 4f^{14} 5d^{10}$
14.	Ac	89	$[\text{Rn}]7s^1 6d^1$
15.	Th	90	$[\text{Rn}]7s^2 6d^2$

MAGNETIC PROPERTIES :

❖ Paramagnetism :

- The substances which are weakly attracted by magnetic field are paramagnetic and this phenomenon is known as paramagnetism.
- Their magnetic character is retained till they are in magnetic field and lose their magnetism when removed from magnetic field.

❖ Diamagnetism :

- The substances which are weakly repelled by magnetic field are diamagnetic and this phenomenon is known as diamagnetism.
- Diamagnetic substances lack unpaired electrons and their spin magnetic moment is zero e.g., NaCl, N₂O₄ etc.

❖ Spin magnetic moment :

The spin magnetic moment of electron (excluding orbit magnetic moment) is given by :

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

Where n is number of unpaired electron in species.

The magnetic moment is expressed in Bohr magneton (B.M.)

Example.6 A compound of vanadium has magnetic moment of 1.73 BM. Work out the electronic configuration of vanadium ion in the compound.

Ans. Vanadium belongs to 3d series with Z = 23. The magnetic moment of 3d series metal is given by spin only formula.

$$\mu = \sqrt{n(n+2)} \text{ BM (BM = Bohr's magneton)}$$

$$\therefore \sqrt{1.73} = \sqrt{3}$$

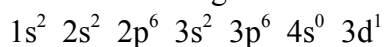
$$\Rightarrow n(n+2) = 3 \Rightarrow n = 1$$

$\Rightarrow$ Magnetic moment correspond to one unpaired electron.

$\Rightarrow$ Electronic configuration of vanadium atom 1s² 2s² 2p⁶ 3s² 3p⁶ 4s² 3d³.

For one unpaired electron 4 electron must be removed in which first 2 electron are lost from 4s orbital (outermost).

Electronic configuration of V²⁺



Nodal Planes of different orbitals :

Nodal plane is a plane at which the probability of finding an electron becomes zero.

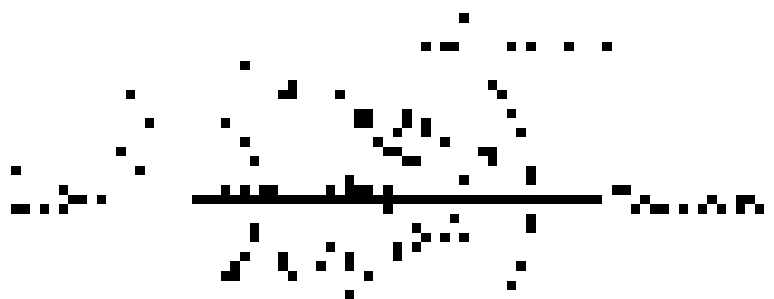
eg.

Orbital	Nodal plane	Orbital	Nodal plane
s	None	d _{xy}	XZ & YZ planes
p _x	YZ plane	d _{yz}	XZ & XY planes
p _y	XZ plane	d _{xz}	XY & YZ planes
p _z	XY plane	d _{x²-y²}	Planes perpendicular to XY plane, passing through origin (nucleus) and inclined at 45° to X & Y axis.
		d _{z²}	None (two nodal cones are available)

SCREENING EFFECT (s) AND EFFECTIVE NUCLEAR CHARGE (Z_{eff})

- Valence shell electron suffer force of attraction due to nucleus and force of repulsion due to inner shell electrons.
- The decrease in force of attraction on valence electron due to inner shell electron is called screening effect or shielding effect. (i.e. total repulsive force is called shielding effect.)
- Due to screening effect, valence shell electron experiences less force of attraction by nucleus.
- Due to screening effect, net attractive force felt by the electron is measured by effective nuclear charge, Z_{eff}
- If nuclear charge = Z , then effective nuclear charge = $Z - \sigma$ (Where σ 'sigma' is called screening constant/shielding constant)

$$\text{So, } Z_{\text{eff}} = Z - \sigma$$

**CALCULATION OF σ (using Slater's rule)**

To calculate the shielding constant (σ) :

- Write the electronic configuration of the element in the following order and groupings :
(1s), (2s, 2p), (3s, 3p), (3d), (4s, 4p), (4d), (4f), (5s, 5p), etc.

For s and p electrons :

- Electrons in any group to the right of the (ns, np) group contribute nothing to the shielding constant.
(n-shell no. of the electron for which σ is calculated)
- All of the other electrons in the (ns, np) group, shield the concern electron to an extent of 0.35 each. (Except for the 1s orbital for which value is 0.30).
- All electrons in the (n - 1) shell shield to an extent of 0.85 each.
- All electrons (n - 2) or lower group shield completely ; that is, their contribution is 1.00 each.

For d and f electrons :

- Electrons in any group to the right of the nd or nf group contribute nothing to the shielding constant.
- All the other electrons in the nd or nf group, shield the valence electron to an extent of 0.35 each.
- All electrons in groups lying to the left of the nd or nf group contribute 1.00.

(Effective Nuclear charge of elements of second period)						
Element	Electronic Configuration	Z	σ of ns & np electron	σ (n-1) orbital	Total Screening Constant	Effective nuclear charge#
			(a)	(b)	(a + b)	$Z^* = Z - \sigma$
${}_3\text{Li}$	$1s^2 2s^1$	3	—	$0.85 \times 2 = 1.70$	1.70	1.30
${}_4\text{Be}$	$1s^2, 2s^2$	4	$1 \times 0.35 = 0.35$	$0.85 \times 2 = 1.70$	2.05	1.95
${}_5\text{B}$	$1s^2, 2s^2, 2p^1$	5	$2 \times 0.35 = 0.70$	$0.85 \times 2 = 1.70$	2.40	2.60
${}_6\text{C}$	$1s^2, 2s^2, 2p^2$	6	$3 \times 0.35 = 1.05$	$0.85 \times 2 = 1.70$	2.75	3.25
${}_7\text{N}$	$1s^2, 2s^2, 2p^3$	7	$4 \times 0.35 = 1.40$	$0.85 \times 2 = 1.70$	3.10	3.90
${}_8\text{O}$	$1s^2, 2s^2, 2p^4$	8	$5 \times 0.35 = 1.75$	$0.85 \times 2 = 1.70$	3.45	4.55
${}_9\text{F}$	$1s^2, 2s^2, 2p^5$	9	$6 \times 0.35 = 2.10$	$0.85 \times 2 = 1.70$	3.80	5.20

Calculated for valence electron.

Key Points :

- (a) From left to right in a period Z_{eff} increases
- (i) For s and p-block elements, Z_{eff} in a period increases by 0.65 where atomic number increases by 1, and hence atomic size decreases considerably.
- (ii) In transition series Z increase by + 1 but screening constant increases by 0.85 So Z_{eff} is increased by 0.15
 $(1 - 0.85 = 0.15)$ [Because e^- enters in $(n - 1)$ orbit which has value of $\sigma = 0.85$]
- (b) From top to bottom in a group Z_{eff} remain constant for s-block elements, after Li and Be.

Element	Li	Na	K	Rb	Cs	Fr
Z_{eff}	1.30	2.20	2.20	2.20	2.20	2.20

Example-7 :

What is the effective nuclear charge at the periphery of nitrogen atom when an extra electron is added during the formation of an anion. Also find the value of Z_{eff} when the atom is ionized to N^+ .

Ans. Ground state electron configuration of $\text{N}(Z = 7) = 1s^2 2s^2 2p^3$

Electron configuration of $\text{N}^- = (1s^2) (2s^2 2p^4)$

Shielding constant for the last 2p electron,

$$\sigma = [(2 \times 0.85) + (5 \times 0.35)] = 3.45$$

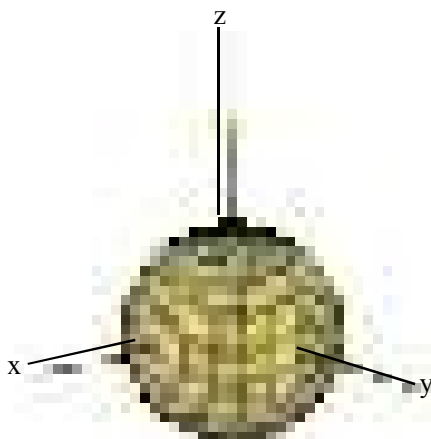
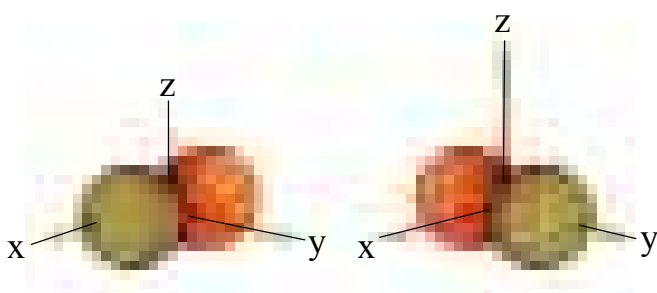
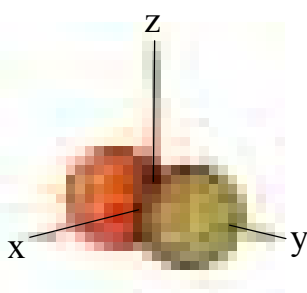
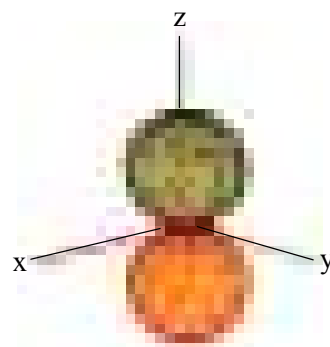
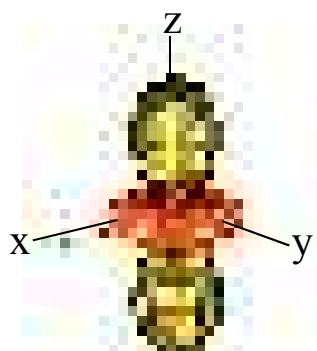
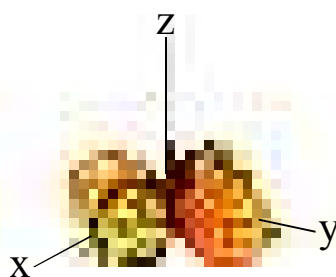
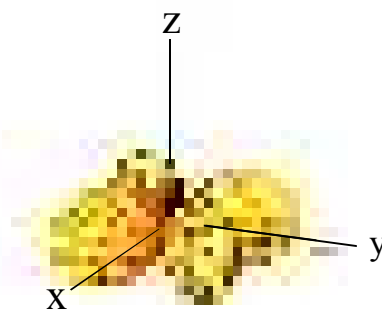
$$\text{So } Z_{\text{eff}} = Z - \sigma = 7 - 3.45 = 3.55$$

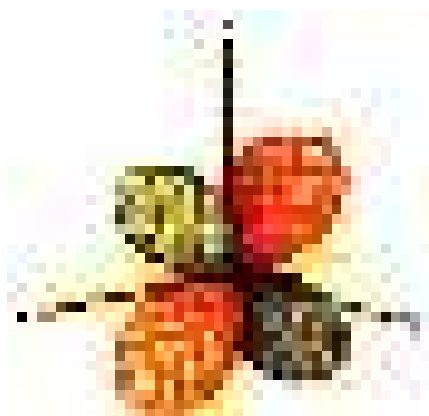
Electron configuration of $\text{N}^+ = (1s^2) (2s^2 2p^2)$

Shielding constant for the last 2p electron,

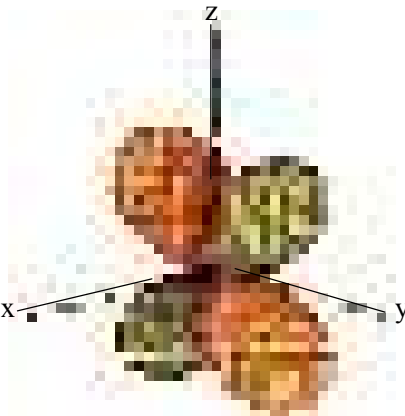
$$\sigma = [(2 \times 0.85) + (3 \times 0.35)] = 2.75$$

$$\text{So } Z_{\text{eff}} \text{ for last electron on } \text{N}^+ = 7 - 2.75 = 4.25$$

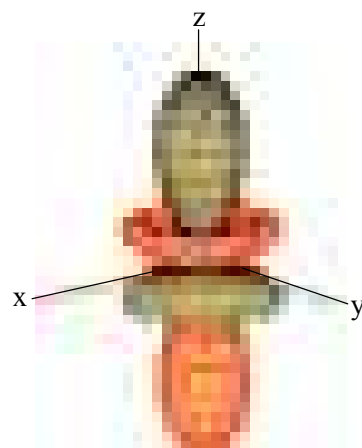
SHAPES OF ATOMIC ORBITALS**S****p_x****p_y****p_z****d_{z²}****d_{x²-y²}****d_{xy}**



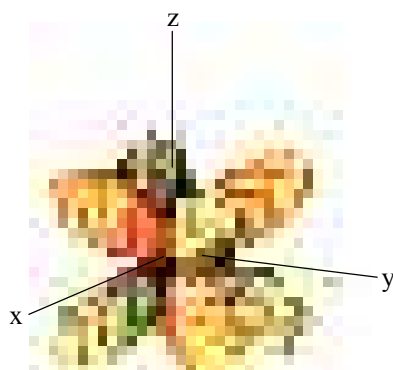
d_{xz}



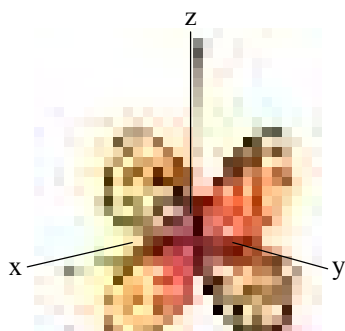
d_{yz}



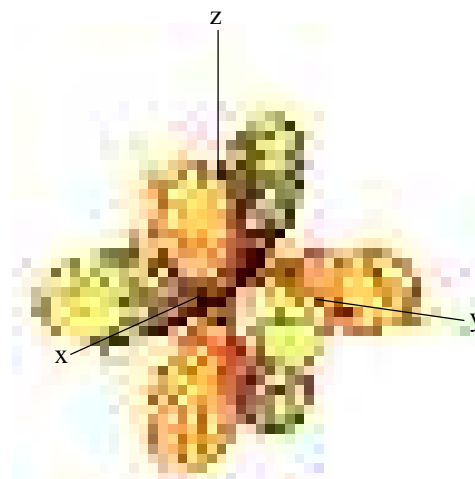
f_z^3



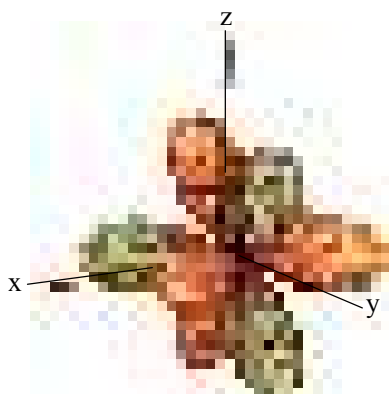
f_{xyz}



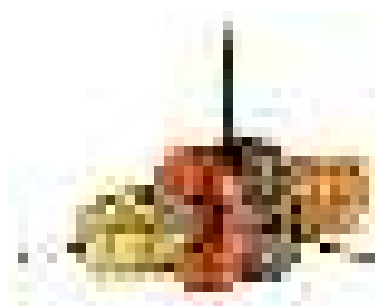
$f_{z(x^2-y^2)}$



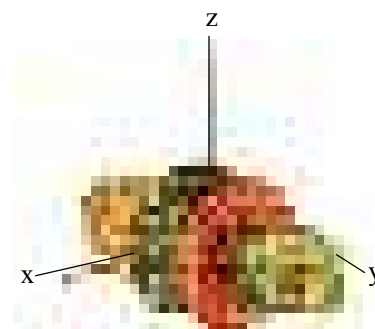
$f_{x(y^2-z^2)}$



$f_{y(z^2-x^2)}$



f_x^3



f_y^3

EXERCISE # O-1

General Introduction :

- The total number of neutrons in dipositive zinc ion with mass number 70 is
(A) 34 (B) 40 (C) 36 (D) 38
- It is known that atom contain protons, neutrons and electrons. If the mass of neutron is assumed to half of its original value where as that of proton is assumed to be twice of its original value then the atomic mass of ${}^{14}_6\text{C}$ will be -
(A) same (B) 25% more (C) 14.28 % more (D) 28.5% less
- Two monoatomic cations $x^{\oplus}$ and y^{2+} are isoelectronic then select the correct statement : (Both elements are consecutive)
(A) Both element x and y have same number of electrons
(B) Total number of valence electrons are more in element x, than element 'y'
(C) Total number of valence electrons are more in element y, than element 'x'
(D) Both (A) and (B) are correct
- Which of the following sets contain only isoelectronic ions?
(A) Zn^{2+} , Ca^{2+} , Ga^{3+} , Al^{3+} (B) K^+ , Ca^{2+} , Sc^{3+} , Cl^-
(C) P^{3-} , S^{2-} , Cl^- , Zn^{+2} (D) Ti^{4+} , Ar , Cr^{3+} , V^{5+}

Quantum number

- Which quantum number will determine the shape of the orbital
(A) Principal quantum number (B) Azimuthal quantum number
(C) Magnetic quantum number (D) Spin quantum number
- In Palladium (Atomic no.-46), number of electron having ($\ell = 2$) will be -
(A) 20 (B) 18 (C) 16 (D) 22
- For an electron present in which of the following orbital for which ($n + l + m + s$) value is maximum. Consider maximum possible value for 'm' and minimum possible value of m_s (where ever applicable).
(A) 3p (B) 5p (C) 4d (D) 5s
- Choose the correct option for the quantum numbers of the last electron of K^+ .
(A) 4, 0, 0, $+1/2$ (B) 3, 1, -1, $-1/2$ (C) 4, 1, 0, $-\frac{1}{2}$ (D) 3, 0, 1, $\frac{1}{2}$
- Find the sum of maximum number of electrons having +1 and -1 value of 'm' in Ti (Atomic number = 22)
(A) 6 (B) 8 (C) 10 (D) 12
- The number of electrons in Ca having minimum value of $\left| \frac{n}{\ell \times m_{\ell}} \right|$ is.
(Consider only non-zero values of ℓ and m)
(A) 6 (B) 3 (C) 4 (D) None of these

Electronic Configuration

11. A neutral atom of an element has two K, eight L, nine M and two N electrons then electronic configuration of the element is
- (A) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^1$ (B) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$
 (C) $1s^2 2s^2 2p^6 3s^2 3d^2 3p^6 4s^1$ (D) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2$
12. The explanation for the presence of three unpaired electrons in the nitrogen atom can be given by
- (A) Pauli's exclusion principle (B) Hund's rule
 (C) Aufbau's principle (D) Uncertainty principle
13. If the nitrogen atom had electronic configuration $1s^7$, it would have energy lower than that of normal ground state configuration $1s^2 2s^2 2p^3$, because the electrons would be closer to the nucleus. Yet $1s^7$ is not observed because it violates :—
- (A) Uncertainty principle (B) Hund's rule
 (C) Pauli's exclusion principle (D) Bohr postulate of stationary orbits

Effective Nuclear charge (Z_{eff}) :

14. The Z_{eff} for (as Slater's rule)
- 3d electron of Cr
 4s electron of Cr
 3d electron of Cr^{3+}
 3s electron of Cr^{3+} are in the order respectively
- (A) 4.6, 2.95, 4.95, 8.05 (B) 4.95, 2.95, 4.6, 8.05
 (C) 4.6, 2.95, 5.3, 12.75 (D) none of these
15. Total number of possible shells in uranium atom (atomic no. $z = 92$)
- (A) 7 (B) 1 (C) 6 (D) None of these
16. Which of the following has the maximum number of unpaired electrons ?
- (A) Mg^{2+} (B) Ti^{3+} (C) V^{3+} (D) Fe^{2+}
17. Gaseous state electronic configuration of nitrogen atom can be represented as :
- (A) $\uparrow\downarrow \uparrow\downarrow \uparrow \uparrow \uparrow$ (B) $\uparrow\downarrow \uparrow\downarrow \uparrow \downarrow \uparrow$
 (C) $\uparrow\downarrow \uparrow\downarrow \uparrow \downarrow \downarrow$ (D) $\uparrow\downarrow \uparrow\downarrow \downarrow \downarrow \uparrow$

EXERCISE : O-2

General Introduction :

1. Isotones are :

- (A) The atoms of different elements
 (B) Have same number of neutrons
 (C) Have same number of (neutrons + protons)
 (D) Have same difference of mass number and atomic number

Quantum number

2. For an electron present in which of the following orbital for which $(n + l)$ value is maximum.

- (A) 3p (B) 5p (C) 4d (D) 5s

3. Correct set of four quantum numbers for valence electron of rubidium ($Z = 37$) is

- (A) $5, 0, 0, +\frac{1}{2}$ (B) $5, 0, 0, -\frac{1}{2}$ (C) $5, 1, 1, +\frac{1}{2}$ (D) $6, 0, 0, +\frac{1}{2}$

4. The correct set of quantum numbers for the unpaired electron of chlorine atom is

- | | n | ℓ | m | | n | ℓ | m |
|-----|----------|----------|----------|-----|----------|----------|----------|
| (A) | 2 | 1 | 0 | (B) | 2 | 1 | 1 |
| (C) | 3 | 1 | 1 | (D) | 3 | 1 | 0 |

5. Which of the following sets of quantum numbers represent an impossible arrangement ?

- | | n | ℓ | m | m_s | | n | ℓ | m | m_s |
|-----|----------|----------|----------|----------------------|-----|----------|----------|----------|----------------------|
| (A) | 3 | 3 | -2 | $\frac{1}{2}$ | (B) | 4 | 0 | 0 | $\frac{1}{2}$ |
| (C) | 3 | 2 | -3 | $\frac{1}{2}$ | (D) | 5 | 3 | 0 | $\frac{1}{2}$ |

6. The quantum numbers for the 19th electron of Cr ($Z = 24$) are

- (A) $n = 3, \ell = 0, m = 0, m_s = +\frac{1}{2}$ (B) $n = 4, \ell = 0, m = 0, m_s = +\frac{1}{2}$
 (C) $n = 3, \ell = 2, m = 2, m_s = +\frac{1}{2}$ (D) $n = 4, \ell = 0, m = 0, m_s = -\frac{1}{2}$

7. The maximum number of electron having $n \times \ell \times m = 0$ in Zn^{2+} is equal to the -

- (A) Atomic number of Mg
 (B) 12
 (C) Total number of electron in Zn which have $n + \ell = 0$
 (D) 'p' electrons in Ar

Electronic Configuration

8. The species which have same number of electrons in outer most and penultimate shell -
 (A) Ca (B) Ar (C) V^{+3} (D) Sc^{3+}
9. Which sub-shell fill completely before the 4f?
 (A) 6s (B) 5p (C) 5d (D) 4d
10. The electronic configuration of a carbon atom is $1s^2, 2s^2, 2p^2$ and consider the following four arrangements of the 2p electrons. Which arrangement have lowest energy ?
 (A) $\boxed{\uparrow\downarrow}\boxed{}\boxed{}$ (B) $\boxed{\uparrow}\boxed{\uparrow\downarrow}\boxed{}$ (C) $\boxed{\uparrow}\boxed{}\boxed{}$ (D) $\boxed{\uparrow}\boxed{}\boxed{\uparrow}$
11. Hund's rule is applicable for :-
 (A) d-subshell (B) p-subshell (C) s-subshell (D) f-subshell
12. Which of the following has maximum number of unpaired electron.
 (A) Fe (B) Fe (II) (C) Fe (III) (D) Mn (II)
13. $Mn (Z = 25) = 1s^3 1p^9 2s^3 2p^9 2d^1$
 Which of the following change is required so that Mn have above ground state electronic configuration :
 (A) Change in the value of ℓ (azimuthal quantum number) for any subshell
 (B) Change in the possible values of ℓ (azimuthal quantum number)
 (C) Change in the Pauli rule
 (D) Change in the $(n + \ell)$ rule
14. The number of d- electrons in Mn^{2+} is equal to that of
 (A) p-electrons in N (B) s-electron in Na
 (C) d-electrons in Fe^{+3} (D) p-electrons in O^{-2}
15. Select incorrect statement(s) :
 (A) d_{z^2} orbital has different shape from rest of all d-orbitals
 (B) For the formation of cation electrons are always removed from 4s.
 (C) Zinc is a p-block element.
 (D) Principal quantum number depend upon the value of azimuthal quantum number

EXERCISE # S-1

Integer Answer Type (0 to 9) :

- Find total no. of orbitals in nickel which have $|m| \leq 1$ and at least one electron is present, where 'm' is magnetic quantum number.

(Given your ans. as sum of digits for example. If your ans is 57 then $5 + 7 = 12$ and $1 + 2 = 3$)

- Minimum number of electrons having $m_s = \left(-\frac{1}{2}\right)$ in Cr is “_____”.
- How many elements are possible for the 1st period of periodic table if azimuthal quantum number can have integral values from 0 to $(n + 1)$. [n = shell number & other rules are remaining same to form periodic table.
- Find number of unpaired electrons when Fe does not follow $(n + \ell)$ rule and filling of electron takes place shell after shell and Hund's rule is also not obeyed.
- Find the maximum number of electrons having same Z_{eff} value for sulphur atom
- Find the sum of maximum unpaired e^- present in one 5g & one 6g orbital.
- Find out the maximum number of electrons that can involve in the shielding of an electron, having quantum numbers : $n = 2$, $\ell = 1$, $m = 0$, $m_s = +\frac{1}{2}$, in an atom.
- Find the sum of minimum and maximum possible value of x in Fe^{+x} ion, if magnetic moment of $\text{Fe}^{+x} = 4.89$ B.M.

EXERCISE # S-2

Paragraph for Question 1 to 3

The general electronic configuration of outer most and penultimate shell is given as $(n-1)s^2 (n-1)p^6 (n-1)d^x ns^2$. Then for an element with $n = 4$ and $x = 6$.

- The number of protons present in the divalent cation of the element of above configuration is :-
(A) 24 (B) 25 (C) 26 (D) 27
- The element is :
(A) Mn (B) Fe (C) Co (D) Li
- The number of unpaired electrons in the divalent cation of the given element in isolated gaseous state is :-
(A) 0 (B) 3 (C) 4 (D) 1

Paragraph for Question 4 & 5

Consider a hypothetical atom where $p_x, p_y, p_z, d_{xy}, d_{xz}, d_{yz}$ and $d_{x^2-y^2}$ orbitals are present for principal quantum number $n = 4$.

- Find the number of other orbital which lobes are fully present in the nodal plane of p_x orbital :-
(A) 2 (B) 4 (C) 3 (D) 5
- Which of the following orbitals lobe is not present at all either in the nodal plane of p_x orbital or in the nodal plane of p_y orbital.
(A) d_{xy} (B) d_{yz} (C) $d_{x^2-y^2}$ (D) p_z

Paragraph for Question 6 & 7

Isotopes, Isobars and Iso-diaphers are some basic definitions related to the atom, which are based upon the variation in no. of electron, proton or neutrons.

- Which of the following pair represents the Isobar species :-
(A) $^{40}_{18}\text{Ar}, ^{12}_6\text{C}$ (B) $^{16}_8\text{O}, ^{14}_7\text{N}$ (C) $\text{N}_2\text{O}, \text{CO}_2$ (D) None of these
- Which of the following pair is correct for isosters :-
(A) $\text{N}_2\text{O}, \text{CH}_4$ (B) $\text{N}_2\text{O}, \text{CO}_2$ (C) CO_2, SO_2 (D) $\text{N}_2\text{O}, \text{HOCl}$

8.	Column-I				Column-II			
	Element				Max. value of n and ℓ respectively ; consider filled subshell only			
	(P) P				(1) 6, 2			
	(Q) N				(2) 3, 1			
	(R) Pb				(3) 6, 3			
	(S) Cs				(4) 2, 1			

Code :

	P	Q	R	S		P	Q	R	S
(A)	4	1	2	3	(B)	2	4	1	3
(C)	3	1	2	4	(D)	2	4	3	1

9. Match the following:

Column-I

- (P) Same number of unpaired electrons are present in (excluding zero)
 (Q) Same number of electrons in s & p subshells.
 (R) Same number of electrons with the $l = 1$
 (S) Same number of total electrons

Column-II

- (1) Na^+ , Mg^{2+} , F^-
 (2) F^- , Mg , O^{2-}
 (3) Mg , Ne , O^{2-}
 (4) Li , Na , K

Code :

	P	Q	R	S
(A)	1, 2	3, 4	1	4
(B)	4, 2	3, 1	2, 3	4, 1
(C)	4	1	1, 2, 3	1
(D)	3	1, 2	3, 4	1, 3, 4

10. Match the following :

Column-I (Orbital)

- (P) s
 (Q) p_x
 (R) d_{xy}
 (S) $d_{x^2-y^2}$

Column-II (Property)

- (1) Have electron density at all three axes
 (2) YZ plane is nodal plane
 (3) dumbell shape
 (4) have azimuthal quantum no. $\ell = 2$

Code :

	P	Q	R	S
(A)	1, 3	2, 3	3	3, 4
(B)	1	2, 3	2, 4	4
(C)	1	2, 3, 4	3, 4	1, 4
(D)	2	3, 4	2, 3	1, 4

Match The Column :

11. Match the following :

Column-I (e^- configuration)

- (A) d^8
 (B) d^{10}
 (C) d^6
 (D) d^5

Column-II (Property)

- (P) Symmetrical distribution
 (Q) Unsymmetrical distribution
 (R) No of exchange pair are maximum among these
 (S) two electrons must be present in $d_{x^2-y^2}$ orbital
 (T) at least one electron is present in orbital having $m = -1$

12. **Column-I**

- (A) Zn^{2+}
 (B) Ga^+
 (C) Fe^{3+}
 (D) Br^-

Column-II

- (P) Diamagnetic
 (Q) Spin magnetic moment = $\sqrt{35}$ BM
 (R) 18 e^- in outer most shell
 (S) 3d subshell is fully filled
 (T) All the orbital of outer most shell are fully filled

MATCHING LIST TYPE 1 × 3 Q. (THREE LIST TYPE Q.)

Column - I	Column - II	Column - III
(A) Paramagnetic set	(i) $\text{Na}^+, \text{Mg}^{+2}, \text{F}^-$	(P) same value of principal quantum number for last electron
(B) Isoelectronic set	(ii) Li, Na, K	(Q) The non zero number(s) of $e^-(s)$ for $n = 3$ and $\ell \geq 1$ is
(C) The set for which value(s) of spin multiplicity is ≥ 1	(iii) $\text{Fe}^{+3}, \text{Co}^{2+}, \text{Ni}^{+2}$	(R) The value of " m_s " must be $+\frac{1}{2}$ for last electron
(D) The set of elements belongs to same period in periodic table	(iv) $\text{S}^{-2}, \text{Cl}^-, \text{P}^{-3}$	(S) Set for which the value of $m = \pm 2$ is possible for electron(s)

13. Which one of the following options is the **CORRECT** combination?
- (A) (A, i, P) (B) (B, iv, S) (C) (D, iii, S) (D) (C, iii, R)
14. Which one of the following options is the **INCORRECT** combination?
- (A) (A, iii, P) (B) (C, ii, P) (C) (B, i, P) (D) (B, iv, Q)
15. Which one of the following options is the **CORRECT** set of species with number of nodal planes for filled/partially filled orbitals is ≤ 1 for all given species in set?
- (A) (B, ii, R) (B) (B, iv, P) (C) (A, i, Q) (D) (D, iii, S)

EXERCISE # JEE-MAIN

- Which of the following groupings represent a collection of isoelectronic species ?
(At. no. Cs = 55, Br = 35) [AIEEE-2003]
(1) N^{3-} , F^- , Na^+ (2) Be, Al^{3+} , Cl^-
(3) Ca^{2+} , Cs^+ , Br (4) Na^+ , Ca^{2+} , Mg^{2+}
- The number of d-electrons retained in Fe^{2+} (At. no. of Fe = 26) ion is : [AIEEE-2003]
(1) 6 (2) 3 (3) 4 (4) 5
- Which of the following sets of quantum number is correct for an electron in 4f orbital ? [AIEEE-2004]
(1) $n = 3, \ell = 2, m = -2, s = +\frac{1}{2}$ (2) $n = 4, \ell = 4, m = -4, s = -\frac{1}{2}$
(3) $n = 4, \ell = 3, m = +1, s = +\frac{1}{2}$ (4) $n = 4, \ell = 3, m = +4, s = +\frac{1}{2}$
- Consider the ground state of Cr atom ($Z = 24$). The numbers of electrons with the azimuthal quantum numbers, $\ell = 1$ and 2 are, respectively. [AIEEE-2004]
(1) 16 and 5 (2) 12 and 5 (3) 16 and 4 (4) 12 and 4
- Which one of the following sets of ions represents the collection of isoelectronic species? [AIEEE-2004]
(1) Na^+ , Mg^{2+} , Al^{3+} , Cl^- (2) Na^+ , Ca^{2+} , Sc^{3+} , F^-
(3) K^+ , Cl^- , Mg^{2+} , Sc^{3+} (4) K^+ , Ca^{2+} , Sc^{3+} , Cl^-
- In a multi-electron atom, which of the following orbitals described by the three quantum members will have the same energy in the absence of magnetic and electric fields ? [AIEEE-2005]
(A) $n = 1, \ell = 0, m = 0$ (B) $n = 2, \ell = 0, m = 0$
(C) $n = 2, \ell = 1, m = 1$ (D) $n = 3, \ell = 2, m = 1$
(E) $n = 3, \ell = 2, m = 0$
(1) (D) and (E) (2) (C) and (D) (3) (B) and (C) (4) (A) and (B)
- Of the following sets which one does not contain isoelectronic species ? [AIEEE-2005]
(1) BO_3^{3-} , CO_3^{2-} , NO_3^- (2) SO_3^{2-} , CO_3^{2-} , NO_3^- (3) CN^- , N_2 , C_2^{2-} (4) PO_4^{3-} , SO_4^{2-} , ClO_4^-
- Which of the following statements in relation to the hydrogen atom is correct ? [AIEEE-2005]
(1) 3s, 3p and 3d orbitals all have the same energy
(2) 3s and 3p orbitals are of lower energy than 3d orbitals
(3) 3p orbital is lower in energy than 3d orbital
(4) 3s orbitals is lower in energy than 3p orbital
- Which one of the following sets of ions represents a collection of isoelectronic species ? [AIEEE-2006]
(1) N^{3-} , O^{2-} , F^- , S^{2-} (2) Li^+ , Na^+ , Mg^{+2} , Ca^{+2}
(3) K^+ , Cl^- , Ca^{+2} , Sc^{+3} (4) Ba^{+2} , Sr^{+2} , K^{+2} , Ca^{+2}
- Which of the following sets of quantum numbers represents the highest energy of an atom ? [AIEEE-2007]
(1) $n = 3, \ell = 1, m = \ell, s = +\frac{1}{2}$
(2) $n = 3, \ell = 2, m = \ell, s = +\frac{1}{2}$
(3) $n = 4, \ell = 0, m = 0, s = +\frac{1}{2}$
(4) $n = 3, \ell = 0, m = 0, s = +\frac{1}{2}$

11. Which one of the following constitutes a group of the isoelectronic species? [AIEEE-2008]
 (1) C_2^{2-}, O_2^-, CO, NO (2) $NO^+, C_2^{2-}, CN^-, N_2$
 (3) $CN^-, N_2, O_2^{2-}, C_2^{2-}$ (4) N_2, O_2^-, NO^+, CO
12. The electrons identified by quantum numbers n and ℓ :- [JEE-1999, AIEEE-2012, JEE-MAIN, (ONLINE)-2012]
 (a) $n = 4, \ell = 1$ (b) $n = 4, \ell = 0$ (c) $n = 3, \ell = 2$ (d) $n = 3, \ell = 1$
 Can be placed in order of increasing energy as
 (1) (a) < (c) < (b) < (d) (2) (c) < (d) < (b) < (a)
 (3) (d) < (b) < (c) < (a) (4) (b) < (d) < (a) < (c)
13. Which of the following paramagnetic ions would exhibit a magnetic moment (spin only) of the order of 5 BM ? [JEE-MAIN, (ONLINE)-2012]
 (At. No : Mn = 25, Cr = 24, V = 23, Ti = 22)
 (1) V^{2+} (2) Ti^{2+} (3) Mn^{2+} (4) Cr^{2+}
14. In an atom how many orbital (s) will have the quantum numbers; $n = 3, l = 2$ and $m_l = +2$? [JEE-MAIN, (ONLINE)-2013]
 (1) 1 (2) 5 (3) 3 (4) 7
15. The numbers of protons, electrons and neutrons in a molecule of heavy water are respectively [JEE-MAIN, (ONLINE)-2013]
 (1) 10, 10, 10 (2) 8, 10, 11
 (3) 10, 11, 10 (4) 11, 10, 10
16. Given [JEE-MAIN, (ONLINE)-2013]
 (a) $n=5, m_l = +1$ (b) $n = 2, l = 1, m_l = -1, m_s = -1/2$
 The maximum number of electron(s) in an atom that can have the quantum numbers as given in (a) and (b) are respectively :
 (1) 8 and 1 (2) 25 and 1 (3) 2 and 4 (4) 4 and 1
17. The correct set of four quantum numbers for the valence electrons of rubidium atom ($Z = 37$) is: [JEE(Main)-2014]
 (1) $5, 1, 1, +\frac{1}{2}$ (2) $5, 0, 1, +\frac{1}{2}$ (3) $5, 0, 0, +\frac{1}{2}$ (4) $5, 1, 0, +\frac{1}{2}$
18. If the principal quantum number $n = 6$, the correct sequence of filling of electrons will be:- [JEE-MAIN, (ONLINE)-2015]
 (1) $ns \rightarrow (n-1)d \rightarrow (n-2)f \rightarrow np$ (2) $ns \rightarrow np \rightarrow (n-1)d \rightarrow (n-2)f$
 (3) $ns \rightarrow (n-2)f \rightarrow (n-1)d \rightarrow np$ (4) $ns \rightarrow (n-2)f \rightarrow np \rightarrow (n-1)d$
19. The total number of orbitals associated with the principal quantum number 5 is : [JEE-MAIN, (ONLINE)-2016]
 (1) 25 (2) 5 (3) 20 (4) 10

EXERCISE # JEE-ADVANCED

1. The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$. This represents its : [JEE 2000]
(A) excited state (B) ground state (C) cationic form (D) none
2. The maximum number of electrons that can have principal quantum number, $n = 3$, and spin quantum number, $m_s = -1/2$, is [JEE 2011]
3. In an atom, the total number of electrons having quantum numbers $n=4$, $|m_l| = 1$ and $m_s = -\frac{1}{2}$ is: [JEE Advanced 2014]
4. Not considering the electronic spin the degeneracy of the second excited state ($n = 3$) of H-atom is 9, where the degeneracy of the second excited state of H^- is [JEE Advanced 2015]

ANSEWR KEY

EXERCISE # (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	C	B	B	A	C	B	C	C
Que.	11	12	13	14	15	16	17			
Ans.	B	B	C	C	D	D	A			

EXERCISE # (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A, B, D	B, C	A, B	C, D	A, C	B, D	A, B, D	B, D	A, B, D	C, D
Que.	11	12	13	14	15					
Ans.	A, B, D	C, D	B, C, D	B, C	B, C, D					

EXERCISE # (S-1)

Que.	1	2	3	4	5	6	7	8
Ans.	4	9	8	0	8	2	9	6

EXERCISE # (S-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	C	C	A	D	B	D	C	B
Que.	11					12				
Ans.	(A)-Q, T (B)-P, R, S, T (C)-Q, T (D)-P, T					(A)-P, R, S, T (B)-P, S (C)-Q (D)-P, S				
Que.	13	14	15							
Ans.	C	B	B							

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	3	2	4	1	2	1	3	2
Que.	11	12	13	14	15	16	17	18	19	
Ans.	2	3	3	1	1	1	3	3	1	

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4
Ans.	B	9	6	3

PART-A (PERIODIC TABLE)

INTRODUCTION :

The arrangement of all the known elements according to their properties in such a way that the elements of similar properties are grouped together in a tabular form is called periodic table.

Development of periodic table :

(A) LAVOISIER CLASSIFICATION OF ELEMENTS

At first Lavoisier classified the elements into two categories.

- (i) Metal (ii) Non-metal

Note : This classification cannot categorise the metalloid variety.

(B) DOBEREINER'S TRIAD RULE [1817]

- (i) He made groups of three elements having similar chemical properties called TRIAD.
 (ii) In Dobereiner triad, atomic weight of middle element is nearly equal to the average atomic weight of first and third element.

e.g.

Cl	Br	I	
35.5	80.0	127	$\frac{35.5 + 127}{2} = 81.25$
Ca	Sr	Ba	
40	87.5	137	$\frac{40 + 137}{2} = 88.5$
Li	Na	K	
7	23	39	At. wt of Na = $\frac{7 + 39}{2} = 23$

- (iii) Other examples – (K, Rb, Cs), (P, As, Sb), (S, Se, Te), (H, F, Cl), (Sc, Y, La)

(C) NEWLAND'S OCTAVE RULE [1865]

- (i) He arranged the elements in the increasing order of their atomic mass and observed that properties of every 8th element was similar to the 1st one, like in the case of musical vowels notation.

Sa	Re	Ga	Ma	Pa	Dha	Ni	Sa
1	2	3	4	5	6	7	8

- (ii) At that time inert gases were not known.

Li	Be	B	C	N	O	H
Na	Mg	Al	Si	P	S	F
K	Ca					Cl

- (iii) The properties of Li are similar to 8th element i.e. Na, Be are similar to Mg and so on.

Drawbacks or Limitations :

- (a) This rule is valid only upto Ca. because after Ca due to filling of d-orbitals there is difference of 18 elements instead of 8 element.
 (b) After the discovery of inert gas this law had to be dropped out.
 (c) He failed in the case of heavier metals as Fe has been placed along with O and S.

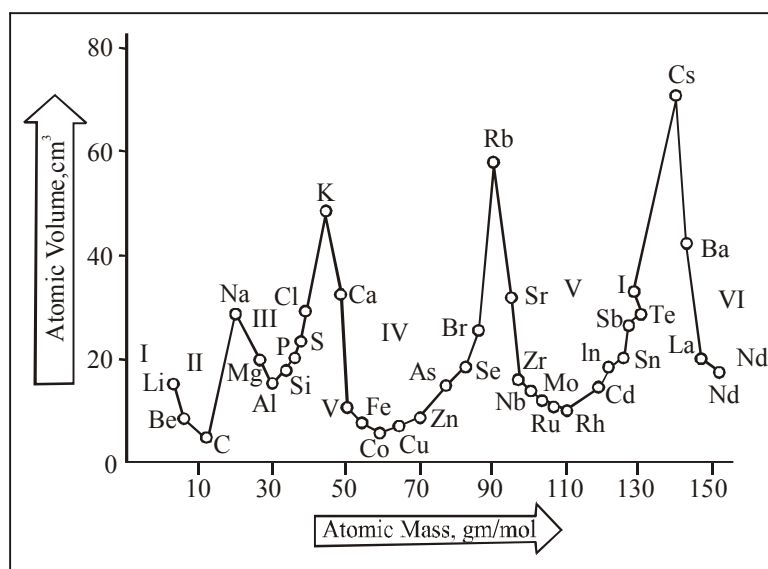
(D) LOTHER MEYER'S CURVE [1869]

- (i) He plotted a curve between atomic weight and atomic volume of different elements.
- (ii) The following observations can be made from the curve –
 - (a) Most electropositive elements i.e. alkali metals (Li, Na, K, Rb, Cs etc.) occupy the peak portions of the curve.
 - (b) Less electropositive i.e. alkaline earth metals (Be, Mg, Ca, Sr, Ba) occupy the descending portions of the curve.
 - (c) Metalloids (B, Si, As, Te, At etc.) and transition metals occupy bottom part of the curve.
 - (d) Most electronegative i.e. halogens (F, Cl, Br, I) occupy the ascending portion of the curve.

Note : Elements having similar properties occupy similar portions of the curve.

Conclusion :

On the basis of this curve, Lotther Meyer proposed that the physical properties of the elements are periodic function of their atomic weight and this became the basis of Mendeleev's periodic table.



(E) MENDELEEV'S PERIODIC TABLE [1869]

- (i) **Mendeleev's periodic law** – The physical and chemical properties of elements are the periodic function of their atomic weight
- (ii) **Characteristic of Mendeleev's periodic table** –
 - (a) It is based on atomic weight
 - (b) 63 elements were known, noble gases were not discovered.
 - (c) He was the first scientist to classify the elements in a systematic manner i.e. in horizontal rows and in vertical columns.
 - (d) Horizontal rows are called periods and there were 7 periods in Mendeleev's Periodic table.
 - (e) Vertical columns are called groups and there were 8 groups in Mendeleev's Periodic table.
 - (f) Each group upto VIIth is divided into A & B subgroups. 'A' sub group elements are called normal elements and 'B' sub group elements are called transition elements.
 - (g) The VIIIth group consists of 9 elements in three rows.
 - (h) The elements belonging to same group exhibit similar properties.

(iii) Merits or advantages of Mendeleev's periodic table –

(a) **Study of elements** – For the first time, all known elements were classified in groups according to their similar properties. So study of the properties of elements became easier.

(b) **Prediction of new elements** – It gave encouragement to the discovery of new elements as some gaps were left in it.

Sc (Scandium), Ga (Gallium), Ge (Germanium), Tc (Technetium) were the elements whose position and properties were well defined by Mendeleev even before their discoveries and he left the blank spaces for them in his table.

e.g. - Blank space at atomic weight 72 in silicon group was called Eka silicon (means properties like silicon) and element (*discovered later*) was named Germanium.

Similarly other elements discovered after mendeleev periodic table were :

Eka aluminium – Gallium(Ga)

Eka Boron – Scandium (Sc)

Eka Silicon – Germanium (Ge)

Eka Manganese – Technetium (Tc)

(c) **Correction of doubtful atomic weights**—Corrections were done in atomic weight of some elements.

$$\text{Atomic Weight} = \text{Valency} \times \text{Equivalent weight.}$$

Initially, it was found that equivalent weight of Be is 4.5 and it is trivalent ($V = 3$), so the weight of Be was 13.5 and there is no space in Mendeleev's table for this element. So, after correction, it was found that Be is actually divalent ($V = 2$). So, the weight of Be became $2 \times 4.5 = 9$ and there was a space between Li and B for this element in Mendeleev's table.

Corrections were done in atomic weight of elements are – U, Be, In, Au, Pt.

(iv) Demerits of Mendeleev's periodic table –

(a) **Position of hydrogen** – Hydrogen resembles both, the alkali metals (IA) and the halogens (VIIA) in properties so Mendeleev could not decide where to place it.

(b) **Position of isotopes** – As atomic weight of isotopes differs, they should have been placed in different position in Mendeleev's periodic table. But there was no such place for isotopes in Mendeleev's table.

(c) **Anomalous pairs of elements** – There were some pair of elements which did not follow the increasing order of atomic wts.

eg : Ar and Co were placed before K and Ni respectively in the periodic table, but having higher atomic weights.

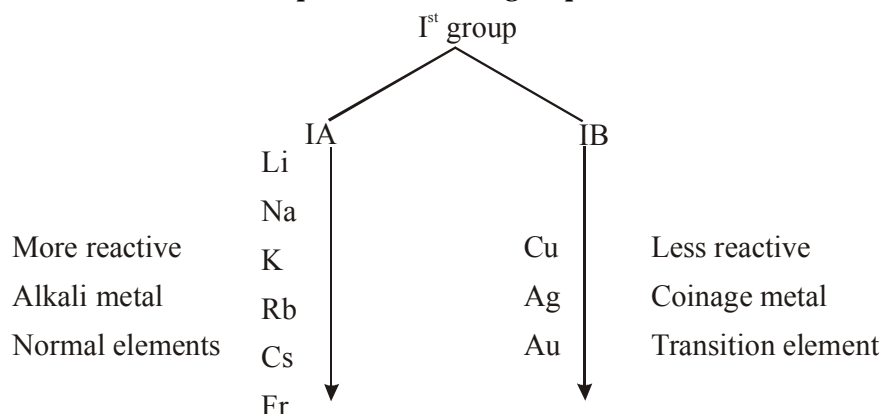
$$\begin{array}{cc} \left(\begin{array}{cc} \text{Ar} & \text{K} \\ 39.9 & 39.1 \end{array} \right) & \left(\begin{array}{cc} \text{Te} & \text{I} \\ 127.5 & 127 \end{array} \right) \\ \text{in modified mendeleev P.T.} & \\ \left(\begin{array}{cc} \text{Co} & \text{Ni} \\ 58.9 & 58.6 \end{array} \right) & \left(\begin{array}{cc} \text{Th} & \text{Pa} \\ 232 & 231 \end{array} \right) \end{array}$$

(d) Like elements were placed in different groups.

There were some elements like Platinum (Pt) and Gold (Au) which have similar properties but were placed in different groups in Mendeleev's table.

Pt	Au
VIII	IB

(e) Unlike elements were placed in same group.



Cu, Ag and Au placed in 1st group along with Na, K etc. While they differ in their properties (Only similar in having ns^1 electronic configuration)

(F) MODERN PERIODIC TABLE (MODIFIED MENDELEEV PERIODIC TABLE)

- (i) It was proposed by **Moseley (1913)**.
- (ii) Modern periodic table is based on atomic number.
- (iii) Moseley did an experiment in which he bombarded high speed electron on different metal surfaces and obtained X-rays.



He found out that $\sqrt{\nu} \propto Z$ (where ν = frequency) of X-rays from this experiment, Moseley concluded that the physical and chemical properties of the elements are periodic function of their atomic number. It means that when the elements are arranged in the increasing order of their atomic number, elements having similar properties gets repeated after a regular interval. This is also known as '**Modern Periodic Law**'.

- (iv) **Modern periodic law** – The physical & chemical properties of elements are a periodic function of their atomic number.
- (v) **Characteristics of modern periodic table** –
 - (a) 9 vertical columns called groups.
 - (b) IA to VIIA, IB to VIIB, VIII and 0
 - (c) Inert gases were introduced in periodic table by Ramsay.
 - (d) 7 horizontal series called periods.

(G) LONG FORM / PRESENT FORM OF MODERN PERIODIC TABLE

(It is also called as 'Bohr-Burry & Rang, Werner Periodic Table.)

- (i) It is based on the Bohr-Burry electronic configuration concept and atomic number.
- (ii) This model was proposed by Rang & Werner
- (iii) It consists of 7 horizontal periods and 18 vertical columns (groups)
- (iv) According to I. U. P. A. C. 18 vertical columns are named as 1st to 18th group.
- (v) The co-relation between the groups in long form of periodic table and in modern form of periodic table are given below.

IA, IIA, IIIB, IVB, VB, VIB, VIIB, VIII, IB, IIB, IIIA, IVA, VA, VIA, VIIA, 0

1 2 3 4 5 6 7 8, 9, 10 11 12 13 14 15 16 17 18

- (vi) Elements belonging to same group have same no. of electrons in the outermost shell so their properties are similar.

Description of periods :

Period	n	Sub shell	No. of elements	Element	Name of Period
1.	1	1s	2	${}_1\text{H} - {}_2\text{He}$	Shortest
2.	2	2s, 2p	8	${}_3\text{Li} - {}_{10}\text{Ne}$	Short
3.	3	3s, 3p	8	${}_{11}\text{Na} - {}_{18}\text{Ar}$	Short
4.	4	4s, 3d, 4p	18	${}_{19}\text{K} - {}_{36}\text{Kr}$	Long
5.	5	5s, 4d, 5p	18	${}_{37}\text{Rb} - {}_{54}\text{Xe}$	Long
6.	6	6s, 4f, 5d, 6p	32	${}_{55}\text{Cs} - {}_{86}\text{Rn}$	Longest
7.	7	7s, 5f, 6d,	26	${}_{87}\text{Fr} - {}_{112}\text{Uub}$	Incomplete

CLASSIFICATION OF ELEMENTS INTO s , p , d & f-BLOCK ELEMENTS :**s – block :**

- (i) Configuration ns^{1-2}
- (ii) Last e^- enters in s orbital
- (iii) Two groups I A or 1 ; II A or 2

p – block :

- (i) Configuration $ns^2 np^{1-6}$
- (ii) Last e^- enters in p orbital
- (iii) Six groups III A , IV A , V A , VI A , VII A , zero or 13 , 14 , 15 , 16 , 17 , 18

d – block : [Transition Elements]

- (i) Configuration $(n-1)d^{1-10} ns^{0-2}$
- (ii) Last e^- enters in d orbital
- (iii) Their two outermost shell are incomplete
- (iv) 10 groups III B , IV B , V B , VI B , VII B , VIII (Triad) , I B , II B or 3 , 4 , 5 , 6 , 7 , (8 , 9 , 10) , 11 , 12 .
- (v) Four series 3 d , 4 d , 5 d , 6 d belong to 4th , 5th , 6th & 7th period respectively in long form of Periodic table.

f – block : [Inner Transition]

- (i) configuration $(n - 2) f^{0-14} (n - 1) d^{0-2} ns^2$
- (ii) last e^- enters in f orbital
- (iii) two series 4 f Lanthanides & 5 f Actinides belong to 6th & 7th period respectively in long form of Periodic table.

Neil Bohr's classification of elements :

Using electronic configuration as the criteria, elements are of four types. The classification of the elements into these groups is dependent on the extent to which the s, p, d and f orbitals are filled.

Inert Gases :

- (a) s – and p-orbitals of the outer most shell of these elements are completely filled. The outermost electronic configuration is ns^2np^6 .
- (b) Helium is also inert gas but its electronic configuration is $1s^2$

Representative or Normal Elements :

- (a) Outermost shell of these elements is incomplete. The number of electrons in the outermost shell is less than eight.
- (b) s-and p-block elements except inert gases are called normal or representative elements.

Transition Elements :

- (a) These metals were placed between s-block metals and p-block elements so, are named transition metals.
- (b) Their outermost electronic configuration is similar to d-block elements i.e. $(n-1) d^{1-10} ns^{1-2}$.
- (c) Last two shells of these elements namely outermost and penultimate shells are incomplete.
- (d) The last shell contains one or two electrons and the penultimate shell may contain more than eight and up to eighteen electrons.
- (e) According to definition of transition elements, those elements which have partly filled d-orbitals in neutral state or in any stable oxidation state are called transition elements. According to this definition Zn, Cd and Hg (IIB group) are d-block elements but not transition elements because these elements have d^{10} configuration in neutral as well as in stable +2 oxidation state.

Inner Transition Elements :

- (a) In these elements last three shells i.e. last, penultimate and prepenultimate shells are incomplete.
- (b) These are related to IIIB i.e. group 3.
- (c) The last shell contains two electrons. Penultimate shell may contain eight or nine electrons and pre-penultimate shell contains more than 18, up to 32 electrons.
- (d) Their outermost electronic configuration is similar to f-block element
i.e. $(n - 2) f^{0-14} (n - 1) d^{0-2} ns^2$

METALS, NON-METALS & METALLOIDS

Apart from classifying elements into s, p, d and f-blocks, there is yet another broad classification of elements based on their properties. The elements can be broadly classified into

(a) Metals:

Majority of the elements in periodic table are metals and appears on the left side of the periodic table.

Properties:

- (i) These are usually solid at room temperature [exception - mercury]
- (ii) They have high melting and boiling point [exception Gallium & Cesium have very low melting point (303 K and 302 K respectively)]
- (iii) They are good conductor of heat and electricity.
- (iv) They are malleable (can be flattened into thin sheets by hammering) and ductile (can be drawn into wires)

(b) Non-Metals:

These are placed at the top right hand side of periodic table. As we move horizontally along a period, the property of elements changes from metallic (on left) to non-metallic (on the right).

Properties :

- (i) These are usually solids or gases at room temperature.
- (ii) They have low melting point and boiling point (exception : Boron, Carbon).
- (iii) Most Non-metallic solids are brittle and are neither malleable nor ductile.

(c) Metalloids (Semi-metals):

Properties of these elements show the characteristics of both metals and non-metals. Silicon (Si), Germanium(Ge), Arsenic(As), Antimony(Sb) and Tellurium(Te) are metalloids.

ESTIMATING POSITION OF AN ELEMENT FROM ITS ELECTRONIC CONFIGURATION

The last electron enters in which subshell gives idea of its block.

 **Think** : $1s^1$ and $1s^2$ belongs to which block]

Period number = Principal quantum number of valence shell electron in ground state electronic configuration.

Group number for s block = number of valence shell electrons

Group number for p block = 10 + number of valence shell electrons

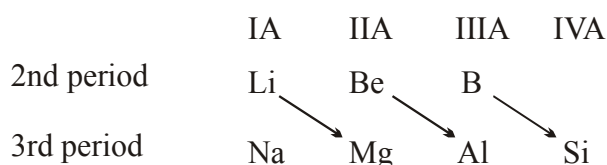
Group number for d block = number of $[ns + (n-1) d]$ electrons

Group number for f-block = 3

 **Use these carefully while locating the position.]**

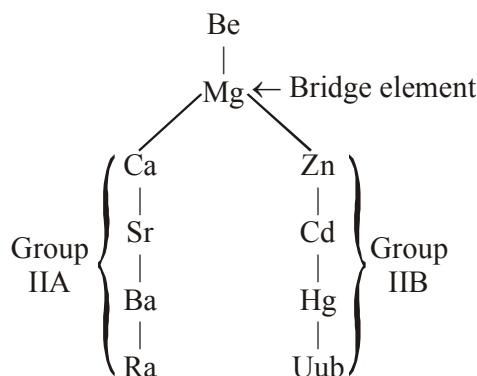
SOME COMMONLY USED TERMS

1. **Noble Gases** : Element of group 18 are called noble gases. These are also called as inert gases because their outermost ns and np orbitals are completely filled (except He and $1s^2$) and these gases are non-reactive in nature under ordinary conditions.
2. **Typical elements** : Elements of second and third period are known as typical elements.
3. **Diagonal relationship** : Properties of elements of second period resemble with the element of third period. This resemblance between properties of 2nd & 3rd period is called diagonal relationship.



4. Bridge elements :

The typical elements of third period are also called bridge elements as the division between two subgroups A and B starts from these elements. In second group Mg acts as a bridge element. The properties of bridge element are some what mixed of the elements of two subgroups as magnesium shows similarities with alkaline earth metals (IIA) on one hand and with zinc metals (IIB) on the other.



IUPAC NOMENCLATURE OF THE ELEMENT :

The names are derived by using roots for the three digits in the atomic number of the element and adding the ending -ium. The roots for the number are

Digit	Name	Abbreviation
0	nil	n
1	un	u
2	bi	b
3	tri	t
4	quad	q
5	pent	p
6	hex	h
7	sept	s
8	oct	o
9	enn	e

Thus element with atomic number 109 will be named as **une** (u for 1, n for 0 and e for 9). Table summarises the names of the elements with atomic number above 100.

PART-B (PERIODIC PROPERTIES)

PERIODICITY :

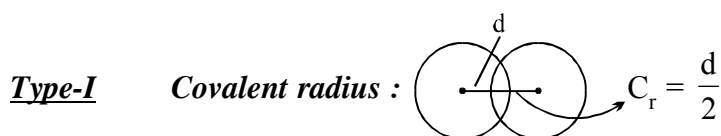
- (a) The regular gradation in properties from top to bottom in a group and from left to right in a period is called periodicity in properties.
- (b) In a period, the ultimate shell remain same, but the number of electrons gradually increases.
- (c) In a group, the number of electrons in the ultimate shell remains same, but the values of n increases.

Cause of periodicity

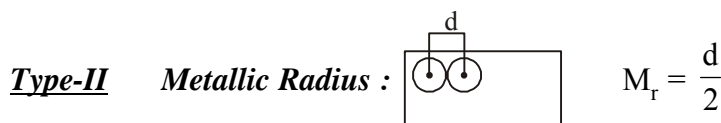
- (a) The cause of periodicity in properties is due to the same outermost shell electronic configuration repeating over regular intervals.
- (b) In the periodic table, elements with similar properties occur at intervals of 2, 8, 8, 18, 18 and 32. These numbers are called magic numbers.

ATOMIC RADIUS :

Since there is a problem in calculating actual size of atom, three types of radii can be defined :

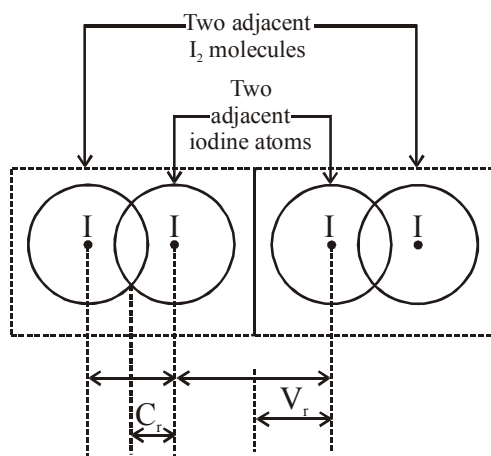


[Used for H_2 , Cl_2 and such molecules]



[Used for metals]

Type-III VanderWaal's Radius or Collision radius



VanderWaal's radius = $\frac{1}{2} \times$ Internuclear distance between nuclei of two neighbouring atoms belonging to nearest molecules.

VanderWaal's radius > Metallic radius > Covalent radius

The VanderWaal's radius and covalent radius of chlorine atom are 1.80 Å and 0.99 Å respectively.

Ionic Radius

A neutral atom changes to a cation by the loss of one or more electrons and to an anion by the gain of one or more electrons. The magnitude of charge on cation and anion is equal to the number of electrons lost or gained respectively. The ionic radii of the ions present in an ionic crystal may be calculated from the internuclear distance between the two ions.

(a) Radius of Cation

Radius of cation is smaller than that of corresponding atom. Since due to removal of electron(s), Z_{eff} increases.

(b) Radius of an Anion

Radius of an anion is invariably larger than that of the corresponding atom, since due to addition of electron(s) Z_{eff} decreases.

Factors affecting atomic radius:

- | | |
|--|--|
| (a) Z_{eff} increases, atomic radius decreases | $\text{Li} > \text{Be} > \text{B} > \text{C} > \text{N} > \text{O} > \text{F}$ |
| (b) Number of shell(n) increases, atomic radius increases | $\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs}$ |
| (c) Screening effect increases, atomic radius increases. | |
| (d) Magnitude of -ve charge increases, atomic radius increases | $\text{O} < \text{O}^- < \text{O}^{2-}$ |
| (e) Magnitude of +ve charge increases, atomic radius decreases | $\text{Mn} > \text{Mn}^{+2} > \text{Mn}^{+3} > \text{Mn}^{+4}$ |
| (f) Bond order increases, atomic radius decreases | $>\text{N}-\text{N}< >-\text{N}=\text{N}- > \text{N}\equiv\text{N}$ |

Periodic Trend :

(a) For normal elements:

- (i) **Across a period :** It decreases from left to right in a period as Z_{eff} increases.

Ex. $\text{Ne} > \text{Li} > \text{Be} > \text{B} > \text{C} > \text{N} > \text{O} > \text{F}$

- (ii) **In a group :** It increases from top to bottom in a group as number of shells increases.

Ex. $\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs}$

 **Note :** In III A group size of Al and Ga is nearly same (transition contraction)

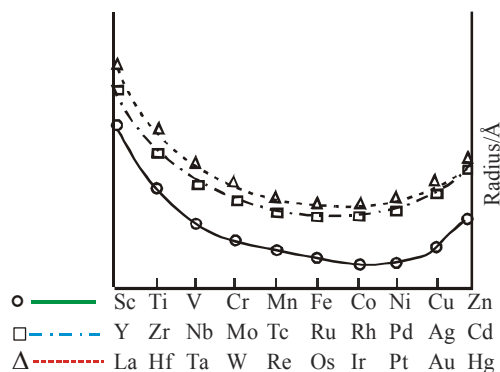
(b) For inert gases:

In respective period generally, the atomic radius of inert gas is largest, because for inert gas VanderWaal's radius is defined. The VanderWaal's radius of inert gases also increases on moving from top to bottom in the group.

(c) For transition elements:

From left to right in a period:

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic radius (Å)	1.44	1.32	1.22	1.17	1.17	1.17	1.16	1.15	1.17	1.25



Trends in atomic radii of transition elements

In a group :

- (i) The atomic radius of elements increases on moving down the first transition series (3d) to second transition series (4d). This is due to the increases in number of shells with the increase in atomic number.
- (ii) The atomic radii of second (4d) and third (5d) transition series in a group is almost same except Y(39) and La(57)

(d) For inner transition elements:

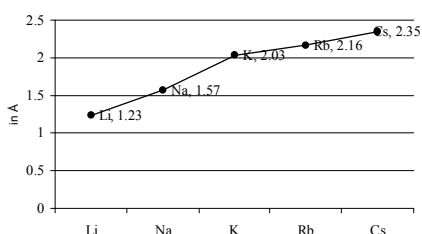
As we move along the lanthanide series, there is a decrease in atomic as well as ionic radius. The decrease in size is regular in ions but not so regular in atoms. This is called lanthanide contraction*.

Exceptions :

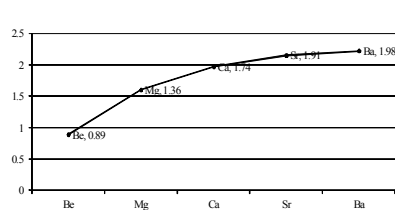
- (1) Noble gases have largest atomic sizes [Vander waal radii]. However, their covalent radii are smaller e.g. Xe.
- (2) Size of Ga and Al are same, [Z_{eff} increasing]
- (3) Size of Hf & Zr are same (lanthanide contraction)

Graphical representation of atomic radius :

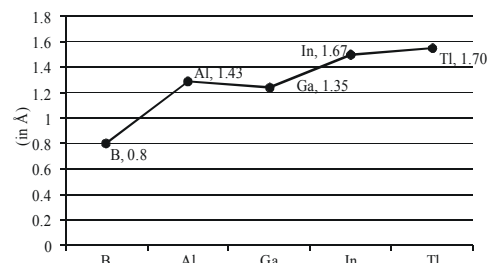
ALKALI METALS



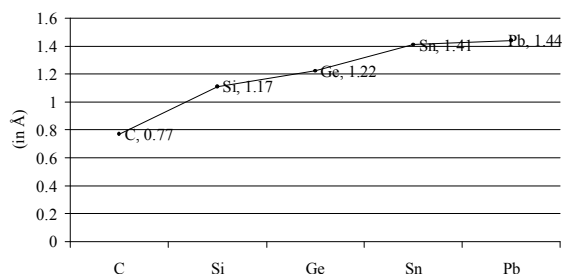
ALKALINE EARTH METALS



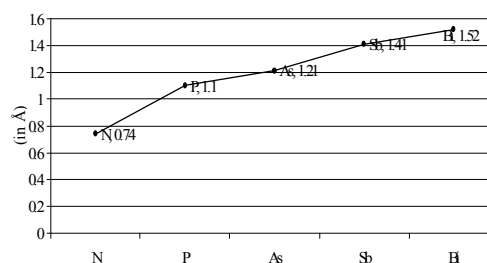
BORON FAMILY



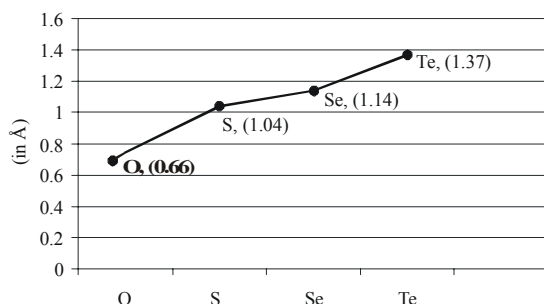
CARBON FAMILY



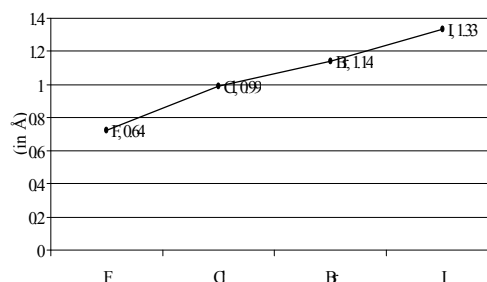
NITROGEN FAMILY (PNICOGENS)



CHALCOGENS



HALOGENS



☺ what can you predict or say about the increment in size along a group and decrement along a period]

ISOELECTRONIC MONOATOMIC SPECIES [$\text{Size} \propto 1/Z$] :

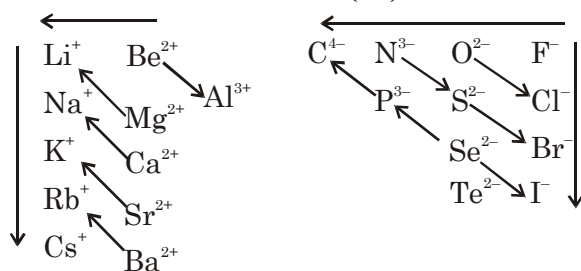
(i) S^{2-} , Cl^- , K^+ , Ca^{2+} , Sc^{3+}

(ii) H^- , He , Li^+

(iii) O^{2-} , F^- , Na^+ , Mg^{2+} , Al^{3+}

(iv) $\frac{\text{radius of cation}}{\text{radius of anion}} = \frac{Z_{\text{eff}} \text{ of Anion}}{Z_{\text{eff}} \text{ of Cation}}$

Note:- In the direction of arrow ($\rightarrow$) ionic size increases.

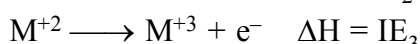
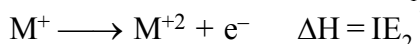
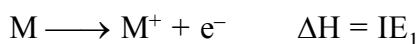


IONISATION ENERGY :

Amount of energy required to remove the most loosely bonded electron from an isolated gaseous atom from its ground state electronic configuration.

Units : kJ mol^{-1} , k cal mol^{-1} , eV per atom .

Ionisation is endothermic (endoergic) i.e. requires energy. Hence $\Delta H_{\text{ionisation}}$ is +ve.



Successive ionisation energy

$\text{IE}_3 > \text{IE}_2 > \text{IE}_1$ (always)

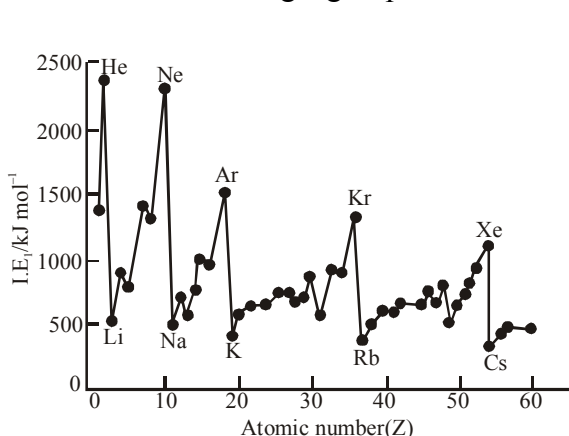
FACTORS AFFECTING IONISATION ENERGY :

- (1) Atomic size : Varies inversely
- (2) Screening effect : Varies inversely
- (3) Nuclear charge (Z) : Varies directly
- (4) Special electronic configuration of outermost electron (half filled / fully filled)
- (5) Type of orbitals involved in Ionisation : $s > p > d > f$.

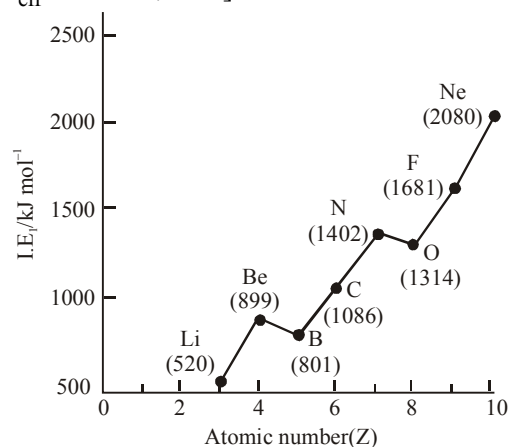
Note : Half filled and full filled inner orbitals, affects d-block and f-block trends.

General Trend: Along period I.E. increases [with some exception] [$Z_{\text{eff}} \uparrow$]

Along a group I.E. decrease [Z_{eff} constant, $n \uparrow$]



Variation of first ionization energy (IE_1) with atomic number for elements with $Z = 1$ to 60



First ionization energy (IE_1) of elements of the second period as a function of atomic number (Z)

Exception :

- (1) Along a period, half filled and fully filled configurations have higher I.E.
e.g. $\text{Be} > \text{B}$ and $\text{N} > \text{O}$.
- (2) Along a group, $\text{Ga} \approx \text{Al}$

PROPERTIES AFFECTED BY IONISATION ENERGY :

- (1) Metallic character (Varies inversely)
- (2) Reducing power (Varies inversely)
- (3) Tendency to stay in which state A^{+1} , A^{+2} or A^{+3}

 **Note :**

(a) Helium (He) has the highest ionisation energy (IE_1) among all the elements and Caesium (Cs) has the least (IE_1) value.

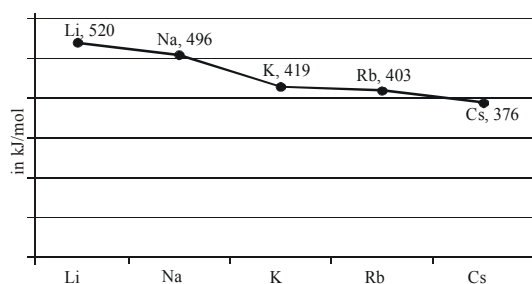
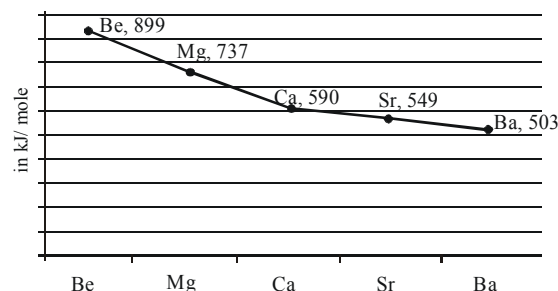
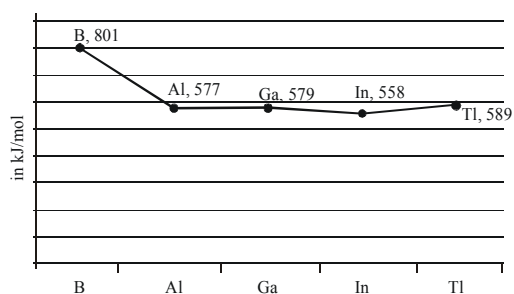
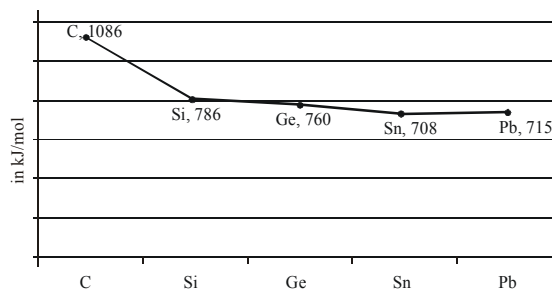
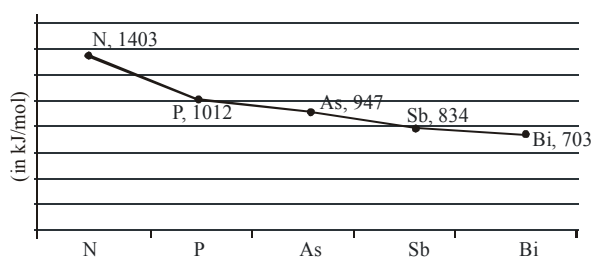
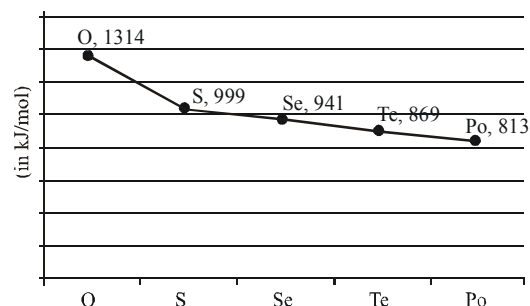
(b) Ionisation potential of inert gases is very high due to most stable s^2p^6 electronic configuration.

Element	He	Ne	Ar	Kr	Xe	Rn
$IE_1(\text{eV})$	24.5	21.6	15.8	14.0	12.1	10.7

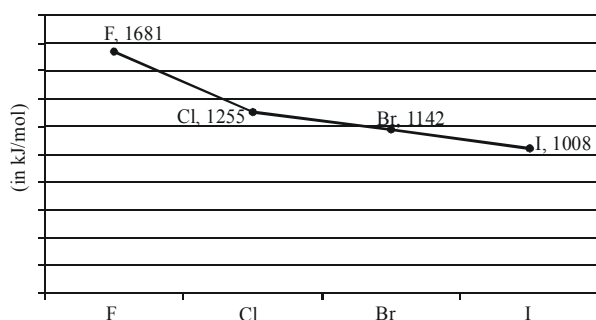
(c) For isoelectronic species I.E. increases with positive charge and decreases with negative charge.

e.g. $Al^{+3} > Mg^{+2} > Na^{+} > F^{-} > O^{-2} > N^{-3}$

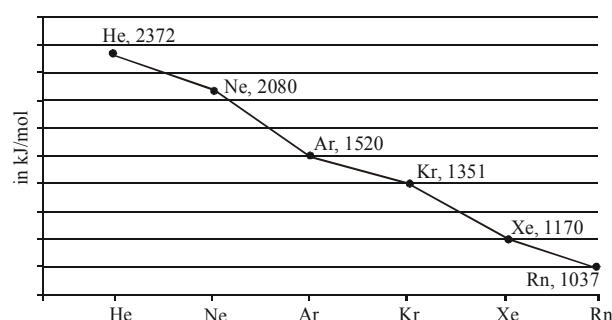
Graphical representation of ionisation energy :

ALKALI METALS**ALKALINE EARTH METALS****BORON FAMILY****CARBON FAMILY****NITROGEN FAMILY (PNICOGENS)****CHALCOGENS**

HALOGENS



NOBLE GASES



Ionisation energy of d-block elements :

1st, 2nd, 3rd IE's are increasing from left to right for 1st Transition series, but not regularly.

For 2nd IE Cr > Fe > Mn and Cu > Zn

For 3rd IE Mn > Cr > Fe and Zn has highest.

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic number	21	22	23	24	25	26	27	28	29	30
Electronic configuration										
M	3d ¹ 4s ²	3d ² 4s ²	3d ³ 4s ²	3d ⁵ 4s ¹	3d ⁵ 4s ²	3d ⁶ 4s ²	3d ⁷ 4s ²	3d ⁸ 4s ²	3d ¹⁰ 4s ¹	3d ¹⁰ 4s ²
M ²⁺	3d ¹	3d ²	3d ³	3d ⁴	3d ⁵	3d ⁶	3d ⁷	3d ⁸	3d ⁹	3d ¹⁰
M ³⁺	[Ar]	3d ¹	3d ²	3d ³	3d ⁴	3d ⁵	3d ⁶	3d ⁷	—	—
Enthalpy of atomisation, Δ _a H ^o /kJ mol ⁻¹	326	473	515	397	281	416	425	430	339	126
Ionisation Enthalpy, Δ ₁ H ^o /kJ mol ⁻¹										
I	631	656	650	653	717	762	758	736	745	906
II	1235	1309	1414	1592	1509	1561	1644	1752	1958	1734
III	2393	2657	2833	2990	3260	2962	3243	3402	3556	3829

ELECTRON AFFINITY & ELECTRON GAIN ENTHALPY (EGE) :

Electron Affinity : Amount of energy released when an electron is added to an isolated gaseous atom.

Units : k J mol⁻¹ , k Cal mol⁻¹ and eV per atom.

Addition of electron results in release of energy in most of the cases but for addition of second electron energy is always required. The sum of EA₁ & EA₂ is +ve (energy required)

$$E A \propto \frac{1}{\text{atomic size}} \propto Z_{\text{eff}} \quad (\text{Cl has the highest E.A.})$$

Electron gain Enthalpy (EGE) : When expressed in terms of enthalpy change (ΔH_{eg}) then it is termed as EGE Remember that ΔH = – ve for exothermic change.

For EA₁, energy is released

$$\therefore \Delta H_{\text{eg1}} = - \text{ve (Not always)}$$

For EA₂, EA₃energy is required

$$\therefore \Delta H_{\text{eg2}} = + \text{ve (always)}$$

In general EA₁ + EA₂ , energy is required.

$$\therefore (\Delta H_{\text{eg1}} + \Delta H_{\text{eg2}}) > 0$$

Note : ΔH_{eg} ≈ – EA

FACTORS AFFECTING ELECTRON AFFINITY :

- (1) **Atomic size** : Varies inversely
- (2) **Nuclear charge** : Varies directly
- (3) For stable electronic configuration i.e. half filled and fully filled shells EA decreases.

General Trend : Along a period, electron affinity increases [with a few exceptions] as $Z_{\text{eff}} \uparrow$.
 Along a group, electron affinity decreases after 3rd period. Between 2nd and 3rd period in p block electron affinity of 2nd period is lesser.

Exception :

- (1) A fully filled and half filled configuration have low values of EA or even sometimes energy is required rather than getting released.
- (2) 2nd period has lower value than 3rd owing to repulsion between electrons in small sized 2nd period elements.

Electron affinity of the main-group elements (in electron volts)

1							18
H							He
+0.754	2	13	14	15	16	17	-0.5
Li	Be	B	C	N	O	F	Ne
+0.618	≤ 0	$\leq +0.277$	+1.263	-0.07	+1.461	+3.399	-1.2
Na	Mg	Al	Si	P	S	Cl	Ar
+0.548	≤ 0	+0.441	+1.385	+0.747	+2.077	+3.617	-1.0
K	Ca	Ga	Ge	As	Se	Br	Kr
+0.502	-	+0.03	+1.2	+0.81	+2.021	+3.365	-1.0
Rb	Sr	In	Sn	Sb	Te	I	Xe
+0.486	-	+0.3	+1.2	+1.07	+1.971	+3.059	-0.8

ELECTRO NEGATIVITY :**Property of an atom in a molecule**

F has highest electronegativity in the periodic table.

Decreasing order $\rightarrow F > O > Cl \approx N > Br > S \approx C > I > H$.

Pauling Scale : $X_A \sim X_B = 0.208\sqrt{\Delta_{A-B}}$ Δ in kcal/mol

$$X_A \sim X_B = 0.102 \sqrt{\Delta_{A-B}} \quad \Delta \text{ in kJ/mol}$$

$$\Delta_{A-B} = E_{A-B} - (E_{A-A} \times E_{B-B})^{1/2}$$

Where, E_{A-B} = Bond energy of A-B; E_{A-A} = Bond energy of A-A; E_{B-B} = Bond energy of B-B

Mulliken's Scale : $X_M = \frac{IP + EA}{2}$ (IP & EA are expressed in eV)

Mulliken's values of EN are about 2.8 times higher than values on Pauling's scale.

$$\text{i.e. } X_P = \frac{X_M}{2.8}$$

$$\text{Allred-Rochow's Scale : } X_{\text{AR}} = \frac{0.359Z_{\text{eff}}}{r^2} ; X_{\text{P}} = X_{\text{AR}} + 0.744$$

r = covalent radius of atom (in Å)

Z_{eff} = Effective nuclear charge on periphery

FACTORS AFFECTING ELECTRO NEGATIVITY :

- (1) **Nuclear attraction** : Varies directly
- (2) **Atomic radius** : Varies inversely
- (3) **Charge on ions** : More positive charge, more electronegativity and more -ve charge, less electronegativity.
- (4) **Hybridisation** : To be discussed later in chemical bonding chapter.

General Trends : Along a period, electronegativity increases

Along a group, electronegativity decreases

Exceptions : Not Noteworthy.

 **Note:** Electronegativity of $\text{F} > \text{Cl}$ but Electron affinity of $\text{Cl} > \text{F}$
Electronegativity of Some Elements (on Pauling's Scale)

H 2.1						
Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
K 0.8	Ca 1.0	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8
Rb 0.8	Sr 1.0	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5
Cs 0.7	Ba 0.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2
Fr 0.7	Ra 0.9					

In **Pauling's Scale**, elements having almost same electronegativity are-

$$\text{C} \approx \text{S} \approx \text{I} \approx 2.5$$

$$\text{N} = \text{Cl} = 3.0$$

$$\text{P} = \text{H} = 2.1$$

$$\text{Cs} = \text{Fr} = 0.7$$

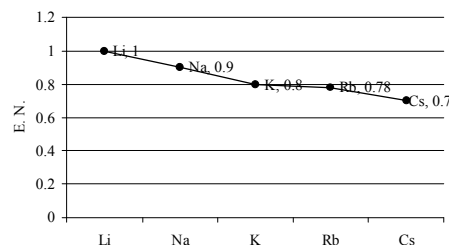
$$\text{Be} = \text{Al} = 1.5$$

PROPERTIES DEPENDENT ON ELECTRO NEGATIVITY :

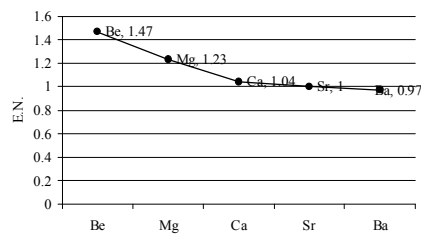
- (1) % ionic character
- (2) Strength of bond
- (3) Bond Length
- (4) Nature of hydrides
- (5) Nature of hydroxide.

Graphical representation of Electronegativity :

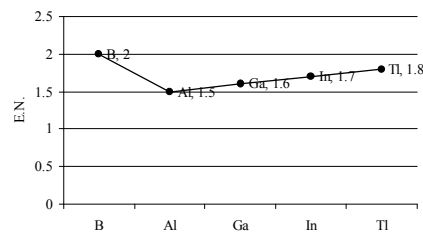
ALKALI METALS



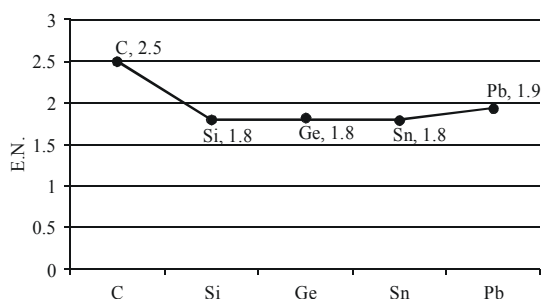
ALKALINE EARTH METALS



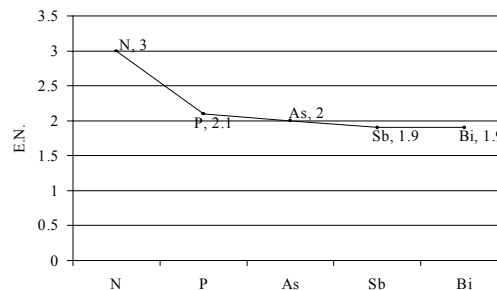
BORON FAMILY



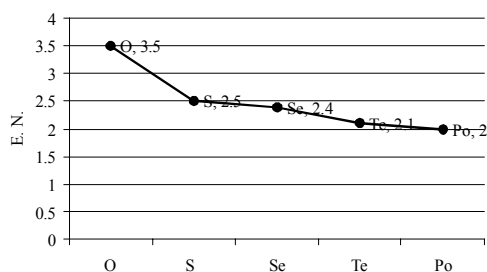
CARBON FAMILY



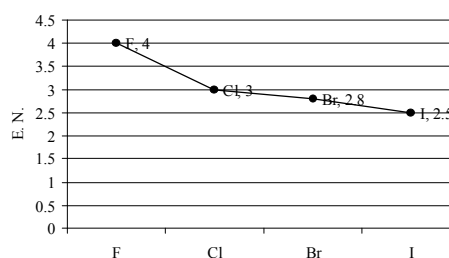
NITROGEN FAMILY (PNICOGENS)



CHALCOGENS



HALOGENS



LATTICE ENERGY :

The amount of energy released during the formation of 1 mole crystal lattice from constituent gaseous ions.



OR

The lattice energy of an ionic compound is the energy required to separate 1 mole of solid ionic substance completely into gaseous ions.



Lattice energies are large and positive because of attraction between positive and negative ions. The potential energy of two interacting charged particles is given by:

$$E = K \frac{Q_1 Q_2}{r}$$

where, $r = r^+ + r^-$; $r^+ \rightarrow$ radius of cation, $r^- \rightarrow$ radius of anion

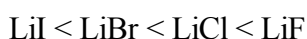
Where Q_1 and Q_2 are the charge on the particles in coulombs, and r is the distance between their centres in meters. The constant K has the value $9.0 \times 10^9 \text{ J} - \text{m/C}^2$.

Factors affecting lattice energy :

- The lattice energy increases as the charge on the ions increases and as their radii decreases.
- The magnitude of lattice energies however depends primarily on the ionic charges because ionic radii do not vary over a wide range.

For Example :

The lattice energy order for some compounds is given below.

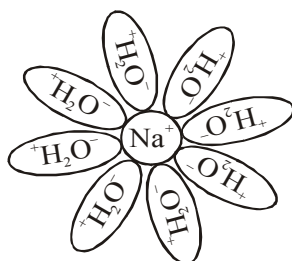
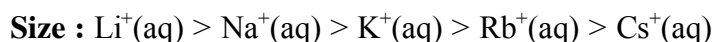


(Due to small size of halide ion F^- alongwith small Li^+)

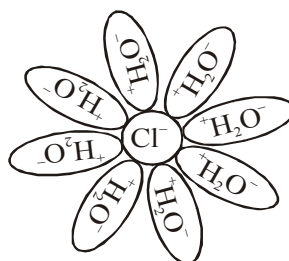
HYDRATION ENERGY :

It is the energy released when 1 mol of gaseous ions are hydrated in water. It is directly proportional to nuclear charge and inversely proportional to size. It always decreases down the group.

- As the charge density of ion increases hydrated size (or aqueous radius) increases.

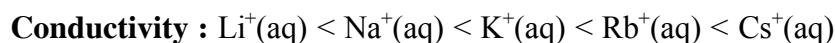
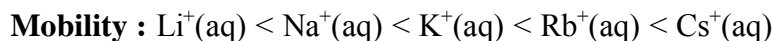


Hydration of Na^+



Hydration of Cl^-

- As the hydrated size of ion increases ionic mobility decreases, which thus, decreases conductivity of ions.



- Hydration energy also affects the solubility of ionic compounds. If hydration energy is greater than lattice energy then ionic compound will be soluble in water. More is the hydration energy, greater is the solubility, whereas, if lattice energy decreases, solubility of ionic compound increases.

MISCELLANEOUS CHEMICAL PROPERTIES :

1. Periodicity of hydra acids :

- (a) Acidic character of hydra acid increases from left to right in a period.
- (b) Acidic character of hydra acid increases from top to bottom in a group.

2. Periodicity of oxy acids :

- (a) Acidic character of oxy acid increases from left to right in a period.
- (b) Acidic character of oxy acid decreases from top to bottom in a group.

3. Periodicity of nature of oxide :

- (a) On moving from left to right in a period acidic nature of oxide generally increases.
e.g. $\text{CO}_2 < \text{P}_2\text{O}_5 < \text{SO}_3 < \text{Cl}_2\text{O}_7$
- (b) On moving from top to bottom in a group acidic nature of oxide generally decreases.

4. General trends

- (a) **Hydration energy** decreases along a group.
- (b) **Lattice energy** decreases along a group.

:: Some points to Remember ::

(without considering radioactive elements)

1. Second most electronegative element is Oxygen
2. Hydrogen is the lightest element and Lithium is lightest metal.
3. Helium has the highest value of I.P.
4. In periodic table metalloids are only in p-block.
5. Total gaseous elements are 11 (He, Ne, Ar, Kr, Xe, Rn, H₂, N₂, O₂, Cl₂, F₂)
6. Liquid metal at room temperature is – Hg.
7. Diamond is hardest natural substance.
8. Halogens have highest electron affinity and amongst them, Cl has the highest amongst them.
9. The largest cation of the periodic table = Cs⁺.
10. The smallest cation of the periodic table = H⁺.
11. The smallest anion of the periodic table = F⁻.
12. The biggest element of periodic table = Fr.
13. The smallest element of periodic table = H.
14. Br is liquid non-metal element at room temperature.
15. Osmium is the heaviest element known.
16. Fluorine is the most electronegative element.

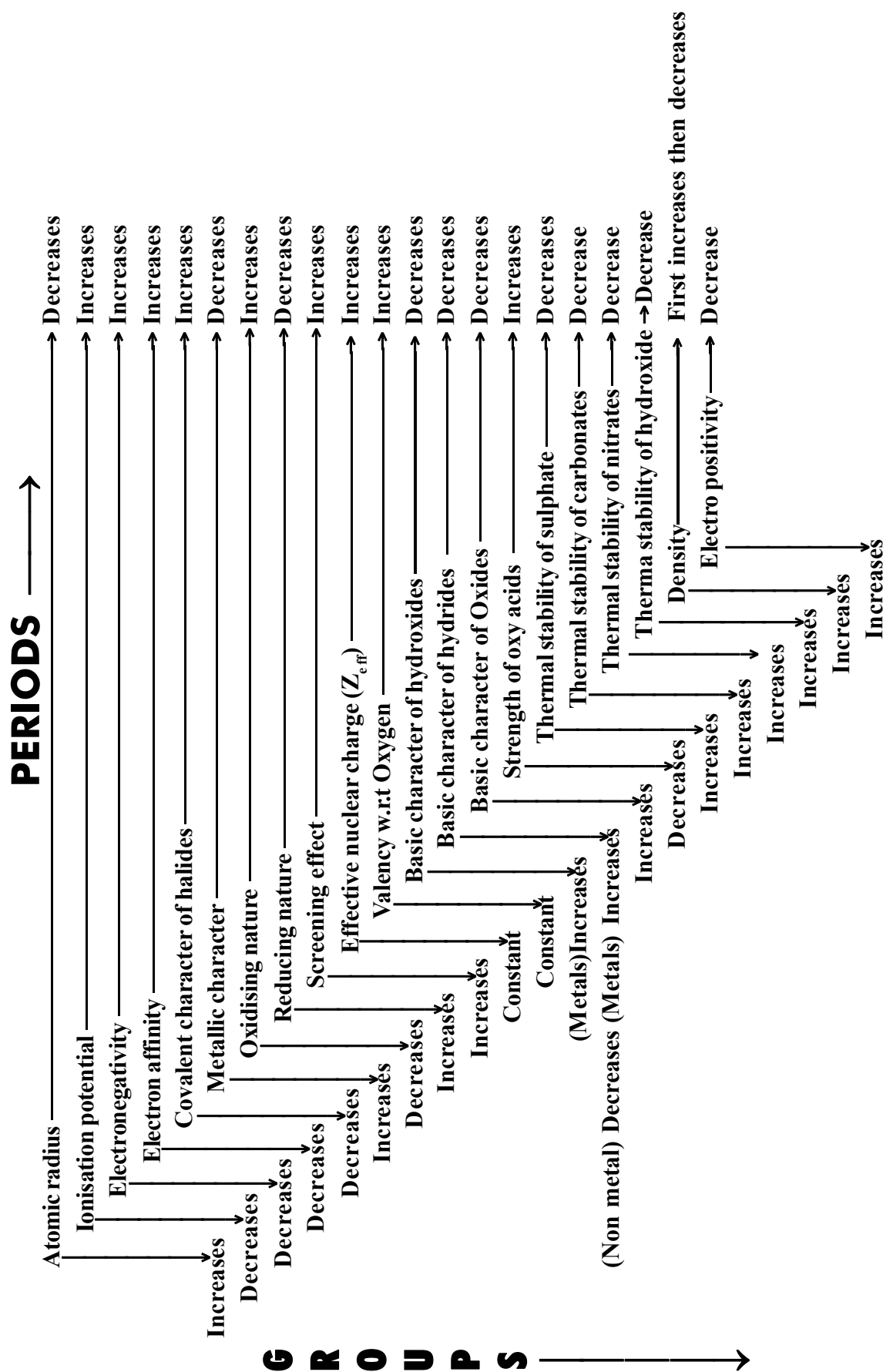
Extended or Long Form of the Periodic Table

Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H 1.0079 Hydrogen																	
2	Li 6.940 Lithium	Be 9.0122 Beryllium											B 10.811 Boron	C 12.011 Carbon	N 14.007 Nitrogen	O 15.999 Oxygen	F 18.998 Fluorine	Ne 20.180 Neon
3	Na 22.990 Sodium	Mg 24.305 Magnesium											Al 26.982 Aluminium	Si 28.086 Silicon	P 30.974 Phosphorus	S 32.066 Sulphur	Cl 35.453 Chlorine	Ar 39.948 Argon
4	K 39.098 Potassium	Ca 40.078 Calcium	Sc 44.956 Scandium	Ti 47.867 Titanium	V 50.941 Vanadium	Cr 51.996 Chromium	Mn 54.938 Manganese	Fe 55.847 Iron	Co 58.933 Cobalt	Ni 58.693 Nickel	Cu 63.546 Copper	Zn 65.39 Zinc	Ga 62.723 Gallium	Ge 72.61 Germanium	As 74.922 Arsenic	Se 78.96 Selenium	Br 79.904 Bromine	Kr 83.80 Krypton
5	Rb 85.468 Rubidium	Sr 87.62 Strontium	Y 88.906 Yttrium	Zr 91.224 Zirconium	Nb 92.906 Niobium	Mo 95.94 Molybdenum	Tc 98 Technetium	Ru 101.07 Ruthenium	Rh 102.91 Rhodium	Pd 106.42 Palladium	Ag 107.87 Silver	Cd 112.41 Cadmium	In 114.82 Indium	Sn 118.71 Tin	Sb 121.76 Antimony	Te 127.60 Tellurium	I 126.90 Iodine	Xe 131.29 Xenon
6	Cs 132.91 Cesium	Ba 137.33 Barium	La* 138.91 Lanthanum	Hf 178.49 Hafnium	Ta 180.95 Tantalum	W 183.84 Tungsten	Re 186.21 Rhenium	Os 190.23 Osmium	Ir 192.22 Iridium	Pt 195.08 Platinum	Au 196.97 Gold	Hg 200.59 Mercury	Tl 204.38 Thallium	Pb 207.2 Lead	Bi 208.98 Bismuth	Po 210 Polonium	At 210 Astatine	Rn 222 Radon
7	Fr 223 Francium	Ra 226 Radium	Ac** 227 Actinium	Unq 261 Unnilquadium	Unp 262 Unnilpentium	Unh 266 Unnilhexium	Uns 264 Unnilseptium	Uno 269 Unniloctium	Une 288 Unnilennium	Unn 269 Unnunnium	Uuu 272 Ununium	Uub 277 Ununbium	The symbols for elements 104-109 used in this table are those proposed by the American Chemical Society and 110-112 proposed by IUPAC					

6	*Lanthanide Series	58 Ce 140.12 Cerium	59 Pr 140.91 Praseodymium	60 Nd 144.24 Neodymium	61 Pm 145 Promethium	62 Sm 150.36 Samarium	63 Eu 151.96 Europium	64 Gd 157.25 Gadolinium	65 Tb 158.93 Terbium	66 Dy 162.50 Dysprosium	67 Ho 164.93 Holmium	68 Er 167.26 Erbium	69 Tm 168.93 Thulium	70 Yb 173.07 Ytterbium	71 Lu 174.97 Lutetium
7	** Actinide Series	90 Th 232.04 Thorium	91 Pa 231.04 Protactinium	92 U 238.03 Uranium	93 Np 237 Neptunium	94 Pu 244 Plutonium	95 Am 243 Americium	96 Cm 247 Curium	97 Bk 247 Berkelium	98 Cf 251 Californium	99 Es 252 Einsteinium	100 Fm 257 Fermium	101 Md 258 Mendelevium	102 No 259 Nobelium	103 Lr 262 Lawrencium

IUPAC designations of groups of elements are given in brackets

GENERAL TRENDS OF DIFFERENT PROPERTIES IN THE PERIOD AND GROUPS



EXERCISE # O-1

Periodic Table

- Which is not anomalous pair of elements in the Mendeleev's periodic table:-
(A) Ar and K (B) Co and Ni (C) Te and I (D) Al and Si
- Representative elements belong to :
(A) s-and p-block (B) d-block (C) d and f-block (D) f-block
- True statement is :-
(A) All the transuranic elements are synthetic elements
(B) Elements of third period are called transition elements
(C) Element of $[\text{Ar}] 3d^{10}4s^2$ configuration is placed in IIA group
(D) Electronic configuration of elements of a group is same
- Which of the following match is correct :-
(A) Last natural element – Uub
(B) General electronic configuration of IA group – ns^2
(C) Inert gas elements lies between 2nd – 6th period
(D) Typical elements – 3rd period elements
- The electronic configuration of elements X and Z are $1s^2 2s^2 2p^6 3s^2 3p^5$ and $1s^2 2s^2 2p^5$ respectively. What is the position of element X with respect to position of Z in the periodic table -
(A) Just below element Z (B) Just above Z
(C) Left to the Z (D) right to the Z
- Which of the following is not a Dobereiner triad :
(A) H, F, Cl (B) N, O, F (C) P, As, Sb (D) S, Se, Te
- Select the incorrect statement for Lothar Meyer's curve :
(A) Curve is plotted between atomic weight and atomic volume
(B) Alkali metals occupy maxima of curve
(C) Halogens occupy descending portions of the curve
(D) Transition metals occupy bottom portions of the curve

Atomic & Ionic Radii

- The size of the following species increases in the order:
(A) $\text{Mg}^{2+} < \text{Na}^+ < \text{F}^-$ (B) $\text{F}^- < \text{Na}^+ < \text{Mg}^{2+}$
(C) $\text{Mg}^{2+} < \text{F}^- < \text{Na}^+$ (D) $\text{Na}^+ < \text{F}^- < \text{Mg}^{2+}$
- Highest size will be of
(A) Br^- (B) I (C) I^- (D) I^+
- Element Cu has two oxidation states Cu^{+1} & Cu^{+2} . the right order of radii of these ions.
(A) $\text{Cu}^{+1} > \text{Cu}^{+2}$ (B) $\text{Cu}^{+2} > \text{Cu}^{+1}$ (C) $\text{Cu}^{+1} = \text{Cu}^{+2}$ (D) $\text{Cu}^{+2} \geq \text{Cu}^{+1}$
- The correct order of increasing atomic size of element N, F, Si & P.
(A) $\text{N} < \text{F} < \text{Si} < \text{P}$ (B) $\text{F} > \text{N} < \text{P} < \text{Si}$ (C) $\text{F} < \text{N} < \text{P} < \text{Si}$ (D) $\text{F} < \text{N} < \text{Si} < \text{P}$
- The correct order of atomic or ionic size
(A) $\text{N} < \text{Li} < \text{B}$ (B) $\text{Cl} < \text{Mg} < \text{Ca}$ (C) $\text{Ca}^{+2} < \text{S}^{-2} < \text{Cl}^-$ (D) $\text{Na}^+ < \text{Mg}^{+2} < \text{Cl}^-$
- In isoelectronic series largest difference between size is observed in N^{3-} , O^{2-} , F^- , Na^+ , Mg^{2+} :
(A) N^{3-} , Mg^{2+} (B) N^{3-} , O^{2-} (C) Mg^{2+} , Na^+ (D) F^- , Na^+

14. Mg, Mg^{2+} , Al and Al^{3+} are arranged in decreasing order of size $1 > 2 > 3 > 4$. Species which are present at 1 and 4 position respectively are :
 (A) Al, Mg^{2+} (B) Mg, Al^{3+} (C) Mg^{2+} , Al (D) Al^{3+} , Mg

Ionization Energy or Potential

15. In which of the following electronic configuration, ionisation energy will be maximum in
 (A) $[\text{Ne}] 3s^2 3p^1$ (B) $[\text{Ne}] 3s^2 3p^2$ (C) $[\text{Ne}] 3s^2 3p^3$ (D) $[\text{Ar}] 3d^{10} 4s^2 4p^3$
16. The correct order of second ionisation potential of C, N, O and F is:
 (A) $\text{C} > \text{N} > \text{O} > \text{F}$ (B) $\text{O} > \text{N} > \text{F} > \text{C}$ (C) $\text{O} > \text{F} > \text{N} > \text{C}$ (D) $\text{F} > \text{O} > \text{N} > \text{C}$
17. The ionization energy will be maximum for which process?
 (A) $\text{Ba} \rightarrow \text{Ba}^+$ (B) $\text{Be} \rightarrow \text{Be}^+$ (C) $\text{Cs} \rightarrow \text{Cs}^+$ (D) $\text{Li} \rightarrow \text{Li}^+$
18. Amongst the following, the incorrect statement is
 (A) $\text{IE}_1(\text{Al}) < \text{IE}_1(\text{Mg})$ (B) $\text{IE}_1(\text{Na}) < \text{IE}_1(\text{Mg})$
 (C) $\text{IE}_2(\text{Mg}) > \text{IE}_2(\text{Na})$ (D) $\text{IE}_3(\text{Mg}) > \text{IE}_3(\text{Al})$
19. Decreasing ionization potential for K, Ca & Ba is
 (A) $\text{Ba} > \text{K} > \text{Ca}$ (B) $\text{Ca} > \text{Ba} > \text{K}$ (C) $\text{K} > \text{Ba} > \text{Ca}$ (D) $\text{K} > \text{Ca} > \text{Ba}$
20. Alkaline earth metals always form dipositive ions due to
 (A) $\text{IE}_2 - \text{IE}_1 > 10 \text{ eV}$ (B) $\text{IE}_2 - \text{IE}_1 = 17 \text{ eV}$
 (C) $\text{IE}_2 - \text{IE}_1 < 10 \text{ eV}$ (D) None of these
21. The correct order of second I.P.
 (A) $\text{Na} < \text{Mg} > \text{Al} < \text{Si}$ (B) $\text{Na} > \text{Mg} < \text{Al} > \text{Si}$
 (C) $\text{Na} > \text{Mg} > \text{Al} < \text{Si}$ (D) $\text{Na} > \text{Mg} > \text{Al} > \text{Si}$

Electron affinity or Electron Gain Enthalpy

22. The process requires absorption of energy is
 (A) $\text{F} \rightarrow \text{F}^-$ (B) $\text{Cl} \rightarrow \text{Cl}^-$ (C) $\text{O}^- \rightarrow \text{O}^{2-}$ (D) $\text{H} \rightarrow \text{H}^-$
23. Of the following elements, which possesses the highest electron affinity?
 (A) As (B) O (C) S (D) Se
24. Electron affinities of O, F, S and Cl are in the order.
 (A) $\text{O} < \text{S} < \text{Cl} < \text{F}$ (B) $\text{O} < \text{S} < \text{F} < \text{Cl}$
 (C) $\text{S} < \text{O} < \text{Cl} < \text{F}$ (D) $\text{S} < \text{O} < \text{F} < \text{Cl}$
25. Increasing order of Electron affinity for following configuration.
 (a) $1s^2, 2s^2 2p^2$ (b) $1s^2, 2s^2 2p^4$
 (c) $1s^2, 2s^2 2p^6 3s^2 3p^4$ (d) $1s^2, 2s^2 2p^6, 3s^2 3p^3$
 (A) $d < a < b < c$ (B) $d < a < c < b$ (C) $a < b < c < d$ (D) $a < b < d < c$
26. Highest electron affinity is shown by
 (A) F^- (B) Cl^- (C) Li^+ (D) Na^+
27. Which of the following statements is not true?
 (A) F atom can hold additional electron more tightly than Cl atom
 (B) Cl atom can hold additional electron more tightly than F atom
 (C) The incoming electron encounters greater repulsion for F atom than for Cl atom
 (D) It is easier to remove an electron from F^- than Cl^- .

Electronegativity

28. The outermost electronic configuration of most electronegative element amongst the following is :
 (A) $ns^2 np^3$ (B) $ns^2 np^4$ (C) $ns^2 np^5$ (D) $ns^2 np^6$
29. In the following, which is the correct representation ?
 (A) $\overset{\delta+}{C} - \overset{\delta-}{F}$ (B) $\overset{\delta+}{C} - \overset{\delta+}{Cl}$ (C) $\overset{\delta+}{F} - \overset{\delta-}{Cl}$ (D) $\overset{\delta-}{O} - \overset{\delta+}{F}$
30. On the Pauling's electronegativity scale, which element is next to F .
 (A) Cl (B) O (C) Br (D) Ne
31. Which one is not correct order of electronegativity.
 (A) $F > Cl > Br > I$ (B) $Si > Al > Mg > Na$
 (C) $Cl > S > P > Si$ (D) None of these
32. The increasing order of acidic nature of Li_2O , BeO , B_2O_3
 (A) $Li_2O > BeO < B_2O_3$ (B) $Li_2O < BeO < B_2O_3$
 (C) $Li_2O < BeO > B_2O_3$ (D) $Li_2O > BeO > B_2O_3$
33. The lowest electronegativity of the element from the following atomic number is.
 (A) 37 (B) 55 (C) 9 (D) 35

Miscellaneous

34. Which of the following does not reflect the periodicity of element
 (A) Bonding behaviour (B) Electronegativity
 (C) Ionisation potential (D) Neutron/ Proton ratio
35. Among the following, which species is/are paramagnetic ?
 (i) Sr^{2+} (ii) Fe^{3+} (iii) Co^{2+} (iv) S^{2-} (v) Pb^{2+}
 (A) i, iv, v (B) i, ii, iii (C) ii, iii (D) iv, v
36. Choose the s-block element from the following:
 (A) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5, 4s^1$ (B) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}, 4s^1$
 (C) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$ (D) all of the above
37. False statement for periodic classification of elements is
 (A) The properties of the elements are periodic function of their atomic numbers.
 (B) No. of non-metallic elements is less than the no. of metallic elements.
 (C) First ionization energy of elements does not increase regularly with the increase in atomic number in a period.
 (D) d-subshell is filled by final electron with increasing atomic number of inner transition elements.
38. Which of the following order is incorrect against the property indicated :
 (A) $Mg < Ar < Na$ (2nd I.E.) (B) $Be < F < Cl$ ($|\Delta H_{eg}|$)
 (C) $Rb < Na < K > Ca$ (atomic radius) (D) $P < S < N$ (electronegativity)
39. If each orbital can hold a maximum of three electrons, the number of elements in 9th period of periodic table (long form) will be
 (A) 48 (B) 162 (C) 50 (D) 75

40. Which of the following element has highest metallic character .
- | Element | IP |
|---------|-------|
| (A) P | 17 eV |
| (B) Q | 2 eV |
| (C) R | 10 eV |
| (D) S | 13 eV |
41. The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^4$. The atomic number and the group number of the element 'X' which is just below the above element in the periodic table are respectively.
- (A) 24 & 6 (B) 24 & 15 (C) 34 & 16 (D) 34 & 8
42. The number of d- electrons in Mn^{2+} is equal to that of
- (A) p-electrons in N (B) s-electrons in Na
(C) d-electrons in Fe^{+2} (D) p-electrons in O^{-2}
43. Which of the following formula has involved all the energy terms used to calculated ΔH_f° of $Na_2O_{(s)}$.
(ΔH_{sb} : Sublimation energy ; $I.E_1$: First ionisation energy ; $I.E_2$: Second ionisation energy ; B.D.E.; Bond dissociation energy ; E.G.E₁ : First electron gain enthalpy ; E.G.E₂ : Second electron gain enthalpy; U : Lattice energy)
- (A) $+ 2\Delta H_{sb} + I.E_1 + I.E_2 + \frac{B.D.E.}{2} + E.G.E_1 + E.G.E_2 + U$
- (B) $+ 2\Delta H_{sb} + 2I.E_1 + \frac{B.D.E.}{2} + E.G.E_1 + E.G.E_2 + U$
- (C) $+ 2\Delta H_{sb} + 2I.E_1 + \frac{B.D.E.}{2} + 2E.G.E_1 + U$
- (D) $+ 2\Delta H_{sb} + I.E_1 + \frac{B.D.E.}{2} + E.G.E_1 + E.G.E_2 + U$
44. EN of the element (A) is E_1 and IP is E_2 . Hence EA will be according to mulliken
- (A) $2E_1 - E_2$ (B) $E_1 - E_2$ (C) $E_1 - 2E_2$ (D) $(E_1 + E_2)/2$
45. Moving from right to left in a periodic table, the atomic size is:
- (A) Increased (B) Decreased (C) Remains constant (D) None of these
46. One element has atomic weight 39. Its electronic configuration is $1s^2, 2s^2 2p^6, 3s^2 3p^6 4s^1$. The true statement for that element is:
- (A) High value of IE (B) Transition element (C) Isotone with $^{38}_{18}Ar$ (D) None
47. The number of paired electrons in oxygen atom is:
- (A) 6 (B) 16 (C) 8 (D) 32
48. The decreasing size of K^+ , Ca^{+2} , Cl^- & S^{2-} follows the order:
- (A) $K^+ > Ca^{+2} > S^{2-} > Cl^-$ (B) $K^+ > Ca^{+2} > Cl^- > S^{2-}$
(C) $Ca^{+2} > K^+ > Cl^- > S^{2-}$ (D) $S^{2-} > Cl^- > K^+ > Ca^{+2}$
49. Which of the following has the maximum number of unpaired electrons
- (A) Mg^{2+} (B) Ti^{3+} (C) V^{3+} (D) Fe^{2+}

EXERCISE : O-2

Atomic & Ionic Radius

- Select correct order of size :
 (A) $\text{Ti}^{2+} < \text{Ti} < \text{Zr}$ (B) $\text{Ti}^{2+} < \text{Ti} < \text{Hf}$ (C) $\text{Zr}^{2+} < \text{Zr} \approx \text{Hf}$ (D) $\text{Hf}^{2+} < \text{Hf} \approx \text{Zr}$
- Which of the following orders of atomic / Ionic radius is correct ?
 (A) $\text{B} < \text{Al} \approx \text{Ga}$ (B) $\text{Sc} > \text{Cu} < \text{Zn}$ (C) $\text{C} < \text{O} < \text{N}$ (D) $\text{Al}^{3+} < \text{Al}^{2+} < \text{Al}^+$

Electron Affinity

- Which of the following is correct order of EA.
 (A) $\text{N} < \text{C} < \text{O} < \text{F}$ (B) $\text{F} > \text{Cl} > \text{Br} > \text{I}$ (C) $\text{Cl} > \text{F} > \text{Br} > \text{I}$ (D) $\text{C} < \text{N} < \text{O} < \text{F}$
- The electron affinity of the members of oxygen family of the periodic table, follows the sequence
 (A) $\text{O} > \text{S} > \text{Se}$ (B) $\text{S} > \text{O} < \text{Se}$ (C) $\text{O} < \text{S} > \text{Se}$ (D) $\text{Se} > \text{O} > \text{S}$

Ionisation Energy

- Considering the following ionisation steps :
 $\text{A(g)} \rightarrow \text{A}^+(\text{g}) + \text{e}^- \quad \Delta H = 100 \text{ eV}$ $\text{A(g)} \rightarrow \text{A}^{2+}(\text{g}) + \text{e}^- \quad \Delta H = 250 \text{ eV}$
 Select the correct statements :
 (A) IE_1 of A(g) is 100 eV (B) IE_1 of $\text{A}^+(\text{g})$ is 150 eV
 (C) IE_2 of A(g) is 150 eV (D) IE_2 of A(g) is 250 eV
- Which of the following are correct ?
 (A) $\text{IE}_2(\text{Mg}) < \text{IE}_2(\text{Na})$ (B) $\text{EA}(\text{N}) < \text{EA}(\text{P})$
 (C) Atomic size $\text{Mg}^{2+} > \text{Atomic size}(\text{Li}^+)$ (D) $\text{IP of Na} < \text{Mg} < \text{Al}$

Electronegativity

- Amongst the following statements, which is / are correct?
 (A) Electronegativity of sulphur is greater than that of oxygen.
 (B) Electron affinity of oxygen is smaller than that of sulphur.
 (C) Electron gain enthalpy of fluorine is most negative
 (D) Electron gain enthalpy of chlorine is most negative
- An element 'E' have $\text{IE} = x \text{ eV/atom}$ and $\text{EA} = y \text{ eV / atom}$ and EN on Pauling scale is 1.2. Find EN of 'E' on Mullikan scale :

$$(A) \frac{x+y}{2} \quad (B) 1.2 \times 2.8 \quad (C) \frac{x+y}{2.8} - 1.2 \quad (D) 1.2 - \frac{x+y}{2}$$

Hydration Energy

- Choose the INCORRECT order of hydrated size of the ions -
 (A) $\text{F}_{(\text{aq.})}^{\ominus} > \text{Cl}_{(\text{aq.})}^{\ominus} > \text{Br}_{(\text{aq.})}^{\ominus} > \text{I}_{(\text{aq.})}^{\ominus}$ (B) $\text{Rb}_{(\text{aq.})}^{\oplus} > \text{K}_{(\text{aq.})}^{\oplus} > \text{Na}_{(\text{aq.})}^{\oplus} > \text{Li}_{(\text{aq.})}^{\oplus}$
 (C) $\text{Na}_{(\text{aq.})}^{\oplus} > \text{Mg}_{(\text{aq.})}^{2+} > \text{Al}_{(\text{aq.})}^{3+}$ (D) $\text{Be}_{(\text{aq.})}^{2+} > \text{Mg}_{(\text{aq.})}^{2+} > \text{Ca}_{(\text{aq.})}^{2+} > \text{Sr}_{(\text{aq.})}^{2+}$
- Find the correct ionic mobility order in aqueous solution from the following options-
 (A) $\text{Li}^+ < \text{Na}^+$ (B) $\text{Mg}^{2+} < \text{Sr}^{2+}$ (C) $\text{Na}^+ < \text{K}^+$ (D) $\text{F}^- < \text{Cl}^-$

Miscellaneous

11. Select the correct statement(s).
 - (A) The value of electron gain enthalpy of an element can be -ve or +ve.
 - (B) In the periodic table, metallic character of the elements increases down the group and decreases across the period
 - (C) The Cl^- & S^{2-} are isoelectronic species but first one is not smaller in size than the second
 - (D) Ionization enthalpy of an atom is equal to electron gain enthalpy of cation
12. In halogens, which of the following properties increase from iodine to fluoroine
 - (A) Ionisation energy
 - (B) Electronegativity
 - (C) Bond length
 - (D) Electron affinity
13. In which of the following set of elements 1st element is more metallic than second.
 - (A) Ba, Ca
 - (B) Sb, Sn
 - (C) Ge, S
 - (D) Na, F
14. Which of the following order(s) is / are **CORRECT** :
 - (A) $\text{Li} < \text{Be} < \text{B} < \text{C}$ (IE_1)
 - (B) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$ (Bond length)
 - (C) $\text{Na}_2\text{O} < \text{MgO} < \text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_5$ (Acidic)
 - (D) $\text{Li}^+(\text{g}) < \text{Na}^+(\text{g}) < \text{K}^+(\text{g}) < \text{Cs}^+(\text{g})$ (Ionic radius)
15. Which of the following order is correct :
 - (A) $\text{P} < \text{Si} < \text{Be} < \text{Mg} < \text{Na}$ (Metallic character)
 - (B) $\text{Mg}^{+2} < \text{Na}^+ < \text{F}^- < \text{O}^{2-}$ (Ionic radius)
 - (C) $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{N} < \text{O}$ (2nd ionization energy)
 - (D) $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$ (Ionic mobility)
16. The ionic compound $\text{A}^+ \text{B}^-$ is formed easily when the
 - (A) electron affinity of B is high
 - (B) ionization energy of A is low
 - (C) lattice energy of AB is high
 - (D) lattice energy of AB is low
17. Which of the following is/are correct?
 - (A) For $\text{A}(\text{g}) + \text{e}^- \longrightarrow \text{A}^-(\text{g})$ ΔH may be negative
 - (B) For $\text{A}^-(\text{g}) + \text{e}^- \longrightarrow \text{A}^{2-}(\text{g})$ ΔH may be negative
 - (C) For $\text{A}^-(\text{g}) + \text{e}^- \longrightarrow \text{A}^{2-}(\text{g})$ ΔH must be positive
 - (D) For $\text{Ne}(\text{g}) + \text{e}^- \longrightarrow \text{Ne}^-(\text{g})$ ΔH may be zero

EXERCISE # S-1

- Find out the atomic number of element whose IUPAC name is Unnilpentium.
Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.
- Select total number of acidic compounds out of given below.
CsOH, OC(OH)₂, SO₂(OH)₂, Sr(OH)₂, Ca(OH)₂, Ba(OH)₂, BrOH, NaOH, O₂NOH
- Total number of enthalpy(s) (out of given eight) of A(g) which is/are **not** associated with conversion of $A_{(g)}^- \longrightarrow A_{(g)}^{+4}$.
IE₁, IE₂, IE₃, IE₄, IE₅, IE₆, EGE₁, EGE₂
(IE = ionization energy, EGE = electron gain enthalpy)
- Write the number of pairs in which size of first element or ion is higher as compared to IInd out of following eight pairs.
(O, S), (He, Ne), (Kr, Ne), (Na, Na⁺), (Cl, Cl⁻), (I⁻, Cl⁻), (Li, Na), (Li⁺, Na⁺)
- Total number of elements which have less IE₁ than that of 'N'.
Be, B, C, F, P, He
- Size of H⁻ is smaller than how many elements among these ?
H, Li, Be, B, C, N, O, F, F⁻
- How many orders are **CORRECT**
(a) Acidic strength $\Rightarrow$ ClOH < BrOH < IOH
(b) Basic strength $\Rightarrow$ MgO < CaO < SrO < BeO
(c) Electronegativity $\Rightarrow$ I < Br < N < O < F
(d) Electron affinity $\Rightarrow$ Mg < Na < Si < S < Cl
(e) % Ionic character $\Rightarrow$ NaF < KF < RbF
- _____ is higher for fluorine as compared to chlorine.
Find the number of properties given below to fill the blank space to make a correct statement.
Atomic mass, Covalent radius, Ionic radius (X⁻), Ionization energy, Electron affinity, Electronegativity, Hydration energy of uninegative ion (X⁻),
- Upto argon find the number of elements which have lower IE₁ as compared to He.
- The number of pairs, in which EA of the second element is more than that of the first element is :
[O, S], [C, N], [O, N], [N, P], [Cl⁺, F⁺], [K⁺, Na⁺]
- For an element the successive ionisation energy values (in eV/atom), are given below.
14.534, 29.601, 47.448, 77.472, 97.888, 552.057, 667.029
Find the number of valence shell electrons in that element.

Subjective :

- Calculate E.N. of chlorine atom on Pauling scale if I.E. of Cl⁻ is 4eV & of E.A. of Cl⁺ is + 13.0 eV.
- Increasing order of ionic size :
N³⁻, Na⁺, F⁻, O²⁻, Mg²⁺

EXERCISE # S-2

Paragraph for Questions 1 to 2

First electron gain enthalpy (in $\frac{\text{kJ}}{\text{mol}}$) of few elements are given below :

Elements	ΔH_{eg}
I	-60
II	-45
III	-328
IV	-295
V	+ 48

Answer the following questions on the basis of above data:

- Which element may be an inert gas
(A) I (B) III (C) IV (D) V
- Which element is most non-metallic among all the elements -
(A) I (B) II (C) III (D) IV

Paragraph for Questions 3 to 4

The IE_1 and the IE_2 in KJ/mol of a few elements designated by U, V, W, X are shown below.

Atom	IE_1	IE_2
U	2464	6110
V	610	7542
W	928	1810
X	1588	3410

Based on the above information answer the following question :-

- Which of the elements represent a noble gas.
(A) U (B) V (C) W (D) X
- Which of the following element belongs to group 1 (IA).
(A) U (B) V (C) W (D) X

Paragraph for Question 5 to 7

Nature of bond can be predicted on the basis of electronegativity of bonded atoms, greater difference in electronegativity (X), more will be the polarity of bond, and polar bond are easily broken in polar solvent like water. For hydroxy acids $\text{X}_\text{O} - \text{X}_\text{A}$ difference predict the nature of oxide formed by the element A.

$|\text{X}_\text{O} - \text{X}_\text{A}| > |\text{X}_\text{O} - \text{X}_\text{H}|$ then A-O-H show basic nature (NaOH)

$|\text{X}_\text{O} - \text{X}_\text{A}| < |\text{X}_\text{O} - \text{X}_\text{H}|$ then A-O-H show acidic nature (H-O-Cl)

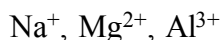
With the help of EN values [$\text{EN}_\text{A} = 1.8$, $\text{EN}_\text{B} = 2.6$, $\text{EN}_\text{C} = 1.6$, $\text{EN}_\text{D} = 2.8$] answer the following questions for the compounds HAO, HBO, HCO, HDO.

- Compounds whose aqueous solution is acidic and order of their acidic strength
(A) AOH, COH ; AOH < COH (B) HDO, HBO ; HDO > HBO
(C) AOH, COH ; AOH > COH (D) HDO, HBO ; HDO < HBO
- Compounds whose aqueous solution is basic and order of their basic strength
(A) AOH, COH ; AOH < COH (B) HDO, HBO ; HDO > HBO
(C) AOH, COH ; AOH > COH (D) HDO, HBO ; HDO < HBO
- Percentage ionic character of compound AB is
(A) 42.42% (B) 24.24% (C) 15.04% (D) None of these

Paragraph for Question 8 to 12

All the simple salt dissolve in water, producing ions and consequently the solution conduct electricity. In this process water molecule surround both the cations and anions & release energy. This process is called hydration & energy released is called hydration energy & it depends on size of gaseous ions.

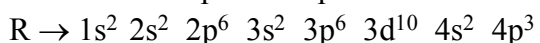
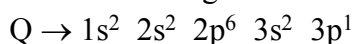
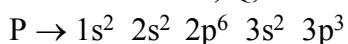
Answer the following questions with respect to given cations.



8. Order of extent of hydration
 (A) $\text{Na}^+ = \text{Mg}^{2+} = \text{Al}^{3+}$ (B) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
 (C) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$ (D) $\text{Al}^{3+} > \text{Mg}^{2+} < \text{Na}^+$
9. Order of hydration energy
 (A) $\text{Na}^+ = \text{Mg}^{2+} = \text{Al}^{3+}$ (B) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
 (C) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$ (D) $\text{Al}^{3+} > \text{Mg}^{2+} < \text{Na}^+$
10. Order of size of hydrated ion.
 (A) $\text{Na}^+ = \text{Mg}^{2+} = \text{Al}^{3+}$ (B) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
 (C) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$ (D) $\text{Al}^{3+} > \text{Mg}^{2+} < \text{Na}^+$
11. Order of ionic mobility
 (A) $\text{Na}^+ = \text{Mg}^{2+} = \text{Al}^{3+}$ (B) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
 (C) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$ (D) $\text{Al}^{3+} > \text{Mg}^{2+} < \text{Na}^+$
12. Order of size of gaseous ions.
 (A) $\text{Na}^+ = \text{Mg}^{2+} = \text{Al}^{3+}$ (B) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
 (C) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$ (D) $\text{Al}^{3+} > \text{Mg}^{2+} < \text{Na}^+$

Paragraph for Question 13 to 14 are based on the following information.

Four elements P, Q, R & S have ground state electronic configuration as:



13. Comment which of the following option represent the correct order of true (T) & false (F) statement.
 I size of P < size of Q II size of R < size of S
 III size of P < size of R (appreciable difference) IV size of Q < size of S (appreciable difference)
 (A) TTTT (B) TTTF (C) FFTT (D) TTFF
14. Order of IE_1 values among the following is
 (A) $\text{P} > \text{R} > \text{S} > \text{Q}$ (B) $\text{P} < \text{R} < \text{S} < \text{Q}$ (C) $\text{R} > \text{S} > \text{P} > \text{Q}$ (D) $\text{P} > \text{S} > \text{R} > \text{Q}$

Matching List

- | | | |
|-----|---|--|
| 15. | Column-I
Element
(P) Si
(Q) Sc
(R) Ga
(S) Tl | Column-II
Period and group number respectively
(1) 4, 3
(2) 3, 14
(3) 6, 13
(4) 4, 13 |
|-----|---|--|

Code :

- | | | | | |
|-----|----------|----------|----------|----------|
| | P | Q | R | S |
| (A) | 2 | 1 | 3 | 4 |
| (B) | 1 | 2 | 4 | 3 |
| (C) | 2 | 1 | 4 | 3 |
| (D) | 4 | 3 | 1 | 2 |

Match the column

16. Match the column :

Column I

- (A) Highest density
- (B) Metallic character
- (C) Lightest Metal
- (D) Liquid at room temperature

Column II

- (P) Lithium
- (Q) Osmium
- (R) Mercury
- (S) Bromine

17. If electrons are filled in the sub shells of an atom in the following order 1s, 2s, 2p, 3s, 3p, 3d, 4s, 4p, 4d, 4f..... then match the following element in List I with block in List II.

List-I

- (A) K(19)
- (B) Fe(26)
- (C) Ga(31)
- (D) Sn(50)

List-II

- (P) s-Block
- (Q) p-Block
- (R) d-Block
- (S) f-block

18. Match the characteristics mentioned in List II with the process in List I.

List I

- (A) $O(g) + e^- \rightarrow O^-(g)$
- (B) $O^-(g) + e^- \rightarrow O^{2-}(g)$
- (C) $Na^-(g) \rightarrow Na(g) + e^-$
- (D) $Mg^+(g) + e^- \rightarrow Mg(g)$

List II

- (P) Positive electron gain enthalpy
- (Q) Negative electron gain enthalpy
- (R) Exothermic
- (S) Endothermic

19. Match the column :

Column I

- (A) Cl
- (B) F
- (C) Cu
- (D) He

Column II

- (P) Metal
- (Q) Highest negative electron gain enthalpy
- (R) Most Electronegative element
- (S) Highest ionisation energy.

EXERCISE # JEE-MAIN

- According to the Periodic law of elements, the variation in properties of elements is related to their :- [AIEEE-2003]
 - (1) Nuclear masses
 - (2) Atomic numbers
 - (3) Nuclear neutron-proton number ratio
 - (4) Atomic masses
- The atomic numbers of vanadium (V), chromium (Cr), manganese (Mn) and iron (Fe) are respectively 23, 24, 25 and 26. Which one of these may be expected to have the highest second ionization enthalpy? [AIEEE-2003]
 - (1) Cr
 - (2) Mn
 - (3) Fe
 - (4) V
- Among Al_2O_3 , SiO_2 , P_2O_5 and SO_3 , the correct order of acid strength is :- [AIEEE-2004]
 - (1) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{SO}_3 < \text{P}_2\text{O}_5$
 - (2) $\text{SiO}_2 < \text{SO}_3 < \text{Al}_2\text{O}_3 < \text{P}_2\text{O}_5$
 - (3) $\text{SO}_3 < \text{P}_2\text{O}_5 < \text{SiO}_2 < \text{Al}_2\text{O}_3$
 - (4) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_5 < \text{SO}_3$
- The formation of the oxide ion $\text{O}^{2-}(\text{g})$ requires first an exothermic and then an endothermic step as shown below :- [AIEEE-2004]

$$\text{O}(\text{g}) + \text{e}^- = \text{O}^-(\text{g}), \quad \Delta H^\circ = -142 \text{ kJ mol}^{-1}$$

$$\text{O}^-(\text{g}) + \text{e}^- = \text{O}^{2-}(\text{g}), \quad \Delta H^\circ = 844 \text{ kJ mol}^{-1}$$

This is because :-

 - (1) O^- ion will tend to resist the addition of another electron
 - (2) Oxygen has high electron affinity
 - (3) Oxygen is more electronegative
 - (4) O^- ion has comparatively larger size than oxygen atom
- In which of the following arrangements, the order is NOT according to the property indicated below? [AIEEE-2005]
 - (1) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^{1+} < \text{F}^-$ – increasing ionic size
 - (2) $\text{B} < \text{C} < \text{N} < \text{O}$ - increasing first ionization enthalpy
 - (3) $\text{I} < \text{Br} < \text{F} < \text{Cl}$ - increasing electron gain enthalpy (with negative sign)
 - (4) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ - increasing metallic radius
- Which of the following oxides is amphoteric in character ? [AIEEE-2005]
 - (1) SnO_2
 - (2) SiO_2
 - (3) CO_2
 - (4) CaO
- Which of the following factors may be regarded as the main cause of lanthanide contraction ? [AIEEE-2005]
 - (1) poor shielding of one of 4f electron by another in the subshell
 - (2) effective shielding of one of 4f electrons by another in the subshell
 - (3) poorer shielding of 5d electrons by 4f electrons
 - (4) greater shielding of 5d electrons by 4f electrons
- The increasing order of the first ionization enthalpies of the elements B, P, S and F (lowest first) is:- [AIEEE-2006]
 - (1) $\text{F} < \text{S} < \text{P} < \text{B}$
 - (2) $\text{P} < \text{S} < \text{B} < \text{F}$
 - (3) $\text{B} < \text{P} < \text{S} < \text{F}$
 - (4) $\text{B} < \text{S} < \text{P} < \text{F}$

9. The set representing the correct order of ionic radius is :- [AIEEE-2009]
 (1) $\text{Li}^+ > \text{Na}^+ > \text{Mg}^{2+} > \text{Be}^{2+}$ (2) $\text{Mg}^{2+} > \text{Be}^{2+} > \text{Li}^+ > \text{Na}^+$
 (3) $\text{Li}^+ > \text{Be}^{2+} > \text{Na}^+ > \text{Mg}^{2+}$ (4) $\text{Na}^+ > \text{Li}^+ > \text{Mg}^{2+} > \text{Be}^{2+}$
10. The correct order of electron gain enthalpy with negative sign of F, Cl, Br and I, having atomic number 9, 17, 35 and 53 respectively, is :- [AIEEE-2011]
 (1) $\text{I} > \text{Br} > \text{Cl} > \text{F}$ (2) $\text{F} > \text{Cl} > \text{Br} > \text{I}$ (3) $\text{Cl} > \text{F} > \text{Br} > \text{I}$ (4) $\text{Br} > \text{Cl} > \text{I} > \text{F}$
11. The increasing order of the ionic radii of the given isoelectronic species is :- [AIEEE-2012]
 (1) $\text{K}^+, \text{S}^{2-}, \text{Ca}^{2+}, \text{Cl}^-$ (2) $\text{Cl}^-, \text{Ca}^{2+}, \text{K}^+, \text{S}^{2-}$
 (3) $\text{S}^{2-}, \text{Cl}^-, \text{Ca}^{2+}, \text{K}^+$ (4) $\text{Ca}^{2+}, \text{K}^+, \text{Cl}^-, \text{S}^{2-}$
12. In which of the following arrangements, the sequence is not strictly according to the property written against it ? [JEE-MAIN 2012] (On-Line)
 (1) $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: increasing oxidising power
 (2) $\text{B} < \text{C} < \text{O} < \text{N}$: increasing first ionisation enthalpy
 (3) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: increasing basic strength
 (4) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: increasing acid strength
13. Which of the following represents the correct order of increasing first ionization enthalpy for Ca, Ba, S, Se and Ar ? [JEE-MAIN-2013]
 (1) $\text{Ca} < \text{S} < \text{Ba} < \text{Se} < \text{Ar}$ (2) $\text{S} < \text{Se} < \text{Ca} < \text{Ba} < \text{Ar}$
 (3) $\text{Ba} < \text{Ca} < \text{Se} < \text{S} < \text{Ar}$ (4) $\text{Ca} < \text{Ba} < \text{S} < \text{Se} < \text{Ar}$
14. The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be :- [JEE-MAIN-2013]
 (1) - 2.55 eV (2) - 5.1 eV (3) - 10.2 eV (4) + 2.55 eV
15. Electron gain enthalpy with negative sign of fluorine is less than that of chlorine due to : [JEE-MAIN 2013 (On-Line)]
 (1) Smaller size of chlorine atom (2) Bigger size of 2p orbital of fluorine
 (3) High ionization enthalpy of fluorine (4) Smaller size of fluorine atom
16. Given [JEE-MAIN 2013 (On-Line)]
- | Reaction | Energy Change (in kJ) |
|---|--------------------------|
| $\text{Li(s)} \longrightarrow \text{Li(g)}$ | 161 |
| $\text{Li(g)} \longrightarrow \text{Li}^+(\text{g})$ | 520 |
| $\frac{1}{2} \text{F}_2(\text{g}) \longrightarrow \text{F(g)}$ | 77 |
| $\text{F(g)} + \text{e}^- \longrightarrow \text{F}^-(\text{g})$ | (Electron gain enthalpy) |
| $\text{Li}^+(\text{g}) + \text{F}^-(\text{g}) \longrightarrow \text{LiF(s)}$ | -1047 |
| $\text{Li(s)} + \frac{1}{2} \text{F}_2(\text{g}) \longrightarrow \text{LiF(s)}$ | -617 |
- Based on data provided, the value of electron gain enthalpy of fluorine would be :
 (1) -300 kJ mol^{-1} (2) -328 kJ mol^{-1} (3) -350 kJ mol^{-1} (4) -228 kJ mol^{-1}

17. The order of increasing sizes of atomic radii among the elements O, S, Se and As is :
[JEE-MAIN 2013 (On-Line)]
- (1) $\text{As} < \text{S} < \text{O} < \text{Se}$ (2) $\text{O} < \text{S} < \text{As} < \text{Se}$
(3) $\text{Se} < \text{S} < \text{As} < \text{O}$ (4) $\text{O} < \text{S} < \text{Se} < \text{As}$
18. Which is the correct order of second ionization potential of C, N, O and F in the following ?
[JEE-MAIN 2013 (On-Line)]
- (1) $\text{O} > \text{F} > \text{N} > \text{C}$ (2) $\text{O} > \text{N} > \text{F} > \text{C}$
(3) $\text{C} > \text{N} > \text{O} > \text{F}$ (4) $\text{F} > \text{O} > \text{N} > \text{C}$
19. Which of the following series correctly represents relations between the elements from X to Y ?
[JEE-MAIN 2014 (On-Line)]
- $\text{X} \longrightarrow \text{Y}$
- (1) $_{18}\text{Ar} \rightarrow _{54}\text{Xe}$ Noble character increases
(2) $_3\text{Li} \rightarrow _{19}\text{K}$ Ionization enthalpy increases
(3) $_6\text{C} \rightarrow _{32}\text{Ge}$ Atomic radii increases
(4) $_9\text{F} \rightarrow _{35}\text{Br}$ Electron gain enthalpy with negative sign increases
20. The ionic radii (in Å) of N^{3-} , O^{2-} and F^- are respectively :- [JEE-MAIN 2015 (Off-Line)]
- (1) 1.71, 1.40 and 1.36 (2) 1.71, 1.36 and 1.40
(3) 1.36, 1.40 and 1.71 (4) 1.36, 1.71 and 1.40
21. In the long form of the periodic table, the valence shell electronic configuration of $5s^2 5p^4$ corresponds to the element present in: [JEE-MAIN 2015 (On-Line)]
- (1) Group 16 and period 5 (2) Group 17 and period 6
(3) Group 17 and period 5 (4) Group 16 and period 6
22. Which of the following atoms has the highest first ionization energy ?
[JEE-MAIN 2016 (Off-Line)]
- (1) Sc (2) Rb (3) Na (4) K
23. The non-metal that does not exhibit positive oxidation state is :
[JEE-MAIN 2016 (On-Line)]
- (1) Oxygen (2) Fluorine (3) Iodine (4) Chlorine
24. The electronic configuration with the highest ionization enthalpy is:- [JEE-MAIN 2017]
- (1) $[\text{Ar}] 3d^{10} 4s^2 4p^3$ (2) $[\text{Ne}] 3s^2 3p^1$
(3) $[\text{Ne}] 3s^2 3p^2$ (4) $[\text{Ne}] 3s^2 3p^3$

25. Consider the following ionization enthalpies of two elements 'A' and 'B' :

Element	Ionization enthalpy (kJ/mol)		
	1 st	2 nd	3 rd
A	899	1757	14847
B	737	1450	7731

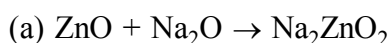
[JEE-MAIN 2017]

Which of the following statements is correct ?

- (1) Both A and B belong to group-2 where A comes below B
- (2) Both A and B belong to group-1 where A comes below B
- (3) Both A and B belong to group-1 where B comes below A
- (4) Both A and B belong to group-2 where B comes below A

26. In the following reactions, ZnO is respectively acting as a/an :

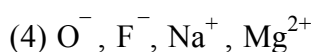
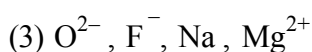
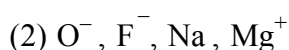
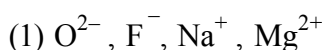
[JEE-MAIN 2017]



- (1) base and acid (2) base and base (3) acid and acid (4) acid and base

27. The group having isoelectronic species is :-

[JEE-MAIN 2017]



EXERCISE # JEE-ADVANCED

1. **Statement-1** : F atom has a less negative electron gain enthalpy than Cl atom. [JEE 2000]
Statement-2 : Additional electron is repelled more efficiently by 3p electron in Cl atom than by 2p electron in F atom.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
2. The correct order of radii is: [JEE 2000]
 (A) $N < Be < B$ (B) $F^- < O^{2-} < N^{3-}$ (C) $Na < Li < K$ (D) $Fe^{3+} < Fe^{2+} < Fe^{4+}$
3. The IE_1 of Be is greater than that of B. [T/F] [JEE 2001]
4. The set representing correct order of IP_1 is [JEE 2001]
 (A) $K > Na > Li$ (B) $Be > Mg > Ca$ (C) $B > C > N$ (D) $Fe > Si > C$
5. Identify the least stable ion amongst the following: [JEE 2002]
 (A) Li^- (B) Be^- (C) B^- (D) C^-
6. The increasing order of atomic radii of the following group 13 elements is [JEE 2016]
 (A) $Al < Ga < In < Tl$ (B) $Ga < Al < In < Tl$
 (C) $Al < In < Ga < Tl$ (D) $Al < Ga < Tl < In$
7. The option(s) with only amphoteric oxides is (are): [JEE 2017]
 (A) Cr_2O_3, CrO, SnO, PbO
 (B) NO, B_2O_3, PbO, SnO_2
 (C) Cr_2O_3, BeO, SnO, SnO_2
 (D) ZnO, Al_2O_3, PbO, PbO_2

ANSWERS KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	A	D	A	B	C	A	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	B	C	C	B	C	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	C	B	A	C	A	C	A	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	B	D	C	C	D	C	D	B
Que.	41	42	43	44	45	46	47	48	49	
Ans.	C	B	B	A	A	C	A	D	D	

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	A, B, D	A, C	B, C	A, B, C	A, B	B, D	A, B	B, C	A,B,C,D
Que.	11	12	13	14	15	16	17			
Ans.	A, B, D	A, B	A, C, D	B, C, D	A, B, D	A, B, C	A, C			

EXERCISE # S-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	6	4	3	3	4	0	3	3	17	4
Que.	11	12	13							
Ans.	5	3.03 (Pauling)	$Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$							

EXERCISE # S-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	A	B	B	A	C	C	C	C
Que.	11	12	13	14	15	16				
Ans.	B	B	B	A	C	(A)→Q;(B)→P,Q,R ;(C)→P;(D)→R, S				
Que.	17					18				
Ans.	(A)→R ; (B)→R ; (C)→Q ; (D)→S					(A) → Q,R ; (B) → P,S ; (C) → S ; (D) → Q,R				
Que.	19									
Ans.	(A) → Q ; (B) → R ; (C) → P ; (D) → S									

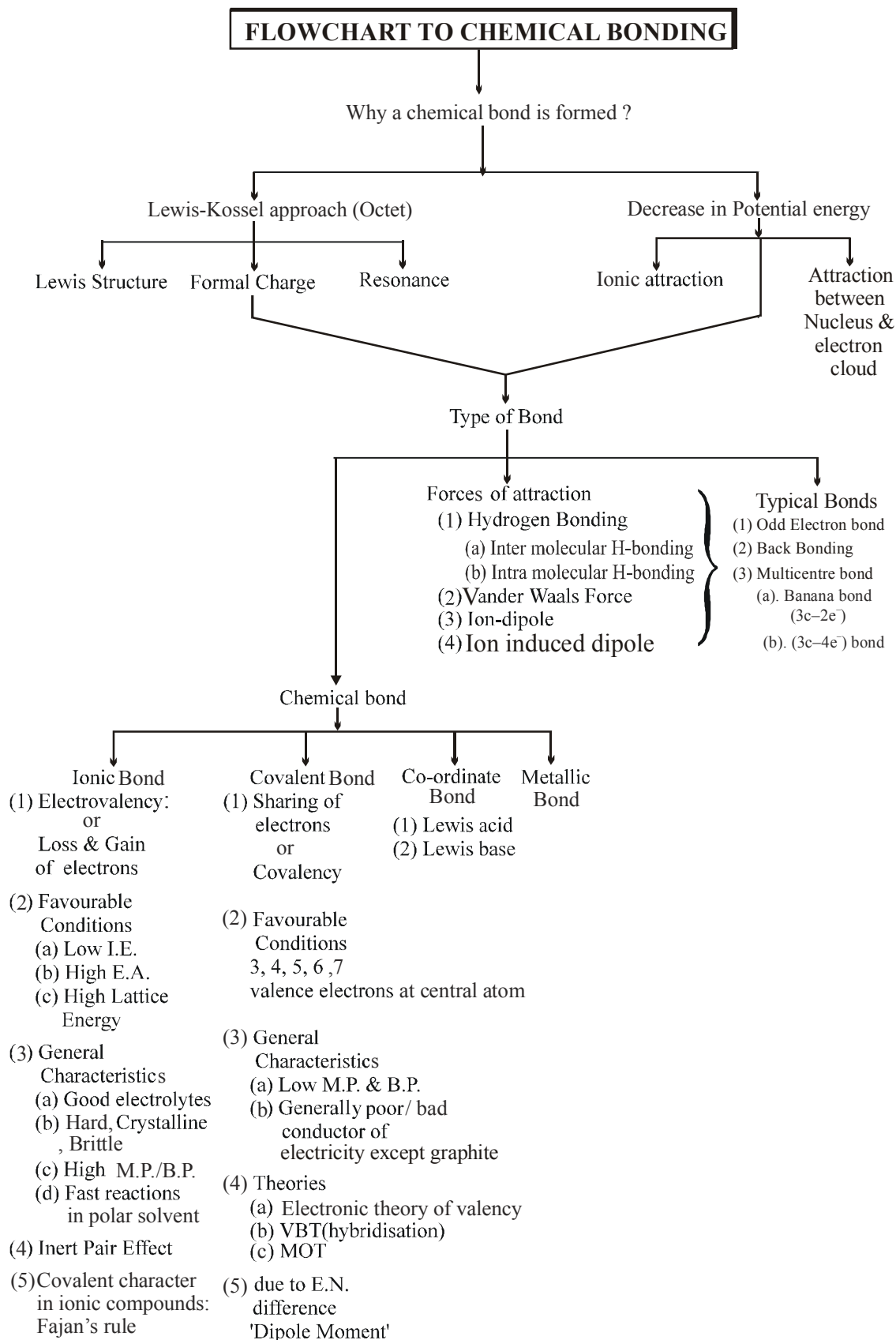
EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	4	1	2	1	3	4	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	3	2	4	2	4	1	3	1
Que.	21	22	23	24	25	26	27			
Ans.	1	1	2	4	4	4	1			

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7			
Ans.	C	B	T	B	B	B	C, D			

CHEMICAL BONDING



KEY CONCEPT

CHEMICAL BOND

- The attractive force which holds various constituents (atoms, ions, etc.) together in different chemical species is called a chemical bond.
- It is the combination of two or more chemical species involving redistribution of e^- among them.
- This process is accompanied with decrease in energy.
- Decrease in energy strengthens the bond.
- Therefore, molecules are more stable than atoms.

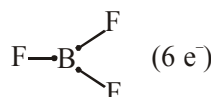
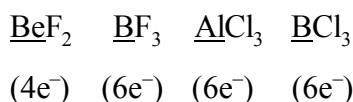
CAUSE OF CHEMICAL COMBINATION

- Tendency to acquire minimum energy :**
- Tendency to acquire noble gas configuration (Octet rule) :**

Atoms combine to complete an octet of electrons in their outer most shell. Hence all atoms have a tendency to acquire octet (ns^2np^6) configuration or to attain nearest noble gas configuration in their outermost orbit. This can be achieved by combining with other atom or ion.

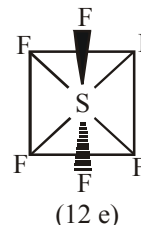
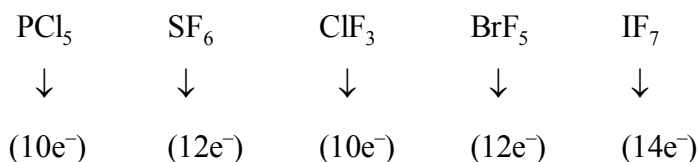
LIMITATION OF OCTET RULE

- Contraction of octet (incomplete octet)**



These compounds are hypovalent.

- Expansion of Octet (due to empty d-orbitals)**



These compounds are hypervalent.

- Odd electron species**

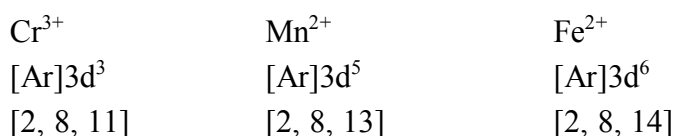
Ex. NO, NO₂, ClO₂ etc.

OTHER EXCEPTIONS OF OCTET RULE

- Compounds of Noble gases**

Noble gases which have already completed their octets (or duplet in case of He.) should not form compounds. However, their compounds like XeF₂, XeF₆ & KrF₂ etc., have been actually prepared.

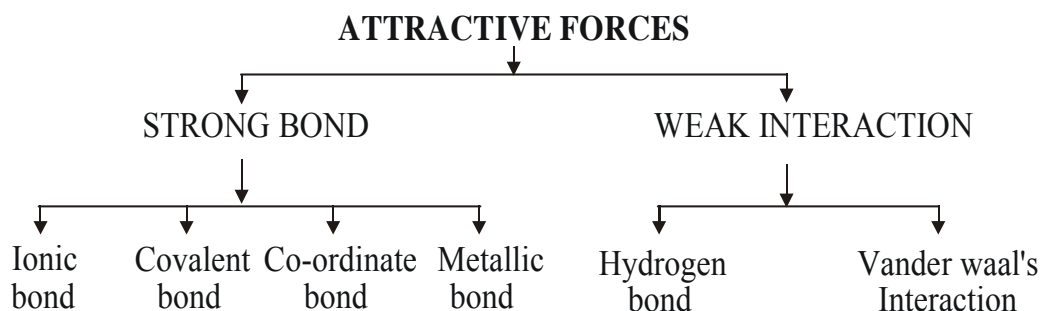
- Transition metal ions**



- Pseudo inert gas configuration $[(n-1)d^{10}ns^2np^6]$**



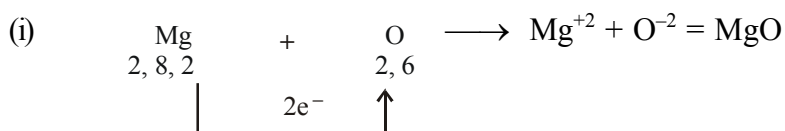
CLASSIFICATION OF BONDS



ELECTROVALENT OR IONIC BOND

- (a) The chemical bond formed between two or more atoms as a result of the complete transfer of one or more electrons from one atom to another is called as Ionic or electrovalent bond.

Example



electrovalency of Mg = 2

electrovalency of O = 2

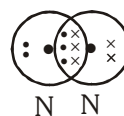
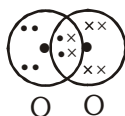
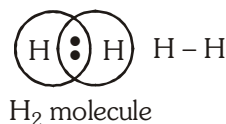
Note : Ionic bond is non-directional.

FACTORS FAVOURING IONIC BONDING

- (a) Less Ionization energy of atom forming cation
(b) Higher electron affinity of atom forming anion
(c) Greater Lattice energy of formed product.
(d) Greater Electronegativity difference between atoms forming cation & anion.

COVALENT BOND

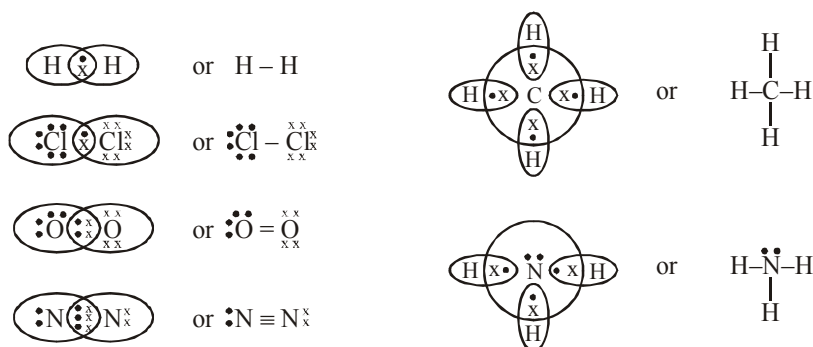
- (a) A covalent bond is formed by the mutual sharing of electrons between two atoms.



- (b) The shared pair of electrons must have opposite spins, and are localised between two atoms concerned.
- (c) On the basis of electrons being shared between two atoms the bonds are of three types –

Covalency : Capacity to form covalent bond is known as covalency

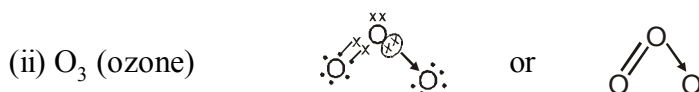
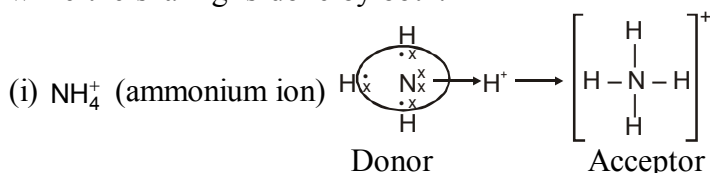
It is represented by (–) small line. – means single bond ; = means double bond ; ≡ means triple bond.



★ Covalent bonds are directional in nature

COORDINATE BOND (DATIVE BOND)

The bond formed between two atom in which contribution of an electron pair is made by one of them while the sharing is done by both.

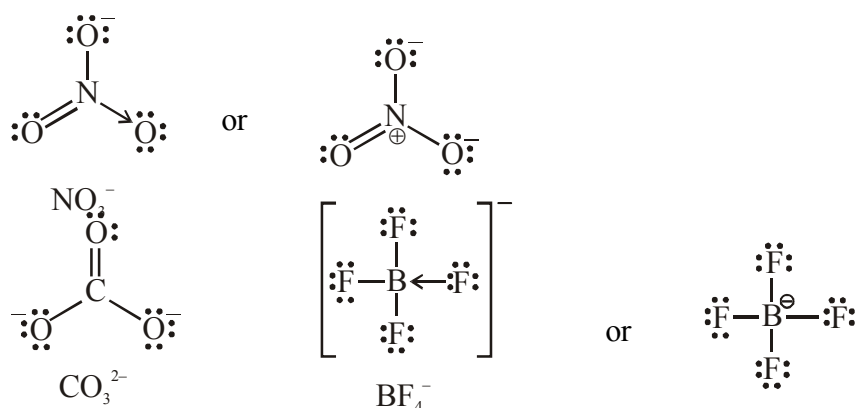


Lewis Dot structures:

- ★ Arrangement of various atoms in a molecule & types of bonding present in it but no idea of geometry of the molecule.
- ★ In most cases we can construct a Lewis structure in three steps :
 - (1) Decide on the number of electrons that are included in the structure by adding together the number of all the valence electrons provided by the atoms.
 - (2) Write the chemical symbols of the atoms in the arrangement that shows which atoms are bonded together.
 - (3) Distribute the electrons in pairs so that there is one pair of electrons forming a single bond between each pair of atom bonded together and then supply electron pairs (to form lone pairs or multiple bonds) until each atom has an octet.
- ★ In oxy acids all 'H' atoms are attached to oxygen as -OH groups except in H_3PO_3 (dibasic), H_3PO_2 (monobasic) & $\text{H}_4\text{P}_2\text{O}_5$ (dibasic). Which are directly bonded to central atom.

Applications :

- ★ To know various linkages present
- ★ To calculate formal charge of various elements.



FORMAL CHARGE

Lewis dot structures, in general, do not represent the actual shapes of the molecules. In case of polyatomic ions, the net charge is possessed by the ion as a whole and not by a particular atom. It is, however, feasible to assign a formal charge on each atom.

$$\text{Formal Charge : } Q_F = N_A - N_{L.P.} - \frac{1}{2} N_{B.P.}$$

Where :

N_A = Total number of valence electron in the free atom

$N_{L.P.}$ = Total number of non bonding (lone pair) electrons

$N_{B.P.}$ = Total number of bonding (shared) electrons

Molecule	Structure	Formal Charge
O_3		$O(1) = 6 - 2 - \frac{1}{2} (6) = +1$ $O(2) = 6 - 4 - \frac{1}{2} (4) = 0$ $O(3) = 6 - 6 - \frac{1}{2} (2) = -1$
CO		$C = 4 - 2 - \frac{1}{2} \times 6 = -1$ $O = 6 - 2 - \frac{1}{2} \times 6 = +1$
NH_4^+		$N = 5 - 0 - \frac{1}{2} (8) = +1$ $\text{On each H} = 1 - 0 - \frac{1}{2} (2) = 0$

RESONANCE*

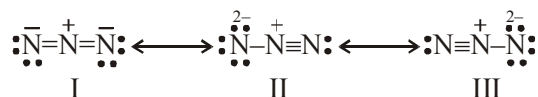
When a molecule cannot be completely represented by a single Lewis structure but its characteristic properties can be described by two or more different structures, with similar energy position of nuclei, bonding and non-bonding pairs of electrons, then the true structure is said to be resonance hybrid of these structures. The phenomenon is called resonance and different contributing structures are called resonating structures or canonical structures.

- ❖ Resonance stabilizes the molecule as the energy of the resonance hybrid is less than the energy of any single canonical structure.
- ❖ Resonance averages the bond characteristics as a whole.
- ❖ The canonical forms have no real existence.

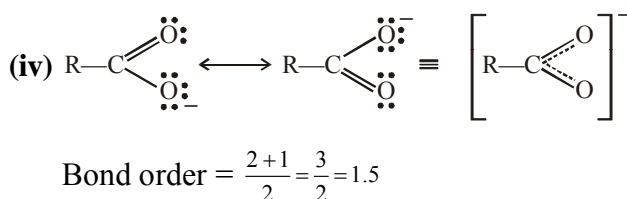
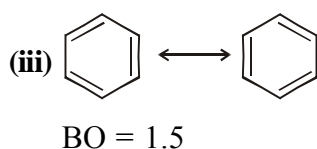
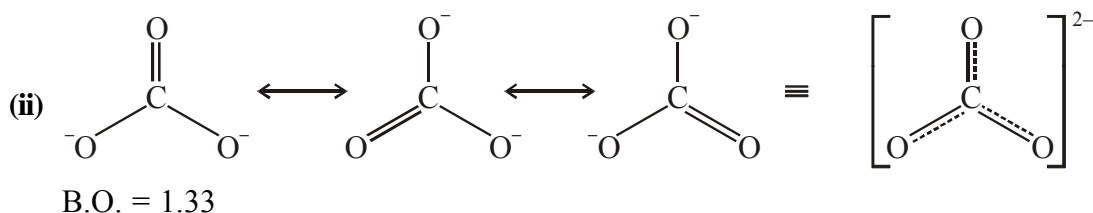
Resonance Structure of Some Molecules/Ions :

(i) Azide ion, N_3^- :

The azide ion may be represented as -



The structures II and III contribute equally and the molecule has almost double bond character in each N-N bond.



*Descriptive discussion of concept of resonance will be done in **Organic Chemistry**.

VALENCE BOND THEORY

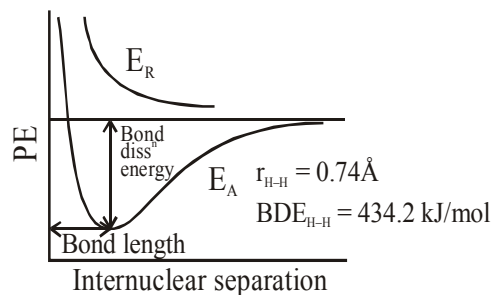
Valence bond theory was introduced by Heitler and London (1927) and developed further by Pauling and others. A discussion of the valence bond theory is based on the knowledge of atomic orbitals, electronic configurations of elements, the overlap criteria of atomic orbitals and the hybridization of atomic orbitals.

Formation of H_2 molecule :

When two 'H'-atoms approaches towards each other for the formation of H_2 molecule. The following interactions takes place.

- (i) e-e repulsion
- (ii) e-p attraction
- (iii) p-p repulsion

Experimentally, it has been found that the magnitude of attractive forces is more than the repulsive forces. As a result, when two atoms approach each other then potential energy decreases. Ultimately a stage is reached where the net force of attraction balances the force of repulsion and system acquires minimum energy.



The potential energy curve for the formation of H_2 molecule as a function of internuclear distance of the H atoms. The minimum in the curve corresponds to the most stable state of H_2 .

At this stage two hydrogen atoms are said to be bonded together to form a stable molecule having the bond length of 74 pm. Since the energy gets released when the bond is formed between two hydrogen atoms, the hydrogen molecule is more stable than that of isolated hydrogen atoms.

Bond Length : Internuclear distance at minimum potential energy is called bond length.

Bond Dissociation Energy : The amount of energy released when one mole of same type of bonds are formed is called B.D.E.

434.2 kJ/mol of energy is required to dissociate one mole of H_2 molecule.



ORBITAL OVERLAP CONCEPT

In the formation of hydrogen molecule, there is a minimum energy state when two hydrogen atoms are so near that their atomic orbitals undergo partial interpenetration. This partial merging of atomic orbitals is called overlapping of atomic orbitals which results in the pairing of electrons. The extent of overlap decides the strength of a covalent bond. In general, greater the overlap the stronger is the bond formed between two atoms. Therefore, according to orbital overlap concept, the formation of a covalent bond between two atoms results by pairing of electrons, present in the valence shell having opposite spins.

Directional Properties of Bonds :

The valence bond theory explains the formation and directional properties of bonds in polyatomic molecules like CH_4 , NH_3 and H_2O , etc. in terms of overlap and hybridisation of atomic orbitals.

Types of Overlapping and Nature of Covalent Bonds

The covalent bond may be classified into following types depending upon the types of overlapping :-

(i) sigma(σ) bond (ii) pi(π) bond (iii) delta(δ) bond

- (i) **Sigma (σ) bond :** This type of covalent bond is formed by the end to end (head on or axial) overlap of bonding orbitals along the internuclear axis. This can be formed by any one of the following types of combinations of atomic orbitals.

- s-s overlapping : In this case, there is overlap of two half filled s-orbitals along the internuclear axis as shown below :



- s-p overlapping: This type of overlap occurs between half filled s-orbital of one atom and half filled p-orbital of another atom.

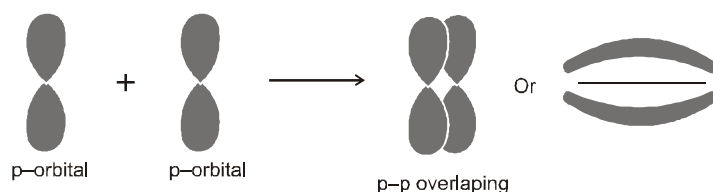


- p-p overlapping : This type of overlap takes place between half filled similar p-orbitals of the two approaching atoms.

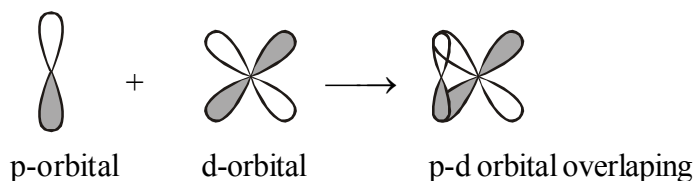


- (ii) **pi(π) bond** : In the formation of π bond the atomic orbitals overlap in such a way that their axis remain parallel to each other and perpendicular to the internuclear axis. The orbitals formed due to sidewise overlapping consists of two saucer type charged clouds above and below the plane of the participating atoms.

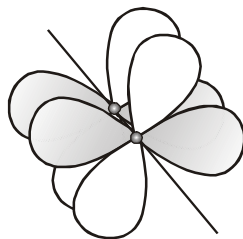
(a) **$p_{\pi}-p_{\pi}$**



(b) **$p_{\pi}-d_{\pi}$**



- (iii) **delta (δ) bond** : are the covalent bonds where four lobes of d-orbital of one atom overlap with four lobes of the similar d-orbital of other atom. Except d_{z^2} all d orbitals form δ bond.



Strength of sigma and pi Bonds :

Basically the strength of a bond depends upon the extent of overlapping- In case of sigma bond, the overlapping of orbitals takes place to a larger extent. Hence, it is stronger as compared to the pi bond where the extent of overlapping occurs to a smaller extent. Further, it is important to note that pi bond between two atoms is formed in addition to a sigma bond. It is always present in the molecules containing multiple bond (double or triple bonds)

Advantages of VBT :

- ❖ It explain various bond characteristics e.g., bond length, bond strength.
- ❖ It explains the quantitative relationship between the extent of overlapping and bond dissociation energy.
- ❖ This theory accounts for shape and nature of bonding of the molecule whose covalency is not according to the number of half-filled orbitals present in the ground state.
- ❖ This theory redefined the stability of molecules e.g. BF_3 , AlCl_3 , PCl_5 , SF_6 etc which are exception to octet rule.

Disadvantages of VBT :

- ❖ According to this theory three bond angle in CH_4 should be 90° , as these are formed by p-p overlapping, but actually it has four $109^\circ 28'$ angles. In NH_3 & H_2O , angle should be 90° . This is in disagreement with the actual bond angles of 107° & 104.5° in NH_3 & H_2O molecules respectively.
- ❖ In order to explain the characteristic geometrical shapes of polyatomic molecules like CH_4 , NH_3 , H_2O etc. Pauling introduced the concept of hybridisation.

HYBRIDISATION

The phenomenon of intermixing of the orbitals of slightly different energies so as to redistribute their energies, resulting in the formation of new set of orbitals of equivalent energies and shape. For example when one 2s and three 2p-orbitals of carbon hybridise, there is the formation of four new sp^3 hybrid orbitals.

Salient features of hybridisation: The main features of hybridisation are as under :

1. The number of hybrid orbitals is equal to the number of the atomic orbitals that get hybridised.
2. The hybridised orbitals are always equivalent in energy and shape.
3. The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals.
4. These hybrid orbitals are directed in space in some preferred direction to have minimum repulsion between electron pairs and thus a stable arrangement. Therefore, the type of hybridisation indicates the geometry of the molecules.

Important conditions for hybridisation

- (i) The orbitals present in the valence shell of the atom are hybridised.
- (ii) The orbitals undergoing hybridisation should have almost equal energy.
- (iii) Promotion of electron is not essential condition prior to hybridisation.
- (iv) It is not necessary that only half filled orbitals participate in hybridisation. In some cases, even filled orbitals of valence shell take part in hybridisation.

Types of Hybridisation

There are various types of hybridisation involving s, p and d orbitals. The different types of hybridisation are as under :

- (i) sp hybridisation
- (ii) sp^2 hybridisation
- (iii) sp^3 hybridisation
- (iv) sp^3d hybridisation:
- (v) sp^3d^2 hybridisation:
- (vi) sp^3d^3 hybridisation:

Determination of hybridisation state –

To predict hybridisation following formula may be used :

Number of hybrid orbital = Steric Number (S.N.) = Number of σ -bond around that atom + Number of lone pair on that atom.

Molecule	Method	Hybridisation
NH_4^+	$S.N. = 4 + 0 = 4$	sp^3 hybridisation.

Number of hybrid orbitals	Hybridisation
two	sp
three	sp^2
four	sp^3
five	sp^3d
six	sp^3d^2
seven	sp^3d^3

Hybridisation in Ionic solid species :

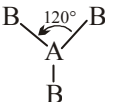
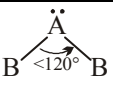
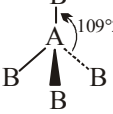
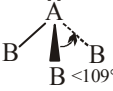
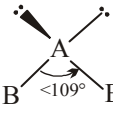
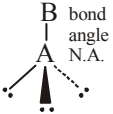
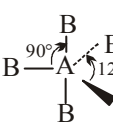
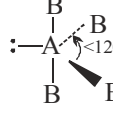
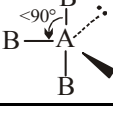
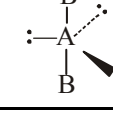
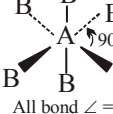
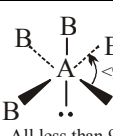
Species	Cationic part	Anionic part
PCl_5	$\text{PCl}_4^+ (\text{sp}^3)$	$\text{PCl}_6^- (\text{sp}^3\text{d}^2)$
PBr_5	$\text{PBr}_4^+ (\text{sp}^3)$	Br^-
XeF_6	$\text{XeF}_5^+ (\text{sp}^3\text{d}^2)$	F^-
N_2O_5	$\text{NO}_2^+ (\text{sp})$	$\text{NO}_3^- (\text{sp}^2)$
I_2Cl_6 (liquid)	$\text{ICl}_2^+ (\text{sp}^3)$	$\text{ICl}_4^- (\text{sp}^3\text{d}^2)$
Cl_2O_6	$\text{ClO}_2^+ (\text{sp}^2)$	$\text{ClO}_4^- (\text{sp}^3)$

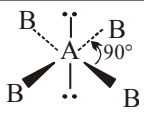
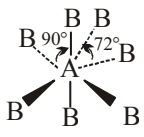
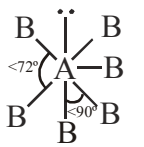
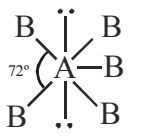
VALENCE SHELL ELECTRON PAIR REPULSION THEORY (VSEPR)

- (a) Molecules having covalent bond have definite geometry as covalent bonding has directional characteristics. A simple theory was given for the molecular shape of the covalent molecules by Gillespie and Nyholm in 1957.
- (b) This theory predicts the shape of the molecule by considering the most stable configuration of the bond angles in the molecule. This theory states
- Electron pairs in the valence shell of the central atom of a molecule, whether bonding or lone pairs are regarded as occupying localised orbitals. These orbitals arrange themselves in so as to minimize the mutual electronic repulsions.
 - The magnitude of the different types of electronic repulsions follows the order given below:

lone pair-lone pair > lone pair - bonded pair > bonded pair - bonded pair
 - The electronic repulsion between two pairs of electrons will be minimum if they are as far apart as possible.
 - The actual shape of the molecules containing lone pairs is a little distorted from the basic shape as in the NH_3 and H_2O molecules, the bond angles are not $109^\circ 28'$ but 107° and 104.5° respectively due to presence of one lone pair in NH_3 and two lone pairs in H_2O .

SHAPES OF MOLECULES BASED ON VSEPR THEORY

Total no. of hybrid orbitals	No. of b.p. (bond pairs)	No. of unshared pair i.e. l_p	General formula	Type of hybridisation	Stereo chemical formula/str.	Shape	Exam.
2	2	0	AB_2	sp		linear	$BeCl_2$
3	3	0	AB_3	sp^2		Trigonal planar	BCl_3 , NO_3^- , GaF_3 , CO_3^{2-}
3	2	1	AB_2	sp^2		V or Bent or angular	$SnCl_2$, O_3 , SO_2
4	4	0	AB_4	sp^3		Tetrahedron	CH_4 , SiF_4 , NH_4^+
4	3	1	AB_3	sp^3		Trigonal pyramidal	NH_3 , CH_3^-
4	2	2	AB_2	sp^3		V or Bent or angular	H_2O , SF_2
4	1	3	AB	sp^3		linear	ClO^-
5	5	0	AB_5	sp^3d		Trigonal bipyramidal	PF_5 , SF_5^+ , $SbBr_5$, XeO_3F_2
5	4	1	AB_4	sp^3d		Seesaw	SF_4
5	3	2	AB_3	sp^3d		T-shaped	ClF_3 , BrF_3
5	2	3	AB_2	sp^3d		Linear	ICl_2^- , XeF_2 , I_3^-
6	6	0	AB_6	sp^3d^2		Octahedral or Square bipyramidal	SF_6 , IF_6^+
6	5	1	AB_5	sp^3d^2		Square pyramidal	IF_5 , $XeOF_4$, BrF_5

6	4	2	AB_4	sp^3d^2		Square planar	IF_4^- , XeF_4 , ICl_4
7	7	0	AB_7	sp^3d^3		Pentagonal bipyramidal	IF_7
7	6	1	AB_6	sp^3d^3		Distorted octahedral	XeF_6 , IF_6^-
7	5	2	AB_5	sp^3d^3		Pentagonal planar	XeF_5^-

BOND PARAMETERS

(I) Bond order

(III) Bond Angle

(I) Bond order :

The Bond Order is given by the number of bonds between the two atoms in a molecule.

The bond order, for example in H_2 (with a single shared electron pair), in O_2 (with two shared electron pairs) and in N_2 (with three shared electron pairs) is 1,2,3 respectively.

Note : Isoelectronic molecules and ions have identical bond orders; for example, F_2 and O_2^{2-} have bond order 1. N_2 , CO and NO^+ have bond order 3.

Note : A general correlation useful for understanding the stabilities of molecules is that: with increase in bond order, bond enthalpy increases and bond length decreases.

(II) Bond Length :-

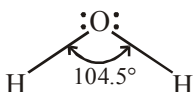
Bond length is defined as the equilibrium distance between the nuclei of two bonded atoms in a molecule. Bond lengths are measured by spectroscopic, X-ray diffraction and electron-diffraction techniques about which you will learn in higher classes.

Factors affecting bond length

- | | |
|-----------------------------|---|
| (a) Size of atoms | (b) Effect of bond order or number of bonds |
| (c) Effect of Resonance | (d) Effect of Electronegativity difference |
| (f) Effect of Hybridisation | |

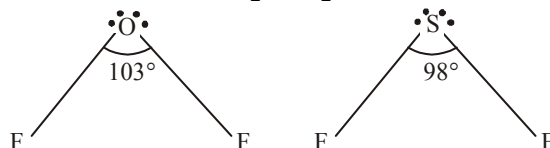
(III) Bond angle :

It is defined as the angle between the orbitals containing bonding electron pairs around the central atom in a molecule/complex ion. Bond angle is expressed in degree which can be experimentally determined by spectroscopic methods. For example $H-\hat{O}-H$ bond angle in water can be represented as under :



Comparison of bond angles.

- (a) If central atoms are in different hybridisations then it can be compared.
 (b) If same hybridisation but different central atoms then bond angle would be more of the molecule in which central atom is more E.N. eg. OF_2 , SF_2



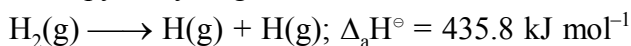
- (c) Molecules having same C.A. but different substituent then bond angle increases as the size of attached atom increases except in symmetrical molecules and H_2O / F_2O , NH_3 / NF_3

Factors affecting bond angle

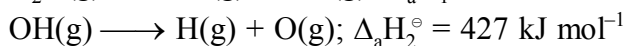
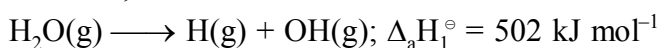
- State of Hybridisation
- Presence of lone pair
- Electronegativity of central atom
- Electronegativity of surrounding atom
- Size of surrounding atom
- Lone pairs may sometimes be transferred from a filled shell of one atom to an unfilled shell of another bonded atom, causing less repulsion.
- Multiple bond orbital repel other orbitals more strongly than single bond orbitals.

(IV) Bond Enthalpy

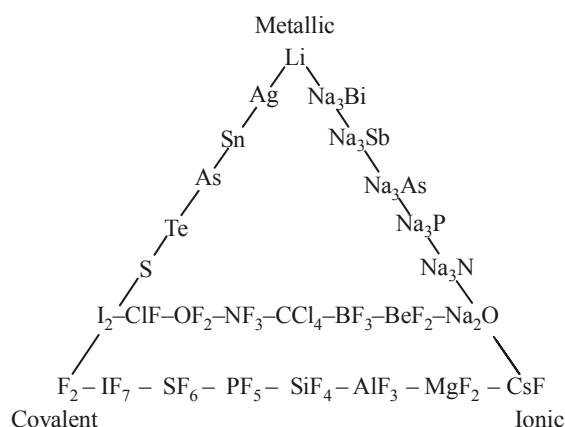
It is defined as the amount of energy required to break one mole of bonds of a particular type between two atoms in a gaseous state. The unit of bond enthalpy is kJ mol^{-1} . For example, the H – H bond enthalpy in hydrogen molecule is $435.8 \text{ kJ mol}^{-1}$.



⇒ In polyatomic molecules the **term mean or average bond enthalpy** is used. It is obtained by dividing total bond dissociation enthalpy by the number of bonds broken as explained below in case of water molecule,

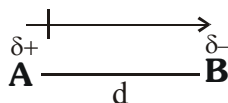


$$\text{Average bond enthalpy} = \frac{502 + 427}{2} = 464.5 \text{ kJ mol}^{-1}$$

SUMMARY OF THE THREE MAIN TYPES OF BONDS

DIPOLE MOMENT

- The degree of polarity of covalent bond is given by the dipole moment (μ)
- It is the product of the either charge and the distance between them.



Electronegativity of A < Electronegativity of B

$$\mu = \delta \times d$$

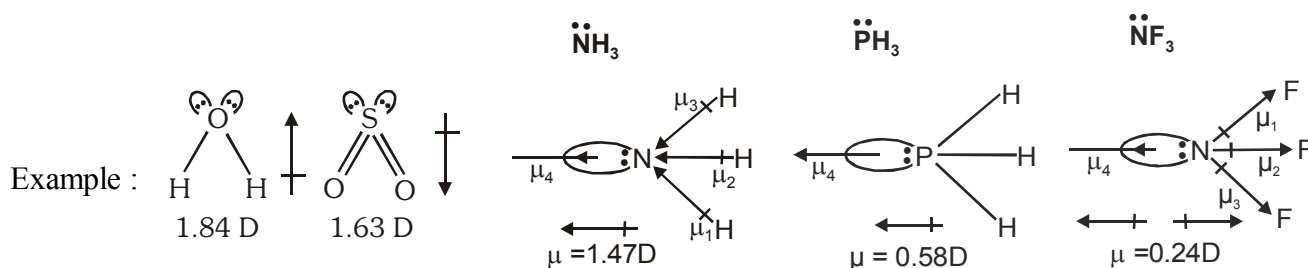
- Dipole moment is a vector quantity.
The direction of dipole moment is represented by $\rightarrow$
- Units = Cm (S.I.) or esu cm (CGS) or Debye (common unit)
- 1 D = 10^{-18} esu cm = 3.33×10^{-30} coulomb m

Dipole moment depends on

- Electronegativity difference between bonded atoms
- Direction of bond dipole moment
- Angle between various bonds
- Influence of unshared e^- pairs
- Magnitude of polarity of the molecule
- Symmetrical / Unsymmetrical shape.

Application of dipole moment

- To determine the polarity and geometry of molecules



- To calculate the percentage of ionic character

$$\% \text{ ionic character} = \frac{\text{observed } \mu}{\text{calculated } \mu \text{ (for 100 \% ionic)}} \times 100 \%$$

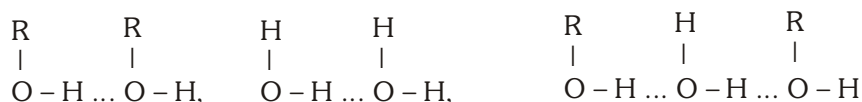
HYDROGEN BONDING

- ★ **Hydrogen bonding:** When a hydrogen atom is bonded to a highly electronegative atom (like F, O or N) comes under the influence of another strongly electronegative atom, then a weak electrostatic force of attraction is developed between them, which is called as hydrogen bond.

Types of H-bonding:

- ★ Intermolecular H-bond

(i) This type of H-bonding takes place between two molecules. **Ex.** ROH, H₂O, R - OH & H₂O

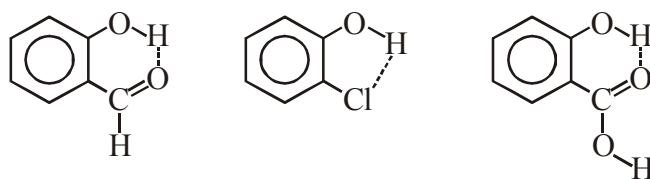


(ii) In such compounds molecular wt., M.P. & B.P. are high.

(iii) Extent of Inter molecular H-bonding $\uparrow$ viscosity & density $\uparrow$.

- ★ Intramolecular H-bond

Normally when 2 hydroxyl groups are present on the same carbon atom i.e. gem diols are unstable, but "chloralhydrate" is a stable molecule due to formation of H-bond.



Properties influenced by hydrogen bonding

- Abnormal behaviour of water.
- Association of molecules eg. dimerisation of CH₃COOH, HCOOH
- Dissociation of a polar species.
- Abnormal melting point & boiling point.
- Enhanced solubility in water.

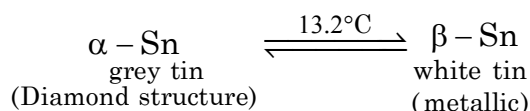
★ **Metallic bonds :**

Electron gas model or sea model, with metal atom existing as kernels along with less firmly held valence electrons & bonds between various kernels (at the lattice site) & valence electrons are known as metallic bonds.

ALLOTROPE OF CARBON FAMILY

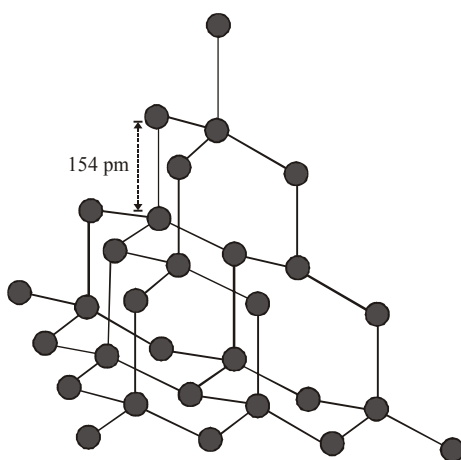
Carbon exhibits many allotropic forms; both crystalline as well as amorphous. Diamond and graphite are two well-known crystalline forms of carbon. In 1985, third form of carbon known as **fullerenes** was discovered by H.W.Kroto, E.Smalley and R.F.Curl. For this discovery they were awarded the Nobel Prize in 1996.

Si, Ge and Sn have a diamond type of structure, though Sn exists as a metallic form. Pb exists only in the metallic form. Ge is unusual because the liquid expands when it forms the solid. This property is unique to Ga, Ge and Bi.



Diamond (*kinetically most stable allotrope of carbon, meta stable phase of carbon*)

It has a crystalline lattice. In diamond each carbon atom undergoes sp^3 hybridisation and linked to four other carbon atoms by using hybridised orbitals in tetrahedral fashion. The C—C bond length is 154 pm. The structure extends in space and produces a rigid three dimensional network of carbon atoms.



The structure of diamond

In this structure, directional covalent bonds are present throughout the lattice. It is very difficult to break extended covalent bonding and, therefore, diamond is a very hard substance. It is used as an abrasive for sharpening hard tools, in making dyes and in the manufacture of tungsten filaments for electric light bulbs.

Problem

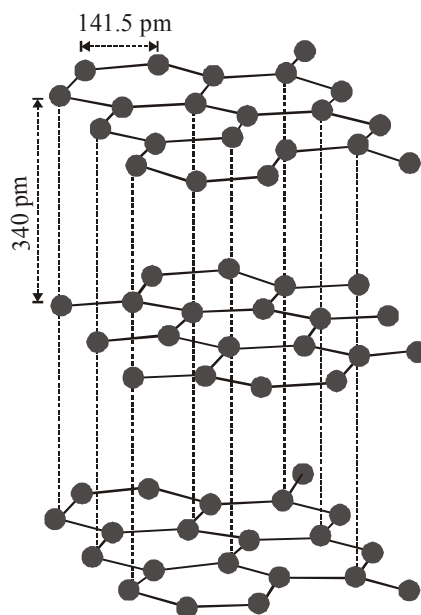
Diamond is covalent, yet it has high melting point. Why ?

Solution

Diamond has a three-dimensional network involving strong C—C bonds, which are very difficult to break and, in turn has high melting point.

Graphite (*Thermodynamically most stable allotrope of carbon*)

Graphite has layered structure. Layers are held by van der Waals forces and distance between two layers is 340 pm. Each layer is composed of planar hexagonal rings of carbon atoms. C—C bond length within the layer is 141.5 pm. Each carbon atom in hexagonal ring undergoes sp^2 hybridisation and makes three sigma bonds with three neighbouring carbon atoms. Fourth electron forms a π bond. The electrons are delocalised over the whole sheet. Thus, graphite is lustrous. Electrons are mobile and,



The structure of graphite

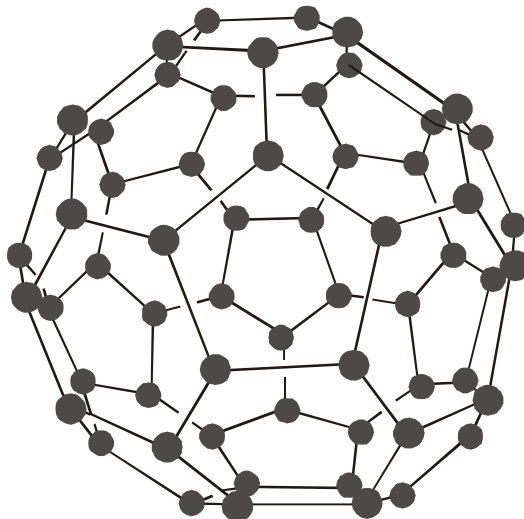
therefore, graphite conducts electricity along the sheet. Graphite cleaves easily between the layers and, therefore, it is very soft and slippery. For this reason graphite is used as a dry lubricant in machines running at high temperature, where oil cannot be used as a lubricant.

Fullerenes

Fullerenes are made by the heating of graphite in an electric arc in the presence of inert gases such as helium or argon. The sooty material formed by condensation of vapourised C_n small molecules consists of mainly C_{60} with smaller quantity of C_{70} and traces of fullerenes consisting of even number of carbon atoms up to 350 or above. Fullerenes are the only pure form of carbon because they have smooth structure without having 'dangling' bonds. Fullerenes are cage like molecules. C_{60} molecule has a shape like soccer ball and called **Buckminsterfullerene**.

It contains twenty, six-membered rings and twelve, five membered rings. A six membered ring is fused with six or five membered rings but a five membered ring can only fuse with six membered rings. All the carbon atoms are equal and they undergo sp^2 hybridisation. Each carbon atom forms three sigma bonds with other three carbon atoms. The remaining electron at each carbon is delocalised in molecular orbitals, which in turn give aromatic character to molecule. However, because of non-planar nature of fullerenes, their aromatic character gets diminished and reactivity increases.

This ball shaped molecule has 60 vertices and each one is occupied by one carbon atom and it also contains both single and double bonds with C–C distances of 143.5 pm and 138.3 pm respectively. Spherical fullerenes are also called *bucky balls* in short.



[The structure of C_{60} , Buckminsterfullerene : Note that molecule has the shape of a soccer ball (football)]

It is very important to know that graphite is thermodynamically most stable allotrope of carbon and, therefore, ΔH_f of graphite is taken as zero. ΔH_f values of diamond and fullerene, C_{60} are 1.90 and 38.1 kJ mol⁻¹, respectively.

Other forms of elemental carbon like carbon black, coke, and charcoal are all impure forms of graphite or fullerenes. Carbon black is obtained by burning hydrocarbons in a limited supply of air. Charcoal and coke are obtained by heating wood or coal respectively at high temperatures in the absence of air.

Uses of Carbon

Graphite fibres embedded in plastic material form high strength, lightweight composites. The composites are used in products such as tennis rackets, fishing rods, aircrafts and canoes. Being good conductor, graphite is used for electrodes in batteries and industrial electrolysis. Crucibles made from graphite are inert to dilute acids and alkalies. Being highly porous, activated charcoal is used in adsorbing poisonous gases; also used in water filters to remove organic contaminants and in airconditioning system to control odour. Carbon black is used as black pigment in black ink and as filler in automobile tyres. Coke is used as a fuel and largely as a reducing agent in metallurgy. Diamond is a precious stone and used in jewellery. It is measured in carats (1 carat = 200 mg).

ALLOTROPE OF OXYGEN FAMILY

Oxygen occurs as two non-metallic forms, dioxygen O_2 and ozone O_3 . Dioxygen O_2 is stable as a diatomic molecule. O_3 molecule is diamagnetic while O_2 is paramagnetic.

Six allotropes of selenium are known.

Tellurium has only one crystalline form, which is silvery white and semimetallic.

This is similar to grey Se, but has stronger metallic interaction.

Sulphur Allotropic forms :

Sulphur forms numerous allotropes of which the yellow rhombic (α -sulphur) and monoclinic (β -sulphur) forms are the most important. The stable form at room temperature is rhombic sulphur, which transforms to monoclinic sulphur when heated above 369 K.

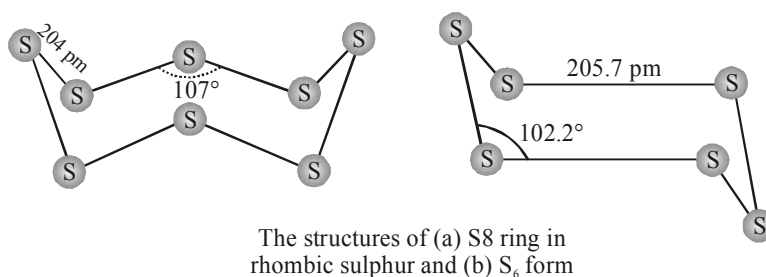
Rhombic sulphur (α -sulphur) (*Most stable sulphur allotrope at room temperature*)

This allotrope is yellow in colour, m.p. 385.8 K and specific gravity 2.06. Rhombic sulphur crystals are formed on evaporating the solution of roll sulphur in CS_2 . It is insoluble in water but dissolves to some extent in benzene, alcohol and ether. It is readily soluble in CS_2 .

Monoclinic sulphur (β -sulphur)

Its m.p. is 393 K and specific gravity 1.98. It is soluble in CS_2 while water insoluble. This form of sulphur is prepared by melting rhombic sulphur in a dish and cooling, till crust is formed. Two holes are made in the crust and the remaining liquid poured out. On removing the crust, colourless needle shaped crystals of β -sulphur are formed. It is stable above 369 K and transforms into α -sulphur below it. Conversely, α -sulphur is stable below 369 K and transforms into β -sulphur above this. At 369 K, both the forms are stable. This temperature is called transition temperature.

Both rhombic and monoclinic sulphur have S_8 molecules. These S_8 molecules are packed to give different crystal structures. The S_8 ring in both the forms is puckered and has a crown shape. The molecular dimensions are given in the figure.



Several other modifications of sulphur containing 6-20 sulphur atoms per ring have been synthesised in the last two decades. In cyclo- S_6 , the ring adopts the chair form and the molecular dimensions are as shown in figure. At elevated temperatures (~ 1000 K), S_2 is the dominant species and is paramagnetic just like O_2 .

Note : Engel's sulphur (ϵ -sulphur) is unstable and contains S_6 rings arranged in the chair conformation. It is made by pouring $Na_2S_2O_3$ solution into concentrated HCl and extracting the S with toluene. Plastic sulphur-(χ) is obtained by pouring liquid sulphur into water.

Example

Which form of sulphur shows paramagnetic behaviour ?

Solution

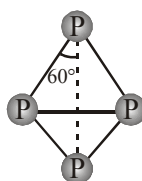
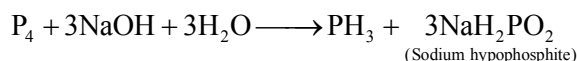
In vapour state sulphur partly exists as S_2 molecule which has two unpaired electrons in the antibonding π^* orbitals like O_2 and, hence, exhibits paramagnetism.

Allotropes of Nitrogen Family :

Solid As, Sb and Bi each exist in several allotropic forms. Arsenic vapour contains tetrahedral As_4 molecules. A reactive yellow form of the solid resembles white phosphorus and is thought to contain tetrahedral As_4 units. Sb also has a yellow form. All three elements have much less reactive metallic or α -forms.

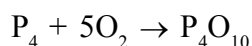
PHOSPHORUS ALLOTROPIC FORMS

Phosphorus is found in many allotropic forms, the important ones being white, red and black. White phosphorus is a translucent, white, waxy solid. It is poisonous, insoluble in water but soluble in carbon disulphide and glows in dark (chemiluminescence). It dissolves in boiling NaOH solution in an inert atmosphere giving PH_3 .



White Phosphorus

White phosphorus is less stable and therefore, more reactive than the other solid phases under normal conditions because of angular strain in the P_4 molecule where the angles are only 60° . It readily catches fire in air to give dense white fumes of P_4O_{10} .

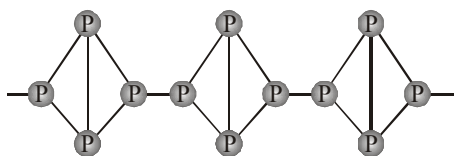


It consists of discrete tetrahedral P_4 molecule as shown in Fig.

Red phosphorus is obtained by heating white phosphorus at 573K in an inert atmosphere for several days. When red phosphorus is heated under high pressure, a series of phases of black phosphorus is formed.

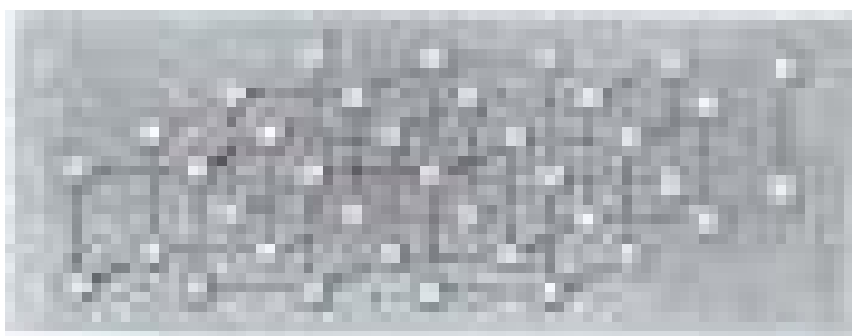
Red phosphorus possesses iron grey lustre. It is odourless, nonpoisonous and insoluble in water as well as in carbon disulphide. Chemically, red phosphorus is much less reactive than white phosphorus. It does not glow in the dark.

It is polymeric, consisting of chains of P_4 tetrahedra linked together in the manner as shown in Fig.



Red Phosphorus

Black phosphorus has two forms α -black phosphorus and β -black phosphorus. α -Black phosphorus is formed when red phosphorus is heated in a sealed tube at 803K. It can be sublimed in air and has opaque monoclinic or rhombohedral crystals. It does not oxidise in air. β -Black phosphorus is prepared by heating white phosphorus at 473 K under high pressure. It does not burn in air upto 673 K.

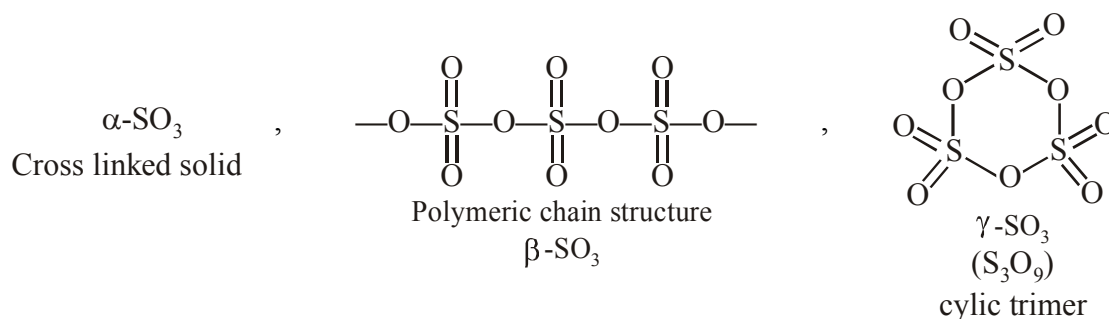


Thermodynamic stability order : Black phosphorus > Red phosphorus > White phosphorus

Reactivity order : Black phosphorus < Red phosphorus < White phosphorus

ALLOTROPIC FORM OF SO_3

SO_3 have three allotropic forms α - SO_3 , β - SO_3 and γ - SO_3



QUESTION OF HYDROGEN BONDING

- Q.1 Explain the structure of Boric acid in solid state.
- Q.2 Boiling point of o-Nitrophenol is less than meta and para nitrophenol. Why?
- Q.3 Maleic acid is more acidic than fumaric acid. Why?
- Q.4 $\text{H}-\text{F}$ is only liquid among halogen acid. Why?
- Q.5 Ammonia is more easily liquefied than HCl , explain.
- Q.6 Why ice floats on water?
- Q.7 Water shows maximum density at 4°C . Why?
- Q.8 HI is the strongest halogen acid, whereas $\text{H}-\text{F}$ is the weakest. Why?
- Q.9 Wood pieces are used to hold ice-cream. Why?
- Q.10 KHF_2 is possible but not KBr_2 or KI_2 . Why?
- Q.11 o-Nitrophenol is less soluble in H_2O than p-Nitrophenol. Why?
- Q.12 o-Hydroxy benzaldehyde is a liquid at room temperature while p-hydroxy benzaldehyde is a high melting solid.
- Q.13 Glycerol is more viscous than ethanol. Explain.
- Q.14 CH_4 and H_2O have nearly same molecular weight. Yet CH_4 has a boiling point 112 K and water 373 K. Explain.
- Q.15 The experimental molecular weight of acetic acid is just double than theoretical molecular weight of acetic acid. Why?
- Q.16 Although chlorine has same electronegativity as nitrogen but the former does not form effective H-bonding. Explain.
- Q.17 Molar entropy change of vapourization of acetic acid is less than that of water. Explain.
- Q.18 Heat of vapourization of water is higher than HF , however strength of H-bond in HF is higher than water. Explain.

EXERCISE # O-1

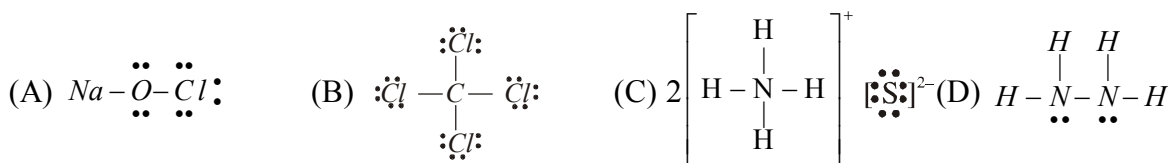
IONIC BOND

(Only one option is correct)

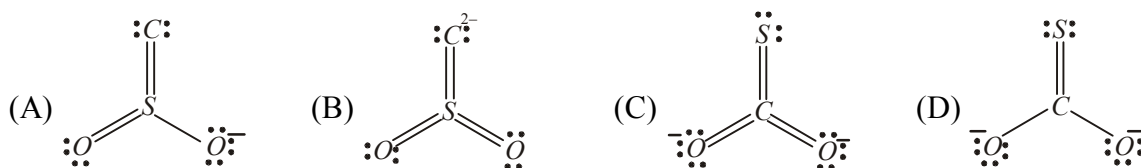
- The compound which contains ionic as well as covalent bonds is
(A) $C_2H_4Cl_2$ (B) CH_3I (C) KCN (D) H_2O_2
- A bond formed between two like atoms cannot be
(A) ionic (B) covalent (C) coordinate (D) metallic
- An ionic bond A^+B^- is most likely to be formed when :
(A) the ionization energy of A is high and the electron gain enthalpy of B is low
(B) the ionization energy of A is low and the electron gain enthalpy of B is high
(C) the ionization energy of A and the electron gain enthalpy of B both are high
(D) the ionization energy of A and the electron gain enthalpy of B both are low

COVALENT BOND, CO-ORDINATE BOND & LEWIS STRUCTURE

- PCl_5 exists but NCl_5 does not because :
(A) Nitrogen has no vacant $2d$ -orbitals (B) NCl_5 is unstable
(C) Nitrogen atom is much smaller than P (D) Nitrogen is highly inert
- Which of the following species are hypervalent?
(1) PCl_5 , (2) BF_3 , (3) XeF_2 , (4) CO_3^{2-}
(A) 1, 2, 3 (B) 1, 3 (C) 3, 4 (D) 1, 2
- The types of bond present in $N_2O_{5(g)}$ are
(A) only covalent (B) only ionic (C) ionic and covalent (D) covalent & coordinate
- Which of the following molecule does not have coordinate bonds?
(A) CH_3-NC (B) CO (C) O_3 (D) CO_3^{2-}
- Which of the following Lewis dot diagrams is(are) incorrect ?



- The possible structure(s) of monothiocarbonate ion is :



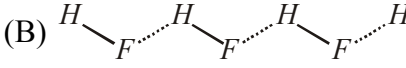
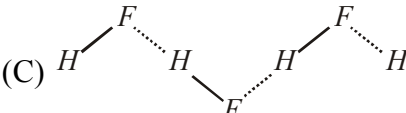
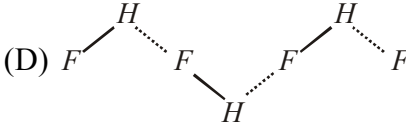
- The valency of sulphur in sulphuric acid is :



V.B.T., HYBRIDISATION & VSEPR THEORY

11. In the following compound $\overset{1}{\text{C}}\text{H}_2 = \overset{2}{\text{C}}\text{H} - \overset{3}{\text{C}}\text{H}_2 - \text{C} \equiv \text{CH}$, the $\text{C}_2 - \text{C}_3$ bond is of the type :
 (A) $\text{sp}-\text{sp}^2$ (B) sp^3-sp^3 (C) $\text{sp}-\text{sp}^3$ (D) sp^2-sp^3
12. Which of the following has a geometry different from the other three species (having the same geometry)?
 (A) BF_4^- (B) SO_4^{2-} (C) XeF_4 (D) PH_4^+
13. Maximum bond energy is in :
 (A) F_2 (B) N_2 (C) O_2 (D) equal
14. The hybridisation and shape of BrF_3 molecule are :
 (A) sp^3d and bent T shape (B) sp^2d^2 and tetragonal
 (C) sp^3d and bent (D) none of these
15. The shape of methyl cation (CH_3^+) is likely to be:
 (A) linear (B) pyramidal (C) planar (D) spherical
16. The structure of XeF_2 involves hybridization of the type :
 (A) sp^3 (B) sp^2 (C) sp^3d (D) sp^3d^2
17. Which of the following has been arranged in increasing order of length of the hybrid orbitals ?
 (A) $\text{sp} < \text{sp}^2 < \text{sp}^3$ (B) $\text{sp}^3 < \text{sp}^2 < \text{sp}$ (C) $\text{sp}^2 < \text{sp}^3 < \text{sp}$ (D) $\text{sp}^2 < \text{sp} < \text{sp}^3$
18. In the context of carbon, which of the following is arranged in the correct order of electronegativity :
 (A) $\text{sp} > \text{sp}^2 > \text{sp}^3$ (B) $\text{sp}^3 > \text{sp}^2 > \text{sp}$ (C) $\text{sp}^2 > \text{sp} > \text{sp}^3$ (D) $\text{sp}^3 < \text{sp} < \text{sp}^2$
19. When $2\text{s}-2\text{s}$, $2\text{p}-2\text{p}$ and $2\text{p}-2\text{s}$ orbitals overlap, the bond strength decreases in the order :
 (A) $\text{p} - \text{p} > \text{s} - \text{s} > \text{p} - \text{s}$ (B) $\text{p} - \text{p} > \text{p} - \text{s} > \text{s} - \text{s}$
 (C) $\text{s} - \text{s} > \text{p} - \text{p} > \text{p} - \text{s}$ (D) $\text{s} - \text{s} > \text{p} - \text{s} > \text{p} - \text{p}$
20. Carbon atoms in $\text{C}_2(\text{CN})_4$ are :
 (A) sp -hybridized (B) sp^2 -hybridized
 (C) sp - and sp^2 hybridized (D) sp , sp^2 and sp^3 - hybridized
21. The bond angle and hybridization in ether (CH_3OCH_3) is :
 (A) $106^\circ 51'$, sp^3 (B) $104^\circ 31'$, sp^3 (C) $> 109^\circ 28'$, sp^3 (D) None of these
22. The structure of XeF_6 in vapour phase is
 (A) pentagonal bipyramidal (B) trigonal bipyramidal
 (C) capped octahedron (D) square bipyramidal

DIPOLE MOMENT & HYDROGEN BONDING

23. Which of the following has been arranged in order of decreasing dipole moment ?
 (A) $\text{CH}_3\text{Cl} > \text{CH}_3\text{F} > \text{CH}_3\text{Br} > \text{CH}_3\text{I}$ (B) $\text{CH}_3\text{F} > \text{CH}_3\text{Cl} > \text{CH}_3\text{Br} > \text{CH}_3\text{I}$
 (C) $\text{CH}_3\text{Cl} > \text{CH}_3\text{Br} > \text{CH}_3\text{I} > \text{CH}_3\text{F}$ (D) $\text{CH}_3\text{F} > \text{CH}_3\text{Cl} > \text{CH}_3\text{I} > \text{CH}_3\text{Br}$
24. The experimental value of the dipole moment of HCl is 1.03 D. The length of the $\text{H}-\text{Cl}$ bond is 1.275 Å. The percentage of ionic character in HCl is nearly :
 (A) 43 (B) 21 (C) 17 (D) 7
25. Ethanol has a higher boiling point than dimethyl ether though they have the same molecular weight. This is due to :
 (A) resonance (B) coordinate bonding
 (C) hydrogen bonding (D) ionic bonding
26. Which of the following compounds would have significant intermolecular hydrogen bonding ?
 HF , CH_3OH , N_2O_4 , CH_4
 (A) HF , N_2O_4 (B) HF , CH_4 , CH_3OH (C) HF , CH_3OH (D) CH_3OH , CH_4
27. For H_2O_2 , H_2S , H_2O and HF , the correct order of decreasing strength of hydrogen bonding is :
 (A) $\text{H}_2\text{O} > \text{HF} > \text{H}_2\text{O}_2 > \text{H}_2\text{S}$ (B) $\text{HF} > \text{H}_2\text{O}_2 > \text{H}_2\text{O} > \text{H}_2\text{S}$
 (C) $\text{HF} > \text{H}_2\text{O} > \text{H}_2\text{O}_2 > \text{H}_2\text{S}$ (D) $\text{H}_2\text{O}_2 > \text{H}_2\text{O} > \text{HF} > \text{H}_2\text{S}$
28. Which one of the following does not have intermolecular H-bonding?
 (A) H_2O (B) *o*-nitro phenol (C) HF (D) CH_3COOH
29. The order of strength of hydrogen bonds is:
 (A) $\text{NH}\cdots\text{N} > \text{OH}\cdots\text{O} > \text{FH}\cdots\text{F}$ (B) $\text{NH}\cdots\text{N} < \text{OH}\cdots\text{O} < \text{FH}\cdots\text{F}$
 (C) $\text{NH}\cdots\text{N} > \text{OH}\cdots\text{O} < \text{FH}\cdots\text{F}$ (D) $\text{NH}\cdots\text{N} < \text{OH}\cdots\text{O} > \text{FH}\cdots\text{F}$
30. The H bond in solid HF can be best represented as:
 (A) $\text{H} - \text{F} \cdots \text{H} - \text{F} \cdots \text{H} - \text{F}$ (B) 
 (C) 
 (D) 

MISCELLANEOUS

31. The formal charges on the three O-atoms in O_3 molecule are
 (A) 0, 0, 0 (B) 0, 0, -1 (C) 0, 0, +1 (D) 0, +1, -1
32. The types of bonds present in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ are
 (A) electrovalent and covalent
 (B) electrovalent, coordinate covalent & H-bond
 (C) covalent, coordinate covalent & H-bonds
 (D) electrovalent, covalent, coordinate covalent & H-bond
33. If ethylene molecule lies in X-Y plane then nodal planes of the π -bond will lie in
 (A) XZ plane (B) YZ plane
 (C) In a plane that bisects C-C axis (D) XY plane
34. Two ice cubes are pressed over each other and unite to form one cube. Which force is responsible for holding them together :
 (A) van der Waal's forces (B) Covalent attraction
 (C) Hydrogen bond formation (D) Dipole-dipole attraction

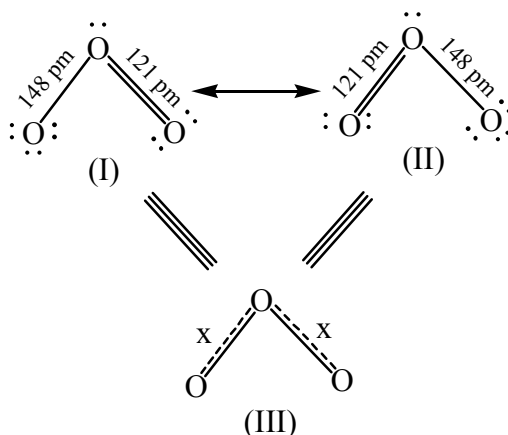
EXERCISE # O-2

IONIC BOND, COVALENT BOND & COORDINATE BOND

- Which of the following compounds contain ionic, covalent and co-ordinate bonds?
(A) NH_4Cl (B) KCN (C) NaBF_4 (D) NaOH
- Which of the following statements is/are true?
(A) Covalent bonds are directional
(B) Ionic bonds are nondirectional
(C) A polar bond is formed between two atoms which have the same electronegativity value.
(D) The presence of polar bonds in a polyatomic molecule suggests that it has zero dipole moment
- To which of the following species octet rule is not applicable :
(A) BrF_5 (B) SF_6 (C) IF_7 (D) CO

LEWIS STRUCTURE & BOND LENGTH

- O_3 (ozone) molecule can be equally represented by the structures I and II shown below



which of the following values of x are not correct

- (A) 148 pm (B) 121 pm (C) less than 121 pm (D) more than 148 pm
- Select the correct statement -
(A) C–O bond length of CO molecule is higher than the C–O bond length of CO_2
(B) C–O bond length of CO molecule is lower than the C–O bond length of CO_2
(C) N–O bond length of NO_3^- ion is higher than the N–O bond length of NO_2^-
(D) N–O bond length of NO_3^- ion is lower than the N–O bond length of NO_2^-

VBT, HYBRIDISATION & VSEPR THEORY

- Which of the following statement(s) is / are not correct?
(A) Hybridization is the mixing of atomic orbitals of large energy difference.
(B) sp^2 -hybrid orbitals are formed from two p - atomic orbitals and one s - atomic orbital
(C) sp^3d^2 -hybrid orbitals are all at 90° to one another
(D) sp^3 -hybrid orbitals are directed towards the corners of a regular tetrahedron
- Which of the following species are linear ?
(A) ICl_2^- (B) I_3^- (C) N_3^- (D) ClO_2^+

8. Which of the following molecules have bent shape.
Where A is central atom, B is surrounding atom and E is lone pair -
(A) AB_2E_2 (B) AB_2E (C) AB_3E (D) AB_2
9. In which of the following process(s) hybridisation of underlined atom does not change -
(A) $\underline{N}H_3 + BF_3 \rightarrow H_3N.BF_3$ (B) $\underline{Si}F_4 + 2F^\ominus \rightarrow [SiF_6]^{2-}$
(C) $BH_3 + \text{cyclopropane} \rightarrow H_3B \leftarrow \text{cyclopropane}$ (D) $H_3\underline{B}O_3 \xrightarrow{OH^-} [B(OH)_4]^-$
10. Which of the following 'd' orbital(s) participate in the hybridisation for Xe in the cationic part of $XeF_6(s)$
(A) d_{xy} (B) $d_{x^2-y^2}$ (C) d_{z^2} (D) d_{y^2}

DIPOLE MOMENT & HYDROGEN BOND

11. Which of the following compounds possesses zero dipole moment?
(A) Water (B) Benzene
(C) Carbon tetrachloride (D) Boron trifluoride
12. Which of the following statements are correct?
(A) The crystal lattice of ice is formed by covalent as well as hydrogen bonds
(B) The density of water increases when heated from 0°C to 4°C
(C) Above 4°C the thermal agitation of water molecules increases. Therefore, intermolecular distance increases and water starts expanding
(D) The density of water decreases from 0°C to a maximum at 4°C
13. In which of the following chemical species hydrogen bonding takes place -
(A) $Na_2HPO_3(s)$ (B) $K_2HPO_4(s)$ (C) $KHCO_3(s)$ (D) $NaHCO_3(s)$

MISCELLANEOUS & STRUCTURE

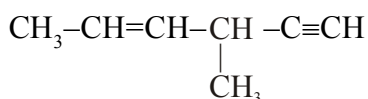
14. Which of the following oxyacids of sulphur contain S-S bonds?
(A) $H_2S_2O_8$ (B) $H_2S_2O_6$ (C) $H_2S_2O_4$ (D) $H_2S_2O_5$
15. Molecule(s) having both polar and non polar bonds is / are
(A) O_2F_2 (B) S_2Cl_2 (C) N_2H_4 (D) S_2F_{10}
16. Which of the following molecule is/are planar -
(A) CH_2Cl_2 (B) $B_3N_3H_6$ (C) SO_3 (D) NF_3

EXERCISE # S-1

Integer Type Answer :

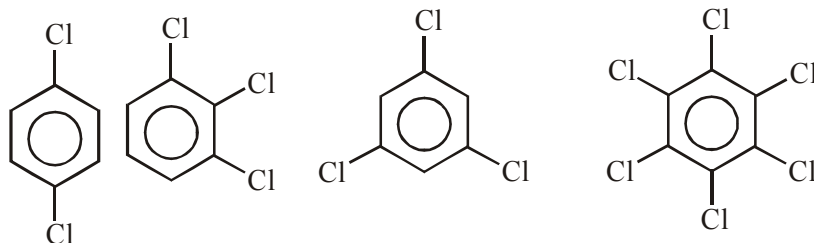
1. In solid NaCl^+ one Na^+ is surrounded by six Cl^- ions find out total number of directional bonds formed by each Na^+ with Cl^- ions.
2. Find out total number of $p\pi-d\pi$ bonds present in SO_2 molecule
3. Find out total number of $p\pi-p\pi$ bonds present in SO_4^{2-}
4. How many number of atomic orbitals are required, so that their mixing produces hybrid orbitals each having % of s-character nearly equal to 17%.
5. Calculate the maximum number of atoms lying in one plane of PCl_3F_2 .
6. Find the lone pair-bond pair ratio for $[\text{SiF}_6]^{2-}$
7. A diatomic molecule has a dipole moment of 1.2 D. If bond length is 1.0 Å, what percentage of an electronic charge exists on each atom.
[Add the digits untill you are getting single digit. e.g. If the answer is 537, then $5 + 3 + 7 = 15$] again $1 + 5 = 6$ and 6 is to be written as answer.

8. Total number of sp^3 hybridised atom(s) in given hydrocarbon.



9. Find the number of lone pair on central atom of ICl_4^-
10. Find the number of acid(s) which are having hypo prefix in it's name from the following.
 H_3PO_4 , H_3BO_3 , H_3PO_3 , H_3PO_2 , HClO_3 , HClO , $(\text{HNO})_2$, $\text{H}_4\text{P}_2\text{O}_6$
11. Find the number of planar molecules/ions which are cyclic –
 $\text{B}_3\text{N}_3\text{H}_6$, $\text{C}_3\text{N}_3\text{Cl}_3$, $(\text{BO}_2^-)_3$, $\text{S}_2\text{O}_8^{2-}$
12. Find the maximum number of plane having maximum number of atoms in CH_4 .

13. Find the number of molecule having dipole moment less than the dipole moment of



14. Find the number of molecules having intramolecular hydrogen bonding in the following :
ice, o-boric acid, o-nitrophenol, o-chlorophenol, chloral hydrate, o-salicylaldehyde
15. Find the number of chemical species which are planar from in the following.
 $\text{XeF}_5^\ominus$, XeF_5^+ , HCN , CO_2 , SO_2

EXERCISE # S-2

(Comprehension) (Q.1 to Q.2)

Bond angle and bond polarity are important parameters associated for bonding in molecule.

- Which of the following bond angle order is incorrect.
 (A) $\text{NH}_3 > \text{PH}_3$ (B) $\text{BF}_3 < \text{BH}_3$
 (C) $\text{O}(\text{CH}_3)_2 < \text{O}(\text{C}_2\text{H}_5)_2$ (D) $\text{CO}_3^{2-} < \text{CO}_2$
- Which of the following molecule is polar and non-planar
 (A) CH_2Cl_2 (B) PCl_5 (C) SO_2 (D) XeF_4

(Comprehension) (Q.3 to Q.4)

During the formation of a covalent bond each participating atom usually acquires electronic configuration to that of nearest noble gas, and the shared electron pair remains localized between the bonded nuclei. Besides these shared electrons there are certain electron pairs which remains localized on C.A. or on substituent or both and are called non bonding or lone pair electrons.

- Which of the following species does / do not follow octet rule:
 (A) Hypovalent (B) Hypervalent
 (C) Odd electron molecules (D) All
- In SO_3^{2-} the total number of lone pairs and bond pairs are respectively
 (A) 18 and 4 (B) 9 and 8 (C) 9 and 4 (D) 7 and 4

(Comprehension) (Q.5 to Q.7)

Hybridisation is a theoretical concept, as state of hybridisation cannot be detected even by spectroscopically; unlike intermediates or transition state in various reactions. but it corrects the predictions which are based simple on overlapping of pure atomic orbitals. VSEPR theory predicts precisely shape and bond angle in a given molecule.

- In which pair of molecules bond angles are not same:
 (A) CCl_4 & SiCl_4 (B) NH_4^+ & NF_4^+ (C) ClF_6^+ & SF_6 (D) None
- The molecules / ions which are planar as well as polar.
 (A) BF_3 , H_2O , HF , NH_2^- (B) SnCl_2 , I_3^+ , NH_2^- , IF_3
 (C) CO_3^{2-} , I_3^- , SF_2 , XeF_2 (D) NO_2^- , XeF_4 , ICl_4^- , NH_2^+
- The correct order of energy levels of hybrid orbitals.
 (A) $sp > sp^2 > sp^3$ (B) $sp < sp^2 < sp^3$ (C) $sp^2 > sp^3 > sp$ (D) $sp^3 > sp > sp^2$

(Comprehension) (Q.8 to Q.10)

As one moves from sp hybridisation to sp^3 . % of s-character in hybrid orbital decreases from 50% to 25% and p-character increases from 50% to 75% and in any hybrid orbital total % of s & p-character remains 100%. By increasing p-character the hybrid orbitals become elongated hence, their overlapping extent decreases that is results into weak bond energy also bond angle decreases.

- Statement-1** : On decreasing s-character in hybrid orbitals, bond angle decreases.
Statement-2 : p-orbitals are at 90° to one another.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

9. The type of overlapping which produces bond of maximum bond energy is:
 (A) $sp^3 - 1s$ (B) $sp^2 - 1s$
 (C) $sp - 1s$ (D) All have same bond energy
10. The molecule having bond pair in which % of s character is minimum for the central atom's hybrid orbital.
 (A) SiF_4 (B) BCl_3 (C) CH_3^+ (D) CO_2

Match the Column :

11. Match the column

Column-I (molecules)

- (A) PCl_5 ($\theta = 120^\circ$, ($\theta' = 90^\circ$)
 (B) NH_3 ($\theta = 107^\circ$)
 (C) SiH_4 ($\theta = 109^\circ 28'$) (R)
 (D) SO_2 ($\theta = 116^\circ$)

Column-II (correct about molecule)

- (P) Number of the given bond angle (θ or θ') is = One
 (Q) Number of the given bond angle (θ or θ') is = Three
 (R) Number of the given bond angle (θ or θ') is = Six
 (S) d-orbital(s) is/are involved in hybridisation of central atom
 (T) Lone pair at underlined atom is present

12. Column I

- (A) BrF_3
 (B) TeF_5^-
 (C) IF_7
 (D) XeF_4

Column II

- (P) Atleast one angle $\leq 90^\circ$
 (Q) Central atom is sp^3d^2 hybridised
 (R) Non planar
 (S) Polar

13. Column I

- (A) $I(CN)_2^-$
 (B) CO_3^{2-}
 (C) XeO_2F_2
 (D) SOF_4

Column II

- (P) Having $p\pi-p\pi$ bond & $\mu_D = 0$
 (Q) Having $p\pi-d\pi$ bond & $\mu_D \neq 0$
 (R) Planar
 (S) Central atom is sp^3d

14. Column I

- (A) $H_2S_2O_5$
 (B) $H_6B_2O_7^{2-}$
 (C) $H_4P_2O_6$ (tetrabasic)
 (D) $H_6Si_2O_7$

Column II

- (P) Central atom is sp^3 hybridised
 (Q) M-O-M i.e. oxo linkage is present
 (R) M-M linkage is present
 (S) Non planar

15. Column-I

- (P) XeF_4
 (Q) BrF_5
 (R) ClF_3
 (S) SO_3^{2-}

Column-II

- (1) sp^3d hybridised with 2 lone pair on central atom
 (2) sp^3d^2 hybridised with 2 lone pair on central atom
 (3) sp^3 hybridised with 1 lone pair on central atom
 (4) sp^3d^2 hybridised with 1 lone pair on central atom

Code :

	P	Q	R	S
(A)	1	2	3	4
(B)	2	4	1	3
(C)	2	4	3	1
(D)	3	1	4	2

16. Column-I

(P) NH_4Cl (Q) $\text{K}[\text{HF}_2]$ (R) $\text{B}_3\text{N}_3\text{H}_6$ (S) SiC

Column-II

(1) diamond like structure

(2) Benzene like structure

(3) Both hydrogen bond and ionic bond exist

(4) Both covalent bond and ionic bond exist

Code :

	P	Q	R	S
(A)	4	3	2	1
(B)	3	4	2	1
(C)	3	4	1	2
(D)	4	1	3	2

MATCHING LIST TYPE 1 × 3 Q. (THREE LIST TYPE Q.)

Column - I	Column - II	Column - III
(P) d_{yz}	(1) p_y	(i) 1 lobe – 1 lobe overlap
(Q) s	(2) p_x	(ii) 2 lobe – 2 lobe overlap
(R) d_{xz}	(3) d_{yz}	(iii) 4 lobe – 4 lobe overlap
(S) p_z	(4) s	(iv) Zero overlap

17. Which of the following combination results in π -bond formation if internuclear axis is z-axis

(A) (P), (1), (ii) (B) (P), (4), (i) (C) (R), (4), (iv) (D) (R), (3), (ii)

18. Which of the following combination results in σ -bond formation if internuclear axis is z-axis

(A) (P), (1), (ii) (B) (Q), (3), (ii) (C) (Q), (4), (i) (D) (R), (2), (ii)

19. Which of the following combination results in δ -bond formation if internuclear axis is x-axis

(A) (P), (1), (i) (B) (P), (3), (iii) (C) (R), (3), (iv) (D) (P), (2), (ii)

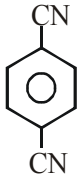
EXERCISE # JEE-MAIN

- Which of the following statements is true ? [AIEEE-2002]
 - HF is less polar than HBr
 - Water does not contain any ions
 - Chemical bond formation takes place when forces of attraction overcome the forces of repulsion
 - In covalent bond, transfer of electrons takes place
- Which one of the following pairs of molecules will have permanent dipole moments for both members [AIEEE-2003]
 - (1) NO_2 and CO_2
 - (2) NO_2 and O_3
 - (3) SiF_4 and CO_2
 - (4) SiF_4 and NO_2
- The pair of species having identical shapes for molecules of both species is [AIEEE-2003]
 - (1) XeF_2 , CO_2
 - (2) BF_3 , PCl_3
 - (3) PF_5 , IF_5
 - (4) CF_4 , SF_4
- The correct order of bond angles (smallest first) in H_2S , NH_3 , BF_3 and SiH_4 is :- [AIEEE-2004]
 - (1) $\text{H}_2\text{S} < \text{NH}_3 < \text{SiH}_4 < \text{BF}_3$
 - (2) $\text{NH}_3 < \text{H}_2\text{S} < \text{SiH}_4 < \text{BF}_3$
 - (3) $\text{H}_2\text{S} < \text{SiH}_4 < \text{NH}_3 < \text{BF}_3$
 - (4) $\text{H}_2\text{S} < \text{NH}_3 < \text{BF}_3 < \text{SiH}_4$
- Lattice energy of an ionic compound depends upon [AIEEE-2005]
 - (1) charge on the ion only
 - (2) size of the ion only
 - (3) packing of the ion only
 - (4) charge and size of the ion
- Of the following sets which one does not contain isoelectronic species ? [AIEEE-2005]
 - (1) PO_4^{3-} , SO_4^{2-} , ClO_4^-
 - (2) CN^- , N_2 , C_2^{2-}
 - (3) SO_3^{2-} , CO_3^{2-} , NO_3^-
 - (4) BO_3^{3-} , CO_3^{2-} , NO_3^-
- In which of the following molecules/ions are all the bonds **not** equal? [AIEEE-2006]
 - (1) XeF_4
 - (2) BF_4^-
 - (3) SF_4
 - (4) SiF_4
- Which of the following hydrogen bonds is the strongest [AIEEE-2007]
 - (1) $\text{F}-\text{H} \cdots \text{F}$
 - (2) $\text{O}-\text{H} \cdots \text{O}$
 - (3) $\text{O}-\text{H} \cdots \text{F}$
 - (4) $\text{O}-\text{H} \cdots \text{N}$
- The hybridisation of orbitals of N atom in NO_3^- , NO_2^+ and NH_4^+ are respectively:- [AIEEE-2011]
 - (1) sp , sp^3 , sp^2
 - (2) sp^2 , sp^3 , sp
 - (3) sp , sp^2 , sp^3
 - (4) sp^2 , sp , sp^3
- The structure of IF_7 is :- [AIEEE-2011]
 - (1) octahedral
 - (2) pentagonal bipyramid
 - (3) square pyramid
 - (4) trigonal bipyramid
- Which of the following has maximum number of lone pairs associated with Xe [AIEEE-2011]
 - (1) XeO_3
 - (2) XeF_4
 - (3) XeF_6
 - (4) XeF_2
- The molecule having smallest bond angle is :- [AIEEE-2012]
 - (1) PCl_3
 - (2) NCl_3
 - (3) AsCl_3
 - (4) SbCl_3
- In which of the following pairs the two species are not isostructural ? [AIEEE-2012]
 - (1) AlF_6^{3-} and SF_6
 - (2) CO_3^{2-} and NO_3^-
 - (3) PCl_4^+ and SiCl_4
 - (4) PF_5 and BrF_5

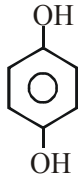
14. The number of S-S bonds in SO_3 , $\text{S}_2\text{O}_3^{2-}$, $\text{S}_2\text{O}_6^{2-}$ and $\text{S}_2\text{O}_8^{2-}$ respectively are :-
[JEE-MAINS-2012]
(1) 1, 0, 1, 0 (2) 0, 1, 1, 0 (3) 1, 0, 0, 1 (4) 0, 1, 0, 1 (On line)
15. Among the following species which two have trigonal bipyramidal shape ? [JEE-MAINS-2012]
(I) NI_3 (II) I_3^- (III) SO_3^{2-} (IV) NO_3^- (On line)
(1) II and III (2) III and IV (3) I and IV (4) I and III
16. Which of the following has the square planar structure :-
[JEE-MAINS-2012]
(1) NH_4^+ (2) CCl_4 (3) XeF_4 (4) BF_4^- (On line)
17. The compound of Xenon with zero dipole moment is :-
[JEE-MAINS-2012]
(1) XeO_3 (2) XeO_2 (3) XeF_4 (4) XeOF_4 (On line)
18. Among the following the molecule with the lowest dipole moment is :- [JEE-MAINS-2012]
(1) CHCl_3 (2) CH_2Cl_2 (3) CCl_4 (4) CH_3Cl (On line)
19. The formation of molecular complex $\text{BF}_3 - \text{NH}_3$ results in a change in hybridisation of boron :-
[JEE-MAINS-2012]
(1) from sp^3 to sp^3d (2) from sp^2 to dsp^2 (On line)
(3) from sp^3 to sp^2 (4) from sp^2 to sp^3
20. Trigonal bipyramidal geometry is shown by:
[JEE-MAINS-2013]
(1) XeO_3F_2 (2) XeOF_2 (3) XeO_3 (4) FXeOSO_2F (On line)
21. Which one of the following molecules is polar?
[JEE-MAINS-2013]
(1) CF_4 (2) SbF_5 (3) IF_5 (4) XeF_4 (On line)
22. Oxidation state of sulphur in anions SO_3^{2-} , $\text{S}_2\text{O}_4^{2-}$ and $\text{S}_2\text{O}_6^{2-}$ increases in the orders :
[JEE-MAINS-2013]
(1) $\text{S}_2\text{O}_6^{2-} < \text{S}_2\text{O}_4^{2-} < \text{SO}_3^{2-}$ (2) $\text{SO}_3^{2-} < \text{S}_2\text{O}_4^{2-} < \text{S}_2\text{O}_6^{2-}$ (On line)
(3) $\text{S}_2\text{O}_4^{2-} < \text{SO}_3^{2-} < \text{S}_2\text{O}_6^{2-}$ (4) $\text{S}_2\text{O}_4^{2-} < \text{S}_2\text{O}_6^{2-} < \text{SO}_3^{2-}$
23. XeO_4 molecule is tetrahedral having :
[JEE-MAINS-2013]
(1) Two $\text{p}\pi\text{-d}\pi$ bonds (2) Four $\text{p}\pi\text{-d}\pi$ bonds (On line)
(3) One $\text{p}\pi\text{-d}\pi$ bond (4) Three $\text{p}\pi\text{-d}\pi$ bonds
24. Bond distance in HF is 9.17×10^{-11} m. Dipole moment of HF is 6.104×10^{-30} Cm. The percent ionic character in HF will be : (electron charge = 1.60×10^{-19} C) [JEE-MAINS-2013]
(1) 61.0% (2) 38.0% (On line)
(3) 35.5% (4) 41.5%
25. The shape of IF_6^- is :
[JEE-MAINS-2013]
(1) distorted octahedron (2) Pyramidal (On line)
(3) Octahedral (4) Square antiprism
26. Which has trigonal bipyramidal shape ?
[JEE-MAINS-2013]
(1) XeOF_4 (2) XeO_3 (3) XeO_3F_2 (4) XeOF_2 (On line)

27. The catenation tendency of C, Si and Ge is in the order $\text{Ge} < \text{Si} < \text{C}$. The bond energies (in kJ mol^{-1}) of C—C, Si—Si and Ge—Ge bonds are respectively : [JEE-MAINS-2013 (On line)]
 (1) 348, 260, 297 (2) 348, 297, 260 (3) 297, 348, 260 (4) 260, 297, 348
28. In which of the following sets, all the given species are isostructural ? [JEE-MAINS-2013 (On line)]
 (1) BF_3 , NF_3 , PF_3 , AlF_3 (2) PCl_3 , AlCl_3 , BCl_3 , SbCl_3
 (3) BF_4^- , CCl_4 , NH_4^+ , PCl_4^+ (4) CO_2 , NO_2 , ClO_2 , SiO_2
29. For which of the following molecule significant $\mu \neq 0$ [JEE-M-2014]
- 

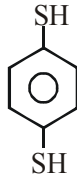
(a)



(b)



(c)



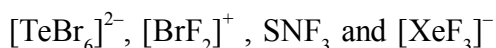
(d)
- (1) Only (c) (2) (c) and (d) (3) Only (a) (4) (a) and (b)
30. Among the following oxoacids, the correct decreasing order of acid strength is :- [JEE-M-2014]
 (1) $\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HOCl}$ (2) $\text{HClO}_2 > \text{HClO}_4 > \text{HClO}_3 > \text{HOCl}$
 (3) $\text{HOCl} > \text{HClO}_2 > \text{HClO}_3 > \text{HClO}_4$ (4) $\text{HClO}_4 > \text{HOCl} > \text{HClO}_2 > \text{HClO}_3$
31. The number and type of bonds in C_2^{2-} ion in CaC_2 are:
 [AIEEE-2005, AIEEE-2011, JEE-MAINS-2014 (On line)]
 (1) Two σ bonds and one π - bond (2) Two σ bonds and two π - bonds
 (3) One σ bond and two π - bonds (4) One σ bond and one π - bond
32. For the compounds [JEE-MAINS-2014]
 CH_3Cl , CH_3Br , CH_3I and CH_3F , (On line)
 the correct order of increasing C-halogen bond length is :
 (1) $\text{CH}_3\text{F} < \text{CH}_3\text{Br} < \text{CH}_3\text{Cl} < \text{CH}_3\text{I}$ (2) $\text{CH}_3\text{F} < \text{CH}_3\text{Cl} < \text{CH}_3\text{Br} < \text{CH}_3\text{I}$
 (3) $\text{CH}_3\text{Cl} < \text{CH}_3\text{Br} < \text{CH}_3\text{F} < \text{CH}_3\text{I}$ (4) $\text{CH}_3\text{F} < \text{CH}_3\text{I} < \text{CH}_3\text{Br} < \text{CH}_3\text{Cl}$
33. Shapes of certain interhalogen compounds are stated below. Which one of them is not correctly stated? [JEE-MAINS-2014]
 (1) IF_7 : Pentagonal bipyramid (2) BrF_5 : Trigonal bipyramid (On line)
 (3) ICl_3 : Planar dimeric (4) BrF_3 : Planar T-shaped
34. Which of the following molecules has two sigma(σ) and two pi(π) bonds :- [JEE-MAINS-2014]
 (1) HCN (2) $\text{C}_2\text{H}_2\text{Cl}_2$ (3) N_2F_2 (4) C_2H_4 (On line)
35. The species in which the N atom is in a state of sp hybridization is [JEE-MAINS(offline)-2016]
 (1) NO_2 (2) NO_2^+ (3) NO_2^- (4) NO_3^-
36. The pair in which phosphorous atoms have a formal oxidation state of + 3 is :- [JEE-MAINS(offline)-2016]
 (1) Pyrophosphorous and pyrophosphoric acids
 (2) Orthophosphorous and pyrophosphorous acids
 (3) Pyrophosphorous and hypophosphoric acids
 (4) Orthophosphorous and hypophosphoric acids

37. The group of molecules having identical shape is : [JEE-MAINS(online)-2016]
 (1) SF_4 , XeF_4 , CCl_4 (2) ClF_3 , XeOF_2 , XeF_3^+
 (3) PCl_5 , IF_5 , XeO_2F_2 (4) BF_3 , PCl_3 , XeO_3
38. **Assertion :** Among the carbon allotropes, diamond is an insulator, whereas, graphite is a good conductor of electricity. [JEE-MAINS(online)-2016]
Reason : Hybridization of carbon in diamond and graphite are sp^3 and sp^2 , respectively.
 (1) Assertion is incorrect statement, but the reason is correct.
 (2) Both assertion and reason are correct, and the reason is the correct explanation for the assertion.
 (3) Both assertion and reason are incorrect.
 (4) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion.
39. Aqueous solution of which salt will not contain ions with the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6$? [JEE-MAINS(online)-2016]
 (1) NaCl (2) CaI_2 (3) KBr (4) NaF
40. The correct sequence of decreasing number of π -bonds in the structures of H_2SO_3 , H_2SO_4 and $\text{H}_2\text{S}_2\text{O}_7$ is:- [JEE-MAINS(online)-2017]
 (1) $\text{H}_2\text{S}_2\text{O}_7 > \text{H}_2\text{SO}_4 > \text{H}_2\text{SO}_3$ (2) $\text{H}_2\text{SO}_3 > \text{H}_2\text{SO}_4 > \text{H}_2\text{S}_2\text{O}_7$
 (3) $\text{H}_2\text{S}_2\text{O}_7 > \text{H}_2\text{SO}_3 > \text{H}_2\text{SO}_4$ (4) $\text{H}_2\text{SO}_4 > \text{H}_2\text{S}_2\text{O}_7 > \text{H}_2\text{SO}_3$
41. The increasing order of the boiling points for the following compounds is:- [JEE-MAINS(online)-2017]
 (I) $\text{C}_2\text{H}_5\text{OH}$ (II) $\text{C}_2\text{H}_5\text{Cl}$
 (III) $\text{C}_2\text{H}_5\text{CH}_3$ (IV) $\text{C}_2\text{H}_5\text{OCH}_3$
 (1) (III) < (II) < (I) < (IV) (2) (II) < (III) < (IV) < (I)
 (3) (IV) < (III) < (I) < (II) (4) (III) < (IV) < (II) < (I)
42. The number of P–OH bonds and the oxidation state of phosphorus atom in pyrophosphoric acid ($\text{H}_4\text{P}_2\text{O}_7$) respectively are:- [JEE-MAINS(online)-2017]
 (1) five and four (2) five and five
 (3) four and five (4) four and four
43. The group having triangular planar structures is :- [JEE-MAINS(online)-2017]
 (1) CO_3^{2-} , NO_3^- , SO_3 (2) NCl_3 , BCl_3 , SO_3
 (3) NH_3 , SO_3 , CO_3^{2-} (4) BF_3 , NF_3 , CO_3^{2-}
44. The number of S=O and S–OH bonds present in peroxodisulphuric acid and pyrosulphuric acid respectively are : [JEE-MAINS(online)-2017]
 (1) (2 and 4) and (2 and 4) (2) (4 and 2) and (2 and 4)
 (3) (2 and 2) and (2 and 2) (4) (4 and 2) and (4 and 2)

EXERCISE # JEE-ADVANCED

Integer Type :

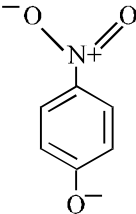
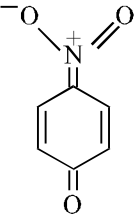
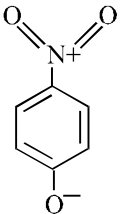
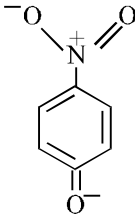
- The number of water molecule(s) directly bonded to the metal centre in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is [JEE 2009]
- Based on VSEPR theory, the number of 90 degree F–Br–F angles in BrF_5 is [JEE 2010]
- The difference in the oxidation numbers of the two types of sulphur atoms in $\text{Na}_2\text{S}_4\text{O}_6$ is. [JEE 2011]
- The total number of lone-pairs of electrons in melamine is [JEE Adv. 2013]
- The sum of the number of lone pairs of electrons on each central atom in the following species is. [JEE 2017]



[Atomic number : N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54]

One or more than one correct :

- The linear structure is assumed by : [IIT-1991]
(A) SnCl_4 (B) NCO^- (C) CS_2 (D) NO_2^+
- Which of the following has zero dipole moment ? [IIT-1993]
(A) ClF (B) PCl_3 (C) SiF_4 (D) CFC_3
- Among the following molecules, which one is planar ? [IIT-1994]
(A) BCl_3 (B) SO_2Cl_2 (C) NH_3 (D) NF_3
- Which of the following species is / are non-linear ? [IIT-1995]
(A) H_2S (B) NH_3 (C) CO_2 (D) SO_2
- Which contains both polar and non-polar bonds ? [IIT-1997]
(A) CH_4 (B) HCN (C) H_2O_2 (D) NH_4Cl
- Which of the following compounds has sp^2 hybridisation ? [IIT-1997]
(A) CO_2 (B) SO_2 (C) N_2O (D) CO
- Which one of the following molecules is planar ? [IIT-1996]
(A) NF_3 (B) NCl_3 (C) BF_3 (D) PH_3
- The number and type of bonds between two carbon atoms in CaC_2 are : [IIT-1996]
(A) 1σ and 1π (B) 1σ and 2π (C) 1σ and 1.5π (D) 1σ
- Among the species NF_3 , NO_3^- , BF_3 , H_3O^+ and HN_3 , identify the isostructural species : [IIT-1996]
(A) (NF_3 , NO_3^-) and (BF_3 , H_3O^+) (B) (NF_3 , HN_3) and (NO_3^- , BF_3)
(C) (NF_3 , H_3O^+) and (NO_3^- , BF_3) (D) (NF_3 , H_3O^+) and (HN_3 , BF_3)
- Among the following the one that is polar and has the central atom with sp^2 hybridisation is : [IIT-1997]
(A) H_2CO_3 (B) SiF_4 (C) BF_3 (D) HClO_2
- The geometry & the type of hybrid orbitals present about the central atom in BF_3 is : [JEE '98]
(A) linear, sp (B) trigonal planar, sp^2
(C) tetrahedral sp^3 (D) pyramidal, sp^3

17. Which one of the following statement (s) is (are) correct? [JEE 1998]
 (A) The electronic configuration of Cr is $[\text{Ar}] 3d^5 4s^1$. (Atomic No. of Cr = 24)
 (B) The magnetic quantum number may have a negative value
 (C) In silver atom, 23 electrons have a spin of one type and 24 of the opposite type. (Atomic No. of Ag = 47)
 (D) The oxidation state of nitrogen in HN_3 is -3.
18. Element X is strongly electronegative and element Y is strongly electropositive. Both are univalent. The compound formed would be : [JEE 1998]
 (A) $\text{X}^+ \text{Y}^-$ (B) $\text{X}^- \text{Y}^+$ (C) $\text{X}^+ \text{Y}^-$ (D) $\text{Y}^+ \text{X}^-$
19. The correct order of increasing C - O bond length of, CO , CO_3^{2-} , CO_2 is [JEE '99]
 (A) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$ (B) $\text{CO}_2 < \text{CO}_3^{2-} < \text{CO}$
 (C) $\text{CO} < \text{CO}_3^{2-} < \text{CO}_2$ (D) $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$
20. The geometry of H_2S and its dipole moment are [JEE '99]
 (A) angular & non zero (B) angular & zero
 (C) linear & non zero (D) linear & zero
21. In compounds type ECl_3 , where $\text{E} = \text{B}, \text{P}, \text{As}$ or Bi , the angles $\text{Cl} - \text{E} - \text{Cl}$ for different E are in the order [JEE '99]
 (A) $\text{B} > \text{P} = \text{As} = \text{Bi}$ (B) $\text{B} > \text{P} > \text{As} > \text{Bi}$
 (C) $\text{B} < \text{P} = \text{As} = \text{Bi}$ (D) $\text{B} < \text{P} < \text{As} < \text{Bi}$
22. The most likely representation of resonance structure of p-nitrophenoxide is:
 (A)  (B)  (C)  (D) 
23. Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is [JEE 2000]
 (A) H_2O because of hydrogen bonding
 (B) H_2Te because of higher molecular weight
 (C) H_2S because of hydrogen bonding
 (D) H_2Se because of lower molecular weight
24. The hybridization of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ are [JEE 2000]
 (A) sp^2 , sp^3 and sp^2 respectively (B) sp , sp^2 and sp^3 respectively
 (C) sp^2 , sp and sp^3 respectively (D) sp^2 , sp^3 and sp respectively
25. The correct order of hybridization of the central atom in the following species NH_3 , PtCl_4^{2-} , PCl_5 and BCl_3 is [JEE 2001]
 (A) dsp^2 , sp^3d , sp^2 and sp^3 (B) sp^3 , dsp^2 , sp^3d , sp^2
 (C) dsp^2 , sp^2 , sp^3 , sp^3d (D) dsp^2 , sp^3 , sp^2 , sp^3d

26. Specify hybridization of N and B atoms in a 1 : 1 complex of BF_3 and NH_3 [JEE 2002]
 (A) N : tetrahedral, sp^3 ; B : tetrahedral, sp^3
 (B) N : pyramidal, sp^3 ; B : pyramidal, sp^3
 (C) N : pyramidal, sp^3 ; B : planar, sp^2
 (D) N : pyramidal, sp^3 ; B : tetrahedral, sp^3
27. The nodal plane in the π -bond of ethene is located in [JEE 2002]
 (A) the molecular plane
 (B) a plane parallel to the molecular plane
 (C) a plane perpendicular to the molecular plane which bisects, the carbon-carbon σ bond at right angle
 (D) a plane perpendicular to the molecular plane which contains, the carbon-carbon bond.
28. Which of the following are isoelectronic and isostructural ? NO_3^- , CO_3^{2-} , ClO_3^- , SO_3 [JEE 2003]
 (A) NO_3^- , CO_3^{2-} (B) SO_3 , NO_3^- (C) ClO_3^- , CO_3^{2-} (D) CO_3^{2-} , SO_3
29. Which species has the maximum number of lone pair of electrons on the central atom ? [JEE 2005]
 (A) ClO_3^- (B) XeF_4 (C) SF_4 (D) I_3^-
30. The percentage of p-character in the orbitals forming P–P bonds in P_4 is [JEE 2007]
 (A) 25 (B) 33 (C) 50 (D) 75
31. The structure of XeO_3 is [JEE 2007]
 (A) linear (B) planar (C) pyramidal (D) T-shaped
32. **Statement-1** : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.
because
Statement-2 : o-Hydroxybenzoic acid has intramolecular hydrogen bonding. [JEE 2007]
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.
33. **Statement-1** : In water, orthoboric acid behaves as a weak monobasic acid.
because
Statement-2 : In water, orthoboric acid acts as a proton donor. [JEE 2007]
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

34. The nitrogen oxide(s) that contain(s) N–N bond(s) is/are [JEE 2009]
(A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5
35. The species having pyramidal shape is/are : [JEE 2010]
(A) SO_3 (B) BrF_3 (C) SiO_3^- (D) OSF_2
36. Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen- [JEE 2012]
(A) HNO_3 , NO , NH_4Cl , N_2 (B) HNO_3 , NO , N_2 , NH_4Cl
(C) HNO_3 , NH_4Cl , NO , N_2 (D) NO , HNO_3 , NH_4Cl , N_2
37. The shape of XeO_2F_2 molecule is : [JEE 2012]
(A) Trigonal bipyramidal (B) Square planar
(C) tetrahedral (D) see-saw
38. The compound(s) with TWO lone pairs of electrons on the central atom is(are) [JEE 2016]
(A) BrF_5 (B) ClF_3 (C) XeF_4 (D) SF_4
39. The crystalline form of borax has [JEE 2016]
(A) Tetranuclear $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ unit
(B) All boron atoms in the same plane
(C) Equal number of sp^2 and sp^3 hybridized boron atoms
(D) One terminal hydroxide per boron atom
40. The order of the oxidation state of the phosphorus atom in H_3PO_2 , H_3PO_4 , H_3PO_3 and $\text{H}_4\text{P}_2\text{O}_6$ is [JEE 2017]
(A) $\text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2$
(B) $\text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6$
(C) $\text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_4$
(D) $\text{H}_3\text{PO}_4 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6$

ANSWERS KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	B	A	B	D	D	A	D	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	B	A	C	C	A	A	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	C	A	C	C	C	C	B	B	C
Que.	31	32	33	34						
Ans.	D	D	D	C						

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A, C	A, B	A, B, C	A, B, C, D	B, C	A, C	A, B, C	A, B	A, C	B, C
Que.	11	12	13	14	15	16				
Ans.	B, C, D	A, B, C	B, C, D	B, C, D	A, B, C, D	B, C				

EXERCISE # S-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	0	1	0	6	4	3	7 (25%)	3	2	4
Que.	11	12	13	14	15					
Ans.	3	10	3	4	4					

EXERCISE # S-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	D	C	D	B	B	B	C	A
Que.	11				12					
Ans.	(A)→Q,R,S;(B)→Q,T;(C)→R;(D)→P,T				(A)→P,S (B)→P,Q,R,S (C)→P,R (D)→P,Q					
Que.	13					14				
Ans.	(A)→P,R,S (B)→P,R (C)→Q,S (D)→Q,S					(A)→P,R,S (B)→P,Q,S (C)→P,R,S (D)→P,Q,S				
Que.	15	16	17	18	19					
Ans.	B	A	A	C	B					

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	1	1	4	3	3	1	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	4	2	BONUS	3	3	3	4	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	2	4	1	3	2	3	2	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	3	2	2	1	2	2	2	4	4	1
Que.	41	42	43	44						
Ans.	4	3	1	4						

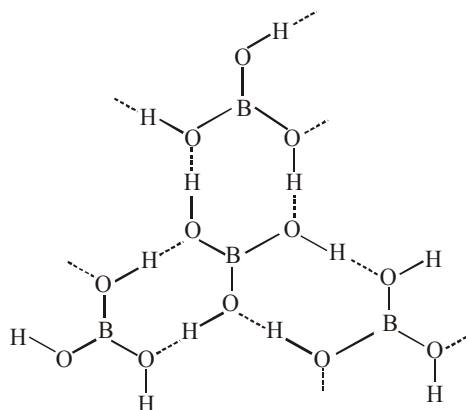
EXERCISE # JEE-ADVANCED

- Ans. 4
- Ans. 0
- Ans. 5
- Ans. 6
- Ans. 6

Que.	6	7	8	9	10	11	12	13	14	15
Ans.	B, C, D	C	A	A, B, D	C	B	C	B	C	A
Que.	16	17	18	19	20	21	22	23	24	25
Ans.	B	A,B,C	B	D	A	B	A	A	B	B
Que.	26	27	28	29	30	31	32	33	34	35
Ans.	A	A	A	D	D	C	D	C	A,B,C or A,C	D
Que.	36	37	38	39	40					
Ans.	B	D	B,C	A,C,D	A					

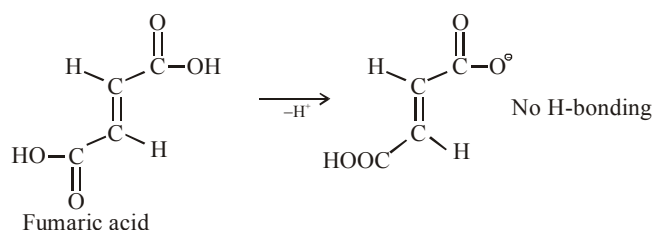
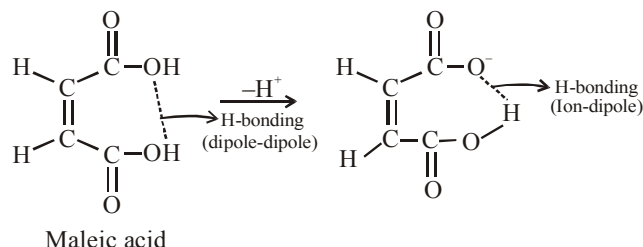
ANSWER OF HYDROGEN BONDING

1.

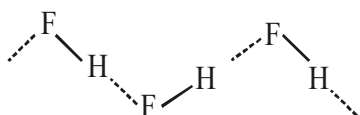


Structure of boric acid; the dotted lines represent hydrogen bonds

- Inter molecular H-Bonding in case of para nitro phenol is strong enough to increase boiling point which is not favoured in o-Nitro phenol.
- Maleic acid anion gets stabilize due to intra molecular H-bonding which is not possible in anion of Fumaric acid. So H^+ releasing tendency is more in case of maleic acid

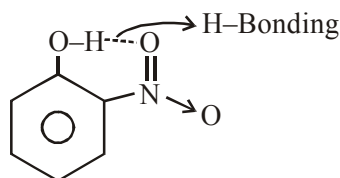


- Due to strong H-Bonding in $H-F$



- Due to H-Bonding in NH_3 .
- Density of ice is higher than that of water due to cage like structure of ice volume of ice increases and density decreases.
- On heating ice in temperature range $0-4^\circ C$ H-bonds of ice break down, which decreases volume, hence density of H_2O increases. On heating ice after $4^\circ C$ thermal vibrations of H_2O molecules increases which increases effective volume of ice, so density of ice again decreases.
- H-I bond is weak as compare to H-F so it can be dissociated easily and can give H^+ easily.
- The organic matter in wood participate in H-bonding with water molecules in ice-cream.
- $K^+[F-H \cdots F]^-$ But $KHBr_2$ & KHI_2 can't form H-Bond.

H-bonding
- In o-Nitrophenol intra molecular H-bond is present which decreases it's solubility in water.



Ortho Nitro-phenol

- In o-hydroxy benzaldehyde intramolecular H-bond is present which is less stronger than intermolecular H-bond in p-hydroxy benzaldehyde.
- Extent of H-bonding in glycerol is more as compare to ethanol so it is more viscous.
- Due to H-bonding in H_2O

**CHEMICAL BONDING,
COORDINATION CHEMISTRY
& METALLURGY**

ALLU
Session 2019-20

CONTENTS

CHEMICAL BONDING

• Theory	01
• EXERCISE-(O-1)	12
• EXERCISE-(O-2)	18
• EXERCISE-(S-I)	22
• EXERCISE-(S-II)	23
• EXERCISE-JEE(Main)	26
• EXERCISE-JEE(Advanced)	30
• ANSWER KEY	33

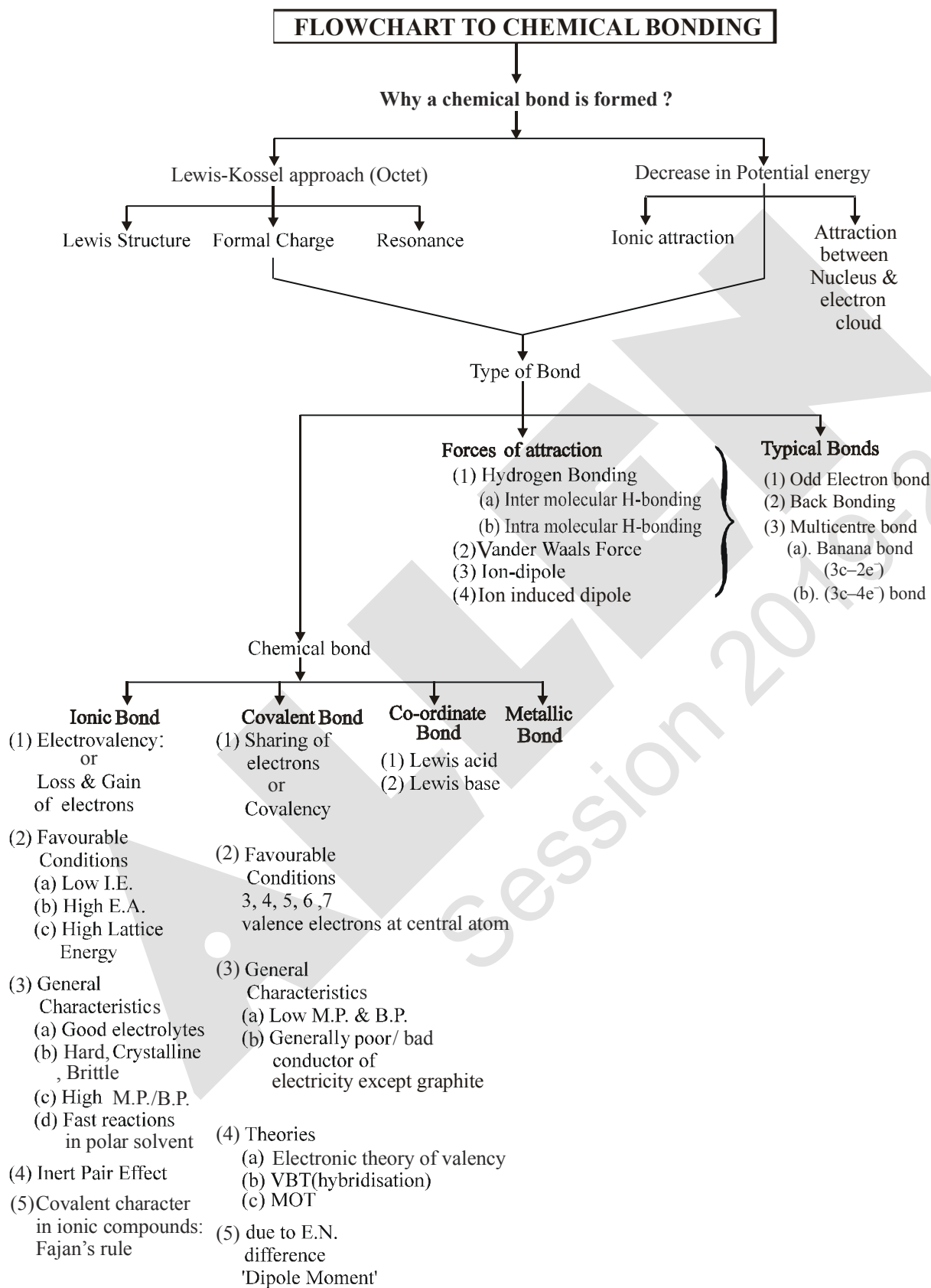
COORDINATION CHEMISTRY

• Theory	35
• EXERCISE-(O-1)	56
• EXERCISE-(O-2)	64
• EXERCISE-(S-I)	66
• EXERCISE-(S-II)	67
• EXERCISE-JEE(Main)	71
• EXERCISE-JEE(Advanced)	79
• ANSWER KEY	85

METALLURGY

• Theory	88
• EXERCISE-(O-1)	101
• EXERCISE-(O-2)	108
• EXERCISE-(S-I)	111
• EXERCISE-(S-II)	112
• EXERCISE-JEE(Main)	115
• EXERCISE-JEE(Advanced)	118
• ANSWER KEY	121

CHEMICAL BONDING



Note : Hydrogen bond is an extreme manifestation of dipole dipole interaction.

VANDER WAAL'S FORCES

- ⇒ These are the weakest type of inter molecular forces that exist among the chemical species which bring significant change in physical properties.
- ⇒ These are non-directional, non-valence cohesive forces. These attractive forces being played between the two molecules are independent of the presence of other molecules.
- ⇒ Solid, liquid or gaseous states of many molecules are explained on the basis of inter molecular forces other than covalent, ionic or metallic bonds. Although inert gases do not form any type of bond but may exist in liquid and solid states. This shows that the atoms of inert gases are attracted by each other through some type of inter molecular forces. These intermolecular forces are called Vander Waals forces.

Types of Vander Waal's Forces

- (1) **Dipole-dipole interaction (Keesom forces) :** The force of attraction between the oppositely charged poles of two polar molecules (for example : H_2S , HCl , PH_3 etc.) is called dipole-dipole attraction.
- (2) **Dipole-induced dipole interaction (Debye forces) :** This type of cohesive forces occurs in a mixture of polar and non polar molecules. The former induce polarity in non polar molecules by disturbing their electron system. for example force of attraction between Cl_2 and H_2O .
- (3) **Instantaneous dipole-Induced dipole interaction (Dispersion forces/London forces) :** The weak intermolecular forces operating in similar non polar gaseous molecules are called London forces. These forces are very weak in nature and exists only at low temperature. For example weak intermolecular forces in F_2 , Cl_2 , N_2 , molecules and in noble gases.

(Note :- London forces present in both polar and non polar species)

Other Weak Interactions

- (1) **Ion-dipole interaction :** Polar molecules are attracted by ions. The negative pole is attracted by cation and positive pole attracted by the anion. This type of attraction is called ion dipole attraction, ion-dipole attraction is observed generally in the process of solvation when sodium chloride ($\text{Na}^+ \text{Cl}^-$) is dissolved in water because negative poles of water aggregate around Na^+ ions and positive poles around Cl^- ions.
- (2) **Ion-induced dipole interaction :** When non polar molecules come in contact with ions, its electron cloud gets polarised and the oppositely charged end of it is attracted by the ion. For example attraction between I^- and I_2 molecule.

HYBRIDISATION

The phenomenon of intermixing of the orbitals of slightly different energies so as to redistribute their energies, resulting in the formation of new set of orbitals of equivalent energies and shape. For example when one 2s and three 2p-orbitals of carbon hybridise, there is the formation of four new sp^3 hybrid orbitals.

Salient features of hybridisation: The main features of hybridisation are as under :

1. The number of hybrid orbitals is equal to the number of the atomic orbitals that get hybridised.
2. The hybridised orbitals are equivalent in energy and shape.
3. The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals.
4. These hybrid orbitals are directed in space in some preferred direction to have minimum repulsion between electron pairs and thus a stable arrangement. Therefore, the type of hybridisation indicates the geometry of the molecules.

Important conditions for hybridisation

- (i) The orbitals present in the valence shell of the atom are hybridised.
- (ii) The orbitals undergoing hybridisation should have almost equal energy.
- (iii) Promotion of electron is not essential condition prior to hybridisation.
- (iv) It is not necessary that only half filled orbitals participate in hybridisation. In some cases, even filled orbitals of valence shell take part in hybridisation.

Types of Hybridisation

There are various types of hybridisation involving s, p and d orbitals. The different types of hybridisation are as under :

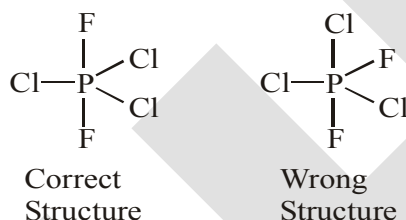
- (i) sp hybridisation (ii) sp^2 hybridisation (iii) sp^3 hybridisation
- (iv) sp^3d hybridisation: (v) sp^3d^2 hybridisation: (vi) sp^3d^3 hybridisation:

BENT'S RULE

- (i) A lone pair of electron prefers to occupy that hybrid orbital which has greater percentage of s-character.
- (ii) A more electronegative atom/group prefers to occupy that hybrid orbital which has smaller percentage of s-character.

Ex. Draw the geometry of PCl_3F_2

Sol.



Because highly electronegative atom occupy axial position (axial position has smaller percentage of s-character).

DRAGO'S RULE

On the basis of experimental bond angles of certain molecules fulfilling the following three conditions,

- (i) Central atom belongs to third or lower period in periodic table
- (ii) Central atom must contain atleast one lone pair of electron
- (iii) Electronegativity of surrounding atom is ≤ 2.5

Drago generalised that in such molecules justification of experimental bond angle can be made satisfactory if one considers no hybridisation, i.e., overlapping of almost pure atomic orbitals from central atom.

In such molecules bond angle is approximately 90° .

Group 15	Bond angle	Group 16	Bond angle
NH_3	$107^\circ 48'$	H_2O	$104^\circ 28'$
PH_3	$93^\circ 36'$	H_2S	92°
AsH_3	$91^\circ 48'$	H_2Se	91°
SbH_3	$91^\circ 18'$	H_2Te	90.5°

- Right order of bond angle.

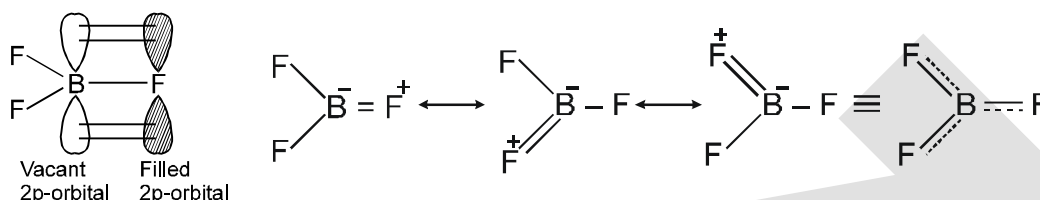
(a) $H_2O > H_2S > H_2Se > H_2Te$

(b) $NH_3 > PH_3 > AsH_3 > SbH_3$

ELECTRON DEFICIENT BONDING**(a) Back Bonding**

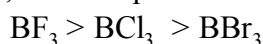
Back bonding generally takes place when out of two bonded atoms one of the atom has vacant orbitals (generally this atom is from second or third period) and the other bonded atom is having some non-bonded electron pair (generally this atom is from the second period). Back bonding increases the bond strength and decreases the bond length.

For example, in BF_3 the boron atom completes its octet by accepting 2p-electrons of fluorine into 2p empty orbital.



- Decrease in B – F bond length is due to delocalised π – π back bonding between filled p-orbital of F atom and vacant p-orbital of B atom.

The extent of back bonding is much larger if the orbitals involved in the back bonding are of same size, for example the extent of back bonding in boron trihalides is as follows :

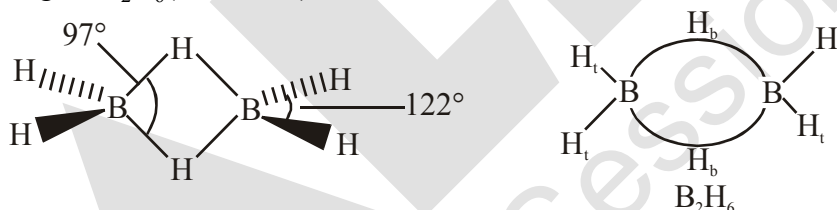


There is π – π back bonding in boron trihalide. The extent of back bonding decreases from BF_3 to BI_3 because of increasing size of p-orbitals participating in back bonding that is from 2p (in F) to 5p (in I).

(b) Bridge Bonding

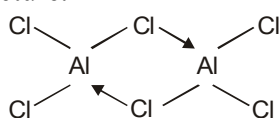
There are many compounds in which some electron deficient bonds are present apart from normal covalent bonds or coordinate bonds which are 2c–2e[–] bonds (two centre two electron bonds). These electron deficient bonds have less number of electrons than the expected such as three centre-two electron bonds (3c–2e) which are present in diborane B_2H_6 , $\text{Al}_2(\text{CH}_3)_6$, $\text{BeH}_2(\text{s})$ etc.

bonding in B_2H_6 (diborane)



The structure of diborane containing four terminal(t) and two bridging(b) hydrogen atoms. The model determined by molecular orbital theory indicates that the bonds between boron and the terminal hydrogen atoms are conventional 2c–2e[–] covalent bonds. The bonding between the boron atoms and the bridging hydrogen atoms is, however different from that in molecules such as hydrocarbons. Having used two electrons in bonding to the terminal hydrogen atoms, each boron has one valence electron remaining for additional bonding. The bridging hydrogen atoms provide one electron each. Thus the B_2H_2 ring is held together by four electrons, an example of 3c–2e bonding. This type of bond is sometimes called as '**banana bond**'. Group 13, gallium is known to form a similar compound, digallane, Ga_2H_6 .

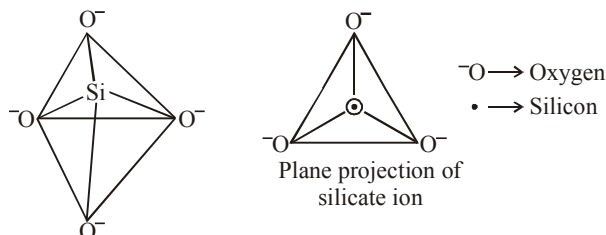
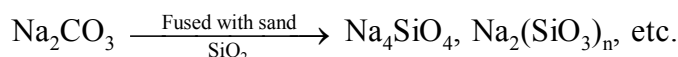
But Al_2Cl_6 have covalent bond only and there is no electron deficient bonding as depicted in the given structure.



(Al – Cl – Al is 3c – 4e bond)

SILICATES

Silicates are metal derivatives of silicic acid, H_4SiO_4 or $\text{Si}(\text{OH})_4$. Silicates are formed by heating metal oxide or metal carbonates with sand, e.g.,



Silicates have basic unit of SiO_4^{4-} , each silicon atom is bonded with four oxide ions tetrahedrally. There are following types of silicates

Silicates	Sharing of O-atom / Basic Tetrahedral unit	Contribution of O-atom/Basic Tetrahedral unit	General formula
Ortho	0	4	SiO_4^{4-}
Pyro	1	3.5	$\text{Si}_2\text{O}_7^{6-}$
Cyclic	2	3	$(\text{SiO}_3)_n^{2n-}$ (n = finite)
Simple chain (pyroxene)	2	3	$(\text{SiO}_3)_n^{2n-}$ (n = infinite)
Double chain (Amphibole)	(3, 2) avg. = 2.5	$\frac{11}{4} = \left(\frac{5.5}{2}\right)$	$(\text{Si}_4\text{O}_{11})_n^{6n-}$
2D or (Sheet)	3	2.5	$(\text{Si}_2\text{O}_5)_n^{2n-}$
3D	4	2	$(\text{SiO}_2)_n$

INERT PAIR EFFECT

In p-block elements the stability of the lower oxidation state increases on descending the group. Because increased effective nuclear charge holds ns electrons tightly due to poor shielding effect of inner d & f orbitals and thereby, restrict their (ns electrons) participation in bonding only np electrons take part in bond formation. As a result of this, +1 oxidation state of Tl is more stable than its +3 oxidation state. Pb shows +2 stable oxidation state and Bi shows +3 stable oxidation state.

For example :

Group 13

B (+3)
Al (+3)
Ga (+3), (+1)
In (+3), (+1)
Tl (+3), (+1)

Group 14

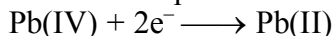
C (+4)
Si (+4)
Ge (+4), (+2)
Sn (+4), (+2)
Pb (+4), (+2)

Order of stability : $\text{Tl}^{+1} > \text{In}^{+1} > \text{Ga}^{+1}$ (due to inert pair effect)

Order of stability : $\text{Pb}^{+2} > \text{Sn}^{+2} > \text{Ge}^{+2}$ (due to inert pair effect)

Ex. PbCl_4 is stable at room temperature whereas PbI_4 doesn't exist.

Sol. Due to inert pair effect $\text{Pb}(+4)$ is less stable than $\text{Pb}(+2)$. Hence it is very good oxidant.



Reducing abilities of halides follows the sequence



MOLECULES THAT DO NOT EXIST

(1) SF_4 , SF_6 & PF_5 exist while OF_4 , OF_6 , NF_5 do not exist

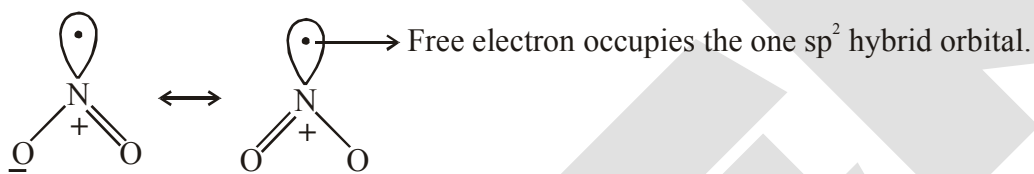
(2) (a) $\text{PI}_5(\text{vap})$ does not exist

(b) PI_5 (Solid) exist

(3) SF_6 , PF_5 , XeF_6 , XeF_4 & XeF_2 exist while SH_6 , PH_5 , XeH_6 , XeH_4 , XeH_2 do not exist

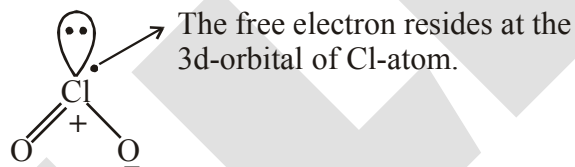
STRUCTURE OF ODD ELECTRONIC SPECIES

(1) NO_2



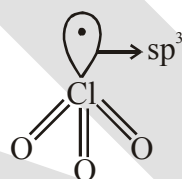
(2) ClO_2

Structure :-



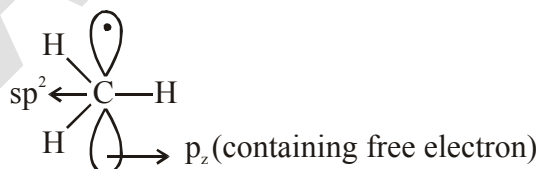
(3) ClO_3 :

Structure :



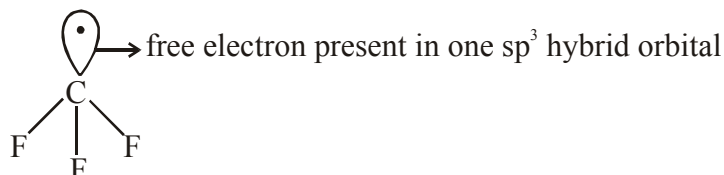
(4) $\dot{\text{C}}\text{H}_3$:

Structure :



(5) $\dot{\text{C}}\text{F}_3$:

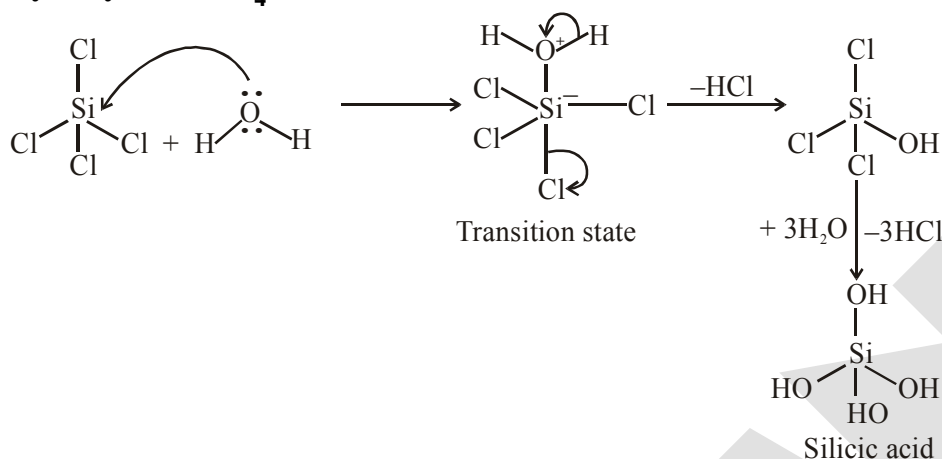
Structure :



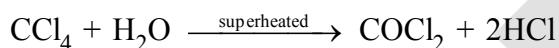
HYDROLYSIS :

In hydrolysis of covalent molecules the nucleophilic centre of molecule is replaced by OH^- group of water generally through nucleophilic substitution reaction.

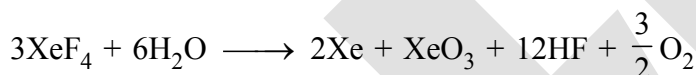
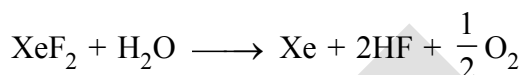
Ex. Hydrolysis of SiCl_4



Note : CCl_4 , NF_3 , is inert towards hydrolysis due to the absence of d orbital, but under drastic condition these molecules under goes hydrolysis.



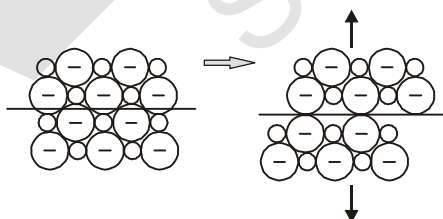
Note : Hydrolysis of XeF_2 & XeF_4 takes place through with redox reaction.



IONIC COMPOUNDS

Properties of ionic compound

(a) **Physical state :** Ionic compounds are hard, crystalline and brittle due to strong electrostatic force of attraction. Brittleness $\rightarrow$ {Same charged ions comes nearer. So they repel each other}



(b) **Isomorphism :** The phenomenon of different ionic compounds, having same crystal arrangement of ions is termed as isomorphism

Condition of Isomorphism :

- (i) Same charge on cation & anion between isomorphs
- (ii) Same radius ratio range of cation & anion between isomorphs
- (iii) Same number of water of crystallization between isomorphs

- Ex.** (i) $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ are isomorphous
 (ii) All alums are isomorphous

(c) Boiling point and melting point : Ionic compounds have high boiling point and melting point due to strong electrostatic force of attraction among oppositely charged ions.

(d) Conductivity : It depends on ionic mobility.

- (i) In solid state - No free ions - Bad conductor of electricity.
 (ii) In fused state or aqueous solution Due to free ions - Good conductor of electricity.

Conductivity order : Solid state < fused state < Aqueous solution

(e) Solubility :

Ionic compounds are more soluble in polar solvents and less soluble in non polar solvents.

- Solubility of ionic compounds in water mainly depends upon hydration energy & lattice energy.

Q. Why does the solubility of alkaline earth metal hydroxides in water increase down the group?

Ans. Among alkaline earth metal hydroxides, the anion being common the cationic radius will influence the lattice enthalpy. Since lattice enthalpy decreases much more than the hydration enthalpy with increasing ionic size, the solubility increases as we go down the group.

Q. Why does the solubility of alkaline earth metal carbonates and sulphates in water decrease down the group?

Ans. The size of anions being much larger compared to cations, the lattice enthalpy will remain almost constant within a particular group. Since the hydration enthalpies decrease down the group, solubility will decrease as found for alkaline earth metal carbonates and sulphates.

FAJAN'S RULE

Just as all the covalent bonds have some partial ionic character, the ionic bonds also have partial covalent character. The partial covalent character of ionic bonds was discussed by Fajans in terms of the following rules :

- The smaller the size of the cation and the larger the size of the anion, the greater the covalent character of an ionic bond.
- The greater the charge on the cation, the greater the covalent character of the ionic bond.
- For cations of the same size and charge, the one, with electronic configuration $(n-1)d^n ns^0$, typical of transition metals, is more polarising than the one with a noble gas configuration, $ns^2 np^6$, typical of alkali and alkaline earth metal cations.

- The cation polarises the anion, pulling the electronic charge toward itself and thereby increasing the electronic charge between the two. This is precisely what happens in a covalent bond, i.e., buildup of electron charge density between the nuclei. The polarising power of the cation, the polarisability of the anion and the extent of distortion (polarisation) of anion are the factors, which determine the per cent covalent character of the ionic bond.

⇒ Polarisation power of a cation is usually called ionic potential or charge density.

$$\text{Ionic potential } \phi (\text{phi}) = \frac{\text{Charge on cation}}{\text{Size of cation}}$$

APPLICATION OF THE CONCEPT OF POLARISATION :

- To compare the covalent and ionic character of molecule
- To compare the nature of oxide
- To compare the electrical conductivity of ionic compounds
- Tendency of the formation of complex compounds
- To compare the thermal stability of metal salts
- To compare the intensity of colour of compounds
- To compare the solubility of heavier metal halide in water.

MOLECULAR ORBITAL THEORY (MOT)

Given by Hunds & Mulliken

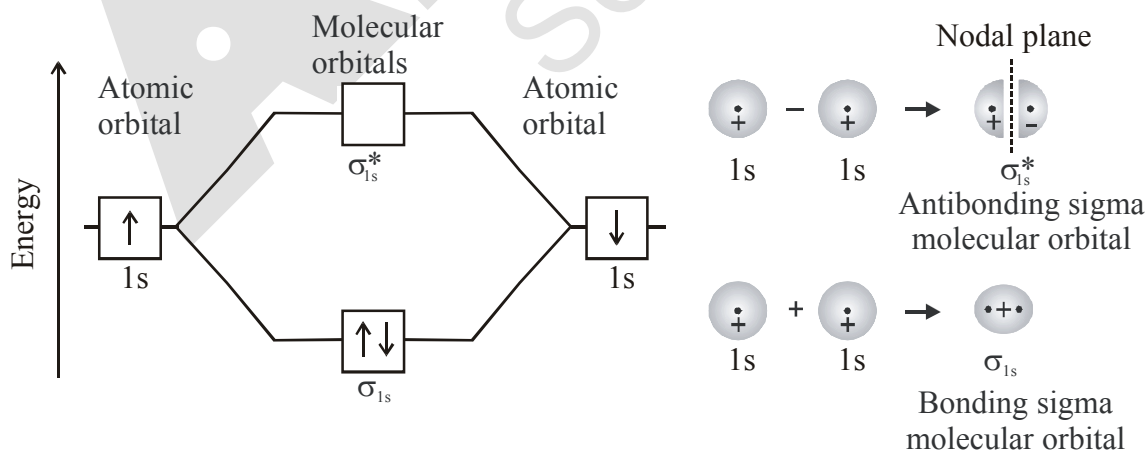
- Two atomic orbital come nearer & then overlap each other to form two molecular orbitals (MO)
- Combination of atomic orbital (AO) forms molecular orbital (MO)

Types of molecular orbitals

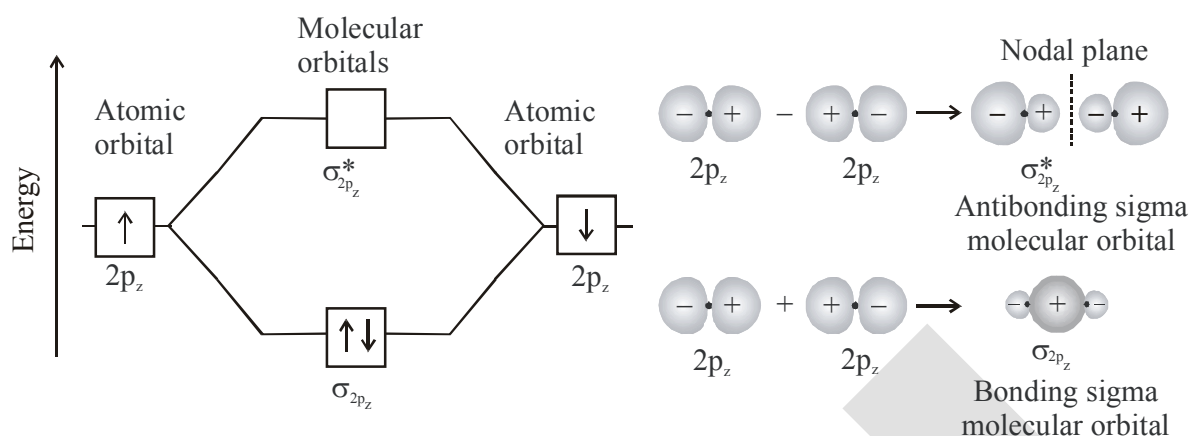
Molecular orbitals of diatomic molecules are designated as σ (sigma), π (pie), δ (delta) etc.

In this nomenclature, the **sigma (σ) molecular orbitals are symmetrical around the inter molecular axis (assumed to be z-axis) while pi (π) molecular orbitals are not symmetrical.**

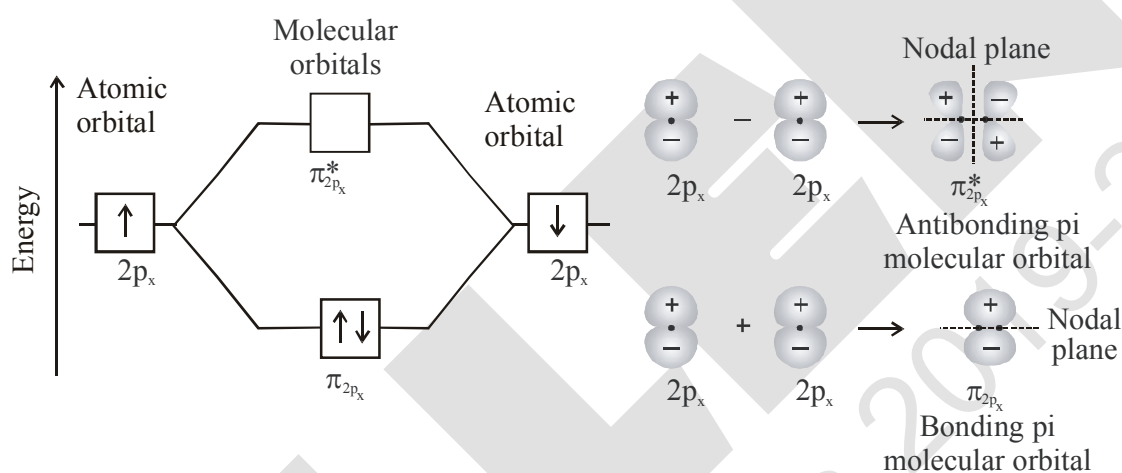
- s-s combination of orbitals



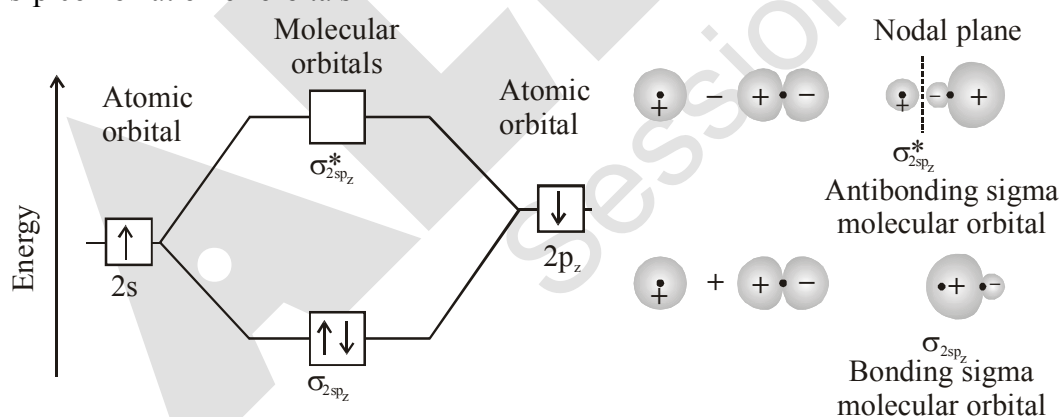
(2) p-p combination of orbitals(end to end overlap)



(3) p-p combination of orbitals (side by side overlap)



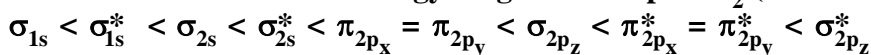
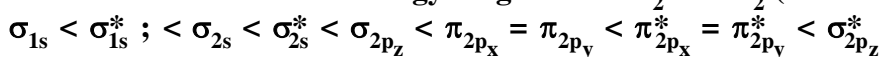
(4) s-p combination of orbitals



(c) Energy of BMO < Energy of ABMO.

(d) Molecular orbitals can be filled by electrons according to Aufbau, Hund's, Pauli's principle.

(e) Energy order of the molecular orbitals of homonuclear di-atomic molecules.

Note : Molecular orbital energy diagram for up to N₂ (molecule having ≤ 14 electrons)**Note : Molecular orbital energy diagram for O₂ and F₂ (molecule having > 14 electrons)**

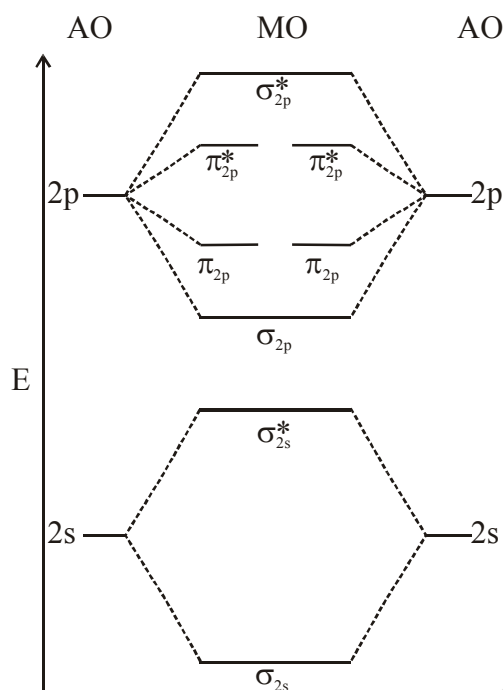
σ*, π* = antibonding molecular orbital

σ, π = bonding molecular orbital

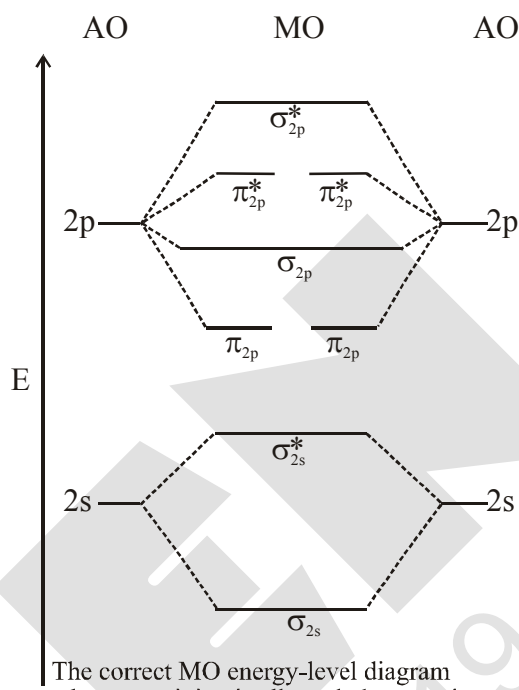
Ex. Why molecular orbitals have different order of energy in N_2 & O_2 ?

Sol. s-p mixing

Hint



The correct MO energy-level diagram when s-p mixing is not allowed.



The correct MO energy-level diagram when s-p mixing is allowed, the energies of the σ_{2p} and π_{2p} orbitals are reversed.

Bond Order

Bond order can be defined as :

$$\text{Bond order} = \frac{N_b - N_a}{2}$$

N_b = No. of electron in bonding MO's

N_a = No. of electron in antibonding MO's

- If bond order = 0, it means species does not exist.
- Bond order of 1, 2 & 3 corresponds to a single bond, double & triple bond respectively.
- Bond order $\uparrow$ stability of molecule $\uparrow$ bond length $\downarrow$

Magnetic behaviour

- If the molecule has one or more unpaired electron, it will be paramagnetic,
- If all the electrons are paired it will be diamagnetic.
- Magnetic strength can be calculated by using spin only formula of magnetic moment (μ).
- $\mu = \sqrt{n(n+2)}$ B.M. (where n = number of unpaired electron)

Ex. H_2 = Configuration : $\sigma_{(1s)}^2, \sigma_{(1s)}^{*0}$

$$\text{Bond order} = \frac{2 - 0}{2} = 1, \quad \text{Hence H - H (diamagnetic)}$$

EXERCISE # (O-1)

WEAK FORCES

- Statement-1 :** The melting point of noble gases increases as its atomic mass increases.
Statement-2 : Instantaneous dipole induced dipole attraction increases with increase in atomic mass of noble gases.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- The critical temperature of water is higher than that of O_2 because the H_2O molecule has :

(A) fewer electrons than O_2 (B) two ionic bonds
 (C) V-shape (D) dipole moment
- Which of the following boiling point order is correct -

(A) $He > T_2 > D_2$ (B) $He < T_2 < D_2$ (C) $T_2 > He > D_2$ (D) $He < D_2 < T_2$
- Which is the incorrect match for the energy distance function for following interaction -

(A) Debye force : r^{-6} (B) Ion-induced dipole interaction : r^{-2}
 (C) London force : r^{-6} (D) Keesom force : r^{-3}
- Identify the incorrect order of boiling point in the following pair.

(A) $B(OH)_3 < B(OCH_3)_3$ (B) $NF_3 < N(CH_3)_3$
 (C) $BF_3 < B(CH_3)_3$ (D) $C_2H_6 < C_2F_6$

BENT'S RULE AND DRAGO'S

- C-H bond distance is the longest in:

(A) C_2H_2 (B) C_2H_4 (C) C_2H_6 (D) $C_2H_2Br_2$
- The bond angle and hybridization in ether (CH_3OCH_3) is :

(A) $106^\circ 51'$, sp^3 (B) $104^\circ 31'$, sp^3 (C) 110° , sp^3 (D) None of these
- Which of the following has been arranged in order of decreasing bond length ?

(A) $P-O > Cl-O > S-O$ (B) $P-O > S-O > Cl-O$
 (C) $S-O > Cl-O > P-O$ (D) $Cl-O > S-O > P-O$
- Which is correct statement ?

As the s-character of a hybrid orbital decreases

(I) The bond angle decreases (II) The bond strength increases
 (III) The bond length increases (IV) Size of orbital increases

(A) (I), (III) and (IV) (B) (II), (III) and (IV)
 (C) (I) and (II) (D) All are correct
- Among the following, the correct statement is :

(A) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional
 (B) Between NH_3 and PH_3 , PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
 (C) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
 (D) Between NH_3 and PH_3 , PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

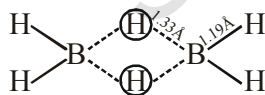
11. In which of the following molecule C—C bond length will be highest ?
 (A) $\text{CF}_3 - \text{CF}_3$ (B) $\text{F}_2\text{CH} - \text{CHF}_2$ (C) $\text{FCH}_2 - \text{CH}_2\text{F}$ (D) $\text{CH}_3 - \text{CF}_3$
12. In BClBrI molecule the maximum % s-character provided from the central atom is in bond :
 (A) B—I (B) B—Cl (C) B—Br (D) Can not predict
13. Find out the % p-character in the orbital occupied by lone pairs in H_2O .
 $[\widehat{\text{HOH}} = 104.5^\circ \text{ and } \cos(104.5) = -0.25]$
 (A) 80 % (B) 20 % (C) 70 % (D) 75 %
14. Which of the following order is correct for increasing p-character in orbital used for bonding by central atom
 (A) $\text{SiH}_4 > \text{CH}_4$ (B) $\text{H}_2\text{S} > \text{H}_2\text{O}$ (C) $\text{PH}_4^+ > \text{PH}_3$ (D) $\text{NH}_3 > \text{PH}_3$

BACK BONDING

15. Boron forms BX_3 type of halides. The correct increasing order of Lewis-acid strength of these halides is
 (A) $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3 > \text{BI}_3$ (B) $\text{BI}_3 > \text{BBr}_3 > \text{BCl}_3 > \text{BF}_3$
 (C) $\text{BF}_3 > \text{BI}_3 > \text{BCl}_3 > \text{BBr}_3$ (D) $\text{BF}_3 > \text{BCl}_3 > \text{BI}_3 > \text{BBr}_3$
16. Select species which is planar at nitrogen :
 (A) $(\text{CH}_3)_3\text{N}$ (B) $(\text{SiH}_3)_3\text{N}$ (C) NF_3 (D) NH_3
17. Type of back bonding in $(\text{SiH}_3)_2\text{O}$ is :
 (A) $p\pi - d\pi$ (B) $p\pi - p\pi$ (C) $d\pi - d\pi$ (D) None of these

MULTICENTERED BOND

18. The type of overlap in the bridge bond existing in $\text{Al}_2(\text{CH}_3)_6$ is :-
 (A) $sp^3-sp^3d-sp^3$ (B) $sp^3-sp^2-sp^3$ (C) sp^3-s-sp^3 (D) $sp^3-sp^3-sp^3$
19. Which one of the following statement is not true regarding diborane?
 (A) It has two bridging hydrogens and four perpendicular to the rest.
 (B) When methylated, the product is $\text{Me}_4\text{B}_2\text{H}_2$
 (C) The bridging hydrogens are in a plane perpendicular to the rest.
 (D) All the B—H bond distances are equal.
20. The molecular shape of diborane, is shown:



Consider the following statements for diborane :

- (1) Boron is approximately sp^3 hybridised
 (2) B—H—B angle is 180°
 (3) There are two terminal B—H bonds for each boron atom
 (4) There are only 12 bonding electrons available

Of these statements:

- (A) 1, 3 and 4 are correct (B) 1, 2 and 3 are correct
 (C) 2, 3 and 4 are correct (D) 1, 2 and 4 are correct

SILICATE

21. The number of corners or O-atoms shared per tetrahedron for pyroxene chain silicate is -
 (A) 3 (B) 2 (C) 2.5 (D) 1
22. The mineral $\text{Na}_2\text{Fe}_3^{\text{II}}\text{Fe}_2^{\text{III}}[\text{Si}_8\text{O}_{22}](\text{OH})_2$ (chrocidolite) is a :
 (A) Pyroxene chain silicate (B) Sheet silicate
 (C) Amphiboles chain silicate (D) 3D-silicate
23. The silicate anion in the mineral kunoite is a chain of three SiO_4 tetrahedral those share corners with adjacent tetrahedral. The mineral also contains Ca^{2+} ions, Cu^{2+} ions, and water molecules in a 1:1:1 ratio mineral is represented as :
 (A) $\text{CaCuSi}_3\text{O}_{10} \cdot \text{H}_2\text{O}$ (B) $\text{CaCuSi}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$
 (C) $\text{Ca}_2\text{Cu}_2\text{Si}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$ (D) none of these

ODD ELECTRON SPECIES

24. Hybridisation related to NO_2 molecule is -
 (A) sp^3 (B) sp
 (C) sp^3d (D) sp^2
25. In which of the following processes, the magnetic behaviour of the species is changed :-
 (A) $2\dot{\text{C}}\text{H}_3 \longrightarrow \text{C}_2\text{H}_6$ (B) $2\text{NO}_2 \longrightarrow \text{N}_2\text{O}_4$
 (C) $2\text{ClO}_3 \longrightarrow \text{Cl}_2\text{O}_6$ (D) All of these

HYDROLYSIS

26. Which of the following statement is correct ?
 (A) BCl_3 is not hydrolysed while SiCl_4 can be hydrolysed
 (B) CCl_4 is hydrolysed under ordinary condition
 (C) XeF_2 produces $\text{Xe}(\text{OH})_2$ on hydrolysis
 (D) hydrolysis of XeF_2 is a redox reaction
27. **Statement-1 :** Between SiCl_4 and CCl_4 only SiCl_4 reacts with water at room temperature.
Statement-2 : SiCl_4 is ionic and CCl_4 is covalent.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

28. **Statement-1** : PCl_3 on hydrolysis gives $\text{OH}-\overset{\text{O}}{\underset{\text{H}}{\text{P}}}-\text{OH}$ and $\text{OH}-\overset{\text{OH}}{\text{P}}-\text{OH}$

Statement-2 : H_3PO_3 exists in two tautomeric forms : $\text{OH}-\overset{\text{OH}}{\text{P}}-\text{OH} \rightleftharpoons \text{OH}-\overset{\text{H}}{\underset{\text{O}}{\text{P}}}-\text{OH}$

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

MOLECULE DOES NOT EXIST

29. Which of the following molecule does not exist.

(A) PH_5 (B) NCl_3 (C) NOF_3 (D) XeF_5^-

30. Select non existing specie

(A) PH_3 (B) PH_4^+
 (C) $[\text{PF}_6]^-$ (D) None of these

INERT PAIR EFFECT

31. **Statement-1** : Boron does not show univalent nature but unipositive nature of thallium is quite stable.

Statement-2 : Inert pair effect predominates in thallium.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

32. **Statement-1** : PbI_4 doesn't exist and converts into PbI_2 and I_2 spontaneously at room temperature but PbCl_4 needs heating to convert into PbCl_2 and Cl_2 .

Statement-2 : Pb^{2+} is more stable than Pb^{4+} due to inert pair effect.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

IONIC COMPOUND

33. The dissolution of ionic compounds involves :
 (A) Evolution of heat (B) Weakening of attractive forces
 (C) Dissociation into ions (D) All of these
34. Select correct order out of given options :
 (A) $\text{BeCO}_3 < \text{BaCO}_3$: Covalent character (B) $\text{BeO} > \text{SrO}$: lattice energy
 (C) $\text{Be}^{2+} < \text{Li}^+$: Hydration energy (D) $\text{Be}^{2+}(\text{aq.}) > \text{Li}^+(\text{aq.})$: Ionic mobility
35. The polarizability of the following ions is/are in the order of
 (A) $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$ (B) $\text{I}^- > \text{Br}^- > \text{F}^- > \text{Cl}^-$
 (C) $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$ (D) $\text{I}^- < \text{Br}^- < \text{F}^- < \text{Cl}^-$
36. Which of the following equilibria would have the highest value of K_p at a common temperature?
 (A) $\text{MgCO}_3 \rightleftharpoons \text{MgO} + \text{CO}_2$ (B) $\text{CaCO}_3 \rightleftharpoons \text{CaO} + \text{CO}_2$
 (C) $\text{SrCO}_3 \rightleftharpoons \text{SrO} + \text{CO}_2$ (D) $\text{BaCO}_3 \rightleftharpoons \text{BaO} + \text{CO}_2$
37. Which of the following set of characteristics lead to the increase in solubility of ionic substances?
 (A) High dipole moment, strong attraction by an ion towards solvent and large solvation energy
 (B) Low dipole moment, weak attraction by an ion towards solvent and high solvation energy
 (C) High dipole moment, strong attraction by an ion towards solvent and low solvation energy
 (D) High dipole moment, weak attraction by an ion towards solvent and large solvation energy
38. The solubility of anhydrous AlCl_3 and hydrated AlCl_3 in diethyl ether are S_1 and S_2 respectively. Then
 (A) $S_1 = S_2$ (B) $S_1 > S_2$ (C) $S_1 < S_2$ (D) $S_1 < S_2$ but not $S_1 = S_2$
39. **Statement-1** : Among alkali metal cations, $\text{Li}^+(\text{aq.})$ has highest electrical conductance.
Statement-2 : $\text{Li}^+(\text{aq.})$ is largest alkali metal cation because of greater degree of hydration.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
40. **Statement-1** : $\text{Al}(\text{OH})_3$ is amphoteric in nature.
Statement-2 : Al-O and O-H bonds can be broken with equal ease in $\text{Al}(\text{OH})_3$.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

MOLECULAR ORBITAL THEORY

41. The bond energy order of He_2^+ and HeH^+ is
 (A) $\text{He}_2^+ > \text{HeH}^+$ (B) $\text{HeH}^+ = \text{He}_2^+$ (C) $\text{HeH}^+ > \text{He}_2^+$ (D) Can't be predicted
42. Among KO_2 , AlO_2^- , BaO_2 and NO_2^+ unpaired electron is present in :
 (A) KO_2 only (B) NO_2^+ and BaO_2 (C) KO_2 and AlO_2^- (D) BaO_2 only
43. During the formation of a molecular orbital from atomic orbitals, probability of electron density is
 (A) minimum in the nodal plane (B) maximum in the nodal plane
 (C) zero in the nodal plane (D) zero on the surface of the lobe

44. Pick out the **incorrect** statement ?
 (A) N_2 has greater dissociation energy than N_2^+
 (B) O_2 has lower dissociation energy than O_2^+
 (C) Bond length in N_2^+ is less than N_2
 (D) Bond length in NO^+ is less than in NO
45. A simplified application of MO theory to the hypothetical molecule 'OF' would give its bond order as :
 (A) 2 (B) 1.5 (C) 1.0 (D) 0.5
46. Which of the following is true ?
 (A) With increasing Bond order, Bond length decreases & Bond energy increases
 (B) With increasing Bond order, Bond length increases & Bond energy decreases
 (C) With increasing Bond order, Bond length decreases & Bond energy decreases
 (D) With increasing Bond order, Bond length increases & Bond energy increases
47. Which of the following has fractional bond order :
 (A) O_2^{2+} (B) O_2^{2-} (C) F_2^{2-} (D) H_2^-
48. **Statement-1:** H_2 molecule is more stable than He-H molecule.
Statement-2 : The antibonding electron in He-H molecule decreases the bond order and there by the stability.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
49. **Statement-1 :** Super oxide ion is paramagnetic whereas peroxide ion is diamagnetic.
Statement-2 : Super oxide ion has one unpaired electron whereas peroxide ion has no unpaired electron.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
50. **Statement-1:** π_b and π^* orbitals obtained from 2p orbital are lying in the same plane.
Statement-2 : Bonding M.O's are formed by constructive interference while antibonding M.O's are formed by destructive interference.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

EXERCISE # (O-2)

WEAK FORCES

- Which of the following factors are responsible for origination of vander Waals forces ?
 (A) Instantaneous dipole-induced dipole interaction
 (B) Dipole-induced dipole interaction
 (C) Dipole-dipole interaction
 (D) Size of molecule
- Which of the following are true ?
 (A) Vander Waals forces are responsible for the formation of molecular crystals
 (B) Branching lowers the boiling points of isomeric organic compounds due to decrease in Vander Waals forces of attraction
 (C) In graphite, vander Waals forces act between the carbon layers
 (D) In diamond, vander Waals forces act between the carbon layers

BENT'S AND DRAGO'S RULE

- Select the correct statement for following molecules :
 (I) $\text{PF}_2(\text{CH}_3)_3$; (II) $\text{PF}_2(\text{CF}_3)_3$
 (A) Both have trigonal bipyramidal structure with respect to P.
 (B) P-F bond length is longer in $\text{PF}_2(\text{CH}_3)_3$ than in $\text{PF}_2(\text{CF}_3)_3$
 (C) F-atoms occupy axial position in both
 (D) P-F bond length is lower in $\text{PF}_2(\text{CH}_3)_3$ than in $\text{PF}_2(\text{CF}_3)_3$

BACK BONDING

- Molecules in which bond angle is changed due to back bonding w.r.t B/O/N.
 (A) H_3BO_3 (B) $\text{B}(\text{OMe})_3$ (C) BF_3 (D) $\text{N}(\text{SiH}_3)_3$
- $3d\pi-2p\pi$ type back bonding is observed in :
 (A) $\text{N}(\text{SiH}_3)_3$ (B) CCl_3 (C) $\text{S}(\text{CH}_3)_2$ (D) BCl_3

MULTICENTERED BOND

- No X-X bond exists in which of the following compounds having general form of X_2H_6 ?
 (A) B_2H_6 (B) C_2H_6 (C) Al_2H_6 (D) Si_2H_6
- Three centre - two electron bonds exist in :
 (A) B_2H_6 (B) $\text{Al}_2(\text{CH}_3)_6$ (C) $\text{BeH}_2(\text{s})$ (D) $\text{BeCl}_2(\text{s})$
- Select correct statement about B_2H_6
 (A) Bridging groups are electron-deficient with 12 valence electrons
 (B) It has $2c - 2e$ B-H bonds
 (C) It has $3c - 2e$ B-H-B bonds
 (D) It has $3c - 4e$ B-H-B bonds

SILICATE

- In which of the following cases the number of corner shared per tetrahedron is '2' -
 (A) Pyroxene chain silicate (B) Amphibole chain silicate
 (C) 5-membered cyclic silicate (D) None of these

ODD ELECTRON SPECIES

10. Select correct about NO_2 :
 (A) It is odd electron specie (B) N–O bond order = 1.5
 (C) Paramagnetic specie (D) Isoelectronic with CO_2
11. The number of specie(s) which are not perfectly planar.
 (A) $\dot{\text{C}}\text{H}_3$ (B) $\dot{\text{C}}\text{F}_3$ (C) $\dot{\text{C}}\text{HF}_2$ (D) $\dot{\text{C}}\text{H}_2\text{F}$
12. Which of the following statement is **CORRECT** :-
 (A) The free electron of ClO_3 molecule is present in d-orbital of Cl-atom
 (B) The free electron of $\dot{\text{C}}\text{F}_3$ is present in sp^3 hybrid orbital
 (C) NO is polar
 (D) The free electron of ClO_2 molecule is present in d-orbital of Cl-atom
13. Which of the following statement(s) is / are **INCORRECT** for
 $\dot{\text{C}}\text{H}_3 = \text{X}$ and $\dot{\text{C}}\text{F}_3 = \text{Y}$
 (A) When X dimerises bond angle decreases
 (B) When X dimerises bond angle increases
 (C) In X–Y molecule C–C bond length less than that in Y–Y molecule
 (D) Bond angle in Y is less than X

HYDROLYSIS

14. Which of the following halide(s) cannot be hydrolysed at room temperature
 (A) SeF_6 (B) SF_6 (C) CCl_4 (D) NF_3
15. Which of the following statement is/are **INCORRECT** ?
 (A) P_4S_{10} gives rise to H_2S gas on hydrolysis
 (B) PCl_5 produces POCl_3 on partial hydrolysis
 (C) H_2SO_5 gives rise to H_2SO_3 on hydrolysis
 (D) d-orbital participates in the hydrolysis of SF_6 at room temperature

MOLECULE DOES NOT EXIST

16. Which of the following do/does not exist ?
 (A) SH_6 (B) HFO_4 (C) FeI_3 (D) HClO_3
17. Which of the following molecule(s) exist-
 (A) SF_6 (B) PH_5 (C) PH_3 (D) PCl_5

INERT PAIR EFFECT

18. Which of the following have $(18 + 2)$ electron configuration ?
 (A) Pb^{2+} (B) Cd^{2+} (C) Bi^{3+} (D) SO_4^{2-}

19. Which of following stability order is/are correct due to inert pair effect.

- (A) $\text{Hg} > \text{Hg}^{2+}$ (B) $\text{Bi}^{3+} < \text{Bi}^{5+}$ (C) $\text{Pb}^{2+} > \text{Pb}^{4+}$ (D) $\text{Fe}^{2+} < \text{Fe}^{3+}$

IONIC COMPOUND

20. Choose the correct order(s) for the given properties.

- (A) $\text{MgSO}_4 < \text{SrSO}_4 < \text{BaSO}_4$: Thermal stability order
 (B) $\text{BeC}_2\text{O}_4 < \text{CaC}_2\text{O}_4 < \text{BaC}_2\text{O}_4$: Solubility in water
 (C) $\text{LiCl} > \text{NaCl} > \text{KCl}$: Melting point order
 (D) $\text{BeF}_2 > \text{CaF}_2 > \text{SrF}_2$: Covalent character order

21. Polarization may be called the distortion of the shape of an anion by an adjacently placed cation. Which of the following statements is/are incorrect :

- (A) Minimum polarization is brought about by a cation of low radius
 (B) A large cation is likely to bring about a large degree of polarization
 (C) Maximum polarization is brought about by a cation of high charge
 (D) A small anion is likely to undergo a large degree of polarization

22. Most ionic compounds have :

- (A) high melting points and low boiling points
 (B) high melting points and nondirectional bonds
 (C) high solubilities in polar solvents and low solubilities in nonpolar solvents
 (D) three-dimensional network structures, and are good conductors of electricity in the molten state

23. Choose the correct order for the given properties.

- (A) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$: covalent character order.
 (B) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$: melting point order
 (C) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$: lattice energy order
 (D) $\text{NaF} > \text{MgF}_2 > \text{AlF}_3$: order of polarising power of cation.

MOLECULAR ORBITAL THEORY

24. Which of the following have identical bond order ?

- (A) O_2^{2+} (B) NO^+ (C) CN^- (D) CN^+

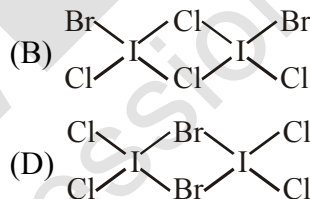
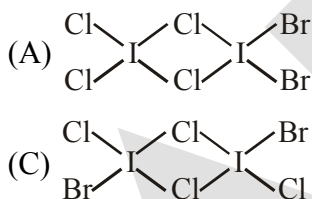
25. Which of the following statement is/are correct

- (A) The peroxide ion has a bond order of 1 while the oxygen molecule has a bond order of 2
 (B) The peroxide ion has a weaker bond than the dioxygen molecule has.
 (C) The peroxide ion as well as the dioxygen molecules are paramagnetic
 (D) The bond length of the peroxide ion is greater than that of the dioxygen molecule

26. Which of the following statements are true for these given species : N_2 , CO , CN^- and NO^+ .
- (A) All species are paramagnetic (B) The species are isoelectronic
(C) All the species have dipole moment (D) All the species are linear
27. Which of the following have unpaired electron(s)
- (A) O_2^+ (B) O_2^- (C) NO (D) H_2^+
28. Which of the following is/are paramagnetic ?
- (A) B_2 (B) O_2 (C) N_2 (D) He_2
29. Which of the following species have a bond order of 3 ?
- (A) CO (B) CN^- (C) NO^+ (D) O_2^+
30. Which of the following is/are correct ?
- (A) During N_2^+ formation, one electron is removed from the bonding molecular orbitals
(B) During O_2^+ formation, one electron is removed from the antibonding molecular orbitals
(C) During O_2^- formation, one electron is added to the bonding molecular orbitals
(D) During CN^- formation, one electron is added to the bonding molecular orbitals

MISCELLANEOUS

31. Rotation around the bond (between the underlined atoms) is restricted in :
- (A) $\underline{C}_2H_4$ (B) $H_2\underline{O}_2$ (C) $\underline{C}_2H_2$ (D) $\underline{C}_2H_6$
32. The experimental result says that two iodine atoms are in different environments and predict the all possible incorrect arrangement for $I_2Cl_4Br_2$.



EXERCISE # (S-1)

- The number of corner of O-atom shared per tetrahedron in 2D-silicate is
- Find the number of angles less than 120° in PH_2F_3 .
- Find the number of molecules which are not having 3c-2e bond from the following.
 $\text{Al}_2(\text{CH}_3)_6$, Si_2H_6 , B_2H_6 , C_2H_6 , Si_2Cl_6 , Al_2Cl_6
- Find the ratio of π -electrons in the C_2 -molecule with that of B_2 molecule according to M.O.T.
- In which of the following silicates structure, the number of corner/oxygen atoms shared per tetrahedron is '2'.
 Pyrosilicate, pyroxene chain silicate,
 2D-silicate, 3D-silicate, 4-membered cyclic silicate
- The total number of bonding and antibonding electrons in O_2^+ are "....." and "....." respectively.
[If the answer is 14 and 7, then represent as 147]
- Find the total number of following molecule(s) which have all bond lengths are same.
 XeF_4 , SF_4 , SH_2 , NO_3^- , SiF_4 , ClF_3 , PF_2Cl_3 , XeO_3F_2
- Among the following total number of planar molecules is / are
 Cl_2O , $\text{P}(\text{CH}_3)_3$, $\text{N}(\text{CH}_3)_3$, ClO_2 , $\cdot\text{CH}_3$, NCl_3
- Calculate the value of "n" in $\text{Zn}_n\text{Ca}_2(\text{Si}_3\text{O}_{10}) \cdot 2\text{H}_2\text{O}$
- How many compound(s) gives diprotic acid on hydrolysis?
 SO_2Cl_2 , SOCl_2 , POCl_3 , XeF_2 , N_2O_5 , P_4O_6 , SF_4 , BCl_3

EXERCISE # (S-2)

Comprehension # 1 (1 to 3 Questions)

B is the first element of IIIrd group. It forms a number of electron deficient halides and hydrides. Among the hydrides diborane is an important compound.

- Which of the following halide is the strongest Lewis acid?
(A) BF_3 (B) BCl_3 (C) BBr_3 (D) BI_3
- Which of the following compounds has $2p\pi - 2p\pi$ bond?
(A) BF_3 (B) BCl_3 (C) BBr_3 (D) BI_3
- In B_2H_6 number of $3c - 2e$ bonds is/are
(A) 1 (B) 2 (C) 3 (D) None

Comprehension # 2 (4 to 6 Questions)

Molecular orbital theory is based on linear combination of atomic orbitals (LCAO). According to LCAO when respective atomic orbitals of the atoms interact, they undergo constructive and destructive interference giving two types of molecular orbital i.e. bonding and antibonding molecular orbitals respectively.

- Which of the following species is paramagnetic?
(A) NO^- (B) O_2^{2-} (C) CN^- (D) CO
- Bond order of Be_2 is :
(A) 1 (B) 2 (C) 3 (D) 0
- Number of anti bonding electrons in N_2 is :
(A) 4 (B) 10 (C) 12 (D) 14

Comprehension # 3 (7 to 8 Questions)

Polarisation of anion in ionic compounds play an important role to influence the various physical and chemical properties of ionic compound.

- Amongst LiCl , RbCl , BeCl_2 and MgCl_2 the compounds with the greatest and the least ionic character, respectively are :
(A) LiCl and RbCl (B) RbCl and BeCl_2 (C) RbCl and MgCl_2 (D) MgCl_2 and BeCl_2
- Compound with maximum ionic character is formed from :
(A) Na and Cl (B) Cs and F (C) Cs and I (D) Na and F

Comprehension # 4 (9 to 10 Questions)

"Hydrolysis is defined as the reaction of water with any substance"

- Choose the correct order of ease of hydrolysis -
(A) $\text{MgCl}_2 > \text{AlCl}_3$ (B) $\text{SF}_6 < \text{SeF}_6$ (C) $\text{SnCl}_2 > \text{SnCl}_4$ (D) None
- Which of the following oxyacids are formed during the stepwise hydrolysis of P_4O_{10}
(A) tetrametaphosphoric acid (B) tetrapolyphosphoric acid
(C) pyrophosphoric acid (D) All of these

MATCHING LIST

11. Match list I with list II and select the correct answer:

List I (species)				List II (O-N-O angle)			
(P)	NO_2^+			(1)	180°		
(Q)	NO_2			(2)	134°		
(R)	NO_2^-			(3)	120°		
(S)	NO_3^-			(4)	115°		
				(5)	109°		

	P	Q	R	S		P	Q	R	S
(A)	5	4	3	2	(B)	5	2	4	3
(C)	1	2	4	3	(D)	1	4	3	2

12. Match list I with list II and select the correct answer:

List-I (Molecule / Species)				List-II (Unpaired electron resides in)			
(P)	NO_2			(1)	d-orbital		
(Q)	ClO_2			(2)	sp^2 -orbital		
(R)	ClO_3			(3)	sp^3 -orbital		
(S)	$\cdot\text{CH}_3$			(4)	p-orbital		

Code :

	P	Q	R	S		P	Q	R	S
(A)	2	4	1	3	(B)	2	1	3	4
(C)	1	4	2	3	(D)	3	1	2	4

13. Match list I with list II and select the correct answer:

List-I (Process)				List-II (Operating interaction involved)			
(P)	Clathrate compound of Xe in ice			(1)	Ion - Dipole		
(Q)	Liquation of Xe gas			(2)	Dipole - Dipole		
(R)	Liquation of HCl gas			(3)	Dipole - Induced dipole		
(S)	Hydration of Na^+			(4)	London forces		

Code :

	P	Q	R	S		P	Q	R	S
(A)	1	3	4	2	(B)	3	4	1	2
(C)	3	4	2	1	(D)	1	3	2	4

MATRIX MATCH

14. Match the column

Column-I (Compounds given)	Column-II (Characteristics associated with given compounds)
(A) BeCl_2	(P) Undergoes partial hydrolysis
(B) SiF_4	(Q) All possible bond angles are identical
(C) SO_2Cl_2	(R) Hydrolysed product of the attacking site is electron deficient and finally produces polymerised product
(D) BF_3	(S) Maximum number of atoms present in one plane is three

15. Match the column

Column-I
(Molecules)

- (A) CH_4
(B) CH_2F_2
(C) CHF_3
(D) CF_4

Column-I

(Characteristics given among the molecules of column-I)

- (P) Molecule is having perfect tetrahedral shape
(Q) C-F bond has maximum p-character
(R) C-H bond has maximum s-character
(S) Molecule is having maximum number of equal angles
(T) Molecule has lowest bond angle

16. Match the column

Column-I
Processes

- (A) $\text{N}_2^+ \longrightarrow \text{N}_2$
(B) $\text{Zn}^{2+} \longrightarrow \text{Zn}$
(C) $\text{O}_2^{2-} \longrightarrow 2\text{O}^{2-}$
(D) $\text{C}_2^{2-} \longrightarrow \text{C}_2$

Column-II

Correct characteristics

- (P) Magnetic moment gets changed
(Q) The process is associated with two electronic change
(R) Magnetic behaviour gets changed
(S) Electron(s) associated in the process enter(s) into π_{2p}^* orbital
(T) Electron(s) associated in the process involve(s) σ_{2p} orbital

Answer Q.17 to Q.19 by appropriately matching the information given in the three columns of the following table

Molecular Orbital	Number of nodal plane	Symmetry of Molecular Orbital
(1) σ_s	(P) 3	(I) BMO, gerade
(2) π_p	(Q) 2	(II) ABMO, ungerade
(3) $\sigma_{p_z}^*$	(R) 1	(III) BMO, ungerade
(4) δ^*	(S) 0	(IV) ABMO, gerade

17. Which of the following matching is **INCORRECT**.

- (A) (1), S, I (B) (2), R, I (C) (3), R, II (D) (4), P, II

18. How many orbitals are occupied with set (1)(S)(I) in F_2

- (A) 0 (B) 1 (C) 2 (D) 3

19. If z axis is the molecular axis then **CORRECT** matching for $d_{xy} + d_{xy}$ orbital is 4, P, II for the subtraction of wave function.

Which of the following matching is **CORRECT** for the addition of wave function for same combination of orbitals.

- (A) P, IV (B) P, III (C) Q, II (D) Q, I

EXERCISE # J-MAIN

- The bond order in NO is 2.5 while that in NO^+ is 3. Which of the following statement is true for these two species ? [AIEEE-2004]
 (1) Bond length in NO^+ is equal to that NO (2) Bond length in NO is greater than NO^+
 (3) Bond length in NO^+ is greater than NO (4) Bond length is unpredictable
- The states of hybridization of boron and oxygen atoms in boric acid (H_3BO_3) are respectively [AIEEE-2004]
 (1) sp^3 and sp^2 (2) sp^2 and sp^3 (3) sp^2 and sp^2 (4) sp^3 and sp^3
- The maximum number of 90° angles between bond pair-bond pair of electrons is observed in :- [AIEEE-2004]
 (1) dsp^2 hybridization (2) sp^3d hybridization
 (3) dsp^3 hybridization (4) sp^3d^2 hybridization
- Which one of the following specie is diamagnetic in nature ? [AIEEE-2005]
 (1) He_2^+ (2) H_2 (3) H_2^+ (4) H_2^-
- Which of the following molecule/ion does not contain unpaired electrons? [AIEEE-2006]
 (1) N_2^+ (2) O_2 (3) O_2^{2-} (4) B_2
- Among the following mixtures, dipole-dipole as the major interaction, is present in [AIEEE-2006]
 (1) KCl and water (2) benzene and carbon tetrachloride
 (3) benzene and ethanol (4) acetonitrile and acetone
- A metal, M forms chlorides in its +2 and +4 oxidation states. Which of the following statement about these chlorides is correct ? [AIEEE-2006]
 (1) MCl_2 is more ionic than MCl_4
 (2) MCl_2 is more easily hydrolysed than MCl_4
 (3) MCl_2 is more volatile than MCl_4
 (4) MCl_2 is more soluble in anhydrous ethanol than MCl_4
- The decreasing values of bond angles from NH_3 (106°) to SbH_3 (91°) down group-15 of the periodic table is due to [AIEEE-2006]
 (1) decreasing lp – bp repulsion (2) increasing electronegativity
 (3) increasing bp – bp repulsion (4) increasing p-orbital character in sp^3
- In which of the following ionization process, the bond order has increased and the magnetic behaviour has changed [AIEEE-2007]
 (1) $\text{NO} \rightarrow \text{NO}^+$ (2) $\text{O}_2 \rightarrow \text{O}_2^+$ (3) $\text{N}_2 \rightarrow \text{N}_2^+$ (4) $\text{C}_2 \rightarrow \text{C}_2^+$
- Which of the following species exhibits the diamagnetic behaviour [AIEEE-2007]
 (1) $+$ (2) O_2 (3) NO (4) $-$
- which one of the following pairs of species have the same bond order? [AIEEE-2008]
 (1) CN^- and NO^+ (2) CN^- and CN^+ (3) O_2^- and CN^- (4) NO^+ and CN^+

12. The bond dissociation energy of B–F in BF_3 is 646 kJ mol^{-1} whereas that of C–F in CF_4 is 515 kJ mol^{-1} . The correct reason for higher B–F bond dissociation energy as compared to that of C–F is :- [AIEEE-2009]
- Significant $p\pi - p\pi$ interaction between B and F in BF_3 whereas there is not possibility of such interaction between C and F in CF_4 .
 - Lower degree of $p\pi - p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4
 - Smaller size of B-atom as compared to that of C-atom
 - Stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4
13. Using MO theory predict which of the following species has the shortest bond length ? [AIEEE-2009]
- O_2^-
 - O_2^{2-}
 - O_2^{2+}
 - O_2^+
14. Among the following the maximum covalent character is shown by the compound :- [AIEEE-2011]
- AlCl_3
 - MgCl_2
 - FeCl_2
 - SnCl_2
15. Which one of the following molecules is expected to exhibit diamagnetic behaviour ?
- C_2
 - N_2
 - O_2
 - S_2 [AIEEE-2013]
16. In which of the following pairs of molecules/ions, both the species are not likely to exist ?
- $\text{H}_2^+, \text{He}_2^{2-}$
 - $\text{H}_2^-, \text{He}_2^{2-}$
 - $\text{H}_2^{2+}, \text{He}_2$
 - $\text{H}_2^-, \text{He}_2^{2+}$ [JEE-M-2013]
17. Stability of the species Li_2 , Li_2^- and Li_2^+ increases in the order of :- [JEE-M-2013]
- $\text{Li}_2 < \text{Li}_2^+ < \text{Li}_2^-$
 - $\text{Li}_2^- < \text{Li}_2^+ < \text{Li}_2$
 - $\text{Li}_2 < \text{Li}_2^- < \text{Li}_2^+$
 - $\text{Li}_2^- < \text{Li}_2 < \text{Li}_2^+$
18. Which one of the following properties is **not** shown by NO ? [JEE-M-2014]
- It combines with oxygen to form nitrogen dioxide
 - It's bond order is 2.5
 - It is diamagnetic in gaseous state
 - It is a neutral oxide
19. The correct order of thermal stability of hydroxides is : [JEE-M-2015 (on line)]
- $\text{Ba}(\text{OH})_2 < \text{Sr}(\text{OH})_2 < \text{Ca}(\text{OH})_2 < \text{Mg}(\text{OH})_2$
 - $\text{Mg}(\text{OH})_2 < \text{Sr}(\text{OH})_2 < \text{Ca}(\text{OH})_2 < \text{Ba}(\text{OH})_2$
 - $\text{Mg}(\text{OH})_2 < \text{Ca}(\text{OH})_2 < \text{Sr}(\text{OH})_2 < \text{Ba}(\text{OH})_2$
 - $\text{Ba}(\text{OH})_2 < \text{Ca}(\text{OH})_2 < \text{Sr}(\text{OH})_2 < \text{Mg}(\text{OH})_2$
20. Which of the alkaline earth metal halides given below is essentially covalent in nature :- [JEE-M-2015 (on line)]
- SrCl_2
 - CaCl_2
 - BeCl_2
 - MgCl_2
21. Which one of the following alkaline earth metal sulphates has its hydration enthalpy greater than its lattice enthalpy ? [JEE-M-2015]
- BaSO_4
 - SrSO_4
 - CaSO_4
 - BeSO_4

22. The intermolecular interaction that is dependent on the inverse cube of distance between the molecules is :- [JEE-M-2015]
 (1) London force (2) Hydrogen bond
 (3) ion-ion interaction (4) ion-dipole interaction
23. Which one has the highest boiling point ? [JEE-M-2015]
 (1) Kr (2) Xe (3) He (4) Ne
24. Which intermolecular force is most responsible in allowing xenon gas to liquefy? [JEE (MAIN) ONLINE 2016]
 (1) Ionic
 (2) Instantaneous dipole- induced dipole
 (3) Dipole - dipole
 (4) Ion - dipole
25. The bond angle H-X-H is the greatest in the compound : [JEE (MAIN) ONLINE 2016]
 (1) NH_3 (2) H_2O (3) PH_3 (4) CH_4
26. Which of the following species is not paramagnetic :- [JEE-MAINS-2017]
 (1) NO (2) CO (3) O_2 (4) B_2
27. Which of the following is paramagnetic ? [JEE-MAINS-2017 (On-line)]
 (1) CO (2) O_2^{2-} (3) NO^+ (4) B_2
28. sp^3d^2 hybridization is not displayed by : [JEE-MAINS-2017 (On-line)]
 (1) $[\text{CrF}_6]^{3-}$ (2) BrF_5 (3) PF_5 (4) SF_6
29. The number of S=O and S-OH bonds present in peroxodisulphuric acid and pyrosulphuric acid respectively are : [JEE-MAINS-2017 (On-line)]
 (1) (2 and 4) and (2 and 4) (2) (4 and 2) and (2 and 4)
 (3) (2 and 2) and (2 and 2) (4) (4 and 2) and (4 and 2)
30. The correct sequence of decreasing number of π -bonds in the structures of H_2SO_3 , H_2SO_4 and $\text{H}_2\text{S}_2\text{O}_7$ is:- [JEE-MAINS-2017 (On-line)]
 (1) $\text{H}_2\text{S}_2\text{O}_7 > \text{H}_2\text{SO}_4 > \text{H}_2\text{SO}_3$ (2) $\text{H}_2\text{SO}_3 > \text{H}_2\text{SO}_4 > \text{H}_2\text{S}_2\text{O}_7$
 (3) $\text{H}_2\text{S}_2\text{O}_7 > \text{H}_2\text{SO}_3 > \text{H}_2\text{SO}_4$ (4) $\text{H}_2\text{SO}_4 > \text{H}_2\text{S}_2\text{O}_7 > \text{H}_2\text{SO}_3$
31. The increasing order of the boiling points for the following compounds is:- [JEE-MAINS-2017 (On-line)]
 (I) $\text{C}_2\text{H}_5\text{OH}$ (II) $\text{C}_2\text{H}_5\text{Cl}$ (III) $\text{C}_2\text{H}_5\text{CH}_3$ (IV) $\text{C}_2\text{H}_5\text{OCH}_3$
 (1) (III) < (II) < (I) < (IV) (2) (II) < (III) < (IV) < (I)
 (3) (IV) < (III) < (I) < (II) (4) (III) < (IV) < (II) < (I)
32. The number of P-OH bonds and the oxidation state of phosphorus atom in pyrophosphoric acid ($\text{H}_4\text{P}_2\text{O}_7$) respectively are :- [JEE-MAINS-2017 (On-line)]
 (1) five and four (2) five and five (3) four and five (4) four and four
33. The group having triangular planar structures is :- [JEE-MAINS-2017 (On-line)]
 (1) CO_3^{2-} , NO_3^- , SO_3 (2) NCl_3 , BCl_3 , SO_3 (3) NH_3 , SO_3 , CO_3^{2-} (4) BF_3 , NF_3 , CO_3^{2-}

34. Which of the following are Lewis acids ? [JEE-MAINS-2018]

- (1) AlCl_3 and SiCl_4 (2) PH_3 and SiCl_4 (3) BCl_3 and AlCl_3 (4) PH_3 and BCl_3

35. According to molecular orbital theory, which of the following will not be a viable molecule ?

[JEE-MAINS-2018]

- (1) He_2^+ (2) H_2^- (3) H_2^{2-} (4) He_2^{2+}

36. Which of the following compounds contain(s) no covalent bond(s) ?

[JEE-MAINS-2018]

KCl , PH_3 , O_2 , B_2H_6 , H_2SO_4

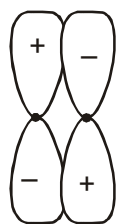
- (1) KCl , H_2SO_4 (2) KCl (3) KCl , B_2H_6 (4) KCl , B_2H_6 , PH_3

37. In KO_2 , the nature of oxygen species and the oxidation state of oxygen atom are, respectively

[JEE-MAINS-2018 (Online)]

- (1) Superoxide and $-1/2$ (2) Oxide and -2
(3) Peroxide and $-1/2$ (4) Superoxide and -1

38. Which of the following best describes the diagram below of a molecular orbital ?



[JEE-MAINS-2018 (Online)]

- (1) An antibonding π orbital (2) An antibonding σ orbital
(3) A non-bonding orbital (4) A bonding π orbital

39. In the molecular orbital diagram for the molecular ion, N_2^+ , the number of electrons in the σ_{2p} molecular orbital is :-

[JEE-MAINS-2018 (Online)]

- (1) 3 (2) 1 (3) 0 (4) 2

40. Xenon hexafluoride on partial hydrolysis produces compounds 'X' and 'Y'. Compounds 'X' and 'Y' and the oxidation state of Xe are respectively :

[JEE-MAINS-2018 (Online)]

- (1) $\text{XeO}_2\text{F}_2(+6)$ and $\text{XeO}_2(+4)$ (2) $\text{XeOF}_4(+6)$ and $\text{XeO}_2\text{F}_2(+6)$
(3) $\text{XeOF}_4(+6)$ and $\text{XeO}_3(+6)$ (4) $\text{XeO}_2(+4)$ and $\text{XeO}_3(+6)$

41. Which of the following is a Lewis acid ?

[JEE-MAINS-2018 (Online)]

- (1) NaH (2) NF_3 (3) PH_3 (4) $\text{B}(\text{CH}_3)_3$

42. A group 13 element 'X' reacts with chlorine gas to produce a compound XCl_3 . XCl_3 is electron deficient and easily reacts with NH_3 to form $\text{Cl}_3\text{X} \leftarrow \text{NH}_3$ adduct; however, XCl_3 does not dimerize. X is :-

[JEE-MAINS-2018 (Online)]

- (1) Ga (2) Al (3) In (4) B

43. Which of the following conversions involves change in both shape and hybridisation ?

[JEE-MAINS-2018 (Online)]

- (1) $\text{BF}_3 \rightarrow \text{BF}_4^-$ (2) $\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$ (3) $\text{CH}_4 \rightarrow \text{C}_2\text{H}_6$ (4) $\text{NH}_3 \rightarrow \text{NH}_4^+$

EXERCISE # J-ADVANCE

- The molecules that will have dipole moment are : [IIT-1992]
 (A) 2, 2-dimethylpropane (B) trans-pent-2-ene
 (C) cis-hex-3-ene (D) 2, 2, 3, 3-tetramethylbutane
- Which of the following have identical bond order ? [IIT-1992]
 (A) CN^- (B) O_2^- (C) NO^+ (D) CN^+
- Among the following the one that is polar and has the central atom with sp^2 hybridisation is :
 (A) H_2CO_3 (B) SiF_4 (C) BF_3 (D) HClO_2 [IIT-1997]
- Which of the following is soluble in water ? [IIT 98]
 (A) CS_2 (B) $\text{C}_2\text{H}_5\text{OH}$ (C) CCl_4 (D) CHCl_3
- The correct order of hybridization of the central atom in the following species NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3 is : [IIT 2001]
 (A) dsp^2 , sp^3d , sp^2 and sp^3 (B) sp^3 , dsp^2 , sp^3d , sp^2
 (C) dsp^2 , sp^2 , sp^3 , sp^3d (D) dsp^2 , sp^3 , sp^2 , sp^3d
- The common features among the species CN^- , CO and NO^+ are : [IIT 2001]
 (A) bond order three and isoelectronic (B) bond order three and weak field ligands
 (C) bond order two and π - acceptors (D) isoelectronic and weak field ligands
- Which of the following molecular specie has unpaired electron(s) ? [JEE 2002]
 (A) N_2 (B) F_2 (C) O_2^- (D) O_2^{2-}
- According to molecular orbital theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+ [JEE 2004]
 (A) Paramagnetic and Bond order $< \text{O}_2$ (B) Paramagnetic and Bond order $> \text{O}_2$
 (C) Diamagnetic and Bond order $< \text{O}_2$ (D) Diamagnetic and Bond order $> \text{O}_2$
- Among the following, the paramagnetic compound is [JEE 2007]
 (A) Na_2O_2 (B) O_3 (C) N_2O (D) KO_2
- The species having bond order different from that in CO is [JEE 2007]
 (A) NO^- (B) NO^+ (C) CN^- (D) N_2
- Statement-1** : In water, orthoboric acid behaves as a weak monobasic acid. [JEE 2007]
Statement-2 : In water, orthoboric acid acts as a proton donor.
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.
- Statement-1** : Pb^{+4} compounds are stronger oxidizing agents than Sn^{+4} compounds [JEE 2008]
Statement-2 : The higher oxidation states for the group 14 elements are more stable for the heavier members of the group due to 'inert pair effect' .
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

13. Match each of the diatomic molecules/ions in Column I with its property / properties in Column II. [JEE 2009]

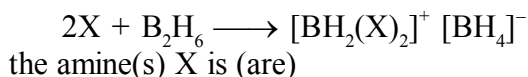
Column I

- (A) B_2
(B) N_2
(C) O_2^-
(D) O_2

Column II

- (P) Paramagnetic
(Q) undergoes oxidation
(R) Undergoes reduction
(S) Bond order ≥ 2
(T) Mixing of 's' and 'p' orbitals

14. In the reaction



- (A) NH_3 (B) CH_3NH_2 (C) $(CH_3)_2NH$ (D) $(CH_3)_3N$

15. The species having pyramidal shape is

- (A) SO_3 (B) BrF_3 (C) SiO_3^{2-} (D) OSF_2

16. Assuming that Hund's rule is violated, the bond order and magnetic nature of the diatomic molecule B_2 is [JEE 2010]

- (A) 1 and diamagnetic (B) 0 and diamagnetic (C) 1 and paramagnetic (D) 0 and paramagnetic

Subjective

17. The value of n in the molecular formula $Be_nAl_2Si_6O_{18}$ is [JEE 2010]

18. The total number of diprotic acids among the following is [JEE 2010]

- H_3PO_4 H_2SO_4 H_3PO_3 H_2CO_3 $H_2S_2O_7$
 H_3BO_3 H_3PO_2 H_2CrO_4 H_2SO_3

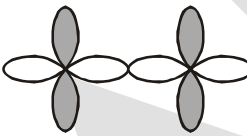
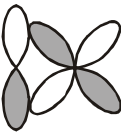
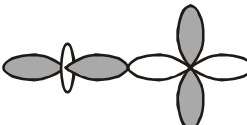
19. Among the following, the number of elements showing only one non-zero oxidation state is [JEE 2010]
O, Cl, F, N, P, Sn, Tl, Na, Ti

20. Assuming 2s-2p mixing is NOT operative, the paramagnetic species among the following is : [JEE Adv. 2014]

- (A) Be_2 (B) B_2 (C) C_2 (D) N_2

21. Match the orbital overlap figures shown in List-I with the description given in List-II and select the correct answer using the code given below the lists. [JEE Adv. 2014]

List-I

- (P) 
(Q) 
(R) 
(S) 

List-II

- (1) p - d π antibonding
(2) d - d σ bonding
(3) p - d π bonding
(4) d - d σ antibonding

Code :

- | | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 2 | 1 | 3 | 4 |
| (C) | 2 | 3 | 1 | 4 |

- | | P | Q | R | S |
|-----|---|---|---|---|
| (B) | 4 | 3 | 1 | 2 |
| (D) | 4 | 1 | 3 | 2 |

22. Three moles of B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is - [JEE Adv. 2015]
23. When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2 . The TRUE, statement (s) regarding this adsorption is (are) [JEE Adv. 2015]
 (A) O_2 is physisorbed (B) heat is released
 (C) occupancy of π_{2p}^* of O_2 is increased (D) bond length of O_2 is increased
24. According to Molecular Orbital Theory, [JEE Adv. 2016]
 (A) C_2^{2-} is expected to be diamagnetic
 (B) O_2^{2+} is expected to have a longer bond length than O_2
 (C) N_2^+ and N_2^- have the same bond order
 (D) He_2^+ has the same energy as two isolated He atoms
25. The colour of the X_2 molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to - [JEE Adv. 2017]
 (A) the physical state of X_2 at room temperature changes from gas to solid down the group
 (B) decrease in HOMO-LUMO gap down the group
 (C) decrease in $\pi^*-\sigma^*$ down the group
 (D) decrease in ionization energy down the group
26. Among H_2 , He_2^+ , Li_2 , Be_2 , B_2 , C_2 , N_2 , O_2^- , and F_2 , the number of diamagnetic species is - (Atomic number) : H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, f = 9) [JEE Adv. 2017]
27. The sum of the number of lone pairs of electrons on each central atom in the following species is. [JEE Adv. 2017]
 $[TeBr_6]^{2-}$, $[BrF_2]^+$, SNF_3 and $[XeF_3]^-$
 [Atomic number : N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54]
28. The option(s) with only amphoteric oxides is (are): [JEE Adv. 2017]
 (A) Cr_2O_3 , CrO , SnO , PbO (B) NO , B_2O_3 , PbO , SnO_2
 (C) Cr_2O_3 , BeO , SnO , SnO_2 (D) ZnO , Al_2O_3 , PbO , PbO_2
29. Among the following, the correct statement(s) is are [JEE Adv. 2017]
 (A) $Al(CH_3)_3$ has the three-centre two-electron bonds in its dimeric structure
 (B) $AlCl_3$ has the three-centre two-electron bonds in its dimeric structure
 (C) BH_3 has the three-centre two-electron bonds in its dimeric structure
 (D) The Lewis acidity of BCl_3 is greater than that of $AlCl_3$
30. Based on the compounds of group 15 elements, the correct statement(s) is (are) [JEE Adv. 2018]
 (A) Bi_2O_5 is more basic than N_2O_5
 (B) NF_3 is more covalent than BiF_3
 (C) PH_3 boils at lower temperature than NH_3
 (D) The N-N single bond is stronger than the P-P single bond

ANSWERS KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	D	B	A	C	C	B	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	C	B	B	B	A	D	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	C	D	D	D	C	A	A	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	B	D	B	A	A	A	B	D	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	A	C	C	B	A	D	A	A	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A, B, C	A, B, C	A, B, C	A, B, D	A, B	A, C	A, B, C	B, C	A, C	A, B, C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B, C, D	B, C, D	B, C	B, C, D	C, D	A, B, C	A, C, D	A, C	A, C	A, D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A, B, D	B, C, D	A, B, C	A, B, C	A, B, D	B, D	A, B, C, D	A, B	A, B, C	A, B, D
Que.	31	32								
Ans.	A, B, C	B, C, D								

EXERCISE # S-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	8	4	2	2	105	4	3	2	4

EXERCISE # S-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	B	A	D	A	B	B	B	D
Que.	11	12	13	14				15		
Ans.	C	B	C	(A)-Q,R,S (B)-P,Q,S (C)-S (D)-P,Q				(A)-P,S (B)-Q,T (C)-R (D)-P,S		
Que.	16					17	18	19		
Ans.	(A)-P,R,T (B)-Q (C)-Q (D)-Q,T					B	C	D		

EXERCISE # J-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	4	2	3	4	1	4	1	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	3	1	2	3	2	3	3	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	2	2	4	2	4	1 & 3	4	1
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	4	3	1	3	3	2	1	1	2	2
Que.	41	42	43							
Ans.	4	4	1							

EXERCISE # J-ADVANCE

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B, C	A, C	A	B	B	A	C	B	D	A
Que.	11	12	13					14	15	16
Ans.	C	C	(A)-P, Q, R, T (B)-Q, R, S, T (C)-P, Q, R (D)-P, Q, R, S					B, C	D	A
Que.	17	18	19	20	21	22	23	24	25	26
Ans.	3	6	2	C	C	6	B, C, D	A, C	B, C	5 or 6
Que.	27	28	29	30						
Ans.	6	C, D	A, C, D	A, B, C						

CO-ORDINATION CHEMISTRY

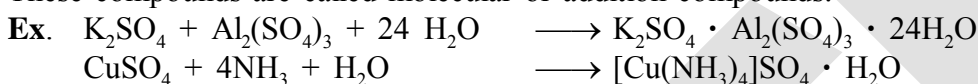
□ INTRODUCTION :

- The concept of co-ordination compounds arises from the complex formation tendency of transition elements.
- These compounds play a vital role in our lives, as chlorophyll of plants, vitamin B₁₂ and haemoglobin of animal blood are the co-ordination compounds of Mg, Co and Fe respectively.
- The co-ordination compounds play important role in analytical chemistry, polymerisation reactions, metallurgy and refining of metals, photography, water purification etc.
- Co-ordination compounds also find many applications in electroplating, textile dyeing and medicinal chemistry.

□ ADDITION COMPOUNDS :

When solutions containing two or more salts in simple molecular proportion are evaporated, crystals of new compound separate out.

These compounds are called molecular or addition compounds.



These addition compounds can be divided into two classes :

(A) DOUBLE SALTS :

Those which lose their identity in solution

In solutions these compounds break down into simpler ions. Such addition compounds which lose their identity in solutions are called **double salts**.

Example :

Potash alum $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ when dissolved in water breaks down into K^+ , SO_4^{2-} , Al^{+3} ions and therefore is an example of **double salt**.

(B) COORDINATION COMPOUNDS :

Those which retain their identity in solution.

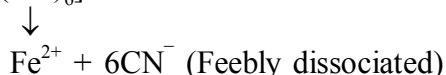
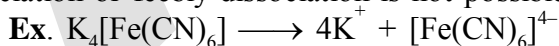
In aqueous solution, these addition compounds do not furnish all simple ions but instead give more complex ions having complicated structure.

Example :

Potassium ferrocyanide $\text{K}_4[\text{Fe}(\text{CN})_6]$ does not furnish simple K^+ , Fe^{2+} and CN^- ions but gives K^+ ions and complex ferrocyanide ions, $[\text{Fe}(\text{CN})_6]^{4-}$. These types of compounds are called **complex compounds or co-ordination compounds**.

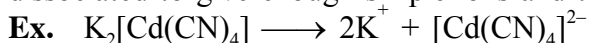
On the basis of stability of complex ion, complex ions are further divided as follows –

- Perfect complexes :** The compounds in which complex ion is fairly stable and further dissociation or feeble dissociation is not possible in solution state.



The ferrocyanide ion $[\text{Fe}(\text{CN})_6]^{4-}$ is so insignificantly dissociated that it can be considered as practically undissociated and does not give the qualitative test of Fe^{2+} or CN^- ions..

- Imperfect complexes :** Those complexes in which complex ion is less stable and is reversibly dissociated to give enough simple ions and thus respond to their usual qualitative test.



□ DEFINITIONS OF TERMS USED IN CO-ORDINATION COMPOUNDS

- (a) **Co-ordination or complex compound** : Co-ordination compounds are those molecular compounds which retain their identity even when dissolved in water or any other solvent and their properties are completely different from those of the constituent ions.
- (b) **Central ion** : The cation to which one or more neutral molecules or ions are attached is called the atom / ion. Since, the central ion acts as an acceptor and thus has to accommodate electron pairs donated by the donor atoms of neutral molecules or ions, it must have empty orbitals of appropriate energy.
- (c) **Complex ion** : A complex ion may be defined as an electrically charged radical which is formed by the combination of a simple cation with one/more neutral molecules or one/more simple.
- (d) **Co-ordination number** : The total number of co-ordinate covalent bond formed by central metal in complex is called the co-ordination number of the central metal ion .

Some common co-ordination number of important metals are as given below.

Metal	Coordination Number	Metal	Coordination Number
Cu^+	2, 4	Ni^{2+}	4, 6
Ag^+	2	Fe^{2+}	4, 6
Au^+	2, 4	Fe^{3+}	6
Hg_2^{2+}	2	Co^{2+}	4, 6
Cu^{2+}	4, 6	Co^{3+}	6
Ag^{2+}	4	Al^{3+}	6
Pt^{2+}	4	Sc^{3+}	6
Pd^{2+}	4	Pt^{4+}	6
Mg^{2+}	6	Pd^{4+}	6

Example. Coordination number of the central metal ions in

- (i) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is four (ii) $[\text{Fe}(\text{EDTA})]^-$ is six

- (e) **Co-ordination sphere** : The part of the complex enclosed in square bracket is known as co-ordination sphere. It is actually combination of central metal and ligands.
- (f) **Ligands** :
- (a) The ions or neutral molecules which combine with central metal ion to form complex are called ligands.
- (b) They act as electron pair donor (i.e. Lewis bases) though certain ligands also accept electron from central metal and such ligands are known as π - acid ligands.

❖ CLASSIFICATION OF LIGANDS

(A) Based on charge

- (i) Neutral ligands : H_2O , NO , CO , C_6H_6 etc.
- (ii) Positive ligands : NO^+ , N_2H_5^+
- (iii) Negative ligands : Cl^- , NO_2^- , CN^- , OH^-

(B) Based on denticity

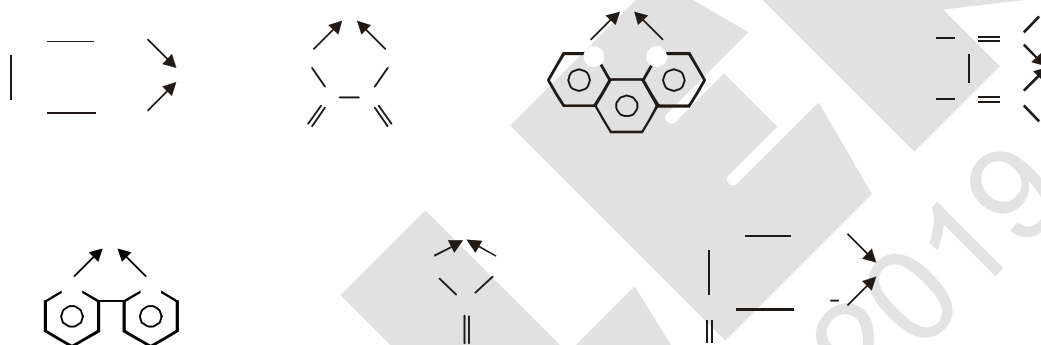
The number of electron pairs donated to central metal by a particular ligand is known as DENTICITY. Depending on number of electron pairs donated, these are classified in following categories.

(a) Unidentate/monodentate ligands

Ligands which donate one pair of electron to the central metal are called unidentate ligands. X^- , CN^- , NO_2^- , NH_3 , Pyridine, OH^- , NO_3^- , H_2O , SO_3^{2-} , CO , NO , OH^- , O^{2-} , $(C_6H_5)_3P$ etc.

(b) Bidentate ligands

Ligands which have two donor atoms and have the ability to link with central metal ion at two positions are called bidentate ligands.



(c) Tridentate ligands

The ligands which donate three pairs of electrons to the central metal are called tridentate ligands

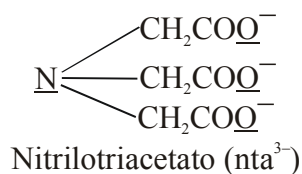
Example.



(d) Tetradentate ligands

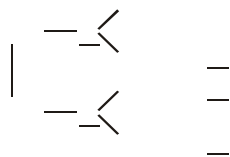
Those ligands which can donate four electron pairs to the central metal are known as tetradentate ligands,

Example : (Underline atoms are donating atom)



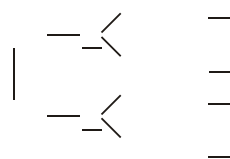
- (e) **Pentadentate ligands** : Those ligands which can give five electron pairs to the central metal are known as pentadentate ligands.

Example : Ethylenediamine triacetate ion. (Underline atoms are donating atom)



- (f) **Hexadentate ligands** : Those ligands which can donate six electron pairs to the central metal are known as hexadentate ligands.

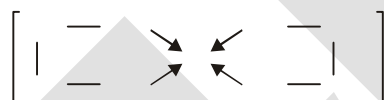
Example : (Underline atoms are donating atom)



- (g) **Chelating ligands**

Polydentate ligands whose structures permit the attachment of two or more donor site to metal ion simultaneously, thus resulting in cyclic structure are called chelating ligands and compound formed is known as **chelate compound**.

Example :



- (h) **Ambidentate ligands**

Ligands which can ligate through two different atoms present in it are called ambidentate ligands. At a time only one atom can donate.



Example :

CN^- can coordinate through either the nitrogen or the carbon atom to central metal ion.

- (i) **Flexidentate ligands**

Ligands which sometimes do not use all the donor sites to get coordinated with central metal ion are known as flexidentate ligands.

Ex. $\text{C}_2\text{O}_4^{2-}$, $\text{C}_2\text{O}_4^{2-}$ etc.

- (C) Based upon bonding interaction between the ligand and the central atom.

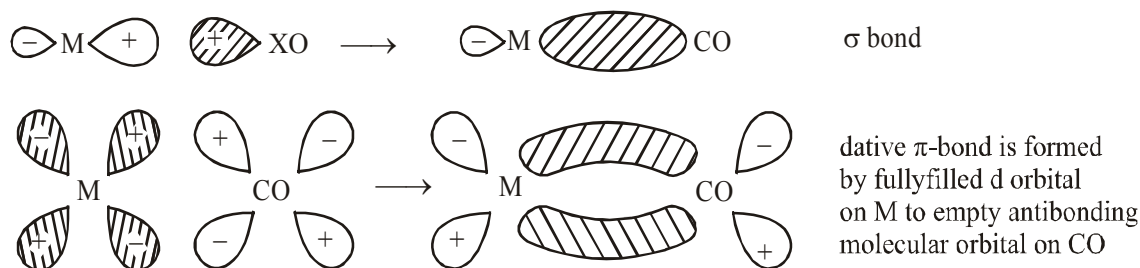
- (i) **Classical or simple ligand** : These ligand only donate the lone pair of electrons to the central atom.

eg. : O^{2-} , OH^- , F^- etc.

- (ii) **Non classical or π -acid or π -acceptor ligand** : These ligand not only donate the lone pair of electrons to central metal but also accept the electron cloud from central atom

eg. : CO , CN^- , NO^+ , PF_3 , PR_3 etc.

BONDING IN METAL CARBONYL : Ex. $[\text{Fe}(\text{CO})_5]$; $[\text{Ni}(\text{CO})_4]$; $[\text{Cr}(\text{CO})_6]$



Schematic diagram of orbital overlaps in metal carbonyls.

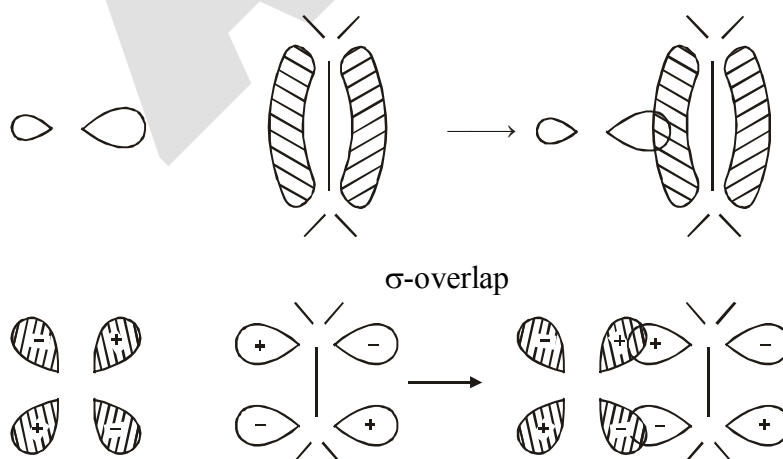
- The metal-carbon bond in metal carbonyls may be represented as the donation of an electron pair from carbon to vacant orbital of metal & form σ bond ($\text{M} \leftarrow \text{CO}$).
- A second bond is formed by back bonding sometimes called dative π -bonding. This arises from side ways overlap of a full orbital on the metal with the empty antibonding π^*p_y/π^*p_z (if x-axis is molecular axis) molecular orbital of the carbon monoxide, thus forming a $\pi \text{ M} \xrightarrow{\pi-\pi} \text{CO}$ bond ($d\pi-p\pi$ back bond).
- The filling or partial filling, of the antibonding orbital on CO reduces the bond order of C–O bond from the triple bond in CO towards a double bond. This shown by the increase in C–O bond length from $1.128\text{\AA}$ in CO to about $1.15\text{\AA}$ in many carbonyls.
- Since CO accept the back donated electrons from the metal atom in to its vacant π^* orbital, CO is called π -acid or π -acceptor ligand or π -bonding ligand. Other such π -acid ligands are— CN^- , RCN , .

Note:- π -acid ligands like PF_3 , PPh_3 , AsCl_3 etc. accept the back donated electrons from the metal atom in to its vacant d-orbital of central atom.

Bonding in π -bonded organo metallic compound. Like zeises salt $\text{K} [\text{PtCl}_3(\pi\text{-C}_2\text{H}_4)]$

The bonding of alkenes to a transition metal to form complexes has two components. First, the π -electron density of the alkene overlaps with a σ -type vacant orbital or the metal atom.

Second is the π back bond formed by the flow of electron density from a filled d-orbital on the metal into the vacant π^* -antibonding molecular orbital on the alkene molecule as shown below :



□ IUPAC NOMENCLATURE OF COORDINATION COMPOUNDS

The main rules of naming of complexes are –

- (a) Like simple salts, the positive part of the coordination compound is named first.

Ex. $K_4[Fe(CN)_6]$ the naming of this complex starts with potassium.

$[Cr(NH_3)_6]Cl_3$ the naming of this complex starts with name of complex ion.

- (b) Naming of coordination sphere :- The names of ligands along with their numerical prefixes (to represent their no) are written first, followed by the name of central metal.

- (c) The ligands can be neutral, anionic or cationic.

- (i) The neutral ligands are named as the molecule **Ex.** C_5H_5N pyridine, $(C_6H_5)_3P$ Triphenyl phosphine.

$H_2N-CH_2-CH_2-NH_2$ ethylene diamine.

The neutral ligands which are not named as the molecule are CO carbonyl, NO nitrosyl, H_2O Aqua, NH_3 ammine.

- (ii) Anionic ligands ending with 'ide' are named by replacing the 'ide' with suffix 'O'.

Symbol	Name as ligand	Symbol	Name as ligand
Cl^-	Chloro/Chlorido	N^{3-}	Nitrido
Br^-	Bromo/Bromido	O_2^{2-}	Peroxo/Peroxido
CN^-	Cyano/Cyanido	O_2H^-	Perhydroxo/Perhydroxido
O^{2-}	Oxo/Oxido	S^{2-}	Sulphido
OH^-	Hydroxo/Hydroxido	NH^{2-}	Imido
H^-	Hydrido/Hydrido	NH_2^-	Amido

Ligands whose names end in 'ite' or 'ate' become 'ito' i.e., by replacing the ending 'e' with 'o' as follows.

Symbol	Name as ligand	Symbol	Name as ligand
CO_3^{2-}	Carbonato	SO_3^{2-}	Sulphito
$C_2O_4^{2-}$	Oxalato	CH_3COO^-	Acetato
SO_4^{2-}	Sulphato	ONO^-	(bonded through oxygen) nitrito
NO_3^-	Nitrato	NO_2^-	(bonded through nitrogen) nitro
$S_2O_3^{2-}$	Thiosulphato		

- (iii) Positive ligands naming ends in 'ium' $NH_2-NH_3^+$ Hydrazinium, NO^+ nitrosonium/nitrosylium.

- (d) If ligands are present more than once, then their number is indicated by prefixes like di, tri, tetra etc.
- (e) If words like di, tri, tetra are already used in the naming of ligand, or if it is polydentate ligand or organic ligand, the prefixes bis-, tris-, tetrakis-, pentakis- etc. are used to specify their number.

Example : $[Pt(en)_2Cl_2]Cl_2$: Dichlorobis(ethylenediamine)platinum(IV) chloride.

- (f) When more than one type of ligands are present in the complex, then the ligands are named in the alphabetical order.
- (g) After naming of ligands the central metal ion is to be named followed by its oxidation state in Roman numbers in brackets. (as per IUPAC)

If the complex is neutral or provides a cationic complex ion, then the central metal ion is to be named as it is.

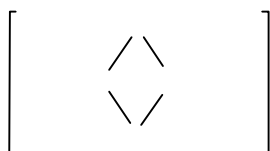
If the complex provides anionic complex ion then the name of central metal ion ends with 'ate'

Example : $(\text{NH}_4)_2[\text{CuCl}_4]$: Ammonium tetrachloridocuprate(II)

- (h) After the naming of central metal ion, anion which is in the outer sphere is to be named. The naming of some of the complexes is done as follows – (as per IUPAC)

Complex Compounds		IUPAC Name
(i)	$\text{K}_4[\text{Fe}(\text{CN})_6]$ (anionic complex) so suffix 'ate' is added with metal name	Potassium hexacyanoferrate(II)
(ii)	$\text{K}_2[\text{Pt Cl}_6]$	Potassium hexachloridoplatinate(IV)
(iii)	$[\text{Co}(\text{NH}_3)_6] \text{Cl}_3$ (Cationic complex) so metal is without any suffix	Hexamminecobalt(III) chloride
(iv)	$[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2] \text{Cl}$	Tetraaquadichloridochromium(III) chloride
(v)	$[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$	Diamminetetrachloridoplatinum(IV)
(vi)	$[\text{Co}(\text{NH}_3)_3 \text{Cl}_3]$ (Neutral complex) So no suffix is used with metal ion	Triamminetrichloridocobalt(III)
(vii)	$\text{K}_3[\text{Co}(\text{NO}_2)_6]$	Potassium hexanitrocobaltate(III)
(viii)	$\text{Na}_3[\text{Fe}(\text{CN})_5 \text{NO}]$	Sodium pentacyanonitrosylferrate(II)
(ix)	$[\text{NiCl}_4]^{-2}$	Tetrachloridonickelate(II) ion
(x)	$[\text{Ru}(\text{NH}_3)_5\text{Cl}]^{+2}$	Pentamminechloridoruthenium(III) ion
(xi)	$[\text{Fe}(\text{en})_3]\text{Cl}_3$	Tris(ethylenediamine)iron(III) chloride
(xii)	$[\text{Ni}(\text{Gly})_2]$	Bis(glycinato)nickel(II)

- (i) If a complex ion has two metal atoms then it is termed polynuclear. The ligand which connects the two metal ions is called as **Bridging ligand or Bridge group**.
A prefix of Greek letter μ , is repeated before the name of each different kind of bridging group.



FORMATION OF CO-ORDINATION COMPOUNDS

It can be explained by number of theories.

- (A) Werner's co-ordination theory
- (B) Sidwick theory or Effective Atomic Number Theory (EAN)
- (C) Valence bond theory
- (D) Crystal field theory

(A) WERNER'S CO-ORDINATION THEORY :

Werner's co-ordination theory was the first attempt to explain the bonding in co-ordination compounds. The main postulates of this theory are :

- (a) Metals possesses two types of valencies - Primary valency and secondary valency.
- (b) Primary valencies are normally ionisable and are exhibited by a metal in the formation of its simple salts such as CoCl_3 , CuSO_4 and AgCl . In these salts the primary valencies of Co, Cu and Ag are 3, 2, 1 respectively. Primary valencies are referred to as oxidation state of their metal ion.
- (c) Secondary valencies are non-ionisable and are exhibited by a metal in the formation of its complex ions such as $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Cu}(\text{NH}_3)_4]^{2+}$ and $[\text{Ag}(\text{NH}_3)_2]^+$. In these complex, the secondary valencies of Co^{3+} , Cu^{2+} , Ag^+ are 6, 4 and 2 respectively. These are referred to as co-ordination number (C.N.) of the metal cation.
- (d) Primary linkages (valencies) are satisfied by negative ions while secondary valencies are satisfied by neutral molecules, negative ions or in some cases positive ions also.
- (e) Every metal atom or ion has a fixed number of secondary valencies. In other words, the co-ordination number of the metal atom is usually fixed.
- (f) Every metal has tendency to satisfy both its primary and secondary valencies.
- (g) The ligands satisfying secondary valency are always directed towards fixed positions in space about the central metal atom or ion. Thus, the co-ordination compounds have a definite geometry. Werner deduced that in $\text{CoCl}_3 \cdot 5\text{NH}_3$ only two of the three chlorine atoms are ionic and 5 NH_3 and one Cl form co-ordinate bonds to Co^{3+} ion.

Formula of some cobalt complexes.

Example :

Old	New	No. of Cl^- ions precipitated	Total No. of ions
(i) $\text{CoCl}_3 \cdot 6 \text{NH}_3$	$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	3	4
(ii) $\text{CoCl}_3 \cdot 5 \text{NH}_3$	$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$	2	3
(iii) $\text{CoCl}_3 \cdot 4 \text{NH}_3$	$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$	1	2

Complex	Modern formula	No. of Cl^- ions precipitated	Total number of ions
$\text{PtCl}_4 \cdot 6\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$	4	5
$\text{PtCl}_4 \cdot 5\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$	3	4
$\text{PtCl}_4 \cdot 4\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$	2	3
$\text{PtCl}_4 \cdot 3\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]\text{Cl}$	1	2
$\text{PtCl}_4 \cdot 2\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$	0	0 (non-electrolyte)

❖ WERNER'S REPRESENTATION OF COMPLEXES

(i)	$\text{Fe}(\text{NH}_3)_6\text{Cl}_3$		$[\text{Fe}(\text{NH}_3)_6]\text{Cl}_3$ Dotted lines indicate primary valency and continuous lines indicate secondary valency of metal ion.
(ii)	$\text{Fe}(\text{NH}_3)_5\text{Cl}_3$		$[\text{Fe}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ In this complex two 'Cl' groups act as primary valencies and one of the 'Cl' acts as secondary valency also.
(iii)	$\text{Fe}(\text{NH}_3)_4\text{Cl}_3$		$[\text{Fe}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ In this complex one 'Cl' group act as primary valency and two of the 'Cl' groups act as secondary valencies also.

(B) SIDWICK THEORY OR EFFECTIVE ATOMIC NUMBER CONCEPT (EAN)

Sidwick proposed effective atomic number theory to explain the stability of the complexes. Total number of electrons on central metal including those transferred from ligands is known as EAN. The EAN generally coincides with the atomic number of next inert gas except in some cases.

EAN can be calculated by the following relation :

EAN = (atomic number of the metal – oxidation state of central metal with sign) + number of electrons gained from the donor atoms of the ligands.

Example Effective atomic number of the metal atom in the following :

- (a) $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$ is 33 (b) $\text{K}_4[\text{Fe}(\text{CN})_6]$ is 36

(C) VALENCE BOND THEORY

The main features of this theory are -

- Every metal ion when it forms a complex compound undergoes formation of coordinate covalent bond.
- During this bond formation, the metal ion acts as electron pair acceptor. For this the metal ion provides vacant orbitals.
- The number of vacant orbitals provided is equal to the coordination number of metal ion.

Example : In the formation of $[\text{Fe}(\text{NH}_3)_6]^{3+}$, Fe^{3+} ion provides six vacant orbitals.

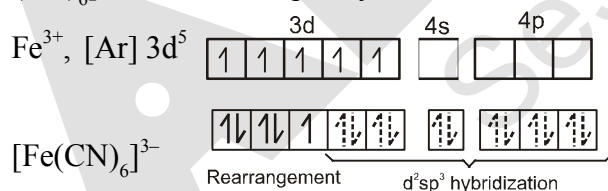
In $[\text{Cu}(\text{NH}_3)_4]^{2+}$, Cu^{+2} ion provides four vacant orbitals.

- The metal provides vacant orbitals only after the process of hybridisation, thus vacant hybrid orbitals are provided by the metal ion.
- The vacant hybrid orbitals of metal ion get overlapped by orbitals of ligands containing lone pair of electrons.
- The number of such overlappings is equal to the coordination number of metal ion.
- The empty 'd' orbitals involved in hybridisation may be inner (n-1)d or outer "nd" orbitals and accordingly complexes are called as **Inner orbital complexes** and **outer orbital complexes** respectively.
- In certain complexes pairing of electrons takes place in ligand field, resulting in decrease in spin only magnetic moment, such complexes are known as Low spin complexes
- Bohr magneton = $\frac{eh}{4\pi mc}$
- Paramagnetism is represented in the term of spin only magnetic moment.

$$\mu = \sqrt{n(n+2)} \text{ B.M.} \quad n = \text{Number of unpaired electron}$$

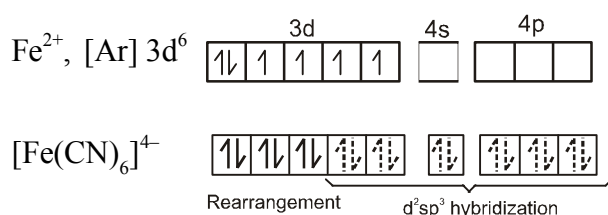
Example $[\text{Fe}(\text{CN})_6]^{3-}$ is weakly paramagnetic while $[\text{Fe}(\text{CN})_6]^{4-}$ is diamagnetic.

Sol. $[\text{Fe}(\text{CN})_6]^{3-}$ involves d^2sp^3 hybridization.



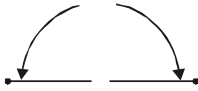
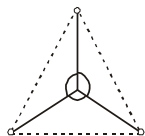
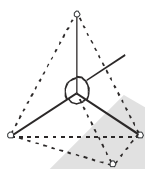
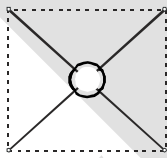
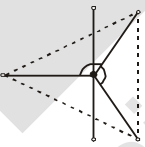
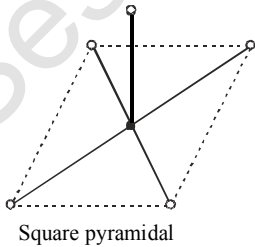
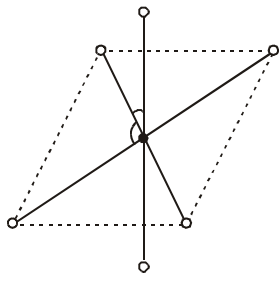
One d-orbital is singly occupied, hence it is weakly paramagnetic in nature.

$[\text{Fe}(\text{CN})_6]^{4-}$ also involves d^2sp^3 hybridization but it has Fe^{2+} ion as central ion.



All electrons are paired, hence it is diamagnetic in nature.

Some Example :

Coordination Number	Hybridised orbitals	Geometrical shape of the Complex	Examples of Complex
2	sp		$[\text{Ag}(\text{NH}_3)_2]^+$ $[\text{Ag}(\text{CN})_2]^-$
3	sp^2		$[\text{HgI}_3]^-$
4	sp^3		$[\text{CuCl}_4]^{2-}$ $[\text{ZnCl}_4]^{2-}$ $[\text{FeCl}_4]^-$ $[\text{Ni}(\text{CO})_4]$ $[\text{Zn}(\text{NH}_3)_4]^{+2}$
4	dsp^2 ($d = d_{x^2-y^2}$)		$[\text{PdCl}_4]^{2-}$ $[\text{Ni}(\text{CN})_4]^{2-}$ $[\text{Pt}(\text{NH}_3)_4]^{+2}$ $[\text{Cu}(\text{NH}_3)_4]^{+2}$ $[\text{PtCl}_4]^{2-}$
5	sp^3d ($d = d_{z^2}$) or dsp^3 ($d = d_{z^2}$)		$[\text{Fe}(\text{CO})_5]$ $[\text{CuCl}_5]^{3-}$
5	sp^3d ($d = d_{x^2-y^2}$) or dsp^3 ($d = d_{x^2-y^2}$)	 Square pyramidal	$[\text{Ni}(\text{CN})_5]^{-3}$
6	d^2sp^3 (inner orbital complex) or sp^3d^2 (outer orbital complex) in both case d-orbitals are d_{z^2} & $d_{x^2-y^2}$.		$[\text{Cr}(\text{NH}_3)_6]^{+3}$ $[\text{Ti}(\text{H}_2\text{O})_6]^{+3}$ $[\text{Fe}(\text{CN})_6]^{-3}$ $[\text{Co}(\text{NH}_3)_6]^{+3}$ $[\text{PtCl}_6]^{2-}, [\text{CoF}_6]^{-3}$

♦ **Drawback of valence bond theory :**

- It describes bonding in co-ordination compounds only qualitatively but not account for the relative stabilities for different co-ordination complexes.
- It does not offer any explanation for optical absorption spectra (coloration) of complexes
- It does not describe the detailed magnetic properties of co-ordination compounds.

(D) CRYSTAL FIELD THEORY : The drawbacks of VBT of coordination compounds are, to a considerable extent, removed by the Crystal Field Theory.

The crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interaction between the metal ion and the ligand. Ligands are treated as point charges in case of anions or dipoles in case of neutral molecules. The five d-orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands (either anions or the negative ends of polar molecules like NH_3 and H_2O) in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lost. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field.

(a) Crystal field splitting in octahedral coordination entities :

In an octahedral coordination entity with six ligands surrounding the metal atom/ion, there will be repulsion between the electrons in d orbitals of metal and the electrons (or negative charges) of the ligands. Such a repulsion is more when the d orbitals of metal are directed towards the ligand than when it is away from the ligand.

Thus, the $d_{x^2-y^2}$ and d_{z^2} orbitals (axial orbitals) which point towards the axis along the direction of the ligand will experience more repulsion and will be raised in energy ; and the d_{xy} , d_{yz} and d_{zx} orbitals (non-axial) orbitals which are directed between the axis will be lowered in energy relative to the average energy in the spherical crystal field.

Thus, the degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the octahedral complex to yield three orbitals of lower energy, t_{2g} set and two orbitals of higher energy, e_g set. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting and the energy separation is denoted by Δ_0 (the subscript o is for octahedral). Thus, the energy of the two e_g orbitals will increase by $(3/5)\Delta_0$ and that of the three t_{2g} will decrease by $(2/5)\Delta_0$.

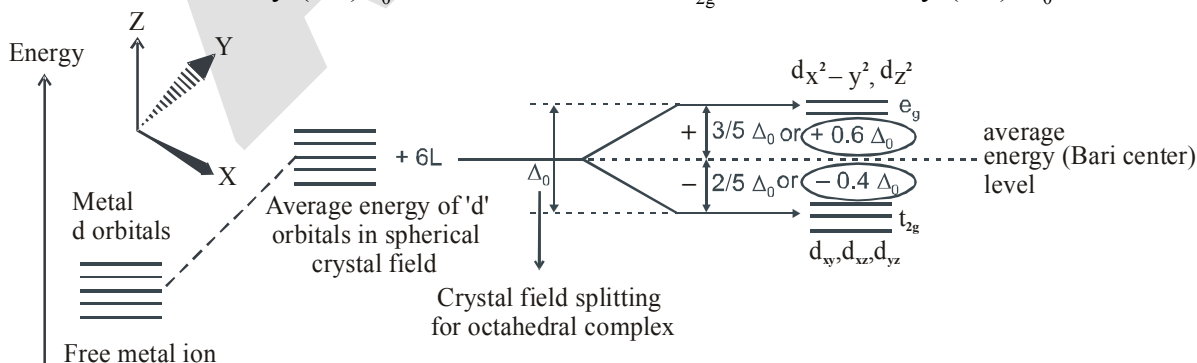


Figure showing crystal field splitting in octahedral complex.

The crystal field splitting, Δ_0 , depends upon the fields produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields in which case, the splitting will be large whereas others produce weak fields and consequently result in small splitting of d orbitals. In general, ligands can be arranged in a series in the order of increasing field strength as given below :



Note : In SCN^- , S is donating atom and in NCS^- , N is donating atom.

Such a series is termed as spectrochemical series. It is an experimentally determined series based on the absorption of light by complexes with different ligands. For d^4 configuration, the fourth electron will singly occupy e_g orbital (according to Hund's rule) or will undergo pairing in t_{2g} orbital, which of these possibilities occurs, depends on the relative magnitude of the crystal field splitting, Δ_0 and the pairing energy, P (P represents the energy required for electron pairing in a single orbital). The two possibilities are :

- (i) If $\Delta_0 < P$, the fourth electron enters in one of the e_g orbitals giving the configuration $t_{2g}^3 e_g^1$. Ligands for which $\Delta_0 < P$ are known as weak field ligands and form high spin complexes.
- (ii) If $\Delta_0 > P$, it becomes more energetically favourable for the fourth electron to occupy a t_{2g} orbital with configuration $t_{2g}^4 e_g^0$. Ligands which produce this effect are known as strong field ligands and form low spin complexes.

Crystal Field stabilising energy in Octahedral field :

Formula : $\text{CFSE} = [-0.4 n_{t_{2g}} + 0.6 n_{e_g}] \Delta_0 + xP$.

Where $n_{t_{2g}}$ & n_{e_g} are number of electron(s) in t_{2g} & e_g orbitals respectively and Δ_0 crystal field splitting energy for octahedral complex. "x" represents the number of electron pairs and P is mean pairing energy.

(b) Crystal field splitting in tetrahedral coordination entities :

In tetrahedral coordination entity formation, the d orbital splitting is inverted and is smaller as compared to the octahedral field splitting. For the same metal, the same ligands and metal-ligand distances, it can be shown that $\Delta_t = (4/9)\Delta_0$. This may be attributed to the following two reasons.

- (i) There are only four ligands instead of six, so the ligand field is only two thirds the size ; as the ligand field splitting is also the two thirds the size and (ii) the direction of the orbitals does not coincide with the direction of the ligands. This reduces the crystal field splitting by roughly further two third. So $\Delta_t = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9} \Delta_0$.

Consequently, the orbital splitting energies are not sufficiently large for forcing pairing and, therefore, low spin configurations are rarely observed.

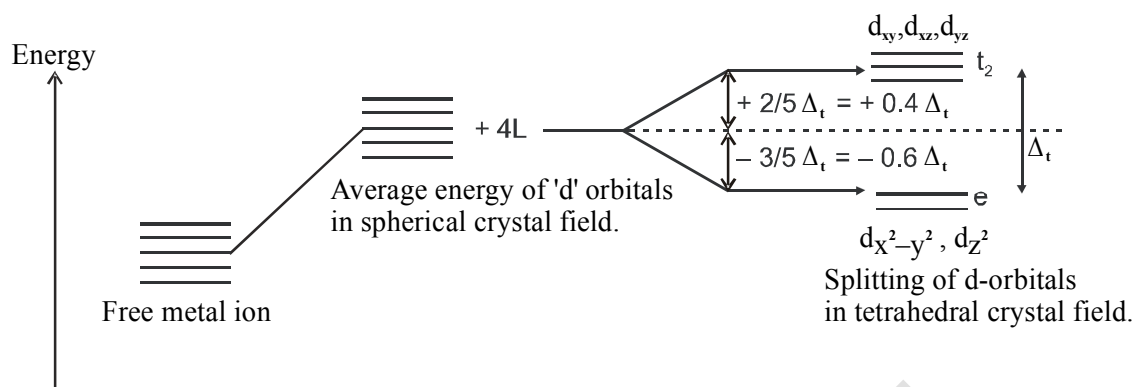


Figure showing crystal field splitting in tetrahedral complex.

Since $\Delta_t < \Delta_o$ crystal field splitting favours the formation of octahedral complexes.

Crystal Field stabilising energy in Tetrahedral field :

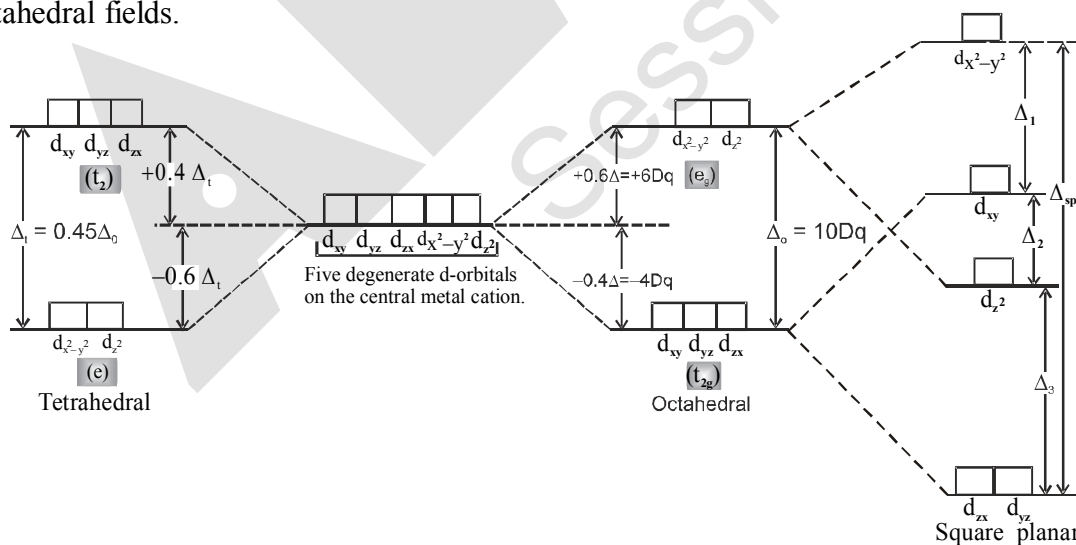
Formula : $CFSE = [-0.6 n_e + 0.4 n_{t_2}] \Delta_t + xP$.

where n_{t_2} & n_e are number of electron(s) in t_2 & e orbitals respectively and Δ_t crystal field splitting energy for tetrahedral complex. "x" represents the number of electron pairs and P is mean pairing energy.

(c) Crystal field splitting in square planar co-ordination entities :

The square planar arrangement of ligands may be considered to be one derived from the octahedral field by removing two trans-ligands located along the Z-axis. In the process, the e_g and t_{2g} sets of orbitals is lifted i.e., these orbitals will no longer be degenerate.

The four ligands in square planar arrangement around the central metal ion are shown in Fig. As the ligands approach through the x and y axis, they would have greatest influence on $d_{x^2-y^2}$ orbital, so the energy of this orbital, will be raised most. The d_{xy} orbital, lying in the same plane, but between the ligands will also have a greater energy though the effect will be less than that on the $d_{x^2-y^2}$ orbitals. On the other hand, due to absence of ligands along Z-axis, the d_{z^2} orbital becomes stable and has energy lower than that of d_{xy} orbital. Similarly d_{yz} and d_{zx} become more stable. The energy level diagram may be represented as shown in figure along with tetrahedral and octahedral fields.



The value of Δ_{sp} has been found larger than Δ_o because of the reason that d_{xz} and d_{yz} orbitals interact with only two ligands in the square planar complexes, while in octahedral complexes the interaction takes place only with four ligands. Δ_{sp} has been found equal to $1.3 \Delta_o$. Thus,

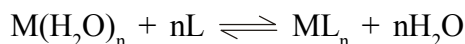
$$\Delta_{sp} = (\Delta_1 + \Delta_2 + \Delta_3) > \Delta_o \quad \text{and} \quad \Delta_{sp} = 1.3 \Delta_o.$$

(E). STABILITY OF COORDINATION COMPOUNDS :

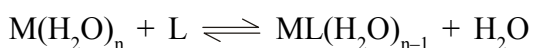
The stability of a coordination compound $[ML_n]$ is measured in terms of the stability constant (equilibrium constant) given by the expression,

$$\beta_n = [ML_n] / [M(H_2O)_n][L]^n$$

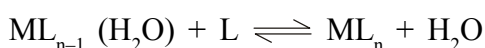
for the overall reaction :



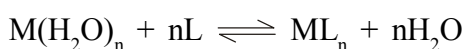
By convention, the water displaced is ignored, as its concentration remains essentially constant. The above overall reaction takes place in steps, with a stability (formation) constant, K_1 , K_2 , K_3 , K_n for each step as represented below :



$$K_1 = [ML(H_2O)_{n-1}] / \{[M(H_2O)_n][L]\}$$



$$K_n = [ML_n] / \{[ML_{n-1}(H_2O)][L]\}$$



$$\beta_n = K_1 \times K_2 \times K_3 \times \dots \times K_n$$

β_n , the stability constant, is related to thermodynamic stability when the system has reached equilibrium. Most of the measurements have been made from aqueous solutions, which implies that the complex is formed by the ligand displacing, water from the aqua complex of the metal ion. Ignoring the charge and taking L as an unidentate ligand, the stepwise formation of the complex is represented as shown above.

K_1 , K_2 , K_3 K_n representing the stepwise stability (or formation) constants.

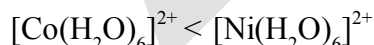
The above is thermodynamic stability criteria, there can be another kind of stability called kinetic stability, which measures the rate of ligand replacement.

(F). FACTORS INFLUENCING THE MAGNITUDE OF C.F.S.E. :

- 1. Different charges on the cation of the same metal :** The cation with a higher oxidation state has a larger value of CFSE than that with lower oxidation state e.g.,



- 2. Same charges on the cation but the number of d-electrons is different :** The metal cation the magnitude of CFSE with the increase of the number of d-electrons, e.g.,



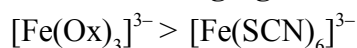
- 3. Quantum number (n) of the d-orbitals of the central metal ion :** As 'n' increase CFSE increases.



- 4. Types of Hybridisation :**

$$\Delta_t = \frac{4}{9} \Delta_o$$

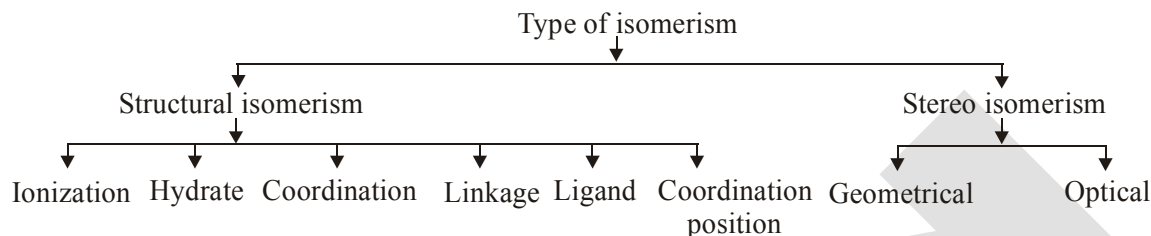
- 5. Presence of cheleting ligand increases CFSE :**



□ ISOMERISM IN COMPLEXES

- (a) Compounds which have the same molecular formula, but differ in their properties due to the difference in structure are called as **Isomers**.
- (b) Isomerism is commonly considered, to be the characteristic of only organic compounds, it is also found although less frequently among inorganic substances.

❖ CLASSIFICATION OF ISOMERISM



(A) Structural isomerism

It arises due to the difference in the type of chemical linkages and distribution of ligands within and outside the coordination sphere.

(a) Ionisation isomerism

The type of isomerism which is due to the exchange of groups or ions between the coordination sphere and the ionisation sphere.

Example. (i) $\text{Co}(\text{NH}_3)_4\text{Br}_2\text{SO}_4$ can be represented as $[\text{Co}(\text{NH}_3)_4\text{Br}_2]\text{SO}_4$ (red violet) and $[\text{Co}(\text{NH}_3)_4\text{SO}_4]\text{Br}_2$ (red). These complexes give sulphate ion and bromide ion respectively.

(ii) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$ and $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$

(iii) $[\text{Co}(\text{NH}_3)_4(\text{NO}_3)_2]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_4\text{SO}_4](\text{NO}_3)_2$

(b) Hydrate isomerism

This type of isomerism is due to presence of different number of water molecules inside a coordination sphere.

Example. $\text{Cr}(\text{H}_2\text{O})_6\text{Cl}_3$ has four possible structures

(i) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ violet

(ii) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ green

(iii) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ dark green.

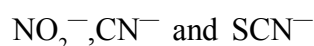
(iv) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$ dark green.

These complexes differ from one another with respect to the number of water molecules acting as ligands. Other hydrate isomers are

$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{H}_2\text{O}$

(c) Linkage isomerism

(i) This type of isomerism arises due to presence of ambidentate ligands like



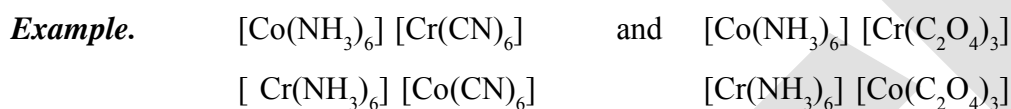
(ii) These ligands have two donor atoms but at a time only one atom is directly linked to the central metal atom of the complex.

(iii) Such type of isomers are distinguished by infra red (I.R.) spectroscopy.

- Example.**
- $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$
 - In NO_2^- ligand, The coordinating sites are nitrogen (i.e., NO_2^- Nitro ligand) or through oxygen (i.e. ONO Nitrito ligand)
 - The nitro isomer is yellow and is stable to acids whereas nitrito isomer is red and is decomposed by acids.

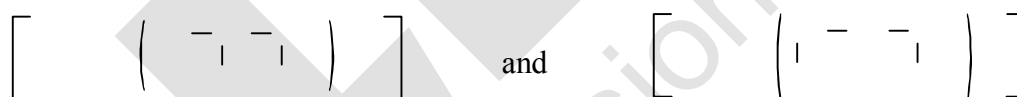
(d) **Coordination isomerism**

- This type of isomerism is exhibited when the complex has two complex ions in it - 'cationic and anionic'.
- This type of isomerism is caused by the interchange of ligands between the two complex ions of the same complex.



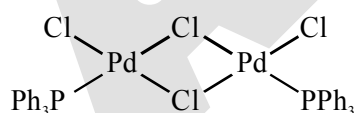
(e) **Ligand isomerism**

- Ligands with $\text{C}_3\text{H}_6(\text{NH}_2)_2$ have two different structures i.e. 1, 3-diamino propane and 1, 2-diaminopropane(propylene diamine).
- Those complexes which have same molecular formula, but differ with respect to their ligands are called as **Ligand isomers**.

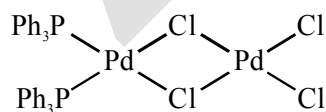


(f) **Co-ordination Position Isomerisation :**

It is shown by polynuclear complexes, due interchange of ligands between the different metal nuclei.

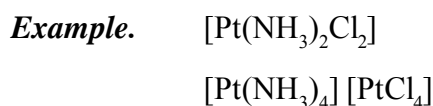


Example.



(g) **Polymerization Isomerism :**

This is not true isomerism because it occurs between compounds having the same empirical formula, but different molecular weights.



(B) Stereo isomerism

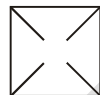
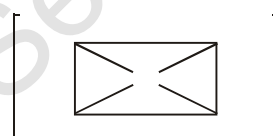
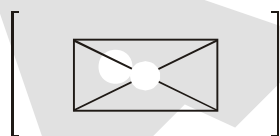
They have same molecular formula, same constitution, they differ only with respect to the spatial orientation of ligands in space around the metal ion. The two stereo isomers which are possible Geometrical and Optical.

(a) Geometrical isomerism

- (i) The ligands occupy different positions around the central metal ion.
- (ii) When two identical ligands are co-ordinated to the metal ion from same side then it is **cis isomer**. (Latin, cis means same).
- (iii) If the two identical ligands are co-ordinated to the metal ion from opposite side then it is **trans isomer** (in Latin, trans means across).

Geometrical isomers with co-ordination number = 4 (Square planar complexes)

- (i) Complexes with general formula, Ma_2b_2 (where both a and b are monodentate) can have cis-and trans isomers.

**Example.**

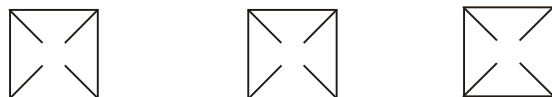
- (ii) Complexes with general formula Ma_2bc can have cis - and trans-isomers.



Example. $[\text{Pt}(\text{NH}_3)_2\text{ClBr}]$



(iii) Complexes with general formula, Mabcd can have three isomers.



Geometrical isomers with Co-ordination number = 6

(i) Complexes with general formula Ma_4b_2 can have cis - and trans-isomers.

Example. $[\text{Fe}(\text{NH}_3)_4\text{Cl}_2]$



(iii) **Facial and Meridional isomerism** (Ma_3b_3)



Other 6-Coordinated geometrical isomers are

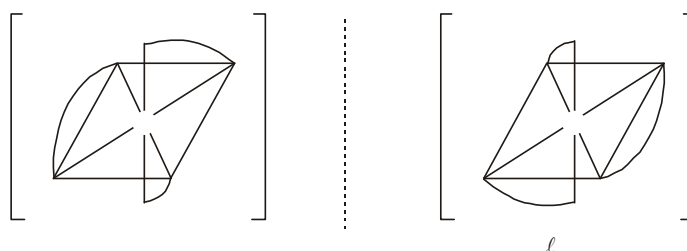
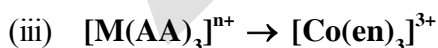
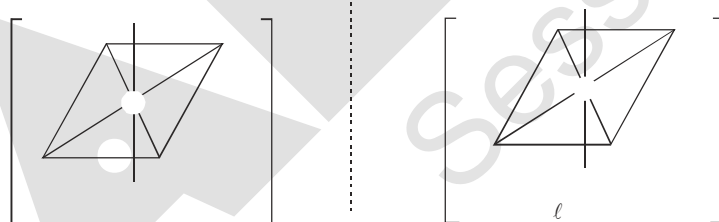
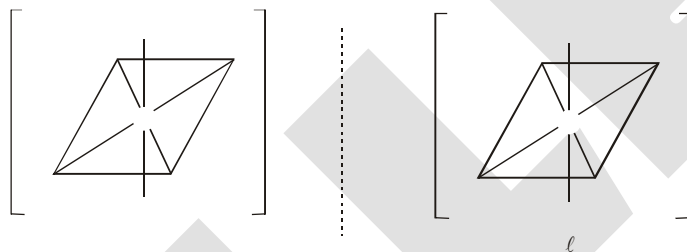
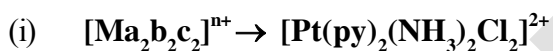
Note :	General formula	Total No. of geometrical isomers
	Mabcdef	15
	Ma_2bcde	9
	$\text{Ma}_2\text{b}_2\text{cd}$	6
	$\text{Ma}_2\text{b}_2\text{c}_2$	5
	Ma_3bcd	4
	$\text{Ma}_3\text{b}_2\text{c}$	3
	Ma_3b_3	2
	Ma_4bc	2
	Ma_4b_2	2
	Ma_5b	Not possible
	Ma_6	Not possible

Here M = central atom a, b, c, d, e, f = Monodentate ligands

(b) **Optical isomers**

- (i) Optically active complexes are those which are nonsuperimposable over the mirror image structure.
- (ii) An optically active complex is one which is asymmetric in nature i.e., not divisible into two identical halves.
- (iii) The complex which rotates plane polarised light to left hand side is **laevo rotatory** i.e. ' ℓ ' or '-' and if the complex rotates the plane polarised light to right hand side then it is **dextro rotatory** 'd' or '+'.

Session 2019-20
- (iv) Thus complexes which have same physical and chemical properties but differ in their action towards plane polarised light are called as **optical isomers**.
- (v) The 'd' and ' ℓ ' isomers of a compound are called as **Enantiomers or Enantiomorphs**.
- (vi) Optical isomerism is expected in tetrahedral complexes of the type Ma_2bcd .

Optical isomers with Co-ordination number = 6

NUMBER OF POSSIBLE ISOMERS FOR SPECIFIC COMPLEXES

Formula	Number of stereoisomers	Pairs of Enantiomers
	2	0
	2	0
	2	0
	5	1
	15	6
	30	15
	6	1
	8	2
	3	0
M(AA)(BC)de	10	5
M(AB)(AB)cd	11	5
M(AB)(CD)ef	20	10
M(AB) ₃	4	2

Note : Uppercase letters represent chelating ligands and lowercase letters represent monodentate ligands.

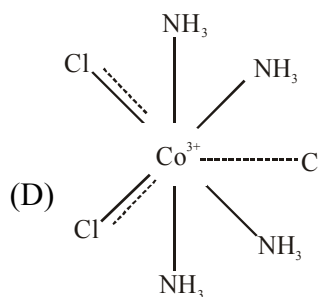
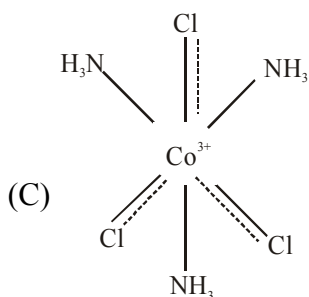
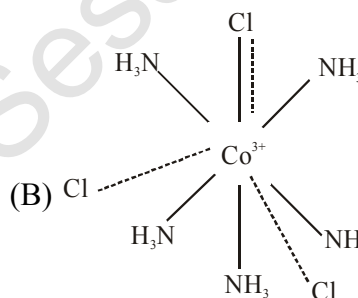
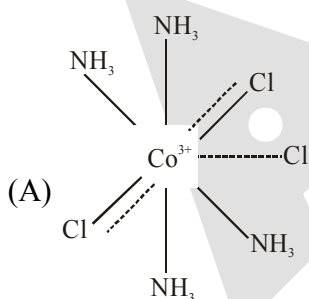
EXERCISE # O-1

SINGLE OPTION CORRECT :**Double salt and complex compound**

- Some salts although containing two different metallic elements give test for one of them in solution. Such salts are
(A) complex salt (B) double salt (C) normal salt (D) none of these
- Aqueous solution of FeSO_4 gives tests for both Fe^{2+} and SO_4^{2-} but after addition of excess of KCN, solution ceases to give test for Fe^{2+} . This is due to the formation of
(A) the double salt $\text{FeSO}_4 \cdot 2\text{KCN} \cdot 6\text{H}_2\text{O}$ (B) $\text{Fe}(\text{CN})_3$
(C) the complex ion $[\text{Fe}(\text{CN})_6]^{4-}$ (D) the complex ion $[\text{Fe}(\text{CN})_6]^{3-}$

Werner theory

- Consider the following statements:
According to Werner's theory.
(a) Ligands are connected to the metal ions by covalent bonds.
(b) Secondary valencies have directional properties
(c) Secondary valencies are non-ionisable
Of these statements:
(A) a, b and c are correct (B) b and c are correct
(C) a and c are correct (D) a and b are correct
- A complex of platinum, ammonia and chloride produces four ions per molecule in the solution. The structure consistent with the observation is:
(A) $[\text{Pt}(\text{NH}_3)_4]\text{Cl}_4$ (B) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (C) $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$ (D) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$
- Which of the following Werner's complex has least electrical conductivity ?



Classification of ligand

6. How many EDTA^{4-} molecules are required to make an octahedral complex with a Ca^{2+} ion ?
(A) Six (B) Three (C) One (D) Two
7. π -bonding is not involved in :
(A) ferrocene (B) dibenzene chromium (C) Zeise's salt (D) Grignard reagent
8. Which of the following is not considered as an organometallic compound ?
(A) Ferrocene (B) Cis-platin (C) Ziese's salt (D) Grignard reagent
9. Diethylene triamine is:
(A) Chelating agent (B) Polydentate ligand
(C) Tridentate ligand (D) All of these
10. Which of the following species is not expected to be a ligand
(A) NO^+ (B) NH_4^+ (C) $\text{NH}_2\text{--NH}_3^+$ (D) CO
11. The disodium salt of ethylene diamine tetraacetic acid can be used to estimate the following ion(s) in the aqueous solution
(A) Mg^{2+} ion (B) Ca^{2+} ion (C) Na^+ ion (D) both Mg^{2+} and Ca^{2+}
12. Which of the following ligand does not act as bidentate ligand
(A) dipy (B) dien (C) gly⁻ (D) dmg⁻

Synergic bonding

13. Which of the following order is correct for the IR vibrational frequency of CO.
(A) $[\text{Fe}(\text{CO})_4]^{2-} < [\text{Co}(\text{CO})_4]^- < [\text{Ni}(\text{CO})_4]$ (B) $[\text{Fe}(\text{CO})_4]^{2-} > [\text{Co}(\text{CO})_4]^- > [\text{Ni}(\text{CO})_4]$
(C) $[\text{Fe}(\text{CO})_4]^{2-} > [\text{Co}(\text{CO})_4]^- < [\text{Ni}(\text{CO})_4]$ (D) $[\text{Fe}(\text{CO})_4]^{2-} < [\text{Co}(\text{CO})_4]^- > [\text{Ni}(\text{CO})_4]$
14. In the isoelectronic series of metal carbonyl, the C–O bond strength is expected to increase in the order.
(A) $[\text{Mn}(\text{CO})_6]^+ < [\text{Cr}(\text{CO})_6] < [\text{V}(\text{CO})_6]^-$ (B) $[\text{V}(\text{CO})_6]^- < [\text{Cr}(\text{CO})_6] < [\text{Mn}(\text{CO})_6]^+$
(C) $[\text{V}(\text{CO})_6]^- < [\text{Mn}(\text{CO})_6]^+ < [\text{Cr}(\text{CO})_6]$ (D) $[\text{Cr}(\text{CO})_6] < [\text{Mn}(\text{CO})_6]^+ < [\text{V}(\text{CO})_6]^-$
15. Which of the following has higher stretching frequency for C–O bond -
(A) $[\text{Ni}(\text{CO})_3\text{PF}_3]$ (B) $[\text{Ni}(\text{CO})_3(\text{PMe}_3)]$
(C) both have equal stretching frequency (D) None of these
16. Which of the following has higher multiple bond character in M–C bond -
(A) $[\text{Ni}(\text{CO})_4]$
(B) $[\text{Co}(\text{CO})_4]^-$
(C) $[\text{Fe}(\text{CO})_4]^{2-}$
(D) (B) and (C) both have equal multiple bond character in M–C bond
17. The V–C distance in $\text{V}(\text{CO})_6$ and $[\text{V}(\text{CO})_6]^-$ are respectively (in pm)-
(A) 200, 200 (B) 193, 200 (C) 200, 193 (D) 193, 193

Co-ordination number and E.A.N.

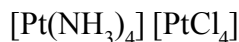
18. Among the following complexes which can act as oxidising agent.
 (A) $[\text{Mn}(\text{CO})_6]$ (B) $[\text{Mn}(\text{CO})_6]^+$ (C) $[\text{Mn}(\text{CO})_5]^-$ (D) $[\text{V}(\text{CO})_6]$
19. Which of the following statement is correct regarding the compound " $[(\text{CO})_3\text{Fe}(\text{CO})_3\text{Fe}(\text{CO})_3]$ ".
 (A) The $d_{\text{C-O}}$ (bridging) is greater than $d_{\text{C-O}}$ (terminal)
 (B) The bond order of bridging C – O bond is greater than that of terminal C – O bond
 (C) The E.A.N. value of each Fe-atom is 35
 (D) The oxidation state of Fe in this complex is (–I)
20. How many π electrons are donated by C_5H_5^- ligand -
 (A) 2 (B) 4 (C) 5 (D) 6
21. Effective atomic number of $\text{Co}(\text{CO})_4$ is 35, hence it is less stable. It attains stability by
 (A) Oxidation of Co (B) Reduction of Co
 (C) Dimerization (D) Both (B) & (C)
22. In the complex $\text{Fe}(\text{CO})_x$, the value of x is:
 (A) 3 (B) 4 (C) 5 (D) 6
23. The EAN of platinum in potassium hexachloroplatinate (IV) is:
 (A) 46 (B) 86 (C) 36 (D) 84
24. The EAN of metal atoms in $\text{Fe}(\text{CO})_2(\text{NO})_2$ and $\text{Co}_2(\text{CO})_8$ respectively are
 (A) 34, 35 (B) 34, 36 (C) 36, 36 (D) 36, 35

Naming of complex compound

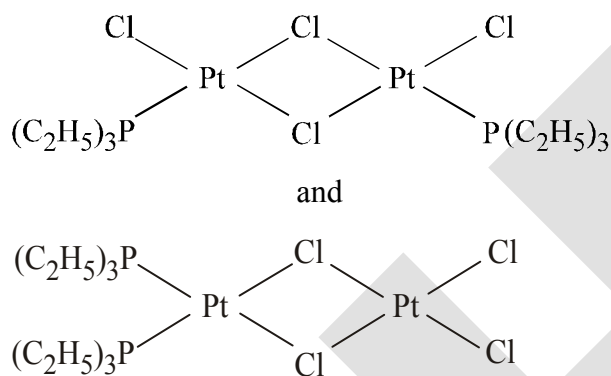
25. The IUPAC name for the coordination compound $\text{Ba}[\text{BrF}_4]_2$ is
 (A) Barium tetrafluorobromate (V) (B) Barium tetrafluorobromate (III)
 (C) Barium bis (tetrafluorobromate) (III) (D) none of these
26. The number of ions formed, when bis (ethane-1,2-diamine) copper (II) sulphate is dissolved in water will be
 (A) 1 (B) 2 (C) 3 (D) 4
27. The IUPAC name of the Wilkinson's catalyst $[\text{Rh Cl}(\text{P Ph}_3)_3]$ is
 (A) Chloridotris(triphenylphosphine)rhodium(I)
 (B) Chloridotris(triphenylphosphine)rhodium(IV)
 (C) Chloridotris(triphenylphosphine)rhodium(0)
 (D) Chloridotris(triphenylphosphine)rhodium(VI)
28. The formula for the compound tris (ethane-1, 2-diamine)cobalt (III) sulphate is
 (A) $[\text{Co}(\text{en})_3]\text{SO}_4$ (B) $[\text{Co}(\text{SO})_4(\text{en})_3]$
 (C) $[\text{Co}(\text{en})_3](\text{SO}_4)_2$ (D) $[\text{Co}(\text{en})_3]_2(\text{SO}_4)_3$

Structural isomerism

29. Which of the following statement is **INCORRECT** regarding the following compound



- (A) It is the polymerisation isomer of $[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]$
 (B) E.A.N. of cationic part is equal to that of anionic part
 (C) It is the co-ordination isomer of $[\text{Pt}(\text{NH}_3)_3\text{Cl}][\text{Pt}(\text{NH}_3)\text{Cl}_3]$
 (D) Synergic bonding is not involved in the complex
30. The type of isomerism present in pentaamminenitrochromium (III) chloride is :
 (A) optical (B) linkage (C) hydrate (D) polymerisation
31. The complexes given below show:



- (A) Optical isomerism (B) Co-ordination isomerism
 (C) Geometrical isomerism (D) Co-ordination position isomerism
32. Which of the following complex shows ionization isomerism
 (A) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (B) $[\text{Cr}(\text{en})_2]\text{Cl}_2$
 (C) $[\text{Cr}(\text{en})_3]\text{Cl}_3$ (D) $[\text{CoBr}(\text{NH}_3)_5]\text{SO}_4$
33. Find the name of the hydrate isomer of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$, which is having lowest electrical conductivity excluding zero value of conductivity.
 (A) Hexaaquachromium(III) chloride
 (B) Tetraaquadichloridochromium(III) chloride dihydrate
 (C) Pentaquachloridochromium(III) chloride monohydrate
 (D) Triaquatrichloridochromium(III) chloride trihydrate

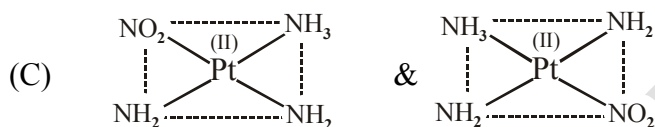
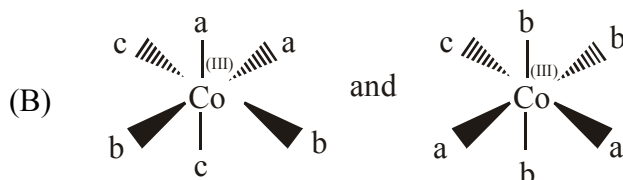
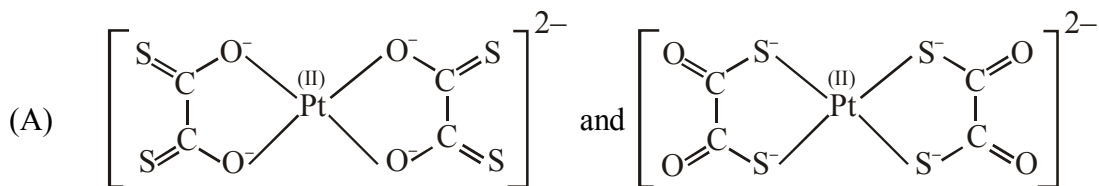
Stereoisomerism

34. Which of the following complex shows optical isomerism -
 (A) $[\text{Cd}(\text{CN})_4]^{2-}$ (B) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
 (C) $[\text{Zn}(\text{gly})_2]^0$ (D) $[\text{Ni}(\text{dmg})_2]^0$

35. How many coordination isomers of $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$ show geometrical isomerism.

- (A) All (B) One (C) Two (D) None

36. Identify the pair of complex which are stereoisomer of each other -



- (D) All of the above

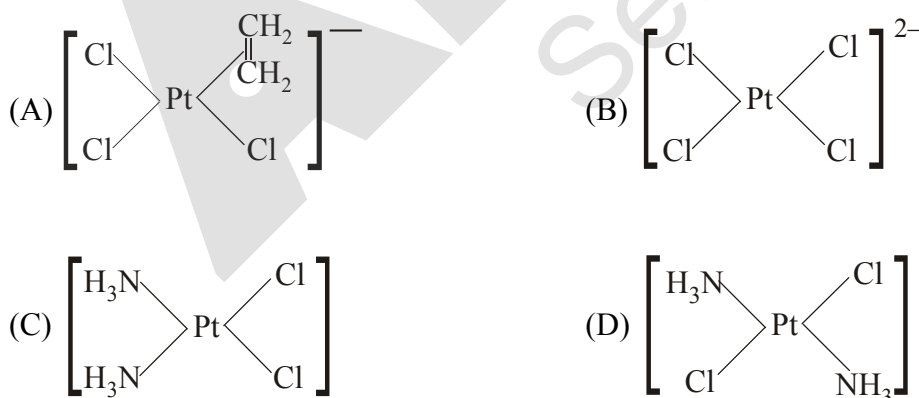
37. Find complex which have maximum number of stereoisomers -

- (A) $[\text{Ma}_3\text{b}_3]$ (B) $[\text{Ma}_3\text{b}_2\text{c}]$ (C) $[\text{Ma}_2\text{b}_2\text{c}_2]$ (D) $[\text{M}(\text{AA})_2\text{a}_2\text{b}_2]$

38. In which of the following pairs both the complexes show optical isomerism ?

- (A) $\text{cis}-[\text{Cr}(\text{C}_2\text{O}_4)_2\text{Cl}_2]^{3-}$, $\text{trans}-[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$
 (B) $[\text{Co}(\text{en})_3]\text{Cl}_3$, $\text{cis}-[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
 (C) $[\text{PtCl}(\text{dien})]\text{Cl}$, $[\text{NiCl}_2\text{Br}_2]^{2-}$
 (D) $[\text{Co}(\text{NO}_3)_3(\text{NH}_3)_3]$, $\text{cis}-[\text{Pt}(\text{en})_2\text{Cl}_2]$

39. Which of the following is considered to be an anticancer species ?



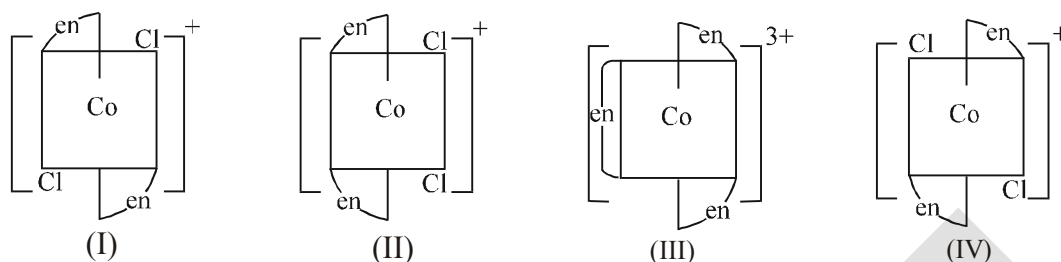
40. Which of the following can exhibit geometrical isomerism ?

- (A) $[\text{MnBr}_4]^{2-}$ (B) $[\text{Pt}(\text{NH}_3)_3\text{Cl}]^+$
 (C) $[\text{PtCl}_2(\text{P}(\text{C}_2\text{H}_5)_3)_2]$ (D) $[\text{Fe}(\text{H}_2\text{O})_5\text{NOS}]^{2+}$

41. The oxidation state of Mo in its oxo-complex species $[\text{Mo}_2\text{O}_4(\text{C}_2\text{H}_4)_2(\text{H}_2\text{O})_2]^{2-}$ is:

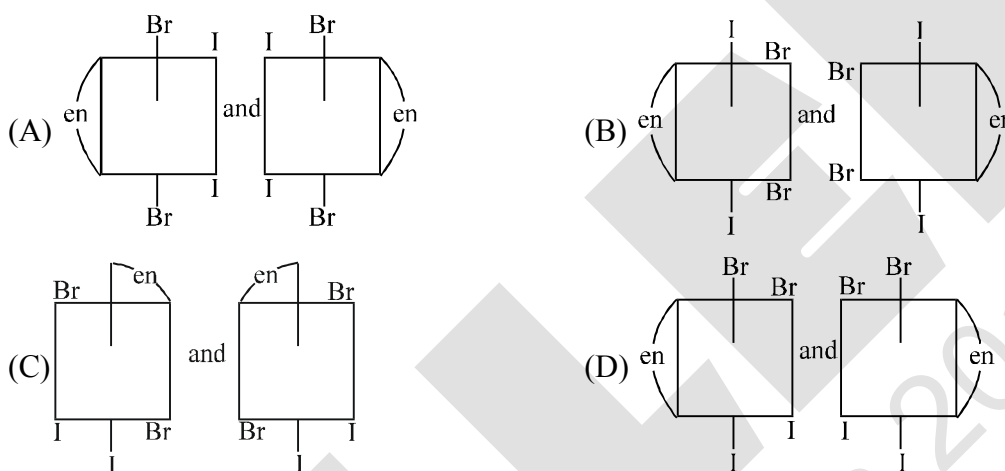
- (A) +2 (B) +3 (C) +4 (D) +5

42. Which of the following ions are optically active?



- (A) I only (B) II only (C) II and III (D) IV only

43. The complex ion has two optical isomers. Their **CORRECT** configurations are:



V.B.T/CFT

44. Which of the following complex is coloured and diamagnetic -

- (A) MnO_4^{2-} (B) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ (C) $[\text{CrCl}_6]^{3-}$ (D) CrO_4^{2-}

45. What is the hybridisation of Fe in $[\text{Fe}(\text{CO})_4]^{-2}$

- (A) d^2sp^3 (B) sp^3d^2 (C) sp^3d (D) sp^3

46. One unknown complex has the spin only magnetic moment is of 1.73 BM. As per the C. F. T., complex is.

- (A) d^7 , Oh-field, with Strong Field Ligand (B) d^9 , sq.planar-field, with Strong Field Ligand
(C) d^9 , Td, field with Weak Field Ligand (D) All of these

47. Identify tetrahedral species which has maximum magnetic moment value :

- (A) $[\text{CuCl}_4]^{2-}$ (B) $[\text{CoCl}_4]^{2-}$ (C) $[\text{FeCl}_4]^{2-}$ (D) $[\text{AlCl}_4]^-$

48. In the co-ordination compound $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$ oxidation state of Fe is -

- (A) +1 (B) +2 (C) +3 (D) +4

49. Which one of the following complex is an outer orbital complex -

- (A) $[\text{Ni}(\text{H}_2\text{O})_6]^{+2}$ (B) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$
(C) $[\text{Fe}(\text{NH}_3)_6]^{+2}$ (D) All of these

50. Which of the following is diamagnetic and sp^3 hybridised -
 (A) $[\text{NiCl}_4]^{2-}$ (B) $[\text{Ni}(\text{CN})_4]^{4-}$ (C) $[\text{Ni}(\text{CN})_4]^{2-}$ (D) $[\text{NiCl}_2(\text{PPh}_3)_2]$
51. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (atomic number of Cr = 24) has a magnetic moment of 3.83 B.M. The **CORRECT** distribution of 3d electrons in the chromium present in the complex is:
 (A) $3d_{xy}^1, 3d_{yz}^1, 3d_{zx}^1$ (B) $3d_{xy}^1, 3d_{yz}^1, 3d_{z^2}^1$
 (C) $3d_{(x^2-y^2)}^1, 3d_{z^2}^1, 3d_{xz}^1$ (D) $3d_{xy}^1, 3d_{(x^2-y^2)}^1, 3d_{yz}^1$
52. $[\text{Fe}(\text{H}_2\text{O})_6]^{+2}$ has Crystal Field Splitting Energy value $10,400 \text{ cm}^{-1}$ and pairing energy value $17,600 \text{ cm}^{-1}$ then it is :
 (A) Low spin complex (B) Paramagnetic in nature
 (C) Diamagnetic in nature (D) None of these
53. In which of the following coordination entities, the magnitude of Δ_0 [CFSE in octahedral field] will be maximum? :
 (A) $[\text{Co}(\text{CN})_6]^{3-}$ (B) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
 (C) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (D) $[\text{Co}(\text{NH}_3)_6]^{3+}$
54. The number of unpaired electrons calculated in $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$ are :
 (A) 4 and 4 (B) 0 and 2 (C) 2 and 4 (D) 0 and 4
55. An ion M^{2+} , forms the complexes $[\text{M}(\text{H}_2\text{O})_6]^{2+}$, $[\text{M}(\text{en})_3]^{2+}$ and $[\text{MBr}_6]^{4-}$, match the complex with the appropriate colour.
 (A) Green, blue and red (B) Blue, red and green
 (C) Green, red and blue (D) Red, blue and green
56. Formula of ferrocene is:
 (A) $[\text{Fe}(\text{CN})_6]^{4-}$ (B) $[\text{Fe}(\text{CN})_6]^{3+}$ (C) $[\text{Fe}(\text{CO})_5]$ (D) $[\text{Fe}(\text{C}_5\text{H}_5)_2]$
57. $\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{NH}_3)_4]^{2+}$ do not differ in
 (A) magnetic moment (B) oxidation number of Ni
 (C) geometry (D) EAN
58. A complex of certain metal has the magnetic moment of 4.91 BM whereas another complex of the same metal with same oxidation state has zero magnetic moment. The metal ion could be
 (A) Co^{2+} (B) Mn^{2+} (C) Fe^{2+} (D) Fe^{3+}
59. The tetrahedral $[\text{CoI}_4]^{2-}$ and square planar $[\text{PdBr}_4]^{2-}$ complex ions are respectively
 (A) low spin, high spin (B) high spin, low spin
 (C) both low spin (D) both high spin
60. Which one of the following species does not represent cationic species of vanadium formed in aqueous solution
 (A) VO_2^+ (B) VO^{2+} (C) $[\text{V}(\text{H}_2\text{O})_6]^{3+}$ (D) VO_2^{2+}

61. On treatment of $[\text{Ni}(\text{NH}_3)_4]^{2+}$ with concentrated HCl , two compounds I and II having the same formula, $[\text{NiCl}_2(\text{NH}_3)_2]$ are obtained, I can be converted into II by boiling with dilute HCl . A solution of I reacts with oxalic acid to form $[\text{Ni}(\text{C}_2\text{O}_4)(\text{NH}_3)_2]$ whereas II does not react. Point out the correct statement of the following
- (A) I cis, II trans; both tetrahedral (B) I cis, II trans; both square planar
(C) I trans, II cis; both tetrahedral (D) I trans, II cis; both square planar
62. Among the following, the compound that is both paramagnetic and coloured is
- (A) $\text{K}_2\text{Cr}_2\text{O}_7$ (B) $(\text{NH}_4)_2[\text{TiCl}_6]$ (C) VOSO_4 (D) $\text{K}_3[\text{Cu}(\text{CN})_4]$
63. The magnetic moment of $[\text{NiX}_4]^{2-}$ ion is found to be zero. Then the metal of the complex ion is (X = monodentate anionic ligand).
- (A) sp^3 hybridised (B) sp^2 hybridised (C) dsp^2 hybridised (D) d^2sp hybridised
64. For which of the following types of d^n configuration, the number of unpaired electrons in octahedral complexes remains same irrespective of the ligand field strength.
- (A) d^3 (B) d^4 (C) d^5 (D) d^6
65. Which of the following electronic arrangement gives the highest value of the magnetic moment?
- (A) d^6 , strong field (B) d^7 , high spin (C) d^4 , weak field (D) d^2 , strong field
66. Select appropriate ligand for given complex
- $[\text{Co}(\text{.....})_6]^{+x}$; $\mu = 0 \text{ BM}$
- (A) $\text{C}_2\text{O}_4^{2-}$ (B) en (C) H_2O (D) F^-
67. According to C.F.T., ligands are treated as -
- (A) Point charges (B) Lewis acids (C) Proton donor (D) All of the above
68. Which of the following is correct electronic configuration of 3d orbital in excited state of central metal ion, when $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ absorbed yellow-green light.
- (A) 3d

			1	
--	--	--	---	--

 (B) t_{2g}^1, e_g^0
(C) t_{2g}^1, e_g^1 (D) t_{2g}^0, e_g^1
69. If $\lambda_{\text{absorbed}}$ for d-d transition is in order $[\text{Ti}(\text{X})_6]^{3+} > [\text{Ti}(\text{Y})_6]^{3+} > [\text{Ti}(\text{Z})_6]^{3+}$.
Select correct order of strength of ligands (X, Y, Z are monodentate ligand)-
- (A) $\text{Z} > \text{Y} > \text{X}$ (B) $\text{X} > \text{Y} > \text{Z}$
(C) $\text{Z} > \text{X} > \text{Y}$ (D) Not predictable

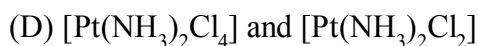
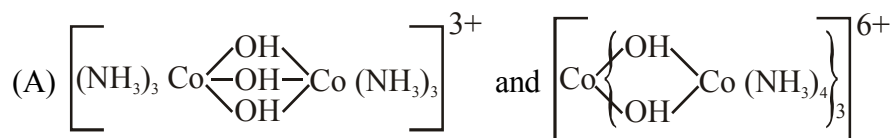
EXERCISE # O-2

MORE THAN ONE MAY BE CORRECT

- Which of the following exhibit geometrical isomerism (M stands for a metal, and a and b are achiral ligands)?
(A) Ma_2b_2 (Sq. Pl.) (B) Ma_4b_2 (C) Ma_5b (D) Ma_6
- Which of the following statement(s) is (are) **CORRECT** ?
(A) The oxidation state of iron in sodium nitroprusside $\text{Na}_2[\text{Fe}(\text{CN})_5(\text{NO})]$ is +II.
(B) $[\text{Ag}(\text{CN})_2]^-$ is linear in shape.
(C) In $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, Fe is d^2sp^3 hybridized.
(D) In $\text{Ni}(\text{CO})_4$, the oxidation state of Ni is zero.
- Which of the following compound(s) show(s) optical isomerism.
(A) $[\text{Pt}(\text{bn})_2]^{2+}$ (B) $[\text{CrCl}_2(\text{en})_2]^+$ (C) $[\text{Co}(\text{en})_3][\text{CoF}_6]$ (D) $[\text{Zn}(\text{gly})_2]$
- Select **INCORRECT** statement(s) for $[\text{Cu}(\text{CN})_4]^{3-}$, $[\text{Cd}(\text{CN})_4]^{2-}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ complex ion.
(A) Both $[\text{Cd}(\text{CN})_4]^{2-}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ have square planar geometry
(B) $[\text{Cu}(\text{CN})_4]^{3-}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ have equal no. of unpaired electron
(C) $[\text{Cu}(\text{CN})_4]^{3-}$ and $[\text{Cd}(\text{CN})_4]^{2-}$ can be separated from the mixture on passing H_2S gas.
(D) All the three complexes have magnetic moment equal to zero.
- Which of the following will have two stereoisomeric forms?
(A) $[\text{Cr}(\text{NO}_3)_3(\text{NH}_3)_3]$ (B) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$
(C) $[\text{CoCl}_2(\text{en})_2]^+$ (D) $[\text{CoBrCl}(\text{Ox})_2]^{3-}$
- Which is / are **NOT** correctly matched.

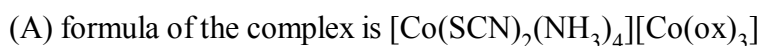
Complex compounds	IUPAC name
(A) $\text{K}[\text{CrF}_4\text{O}]$	Potassium tetrafluoridooxidochromate(V)
(B) $\text{Na}[\text{BH}(\text{OCH}_3)_3]$	Sodium hydridotrimethoxyborate(III)
(C) $[\text{Be}(\text{CH}_3\text{—CO—CH}_2\text{—CO—C}_6\text{H}_5)_2]^\circ$	Bis(benzoylacetato)beryllium(III)
(D) $\text{H}[\text{AuCl}_4]$	Hydrogen tetrachloroaurate(III)
- Which of the following statement(s) is/are **INCORRECT**
(A) In $[\text{CoBrCl}(\text{en})_2]^+$ geometrical isomerism exists, while optical isomerism does not exist
(B) Potassium aquadicyanidosuperoxidoperoxidochromate(III) is IUPAC name for $\text{K}_2[\text{Cr}(\text{CN})_2\text{O}_2(\text{O}_2)(\text{H}_2\text{O})]$
(C) There are 3 geometrical isomers and 15 stereoisomers possible for $[\text{Pt}(\text{NO}_2)(\text{NH}_3)(\text{NH}_2\text{OH})(\text{py})]^+$ and $[\text{PtBrCl}(\text{NO}_2)(\text{NH}_3)(\text{py})]$ respectively
(D) cis and trans forms are not diastereomers of each other

8. Which of the following complexes are polymerisation isomers :



9. Which of the following is **CORRECT** about

Tetraamminedithiocyanato-Scobalt(III) tris(oxalato)cobaltate(III)

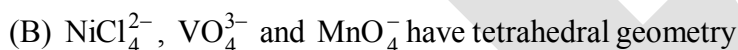
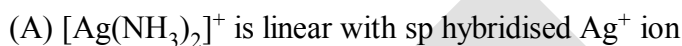


(B) It is a chelating complex and show linkage isomerism.

(C) It shows optical isomerism.

(D) It shows geometrical isomerism.

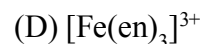
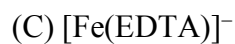
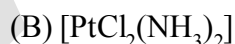
10. Which is **CORRECT** statement(s)?



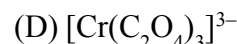
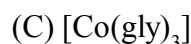
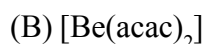
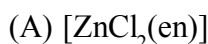
11. In which of the following complex(s) spin only magnetic moment is independent, from the nature of ligand. (L = monodentate ligand) -



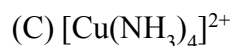
12. Which of the following compound(s) can show optical isomerism ?



13. Which of the following compounds are resolvable into d or ℓ -forms?



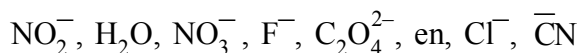
14. Which of the following ion is/are diamagnetic and non planar ?



EXERCISE # S-1

INTEGER TYPE

1. Find number of ligands which is / are stronger ligand as compared to NH_3



2. If crystal field stabilization energy of $[\text{ML}_6]^{+n}$ is $-0.8 \Delta_0$.

Find minimum number of electrons in t_{2g} orbitals of metal ion ?

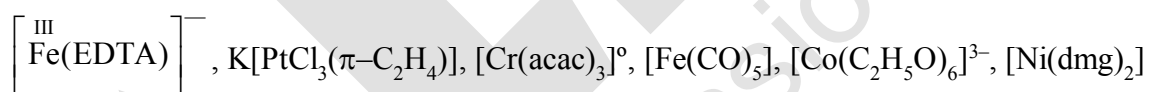
3. Find number of Co–N linkage in,

Pentaamminecobalt(III)– μ –amidodiamminetriaquacobalt(III) chloride.

4. Find the EAN value of central atom of $[\text{Fe}(\pi\text{-C}_4\text{H}_4)(\text{CO})_3]$

5. Find the maximum number of atoms lying in one plane for $[\text{Cr}(\text{CN})_6]^{3-}$

6. Select complex in which metal–carbon linkage(s) is / are present :



7. Find out the total number of geometrical isomers of $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3]$.

8. Find the value of E.A.N of $[\text{Pd}(\text{CO})_4]^{+2}$ (atomic number = 46) :

9. A co-ordination compound have magnetic moment 5.92 B.M. Find out the number of unpaired electron(s) in the compound.

10. Find the number of optically active isomers for $[\text{Pd}(\text{en})_2(\text{NH}_3)(\text{H}_2\text{O})]^{4+}$ cation.

EXERCISE # S-2

MATCH THE COLUMN :

1. Match the complexes in column-I with the EAN of central atom in column-II:

Column-I	Column-II
(A) $[\text{Fe}(\text{CO})_4]^{2-}$	(P) 34
(B) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$	(Q) 35
(C) $\text{K}_2[\text{Ni}(\text{CN})_4]$	(R) 36
(D) $[\text{Cu}(\text{NH}_3)_4]^{2+}$	(S) 37

2. Match the complexes in column-I with their properties in column-II:

Column-I	Column-II
(A) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$	(P) $\mu = 0$ B.M.
(B) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$	(Q) octahedral
(C) $[\text{Ag}(\text{CN})_2]^-$	(R) $\mu = \sqrt{15}$ B.M.
(D) $\text{K}_4[\text{Fe}(\text{CN})_6]$	(S) NO^+ ligand

3. Match the complexes in column-I with their stereoproperties in column-II:

Column-I	Column-II
(A) $[\text{CoCl}_3(\text{NH}_3)_3]$	(P) Has a facial isomer
(B) $[\text{Cr}(\text{ox})_3]^{3-}$	(Q) Cis form is optically active
(C) $[\text{CrCl}_2(\text{ox})_2]$	(R) Trans form is optically inactive
(D) $[\text{RhCl}_3(\text{Py})_3]$	(S) Has a meridional form
	(T) Two optically active isomer

4. Match each coordination compound in List-I with co-ordination number of central metal/ion from List-II and select the correct answer using the code given below the lists.

List-I	List-II
(P) $[\text{Co}(\text{en})_3]^{3+}$	(1) 6
(Q) $[\text{Ca}(\text{EDTA})]^{2-}$	(2) 4
(R) $[\text{Ni}(\text{CO})_4]$	(3) 2
(S) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$	(4) 5

Code :

	P	Q	R	S
(A)	2	1	2	3
(B)	1	1	2	3
(C)	1	4	2	3
(D)	1	1	3	2

5. Match the List-I with List-II :

List-I

- (P) Ferrocene
(Q) $\text{Mn}_2(\text{CO})_{10}$
(R) Vitamine B_{12}
(S) Haemoglobin

List-II

- (1) Iron present
(2) Cobalt
(3) Metal-Metal bonding
(4) Sandwich structure

Code :

	P	Q	R	S
(A)	4	3	1	2
(B)	1	3	1	2
(C)	1,4	3	2	1
(D)	1	3	4	2

6. Match the List-I with List-II :

List-I

- (P) EDTA^{4-}
(Q) en
(R) gly^-
(S) amide

List-II

- (1) N-donor atom
(2) Chelate ligand with same donor site
(3) Bidentate with different donor atom
(4) Hexadentate

Code :

	P	Q	R	S
(A)	4	2	1	3
(B)	2	1	3	4
(C)	4	1	2	3
(D)	4	2	3	1

Assertion Reason :

7. **Statement-1:** Complexes containing three bidentate groups such as $[\text{Cr}(\text{ox})_3]^{3-}$ and $[\text{Co}(\text{en})_3]^{3+}$ do not show optical activity.
Statement-2: Octahedral complex, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ shows geometrical isomerism.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
8. **Statement-1:** After splitting of d-orbitals during complex formation, the orbitals form two sets of orbitals t_{2g} and e_g in an octahedral field.
Statement-2: Splitting of d-orbitals occur only in the case of strong field ligands such as CN^- .
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

9. **Statement-1:** $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is coloured while $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ is colourless.
Statement-2: d-d transition is not possible in $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ because no d-electron is present while possible for Ti^{3+} having d^1 system.
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

Comprehension (10 to 12)

Ligands are neutral or ionic species capable of donating at least one electron pair to central metal. Hence ligands can be of different denticities.

10. For a given metal M^{3+} coordination number is six, then for which set of ligands, complex will be most stable-
- (A) $6\text{H}_2\text{O}$ (B) 6F^- (C) EDTA^{4-} (D) $2\text{H}_2\text{O}$ and $2\text{C}_2\text{O}_4^{2-}$
11. $[\text{Mn}(\text{CO})_5]$ can attain more stability by :
- (A) Oxidation of itself (B) Reduction of itself
 (C) Dimerization (D) Both (B) and (C)
12. The metal cation that has least tendency to accept electron pair from NH_3 is
- (A) Fe^{3+} (B) Rh^{3+} (C) Zn^{2+} (D) Ba^{++}

Comprehension (13 to 15)

Complex compounds are molecular compounds which retain their identities even when dissolved in water. They do not give all the simple ions in solution but instead furnish complex ions with complicated structures. The complex compounds are often called coordination compounds because certain groups called ligands are attached to the central metal ion by coordinate or dative bonds. Coordination compounds exhibit isomerism, both structural and stereoisomerism. The structure, magnetic property, colour and electrical properties of complexes are explained by various theories.

13. Arrange the following compounds in order of their Molar conductance:
- (I) $\text{K}[\text{Co}(\text{NO}_2)_4(\text{NH}_3)_2]$ (II) $[\text{Cr}(\text{ONO})_3(\text{NH}_3)_3]$
 (III) $[\text{Cr}(\text{NO}_2)(\text{NH}_3)_5]_3[\text{Co}(\text{NO}_2)_6]_2$ (IV) $\text{Mg}[\text{Cr}(\text{NO}_2)_5(\text{NH}_3)]$
 (A) $\text{II} < \text{I} < \text{IV} < \text{III}$ (B) $\text{I} < \text{II} < \text{III} < \text{IV}$
 (C) $\text{II} < \text{I} < \text{III} < \text{IV}$ (D) $\text{IV} < \text{III} < \text{II} < \text{I}$
14. The oxidation number and coordination number of chromium in the following complex is $[\text{Cr}(\text{C}_2\text{O}_4)_2(\text{NH}_3)_2]^{1-}$
- (A) O.N. = +4, C.N. = 4 (B) O.N. = +3, C.N. = 4
 (C) O.N. = -1, C.N. = 4 (D) O.N. = +3, C.N. = 6
15. In which of the following pairs, both the complexes have the same geometry but different hybridisation
- (A) $[\text{NiCl}_4]^{2-}$, $[\text{Ni}(\text{CN})_4]^{2-}$ (B) $[\text{CoF}_6]^{3-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (C) $[\text{Ni}(\text{CO})_4]$, $[\text{Ni}(\text{CN})_4]^{2-}$ (D) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{Ni}(\text{NH}_3)_6]^{2+}$

Matching list type 1 × 3Q. (Three list type Question)

The following column I, II, III represent the different type of observations based on CFT in complex compounds.

Answer the questions that follow

Column-I - Crystal field stabilization energy (CFSE) (neglecting PE in all cases)

Column-II - Electronic configuration

Column-III - Type of complex

Column - I CFSE (neglecting PE in all cases)	Column - II Electronic Configuration	Column - III Type of Complex
(I) $-0.4 \Delta_o$	(i) t_{2g}^5, e_g^0	(P) High spin & Paramagnetic
(II) $-2.0 \Delta_o$	(ii) t_{2g}^4, e_g^0	(Q) Low spin & Paramagnetic
(III) $-2.4 \Delta_o$	(iii) t_{2g}^6, e_g^0	(R) High spin & Diamagnetic
(IV) $-1.2 \Delta_o$	(iv) t_{2g}^4, e_g^2	(S) Low spin & Diamagnetic

16. For sodium nitroprusside complex the only **CORRECT** combination is
 (A) (III), (iv), (Q) (B) (III), (iii), (S) (C) (III), (iii), (R) (D) (II), (iii), (Q)
17. For $[\text{Co}(\text{H}_2\text{O})_3\text{F}_3]$ complex the only **CORRECT** combination is.
 (A) (I), (iv), (Q) (B) (II), (iv), (S) (C) (III), (ii), (R) (D) (I), (iv), (P)
18. For $[\text{Mn}(\text{CN})_6]^{4-}$ complex the only **CORRECT** combination is.
 (A) (IV), (i), (S) (B) (II), (i), (R) (C) (I), (i), (S) (D) (II), (i), (Q)

EXERCISE # JEE-MAIN

- In $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$, the isomerism shown is - [AIEEE-2002]
 (1) Ligand (2) Optical (3) Geometrical (4) Ionization
- In the complexes $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Fe}(\text{SCN})_6]^{3-}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{FeCl}_6]^{3-}$, more stability is shown by - [AIEEE-2002]
 (1) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (2) $[\text{Fe}(\text{SCN})_6]^{3-}$ (3) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (4) $[\text{FeCl}_6]^{3-}$
- One mole of the complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of $\text{AgCl}(\text{s})$. The structure of the complex is - [AIEEE-2003]
 (1) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$ (2) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{NH}_3$
 (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2 \cdot \text{NH}_3$ (4) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
- In the coordination compound $\text{K}_4[\text{Ni}(\text{CN})_4]$, the oxidation state of nickel is - [AIEEE-2003]
 (1) 0 (2) +1 (3) +2 (4) -1
- The number of 3d-electrons remained in Fe^{2+} (At.no. of Fe = 26) ion is - [AIEEE-2003]
 (1) 4 (2) 5 (3) 6 (4) 3
- Ammonia forms the complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ with copper ions in alkaline solutions but not in acidic solution. What is the reason for it :- [AIEEE-2003]
 (1) In acidic solutions hydration protects copper ions
 (2) In acidic solutions protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available
 (3) In alkaline solutions insoluble $\text{Cu}(\text{OH})_2$ is precipitated which is soluble in excess of any alkali
 (4) Copper hydroxide is an amphoteric substance
- Among the properties (a) reducing (b) oxidising (c) complexing, the set of properties shown by CN^- ion towards metal species is :- [AIEEE-2004]
 (1) c, a (2) b, c (3) a, b (4) a, b, c
- The coordination number of a central metal atom in a complex is determined by :- [AIEEE-2004]
 (1) The number of ligands around a metal ion bonded by sigma and pi-bonds both
 (2) The number of ligands around a metal ion bonded by pi-bonds
 (3) The number of ligands around a metal ion bonded by sigma bonds
 (4) The number of only anionic ligands bonded to the metal ion
- Which one of the following complexes is an outer orbital complex :- [AIEEE-2004]
 (1) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (2) $[\text{Mn}(\text{CN})_6]^{4-}$
 (3) $[\text{Fe}(\text{CN})_6]^{4-}$ (4) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 (Atomic nos.: Mn=25 ; Fe=26 ; Co=27 ; Ni = 28)

10. Coordination compounds have great importance in biological systems. In this context which of the following statements is **INCORRECT** ? [AIEEE-2004]
- (1) Cyanocobalamin is vitamin B₁₂ and contains cobalt
 - (2) Haemoglobin is the red pigment of blood and contains iron
 - (3) Chlorophylls are green pigments in plants and contain calcium
 - (4) Carboxypeptidase - A is an enzyme and contains zinc
11. The **CORRECT** order of magnetic moments (spin only values in B.M.) among is :- [AIEEE-2004]
- (1) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$
 - (2) $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
 - (3) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$
 - (4) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
- (Atomic nos. : Mn = 25, Fe = 26, Co = 27)
12. For octahedral complex, the value of the 'spin only' magnetic moment for one of the following configurations is 2.84 BM. The **CORRECT** one is [AIEEE-2005]
- (1) d⁴ (in strong ligand field)
 - (2) d⁴ (in weak ligand field)
 - (3) d³ (in weak as well as in strong field)
 - (4) d⁵ (in strong ligand field)
13. The IUPAC name for the complex $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ is - [AIEEE-2006]
- (1) pentaammine nitrito-N- cobalt (II) chloride
 - (2) pentaammine nitrito-N- cobalt (III) chloride
 - (3) nitrito-N- pentaamminecobalt (III) chloride
 - (4) nitrito-N- pentaamminecobalt (II) chloride
14. Nickel (Z=28) combines with a uninegative monodentate ligand X⁻ to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electrons in the nickel and geometry of this complex ion are, respectively. [AIEEE-2006]
- (1) one, square planar
 - (2) two, square planar
 - (3) one, tetrahedral
 - (4) two, tetrahedral
15. In $\text{Fe}(\text{CO})_5$, the Fe-C bond possesses [AIEEE-2006]
- (1) ionic character
 - (2) σ - character only
 - (3) π -character only
 - (4) both σ and π character
16. How many EDTA (ethylenediaminetetraacetate) molecules are required to make an octahedral complex with a Ca²⁺ ion ? [AIEEE-2006]
- (1) One
 - (2) Two
 - (3) Six
 - (4) Three
17. The "spin-only" magnetic moment [in units of Bohr magneton, (μ_B)] of Ni²⁺ in aqueous solution would be (At. No. Ni= 28)- [AIEEE-2006]
- (1) 0
 - (2) 1.73
 - (3) 2.84
 - (4) 4.90

18. Which one of the following has a square planar geometry :-
(Co = 27, Ni = 28, Fe = 26, Pt = 78) [AIEEE-2007]
(1) $[\text{CoCl}_4]^{2-}$ (2) $[\text{FeCl}_4]^{2-}$ (3) $[\text{NiCl}_4]^{2-}$ (4) $[\text{PtCl}_4]^{2-}$
19. The coordination number and the oxidation state of the element 'E' in the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4^{2-})]\text{NO}_2^0$ (where en) is ethylene diamine) are, respectively - [AIEEE-2008]
(1) 6 and 2 (2) 4 and 2 (3) 4 and 3 (4) 6 and 3
20. In which of the following octahedral complexes of Co (at. no. 27), will the magnitude of Δ be the highest ? [AIEEE-2008]
(1) $[\text{Co}(\text{CN})_6]^{3-}$ (2) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (3) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (4) $[\text{Co}(\text{NH}_3)_6]^{3+}$
21. Which of the following pairs represent linkage isomers ? [AIEEE-2009]
(1) $[\text{Co}(\text{NH}_3)_5\text{NO}_3]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{NO}_3$
(2) $[\text{PtCl}_2(\text{NH}_3)_4]\text{Br}_2$ and $[\text{PtBr}_2(\text{NH}_3)_4]\text{Cl}_2$
(3) $[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4]$
(4) $[\text{Pd}(\text{PPh}_3)_2(\text{NCS})_2]$ and $[\text{Pd}(\text{PPh}_3)_2(\text{SCN})_2]$
22. Which of the following has an optical isomer? [AIEEE-2009]
(1) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$ (2) $[\text{Co}(\text{en})_2(\text{NH}_3)_2]^{3+}$
(3) $[\text{Co}(\text{NH}_3)_3\text{Cl}]^+$ (4) $[\text{Co}(\text{en})(\text{NH}_3)_2]^{2+}$
23. Which one of the following has an optical isomer ? [AIEEE-2010]
(1) $[\text{Zn}(\text{en})_2]^{2+}$ (2) $[\text{Zn}(\text{en})(\text{NH}_3)_2]^{2+}$ (3) $[\text{Co}(\text{en})_3]^{3+}$ (4) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$
(en = ethylenediamine)
24. A solution containing 2.675 g of $\text{CoCl}_3 \cdot 6\text{NH}_3$ (molar mass = 267.5 g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of AgNO_3 to give 4.78 g of AgCl (molar mass = 143.5 g mol^{-1}). The formula of the complex is :- [AIEEE-2010]
(At. mass of Ag = 108 u)
(1) $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$ (2) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (3) $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$ (4) $[\text{CoCl}_3(\text{NH}_3)_3]$
25. Which of the following facts about the complex $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ is **wrong** ? [AIEEE-2011]
(1) The complex is an outer orbital complex
(2) The complex gives white precipitate with silver nitrate solution
(3) The complex involves d^2sp^3 hybridisation and is octahedral in shape
(4) The complex is paramagnetic
26. The magnetic moment (spin only) of $[\text{NiCl}_4]^{2-}$ is :- [AIEEE-2011]
(1) 2.82 BM (2) 1.41 BM (3) 1.82 BM (4) 5.46 BM
27. Among the ligands NH_3 , en, CN^- and CO the **CORRECT** order of their increasing field strength, is :- [AIEEE-2011]
(1) $\text{CO} < \text{NH}_3 < \text{en} < \text{CN}^-$ (2) $\text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$
(3) $\text{CN}^- < \text{NH}_3 < \text{CO} < \text{en}$ (4) $\text{en} < \text{CN}^- < \text{NH}_3 < \text{CO}$
28. Which one of the following complex ions has geometrical isomers ? [AIEEE-2011]
(1) $[\text{Co}(\text{en})_3]^{3+}$ (2) $[\text{Ni}(\text{NH}_3)_5\text{Br}]^+$
(3) $[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{3+}$ (4) $[\text{Cr}(\text{NH}_3)_4(\text{en})]^{3+}$
29. Which among the following will be named as dibromidobis (ethylene diamine) chromium (III) bromide ? [AIEEE-2012]
(1) $[\text{Cr}(\text{en})\text{Br}_2]\text{Br}$ (2) $[\text{Cr}(\text{en})_3]\text{Br}_3$ (3) $[\text{Cr}(\text{en})_2\text{Br}_2]\text{Br}$ (4) $[\text{Cr}(\text{en})\text{Br}_4]^-$

30. The complex ion $[\text{Pt}(\text{NO}_2)(\text{Py})(\text{NH}_3)(\text{NH}_2\text{OH})]^+$ will give :- [J-MAIN-2012, Online]
 (1) 4 isomers (Geometrical) (2) 2 isomers (Geometrical)
 (3) 3 isomers (Geometrical) (4) 6 isomers (Geometrical)
31. Which of the following complex ions will exhibit optical isomerism? [J-MAIN-2012, Online]
 (en = 1, 2-diamine ethane)
 (1) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (2) $[\text{Zn}(\text{en})_2]^{2+}$
 (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ (4) $[\text{Cr}(\text{NH}_3)_2\text{Cl}_2]^+$
32. Which of the following complex species is **NOT** expected to exhibit optical isomerism ? [J-MAIN-2013]
 (1) $[\text{Co}(\text{en})_3]^{3+}$ (2) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$
 (3) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (4) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$
33. Type of isomerism which exists between $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{SCN})_2]$ and $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{NCS})_2]$ is : [J-MAIN-2013, Online]
 (1) Solvate isomerism (2) Ionisation isomerism
 (3) Linkage isomerism (4) Coordination isomerism
34. Which of the following is diamagnetic ? [J-MAIN-2013, Online]
 (1) $[\text{CoF}_6]^{3-}$ (2) $[\text{FeF}_6]^{3-}$ (3) $[\text{Fe}(\text{CN})_6]^{3-}$ (4) $[\text{Co}(\text{Ox})_3]^{3-}$
35. The magnetic moment of the complex anion $[\text{Cr}(\text{NO})(\text{NH}_3)(\text{CN})_4]^{2-}$ is : [J-MAIN-2013, Online]
 (1) 2.82 BM (2) 5.91 BM (3) 1.73 BM (4) 3.87 BM
36. The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1 , L_2 , L_3 and L_4 absorb wavelength in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is: [J-MAIN-2014]
 (1) $\text{L}_3 < \text{L}_2 < \text{L}_4 < \text{L}_1$ (2) $\text{L}_1 < \text{L}_2 < \text{L}_4 < \text{L}_3$
 (3) $\text{L}_4 < \text{L}_3 < \text{L}_2 < \text{L}_1$ (4) $\text{L}_1 < \text{L}_3 < \text{L}_2 < \text{L}_4$
37. The equation which is balanced and represents the **CORRECT** product (s) is : [J-MAIN-2014]
 (1) $[\text{Mg}(\text{H}_2\text{O})_6]^{2+} + (\text{EDTA})^{4-} \xrightarrow{\text{excess NaOH}} [\text{Mg}(\text{EDTA})]^{2-} + 6\text{H}_2\text{O}$
 (2) $\text{CuSO}_4 + 4\text{KCN} \rightarrow \text{K}_2[\text{Cu}(\text{CN})_4] + \text{K}_2\text{SO}_4$
 (3) $\text{Li}_2\text{O} + 2\text{KCl} \rightarrow 2\text{LiCl} + \text{K}_2\text{O}$
 (4) $[\text{CoCl}(\text{NH}_3)_5]^+ + 5\text{H}^+ \rightarrow \text{Co}^{2+} + 5\text{NH}_4^+ + \text{Cl}^-$
38. The **CORRECT** statement about the magnetic properties of $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{FeF}_6]^{3-}$ is : ($Z = 26$). [J-MAIN-2014, Online]
 (1) $[\text{Fe}(\text{CN})_6]^{3-}$ is paramagnetic, $[\text{FeF}_6]^{3-}$ is diamagnetic.
 (2) both are diamagnetic.
 (3) $[\text{Fe}(\text{CN})_6]^{3-}$ is diamagnetic, $[\text{FeF}_6]^{3-}$ is paramagnetic.
 (4) both are paramagnetic
39. An octahedral complex of Co^{3+} is diamagnetic. The hybridisation involved in the formation of the complex is : [J-MAIN-2014]
 (1) d^2sp^3 (2) dsp^3d (3) dsp^2 (4) sp^3d^2

40. Which of the following name formula combinations is **NOT CORRECT**? [J-MAIN-2014, Online]

Formula

Name

- | | |
|-------------------------------|--|
| (1) $K[Cr(NH_3)_2Cl_4]$ | Potassium diammine Tetrachlorochromate (III) |
| (2) $[Co(NH_3)_4(H_2O)I]SO_4$ | Tetraammine aquaiodo cobalt (III) sulphate |
| (3) $[Mn(CN)_5]^{2-}$ | Pentacyanomagnate (II) ion |
| (4) $K_2[Pt(CN)_4]$ | Potassium tetracyanoplatinate(II) |

41. Consider the coordination compound, $[Co(NH_3)_6]Cl_3$. In the formation of this complex, the species which acts as the Lewis acid is : [J-MAIN-2014, Online]

- (1) $[Co(NH_3)_6]^{3+}$ (2) NH_3 (3) Co^{3+} (4) Cl^-

42. Among the following species the one which causes the highest CFSE, Δ_0 as a ligand is :- [J-MAIN-2014, Online]

- (1) CN^- (2) NH_3 (3) CO (4) F^-

43. Which one of the following complexes will most likely absorb visible light ? [J-MAIN-2014, Online] (At nos. Sc = 21, Ti = 22, V = 23, Zn = 30) :-

- (1) $[Ti(NH_3)_6]^{4+}$ (2) $[V(NH_3)_6]^{3+}$ (3) $[Zn(NH_3)_6]^{2+}$ (4) $[Sc(H_2O)_6]^{3+}$

44. Nickel (Z = 28) combines with a uninegative monodentate ligand to form a diamagnetic complex $[NiL_4]^{2-}$. The hybridisation involved and the number of unpaired electrons present in the complex are respectively : [J-MAIN-2014, Online]

- (1) sp^3 , zero (2) sp^3 , two (3) dsp^2 , one (4) dsp^2 , zero

45. The number of geometrical isomers that can exist for square planar $[Pt(Cl)(py)(NH_3)(NH_2OH)]^+$ is (py = pyridine) : [J-MAIN-2015]

- (1) 4 (2) 6 (3) 2 (4) 3

46. The color of $KMnO_4$ is due to : [J-MAIN-2015]

- (1) $L \rightarrow M$ charge transfer transition (2) $\sigma - \sigma^*$ transition
(3) $M \rightarrow L$ charge transfer transition (4) $d - d$ transition

47. Which of the following complex ions has electrons that are symmetrically filled in both t_{2g} and e_g orbitals ? [J-MAIN-2015, Online]

- (1) $[CoF_6]^{3-}$ (2) $[Mn(CN)_6]^{4-}$ (3) $[FeF_6]^{3-}$ (4) $[Co(NH_3)_6]^{2+}$

48. When concentrated HCl is added to an aqueous solution of $CoCl_2$, its colour changes from reddish pink to deep blue. Which complex ion gives blue colour in this reaction ?:- [J-MAIN-2015, Online]

- (1) $[Co(H_2O)_6]^{2+}$ (2) $[CoCl_6]^{3-}$
(3) $[CoCl_4]^{2-}$ (4) $[CoCl_6]^{4-}$

49. The **CORRECT** statement on the isomerism associated with the following complex ions, [J-MAIN-2015, Online]

- (a) $[Ni(H_2O)_5NH_3]^{2+}$
(b) $[Ni(H_2O)_4(NH_3)_2]^{2+}$ and
(c) $[Ni(H_2O)_3(NH_3)_3]^{2+}$ is :

- (1) (a) and (b) show geometrical and optical isomerism
(2) (b) and (c) show geometrical and optical isomerism
(3) (a) and (b) show only geometrical Isomerism
(4) (b) and (c) show only geometrical Isomerism

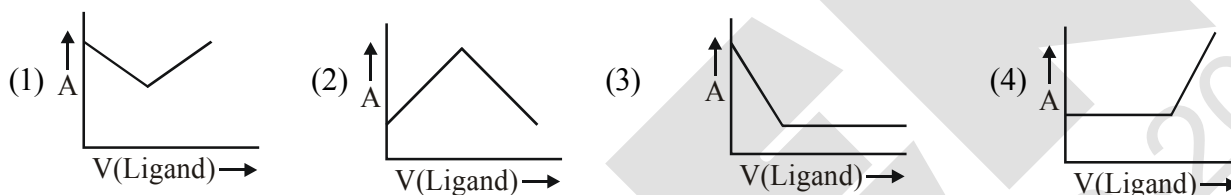
50. Which one of the following complexes shows optical isomerism :- [J-MAIN-2016]
 (1) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (2) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
 (3) $\text{cis}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (4) $\text{trans}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
 (en = ethylenediamine)
51. The pair having the same magnetic moment is:- [J-MAIN-2016]
 [At. No.: Cr = 24, Mn = 25, Fe = 26, Co = 27]
 (1) $[\text{CoCl}_4]^{2-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2-}$
 (3) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (4) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$
52. Which one of the following complexes will consume more equivalents of aqueous solution of $\text{Ag}(\text{NO}_3)$? [J-MAIN-2016, Online]
 (1) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (2) $\text{Na}_2[\text{CrCl}_5(\text{H}_2\text{O})]$ (3) $\text{Na}_3[\text{CrCl}_6]$ (4) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$
53. Identify the **CORRECT** trend given below : [J-MAIN-2016, Online]
 (Atomic No. = Ti : 22, Cr : 24 and Mo : 42)
 (1) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} < [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} < [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
 (2) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} < [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} > [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
 (3) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} > [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} > [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
 (4) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} > [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} < [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
54. $[\text{Co}_2(\text{CO})_8]$ displays:- [J-MAIN-2017, Online]
 (1) no Co-Co bond, four terminal CO and four bridging CO
 (2) one Co-Co bond, six terminal CO and two bridging CO
 (3) no Co-Co bond, six terminal CO and two bridging CO
 (4) one Co-Co bond, four terminal CO and four bridging CO
55. On treatment of 100 mL of 0.1 M solution of $\text{CoCl}_3 \cdot 6\text{H}_2\text{O}$ with excess AgNO_3 ; 1.2×10^{22} ions are precipitated. The complex is :- [J-MAIN-2017, Offline]
 (1) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ (2) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
 (3) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$ (4) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
56. The pair of compounds having metal in their highest oxidation state is : [J-MAIN-2017, Online]
 (1) $[\text{NiCl}_4]^{2-}$ and $[\text{CoCl}_4]^{2-}$ (2) $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Cu}(\text{CN})_4]^{2-}$
 (3) $[\text{FeCl}_4]^-$ and Co_2O_3 (4) MnO_2 and CrO_2Cl_2
57. The oxidation states of Cr in $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$, $[\text{Cr}(\text{C}_6\text{H}_6)_2]$, and $\text{K}_2[\text{Cr}(\text{CN})_2(\text{O})_2(\text{O}_2)(\text{NH}_3)]$ respectively are : [J-MAIN-2018, Offline]
 (1) +3, +2, and +4 (2) +3, 0, and +6
 (3) +3, 0, and +4 (4) +3, +4, and +6

58. Consider the following reaction and statements : [J-MAIN-2018, Offline]
 $[\text{Co}(\text{NH}_3)_4\text{Br}_2]^+ + \text{Br}^- \rightarrow [\text{Co}(\text{NH}_3)_3\text{Br}_3] + \text{NH}_3$
 (I) Two isomers are produced if the reactant complex ion is a *cis*-isomer.
 (II) Two isomers are produced if the reactant complex ion is a *trans*-isomer.
 (III) Only one isomer is produced if the reactant complex ion is a *trans*-isomer.
 (IV) Only one isomer is produced if the reactant complex ion is a *cis*-isomer.
 The correct statements are :
 (1) (I) and (III) (2) (III) and (IV) (3) (II) and (IV) (4) (I) and (II)
59. For 1 molal aqueous solution of the following compounds, which one will show the highest freezing point ? [J-MAIN-2018, Offline]
 (1) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
 (2) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
 (3) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
 (4) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$
60. The total number of possible isomers for square-planar $[\text{Pt}(\text{Cl})(\text{NO}_2)(\text{NO}_3)(\text{SCN})]^{2-}$ is :- [J-MAIN-2018, Online]
 (1) 16 (2) 8 (3) 24 (4) 12
61. The correct order of spin-only magnetic moments among the following is : [J-MAIN-2018, Online]
 (Atomic number : Mn = 25, Co = 27, Ni = 28, Zn = 30)
 (1) $[\text{ZnCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
 (2) $[\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
 (3) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
 (4) $[\text{NiCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
62. The correct combination is [J-MAIN-2018, Online]
 (1) $[\text{Ni}(\text{CN})_4]^{2-}$ – tetrahedral;
 $[\text{Ni}(\text{CO})_4]$ – paramagnetic
 (2) $[\text{NiCl}_4]^{2-}$ – paramagnetic;
 $[\text{Ni}(\text{CO})_4]$ – tetrahedral
 (3) $[\text{NiCl}_4]^{2-}$ – diamagnetic;
 $[\text{Ni}(\text{CO})_4]$ – square-planar
 (4) $[\text{NiCl}_4]^{2-}$ – square-planar;
 $[\text{Ni}(\text{CN})_4]^{2-}$ – paramagnetic

63. Which of the following complexes will show geometrical isomerism ? [J-MAIN-2018, Online]

- (1) Potassium amminetrichloroplatinate(II)
- (2) Aquachlorobis (ethylenediamine) cobalt(II) chloride
- (3) Potassium tris(oxalato) chromate(III)
- (4) Pentaquachlorochromium(III) chloride

64. In a complexometric titration of metal ion with ligand $M(\text{Metal ion}) + L(\text{Ligand}) \rightarrow C(\text{Complex})$ end point is estimated spectrophotometrically (through light absorption). If 'M' and 'C' do not absorb light and only 'L' absorbs, then the titration plot between absorbed light (A) versus volume of ligand 'L' (v) would look like :- [J-MAIN-2018, Online]



65. In Wilkinson's catalyst, the hybridization of central metal ion and its shape are respectively

- (1) dsp^2 , square planar
- (2) sp^3d , trigonal bipyramidal
- (3) sp^3 , tetrahedral
- (4) d^2sp^3 , octahedral

[J-MAIN-2018, Online]

EXERCISE # JEE-ADVANCED

- The complex ion which has no 'd' electrons in the central metal atom is : [JEE 2001]
[At No. Cr = 24, Mn = 25, Fe = 26, Co = 27]
(A) $[\text{MnO}_4]^-$ (B) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (C) $[\text{Fe}(\text{CN})_6]^{3-}$ (D) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
- The **CORRECT** order of hybridisation of the central atom in the following species. [JEE 2001]
 NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3 is [At No. Pt = 78]
(A) $\text{dsp}^2, \text{sp}^3\text{d}, \text{sp}^2$ and sp^3 (B) $\text{sp}^3, \text{dsp}^2, \text{sp}^3\text{d}$, sp^2
(C) $\text{dsp}^2, \text{sp}^2, \text{sp}^3$ and sp^3d (D) dsp^2 , sp^3, sp^2 and sp^3d
- The species having tetrahedral shape is : [JEE 2004]
(A) $[\text{PdCl}_4]^{2-}$ (B) $[\text{Ni}(\text{CN})_4]^{2-}$ (C) $[\text{Pd}(\text{CN})_4]^{2-}$ (D) $[\text{NiCl}_4]^{2-}$
- The pair of compounds having metals in their highest oxidation state is [JEE 2004]
(A) MnO_2 , FeCl_3 (B) $[\text{MnO}_4]^-$, CrO_2Cl_2
(C) $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{CN})_3]$ (D) $[\text{NiCl}_4]^{2-}$, $[\text{CoCl}_4]^-$
- Spin only magnetic moment of the compound $\text{Hg}[\text{Co}(\text{SCN})_4]$ is [JEE 2004]
(A) $\sqrt{3}$ (B) $\sqrt{15}$ (C) $\sqrt{24}$ (D) $\sqrt{8}$
- Which of the following pair is expected to exhibit same colour in solution? [JEE 2005]
(A) VOCl_2 ; FeCl_2 (B) CuCl_2 ; VOCl_2 (C) MnCl_2 ; FeCl_2 (D) FeCl_2 ; CuCl_2
- Which type of isomerism is shown by $\text{Co}(\text{NH}_3)_4\text{Br}_2\text{Cl}$? [JEE 2005]
(A) Geometrical and Ionisation (B) Optical and Ionisation
(C) Geometrical and Optical (D) Geometrical only

Question No. 8 to 10 (3 questions)

[JEE 2006]

The coordination number of Ni^{2+} is 4.

$\text{NiCl}_2 + \text{KCN} (\text{excess}) \rightarrow \text{A} (\text{cyano complex})$

$\text{NiCl}_2 + \text{KCl} (\text{excess}) \rightarrow \text{B} (\text{chloro complex})$

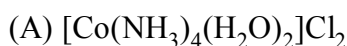
- The IUPAC name of A and B are
(A) Potassium tetracyanonickelate (II), potassium tetrachloridonickelate (II)
(B) Tetracyanidopotassiumnickelate (II), tetrachloridopotassiumnickelate (II)
(C) Tetracyanidonickel (II), tetrachloridonickel (II)
(D) Potassium tetracyanonickel (II), potassium tetrachloridonickel (II)
- Predict the magnetic nature of A and B.
(A) Both are diamagnetic.
(B) A is diamagnetic and B is paramagnetic with one unpaired electron.
(C) A is diamagnetic and B is paramagnetic with two unpaired electrons.
(D) Both are paramagnetic.
- The hybridization of A and B are
(A) dsp^2 , sp^3 (B) sp^3 , sp^3 (C) dsp^2 , dsp^2 (D) sp^3d^2 , d^2sp^3

11. If the bond length of CO bond in carbon monoxide is $1.128\text{\AA}$, then what is the value of CO bond length in $\text{Fe}(\text{CO})_5$? [JEE 2006]
 (A) $1.15\text{\AA}$ (B) $1.128\text{\AA}$ (C) $1.72\text{\AA}$ (D) $1.118\text{\AA}$
12. Among the following metal carbonyls, the C–O bond order is lowest in [JEE 2007]
 (A) $[\text{Mn}(\text{CO})_6]^+$ (B) $[\text{Fe}(\text{CO})_5]$ (C) $[\text{Cr}(\text{CO})_6]$ (D) $[\text{V}(\text{CO})_6]^-$
13. Match the complexes in Column I with their properties listed in Column II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [JEE 2007]

Column I

Column II

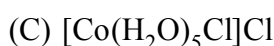
[JEE 2007]



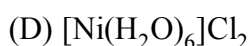
(P) Geometrical isomers



(Q) Paramagnetic



(R) Diamagnetic



(S) Metal ion with 2+ oxidation state

14. Among the following, the coloured compound is [JEE 2008]
 (A) CuCl (B) $\text{K}_3[\text{Cu}(\text{CN})_4]$ (C) CuF_2 (D) $[\text{Cu}(\text{CH}_3\text{CN})_4]\text{BF}_4$

15. The IUPAC name of $[\text{Ni}(\text{NH}_3)_4][\text{NiCl}_4]$ is [JEE 2008]

- (A) Tetrachloronickel(II)-tetraamminenickel(II)
 (B) Tetraamminenickel(II)-tetrachloronickel(II)
 (C) Tetraamminenickel(II)-tetrachloronickelate(II)
 (D) Tetrachloronickel(II)-tetraamminenickelate(0)

16. Both $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic. The hybridisations of nickel in these complexes, respectively, are [JEE 2008]

- (A) sp^3 , sp^3 (B) sp^3 , dsp^2 (C) dsp^2 , sp^3 (D) dsp^2 , dsp^2

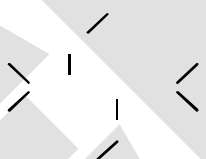
17. Statement-1 : The geometrical isomers of the complex $[\text{M}(\text{NH}_3)_4\text{Cl}_2]$ are optically inactive.
 Statement-2 : Both geometrical isomers of the complex $[\text{M}(\text{NH}_3)_4\text{Cl}_2]$ possess axis of symmetry.
 (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True [JEE 2008]

18. Statement-1 : $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ is paramagnetic [JEE 2008]

Statement-2 : The Fe in $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ has three unpaired electrons.

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True

19. The spin only magnetic moment value (in Bohr magneton units) of $\text{Cr}(\text{CO})_6$ is [JEE 2009]
 (A) 0 (B) 2.84 (C) 4.90 (D) 5.92

20. The compound(s) that exhibit(s) geometrical isomerism is (are) : [JEE 2009]
 (A) $[\text{Pt}(\text{en})\text{Cl}_2]$ (B) $[\text{Pt}(\text{en})_2]\text{Cl}_2$ (C) $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{Cl}_2$ (D) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
21. The number of water molecule(s) directly bonded to the metal centre in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is. [JEE 2009]
22. The ionization isomer of $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}(\text{NO}_2)]\text{Cl}$ is – [JEE 2010]
 (A) $[\text{Cr}(\text{H}_2\text{O})_4(\text{O}_2\text{N})]\text{Cl}_2$ (B) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2](\text{NO}_2)$
 (C) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}(\text{ONO})]\text{Cl}$ (D) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2(\text{NO}_2)] \cdot \text{H}_2\text{O}$
23. Total number of geometrical isomers for the complex $[\text{RhCl}(\text{CO})(\text{PPh}_3)(\text{NH}_3)]$ is. [JEE 2010]
24. The correct structure of ethylenediaminetetraacetic acid (EDTA) is – [JEE 2010]
 (A)  (B) 
 (C)  (D) 
25. Geometrical shapes of the complexes formed by the reaction of Ni^{2+} with Cl^- , CN^- and H_2O respectively, are - [JEE 2011]
 (A) octahedral, tetrahedral and square planar (B) tetrahedral, square planar and octahedral
 (C) square planar, tetrahedral and octahedral (D) octahedral, square planar and octahedral
26. Among the following complexes (**K–P**) [JEE 2011]
 $\text{K}_3[\text{Fe}(\text{CN})_6]$ (**K**), $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (**L**), $\text{Na}_3[\text{Co}(\text{oxalate})_3]$ (**M**), $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$ (**N**),
 $\text{K}_2[\text{Pt}(\text{CN})_4]$ (**O**) and $[\text{Zn}(\text{H}_2\text{O})_6](\text{NO}_3)_2$ (**P**)
 The diamagnetic complex are -
 (A) K, L, M, N (B) K, M, O, P (C) L, M, O, P (D) L, M, N, O
27. The volume (in mL) of 0.1M AgNO_3 required for complete precipitation of chloride ions present in 30 mL of 0.01M solution of $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$, as silver chloride is close to. [JEE 2011]
28. As per IUPAC nomenclature, the name of the complex $[\text{Co}(\text{H}_2\text{O})_4(\text{NH}_3)_2]\text{Cl}_3$ is : [JEE 2012]
 (A) Tetraaquadiamminecobalt(III) chloride (B) Tetraaquadiamminecobalt(III) chloride
 (C) Diaminetetraaquacobalt(III) chloride (D) Diamminetetraaquacobalt(III) chloride
29. The colour of light absorbed by an aqueous solution of CuSO_4 is - [JEE 2012]
 (A) orange-red (B) blue-green (C) yellow (D) violet
30. $\text{NiCl}_2 \cdot \text{P}(\text{C}_2\text{H}_5)_2(\text{C}_6\text{H}_5)_2$ exhibits temperature dependent magnetic behavior (paramagnetic/diamagnetic). The coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively : [JEE 2012]
 (A) tetrahedral and tetrahedral (B) square planar and square planar
 (C) tetrahedral and square planar (D) square planar and tetrahedral

31. Consider the following complex ions P, Q and R ,
 $P = [FeF_6]^{3-}$, $Q = [V(H_2O)_6]^{2+}$ and $R = [Fe(H_2O)_6]^{2+}$
 The **CORRECT** order of the complex ions, according to their spin-only magnetic moment values (in B.M.) is - **[JEE 2013]**
 (A) $R < Q < P$ (B) $Q < R < P$ (C) $R < P < Q$ (D) $Q < P < R$
32. $EDTA^{4-}$ is ethylenediaminetetraacetate ion. The total number of N–Co–O bond angles in $[Co(EDTA)]^{-1}$ complex ion is **[JEE 2013]**
33. The pair(s) of coordination complex/ion exhibiting the same kind of isomerism is(are) - **[JEE 2013]**
 (A) $[Cr(NH_3)_5Cl]Cl_2$ and $[Cr(NH_3)_4Cl_2]Cl$ (B) $[Co(NH_3)_4Cl_2]^+$ and $[Pt(NH_3)_2(H_2O)Cl]^+$
 (C) $[CoBr_2Cl_2]^{2-}$ and $[PtBr_2Cl_2]^{2-}$ (D) $[Pt(NH_3)_3(NO_3)]Cl$ and $[Pt(NH_3)_3Cl]Br$
34. Match each coordination compound in List-I with an appropriate pair of characteristics from List-II and select the correct answer using the code given below the lists. **[JEE Adv. 2014]**
 {en = $H_2NCH_2CH_2NH_2$ ' atomic numbers ; Ti = 22 ; Cr = 24 ; Co = 27 ; Pt = 78}

List-I

- (P) $[Cr(NH_3)_4Cl_2]Cl$
 (Q) $[Ti(H_2O)_5Cl](NO_3)_2$
 (R) $[Pt(en)(NH_3)Cl]NO_3$
 (S) $[Co(NH_3)_4(NO_3)_2]NO_3$

List-II

- (1) Paramagnetic and exhibits ionisation isomerism
 (2) Diamagnetic and exhibits *cis-trans* isomerism
 (3) Paramagnetic and exhibits *cis-trans* isomerism
 (4) Diamagnetic and exhibits ionisation isomerism

Code :

- | | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 4 | 2 | 3 | 1 |
| (C) | 2 | 1 | 3 | 4 |

- | | P | Q | R | S |
|-----|---|---|---|---|
| (B) | 3 | 1 | 4 | 2 |
| (D) | 1 | 3 | 4 | 2 |

35. A list of species having the formula XZ_4 is given below : **[JEE Adv. 2014]**
 XeF_4 , SF_4 , SiF_4 , BF_4^- , BrF_4^- , $[Cu(NH_3)_4]^{2+}$, $[FeCl_4]^{2-}$, $[CoCl_4]^{2-}$ and $[PtCl_4]^{2-}$.
 Defining shape on the basis of the location of X and Z atoms, the total number of species having a square planar shape is

Subjective

36. Draw the structures of $[Co(NH_3)_6]^{3+}$, $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$. Write the hybridisation of atomic orbitals of the transition metal in each case. **[JEE 2000]**
37. A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms A and B. The form A reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia, whereas B gives a pale yellow precipitate soluble in concentrated ammonia. Write the formula of A and B and state the hybridisation of chromium in each. Calculate their magnetic moments (spin-only value). **[JEE 2001]**
38. Deduce the structures of $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ considering the hybridisation of the metal ion. Calculate the magnetic moment (spin only) of the species. **[JEE 2002]**

39. Write the IUPAC name of the compound $K_2[Cr(NO)(CN)_4(NH_3)]$. Spin magnetic moment of the complex $\mu = 1.73$ BM. Give the structure of anion. [JEE 2003]
40. $NiCl_2$ in the presence of dimethyl glyoxime (DMG) gives a complex which precipitates in the presence of NH_4OH , giving a bright red colour. [JEE 2004]
- (a) Draw its structure and show H-bonding
- (b) Give oxidation state of Ni and its hybridisation
- (c) Predict whether it is paramagnetic or diamagnetic
41. For the octahedral complexes of Fe^{3+} in SCN^- (thiocyanato-S) and in CN^- ligand environments, the difference between the spin only magnetic moments in Bohr magnetons (when approximated to the nearest integer) is : [Atomic number of Fe = 26] [JEE Ad. 2015]
42. In the complex acetylbromidodicarbonylbis(triethylphosphine)iron(II), the number of Fe–C bond(s) is- [JEE Ad. 2015]
43. Among the complex ions, $[Co(NH_2-CH_2-CH_2-NH_2)_2Cl_2]^+$, $[CrCl_2(C_2O_4)_2]^{3-}$, $[Fe(H_2O)_4(OH)_2]^+$, $[Fe(NH_3)_2(CN)_4]^-$, $[Co(NH_2-CH_2-CH_2-NH_2)_2(NH_3)Cl]^{2+}$ and $[Co(NH_3)_4(H_2O)Cl]^{2+}$, the number of complex ion(s) that show(s) *cis-trans* isomerism is - [JEE Ad. 2015]
44. Among $[Ni(CO)_4]$, $[NiCl_4]^{2-}$, $[Co(NH_3)_4Cl_2]Cl$, $Na_3[CoF_6]$, Na_2O_2 and CsO_2 , the total number of paramagnetic compounds is - [JEE Ad. 2016]
- (A) 2 (B) 3 (C) 4 (D) 5
45. The number of geometric isomers possible for the complex $[CoL_2Cl_2]^-$ ($L = H_2NCH_2CH_2O^-$) is [JEE Ad. 2016]
46. The geometries of the ammonia complexes of Ni^{2+} , Pt^{2+} and Zn^{2+} , respectively are : [JEE Ad. 2016]
- (A) octahedral, square planar and tetrahedral
- (B) square planar, octahedral and tetrahedral
- (C) tetrahedral, square planar and octahedral
- (D) octahedral, tetrahedral and square planar
47. Addition of excess aqueous ammonia to a pink coloured aqueous solution of $MCl_2 \cdot 6H_2O$ (X) and NH_4Cl gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as 1 : 3 electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 B.M., whereas it is zero for complex Y. [JEE Ad. 2017]
- Among the following options, which statements is(are) **CORRECT** ?
- (A) The hybridization of the central metal ion in Y is d^2sp^3
- (B) Z is tetrahedral complex
- (C) Addition of silver nitrate to Y gives only two equivalents of silver chloride
- (D) When X and Z are in equilibrium at $0^\circ C$, the colour of the solution is pink

48. The correct statement(s) regarding the binary transition metal carbonyl compounds is (are)
(Atomic numbers : Fe = 26, Ni = 28) [JEE Ad. 2018]
(A) Total number of valence shell electrons at metal centre in $\text{Fe}(\text{CO})_5$ or $\text{Ni}(\text{CO})_4$ is 16
(B) These are predominantly low spin in nature
(C) Metal - carbon bond strengthens when the oxidation state of the metal is lowered
(D) The carbonyl C-O bond weakens when the oxidation state of the metal is increased
49. The correct option(s) regarding the complex $[\text{Co}(\text{en})(\text{NH}_3)_3(\text{H}_2\text{O})]^{3+}$:- [JEE Ad. 2018]
(en = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$) is (are)
(A) It has two geometrical isomers
(B) It will have three geometrical isomers if bidentate 'en' is replaced by two cyanide ligands
(C) It is paramagnetic
(D) It absorbs light at longer wavelength as compared to $[\text{Co}(\text{en})(\text{NH}_3)_4]^{3+}$

50. Match each set of hybrid orbitals from LIST-I with complex (es) given in LIST-II.

LIST-I

P. dsp^2 Q. sp^3 R. sp^3d^2 S. d^2sp^3

LIST-II

[JEE Ad. 2018]

1. $[\text{FeF}_6]^{4-}$ 2. $[\text{Ti}(\text{H}_2\text{O})_3\text{Cl}_3]$ 3. $[\text{Cr}(\text{NH}_3)_6]^{3+}$ 4. $[\text{FeCl}_4]^{2-}$ 5. $\text{Ni}(\text{CO})_4$ 6. $[\text{Ni}(\text{CN})_4]^{2-}$

The correct option is

- (A) $\text{P} \rightarrow 5$; $\text{Q} \rightarrow 4,6$; $\text{R} \rightarrow 2,3$; $\text{S} \rightarrow 1$
(B) $\text{P} \rightarrow 5,6$; $\text{Q} \rightarrow 4$; $\text{R} \rightarrow 3$; $\text{S} \rightarrow 1,2$
(C) $\text{P} \rightarrow 6$; $\text{Q} \rightarrow 4,5$; $\text{R} \rightarrow 1$; $\text{S} \rightarrow 2,3$
(D) $\text{P} \rightarrow 4,6$; $\text{Q} \rightarrow 5,6$; $\text{R} \rightarrow 1,2$; $\text{S} \rightarrow 3$
51. Among the species given below, the total number of diamagnetic species is _____.
H atom, NO_2 monomer, O_2^- (superoxide), dimeric sulphur in vapour phase, [JEE Ad. 2018]
 Mn_3O_4 , $(\text{NH}_4)_2[\text{FeCl}_4]$, $(\text{NH}_4)_2[\text{NiCl}_4]$, K_2MnO_4 , K_2CrO_4

ANSWERS KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	B	C	C	C	D	B	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	A	B	A	C	C	D	A	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	C	B	C	B	B	A	D	A	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	D	B	C	D	C	C	B	C	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	C	D	D	D	D	C	B	D	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	B	A	D	B	D	A	C	B	D
Que.	61	62	63	64	65	66	67	68	69	
Ans.	B	C	C	A	C	C	A	D	A	

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,B,D	A,B,C,D	A,B,D	A,B	B,C,D	A,B,C,D	A,B,C	B,C,D	A,B,C,D
Que.	11	12	13	14						
Ans.	B,D	C,D	C,D	B,D						

EXERCISE # S-1

1. (3)	2. (2)	3. (9)	4. (36)
5. (9)	6. (2)	7. (2)	8. (52)
9. (5)	10. (2)		

EXERCISE # S-2

Que.	1				2			
Ans.	(A)-R,	(B)-R	(C)-P	(D)-Q	(A)-P,Q,S	(B)-Q,R,S	(C)-P	(D)-P,Q
Que.	3							
Ans.	(A)-P,S	(B)-T	(C)-Q,R,T	(D)-P,S				

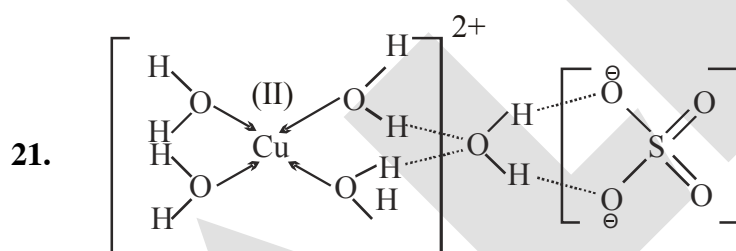
Que.	4	5	6	7	8	9	10	11	12	13
Ans.	B	C	D	D	C	A	C	D	D	A
Que.	14	15	16	17	18					
Ans.	D	B	B	D	D					

EXERCISE # J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	4	1	3	2	1	3	4	3
Que.	11	12	13	14	15	16	17	18	19	20
	3	1	2	4	4	1	3	4	4	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	3	2	1	1	2	3	3	3
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	1	3	3	4	3	4	4	4	1	3
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	3	3	2	4	4	1	3	3	4	3
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	3	1	2	2	1	2	2	1	3	4
Que.	61	62	63	64	65					
Ans.	3	2	2	4	1					

EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	D	B	B	B	A	A	C	A
Que.	11	12	13				14	15	16	17
Ans.	A	D	(A)-P,Q,S ; (B)-PR,S ; (C)-Q,S ; (D)-Q,S				C	C	B	B
Que.	18	19	20							
Ans.	A	A	C, D							



Que.	22	23	24	25	26	27	28	29	30	31
Ans.	B	3	C	B	C	6	D	A	C	B
Que.	32	33	34	35						
Ans.	8	B, D	B	4						

36. d^2sp^3 , dsp^2 and sp^3

37. $A \rightarrow [Cr(NH_3)_4ClBr]Cl$

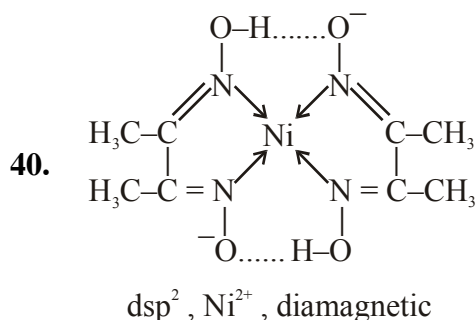
$B \rightarrow [Cr(NH_3)_4Cl_2]Br$

In both Cr is $d^2 sp^3$ hybridised and magnetic moment is $\sqrt{15}$ BM

38. $[NiCl_4]^{2-} \rightarrow sp^3$, $\sqrt{8}$ BM

$[Ni(CN)_4]^{2-} \rightarrow dsp^2$, 0

39. Potassium amminetetrayanidonitrosoniumchromate(I) $\rightarrow d^2sp^3$, octahedral



41. Ans. (4)

42. Ans. (3)

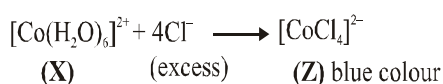
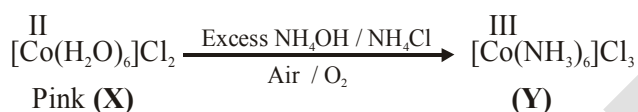
43. Ans. (6)

44. Ans. (B)

45. Ans. (5)

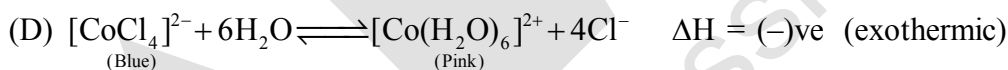
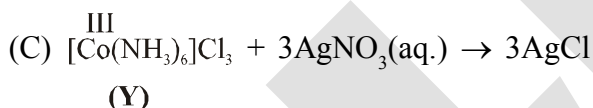
46. Ans. (A)

47. Ans. (A,B,D)



(A) Hybridisation of (Y) is d^2sp^3 as NH_3 is strong field ligand

(B) $[\text{CoCl}_4]^{2-}$ have sp^3 hybridisation as Cl^- is weak field ligand



When ice is added to the solution the equilibrium shifts right hence pink colour will remain predominant

So, correct answer is (A,B& D)

48. Ans.(B,C)

49. Ans.(A,B,D)

50. Ans.(C)

51. Ans.(1)

METALLURGY

□ INTRODUCTION :

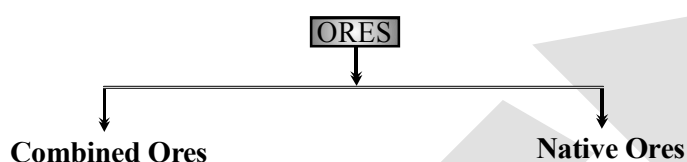
The compound of a metal found in nature is called a **mineral**.

The minerals from which metal can be economically and conveniently extracted are called **ores**.

An ore is usually contaminated with earthy or undesired materials known as **gangue**.

Note : All minerals are not ores but all ores are minerals.

Ores may be classified mainly into following two classes.



(a) **Native ores** : Silver, gold, platinum etc, occur as native ores.

(b) **Combined ores** : They contain the metal in combined form.

(i) **Oxidised ores** : Oxide ores , Carbonate ores, Sulphate ores, Phosphate ores, Silicate ores.

(ii) **Sulphurised ores** : These ores consist of sulphides of metals like iron, lead, zinc, mercury etc.

(iii) **Halide ores** : These ores consist of halides of metals

□ METALLURGY :

The scientific and technological process used for the extraction/isolation of the metal from its ore is called as metallurgy.

The isolation and extraction of metals from their ores involve the following major steps:

(A) Crushing of the ore.

(B) Dressing or concentration of the ore.

(C) Isolation of the crude metal from its ore

(D) Purification or refining of the metal.

(A) **Crushing and Grinding** : The ore is first crushed by crushers and ground to a powder.

(B) **Concentration of the ore** : The removal of unwanted, useless impurities from the ore is called **dressing, concentration or benefaction of ore**.

There are several steps for the concentration of Ores.

(I) By physical separation

(a) **Gravity separation (Levigation)**

(b) **Froth Floatation method**

(c) **Magnetic separation**

(i) Gravity separation or Levigation method :

It is based on the difference in the densities of the gangue and ore particles. This method is generally used for the concentration of oxide and native ores.

(ii) Electromagnetic separation :

It is based on differences in magnetic properties of the ore components.

It is used when either the ore or the impurities associated with it are magnetic in nature.

Examples :

Chromite ore($\text{FeO} \cdot \text{Cr}_2\text{O}_3$) is separated from non-magnetic silicious impurities and cassiterite ore(SnO_2) is separated from magnetic Wolframite ($\text{FeWO}_4 + \text{MnWO}_4$).

(iii) Froth floatation process : This process is based on differential wetting of the ore by oil and gangue by water.

Examples : Galena, PbS (ore of Pb) ; copper pyrites $\text{Cu}_2\text{S} \cdot \text{Fe}_2\text{S}_3$ or CuFeS_2 (ore of copper) ; zinc blende, ZnS (ore of zinc) etc.

(a) **Frothers :** Oil like pine oil, camphor oil etc., are used as frothers.

(b) **Frothers stablizer :** Aneline & Cressol

(c) **Collectors :** Potassium or sodium ethyl xanthate is used as a collector.

(d) **Activating and depressing agents :** For example galena (PbS) usually contains the minerals namely zinc blende (ZnS) and pyrites (FeS_2) as impurities. Floatation is carried out by using potassium ethyl xanthate (used as a collector) along with NaCN and Na_2CO_3 (used as depressing agent).

(II) By Chemical separation

Some of the ores are concentrated by means of chemical treatment.

Leaching : It involves the treatment of the ore with a suitable reagent. as to make it soluble while impurity remain insoluble. The ore is recovered from the solution by suitable chemical method.

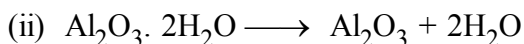
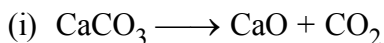
(i) **Bayer's process**

(ii) **Cyanide process**

CALCINATION

Calcination is a process in which ore is heated, generally in the **absence of air**, to expel water from a hydrated oxide or carbon dioxide from a carbonate at temperature below their melting points.

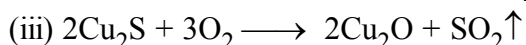
For Example



□ ROASTING

The removal of the excess sulphur contained in sulphide ores by heating **in an excess of air** is called roasting.

(Metal sulphides $\longrightarrow$ Metal oxide + SO_2)



□ REDUCTION OF ORE TO THE METAL

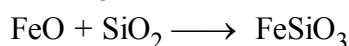
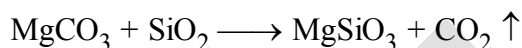
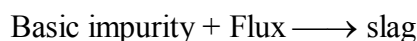
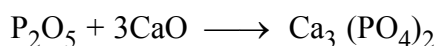
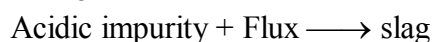
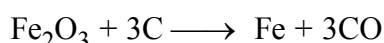
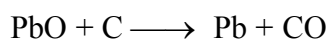
The calcined or roasted ore is then reduced to the metallic state in either of the following ways.

(i) Reduction by Carbon (Smelting)

- Concentrate ore (ore + gangue) + R.A. (carbon) + Flux $[\because \text{R.A.} \Rightarrow \text{Reducing agent}]$



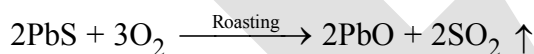
Metal + Slag + gases



(ii) Self Reduction

Sulphides of certain metals are reduced to metal without using any additional reducing agent. ores of Cu, Pb, Hg etc.

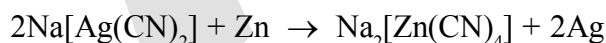
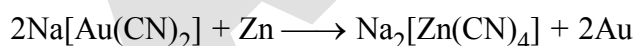
Self Reduction for Pb



(Galena) (air)



(iii) Metal Displacement Method



Sodium tetra cyanozincate

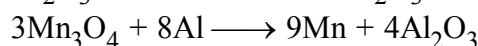
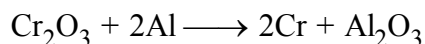
(iv) Electrolytic Reduction

This process is mainly used for the extraction of **highly electropositive metals**.

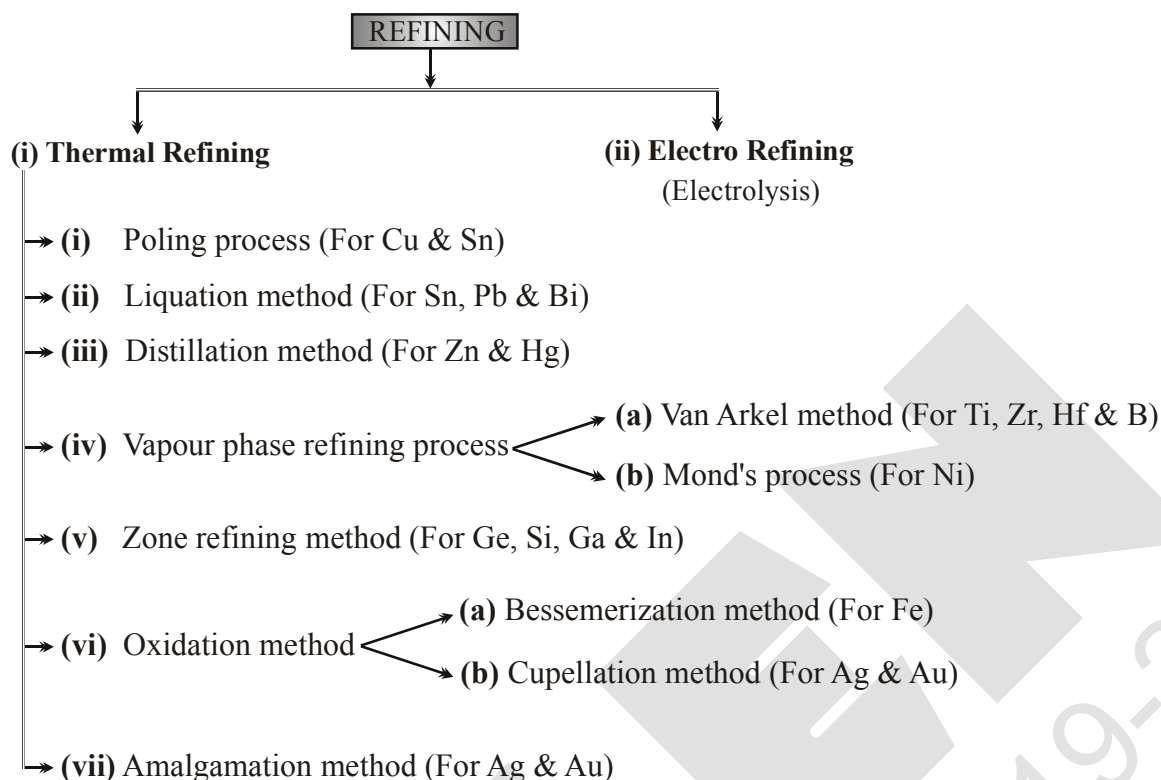
Ex. Na, K, Mg, Ca, Al, etc.

(v) Thermite Reduction or Thermite Process

Al is used as reducing agent in this process. This process is employed in the case of those metals which have very high melting points and are to be extracted from their oxides



REFINING OF METALS



EXTRACTION OF SOME INDIVIDUAL METALS

EXTRACTION OF COPPER

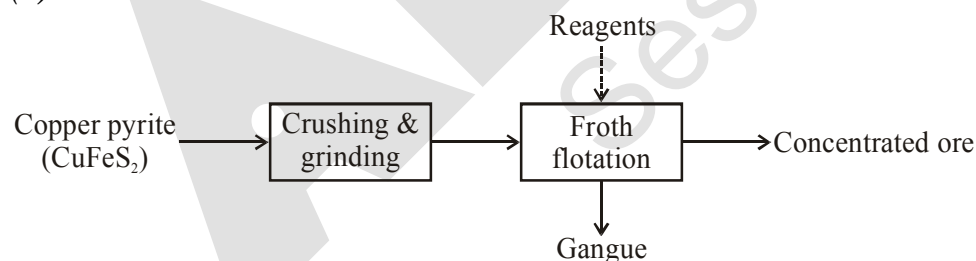
Main Ore : Copper Pyrite (CuFeS_2)

Extraction from pyrites by pyrometallurgical process (Smelting Process)

Some following steps are involved :

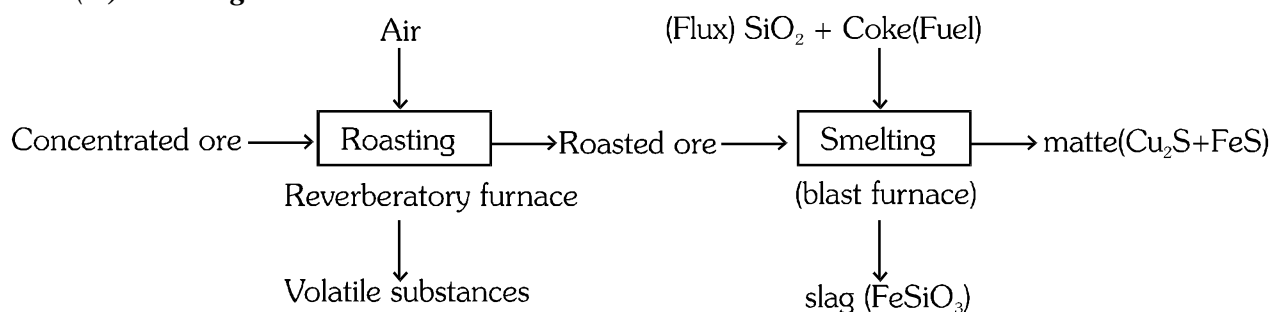
(i) Crushing & Grinding

(ii) Concentration



(iii) Roasting

(iv) Smelting

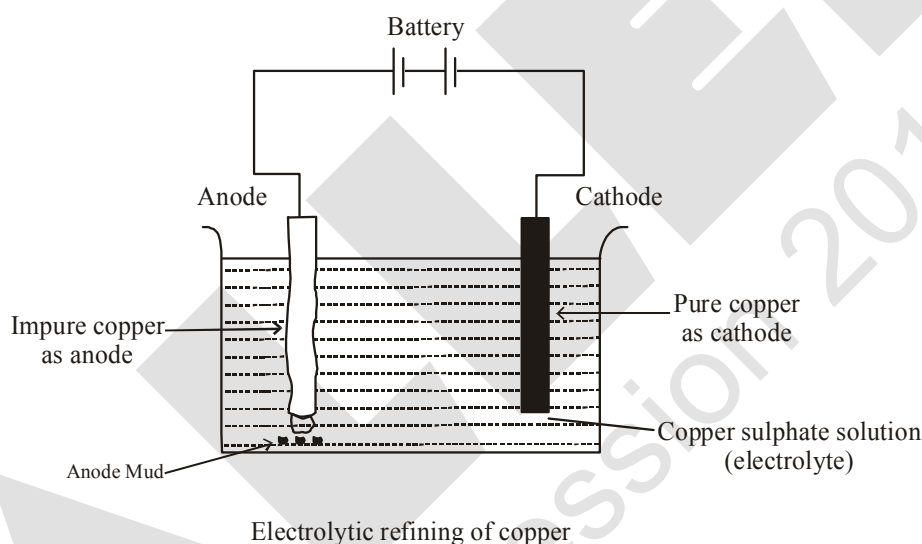
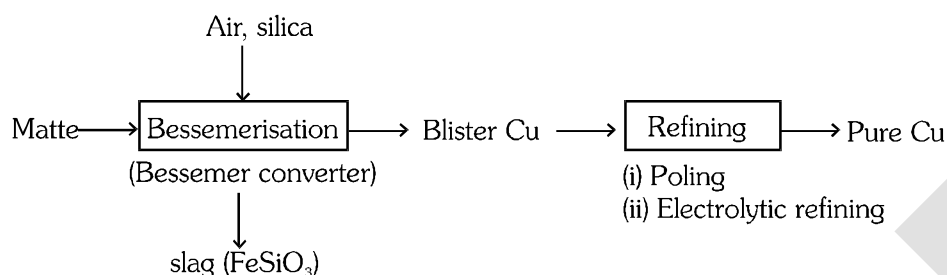


(v) *Self reduction in Bessemer convertor :*

(vi) *Refining of copper :*

(a) Poling

(b) Electrolytic refining



Uses :

Copper is the second most useful metal (the first being Iron) because of its stability in air and water and excellent conductivity.

It is used :

- (i) Copper is used for making wires used in electrical industry and for water and steam pipes.
- (ii) For electroplating.
- (iii) As a coinage alloy (with nickel) and in ornaments and jewellery.
- (iv) For the manufacture of alloys like brass (Cu + Zn), bronze (Cu + Sn) German silver (Cu + Zn + Ni) bell metal (Cu + Sn) gun metal (Cu + Sn + Zn), copper coins (Cu + Zn + Sn) etc.

EXTRACTION OF LEAD

Main Ore : Galena (PbS) - There are mainly two types of process used in the extraction of Lead.

Ore - Galena PbS $\begin{cases} \rightarrow \text{(a) Carbon reduction process} \\ \rightarrow \text{(b) Self reduction process} \end{cases}$

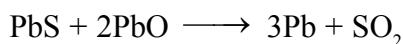
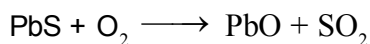
(a) Carbon reduction process (When impurity content is high)

- (i) Crushing & Grinding
- (ii) Concentration by Froth floatation method
- (iii) Roasting process with Lime stone (CaCO_3)
- (iv) Smelting (Carbon reduction method with coke + Fe_2O_3)

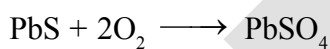
(b) Self reduction process :- (When the impurity content is less)

- (i) Crushing & Grinding
- (ii) Concentration by Froth floatation method
- (iii) Self reduction process

Reaction involved in self reduction :-



Parallel reaction



(c) Refining process :

(a) Liquation

(b) Bett's electrolyrefining

Anode $\rightarrow$ Impure Pb

Cathode $\rightarrow$ Pure Pb

Electrolyte $\rightarrow$ $\text{Pb}[\text{SiF}_6] + \text{H}_2\text{SiF}_6 + \text{Gelatin}$ (to adjust viscosity)

on the electrolysis Pb is deposited at cathode which give 99.95% pure metal.

EXTRACTION OF ZINC

Occurrence :

Its important minerals are :

- | | |
|-------------------------------|-----------------|
| (i) Zinc blende or black jack | ZnS |
| (ii) Zincite | ZnO |
| (iii) Calamine | ZnCO_3 |

Some following steps are involved :

- (i) Concentration by froth floatation method
- (ii) Roasting/calcination
- (iii) Smelting using carbon reduction

(iv) Electrolytic refining

Anode $\rightarrow$ Impure ZnCathode $\rightarrow$ Thin Al-rodElectrolyte $\rightarrow$ Solution of $\text{ZnSO}_4 + \text{H}_2\text{SO}_4$ (dil.)

on the electrolysis zinc is deposited at cathode. The metal is scrapped off and melted to give 99.95% pure metal.

Uses :

- (i) In making alloys e.g. brass, german silver, elektron (Alloy of Mg with smaller amount of Al, Y, Ag, Gd, Zn) etc.
- (ii) In the extraction of silver and gold by cyanide process.
- (iii) It is also used in large quantities in batteries and dry cells for making cathode container.
- (iv) Zn-Cu couple, Zn-Hg, zinc dust etc. are used as reducing agent in organic reactions.
- (v) large amount of zinc is used for galvanizing iron. Zinc is deposited on the surface of iron articles. This process is called galvanization.
- (vi) It is also used in large quantities in batteries, as a constituent of many alloys, e.g., brass, (Cu 60%, Zn 40%) and german silver (Cu 25-30%, Zn 25-30%, Ni 40-50%).
- (vii) Zinc dust is used as a reducing agent in the manufacture of dye-stuffs, paints, etc.

EXTRACTION OF TIN**Main Ore :**

Cassiterite or Tinstone (SnO_2) + Major impurities [(i) SiO_2 , (ii) Sulphides of Fe & Cu, (iii) $\text{FeWO}_4 + \text{MnWO}_4$]

Some following steps are involved :

(i) **Crushing & Grinding :**(ii) **Concentration :** By gravity separation method.(iii) **Roasting :** Followed by washing(iv) **Electromagnetic separation :**

Thus obtained ore contains 60 – 70% SnO_2 and is called as black tin.

(vi) **Carbon reduction method :** Coke & Lime stone (flux) is used.(vii) **Refining method :** (a) Poling (b) ElectrorefiningAnode $\rightarrow$ Impure SnCathode $\rightarrow$ Pure SnElectrolyte $\rightarrow$ (SnSO_4 solution + dil. H_2SO_4)**EXTRACTION OF IRON****Main Ore : Haematite (Fe_2O_3)**

Some following steps are involved :

(i) **Crushing & Grinding :**(ii) **Concentration :** By gravity separation method.(iii) **Roasting :**(iv) **Carbon reduction (Blast furnace) :** Pig iron is obtained from this process(v) **Refining :** Purification of Fe can be done by different method which are as follows :

(a) Puddling Process

(b) Bessemerisation Process

(c) Open hearth Process

(d) L. D. Process

Types of Iron

Cast iron or pig iron

Wrought iron (Fibrous iron) or malleable iron

Steel

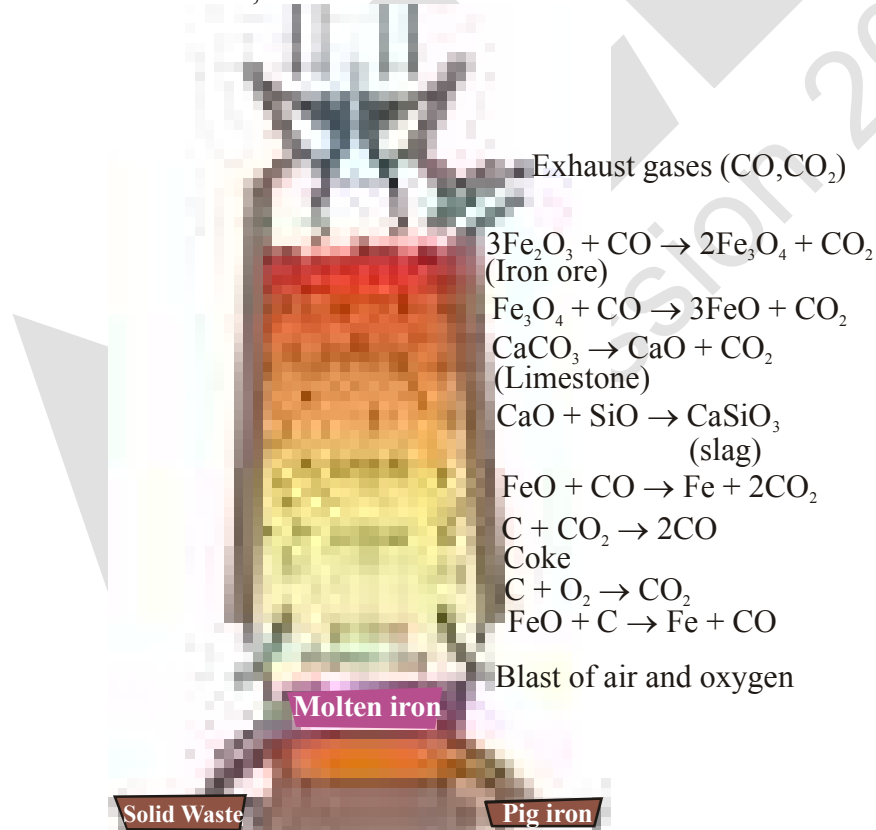
Thus all the three forms of iron differ in their carbon contents, both iron and steel are obtained from cast iron.

Order of M.P. Wrought Iron > Steel > Cast Iron or Pig Iron

% of Carbon in different type of Iron

	<i>NAME</i>	<i>% of C</i>
(1)	Wrought iron	0.1 to 0.25
(2)	Steel	0.25 to 2.0
(3)	Cast Iron	2.6 to 4.3
(4)	Pig Iron	2.3 to 4.6

Ore, limestone and coke



Blast Furnace

Manufacture of Steel : The addition of different desired impurities into molten pure iron is known as steel making

Heat Treatment of Steel

- (i) **Quenching or hardening** : Steel is heated to red hot temperature (700 to 800°C) and is then cooled suddenly by plunging into either cold water or oil. It makes steel hard and brittle.
- (ii) **Annealing** : The steel is heated to red hot temperature (700 to 800°C) and then cooled slowly. It makes steel soft.
- (iii) **Tempering** : If quenched steel is heated to temperature between 500 to 575 K and then cooled slowly, it becomes quite hard but brittleness disappears. The process is called tempering.

Surface treatment of steel

- (i) **Nitriding** : Process of heating steel at 1000 K in an atmosphere of NH_3 . This gives hard coating of iron nitride on the surface.
- (ii) **Case hardening** : Process of giving a thin coating of hardened steel, by heating steel in contact with charcoal followed quenching in oil.
– It is used for axles of railway wagons.

Uses

- (i) **Cast iron** : It is the most important form of iron, is used for casting stoves, railway sleepers, gutter pipes, toys, etc.
- (ii) It is used in the manufacture of wrought iron and steel.
- (iii) **Wrought iron** : It is used in making anchors, wires, bolts, chains and agricultural implements.
- (iv) **Steel finds a number of uses** : Alloy steel is obtained when other metals are added to it. Nickel steel is used for making cables, automobiles and aeroplane parts, pendulum, measuring tapes, chrome steel for cutting tools and crushing machines, and stainless steel for cycles, automobiles, utensils, pens, etc.

EXTRACTION OF SILVER & GOLD**Extraction of silver :****Occurrence :**

Ag found in free and combined state in nature.

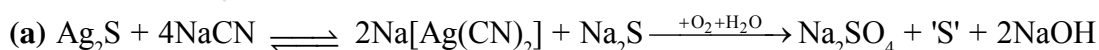
Its main ore is Argentite Ag_2S .

Other ores are

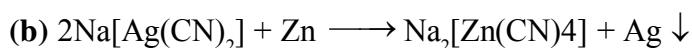
Copper silver glance	—	$\text{Cu}_2\text{S} \cdot \text{Ag}_2\text{S}$
Horn silver	—	AgCl
Argentiferous lead	—	$\text{PbS}(0.01 - 0.1\% \text{ Ag})$

Some steps are involved in the extraction of Silver metal.**(i) Crushing & Grinding :**

(ii) Leaching process : Silver are extracted by the cyanide process (Mc Arthur - Forest process).

Reaction involved :

[$\because \text{O}_2$ is used to make reaction irreversible which remove Na_2S as $\text{Na}_2\text{SO}_4 + S$]

**(iii) Refining Process :**

Anode	→	Impure Ag
Cathode	→	Pure Ag
Electrolyte	→	$(\text{AgNO}_3 \text{ solution} + \text{dil. HNO}_3)$

Uses :

- (i) It is used in silver plating.
- (ii) Silver foils are used in medicine.
- (iii) Silver amalgam is used for dental filling.
- (iv) Compounds of silver are used in silvering of mirrors ($\text{AgNO}_3 + \text{HCHO} + \text{Red Pb}$), in photography, as laboratory reagents etc.
- (v) Silver is easily alloyed with copper, so it is used in making coins, ornaments, silver ware etc.
- (vi) It gives black spot on skin due to decomposition so it is also used as hair dye and ink.

Extraction of Gold :

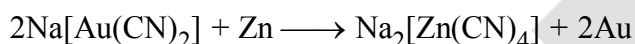
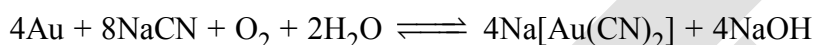
Occurrence : Au found in free (native) state in nature.

Same steps as are involved in the extraction of Silver metal.

(i) **Crushing & Grinding :**

(ii) **Leaching process :** Gold are extracted by the cyanide process (Mc Arthur - Forest process).

Reaction involved :



(iii) **Refining Process :**

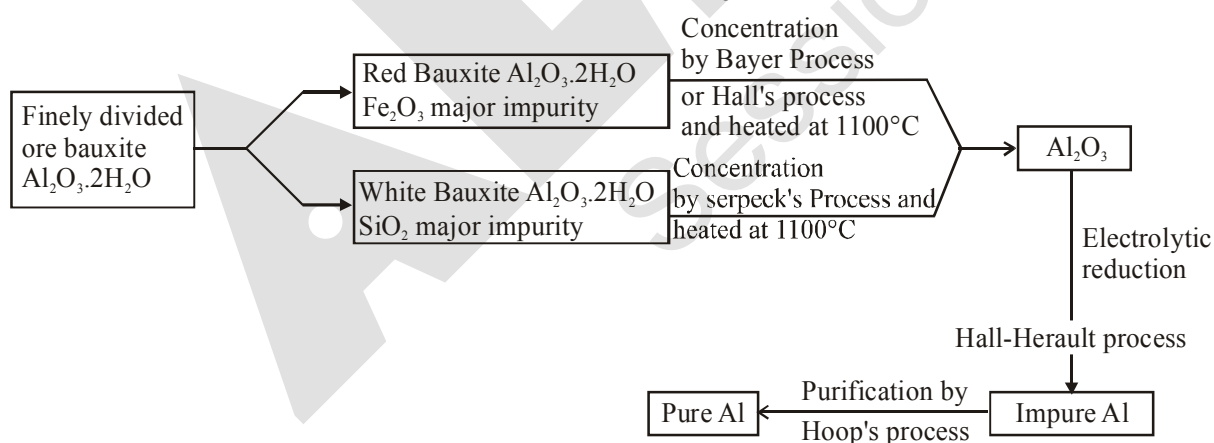
Anode	→	Impure Au
Cathode	→	Pure Au
Electrolyte	→	(AuCl_3 solution + dil. HCl)



EXTRACTION OF ALUMINIUM :

Ore - Bauxite $\text{AlO}_x(\text{OH})_{3-2x}$ (where $0 < x < 1$)

Flow chart of Al from $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ (Bauxite)



Uses

- (i) Aluminium foils are used as wrappers for chocolates.
- (ii) The fine dust of the metal is used in paints and lacquers.
- (iii) Aluminium, being highly reactive, is also used in the extraction of chromium and manganese from their oxides.
- (iv) Wires of aluminium are used as electricity conductors.
- (v) Alloys containing aluminium, being light, are very useful.

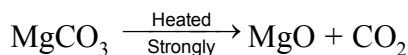
EXTRACTION OF MAGNESIUM :

(i) From Carnallite :

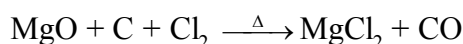
The ore is dehydrated in a current of hydrogen chloride and the mixture of fused chlorides is electrolysed.

(ii) From Magnesite :

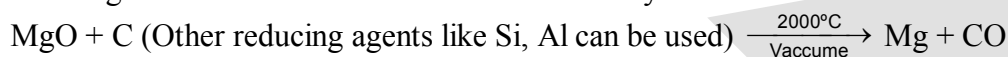
The concentrated ore is calcined at higher temperature



The calcined ore is heated with coke in a current of dry chlorine gas.



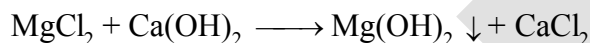
The magnesium chloride is fused and then electrolysed.



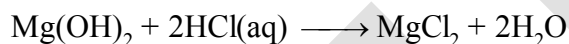
(iii) From Sea water (Dow's process) :

Sea water contains 0.13% magnesium as chloride and sulphate. It involves following steps.

(a) Precipitation of magnesium as magnesium hydroxide by slaked lime :

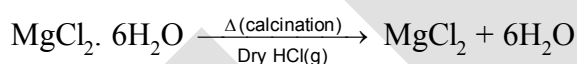


(b) Preparation of hexahydrated magnesium chloride



The solution on concentration and crystallisation gives the crystals of $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$

(c) Preparation of anhydrous magnesium chloride



It is not made anhydrous by simple heating because it gets hydrolysed



(d) Electrolysis of fused anhydrous MgCl_2

(i) Electrolyte : Molten $\text{MgCl}_2 + \text{NaCl} + \text{CaCl}_2$

(ii) Anode : Graphite electrode

(iii) Cathode : Iron cell (steel container)

Reaction occurs :

The molten mixture is electrolysed. Magnesium is liberated at the cathode (iron pot) and chlorine is evolved at graphite anode.

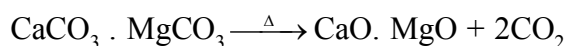


At cathode : $\text{Mg}^{2+} + 2\text{e}^- \longrightarrow \text{Mg(99\% pure)}$;

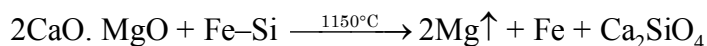
At anode : $2\text{Cl}^- \longrightarrow \text{Cl}_2 + 2\text{e}^-$

(iv) From Dolomite : In the Pidgeon Process Mg is Produced.

The concentrated ore is calcined at higher temperature



It is then reduced by ferrosilicon at 1273 K under reduced pressure.



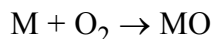
❑ THERMODYNAMICS OF REDUCTION PROCESSES (ELLINGHAM DIAGRAM)

The extraction of metals from their oxides using carbon or other metals, and by thermal decomposition, involves a number of points which merit detailed discussion.

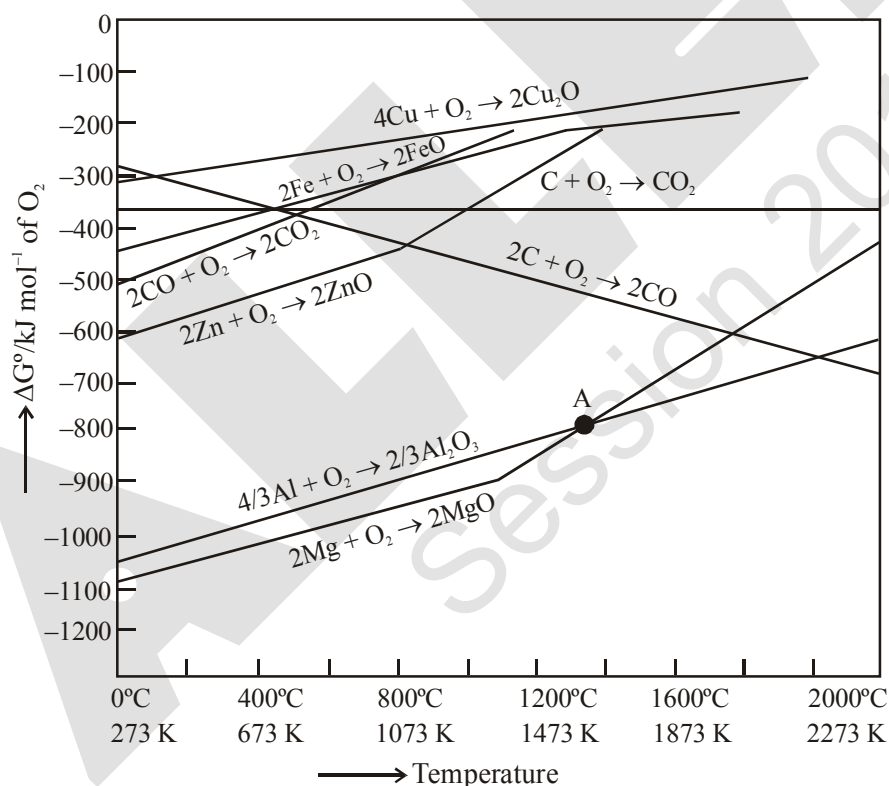
For a spontaneous reaction, the free energy change ΔG must be negative.

$$\Delta G = \Delta H - T\Delta S$$

ΔH is the enthalpy change during the reaction, T is the absolute temperature, and ΔS is the change in entropy during the reaction. Consider a reaction such as the formation of an oxide:



Dioxygen is used up in the course of this reaction. Gases have a more random structure (less ordered) than liquids or solids. Consequently gases have a higher entropy than liquids or solids. In this reaction S the entropy or randomness decreases, the hence ΔS is negative. Thus if the temperature is raised then $T\Delta S$ becomes more negative. Since $T\Delta S$ is subtracted in the equation, then ΔG becomes less negative. Thus the free energy changed increases with an increase of temperature.



The free energy changes that occur when one gram molecule of a common reactant (in this case dioxygen) is used may be plotted graphically against temperature for a number of reactions of metals of their oxides. This graph is shown in figure and is called an Ellingham diagram (for oxides). Similar diagrams can be produced for one gram molecule of sulphur, giving an Ellingham diagram for sulphides, and similarly for halides.

The Ellingham diagram for oxides shows several important features:

- (i) The graph for metal oxide all slope upwards, because the free energy change increases with an increase of temperature as discussed above.
- (ii) The free energy changes all follows a straight line unless the materials metal or vaporize.
- (iii) When the temperature is raised, a point will be reached where the graph crosses the $\Delta G = 0$ line. Below this temperature the free energy of formation of the oxide is negative, so the oxide is stable. Above this temperature the free of formation of the oxide is positive, and the oxide becomes unstable, and should decompose into the metal and dioxygen.
- (iv) Any metal will reduce the oxide of other metals which lie above it in the Ellingham diagram because the free energy will become more negative by an amount equal to the different between the two graphs at that particular temperature.

Limitations of Ellingham Diagram

- (i) The graph simply indicates whether a reaction is possible or not i.e., the tendency of reduction with a reducing agent is indicated. This is so because it is based only on the thermodynamic concepts. It does not say about the kinetics of the reduction process (Cannot answer questions like how fast it could be ?).
- (ii) The interpretation of $\Delta G^\ominus$ is based on $K(\Delta G^\ominus = -RT \ln K)$. Thus it is presumed that the reactants of products are in equilibrium.

Aluminium	1. Bauxite, $\text{Al}_2\text{O}_3 \cdot x \text{H}_2\text{O}$ 2. Cryolite, Na_3AlF_6	Electrolysis of Al_2O_3 dissolved in molten Na_3AlF_6	For the extraction, a good source of electricity is required.
Iron	1. Haematite, Fe_2O_3 2. Magnetite, Fe_3O_4	Reduction of the oxide with CO and coke in Blast furnace	Temperature approaching 2170 K is required.
Copper	1. Copper pyrites, CuFeS_2 2. Copper glance, Cu_2S 3. Malachite, $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ 4. Cuprite, Cu_2O	Roasting of sulphide partially and reduction	It is self reduction in a specially designed converter. The reduction takes place easily. Sulphuric acid leaching is also used in hydrometallurgy from low grade ores.
Zinc	1. Zinc blende or Sphalerite, ZnS 2. Calamine, ZnCO_3 3. Zincite, ZnO	Roasting followed by reduction with coke	The metal may be purified by fractional distillation.

GENERAL PRINCIPLES & PROCESSES OF ISOLATION OF ELEMENTS

EXERCISE # 0-1

ONLY ONE OPTION IS CORRECT.

ORES

- Which of the following does not contain Mg:
(A) magnetite (B) magnesite (C) asbestos (D) carnallite
- Which of the following is not an ore:
(A) malachite (B) calamine (C) stellite (D) cerussite
- Carnallite does not contain
(A) K (B) Ca (C) Mg (D) Cl
- Among the following statements, the incorrect one is
(A) calamine and siderite are carbonate ores (B) argentite and cuprite are oxide ores
(C) zinc blende and pyrites are sulphide ores (D) malachite and azurite are ores of copper
- Select the correct statement :
(A) Magnetite is an ore of manganese (B) Pyrolusite is an ore of lead
(C) Siderite is carbonate ore of iron (D) FeS_2 is rolled gold
- "Fool's gold" is
(A) iron pyrites (B) horn silver (C) copper pyrites (D) bronze
- Assertion :** Platinum and gold occur in native state in nature.
Reason : Platinum and gold are noble metals.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CONCENTRATION METHODS

8. $\text{Ag}_2\text{S} + \text{NaCN} + \text{Zn} \longrightarrow \text{Ag}$
This method of extraction of Ag by complex formation and then its displacement is called:
(A) Parke's method (B) McArthur-Forest method
(C) Serpeck method (D) Hall's method
9. Which one of the following is not a method of concentration of ore?
(A) gravity separation (B) froth floating process
(C) electromagnetic separation (D) smelting
10. Chemical leaching is useful in the concentration of:
(A) copper pyrites (B) bauxite (C) galena (D) cassiterite
11. In froth-floatation process, pine oil functions as
(A) activator (B) frother (C) collector (D) agitator

12. Collectors are the substances which help in attachment of an ore particle to air bubble in froth. A popular collector used industrially is
(A) sodium ethyl xanthate (B) sodium xenate
(C) sodium pyrophosphate (D) sodium nitroprusside
13. In the cyanide process involving extraction of silver, zinc is used industrially as a(an)
(A) oxidising agent (B) reducing agent
(C) solvent (D) solvating agent
14. During initial treatment, preferential wetting of ore by oil and gangue by water takes place in
(A) Levigation (gravity separation) (B) Froth floatation
(C) Leaching (D) Bessemerisation
15. An non-magnetic ore containing the impurity of FeCr_2O_4 is concentrated by
(A) magnetic-separation (B) gravity separation
(C) froth-floatation method (D) electrostatic method
16. The beneficiation of the sulphide ores is usually done by
(A) Electrolysis (B) Smelting process
(C) Metal displacement method (D) Froth flotation method
17. The process of the isolation of a metal by dissolving the ore in aqueous solution of suitable chemical reagent followed by precipitation of the metal by a more electropositive metal is called:
(A) hydrometallurgy (B) electrometallurgy
(C) zone refining (D) electrorefining
18. Froth floatation process for concentration of ores is an illustration of the practical application of:
(A) Adsorption (B) Absorption
(C) Coagulation (D) Sedimentation
19. **Assertion :** Sulphide ores are concentrated by froth floatation process.
Reason : Pine oil acts as a frothing agent in froth floatation process.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
20. **Assertion :** Wolframite impurities are separated from cassiterite by electromagnetic separation.
Reason : Cassiterite being magnetic is attracted by the magnet and forms a separate heap.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CALCINATION/ROASTING

21. Calcination is the process of heating the ore:
 (A) in inert gas (B) in the presence of air
 (C) in the absence of air (D) in the presence of CaO and MgO
22. When roasting is carried out :
 (i) Sulphide ore is converted into oxide and sulphate
 (ii) remove water of hydration
 (iii) the ore melts (iv) arsenic and sulphur impurities are removed
 Of these statements:
 (A) (i), (ii) and (iii) are correct (B) (i) and (iv) are correct
 (C) (i), (ii) and (iv) are correct (D) (ii), (iii) and (iv) are correct

REDUCTION PROCESS

23. In the aluminio thermite process, Al acts as
 (A) An oxidising agent (B) A flux
 (C) A reducing agent (D) A solder
24. **Assertion :** Al is used as a reducing agent in aluminothermy.
Reason : Al has a lower melting point than Fe, Cr and Mn.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
25. Formation of metallic copper from the sulphide ore in the commercial thermo-metallurgical process essentially involves which one of the following reaction:
 (A) $\text{Cu}_2\text{S} + \frac{3}{2}\text{O}_2 \longrightarrow \text{Cu}_2\text{O} + \text{SO}_2$; $\text{CuO} + \text{C} \longrightarrow \text{Cu} + \text{CO}$
 (B) $\text{Cu}_2\text{S} + \frac{3}{2}\text{O}_2 \longrightarrow \text{Cu}_2\text{O} + \text{SO}_2$; $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \longrightarrow 6\text{Cu} + \text{SO}_2$
 (C) $\text{Cu}_2\text{S} + 2\text{O}_2 \longrightarrow \text{CuSO}_4$; $\text{CuSO}_4 + \text{Cu}_2\text{S} \longrightarrow 3\text{Cu} + 2\text{SO}_2$
 (D) $\text{Cu}_2\text{S} + \frac{3}{2}\text{O}_2 \longrightarrow \text{Cu}_2\text{O} + \text{SO}_2$; $\text{Cu}_2\text{O} + \text{CO} \longrightarrow 2\text{Cu} + \text{CO}_2$
26. The element which could be extracted by electrolytic reduction of its oxide dissolved in a high temperature melt is:
 (A) sodium (B) magnesium (C) fluorine (D) aluminium
27. In which of the following isolations no reducing agent is required:
 (A) iron from haematite (B) Tin from cassiterite
 (C) mercury from cinnabar (D) zinc from zinc blende

PURIFICATION METHODS

28. A metal has a high concentration into the earth crust and whose oxides cannot be reduced by carbon. The most suitable method for the extraction of such metal is
 (A) Alumino thermite process (B) Electrolysis process
 (C) Van-Arkel's process (D) Cupellation
29. **Assertion :** Alkali metals can not be prepared by the electrolysis of their chlorides in aqueous solution
Reason : Reduction potentials of alkali metals cations is much lower than that of H_2O .
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
30. **Assertion :** Magnesium can be prepared by the electrolysis of aq. $MgCl_2$.
Reason : The reduction potential of Mg^{2+} is much lower than that of H_2O .
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
31. Bessemerisation is carried out for
 I : Fe, II : Cu, III : Al, IV : silver
 (A) I, II (B) II, III (C) III, IV (D) I, III
32. In the extraction of nickel by Mond process, the metal is obtained by:
 (A) electrochemical reduction (B) thermal decomposition
 (C) chemical reduction by aluminium (D) reduction by carbon
33. Formation of $Ni(CO)_4$ and subsequent its decomposition into Ni and CO (recycled) makes basis of Mond's process

$$Ni + 4CO \xrightarrow{T_1} Ni(CO)_4 \xrightarrow{T_2} Ni + 4CO$$
 T_1 and T_2 are:
 (A) $100^\circ C$, $50^\circ C$ (B) $50^\circ C$, $100^\circ C$ (C) $50^\circ C$, $230^\circ C$ (D) $230^\circ C$, $50^\circ C$
34. Zone refining is based on the principle of
 (A) fractional distillation (B) fractional crystallisation
 (C) partition coefficient (D) chromatographic separation
35. Si and Ge used for semiconductors are required to be of high purity and hence purified by
 (A) zone-refining (B) electrorefining
 (C) Van-Arkel's process (D) cupellation process
36. Which process of purification is represented by the following equation :

$$Ti(\text{Impure}) + 2I_2 \xrightarrow{250^\circ C} TiI_4 \xrightarrow{1400^\circ C} Ti(\text{Pure}) + 2I_2$$
 (A) Cupellation (B) Poling (C) Van-Arkel Process (D) Zone refining
37. Which of the following employ(s) thermal decomposition of volatile iodide compounds?
 (A) Thermite process (B) Hall's process (C) Van-Arkel's process (D) Mond's process

38. The method of zone refining of metals is based on the principle of:
 (A) Greater mobility of the pure metal than that of impurity.
 (B) Higher melting point of the impurity than that of the pure metal.
 (C) Greater noble character of the solid metal than that of the impurity
 (D) Greater solubility of the impurity in the molten state than in the solid
39. **Assertion :** Titanium is purified by Van-Arkel method.
Reason : Ti reacts with I_2 to form volatile TiI_4 which decomposes at 1673 K to give pure Ti.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
40. **Assertion :** Nickel is purified by the thermal decomposition of nickel tetracarbonyl.
Reason : Nickel is a transition element.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
41. Refining of silver is done by:
 (A) liquation (B) poling (C) cupellation (D) van Arkel method
42. Mercury is purified by:
 (A) Passing through dilute HNO_3 (B) Distillation
 (C) Distribution (D) Vapour phase refining
43. **Assertion :** Lead, tin and bismuth are purified by liquation method.
Reason : Lead, tin and bismuth have low m.p. as compared to impurities.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
44. When an impurity in a metal has greater affinity for oxygen and is more easily oxidised than the metal itself. Then, the metal is refined by
 (A) cupellation (B) zone-refining (C) distillation (D) electrolytic process

EXTRACTION OF METALS

45. Which of the following process is not associated with recovery of the silver -
 (A) As a side product in electrolytic refining of copper
 (B) Parke's process in which Zn is used to extract silver by solvent extraction from molten lead
 (C) By reaction of silver sulphide with KCN and then reaction of soluble complex with Zn
 (D) By boiling $Na[Ag(CN)_2]$ aq.
46. Blister Cu is about:
 (A) 60% Cu (B) 90% Cu (C) 98% Cu (D) 100% Cu

47. Iron obtained from blast furnace is:
(A) wrought iron (B) cast iron (C) pig iron (D) steel
48. Which of the following term is not related to Al-extraction
(A) Serpek's process (B) Hall-Heroult process
(C) Thermite process (D) Hoop's process
49. Dow's process
(A) involves purification of copper (B) involves extraction of magnesium
(C) gives metal chloride as product (D) gives pure Na as product
50. Silica is added to roasted copper ores during extraction in order to remove
(A) cuprous sulphide (B) ferrous oxide (C) ferrous sulphide (D) cuprous oxide
51. Addition of high proportions of manganese makes steel useful in making rails of railroads, because manganese
(A) gives hardness to steel (B) helps the formation of oxides of iron
(C) can remove oxygen and sulphur (D) can show highest oxidation state of +7
52. In the commercial electrochemical process for aluminium extraction the electrolyte used is
(A) $\text{Al}(\text{OH})_3$ in NaOH solution (B) an aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
(C) a molten mixture of Al_2O_3 , Na_3AlF_6 & CaF_2 (D) a molten mixture of Al_2O_3 and $\text{Al}(\text{OH})_3$
53. Blister copper is refined by stirring molten impure metal with green logs of wood because such a wood liberates hydrocarbon gases (like CH_4). This process X is called _____ and the metal contains impurities of Y is _____.
(A) X = cupellation, Y = CuO_2 (B) X = poling, Y = Cu_2O
(C) X = poling, Y = CuO (D) X = cupellation, Y = CuO
54. A piece of steel is heated until redness and then plunged into cold water or oil. This treatment of steel makes it
(A) soft and malleable (B) hard but not brittle
(C) more brittle (D) hard and brittle
55. Modern method of steel manufacturing is
(A) open hearth process (B) L.D. Process
(C) Bessemerisation (D) Cupellation
56. During electrolytic reduction of alumina, two auxiliary electrolytes X and Y are added to increase the electrical conductance and lower the temperature of melt in order to making fused mixture very conducting. X and Y are
(A) cryolite and flourspar (B) cryolite and alum
(C) alum and flourspar (D) flourspar and bauxite
57. For extraction of sodium from NaCl, the electrolytic mixture $\text{NaCl} + \text{KCl} + \text{CaCl}_2$ is used. During extraction process, only sodium is deposited on cathode but K and Ca do not because
(A) Na is more reactive than K and Ca
(B) Na is less reactive than K and Ca
(C) NaCl is less stable than Na_3AlF_6 and CaCl_2
(D) the discharge potential of Na^+ is less than that of K^+ and Ca^{2+} ions.

58. Railway wagon axles are made by heating iron rods embedded in charcoal powder. This process is known as:
 (A) Sherardising (B) Annealing (C) Tempering (D) Case hardening
59. In the extraction of copper from its sulphide ore the metal is formed by the reduction of Cu_2O with:
 (A) FeS (B) CO (C) Cu_2S (D) SO_2
60. Carnallite on electrolysis gives:
 (A) Ca and Cl_2 (B) Na and CO_2 (C) Al and Cl_2 (D) Mg and Cl_2

MISCELLANEOUS

61. Which of the following statement is correct regarding Cu-extraction
 (A) In the smelting step carbon reduction takes places
 (B) During partial roasting Cu_2S remains almost unaffected
 (C) In Bessemer converter, only self reduction occur, not slag formation
 (D) Blister forms in the blister Cu is due to dissolved CO_2
62. Refractory materials are generally used in furnaces because
 (A) they are chemically inert (B) they can withstand high temperature
 (C) they do not contain impurities (D) they decrease melting point of ore
63. Which of the following statements is correct regarding the slag formation during the extraction of a metal like copper or iron.
 (A) The slag is lighter and has lower melting temperature than the metal
 (B) The slag is heavier and has lower melting temperature than the metal
 (C) The slag is lighter and has higher melting temperature than the metal
 (D) The slag is heavier and has higher melting temperature than the metal
64. **Assertion :** Generally in smelting, roasted/cacinated ore is heated with powdered coke in presence of a flux.
Reason : Oxides are reduced to metals by C or CO. Impurities are removed as slag.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
65. **Assertion :** Magnesita and quick lime are used as basic flux.
Reason : MgO and CaO can withstand very high temperatures.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
66. **Assertion :** Wolframite impurity is separated from SnO_2 by magnetic separation
Reason : Tin stone is ferromagnetic, therefore attracted by magnet.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

EXERCISE # O-2

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

- Which of the following is(are) sulphide ores?
(A) Argentite (B) Galena (C) Anglesite (D) Copper glance
- Which of the following is(are) regarded as iron ores?
(A) Haematite (B) Magnetite (C) Limonite (D) Copper pyrites

CONCENTRATION

- Which of the following ores is(are) concentrated by froth floatation?
(A) haematite (B) galena (C) copper pyrite (D) azurite
- Which of the following ores is (are) concentrated industrially by froth floatation?
(A) Copper pyrites (B) Galena (C) Dolomite (D) Carnallite
- Leaching is used for the concentration of:
(A) Red bauxite (B) Haematite (C) Gold ore (D) Silver ore

CALCINATION/ROASTING

- Calcination and roasting processes of ores to form their oxides are beneficial
(A) to convert ores into porous form so that their reduction becomes easier
(B) as impurities like S, As, Sb, are removed
(C) as organic impurities are removed.
(D) as the ores are converted into oxide form which makes the reduction easier
- Which of the following reaction(s) occur during calcination?
(A) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (B) $4\text{FeS}_2 + 11\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$
(C) $2\text{Al}(\text{OH})_3 \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2\text{O}$ (D) $\text{CuS} + \text{CuSO}_4 \rightarrow 2\text{Cu} + 2\text{SO}_2$
- Which of the following is true for calcination of a metal ore?
(A) It makes the ore more porous
(B) The ore is heated to a temperature when fusion just begins
(C) Hydrated salts lose their water of crystallisation
(D) Impurities of S, As and Sb are removed in the form of their volatile oxides.
- Roasting can be performed in
(A) blast furnace (B) reverberatory furnace
(C) electric furnace (D) None of these

REDUCTION

- Carbon reduction method is employed for commercial extraction of metal from amongst these :
(A) haematite (B) cassiterite (C) iron pyrite (D) corundum
- Auto reduction process is used in extraction of
(A) Cu (B) Hg (C) Al (D) Fe
- Which of the following reduction reactions are actually employed in commercial extraction of metals?
(A) $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$
(B) $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$
(C) $2\text{Na}[\text{Au}(\text{CN})_2] + \text{Zn} \rightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Au}$
(D) $\text{Cu}_2\text{S} + \text{Pb} \rightarrow \text{Cu} + \text{PbS} \downarrow$

PURIFICATION

13. In the manufacturing of metallic sodium by fused salt-electrolysis method (Down's process), small amount of CaCl_2 that added is known as auxiliary electrolyte and is used to
 (A) improve the electrical conductance (B) decrease the melting point of electrolyte
 (C) stabilise the metallic sodium (D) increase the temperature of electrolysis
14. Poling is employed in refining of
 (A) iron (B) copper (C) tin (D) lead
15. Zone refining is used for purification of
 (A) Ge (B) Si (C) Ga (D) In
16. Metal(s) which does/do not form amalgam is/are
 (A) Fe (B) Pt (C) Zn (D) Au
17. Metals which can be commercially extracted by smelting process
 (A) Pb (B) Fe (C) Zn (D) Mg

EXTRACTION OF METALS

18. Hoop's process of purification of aluminium involves formation of layers during electrolysis. It involves
 (A) the three layers have same densities but different materials.
 (B) the three layers have different densities
 (C) the upper layer is of pure aluminium which acts as a cathode
 (D) the bottom layer is of impure aluminium which acts as an anode and middle layer consists of cryolite and BaF_2 .
19. Metallurgical process of zinc involves roasting of zinc sulphide followed by reduction. Metallic zinc distills over as it is volatile and impurities like Cu, Pb and Fe gets condensed. The crude metal obtained is called spelter, which may be purified by
 (A) electrolysis process (B) fractional distillation
 (C) polling (D) heating with iodine
20. Which of the following process (es) are used for purification of Bauxite ore?
 (A) Hall's process (B) Serpeck's process (C) Baeyer's process (D) Mond's process
21. Common impurities present in Bauxite are
 (A) CuO (B) ZnO (C) Fe_2O_3 (D) SiO_2
22. Calcium silicate slag formed in extraction of iron
 (A) prevents the reoxidation of molten iron.
 (B) catalyses the combustion of carbon.
 (C) reduces CO_2 to CO at the bottom of the furnace.
 (D) is used in cement industry.
23. Amphoteric nature of aluminium is employed in which of the following process for extraction of aluminium?
 (A) Baeyer's process (B) Hall's process
 (C) Serpeck's process (D) Dow's process
24. The chief reaction(s) occurring in blast furnace during extraction of iron from haematite is(are)
 (A) $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ (B) $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
 (C) $\text{Fe}_2\text{O}_3 + \text{C} \rightarrow 2\text{Fe} + 3\text{CO}$ (D) $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$

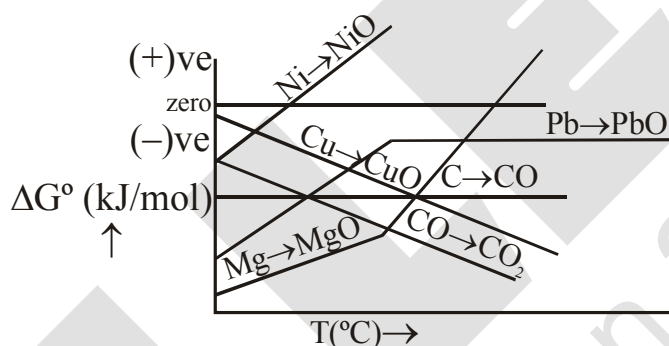
25. Which of the following are true for electrolytic extraction of aluminium
(A) cathode material contains graphite (B) anode material contains graphite
(C) cathode reacts away forming CO_2 (D) anode reacts away forming CO_2
26. During extraction of copper, it is obtained in the form of molten *matte*. Which of the following is **not true**?
(A) *matte* is further treated in Bessemer's converter
(B) molten *matte* is electrolysed
(C) It is treated with a blast of air and sand
(D) It is dissolved in CuSiF_6 and crystallised.
27. The major role of fluorspar (CaF_2) which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6) is
(A) as a catalyst
(B) to make the fused mixture very conducting
(C) to lower the melting temperature of the mixture
(D) to decrease the rate of oxidation of carbon at the anode.
28. Which of the following reaction does not occur in blast furnace during extraction of iron :
(A) $\text{CaO} + \text{SiO}_2 \longrightarrow \text{CaSiO}_3$ (B) $\text{Fe}_2\text{O}_3 + 3\text{CO} \longrightarrow 2\text{Fe} + 3\text{CO}_2$
(C) $\text{FeO} + \text{SiO}_2 \longrightarrow \text{FeSiO}_3$ (D) $\text{FeO} \longrightarrow \text{Fe} + \frac{1}{2}\text{O}_2$

MISCELLANEOUS

29. Which of the following employ downward movement of ore due to gravity?
(A) Gravity separation (B) Froth floatation
(C) Blast furnace (D) Bessemer's converter
30. The **CORRECT** statements are :
(A) generally the calcination and roasting is done in blast furnace
(B) the sandy and rocky materials associated with ore are called matrix
(C) froth floatation process is suitable for sulphide ores
(D) substance that reacts with gangue to form fusible mass is called slag

EXERCISE # S-1

- Find the number of ore which are concentrated by magnetic separation method.
Haemetite, Cassiterite, Copper Glance, Chromite, Cinnabar
- Find the number of metals which are commercially extracted by carbon reduction method
Pb, Fe, Zn, Mg, Al, Na, Au, Ag
- The number of following pairs is correctly matched
 - Van Arkel method – Zirconium
 - Mond Process – Titanium
 - Froth Floatation Method – Cerussite
 - Distillation method – Zinc
 - Poling Process – Copper
 - amalgamation – Gold
- Find the number of curves which are wrongly presented in the Ellingham diagram.

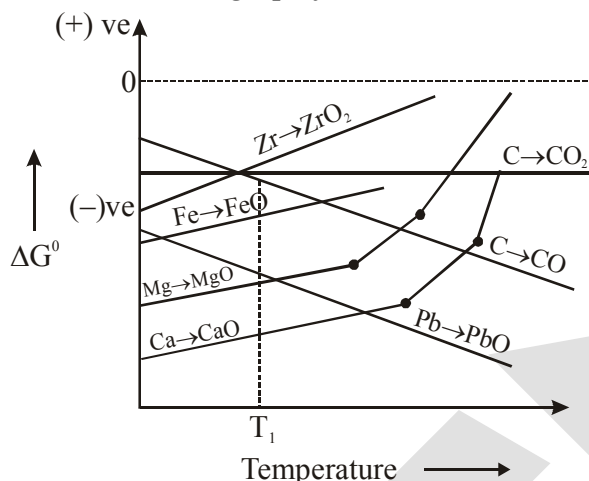


- How many of the following minerals containing Mg.
Magnetite, Carnallite, Epsom salt, Siderite
- Find out the number of minerals given below contain iron as Fe(II).
Haematite, Magnetite, Limonite, Siderite, Chromite, Wolframite
- Amongst the following ores, the total number of oxide ores are
Siderite, Magnetite, Haematite, Malachite, Zincite, Cuprite
- Amongst the following, total number of sulphide ores are
Calamine, Sphalertie, Copper pyrites, Copper glance, Iron pyrites, Bauxite
- How many of the following ores of silver ?
Hornsilver, Cerrusite, Chalcopyrite, Galena, Anglesite, Argentite

EXERCISE # S-2

COMPREHENSION AND MATCH THE COLUMN
ELLINGHAM DIAGRAM

Paragraph for 1 to 3



- Which of the above curve is wrongly presented -
(A) $C \rightarrow CO_2$ (B) $Pb \rightarrow PbO$ (C) $Zr \rightarrow ZrO_2$ (D) $Mg \rightarrow MgO$
- Which of the above metal oxide is having minimum thermal decomposition temperature.
(A) CaO (B) FeO (C) ZrO_2 (D) MgO
- Which of the following metal's oxide can be reduced by Fe as reducing agent at temperature (T_1)
(A) Zr (B) Ca (C) Mg (D) None of these

PURIFICATION METHOD

Paragraph for 4 to 5

At high temperature carbon reacts with water to produce a mixture of carbon monoxide, CO and hydrogen, H_2 .



CO is separated from H_2 and then used to separate nickel from cobalt by forming a volatile compound, nickel tetracarbonyl, $Ni(CO)_4$.



- How many moles of $Ni(CO)_4$ could be obtained from the CO produced by the reaction of 75.0g of carbon? Assume 100% reaction and 100% recovery in both steps.
(A) 6.25 (B) 1.563 (C) 3.125 (D) 25.0
- Formation of volatile $Ni(CO)_4$ and its subsequent heating gives pure Ni. Process is called -
(A) Hall (B) Dow (C) Serpeck (D) Mond

MISCELLANEOUS

Match Column

- Match Column-I with Column-II

Column-I (Metals)	Column-II (Method used for refining)
(A) Iron & copper	(P) Poling
(B) Zirconium & Titanium	(Q) Bessemerisation
(C) Lead & Tin	(R) Van-Arkel
(D) Copper & Tin	(S) Liquation

7. Match the following choosing one item from column X and the appropriate item from column Y.

Column -X

- (A) Zinc from ZnCO_3
(B) Lead from PbS
(C) Cu from CuFeS_2
(D) Tin from cassiterite

Column-Y

- (P) Calcination
(Q) Removal of iron
(R) Froth floatation process
(S) Poling

8. Match column (I) (process) with column (II) (electrolyte)

Column (I) (process)

- (A) Downs cell
(B) Dow's sea water process
(C) Hall-Heroult

Column (II) (electrolyte)

- (P) fused MgCl_2
(Q) fused $(\text{Al}_2\text{O}_3 + \text{Na}_3\text{AlF}_6 + \text{CaF}_2)$
(R) fused $(40\% \text{NaCl} + 60\% \text{CaCl}_2)$
(S) $(\text{Al}/\text{N} + \text{C} + \text{N}_2)$

9. Match column - I with column - II

Column - I (Property)

- (A) Explosive
(B) Self-reduction
(C) Ferrimagnetic material
(D) Verdigris

Column - II (Element/compound)

- (P) Cu
(Q) Fe_3O_4
(R) $\text{Cu}(\text{CH}_3\text{COO})_2 \cdot \text{Cu}(\text{OH})_2$
(S) $\text{Pb}(\text{NO}_3)_2$

10. Match column - I and column - II and select the correct answer using the codes given below the lists:

Column - I

- (A) Cyanide process
(B) Floatation process
(C) Electrolytic reduction
(D) Zone refining

Column - II

- (P) Ultrapure Ge
(Q) Dressing of HgS
(R) Extraction of Al
(S) Extraction of Au

11. Match the items of Column I with items of Column II and assign the correct code :

Column I

- (P) Blistered Cu
(Q) Blast furnace
(R) Reverberatory furnace
(S) Hall-Heroult process

Column II

- (1) Aluminium
(2) $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6\text{Cu} + \text{SO}_2$
(3) Iron
(4) $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
(5) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$

- Code :** (A) $\text{P} \rightarrow (2)$; $\text{Q} \rightarrow (3)$; $\text{R} \rightarrow (4)$; $\text{S} \rightarrow (1)$ (B) $\text{P} \rightarrow (1)$; $\text{Q} \rightarrow (2)$; $\text{R} \rightarrow (3)$; $\text{S} \rightarrow (5)$
(C) $\text{P} \rightarrow (5)$; $\text{Q} \rightarrow (4)$; $\text{R} \rightarrow (3)$; $\text{S} \rightarrow (2)$ (D) $\text{P} \rightarrow (4)$; $\text{Q} \rightarrow (5)$; $\text{R} \rightarrow (3)$; $\text{S} \rightarrow (2)$

Answer Q.12, Q.13 and Q.14 by appropriately matching the information given in the three columns of the following table.

Column - I Extraction of metal	Column - II Methods for Reduction	Column - III Refining Method
(1) $\text{SnO}_2 \rightarrow \text{Sn}$	(i) Carbon Reduction	(P) Poling
(2) $\text{Al}_2\text{O}_3 \rightarrow \text{Al}$	(ii) Hydrometallurgic Reduction	(Q) Electrolytic Refining
(3) $\text{Cu}_2\text{S} \rightarrow \text{Cu}$	(iii) Electrolytic Reduction	(R) Distillation
(4) $\text{ZnS} \rightarrow \text{Zn}$	(iv) Self-Reduction	(S) Puddling Process

12. Which of the following is **NOT** correctly matched?
 (A) (1), (iv), (S) (B) (3), (iv), (P) (C) (4), (i), (R) (D) (2), (iii), (Q)
13. Which of the following match is the **CORRECT** reduction and purification methods for Zn
 (A) (i), (R) (B) (iv), (S) (C) (iv), (P) (D) None of these
14. Which of the following set of code shows the **CORRECT** similarity with the extraction processes for Pb?
 (A) (1), (ii), (S) (B) (4), (iii), (P) (C) (2), (iii), (Q) (D) (3), (iv), (Q)

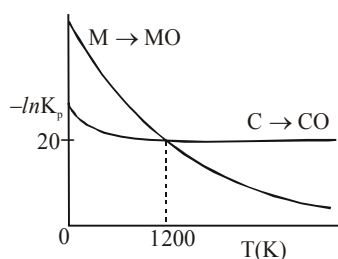
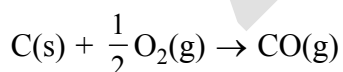
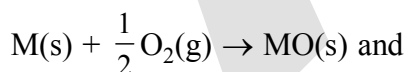
EXERCISE # JEE-MAINS

- Aluminium is extracted by the electrolysis of :- [AIEEE-2002]
 (1) Bauxite (2) Alumina
 (3) Alumina mixed with molten cryolite (4) Molten cryolite
- Pyrolusite is a/an :- [AIEEE-2002]
 (1) Oxide ore (2) Sulphide ore (3) Carbide ore (4) Not an ore
- Which one of the following ores is best concentrated by froth-flotation method : [AIEEE-2004]
 (1) Galena (2) Cassiterite (3) Magnetite (4) Malachite
- Which of the following factors is of no significance for roasting sulphide ores to the oxides and not subjecting the sulphide ores to carbon reduction directly ? [AIEEE-2008]
 (1) Metal sulphides are thermodynamically more stable than CS_2
 (2) CO_2 is thermodynamically more stable than CS_2
 (3) Metal sulphides are less stable than the corresponding oxides
 (4) CO_2 is more volatile than CS_2
- Which method of purification is represented by the following equation : [AIEEE-2012]

$$Ti(s) + 2I_2(g) \xrightarrow{523K} TiI_4(g) \xrightarrow{1700K} Ti(s) + 2I_2(g)$$

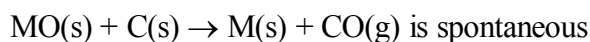
 (1) Van Arkel (2) Zone refining (3) Cupellation (4) Poling
- The substance used as froth stabilisers in froth-flotation process is : [J-Mains-2012 (On line)]
 (1) Copper sulphate (2) Aniline
 (3) Sodium cyanide (4) Potassium ethyl xanthate
- Which of the oxide groups among the following cannot be reduced by carbon :- [J-Mains-2012 (On line)]
 (1) Fe_3O_4 , ZnO (2) PbO, Fe_3O_4 (3) Cu_2O , SnO_2 (4) CaO, K_2O
- In Goldschmidt aluminothermic process which of the following reducing agents is used : [J-Mains-2013 (On line)]
 (1) Calcium (2) Coke (3) Sodium (4) Al-powder
- Calcination is the process in which :
 (1) Ore is heated strongly below its melting point in the presence of excess of air and is used for the conversion of carbonates and hydrated oxide ores to their respective oxides.
 (2) Ore is heated strongly below its melting point in the absence or limited supply of air and is used for conversion of sulphide ores to their respective oxides
 (3) Ore is heated strongly below its melting point either in the limited supply or absence of air and is used to convert carbonates and hydrated oxide ores to their respective oxides
 (4) Ore is heated strongly above its melting point in the limited supply of air to convert sulphide ores to their respective oxides.

10. The metal that cannot be obtained by electrolysis of an aqueous solution of its salts is :
[JEE-MAINS-2014]
- (1) Cu (2) Cr (3) Ag (4) Ca
11. The form of iron obtained from blast furnace is :
[J-Mains-2014 (On line)]
- (1) Steel (2) Wrought Iron (3) Cast Iron (4) Pig iron
12. In the context of the Hall-Heroult process for the extraction of Al, which of the following statements is false ?
[JEE-MAINS-2015]
- (1) Al^{3+} is reduced at the cathode to form Al
(2) Na_3AlF_6 serves as the electrolyte
(3) CO and CO_2 are produced in this process
(4) Al_2O_3 is mixed with CaF_2 which lowers the melting point of the mixture and brings conductivity
13. Galvanization is applying a coating of :-
[JEE-MAINS-2016]
- (1) Zn (2) Pb (3) Cr (4) Cu
14. Which one of the following ores is best concentrated by froth floatation method ?
[JEE-MAINS-2016]
- (1) Malachite (2) Magnetite (3) Siderite (4) Galena
15. What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M ZnSO_4 ?
[JEE-MAINS (Online) - 2016]
- (1) The copper metal will dissolve and zinc metal will be deposited
(2) No reaction will occur
(3) The copper metal will dissolve with evolution of oxygen gas
(4) The copper metal will dissolve with evolution of hydrogen gas
16. The plot shows the variation of $-\ln K_p$ versus temperature for the two reactions ?



Identify the correct statement :

- (1) At $T > 1200\text{ K}$, carbon will reduce MO(s) to M(s)
- (2) At $T < 1200\text{ K}$, oxidation of carbon is unfavourable.
- (3) Oxidation of carbon is favourable at all temperature
- (4) At $T < 1200\text{ K}$, the reaction



17. In the leaching method, bauxite ore is digested with a concentrated solution of NaOH that produces 'X'. When CO_2 gas is passed through the aqueous solution of 'X', a hydrated compound 'Y' is precipitated. 'X' and 'Y' respectively are :- **[JEE MAIN ONLINE. 2018]**

- (1) $\text{Na[Al(OH)}_4\text{]}$ and $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- (2) Al(OH)_3 and $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- (3) $\text{Na[Al(OH)}_4\text{]}$ and $\text{Al}_2(\text{CO}_3)_3 \cdot x\text{H}_2\text{O}$
- (4) Na AlO_2 and $\text{Al}_2(\text{CO}_3)_3 \cdot x\text{H}_2\text{O}$

18. When metal 'M' is treated with NaOH , a white gelatinous precipitate 'X' is obtained, which is soluble in excess of NaOH . Compound 'X' when heated strongly gives an oxide which is used in chromatography as an adsorbent. The metal 'M' is **[JEE MAIN OFFLINE. 2018]**

- (1) Ca
- (2) Al
- (3) Fe
- (4) Zn

19. In the extraction of copper from its sulphide ore, metal is finally obtained by the oxidation of cuprous sulphide with :- **[JEE MAIN ONLINE. 2018]**

- (1) CO
- (2) Cu_2O
- (3) Fe_2O_3
- (4) SO_2

EXERCISE # JEE-ADVANCED

- Carnallite does not contain
(A) K (B) Ca (C) Mg (D) Cl
- During initial treatment, preferential wetting of ore by oil and gangue by water takes place in
(A) Levigation (gravity separation) (B) Froth floatation
(C) Leaching (D) Bessemerisation
- Which of the following is true for calcination of a metal ore?
(A) It makes the ore more porous
(B) The ore is heated to a temperature when fusion just begins
(C) Hydrated salts lose their water of crystallisation
(D) Sulphur in sulphides is oxidised to SO_2
(E) Heating with carbon leads to better calcination
- In the commercial electrochemical process for aluminium extraction, the electrolyte used as :
[JEE-1999]
(A) $\text{Al}(\text{OH})_3$ in NaOH solution (B) an aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
(C) a molten mixture of Al_2O_3 and Na_3AlF_6 (D) a molten mixture of $\text{AlO}(\text{OH})$ and $\text{Al}(\text{OH})_3$
- The chemical process in the production of steel from haematite ore involve: [2000 Qualifying]
(A) reduction (B) oxidation
(C) reduction followed by oxidation (D) oxidation followed by reduction
- Electrolytic reduction of alumina to aluminium by Hall-Heroult process is carried out:
(A) in the presence of NaCl [2000 Qualifying]
(B) in the presence of fluorite
(C) in the presence of cryolite which forms a melt with lower melting temperature
(D) in the presence of cryolite which forms a melt with higher melting temperature
- The chemical composition of "slag" formed during the smelting process in the extraction of copper is :
[2001 Qualifying]
(A) $\text{Cu}_2\text{O} + \text{FeS}$ (B) FeSiO_3 (C) CuFeS_2 (D) $\text{Cu}_2\text{S} + \text{FeO}$
- Which of the following processes is used in extractive metallurgy of magnesium? [2002 Qualifying]
(A) Fused salt electrolysis (B) Self reduction
(C) Aqueous solution electrolysis (D) Thermite reduction
- In the process of extraction of gold, [2003 Qualifying]
Roasted gold ore + $\text{CN}^- + \text{H}_2\text{O} \xrightarrow{\text{O}_2} [\text{X}] + \text{OH}^-$
 $[\text{X}] + \text{Zn} \longrightarrow [\text{Y}] + \text{Au}$
Identify the complexes [X] and [Y] :
(A) $\text{X} = [\text{Au}(\text{CN})_2]^-$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$ (B) $\text{X} = [\text{Au}(\text{CN})_4]^{3-}$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
(C) $\text{X} = [\text{Au}(\text{CN})_2]^-$, $\text{Y} = [\text{Zn}(\text{CN})_6]^{4-}$ (D) $\text{X} = [\text{Au}(\text{CN})_4]^-$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
- The methods chiefly used for the extraction of lead and tin from their ores are respectively :
[JEE-2004]
(A) self reduction and carbon reduction (B) self reduction and electrolytic reduction
(C) carbon reduction and self reduction (D) cyanide process and carbon reduction

11. Which ore contains both iron and copper ? [JEE-2004]
 (A) Cuprite (B) Chalcocite (C) Chalcopyrite (D) Malachite
12. Extraction for zinc from zinc blende is achieved by : [JEE-2007]
 (A) electrolytic reduction
 (B) roasting followed by reduction with carbon
 (C) roasting followed by reduction with another metal
 (D) roasting followed by self-reduction
13. Native silver metal forms a water soluble complex with a dilute aqueous solution of NaCN in the presence of :- [JEE-2008]
 (A) nitrogen (B) oxygen
 (C) carbon dioxide (D) argon

Paragraph for questions 14 to 16

Copper is the most noble of the first row transition metals and occurs in small deposits in several countries. Ores of copper include chalcanthite ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$), atacamite ($\text{Cu}_2\text{Cl}(\text{OH})_3$), cuprite (Cu_2O), copper glance (Cu_2S) and malachite ($\text{Cu}_2(\text{OH})_2\text{CO}_3$). However, 80% of the world copper production comes from the ore chalcopyrite (CuFeS_2). The extraction of copper from chalcopyrite involves partial roasting, removal of iron and self-reduction. [JEE-2010]

14. Partial roasting of chalcopyrite produces :-
 (A) Cu_2S and FeO (B) Cu_2O and FeO (C) CuS and Fe_2O_3 (D) Cu_2O and Fe_2O_3
15. Iron is removed from chalcopyrite as :-
 (A) FeO (B) FeS (C) Fe_2O_3 (D) FeSiO_3
16. In self-reduction, the reducing species is :-
 (A) S (B) O^{2-} (C) S^{2-} (D) SO_2
17. Match the extraction processes listed in column I with metals listed in column II. [JEE-2006]

Column I

- (A) Self reduction
 (B) Carbon reduction
 (C) Complex formation and displacement by metal
 (D) Decomposition of iodide

Column II

- (P) Lead
 (Q) Silver
 (R) Copper
 (S) Boron

18. Match the conversions in **Column I** with the type(s) of reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS. [JEE-2008]

Column I

- (A) $\text{PbS} \rightarrow \text{PbO}$
 (B) $\text{CaCO}_3 \rightarrow \text{CaO}$
 (C) $\text{ZnS} \rightarrow \text{Zn}$
 (D) $\text{Cu}_2\text{S} \rightarrow \text{Cu}$

Column II

- (P) Roasting
 (Q) Calcination
 (R) Carbon reduction
 (S) Self reduction

19. In extractive metallurgy of zinc partial fusion of ZnO with coke is called _____ and reduction of the ore to the molten metal is called _____ (smelting, calcining, roasting, sintering). [JEE-1988]

20. Extraction of metal from the ore cassiterite involves [JEE-2011]
 (A) carbon reduction of an oxide ore (B) self-reduction of a sulphide ore
 (C) removal of copper impurity (D) removal of iron impurity
21. Oxidation states of the metal in the minerals haematite and magnetite, respectively, are [JEE-2011]
 (A) II, III in haematite and III in magnetite (B) II, III in haematite and II in magnetite
 (C) II in haematite and II, III in magnetite (D) III in haematite and II, III in magnetite
22. In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used are : [JEE-2012]
 (A) O_2 and CO respectively. (B) O_2 and Zn dust respectively.
 (C) HNO_3 and Zn dust respectively. (D) HNO_3 and CO respectively.
23. Sulfide ores are common for the metals - [JEE-2013]
 (A) Ag, Cu and Pb (B) Ag, Cu and Sn (C) Ag, Mg and Pb (D) Al, Cu and Pb
24. The carbon-based reduction method is **NOT** used for the extraction of [JEE-2013]
 (A) tin from SnO_2 (B) Iron from Fe_2O_3
 (C) aluminium from Al_2O_3 (D) magnesium from $MgCO_3 \cdot CaCO_3$
25. Upon heating with Cu_2S , the reagent(s) that give copper metal is/are [JEE Adv. 2014]
 (A) $CuFeS_2$ (B) CuO (C) Cu_2O (D) $CuSO_4$
26. Copper is purified by electrolytic refining of blister copper. The correct statement(s) about this process is (are) [JEE Adv. 2015]
 (A) Impure Cu strip is used as cathode
 (B) Acidified aqueous $CuSO_4$ is used as electrolyte
 (C) Pure Cu deposits at cathode
 (D) Impurities settle as anode-mud
27. Match the anionic species given in Column-I that are present in the ore(s) given in Column-II [JEE Adv. 2015]
- | Column-I | Column-II |
|---------------|---------------|
| (A) Carbonate | (P) Siderite |
| (B) Sulphide | (Q) Malachite |
| (C) Hydroxide | (R) Bauxite |
| (D) Oxide | (S) Calamine |
| | (T) Argentite |
28. Extraction of copper from copper pyrite ($CuFeS_2$) involves [JEE Adv. 2016]
 (A) crushing followed by concentration of the ore by froth-flotation
 (B) removal of iron as slag
 (C) self-reduction step to produce 'blister copper' following evolution of SO_2
 (D) refining of 'blister copper' by carbon reduction
29. Galena (an ore) is partially oxidized by passing air through it at high temperature. After some time, the passage of air is stopped, but the heating is continued in a closed furnace such that the contents undergo self-reduction. The weight (in kg) of Pb produced per kg of O_2 consumed is _____. [JEE ADV. 2018]
 (Atomic weights in $g\ mol^{-1}$: O = 16, S = 32, Pb = 207)

ANSWER KEY**EXERCISE # O-I**

1. (A)	2. (C)	3. (B)	4. (B)	5. (C)	6. (A)
7. (A)	8. (B)	9. (D)	10. (B)	11. (B)	12. (A)
13. (B)	14. (B)	15. (A)	16. (D)	17. (A)	18. (A)
19. (B)	20. (C)	21. (C)	22. (C)	23. (C)	24. (B)
25. (B)	26. (D)	27. (C)	28. (B)	29. (A)	30. (D)
31. (A)	32. (B)	33. (C)	34. (B)	35. (A)	36. (C)
37. (C)	38. (D)	39. (A)	40. (B)	41. (C)	42. (B)
43. (A)	44. (A)	45. (D)	46. (C)	47. (C)	48. (C)
49. (B)	50. (B)	51. (A)	52. (C)	53. (B)	54. (D)
55. (B)	56. (A)	57. (D)	58. (D)	59. (C)	60. (D)
61. (B)	62. (B)	63. (A)	64. (A)	65. (B)	66. (C)

EXERCISE # O-II

1. (A,B,D)	2. (A,B,C)	3. (B,C)	4. (A,B)	5. (A,C,D)	6. (A,B,C,D)
7. (A, C)	8. (A, C)	9. (A, B)	10. (A, B)	11. (A,B)	12. (B, C)
13. (A,B)	14. (B,C)	15. (A,B,C, D)	16. (A,B)	17. (A,B, C)	18. (B,C,D)
19. (A,B)	20. (A,B,C)	21. (C,D)	22. (A, D)	23. (A,B)	24. (A, D)
25. (A,B,D)	26. (B,D)	27. (B,C)	28. (C,D)	29. (A,C)	30. (B, C)

EXERCISE # S-1

1. (3)	2. (3)	3. (4)	4. (4)	5. (2)	6. (4)
7. (4)	8. (4)	9. (2)			

EXERCISE # S-2

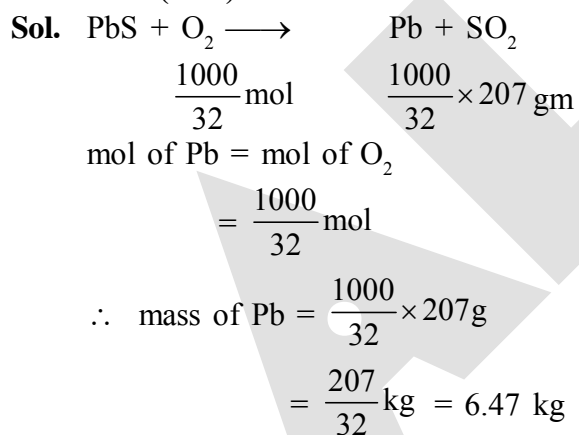
1. (B)	2. (C)	3. (A)	4. (B)	5. (D)	
6. (A) Q; (B) R; (C) S; (D) P	7. (A) $\rightarrow$ P, (B) $\rightarrow$ R ; (C) $\rightarrow$ Q, R, S (D) $\rightarrow$ Q, S				
8. (A) R ; (B) P; (C) Q	9. (A) S ; (B) P; (C) Q; (D) R				
10. (A) S ; (B) Q ; (C) R ; (D) P	11. (A)	12. (A)	13. (A)	14. (D)	

EXERCISE # JEE-MAINS

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 1. (3) | 2. (1) | 3. (1) | 4. (3) | 5. (1) | 6. (2) |
| 7. (4) | 8. (4) | 9. (3) | 10. (4) | 11. (4) | 12. (2) |
| 13. (1) | 14. (4) | 15. (2) | 16. (3) | 17. (1) | 18. (2) |
| 19. (2) | | | | | |

EXERCISE # JEE-ADVANCED

- | | | | | | |
|---|---------|-------------------------|---|---------------|---------|
| 1. (B) | 2. (B) | 3. (A,C) | 4. (C) | 5. (C) | 6. (C) |
| 7. (B) | 8. (A) | 9. (A) | 10. (A) | 11. (C) | 12. (B) |
| 13. (B) | 14. (A) | 15. (D) | 16. (C) | | |
| 17. (A) – P, R ; (B) – P ; (C) – Q ; (D) – S | | | 18. (A) – P ; (B) – Q ; (C) – P, R ; (D) – P, S | | |
| 19. Sintering , Smelting | | 20. (A, D) or (A, C, D) | | 21. (D) | |
| 22. (B) | 23. (A) | 24. (C, D) | 25. (B, C, D) | 26. (B, C, D) | |
| 27. (A) – P , Q , S ; (B) – T ; (C) – Q , R ; (D) – R | | | | 28. (A, B, C) | |

29. Ans. (6.47)

NURTURE COURSE
IONIC EQUILIBRIUM

IONIC EQUILIBRIUM

1. INTRODUCTION

Ionic equilibrium deals with the equilibrium of any substance with its ions in solution. The substance producing ions are called electrolytes.

According to conductivity, substances are of two types :

(i) Non-Conductor :

Those substances which do not show the flow of current or electricity.

Ex. Non - metals, plastic rubber, wood, etc.

Exception – Graphite is a non-metal but shows conductivity due to motion of free electrons.

(ii) Conductors :

Those substances which show conductivity or flow of current are called conductors. These are of 2 types :

(a) Metallic or electronic conductors :

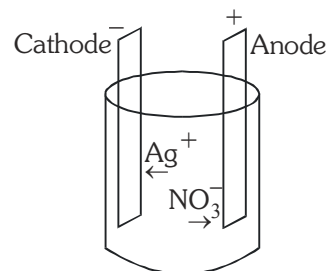
Those conductor which show conductivity due to motion of free electrons. Resistance increases with temperature.

Ex. All metals, Graphite

(b) Ionic or electrolytic conductors :

Those conductors which show conductivity due to movement of free ions. Ions are in free state in the solutions of ionic compounds. On passing electric current through the solution, ions move towards oppositely charged electrodes, i.e., the cation moves towards cathode (negative electrode) and the anion moves towards anode (positive electrode).

The current flows through the solution due to the movement of the ions. Resistance decrease with temperature.



According to strength, ionic conductors are of two types :

(i) Strong electrolyte :

Those ionic conductors which are completely ionized in aqueous solution are called as strong electrolyte.

For strong electrolyte the value of degree of dissociation is 100%.

i.e. : $\alpha = 1$

- Ex.** (a) Strong acid $\rightarrow$ H_2SO_4 , HCl , HNO_3 , HClO_4 , H_2SO_5 , HBr , HI , HBrO_4 , HIO_4 , RSO_3H
 (b) Strong base $\rightarrow$ KOH , NaOH , $\text{Ba}(\text{OH})_2$, CsOH , RbOH
 (c) All soluble salts $\rightarrow$ NaCl , KCl , CuSO_4 ,

(ii) Weak electrolytes :

Those electrolytes which are partially ionized in aqueous solution are called as weak electrolytes. For weak electrolytes the value of α is less than one.

- Ex.** (a) Weak acid : HCN , CH_3COOH , HCOOH , H_2CO_3 , H_3PO_3 , H_3PO_2 , etc.
 (b) Weak base : NH_4OH , $\text{Cu}(\text{OH})_2$, $\text{Zn}(\text{OH})_2$, $\text{Fe}(\text{OH})_3$, $\text{Al}(\text{OH})_3$, etc.

1.2 DEGREE OF DISSOCIATION / IONISATION

- When an electrolyte is dissolved in a solvent (H_2O), it spontaneously dissociates into ions.
- It may dissociate partially ($\alpha < 1$) or sometimes completely ($\alpha \cong 1$)
- The degree of dissociation (α) of an electrolyte is the fraction of mole of the electrolyte that has dissociated under the given conditions.

$$\alpha = \frac{\text{No. of moles dissociated}}{\text{No. of moles taken initially}}$$

1.3 FACTORS AFFECTING THE VALUE OF DEGREE OF DISSOCIATION:

- (i) **Dilution** : $\alpha \propto \sqrt{V}$. So on dilution, α increases
- (ii) **Temperature** : On increasing temperature, ionization increases so, α increases
- (iii) **Nature of electrolyte** :

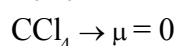
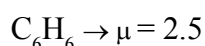
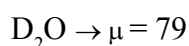
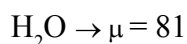
(a) Strong electrolyte

$$\alpha = 100 \%$$

(b) Weak electrolyte

$$\alpha \ll 100 \%$$

- (iv)
- Nature of solvent**
- :

If dielectric constant, μ , of solvent increases, then the value of α increases.

Ex.1 Which one has greater α_1 or α_2 for the following equation :



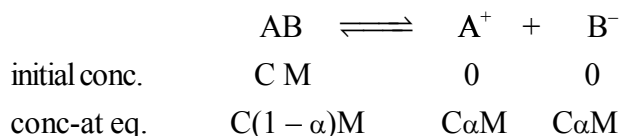
Sol. Dielectric constant of H_2O is more than that of D_2O , so $\alpha_1 > \alpha_2$

- (v)
- Mixing of Ions**
- :

Common ion Effect	Odd ion Effect
$NH_4OH \rightleftharpoons NH_4^+ + OH^-$ On mixing NH_4Cl $NH_4Cl \rightarrow NH_4^+ + Cl^-$ Due to mixing of common ion, concentration of ammonium ion will increase therefore equilibrium will shift in backward direction means α decreases.	$NH_4OH \rightleftharpoons NH_4^+ + OH^-$ On mixing HCl $HCl \rightarrow H^+ + Cl^-$ Due to reaction of OH^- ions with H^+ ion, concentration of OH^- will decrease $\therefore$ Equilibrium will shift in forward direction means α increases.

1.4 OSTWALD'S DILUTION LAW (FOR WEAK ELECTROLYTE)

- For a weak electrolyte A^+B^- dissolved in water, if α is the degree of dissociation then



Then according to law of mass action,

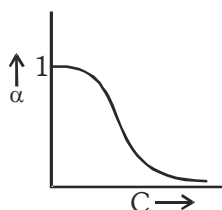
$$K_{\text{diss}} = \frac{[A^+][B^-]}{[AB]} = \frac{C\alpha \cdot C\alpha}{C(1-\alpha)} = \frac{C\alpha^2}{(1-\alpha)} = \text{dissociation constant of the weak electrolyte.}$$

$$\left[C = \frac{1}{V}, \text{ then } V = 1/C \text{ (volume of solution in which 1 mole is present) is called dilution, so } K_{\text{diss}} = \frac{\alpha^2}{(1-\alpha)V} \right]$$

If α is negligible in comparison to unity, $1 - \alpha \approx 1$. so $K_{\text{diss}} = \alpha^2 C \Rightarrow \alpha = \sqrt{\frac{K_{\text{diss}}}{C}} = \sqrt{K_{\text{diss}} V}$

$$\alpha \propto \frac{1}{\sqrt{\text{concentration}}}$$

- As concentration increases $\Rightarrow \alpha$ decreases
- At infinite dilution α reaches its maximum value, unity.



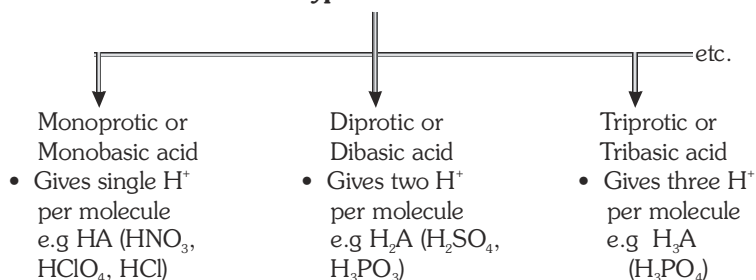
2. ACIDS BASES AND SALTS

2.1 Arrhenius concept :

- (i) **Arrhenius Acid** : Substance which gives H^+ ion on dissolving in water (H^+ donor)

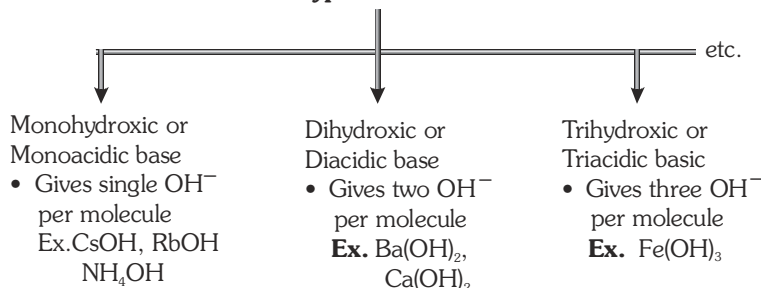
Ex. HNO_3 , $HClO_4$, HCl , HI , HBr , H_2SO_4 , H_3PO_4 etc.

Types of acids



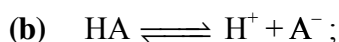
- (ii) **Arrhenius base** : Any substance which releases OH^- (hydroxyl) ion in water (OH^- ion donor)

Types of bases



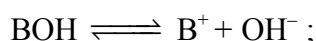
(iii) **Strength of Acid or Base :**

- (a) Strength of acids or bases depends on the extent of its ionisation. Hence equilibrium constant K_a or K_b respectively of the following equilibria give a quantitative measurement of the strength of the acid or base.



$$K_a = \frac{[H^+][A^-]}{[HA]} = \text{dissociation or ionisation constant of acid.}$$

- (c) Similarly

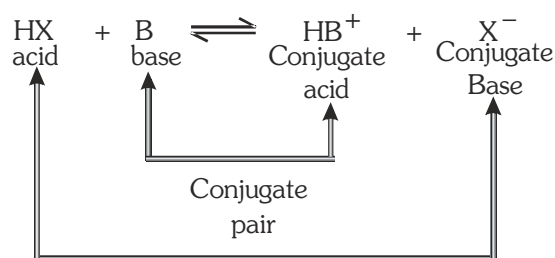


$$K_b = \frac{[B^+][OH^-]}{[BOH]} = \text{dissociation or ionisation constant of base}$$

- (d) Larger the value of K_a or K_b , stronger is the acid or base respectively.

2.2 **Bronsted - Lowry concept : (Conjugate acid - base concept) (Protonic concept)**

- (i) **Acid** : substances which donate H^+ are Bronsted Lowry acids (H^+ donor)
 (ii) **Base** : substances which accept H^+ are Bronsted Lowry bases (H^+ acceptor)
 (iii) **Conjugate acid - base pairs** : In a typical acid base reaction, $HX + B \rightleftharpoons X^- + HB^+$

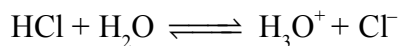


- Forward reaction - Here HX being a proton donor is an acid
B being a proton acceptor is a base.
- Backward reaction - Here HB^+ being a proton donor is an acid
 X^- being a proton acceptor is a base.

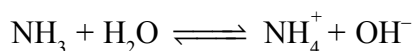
Acid	Base		Conjugate Acid		Conjugate Base
• HCl	+ H ₂ O	$\rightleftharpoons$	H ₃ O ⁺	+	Cl ⁻
• HSO ₄ ⁻	+ NH ₃	$\rightleftharpoons$	NH ₄ ⁺	+	SO ₄ ⁻²
• [Fe(H ₂ O) ₆] ³⁺	+ H ₂ O	$\rightleftharpoons$	H ₃ O ⁺	+	[Fe(H ₂ O) ₅ (OH)] ²⁺
• Conjugate acid - base pair differ by only one proton.					
• Strong acid will have weak conjugate base and vice versa.					

Acid	Conjugate base	Base	Conjugate acid
HCl	Cl ⁻	NH ₃	NH ₄ ⁺
H ₂ SO ₄	HSO ₄ ⁻	H ₂ O	H ₃ O ⁺
HSO ₄ ⁻	SO ₄ ²⁻	RNH ₂	RNH ₃ ⁺
H ₂ O	OH ⁻		

(iv) **Amphoteric (amphiprotic)** : Substances which can act as acid as well as base are known as amphoteric



base



acid

(v) **Classification of Bronsted - Lowry Acids and Bases :**

Bronsted - Lowry acids and bases can be

- (i) Molecular (ii) Cationic and (iii) Anionic

Table - 1

Type	Acid	Base
Molecular	HCl, HNO ₃ , HClO ₄ , H ₂ SO ₄ , H ₃ PO ₄ , H ₂ O etc.	NH ₃ , N ₂ H ₄ , Amines, H ₂ O, Alcohol, Ethers, etc.
Cationic	NH ₄ ⁺ , N ₂ H ₅ ⁺ , PH ₄ ⁺ , [Fe(H ₂ O) ₆] ³⁺ , [Al(H ₂ O) ₆] ³⁺ etc.	[Fe(H ₂ O) ₅ OH] ²⁺ [Al(H ₂ O) ₅ OH] ²⁺ etc.
Anionic	HS ⁻ , HSO ₃ ⁻ , H ₂ PO ₄ ⁻ , HSO ₄ ⁻ HCO ₃ ⁻ , HPO ₄ ²⁻ , etc. all amphiprotic anions	Cl ⁻ , Br ⁻ , OH ⁻ , HSO ₄ ⁻ , CN ⁻ , CO ₃ ²⁻ , SO ₄ ²⁻ , NH ₂ ⁻ , CH ₃ COO ⁻ , etc. all amphiprotic anions

2.3 **Lewis concept (electronic concept) :**

- (i) **Acid** : An acid is a molecule/ion which can accept an electron pair with the formation of a coordinate bond.

Ex. Electron deficient molecules : BF₃, AlCl₃, etc.

Cations : H⁺, Fe²⁺, Na⁺, etc.

Molecules with vacant orbitals : SF₄, PF₃

- (ii) **Base** : A base is any molecule/ion which has a pair of electrons which can be donated.

Ex. Molecules with lone pairs : NH₃, PH₃, H₂O, CH₃OH

Anions : OH⁻, H⁻, NH₂⁻, etc.

3. PROPERTIES OF WATER

(i) Molar concentration / Molarity of water :

$$\text{Molarity} = \text{No. of moles/litre} = \frac{1000 \text{ g/litre}}{18 \text{ g/mole}} = 55.55 \text{ mole/litre} = \mathbf{55.55 \text{ M}} \text{ (density} = 1 \text{ g/cc)}$$

(ii) Ionic product of water :

According to arrhenius concept, $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$

So, ionic product of water, $K_w = [\text{H}^+][\text{OH}^-] = 10^{-14}$ at 25° (experimental)

Dissociation of water is endothermic, so on increasing temperature K_w increases.

(iii) Degree of dissociation of water :

$$\begin{aligned} \text{H}_2\text{O} &\rightleftharpoons \text{H}^+ + \text{OH}^- \Rightarrow \alpha = \frac{\text{decrease in concentration}}{\text{initially concentration}} \\ &= \frac{10^{-7}}{55.55} = 18 \times 10^{-10} \text{ or } 1.8 \times 10^{-7} \% \quad [\text{at } 25^\circ\text{C}] \end{aligned}$$

(iv) Dissociation or ionisation constant of water :

$$\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^- \quad K_a = K_b = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]} = \frac{10^{-7} \times 10^{-7}}{55.55} = 1.8 \times 10^{-16}$$

$$\text{So, } pK_a = pK_b = -\log(1.8 \times 10^{-16}) = 16 - \log 1.8 = 15.74$$

Ex.2. At dissociation constant of heavy water is 4×10^{-15} at 35°C . If its density is 1.04 g/mL . Calculate its ionic product & degree of dissociation.

$$\text{Sol. } K_w = K_d[D_2O] = \left(4 \times 10^{-15} \times \frac{1040}{20}\right) = 2.08 \times 10^{-13}$$

$$d = \sqrt{\frac{K_w}{C}} = \sqrt{\frac{2.08 \times 10^{-13}}{52}} = 12.64 \times 10^{-8}$$

Ex.3 Calculate ionic product of H_2O at 50°C .

$$\text{Sol. } \Delta H = 13.7 \times 10^3 \text{ cal}$$

$$\log \frac{K_2}{K_1} = \frac{\Delta H}{2} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\log \frac{K_2}{10^{-14}} = \frac{13.7 \times 10^3}{2} \left(\frac{1}{298} - \frac{1}{323} \right)$$

Ex.4 The hydronium ion conc. in an aq. H_2CO_3 solution is $4 \times 10^{-4} \text{ M}$ at 25°C OH^- ion conc. in the solution is :

- (A) 0 (B) 2.5×10^{-10} (C) 2.5×10^3 (D) $2.5 \times 10^{-11} \text{ M}$

Answer : (D)

Ex.5 Select the correct option from the following ?

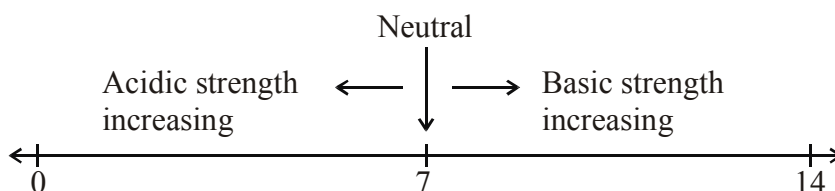
- (A) pK_w increases with increase of temperature
 (B) pK_w decreases with increase of temperature
 (C) $pK_w = 14$ at all temperatures
 (D) $pK_w = pH$ at all temperatures

Sol. (B)

4. Acidity and pH scale :

- (i) Acidic strength means the tendency of an acid to give H_3O^+ or H^+ ions in water.
So greater then tendency to give H^+ , more will be the acidic strength of the substance.
- (ii) Basic strength means the tendency of a base to give OH^- ions in water.
So greater the tendency to give OH^- ions, more will be basic strength of the substance.
- (iii) The concentration of H^+ ions is written in a simplified form introduced by **Sorenson** known as pH scale. pH is defined as negative logarithm of activity of H^+ ions.
 $\therefore \text{pH} = -\log a_{\text{H}^+}$ (where a_{H^+} is the activity of H^+ ions)
- (iv) Activity of H^+ ions is the molar concentration of free H^+ ions or H_3O^+ ions in a dilute solution, but unitless.
- (v) Now $\text{pH} = -\log[\text{H}^+] = 7$ and $\text{pOH} = -\log[\text{OH}^-] = 7$ for water at 25°C (experimental)

$$\left. \begin{array}{l} \text{pH} = 7 = \text{pOH} \Rightarrow \text{neutral} \\ \text{pH} < 7 \text{ or } \text{pOH} > 7 \Rightarrow \text{acidic} \\ \text{pH} > 7 \text{ or } \text{pOH} < 7 \Rightarrow \text{Basic} \end{array} \right\} \text{ at } 25^\circ\text{C}$$

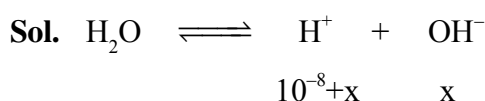


4.1 pH Calculation of different Types of solutions :

(a) Strong acid solution :

- (i) If concentration of H^+ ions is greater than 10^{-6}M , H^+ ions coming from water can be neglected,
So $[\text{H}^+] = \text{normality of strong acid solution}$
- (ii) If concentration is less than 10^{-6}M , H^+ ions coming from water cannot be neglected. So
 $[\text{H}^+] = \text{normality of strong acid} + \text{H}^+$ ions coming from water in presence of this strong acid

Ex.6 Calculate pH of 10^{-8}M HCl solution.



$$K_w = [\text{H}^+][\text{OH}^-]$$

$$10^{-14} = x(x + 10^{-8})$$

$$\Rightarrow x^2 + x \times 10^{-8} - 10^{-14} = 0$$

$$x = \frac{-10^{-8} \pm \sqrt{10^{-16} + 4 \times 10^{-14}}}{2} = \frac{-10^{-8} + 10^{-7} \sqrt{4 + \frac{1}{100}}}{2} = \frac{(\sqrt{401} - 1)10^{-8}}{2} = 0.95 \times 10^{-7}$$

$$[\text{H}^+] = 10.5 \times 10^{-8} = 1.05 \times 10^{-7}$$

$$\text{pH} = -\log [\text{H}^+]$$

$$\text{pH} = 7 - \log 1.05 \approx 6.98$$

4.2 Strong base solution :

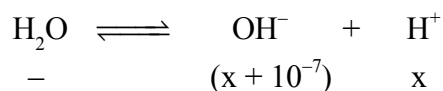
Calculate the $[\text{OH}^-]$ which will be equal to normality of the strong base solution and then use

$$K_w = [\text{H}^+] \times [\text{OH}^-] = 10^{-14}, \text{ to calculate } [\text{H}^+]$$

Ex.7 Calculate pH of 10^{-7} M of NaOH solution

Sol. $[\text{OH}^-]$ from NaOH = 10^{-7}

$[\text{OH}^-]$ from water = $x < 10^{-7}$ M (due to common ion effect)



$$K_w = [\text{H}^+] [\text{OH}^-] = 10^{-14} = x (x + 10^{-7})$$

$$x^2 + 10^{-7}x - 10^{-14} = 0$$

$$\Rightarrow x = \frac{\sqrt{5}-1}{2} \times 10^{-7} = 0.618 \times 10^{-7} \quad (\sqrt{5} = 2.236)$$

$$[\text{OH}^-] = 10^{-7} + 0.618 \times 10^{-7} = 1.618 \times 10^{-7}$$

$$\text{pOH} = 7 - \log(1.618) = 6.79$$

$$\text{pH} = 14 - 6.79 = 7.21$$

4.3 pH of mixture of two strong acids : If V_1 volume of a strong acid solution of normality N_1 is mixed with V_2 volume of another strong acid solution of normality N_2 , then

$$\text{Number of } \text{H}^+ \text{ ions from I-solution} = N_1 V_1$$

$$\text{Number of } \text{H}^+ \text{ ions from II-solution} = N_2 V_2$$

If final normality is N and final volume is V , then

$$NV = N_1 V_1 + N_2 V_2$$

[dissociation equilibrium of none of these acids will be disturbed as both are strong acid]

$$[\text{H}^+] = N = \frac{N_1 V_1 + N_2 V_2}{V_1 + V_2} \quad \left[\begin{array}{l} \text{where } N = M \times n \\ n = \text{Basicity of acid} \end{array} \right]$$

4.4 pH of mixture of two strong bases :

Similar to above calculation

$$[\text{OH}^-] = N = \frac{N_1 V_1 + N_2 V_2}{V_1 + V_2} \quad [\text{H}^+] = \frac{10^{-14}}{[\text{OH}^-]}$$

$$\left[\begin{array}{l} \text{where } N = M \times n \\ n = \text{Acidity of base} \end{array} \right]$$

Ex.8 Calculate pH of mixture of $(400 \text{ mL}, \frac{1}{200} \text{ M H}_2\text{SO}_4) + (400 \text{ mL}, \frac{1}{100} \text{ M HCl}) + (200 \text{ mL of water})$

Sol. $N_1 V_1 = \frac{1}{200} \times \frac{400}{1000} \times 2 = \frac{4}{1000}$, $N_2 V_2 = \frac{4}{1000}$, H^+ ions from water will be neglected

$$N_1 V_1 + N_2 V_2 = 8 \times 10^{-3} \quad [\text{H}^+] = \frac{8 \times 10^{-3}}{1} = 8 \times 10^{-3}$$

$$\text{pH} = 3 - \log 8 = 2.1$$

Ex.9 500 mL of 10^{-5} M NaOH is mixed with 500 mL of 2.5×10^{-5} M of $\text{Ba}(\text{OH})_2$. To the resulting solution 99 L water is added. Calculate pH.

Sol. $[\text{OH}^-] = \frac{500 \times 10^{-5} + 500 \times 2 \times 2.5 \times 10^{-5}}{1000} = 3 \times 10^{-5} \text{ M}$

$$M_1 = 3 \times 10^{-5} \text{ M}$$

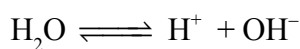
$$V_2 + V_1 = 1 \text{ L}$$

$$V_F = 100 \text{ L}$$

no. of moles of $[\text{OH}^-]$ initially = no. of moles of $[\text{OH}^-]$

$$3 \times 10^{-5} = M_2 \times 100$$

$$\therefore M_2 = 3 \times 10^{-7} < 10^{-6}$$



$$x \quad (x + 3 \times 10^{-7})$$

$$K_w = x(x + 3 \times 10^{-7}) = 10^{-14}$$

$$\therefore x = \left(\frac{\sqrt{13} - 3}{2} \right) \times 10^{-7}$$

$$x = 0.302 \times 10^{-7}$$

$$[\text{OH}^-]_{\text{Net}} = 3.302 \times 10^{-7}$$

4.5 pH of mixture of a strong acid and a strong base :

- Acid Base neutralisation reaction will take place.
- The solution will be acidic or basic depending on which component has been taken in excess.
- If V_1 volume of a strong acid solution of normality N_1 is mixed with V_2 volume of a strong base solution of normality N_2 , then

$$\text{Number of } \text{H}^+ \text{ ions from I-solution} = N_1 V_1$$

$$\text{Number of } \text{OH}^- \text{ ions from II-solution} = N_2 V_2$$

If $N_1 V_1 > N_2 V_2$

$$[\text{H}^+] = N = \frac{N_1 V_1 - N_2 V_2}{V_1 + V_2}$$

Solution will
be acidic in nature

If $N_2 V_2 > N_1 V_1$

$$[\text{OH}^-] = N = \frac{N_2 V_2 - N_1 V_1}{V_1 + V_2}$$

Solution will
be basic in nature

$$[\text{H}^+] = \frac{10^{-14}}{[\text{OH}^-]}$$

Ex.10 Calculate pH of mixture of (400 mL, $\frac{1}{200}$ M Ba(OH)₂) + (400 mL, $\frac{1}{50}$ M HCl) + (200 mL of water)

Sol. $[H^+] = \frac{400 \times \frac{1}{50} - 400 \times \frac{1}{200} \times 2}{1000} = 4 \times 10^{-3}$, so $pH = 3 - 2 \log 2 = 2.4$

Ex.11 What will be the resultant pH when 150 mL of an aqueous solution of HCl (pH = 2.0) is mixed with 350 mL of an aqueous solution of NaOH (pH = 12.0) ?

Sol. pH of HCl = 2

$\therefore [HCl] = 10^{-2}$ M

pH of NaOH = 12, pOH = 2 $\therefore [NaOH] = 10^{-2}$ M

	HCl	+	NaOH	→	NaCl	+	H ₂ O
Meq. initial	150×10^{-2}		350×10^{-2}		0		0
	= 1.5		= 3.5				
Meq. final	0		2		1.5		1.5

$\therefore [OH^-] \text{ from NaOH} = \frac{2}{500} = 4 \times 10^{-3}$ M

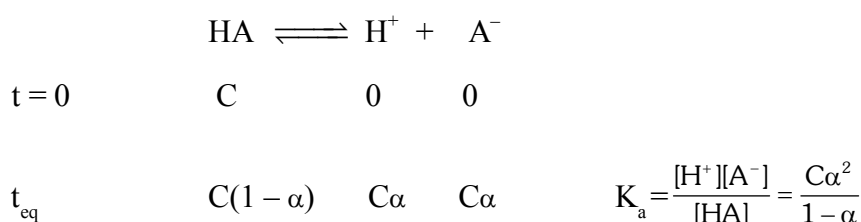
pOH = $-\log[OH^-] = -\log(4 \times 10^{-3})$

$\therefore$ pOH = 2.3979

$\therefore$ pH = 14 - pOH = 14 - 2.3979 = 11.6021

4.6 pH of a weak acid or weak base (monoprotic) Solution :

- Weak acid does not dissociated 100 % therefore we have to calculate the percentage dissociation using K_a dissociation constant of the acid.
- We have to use Ostwald's Dilution law (as have been derived earlier)



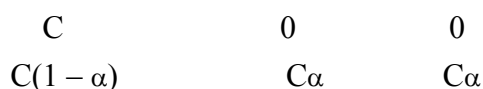
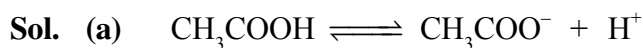
If $\alpha \ll 1 \Rightarrow (1 - \alpha) \approx 1 \Rightarrow K_a \approx C\alpha^2 \Rightarrow \alpha = \sqrt{\frac{K_a}{C}}$ (is valid if $\alpha < 0.1$ or 10%)

$[H^+] = C\alpha = C\sqrt{\frac{K_a}{C}} = \sqrt{K_a \times C}$ So $pH = \frac{1}{2}(pK_a - \log C)$

On increasing the dilution $\Rightarrow C \downarrow = \alpha \uparrow$ and $[H^+] \downarrow \Rightarrow pH \uparrow$

Ex.12 Calculate pH of : (a) 10^{-1} M CH_3COOH (b) 10^{-3} M CH_3COOH (c) 10^{-6} M CH_3COOH

Take $K_a = 2 \times 10^{-5}$



$$K_a = \frac{\text{C}\alpha^2}{1-\alpha} \Rightarrow \alpha = \sqrt{\frac{K_a}{\text{C}}} = \sqrt{\frac{2 \times 10^{-5}}{10^{-1}}} = \sqrt{2 \times 10^{-4}} \quad (\alpha \ll 0.1)$$

$$\text{So, } [\text{H}^+] = 10^{-1} \times \sqrt{2} \times 10^{-2} \Rightarrow \text{pH} = 3 - \frac{1}{2} \log 2 = 2.85$$

(b) $\alpha = \sqrt{\frac{K_a}{\text{C}}} \Rightarrow \alpha = \sqrt{\frac{K_a}{\text{C}}} = \sqrt{\frac{2 \times 10^{-5}}{10^{-3}}} = \sqrt{2 \times 10^{-2}} \quad (\alpha > 0.1)$

So we have to do the exact calculations

$$K_a = \frac{\text{C}\alpha^2}{1-\alpha} \Rightarrow 2 \times 10^{-5} = \frac{10^{-3} \times \alpha^2}{1-\alpha} \Rightarrow \alpha = 13.14 \%$$

$$[\text{H}^+] = 10^{-3} \times 0.1314 = 1.314 \times 10^{-4} \Rightarrow \text{pH} = 4 - \log(1.314) \approx 3.8$$

(c) If approximation is used the, $\alpha = \sqrt{\frac{2 \times 10^{-5}}{10^{-6}}} = \sqrt{20} > 1$,

So we have to do the exact calculation, $2 \times 10^{-5} = 10^{-6} \frac{\alpha^2}{1-\alpha} \Rightarrow \alpha \approx 0.95$ or 95%

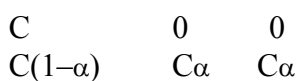
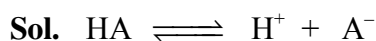
$$[\text{H}^+] = 0.95 \times 10^{-6} = 9.5 \times 10^{-7} \Rightarrow \text{pH} = 7 - \log(9.5) = 6.022$$

- At very low concentration (at infinite dilution) weak electrolyte will be almost 100% dissociate, so behave as strong electrolyte.

(pH) of 10^{-6} M HCl $\approx$ pH of 10^{-6} M $\text{CH}_3\text{COOH} \approx 6$

Ex.13 K_a for acid HA is 2.5×10^{-8} calculate for its decimolar solution at 25°C .

(i) % dissociation (ii) pH (iii) OH^- ion concentration



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} \Rightarrow K_a = \frac{\text{C}\alpha \cdot \text{C}\alpha}{\text{C}(1-\alpha)} = \frac{\text{C}\alpha^2}{(1-\alpha)} \approx \text{C}\alpha^2$$

(i) $\therefore \alpha = \sqrt{\frac{K_a}{\text{C}}} = \sqrt{\frac{2.5 \times 10^{-8}}{1/10}} \quad (\text{C} = 1/10 \text{ M})$
 $= 5 \times 10^{-4} = 0.05\%$

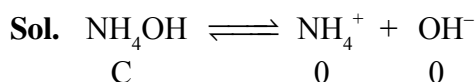
(ii) $[\text{H}^+] = \text{C}\alpha = \frac{1}{10} \times 5 \times 10^{-4} = 5 \times 10^{-5} \text{ mol/L}$

So $\text{pH} = 5 - \log 5 = 4.30$

(iii) $[\text{H}^+][\text{OH}^-] = 1 \times 10^{-14}$

$$\therefore [\text{OH}^-] = \frac{10^{-14}}{5 \times 10^{-5}} = 2 \times 10^{-10} \text{ mol/L}$$

Ex.14 Determine the degree of dissociation of $0.05\text{ M NH}_4\text{OH}$ at 25°C in a solution of $\text{pH} = 10$.



Given, $\text{pH} = 10$

$$[\text{H}^+] = 10^{-10}$$

$$[\text{H}^+][\text{OH}^-] = 1 \times 10^{-14}$$

$$\therefore [\text{OH}^-] = \frac{1 \times 10^{-14}}{10^{-10}} = 10^{-4} = C\alpha$$

$$\therefore \alpha = \frac{[\text{OH}^-]}{C} = \frac{10^{-4}}{0.05} = 2 \times 10^{-3} \text{ or } 0.2\%$$

Ex.15 The concentration of $[\text{H}^+]$ and $[\text{OH}^-]$ of the 10^{-1} M aqueous solution of 2% ionised weak acid is :

(A) $2 \times 10^{-3}\text{ M}$ and $5 \times 10^{-12}\text{ M}$

(B) $1 \times 10^{-3}\text{ M}$ and $3 \times 10^{-11}\text{ M}$

(C) $2 \times 10^{-4}\text{ M}$ and $5 \times 10^{-11}\text{ M}$

(D) $3 \times 10^{-2}\text{ M}$ and $4 \times 10^{-13}\text{ M}$

Sol. (A)

$$[\text{H}^+] = C\alpha = 2 \times 10^{-3}\text{ M} \text{ or } [\text{OH}^-] = \frac{10^{-14}}{[\text{H}^+]} = 5 \times 10^{-12}\text{ M}$$

Ex.16 When a 0.1 N solution of an acid at 25°C has a degree of ionisation of 4%, the concentration of OH^- present is :

(A) 2.5×10^{-3}

(B) 2.5×10^{-11}

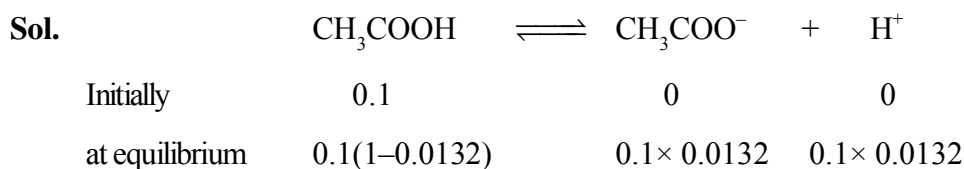
(C) 2.5×10^{-12}

(D) 2.5×10^{-13}

Sol. (C)

$$[\text{H}^+] = C\alpha = 0.1 \times 4 \times 10^{-2} = 4 \times 10^{-3}\text{ M} \text{ or } [\text{OH}^-] = \frac{10^{-14}}{[\text{H}^+]} = 2.5 \times 10^{-12}\text{ N}$$

Ex.17 The degree of dissociation of acetic acid in a 0.1 M solution is 1.32×10^{-2} . Calculate dissociation constant of acid and its pK_a value :

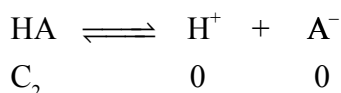


$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]} = \frac{0.1 \times 0.0132 \times 0.1 \times 0.0132}{0.1(1-0.0132)} = 1.76 \times 10^{-5}$$

$$\text{pK}_a = -\log K_a = -\log (1.76 \times 10^{-5}) = 4.75$$

4.7 pH of a mixture of weak acid (monoprotic) and a strong acid solution :

- Weak acid and Strong acid both will contribute H^+ ion.
- For the first approximation we can neglect the H^+ ions coming from the weak acid solution and calculate the pH of the solution from the concentration of the strong acid only.
- To calculate exact pH, we have to take the effect of presence of strong acid on the dissociation equilibrium of the weak acid.
- If $[SA] = C_1$ and $[WA] = C_2$, then $[H^+]$ from $SA = C_1$ the weak acid will dissociate as follows.



$$C_2(1 - \alpha) \quad C_2\alpha + C_1 \quad C_2\alpha \quad K_a = \frac{(C_2\alpha + C_1)C_2\alpha}{C_2(1 - \alpha)} \quad (\alpha \ll 1)$$

(The weak acids dissociation will be further suppressed because of presence of strong acid, common ion effect)

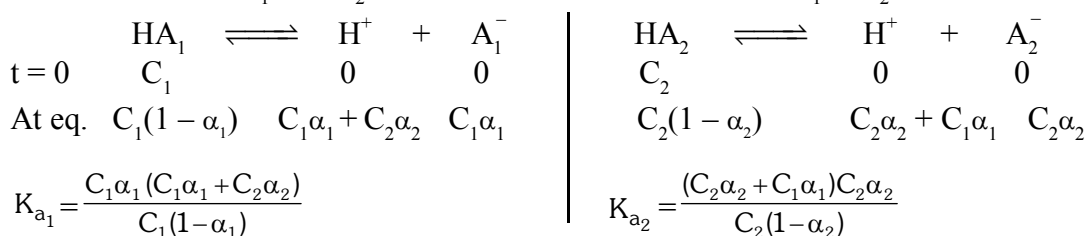
$$K_a = (C_2\alpha + C_1)\alpha$$

Total H^+ ion concentration = $C_1 + C_2\alpha$

- If the total $[H^+]$ from the acid is more than $10^{-6} M$, then contribution from the water can be neglected, if not then we have to take $[H^+]$ from the water also.

4.8 pH of a mixture of two weak acid (both monoprotic) solution :

- Both acids will dissociate partially.
- Let the acid are HA_1 & HA_2 and their final concentrations are C_1 & C_2 respectively, then



(Since α_1, α_2 both are small in comparison to unity)

$$K_{a1} = (C_1\alpha_1 + C_2\alpha_2)\alpha_1 ; K_{a2} = (C_1\alpha_1 + C_2\alpha_2)\alpha_2 \Rightarrow \frac{K_{a1}}{K_{a2}} = \frac{\alpha_1}{\alpha_2}$$

$$\begin{aligned} [H^+] = C_1\alpha_1 + C_2\alpha_2 &= \frac{C_1K_{a1}}{\sqrt{C_1K_{a1} + C_2K_{a2}}} + \frac{C_2K_{a2}}{\sqrt{C_1K_{a1} + C_2K_{a2}}} \Rightarrow [H^+] \\ &= \sqrt{C_1K_{a1} + C_2K_{a2}} \end{aligned}$$

- If the dissociation constant of one of the acid is very much greater than that of the second acid then contribution from the second acid can be neglected.

$$\text{So, } [H^+] = C_1\alpha_1 + C_2\alpha_2 \approx C_1\alpha_1$$

Ex.18 Calculate pH of solution obtained by mixing equal vol. of 0.02 M HOCl & 0.2 M CH₃COOH solution given that $K_{a_1}(\text{HOCl}) = 2 \times 10^{-4}$, $K_{a_2}(\text{CH}_3\text{COOH}) = 2 \times 10^{-5}$

Also calculate OH⁻, OCl⁻, CH₃COO⁻

Sol. Final solution volume become double

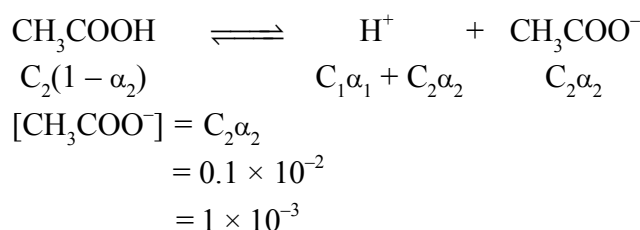
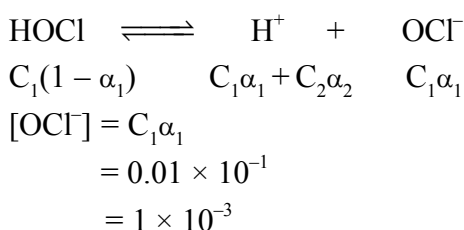
$$C_1 = 0.01, \quad C_2 = 0.1$$

$$[\text{H}^+] = \sqrt{K_{a_1}C_1 + K_{a_2}C_2} = \sqrt{2 \times 10^{-4} \times 0.01 + 2 \times 10^{-5} \times 0.1}$$

$$= \sqrt{2 \times 10^{-6} + 2 \times 10^{-6}} = 2 \times 10^{-3}$$

$$\text{pH} = 3 - \log 2 = 3 - 0.3010 = 2.69$$

$$\alpha_1 = \frac{2 \times 10^{-4}}{2 \times 10^{-3}} = 10^{-1} \quad \alpha_2 = \frac{2 \times 10^{-5}}{2 \times 10^{-3}} = 10^{-2}$$



$$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} = \frac{10^{-14}}{2 \times 10^{-3}} = 0.5 \times 10^{-11} = 5 \times 10^{-12} \text{ M}$$

$$[\text{HOCl}] = 10^{-2}(1 - 0.1) = 9 \times 10^{-3} \text{ M}$$

$$[\text{CH}_3\text{COOH}] = 10^{-1}(1 - 0.01) \approx 10^{-1}$$

4.9 pH of a solution of a polyprotic weak acid :

- Diprotic acid is the one, which is capable of giving 2 protons per molecule in water. Let us take a weak diprotic acid (H₂A) in water whose concentration is c M.

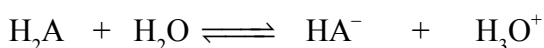
In an aqueous solution, following equilibria exist.

If

α_1 = degree of ionization of H₂A in presence of HA⁻ K_{a_1} = first ionisation constant of H₂A

α_2 = degree of ionisation of HA⁻ in presence of H₂A K_{a_2} = second ionisation constant of H₂A

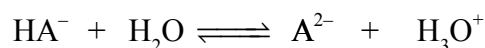
I step



at eq. $c(1 - \alpha_1)$ $c\alpha_1(1 - \alpha_2)$ $(c\alpha_1 + c\alpha_1\alpha_2)$

$$(K_{\text{eq}})_1 [\text{H}_2\text{O}] = \frac{[\text{H}_3\text{O}^+][\text{HA}^-]}{[\text{H}_2\text{A}]} = K_{a_1}$$

II step



at eq. $c\alpha_1(1 - \alpha_2)$ $c\alpha_1\alpha_2$ $(c\alpha_1 + c\alpha_1\alpha_2)$

$$(K_{\text{eq}})_2 [\text{H}_2\text{O}] = \frac{[\text{H}_3\text{O}^+][\text{A}^{2-}]}{[\text{HA}^-]} = K_{a_2}$$

$$\therefore K_{a_1} = \frac{(c\alpha_1 + c\alpha_1\alpha_2)[c\alpha_1(1-\alpha_2)]}{c(1-\alpha_1)} \quad K_{a_2} = \frac{(c\alpha_1 + c\alpha_1\alpha_2)[c\alpha_1\alpha_2]}{c\alpha_1(1-\alpha_2)}$$

$$= \frac{[c\alpha_1(1+\alpha_2)][\alpha_1(1-\alpha_2)]}{1-\alpha_1} \dots\dots (i) \quad = \frac{[c\alpha_1(1+\alpha_2)]\alpha_2}{1-\alpha_2} \dots\dots (ii)$$

Knowing the values of K_{a_1} , K_{a_2} and c , the values of α_1 and α_2 can be calculated using equations (i)

and (ii) After getting the values of α_1 and α_2 , $[H_3O^+]$ can be calculated as

$$[H_3O^+]_T = c\alpha_1 + c\alpha_1\alpha_2$$

Finally, for calculation of pH

- If the total $[H_3O^+] < 10^{-6}$ M, the contribution of H_3O^+ from water should be added.
- If the total $[H_3O^+] > 10^{-6}$ M, then $[H_3O^+]$ contribution from water can be ignored.

Using this $[H_3O^+]$, pH of the solution can be calculated.

❖ Approximation :

For diprotic acids, $K_{a_2} \ll K_{a_1}$ and α_2 would be even smaller than α_1

$$\therefore 1 - \alpha_2 \approx 1 \text{ and } 1 + \alpha_2 \approx 1$$

Thus, equation (i) can be reduced to $K_{a_1} = \frac{C\alpha_1 \times \alpha_1}{1 - \alpha_1}$

This is expression similar to the expression for a weak monoprotic acid.

- Hence, for a diprotic acid (or a polyprotic acid) the $[H_3O^+]$ can be calculated from its first equilibrium constant expression alone provided $K_{a_2} \ll K_{a_1}$.

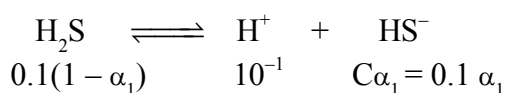
Ex.19 Calculate pH of $[HS^-]$, $[S^{2-}]$, $[Cl^-]$ in a solution which is 0.1 M HCl & 0.1 M H_2S given that $K_{a_1}(H_2S) = 10^{-7}$, $K_{a_2}(H_2S) = 10^{-14}$ also calculate α_1 & α_2 .

Sol. $HCl + H_2S$

$$0.1 \quad 0.1$$

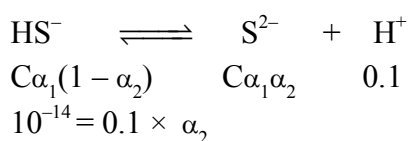
$$C_1 = C_2 = 0.1$$

$$\therefore pH = 1 \quad (\text{most of } [H^+] \text{ comes from HCl})$$



$$K_{a_1} = \frac{C\alpha_1 \times 10^{-1}}{C(1 - \alpha_1)} = \frac{10^{-7}}{10^{-1}} = \alpha_1 \quad (\because 1 - \alpha_1 = 1)$$

$$\Rightarrow \alpha_1 = 10^{-6}$$



$$\Rightarrow \alpha_2 = 10^{-13}$$

$$[S^{2-}] = C\alpha_1\alpha_2$$

$$= 10^{-6} \times 10^{-1} \times 10^{-13} = 10^{-20} \text{ M}$$

4.10 ISOHYDRIC SOLUTIONS

- (i) Solutions of electrolytes are said to be isohydric if the concentration of the common ion present in them is the same and on mixing such solutions, there occurs no change in the degree of dissociation of either of the electrolyte.
- (ii) Let the isohydric solution is made by HA_1 and HA_2 acids, then $[H^+]$ of both acids should be equal i.e.

$$\sqrt{K_{a_1} C_1} = \sqrt{K_{a_2} C_2} \quad \text{or} \quad \frac{K_{a_1}}{K_{a_2}} = \frac{C_2}{C_1}$$

4.11 RELATIVE STRENGTH OF WEAK ACIDS AND BASES

For two acids of equimolar concentrations.

$$\frac{\text{Strength of acid (I)}}{\text{Strength of acid (II)}} = \sqrt{\frac{K_{a_1}}{K_{a_2}}}$$

Similarly for bases, $\frac{\text{Strength of base (I)}}{\text{Strength of base (II)}} = \sqrt{\frac{K_{b_1}}{K_{b_2}}}$

The modern method is to convert K_a as a power of 10 and express acid strength by power of 10 with sign changed and call this new unit pK_a . Thus, if K_a for acid is equal to 10^{-4} , $pK_a = 4$. So higher pK_a value means lower acid strength, that is, $pK_a = -\log K_a$

Also, $pK_b = -\log K_b$

5. SALTS

- (i) Salts are the ionic compounds formed when its positive part (Cation) come from a base and its negative part (Anion) come from an acid.
- (ii) Salts may taste salty, bitter or sweet or tasteless.
- (iii) Solution of salts may be acidic, basic or neutral.
- (iv) Fused salts and their aqueous solutions conduct electricity and undergo electrolysis.
- (v) The salts are generally crystalline solids.

5.1 Classification of salts :

The salts may be classified into four categories.

(a) Normal salt :

- (i) The salt formed by the loss of all possible protons (replaceable H^+ ions)

Ex. $NaCl$, $NaNO_3$, K_2SO_4 , $Ca_3(PO_4)_2$, Na_3BO_3 , Na_2HPO_3 , NaH_2PO_2 etc.

(b) Acid salts :

- (i) Salts formed by incomplete neutralisation of polybasic acids. Such salts contain one or more replaceable H atom.

Ex. $NaHCO_3$, $NaHSO_4$, NaH_2PO_4 , Na_2HPO_4 etc.

- (ii) Above salts when neutralized by base form normal salts.

(c) **Basic salts :**

- (i) Salts formed by in complete neutralisation of poly acidic bases are called basic salts. These salt contain one or more hydroxyl groups.

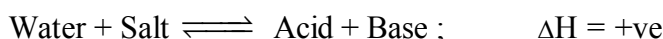
Ex. Zn(OH)Cl , Mg(OH)Cl , $\text{Fe(OH)}_2\text{Cl}$, $\text{Bi(OH)}_2\text{Cl}$ etc.

- (ii) Above salts when neutralised by acids form normal salts.

5.2 HYDROLYSIS OF SALTS

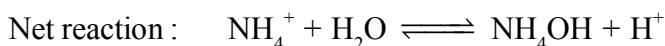
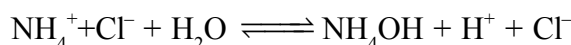
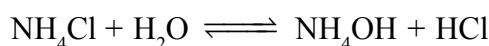
Salt hydrolysis is defined as the process in which water reacts with cation or anion or both of a salt to change the concentration of H^+ and OH^- ions of water.

Salt hydrolysis is reverse process of neutralization.

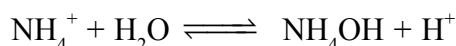


5.2.1 Hydrolysis of strong acid - weak base [SA - WB] type salt –

Ex. CaSO_4 , NH_4Cl , $(\text{NH}_4)_2\text{SO}_4$, $\text{Ca(NO}_3)_2$, ZnCl_2 , CuCl_2 , CaCl_2



- (i) In this type of salt hydrolysis, cation reacts with H_2O , therefore called as **cationic hydrolysis**.
 (ii) Solution is acidic in nature (SAWB) as $[\text{H}^+]$ is increased.
 (iii) pH of the solution is less than 7.
 (iv) **Relation between K_h , K_w & K_b**



$$\text{Hydrolysis constant } K_h = \frac{[\text{NH}_4\text{OH}][\text{H}^+]}{[\text{NH}_4^+]} \quad \dots(i)$$



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} \quad \dots(ii)$$



$$K_w = [\text{OH}^-][\text{H}^+] \quad \dots(iii)$$

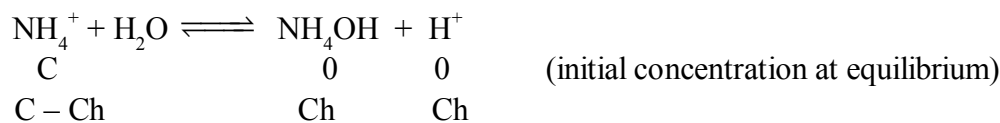
Now multiplying Eq. (1) & (2) = Eq. (3)

$$\frac{[\text{NH}_4\text{OH}][\text{H}^+]}{[\text{NH}_4^+]} \times \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} = [\text{H}^+][\text{OH}^-]$$

i.e. $K_h \times K_b = K_w$

$$\boxed{K_h = \frac{K_w}{K_b}}$$

(v) Degree of hydrolysis – (Represented by h)



$$K_h = \frac{[\text{NH}_4\text{OH}][\text{H}^+]}{[\text{NH}_4^+]} = \frac{\text{Ch}^2}{(1-h)}$$

Since $h \ll 1$ then $(1-h) \approx 1$

$$\therefore K_h = \text{Ch}^2$$

$$\Rightarrow h = \sqrt{\frac{K_h}{C}}$$

$$\therefore \Rightarrow h = \sqrt{\frac{K_w}{K_b C}} \Rightarrow \boxed{h = \sqrt{\frac{K_w}{K_b \times C}}}$$

(vi) pH of the solution :

$$\text{pH} = -\log [\text{H}^+]$$

$$\Rightarrow [\text{H}^+] = \sqrt{\frac{K_w \times C}{K_b}}$$

On taking $-\log$ on both sides

$$\text{pH} = -\log \left(\frac{K_w \times C}{K_b} \right)^{1/2}$$

$$\text{pH} = -\frac{1}{2} \log K_w - \frac{1}{2} \log C - \frac{1}{2} (-\log K_b)$$

$$\text{pH} = 7 - \frac{1}{2} \text{p}K_b - \frac{1}{2} \log C$$

Ex.20 Find out the K_h of centi normal $[10^{-2} \text{ N}]$ solution of NH_4Cl (SA - WB) if dissociation constant of NH_4OH is 10^{-6} and $K_w = 10^{-14}$. Find out degree of hydrolysis and also find $[\text{H}^+]$ and pH of solution?

(Given : $K_w = 10^{-14}$; $K_b = 10^{-6}$)

Sol. (1) $K_h = \frac{K_w}{K_b} = \frac{10^{-14}}{10^{-6}} = 10^{-8}$

(2) $h = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{10^{-8}}{10^{-2}}} = \sqrt{10^{-6}} = 10^{-3}$

(3) $[\text{H}^+] = \text{Ch}$
 $= 10^{-2} \times 10^{-3} = 10^{-5}$

(4) $\text{pH} = -\log [\text{H}^+] = -\log [10^{-5}] = +5 \log 10 = +5 \times 1 = 5$

Ex.21 How many grams of NH_4Cl should be dissolved per litre of solution to have a pH of 5.13 ? K_b for NH_3 is 1.8×10^{-5} .

Sol. NH_4Cl is a salt of strong acid and weak base for solutions of such salts.

$$\text{pH} = \frac{1}{2} [\text{p}K_w - \log C - \text{p}K_b]$$

$$\Rightarrow 10.26 = 14 - \log C - 4.74$$

$$\Rightarrow \log C = 9.26 - 10.26 = -1.0$$

$$\therefore C = 10^{-1} \text{ M}$$

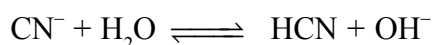
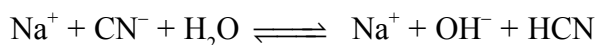
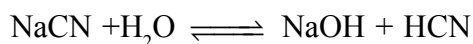
$$[\text{NH}_4\text{Cl}] = 10^{-1} \text{ M}$$

$$W_{\text{NH}_4\text{NO}_3} = 10^{-1} \times 53.5 \text{ gL}^{-1}$$

$$= 5.35 \text{ gL}^{-1}$$

5.2.2 Hydrolysis of [WA – SB] type salt :

Ex. KCN, NaCN, K_2CO_3 , BaCO_3 , K_3PO_4

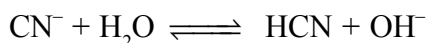


(i) In this type of salt hydrolysis anion reacts with water therefore called as anionic hydrolysis.

(ii) Solution is basic in nature as $[\text{OH}^-]$ increases.

(iii) pH of the solution is greater than 7.

(iv) **Relation between K_h , K_w , K_a**

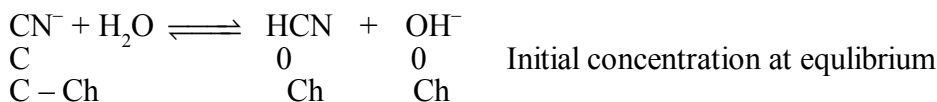


$$K_h = \frac{[\text{HCN}][\text{OH}^-]}{[\text{CN}^-]} \quad \dots (i)$$

$$\frac{[\text{HCN}][\text{OH}^-]}{[\text{CN}^-]} \times \frac{[\text{CN}^-][\text{H}^+]}{[\text{HCN}]} = [\text{H}^+][\text{OH}^-]$$

$$\boxed{K_h = \frac{K_w}{K_a}}$$

(v) **Degree of hydrolysis :**



$$K_h = \frac{[\text{HCN}][\text{OH}^-]}{[\text{CN}^-]}$$

$$K_h = \frac{\text{Ch}^2}{(1 - h)}$$

Since $h \ll 1$, therefore $(1 - h) \approx 1$

$$\therefore K_h = \text{Ch}^2$$

$$h^2 = \frac{K_h}{C}$$

$$\Rightarrow \boxed{h = \sqrt{\frac{K_h}{C}}}$$

$$\boxed{h = \sqrt{\frac{K_w}{K_a \times C}}}$$

(vi) pH of the solution

$$[\text{OH}^-] = \text{Ch}$$

$$[\text{OH}^-] = \sqrt{\frac{K_w \times C}{K_a}}$$

$$[\text{H}^+] = \frac{K_w}{\sqrt{\frac{K_w \times C}{K_a}}} \Rightarrow [\text{H}^+] = \sqrt{\frac{K_w \times K_a}{C}}$$

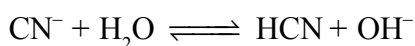
On taking $-\log$ on both sides

$$\text{pH} = -\frac{1}{2} [\log K_w + \log K_a - \log C]$$

$$\text{pH} = 7 + \frac{1}{2} \text{p}K_a + \frac{1}{2} \log C$$

Ex.22 Calculate the pH and degree of hydrolysis of 0.01 M solution of NaCN, K_a for HCN is 6.2×10^{-12} .

Sol. NaCN is a salt of strong base NaOH and weak acid HCN. Na^+ does not react with water whereas CN^- reacts with water as here under



$$K_h = \frac{[\text{HCN}][\text{OH}^-]}{[\text{CN}^-]} = \frac{K_w}{K_a} = \frac{10^{-14}}{6.2 \times 10^{-12}} = 1.6 \times 10^{-3}$$

Let, x moles of salt undergo hydrolysis then concentrations of various species would be

$$[\text{CN}^-] = (0.01 - x) \approx 0.01, [\text{HCN}] = x$$

$$[\text{OH}^-] = x$$

$$\therefore K_h = \frac{x \cdot x}{0.01} = 1.6 \times 10^{-3}$$

$$\therefore x^2 = 1.6 \times 10^{-5}$$

$$\therefore x = 4 \times 10^{-3}$$

$$[\text{OH}^-] = x = 4 \times 10^{-3} \text{ M}$$

$$[\text{H}_3\text{O}^+] = \frac{K_w}{[\text{OH}^-]} = \frac{10^{-14}}{4 \times 10^{-3}} = 0.25 \times 10^{-11}$$

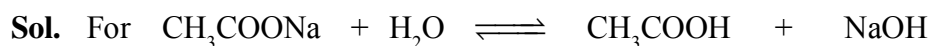
$$\text{pH} = -\log(0.25 \times 10^{-11}) = 11.6020$$

$$\text{Degree of hydrolysis} = \frac{x}{0.01} = \frac{4 \times 10^{-3}}{0.01} = 4 \times 10^{-1}$$

Ex.23. Calculate for 0.01 N solution of sodium acetate -

- (i) *Hydrolysis constant*
- (ii) *Degree of hydrolysis*
- (iii) *pH*

Given K_a of $CH_3COOH = 1.9 \times 10^{-5}$.



Initial	C	0	0
After	C(1-h)	Ch	Ch

$$(i) \quad K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{1.9 \times 10^{-5}} = 5.26 \times 10^{-10}$$

$$(ii) \quad h = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{5.26 \times 10^{-10}}{0.01}} = 2.29 \times 10^{-4} \text{ M}$$

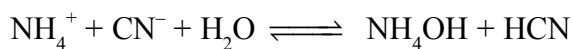
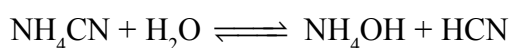
(iii) $[\text{OH}^-]$ from NaOH, a strong base = Ch = $0.01 \times 2.29 \times 10^{-4} = 2.29 \times 10^{-6} \text{ M}$

$$\text{pOH} = 5.64$$

$$\therefore \text{pH} = 14 - 5.64 = 8.36$$

5.2.3 Hydrolysis of (WA - WB) type salt :

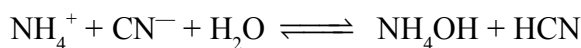
Ex. NH_4CN , CaCO_3 , $(\text{NH}_4)_2\text{CO}_3$, ZnHPO_3



Solution is almost neutral but it may be acidic or basic depending upon the nature of acid & base & pH of the solution is near to 7.

For WA - WB types of salt :

	$K_a > K_b$	$K_b > K_a$	$K_a = K_b$
1. Hydrolysis	Cationic-anionic	Anionic-cationic	Neutral hydrolysis
2. Nature	Acidic	Basic	Neutral
3. pH	$pH < 7$	$pH > 7$	$pH = 7$

(i) Relation between K_h , K_w , K_a & K_b 

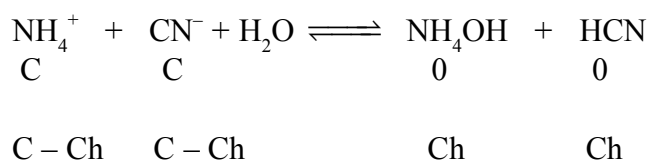
$$K_h = \frac{[\text{NH}_4\text{OH}][\text{HCN}]}{[\text{NH}_4^+][\text{CN}^-]} \quad \dots (i)$$

$$\frac{[\text{NH}_4\text{OH}][\text{HCN}]}{[\text{NH}_4^+][\text{CN}^-]} \times \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} \times \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]} = [\text{H}^+][\text{OH}^-]$$

$$K_h \times K_b \times K_a = K_w$$

$$\boxed{K_h = \frac{K_w}{K_a \times K_b}}$$

(ii) Degree of Hydrolysis :



Initial concentration
at equilibrium

$$K_h = \frac{[\text{NH}_4\text{OH}][\text{HCN}]}{[\text{NH}_4^+][\text{CN}^-]}$$

Since $h \ll \ll \ll 1$

Then $(1 - h) \approx 1$

$$\therefore \boxed{K_h = h^2} \quad \text{or} \quad h^2 = \frac{K_w}{K_a \times K_b}$$

$$h = \sqrt{\frac{K_w}{K_a \times K_b}} \quad \dots (v)$$

(iii) pH of the solution

From eq. (iii)

$$K_a = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]}$$

$$[\text{H}^+] = \frac{K_a \times [\text{HCN}]}{[\text{CN}^-]}$$

$$[\text{H}^+] = \frac{K_a \times \text{Ch}}{\text{C} - \text{Ch}} = \frac{K_a \times h}{1 - h}$$

Since $h \ll \ll \ll 1$, $(1 - h) \approx 1$

$$[\text{H}^+] = K_a \times h \quad [\text{Now put the value of } h \text{ from eq. (5)}]$$

$$= K_a \times \sqrt{\frac{K_w}{K_a \times K_b}}$$

$$\boxed{[\text{H}^+] = \sqrt{\frac{K_w \times K_a}{K_b}}}$$

On taking $-\log$ on both sides

$$-\log [H^+] = -\log \left(\frac{K_w \times K_a}{K_b} \right)^{1/2}$$

$$pH = -\frac{1}{2} [\log K_w + \log K_a - \log K_b]$$

$$pH = 7 + \frac{1}{2} pK_a - \frac{1}{2} pK_b$$

Note : Degree of hydrolysis of [WA – WB] type salt does not depend on the concentration of salt.

Ex.24 Salt of weak acid and weak base

(i) Calculate pH of the mixture (25 mL of 0.1 M NH_4OH + 25 mL of 0.1 M CH_3COOH).

Given that $K_a : 1.8 \times 10^{-5}$, and $K_b = 1.8 \times 10^{-5}$

Sol.	NH_4OH	+	CH_3COOH	$\rightarrow$	CH_3COONH_4	+	H_2O
Initial milli moles	25×0.1		25×0.1		0		0
	= 2.5		= 2.5		–		
Final milli moles	0		0		2.5		2.5

As salt is formed (salt of weak acid and weak base) and pH will be decided by salt hydrolysis

$$pH = \frac{pK_w + pK_a - pK_b}{2} = \frac{1}{2} (-\log 10^{-14} - \log 1.8 \times 10^{-5} + \log 1.8 \times 10^{-5}) = 7$$

Ex.25 In the following which one has highest / maximum degree of hydrolysis.

(1) 0.01 M – NH_4Cl

(2) 0.1 M – NH_4Cl

(3) 0.001 M – NH_4Cl

(4) Same

Sol. [3]

$$\left(h = \sqrt{\frac{K_h}{C}} \quad \text{if } C \text{ decreases, } h \text{ increases} \right)$$

Ex.26 In the following which one has lowest value of degree of hydrolysis.

(1) 0.01 M – CH_3COONH_4

(2) 0.1 M – CH_3COONH_4

(3) 0.001 M – CH_3COONH_4

(4) Same

Sol. [4]

Ex.27 Find out the concentration of $[H^+]$ in 0.1M CH_3COONa solution ($K_a = 10^{-5}$)

Sol. Salt is [WA – SB] type

$$\therefore [H^+] = \sqrt{\frac{K_w \times K_a}{C}} = \sqrt{\frac{10^{-14} \times 10^{-5}}{10^{-1}}} = \sqrt{10^{-19} \times 10^{+1}} = \sqrt{10^{-18}} = 10^{-9}$$

Ex.28 Calculate the degree of hydrolysis of a mixture containing 0.1N NH_4OH and 0.1N HCN

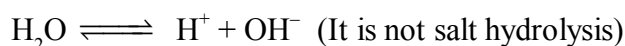
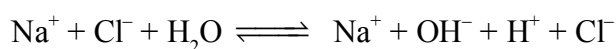
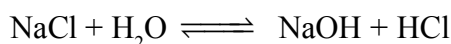
$$K_a = 10^{-5} \quad \& \quad K_b = 10^{-5}$$

Sol. Salt is [WA – WB]

$$h = \sqrt{\frac{K_w}{K_a \times K_b}} = \sqrt{\frac{10^{-14}}{10^{-5} \times 10^{-5}}} \\ = \sqrt{10^{-14} \times 10^{+10}} = \sqrt{10^{-4}} = 10^{-2}$$

5.2.4 Hydrolysis of [SA – SB] type salt :

Ex. NaCl , BaCl_2 , Na_2SO_4 , KClO_4 etc.



- (1) Hydrolysis of salt of [SA – SB] is not possible
- (2) Solution is neutral in nature ($\text{pH} = \text{pOH} = 7$)
- (3) pH of the solution is 7

5.2.5 Hydrolysis of Amphotropic Anion :

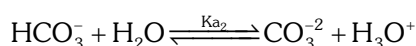
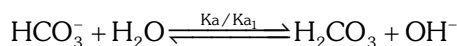
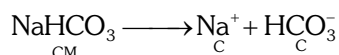
NaHCO_3 , NaHS , etc., can undergo ionisation to form H^+ ion and can undergo hydrolysis to form OH^- (Na^+ ion is not hydrolysed)

- (a) (i) $\text{HCO}_3^- + \text{H}_2\text{O} \xrightleftharpoons{\text{ionisation}} \text{CO}_3^{2-} + \text{H}_3\text{O}^+$ (acid)
- (ii) $\text{HCO}_3^- + \text{H}_2\text{O} \xrightleftharpoons{\text{hydrolysis}} \text{H}_2\text{CO}_3 + \text{OH}^-$ (base)

$$\text{pH}(\text{HCO}_3^-) = \left(\frac{\text{p}K_{a_1} + \text{p}K_{a_2}}{2} \right)$$

- (b) Similarly for H_2PO_4^- and HPO_4^{2-} amphotropic anions.

$$\text{pH}_{(\text{H}_2\text{PO}_4^-)} = \left(\frac{\text{p}K_{a_1} + \text{p}K_{a_2}}{2} \right) \quad \text{and} \quad \text{pH}_{(\text{HPO}_4^{2-})} = \left(\frac{\text{p}K_{a_2} + \text{p}K_{a_3}}{2} \right)$$



$\therefore \text{H}^+$ and OH^- also react

$\therefore$ We can safely assume that both reactions have nearly same degree of dissociation

$$\therefore [\text{H}_2\text{CO}_3] \approx [\text{CO}_3^{2-}] \quad \dots(1)$$

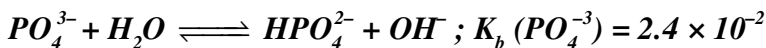
$$\frac{K_w}{K_{a_1}} = \frac{[\text{H}_2\text{CO}_3][\text{OH}^-]}{[\text{HCO}_3^-]} \Rightarrow \frac{1}{K_{a_1}} = \frac{[\text{H}_2\text{CO}_3]}{[\text{H}^+][\text{HCO}_3^-]} \quad \dots(2)$$

$$\frac{[\text{CO}_3^{2-}][\text{H}^+]}{[\text{HCO}_3^-]} = K_{a_2} \quad \dots(3)$$

Divide (2) by (3)

$$[\text{H}^+] = \sqrt{K_{a_1}K_{a_2}} \Rightarrow \text{pH} = \frac{\text{p}K_{a_1} + \text{p}K_{a_2}}{2}$$

Ex.29 Calculate the pH of 0.5 M Na_3PO_4 in aqueous solution ?



Sol. HPO_4^{2-} and PO_4^{3-} are conjugate acid and base so $K_a \times K_b = 10^{-14}$

$$K_a(\text{HPO}_4^{2-}) = \frac{10^{-14}}{2.4 \times 10^{-2}} = 4.17 \times 10^{-13}$$

$$\text{p}K_a = -\log K_a = 12.38$$

$$\text{or } \text{pH} = 7 + \frac{1}{2} \text{p}K_a + \frac{1}{2} \log C$$

$$\text{pH} = 13.04$$

6 BUFFER SOLUTIONS

A solution that resists change in pH value upon addition of small amount of strong acid or base or when solution is diluted is called buffer solution.

The capacity of a solution to resist alteration in its pH value is known as buffer capacity and the mechanism of buffer solution is called buffer action.

6.1 Types of buffer solutions

(A) Simple buffer solution

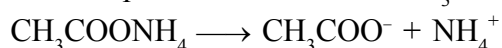
(B) Mixed buffer solution

6.2 SIMPLE BUFFER SOLUTION :

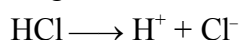
A salt of weak acid and weak base in water e.g. $\text{CH}_3\text{COONH}_4$, HCOONH_4 , AgCN , NH_4CN .

Buffer action of simple buffer solution

Consider a simple buffer solution of $\text{CH}_3\text{COONH}_4$, since it is a salt will dissociated completely.



If a strong acid such as HCl is added then

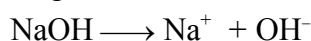


The H^+ ions from the added acid (HCl) combine with CH_3COO^- ions to form CH_3COOH , which is a weak acid so will not further ionized.

Thus there is no rise in H^+ ion concentration and the pH remains constant.



• If a strong base is added as NaOH



Thus change in OH^- ion concentration is resisted by NH_4^+ ions by forming NH_4OH which is a weak base. So it will not further ionized and pH remains constant.

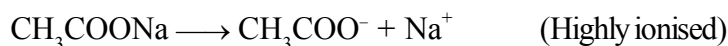
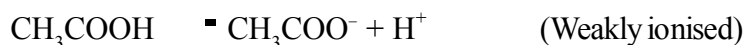
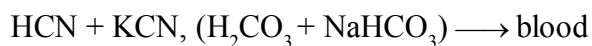
pH of a simple buffer solution :-

$$\text{pH} = 7 + \frac{1}{2} \text{p}K_a - \frac{1}{2} \text{p}K_b$$

6.3 MIXED BUFFER SOLUTIONS :

6.3.1 Acidic buffer solution :

An acidic buffer solution consists of solution of a weak acid and its salt with strong base. The best known example is a mixture of solution of acetic acid and its salt with strong base (CH_3COONa). Other example :



When a few drops of an acid (HCl) are added to it, the H^+ ions from the added acid (HCl) combine with the CH_3COO^- ions to form CH_3COOH . Thus there is no rise in H^+ ion concentration and the pH of solution remains constant. On the other hand, when a few drops of base (NaOH) are added, the OH^- of the added base reacts with acetic acid to form unionise water and acetate ions.



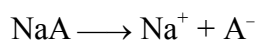
Thus there is no increase in OH^- ion concentration and hence the pH of the solution remains constant.

pH of a acidic buffer solution (Henderson equation) :

Consider a buffer mixture (acidic buffer)



where $\text{A} = \text{CH}_3\text{COO}^-$, $\text{A}^- = \text{CH}_3\text{COO}^-$



Applying law of mass action to dissociation equilibrium of HA

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} ; \text{ so } [\text{H}^+] = \frac{K_a[\text{HA}]}{[\text{A}^-]}$$

taking log, $\log [\text{H}^+] = \log K_a + \log \frac{[\text{HA}]}{[\text{A}^-]}$

$$-\log [\text{H}^+] = -\log K_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

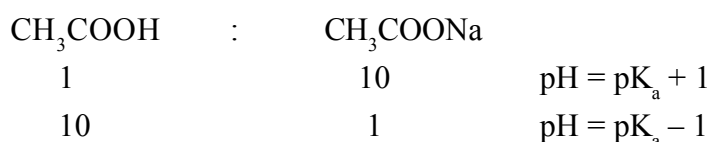
$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$[\text{A}^-]$ = Initial concentration of salt as it is mainly comes from salt.

$[\text{HA}]$ = Initial concentration of the acid.

$$\text{pH} = \text{p}K_a + \log \frac{[\text{Salt}]}{[\text{Acid}]} \quad (\text{it is known as Henderson-Hasselbalch equation.})$$

Note : A solution can act as buffer only if ratio of concentration of salt to acid is between 0.1 to 10.



Thus pH range of an acidic buffer solution is $(pK_a + 1)$ to $(pK_a - 1)$

$$\text{pH range} = pK_a \pm 1$$

Maximum buffer action will be only when ratio of concentration of acid and salt is 1. So for maximum buffer action, $\text{pH} = pK_a$

Ex.30 How much volume of 0.2 M solution of acetic acid should be added to 100 mL of 0.2 M solution of sodium acetate to prepare a buffer solution of $\text{pH} = 6.00$? (pK_a for acetic acid is 4.74)

Sol. $\text{pH} = pK_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$

$$\log \frac{[\text{Salt}]}{[\text{Acid}]} = \text{pH} - pK_a = 6.00 - 4.74 = 1.26 \quad \therefore \quad \frac{[\text{Salt}]}{[\text{Acid}]} = 18.2$$

$$\text{Moles of } \text{CH}_3\text{COONa} \text{ in solution} = \frac{100 \times 0.2}{1000} = 0.02$$

Let, volume of 0.2 acetic acid added = V mL

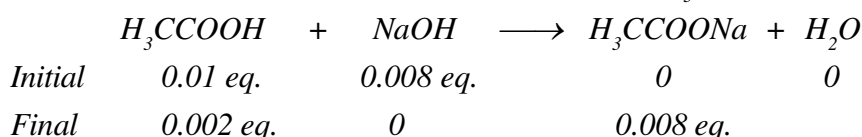
$$\therefore \quad \text{Moles of acetic acid} = \frac{V \times 0.2}{1000}$$

$$\therefore \quad \frac{0.02}{V \times \frac{0.2}{1000}} = 18.2$$

$$\therefore \quad V = 5.49 \text{ mL}$$

Ex.31 Calculate the pH after the addition of 80 mL and 100 mL respectively of 0.1 N NaOH to 100 mL, 0.1 N CH_3COOH . (Given pK_a for $\text{CH}_3\text{COOH} = 4.74$)

Sol. If 80 mL of 0.1 N NaOH is added to 100 mL of 0.1 N CH_3COOH , acidic buffer will form as



$$\text{pH} = pK_a + \log \frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 4.74 + \log \frac{0.008}{0.002} = 5.342$$

If 100 mL of 0.1 N NaOH is added to 100 mL of 0.1 N CH_3COOH , complete neutralization takes place and the concentration of $\text{H}_3\text{CCOONa} = \frac{0.1}{2} \text{ M} = 0.05 \text{ M}$

$$\text{Now, } \text{pH} = 7 + \frac{1}{2} pK_a + \frac{1}{2} \log C = 8.72$$

Ex.32 Calculate the pH of a solution when 0.20 moles of HCl is added to one litre solution containing -
(a) 1 M each of acetic acid and acetate ion ?

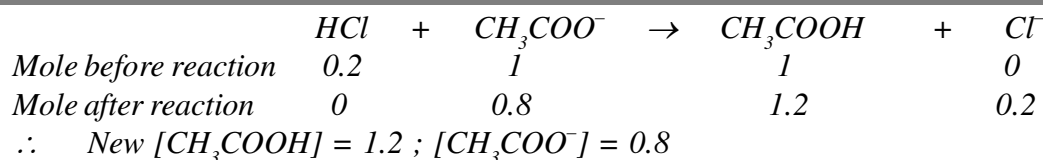
(b) 0.1 M each of acetic acid and acetate ion ?

Given K_a for acetic acid is 1.8×10^{-5} .

Sol. (a) Initially $[\text{Acetic acid}] = 1 \text{ M}$

$$[\text{Acetate}] = 1 \text{ M}$$

Now 0.2 moles of HCl are added to it.



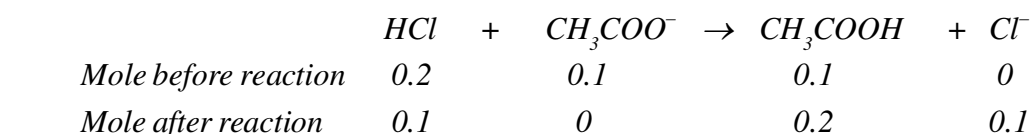
$$\therefore \text{pH} = \text{p}K_a + \log \frac{[\text{conjugate}]}{[\text{acid}]}$$

$$\therefore \text{pH} = -\log 1.8 \times 10^{-5} + \log \frac{0.8}{1.2} = 4.5686$$

(b) In II case initially [Acetic acid] = 0.1 M

[Acetate] = 0.1 M

Now 0.2 mole of HCl are added to it



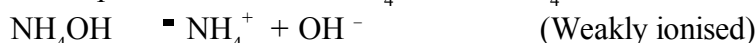
$$\therefore [\text{H}^+] \text{ from free HCl} = 0.1 \text{ M}$$

$$\therefore \text{pH} = 1$$

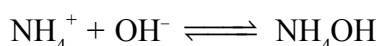
Note: CH_3COOH no doubt gives H^+ but being weak acid as well as in presence of HCl does not dissociate appreciably and thus, H^+ from CH_3COOH may be neglected.

6.3.2 Basic buffer solution :

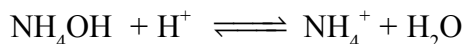
A basic buffer solution consists of a mixture of a weak base and its salt with strong acid. The best known example is a mixture of NH_4OH and NH_4Cl .



When a few drops of a base (NaOH) are added, the OH^- ions from NaOH combine with NH_4^+ ions to form feebly ionised NH_4OH thus there is no rise in the concentration of OH^- ions and hence the pH value remains constant.

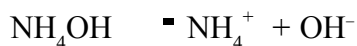


If a few drops of an acid (HCl) are added the H^+ from acid combine with NH_4OH to form H_2O and NH_4^+ ions.



Thus the addition of acid does not increase the H^+ ion concentration and hence pH remains unchanged.

• pH of basic buffer solution :



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]}$$

$$[\text{OH}^-] = \frac{K_b [\text{NH}_4\text{OH}]}{[\text{NH}_4^+]} = \frac{K_b [\text{Base}]}{[\text{Salt}]}$$

(NH_4^+ mainly comes from salt)

taking $-\log$ on both side

$$-\log \text{OH}^- = -\log \frac{K_b [\text{Base}]}{[\text{Salt}]} \Rightarrow \text{pOH} = -\log K_b - \log \frac{[\text{Base}]}{[\text{Salt}]}$$

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{Salt}]}{[\text{Base}]} \Rightarrow \text{pH} = 14 - \text{pOH}$$

❖ **pOH range :**

A solution can act as buffer solution only if ratio of concentration of salt to base is from 0.1 to 10.

NH ₄ OH	:	NH ₄ Cl	
1		10	pOH = pK _b + 1
10		1	pOH = pK _b - 1

So pOH range is pK_b ± 1

❖ **Condition for maximum buffer action :**

[NH ₄ OH]	:	[NH ₄ Cl]	
1		1	

$$\text{pOH} = \text{p}K_b + \log \frac{1}{1}$$

$$\text{pOH} = \text{p}K_b \quad \text{and} \quad \text{pH} = 14 - \text{p}K_b$$

Maximum buffer action because pH remains constant.

Ex.33 An organic base *B* has *K_b* value equal to 1×10^{-8} . In what amounts should 0.01 M HCl and 0.01 M solution of *B* be mixed to prepare 1 L of a buffer solution having pH = 7.0 ?

Sol. $\text{B} + \text{H}_2\text{O} \rightleftharpoons \text{BH}^+ + \text{OH}^-$

$$K_b = \frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]} = 1 \times 10^{-8}$$

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{BH}^+]}{[\text{B}]}$$

$$\Rightarrow 7 = -\log (10^{-8}) + \log \frac{[\text{BH}^+]}{[\text{B}]} \Rightarrow 7 = 8 + \log \frac{[\text{BH}^+]}{[\text{B}]}$$

$$\log \frac{[\text{BH}^+]}{[\text{B}]} = -1$$

$$\therefore \frac{[\text{BH}^+]}{[\text{B}]} = 10^{-1} = 0.1$$

Let, volume of HCl taken = xL

$\therefore$ Volume of base taken = (1 - x) L

After the reaction, millimole of BH^+ formed = $0.01 \times (x)$

Millimoles of base left = $0.01 (1 - 2x)$

$$\therefore \frac{[\text{BH}^+]}{[\text{B}]} = \frac{x}{[1 - 2x]} = 0.1$$

$\therefore x = 0.083 \text{ L} = \text{Volume of HCl}$

$\therefore$ Volume of base = 0.917 L

Ex.34 Which of the following buffers containing NH_4OH and NH_4Cl show the lowest pH value?

	conc. of $\text{NH}_4\text{OH} (\text{mol L}^{-1})$	conc. of $\text{NH}_4\text{Cl} (\text{mol L}^{-1})$
(A)	0.50	0.50
(B)	0.10	0.50
(C)	0.50	1.50
(D)	0.50	0.10

Sol. (B) $\text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]}$ for $\text{NH}_4\text{Cl} = 0.5$ and $\text{NH}_4\text{OH} = 0.1$

pOH will be maximum and so pH will be minimum.

Ex.35 A solution of weak base BOH was titrated with 0.1 N HCl. The pH of the solution was found to be 10.04 and 9.14 after the addition of 5 mL and 20 mL of the acid respectively. Find the dissociation constant of the base.

Sol. Case I :

	BOH	+	HCl	$\longrightarrow$	BCl	+	H ₂ O
Millimole before reac.	a		$0.1 \times 5 = 0.5$		0		0
Millimole after reac.	(a-0.5)		0		0.5		0.5

$$\therefore \text{pOH} = -\log K_b + \log \frac{[\text{BCl}]}{[\text{BOH}]} \dots\dots\dots(\text{i})$$

$$\therefore \text{pH} = 10.04 \quad \text{so} \quad \text{pOH} = 3.96$$

$$\therefore 3.96 = -\log K_b + \log \frac{0.5}{(a-0.5)} \dots\dots\dots(\text{ii})$$

Case II :

	BOH	+	HCl	$\longrightarrow$	BCl	+	H ₂ O
Millimole before reac.	a		$0.1 \times 20 = 2$		0		0
Millimole after reac.	(a-2)		0		2		2

$$\therefore \text{pOH} = -\log K_b + \log \frac{[\text{BCl}]}{[\text{BOH}]} \dots\dots\dots(\text{iii})$$

$$\therefore \text{pH} = 9.14 \quad \therefore \text{pOH} = 4.86$$

$$\therefore 4.86 = -\log K_b + \log \frac{2}{(a-2)} \quad \dots\dots\dots(\text{iv})$$

$$\text{Solving Eqs. (ii) and (iv), } K_b = 1.81 \times 10^{-5}$$

7. INDICATORS

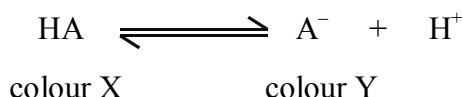
The stage of titration when complete reaction occur between the solution is called **equivalent point**.

The stage of titration when sudden change in colour of solution is observed is called **end point**. A

perfect indicator response sudden colour change exactly on completion of reaction. An **indicator** is a substance which response sudden change in colour of solution at the end point or neutral point of the acid-base titration. **At end point** $N_1 V_1 = N_2 V_2$

At end point $N_1 V_1 = N_2 V_2$

- (i) The indicators in acid-base titration changes colour on changing the pH of solution.
- (ii) All the acid-base indicators are either weak organic acid or base and having different colour for unionized and ionised form.
- (iii) A mixture of two colour is recognized in a single colour if the conc. of one is 10 times or more than that of others. (This 10 time is flexible)



$$\text{Diss. const. or (Ionisation const)} = K_a = K_{in} = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

$$\text{pH} = \text{pK}_{in} + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$(a) \quad \text{The solution will appear only of colour Y, if } \frac{[\text{A}^-]}{[\text{HA}]} \geq 10 \Rightarrow \text{pH} \geq (\text{pK} + 1)$$

$$(b) \quad \text{The solution will appear only of colour X, if } \frac{[\text{A}^-]}{[\text{HA}]} \leq \frac{1}{10} \Rightarrow \text{pH} \leq (\text{pK} - 1)$$

pH of solution below and above which solution appears in a single colour is called pH range of indicator.

Indicator	pH range	Colour change	pK _a
Methyl orange	3.2 – 4.5	Pink to yellow	3.7
Methyl red	4.4 – 6.5	Red to yellow	5.1
Litmus	5.5 – 7.5	Red to blue	7.0
Phenol red	6.8 – 8.4	Yellow to red	7.8
Phenolphthalein	8.3 – 10.5	Colourless to pink	9.6

Ex.36 The disso. const. of a basic indicator is 2×10^{-7} . Calculate its pH range.

Sol. $5.7 - 7.7 = \text{pOH} \quad \therefore \quad \text{pH} = 6.3 - 8.3$

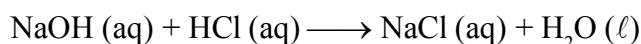
Ex.37 The pH range of an acidic indicator HIn is $4.0 - 5.2$. Calculate dissociation constant. Also

calculate $\frac{\text{In}^-}{\text{HIn}}$ for the appearance of solution in single colour.

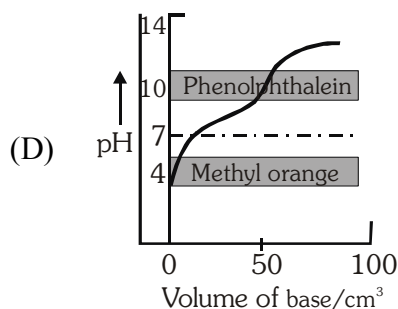
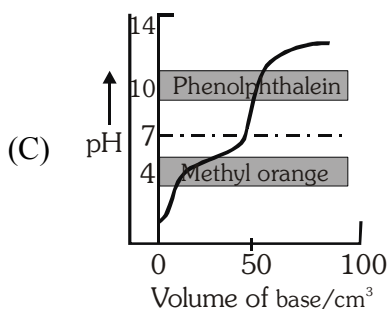
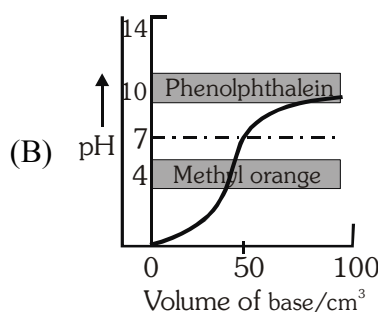
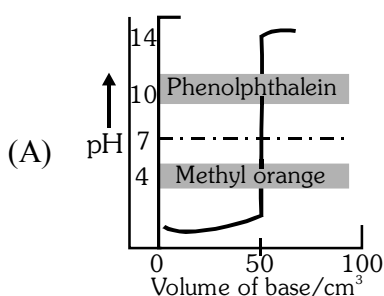
Sol. Diss. constant $= 2.5 \times 10^{-5}$, 4

9.1 TITRATION OF STRONG ACID AGAINST STRONG ALKALI :

The graph (A) shows how pH changes during the titration of 50 cm^3 of 0.1 M HCl with 0.1 M NaOH .



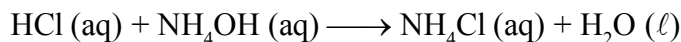
The pH of 0.1 M solution of HCl in the beginning would be 1. As alkali is added, the pH changes slowly in the beginning. However, at the equivalence point pH changes rapidly from about 3.5 to 10. It can be shown by simple calculations that pH of the solution is 3.7 when 49.8 cm^3 of 0.1 M NaOH solution have been added. The pH suddenly changes to 10 after addition of 50.1 cm^3 of the NaOH solution. Thus, any indicator having pH range between 3.5 to 10 will identify the equivalence point. This means that any one of phenolphthalein, methyl orange or bromothymol blue could be used as an indicator.



Titration curves : (A) strong base with strong acid ; (B) weak base with strong acid ; (C) strong base with weak acid ; (D) weak base with weak acid.

7.2 TITRATION OF STRONG ACID AGAINST WEAK ALKALI :

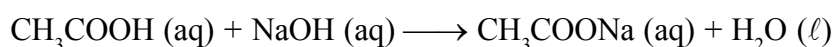
The graph (B) shows how pH changes during titration of 50 cm³ of 0.1 M HCl with 0.1 M NH₃.



In this case, the pH changes rapidly from 3.5 to 7.0 at the equivalence point. Methyl orange, methyl red and bromocresol green are suitable indicators for this type of titration. Phenolphthalein is unsuitable because its pH range lies outside the vertical portion of the curve.

7.3 TITRATION OF WEAK ACID AGAINST STRONG BASE :

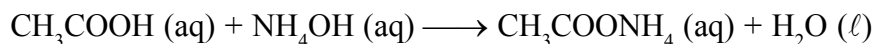
The graph (C) shows how pH changes during titration of 50 cm³ of 0.1 M CH₃COOH with 0.1 M NaOH.



The vertical portion of this titration curve lies between pH range 7 to 10.6. Phenolphthalein is suitable indicator for this titration. Methyl orange is not suitable for this titration because its pH range lies on the flat portion of the curve.

7.4 TITRATION OF WEAK ACID AGAINST WEAK BASE :

The graph (D) represents the titration curve obtained for titration of 50 cm³ of 0.1 M CH₃COOH with 0.1 M NH₃.



For this type of titration there is no sharp increase in pH at the equivalence point. No indicator is suitable for this type of titration.

Ex.38 Bromophenol blue is an indicator with a value of $K_a = 6.84 \times 10^{-6}$. At what pH it will work as an indicator? Also report the % of this indicator in its basic form at a pH of 5.84.

Sol. $\text{HBPh} \rightleftharpoons \text{H}^+ + \text{BPh}^-$

$$K_a = \frac{[\text{H}^+][\text{BPh}^-]}{[\text{HBPh}]}, \text{ when } \text{BPh}^- = \text{HBPh}, \text{ indicator will work. Thus}$$

$$[\text{H}^+] = 6.84 \times 10^{-6}$$

$$\therefore \text{pH} = 5.165$$

Also if pH = 5.84

$$\text{or } [\text{H}^+] = 1.44 \times 10^{-6}, \text{ then}$$

$$K_a = \frac{[\text{H}^+][\text{BPh}^-]}{[\text{HBPh}]} \quad \text{or} \quad 6.84 \times 10^{-6} = \frac{1.44 \times 10^{-6} \cdot C\alpha}{C(1-\alpha)} \quad \text{or} \quad \alpha = 0.83 \text{ or } 83 \%$$

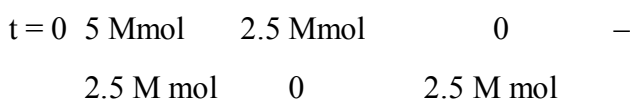
7.5 Titration of Poly protic acid v/s S.B. :

50 mL of 0.1 M H_3PO_4 agent 0.1 M NaOH. Calculate pH when vol. of NaOH added is

- (a) 0 mL (b) 25 mL (c) 50 mL (d) 75 mL (e) 100 mL
 (f) 125 mL (g) 150 mL (h) 200 mL (i) 90 mL

$$pK_{a1} = 3 \quad pK_{a2} = 7, pK_{a3} = 11$$

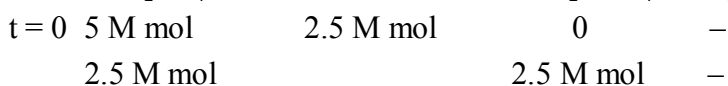
(a) $\text{pH} = 1/2 (pK_{a1} - \log C) = 1/2 (3 + 1) = 2$



$$\text{pH} = pK_{a1} + \log 1/1 = pK_{a1} = 3$$

(c) Solution of H_2PO_4^- amphoteric species

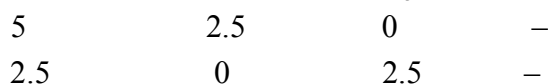
$$\text{pH} = \frac{pK_{a1} + pK_{a2}}{2} = \frac{3 + 7}{2} = 5$$



$$\text{pH} = \log 1/1 + pK_{a2} = 7$$

(e) HPO_4^{2-} solution (amphoteric species)

$$\text{pH} = \frac{pK_{a2} + pK_{a3}}{2} = 9$$



$$\text{pH} = pK_{a3} + \log 1/1 = 11$$

(g) 3rd eq. pt Na_3PO_4 solution

$$[\text{Na}_3\text{PO}_4] = 5/200 = 1/40$$

$$\text{pH} = 1/2 \{pK_w + pK_{a3} + \log C\} = 1/2 (14 + 11 - 2 + 0.4) = 11.7$$

(h) 200 mL

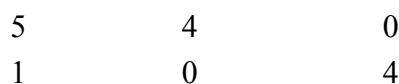
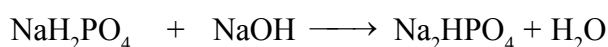
hydrolysis of Na_3PO_4 can be neglected in presence of NaOH

$$[\text{NaOH}] = 5/250 = 1/50$$

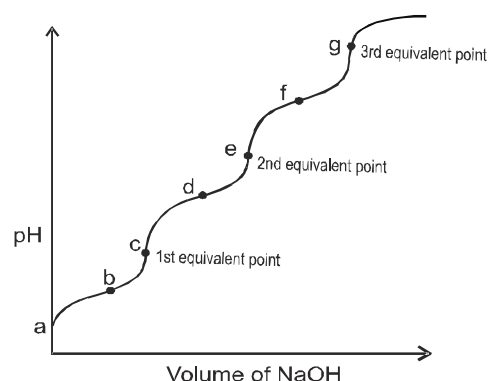
$$\text{pOH} = 1.7$$

$$\text{pH} = 12.3$$

(i) 90 mL



$$\text{pH} = pK_{a2} + \log 4 = 7.6$$



10. SOLUBILITY (s) & SOLUBILITY PRODUCT (K_{sp})

10.1 SOLUBILITY :

At constant temperature, the maximum number of moles of solute which can be dissolved in a solvent to obtain 1 litre of saturated solution is called solubility.

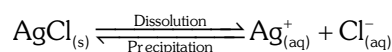
Solubility depends on the following –

- Temperature
- Presence of common ion
- Nature of solvent

10.2 SOLUBILITY PRODUCT(K_{sp}) :

When a sparingly soluble salt such as AgCl is put into water, a very small amount of AgCl dissolves in water and most of the salt remains undissolved in its saturated solution.

- A solution which remains in contact with undissolved solute is said to be saturated.
- The salt AgCl is an electrolyte, its dissociation occurs in solution. Hence, the quantity of AgCl that dissolves in water dissociates into Ag^+ and Cl^- ions. Thus, in the saturated solution of AgCl an equilibrium exists between undissolved solid AgCl and its ions, Ag^+ and Cl^- ions.



according to law of mass action

$$K = \frac{[Ag^+].[Cl^-]}{[AgCl]}$$

Since, the concentration of undissolved solid AgCl is constant. Thus, the product $K.[AgCl]$ gives another constant which is designated as K_{sp}

So, $K.[AgCl] = [Ag^+].[Cl^-]$

$\therefore K_{sp} = [Ag^+] \cdot [Cl^-]$

- K_{sp} for $CaCl_2$** $CaCl_2(s) \rightleftharpoons Ca^{+2}(aq) + 2Cl^-(aq)$

Solubility product in terms of concentration of ions

$$K_{sp} = [Ca^{+2}] [Cl^-]^2$$

- K_{sp} for $AlCl_3$** $AlCl_3(s) \rightleftharpoons Al^{+3}(aq) + 3Cl^-(aq)$

Solubility product in terms of concentration of ions $K_{sp} = [Al^{+3}] [Cl^-]^3$

- General form** $A_xB_y(s) \rightleftharpoons xA^{+y}(aq) + yB^{-x}(aq)$

$$K_{sp} = [A^{+y}]^x [B^{-x}]^y$$

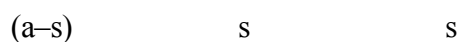
Thus, solubility product is defined as the product of concentrations of the ions raised to a power equal to the number of ions given by the dissociation of electrolyte at a given temperature when the solution is saturated.

10.3 APPLICATION OF SOLUBILITY PRODUCT (K_{sp}) :

10.3.1 To find out the solubility (S) :

- K_{sp} of AB (Mono-mono, di-di, tri-tri valency) type salt –**

Ex. NaCl, $BaSO_4$, CH_3COONa , $CaCO_3$, NaCN, KCN, NH_4CN , NH_4Cl etc.

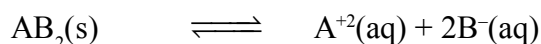


$$K_{sp} = [A^+] [B^-]$$

$$K_{sp} = s^2 \quad \text{or} \quad s = \sqrt{K_{sp}}$$

(ii) **K_{sp} of AB_2 or A_2B (Mono-di or di-mono valency) type salt –**

Ex. $CaCl_2$, $CaBr_2$, K_2S , $(NH_4)_2SO_4$, K_2SO_4 , K_2CO_3 etc.



$$K_{sp} = [A^{+2}] [B^-]^2$$

$$K_{sp} = s \times (2s)^2 = s \times 4s^2 = 4s^3$$

$$s = \left(\frac{K_{sp}}{4} \right)^{1/3}$$

(iii) **General form :**



$$K_{sp} = [A^{+y}]^x \cdot [B^{-x}]^y$$

$$K_{sp} = (xs)^x \cdot (ys)^y$$

$$K_{sp} = x^x \cdot y^y \cdot s^{(x+y)}$$

10.4 COMMON ION EFFECT ON SOLUBILITY :

Solubility of substances always decreases in the presence of common ion. According to Le-Chatelier's principle, on increasing common ion concentration equilibrium shifts in backward direction until the equilibrium is reestablished so, the solubility of substances decreases.

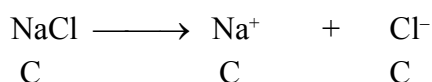
Ex. Find out the solubility of $AgCl$ in water and in the presence of CM – $NaCl$ solution?



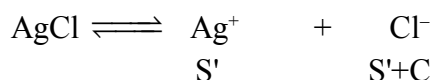
$$K_{sp} = [Ag^+] [Cl^-]$$

$$K_{sp} = S^2$$

In $NaCl$ solution



Let solubility of AgCl in the presence of NaCl solution is S' mol L^{-1}



$$K_{sp} = [\text{Ag}^+][\text{Cl}^-]$$

$$K_{sp} = S' (S' + C) = S'^2 + S'C \quad (\text{Neglecting the higher power terms of } S')$$

$$K_{sp} = S' C$$

$$\boxed{S' = \frac{K_{sp}}{C}}$$

10.5 SIMULTANEOUS SOLUBILITY :

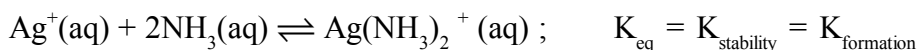
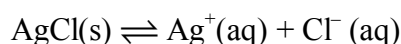
When two sparingly soluble salts are added in water simultaneously, there will be two simultaneous equilibria in the solution.

10.6 SOLUBILITY IN APPROPRIATE BUFFER SOLUTIONS :

Appropriate buffer means that the components of buffer should not interfere with the salt or only H^+ or OH^- ions should be interacting with the ions of the salt.

10.7 EFFECT ON SOLUBILITY BECAUSE OF COMPLEX FORMATION :

Solubility of AgCl in aqueous NH_3 is roughly 10,000 times as its solubility in water, due to complex formation.



$$\text{and } \frac{1}{K_{stability}} = K_{dissociation} = K_{instability}$$

10.8 CONDITION OF PRECIPITATION / IONIC PRODUCT (IP OR Q_{sp}) :

- Ionic product (IP) of an electrolyte is defined in the same way as K_{sp} . The only difference is that ionic product expression contains the initial concentration of ions or the concentration at any time whereas the expression of K_{sp} contains only equilibrium concentration. Thus, for AgCl.

$$IP = [\text{Ag}^+]_i [\text{Cl}^-]_i \quad \text{and} \quad K_{sp} = [\text{Ag}^+]_{eq} [\text{Cl}^-]_{eq}$$

- Ionic product changes with concentration but K_{sp} does not.
- To decide whether an ionic compound will precipitate, its K_{sp} is compared with the value of ionic product.

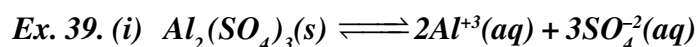
The following three cases arise :

- $IP < K_{sp}$: The solution is unsaturated and precipitation will not occur.
- $IP = K_{sp}$: The solution is saturated and solubility equilibrium exists.
- $IP > K_{sp}$: The solution is supersaturated and hence precipitation of the compound will occur.

Thus, a salt is precipitated when its ionic product exceeds the solubility product of the salt.

10.9 SELECTIVE PRECIPITATION :

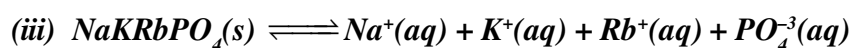
When the K_{sp} values differ then one of the salt can be selectively precipitated.



$$K_{sp} = 2^2 \times 3^3 \times (S)^{2+3} = 4 \times 27 \times S^5 = 108 S^5$$



$$K_{sp} = 2^2 \times 1^1 \times 1^1 (S)^{2+1+1} = 4S^4$$



$$K_{sp} = 1^1 \times 1^1 \times 1^1 \times 1^1 \times (S)^{1+1+1+1} = S^4$$

Ex. 40. If solubility product of the base $M(OH)_3$ is 2.7×10^{-11} , the concentration of OH^- will be

(1) 3×10^{-3}

(2) 3×10^{-4}

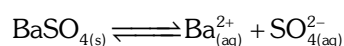
(3) 10^{-3}

(4) 10^{-11}

Answer : (3)

Ex. 41. The solubility of $BaSO_4$ in water is $1.07 \times 10^{-5} \text{ mol dm}^{-3}$. Estimate its solubility product.

Sol. Solubility equilibrium for $BaSO_4$ is

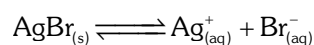


Now, $S = 1.07 \times 10^{-5} \text{ M}$

Hence, $K_{sp} = (1.07 \times 10^{-5})^2 = 1.145 \times 10^{-10}$

Ex. 42. The solubility product of $AgBr$ is 5.2×10^{-13} . Calculate its solubility in mol dm^{-3} and g dm^{-3} . (Molar mass of $AgBr$ = 187.8 g mol^{-1})

Sol. The solubility equilibrium of $AgBr$ is



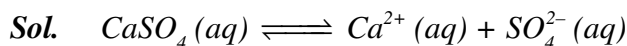
The molar solubility S of $AgBr$ is given by

$$S = \sqrt{K_{sp}} = \sqrt{5.2 \times 10^{-13}} = 7.2 \times 10^{-7} \text{ mol dm}^{-3}$$

The solubility in g dm^{-3} = molar solubility (mol dm^{-3}) $\times$ molar mass (g mol^{-1})

$$= 7.2 \times 10^{-7} \times 187.8 = 1.35 \times 10^{-4} \text{ g dm}^{-3}$$

Ex. 43. What is the maximum volume of water required to dissolve 1 g of calcium sulphate at 25°C. For calcium sulphate, $K_{sp} = 9.0 \times 10^{-6}$.



If S is the solubility of CaSO_4 in moles L^{-1}

$$K_{sp} = [\text{Ca}^{2+}] \times [\text{SO}_4^{2-}] = S^2$$

$$\therefore S = \sqrt{K_{sp}} = \sqrt{9.0 \times 10^{-6}}$$

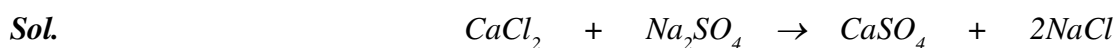
$$= 3 \times 10^{-3} \text{ mol L}^{-1}$$

$$= 3 \times 10^{-3} \times 136 \text{ g L}^{-1} = 0.408 \text{ g L}^{-1}$$

For dissolving 0.408 g of CaSO_4 water required = 1 L

$$\therefore \text{For dissolving 1 g } \text{CaSO}_4 \text{ water required} = \frac{1}{0.408} \text{ L} = 2.45 \text{ L}$$

Ex. 44. Equal volumes of 0.04 M CaCl_2 and 0.0008 M Na_2SO_4 are mixed. Will a precipitate form? K_{sp} for $\text{CaSO}_4 = 2.4 \times 10^{-5}$



Millimole added	0.04 V	0.0008 × V	0	0
-----------------	--------	------------	---	---

Suppose V mL of both are mixed

$$\therefore [\text{Ca}^{2+}] = \frac{0.04V}{2V}$$

$$[\text{SO}_4^{2-}] = \frac{0.0008V}{2V}$$

$$\therefore [\text{Ca}^{2+}][\text{SO}_4^{2-}] = \frac{0.04V}{2V} \times \frac{0.0008V}{2V} = 8 \times 10^{-6}$$

Thus, $[\text{Ca}^{2+}][\text{SO}_4^{2-}]$ in solution $< K_{sp}$

$$8 \times 10^{-6} < 2.4 \times 10^{-5}$$

$\therefore \text{CaSO}_4$ will not precipitate.

Ex. 45. Calculate simultaneous solubility of silverthiocyanate and silver bromide in water given that k_{sp} of silver thiocyanate = 10^{-12} and k_{sp} of silver bromide = 5×10^{-13} respectively.

Sol. Let the solubility of AgSCN be x and that of AgBr is y , then



$$10^{-12} = x(x + y) \quad \dots\dots (i)$$

$$5 \times 10^{-13} = y(x + y) \quad \dots\dots (ii)$$

On solving we get, $x = 2y$

So $y = 4.08 \times 10^{-7}$ and $x = 8.16 \times 10^{-7}$

Ex. 46. What $[H^+]$ must be maintained in saturated $H_2S(0.1\text{ M})$ to precipitate CdS but not ZnS , if $[Cd^{2+}] = [Zn^{2+}] = 0.1$ initially ?

$$K_{sp} (CdS) = 8 \times 10^{-27}$$

$$K_{sp} (ZnS) = 1 \times 10^{-21}$$

$$K_a (H_2S) = 1.1 \times 10^{-21}$$

Sol. In order to prevent precipitation of ZnS

$$[Zn^{2+}] [S^{2-}] < K_{sp} (ZnS) = 1 \times 10^{-21}$$

(ionic product)

$$\text{or } (0.1) [S^{2-}] < 1 \times 10^{-21}$$

$$\text{or } [S^{2-}] < 1 \times 10^{-20}$$

This is the maximum value of $[S^{2-}]$ before ZnS will precipitate. Let $[H^+]$ to maintain this $[S^{2-}]$ be x .

Thus for $H_2S \rightleftharpoons 2H^+ + S^{2-}$

$$K_a = \frac{[H^+]^2 [S^{2-}]}{[H_2S]} = \frac{x^2 (1 \times 10^{-20})}{0.1} = 1.1 \times 10^{-21}$$

$$\text{or } x = [H^+] = 0.1\text{ M}$$

$\therefore$ No ZnS will precipitate at a concentration of H^+ greater than 0.1 M

Ex. 47. What must be the concentration of aq. $NH_3(\text{eq.})$ which must be added to a solution containing $4 \times 10^{-3}\text{ M Ag}^+$ and 0.001 M NaCl , to prevent the precipitation of $AgCl$.

Given that $K_{sp}(AgCl) = 1.8 \times 10^{-10}$ and the formation constant of $[Ag(NH_3)_2]^+$ is

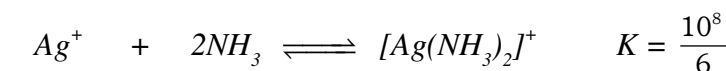
$$K_{\text{formation}} = \frac{10^8}{6}.$$

Sol. Calculate silver ion concentration which can be allowed to remain in the solution,

$$1.8 \times 10^{-10} = [Ag^+][Cl^-]$$

$$[Ag^+] = \frac{1.8 \times 10^{-10}}{0.001} = 1.8 \times 10^{-7}\text{ M},$$

This quantity is so small that almost all the Ag^+ ion will be consumed.



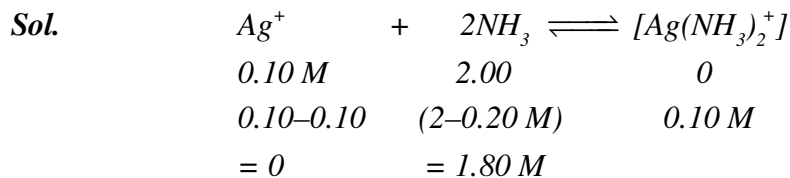
$$4 \times 10^{-3} \quad b \quad 0$$

$$1.8 \times 10^{-7} \quad (b - 8 \times 10^{-3}) \quad 4 \times 10^{-3} \quad K = \frac{10^8}{6} = \frac{4 \times 10^{-3}}{1.8 \times 10^{-7} \times (b - 8 \times 10^{-3})^2}$$

$$\Rightarrow b = 0.0445$$

Ex. 48. 0.10 mol sample of AgNO_3 is dissolved in one litre of 2.00 M NH_3 . Is it possible to form $\text{AgCl}(s)$ in the solution by adding 0.010 mol of NaCl ?

$$(K_{sp}(\text{AgCl}) = 1.8 \times 10^{-10}, K_f[\text{Ag}(\text{NH}_3)_2^+] = 1.6 \times 10^7)$$



It is assumed that all Ag^+ ions have been complexed and only x amount is left

$$K_f = \frac{[\text{Ag}(\text{NH}_3)_2^+]}{[\text{Ag}^+][\text{NH}_3]^2} \Rightarrow 1.6 \times 10^7 = \frac{0.10}{x(1.80)^2}$$

$$\therefore x = 1.93 \times 10^{-9} \text{ M} = [\text{Ag}^+] \text{ undissolved}$$

$$[\text{Cl}^-] = 1.0 \times 10^{-2} \text{ M}$$

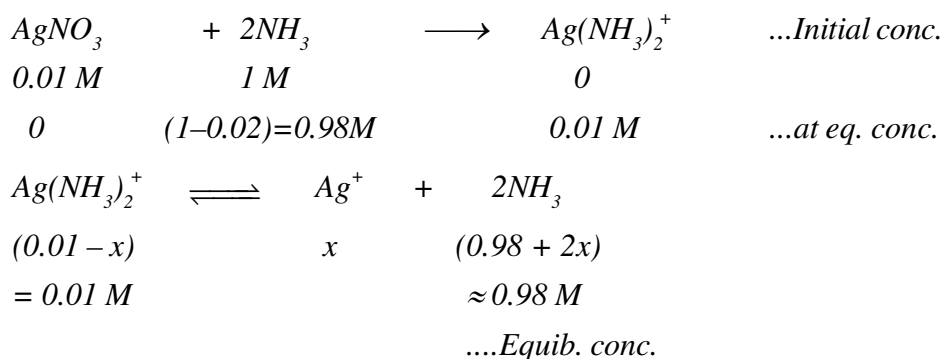
$$\therefore [\text{Ag}^+][\text{Cl}^-] = 1.93 \times 10^{-9} \times 1.0 \times 10^{-2} = 1.93 \times 10^{-11} < 1.8 \times 10^{-10} [K_{sp}(\text{AgCl})]$$

Hence, $\text{AgCl}(s)$ will not precipitate.

Ex. 49. What is the concentration of Ag^+ ions in 0.01 M AgNO_3 that is also 1.0 M NH_3 ? Will AgCl precipitate from a solution that is 0.01 M AgNO_3 , 0.01 M NaCl and 1 M NH_3 ?

$$K_d(\text{Ag}[\text{NH}_3)_2^+) = 5.88 \times 10^{-8}; K_{sp}(\text{AgCl}) = 1.8 \times 10^{-10}.$$

Sol. Let us first assume that 0.01 M AgNO_3 shall combine with 0.02 NH_3 to form 0.01 M $\text{Ag}(\text{NH}_3)_2^+$ and then consider its dissociation.



Since $x \ll 1$

$$K_d = \frac{[\text{Ag}^+][\text{NH}_3]^2}{[\text{Ag}(\text{NH}_3)_2^+]} = 5.88 \times 10^{-8}$$

$$\therefore [\text{Ag}^+] = \frac{5.88 \times 10^{-8} \times 0.01}{(0.98)^2} = 6.12 \times 10^{-10} \text{ M}$$

Further, ionic product of $\text{AgCl} = [\text{Ag}^+][\text{Cl}^-] = (6.12 \times 10^{-10})(0.01) = 6.12 \times 10^{-12}$

Because the ionic product is smaller than $K_{sp} = 1.8 \times 10^{-10}$, no precipitate should form.

EXERCISE # S-I

IONIZATION CONSTANTS AND pH

- Q.1 Calculate the number of H^+ present in one ml of solution whose pH is 13.
- Q.2 (i) K_w for H_2O is 9×10^{-14} at $60^\circ C$. What is pH of water at $60^\circ C$. ($\log 3 = 0.47$)
 (ii) What is the nature of solution at $60^\circ C$ whose
 (a) pH = 6.7 (b) pH = 6.35
- Q.3 The value of K_w at the physiological temperature ($37^\circ C$) is 2.56×10^{-14} . What is the pH at the neutral point of water at this temperature? ($\log 2 = 0.3$)
- Q.4 Calculate pH of following solutions :
- 0.1 M HCl
 - 0.1 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) ($\log \sqrt{1.8} = 0.13$)
 - 0.1 M NH_4OH ($K_b = 1.8 \times 10^{-5}$)
 - 10^{-8} M HCl [$\sqrt{401} = (20.02)$] [$\log 1.051 = 0.03$]
 - 10^{-10} M NaOH
 - 10^{-6} M CH_3COOH ($K_a = 1.8 \times 10^{-5}$)
 - 10^{-8} M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) [$\sqrt{401} = (20.02)$] [$\log 1.051 = 0.03$]
 - Decimolar solution of Baryta ($Ba(OH)_2$), diluted 100 times. ($\log 2 = 0.3$)
 - 10^{-3} mole of KOH dissolved in 100 L of water.
 - Equal volume of HCl solution (PH = 4) + 0.0019 N HCl solution
- Q.5 Calculate :
- K_a for a monobasic acid whose 0.10 M solution has pH of 4.50.
 - K_b for a monoacidic base whose 0.10 M solution has a pH of 10.50.
- Q.6 Calculate the ratio of degree of dissociation (α_2/α_1) when an acetic acid solution is diluted 100 times. Assume $\alpha \ll 1$, even on dilution. [Given $K_a = 10^{-5}$ M]
- Q.7 Calculate the ratio of degree of dissociation of acetic acid and hydrocyanic acid (HCN) in 1 M their respective solution of acids. [Given $K_{a(CH_3COOH)} = 1.8 \times 10^{-5}$; $K_{a(HCN)} = 6 \times 10^{-10}$]
- Q.8 How many moles of HCl must be removed from 1 litre of aqueous HCl solution to change its pH from 2 to 3 ?
- Q.9 pH of a dilute solution of HCl is 6.95. Calculate molarity of HCl solution.

$$\left[\begin{array}{l} 10^{-6.95} = 11.22 \times 10^{-8} \\ 10^{-7.05} = 8.90 \times 10^{-8} \end{array} \right]$$

- Q.10 The pH of aqueous solution of ammonia is 10. Find molarity of solution. $K_b(\text{NH}_4\text{OH}) = 10^{-5}$.
- Q.11 The solution of weak monoprotic acid which is 0.01 M, has pH = 3. Calculate K_a of weak acid.
- Q.12 Boric acid is a weak monobasic acid. It ionizes in water as



Calculate pH of 0.5 M boric acid.

MIXTURE OF TWO OR MORE ACIDS / BASES

- Q.13 The pH of the solution produced when an aqueous solution of strong acid pH 5 is mixed with equal volume of an aqueous solution of strong acid of pH 3 is :-
- Q.14 Calculate pH of following solutions : $[\log 0.3 = -0.522]$
- (a) 0.1 M H_2SO_4 (50 ml) + 0.4 M HCl 50 (ml)
- (b) 0.1 M HA + 0.1 M HB $[K_a(\text{HA}) = 5 \times 10^{-5} ; K_a(\text{HB}) = 4 \times 10^{-5}]$
- Q.15 Calculate pH of a solution containing 0.1M HA ($K_a = 10^{-5}$) & 0.1 M HCl.
- Q.16 Calculate $[\text{H}^+]$ and $[\text{CHCl}_2\text{COO}^-]$ in a solution that is 0.01 M in HCl and 0.01 M in CHCl_2COOH .
Take ($K_a = 3 \times 10^{-2}$) ($\sqrt{28} = 5.3$)
- Q.17 Calculate $[\text{H}^+]$, $[\text{CH}_3\text{COO}^-]$ and $[\text{C}_7\text{H}_5\text{O}_2^-]$ in a solution that is 0.02 M in acetic acid and 0.01M in benzoic acid. $K_a(\text{acetic}) = 1.8 \times 10^{-5}$, $K_a(\text{benzoic}) = 6.4 \times 10^{-5}$.

POLYPROTIC ACIDS & BASES

- Q.18 What are the concentration of H^+ , $\text{H}_2\text{C}_2\text{O}_4$, HC_2O_4^- and $\text{C}_2\text{O}_4^{2-}$ in a 0.1 M solution of oxalic acid ?

$$[K_1 = 10^{-2} \text{ M and } K_2 = 10^{-5} \text{ M}] \left[\sqrt{41} = 6.4 \right]$$

- Q.19 Calculate $[\text{H}^+]$, $[\text{H}_2\text{PO}_4^-]$, $[\text{HPO}_4^{2-}]$ and $[\text{PO}_4^{3-}]$ in a 0.01M solution of H_3PO_4 .

$$\text{Take } K_1 = 10^{-3}, K_2 = 10^{-8}, K_3 = 10^{-13}, \sqrt{41} = 6.4$$

- Q.20 Calculate pH of 0.2 M – B(OH)_2 solution.

$$(K_{b1} = 2 \times 10^{-5} ; K_{b2} = 4 \times 10^{-11}, \log 2 = 0.3)$$

HYDROLYSIS

- Q.21 What is the OH^- concentration of a 0.18 M solution of CH_3COONa . $[K_a(\text{CH}_3\text{COOH}) = 1.8 \times 10^{-5}]$
- Q.22 Calculate the pH of a 2.0 M solution of NH_4Cl . $[K_b(\text{NH}_3) = 2 \times 10^{-5}]$
- Q.23 0.25 M solution of pyridinium chloride $\text{C}_5\text{H}_6\text{N}^+\text{Cl}^-$ was found to have a pH of 2.699. What is K_b for pyridine, $\text{C}_5\text{H}_5\text{N}$? ($\log 2 = 0.3010$)
- Q.24 Calculate the extent of hydrolysis & the pH of 0.02 M $\text{CH}_3\text{COONH}_4$.
 $[K_b(\text{NH}_3) = 1.8 \times 10^{-5}, K_a(\text{CH}_3\text{COOH}) = 1.8 \times 10^{-5}]$
- Q.25 Calculate the percent hydrolysis in a 0.06 M solution of KCN. $[K_a(\text{HCN}) = 6 \times 10^{-10}]$

Q.26 Calculate the extent of hydrolysis of 0.005 M K_2CrO_4 . [$K_2 = 3.2 \times 10^{-7}$ for H_2CrO_4]

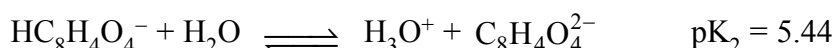
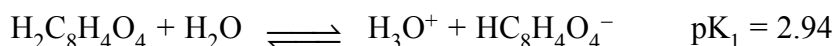
(It is essentially strong for first ionization).

Q.27 A 0.010 M solution of $PuO_2(NO_3)_2$ was found to have a pH of 4.0. What is the hydrolysis constant,

K_h , for PuO_2^{2+} , and what is K_b for PuO_2OH^+ ?

Q.28 What is the pH of 0.1M $NaHCO_3$? $K_1 = 5 \times 10^{-7}$, $K_2 = 5 \times 10^{-11}$ for carbonic acids.

Q.29 Calculate pH of 0.05M potassium hydrogen phthalate, $KHC_8H_4O_4$.



Q.30 The acid ionization (hydrolysis) constant of Zn^{2+} is 1.0×10^{-9}

(a) Calculate the pH of a 0.001 M solution of $ZnCl_2$

(b) What is the basic dissociation constant of $Zn(OH)^+$?

BUFFER SOLUTION

Q.31 Calculate the pH of solution containing 0.1M - HCN and 0.1M - NaCN. K_a of HCN = 10^{-9}

Q.32 Calculate the pH of solution containing 0.2 M - NH_4OH and 0.1 M - NH_4Cl . K_b of NH_4OH = 1.8×10^{-5} . ($\log 2 = 0.3$, $\log 1.8 = 0.26$)

Q.33 0.4 mole CH_3COONa is added in 500 ml 0.4 M - CH_3COOH solutions. What is the pH of final solution? K_a of $CH_3COOH = 1.8 \times 10^{-5}$. ($\log 2 = 0.3$, $\log 1.8 = 0.26$).

Q.34 A buffer of pH 9.26 is made by dissolving x moles of ammonium sulphate and 0.1 mole of ammonia into 100 mL solution. If pK_b of ammonia is 4.74, calculate value of x.

Q.35 Determine $[OH^-]$ of a 0.050 M solution of ammonia to which sufficient NH_4Cl has been added to make the total $[NH_4^+]$ equal to 0.100. [$K_{b(NH_3)} = 1.8 \times 10^{-5}$, $pK_b = 4.74$]

Q.36 Calculate the pH of a solution containing 0.2 M HCO_3^- and 0.1 M CO_3^{2-}

$[K_1(H_2CO_3) = 4 \times 10^{-7}$; $K_2(HCO_3^-) = 4 \times 10^{-11}]$

Q.37 Calculate the pH of a solution prepared by mixing 50.0 mL of 0.200 M $HC_2H_3O_2$ and 50.0 mL of 0.100 M NaOH. [$K_a(CH_3COOH) = 1.8 \times 10^{-5}$, $pK_a = 4.74$]

Q.38 50 mL of 0.1 M NaOH is added to 75 mL of 0.1 M NH_4Cl to make a basic buffer. If pK_a of NH_4^+ is 9.26, calculate pH.

Q.39 Calculate the pH of a solution which results from the mixing of 50.0 ml of 0.3 M HCl with 50.0 ml of 0.4 M NH_3 . [$K_b(NH_3) = 1.8 \times 10^{-5}$, $pK_b = 4.74$]

Q.40 In 100 ml buffer solution of 0.1M CH_3COOH & 0.1M CH_3COONa , how many millimoles of NaOH should be added to increase its pH by 0.3.

Given ($\log 2 = 0.3$)

ACID BASE REACTIONS & TITRATIONS

Q.41 Calculate OH^- concentration at the equivalent point when a solution of 0.2 M acetic acid is titrated with a solution of 0.2 M NaOH. K_a for the acid = 10^{-5} .

Q.42 Calculate the hydronium ion concentration and pH at the equivalence point in the reaction of 22.0 mL of 0.10M acetic acid, CH_3COOH , with 22.0 mL of 0.10 M NaOH. [$K_a = 2 \times 10^{-5}$]

Q.43 Calculate the hydronium ion concentration and the pH at the equivalence point in a titration of 50.0 mL of 0.40 M NH_3 with 0.40M HCl. [$K_b = 2 \times 10^{-5}$]

Q.44 CH_3COOH (50 ml, 0.1 M) is titrated against 0.1 M NaOH solution. Calculate the pH at the addition of 0 ml, 10 ml 20 ml, 25 ml, 40 ml, 50 ml of NaOH. K_a of CH_3COOH is 2×10^{-5} .
[$\log 2 = 0.3010$, $\log 3 = 0.4771$]

INDICATORS

Q.45 For the acid indicator thymol blue, pH is 3 when half the indicator is in unionised form. Find the % of indicator in unionised form in the solution with $[\text{H}^+] = 4 \times 10^{-3}$ M.

Q.46 Bromophenol blue is an acid indicator with a K_a value of 6×10^{-5} . What % of this indicator is in its basic form at a pH of 5 ?

Q.47 At what pH does an indicator change colour if the indicator is a weak acid with $K_{\text{ind}} = 4 \times 10^{-4}$. For which one(s) of the following neutralizations would the indicator be useful ? Explain.

(a) $\text{NaOH} + \text{CH}_3\text{COOH}$ (b) $\text{HCl} + \text{NH}_3$ (c) $\text{HCl} + \text{NaOH}$

Q.48 An acid indicator has a K_a of 3×10^{-5} . The acid form of the indicator is red & the basic form is blue. By how much must the pH change in order to change the indicator form 75% red to 75 % blue? [$\log 3 = 0.4771$]

SOLUBILITY & SOLUBILITY PRODUCT'S

Q.49 The values of K_{sp} for the slightly soluble salts MX and QX_2 are each equal to 4.0×10^{-18} . Which salt is more soluble? Explain your answer fully.

Q.50 The solubility of PbSO_4 in water is 0.0608 g/L. Calculate the solubility product constant of PbSO_4 .
Molar mass $\text{PbSO}_4 = 304$ g/mole

Q.51 How many mole CuI ($K_{\text{sp}} = 5 \times 10^{-12}$) will dissolve in 1.0 L of 0.10 M NaI solution ?

Q.52 A solution of saturated CaF_2 is found to contain 4×10^{-4} M fluoride ion. Calculate the K_{sp} of CaF_2 .
Neglect hydrolysis.

Q.53 The solubility of ML_2 (formula weight = 60 g/mol) in water is 2.4×10^{-5} g/100 mL solution. Calculate the solubility product constant for ML_2 .

- Q.54 Calculate the solubility of A_2X_3 in pure water, assuming that neither kind of ion reacts with water. For A_2X_3 , $K_{sp} = 1.08 \times 10^{-23}$
- Q.55 Determine the solubility of AgCl in 0.1 M $BaCl_2$. [K_{sp} for AgCl = 1×10^{-10}]
- Q.56 Calculate solubility of $Ca_3(PO_4)_2$ ($K_{sp} = 10^{-15}$) in presence of 0.1 M $CaCl_2$ solution.

SIMULTANEOUS SOLUBILITY

- Q.57 Calculate the Simultaneous solubility of AgSCN and AgBr. K_{sp} (AgSCN) = 3.2×10^{-12} , K_{sp} (AgBr) = 8×10^{-13} .
- Q.58 Calculate F^- in a solution saturated with respect of both MgF_2 and SrF_2 . $K_{sp}(MgF_2) = 9.5 \times 10^{-9}$, $K_{sp}(SrF_2) = 4 \times 10^{-9}$.

COMPLEX FORMATION

- Q.59 Calculate the solubility of AgCl in 0.2 M - NH_3 solution.
Given : K_{sp} of AgCl = 2×10^{-10} , K_f of $Ag(NH_3)_2^+ = 8 \times 10^6$.
- Q.60 Calculate the solubility of AgCN in 0.4 M - KCN solution
(i) neglecting complex formation
(ii) considering complex formation. Given : K_{sp} of AgCN = 8×10^{-10} , K_d of $Ag(CN)_2^- = 4 \times 10^{-8}$.

SOLUBILITY, CONSIDERING HYDROLYSIS

- Q.61 Calculating the solubility of MX in water. Also calculate pH of solution.
Given : K_{sp} of MX = 4×10^{-8} ; K_a of HX = 2×10^{-6} and MOH is strong base.
- Q.62 Calculate the solubility of AgCN in a buffer solution at pH = 3.0. .
Given : K_{sp} of AgCN = 8×10^{-10} , K_a of HCN = 5×10^{-10} .

PRECIPITATION

- Q.63 A solution has a Mg^{2+} concentration of 0.0010 mol/L. Will $Mg(OH)_2$ precipitate if the OH^- concentration of the solution is [$K_{sp} = 1.2 \times 10^{-11}$]
(a) 10^{-5} mol/L (b) 10^{-3} mol/L ?
- Q.64 200 ml of 2×10^{-4} M - $AgNO_3$ solution is mixed with 400 ml of 1.2×10^{-6} M - NaCl solution. Predict whether precipitation of AgCl will occur or not. K_{sp} of AgCl = 2×10^{-10} .
- Q.65 Calculate the minimum mass of Na_2SO_4 needed to just start precipitation of $BaSO_4$ from 500 ml of 2×10^{-5} M - $BaCl_2$ solution. K_{sp} of $BaSO_4 = 8 \times 10^{-8}$.

EXERCISE # S-II

- Q.1 What are the concentrations of H^+ , HSO_4^- , SO_4^{2-} and H_2SO_4 in a 0.20 M solution of sulphuric acid?

Given : $\text{H}_2\text{SO}_4 \longrightarrow \text{H}^+ + \text{HSO}_4^-$; strong



- Q.2 Calculate the pH of a 0.1M solution of $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$; ethylenediamine (en). Determine the H_2^{2+} concentration in the solution. K_{b1} and K_{b2} values of ethylenediamine are 9×10^{-5} and 7.1×10^{-8} respectively.
- Q.3 Nicotine, $\text{C}_{10}\text{H}_{14}\text{N}_2$, has two basic nitrogen atoms and both can react with water to give a basic solution
- $$\text{Nic (aq)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{NicH}^+ \text{ (aq)} + \text{OH}^- \text{ (aq)}$$
- $$\text{NicH}^+ \text{ (aq)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{NicH}_2^{2+} \text{ (aq)} + \text{OH}^- \text{ (aq)}$$
- K_{b1} is 8×10^{-7} and K_{b2} is 10^{-10} . Calculate the approximate pH of a 0.20 M solution.
- Q.4 Determine the $[\text{S}^{2-}]$ in a saturated (0.1M) H_2S solution to which enough HCl has been added to produce a $[\text{H}^+]$ of 2×10^{-4} . $K_1 = 10^{-7}$, $K_2 = 10^{-14}$.
- Q.5 An aqueous solution contains 0.01 M RNH_2 ($K_b = 2 \times 10^{-6}$) & 10^{-4} M NaOH. The concentration of OH^- is nearly :
- Q.6 Calculate the pH of 1.0×10^{-3} M sodium phenoxide, NaOC_6H_5 . K_a for HOC_6H_5 is 0.6×10^{-10} .
- Q.7 Calculate the OH^- concentration and the H_3PO_4 concentration of a solution prepared by dissolving 0.1 mol of Na_3PO_4 in sufficient water to make 1L of solution. $K_1 = 7.1 \times 10^{-3}$, $K_2 = 6.3 \times 10^{-8}$, $K_3 = 4.5 \times 10^{-13}$.
- Q.8 Calculate the pH of 0.1 M solution of (i) NaHCO_3 , (ii) Na_2HPO_4 and (iii) NaH_2PO_4 . Given that:



($\log 4.2 = 0.62$, $\log 4.8 = 6.8$, $\log 6.2 = 0.80$, $\log 7.5 = 0.88$)

- Q.9 An ammonia-ammonium chloride buffer has a pH value of 9 with $[\text{NH}_3] = 0.25$. What will be the new pH if 500 ml 0.1 M KOH is added to 200 ml buffer solution ($K_b = 2 \times 10^{-5}$) [$\log 2 = 0.3$]
- Q.10 A weak base (50.0mL) was titrated with 0.1 M HCl. The pH of the solution after the addition of 10.0 mL and 25.0 mL were found to be 9.84 and 9.24, respectively. Calculate K_b of the base and pH at the equivalence point. [$\log 2 = 0.3$]
- Q.11 A weak acid (50.0mL) was titrated with 0.1 M NaOH. The pH values when 10.0 mL and 25.0 mL of base have been added are found to be 4.16 and 4.76, respectively. Calculate K_a of the acid and pH at the equivalence point. [$\log 2 = 0.3$]

- Q.12 10 ml of 0.1M weak acid $\text{HA}(K_a = 10^{-5})$ is mixed with 10 ml 0.2M HCl and 10 ml 0.1M NaOH . Find the value of $[\text{A}^-]$ in the resulting solution.
- Q.13 150 ml of 0.5 M HCN ($K_a = 3.75 \times 10^{-9}$) was reacted with 1.5 M KOH for complete neutralisation. What will be molarity of HCN at equilibrium.
- Q.14 The indicator phenol red is half in the ionic form when pH is 7.2. If the ratio of the undissociated form to the ionic form is 1 : 5, find the pH of the solution. With the same pH for solution, if indicator is altered such that the ratio of undissociated form to dissociated form becomes 1 : 4, find the pH when 50 % of the new indicator is in ionic form. $[\log 2 = 0.3]$
- Q.15 How much AgBr could dissolve in 1.0 L of 0.40 M NH_3 ? Assume that $\text{Ag}(\text{NH}_3)_2^+$ is the only complex formed. $[K_f(\text{Ag}(\text{NH}_3)_2^+) = 1 \times 10^8; K_{sp}(\text{AgBr}) = 5 \times 10^{-13}]$
 $[\sqrt{50} \approx 7]$
- Q.16 Calculate solubility of PbI_2 ($K_{sp} = 1.4 \times 10^{-8}$) in water at 25° , which is 90% dissociated.
- $$\left(\frac{1.4}{(0.81)(3.6)} \right)^{1/3} = 0.78$$
- Q.17 A recent investigation of the complexation of SCN^- with Fe^{3+} led to 130, 16, and 1.0 for K_1 , K_2 , and K_3 , respectively. What is the overall formation constant of $\text{Fe}(\text{SCN})_3$ from its component ions, and what is the dissociation constant of $\text{Fe}(\text{SCN})_3$ into its simplest ions on the basis of these data ?

EXERCISE # O-I

Single correct

- Q.1 The conjugate acid of NH_2^- is
 (A) NH_3 (B) NH_2OH (C) NH_4^+ (D) N_2H_4
- Q.2 Which of the following is not a Bronsted acid:-
 (A) CH_3NH_4^+ (B) CH_3COO^- (C) H_2O (D) HSO_4^-
- Q.3 In the reaction
 $\text{HNO}_3 + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{NO}_3^-$, the conjugate base of HNO_3 is :-
 (A) H_2O (B) H_3O^+ (C) NO_3^- (D) H_3O^+ and NO_3^-
- Q.4 Out of the following, amphiprotic species in aqueous medium are
 I: HPO_3^{2-} II OH^- III H_2PO_4^- IV HCO_3^-
 (A) I, III, IV (B) I and III (C) III and IV (D) All
- Q.5 When ammonia is added to water, it decreases the concentration of which of the following ion
 (A) OH^- (B) H_3O^+ (C) NH_4^+ (D) NH_4^+ & OH^-
- Q.6 Which of the following pair is Lewis acid & Lewis base & Product of these is also Lewis base
 (A) BF_3 , NH_3 (B) SiCl_4 , 2Cl^- (C) CH_3^+ , $^-\text{OC}_2\text{H}_5$ (D) All of these
- Q.7 Ionic product of water will increase, if :-
 (A) Pressure is decreased (B) H^+ is added
 (C) OH^- is increased (D) Temperature is increased
- Q.8 At 60°C , pure water has $[\text{H}_3\text{O}^+] = 10^{-6.7} \text{ mol/lit.}$ what is the value of K_w at 60°C :-
 (A) 10^{-6} (B) 10^{-12} (C) 10^{-67} (D) $10^{-13.4}$
- Q.9 Liquid NH_3 ionises to a slight extent. At a certain temperature its self ionization constant $K_{\text{SIC}(\text{NH}_3)} = 10^{-30}$. The number of NH_4^+ ions present per 100 cm^3 of pure liquid are
 (A) 10^{-15} (B) 6.022×10^8 (C) 6.022×10^7 (D) None
- Q.10 The pH of solution is increased from 3 to 6. Its H^+ ion conc. will be :-
 (A) Reduced to half (B) Doubled
 (C) Reduced by 1000 times (D) Increased by 1000 times
- Q.11. pOH of $[1/200] \text{ mol/m}^3 \text{ H}_2\text{SO}_4 \text{ (aq.)}$ solution at 25°C is-
 (A) 2 (B) 5 (C) 9 (D) 12
- Q.12 Degree of dissociation of 0.1 N CH_3COOH is :- (Dissociation constant = 1×10^{-5})
 (A) 10^{-5} (B) 10^{-4} (C) 10^{-3} (D) 10^{-2}
- Q.13 The pH of a 0.02 M ammonia solution which is 5% ionised will be :-
 (A) 2 (B) 11 (C) 5 (D) 7
- Q.14 The pH of an aqueous solution of 1.0 M solution of a weak monoprotic acid which is 1% ionised is
 (A) 1 (B) 2 (C) 3 (D) 11
- Q.15 The concentration of $[\text{H}^+]$ and concentration of $[\text{OH}]^-$ of a 0.1 M aqueous solution of 2% ionised weak acid is [ionic product of water = 1×10^{-14}]
 (A) $0.02 \times 10^{-3} \text{ M}$ and $5 \times 10^{-11} \text{ M}$ (B) $1 \times 10^{-3} \text{ M}$ and $3 \times 10^{-11} \text{ M}$
 (C) $2 \times 10^{-3} \text{ M}$ and $5 \times 10^{-12} \text{ M}$ (D) $3 \times 10^{-2} \text{ M}$ and $4 \times 10^{-13} \text{ M}$

- Q.16 What is the quantity of NaOH present in 250 cc of the solution, so that it gives a pH = 13 :-
 (A) 10^{-13} g (B) 10^{-1} g (C) 1.0 g (D) 4.0 g
- Q.17 An aqueous solution of HCl is 10^{-9} M HCl. The pH of the solution should be:-
 (A) 9 (B) Between 6 and 7 (C) 7 (D) Unpredictable
- Q.18. The moles of H^+ from H_2O in a 1l, $\sqrt{5} \times 10^{-7}$ M HCl solution at $25^\circ C$, is
 ($\sqrt{5} = 2.23$)
 (A) 10^{-7} (B) 6.85×10^{-8} (C) 3.85×10^{-8} (D) 10^{-8}
- Q.19 Which one of the following has highest pH:-
 (A) Distilled water (B) 1 M NH_3
 (C) 1 M NaOH (D) Water saturated with chlorine
- Q.20 8 gm NaOH and 4.9 gm H_2SO_4 are present in one litre of the solution. What is its pH
 (A) 1 (B) 13 (C) 12 (D) 2
- Q.21 10 ml of $\frac{M}{200} H_2SO_4$ is mixed with 40 ml of $\frac{M}{200} H_2SO_4$. The pH of the resulting solution is
 (A) 1 (B) 2 (C) 2.3 (D) none of these
- Q.22 Which of the following solution will have pH close to 1.0?
 (A) 100 ml of M/100 HCl + 100 ml of M/10 NaOH
 (B) 55 ml of M/10 HCl + 45 ml of M/10 NaOH
 (C) 10 ml of M/10 HCl + 90 ml of M/10 NaOH
 (D) 75 ml of M/5 HCl + 25 ml of M/5 NaOH
- Q.23 A solution with pH 2.0 is more acidic than the one with pH 6.0 by a factor of:
 (A) 3 (B) 4 (C) 3000 (D) 10000
- Q.24 The first and second dissociation constants of an acid H_2A are 1.0×10^{-5} and 5.0×10^{-10} respectively. The overall dissociation constant of the acid will be :
 (A) 5.0×10^{-5} (B) 5.0×10^{15} (C) 5.0×10^{-15} (D) 0.2×10^5
- Q.25 If pK_b for fluoride ion at $25^\circ C$ is 10.4, the ionisation constant of hydrofluoric acid in water at this temperature is :
 (A) 4×10^{-11} (B) 3×10^{-3} (C) 2.5×10^{-4} (D) 2×10^{-2}
- Q.26 pH of an aqueous solution of NaCl at $85^\circ C$ should be
 (A) 7 (B) > 7 (C) < 7 (D) 0
- Q.27 1 cc of 0.1 N HCl is added to 99 cc solution of NaCl. The pH of the resulting solution will be
 (A) 7 (B) 3 (C) 4 (D) 1
- Q.28 The degree of hydrolysis of a salt of weak acid and weak base in its 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be
 (A) 100% (B) 50% (C) 25% (D) none of these
- Q.29 What is the percentage hydrolysis of NaCN in N/80 solution when the dissociation constant for HCN is 2×10^{-9} and $K_w = 1.0 \times 10^{-14}$
 (A) 2 (B) 5.26 (C) 8.2 (D) 9.6

- Q.30 The compound whose 0.1 M solution is basic is
 (A) Ammonium acetate (B) Ammonium chloride
 (C) Ammonium sulphate (D) Sodium acetate
- Q.31 If equilibrium constant of
 $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$
 is 1.8×10^{-5} , equilibrium constant for
 $\text{CH}_3\text{COOH} + \text{OH}^- \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_2\text{O}$ is
 (A) 1.8×10^{-9} (B) 1.8×10^9 (C) 5.55×10^{-9} (D) 5.55×10^{10}
- Q.32 The pK_a of a weak acid, HA, is 4.80. The pK_b of a weak base, BOH, is 4.78. The pH of an aqueous solution of the corresponding salt, BA, will be :
 (A) 8.58 (B) 4.79 (C) 7.01 (D) 9.22
- Q.33 The highest pH value is of :-
 (A) 0.1 M NaCl (B) 0.1 M NH_4Cl
 (C) 0.1 M CH_3COONa (D) 0.1 M $\text{CH}_3\text{COONH}_4$
- Q.34 pH of K_2S solution is:-
 (A) 7 (B) Less than 7 (C) More than 7 (D) 0
- Q.35 Degree of Hydrolysis of $\frac{N}{100}$ solution of KCN is (Given $K_a = 1.6 \times 10^{-9}$)
 (A) 2.5×10^{-3} (B) 2.5×10^{-2} (C) 2.5×10^{-4} (D) 2.5×10^{-5}
- Q.36 A solution of FeCl_3 in water acts as acidic due to :-
 (A) Acidic impurities (B) Ionisation (C) Hydrolysis of Fe^{3+} (D) Dissociation
- Q.37 If 40 ml of 0.2 M KOH is added to 160 ml of 0.1 M HCOOH [$K_a = 2 \times 10^{-4}$], the pOH of the resulting solution is
 (A) 3.4 (B) 3.7 (C) 7 (D) 10.3
- Q.38 1 M NaCl and 1M HCl are present in an aqueous solution. The solution is
 (A) not a buffer solution and with $\text{pH} < 7$
 (B) not a buffer solution with $\text{pH} > 7$
 (C) a buffer solution with $\text{pH} < 7$
- Q.39 The pK_a of a weak acid (HA) is 4.5. The pOH of an aqueous buffered solution of HA in which 50% of the acid is ionized is :
 (A) 4.5 (B) 2.5 (C) 9.5 (D) 7.0
- Q.40 To a 50 ml. of 0.05M formic acid, how much volume of 0.10M sodium formate must be added to get a buffer solution of $\text{pH} = 4.0$?
 (pK_a of the acid is 3.7) ($\log 2 = 0.3$)
 (A) 40 ml. (B) 4 ml. (C) 50 ml. (D) 100 ml.
- Q.41 Which can act as buffer :-
 (A) $\text{NH}_4\text{OH} + \text{NaOH}$
 (B) $\text{HCOOH} + \text{HCl}$
 (C) 40 ml. of 0.1 M NaCN + 20 ml. of 0.1 M HCl
 (D) All of them

- Q.42 If equal volume of 0.05 M ammonium hydroxide solution is dissolved in 0.001 M ammonium chloride solution. What will be the OH^- ion concentration of this solution :
 $K_b(\text{NH}_4\text{OH}) = 1.8 \times 10^{-5}$
 (A) 3.0×10^{-3} (B) 4.6×10^{-4} (C) 9.0×10^{-3} (D) 9.0×10^{-4}
- Q.43 Calculate the pH of a buffer prepared by mixing 600 cc of 0.6 M NH_3 and 400 cc of 0.5 M NH_4Cl .
 K_b for $\text{NH}_3 = 1.8 \times 10^{-5}$, ($\log 1.8 = 0.26$)
 (A) 11.3 (B) 9.0 (C) 9.52 (D) 5
- Q.44 pK_b for NH_4OH at certain temperature is 4.74. The pH of basic buffer containing equimolar concentration of NH_4OH and NH_4Cl will be:-
 (A) 7.74 (B) 4.74 (C) 2.37 (D) 9.26
- Q.45 On addition of NaOH to CH_3COOH solution, 60% of the acid is neutralised. If pK_a of CH_3COOH is 4.7 then the pH of the resulting solution is :-
 (A) More than 4.7 but less than 5.0 (B) Less than 4.7 but more than 4.0
 (C) More than 5.0 (D) Remains unchanged
- Q.46 Henderson equation $\text{pH} - \text{pK}_a = 5$ will be applicable to an acidic buffer when :-
 (A) $[\text{Acid}] = [\text{Conjugate base}]$ (B) $[\text{Acid}] \times 10^5 = [\text{Conjugate base}]$
 (C) $[\text{Acid}] = [\text{Conjugate base}] \times 10^5$ (D) $[\text{acid}] = 2 [\text{conjugate base}]$
- Q.47 What amount of sodium propanoate should be added to one litre of an aqueous solution containing 0.02 mole of propanoic acid ($K_a = 3 \times 10^{-5}$ at 25°C) to obtain a buffer solution of pH 4.7
 (A) 4.52×10^{-2} mol (B) 3.52×10^{-2} mol
 (C) 2.52×10^{-2} mol (D) 3×10^{-2} mol
- Q.48 In a buffer solution the ratio of concentration of NH_4Cl and NH_4OH is 1 : 1. When it changes in 2 : 1, what will be the value of pH of buffer ?
 (A) Increase (B) Decrease (C) No effect (D) None
- Q.49 The buffer solution play an important role in :-
 (A) Increasing the pH value (B) Decreasing the pH value
 (C) Keeping the pH constant (D) Solution will be neutral
- Q.50 The total number of different kind of acidic buffers obtained during the titration of H_3PO_4 with NaOH are :
 (A) 3 (B) 1 (C) 2 (D) 0
- Q.51 Which of the following solutions does not act as buffer :
 (A) $\text{H}_3\text{PO}_4 + \text{NaH}_2\text{PO}_4$ (B) $\text{NaHCO}_3 + \text{H}_2\text{CO}_3$
 (C) $\text{NH}_4\text{Cl} + \text{HCl}$ (D) $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$
- Q.52 Half of the formic acid solution is neutralised on addition of a KOH solution to it. If $K_a(\text{HCOOH}) = 2 \times 10^{-4}$ then pH of the solution is : - ($\log 2 = 0.3010$)
 (A) 3.6990 (B) 10.3010 (C) 3.85 (D) 4.3010
- Q.53 When 0.02 moles of NaOH are added to a litre of buffer solution, its pH changes from 5.75 to 5.80. What is its buffer capacity :-
 (A) 0.4 (B) 0.05 (C) - 0.05 (D) 2.5

- Q.54 Calculate pH when 100 ml of 0.2M NaOH is reacted with 100 ml of 0.2 M CH_3COOH ($K_a = 10^{-5}$)
 (A) 9 (B) 7 (C) 5 (D) 2
- Q.55 At 90°C , pure water has $[\text{H}^+] = 10^{-6}\text{ M}$, if 100 ml of 0.2 M HNO_3 is added to 20 ml of 1 M NaOH at 90°C then pH of the resulting solution will be
 (A) 5 (B) 6 (C) 7 (D) None of these
- Q.56 When 20 ml of $\frac{\text{M}}{20}$ NaOH are added to 10 ml of $\frac{\text{M}}{10}$ HCl, the resulting solution will:-
 (A) Turn blue litmus red
 (B) Turn phenolphthalein solution pink colour
 (C) Turn methyl orange red
 (D) Will have no effect on either red or blue litmus
- Q.57 The rapid change of pH near the stoichiometric point of an acid-base titration is the basis of indicator detection. pH of the solution is related to ratio of the concentrations of the conjugate acid (HIn) and base (In^-) forms of the indicator by the expression :-
 (A) $\log \frac{[\text{HIn}]}{[\text{In}^-]} = \text{pK}_{\text{In}} - \text{pH}$ (B) $\log \frac{[\text{HIn}]}{[\text{In}^-]} = \text{pH} - \text{pK}_{\text{In}}$
 (C) $\log \frac{[\text{In}^-]}{[\text{HIn}]} = \text{pH} + \text{pK}_{\text{In}}$ (D) $\log \frac{[\text{In}^-]}{[\text{HIn}]} = \text{pK}_{\text{In}} - \text{pH}$
- Q.58 Calculate the pH range in which an acid indicator with $K_{\text{acid}} (\text{indicator}) = 1.0 \times 10^{-5}$ changes colour when the concentration of the indicator is $1 \times 10^{-3}\text{ M}$.
 (A) 5 ± 1 (B) 11 ± 1 (C) 3 ± 1 (D) 8 ± 1
- Q.59 In what pH range will a $1 \times 10^{-4}\text{ M}$ solution of an indicator will $K_b (\text{indicator}) = 1 \times 10^{-11}$ change colour?
 (A) 7.0 ± 1 (B) 3.0 ± 1 (C) 5.5 ± 1 (D) 11.0 ± 1
- Q.60 Indicator which is used in the titration of CH_3COOH & NaOH :-
 (A) Methyl orange (B) Methyl red (C) Phenolphthalein (D) Litmus
- Q.61 Phenolphthalein is a :-
 (A) Strong acid (B) Strong base (C) Weak base (D) Weak acid
- Q.62 pH-range of Methyl red indicator is :-
 (A) 4.2–6.2 (B) 6.8–10.8 (C) 8 – 9.6 (D) 6.8 – 8.2
- Q.63 In the volumetric estimation of HCl, if we make use of phenolphthalein as an indicator, which base is unsuitable for the titration :-
 (A) NaOH (B) RbOH (C) KOH (D) NH_4OH
- Q.64 Phenolphthalein does not act as an indicator for the titration between :-
 (A) KOH and H_2SO_4 (B) NaOH and CH_3COOH
 (C) Oxalic acid and KMnO_4 (D) $\text{Ba}(\text{OH})_2$ and HCl
- Q.65 For weak acid and strong base titration, the indicator used is :-
 (A) Potassium di-chromate (B) Methyl orange
 (C) Litmus (D) Phenolphthalein

- Q.66 The solubility of A_2X_3 is $y \text{ mol dm}^{-3}$. Its solubility product is
 (A) $6y^2$ (B) $64y^4$ (C) $36y^5$ (D) $108y^5$
- Q.67 If K_{sp} for $HgSO_4$ is 6.4×10^{-5} , then solubility of this substance in mole per m^3 is
 (A) 8×10^{-3} (B) 6.4×10^{-5} (C) 8×10^{-6} (D) 8
- Q.68 If the solubility of $AgCl$ (formula mass=143) in water at $25^\circ C$ is $1.43 \times 10^{-4} \text{ gm/100 ml}$ of solution then the value of K_{sp} will be :-
 (A) 1×10^{-5} (B) 2×10^{-5} (C) 1×10^{-10} (D) 2×10^{-10}
- Q.69 One litre of saturated solution of $CaCO_3$ is evaporated to dryness, 7.0 g of residue is left. The solubility product for $CaCO_3$ is:-
 (A) 4.9×10^{-3} (B) 4.9×10^{-5} (C) 4.9×10^{-9} (D) 4.9×10^{-7}
- Q.70 A_3B_2 is a sparingly soluble salt of molar mass $M \text{ (g mol}^{-1}\text{)}$ and solubility $x \text{ g lit}^{-1}$. The ratio of the molar concentration of B^{3-} to the solubility product of the salt is
 (A) $108 \frac{x^5}{M^5}$ (B) $\frac{1}{108} \frac{M^4}{x^4}$ (C*) $\frac{1}{54} \frac{M^4}{x^4}$ (D) None
- Q.71. Solubility of Ag_2CrO_4 ($K_{sp} = 4 \times 10^{-13}$) in $0.1 \text{ M } K_2CrO_4$ solution will be :-
 (A) 10^{-3} M (B) 10^{-6} M (C) $4 \times 10^{-6} \text{ M}$ (D) $5 \times 10^{-7} \text{ M}$
- Q.72. How many times solubility of CaF_2 is decreased in $4 \times 10^{-3} \text{ M KF (aq.)}$ solution as compare to pure water at $25^\circ C$. Given $K_{sp} (CaF_2) = 3.2 \times 10^{-11}$
 (A) 50 (B*) 100 (C) 500 (D) 1000
- Q.73 At $30^\circ C$, In which of the one litre solution, the solubility of Ag_2CO_3 (solubility product = 8×10^{-12}) will be maximum :-
 (A) $0.05 \text{ M } Na_2CO_3$ (B) Pure water (C) $0.05 \text{ M } AgNO_3$ (D) $0.05 \text{ M } NH_3$
- Q.74 What will happen if the pH of the solution of $0.001 \text{ M } Mg(NO_3)_2$ solution is adjusted to $pH = 9$ (K_{sp} of $Mg(OH)_2 = 8.9 \times 10^{-12}$)
 (A) ppt will take place (B) ppt will not take place
 (C) Solution will be saturated (D) None of these
- Q.75 Na_3PO_4 which should be added in 10 L of $1.0 \times 10^{-5} \text{ M } BaCl_2$ solution without any precipitation of $Ba_3(PO_4)_2$ is [K_{sp} of $Ba_3(PO_4)_2$] = 4×10^{-23}
 (A) $2 \times 10^{-4} \text{ gm}$ (B) 0.328 gm (C) 0.164 gm (D) 0.82 gm

EXERCISE # O-II

Single correct :

- Q.1 The pH of the solution produced when an aqueous solution of strong acid pH 5 is mixed with equal volume of an aqueous solution of strong acid of pH 3 is :-
 (A) 3.3 (B) 3.5 (C) 4.5 (D) 4.0
- Q.2 How many moles of HCl must be removed from 1 litre of aqueous HCl solution to change its pH from 2 to 3 :-
 (A) 1 (B) 0.02 (C) 0.009 (D) 0.01
- Q.3 Which of the following is most soluble in water?
 (A) MnS ($K_{sp} = 8 \times 10^{-37}$) (B) ZnS ($K_{sp} = 7 \times 10^{-16}$)
 (C) Bi_2S_3 ($K_{sp} = 1 \times 10^{-72}$) (D) $\text{Ag}_3(\text{PO}_4)$ ($K_{sp} = 1.8 \times 10^{-18}$)
- Q.4 Solubility of AgBr will be minimum in :-
 (A) Pure water (B) 0.1 M CaBr_2 (C) 0.1 M NaBr (D) 0.1 M AgNO_3
- Q.5 pH of solution at first $1/4^{\text{th}}$ equivalence point of Na_2CO_3 when titrated with HCl will be (for H_2CO_3 $K_{a1} = 10^{-7}$; $K_{a2} = 10^{-11}$)
 (A) $7 + \log 3$ (B) $7 - \log 3$ (C) $11 + \log 3$ (D) $11 - \log 3$
- Q.6 An acid-base indicator has a $K_a = 1.0 \times 10^{-5}$. The acid form of the indicator is red and the basic form is blue. Calculate the pH change required to change the colour of the indicator from 80% red to 80% blue.
 (A) 1.20 (B) 0.80 (C) 0.20 (D) 1.40

Assertion / Reason :

- Q.7 **Statement-1** pH of 10^{-7} M NaOH solution is exist between 7 to 7.3 at 25°C .
Statement-2 Due to common ion effect ionization of water is reduced.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- Q.8 **Statement-1** In general phenolphthalein is used as an indicator for the titration of weak acid (HA) against strong base (NaOH)
Statement-2 At equivalent point solution is basic.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- Q.9 **Statement-1** : Moles of Sr^{2+} furnished by sparingly soluble substance $\text{Sr}(\text{OH})_2$ decreases due to dilution.
Statement-2 : Solubility product constant of $\text{Sr}(\text{OH})_2$ is not affected by dilution.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

- Q.10 **Statement-1** : On dilution of a concentrated solution of CH_3COOH , the concentration of $[\text{H}^+]$ decreases.
Statement-2 : Increase in volume is more than the increase in degree of ionisation.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

Multiple correct :

- Q.11 Which of the following is correct for 0.1 M BOH solution ($K_b = 10^{-5}$)
 (A) pH of solution is 11
 (B) OH^- concentration is 10^{-3} mol/L
 (C) its salt with HCl (i.e. BCl) form the acidic solution in water
 (D) Phenolphthalein indicator can be used during the titration of BOH with HCl
- Q.12 For weak monobasic acid, HA, the dissociation constant is 2×10^{-6} , at 25°C . Which of the following is/are correct regarding this acid? [$\log 2 = 0.3$]
 (A) $\text{A}^- + \text{H}_2\text{O} \rightleftharpoons \text{HA} + \text{OH}^-$; $K_{\text{eq}} = 5 \times 10^{-9}$
 (B) The equilibrium constant for the reaction of HA with aq. NaOH is 2×10^8
 (C) The pH of 0.1 M, HA solution is 3.35
 (D) solution of A^- is basic
- Q.13 Select correct statement for 50ml 0.1M $\text{H}_2\text{A}(\text{aq.})$ solution ; $K_{a_1} = 10^{-5}$; $K_{a_2} = 10^{-8}$
 (A) $[\text{H}^+] = 2[\text{A}^{2-}]$
 (B) pH of 0.1M H_2A solution is 3
 (C) In above H_2A solution when 5 millimoles of NaHA are added then pH increases by 2 units
 (D) 50 ml of 0.1M NaOH required to neutralised completely 50 ml of 0.1M H_2A solution
- Q.14 A solution containing 0.01M each of Pb^{2+} , Ag^+ , Zn^{2+} & Cr^{3+} ion. If solid Na_2S is added slowly to the solution then correct statement is based on given data -

Precipitate	PbS	Ag_2S	ZnS	Cr_2S_3
K_{sp}	10^{-11}	10^{-12}	10^{-8}	10^{-10}

- (A) Pb^{2+} will start precipitating first
 (B) Cr^{3+} will start precipitating last
 (C) Zn^{2+} will start precipitating before Ag^+
 (D) When Zn^{2+} just starts precipitating then Pb^{2+} ion gets 99.9% precipitated
- Q.15 If K_1 & K_2 be first and second ionisation constant of H_3PO_4 and $K_1 \gg K_2$ which is/are incorrect.
 (A) $[\text{H}^+] = [\text{H}_2\text{PO}_4^-]$ (B) $[\text{H}^+] = \sqrt{K_1[\text{H}_3\text{PO}_4]}$
 (C) $K_2 = [\text{HPO}_4^{2-}]$ (D) $[\text{H}^+] = 3[\text{PO}_4^{3-}]$
- Q.16 10 ml. of a solution contains 0.1 M NH_4Cl + 0.01M NH_4OH . Which addition would not change the pH of solution :-
 (A) Adding 1 ml. water (B) Adding 5 ml. of 0.1 M NH_4Cl
 (C) Adding 5 ml. of 0.1 M NH_4OH (D) Adding 10 ml. of 0.1 M NH_4Cl

- Q.17 When equal volumes of the following solutions are mixed, precipitation of AgCl ($K_{sp} = 1.8 \times 10^{-10}$) will occur only with:
- (A) $10^{-4} \text{ M (Ag}^+) \text{ and } 10^{-4} \text{ M (Cl}^-)$ (B) $10^{-5} \text{ M (Ag}^+) \text{ and } 10^{-5} \text{ M (Cl}^-)$
 (C) $10^{-6} \text{ M (Ag}^+) \text{ and } 10^{-6} \text{ M (Cl}^-)$ (D) $10^{-10} \text{ M (Ag}^+) \text{ and } 10^{-10} \text{ M (Cl}^-)$

Paragraph for Q.18 to Q.20

8 gm weak acid HX (molecular mass = 80) is dissolved in 100 ml water. ($K_a = 10^{-4}$)

- Q.18 Find pH of solution-
- (A) 3.3 (B) 2 (C) 2.3 (D) 3
- Q.19 If it is titrated with 0.25 M NaOH find pH at equivalence point ($\log 5 = 0.7$)
- (A) 9.15 (B) 8.65 (C) 4.65 (D) 4.85
- Q.20 Find $[\text{H}^+]$ if 10^{-3} mol HCl is added to 100 ml original solution ($\sqrt{41} = 6.4$; $\sqrt{5} = 2.24$)
- (A) 0.62×10^{-2} (B) 1.62×10^{-2} (C) 2.7×10^{-2} (D) 0.27×10^{-2}

MATCH THE COLUMN :

- Q.21 Match the effect of addition of 1 M NaOH to 100 mL 1 M CH_3COOH (in Column I) with pH (in Column II):

Column-I

- (A) 25 mL of NaOH
 (B) 50 mL of NaOH
 (C) 75 mL of NaOH
 (D) 100 mL of NaOH

Column-II

- (P) pK_a
 (Q) $\text{pK}_a + \log 3$
 (R) $\text{pK}_a - \log 3$
 (S) $\frac{1}{2} [\text{pK}_w + \text{pK}_a - \log 2]$

- Q.22 **Column-I**
PH

- (A) 3
 (B) 5
 (C) 8
 (D) 9

Column-II
Solution

- (P) When equal volumes of 0.2M NH_4OH ($K_b = 10^{-5}$) & 0.2M HCl are mixed
 (Q) When equal volumes of 0.2M CH_3COONa & 0.2M HCl are mixed ($K_{a(\text{CH}_3\text{COOH})} = 10^{-5}$)
 (R) 0.1M Na_2HPO_4
 (for H_3PO_4 ; $K_{a1} = 10^{-4}$; $K_{a2} = 10^{-6}$; $K_{a3} = 10^{-10}$)
 (S) At 1st half equivalence point of H_2CO_3 when titrated against 0.1M NaOH , $K_{a1} = 10^{-5}$, $K_{a2} = 10^{-9}$
 (T) Mg(OH)_2 ; $K_{sp} = 5 \times 10^{-16}$

TABLE TYPE QUESTION :

Column-I (solution)	Column-II (pH of solution)	Column-III (Introduction about solution)
(A) $\text{CH}_3\text{COOH}(0.2\text{M}, 1\text{L}) + \text{NaOH}(0.1\text{M}, 1\text{L})$	(P) 1.3	(1) pH is determined by strong acid
(B) $\text{CH}_3\text{COOH}(0.1\text{M}, 1\text{L}) + \text{HCl}(0.1\text{M}, 1\text{L})$	(Q) 7	(2) Buffer solution at its maximum buffer capacity
(C) $\text{CH}_3\text{COOH}(0.1\text{M}, 1\text{L}) + \text{NH}_4\text{OH}(0.1\text{M}, 1\text{L})$	(R) 9	(3) pH is determined by salt hydrolysis.
(D) $\text{NH}_4\text{Cl}(200\text{ ml}, 0.1\text{M}) + \text{NaOH}(100\text{ ml}, 0.1\text{M})$	(S) 5	(4) pH is determined by buffer solution

(Given : $(K_a)_{\text{CH}_3\text{COOH}} = 10^{-5}$, $(K_b)_{\text{NH}_4\text{OH}} = 10^{-5}$)

Q.23 Which of the following is incorrectly matched

- (A) A - S - 4 (B) B - P - 1 (C) D - R - 2 (D) C - Q - 1

Q.24 If 0.15 mole NaOH is added in solution (B) of column-I then which of the following is correct

- (A) S - 3 (B) S - 4 (C) R - 1 (D) P - 3

Q.25 If 0.1mole HCl is added in solution (A) of column-I then pH of the resulting solution will be

- (A) 7 (B) 13 (C) 3.0 (D) 1

EXERCISE # J-MAIN

- Q.1 The solubility in water of a sparingly soluble salt AB_2 is $1.0 \times 10^{-5} \text{ mol L}^{-1}$. Its solubility product will be [AIEEE-2003]
 (1) 1×10^{-15} (2) 1×10^{-10} (3) 4×10^{-15} (4) 4×10^{-10}
- Q.2 The solubility of $Mg(OH)_2$ is x mole/lit. then its solubility product is- [AIEEE-2002]
 (1) x^3 (2) $5x^3$ (3) $4x^3$ (4) $2x^2$
- Q.3 The molar solubility in mol L^{-1} of a sparingly soluble salt MX_4 is 's'. The corresponding solubility product is K_{sp} . 's' is given in terms of K_{sp} by relation : [AIEEE-2004]
 (1) $s = (K_{sp} / 128)^{1/4}$ (2) $s = (128K_{sp})^{1/4}$
 (3) $s = (256K_{sp})^{1/5}$ (4) $s = (K_{sp}/256)^{1/5}$
- Q.4 The solubility product of a salt having general formula MX_2 , in water is : 4×10^{-12} . The concentration of M^{2+} ions in the aqueous solution of the salt is - [AIEEE-2005]
 (1) $1.0 \times 10^{-4} \text{ M}$ (2) $2.0 \times 10^{-6} \text{ M}$ (3) $4.0 \times 10^{-10} \text{ M}$ (4) $1.6 \times 10^{-4} \text{ M}$
- Q.5 Hydrogen ion concentration in mol/L in a solution of $\text{pH} = 5.4$ will be - [AIEEE-2005]
 (1) 3.88×10^6 (2) 3.98×10^8 (3) 3.98×10^{-6} (4) 3.68×10^{-6}
- Q.6 In a saturated solution of the sparingly soluble strong electrolyte $AgIO_3$ (molecular mass = 283) the equilibrium which sets in is - [AIEEE-2007]

$$AgIO_3 \rightleftharpoons Ag^+_{(aq)} + IO_3^-_{(aq)}$$

 If the solubility product constant K_{sp} of $AgIO_3$ at a given temperature is 1.0×10^{-8} , what is the mass of $AgIO_3$ contained in 100 ml of its saturated solution ?
 (1) $28.3 \times 10^{-2} \text{ g}$ (2) $2.83 \times 10^{-3} \text{ g}$ (3) $1.0 \times 10^{-7} \text{ g}$ (4) $1.0 \times 10^{-4} \text{ g}$
- Q.7 The pK_a of a weak acid, HA, is 4.80. The pK_b of a weak base, BOH, is 4.78. The pH of an aqueous solution of the corresponding salt, BA, will be - [AIEEE-2008]
 (1) 9.58 (2) 4.79 (3) 7.01 (4) 9.22
- Q.8 Solid $Ba(NO_3)_2$ is gradually dissolved in a $1.0 \times 10^{-4} \text{ M Na}_2\text{CO}_3$ solution. At what concentration of Ba^{2+} will a precipitate begin to form? [AIEEE-2009]
 $(K_{sp} \text{ for } BaCO_3 = 5.1 \times 10^{-9})$
 (1) $8.1 \times 10^{-8} \text{ M}$ (2) $8.1 \times 10^{-7} \text{ M}$ (3) $4.1 \times 10^{-5} \text{ M}$ (4) $5.1 \times 10^{-5} \text{ M}$
- Q.9 Solubility product of silver bromide is 5.0×10^{-13} . The quantity of potassium bromide (molar mass taken as 120 g mol^{-1}) to be added to 1 litre of 0.05 M solution of silver nitrate to start the precipitation of AgBr is :- [AIEEE-2010]
 (1) $5.0 \times 10^{-8} \text{ g}$ (2) $1.2 \times 10^{-10} \text{ g}$ (3) $1.2 \times 10^{-9} \text{ g}$ (4) $6.2 \times 10^{-5} \text{ g}$
- Q.10 In aqueous solution the ionization constants for carbonic acid are
 $K_1 = 4.2 \times 10^{-7}$ and $K_2 = 4.8 \times 10^{-11}$ [AIEEE-2010]
 Select the correct statement for a saturated 0.034 M solution of the carbonic acid :-
 (1) The concentration of H^+ is double that of CO_3^{2-}
 (2) The concentration of CO_3^{2-} is 0.034 M
 (3) The concentration of CO_3^{2-} is greater than that of HCO_3^-
 (4) The concentrations of H^+ and HCO_3^- are approximately equal

Q.11 At 25°C, the solubility product of $\text{Mg}(\text{OH})_2$ is 1.0×10^{-11} . At which pH, will Mg^{2+} ions start precipitating in the form of $\text{Mg}(\text{OH})_2$ from a solution of 0.001 M Mg^{2+} ions ? [AIEEE-2010]

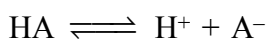
- (1) 8 (2) 9 (3) 10 (4) 11

Q.12 The K_{sp} for $\text{Cr}(\text{OH})_3$ is 1.6×10^{-30} . The molar solubility of this compound in water is :-

[AIEEE-2011]

- (1) $\sqrt[3]{1.6 \times 10^{-30}}$ (2) $\sqrt[4]{1.6 \times 10^{-30}}$ (3) $\sqrt[4]{1.6 \times 10^{-30}/27}$ (4) $1.6 \times 10^{-30}/27$

Q.13 An acid HA ionises as



The pH of 1.0 M solution is 5. Its dissociation constant would be :-

[AIEEE-2011]

- (1) 1×10^{-10} (2) 5 (3) 5×10^{-8} (4) 1×10^{-5}

Q.14 If K_{sp} of CaF_2 at 25°C is 1.7×10^{-10} , the combination amongst the following which gives a precipitate of CaF_2 is :- [JEE-MAIN(online)-2012]

- (1) 1×10^{-2} M Ca^{2+} and 1×10^{-5} M F^- (2) 1×10^{-4} M Ca^{2+} and 1×10^{-4} M F^-
(3) 1×10^{-3} M Ca^{2+} and 1×10^{-5} M F^- (4) 1×10^{-2} M Ca^{2+} and 1×10^{-3} M F^-

Q.15 The pH of a 0.1 molar solution of the acid HQ is 3. The value of the ionization constant, K_a of this acid is :- [AIEEE-2012]

- (1) 1×10^{-7} (2) 3×10^{-7} (3) 1×10^{-3} (4) 1×10^{-5}

Q.16 How many litres of water must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2 ? [AIEEE-2013]

- (1) 0.1 L (2) 0.9 L (3) 2.0 L (4) 9.0 L

Q.17 Solid $\text{Ba}(\text{NO}_3)_2$ is gradually dissolved in a 1.0×10^{-4} M Na_2CO_3 solution. At which concentration of Ba^{2+} , precipitate of BaCO_3 begins to form ? (K_{sp} for $\text{BaCO}_3 = 5.1 \times 10^{-9}$)

- (1) 5.1×10^{-5} M (2) 8.1×10^{-7} M [JEE-MAIN(Online)-2013]
(3) 4.1×10^{-5} M (4) 7.1×10^{-8} M

Q.18 NaOH is a strong base. What will be pH of 5.0×10^{-2} M NaOH solution ? ($\log 2 = 0.3$)

[JEE-MAIN(Online)-2013]

- (1) 13.70 (2) 13.00 (3) 14.00 (4) 12.70

Q.19 Which one of the following arrangements represents the correct order of solubilities of sparingly soluble salts Hg_2Cl_2 , $\text{Cr}_2(\text{SO}_4)_3$, BaSO_4 and CrCl_3 respectively ? [JEE-MAIN(Online)-2013]

- (1) $\left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}}$, $\left(\frac{K_{sp}}{108}\right)^{\frac{1}{5}}$, $(K_{sp})^{\frac{1}{2}}$, $\left(\frac{K_{sp}}{27}\right)^{\frac{1}{4}}$ (2) $(K_{sp})^{\frac{1}{2}}$, $\left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}}$, $\left(\frac{K_{sp}}{27}\right)^{\frac{1}{4}}$, $\left(\frac{K_{sp}}{108}\right)^{\frac{1}{5}}$
(3) $(K_{sp})^{\frac{1}{2}}$, $\left(\frac{K_{sp}}{108}\right)^{\frac{1}{5}}$, $\left(\frac{K_{sp}}{27}\right)^{\frac{1}{4}}$, $\left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}}$ (4) $\left(\frac{K_{sp}}{108}\right)^{\frac{1}{5}}$, $\left(\frac{K_{sp}}{27}\right)^{\frac{1}{4}}$, $(K_{sp})^{\frac{1}{2}}$, $\left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}}$

Q.20 What would be the pH of a solution obtained by mixing 5 g of acetic acid and 7.5 g of sodium acetate and making the volume equal to 500 mL? [JEE-MAIN(Online)-2013]

($K_a = 1.75 \times 10^{-5}$, $pK_a = 4.76$)

- (1) $4.76 < \text{pH} < 5.0$
- (2) $\text{pH} < 4.70$
- (3) pH of solution will be equal to pH of acetic acid
- (4) $\text{pH} = 4.70$

Q.21 In some solutions, the concentration of H_3O^+ remains constant even when small amounts of strong acid or strong base are added to them. These solutions are known as :-[JEE-MAIN(Online)-2014]

- (1) Colloidal solutions
- (2) True solutions
- (3) Ideal solutions
- (4) Buffer solutions

Q.22 Zirconium phosphate $[\text{Zr}_3(\text{PO}_4)_4]$ dissociates into three zirconium cations of charge +4 and four phosphate anions of charge -3. If molar solubility of zirconium phosphate is denoted by S and its solubility product by K_{sp} then which of the following relationship between S and K_{sp} is correct?

- (1) $S = \{K_{sp}/144\}^{1/7}$
- (2) $S = \{K_{sp}/(6912)^{1/7}\}$ [JEE-MAIN(Online)-2014]
- (3) $S = (K_{sp}/6912)^{1/7}$
- (4) $S = \{K_{sp}/6912\}^7$

Q.23 pK_a of a weak acid (HA) and pK_b of a weak base (BOH) are 3.2 and 3.4, respectively. The pH of their salt (AB) solution is [JEE-MAIN(Offline)-2017]

- (1) 7.2
- (2) 6.9
- (3) 7.0
- (4) 1.0

Q.24 Addition of sodium hydroxide solution to a weak acid (HA) results in a buffer of pH 6. If ionisation constant of HA is 10^{-5} , the ratio of salt to acid concentration in the buffer solution will be :

[JEE-MAIN(Online)-2017]

- (1) 4 : 5
- (2) 1 : 10
- (3) 10 : 1
- (4) 5 : 4

Q.25 50 mL of 0.2 M ammonia solution is treated with 25 mL of 0.2 M HCl. If pK_b of ammonia solution is 4.75, the pH of the mixture will be:- [JEE-MAIN(Online)-2017]

- (1) 8.25
- (2) 4.75
- (3) 9.25
- (4) 3.75

Q.26 Which of the following salts is the most basic in aqueous solution? [JEE-MAIN(Offline)-2018]

- (1) CH_3COOK
- (2) FeCl_3
- (3) $\text{Pb}(\text{CH}_3\text{COO})_2$
- (4) $\text{Al}(\text{CN})_3$

Q.27 An alkali is titrated against an acid with methyl orange as indicator, which of the following is a correct combination ?

[JEE-MAIN(Offline)-2018]

- | Base | Acid | End point |
|------------|--------|-----------------------|
| (1) Strong | Strong | Pinkish red to yellow |
| (2) Weak | Strong | Yellow to pinkish red |
| (3) Strong | Strong | Pink to colourless |
| (4) Weak | Strong | Colourless to pink |

Q.28 An aqueous solution contains 0.10 M H_2S and 0.20 M HCl. If the equilibrium constants for the formation of HS^- from H_2S is 1.0×10^{-7} and that of S^{2-} from HS^- ions is 1.2×10^{-13} then the concentration of S^{2-} ions in aqueous solution is : [JEE-MAIN(Offline)-2018]

- (1) 3×10^{-20}
- (2) 6×10^{-21}
- (3) 5×10^{-19}
- (4) 5×10^{-8}

Q.29 A aqueous solution contains an unknown concentration of Ba^{2+} . When 50 mL of a 1 M solution of Na_2SO_4 is added, BaSO_4 just begins to precipitate. The final volume is 500 mL. The solubility product of BaSO_4 is 1×10^{-10} . What is the original concentration of Ba^{2+} ? [JEE-MAIN(Offline)-2018]

- (1) 2×10^{-9} M (2) 1.1×10^{-9} M (3) 1.0×10^{-10} M (4) 5×10^{-9} M

Q.30 Following four solutions are prepared by mixing different volumes of NaOH and HCl of different concentrations, pH of which one of them will be equal to 1? [JEE-MAIN(Online)-2018]

- (1) $75\text{mL } \frac{\text{M}}{5} \text{HCl} + 25\text{mL } \frac{\text{M}}{5} \text{NaOH}$ (2) $100\text{mL } \frac{\text{M}}{10} \text{HCl} + 100\text{mL } \frac{\text{M}}{10} \text{NaOH}$

- (3) $55\text{mL } \frac{\text{M}}{10} \text{HCl} + 45\text{mL } \frac{\text{M}}{10} \text{NaOH}$ (4) $60\text{mL } \frac{\text{M}}{10} \text{HCl} + 40\text{mL } \frac{\text{M}}{10} \text{NaOH}$

Q.31 The minimum volume of water required to dissolve 0.1 g lead (II) chloride to get a saturated solution (K_{sp} of $\text{PbCl}_2 = 3.2 \times 10^{-8}$; atomic mass of Pb = 207 u) is : [JEE-MAIN(Online)-2018]

- (1) 0.36 L (2) 0.18 L (3) 17.98 L (4) 1.798 L

EXERCISE # J-ADVANCE

- Q.1 What will be the resultant pH when 200 ml of an aqueous solution of HCl (pH = 2.0) is mixed with 300 ml of an aqueous solution of NaOH (pH = 12.0) ? [JEE '1998]
- Q.2 The pH of 0.1 M solution of the following salts increases in the order [JEE 1999]
 (A) $\text{NaCl} < \text{NH}_4\text{Cl} < \text{NaCN} < \text{HCl}$
 (B) $\text{HCl} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{NaCN}$
 (C) $\text{NaCN} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{HCl}$
 (D) $\text{HCl} < \text{NaCl} < \text{NaCN} < \text{NH}_4\text{Cl}$
- Q.3 A buffer solution can be prepared from a mixture of [JEE 1999]
 (A) sodium acetate and acetic acid in water
 (B) sodium acetate and hydrochloric acid in water
 (C) ammonia and ammonium chloride in water
 (D) ammonia and sodium hydroxide in water.
- Q.4 The solubility of $\text{Pb}(\text{OH})_2$ in water is $6.7 \times 10^{-6} \text{ M}$. Calculate the solubility of $\text{Pb}(\text{OH})_2$ in a buffer solution of pH = 8. [JEE '1999]
- Q.5 The average concentration of SO_2 in the atmosphere over a city on a certain day is 10 ppm, when the average temperature is 298 K. Given that the solubility of SO_2 in water at 298 K is 1.3653 moles litre^{-1} and the pK_a of H_2SO_3 is 1.92, estimate the pH of rain on that day. [JEE 2000]
 [Given : $10^{-1.92} = 1.2 \times 10^{-2}$, $\sqrt{5.5678} = 2.5627$, $\log(1.2213) = 0.0868$]
- Q.6 For sparingly soluble salt ApBq , the relationship of its solubility product (L_s) with its solubility (S) is - [JEE 2001]
 (A) $L_s = S^{p+q} \cdot p^p \cdot q^q$ (B) $L_s = S^{p+q} \cdot p^p \cdot q^p$
 (C) $L_s = S^{pq} \cdot p^p \cdot q^q$ (D) $L_s = S^{pq} \cdot (p \cdot q)^{p+q}$
- Q.7 500 ml of 0.2 M aqueous solution of acetic acid is mixed with 500 mL of 0.2 M HCl at 25°C.
 (a) Calculate the degree of dissociation of acetic acid in the resulting solution and pH of the solution.
 (b) If 6 g of NaOH is added to the above solution, determine final pH. Assume there is no change in volume on mixing. K_a of acetic acid is $1.75 \times 10^{-5} \text{ M}$. [JEE 2002]
- Q.8 A solution which is 10^{-3} M each in Mn^{2+} , Fe^{2+} , Zn^{2+} and Hg^{2+} is treated with 10^{-16} M sulphide ion. If K_{sp} , MnS , FeS , ZnS and HgS are 10^{-15} , 10^{-23} , 10^{-20} and 10^{-54} respectively, which one will precipitate first ? [JEE 2003]
 (A) FeS (B) MnS (C) HgS (D) ZnS
- Q.9 Will the pH of water be same at 4°C and 25°C ? Explain. [JEE 2003]

- Q.10 0.1 M of HA is titrated with 0.1 M NaOH, calculate the pH at end point. Given $K_a(\text{HA}) = 5 \times 10^{-6}$ and $\alpha \ll 1$. [JEE 2004]
- Q.11 HX is a weak acid ($K_a = 10^{-5}$). It forms a salt NaX (0.1 M) on reacting with caustic soda. The degree of hydrolysis of NaX is
(A) 0.01% (B) 0.0001% (C) 0.1% (D) 0.5% [JEE 2004]
- Q.12 CH_3NH_2 (0.1 mole, $K_b = 5 \times 10^{-4}$) is added to 0.08 moles of HCl and the solution is diluted to one litre, resulting hydrogen ion concentration is
(A) 1.6×10^{-11} (B) 8×10^{-11} (C) 5×10^{-5} (D) 2×10^{-2} [JEE 2005]
- Q.13 If $\text{Ag}^+ + \text{NH}_3 \rightleftharpoons [\text{Ag}(\text{NH}_3)]^+$; $K_1 = 1.6 \times 10^3$ and
 $[\text{Ag}(\text{NH}_3)]^+ + \text{NH}_3 \rightleftharpoons [\text{Ag}(\text{NH}_3)_2]^+$; $K_2 = 6.8 \times 10^3$.
The formation constant of $[\text{Ag}(\text{NH}_3)_2]^+$ is :
(A) 6.08×10^{-6} (B) 6.8×10^{-6} (C) 1.6×10^3 (D) 1.088×10^7 [JEE 2006]
- Q.14 The species present in solution when CO_2 is dissolved in water :
(A) CO_2 , H_2CO_3 , HCO_3^- , CO_3^{2-} (B) H_2CO_3 , CO_3^{2-} [JEE 2006]
(C) CO_3^{2-} , HCO_3^- (D) CO_2 , H_2CO_3
- Q.15 2.5 mL of $\frac{2}{5}$ M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25°C) is titrated with $\frac{2}{15}$ M HCl in water at 25°C . The concentration of H^+ at equivalence point is [JEE 2008]
($K_w = 1 \times 10^{-14}$ at 25°C)
(A) 3.7×10^{-13} M (B) 3.2×10^{-7} M
(C) 3.2×10^{-2} M (D) 2.7×10^{-2} M
- Q.16 Solubility product constants (K_{sp}) of salts of types MX, MX_2 and M_3X at temperature 'T' are 4.0×10^{-8} , 3.2×10^{-14} and 2.7×10^{-15} , respectively. Solubilities (mol dm^{-3}) of the salts at temperature 'T' are in the order :
(A) $\text{MX} > \text{MX}_2 > \text{M}_3\text{X}$ (B) $\text{M}_3\text{X} > \text{MX}_2 > \text{MX}$ [JEE 2008]
(C) $\text{MX}_2 > \text{M}_3\text{X} > \text{MX}$ (D) $\text{MX} > \text{M}_3\text{X} > \text{MX}_2$
- Q.17 The dissociation constant of a substituted benzoic acid at 25°C is 1.0×10^{-4} . The pH of a 0.01 M solution of its sodium salt is [JEE 2009]
- Q.18 Aqueous solutions of HNO_3 , KOH, CH_3COOH and CH_3COONa of identical concentrations are provided. The pair(s) of solutions which form a buffer upon mixing is(are) –
(A) HNO_3 and CH_3COOH
(B) KOH and CH_3COONa
(C) HNO_3 and CH_3COONa
(D) CH_3COOH and CH_3COONa

- Q.19 In 1 L saturated solution of AgCl [$K_{sp}(\text{AgCl}) = 1.6 \times 10^{-10}$], 0.1 mol of CuCl [$K_{sp}(\text{CuCl}) = 1.0 \times 10^{-6}$] is added. The resultant concentration of Ag^+ in the solution is 1.6×10^{-x} . The value of 'x' is. **[JEE -2011]**
- Q.20 The K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K. The solubility (in mol/L) of Ag_2CrO_4 in a 0.1 M AgNO_3 solution is **[JEE 2013]**
- (A) 1.1×10^{-11} (B) 1.1×10^{-10}
(C) 1.1×10^{-12} (D) 1.1×10^{-9}
- Q.21 The solubility of a salt of weak acid (AB) at pH 3 is $Y \times 10^{-3} \text{ mol L}^{-1}$. The value of Y is ____.
(Given that the value of solubility product of AB (K_{sp}) = 2×10^{-10} and the value of ionization constant of HB (K_a) = 1×10^{-8}) **[JEE 2018]**

ANSWER KEY

EXERCISE # S-I

- Q.1 Ans. 6.022×10^7 Q.2 Ans. (i) 6.53 ; (ii) (a) Basic , (b) Acidic
- Q.3 Ans. 6.8
- Q.4 Ans. (a) 1, (b) 2.87, (c) 11.13 (d) 6.97, (e) 7, (f) 6, (g) 6.97, (h) 11.30 (i) 9, (j) 3
- Q.5 Ans. (a) $K_a = 10^{-8}$, (b) $K_b = 10^{-6}$ Q.6 Ans. 10
- Q.7 Ans. 173.2 : 1 Q.8 Ans. 0.009
- Q.9 Ans. 2.32×10^{-8} M Q.10 Ans. 1.1×10^{-3} M
- Q.11 Ans. 1.11×10^{-4} Q.12 Ans. 4.7
- Q.13 Ans. 3.3 Q.14 Ans. (a) 0.522 , (b) 2.522
- Q.15. Ans. (1)
- Q.16 Ans. $[H^+] = 1.65 \times 10^{-2}$ M, $[CHCl_2COO^-] = 6.5 \times 10^{-3}$ M
- Q.17 Ans. $[H^+] = 10^{-3}$ M, $[CH_3COO^-] = 3.6 \times 10^{-4}$ M, $[C_7H_5O_2^-] = 6.4 \times 10^{-4}$ M
- Q.18 Ans. 0.027 M, 0.073 M, 0.027 M, 10^{-5} M
- Q.19 Ans. $[H^+] = [H_2PO_4^-] = 2.7 \times 10^{-3}$ M, $[HPO_4^{2-}] = 10^{-8}$ M, $[PO_4^{3-}] = 3.7 \times 10^{-19}$ M
- Q.20 Ans. 11.3 Q.21 Ans. 10^{-5} M
- Q.22 Ans. pH = 4.5 Q.23 Ans. $K_b = 6.25 \times 10^{-10}$
- Q.24 Ans. 0.56%, pH = 7 Q.25 Ans. 1.667%
- Q.26 Ans. 0.25 % Q.27 Ans. 10^{-6} ; 10^{-8}
- Q.28 Ans. 8.3 Q.29 Ans. 4.19
- Q.30 Ans. (a) 6, (b) 1×10^{-5} Q.31 Ans. 9.0
- Q.32 Ans. 9.56 Q.33 Ans. 5.04
- Q.34 Ans. 0.05 mol Q.35 Ans. $[OH^-] = 9.0 \times 10^{-6}$ M
- Q.36 Ans. (10.1) Q.37 Ans. 4.74
- Q.38 Ans. 9.56 Q.39 Ans. 8.7782
- Q.40 Ans. (3.33) Q.41 Ans. 10^{-5} M
- Q.42 Ans. 8.7, $[H_3O^+] = 2 \times 10^{-9}$ M Q.43 Ans. 5, 10^{-5} M
- Q.44 Ans. (i) 2.85, (ii) 4.0969, (iii) 4.5229, (iv) 4.699, (v) 5.301, (vi) 8.699
- Q.45 Ans. $[HI_n] = 80$ % Q.46 Ans. 85.71%
- Q.47 Ans. (b), (c) Q.48 Ans. $\Delta pH = 0.954$
- Q.49 Ans. QX_2 is more soluble Q.50 Ans. 4×10^{-8}
- Q.51 Ans. 5×10^{-11} Q.52 Ans. 3.2×10^{-11}
- Q.53 Ans. 2.56×10^{-16} Q.54 Ans. 1.0×10^{-5} mol/lit
- Q.55 Ans. 5×10^{-10} M
- Q.56 Ans. 5×10^{-7}
- Q.57 Ans. 4×10^{-7} mol/L AgBr, 1.6×10^{-6} mol/L AgSCN
- Q.58 Ans. $[F^-] = 3 \times 10^{-3}$ M Q.59 Ans. 8×10^{-3} M
- Q.60 Ans. (i) 2×10^{-9} , (ii) 8×10^{-3} Q.61 Ans. $S = 2 \times 10^{-4}$ M, pH = 8.0
- Q.62 Ans. 4×10^{-2} M

Q.63 Ans. (a) no precipitation will occur, (b) a precipitate will form

Q.64 Ans. No.

Q.65 Ans. 0.284 gm

EXERCISE # S-II

Q.1 Ans. 0.209 M, 0.191 M, 9.13×10^{-3} M, 0 Q.2 Ans. pH = 11.48, $[\text{enH}_2^{2+}] = 7.1 \times 10^{-8}$ M

Q.3 Ans. 10.6

Q.4 Ans. $[\text{S}^{2-}] = 2.5 \times 10^{-15}$ M

Q.5 Ans. (2×10^{-4}) M

Q.6 Ans. pH = 10.52

Q.7 Ans. $[\text{OH}^-] = 3.73 \times 10^{-2}$ M, $[\text{H}_3\text{PO}_4] = 6 \times 10^{-18}$ M

Q.8 Ans. 8.35, 9.60, 4.66

Q.9 Ans. (9.6)

Q.10 Ans. $K_b \approx 1.73 \times 10^{-5}$, 5.27

Q.11 Ans. $K_a \approx 1.73 \times 10^{-5}$, 8.73

Q.12 Ans. (10^{-5}) M

Q.13 Ans. (10^{-3}) M

Q.14 Ans. pH = 7.9, pH = 7.3

Q.15 Ans. 2.8×10^{-3} mole

Q.16 Ans. 1.6×10^{-3}

Q.17 Ans. $K_d = 1/K_f = 4.8 \times 10^{-4}$

EXERCISE # O-I

Q.1 Ans. (A)

Q.2 Ans. (B)

Q.3 Ans. (C)

Q.4 Ans. (C)

Q.5 Ans. (B)

Q.6 Ans. (C)

Q.7 Ans. (D)

Q.8 Ans. (D)

Q.9 Ans. (C)

Q.10 Ans. (C)

Q.11 Ans. (C)

Q.12 Ans. (D)

Q.13 Ans. (B)

Q.14 Ans. (B)

Q.15 Ans. (C)

Q.16 Ans. (C)

Q.17 Ans. (B)

Q.18 Ans. (C)

Q.19 Ans. (C)

Q.20 Ans. (B)

Q.21 Ans. (B)

Q.22 Ans. (D)

Q.23 Ans. (D)

Q.24 Ans. (C)

Q.25 Ans. (C)

Q.26 Ans. (C)

Q.27 Ans. (B)

Q.28 Ans. (B)

Q.29 Ans. (A)

Q.30 Ans. (D)

Q.31 Ans. (B)

Q.32 Ans. (C)

Q.33 Ans. (C)

Q.34 Ans. (C)

Q.35 Ans. (B)

Q.36 Ans. (C)

Q.37 Ans. (D)

Q.38 Ans. (A)

Q.39 Ans. (C)

Q.40 Ans. (C)

Q.41 Ans. (C)

Q.42 Ans. (B)

Q.43 Ans. (C)

Q.44 Ans. (D)

Q.45 Ans. (A)

Q.46 Ans. (B)

Q.47 Ans. (D)

Q.48 Ans. (B)

Q.49 Ans. (C)

Q.50 Ans. (A)

Q.51 Ans. (C)

Q.52 Ans. (A)

Q.53 Ans. (A)

Q.54 Ans. (A)

Q.55 Ans. (B)

Q.56 Ans. (D)

Q.57 Ans. (A)

Q.58 Ans. (A)

Q.59 Ans. (B)

Q.60 Ans. (C)

Q.61 Ans. (D)

Q.62 Ans. (A)

Q.63 Ans. (D)

Q.64 Ans. (C)

Q.65 Ans. (D)

Q.66 Ans. (D)

Q.67 Ans. (D)

Q.68 Ans. (C)

Q.69 Ans. (A)

Q.70 Sol. (C)

Q.71 Ans. (B)

Q.72 Ans. (B)

Q.73 Ans. (D)

Q.74 Ans. (B)

Q.75 Ans. (B)

EXERCISE # O-II

Q.1 Ans.(A)	Q.2 Ans.(C)	Q.3 Ans.(D)
Q.4 Ans.(B)	Q.5 Ans (C)	Q.6 Ans.(A)
Q.7 Ans.(A)	Q.8 Ans.(B)	Q.9 Ans.(D)
Q.10 Ans.(A)	Q.11 Ans. (A,B,C)	Q.12 Ans.(A,B,C,D)
Q.13 Ans.(B,C)	Q.14 Ans. (A, B, D)	Q.15 Ans. (D)
Q.16 Ans.(A)	Q.17 Ans. (A)	Q.18 Ans (B)
Q.19 Ans (B)	Q.20 Ans (B)	
Q.21 Ans. A - (R), B - (P), C - (Q), D - (S)		
Q.22 Ans (A) - Q; (B) - P, S; (C) - R; (D) - T		
Q.23 Ans.(D)	Q.24 Ans.(B)	Q.25 Ans.(C)

EXERCISE # J-MAIN

Q.1 Ans.(3)	Q.2 Ans.(3)	Q.3 Ans.(4)
Q.4 Ans.(1)	Q.5 Ans.(3)	Q.6 Ans.(2)
Q.7 Ans.(3)	Q.8 Ans.(4)	Q.9 Ans.(3)
Q.10 Ans.(4)	Q.11 Ans.(3)	Q.12 Ans.(3)
Q.13 Ans.(1)	Q.14 Ans.(4)	Q.15 Ans.(4)
Q.16 Ans.(4)	Q.17 Ans.(1)	Q.18 Ans.(4)
Q.19 Ans.(1)	Q.20 Ans.(1)	Q.21 Ans.(4)
Q.22 Ans.(3)	Q.23 Ans. (2)	Q.24 Ans.(3)
Q.25 Ans.(3)	Q.26 Ans.(1)	Q.27 Ans.(2)
Q.28 Ans.(1)	Q.29 Ans.(2)	Q.30 Ans.(1)
Q.31 Ans.(2)		

EXERCISE # J-ADVANCE

Q.1 Ans. pH = 11.3010	Q.2 Ans.(B)	Q.3 Ans.(A, B, C)
Q.4 Ans. $s = 1.203 \times 10^{-3} \text{M}$	Q.5 Ans. pH = 0.91325	Q.6 Ans.(A)
Q.7 Ans.(a) 0.0175% , (b) 4.757	Q.8 Ans.(C)	
Q.9 Ans.No, it will be > 7 at 0°C .	Q.10 Ans.pH = 9	Q.11 Ans.(A)
Q.12 Ans.(B)	Q.13 Ans.(D)	Q.14 Ans.(A)
Q.15 Ans.(D)	Q.16 Ans.(D)	Q.17 Ans.(8)
Q.18 Ans.(C), (D)	Q.19 Ans.(7)	Q.20 Ans.(B)
Q.21 Ans. (4.47)		

NURTURE COURSE
CHEMICAL EQUILIBRIUM

CHEMICAL EQUILIBRIUM

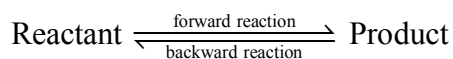
1. TYPES OF CHEMICAL REACTION :

The chemical reactions are classified on the basis of the extent to which they proceed, into the following two classes ;

I. Reversible reactions : Reaction in which entire amount of the reactants is not converted into products is termed as reversible reaction.

(i) Characteristics of reversible reactions :

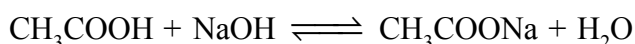
- (a) These reactions can be started from either side.
- (b) These reactions are never complete.
- (c) This sign ($\rightleftharpoons$) represents the reversibility of the reaction.



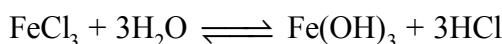
- (d) These reactions attain equilibrium and all the times, reaction mixture contains both reactant and product molecules. When reaction attains equilibrium the concentrations of reactants and product become constant (not same necessarily).

(ii) Examples of reversible reactions :

- (a) Neutralisation between an acid and a base either one or both are weak.



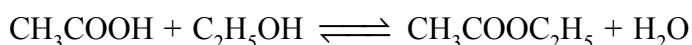
- (b) Salt hydrolysis



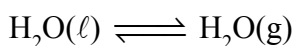
- (c) Thermal decomposition



- (d) Esterification



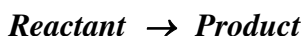
- (e) Evaporation of water in a closed vessel



II. Irreversible reactions : Reaction in which entire amount of the reactants is converted into products is termed as irreversible reaction.

(i) Characteristics of irreversible reactions :

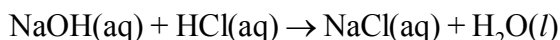
- (a) These reactions proceed only in one direction (forward direction)
- (b) These reactions can proceed to completion and reactant is completely converted into product.
- (c) The arrow ($\rightarrow$) is placed between reactants and products



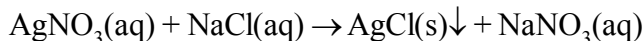
- (d) These reactions never attain equilibrium.

(ii) Examples of irreversible reactions :

- (a) Neutralisation between strong acid and strong base



- (b) Precipitation reactions

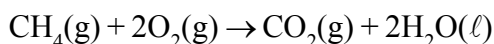


- (c) Thermal decomposition



However, if the above reaction is carried out in closed container, i.e., leaving no scope for gas to escape out, the reaction shows reversible nature.

- (d) Combustion reactions

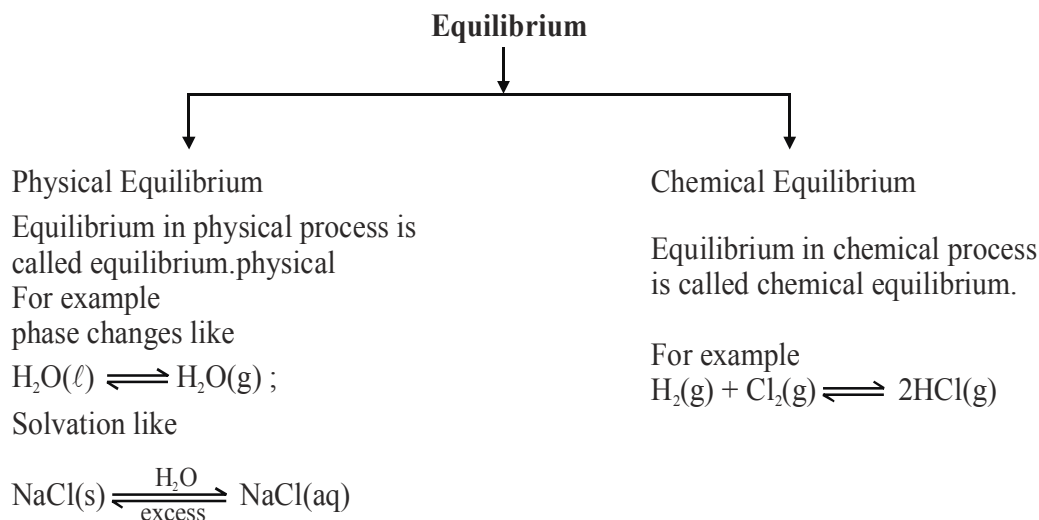


2. EQUILIBRIUM :

In the state of equilibrium, system loses its tendency for a change and all the properties associated with system like pressure, temperature, composition, etc become constant and do not vary without external stimulation. On the basis of nature of process in which state of equilibrium is attained, it may be of two types :

- (A) Physical equilibrium

- (B) Chemical equilibrium

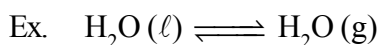


2.1 Physical equilibrium :

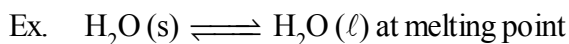
If in a system only physical state (phase) is changed and equilibrium is established, (i.e. there is no chemical change), the equilibrium is called **physical equilibrium**. Ex : Fusion of ice, evaporation of water, dissolution of salts, absorption of gases in liquid, etc.

Following are the types of common physical equilibria :

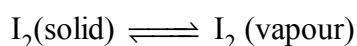
- (i) **Liquid-Vapour equilibria** : In a closed vessel, the vapours above the liquid are in equilibrium at given temperature.



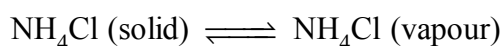
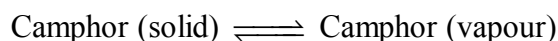
- (ii) **Solid-Liquid equilibria** : This equilibrium can be established only at melting point of solid. At this stage solid and liquid phases exist simultaneously in equilibrium.



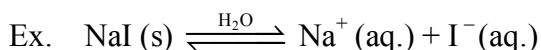
- (iii) **Solid-Vapour equilibria** : Let us now consider the systems where solids sublime to vapour phase. If we place solid iodine in a closed vessel, after sometime the vessel gets filled up with violet vapour and the intensity of colour increases with time. After certain time the intensity of colour becomes constant and at this stage equilibrium is attained. Hence solid iodine sublimes to give iodine vapour and the iodine vapour condenses to give solid iodine. The equilibrium can be represented as,



Other examples showing this kind of equilibrium are,



- (iv) **(Solute-Solvent) Saturated solution equilibria** : If the rate of dissolution of solids in liquid is equal to the rate of crystallization of solid from solution i.e. solution is saturated with respect to solid then saturated solution equilibria established, provided temperature is constant.



- (v) **(Gas + Solvent) Saturated solution equilibria** : In such equilibria, solvent is saturated with respect to gas i.e. rate of entering of gas molecules in solvent is equal to rate of escaping of gas molecules from solvents. Above phenomenon can be observed in closed container at definite temperature. Ex : Dissolved CO_2 in cold drinks, dissolved O_2 in water, etc.

Note:(i) The solubilities of gases in liquid is a function of pressure of gas over liquid.

- (ii) **Henry's law** can be applied on such system, that states, the mass of gas dissolved in a given mass of solvent at any temperature is proportional to the pressure of the gas above the solvent.

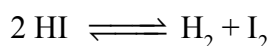
- (iii) One should not compare it with liquid vapour equilibria.

2.2 Chemical equilibrium :

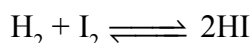
Analogous to the physical systems, chemical reactions also attain a state of equilibrium. These reactions can occur both in forward and backward directions. When the rates of the forward and reverse reactions become equal, the concentrations of the reactants and the products become constant. This is the stage of chemical equilibrium. This equilibrium consists of a forward reaction in which the reactants give product(s) and reverse reaction in which product(s) gives the original reactants.

2.2.1 Characteristics of Chemical equilibrium :

- (i) It is attained in reversible chemical reactions only.
- (ii) Equilibrium is possible only in closed system.
- (iii) In this state, all the measurable properties of the system like temperature, concentration, colour, density etc. don't undergo any change with time, i.e. constant.
- (iv) Equilibrium is dynamic in nature i.e., at microscopic level reaction is not stopped. It appears that no change is occurring but both the opposing reactions are proceeding at the same rate. So there is no net change. Thus equilibrium is not static in nature.
- (v) Chemical equilibrium can be approached from both sides



or

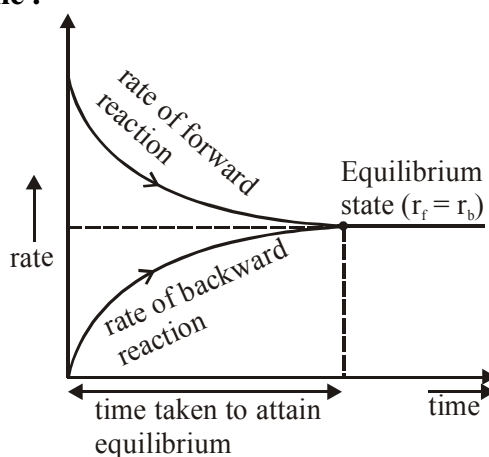


At equilibrium, reactants and products have fixed composition and this is independent of the fact whether the reaction start with the reactant or with the product.

- (vi) Equilibrium is not affected by the presence of catalyst. The catalyst only helps in attaining equilibrium rapidly.
- (vii) At equilibrium, opposing reactions (i.e., forward and backward) proceeds with equal rates.
i.e., rate of forward reaction = rate of backward reaction.

- Note:**
- (i) Whenever question doesn't ask about direction, then we take forward direction only.
 - (ii) In a reversible reaction if forward reaction is exothermic then the backward reaction will be endothermic and vice versa.

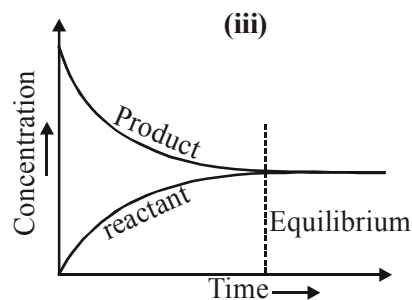
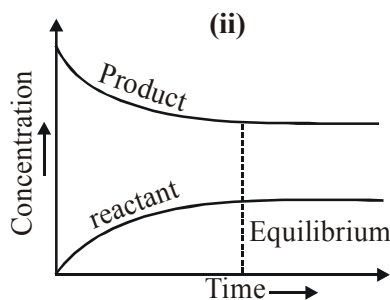
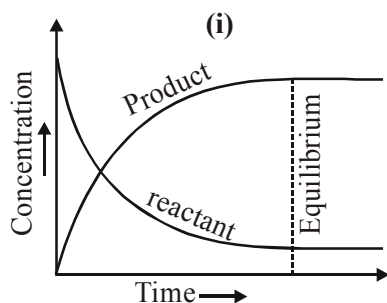
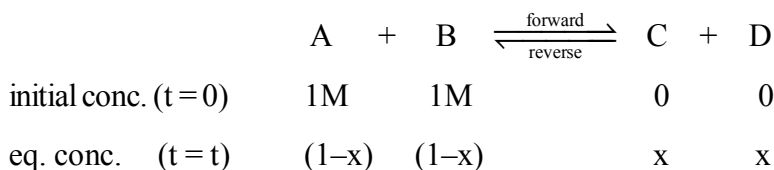
2.2.2 Variation of Rate v/s Time :



From above graph, initially rate of forward reaction decreases and the rate of backward reaction increases. At certain stage, rate of forward reaction becomes equal to rate of backward reaction called **equilibrium state**.

2.2.3 Variation of Concentration v/s Time :

Let us consider a reversible reaction,



In the beginning (at time $t = 0$), the container has only the reacting molecules A and B while the products C and D are nil. With the passage of time, the reactants A and B will be used up or consumed to form the products, C and D. It means that the concentration or molar concentration of the reactants will decrease while those of the products will increase. A stage will be ultimately reached when their concentration becomes constant i.e., there will be no further change in concentration of either of the reactants A and B or of the products C and D. This represents a state of equilibrium.

Note: At equilibrium, the concentration of reactant and product will be constant. It means, it may be different as above graph (i) & (ii) or same as above graph (iii) but the rate of forward reaction and the backward reaction will be always same.

Ex.1 Chemical equilibrium is a condition :

- (A) where all species have same concentration
- (B) where all species have constant concentration with respect to time.
- (C) where all species have unit concentration
- (D) all of above

Sol. (B)

Ex.2 Example of physical equilibria, is :

- (A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
- (B) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
- (C) $H_2O(s) \rightleftharpoons H_2O(l)$
- (D) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Sol. (C)

Physical equilibria does not include any chemical change.

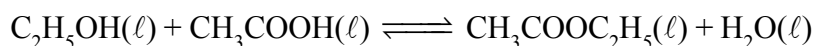
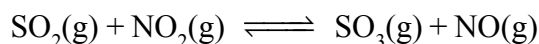
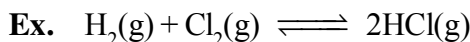
3. TYPES OF CHEMICAL EQUILIBRIUM :

I. Homogeneous equilibrium

II. Heterogeneous equilibrium

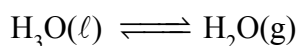
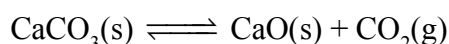
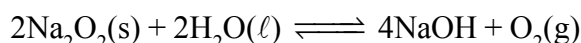
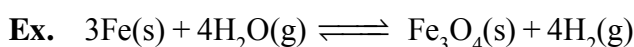
I. Homogeneous equilibrium :

It is the equilibrium when all reactants and products are in same phase.



II. Heterogeneous equilibrium :

It is the equilibrium the reactants and the products are present in different phases. All physical equilibria are heterogeneous.



4. ACTIVE MASS :

The mass of a substance which effect the rate of reaction i.e. mass of substance which take a part actively in a reaction. Active mass depends on state of substance.

(i) **Solution state :** In this state, active mass of a substance is represented by concentration (molarity).

Active mass is usually expressed in concentration by enclosing the symbol of the reactant in square bracket [].

$$\text{Active mass} = \frac{\text{moles}}{\text{Volume in litres}} = \frac{\text{grams(w)}}{\text{mol.wt.}(M_w) \times \text{Volume in litres}(V)} = \frac{w \times 1000}{M_w \times V(\text{mL})}$$

(ii) **Gaseous state :** In this state, active mass of a substance may be represented as concentration (molarity) or partial pressure.

(iii) **Pure solid & pure liquid :** In this state, active mass of solids, pure liquids and solvents in large excess is a constant quantity because there is no change in activity with the change in quantity or volume of system.

$$\text{Molar concentration (M)} = \frac{w}{M \times V} = \frac{d}{M} = \frac{\text{density of the substance}}{\text{molar mass of the substance}} = \text{constant}$$

as density of pure solids and liquids is constant and molar mass is also constant.

5. LAW OF MASS ACTION AND LAW OF EQUILIBRIUM CONSTANT :

The law of mass action is given by **Guldberg** and **Waage**. According to it, "the rate of a chemical reaction at a particular temperature is proportional to the product of active masses of reactants raised to the powers equal to their stoichiometric coefficients".

Consider a reversible reaction : $m_1A + m_2B \rightleftharpoons n_1C + n_2D$

According to law of mass action

rate of forward reaction (r_f) $\propto (a_A)^{m_1} (a_B)^{m_2}$

$$r_f = k_f(a_A)^{m_1} (a_B)^{m_2}$$

rate of backward reaction (r_b) $\propto (a_C)^{n_1} (a_D)^{n_2}$

$$r_b = k_b(a_C)^{n_1} (a_D)^{n_2}$$

At equilibrium $r_f = r_b$

$$k_f(a_A)^{m_1} (a_B)^{m_2} = k_b(a_C)^{n_1} (a_D)^{n_2}$$

$$K_{eq} = \frac{k_f}{k_b} = \frac{(a_C)^{n_1} (a_D)^{n_2}}{(a_A)^{m_1} (a_B)^{m_2}}$$

where, K_{eq} = equilibrium constant

k_f = forward rate (velocity) constant

k_b = backward rate (velocity) constant

a_A, a_B = active mass of reactant A & B

a_C, a_D = active mass of product C & D

m_1, m_2 = stoichiometry coefficient of reactant A & B

n_1, n_2 = stoichiometry coefficient of product C & D

At a given temperature, the product of molar concentrations of the reaction products raised to the respective stoichiometric coefficient in the balanced chemical equation divided by the product of molar concentrations of the reactants raised to their individual stoichiometric coefficients has a constant value. This is known as the **Equilibrium Law or Law of Chemical Equilibrium**.

5.1 Types of equilibrium constant (K_{eq}) :

(i) **Equilibrium constant in terms of concentration (K_C)** : K_C is defined for reactions in gas phase & solution phase. Molar concentrations are used to express amounts.

(ii) **Equilibrium constant in terms of partial pressure (K_P)** : K_P is defined for reactions in gas phase. Partial pressures are used to express amounts.

Ex. Consider a reversible reaction in gas phase : $m_1A + m_2B \rightleftharpoons n_1C + n_2D$

$$K_C = \frac{[C]^{n_1} [D]^{n_2}}{[A]^{m_1} [B]^{m_2}}$$

$$K_P = \frac{(P_C)^{n_1} (P_D)^{n_2}}{(P_A)^{m_1} (P_B)^{m_2}}$$

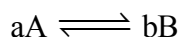
here,

$[A], [B], [C], [D]$ = Molar concentration of A, B, C, D respectively at equilibrium .

$(P_A), (P_B), (P_C), (P_D)$ = partial pressure of A, B, C, D respectively at equilibrium .

5.2 Units of equilibrium constant :

5.2.1 The value of equilibrium constant K_c can be calculated by substituting the concentration terms in mol/L or M and for K_p partial pressure is substituted in Pa, kPa, bar or atm. This results in units of equilibrium constant based on molarity or pressure, unless the exponents of both the numerator and denominator are same. For general reactions,



$$(i) \quad K_c = \frac{[B]^b}{[A]^a}$$

$$\text{unit of } K_c = \left(\frac{\text{mol}}{\text{L}} \right)^{b-a} = [M]^{b-a} = M^{\Delta n_g}$$

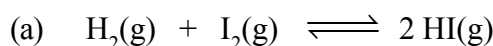
here, $\Delta n_g = b - a = \text{moles of (Product - Reactant)}$

$$(ii) \quad K_p = \frac{(P_B)^b}{(P_A)^a}$$

$$\text{unit of } K_p = (\text{atm})^{b-a} = (\text{atm})^{\Delta n_g} \text{ or } (\text{Pa})^{\Delta n_g} \text{ or } (\text{bar})^{\Delta n_g}$$

here, $\Delta n_g = b - a = \text{moles of (Product - Reactant)}$

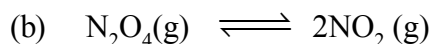
Ex.3 Write down the expression and unit for following reaction :



$$K_c = \frac{[HI]^2}{[H_2][I_2]} \quad \left(\frac{\text{mol}}{\text{L}} \right)^0$$

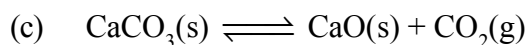
$$K_p = \frac{P_{HI}^2}{P_{H_2} P_{I_2}} \quad (\text{atm})^0$$

$\therefore K_c$ and K_p have no unit.



$$K_c = \frac{[NO_2]^2}{[N_2O_4]} \quad \left(\frac{\text{mol}}{\text{L}} \right)^1$$

$$K_p = \frac{P_{NO_2}^2}{P_{N_2O_4}} \quad (\text{atm})^1$$

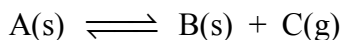


$$K_c = [CO_2] \quad \left(\frac{\text{mol}}{\text{L}} \right)^1$$

$$K_p = P_{CO_2} \quad (\text{atm})^1$$

5.2.2

Equilibrium constants is expressed in standard state that is called standard state equilibrium constant or thermodynamic equilibrium constant, which is dimensionless quantity. It is denoted by K_c^0 and K_p^0 . For a pure gas, the standard state is 1 bar. Therefore a pressure of 4 bar in standard state can be expressed as $4 \text{ bar}/1 \text{ bar} = 4$, which is a dimensionless number and concentration of 3 M solution in standard state can be expressed as $\frac{3\text{M}}{1\text{M}} = 3$, which is a dimensionless number. The numerical value of equilibrium constant depends on the standard state chosen. Thus, in this system both K_c^0 and K_p^0 are dimensionless quantities but have different numerical values due to different standard states. For example,



$$K_c = [\text{C}] \quad (\text{mol/L})$$

$$\text{but } K_c^0 \text{ is a dimensionless quantity so } K_c^0 = \frac{[\text{C}]}{1\text{M}}$$

$$K_p = P_c \quad (\text{bar})^1$$

$$\text{but } K_p^0 \text{ is a dimensionless quantity so } K_p^0 = \frac{P_c}{1 \text{ bar}}$$

5.3 Relation between K_p & K_c :

For the reaction, $a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$

we can write

$$K_c = \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b} \quad \dots\dots\dots(\text{i})$$

Assuming the gaseous components to behave ideally ,

$$P_i V_i = n_i RT$$

$$P_i = (n_i/V_i) RT = C_i RT = [i] RT$$

where $[i]$ is the molar concentration of the species 'i'.

$$P_A = [A]RT, P_B = [B]RT, P_C = [C]RT, P_D = [D]RT$$

$$K_p = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b} = \frac{([C]RT)^c ([D]RT)^d}{([A]RT)^a ([B]RT)^b} = \frac{[C]^c [D]^d}{[A]^a [B]^b} \times (RT)^{(c+d)-(a+b)}$$

from eq..(i)

$$K_p = K_c (RT)^{(c+d)-(a+b)}$$

$$\boxed{K_p = K_c (RT)^{\Delta n_g}}$$

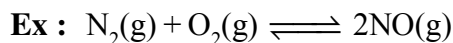
$\Delta n_g = (\text{Number of moles of gaseous products}) - (\text{Number of moles of gaseous reactants}).$

$$\Delta n_g = (c + d) - (a + b)$$

The units of K_p & K_c are not fixed and depend on stoichiometry of the reaction. In case the number of moles of the reactant & that of the product are same K_p & K_c do not have any unit.

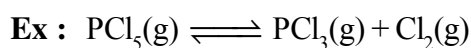
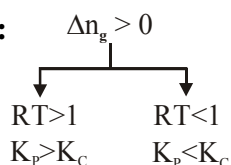
5.4 Different cases for $K_p = K_c(RT)^{\Delta n_g}$:

Case-I : If $\Delta n_g = 0$ then $K_p = K_c$



$K_p = \text{unit less} ; K_c = \text{unit less} ;$

Case-II :



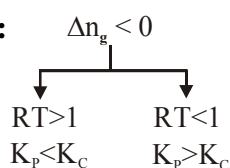
$$\Delta n_g = 2 - 1 = 1 \text{ then } K_p = K_c(RT)^1$$

if $RT > 1$ then $K_p > K_c$

if $RT < 1$ then $K_p < K_c$

Unit : $K_p = \text{atm}^1 ; K_c = \text{conc.}^1$

Case-III :



$$\Delta n_g = 2 - 4 = -2 \text{ then } K_p = K_c(RT)^{-2}$$

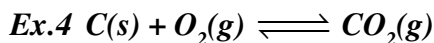
if $RT > 1$ then $K_p < K_c$

if $RT < 1$ then $K_p > K_c$

$K_p = \text{atm}^{-2} ; K_c = \text{conc.}^{-2}$

Case-IV : If $T = \frac{1}{R} = \frac{1}{0.821} \approx 12.2 \text{ K}$

$[K_p = K_c ; \text{for any value of } \Delta n_g]$



Mole of O_2 and CO_2 are 5 mole and 10 mole at equilibrium respectively, then find K_p & K_c for above reaction.

Sol. Partial pressure = Mole fraction $\times$ Total pressure

$$P_{CO_2} = X_{CO_2} \cdot P_T = \frac{n_{CO_2}}{n_T} \times P_T = \frac{10}{15} \times P_T$$

$$P_{O_2} = X_{O_2} \cdot P_T = \frac{n_{O_2}}{n_T} \times P_T = \frac{5}{15} \times P_T$$

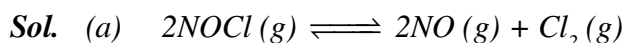
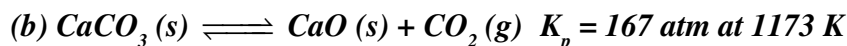
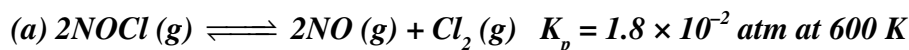
$$K_p = \frac{P_{CO_2}}{P_{O_2}}$$

$$K_p = \frac{\frac{10}{15} \times P_T}{\frac{5}{15} \times P_T} = 2$$

$$\therefore K_p = K_c (RT)^{\Delta n_g}$$

$$K_c = \frac{K_p}{(RT)^{\Delta n_g}} = \frac{2}{(RT)^0} = 2$$

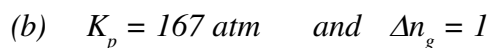
Ex.5 Find the values of K_c for each of the following equilibria from the value of K_p .



$$K_p = 1.8 \times 10^{-2} \text{ atm} \quad \text{and} \quad \Delta n_g = 3 - 2 = 1$$

$$K_p = K_c (RT)^{\Delta n_g} = K_c \times (RT)$$

$$\therefore K_c = \frac{K_p}{RT} = \frac{1.8 \times 10^{-2}}{0.0821 \times 600} = 3.65 \times 10^{-4} M$$



$$K_p = K_c (RT)^{\Delta n_g} = K_c \times (RT)$$

$$\therefore K_c = \frac{K_p}{RT} = \frac{167}{0.0821 \times 1173} = 1.734 M$$

Ex.6 In the reaction, $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ the concentration of H_2 , I_2 and HI at equilibrium are 10.0, 6.0 and 28 moles per litre respectively. What will be the equilibrium constant?

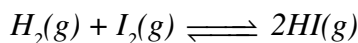
(A) 30.61

(B) 13.066

(C) 29.40

(D) 20.90

Sol.(B)



Applying law of mass action,

$$K_c = \frac{[HI]^2}{[H_2][I_2]}$$

Given $[H_2] = 10 \text{ mol L}^{-1}$
 $[I_2] = 6.0 \text{ mol L}^{-1}$
 $[HI] = 28.0 \text{ mol L}^{-1}$

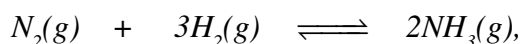
So, $K_c = \frac{(28.0)^2}{(10) \times (6.0)} = 13.066$

Ex.7 For a gas phase reaction at equilibrium,

$3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g)$, the partial pressures of H_2 and N_2 are 0.4 and 0.8 atmosphere respectively. The total pressure of the entire system is 2.4 atmosphere. What will be the value of K_p if all the pressures are given in atmosphere ?

- (A) 32 atm^{-2} (B) 20 atm^{-2} (C) 28.125 atm^{-2} (D) 80 atm^{-2}

Sol.(C)



Partial pressures at equilibrium

$$0.8 \quad 0.4 \quad [2.4 - (0.8 + 0.4) = 1.2]$$

Applying law of mass action,

$$K_p = \frac{[P_{NH_3}]^2}{[P_{N_2}][P_{H_2}]^3} = \frac{1.2 \times 1.2}{0.8 \times 0.4 \times 0.4 \times 0.4} \Rightarrow K_p = 28.125 \text{ atm}^{-2}$$

Ex.8. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; Correct option for reaction is/are :

- (a) $K_p = K_c$
 (b) $K_p > K_c$
 (c) $K_p < K_c$
 (d) any of these, depending on temperature.

Sol.(D)

If (d) option is not given, then the answer is (b) because $\frac{1}{R} \approx 12.2K$ which is very low relative to room temperature.

Ex.9. For which of the following reactions, $K_p > K_c$ at 25°C (298K)

- (a) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
 (b) $H_2(g) + I_2(s) \rightleftharpoons 2HI(g)$
 (c) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
 (d) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
 (e) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

Sol. (b, d, e)

Ex.10. For a gaseous reaction $K_p = 0.4 \text{ atm}^3$ at 27°C . Calculate K_c .

Sol. $K_p = K_c (RT)^{\Delta n}$

$$0.4 = K_c (0.0821 \times 300)^3$$

$$K_c = 2.6 \times 10^{-8} \text{ M}^3$$

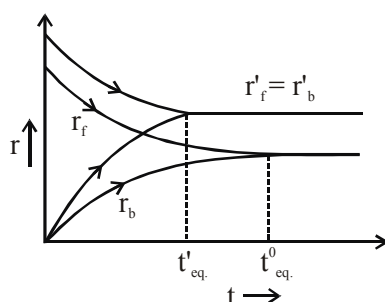
6. Characteristics of equilibrium constant :

- The expression for equilibrium constant K is applicable only when concentrations of the reactants and products have attained their equilibrium values and do not change with time.
- The value of equilibrium constant is independent of initial concentration of the reactants and product.
- Equilibrium constant has one unique value for a particular reaction represented by a balanced equation at a given temperature.
- Value of equilibrium constant is not affected by catalyst. Catalyst simply helps in attaining equilibrium earlier.

The relative increase in the rate of forward as well as backward reaction remain same on using the catalyst.

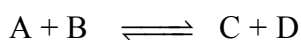
$$K_{\text{eq. uncatalyst}} = \frac{K_f}{K_b}$$

$$K_{\text{eq. catalyst}} = \frac{K_f \times x}{K_b \times x} = \frac{K_f}{K_b} = K_{\text{eq. (uncatalyst)}}$$



6.1 Factor affecting the equilibrium constant :

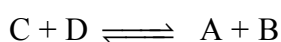
(A) Mode of representation of the reaction :



The equilibrium constant for the reaction

$$K_c = \frac{[C][D]}{[A][B]}$$

If the reaction is reversed



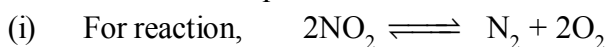
$$\text{then, } K'_c = \frac{[A][B]}{[C][D]}$$

The two equilibrium constant related as ; $K_c = \frac{1}{K'_c}$

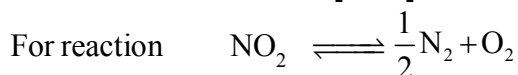
If the reaction is reversed, the value of the equilibrium constant is inverted.

(B) Stoichiometry of the reaction :

When a reversible reaction can be written with the help of two or more stoichiometric equation, the value of equilibrium constant will be numerically different.

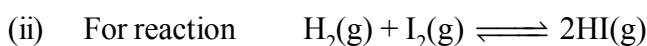


$$K_c = \frac{[\text{N}_2][\text{O}_2]^2}{[\text{NO}_2]^2}$$

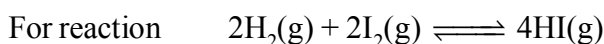


$$K'_c = \frac{[\text{N}_2]^{\frac{1}{2}}[\text{O}_2]}{[\text{NO}_2]}$$

The two constants are related as $K'_c = \sqrt{K_c}$

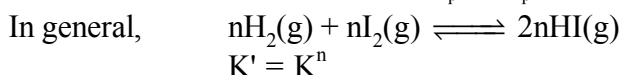


$$K_p = \frac{P_{\text{HI}}^2}{P_{\text{H}_2} \cdot P_{\text{I}_2}}$$



$$K'_p = \frac{P_{\text{HI}}^4}{P_{\text{H}_2}^2 \cdot P_{\text{I}_2}^2}$$

The two constants are related as $K'_p = K_p^2$.



When the coefficient of a balanced equation are multiplied by a common factor 'n', the equilibrium constant must be raised to the respective factor, K_{eq}^n .

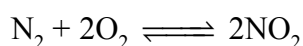
(C) Addition the reaction :



For the 1st step, $K_1 = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$

For the 2nd step, $K_2 = \frac{[\text{NO}_2]^2}{[\text{NO}]^2[\text{O}_2]}$

from eq. (i) + (ii)

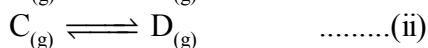
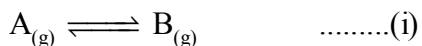


For this reaction, $K = \frac{[\text{NO}_2]^2}{[\text{N}_2][\text{O}_2]^2}$
 the above reaction is related as

$$\therefore K_1 \times K_2 = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]} \times \frac{[\text{NO}_2]^2}{[\text{NO}]^2[\text{O}_2]} = \frac{[\text{NO}_2]^2}{[\text{N}_2][\text{O}_2]^2} = K$$

If there is addition of two reaction then the equilibrium constant is multiplied.

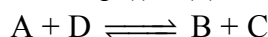
(D) Subtract the reaction :



For the (i) step, $K_1 = \frac{[B]}{[A]}$

For the (ii) step, $K_2 = \frac{[D]}{[C]}$

from eq. (i) – (ii)



For this reaction, $K = \frac{[B][C]}{[A][D]}$

the above reaction is related as, $K = \frac{K_1}{K_2} = \frac{[B]}{[A]} \times \frac{[C]}{[D]} = K$

If we subtract of two reaction, then the equilibrium constant will be divided.

Ex.11 What should be the equilibrium constant for the reaction $HI \rightleftharpoons \frac{1}{2}H_2 + \frac{1}{2}I_2$ if the equilibrium constant for the reaction $H_2 + I_2 \rightleftharpoons 2HI$ at $444^\circ C$ is 64 ?

(A) 64

(B) 8

(C) 0.12

(D) 0.81



$$K_1 = \frac{[HI]^2}{[H_2][I_2]} \quad K_2 = \frac{[H_2]^{1/2}[I_2]^{1/2}}{[HI]}$$

Therefore, $K_2 = \frac{1}{\sqrt{K_1}} = \frac{1}{\sqrt{64}} = \frac{1}{8} = 0.12$

(E) Equilibrium constant depends upon the temperature.

The equilibrium constant of a particularly balanced reaction depends only on temperature. It is independent from all other factor like amount of components, concentration, pressure, volume etc.

$$K_{eq.} = \frac{K_f}{K_b}$$

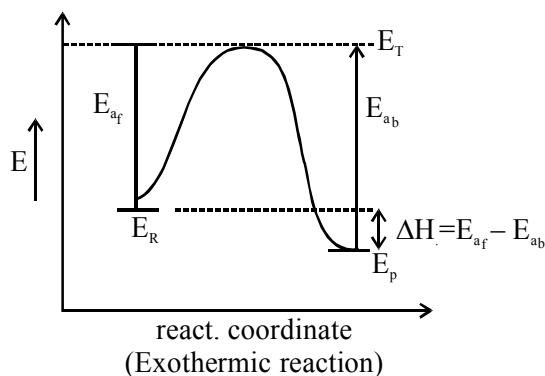
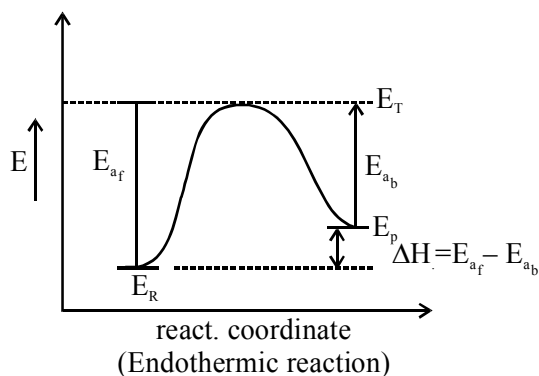
And rate constant, $K = A \cdot e^{-E_a/RT}$

Now, $K_{eq.} = \frac{K_f}{K_b} = \frac{A_f \cdot e^{-E_{af}/RT}}{A_b \cdot e^{-E_{ab}/RT}} = \frac{A_f}{A_b} \cdot e^{-(E_{af}-E_{ab})/RT}$

$$K_{eq.} = A \cdot e^{-\Delta H/RT} \quad (\text{Van't Hoff's equation})$$

where $A = \frac{A_f}{A_b} = \text{constant}$

and $\Delta H = E_{af} - E_{ab}$



If K_1 and K_2 be the equilibrium constants of a reaction at absolute temperatures T_1 and T_2 , then

$$\ln \frac{K_2}{K_1} = \frac{\Delta H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

On increasing the temperature the rate constant of reaction increases. But the reaction with higher activation energy is more sensitive towards temperature change. For the reaction of higher activation energy the value of rate constant increases largely on increasing temperature as well as the rate constant decreases largely as decrease in temperature.

For endothermic reaction

$$\begin{aligned} \Delta H &= E_{a_f} - E_{a_b} = +ve \\ \Rightarrow T \uparrow ; K_{eq.} &= \frac{K_f \uparrow}{K_b \uparrow} \quad (K_f \text{ larger increasing and hence } K_{eq.} \text{ increase}) \end{aligned}$$

For exothermic reaction

$$\begin{aligned} \Delta H &= E_{a_f} - E_{a_b} = -ve \\ \Rightarrow T \uparrow ; K_{eq.} &= \frac{K_f \uparrow}{K_b \uparrow} \quad (K_b \text{ larger increasing and hence } K_{eq.} \text{ decreases}) \end{aligned}$$

Ex.12 The equilibrium constant for the reaction $H_2(g) + S(s) \rightleftharpoons H_2S(g)$ is 18.5 at 925 K and 9.25 at 1000 K respectively. The enthalpy of the reaction will be :

(A) $-68000.05 \text{ J mol}^{-1}$

(B) $-71080.57 \text{ J mol}^{-1}$

(C) $-80071.75 \text{ J mol}^{-1}$

(D) $57080.75 \text{ J mol}^{-1}$

Sol.(B)

Using the relation,

$$\log \frac{K_2}{K_1} = \frac{\Delta H}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

$$\log \frac{9.25}{18.5} = \frac{\Delta H}{2.303 \times 8.314} \times \frac{75}{925 \times 1000}$$

$$-0.301 = \frac{\Delta H \times 75}{2.303 \times 8.314 \times 925 \times 1000} \Rightarrow \Delta H = -71080.57 \text{ J mol}^{-1}.$$

Ex.13. The equilibrium $2A + B_2 \rightleftharpoons 2AB$ involves the two component equilibria,

$A + B_2 \rightleftharpoons AB + B$ and $A + B \rightleftharpoons AB$ with respective equilibrium constants K_1 and K_2 . How is the equilibrium constant, K , for the overall equilibrium related to K_1 and K_2 ?

- (A) $K = K_1 + K_2$ (B) $K = K_1 K_2$ (C) $K = K_1 / K_2$ (D) $K / 2 = K_1 K_2$

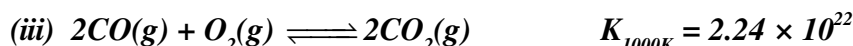
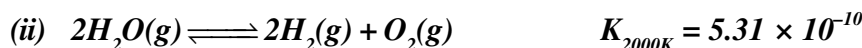
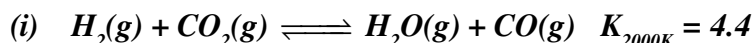
Sol. (B)

Ex.14. For the reactions :



Sol. 9.04×10^{-16}

Ex.15. From the following data :



State whether the reaction (iii) is exothermic or endothermic?

Sol. Equation (iii) = $-[2 \times (i) + (ii)]$

$$\therefore K_{2000(iii)} = \frac{1}{K_1^2 K_2} = \frac{1}{(4.4)^2 \times 5.31 \times 10^{-10}} = 9.7 \times 10^7$$

$$\therefore T \uparrow K \downarrow \Rightarrow \text{reaction is exothermic.}$$

7. APPLICATION OF EQUILIBRIUM CONSTANT :

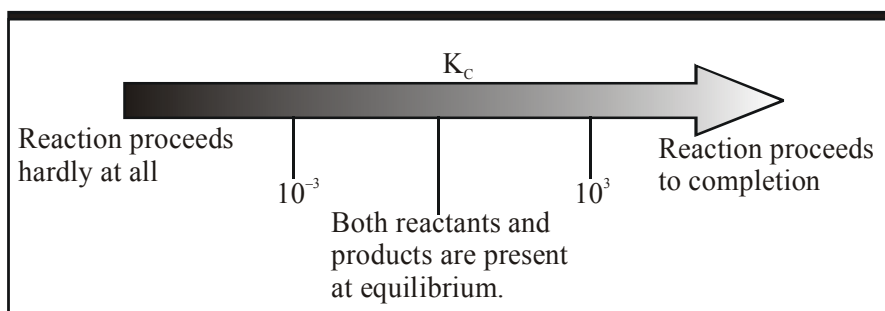
- (I) Predicting the extent of a reaction
- (II) Predicting the direction of the reaction and
- (III) Calculating equilibrium composition.

7.1 Predicting the extent of reaction :

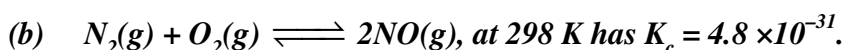
The numerical value of the equilibrium constant for a reaction indicates the extent of the reaction. But it is important to note that an equilibrium does not give any information about the rate at which the equilibrium is reached. The magnitude of K_c or K_p is directly proportional to the concentrations of products (as these appear in the numerator of equilibrium constant expression) and inversely proportional to the concentrations of the reactants (these appear in the denominator). This implies that a high value of K is suggestive of a high concentration of products and vice-versa.

We can make the following generalisations concerning the composition of equilibrium mixtures:

- (i) If $K_c > 10^3$, products predominate over reactants. If K_c is very large, the reaction proceeds almost all the way to completion.
- (ii) If $K_c < 10^{-3}$, reactants predominate over products. If K_c is very small, the reaction proceeds hardly at all.
- (iii) If K_c is in the range 10^{-3} to 10^3 , appreciable concentration of both reactants and products are present.



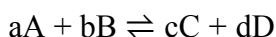
Ex.16 In the following cases which is predominant (reactant or product) for the given value of K_c .



- Sol.**
- (a) If $K_c > 10^3$, products predominate over reactants. If K_c is very large, the reaction proceeds almost all the way to completion.
 - (b) If $K_c < 10^{-3}$, reactants predominate over products. If K_c is very small, the reaction proceeds hardly at all.
 - (c) Also, gas phase decomposition of N_2O_4 to NO_2 is another reaction with a value of $K_c = 4.64 \times 10^{-3}$ at $25^\circ C$ which is neither too small nor too large. Hence, equilibrium mixtures contain appreciable concentrations of both N_2O_4 and NO_2 .

7.2 Predicting the direction of the Reaction :

The equilibrium constant is also used to find in which direction, the reaction mixture of reactants and products will proceed. For this purpose, we calculate the reaction quotient, Q . The reaction quotient is defined in the same way as the equilibrium constant (with molar concentrations to give Q_c , or with partial pressure to give Q_p) at any stage of reaction. For a general reaction:



$$Q_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Then, if $Q_c > K_c$, the reaction will proceed in the direction of reactants (reverse reaction).

if $Q_c < K_c$, the reaction will move in the direction of the products

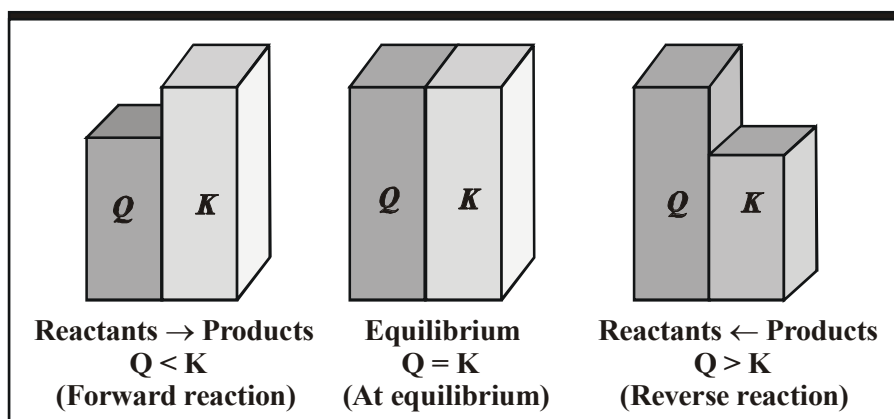
if $Q_c = K_c$, the reaction mixture is already at equilibrium.

In the reaction, $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, if the molar concentrations of H_2 , I_2 and HI are 0.1 M, 0.2 M and 0.4 M, respectively at 783 K, then reaction quotient at this stage of the reaction is

$$Q_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(0.4)^2}{(0.1)(0.2)} = 8$$

K_c for this reaction at 783 K is 46 and we find that $Q_c < K_c$. The reaction, therefore, will move to right i.e. more $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ will react to form more $\text{HI}(\text{g})$ and their concentration will decrease till $Q_c = K_c$.

Thus, a reaction has a tendency to form products if $Q < K$ and to form reactants if $Q > K$.



Ex.17. For the reaction $\text{NOBr}(\text{g}) \rightleftharpoons \text{NO}(\text{g}) + \frac{1}{2} \text{Br}_2(\text{g})$

$K_p = 0.15$ atm at 90°C . If NOBr , NO and Br_2 are mixed at this temperature having partial pressures 0.5 atm, 0.4 atm and 2.0 atm respectively, will Br_2 be consumed or formed ?

Sol.
$$Q_p = \frac{[\text{P}_{\text{Br}_2}]^{1/2} [\text{P}_{\text{NO}}]}{[\text{P}_{\text{NOBr}}]} = \frac{[0.20]^{1/2} [0.4]}{[0.50]} = 0.36$$

$\therefore K_p = 0.15$

Hence, reaction will shift in backward direction

$\therefore \text{Br}_2$ will be consumed

Ex.18. Predict whether 1% CO_2 in air be sufficient to prevent any loss in weight of MgCO_3 or not.

Sol. $\text{MgCO}_3(\text{s}) \rightleftharpoons \text{MgO}(\text{s}) + \text{CO}_2(\text{g}) \quad K_p = 0.095 \text{ atm}$

$$Q = \frac{P_{\text{CO}_2}}{1} = P_{\text{CO}_2} = X_{\text{CO}_2} \cdot 1_{\text{total}}$$

$$\frac{1}{100} \times 1 \text{ atm} = 0.01 \text{ atm} < K_p$$

It will go in forward direction

0.1% CO_2 is not sufficient

Ex.19. Predict whether $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$ be efflorescent or $\text{CuSO}_4 \cdot 3\text{H}_2\text{O}(\text{s})$ be hygroscopic and vapour pressure of H_2O at $25^\circ\text{C} = 0.04 \text{ atm}$.

Sol. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{CuSO}_4 \cdot 3\text{H}_2\text{O}(\text{s}) + 2\text{H}_2\text{O}(\text{g})$ $K_p = 4 \times 10^{-4} \text{ atm}$

Partial pressure of H_2O at eq. = $P_{\text{H}_2\text{O}}$

$$K_p = P_{\text{H}_2\text{O}}^2$$

$$P_{\text{H}_2\text{O}} = \sqrt{4 \times 10^{-4}} = 2 \times 10^{-2} \text{ atm}$$

$$Q = 4 \times 10^{-2} \text{ atm}$$

So, $\text{CuSO}_4 \cdot 3\text{H}_2\text{O}$ be hygroscopic in this condition.

Ex.20. In above problem what maximum humidity for which $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$ work as efflorescent.

Sol. $\left(0.04 \times \frac{x}{100}\right)^2 < 4 \times 10^{-4}$

$$0.16 \times 10^{-4} \times \frac{x^2}{10000} < 4 \times 10^{-4}$$

$$x^2 < 50$$

$$x < 50\%$$

Max. R.H. = 50% Ans.

Ex.21. Study the following reaction



(i) Which of the following is best dehydrating drying agent.

- | | |
|--|----------|
| (A) A. $2\text{H}_2\text{O}(\text{g})$ | (D) A(s) |
| (B) B. $3\text{H}_2\text{O}(\text{g})$ | (E) B(s) |
| (C) C. $4\text{H}_2\text{O}(\text{g})$ | (F) C(s) |

$$A \rightarrow K_p = P_{\text{H}_2\text{O}}^2 \Rightarrow P_{\text{H}_2\text{O}} = 0.02 \text{ atm}$$

$$B \rightarrow K_p = P_{\text{H}_2\text{O}}^3 \Rightarrow P_{\text{H}_2\text{O}} = 0.03 \text{ atm}$$

$$C \rightarrow K_p = P_{\text{H}_2\text{O}}^4 \Rightarrow P_{\text{H}_2\text{O}} = 0.04 \text{ atm}$$

For the best dehydrating agent the moisture remained must be minimum.

(ii) If A(s), B(s) and C(s) are placed in a room where $P_{\text{H}_2\text{O}}(\text{g}) = 0.025 \text{ atm}$, which will act as dehydrating agent.

- | | | | |
|----------|-----------------|----------|---------|
| (A) A(s) | (B) B(s) & C(s) | (C) B(s) | (D) All |
|----------|-----------------|----------|---------|

(iii) Which of the following will be efflorescent at 7.5% R.H. if aqueous terim = 0.04 atm.

- | | |
|--|--|
| (a) Only A. $2\text{H}_2\text{O}(\text{s})$ | (b) Only B. $3\text{H}_2\text{O}(\text{s})$ |
| (c) Only C. $4\text{H}_2\text{O}(\text{s})$ | (d) Only A. $2\text{H}_2\text{O}(\text{s})$ & B. $3\text{H}_2\text{O}(\text{s})$ |
| (e) Only B. $3\text{H}_2\text{O}(\text{s})$ & C. $4\text{H}_2\text{O}(\text{s})$ | |

Sol. (i) D (ii) A (iii) C

7.3

Calculating equilibrium concentrations :

In case of a problem in which we know the initial concentrations but do not know any of the equilibrium concentrations, the following three steps shall be followed :

Step 1. Write the balanced equation for the reaction.

Step 2. Under the balanced equation, make a table that lists for each substance involved in the reaction :

- (a) the initial concentration,
- (b) the change in concentration on going to equilibrium, and
- (c) the equilibrium concentration.

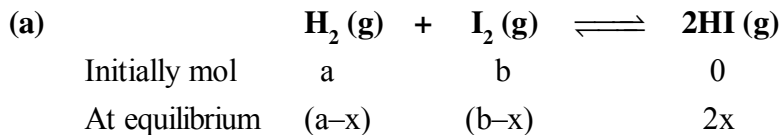
In constructing the table, define x as the concentration (mol/L) of one of the substances that reacts on going to equilibrium, then use the stoichiometry of the reaction to determine the concentrations of the other substances in terms of x .

Step 3. Substitute the equilibrium concentrations into the equilibrium equation for the reaction and solve for x . If you are to solve a quadratic equation choose the mathematical solution that makes chemical sense.

Step 4. Calculate the equilibrium concentrations from the calculated value of x .

Step 5. Check your results by substituting them into the equilibrium equation.

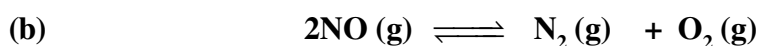
7.3.1 Some General Equilibrium Expressions :



$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{a-x}{V}\right)\left(\frac{b-x}{V}\right)} = \frac{4x^2}{(a-x)(b-x)}$$

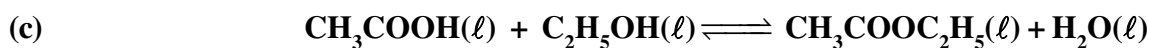
$$K_p = \frac{(p_{\text{HI}})^2}{p_{\text{H}_2} \times p_{\text{I}_2}} = \frac{\frac{(2x)^2}{(a+b)^2} P^2}{\left(\frac{a-x}{a+b} \cdot P\right)\left(\frac{b-x}{a+b} \cdot P\right)} = \frac{4x^2}{(a-x)(b-x)}$$

So $K_c = K_p$ ($\Delta n_g = 0$)



Initially mol	a	0	0
At equilibrium	(a-x)	x/2	x/2

$$K_c = \frac{[\text{N}_2][\text{O}_2]}{[\text{NO}]^2} = \frac{x/2 \times x/2}{(a-x)^2} = \frac{x^2}{4(a-x)^2} = K_p \quad (\Delta n_g = 0)$$



Initially mol	a	b	0	0
At equilibrium	(a-x)	(b-x)	x	x

$$K_c = \frac{[\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}]}{[\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}]} = \frac{x^2}{(a-x)(b-x)}$$

K_p should not be given for this reaction ,



Initially mol	a	0	0
At equilibrium	(a-x)	x	x
Active mass	$\frac{(a-x)}{v}$	$\frac{x}{v}$	$\frac{x}{v}$

$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{\frac{x}{v} \times \frac{x}{v}}{\frac{(a-x)}{v}} = \frac{x^2}{(a-x)v}$$

$$K_p = \frac{p_{\text{PCl}_3} \times p_{\text{Cl}_2}}{p_{\text{PCl}_5}} = \frac{\left(\frac{x}{a+x} \cdot P\right) \times \left(\frac{x}{a+x} \cdot P\right)}{\left(\frac{a-x}{a+x}\right) P} = \frac{x^2 P}{(a+x)(a-x)} = \frac{x^2 P}{a^2 - x^2}$$

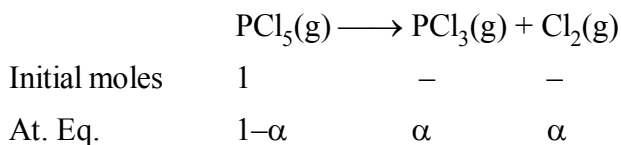


Initially mol	a	b	0
At equilibrium	(a-x)	(b-3x)	2x
Active mass	$\frac{(a-x)}{v}$	$\left(\frac{b-3x}{v}\right)$	$\left(\frac{2x}{v}\right)$

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = \frac{\left(\frac{2x}{v}\right)^2}{\left(\frac{a-x}{v}\right)\left(\frac{b-3x}{v}\right)^3} = \frac{4x^2 V^2}{(a-x)(b-3x)^3}$$

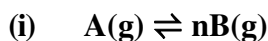
$$K_p = \frac{(p_{\text{NH}_3})^2}{p_{\text{N}_2} \times (p_{\text{H}_2})^3} = \frac{\left[\frac{2xP}{a+b-2x}\right]^2}{\left[\frac{(a-x)P}{(a+b-2x)}\right]\left[\frac{(b-3x)P}{(a+b-2x)}\right]^3} = \frac{4x^2(a+b-2x)^2}{(a-x)(b-3x)^3 P^2}$$

(g) Equilibrium constant expressions in term of ' α '



$$K_C = \frac{\alpha^2}{1-\alpha} \left(\frac{1}{V} \right) \quad \text{and} \quad K_P = \frac{\alpha^2}{1-\alpha^2} P_T$$

(h) Degree of dissociation in terms of molar mass and vapour density

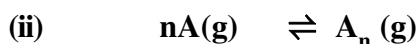


$$\alpha = \frac{M_T - M_O}{M_O(n-1)} \quad \text{or} \quad \alpha = \frac{D_T - D_O}{D_O(n-1)}$$

M_T = Theoretical molar mass of reactant

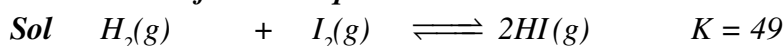
M_O = Observed molar mass of mixture

$$D_T = \frac{M_T}{2} \quad \text{and} \quad D_O = \frac{M_O}{2}$$



$$\alpha = \frac{M_T - M_O}{M_O \left(\frac{1}{n} - 1 \right)}$$

Ex.22 5 moles H_2 gas and 5 moles iodine-vapours are taken in vessel of 10 L capacity. Determine the moles of each at equilibrium.



$$49 = \frac{(2x)^2 v^2}{(5-x)(5-x)v^2}$$

$$49 = \frac{(2x)^2}{(5-x)^2}$$

$$7 = \frac{2x}{5-x}$$

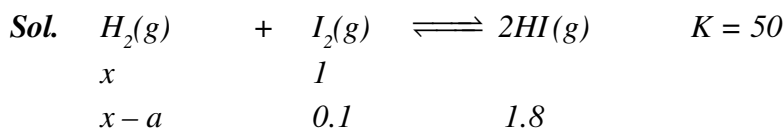
$$35 - 7x = 2x$$

$$35 = 9x$$

$$x = \frac{35}{9}$$

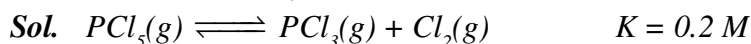
$$\text{Moles at equilibrium} = \frac{10}{9}, \frac{10}{9}, \frac{70}{9}$$

Ex.23 How many moles of $H_2(g)$ should be mixed with each mole I_2 vapours in order to convert 90% of it into HI .



$$50 = \frac{3.24}{(x-0.9)(0.1)} \Rightarrow x = 1.548$$

Ex.24 2 moles of $PCl_5(g)$ is taken in 10 l vessel. Calculate its concentration at equilibrium.



$$2 - x \quad x \quad x$$

$$\frac{2-x}{V} \quad \frac{x}{V} \quad \frac{x}{V}$$

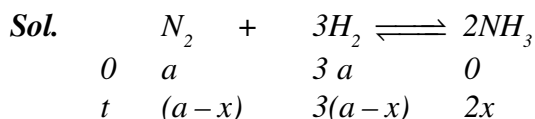
$$0.2 = \frac{\left(\frac{x}{V}\right)^2}{\left(\frac{2-x}{V}\right)} \Rightarrow x = 1.24$$

$$[PCl_5] = 0.076 M$$

$$[PCl_3] = 0.124 M$$

$$[Cl_2] = 0.124 M$$

Ex.25 1 : 3 molar ratio mixture of N_2 and H_2 yields 20% by mole NH_3 at 30 atm; calculate K_p for the equilibrium represented by



$$\text{Total moles} = 4a - 2x$$

$$X_{NH_3} = 0.2 = \frac{2x}{4a-2x} \Rightarrow 2x = 0.8a - 0.4x \Rightarrow x = a/3$$

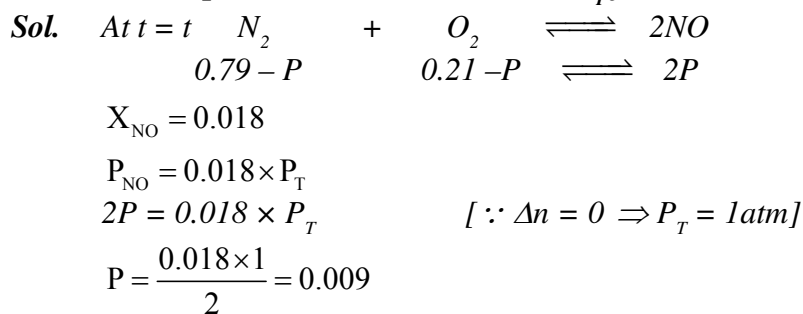
$$P_{NH_3} = 0.2 \times 30 = 6 \text{ atm}$$

$$P_{NH_3} = X_{N_2} \times P_T = \frac{a-x}{4a-2x} \times 30 = 0.2 \times 30 = 6 \text{ atm}$$

$$P_{H_2} = \frac{3(a-x)}{4a-2x} = 0.6 \times 30 = 18 \text{ atm}$$

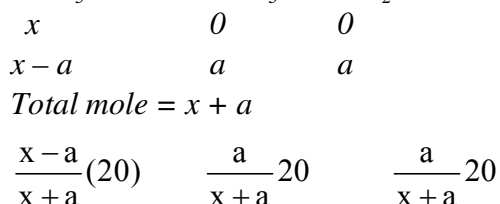
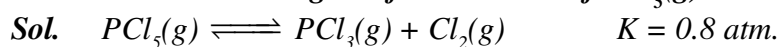
$$\therefore K_p = \frac{[P_{NH_3}]^2}{[P_{N_2}][P_{H_2}]^3} = \frac{6}{6 \times (18)^2} = 1.03 \times 10^{-3} \text{ atm}^{-2}$$

Ex.27 An air sample containing 21:79 (mole ratio) of O_2 & N_2 is heated to $2400^\circ C$. If the mole percent of NO at equilibrium is 1.8%, calculate K_p for the reaction $N_2 + O_2 \rightleftharpoons 2NO$.



$$K_p = \frac{[P_{NO}]^2}{P_{N_2} \times P_{O_2}} = \frac{(0.018)^2}{(0.781)(0.201)} = 2.06 \times 10^{-3}$$

Ex.28 Calculate the degree of dissociation of $PCl_5(g)$ at 20 atm.

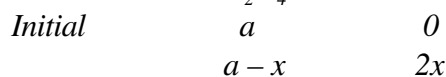


$$0.8 = \frac{\left(\frac{a}{x+a}\right)^2 (20)^2}{\left(\frac{x-a}{x+a}\right) 20} \Rightarrow 0.04 = \frac{a^2}{x^2 - a^2}$$

$$\frac{x}{a} = \frac{1}{\sqrt{26}} = 0.2$$

$$\alpha = 0.2$$

Ex.28 The vapour density a sample of $N_2O_4(g)$ is 40 at 10 atm. Calculate K_p for the reaction.



$$\text{Total mole} = (a - x) + 2x = a + x$$

$$\frac{a-x}{a+x}P \quad \frac{2x}{a+x}P$$

$$V.D. = \frac{M_{avg}}{2}$$

$$M_{avg} = 2 \times 40 = 80$$

$$M_{avg} = \frac{\text{Total mass}}{\text{Total moles}} = \frac{92a}{a+x} = 80$$

$$x = \frac{12}{80}a = 0.15a$$

$$K_p = \frac{4x^2}{a^2 - x^2} = 9.2 \times 10^{-2}$$

Ex.29 At what pressure an equimolar mixture of $\text{PCl}_5(\text{g})$ and $\text{Cl}_2(\text{g})$ should be taken in order to convert 75% of PCl_5 into PCl_3 .

Sol. $\text{PCl}_5(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) \quad K_p = 0.4 \text{ atm}$

Initial	a	0	
Final	$a - 0.75a$ $= 0.25a$	$a - 0.75a$ $= 0.25a$	$0.75a$

Total mole = $1.25a$

Equ. par. pr.	$\frac{0.25a \times P}{1.25a}$	$\frac{0.25a \times P}{1.25a}$	$\frac{0.75a \times P}{1.25}$
	$\frac{P}{5}$	$\frac{P}{5}$	$\frac{3P}{5}$

$$K_p = \frac{P_{\text{PCl}_3}}{P_{\text{PCl}_5} P_{\text{Cl}_2}} \Rightarrow 0.4 = \frac{\frac{3P}{5}}{\frac{P}{5} \times \frac{P}{5}} \Rightarrow P = 37.5$$

But this is pressure at equilibrium but we have to find initial pressure

Now, from $PV = nRT$

$$\frac{P_1}{P_2} = \frac{n_1}{n_2} \Rightarrow \frac{P_1}{37.5} = \frac{2}{1.25} \Rightarrow P = 60 \text{ atm}$$

Alternate method :

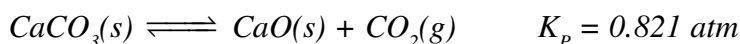
	$\text{PCl}_5(\text{g})$	+	$\text{Cl}_2(\text{g})$	$\rightleftharpoons$	$\text{PCl}_3(\text{g})$
Initial partial pressure	P_0		P_0		0
E.P.P.	$P_0 - 0.75P_0$ $= 0.25P_0$		$P_0 - 0.75P_0$ $= 0.25P_0$		$0.75P_0$

$$0.4 = \frac{0.75P_0}{0.25P_0 \times 0.25P_0} \Rightarrow P_0 = 30 \text{ atm}$$

$\therefore$ Initial total pressure of mixture = $P_0 + P_0 = 60 \text{ atm}$

Note : In any problem related with relative extent of reaction, initial mole may be taken as 1 but the initial pressure or concentration should not be taken as 1.

Ex.30 100 gm CaCO_3 is taken in 30 l empty vessel and the vessel is sealed and the sample is heated to 627°C . Calculate the mass % of CaCO_3 decomposed.



Sol. $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

Initial	1 mole	0	0
Final	$1 - x$	x	x

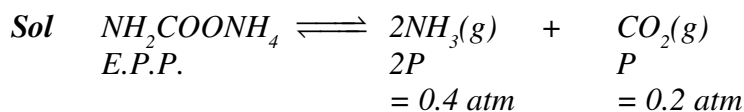
Total mole = $1 + x$

$$K_p = \frac{P_{\text{CO}_2}}{1} = P_{\text{CO}_2} \Rightarrow P_{\text{CO}_2} = \frac{nRT}{V}$$

$$0.821 = \frac{x \times 0.0821 \times 900}{30} \Rightarrow x = \frac{1}{3}$$

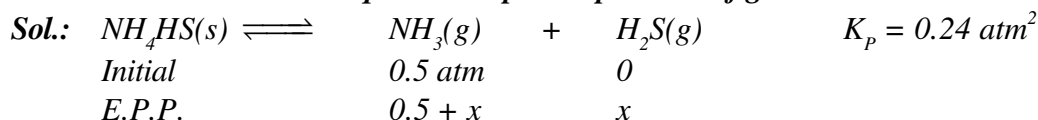
$$\% \text{ of } \text{CaCO}_3 \text{ decomposed} = \frac{1}{3} \times 100 = \frac{100}{3} \%$$

Ex.31 Some solid ammonium carbamate $\text{NH}_4\text{COONH}_4$ is taken in a evacuated vessel and the vessel is sealed after a very long time a constant pressure of 0.6 atm is observed due to dissociation of the solid into NH_3 and CO_2 gas. Calculate the dissociation constant of solid.



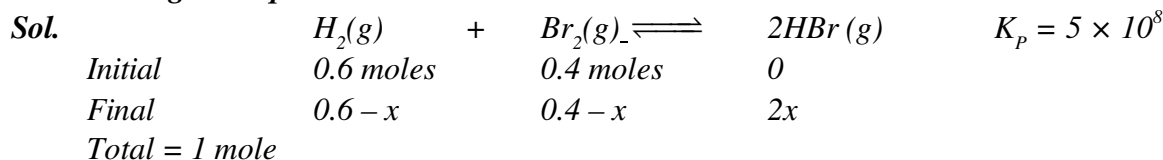
$$K_p = \frac{[0.4]^2 [0.2]}{1} = 0.032 \text{ atm}^3$$

Ex.32 Some solid ammonium hydrogen sulphide NH_4HS is in a vessel containing ammonium gas at 0.5 atm. Calculate the equilibrium partial pressure of gas.



$$K_p = \frac{P_{\text{NH}_3}}{1} \cdot P_{\text{H}_2\text{S}} \Rightarrow 0.24 = (0.5 + x)x \Rightarrow x = 0.3 \text{ atm.}$$

Ex.33 0.6 moles of $\text{H}_2(\text{g})$ and 0.4 moles of Br_2 vapour are allowed to react. Calculate the moles of each gas at equilibrium.



$$K = \frac{(2x)^2}{(0.6-x)(0.4-x)} = \frac{4x^2}{x^2 - x + 0.24}$$

$$(5 \times 10^8 - 4)x^2 - 5 \times 10^8 x + 1.2 \times 10^8 = 0$$

$$5 \times 10^8 x^2 - 5 \times 10^8 x + 1.2 \times 10^8 = 0$$

$$8x^2 - x + 0.24 = 0$$

$$x = 0.4, 0.6 (\text{Not acceptable})$$

As K_p is much larger than 1000 the reaction almost completes.

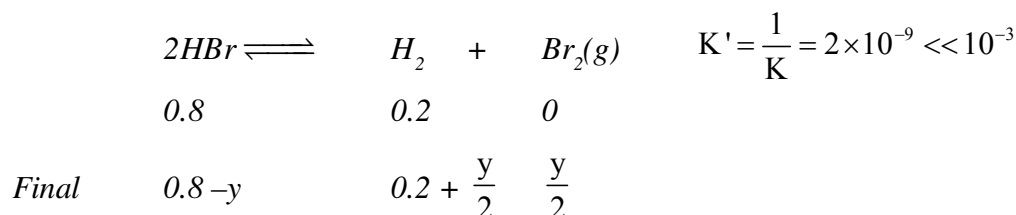
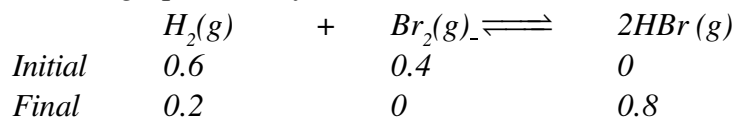
$$\text{Final moles of HBr} = 0.8$$

$$\text{Final moles of H}_2 = 0.2$$

But Final moles of $\text{Br}_2 \neq 0$

Note :

The final moles of Br_2 may be calculated by first completing the reaction in forwards direction and then achieving equilibrium from backward direction.



$$K' = \frac{[H_2][Br_2]}{[HBr]^2} = 2 \times 10^{-9} = \frac{\left(0.2 + \frac{y}{2}\right) \frac{y}{2}}{(0.8 - y)^2}$$

As K' is very small we may assume that $0.8 - y \simeq 0.8$

and, $0.2 + \frac{y}{2} \simeq 0.2$

$$\therefore 2 \times 10^{-9} = \frac{0.2 \times \frac{y}{2}}{0.64}$$

$$y = 1.28 \times 10^{-8}$$

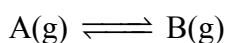
Final moles $HBr = 0.8 - y \simeq 0.8$

$$H_2 = 0.2 + \frac{y}{2} \simeq 0.2$$

$$Br_2 = \frac{y}{2} = 6.4 \times 10^{-9}$$

Ex.34 Calculate concentration of A at equilibrium for a reaction, $A(g) \rightleftharpoons B(g)$; $K_C = 10^{50}$, if the initial concentration of A is 2 M.

Sol.



initial conc.	2	0
Change in conc.	-x	x
equib. conc.	2-x	0+x

K_C is very-very high so maximum reactant converted into product.

$$\therefore \frac{K_C}{\text{initial conc.}} \geq 10^3 \text{ then } \frac{10^{50}}{2} \geq 10^3.$$

So $2-x \simeq 0 \Rightarrow x \simeq 2$ but reactant never zero.

equib. conc. $2-x \simeq y$ $x \simeq 2$

$$K_C = \frac{[B]}{[A]}$$

$$10^{50} = \frac{x}{2-x}$$

$$10^{50} = \frac{2}{y} \Rightarrow y = 2 \times 10^{-50} \text{ M}$$

$$[A] = 2 \times 10^{-50} \text{ M}$$

8. LE CHATELIER'S PRINCIPLE :

Le Chatelier's principle is applicable for any system in equilibrium which states as :

When a system in equilibrium is disturbed by external agency, the system tends to attain again an equilibrium by adjusting itself. These are these some factors by which a system in equilibrium can be disturbed.

- (i) Changing the concentration of reactant and product.
- (ii) Changing the pressure (or volume) of the system.
- (iii) Changing the temperature.

For a chemical reaction in equilibrium, Le Chatelier's principle can be stated as,

If we change concentration, pressure or temperature of a chemical reaction in equilibrium, the equilibrium will shift to the right or the left so as to minimise the change.

8.1 Effect of a change in concentration :

- (a) If more reactant is added or some product is removed from an equilibrium mixture having equilibrium constant K then the reaction moves in the forward direction (as $Q_c < K_c$) to give a new equilibrium and more products are produced.
- (b) If more product are added to or some reactant are removed from an equilibrium mixture, the reaction moves in the reverse direction (as $Q_c > K_c$) to give a new equilibrium and more reactant are produced.

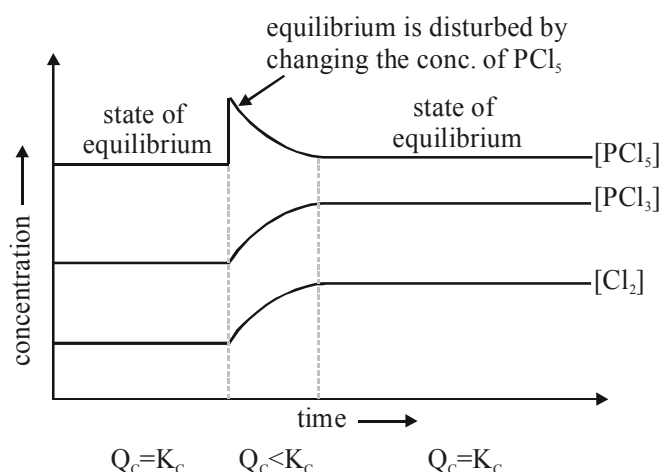
Ex. The decomposition of gaseous PCl_5 is a reversible reaction, $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$. Let the equilibrium concentrations of PCl_5 , PCl_3 and Cl_2 are respectively $[\text{PCl}_5]$, $[\text{PCl}_3]$ and $[\text{Cl}_2]$. The K_c for this reaction can be written as :

$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} \text{ Also we know that at equilibrium } K_c = Q_c.$$

Case-I : Let the concentration of PCl_5 is doubled to disturb the equilibrium. This will change the reaction quotient, Q_c to :

$$Q_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[2\text{PCl}_5]} = \frac{1}{2} \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{K_c}{2}$$

After disturbing the equilibrium, the value of Q_c becomes less than K_c . In order to restore the Q_c value to K_c , the concentration of PCl_5 must be decreased while the concentrations of PCl_3 and Cl_2 are to be increased. This is achieved by favoring the forward reaction.

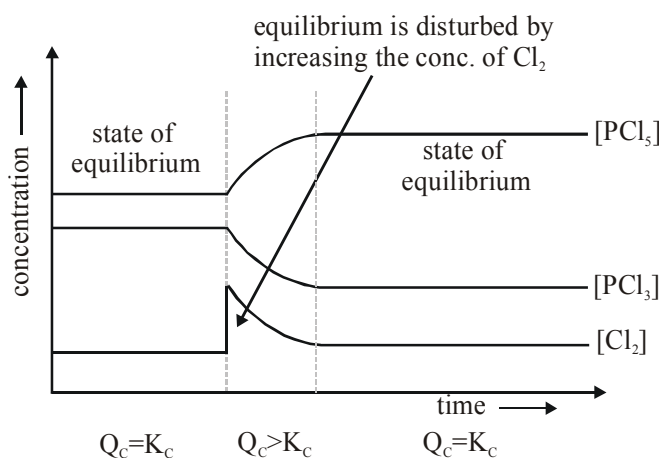


The forward reaction is also favored by removing the products from the reaction mixture (decrease in the concentration of products). Upon removal of products, the rate of forward reaction becomes greater than that of backward reaction momentarily. This will also decrease the reaction quotient. Hence the system tries to reestablish the equilibrium by converting more reactants to products so as to make the rates of both forward and backward reactions become equal again.

Case-II : For example, in case of the decomposition of PCl_5 , if the concentration of Cl_2 is increased by two times at equilibrium, the Q_c value becomes greater than the K_c value.

$$Q_c = \frac{[\text{PCl}_3][2\text{Cl}_2]}{[\text{PCl}_5]} = 2 \times \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = 2K_c$$

Hence the system tries to restore the value of Q_c to K_c again. The backward reaction is favored to decrease the concentration of Cl_2 . However the concentration of PCl_3 also decreases automatically while the concentration of PCl_5 increases while doing so.



8.2 Effect of a change in pressure :

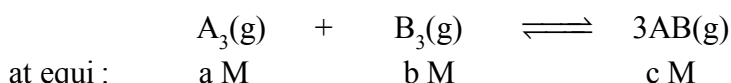
We know that, if we increase the volume of container then the pressure of the container decreases and if we decrease the volume of container then the pressure of the container increases.

$$\text{i.e., Pressure (P)} \propto \frac{1}{\text{Volume (V)}}$$

Hence, if the pressure is decreased by increasing the volume of a reaction mixture, the reaction shifts in the direction of more moles of gas while if the pressure is increased by decreasing the volume of a reaction mixture, the reaction shifts in the direction of fewer moles of gas.

Here moles of reactant or product means co-efficients of reactant or product.

(a) Suppose a general equation



For this reaction, moles of product > moles of reactant i.e., $[\Delta n = (3 - 2) > 0]$

$$K_c = \frac{(c)^3}{a \times b}$$

If we increase the pressure of the system from 0.5 atm pressure to 1 atm pressure by decreasing the volume of container from 5 litre to 2.5 litre. Hence, concentration of all the reactant and product will change and it will be greater than the initial value. Hence we can say that,

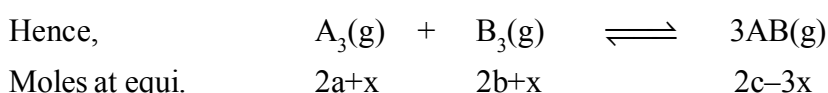
If pressure becomes double then volume becomes half and hence concentration becomes double as

$$(\text{Concentration} \propto \frac{\text{Moles}}{\text{Volume}})$$

$\therefore$ Concentration $\propto$ Pressure

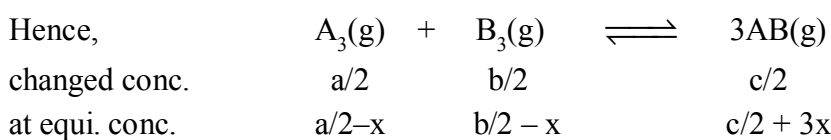
$$Q_c = 2 \times \frac{c^3}{a \times b} > K_c$$

Hence reaction will move from right to left i.e., in the backward direction.



It can be simply said that, reaction will move from right to left by increasing the pressure of the system as moles of product > moles of reactant i.e., move in the direction of fewer moles.

(b) If pressure decreases from 1 atm to 0.5 atm. Volume will increase by factor 2 concentration will decrease by factor 1/2



$$Q_c = \frac{(c/2)^3}{a/2 \times b/2} = \frac{1}{2} \frac{c^3}{a \times b} < K_c$$

$$\text{i.e., } Q_c < K_c$$

Therefore, reaction will move from left to right and as moles of product > moles of reactants, therefore reaction will move from left to right by decreasing the pressure of the system i.e. move in the direction of greater moles.

- (c) If moles of reactant = moles of product i.e., $\Delta n = 0$ then change in pressure of equilibrium mixture has no effect i.e. at this position $Q_c = K_c$ will always exist.

8.3 Effect of a change in temperature :

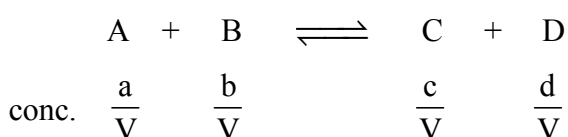
On increasing the temperature equilibrium shifts in the direction of absorption of heat (endothermic direction)

8.4 EFFECT OF INERT GAS AT THE EQUILIBRIUM :

(i) Effect of inert gas at constant volume :

When the addition of inert gas (non-reacting gas) is carried out at constant volume (V), the equilibrium remains unaffected for reactions whether they have $\Delta n = 0$ or $\Delta n \neq 0$.

Let us consider a general equation at temperature T K, and a, b, c and d are the moles of A, B, C and D respectively,



Since, V = constant, so addition of inert gas has no effect on equilibrium

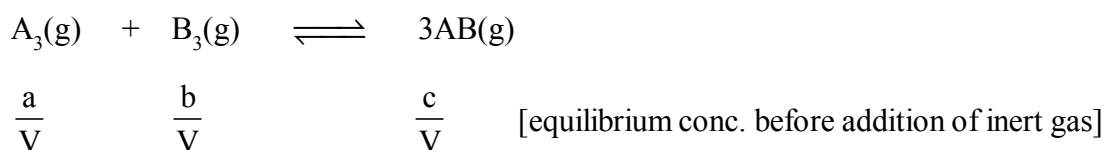
$\therefore \frac{a}{V}, \frac{b}{V}, \frac{c}{V}$ and $\frac{d}{V}$ will not change

$$Q_c = K_c \quad \text{or} \quad Q_p = K_p$$

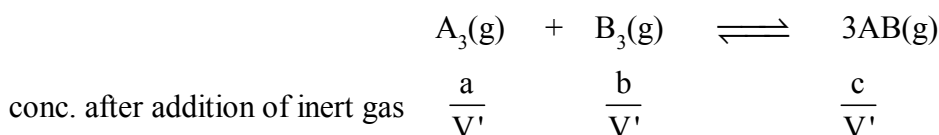
(ii) Effect of inert gas at constant pressure :

- (a) The addition of an inert gas at constant pressure to an equilibrium, the equilibrium shifts in the direction of greater number of moles.

At constant pressure, addition of inert gas or non-reacting gas means increase in volume of the system. Suppose a reaction having $\Delta n > 0$ and a, b, c and d are the moles of A, B, C and D respectively and V be the volume before addition of inert gas.



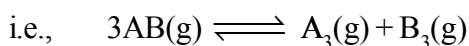
Let us consider V' be the total volume of the system after addition of inert gas So, $V' > V$



$\therefore \frac{a}{V'} < \frac{a}{V}, \frac{b}{V'} < \frac{b}{V}$ and $\frac{c}{V'} < \frac{c}{V}$

$$\text{Now, } K_c = \frac{\left(\frac{c}{V}\right)^3}{\left(\frac{a}{V}\right)\left(\frac{b}{V}\right)} = \frac{1}{V} \times \frac{c^3}{a \times b} \quad \text{and} \quad Q_c = \frac{\left(\frac{c}{V'}\right)^3}{\left(\frac{a}{V'}\right)\left(\frac{b}{V'}\right)} = \frac{1}{V'} \times \frac{c^3}{a \times b}$$

$K_c > Q_c$, hence reaction will move from left to right (forward direction) as $\Delta n > 0$ and similarly we can prove that for a reaction having $\Delta n < 0$

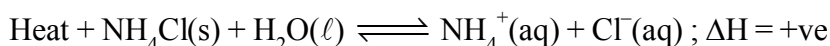


For this reaction $K_c < Q_c$, after addition of inert gas. Hence, reaction will move from right to left (reverse direction) as $\Delta n < 0$.

8.5 APPLICATION OF LE-CHATELIER'S PRINCIPLE :

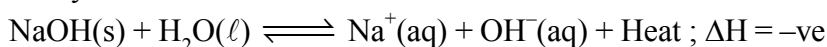
The principle is applicable not only to chemical equilibria but also to physical equilibria in similar way.

- (a) Dissolution of ammonium chloride in water



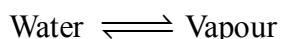
With rise in temperature, equilibrium will shift in direction which will lower the temperature counteract the effect. So, equilibrium shifts in forward direction which is endotherm reaction. Hence, solubility of NH_4Cl increases with rise in temperature.

- (b) Dissolution of sodium hydroxide in water



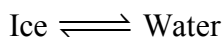
Sodium hydroxide dissolve in water with evolution of heat. Therefore, a rise in temperature will decrease its solubility. On the other hand, a decrease in temperature will increase the solubility of sodium hydroxide and reaction will shift in forward direction.

- (c) Effect of pressure on boiling point :



An increase in pressure will favour backward reaction i.e., the reaction in which volume decreases ($V_{\text{vap.}} > V_{\text{w.}}$). thus more water will exist at equilibrium (B.pt of solvent increase with increase in pressure). So decrease in pressure will shift the reaction forward.

- (d) Effect of pressure on melting point :

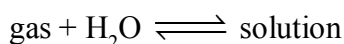


increase in pressure will favour forward reaction because $V_{\text{ice}} > V_{\text{water}}$. Thus more ice melt or the m.pt of ice is lowered with pressure.

For, $\text{Solid} \rightleftharpoons \text{Liquid}$

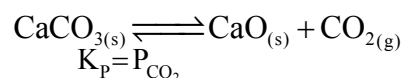
An increase in pressure will favour backward reaction because volume of liquid is more than solid thus more solid will exist at equilibrium (m.pt of solid increases with pressure).

- (e) Effect of pressure on solubility of gases :



increase in pressure favour forward reaction. [Henry's law]

- (f) In solid reactants equilibrium will not shift to the right side even if the more reactants are added.

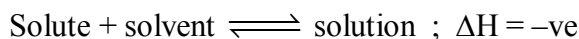


Equilibrium will not be affected by adding CaO or CaCO_3 at that temperature. But if volume is increased the equilibrium will shift to the right side to keep the pressure of CO_2 constant.

- (g) Effect of temperature on solubility



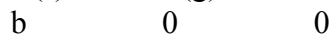
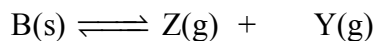
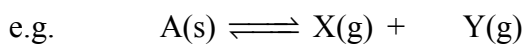
An increase in temperature favour endothermic or forward reaction i.e., solubility increases (urea, glucose).



In exothermic dissolution solubility decreases with temperature i.e., reaction will be in backward direction.

9. SIMULTANEOUS EQUILIBRIUM

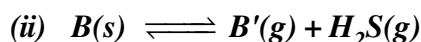
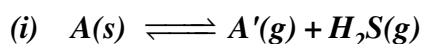
If in any container there are two or more equilibria existing simultaneously involving one or more than one common species, then in both/all the equilibrium, the concentration of common species is the total concentration of that species due to all the equilibria under consideration.



$$K_{C_1} = t(u + t)$$

$$K_{C_2} = (u + t)u$$

Ex.35 Two solid compounds A and B dissociate into gaseous products at 20°C as

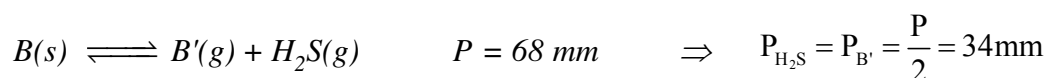
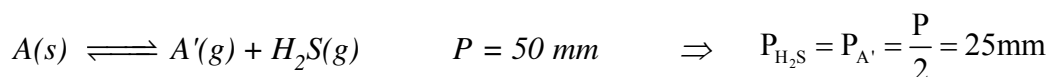
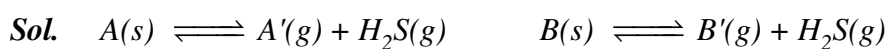


At 20°C, pressure over excess solid A is 50 mm and that over excess solid B is 68 mm find

(a) The dissociation constant of A and B

(b) Relative no. of moles of A and B in the vapour phase over a mixture of solid A and B.

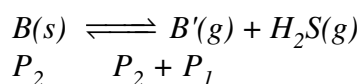
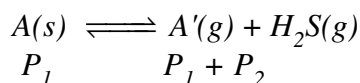
(c) Show that the total pressure of the gas over the solid mixture would be 84.4 mm



(i) $K_{P_1} = (25)^2 = 625 \text{ mm}^2$

$K_{P_2} = (34)^2 = 1156 \text{ mm}^2$

(ii) Ratio of moles is same as that of partial pressure so,



$K_{P_1} = P_{A'} \times P_{H_2S} = P_1(P_1 + P_2)$... (i)

$K_{P_2} = P_{B'} \times P_{H_2S} = P_2(P_1 + P_2)$... (ii)

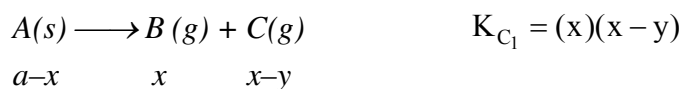
$$\frac{K_{P_1}}{K_{P_2}} = \frac{P_1}{P_2} = \frac{625}{1156}$$

(iii) total pressure = $P_1 + P_2 + (P_1 + P_2) = 2(P_1 + P_2)$

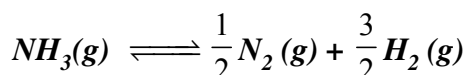
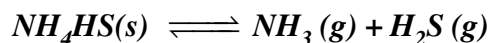
(i) + (ii) = $(P_1 + P_2)^2$

$\sqrt{K_{P_1} + K_{P_2}} = P_1 + P_2 \Rightarrow P_T = \sqrt{2(K_{P_1} + K_{P_2})} = 84.4 \text{ mm}$

10. SEQUENTIAL EQUILIBRIUM

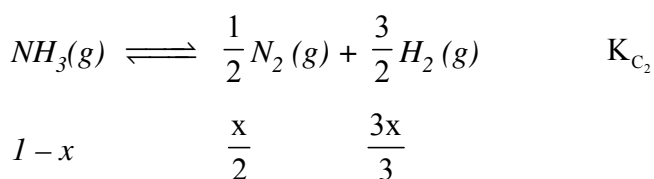
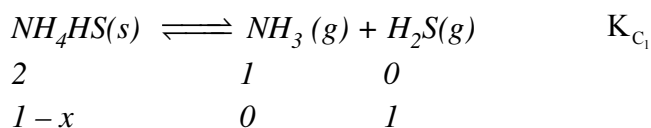


Ex36. 102 g of solid NH_4HS is taken in the 2L evacuated flask at $57^\circ C$. Following two equilibrium exist simultaneously



one mole of the solid decomposes to maintain both the equilibrium and 0.75 mole of H_2 was found at the equilibrium then find the equilibrium concentration of all the species and K_C for both the reaction.

Sol. Moles of $NH_4HS = \frac{102}{51} = 2$

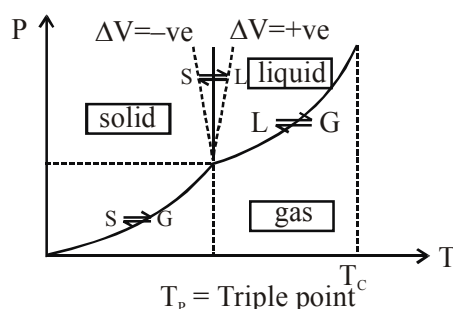


$$\text{Given that moles of } H_2 = \frac{3x}{2} = 0.75 \Rightarrow x = \frac{1}{2}$$

$$K_{C_1} = \frac{1}{2} \frac{(1-x)}{2} = \frac{1}{8} \quad [\text{Since } V = 2 L]$$

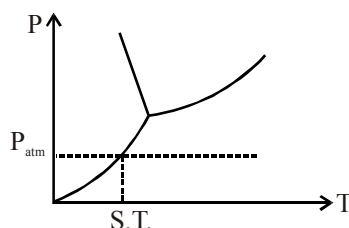
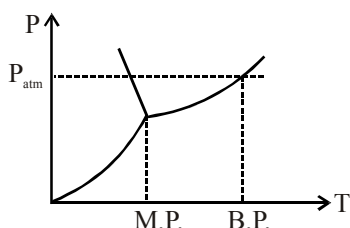
$$K_{C_2} = \frac{\left(\frac{3x}{2}\right)^{3/2} \left(\frac{x}{2}\right)^{1/2}}{\left(\frac{1-x}{2}\right)} = \frac{\left(\frac{3}{2}\right)^{3/2} \left(\frac{1}{2}\right)^{1/2}}{\left(\frac{1}{2}\right)} = (3)^{3/2} \frac{1}{64} \times \frac{4}{1} = \frac{(3)^{3/2}}{16}$$

11. PHYSICAL EQUILIBRIUM



- (i) The combination of temperature and pressure at which all the three physical state of matter co-exist is called triple point of that matter.
- (ii) The vapour pressure solid is minimum (0) at absolute zero and maximum at triple point.
- (iii) The vapour pressure of liquid is minimum at triple point and maximum at critical temperature.
- (iv) On each line in the graph, the matter exist in two physical states is equilibrium.
- (v) In between two lines, the matter exist in only one physical state.
- (vi) The melting point of solid and the triple point of matter differs slightly.
- (vii) Any solid may be directly converted into gaseous state or through liquid state by adjusting the external pressure relative to triple point pressure.

(viii)



EXERCISE # S-I

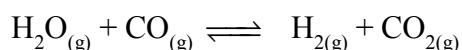
- Which of the following reactions goes almost all the way to completion, and which proceeds hardly at all?
 - $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}); \quad K_c = 2.7 \times 10^{-18}$
 - $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g}); \quad K_c = 6.0 \times 10^{13}$
- Write down the expression for equilibrium constant K_c and K_p for the following reactions :
 - $\text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$
 - $\text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_2\text{O}(\text{g})$
 - $\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g})$
 - $\text{CH}_3\text{COOH}(\ell) + \text{C}_2\text{H}_5\text{OH}(\ell) \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5(\ell) + \text{H}_2\text{O}(\ell)$
 - $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$
(excess)
 - $\text{Zn}(\text{s}) + 2\text{H}^+(\text{aq}) \rightleftharpoons \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
- Write the relationship between K_p and K_c for the following reactions :
 - $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
 - $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
 - $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 - $2\text{H}_2\text{O}(\text{g}) \rightleftharpoons 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$
 - $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$
- Find stability of reactants for the given equilibrium reaction :

$$2\text{XO}(\text{g}) \rightleftharpoons \text{X}_2(\text{g}) + \text{O}_2(\text{g}) \quad ; \quad K_1 = 1 \times 10^{24}$$

$$2\text{XO}_2(\text{g}) \rightleftharpoons \text{X}_2(\text{g}) + 2\text{O}_2(\text{g}) \quad ; \quad K_2 = 2.5 \times 10^{10}$$
- The value of K_c for the reaction $2\text{A} \rightleftharpoons \text{B} + \text{C}$ is 2×10^{-3} . At a given time, the composition of reaction mixture is $[\text{A}] = [\text{B}] = [\text{C}] = 3 \times 10^{-4} \text{ M}$. In which direction the reaction will proceed?
- The ester or ethyl acetate is formed by the reaction between ethanol and acetic acid and equilibrium is represented as :

$$\text{CH}_3\text{COOH}_{(\text{l})} + \text{C}_2\text{H}_5\text{OH}_{(\text{l})} \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5_{(\text{l})} + \text{H}_2\text{O}_{(\text{l})}$$
 - Write the concentration ratio (reaction quotient), Q_c for this reaction. (Note that water is not in excess and is not a solvent in this reaction).
 - At 293 K, if starts with 1.00 mole of acetic acid and 0.180 moles of ethanol, there is 0.171 moles of ethyl acetate in the final equilibrium mixture. Calculate the equilibrium constant.
 - Starting with 0.500 moles of ethanol and 1.000 mole of acetic acid and maintaining it at 293 K, 0.214 moles of ethyl acetate is found after some time. Has equilibrium been reached ?

7. If $K_c = 0.5 \times 10^{-9}$ at 1000 K for the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, what is K_c at 1000 K for the reaction $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$?
8. An equilibrium mixture of PCl_5 , PCl_3 and Cl_2 at a certain temperature contains 0.5×10^{-3} M PCl_5 , 1.5×10^{-2} M PCl_3 , and 3×10^{-2} M Cl_2 . Calculate the equilibrium constant K_c for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$.
9. A sample of HI (9.6×10^{-3} mol) was placed in an empty 2.00 L container at 1000 K. After equilibrium was reached, the concentration of I_2 was 4×10^{-4} M. Calculate the value of K_c at 1000 K for the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$.
10. The vapour pressure of water at 27°C is 0.2463 atm. Calculate the values of K_p and K_c at 27°C for the equilibrium $H_2O(l) \rightleftharpoons H_2O(g)$.
11. The value of K_c for the reaction $3O_2(g) \rightleftharpoons 2O_3(g)$ is 8×10^{-55} at 25°C . Do you expect pure air at 25°C to contain much O_3 (ozone) when O_2 and O_3 are in equilibrium? If the equilibrium concentration of O_2 in air at 25°C is 8×10^{-3} M, what is the equilibrium concentration of O_3 ?
12. One mole of H_2O and one mole of CO are taken in a 10 L vessel and heated to 725 K. At equilibrium 40 percent of water (by mass) reacts with carbon monoxide according to the equation.



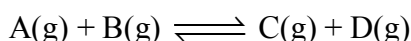
Calculate the equilibrium constant for the reaction.

13. A vessel at 1000 K contains CO_2 with a pressure of 0.6 atm. some of the CO_2 is converted into CO on addition of graphite. Calculate the value of K , if total pressure at equilibrium is 0.9 atm.
14. Calculate the equilibrium concentration of H_2 , I_2 and HI at 300 K if initially 2 mol of H_2 and I_2 are taken in a closed container of having volume 10 lit. [Given : $H_2 + I_2 \rightleftharpoons 2HI$; $K = 100$ at 300 K]
15. A mixture of SO_3 , SO_2 and O_2 gases is maintained at equilibrium in 10 litre flask at a particular temperature in which K_c for the reaction,
 $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is $100 \text{ mol}^{-1} \text{ litre}$.

At equilibrium,

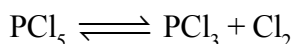
- (a) If number of mole of SO_3 and SO_2 in flask are same, how many mole of O_2 are present ?
- (b) If number of mole of SO_3 in flask are twice the number of mole of SO_2 , how many mole of O_2 are present ?

16. For the given reaction,



the number of moles at equilibrium was found to be 4, 4, 2 & 8 respectively in 1 L container. Find the new equilibrium concentration if 18 moles of D are added to above 1 L container.

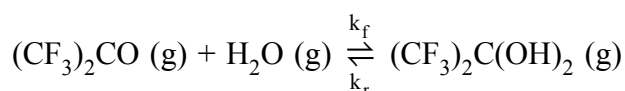
17. For the given reaction,



the number of moles at equilibrium was found to be 2, 6 & 4 respectively at 10 atm pressure. Find the new equilibrium moles & partial pressure, if it is reestablished at an equilibrium pressure 50 atm.

18. For the dissociation reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ derive the expression for the degree of dissociation in terms of K_p and total pressure P .
19. For the reaction $2\text{SO}_{3(\text{g})} \rightleftharpoons 2\text{SO}_{2(\text{g})} + \text{O}_{2(\text{g})}$
Initially 5 moles of SO_3 is taken and at equilibrium 4 moles of SO_3 is dissociated then find out α and the value of K_c (Volume = 2 litre)
20. 1.1 mole of A mixed with 2.2 mole of B and the mixture is kept in a 1 litre flask at the equilibrium, $\text{A} + 2\text{B} \rightleftharpoons 2\text{C} + \text{D}$ is reached. At equilibrium 0.2 mole of C is formed then the value of K_c will be.
21. The air pollutant NO is produced in automobile engines from the high temperature reaction $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$; $K_c = 16$ at 2300 K. If the initial concentrations of N_2 and O_2 at 2300 K are both 1.5 M, what are the concentrations of NO, N_2 , and O_2 when the reaction mixture reaches equilibrium?
22. At 700 K, $K_p = 0.2$ for the reaction $\text{ClF}_3(\text{g}) \rightleftharpoons \text{ClF}(\text{g}) + \text{F}_2(\text{g})$. Calculate the equilibrium partial pressure of ClF_3 , ClF , and F_2 if only ClF_3 is present initially, at a partial pressure of 0.150 atm.
23. The degree of dissociation of N_2O_4 into NO_2 at 1.5 atmosphere and 40°C is 0.25. Calculate its K_p at 40°C . Also report degree of dissociation at 10 atmospheric pressure at same temperature.
24. At 46°C , K_p for the reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ is 0.667 atm. Compute the percent dissociation of N_2O_4 at 46°C at a total pressure of 380 Torr.
25. $2\text{NOBr}(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{Br}_2(\text{g})$. If nitrosyl bromide (NOBr) is 33.33% dissociated at 25°C & a total pressure of 0.28 atm. Calculate K_p for the dissociation at this temperature.
26. The system $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ maintained in a closed vessel at 60°C & a pressure of 5 atm has an average (i.e. observed) molecular weight of 69, calculate K_p . At what pressure at the same temperature would the observed molecular weight be (230/3)?
27. In the esterification $\text{C}_2\text{H}_5\text{OH}(\text{l}) + \text{CH}_3\text{COOH}(\text{l}) \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5(\text{l}) + \text{H}_2\text{O}(\text{l})$ an equimolar mixture of alcohol and acid taken initially yields under equilibrium, the water with mole fraction = 0.333. Calculate the equilibrium constant.
28. The equilibrium constant for the reaction is 9 at 900°C $\text{S}_2(\text{g}) + \text{C}(\text{s}) \rightleftharpoons \text{CS}_2(\text{g})$. Calculate the pressure of two gases at equilibrium, when 2 atm of S_2 and excess of $\text{C}(\text{s})$ come to equilibrium.
29. A sample of $\text{CaCO}_3(\text{s})$ is introduced into a sealed container of volume 0.821 litre & heated to 1000K until equilibrium is reached. The equilibrium constant for the reaction $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ is 4×10^{-2} atm at this temperature. Calculate the mass of CaO present at equilibrium.

30. Anhydrous calcium chloride is often used as a desiccant. In the presence of excess of CaCl_2 , the amount of the water taken up is governed by $K_p = 6.4 \times 10^{85}$ for the following reaction at room temperature, $\text{CaCl}_2(\text{s}) + 6\text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CaCl}_2 \cdot 6\text{H}_2\text{O}(\text{s})$. What is the equilibrium vapour pressure of water in a closed vessel that contains $\text{CaCl}_2(\text{s})$?
31. 20.0 grams of $\text{CaCO}_3(\text{s})$ were placed in a closed vessel, heated & maintained at 727°C under equilibrium $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ and it is found that 75 % of CaCO_3 was decomposed. What is the value of K_p ? The volume of the container was 15 litres.
- 32.(a) Methanol, a liquid fuel that could possibly replace gasoline, can be prepared from water gas and additional hydrogen at high temperature and pressure in the presence of a suitable catalyst. Write the expression for the equilibrium constant for the reversible reaction.
- $$2\text{H}_2(\text{g}) + \text{CO}(\text{g}) \rightleftharpoons \text{CH}_3\text{OH}(\text{g}) \quad \Delta H = -90.2 \text{ kJ}$$
- (b) Assume that equilibrium has been established and predict how the concentration of H_2 , CO and CH_3OH will differ at a new equilibrium if (1) more H_2 is added. (2) CO is removed. (3) CH_3OH is added. (4) the pressure on the system is increased. (5) the temperature of the system is increased. (6) more catalyst is added.
33. Ammonia is a weak base that reacts with water according to the equation
- $$\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$$
- Will any of the following increase the percent of ammonia that is converted to the ammonium ion in water?
- (a) Addition of NaOH . (b) Addition of HCl . (c) Addition of NH_4Cl .
34. Consider the gas-phase hydration of hexafluoroacetone, $(\text{CF}_3)_2\text{CO}$:



At 76°C , the forward and reverse rate constants are $k_f = 0.15 \text{ M}^{-1}\text{s}^{-1}$ and $k_r = 6 \times 10^{-4}\text{s}^{-1}$. What is the value of the equilibrium constant K_c ?

35. Listed in the table are forward and reverse rate constants for the reaction $2\text{NO}(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + \text{O}_2(\text{g})$

Temperature (K)	$k_f (\text{M}^{-1}\text{s}^{-1})$	$k_r (\text{M}^{-1}\text{s}^{-1})$
1400	0.29	1.1×10^{-6}
1500	1.3	1.4×10^{-5}

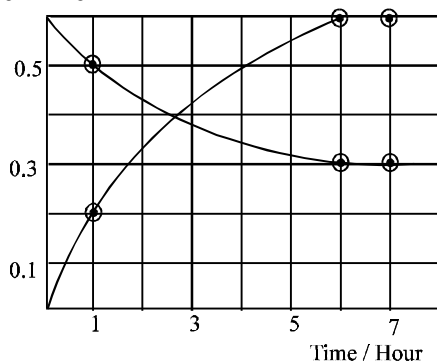
Is the reaction endothermic or exothermic? Explain in terms of kinetics.

EXERCISE # S-II

- One mole of $\text{N}_2\text{O}_4(\text{g})$ at 300 K is left in a closed container under one atm. It is heated to 600 K when 20 % by mass of $\text{N}_2\text{O}_4(\text{g})$ decomposes to $\text{NO}_2(\text{g})$. Calculate resultant pressure.
- Solid Ammonium carbamate dissociates as: $\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g})$. In a closed vessel solid ammonium carbamate is in equilibrium with its dissociation products. At equilibrium, ammonia is added such that the partial pressure of NH_3 at new equilibrium now equals the original total pressure. Calculate the ratio of total pressure at new equilibrium to that of original total pressure.
- A definite amount of solid NH_4HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. NH_4HS decomposes to give NH_3 and H_2S and at equilibrium total pressure in flask is 0.84 atm. Calculate equilibrium constant for the reaction :
- For the reaction $\text{A}(\text{g}) + 2\text{B}(\text{g}) \rightleftharpoons \text{C}(\text{g}) + \text{D}(\text{g})$; $K_c = 10^{12}$.
If the initial moles of A,B,C and D are 0.5, 1, 0.5 and 3.5 moles respectively in a one litre vessel. What is the equilibrium concentration of B?
- Calculate the equilibrium concentration of PCl_3 and Cl_2 at 300 K if initially 1 mol of PCl_5 is taken in a closed container of having volume 10 lit.
[Given : $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$; $K_c = 10^{-2}$ at 300 K]
- When 36.8g $\text{N}_2\text{O}_4(\text{g})$ is introduced into a 1.0-litre flask at 27°C . The following equilibrium reaction occurs : $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$; $K_p = 0.1642$ atm.
 - Calculate K_c of the equilibrium reaction.
 - What are the number of moles of N_2O_4 and NO_2 at equilibrium?
 - What is the total gas pressure in the flask at equilibrium?
 - What is the percent dissociation of N_2O_4 ?
- The progress of the reaction

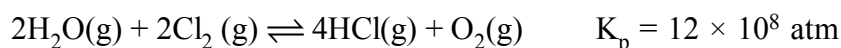
$\text{A} \rightleftharpoons n\text{B}$ with time, is presented in figure. Determine

- the value of n.
- the equilibrium constant K.
- the initial rate of conversion of A.



- A mixture of hydrogen & iodine in the mole ratio 1.5 : 1 is maintained at 450°C . After the attainment of equilibrium $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, it is found on analysis that the mole ratio of I_2 to HI is 1 : 18. Calculate the equilibrium constant and the number of moles of each species present under equilibrium, if initially, 127 grams of iodine were taken.

9. At certain temperature (T) for the gas phase reaction



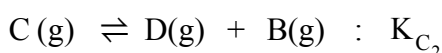
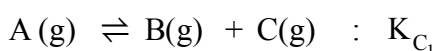
If Cl_2 , HCl & O_2 are mixed in such a manner that the partial pressure of each is 2 atm and the mixture is brought into contact with excess of liquid water. What would be approximate partial pressure of Cl_2 when equilibrium is attained at temperature (T)?

[Given : Vapour pressure of water is 380 mm Hg at temperature (T)]

10. Two solids X and Y dissociate into gaseous products at a certain temperature as follows:

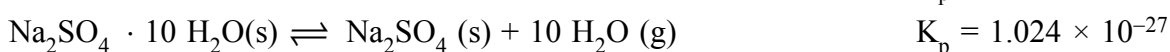
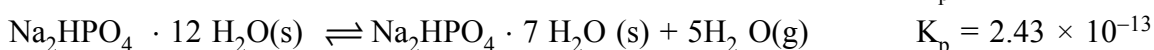
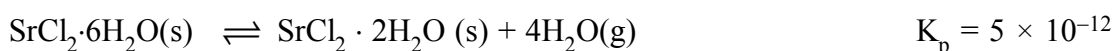
$\text{X(s)} \rightleftharpoons \text{A(g)} + \text{C(g)}$, and $\text{Y(s)} \rightleftharpoons \text{B(g)} + \text{C(g)}$. At a given temperature, pressure over excess solid X is 40 mm and total pressure over solid Y is 60 mm. When they are preset in separate containers. Calculate

- the values of K_p for two reactions (in mm^2)
 - the ratio of moles of A and B in the vapour state over a mixture of X and Y.
 - the total pressure of gases over a mixture of X and Y.
11. When 1 mole of A(g) is introduced in a closed rigid 1 litre vessel maintained at constant temperature the following equilibria are established.



The pressure at equilibrium is twice the initial pressure. Calculate the value of $\frac{K_{C_2}}{K_{C_1}}$ if $\frac{[\text{C}]_{\text{eq}}}{[\text{B}]_{\text{eq}}} = \frac{1}{5}$

12. Equilibrium constants are given (in atm) for the following reactions at 0°C :



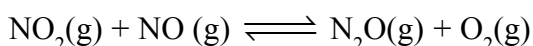
$$(\% \text{ Relative humidity} = \frac{P_{\text{H}_2\text{O}}}{\text{Vapour pressure of H}_2\text{O}} \times 100)$$

The vapor pressure of water at 0°C is 4.56 torr.

Out of $\text{SrCl}_2 \cdot 2\text{H}_2\text{O(s)}$, $\text{Na}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O(s)}$ and $\text{Na}_2\text{SO}_4(\text{s})$,

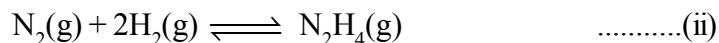
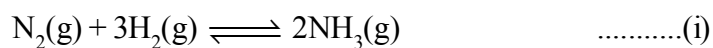
Which is the most effective drying agent at 0°C ?

13. At certain temperature, $K_C = 1.0$ for reaction :



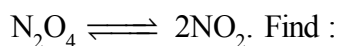
equal moles of NO and NO_2 are to be placed in 5 litre container until N_2O concentration at equilibrium is 0.5M. How many mole of $(\text{NO} + \text{NO}_2)$ must be placed in the container.

14. In a vessel, two equilibrium are simultaneously established at same temperature as follows,



Initially the vessel contains N_2 and H_2 in the molar ratio of 9 : 13. The equilibrium pressure is $7 P_0$ in which pressure due to ammonia is P_0 and due to hydrogen is $2P_0$. Find the values of equilibrium constants (K_p 's) for both the reactions.

15. If vapour density of equilibrium mixture NO_2 and N_2O_4 is found to be 42 for the reaction,



- (a) Abnormal molecular weight
- (b) Degree of dissociation
- (c) Percentage of NO_2 in the mixture

EXERCISE # O-I

Only one is correct

- $x \rightleftharpoons y$ reaction is said to be in equilibrium, when :-

(A) Only 10% conversion of x to y takes place
 (B) Complete conversion of x to y has taken place
 (C) Conversion of x to y is only 50% complete
 (D) The rate of change of x to y is just equal to the rate of change of y to x in the system
- For which reaction is $K_p = K_c$:-

(A) $2\text{NOCl(g)} \rightleftharpoons 2\text{NO(g)} + \text{Cl}_2\text{(g)}$ (B) $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons 2\text{NH}_3\text{(g)}$
 (C) $\text{H}_2\text{(g)} + \text{I}_2\text{(g)} \rightleftharpoons 2\text{HI(g)}$ (D) $2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{SO}_3\text{(g)}$
- For the reaction

$$\text{CuSO}_4 \cdot 5\text{H}_2\text{O(s)} \rightleftharpoons \text{CuSO}_4 \cdot 3\text{H}_2\text{O(s)} + 2\text{H}_2\text{O(g)}$$

Which one is correct representation :-

(A) $K_p = (P_{\text{H}_2\text{O}})^2$ (B) $K_c = [\text{H}_2\text{O}]^2$ (C) $K_p = K_c(RT)^2$ (D) All
- $\log \frac{K_p}{K_c} + \log RT = 0$ is true relationship for the following reaction:-

(A) $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ (B) $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$
 (C) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (D) (B) and (C) both
- For a reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, the value of K_c does not depends upon :-

(a) Initial concentration of the reactants (b) Pressure
 (c) Temperature (d) Catalyst
 (A) Only c (B) a, b, c (C) a, b, d (D) a, b, c, d
- If some He gas is introduced into the equilibrium $\text{PCl}_{5\text{(g)}} \rightleftharpoons \text{PCl}_{3\text{(g)}} + \text{Cl}_{2\text{(g)}}$ at constant pressure and temperature then equilibrium constant of reaction:

(A) Increase (B) Decrease (C) Unchange (D) Nothing can be said
- The equilibrium constant for the given reaction :

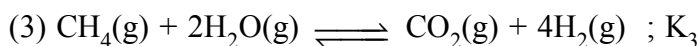
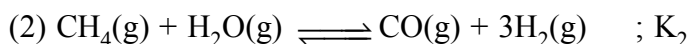
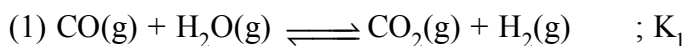
$$\text{SO}_3\text{(g)} \rightleftharpoons \text{SO}_2\text{(g)} + \frac{1}{2} \text{O}_2\text{(g)} ; K_c = 5 \times 10^{-2}$$

The value of K_c for the reaction :

$$2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{SO}_3\text{(g)}, \text{ will be}$$

(A) 400 (B) 2.40×10^{-3} (C) 9.8×10^{-2} (D) 4.9×10^{-2}

8. For the following three reactions, 1, 2 and 3 equilibrium, constants are given :



Which of the following relations is correct ?

(A) $K_1\sqrt{K_2} = K_3$ (B) $K_2K_3 = K_1$ (C) $K_3 = K_1K_2$ (D) $K_3 = K_2^3 K_1^2$

9. Sulfide ion in alkaline solution reacts with solid sulfur to form polysulfide ions having formulas S_2^{2-} , S_3^{2-} , S_4^{2-} and so on. The equilibrium constant for the formation of S_2^{2-} is 12 (K_1) & for the formation of S_3^{2-} is 132 (K_2), both from S and S^{2-} . What is the equilibrium constant for the formation of S_3^{2-} from S_2^{2-} and S?

(A) 11 (B) 12 (C) 132 (D) None of these

10. If 0.5 mole H_2 is reacted with 0.5 mole I_2 in a ten-litre container at 444°C and at same temperature value of equilibrium constant K_c is 49, the ratio of $[\text{HI}]$ and $[\text{I}_2]$ will be :-

(A) 7 (B) $\frac{1}{7}$ (C) $\sqrt{\frac{1}{7}}$ (D) 49

11. In the reaction, $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$, the moles/litre of N_2 , O_2 and NO respectively 0.25, 0.05 and 1.0 at equilibrium. The initial concentrations of N_2 and O_2 will respectively be

(A) 0.75 mole/lit., 0.55 mole/lit. (B) 0.50 mole/lit., 0.75 mole/lit.
(C) 0.25 mole/lit., 0.50 mole/lit. (D) 0.25 mole/lit., 1.0 mole/lit.

12. How many moles per litre of PCl_5 has to be taken to obtain 0.1 mole Cl_2 , if the value of equilibrium constant K_c is 0.04 ?

(A) 0.15 (B) 0.25 (C) 0.35 (D) 0.05

13. In the beginning of the reaction, $\text{A} \rightleftharpoons \text{B} + \text{C}$, 2 moles of A are taken, out of which 0.5 mole gets dissociated. What is the amount of dissociation of A ?

(A) 0.5 (B) 1 (C) 0.25 (D) 4.2

14. In the reaction, $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$, the amount of each PCl_5 , PCl_3 and Cl_2 is 2 mole at equilibrium and total pressure is 3 atmosphere. The value of K_p will be

(A) 1.0 atm. (B) 3.0 atm. (C) 2.9 atm. (D) 6.0 atm.

15. 4 moles of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 calculate total number of moles at equilibrium :-

(A) 4.5 (B) 6 (C) 3 (D) 4

16. For the reaction $A + 2B \rightleftharpoons 2C + D$, initial concentration of A is a and that of B is 1.5 times that of A. Concentration of A and D are same at equilibrium. What should be the concentration of B at equilibrium?
- (A) $\frac{a}{4}$ (B) $\frac{a}{2}$ (C) $\frac{3a}{4}$ (D) All of the above.
17. For the reaction $3A(g) + B(g) \rightleftharpoons 2C(g)$ at a given temperature, $K_c = 9.0$. What must be the volume of the flask, if a mixture of 2.0 mol each of A, B and C exist in equilibrium?
- (A) 6L (B) 9L (C) 36 L (D) None of these
18. For the following gases equilibrium. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
 K_p is found to be equal to K_c . This is attained when temperature is
- (A) 0°C (B) 273 K (C) 1 K (D) 12.19 K
19. The degree of dissociation of SO_3 is α at equilibrium pressure p^0 .
 K_p for $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$
- (A) $\frac{p^0 \alpha^3}{2(1-\alpha)^3}$ (B) $\frac{p^0 \alpha^3}{(2+\alpha)(1-\alpha)^2}$ (C) $\frac{p^0 \alpha^2}{2(1-\alpha)^2}$ (D) None of these
20. For the reaction: $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$, the degree of dissociated (α) of $HI(g)$ is related to equilibrium constant K_p by the expression
- (A) $\frac{1+2\sqrt{K_p}}{2}$ (B) $\sqrt{\frac{1+2K_p}{2}}$ (C) $\sqrt{\frac{2K_p}{1+2K_p}}$ (D) $\frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$
21. The equilibrium constant for the reaction
 $A(g) + 2B(g) \rightleftharpoons C(g)$
 is $0.25 \text{ dm}^6\text{mol}^{-2}$. In a volume of 5 dm^3 , what amount of A must be mixed with 4 mol of B to yield 1 mol of C at equilibrium.
- (A) 3 moles (B) 24 moles (C) 26 moles (D) None of these
22. A 20.0 litre vessel initially contains 0.50 mole each of H_2 and I_2 gases. These substances react and finally reach an equilibrium condition. Calculate the equilibrium concentration of HI if $K_{eq} = 49$ for the reaction $H_2 + I_2 \rightleftharpoons 2HI$.
- (A) 0.78 M (B) 0.039 M (C) 0.033 M (D) 0.021 M
23. At 675 K, $H_2(g)$ and $CO_2(g)$ react to form $CO(g)$ and $H_2O(g)$, K_p for the reaction is 0.16. If a mixture of 0.25 mole of $H_2(g)$ and 0.25 mol of CO_2 is heated at 675 K, mole % of $CO(g)$ in equilibrium mixture is :
- (A) 7.14 (B) 14.28 (C) 28.57 (D) 33.33
24. The vapour density of N_2O_4 at a certain temperature is 30. What is the % dissociation of N_2O_4 at this temperature?
- (A) 53.3% (B) 106.6% (C) 26.7% (D) None

25. The equilibrium constant K_p (in atm) for the reaction is 9 at 7 atm and 300 K.

$$A_2(g) \rightleftharpoons B_2(g) + C_2(g)$$
 Calculate the average molar mass (in gm/mol) of an equilibrium mixture.
Given : Molar mass of A_2 , B_2 and C_2 are 70, 49 & 21 gm/mol respectively.
 (A) 50 (B) 45 (C) 40 (D) 37.5
26. Vapour density of the equilibrium mixture of the reaction

$$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$$
 is 6.0
 Percent dissociation of ammonia gas is:
 (A) 13.88 (B) 58.82 (C) 41.66 (D) None of these
27. The equilibrium concentration of B $[(B)_e]$ for the reversible reaction $A \rightleftharpoons B$ can be evaluated by the expression:-
 (A) $K_c[A]_e^{-1}$ (B) $\frac{k_f}{k_b}[A]_e^{-1}$ (C) $k_f k_b^{-1} [A]_e$ (D) $k_f k_b [A]^{-1}$
28. The equilibrium $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ shifts forward if :-
 (A) A catalyst is used.
 (B) An adsorbent is used to remove SO_3 as soon as it is formed.
 (C) Small amounts of reactants are removed.
 (D) None of these
29. In manufacture of NO, the reaction $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$, ΔH +ve is favourable if :-
 (A) Pressure is increased (B) Pressure is decreased
 (C) Temperature is increased (D) Temperature is decreased
30. In which of the following reactions, increase in the pressure at constant temperature does not affect the moles at equilibrium :
 (A) $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$ (B) $C(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO(g)$
 (C) $H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g)$ (D) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
31. Change in volume of the system does not alter the number of moles in which of the following equilibrium
 (A) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ (B) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
 (C) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ (D) $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$
32. The conditions favourable for the reaction :

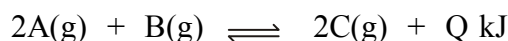
$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) ; \Delta H^\circ = -198 \text{ kJ}$$
 are :
 (A) low temperature, high pressure
 (B) any value of T and P
 (C) low temperature and low pressure
 (D) high temperature and high pressure

33. Densities of diamond and graphite are 3.5 and 2.3 gm/mL.



favourable conditions for formation of diamond are

- (A) high pressure and low temperature
 - (B) low pressure and high temperature
 - (C) high pressure and high temperature
 - (D) low pressure and low temperature
34. The equilibrium $\text{SO}_2\text{Cl}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$ is attained at 25°C in a closed rigid container and an inert gas, helium is introduced. Which of the following statements is/are correct.
- (A) concentrations of SO_2 , Cl_2 and SO_2Cl_2 do not change
 - (B) more chlorine is formed
 - (C) concentration of SO_2 is reduced
 - (D) more SO_2Cl_2 is formed
35. The yield of product in the reaction



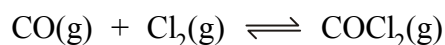
would be lower at :

- (A) low temperature and low pressure
- (B) high temperature & high pressure
- (C) low temperature and to high pressure
- (D) high temperature & low pressure

EXERCISE # O-II

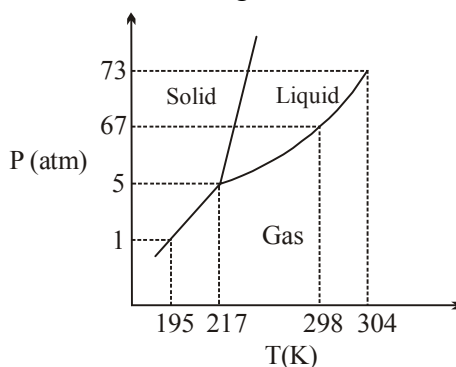
More than one may be correct

1. Following two equilibrium is simultaneously established in a container



If some Ni(s) is introduced in the container forming $\text{Ni}(\text{CO})_4(\text{g})$ then at new equilibrium

- (A) PCl_3 concentration will increase (B) PCl_3 concentration will decrease
(C) Cl_2 concentration will remain same (D) CO concentration will remain same
2. For the reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$, the forward reaction at constant temperature is favoured by
(A) introducing an inert gas at constant volume
(B) introducing chlorine gas at constant volume
(C) introducing an inert gas at constant pressure
(D) introducing PCl_5 at constant volume.
3. When NaNO_3 is heated in a closed vessel, oxygen is liberated and NaNO_2 is left. At equilibrium
(A) addition of NaNO_2 favours reverse reaction
(B) addition of NaNO_3 favours forward reaction
(C) increasing temperature favours forward reaction
(D) increasing pressure favours reverse reaction
4. For the gas phase reaction, $\text{C}_2\text{H}_4 + \text{H}_2 \rightleftharpoons \text{C}_2\text{H}_6$ ($\Delta H = -32.7 \text{ kcal}$), carried out in a closed vessel, the equilibrium moles of C_2H_4 can be increased by
(A) increasing the temperature (B) decreasing the pressure
(C) removing some H_2 (D) adding some C_2H_6
5. Phase diagram of CO_2 is shown as following



Based on above find the correct statement(s)

- (A) 298K is the normal boiling point of liquid CO_2
(B) At 1 atm & 190 K CO_2 will exist as gas.
(C) $\text{CO}_2(\text{s})$ will sublime above 195K under normal atmospheric pressure
(D) Melting point & boiling point of CO_2 will increase on increasing pressure

12. **Statement-1** : Catalyst does not alter the equilibrium constant
Statement-2 : Because for the catalysed reaction and uncatalysed reaction ΔH remain same an equilibrium constant depend on ΔH .
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
13. **Statement-1** : Equilibrium constant may show higher or lower values with increase in temperature
Statement-2 : The change depend on heat of reaction.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

Comprehension :

Paragraph for Question Nos. 12 to 15

In a 7.0 L evacuated chamber, 0.50 mol H_2 and 0.50 mol I_2 react at $427^\circ C$.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$. At the given temperature, $K_C = 49$ for the reaction.

14. What is the value of K_p ?
 (A) 7 (B) 49 (C) 24.5 (D) None
15. What is the total pressure (atm) in the chamber?
 (A) 83.14 (B) 831.4 (C) 8.21 (D) None
16. How many moles of the iodine remain unreacted at equilibrium?
 (A) 0.388 (B) 0.112 (C) 0.25 (D) 0.125
17. What is the partial pressure (atm) of HI in the equilibrium mixture?
 (A) 6.385 (B) 12.77 (C) 40.768 (D) 646.58

Paragraph for Question Nos. 16 to 18

Influence of pressure, temperature, concentration and addition of inert gas on a reversible chemical reaction in equilibrium can be explained by formulating the expression for equilibrium constant K_c or K_p for the equilibrium. On the other hand Le Chatelier principle can be theoretically used to explain the effect of P, T or concentration on the physical or chemical equilibrium both.

18. For the reaction : $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ increase of pressure shows :
 (A) An increase in degree of dissociation and a decrease in K_c
 (B) A decrease in degree of dissociation and a decrease in K_c
 (C) An increase in degree of dissociation but K_c remains constant
 (D) A decrease in degree of dissociation but K_c remains constant
19. For the reaction : $2SO_2 + O_2 \rightleftharpoons 2SO_3$; $\Delta H = -ve$, An increase in temperature shows :
 (A) More dissociation of SO_3 and a decrease in K_c
 (B) Less dissociation of SO_3 and an increase in K_c
 (C) More dissociation of SO_3 and an increase in K_c
 (D) Less dissociation of SO_3 and an decrease in K_c

20. For the reaction: $\underset{\text{Yellow}}{\text{Fe}^{3+}(\text{aq.})} + \text{SCN}^{-}(\text{aq.}) \rightleftharpoons \underset{\text{Red}}{[\text{Fe}(\text{NCS})]^{2+}(\text{aq.})}$; in equilibrium if little more aqueous solution

of FeCl_3 is added, then :

- (A) Forward reaction becomes more and red colour is deepened
 (B) Backward reaction becomes more and red colour faintens
 (C) Solution becomes colourless
 (D) None of these

Match the column :

- | 21. | Column I | Column II |
|-----|---|--|
| (A) | $K_p < K_c$ | (P) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ |
| (B) | Introduction of inert gas at constant pressure will decrease the concentration of reactants | (Q) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ |
| (C) | K_p° is dimensionless | (R) $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$ |
| (D) | Temperature increase will shift the reaction on product side. | (S) $\text{NH}_3(\text{g}) + \text{HI}(\text{g}) \rightleftharpoons \text{NH}_4\text{I}(\text{s})$ |

- | 22. | Column-I
(Reactions) | Column-II
(Favourable conditions) |
|-----|---|--|
| (A) | Oxidation of nitrogen
$\text{N}_2(\text{g}) + \text{O}_2(\text{g}) + 180.5 \text{ kJ} \rightleftharpoons 2\text{NO}(\text{g})$ | (p) Addition of inert gas at constant pressure |
| (B) | Dissociation of $\text{N}_2\text{O}_4(\text{g})$
$\text{N}_2\text{O}_4(\text{g}) + 57.2 \text{ kJ} \rightleftharpoons 2\text{NO}_2(\text{g})$ | (q) Decrease in pressure |
| (C) | Oxidation of $\text{NH}_3(\text{g})$
$4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g}) + 905.6 \text{ kJ}$ | (r) Decrease in temperature |
| (D) | Formation of $\text{NO}_2(\text{g})$
$\text{NO}(\text{g}) + \text{O}_3(\text{g}) \rightleftharpoons \text{NO}_2(\text{g}) + \text{O}_2(\text{g}) + 200 \text{ kJ}$ | (s) Increase in temperature |

Matching list type :

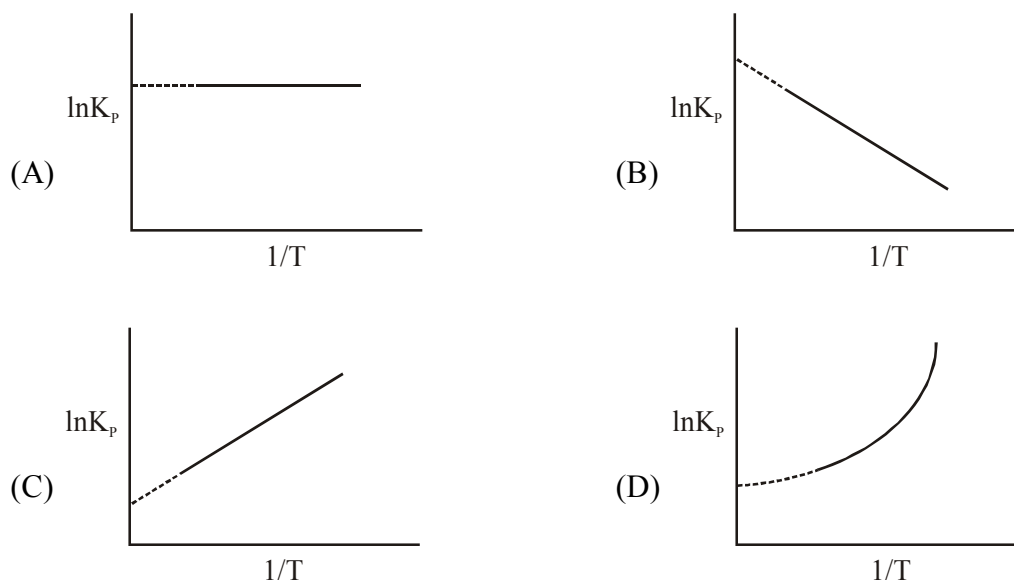
Answer Q.No. 23, 24 and 25 by appropriately matching the information given in the three columns of the following table.

Column-I (Reaction)	Column-II (If α is negligible w.r.t. 1)	Column-III
(A) $2X(g) \rightleftharpoons Y(g) + Z(g), \Delta H = -ve$	(P) $\alpha = 2 \times \sqrt{K_p}$	(I) K increase with increase in temperature
(B) $X(g) \rightleftharpoons Y(g) + Z(g), \Delta H = +ve$	(Q) $\alpha = 3 \times \sqrt{K_p}$	(II) K decrease with increase in temperature
(C) $3X(g) \rightleftharpoons Y(g) + Z(g), \Delta H = +ve$	(R) $\alpha = (2K_p)^{1/3}$	(III) Pressure has no effect
(D) $2X(g) \rightleftharpoons Y(g) + 2Z(g), \Delta H = +ve$	(S) $\alpha = \sqrt{K_p}$	(IV) Addition of inert gas at constant pressure shift equilibrium in forward direction

23. Which of the following option is correctly matched ?

- (A) A-Q-III (B) B-S-II (C) C-Q-II (D) D-R-IV

24. Correct representation of reaction in option (A) is represented by graph



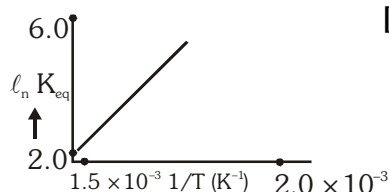
25. For the reaction of option (B) if degree of dissociation varies inversely as square root of pressure of the system. Suppose at constant temperature volume is increased 16 times of its initial volume the degree of dissociation (α) for this reaction will become :

- (A) 4 times (B) 2 times (C) 1/4 times (D) 1/2 times

EXERCISE : J-MAIN

1. One of the following equilibrium is not affected by change in volume of the flask - [AIEEE-2002]
 (1) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ (2) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
 (3) $\text{N}_2(\text{g}) + \text{O}_2 \rightleftharpoons 2\text{NO}(\text{g})$ (4) $\text{SO}_2\text{Cl}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$
2. Reaction $\text{CO}_{(\text{g})} + \frac{1}{2} \text{O}_{2(\text{g})} \rightleftharpoons \text{CO}_{2(\text{g})}$. The value of $\frac{K_p}{K_c}$ is - [AIEEE-2002]
 (1) $\frac{1}{RT}$ (2) $\sqrt{RT}$ (3) $\frac{1}{\sqrt{RT}}$ (4) RT
3. Consider the reaction equilibrium [AIEEE-2003]
 $2\text{SO}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightleftharpoons 2\text{SO}_{3(\text{g})}$; $\Delta H^\circ = -198 \text{ kJ}$
 On the basis of Le-Chatelier's principle, the condition favourable for the forward reaction is -
 (1) Lowering the temperature and increasing the pressure
 (2) Any value of temperature as well as pressure
 (3) Lowering of temperature as well as pressure
 (4) Increasing temperature as well as pressure
4. For the reaction equilibrium,
 $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ the concentration of N_2O_4 and NO_2 at equilibrium are 4.8×10^{-2} and $1.2 \times 10^{-2} \text{ mol L}^{-1}$ respectively. The value of K_c for the reaction is- [AIEEE-2003]
 (1) $3 \times 10^{-3} \text{ mol L}^{-1}$ (2) $3 \times 10^3 \text{ mol L}^{-1}$
 (3) $3.3 \times 10^2 \text{ mol L}^{-1}$ (4) $3 \times 10^{-1} \text{ mol L}^{-1}$
5. The equilibrium constant for the reaction
 $\text{N}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightleftharpoons 2\text{NO}_{(\text{g})}$ at temperature T is 4×10^{-4} . The value of K_c for the reaction
 $\text{NO}_{(\text{g})} \rightleftharpoons \frac{1}{2}\text{N}_{2(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})}$ [AIEEE-2004]
 (1) 2.5×10^2 (2) 50 (3) 4×10^{-4} (4) 0.02
6. For the reaction $\text{CO}_{(\text{g})} + \text{Cl}_{2(\text{g})} \rightleftharpoons \text{COCl}_{2(\text{g})}$ the $\frac{K_p}{K_c}$ is equal to - [AIEEE-2004]
 (1) $\frac{1}{RT}$ (2) RT (3) $\sqrt{RT}$ (4) 1.0
7. What is the equilibrium expression for the reaction $\text{P}_{4(\text{s})} + 5\text{O}_{2(\text{g})} \rightleftharpoons \text{P}_4\text{O}_{10(\text{s})}$? [AIEEE-2004]
 (1) $K_c = [\text{P}_4\text{O}_{10}] / [\text{P}_4] [\text{O}_2]^5$ (2) $K_c = [\text{P}_4\text{O}_{10}] / 5 [\text{P}_4] [\text{O}_2]$
 (3) $K_c = [\text{O}_2]^5$ (4) $K_c = 1 / [\text{O}_2]^5$

8. A schematic plot of $\ln K_{eq}$ versus inverse of temperature for a reaction is shown below. The reaction must be [AIEEE-2005]



- (1) endothermic
(2) exothermic
(3) highly spontaneous at ordinary temperature
(4) one with negligible enthalpy change

9. The exothermic formation of ClF_3 is represented by the equation [AIEEE-2005]



Which of the following will increase the quantity of ClF_3 in an equilibrium mixture of Cl_2 , F_2 and ClF_3 ?

- (1) Removing Cl_2 (2) Increasing the temperature
(3) Adding F_2 (4) Increasing the volume of the container

10. For the reaction $2\text{NO}_{2(g)} \rightleftharpoons 2\text{NO}_{(g)} + \text{O}_{2(g)}$, ($K_c = 1.8 \times 10^{-6}$ at 184°C) ($R = 0.831 \text{ kJ}(\text{mol.K})$)
When K_p and K_c are compared at 184°C it is found that [AIEEE-2005]

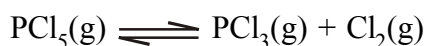
- (1) K_p is less than K_c
(2) K_p is greater than K_c
(3) Whether K_p is greater than, less than or equal to K_c depends upon the total gas pressure
(4) $K_p = K_c$

11. The equilibrium constant for the reaction $\text{SO}_3(g) \rightleftharpoons \text{SO}_2(g) + \frac{1}{2}\text{O}_2(g)$ is $K_c = 4.9 \times 10^{-2}$. The value of K_c for the reaction [AIEEE-2006]



- (1) 2.40×10^{-3} (2) 9.8×10^{-2} (3) 4.9×10^{-2} (4) 416

12. Phosphorus pentachloride dissociates as follows, in a closed reaction vessel, [AIEEE-2006]



If total pressure at equilibrium of the reaction mixture is P and degree of dissociation of PCl_5 is x , the partial pressure of PCl_3 will be-

- (1) $\left(\frac{2x}{1-x}\right)P$ (2) $\left(\frac{x}{x-1}\right)P$ (3) $\left(\frac{x}{1-x}\right)P$ (4) $\left(\frac{x}{x+1}\right)P$

13. The equilibrium constants K_{p1} and K_{p2} for the reaction $X \rightleftharpoons 2Y$ and $Z \rightleftharpoons P + Q$, respectively are in the ratio of 1 : 9. If the degree of dissociation of X and Z be equal then the ratio of total pressure at these equilibria is [AIEEE-2008]

- (1) 1 : 36 (2) 1 : 1 (3) 1 : 3 (4) 1 : 9

14. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K is :- [AIEEE-2011]

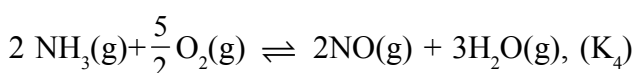
- (1) 0.3 atm (2) 0.18 atm (3) 1.8 atm (4) 3 atm

15. The equilibrium constant (K_C) for the reaction $N_2(g) + O_2(g) \longrightarrow 2NO(g)$ at temperature T is 4×10^{-4} . The value of K_C for the reaction $NO(g) \longrightarrow \frac{1}{2}N_2(g) + \frac{1}{2}O_2(g)$ at the same temperature is :- [AIEEE-2012]
 (1) 50.0 (2) 0.02 (3) 2.5×10^2 (4) 4×10^{-4}
16. 8 mol of $AB_3(g)$ are introduced into a 1.0 dm^3 vessel. If it dissociates as $2AB_3(g) \rightleftharpoons A_2(g) + 3 B_2(g)$ At equilibrium, 2mol of A_2 are found to be present. The equilibrium constant of this reaction is :- [JEE-MAINS(online)-12]
 (1) 36 (2) 3 (3) 27 (4) 2
17. The value of K_p for the equilibrium reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is 2. The percentage dissociation of $N_2O_4(g)$ at a pressure of 0.5 atm is [JEE-MAINS(online)-12]
 (1) 71 (2) 50 (3) 88 (4) 25
18. K_1 , K_2 and K_3 are the equilibrium constants of the following reactions (I), (II) and (III), respectively [JEE-MAINS(online)-12]
 (I) $N_2 + 2O_2 \rightleftharpoons 2NO_2$ (II) $2NO_2 \rightleftharpoons N_2 + 2O_2$
 (III) $NO_2 \rightleftharpoons \frac{1}{2}N_2 + O_2$

The correct relation from the following is :

- (1) $K_1 = \sqrt{K_2} = K_3$ (2) $K_1 = \frac{1}{K_2} = \frac{1}{K_3}$ (3) $K_1 = \frac{1}{K_2} = K_3$ (4) $K_1 = \frac{1}{K_2} = \frac{1}{(K_3)^2}$
19. One mole of $O_2(g)$ and two moles of $SO_2(g)$ were heated in a closed vessel of one litre capacity at 1098 K. At equilibrium 1.6 moles of $SO_3(g)$ were found. The equilibrium constant K_C of the reaction would be :- [JEE-MAINS(online)-12]
 (1) 60 (2) 80 (3) 30 (4) 40
20. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, K_1 (1) [JEE-MAINS(online)-13]
 $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, K_2 (2)
 $H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g)$, K_3 (3)

The equation for the equilibrium constant of the reaction

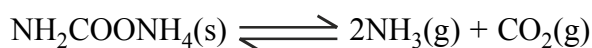


in terms of K_1 , K_2 and K_3 is :

- (1) $\frac{K_1 K_3^2}{K_2}$ (2) $\frac{K_2 K_3^3}{K_1}$ (3) $\frac{K_1 K_2}{K_3}$ (4) $K_1 K_2 K_3$
21. In reaction $A + 2B \rightleftharpoons 2C + D$, initial concentration of B was 1.5 times of $|A|$, but at equilibrium the concentrations of A and B became equal. The equilibrium constant for the reaction is : [JEE-MAINS(online)-13]
 (1) 4 (2) 6 (3) 12 (4) 8

22. For the decomposition of the compound, represented as

[JEE-MAINS(online)-14]



the $K_p = 2.9 \times 10^{-5} \text{ atm}^3$.

If the reaction is started with 1 mol of the compound, the total pressure at equilibrium would be

- (1) $38.8 \times 10^{-2} \text{ atm}$ (2) $1.94 \times 10^{-2} \text{ atm}$
 (3) $5.82 \times 10^{-2} \text{ atm}$ (4) $7.66 \times 10^{-2} \text{ atm}$
23. For the reaction $\text{SO}_{2(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})} \rightleftharpoons \text{SO}_{3(\text{g})}$, if $K_p = K_c (\text{RT})^x$ where the symbols have usual meaning

then the value of x is : (assuming ideality)

[JEE-MAINS-14]

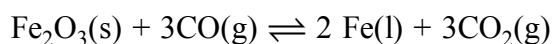
- (1) $\frac{1}{2}$ (2) 1 (3) -1 (4) $-\frac{1}{2}$

24. The equilibrium constants at 298 K for a reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ is 100. If the initial concentration of all the four species were 1 M each, then equilibrium concentration of D (in mol L^{-1}) will be :

[JEE-MAINS-16]

- (1) 1.182 (2) 0.182 (3) 0.818 (4) 1.818

25. The following reaction occurs in the Blast Furnace where iron ore is reduced to iron metal:



[JEE-MAINS(online)-17]

Using the Le Chatelier's principle, predict which one of the following will **not** disturb the equilibrium?

- (1) Removal of CO_2 (2) Addition of Fe_2O_3 (3) Addition of CO_2 (4) Removal of CO

EXERCISE : J-ADVANCE

- For the reaction $\text{CO}_{(g)} + 2\text{H}_{2(g)} \rightleftharpoons \text{CH}_3\text{OH}_{(g)}$; H_2 is introduced into a five litre flask at 327°C , containing 0.2 mole of $\text{CO}_{(g)}$ and a catalyst till the pressure is 4.92 atmosphere. At this point 0.1 mole of $\text{CH}_3\text{OH}_{(g)}$ is formed. Calculate K_c and K_p . [JEE 1990]
- At temperature T , a compound $\text{AB}_{2(g)}$ dissociates according to the reaction $2\text{AB}_{2(g)} \rightleftharpoons 2\text{AB}_{(g)} + \text{B}_{2(g)}$; with a degree of dissociation 'x', which is small compared to unity. Deduce the expression for 'x' in terms of the equilibrium constant K_p and the total pressure P . [JEE 1994]
- A sample of air consisting of N_2 and O_2 was heated to 2500 K until the equilibrium

$$\text{N}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{NO}_{(g)}$$
 was established with an equilibrium constant $K_c = 2.1 \times 10^{-3}$. At equilibrium, the mole% of NO was 1.8. Estimate the initial composition of air in mole fraction of N_2 and O_2 . [JEE 1997]
- The degree of dissociation is 0.4 at 400K & 1.0 atm for the gasoeus reaction $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_{2(g)}$. Assuming ideal behaviour of all gases. Calculate the density of equilibrium mixture at 400K & 1.0 atm pressure. [JEE 1999]
- For the reversible reaction : [JEE 2000]

$$\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g)$$
 at 500°C . The value of K_p is 1.44×10^{-5} , when partial pressure is measured in atmospheres. The corresponding value of K_c with concentration in mol L^{-1} is :

(A) $1.44 \times 10^{-5} / (0.082 \times 500)^2$ (B) $1.44 \times 10^{-5} / (8.314 \times 773)^2$
 (C) $1.44 \times 10^{-5} / (0.082 \times 500)^2$ (D) $1.44 \times 10^{-5} / (0.082 \times 773)^{-2}$
- When two reactants A and B are mixed to give products C and D, the reaction quotient Q , at the initial stages of the reaction : [JEE 2000]

(A) is zero (B) decrease with time
 (C) independent of time (D) increases with time
- When 3.06g of solid NH_4HS is introduced into a two litre evacuated flask at 27°C , 30% of the solid decomposes into gaseous ammonia and hydrogen sulphide. [JEE 2000]

(i) Calculate K_c & K_p for the reaction at 27°C .
 (ii) What would happen to the equilibrium when more solid NH_4HS is introduced into the flask?
- At constant temperature, the equilibrium constant (K_p) for the decomposition reaction. $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ is expressed by $K_p = 4x^2P/(1-x^2)$ where P is pressure, x is extent of decomposition. Which of the following statement is true ? [JEE 2001]

(A) K_p increases with increase of P (B) K_p increases with increase of x
 (C) K_p increases with decrease of x (D) K_p remains constant with change in P or x

9. Consider the following equilibrium in a closed container : $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$.

At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements holds true regarding the equilibrium constant (K_p) and degree of dissociation (α) : [JEE 2002]

- (A) Neither K_p nor α changes
 - (B) Both K_p and α change
 - (C) K_p changes, but α does not change
 - (D) K_p does not change, but α changes
10. The thermal dissociation equilibrium of $\text{CaCO}_3(\text{s})$ is studied under different conditions.



For this equilibrium, the correct statement(s) is(are)

- (A) ΔH is dependent on T
- (B) K is independent of the initial amount of CaCO_3
- (C) K is dependent on the pressure of CO_2 at a given T
- (D) ΔH is independent of the catalyst, if any

ANSWER KEY

EXERCISE # S-I

1. (a) incomplete (b) almost complete

$$2. (a) K_p = \frac{(P_{SO_3})}{(P_{SO_2})(P_{O_2})^{\frac{1}{2}}} ; K_c = \frac{[SO_3]}{[SO_2][O_2]^{\frac{1}{2}}}$$

(b) $K_c = [H_2O]$ [here $H_2O(l)$ is a pure liquid so its concentration is 1.]

$$K_p = P_{H_2O}$$

= vapour pressure

= aqueous tension

(c) $K_c = [NH_3]^2 [CO_2]$ ($\because$ active mass of solid is 1)

$$K_p = (P_{NH_3})^2 P_{CO_2}$$

(d) $K_c = \frac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]}$ [here H_2O is not in excess]

K_p is not defined for liquid phase.

(e) $K_c = \frac{[NH_4^+][OH^-]}{[NH_3]}$ [here $H_2O(l)$ is in excess (solvent) so its concentration is 1]

(f) $K_{PC} = \frac{P_{H_2} \times [Zn^{2+}]}{[H^+]^2}$ [$\because$ active mass of solid is 1]

3. (a) Here, $\Delta n_g = 1 + 1 - 1 = 1$

$$K_p = K_c (RT)^1 = K_c RT$$

(b) Here, $\Delta n_g = 2 - (3+1) = -2$

$$K_p = K_c (RT)^{-2}$$

(c) Here, $\Delta n_g = 2 - (1+1) = 0$

$$K_p = K_c (RT)^0 = K_c$$

(d) Here, $\Delta n_g = 2 + 1 - 2 = 1$

$$K_p = K_c (RT)^1 = K_c RT$$

(d) Here, $\Delta n_g = 2 - (2 + 1) = -1$

$$K_p = K_c (RT)^{-1} = K_c / RT$$

4. $\therefore K_1 > K_2$

So the stability of $XO < XO_2$

For reactants, stability increases when value of K decreases.

5. $Q_c > K_c$ so the reaction will proceed in the reverse direction.

6. (a) $Q_c = \frac{[CH_3COOC_2H_5][H_2O]}{[CH_3COOH][C_2H_5OH]}$ [$\therefore$ Water is not a solvent here]

(b) $K_c = \frac{(0.171) \times (0.171)}{(1 - 0.171) \times (0.180 - 0.171)} = 3.92$

(c) $Q_c = \frac{(0.214) \times (0.214)}{(0.786) \times (0.286)} = 0.204 \neq K_c$
 $\therefore$ Equilibrium has not been reached.

7. 2×10^9

8. 0.9

9. $K_c = 100$

10. $K_p = 0.2463$

11. 64×10^{-32}

12. $K_c = 0.44$

13. $K_p = 1.2 \text{ atm.}$

14. $[HI] = \frac{1}{3} M$

$[I_2] = [H_2] = \frac{1}{30} M$

15. (a) 0.1 (b) 0.4

16. $[A] = 5 M, [B] = 5 M, [C] = 1 M, [D] = 25 M$

17. new equilibrium	PCl_5	PCl_3	Cl_2
Mole	4	4	2
Partial pressure	20 atm	20 atm	10 atm

18. $\alpha = \sqrt{\frac{K_p}{4P + K_p}}$

19. (A) $\alpha = \frac{4}{5} = 0.8$ (B) $K_c = 16$

20. $K_c = 0.001$

21. $[NO] = 2M, [N_2] = [O_2] = 0.5M$

22. $P_{ClF} = P_{F_2} = 0.1 \text{ atm}, P_{ClF_3} = 0.5 \text{ atm}$

23. $K_p = 0.4, a \sim 0.1$

24. 50%

25. $K_p = 0.01 \text{ atm}$

26. $K_p = 2.5 \text{ atm}, P = 15 \text{ atm}$

27. $K = 4$

28. $P_{\text{CS}_2} = 1.8 \text{ atm}$, $P_{\text{S}_2} = 0.2 \text{ atm}$ 29. 22.4 mg
30. $P_{\text{H}_2\text{O}} = 5 \times 10^{-15} \text{ atm}$ 31. 0.821 atm
32. (a) $K = [\text{CH}_3\text{OH}]/[\text{H}_2]^2[\text{CO}]$,
 (b) 1. $[\text{H}_2]$ increase, $[\text{CO}]$ decrease, $[\text{CH}_3\text{OH}]$ increase ; 2. $[\text{H}_2]$ increase, $[\text{CO}]$ decrease, $[\text{CH}_3\text{OH}]$ decrease ; 3. $[\text{H}_2]$ increase, $[\text{CO}]$ increase, $[\text{CH}_3\text{OH}]$ increase ; 4. $[\text{H}_2]$ increase, $[\text{CO}]$ increase, $[\text{CH}_3\text{OH}]$ increase ; 5. $[\text{H}_2]$ increase, $[\text{CO}]$ increase, $[\text{CH}_3\text{OH}]$ decrease ; 6. no change
33. 2
34. 250
35. k_t increase more than k_p this means that E_a (reverse) is greater than E_a (forward). Hence exothermic reaction.

EXERCISE # S-II

1. Ans.(2.4 atm) 2. Ans.31/27
3. Ans.(0.11) 4. Ans.(2×10^{-4})
5. $[\text{PCl}_3] = [\text{Cl}_2] = \frac{-5 \pm \sqrt{15}}{10} \text{ M}$
6. (a) $6.667 \times 10^{-3} \text{ mol L}^{-1}$;
 (b) $n(\text{N}_2\text{O}_4) = 0.374 \text{ mol}$; $n(\text{NO}_2) = 0.052 \text{ mol}$;
 (c) 10.49 atm (d) 6.44 %
7. (i) 2; (ii) 1.2 mol/L; (iii) 0.1 moles/hr
8. Ans. $K_c=54$, $n_{\text{HI}}=0.9 \text{ mol}$, $n_{\text{I}_2}=0.05 \text{ mol}$, $n_{\text{H}_2}=0.3 \text{ mol}$
9. Ans. $3.6 \times 10^{-3} \text{ atm}$
10. Ans.(a) 400mm², 900mm² (b) 4: 9, (c) 72.15 mm Hg
11. Ans. (4) 12. Ans. ($\text{SrCl}_2 \cdot 2\text{H}_2\text{O}$)
13. Ans. (10) 14. $K_{p_1} = \frac{1}{20P_0^2}$; $K_{p_2} = \frac{3}{20P_0^2}$
15. (a) 84 ; (b) 0.095 ; (c) 17.35 %

EXERCISE # O-I

- | | | | |
|---------|---------|---------|---------|
| 1. (D) | 2. (C) | 3. (D)] | 4. (B) |
| 5. (C) | 6. (C) | 7. (A) | 8. (C) |
| 9. (A) | 10. (A) | 11. (A) | 12. (C) |
| 13. (C) | 14. (A) | 15. (B) | 16. (B) |
| 17. (A) | 18. (D) | 19. (B) | 20. (D) |
| 21. (C) | 22. (B) | 23. (B) | 24. (A) |
| 25. (C) | 26. (C) | 27. (C) | 28. (B) |
| 29. (C) | 30. (D) | 31. (A) | 32. (A) |
| 33. (C) | 34. (A) | 35. (D) | |

EXERCISE # O-II

- | | | | |
|----------|------------|------------|--------------|
| 1. (B) | 2. (C,D) | 3. (C,D) | 4. (A,B,C,D) |
| 5. (C,D) | 6. (A,C,D) | 7. (A,B,C) | 8. (C,D) |
| 9. (A,C) | 10. (B) | 11. (D) | 12. (A) |
| 13. (A) | 14. (B) | 15. (C) | 16. (B) |
| 17. (A) | 18. (D) | 19. (A) | 20. (A) |

Match the column :

21. (A) - P,R,S (B) - P,Q,R,S (C) - P, Q, R, S ; (D) - Q
 22. A - (s), B - (p,q,s), C - (p,q,r), D - (r)
 23. (D) 24. (C) 25. (A)

EXERCISE # J-MAIN

- | | | | |
|-------------|-------------|-------------|-------------|
| 1. Ans.(3) | 2. Ans.(3) | 3. Ans.(1) | 4. Ans.(1) |
| 5. Ans.(2) | 6. Ans.(1) | 7. Ans.(4) | 8. Ans.(2). |
| 9. Ans.(3) | 10. Ans.(2) | 11. Ans.(4) | 12. Ans.(4) |
| 13. Ans.(1) | 14. Ans.(3) | 15. Ans.(1) | 16. Ans(3) |
| 17. Ans(1) | 18. Ans(4) | 19. Ans(2) | 20. Ans(2) |
| 21. Ans(1) | 22. Ans(3) | 23. Ans.(4) | 24. Ans.(4) |
| 25. Ans.(2) | | | |

EXERCISE # J-ADVANCE

1. $K_c = 279.64 \text{ litre}^2 \text{ mole}^{-2}$; $K_p = 0.115 \text{ atm}^{-2}$
 2. $x = \sqrt[3]{\frac{2K_p}{P}}$ 3. 79%, 21% 4. Ans. 4.54 g dm^{-3}
 5. Ans.(D) 6. Ans.(D)
 7. (i) $K_c = 8.1 \times 10^{-5} \text{ mol}^2 \text{ L}^2$; $K_p = 4.91 \times 10^{-2} \text{ atm}^2$ (ii) No effect;
 8. Ans.(D) 9. Ans.(D) 10. Ans.(A,B,D)

CHEMISTRY IN EVERYDAY LIFE**1. CHEMICALS IN FOOD**

Chemicals are added to food for (i) their preservation, (ii) enhancing their appeal, and (iii) adding nutritive value in them. Main categories of food additives are as follows:

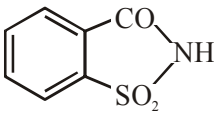
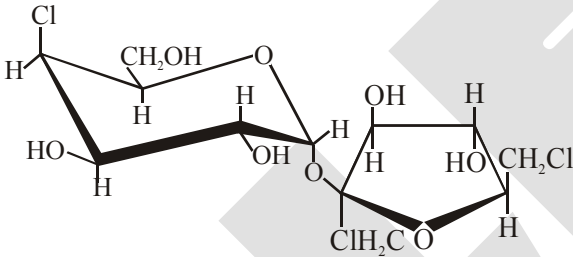
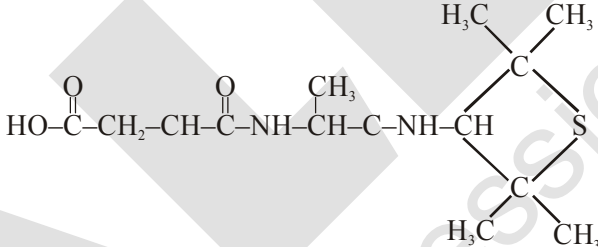
- (i) Food colours
- (ii) Flavours and sweeteners
- (iii) Fat emulsifiers and stabilising agents
- (iv) Flour improvers - antistaling agents and bleaches
- (v) Antioxidants
- (vi) Preservatives
- (vii) Nutritional supplements such as minerals, vitamins and amino acids.

Except for chemicals of category (vii), none of the above additives have nutritive value. These are added either to increase the shelf life of stored food or for cosmetic purposes. In this Section we will discuss only sweeteners and food preservatives.

Artificial Sweetening Agents :

Natural sweeteners, e.g., sucrose add to calorie intake and therefore many people prefer to use artificial sweeteners. Ortho-sulphobenzimide, also called saccharin, is the first popular artificial sweetening agent. It has been used as a sweetening agent ever since it was discovered in 1879. It is about 550 times as sweet as cane sugar. It is excreted from the body in urine unchanged. It appears to be entirely inert and harmless when taken. Its use is of great value to diabetic persons and people who need to control intake of calories. Some other commonly marketed artificial sweeteners are given in Table 16.1.

Table: Artificial Sweeteners

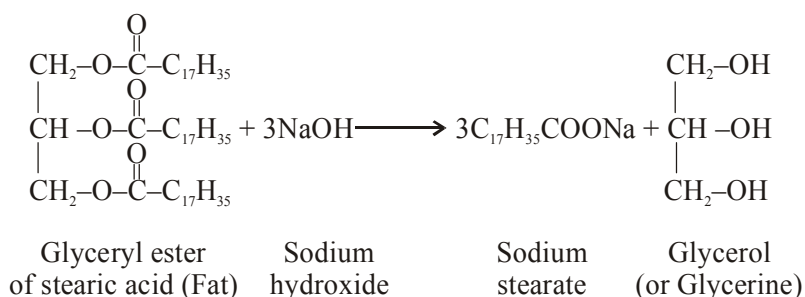
Artificial Sweetner	Artificial Sweetner	Seetness value in Comparisonh to cane sugar
Aspartame	$\text{HO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\underset{\text{NH}_2}{\text{CH}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}-\underset{\text{CH}_2}{\text{CH}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OCH}_3$ Aspartic acid part Phenylalanine methyl ester part	100
Saccharin		550
Sucralose		600
Alitame		2000

Aspartame is the most successful and widely used artificial sweetener. It is roughly 100 times as sweet as cane sugar. It is methyl ester of dipeptide formed from aspartic acid and phenylalanine. Use of aspartame is limited to cold foods and soft drinks because it is unstable at cooking temperature.

Alitame is high potency sweetener, although it is more stable than aspartame, the control of sweetness of food is difficult while using it.

Sucralose is trichloro derivative of sucrose. Its appearance and taste are like sugar. It is stable at cooking temperature. It does not provide calories.

Soaps :



Soaps are the detergents used since long. Soaps used for cleaning purpose are sodium or potassium salts of long chain fatty acids, e.g., stearic, oleic and palmitic acids. Soaps containing sodium salts are formed by heating fat (i.e., glyceryl ester of fatty acid) with aqueous sodium hydroxide solution. This reaction is known as saponification.

In this reaction, esters of fatty acids are hydrolysed and the soap obtained remains in colloidal form. It is precipitated from the solution by adding sodium chloride. The solution left after removing the soap contains glycerol, which can be recovered by fractional distillation. Only sodium and potassium soaps are soluble in water and are used for cleaning purposes. Generally potassium soaps are soft to the skin than sodium soaps. These can be prepared by using potassium hydroxide solution in place of sodium hydroxide.

Types of soaps :

Basically all soaps are made by boiling fats or oils with suitable soluble hydroxide. Variations are made by using different raw materials.

Toilet soaps are prepared by using better grades of fats and oils and care is taken to remove excess alkali. Colour and perfumes are added to make these more attractive.

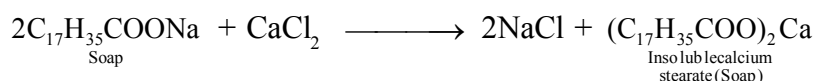
Soaps that float in water are made by beating tiny air bubbles before their hardening. Transparent soaps are made by dissolving the soap in ethanol and then evaporating the excess solvent.

In medicated soaps, substances of medicinal value are added. In some soaps, deodorants are added. Shaving soaps contain glycerol to prevent rapid drying. A gum called, rosin is added while making them. It forms sodium rosinate which lathers well. Laundry soaps contain fillers like sodium rosinate, sodium silicate, borax and sodium carbonate.

Soap chips are made by running a thin sheet of melted soap onto a cool cylinder and scraping off the soaps in small broken pieces. Soap granules are dried miniature soap bubbles. Soap powders and scouring soaps contain some soap, a scouring agent (abrasive) such as powdered pumice or finely divided sand, and builders like sodium carbonate and trisodium phosphate. Builders make the soaps act more rapidly. The cleansing action of soap has been discussed in Unit 5.

Why do soaps not work in hard water ?

Hard water contains calcium and magnesium ions. These ions form insoluble calcium and magnesium soaps respectively when sodium or potassium soaps are dissolved in hard water.



These insoluble soaps separate as scum in water and are useless as cleansing agent. In fact these are hinderance to good washing, because the precipitate adheres onto the fibre of the cloth as gummy mass. Hair washed with hard water looks dull because of this sticky precipitate. Dye does not absorb evenly on cloth washed with soap using hard water, because of this gummy mass.

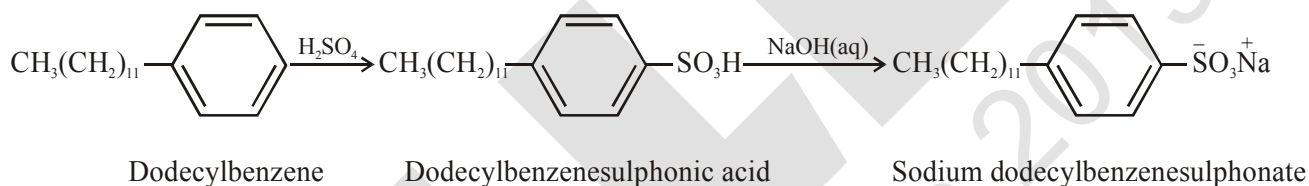
Synthetic Detergents :

Synthetic detergents are cleansing agents which have all the properties of soaps, but which actually do not contain any soap. These can be used both in soft and hard water as they give foam even in hard water. Some of the detergents give foam even in ice cold water.

Synthetic detergents are mainly classified into three categories:

- (i) Anionic detergents (ii) Cationic detergents and (iii) Non-ionic detergents

- (i) Anionic Detergents: Anionic detergents are sodium salts of sulphonated long chain alcohols or hydrocarbons. Alkyl hydrogensulphates formed by treating long chain alcohols with concentrated sulphuric acid are neutralised with alkali to form anionic detergents. Similarly alkyl benzene sulphonates are obtained by neutralising alkyl benzene sulphonic acids with alkali.

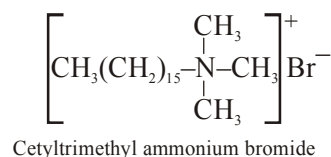


In anionic detergents, the anionic part of the molecule is involved in the cleansing action. Sodium salts of alkylbenzenesulphonates are an important class of anionic detergents.

They are mostly used for household work. Anionic detergents are also used in toothpastes.

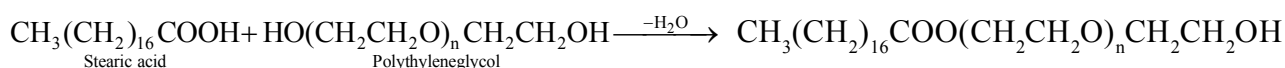
- (ii) Cationic Detergents: Cationic detergents are quaternary ammonium salts of amines with acetates, chlorides or bromides

as anions. Cationic part possess a long hydrocarbon chain and a positive charge on nitrogen atom. Hence, these are called cationic detergents. Cetyltrimethylammonium bromide is a popular cationic detergent and is used in hair conditioners.



Cationic detergents have germicidal properties and are expensive, therefore, these are of limited use.

- (iii) Non-ionic Detergents: Non-ionic detergents do not contain any ion in their constitution. One such detergent is formed when stearic acid reacts with polyethyleneglycol.



Liquid dishwashing detergents are non-ionic type. Mechanism of cleansing action of this type of detergents is the same as that of soaps. These also remove grease and oil by micelle formation. Main problem that appears in the use of detergents is that if their hydrocarbon chain is highly branched, then bacteria cannot degrade this easily. Slow degradation of detergents leads to their accumulation. Effluents containing such detergents reach the rivers, ponds, etc.

These persist in water even after sewage treatment and cause foaming in rivers, ponds and streams and their water gets polluted.

These days the branching of the hydrocarbon chain is controlled and kept to the minimum. Unbranched chains can be biodegraded more easily and hence pollution is prevented.

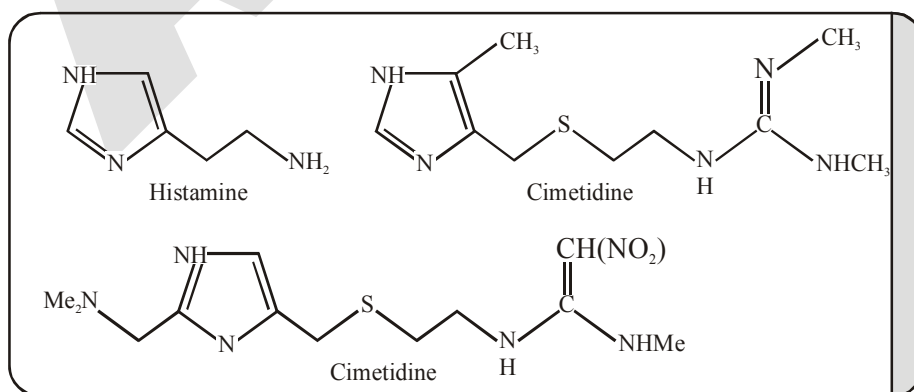
Therapeutic Action of Different Classes of Drugs :

In this Section, we shall discuss the therapeutic action of a few important classes of drugs.

Antacids :

Over production of acid in the stomach causes irritation and pain. In severe cases, ulcers are developed in the stomach. Until 1970, only treatment for acidity was administration of antacids, such as sodium hydrogencarbonate or a mixture of aluminium and magnesium hydroxide. However, excessive hydrogencarbonate can make the stomach alkaline and trigger the production of even more acid. Metal hydroxides are better alternatives because of being insoluble, these do not increase the pH above neutrality. These treatments control only symptoms, and not the cause. Therefore, with these metal salts, the patients cannot be treated easily. In advanced stages, ulcers become life threatening and its only treatment is removal of the affected part of the stomach.

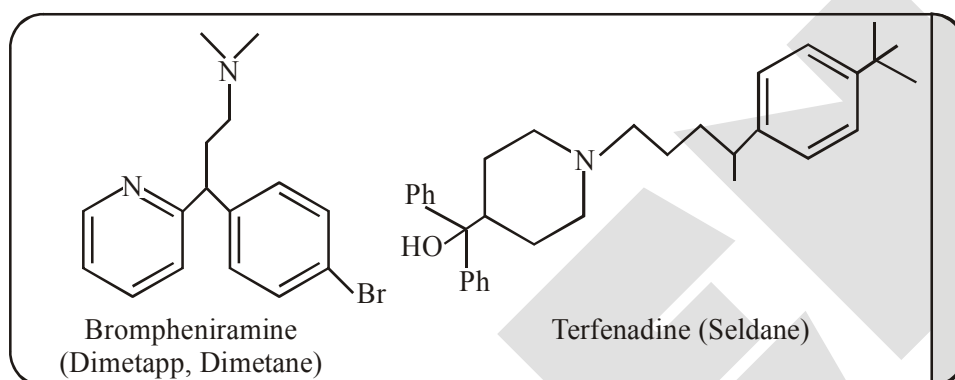
A major breakthrough in the treatment of hyperacidity came through the discovery according to which a chemical, histamine, stimulates the secretion of pepsin and hydrochloric acid in the stomach. The drug cimetidine (Tegamet), was designed to prevent the interaction of histamine with the receptors present in the stomach wall. This resulted in release of lesser amount of acid. The importance of the drug was so much that it remained the largest selling drug in the world until another drug, ranitidine (Zantac), was discovered.



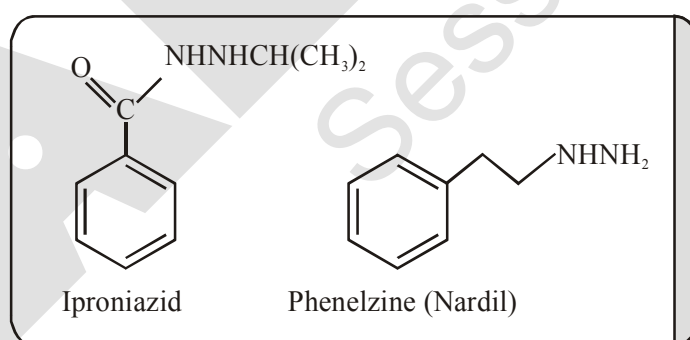
Antihistamine :

Histamine is a potent vasodilator. It has various functions. It contracts the smooth muscles in the bronchi and gut and relaxes other muscles, such as those in the walls of fine blood vessels. Histamine is also responsible for the nasal congestion associated with common cold and allergic response to pollen. Synthetic drugs, brompheniramine (Dimetapp) and terfenadine (Seldane), act as antihistamines. They interfere with the natural action of histamine by competing with histamine for binding sites of receptor where histamine exerts its effect.

Now the question that arises is, "Why do above mentioned antihistamines not affect the secretion of acid in stomach?" The reason is that antiallergic and antacid drugs work on different receptors.

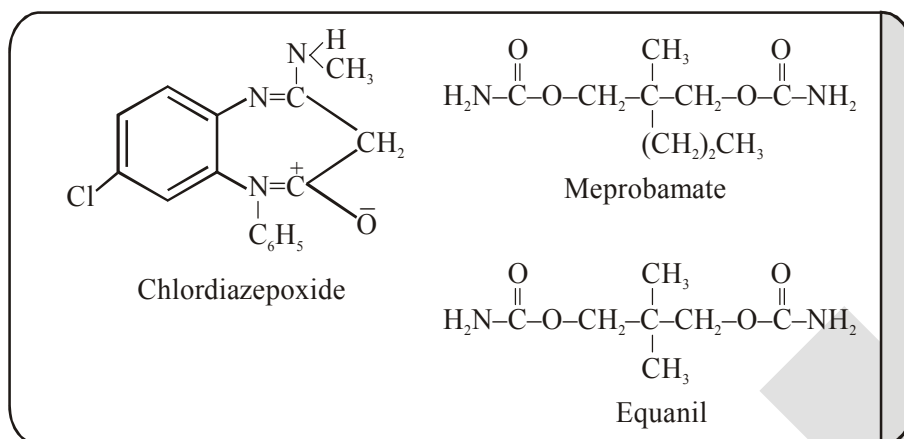
**Neurologically Active Drugs****(a) Tranquilizers**

Tranquilizers and analgesics are neurologically active drugs. These affect the message transfer mechanism from nerve to receptor. Tranquilizers are a class of chemical compounds used for the treatment of stress, and mild or even severe mental diseases. These relieve anxiety, stress, irritability or excitement by inducing a sense of well-being. They form an essential component of sleeping pills. There are various types of tranquilizers. They function by different mechanisms. For example, noradrenaline is one of the neurotransmitters that plays a role in mood changes.

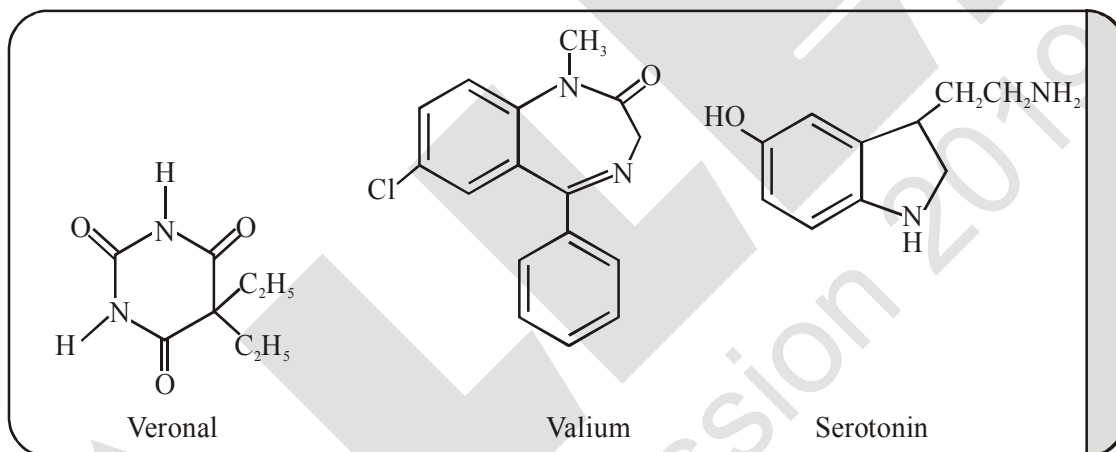


If the level of noradrenaline is low for some reason, then the signal-sending activity becomes low, and the person suffers from depression. In such situations, antidepressant drugs are required. These drugs inhibit the enzymes which catalyse the degradation of noradrenaline. If the enzyme is inhibited, this important neurotransmitter is slowly metabolised and can activate its receptor for longer periods of time, thus counteracting the effect of depression. Iproniazid and phenelzine are two such drugs. Some tranquilizers namely, chlordiazepoxide and meprobamate, are relatively mild tranquilizers suitable for relieving tension.

Equanil is used in controlling depression and hypertension.



Derivatives of barbituric acid viz., veronal, amytal, nembutal, luminal and seconal constitute an important class of tranquilizers. These derivatives are called barbiturates. Barbiturates are hypnotic, i.e., sleep producing agents. Some other substances used as tranquilizers are valium and serotonin.



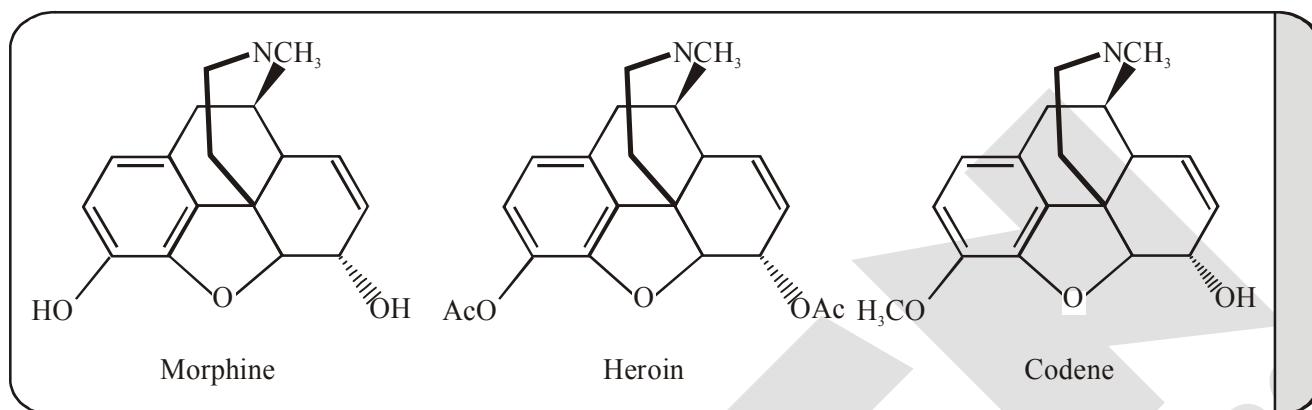
(b) Analgesics

Analgesics reduce or abolish pain without causing impairment of consciousness, mental confusion, incoordination or paralysis or some other disturbances of nervous system. These are classified as follows:

- (i) Non-narcotic (non-addictive) analgesics
- (ii) Narcotic drugs
- (i) **Non-narcotic (non-addictive) analgesics:** Aspirin and paracetamol belong to the class of non-narcotic analgesics. Aspirin is the most familiar example. Aspirin inhibits the synthesis of chemicals known as prostaglandins which stimulate inflammation in the tissue and cause pain. These drugs are effective in relieving skeletal pain such as that due to arthritis. These drugs have many other effects such as reducing fever (antipyretic) and preventing platelet coagulation. Because of its anti blood clotting action, aspirin finds use in prevention of heart attacks.

- (ii) **Narcotic analgesics:** Morphine and many of its homologues, when administered in medicinal doses, relieve pain and produce sleep. In poisonous doses, these produce stupor, coma, convulsions and ultimately death. Morphine narcotics are sometimes referred to as opiates, since they are obtained from the opium poppy.

These analgesics are chiefly used for the relief of postoperative pain, cardiac pain and pains of terminal cancer, and in child birth.



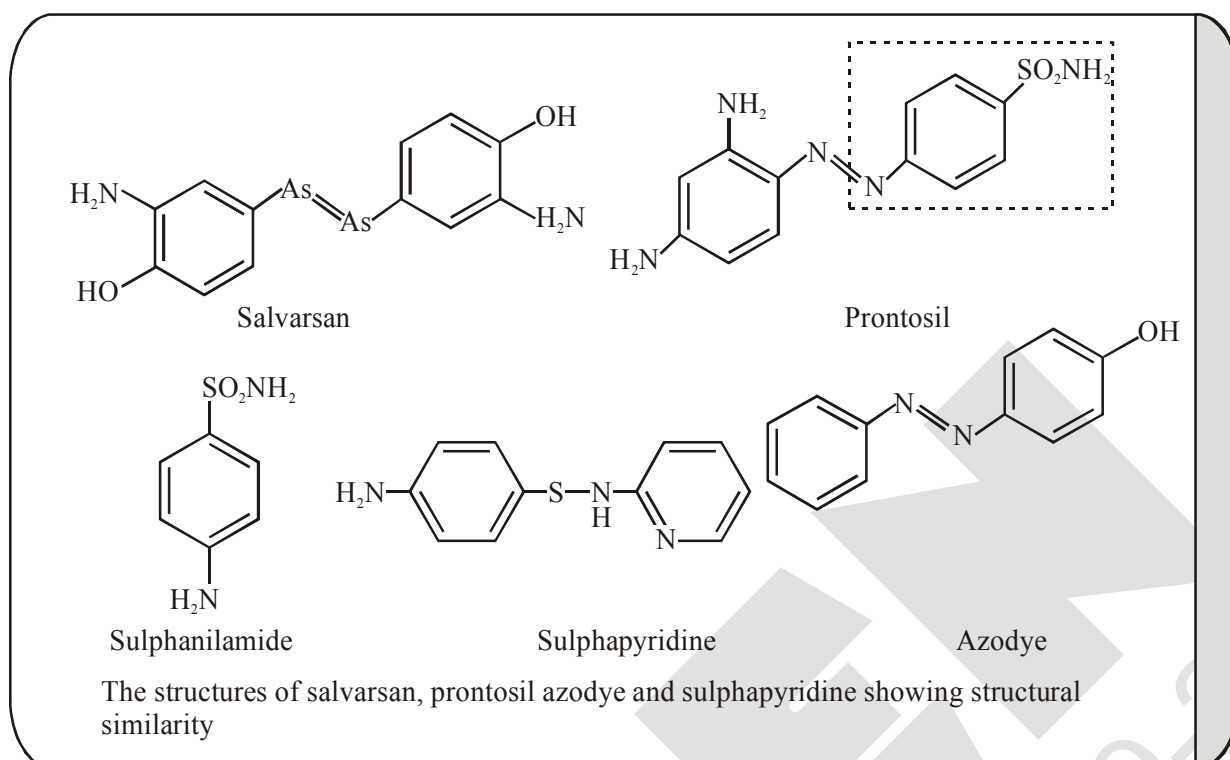
Antimicrobials

Diseases in human beings and animals may be caused by a variety of microorganisms such as bacteria, virus, fungi and other pathogens. An antimicrobial tends to destroy/prevent development or inhibit the pathogenic action of microbes such as bacteria (antibacterial drugs), fungi (antifungal agents), virus (antiviral agents), or other parasites (antiparasitic drugs) selectively. Antibiotics, antiseptics and disinfectants are antimicrobial drugs.

(a) Antibiotics

Antibiotics are used as drugs to treat infections because of their low toxicity for humans and animals. Initially antibiotics were classified as chemical substances produced by microorganisms (bacteria, fungi and molds) that inhibit the growth or even destroy microorganisms. The development of synthetic methods has helped in synthesising some of the compounds that were originally discovered as products of microorganisms. Also, some purely synthetic compounds have antibacterial activity, and therefore, definition of antibiotic has been modified. An antibiotic now refers to a substance produced wholly or partly by chemical synthesis, which in low concentrations inhibits the growth or destroys microorganisms by intervening in their metabolic processes.

The search for chemicals that would adversely affect invading bacteria but not the host began in the nineteenth century. Paul Ehrlich, a German bacteriologist, conceived this idea. He investigated arsenic based structures in order to produce less toxic substances for the treatment of syphilis. He developed the medicine, arsphenamine, known as salvarsan. Paul Ehrlich got Nobel prize for Medicine in 1908 for this discovery. It was the first effective treatment discovered for syphilis. Although salvarsan is toxic to human beings, its effect on the bacteria, spirochete, which causes syphilis is much greater than on human beings. At the same time, Ehrlich was working on azodyes also. He noted that there is similarity in structures of salvarsan and



azodyes. The $-\text{As} = \text{As}-$ linkage present in arsphenamine resembles the $-\text{N} = \text{N}-$ linkage present in azodyes in the sense that arsenic atom is present in place of nitrogen. He also noted tissues getting coloured by dyes selectively. Therefore, Ehrlich began to search for the compounds which resemble in structure to azodyes and selectively bind to bacteria. In 1932, he succeeded in preparing the first effective antibacterial agent, prontosil, which resembles in structure to the compound, salvarsan. Soon it was discovered that in the body prontosil is converted to a compound called sulphanilamide, which is the real active compound. Thus the sulpha drugs were discovered. A large range of sulphonamide analogues was synthesised. One of the most effective is sulphapyridine.

H.W. Florey and Alexander Fleming shared the Noble prize for Medicine in 1945 for their independent contributions to the development of penicillin

Despite the success of sulfonamides, the real revolution in antibacterial therapy began with the discovery of Alexander Fleming in 1929, of the antibacterial properties of a *Penicillium* fungus. Isolation and purification of active compound to accumulate sufficient material for clinical trials took thirteen years. Antibiotics have either cidal (killing) effect or a static (inhibitory) effect on microbes. A few examples of the two types of antibiotics are as follows:

Bactericidal

Penicillin
Aminoglycosides
Ofloxacin

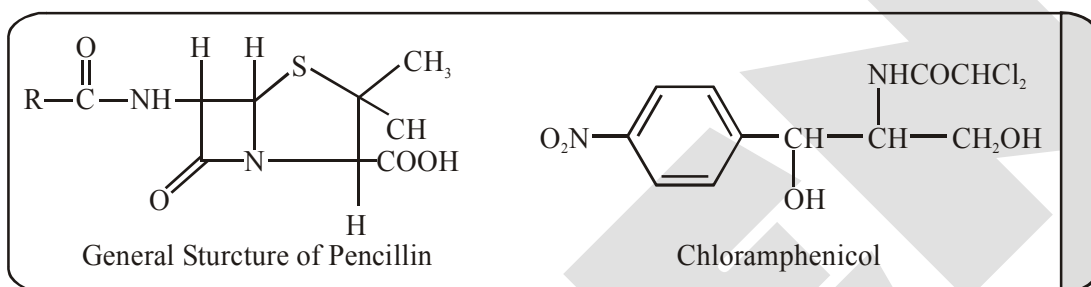
Bacteriostatic

Erythromycin
Tetracycline
Chloramphenicol

The range of bacteria or other microorganisms that are affected by a certain antibiotic is expressed as its spectrum of action. Antibiotics which kill or inhibit a wide range of Gram-positive and Gram-negative bacteria are said to be broad spectrum antibiotics.

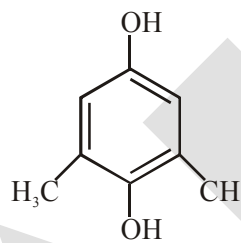
Those effective mainly against Gram-positive or Gram-negative bacteria are narrow spectrum antibiotics. If effective against a single organism or disease, they are referred to as limited spectrum antibiotics. Penicillin G has a narrow spectrum. Ampicillin and Amoxycillin are synthetic modifications of penicillins. These have broad spectrum. It is absolutely essential to test the patients for sensitivity (allergy) to penicillin before it is administered. In India, penicillin is manufactured at the Hindustan Antibiotics in Pimpri and in private sector industry.

Chloramphenicol, isolated in 1947, is a broad spectrum antibiotic. It is rapidly absorbed from the gastrointestinal tract and hence can be given orally in case of typhoid, dysentery, acute fever, certain form of urinary infections, meningitis and pneumonia. Vancomycin and ofloxacin are the other important broad spectrum antibiotics. The antibiotic dysidazirine is supposed to be toxic towards certain strains of cancer cells.

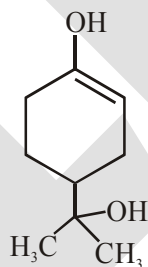


(b) Antiseptics and disinfectants

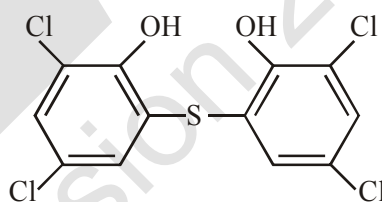
Antiseptics and disinfectants are also the chemicals which either kill or prevent the growth of microorganisms.



Chloroxylenol



Terpineol



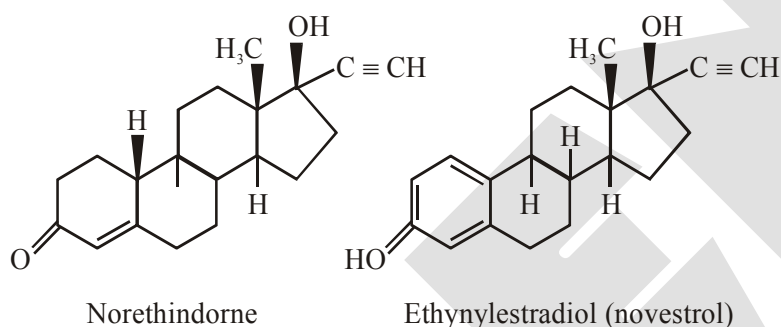
Bithionol

Antiseptics are applied to the living tissues such as wounds, cuts, ulcers and diseased skin surfaces. Examples are furacine, soframycin, etc. These are not ingested like antibiotics. Commonly used antiseptic, dettol is a mixture of chloroxylenol and terpineol. Bithionol (the compound is also called bithional) is added to soaps to impart antiseptic properties. Iodine is a powerful antiseptic. Its 2-3 per cent solution in alcohol-water mixture is known as tincture of iodine. It is applied on wounds. Iodoform is also used as an antiseptic for wounds. Boric acid in dilute aqueous solution is weak antiseptic for eyes.

Disinfectants are applied to inanimate objects such as floors, drainage system, instruments, etc. Some substances can act as an antiseptic as well as disinfectant by varying the concentration. For example, 0.2 per cent solution of phenol is an antiseptic while its one percent solution is disinfectant. Chlorine in the concentration of 0.2 to 0.4 ppm in aqueous solution and sulphur dioxide in very low concentrations, are disinfectants.

Antifertility Drugs

Antibiotic revolution has provided long and healthy life to people. The life expectancy has almost doubled. The increased population has caused many social problems in terms of food resources, environmental issues, employment, etc. To control these problems, population is required to be controlled. This has led to the concept of family planning. Antifertility drugs are of use in this direction. Birth control pills essentially contain a mixture of synthetic estrogen and progesterone derivatives. Both of these compounds are hormones. It is known that progesterone suppresses ovulation. Synthetic progesterone derivatives are more potent than progesterone. Norethindrone is an example of synthetic progesterone derivative most widely used as antifertility drug. The estrogen derivative which is used in combination with progesterone derivative is ethynylestradiol (novestrol).



SUMMARY

Medicine or Drugs

Chemical substances helping to a human body or an animal either for treatment of diseases or to reduce suffering from pain are called medicine or drug.

The treatment of disease by chemical compound which destroy the micro organism without attacking the tissue of the human body is known as chemotherapy, and the compounds used are called chemotherapeutic agent.

Various type of medicinal compounds are -

(A) Antiseptics :

Which prevent or destroy the growth of the harmful micro organism, common antiseptics are- Dettol, Savlon, Cetavlon, acriflavin, Iodine, methylene blue, mercurochrome & KMnO_4 .

Dettol is a mixture of chloroxylenol and terpineol. Its dilute solution is used to clean wounds.

Bithional -It is added to soap to impart antiseptic properties

(B) Disinfectants :

The chemical compounds capable of completely destroying the micro organism are termed as disinfectants. These are toxic to living tissues.

These are utilized for sterilization of floor, toilets instruments & cloths.

eg. 1% solution of phenol in disinfectant while 0.2% solution of phenol is antiseptic.

(C) Analgesics :

The substance which are used to get relief from pain. These are of two types -

- (a) Narcotics or habit forming drugs
- (b) Non-narcotics

(a) Narcotics : These are alkaloids and mostly opium products, causes sleep and unconsciousness when taken in higher doses.

e.g. Morphine, codeine, heroine

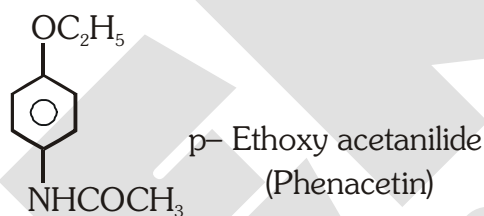
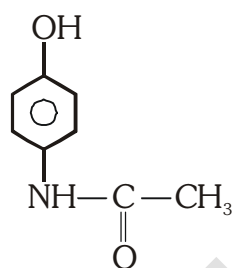
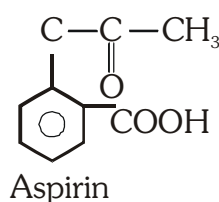
(b) Non-narcotics : Analgesics belonging to this category are effective antipyretics also.

e.g. Aspirin & novalgin, Ibuprofen, Naproxen

(D) Antipyretics :

To bring down the body temp. in high fever are called antipyretics.

e.g : (a) Aspirin, (b) Analgin (Novalgin), (c) Paracetamol, (d) Phenacetin

**(E) Antimalarials :**

To bring down the body temperature during malarial fever.

e.g. Quinine, Chloroquine, Paraquine and Primaquine etc.

(F) Tranquilizers :

The chemical substances which acts on the central nervous system and has a calming effect. Since these are used for mental diseases so are known as psychotherapeutic drugs.

They are of two types - (a) Sedative or hypnotics (b) Mood elevators

(a) **Sedative** : Reduce nervous tension and promote relaxation. e.g. Reserpine, barbituric acid and its

derivatives as luminal & seconal.

- (b) **Mood elevators or Antidepressants** : A drug used for treatment of highly depressed patient, who has lost his confidence.

Example : Benzedrine (amphetamine)

(G) Anaesthetics :

These are chemical substances helping for producing general or local insensibility to pain and other sensation.

These are of two types (a) General (b) Local

- (a) **General** : - Produce unconsciousness and are given at the time of major surgical operations.

Example : Gaseous form → Nitrous oxide, ethylene, cyclopropane etc.

Liquid form → Chloroform, divinyl ether and sodium pentothal etc.

- (b) **Local anaesthetics:** Produce loss of sensation on a small portion of the body. It is used for minor operations.

Example : Jelly form → Oxylocain

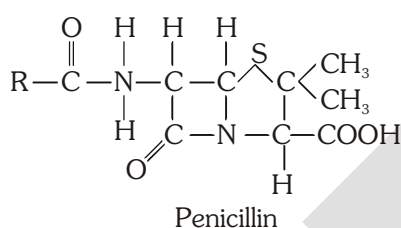
Spray form → Ethyl chloride

Injection form → Procain

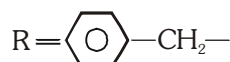
- (H) **Antibiotics :** The chemical substances produced from some micro organism (fungi bacteria or mold) and are used to inhibits the growth of other micro organism.

These are effective in the treatment of infections diseases.

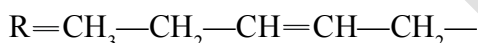
Example : Penicillin - It is highly effective drug for pneumonia, Bronchitis, abcesses, sore throat etc. For other naturally occuring penicillin -



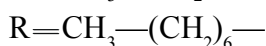
R - May be -



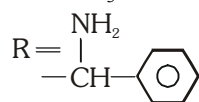
(Penicillin G)



(Penicillin-F)



(Penicillin-K)



(Ampicillin)

Synthetic antibiotics are Streptomycin - (Tuberculosis),

Chloromycetin - (Typhoid, Meningitis, Pneumonia, diarrhoea, dysentary etc.)

Tetracyclins - (Acute fever, trachoma, dysentery & urinary tract infection)

- (I) **Sulpha drugs :** Having great antibacterial powers. These are a group of drugs which are derivatives of sulphanilamide.

Other sulpha drugs are - (a) **Sulphathiazole** -Mainly used in severe infections.

(b) **Sulpha guanidine** - Used in bacillary dysentery

(c) **Sulpha pyridine** - Used in pneumoina

(d) **Sulpha diazine** - Used in dysentery, urinary infection and respiratory infection.

EXERCISE

- An antipyretic is –
(A) Quinine (B) Paracetamol (C) Luminal (D) Piperazine
- Medicine which is an antibiotic is –
(A) Ampicillin (B) Aspirin (C) Chloroquine (D) None of these
- Paracetamol is a/an –
(A) Both antipyretic and analgesic (B) Analgesic
(C) Antipyretic (D) Antimalarial
- Sulpha drugs are derivatives of –
(A) Benzene sulphonic acid (B) Sulphanillic acid
(C) Sulphanilamide (D) p - aminobenzoic acid
- Aspirin is called –
(A) Pyretic (B) Antiseptic (C) Antibiotic (D) Antipyretic
- Which of the following is an antidiabetic drug –
(A) Insulin (B) Penicillin (C) Chloroquine (D) Aspirin
- 2-Acetoxybenzoic acid is called –
(A) Antiseptic (B) Aspirin (C) Antibiotic (D) Mordant dye
- Match List-I with List-II and select the correct answer using the codes given below the lists –

List I	List II
I. Iodoform	A. Anaesthetic
II. Methyl salicylate	B. Antiseptic
III. Diethyl ether	C. Insecticide
IV. Hexachlorocyclohexane	D. Detergent
	E. Pain Balm
(A) I-B, II-E, III-C, IV-D	(B) I-D, II-B, III-A, IV-C
(C) I-B, II-E, III-A, IV-C	(D) I-C, II-A, III-D IV-B
- Arsenic drugs are mainly used in the treatment of –
(A) Jaundice (B) Typhoid (C) Syphilis (D) Cholera
- Aspirin is an acetylation product of –
(A) p-dihydroxybenzene (B) o-hydroxybenzoic acid
(C) o-dihydroxy benzene (D) m-hydroxybenzoic acid
- Chloramphenicol is an –
(A) Analgesic (B) Anaesthetic (C) Antibiotic (D) Antiseptic
- Substances which bring body temperature down are known as –
(A) Antipyretics (B) Analgin (C) Antibiotics (D) Hypnotics
- The drug given during hypertension is –
(A) Streptomycin (B) Chloroxylenol (C) Equanil (D) Aspirin
- Which of the following is known as broad spectrum antibiotic –
(A) Streptomycin (B) Ampicillin (C) Chloramphenicol (D) Penicillin
- Phenol is used as –
(A) An antiseptic (B) A disinfectant (C) Both (A) and (B) (D) None of these

16. The correct match between Item(I) and Item(II) is :

[JEE-MAIN-(Jan)-2019]

Item-I

Item-II

- (A) Nortehindrone (P) Anti-biotic
(B) Ofloxacin (Q) Anti-fertility
(C) Equanil (R) Hypertension
(S) Analgesics

- (A) A-R, B-P, C-S (B) A-Q, B-P, C-R (C) A-R, B-P, C-R (D) A-Q, B-R, C-S

17. The correct match between Item -I and Item-II is :

[JEE-MAIN-(Jan)-2019]

Item – I (drug)		Item – II (test)	
(A)	Chloroxylenol	(P)	Carbylamine Test
(B)	Norethindrone	(Q)	Sodium Hydrogen carbonate Test
(C)	Sulphapyridine	(R)	Ferric chloride test
(D)	Penicillin	(S)	Bayer's test

- (A) A→Q ; B→P ; C→S ; D→R
(B) A→R ; B→P ; C→S ; D→Q
(C) A→R ; B→S ; C→P ; D→Q
(D) A→Q ; B→S ; C→P ; D→R

18. Noradrenaline is a /an

[JEE-MAIN-(April)-2019]

- (A) Neurotransmitter (B) Antidepressant (C) Antihistamine (D) Antacid

ANSWER-KEY

- | | | | |
|------------|------------|------------|------------|
| 1. Ans. B | 2. Ans. A | 3. Ans. A | 4. Ans. C |
| 5. Ans. D | 6. Ans. A | 7. Ans. B | 8. Ans. C |
| 9. Ans. C | 10. Ans. B | 11. Ans. C | 12. Ans. A |
| 13. Ans. C | 14. Ans. C | 15. Ans. C | 16. Ans. B |
| 17. Ans. C | 18. Ans. A | | |

ALLEN
Session 2019-20

NURTURE COURSE
CONCENTRATION TERMS

CONCENTRATION TERMS

1. SOLUTIONS

A solution is a homogenous mixture of two or more pure substances whose composition may be altered within certain limits. Though the solution is homogenous in nature, yet it retains the properties of its constituents.

Generally solution is composed of two components, solute and solvent. Such type of solution is known as binary solutions.

Solvent is that component in solution whose physical state is the same as that of the resulting solution while other component is called as solute. If the physical state of both component is same, than the component in excess is known as solvent and other one is called as solute. Each component in a binary solution can be in any physical state such as liquid, solid and gaseous state.

Table 2.1: Types of Solutions

Type of Solutions	Solute	Solvent	Common Example
Gaseous Solutions	Gas	Gas	Mixture of oxygen and nitrogen gases
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
Liquid Solutions	Gas	Liquid	Oxygen dissolved in water
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
Solid Solutions	Gas	Solid	Solution of hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

2. CONCENTRATION TERMS :

The concentration of a solution is the amount of solute dissolved in a known amount of the solvent or solution. Solution can be described as dilute or concentrated solution as per their concentration. A dilute solution has a very small quantity of solute while concentrated solution has a large quantity of solute in solution. Various concentration terms are as follows.

2.1 Mass percentage :

It may be defined as the number of parts of mass of solute per hundred parts by mass of solution.

$$\% \text{ by mass } \left(\frac{w}{W} \right) : = \frac{\text{wt. of solute}}{\text{wt. of solution}} \times 100$$

[X % by mass means 100 gm solution contains X gm solute ; $\therefore$ (100 – X) gm solvent]

2.2 Mass-volume percentage (W/V %) :

It may be defined as the mass of solute present in 100 cm³ of solution. For example, If 100 cm³ of solution contains 5 g of sodium hydroxide, then the mass-volume percentage will be 5% solution.

$$\% \left(\frac{w}{V} \right) = \frac{\text{wt. of solute}}{\text{volume of solution}} \times 100 \text{ [for liq. solution]}$$

$$[X \% \left(\frac{w}{V} \right) \text{ means } 100 \text{ ml solution contains } X \text{ gm solute}]$$

2.3 Volume Percent :

It can be represented as % v/v or % volume and used to prepare such solutions in which both components are in liquids state. It is the number of parts of by volume of solute per hundred parts by volume of solution. Therefore,

$$\% \left(\frac{v}{V} \right) = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100$$

$$\text{2.4 Mole \%} = \frac{\text{Moles of solute}}{\text{Total moles}} \times 100$$

- For gases **% by volume** is same as **mole %**

2.5 Mole Fraction (X) :

Mole fraction may be defined as the ratio of number of moles of one component to the total number of moles of all the components (solute and solvent) present in solution. It is denoted by letter X and the sum of all mole fractions in a solution always equals one.

$$\text{Mole fraction (X)} = \frac{\text{Moles of solute}}{\text{Total moles}}$$

Mole fraction does not depend upon temperature and can be extended to solutions having more than two components.

2.6 Molarity (M) :

Molarity is most common unit for concentration of solution. It is defined as the number of moles of solute present in one litre or one dm³ of the solution or millimol of solute present in one mL of solution.

$$\text{Molarity (M)} = \frac{\text{Mole of solute}}{\text{volume of solution in litre}}$$

- 2.7 Molality (m) :** The number of gram mole of the solute present in 1000 g of the solvent is known as molality of solution. It represented by letter 'm'.

$$\text{Molality (m)} = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$$

The unit of molality is mol/kg and it does not effect by temperature.

- 2.8 Parts per million (ppm) :** The very low concentration of solute in solution can be expressed in ppm. It is the numbers of parts by mass of solute per million parts by mass of the solution.

$$\text{Parts per million (ppm)} = \frac{\text{Mass of solute}}{\text{Mass of solvent}} \times 10^6 \cong \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$$

- ◆ Get yourselves very much comfortable in their inter conversion. It is very handy.

Concentration Type	Mathematical Formula	Concept
Percentage by mass	$\% \left(\frac{w}{w} \right) = \frac{\text{Mass of solute} \times 100}{\text{Mass of solution}}$	Mass of solute present in 100 gm of solution.
Volume percentage	$\% \left(\frac{v}{v} \right) = \frac{\text{Volume of solute} \times 100}{\text{Volume of solution}}$	Volume of solute present in 100 cm ³ of solution.
Mass-volume percentage	$\% \left(\frac{w}{v} \right) = \frac{\text{Mass of solute} \times 100}{\text{Volume of solution}}$	Mass of solute present in 100 cm ³ of solution.
Parts per million	$\text{ppm} = \frac{\text{Mass of solute} \times 10^6}{\text{Mass of solution}}$	Parts by mass of solute per million parts by mass of the solution
Mole fraction	$X_A = \frac{\text{Mole of A}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + \dots}$ $X_B = \frac{\text{Mole of B}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + \dots}$	Ratio of number of moles of one component to the total number of moles.
Molarity	$M = \frac{\text{Mole of solute}}{\text{Volume of solution (in L)}}$	Moles of solute in one litre of solution.
Molality	$m = \frac{\text{Mass of solute} \times 1000}{\text{Molar mass of solute} \times \text{Mass of solvent (g)}}$	Moles of solute in one kg of solvent

Ex.1 Calculate the mole fractions of the components of the solution composed by 92 g glycerol and 90 g water ? ($M(\text{water}) = 18$; $M(\text{glycerol}) = 92$)

Ans. Moles of water = $90 \text{ g} / 18 \text{ g} = 5 \text{ mol}$ water
 Moles of glycerol = $92 \text{ g} / 92 \text{ g} = 1 \text{ mol}$ glycerol
 Total moles in solution = $5 + 1 = 6 \text{ mol}$
 Mole fraction of water = $5 \text{ mol} / 6 \text{ mol} = 0.833$
 Mole fraction of glycerol = $1 \text{ mol} / 6 \text{ mol} = 0.167$

Ex.2 What will be the Molarity of solution when water is added to 10 g CaCO_3 to make 100 mL of solution?

Ans. Mol of $\text{CaCO}_3 = 10 / 100 = 0.1$
 Molarity = Mole of solute / Volume of solution (L) = $0.10 \text{ mol} / 0.10 \text{ L}$
 Therefore ; Molarity of given solution = 1.0 M

Ex.3 Calculate the molality of a solution containing 20 g of sodium hydroxide (NaOH) in 250 g of water?

Ans. Moles of sodium hydroxide = $20 / 40 = 0.5 \text{ mol}$ NaOH
 250 gm = 0.25 kg of water
 Hence molality of solution = Mole of solute / Mass of solvent (kg) = $0.5 \text{ mol} / 0.25 \text{ kg}$
 or Molality(m) = 2.0 m

Ex.4 Calculate the grams of copper sulphate (CuSO_4) needed to prepare 250.0 mL of 1.00 M CuSO_4 ?

Ans. Moles of $\text{CuSO}_4 = M \times V = 1 \times \frac{250}{1000}$
 Molar mass of copper sulphate = 159.6 g/mol
 Hence Mass of copper sulphate (gm) = Moles of $\text{CuSO}_4 \times$ Molar mass of copper sulphate.

$$= 1 \times \frac{250}{1000} \times 159.6 \text{ g/mol}$$

$$= 39.9 \text{ gm of Copper sulphate}$$

Ex.5 How many grams of H_2SO_4 are present in 500 ml of 0.2M H_2SO_4 solution ?

Ans. $M = \frac{\text{moles}}{\text{vol.}} \Rightarrow \text{moles of } \text{H}_2\text{SO}_4 = M \times V = 0.2 \times \frac{500}{1000} \text{ L} = 0.1$
 Mass of $\text{H}_2\text{SO}_4 = 0.1 \times 98 = 9.8 \text{ g}$

Ex.6 Calculate the ppm of mercury in water in given sample contain 30 mg of Hg in 500 ml of solution.

Ans. Parts per million = $\frac{\text{Mass of solute} \times 10^6}{\text{Mass of solution}}$

Mass of Hg = 30 mg

Mass of water = 500/1 = 500g = 50×10^4 mg

(density = mass / volume ; density of water 1 g / ml) $w = \frac{v}{d}$

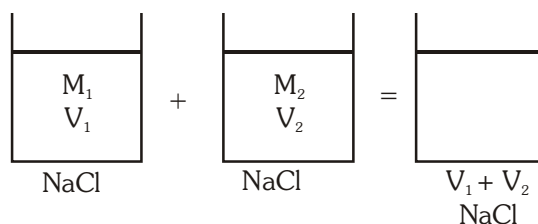
Therefore, ppm of mercury = $\frac{30 \times 10^6}{50 \times 10^4} = 60$ ppm of mercury

3. MIXING OF SOLUTIONS :

It is based on law of conservation of moles.

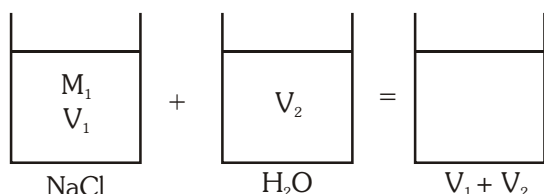
(i) Two solutions having same solute

Final molarity = $\frac{\text{Total moles}}{\text{Total volume}} = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$



(ii) **Dilution Effect :** When a solution is diluted, the moles of solute do not change but molarity changes while on taking out a small volume of solution from a larger volume, the molarity of solution do not change but moles change proportionately.

Final molarity = $\frac{M_1 V_1}{V_1 + V_2}$



n-fold or n-times dilution

$\Rightarrow$ **Final volume = $V_1 + V_2 = n(V_1)$**

Ex.7 50 ml 0.2 M H_2SO_4 is mixed with 50 ml 0.3M H_2SO_4 . Find molarity of final solution.

Ans. $M_f = \frac{\text{Total moles of } H_2SO_4}{\text{Total volume}} = \frac{50 \times 0.2 \times 10^{-3} + 50 \times 10^{-3} \times 0.3}{(50 + 50) \times 10^{-3}} = \boxed{0.25 M}$

Ex.8 Find final molarity in each case :

Ans. (i) 500 ml 0.1 M HCl + 500 ml 0.2M HCl

$M_f = \frac{500 \times 0.1 + 500 \times 0.2}{500 + 500} = \boxed{0.15 M}$

(ii) 50 ml 0.1M HCl + 150 ml 0.3MHCl + 300 ml H_2O

$M_f = \frac{50 \times 0.1 + 150 \times 0.3}{50 + 150 + 300} = \frac{50}{500} = 0.1 M$



$$M_f = \frac{\frac{4.9}{98} + \frac{250}{1000} \times 0.1}{\left(\frac{250 + 250}{1000}\right)} = \frac{50 + 25}{500} = \boxed{0.15 \text{ M}}$$

Ex.9 How much water should be added to 2M HCl solution to form 1 litre of 0.5 M HCl ?

Ans. Let V be initial volume

Then mol of HCl = constant

$$2 \times V = 1 \times 0.5 \Rightarrow V = 0.25 \text{ L}$$

$$\text{Volume of water added} = 1 - 0.25 = 0.75 \text{ L}$$

Ex.10 Find number of Na^+ & PO_4^{3-} ions in 250 ml of 0.2M Na_3PO_4 solution.

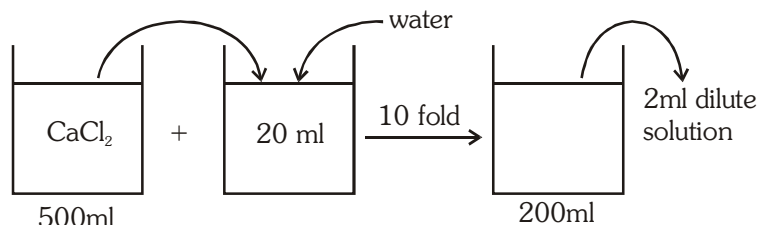
Ans. $\text{Na}_3\text{PO}_4 + \text{aq.} \longrightarrow 3\text{Na}^+(\text{aq}) + \text{PO}_4^{3-}(\text{aq})$ [Ionic compound when added to water ionize completely].

50 millimoles (m.m.) 150 mm 50 mm

$$\text{No. of } \text{Na}^+ \text{ ions} = 150 \times 10^{-3} \times N_A ; \text{No. of } \text{PO}_4^{3-} \text{ ions} = 50 \times 10^{-3} \times N_A$$

Ex.11 1.11g CaCl_2 is added to water forming 500 ml of solution. 20 ml of this solution is taken and diluted 10 folds. Find moles of Cl^- ions in 2 ml of diluted solution.

Ans. $\frac{1.11}{111} = 0.01 \text{ mol CaCl}_2$



$$\text{Moles of CaCl}_2 \text{ in 20ml solution} = \frac{0.01}{500} \times 20 = \frac{0.01}{25}$$

$$\text{In 200 ml solution moles of CaCl}_2 = \frac{0.01}{25} \quad [\text{Note : Dilution does not change moles of solute}]$$

$$\text{In 2 ml of dilute solution moles of CaCl}_2 = \frac{0.01}{25} \times \frac{2}{200} = \frac{0.01}{2500} = 8 \times 10^{-6}$$

$$\therefore \text{moles of } \text{Cl}^- = 2 \times 8 \times 10^{-6} = 1.6 \times 10^{-5}$$

Ex.12 What volumes of 1M & 2M H_2SO_4 solution are required to produce 2L of 1.75M H_2SO_4 solution?

Ans. Let XL be vol. of 1M solution.

$\therefore (2 - X)\text{L}$ is vol. of 2M solution.

$$\text{Moles of H}_2\text{SO}_4 = 2 \times 1.75 = 1(X) + (2 - X)2$$

$$3.5 = 4 - X ; X = 0.5 \text{ L}$$

i.e. 0.5L of 1M & 1.5 L of 2M solution required.

Ex.13 80g NaOH was added to 2L water. Find molality of solution if density of water = 1g/mL

Ans. $m = \frac{\text{moles of NaOH}}{\text{mass of H}_2\text{O}} \times 1000 = \frac{80/40}{2 \times 1000} \times 1000 = \boxed{1 \text{ molal}}$

Ex.14 A 100g NaOH solution has 20g NaOH. Find molality.

Ans. $m = \frac{20/40}{100-20} \times 1000 = \frac{500}{80} = \boxed{6.25 \text{ mol/kg}}$

Ex.15 Find molality of aqueous solution of CH_3COOH whose molarity is 2M and density $d = 1.2 \text{ g/mL}$.

Hint : $m = \frac{M}{d - M M_s} \times 1000$

where d = density in g L^{-1} , M = Molarity, m = molality, M_s = molar mass of solute.

Ans. $m = \frac{2}{1200 - 2 \times 60} \times 1000 = \boxed{1.85 \text{ m}}$

Ex.16 A solution is made by mixing 300 ml 1.5M $\text{Al}_2(\text{SO}_4)_3$ + 300 ml 2M CaSO_4 + 400 ml 3.5M CaCl_2

Find final molarity of (1) SO_4^{2-} , (2) Ca^{2+} , (3) Cl^- . [Assume complete dissociation of these compounds].

Ans. (1) $[\text{SO}_4^{2-}]_f = \frac{\text{Total moles}}{\text{Total volume}} = \frac{300 \times 1.5 \times 10^{-3} \times 3 + 300 \times 2 \times 10^{-3}}{(300 + 300 + 400) \times 10^{-3}} = 1.95 \text{ M}$

(2) $[\text{Ca}^{2+}]_f = \frac{300 \times 2 + 400 \times 3.5}{1000} = 2 \text{ M}$

(3) $[\text{Cl}^-]_f = \frac{400 \times 3.5 \times 2}{1000} = 2.8 \text{ M}$

Ex.17 A solution has 80% $\frac{w}{w}$ NaOH with density 2 g L^{-1} . Find (a) Molarity (b) Molality of solution.

Ans. Let V_{lit} be vol. of solution

Mass of solute = $(d \times V) \times \left(\frac{\% w}{w} \right) = 2 \times V \times \frac{80}{100} = 1.6V$

(a) $M = \frac{1.6V/40}{V} = \boxed{0.04 \text{ M}}$ (b) $m = \frac{1.6V/40}{2V - 1.6V} \times 1000 = \boxed{100 \text{ mol kg}^{-1}}$

Ex.18 4.450 g 100 per cent sulphuric acid was added to 82.20 g water and the density of the solution was found to be 1.029 g/cc at 25°C and 1 atm pressure. Calculate (a) the weight percent, (b) the mole fraction, (c) the mole percent, (d) the molality, (e) the molarity of sulphuric acid in the solution under these conditions.

Ans. Sulphuric acid = 4.450 g, Water = 82.20 g $\Rightarrow$ Wt. of solution = 86.65 g
 $\therefore$ Density of solution = 1.029 g/cc.

$$(a) \quad \text{Weight percent} = \frac{\text{wt. of solute}}{\text{wt. of solution}} \times 100 = \frac{4.450}{86.65} \times 100 = 5.14$$

(b) Mole fraction :

$$\text{Mole of solute} = \frac{\text{wt. of solute}}{\text{mol wt. of solute}} = \frac{4.45}{98} = 0.0454$$

$$\text{Mole of solvent} = \frac{82.20}{18} = 4.566$$

$$\text{Total moles in solution} = 0.0454 + 4.566 = 4.6114$$

$$\text{Mole fraction of solute} = \frac{0.0454}{4.6114} = 0.0098$$

$$(c) \quad \text{Mole percent} = \frac{\text{moles of solute}}{\text{Total moles in solution}} \times 100$$

$$= \text{mole fraction of solute} \times 100 = 0.0098 \times 100 = 0.98$$

$$(d) \quad \text{Molality} = \frac{\text{moles of solute}}{\text{mass of solvent (in gm)}} \times 1000$$

$$= \frac{0.0454 \times 1000}{82.2} = 0.552$$

$$(e) \quad \text{Molarity} = \frac{\text{moles of solute}}{\text{litre of solution}}$$

$$\text{Volume of solution} = \frac{\text{Mass}}{\text{Density}} = \frac{86.65}{1.029} \text{ ml}$$

$$= \frac{86.65}{1.029 \times 1000} \text{ litre}$$

$$\text{Molarity} = \frac{0.0454}{\frac{86.65}{1.029 \times 1000}} = \frac{0.0454 \times 1000 \times 1.029}{86.65} = 0.539$$

Ex.19 A solution of KCl has a density of 1.69 g mL^{-1} and is 67% by weight. Find the density of the solution if it is diluted so that the percentage by weight of KCl in the diluted solution is 30%.

Ans. Let the volume of the KCl solution be 100 mL,

$$\text{Weight of KCl solution} = 100 \times 1.69 = 169 \text{ g}$$

$$100 \text{ g of solution contains} = 67 \text{ g of KCl}$$

$$169 \text{ g of solution} = \frac{67}{100} \times 169 = 113.23 \text{ g}$$

Let x mL of H_2O be added.

$$\text{New volume of solution} = (100 + x) \text{ mL}$$

$$\text{New weight of solution} = (169 + x) \text{ g}$$

$$(\text{Since } x \text{ mL of } \text{H}_2\text{O} = x \text{ g of } \text{H}_2\text{O}, d_{\text{H}_2\text{O}} = 1)$$

$$\text{New percentage of the solution} = 30\%$$

$$\% \text{ by weight} = \frac{\text{weight of solute} \times 100}{\text{weight of solution}}$$

$$30 = \frac{113.23}{(169 + x)} \times 100$$

$$x = 208.43 \text{ mL} = 208.43 \text{ g}$$

$$\text{New density} = \frac{\text{New weight of solution}}{\text{New volume of solution}}$$

$$= \frac{(169 + x)}{(100 + x)}$$

$$\frac{(169 + 208.43)}{(100 + 208.43)} = \frac{377.43}{308.43}$$

$$\therefore d = 1.224$$

4. SOME TYPICAL CONCENTRATION TERMS

4.1 PERCENTAGE LABELLING OF OLEUM :

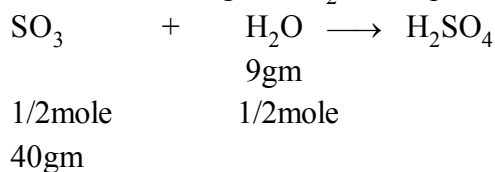
Labelled as '% oleum', it means maximum amount of H_2SO_4 that can be obtained from 100 gm of such oleum (mix of H_2SO_4 and SO_3) by adding sufficient water. For ex. 109 % oleum sample means, with the addition of sufficient water to 100 gm oleum sample 109 gm H_2SO_4 is obtained.

$$\% \text{ labelling of oleum sample} = (100 + x)\%$$

x = mass of H_2O required for the complete conversion of SO_3 in H_2SO_4

Ex.20 Find the mass of free SO_3 present in 100 gm, 109 % oleum sample.

Sol. 109 % means, 9 gm of H_2O is required.

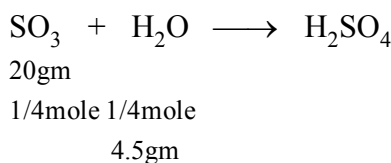


$\therefore$ Mass of free SO_3 = 40 gm, Mass of H_2SO_4 = 60 gm

Note: Work out, what are the maximum and minimum value of the % labelling.

Ex.21 Find the % labelling of 100 gm oleum sample if it contains 20 gm SO_3 .

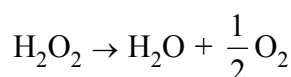
Sol. % labelling of oleum sample = $(100 + x)\%$



$\therefore$ % labelling of oleum sample = $(100 + 4.5)\% = 104.5\%$

II. VOLUME STRENGTH OF H_2O_2 SOLUTION :

Labelled as 'volume H_2O_2 ', it means volume of O_2 (in litre) at STP that can be obtained from 1 litre of such a sample when it decomposes according to



Volume Strength of H_2O_2 Solution = $11.35 \times \text{molarity}$

Ex.22 Find the % w/v of "10 V" H_2O_2 solution-

Sol. Molarity (M) of solution = $\frac{\text{volume strength}}{11.35} = \frac{10}{11.35}$

$$\% \left(\frac{w}{v} \right) = \frac{M \times \text{mol. wt. of solute}}{10} = \frac{10}{11.35} \times \frac{34}{10} = 3\%$$

EXERCISE # S-I

CONCENTRATION TERMS

- Q.1** Calculate the molarity of the following solutions :
- 4g of caustic soda is dissolved in 200 mL of the solution.
 - 5.3 g of anhydrous sodium carbonate is dissolved in 100 mL of solution.
 - 0.365 g of pure HCl gas is dissolved in 50 mL of solution.
- Q.2** Density of a solution containing 13% by mass of sulphuric acid is 0.98 g/mL. Then molarity of solution will be
- Q.3** The density of a solution containing 7.3% by mass of HCl is 1.2 g/mL. Calculate the molarity of the solution.
- Q.4** 15 g of methyl alcohol is present in 100 mL of solution. If density of solution is 0.90 g mL^{-1} . Calculate the mass percentage of methyl alcohol in solution
- Q.5** Units of parts per million (ppm) or per billion (ppb) are often used to describe the concentrations of solutes in very dilute solutions. The units are defined as the number of grams of solute per million or per billion grams of solvent. Bay of Bengal has 2.1 ppm of lithium ions. What is the molality of Li^+ in this water ? ($\text{Li} = 7$)
- Q.6** A 7.0 M solution of KOH in water contains 28% by mass of KOH. What is density of solution in gm/ml ?
- Q.7** The average concentration of Na^+ ion in human body is 3.0 to 3.9 gm per litre. The molarity of Na^+ ion is about.
- Q.8** What is the concentration of chloride ion, in molarity, in a solution containing 10.56 gm $\text{BaCl}_2 \cdot 8\text{H}_2\text{O}$ per litre of solution ? ($\text{Ba} = 137$)
- Q.9** The concentration of a solution is 8% (w/w) and 10% (w/v). Calculate density (in gm/mL) of solution?
- Q.10** The mole fraction of solute in aqueous urea solution is 0.2. Calculate the mass percent of solute ?
- Q.11** The concentration of $\text{Ca}(\text{HCO}_3)_2$ in a sample of hard water is 405 ppm. The density of water sample is 1.0 gm/mL. Calculate the molarity of solution ?
- Q.12** 0.115 gm of sodium metal was dissolved in 500 ml of the solution in distilled water. Calculate the molarity of the solution?
- Q.13** How much BaCl_2 (in gm) would be needed to make 250 ml of a solution having the same concentration of Cl^- as one containing 1.825 gm HCl per 100 ml ? ($\text{Ba} = 137$)
- Q.14** Calculate **molality (m)** of each ion present in the aqueous solution of **2M NH_4Cl** assuming 100% dissociation according to reaction.
- $$\text{NH}_4\text{Cl (aq)} \longrightarrow \text{NH}_4^+ \text{ (aq)} + \text{Cl}^- \text{ (aq)}$$
- Given :** Density of solution = 3.107 gm / ml.
- Q.15** 1200gm aqueous solution contains 200gm calcium bromide (CaBr_2). Calculate molality of solution.

PROBLEMS RELATED WITH MIXING & DILUTION

- Q.16** Find out the volume of 98% w/w H_2SO_4 (density = 1.8 gm/ml), must be diluted to prepare 12.6 litres of 2.0 M sulphuric acid solution.
- Q.17** Determine the volume (in mL) of diluted nitric acid ($d = 1.11 \text{ g mL}^{-1}$, 20% w/v HNO_3) that can be prepared by diluting 50 mL of conc. HNO_3 with water ($d = 1.42 \text{ g mL}^{-1}$, 70% w/v).
- Q.18** 500 ml of 2 M NaCl solution was mixed with 200 ml of 2 M NaCl solution. Calculate the molarity of NaCl in final solution.
- Q.19** Calculate the amount of the water "in mL" which must be added to a given solution of concentration of 40 mg silver nitrate per ml, to yield a solution of concentration of 16 mg silver nitrate per ml?
- Q.20** A mixture containing equimolar amounts of $\text{Ca}(\text{OH})_2$ and $\text{Al}(\text{OH})_3$ requires 0.5 L of 4.0 M HCl to react with it completely. Total moles of the mixture are :
- Q.21** 500 gm of urea solution of mole fraction 0.2 is diluted to 1500 gm. Calculate the mole fraction of solute in the diluted solution?
- Q.22** When V ml of 2.2 M H_2SO_4 solution is mixed with 10 V ml of water, the volume contraction of 2% take place. Calculate the molarity of diluted solution?
- Q.23** What volume (in mL) of 0.8 M AlCl_3 solution should be mixed with 50 ml of 0.2 M CaCl_2 solution to get solution of chloride ion concentration equal to 0.6 M?
- Q.24** A solution containing 200 ml 0.5 M KCl is mixed with 50 ml 19% w/v MgCl_2 and resulting solution is diluted 8 times. Molarity of chloride ion in final solution is :
- Q.25** 100 mL, 3% (w/v) NaOH solution is mixed with 100 ml, 9% (w/v) NaOH solution. The molarity of final solution is-

SOME TYPICAL CONCENTRATION TERMS

- Q.26** An oleum sample is labelled as 118 %, Calculate
- Mass of H_2SO_4 in 100 gm oleum sample.
 - Maximum mass of H_2SO_4 that can be obtained if 30 gm sample is taken.
 - Composition of mixture (mass of components) if 40 gm water is added to 30 gm given oleum sample.
- Q.27** A mixture is prepared by mixing 10 gm H_2SO_4 and 40 gm SO_3 calculate,
- mole fraction of H_2SO_4
 - % labelling of oleum
- Q.28** 500 ml of a H_2O_2 solution on complete decomposition produces 2 moles of H_2O . Calculate the volume strength of H_2O_2 solution?
- Q.29** $2\text{H}_2\text{O}_2(\text{aq}) \longrightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
- Under conditions where 1 mole of gas occupies 24 dm^3 , X L of $\frac{1}{24}$ M solution of H_2O_2 produces 3 dm^3 of O_2 . Thus X is :-
- Q.30** The volume strength of 100 ml H_2O_2 solution which produce 5.6 litre of oxygen gas at 1 bar & 0°C .

EXERCISE # S-II

- Q.1** What volume of 0.2 M NaOH (in ml) solution should be mixed to 500 ml of 0.5 M NaOH solution so that 300 ml of final solution is completely neutralised by 20 ml of 2 M H_3PO_4 solution.
[Assuming 100% dissociation]
- Q.2** How much minimum volume (in ml) of $\left(\frac{5}{51}\right)$ M aluminium sulphate solution should be added to excess calcium nitrate to obtain at least 1 gm of each salt in the reaction.

$$\text{Al}_2(\text{SO}_4)_3 + 3\text{Ca}(\text{NO}_3)_2 \longrightarrow 2\text{Al}(\text{NO}_3)_3 + 3\text{CaSO}_4$$
- Q.3** One litre of milk weighs 1.035 kg. The butter fat is 4% (v/v) of milk and has density of 875 kg/m^3 . If the density of fat free skimmed milk is 'x' kg/m^3 , the value of (4.8x) is ?
- Q.4** 100 ml of 0.1 M solution of AB (d = 1.5 gm/ml) is mixed with 100 ml of 0.2 M solution of CB_2 (d = 2.5 gm/ml). Calculate the molarity of B^- in final solution if the density of final solution is 4 gm/ml. Assuming AB and CB_2 are non reacting & dissociates completely into A^+ , B^- , C^{+2} .
- Q.5** 60 ml of a "x" % w/w alcohol by weight (d = 0.6 g/cm^3) must be used to prepare 200 cm^3 of 12% alcohol by weight (d = 0.90 g/cm^3). Calculate the value of "x"?
- Q.6** If 0.5 M methanol undergo self dissociation like $\text{CH}_3\text{OH} \rightleftharpoons \text{CH}_3\text{O}^- + \text{H}^+$ & if concentration of H^+ is $2.5 \times 10^{-4} \text{ M}$ then calculate % dissociation of methanol.
- Q.7** 1120 gm of 2 'm' urea solution is mixed with 2480 gm of 4 'm' urea solution. Calculate the molality of the resulting solution?
- Q.8** 50 ml of '20V' H_2O_2 is mixed with 200 ml, '10V' H_2O_2 . The volume strength of resulting solution is
- Q.9** 500 ml of 2M CH_3COOH solution is mixed with 600 ml 12% w/v CH_3COOH solution then calculate the final molarity of solution.
- Q.10** 45.4 V H_2O_2 solution (500 ml) when exposed to atmosphere loses 11.2 litre of O_2 at 1 atm, & 273 K. New molarity of H_2O_2 solution. (Assume no change in volume)

EXERCISE # O-I

- Q.1** 125 ml of 8% w/w NaOH solution (sp. gravity 1) is added to 125 ml of 10% w/v HCl solution. The nature of resultant solution would be _____.
(A) Acidic (B) Basic (C) Neutral (D) Can not be predicted
- Q.2** 8 g NaOH is dissolved in one litre of solution, its molarity is :
(A) 0.8 M (B) 0.4 M (C) 0.2 M (D) 0.1 M
- Q.3** If 18 g of glucose is present in 1000 g of solvent, the solution is said to be :
(A) 1 molar (B) 0.1 molar (C) 0.5 molar (D) 0.1 molal
- Q.4** The molarity of pure water is :
(A) 100 M (B) 55.6 M (C) 50 M (D) 18M
- Q.5** Mole fraction of $C_3H_5(OH)_3$ (glycerine) in a solution of 36 g of water and 46 g of glycerine is :
(A) 0.46 (B) 0.36 (C) 0.20 (D) 0.40
- Q.6** A molal solution is one that contains one mole of a solute in
(A) 1000 g of the solvent (B) one litre of the solution
(C) one litre of the solvent (D) 22.4 litres of the solution
- Q.7** The mole fraction of oxygen in a mixture of 7g of nitrogen and 8g of oxygen is :
(A) $\frac{8}{15}$ (B) 0.5 (C) 0.25 (D) 1.0
- Q.8** The molarity of a solution of sodium chloride (mole wt. = 58.5) in water containing 5.85 gm of sodium chloride in 500 ml of solution is :-
(A) 0.25 (B) 2.0 (C) 1.0 (D) 0.2
- Q.9** The molarity of 98% by wt. H_2SO_4 ($d = 1.8$ g/ml) is
(A) 6 M (B) 18 M (C) 10 M (D) 4 M
- Q.10** Which one of the following modes of expressing concentration of solution is independent of temperature -
(A) Molarity (B) Molality (C) % w/v (D) Grams per litre
- Q.11** For preparing 0.1 M solution of H_2SO_4 in one litre, we need H_2SO_4 :
(A) 0.98 g (B) 4.9 g (C) 49.0 g (D) 9.8 g
- Q.12** 1000 g aqueous solution of $Ca(NO_3)_2$ contains 10 g of calcium nitrate. Concentration of the solution is :
(A) 10 ppm (B) 100 ppm (C) 1000 ppm (D) 10,000 ppm
- Q.13** How much volume of 3.0 M H_2SO_4 is required for the preparation of 1.0 litre of 1.0 M solution?
(A) 300 ml (B) 320 ml (C) 333.3 ml (D) 350.0 ml
- Q.14** Equal weight of NaCl and KCl are dissolved separately in equal volumes of solutions. Molarity of the solutions will be –
(A) Equal (B) Greater for NaCl
(C) Greater for KCl (D) Uncomparable.

- Q.15** How much water should be added to 200 cc of semimolar solution of NaOH to make it exactly decimolar :-
 (A) 1000 cc (B) 400 cc (C) 800 cc (D) 600 cc
- Q.16** 100 ml of 0.3 M HCl solution is mixed with 200 ml of 0.3 M H_2SO_4 solution. What is the molarity of H^+ in resultant solution ?
 (A) 0.9 (B) 0.6 (C) 0.4 (D) 0.5
- Q.17** H_2O_2 solution used for hair bleaching is sold as a solution of approximately 5.0 g H_2O_2 per 100 mL of the solution. The molecular mass of H_2O_2 is 34. The molarity of this solution is approximately:-
 (A) 0.15 M (B) 1.5 M (C) 3.0 M (D) 3.4 M
- Q.18** 171 g of cane sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is dissolved in 1 litre of water. The molarity of the solution is :
 (A) 2.0 M (B) 1.0 M (C) 0.5 M (D) 0.25 M
- Q.19** How much grams of CH_3OH should be dissolved in water for preparing 150 ml of 2.0 M CH_3OH solution ?
 (A) 9.6 (B) 2.4 (C) 9.6×10^3 (D) 4.3×10^2
- Q.20** Molality of 20% (w/w) aq. glucose solution is :
 (A) $\frac{25}{18}$ m (B) $\frac{10}{9}$ m (C) $\frac{25}{9}$ m (D) $\frac{5}{18}$ m
- Q.21** Molarity of liquid HCl, if density is 1.17 g/cc. :
 (A) 36.5 M (B) 18.25 M (C) 32.05 M (D) 42.10 M
- Q.22** The molarity of a solution made by mixing 50 ml of conc. H_2SO_4 (18 M) with 50 ml of water, is:
 (A) 36 M (B) 18 M (C) 9 M (D) 6M
- Q.23** Equal volumes of 10% (w/v) of HCl is mixed with 10% (w/v) NaOH solution. The resultant solution be.
 (A) basic (B) neutral
 (C) acidic (D) can't be predicted.
- Q.24** What volume of 0.2 M NaOH solution is needed for complete neutralisation of 0.49 gm orthophosphoric acid -
 (A) 75 ml (B) 300 ml (C) 0.075 ml (D) 50 ml
- Q.25** 34 g of hydrogen peroxide is present in 1135 mL of solution. Volume strength of solution is:
 (A) 10 V (B) 20 V (C) 30 V (D) 32 V
- Q.26** Label an oleum sample which has mass fraction of SO_3 equal to 0.6 :
 (A) 115 % (B) 109 % (C) 104.5 % (D) 113.5 %
- Q.27** If 50 gm oleum sample rated as 118% is mixed with 18 gm water, then the correct option is
 (A) The resulting solution contains 18 gm of water and 118 gm H_2SO_4
 (B) The resulting solution contains 9 gm water and 59 gm H_2SO_4
 (C) The resulting solution contains only 118 gm pure H_2SO_4
 (D) The resulting solution contains 68 gm of pure H_2SO_4

Q.28 12.5gm of fuming H_2SO_4 (labelled as 112%) is mixed with 100 lit water. Molar concentration of H^+ in resultant solution is :

[Note : Assume that H_2SO_4 dissociate completely and there is no change in volume on mixing]

- (A) $\frac{2}{700}$ (B) $\frac{2}{350}$ (C) $\frac{3}{350}$ (D) $\frac{3}{700}$

Q.29 20 ml of '20 vol' H_2O_2 solution is diluted to 80 ml. The final volume strength of solution is -

- (A) '80 vol' (B) '25 vol' (C) '5 vol' (D) '8 vol'

Q.30 Assuming complete precipitation of AgCl , calculate the sum of the molar concentration of all the ions if 2 lit of 2M Ag_2SO_4 is mixed with 4 lit of 1 M NaCl solution is :

- (A) 4M (B) 2M (C) 3 M (D) 2.5 M

Q.31 Molarity and Molality of a solute (M. wt = 50) in aqueous solution is 9 and 18 respectively. What is the density of solution.

- (A) 1 g/cc (B) 0.95 g/cc (C) 1.05 g/cc (D) 0.662 g/cc

Q.32 The relationship between mole fraction (X_A) of the solute & molality 'm' of its solution in ammonia would be

(A) $\frac{55.56(X_A)}{1 - X_A} = m$ (B) $\frac{58.82(X_A)}{1 - X_A} = m$

(C) $\frac{58.82(1 - X_A)}{X_A} = m$ (D) $\frac{55.56(1 - X_A)}{X_A} = m$

Q.33 3.0 molal NaOH solution has a density of 1.12 g/mL. The molarity of the solution is-

- (A) 2.97 (B) 3 (C) 3.05 (D) 3.5

EXERCISE # O-II

Q.1 Statement -1 : Molality of pure ethanol is lesser than pure water.

Statement -2 : As density of ethanol is lesser than density of water.

[Given : $d_{\text{ethanol}} = 0.789 \text{ gm/ml}$; $d_{\text{water}} = 1 \text{ gm/ml}$]

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is false, statement-2 is true.
- (D) Statement-1 is true, statement-2 is false.

Q.2 Statement-1 : Molarity and molality have almost same value for a very dilute aqueous solution.

Statement-2 : In all very dilute solution, the mass of solvent (in gm) is equal to the volume of solution (in ml).

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

Q.3 Statement-1 : Molarity of a solution depends on temperature but molality is independent of temperature.

Statement-2 : Molarity depends on volume of solution but molality depends on mass of solvent.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

Q.4 Statement-1 : The mass fraction of solute in a solution is always greater than its mole fraction.

Statement-2 : Mole fraction of solvent in an aqueous solution of ethanol must be greater than that of solute.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

- Q.5 Statement-1 :** 0.5 M - aq. NaOH solution is identical to 2% (w/v) aq. NaOH solution.
Statement-2 : Concentration in % (w/v) is 4 times the molar concentration for all aqueous solution.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- Q.6** The molar concentration of HCl (aq.) is 10^{-5} M. Which of the following statements are correct.
 ($d_{\text{solution}} = 1 \text{ gm/cc}$)
 (A) The mole fraction of HCl $\cong 1.8 \times 10^{-7}$
 (B) The concentration of HCl in ppm is 3.65 ppm
 (C) The molality of HCl solution is approximately 10^{-5} m
 (D) The (w/v)% of solution is $3.65 \times 10^{-5} \%$
- Q.7** Solution(s) containing 40 gm NaOH is/are
 (A) 50 gm of 80% (w/w) NaOH
 (B) 50 gm of 80% (w/v) NaOH [$d_{\text{soln.}} = 1.2 \text{ gm/ml}$]
 (C) 50 gm of 20 M NaOH [$d_{\text{soln.}} = 1 \text{ gm/ml}$]
 (D) 50 gm of 5m NaOH
- Q.8** The **incorrect** statement(s) regarding 2M MgCl_2 aqueous solution is/are ($d_{\text{solution}} = 1.09 \text{ gm/ml}$)
 (A) Molality of Cl^- is **4.44 m**
 (B) Mole fraction of MgCl_2 is exactly **0.035**
 (C) The conc. of MgCl_2 is **19% w/v**
 (D) The conc. of MgCl_2 is **19×10^4 ppm**
- Q.9** A sample of H_2O_2 solution labelled as 56.75 volume has density of 530 gm/L. Mark the correct option(s) representing concentration of same solution in other units. (Solution contains only H_2O and H_2O_2)
 (A) $M_{\text{H}_2\text{O}_2} = 6$ (B) $\% \frac{w}{v} = 17$
 (C) Mole fraction of $\text{H}_2\text{O}_2 = 0.25$ (D) $m_{\text{H}_2\text{O}_2} = \frac{1000}{72}$
- Q.10** 100 mL of 0.06 M $\text{Ca}(\text{NO}_3)_2$ is added to 50 mL of 0.06 M $\text{Na}_2\text{C}_2\text{O}_4$. After the reaction is complete (CaC_2O_4 is precipitated)
 (A) 0.003 moles of calcium oxalate will get precipitated
 (B) 0.003 M Ca^{2+} will remain in excess
 (C) $\text{Na}_2\text{C}_2\text{O}_4$ is the limiting reagent
 (D) Oxalate ion ($\text{C}_2\text{O}_4^{2-}$) concentration in final solution is 0.003 M

Comprehension Q.11 and Q.12 (2 questions)

2 litre of 9.8 % w/w H_2SO_4 ($d = 1.5 \text{ gm/ml}$) solution is mixed with 3 litre of 1 M KOH solution.

Q.11 The number of moles H_2SO_4 added are

- (A) 1 (B) 2 (C) 3 (D) 0.5

Q.12 The concentration of H^+ if solution is acidic or concentration of OH^- if solution is basic in the final solution is

- (A) 0 (B) $\frac{3}{10}$ (C) $\frac{3}{5}$ (D) $\frac{2}{5}$

Comprehension Q.13 and Q.14 (2 questions)

30 gm H_2SO_4 is mixed with 20 gram SO_3 to form mixture.

Q.13 Find mole fraction of SO_3 .

- (A) 0.2 (B) 0.45 (C) 0.6 (D) 0.8

Q.14 Determine % labelling of oleum solution.

- (A) 104.5 (B) 106 (C) 109 (D) 110

Comprehension Q.15 and Q.16 (2 questions)**Estimation of halogens :**

Carius method : A known mass of compound is heated with conc. HNO_3 in the presence of AgNO_3 contained in a hard glass tube known as carius tube in a furnace. C and H are oxidised to CO_2 and H_2O . The halogen forms the corresponding AgX . It is filtered, dried, and weighed.

Estimation of sulphur : A known mass of compound is heated with fuming HNO_3 or sodium peroxide (Na_2O_2) in the presence of BaCl_2 solution in Carius tube. Sulphur is oxidised to H_2SO_4 and precipitated as BaSO_4 . It is filtered, dried and weighed.

Q.15 0.15 gm of an organic compound gave 0.12 gm of silver bromide by the Carius method. Find the percentage of bromine in the compound. ($\text{Ag} = 108$, $\text{Br} = 80$)

- (A) 34.0 (B) 46.0 (C) 80.0 (D) 50.0

Q.16 0.32 gm of an organic substance when treated by Carius method gave 0.466 gm of BaSO_4 . Calculate the percentage of sulphur in the compound. ($\text{Ba} = 137$)

- (A) 10.0 (B) 34.0 (C) 20.0 (D) 30.0

Comprehension Q.17 and Q.18 (2 questions)**(d) Estimation of phosphorous :**

A known mass of compound is heated with fuming HNO_3 or sodium peroxide (Na_2O_2) in Carius tube which converts phosphorous to H_3PO_4 . Magnesia mixture ($\text{MgCl}_2 + \text{NH}_4\text{Cl}$) is then added, which gives the precipitate of magnesium ammonium phosphate (MgNH_4PO_4) which on heating gives magnesium pyrophosphate ($\text{Mg}_2\text{P}_2\text{O}_7$), which is weighed.

Q.17 0.124 gm of an organic compound containing phosphorus gave 0.222 gm of $\text{Mg}_2\text{P}_2\text{O}_7$ by the usual analysis. Calculate the percentage of phosphorous in the compound. ($\text{Mg} = 24$, $\text{P} = 31$)

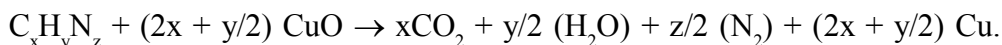
- (A) 25 (B) 75 (C) 62 (D) 50

- Q.18** An organic compound has 6.2 % of phosphorus. On sequence of reaction, the phosphorous present in the 10gm of organic compound is converted to $\text{Mg}_2\text{P}_2\text{O}_7$. Find the weight of $\text{Mg}_2\text{P}_2\text{O}_7$ formed.
- (A) 2.22 gm (B) 10.0 gm (C) 4.44 gm (D) 1.11 gm

Comprehension Q.19 and Q.22 (4 questions)

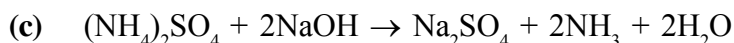
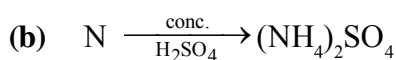
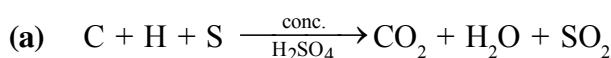
Estimation of nitrogen : There are two methods for the estimation of nitrogen (i) Dumas method and (ii) Kjeldahl's method.

- i. Dumas method :** A known mass of compound is heated with copper oxide (CuO) in an atmosphere of CO_2 , which gives free nitrogen along with CO_2 and H_2O .



The gaseous mixture is passed over a heated copper gauze which converts traces of nitrogen oxides formed to N_2 . The gaseous mixture is collected over an aqueous solution of KOH which absorbs CO_2 , and nitrogen is collected in the upper part of the graduated tube.

- ii. Kjeldahl's method :** A known mass of organic compound (0.5 gm) is mixed with K_2SO_4 (10 gm) and CuSO_4 (1.0 gm) or a drop of mercury (Hg) and conc. H_2SO_4 (25 ml), and heated in Kjeldahl's flask. CuSO_4 or Hg acts as a catalyst, while K_2SO_4 raises the boiling point of H_2SO_4 . The nitrogen in the organic compound is quantitatively converted to ammonium sulphate. The resulting mixture is then distilled with excess of NaOH solution and the NH_3 evolved is passed into a known but excess volume of standard HCl or H_2SO_4 . The acid left unused is estimated by titration with some standard alkali. The amount of acid used against NH_3 can thus be known and from this the percentage of nitrogen is calculated.



- iii.** This method is not applicable to compounds containing N in nitro and azo groups, and N present in the ring (e.g., pyridine) as N of these compounds does not change to $(\text{NH}_4)_2\text{SO}_4$ (ammonium sulphate) under these reaction conditions.

- Q.19** 0.30 gm of an organic compound gave 82.1 ml of nitrogen collected at 300K and 775 mm pressure in Dumas method. Calculate the percentage of nitrogen in the compound. (Vapour pressure of water or aqueous tension of water at 300K is 15 mm.

(A) 31.11 (B) 15.56 (C) 28.0 (D) 31.72

- Q.20** 0.50 gm of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 ml of 0.5M H_2SO_4 . The residual acid required 60 ml of M/2 NaOH solution. Find the percentage of nitrogen in the compound.

(A) 50 (B) 56 (C) 66 (D) 40

- Q.21** 0.4 gm of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 ml of 0.5M H_3PO_3 . The residual acid required 30 ml of 0.5M $\text{Ca}(\text{OH})_2$. Find the percentage of N_2 in the compound.

(A) 20 (B) 50 (C) 70 (D) 45

Q.22 0.002 gm of an organic compound was treated according to Kjeldahl's method. 0.2×10^{-4} mol of H_2SO_4 was required to neutralise NH_3 . Calculate the percentage of N_2 .

- (A) 50 (B) 28 (C) 70 (D) 18

TABLE TYPE QUESTION

Column-I	Column-II	Column-III
(A) 2 M - aqueous NaOH solution (density = 1.25 gm/ml)	(P) 2 mole solute/litre solution	(I) 6 % (w/v) solution
(B) 1.5 m - aqueous NaOH solution (density = 1.06 gm/ml)	(Q) 1.5 mole solute/litre solution	(II) 8 % (w/v) solution
(C) 0.5 M aqueous Glucose solution (density = 1.09 gm/ml)	(R) 0.5 mole solute/litre solution	(III) 9 % (w/v) solution
(D) 1.5 M aqueous Urea solution (density = 1.15 gm/ml)	(S) 1.5 mole solute/kg solvent	(IV) 9 gm solute per 100 gm solvent

Q.23 Which of the following is correct match ?

- (A) A – P – II (B) B – Q – I (C) C – R – IV (D) D – S – III

Q.24 Which of the following is correct match ?

- (A) A – P – II (B) B – S – I (C) C – R – I (D) D – Q – I

Q.25 Which of the following is correct match ?

- (A) A – Q – III (B) B – Q – III (C) C – Q – III (D) D – Q – III

MATCH THE COLUMN :

Q.26 Match the column-

Column-I (Concentration of aqueous solution)	Column-II (Density of given solutions is 1.2 g/ml)
(A) 2M NaOH solution	(P) 16gm solute in 240gm solution
(B) $8\% \left(\frac{w}{v} \right)$ KOH solution	(Q) 60gm solute in 240 gm solution
(C) $25\% \left(\frac{w}{W} \right)$ CaCO_3 solution	(R) 8gm solute in 100 ml solution
(D) $X_{\text{C}_3\text{H}_7\text{OH}} = \frac{1}{11}$	(S) 30 gm solute in 100 ml solution
	(T) 1 mole solute in 400 gm solution

Q.27 Match the column:

Column I	Column II
(A) 20 V H_2O_2	(P) 2.5 M
(B) 24.5 % w/v H_2SO_4	(Q) 1.76 M
(C) Pure water	(R) 1.5 M
(D) 5% w/w NaOH ($d_{\text{solution}} = 1.2 \text{ gm/ml}$)	(S) 55.5 M

Q.28

Column-I	Column-II
(A) 120 g CH_3COOH in 1 L solution ($d_{\text{sol}} = 1.2 \text{ g/mL}$)	(P) $M = 2$
(B) 120 g glucose dissolved in 1 L solution ($d_{\text{sol}} = 1.2 \text{ g/mL}$)	(Q) 10% w/w solution
(C) $X_{\text{NH}_2\text{CONH}_2} = 1/31$ (aqueous solution)	(R) 12% w/v solution
(D) 19.6% (w/v) H_2SO_4 solution $\rightarrow$ ($d_{\text{solution}} = 1.2 \text{ g/mL}$)	(S) $m = 1.85$ (T) $m = 0.617$

EXERCISE : J-MAINS

1. 6.02×10^{21} molecules of urea are present in 100 ml of its solution. The concentration of urea solution is - [AIEEE-2004]
(1) 0.001 M (2) 0.01 M (3) 0.02 M (4) 0.1 M
2. A 5.2 molal aqueous solution of methyl alcohol, CH_3OH , is supplied. What is the mole fraction of methyl alcohol in the solution ? [AIEEE-2011]
(1) 0.086 (2) 0.050 (3) 0.100 (4) 0.190
3. The concentrated sulphuric acid that is peddled commercially is 95% H_2SO_4 by weight. If the density of this commercial acid is 1.834 g cm^{-3} , the molarity of this solution is :- [JEE-(Main)-2012]
(1) 17.8 M (2) 15.7 M (3) 10.5 M (4) 12.0 M
4. The density of a solution prepared by dissolving 120 g of urea (mol. mass = 60 u) in 1000 g of water is 1.15 g/mL . The molarity of this solution is [JEE-(Main)-2012]
(1) 2.05 M (2) 0.50 M (3) 1.78 M (4) 1.02 M
5. 10 mL of 2(M) NaOH solution is added to 200 mL of 0.5 (M) of NaOH solution. What is the final concentration ? [JEE(Main-online)-2013]
(1) 0.57 M (2) 5.7 M (3) 11.4 M (4) 1.14 M
6. The density of 3M solution of sodium chloride is 1.252 g mL^{-1} . The molality of the solution will be (molar mass, $\text{NaCl} = 58.5 \text{ g mol}^{-1}$) [JEE(Main-online)-2013]
(1) 2.18 m (2) 3.00 m (3) 2.60 m (4) 2.79 m
7. The amount of BaSO_4 formed upon mixing 100 mL of 20.8% BaCl_2 solution with 50 mL of 9.8% H_2SO_4 solution will be : [JEE(Main-online)-2014]
(Ba = 137, Cl = 35.5, S=32, H = 1 and O = 16)
(1) 33.2 g (2) 11.65 g (3) 23.3 g (4) 30.6 g
8. For the estimation of nitrogen, 1.4 g of an organic compound was digested by Kjeldahl method and the evolved ammonia was absorbed in 60 mL of $\frac{\text{M}}{10}$ sulphuric acid. The unreacted acid required 20 mL of $\frac{\text{M}}{10}$ sodium hydroxide for complete neutralization. The percentage of nitrogen in the compound is : [JEE(Main-online)-2014]
(1) 3% (2) 5% (3) 6% (4) 10%

EXERCISE # J-ADVANCE

- Q.1** Calculate the molarity of pure water using its density to be 1000 kg m^{-3} . [JEE'2003]
- Q.2** Dissolving 120 g of urea (mol. wt. 60) in 1000 g of water gave a solution of density 1.15 g/mL . The molarity of the solution is
(A) 1.78 M (B) 2.00 M (C) 2.05 M (D) 2.22 M [JEE 2011]
- Q.3** A compound H_2X with molar weight of 80 g is dissolved in a solvent having density of 0.4 g /ml , Assuming no change in volume upon dissolution, the **molality** of a 3.2 molar solution is. [JEE 2014]

ANSWER-KEY

EXERCISE # S-I

Q.1 (a) 0.5 M, (b) 0.5 M, (c) 0.2 M	Q.2 1.3 M	Q.3 2.4M
Q.4 16.66%	Q.5 3.0×10^{-4}	Q.6 1.4
Q.7 0.15 M	Q.8 0.06 M	Q.9 1.25 gm/ml
Q.10 45.45%	Q.11 2.5×10^{-3} M	Q.12 0.01 M
Q.13 13 gm	Q.14 0.6667, 0.6667	Q.14 (1)
Q.16 1.4 litre	Q.17 175 ml	Q.18 2 M
Q.19 1.5 ml	Q.20 (0.8)	Q.21 0.05
Q.22 0.204 M	Q.23 5.56 ml	Q.24 (0.15)
Q.25 (1.5)		
Q.26 (i) 20 gm ; (ii) 35.4 gm ; (iii) $\text{H}_2\text{SO}_4 = 35.4 \text{ gm}$, $\text{H}_2\text{O} = 34.6 \text{ gm}$		
Q.27 (a) 0.169; (b) 118 %	Q.28 45.4 V	Q.29 (6)
Q.30 (56)		

EXERCISE # S-II

Q.1 Ans.250	Q.2 Ans.25 ml	Q.3 Ans.5
Q.4 Ans.0.5	Q.5 Ans.60	Q.6 Ans.0.05
Q.7 Ans.3.33 m	Q.8 Ans.(12)	Q.9 Ans.(2)
Q.10 Ans. (2)		

EXERCISE # O-I

Q.1 Ans.(A)	Q.2 Ans.(C)	Q.3 Ans.(D)
Q.4 Ans.(B)	Q.5 Ans.(C)	Q.6 Ans.(A)
Q.7 Ans.(B)	Q.8 Ans.(D)	Q.9 Ans.(B)
Q.10 Ans.(B)	Q.11 Ans.(D)	Q.12 Ans.(D)
Q.13 Ans.(C)	Q.14 Ans.(B)	Q.15 Ans.(C)
Q.16 Ans.(D)	Q.17 Ans.(B)	Q.18 Ans.(C)
Q.19 Ans.(A)	Q.20 Ans. (A)	Q.21 Ans.(C)
Q.22. Ans.(C)	Q.23 Ans.(C)	Q.24 Ans.(A)
Q.25 Ans.(A)	Q.26 Ans.(D)	Q.27 Ans.(B)
Q.28 Ans.(A)	Q.29 Ans.(C)	Q.30 Ans.(B)
Q.31 Ans.(B)	Q.32 Ans.(B)	
Q.33 Ans.(B)		

EXERCISE # O-II

- | | | |
|---|---------------|------------------|
| Q.1 Ans.(B) | Q.2 Ans. (C) | Q.3 Ans. (A) |
| Q.4 Ans. (D) | Q.5 Ans. (C) | Q.6 Ans. (A,C,D) |
| Q.7 Ans.(A,C) | Q.8 Ans.(B,D) | Q.9 Ans.(B,D) |
| Q.10 Ans.(A,C) | Q.11 Ans.(C) | Q.12 Ans.(C) |
| Q.13 Ans.(B) | Q.14 Ans.(C) | Q.15 Ans.(A) |
| Q.16 Ans.(C) | Q.17 Ans.(D) | Q.18 Ans.(A) |
| Q.19 Ans.(A) | Q.20 Ans.(B) | Q.21 Ans.(C) |
| Q.22 Ans.(B) | Q.23 Ans.(A) | Q.24 Ans.(B) |
| Q.25 Ans.(D) | | |
| Q.26 Ans.(A) - P, R ; (B) - P, R ; (C) - Q, S, T ; (D) - S, Q | | |
| Q.27 Ans.(A) - Q ; (B) - P ; (C) - S ; (D) - R | | |
| Q.28. Ans.(A) - P,Q,R,S ; (B) - Q,R,T ; (C) - Q,S ; (D) - P | | |

EXERCISE # J-MAINS

- | | | |
|------------|------------|------------|
| 1. Ans.(4) | 2. Ans.(1) | 3. Ans.(1) |
| 4. Ans.(1) | 5. Ans.(1) | 6. Ans.(4) |
| 7. Ans.(2) | 8. Ans.(4) | |

EXERCISE # J-ADVANCE

- | | | |
|----------------------------------|-------------|-------------|
| Q.1 Ans.55.5 mol L ⁻¹ | Q.2 Ans.(C) | Q.3 Ans.(8) |
|----------------------------------|-------------|-------------|

Important Notes

NURTURE COURSE

EUDIOMETRY

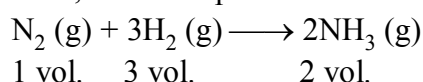
EUDIOMETRY

KEY CONCEPTS

EUDIOMETRY :

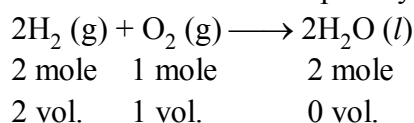
Eudiometry or gas analysis involves the calculations based on gaseous reactions or the reactions in which at least two components are gaseous, in which the amounts of gases are represented by their volumes, measured at the same pressure and temperature. Some basic assumptions related with calculations are:

1. Gay-Lussac's law of volume combination holds good. According to this law, the volumes of gaseous reactants reacted and the volumes of gaseous products formed, all measured at the same temperature and pressure, bear a simple ratio.



Problem may be solved directly in terms of volume, in place of mole. The stoichiometric coefficients of a balanced chemical reaction gives the ratio of volumes in which gaseous substances are reacting and products are formed, at same temperature and pressure.

2. The volumes of solids or liquids is considered to be negligible in comparison to the volume of gas. It is due to the fact that the volume occupied by any substance in gaseous state is even more than thousand times the volume occupied by the same substance in solid or liquid states.



3. Air is considered as a mixture of oxygen and nitrogen gases only. It is due to the fact that about 99% volume of air is composed of oxygen and nitrogen gases only.
4. Nitrogen gas is considered as a non-reactive gas. It is due to the fact that nitrogen gas reacts only at very high temperature due to its very high thermal stability. Eudiometry is performed in an eudiometer tube and the tube can not withstand very high temperature. This is why, nitrogen gas can not participate in the reactions occurring in the eudiometer tube.
5. The total volume of non-reacting gaseous mixture is equal to sum of partial volumes of the component gases (*Amagat's law*).

$$V = V_1 + V_2 + \dots$$

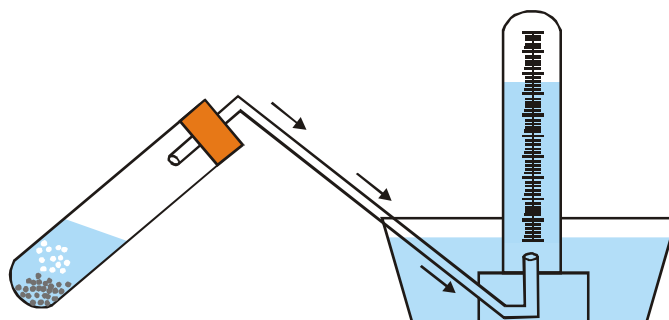
Partial volume of gas in a non-reacting gaseous mixture is its volume when the entire pressure of the mixture is supposed to be exerted only by that gas.

6. The volume of gases produced is often given by certain solvent which absorb contain gases.

Solvent	Gases absorb
KOH	CO ₂ , SO ₂ , Cl ₂
Ammonical Cu ₂ Cl ₂	CO
Turpentine oil	O ₃
Alkaline pyrogallol	O ₂
water	NH ₃ , HCl
CuSO ₄ /CaCl ₂	H ₂ O

EUDIOMETER

An eudiometer is a laboratory device that measures the change in volume of a gas mixture following a physical or chemical change.

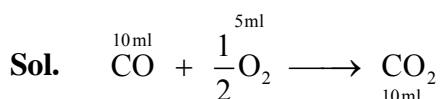


Scheme of eudiometer

To use a eudiometer, it is filled with water, inverted so that its open end is facing the ground (while holding the open end so that no water escapes), and then submersed in a basin of water. A chemical reaction is taking place through which gas is created. One reactant is typically at the bottom of the eudiometer (which flows downward when the eudiometer is inverted) and the other reactant is suspended on the rim of the eudiometer, typically by means of a platinum or copper wire (due to their low reactivity). When the gas created by the chemical reaction is released, it should rise into the eudiometer so that the experimenter may accurately read the volume of the gas produced at any given time. Normally a person would read the volume when the reaction is completed

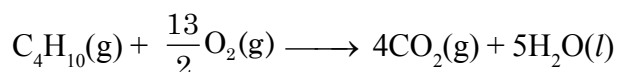
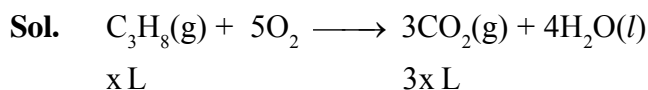
SOLVED EXAMPLE

Ex.1 10 ml of CO is mixed with 25 ml air (20% O_2 by volume) in a container at 1 atm. Find final volume (in ml) of container at 1 atm after complete combustion. (Assume that temperature remain constant).



$$V_f = V_{CO_2} + \text{Volume of remaining air} = 10 + 20 = 30 \text{ ml}$$

Ex.2 A 3 L gas mixture of propane (C_3H_8) and butane (C_4H_{10}) on complete combustion at $25^\circ C$ produced 10 L CO_2 . Assuming constant P and T conditions what was volume of butane present in initial mixture ?

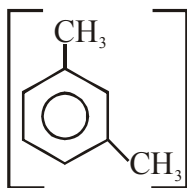


$$(3-x) \text{ L} \qquad \qquad \qquad 4(3-x) \text{ L}$$

$$\text{from question } 3x + 4(3-x) = 10 \Rightarrow x = 2$$

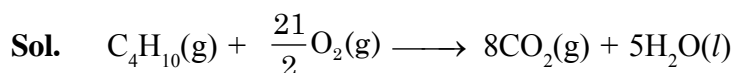
$$\therefore \text{Volume of butane, } C_4H_{10} = (3-x) = 1 \text{ L}$$

Ex.3 100 ml gaseous meta Xylene



undergoes combustion with excess of oxygen at room

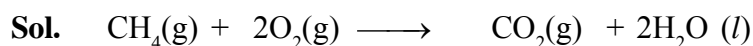
temperature and pressure. Volume contraction / expansion (in ml) during reaction is :



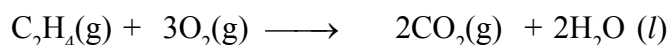
$$\begin{array}{ccccccc} 100 \text{ ml} & & \frac{21}{2} \times 100 & & 800 \text{ ml} & & 0 \\ & & = 1050 \text{ ml} & & & & \end{array}$$

$$\therefore \text{Contraction in volume} = (100 + 1050) - 800 = 350 \text{ ml}$$

Ex.4 30 ml gaseous mixture of methane and ethylene in volume ratio $X : Y$ requires 350 ml air containing 20% of O_2 by volume for complete combustion. If ratio of methane and ethylene changed to $Y : X$. What will be volume of air (in ml) required for complete reaction under similar condition of temperature and pressure.



$$V_1 \text{ ml} \quad 2V_1 \text{ ml} \quad V_1 \text{ ml} \quad 0$$



$$V_2 \text{ ml} \quad 3V_2 \text{ ml} \quad 2V_2 \text{ ml} \quad 0$$

$$\text{For given data : } V_1 + V_2 = 30$$

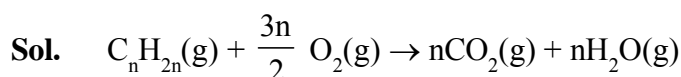
$$\text{and } 2V_1 + 3V_2 = 350 \times \frac{20}{100} = 70$$

$$\therefore V_1 = 20, V_2 = 10$$

$$\text{For required data : } V_1 = 10 \text{ and } V_2 = 20$$

$$\therefore \text{Volume of } \text{O}_2 \text{ required} = 2V_1 + 3V_2 = 80 \text{ ml} \text{ and volume of air required} = 80 = \frac{100}{20} = 400 \text{ ml}$$

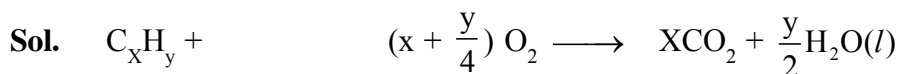
Ex.5 An alkene upon combustion produces $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$. In this combustion process if there is no volume change occurs then the no. of C atoms per molecule of alkene will be :



if there no volume changes i.e. $\Delta_{\text{ng}} = 0$

$$(n + n) - \left(1 + \frac{3n}{2}\right) = 0 \Rightarrow n = 2$$

Ex.6 A gaseous hydrocarbon (C_xH_y) requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times of its volume of CO_2 . Find out the volume of $x + y$.



Vol a $a\left(x + \frac{y}{4}\right)$ ax

Given that : $a(x + y/4) = 6a$

vol of $CO_2 = 4$ vol of C_xH_y

$ax = 4(a)$

$x = 4$

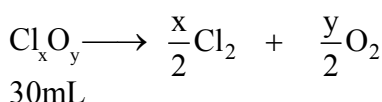
...(2)

from (1) $x + \frac{y}{4} = 6$

$\therefore x + y = 4 + 8 = 12$

Ex.7 On heating 60 ml mixture containing equal volume of chlorine gas and it's gaseous oxide, volume becomes 75 ml due complete decomposition of oxide. On treatment with KOH volume becomes 15 ml. What is the formula of oxide of chlorine ?

Sol. Let oxide of Cl is Cl_xO_y
So in 60 mL $\Rightarrow$ 30 mL Cl_xO_y and 30 mL Cl_2 .
Now,



$\frac{30.x}{2}$ mL $\frac{30.y}{2}$ mL

Given :

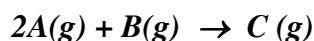
$75 = 30 + \frac{30x}{2} + \frac{30y}{2} \Rightarrow x + y = 3 \dots\dots\dots(i)$

KOH absorbs Cl_2 and volume becomes 15 mL so,

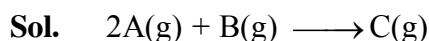
$(75 - 15) = V_{Cl_2} = 30 + \frac{30x}{2} \Rightarrow x = 2 \text{ and } y = 1$

So the oxide : Cl_2O

Ex.8 5 L of A (g) & 3 L of B(g) measured at same T & P are mixed together which react as follows



What will be the total volume (in litre) after the completion of the reaction at same T & P.



5L 3L

L.R. is A

So, volume of C produced = $\frac{1}{2} \times 5 = 2.5$ L

and, volume of B reacted = $\frac{1}{2} \times 5 = 2.5$ L

So, volume fo B remained = $3 - 2.5 = 0.5$ L

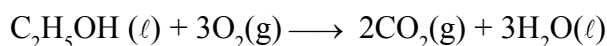
Hence, $V_{\text{total}} = V_C + V_B = 2.5 + 0.5 = 3$ L

EXERCISE # S-I

1. 20 ml propane gas (C_3H_8) is burnt completely in excess of air. The volume of CO_2 gas formed is.
2. What volume of $O_2(g)$ is needed for complete combustion of 40 ml ethane gas (C_2H_6) ?
3. 10 ml of CO is mixed with 25 ml air having 20% O_2 by volume. What would be the final volume if none of CO and O_2 is left after the reaction?
4. Calculate the volume of CO_2 evolved by the combustion of 50 ml of a mixture containing 40% C_2H_4 and 60% CH_4 (by volume)
5. 10 moles of a mixture of CO (g) and $CH_4(g)$ was mixed with 22 moles of O_2 gas and subjected to sparking. Find the moles of gas absorbed when the residual gases are passed through alc. KOH.
6. 60 ml of a mixture of nitrous oxide and nitric oxide was exploded with excess of hydrogen. If 38 ml of N_2 was formed, calculate the volume of NO gas in the mixture.
7. When 100 ml of a $O_2 - O_3$ mixture was passed through turpentine, there was reduction of volume by 20 ml. If 100 ml of such a mixture is heated, what will be the increase in volume?
8. 20 ml of a mixture of C_2H_2 and CO was exploded with 30 ml of oxygen. The gases after the reaction had a volume of 34 ml. On treatment with KOH, 8 ml of oxygen remained. Calculate the volume of C_2H_2 in the mixture.
9. 10 ml of a mixture of CO, CH_4 and N_2 exploded with excess of oxygen gave a contraction of 6.5 ml. There was a further contraction of 7 ml, when the residual gas treated with KOH. Volume of CO, CH_4 and N_2 respectively is
10. 10 ml of a mixture of CH_4 , C_2H_4 and CO_2 were exploded with excess of air. After explosion and further cooling, there was contraction of 17 ml and after treatment with KOH, there was further reduction of 14 ml. What is the composition of the mixture?
11. 10 ml of an oxide of nitrogen produce 20 ml NO_2 and 5 ml O_2 on complete decomposition. The oxide of nitrogen is-
12. Find the nature of hydrocarbon for which volume of oxygen required for combustion is 1.5 times volume of carbon dioxide produced.
13. A gaseous alkane is exploded with O_2 . The volume of O_2 required for complete combustion and the volume of CO_2 formed after combustion are in 7 : 4 ratio. What is the molecular formula of alkane ?
14. 20 ml of a gaseous hydrocarbon (A) was exploded with excess of O_2 in an eudiometer tube. On cooling, the volume was reduced by 50 ml. On further treatment with KOH, there was a further contraction of 40 ml. The molecular formula of hydrocarbon A is :
15. When a certain quantity of oxygen was ozonised in a suitable apparatus, the volume decreased by 4 ml. On addition of turpentine the volume further decreased by 8 ml. All volumes were measured at the same temperature and pressure. From these data, if formula of ozone is O_x then find x.

EXERCISE # S-II

1. A 20 ml mixture of C_2H_4 and C_2H_2 undergoes sparking in gas eudiometer with just sufficient amount of O_2 and shows contraction of 37.5 ml. Volume (in ml) of C_2H_2 in the mixture is.
2. 1120 ml of ozonised oxygen ($O_2 + O_3$) at 1 atm & 273K weighs 1.76 gm. Find the reduction in volume on passing this through alkaline pyrogallol solution is -
3. Balanced chemical equation is given as following :



What value of the volume in litre of CO_2 (g) measured at 200 K and 1 atm, produced from the combustion of 0.25 mole of C_2H_5OH (l) ? [Use $R = 0.08 \text{ L atm mol}^{-1} \text{ K}^{-1}$]

4. 40 ml of a mixture of C_2H_2 and CO is mixed with 100 ml of O_2 gas and the mixture is exploded. The residual gases occupied 104 ml and when these are passed through KOH solution, the volume becomes 48 ml. All the volume are at same temperature and pressure. Determine the composition of original mixture.
5. 10 mL of gaseous organic compound containing C, H and O only was mixed with 100 mL of O_2 and exploded under identical conditions and then cooled. The volume left after cooling was 90 mL. On treatment with KOH a contraction of 20 mL was observed. If vapour density of compound is 23, if molecular formula of the compound is $C_xH_yO_z$, then find $(x + y + z)$.

10. One litre of CO_2 passed over hot coke the volume becomes 1.4 litres then the composition of products will not be (At NTP)
- (A) $V_{\text{CO}_2} : V_{\text{CO}} = 3 : 4$ (B) $V_{\text{CO}_2} = 1.6 \text{ ltr.}$ (C) $n_{\text{CO}_2} : n_{\text{CO}} = 3 : 4$ (D) $\% V \text{ of CO} = \frac{400}{7}$
11. 10 ml of a compound containing 'N' and 'O' is mixed with 30 ml of H_2 to produce $\text{H}_2\text{O} (l)$ and 10 ml of $\text{N}_2 (g)$. Molecular formula of compound if both reactants reacts completely, is
- (A) N_2O (B) NO_2 (C) N_2O_3 (D) N_2O_5
12. An alkane on complete combustion with O_2 shows 50% of volume contraction. Molecular formula of alkane is -
- (A) $\text{C}_4\text{H}_{10} (g)$ (B) $\text{CH}_4 (g)$ (C) $\text{C}_2\text{H}_6 (g)$ (D) $\text{C}_3\text{H}_8 (g)$
13. 20 mL of a gaseous hydrocarbon was exploded with 120 mL of oxygen. A contraction of 60 mL was observed, and a further contraction of 60 mL took place when KOH was added. What is the formula of the hydrocarbon :
- (A) C_3H_6 (B) C_3H_8 (C) C_2H_6 (D) C_4H_{10}
14. When a definite volume of a gaseous alkyne ($\text{C}_n\text{H}_{2n-2}$) is burnt completely in excess of air, a contraction in volume equal to twice the volume of alkyne burnt occurred. The value of 'n' is -
- (A) 10 (B) 6 (C) 3 (D) 20
15. Each volume of a gaseous organic compound containing C, H and S only produce 1 volume CO_2 , 2 volume H_2O vapours and 1 volume SO_2 gases on complete combustion. The molecular formula of compound is -
- (A) CH_2S (B) CH_4S (C) $\text{C}_2\text{H}_4\text{S}$ (D) $\text{C}_2\text{H}_6\text{S}$

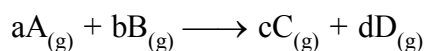
EXERCISE # O-II

Single correct :

1. A 2 L sample of a gaseous hydrocarbon is burnt in excess oxygen. The only products of the reaction are 8L of $\text{CO}_2(\text{g})$ and 10L of $\text{H}_2\text{O}(\text{g})$, all at 100°C and 1 atm pressure. The formula of the hydrocarbon is -
 (A) C_5H_{12} (B) C_4H_5 (C) C_4H_{10} (D) C_8H_{10}
2. 1120 ml of ozonised oxygen ($\text{O}_2 + \text{O}_3$) at 1 atm & 273K weighs 1.76 gm. The reduction in volume on passing this through alkaline pyrogallol solution is -
 (A) 896 ml (B) 224 ml (C) 448 ml (D) 672 ml
3. **Statement-1 :** When a gaseous hydrocarbon is burnt in excess of oxygen and the products of combustion are cooled to the original temperature and pressure, a contraction in volume occurs.
Statement-2 : The contraction in volume is solely due to the liquifaction of water vapour.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
4. **Statement-1 :** For the same volume of gaseous alkane, alkene and alkyne having same number of carbon atoms, the volume of oxygen needed for complete combustion is maximum for alkane.
Statement-2 : The number of hydrogen atoms is maximum in alkane.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

More than one correct:

5. Two gases A and B which react according to the equation



to give two gases C and D are taken (amount not known) in an Eudiometer tube (operating at a constant Pressure and temperature) to cause the above.

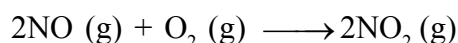
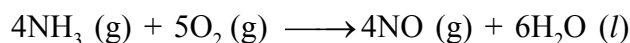
If on causing the reaction there is no volume change observed then which of the following statement is/are correct.

- (A) $(a + b) = (c + d)$
- (B) average molecular mass may increase or decrease if either of A or B is present in limited amount.
- (C) Vapour Density of the mixture will remain same throughout the course of reaction.
- (D) Total moles of all the component of mixture will change.

6. 20 ml mixture of C_3H_8 and CO gas when burnt in excess of oxygen produce 40 ml CO_2 gas. Choose the correct statement(s). (Volume of gases measured under same T & P)
- (A) Volume of C_3H_8 in the mixture is 15 ml
 (B) Volume of CO in the mixture is 10 ml
 (C) Total volume contraction due to combustion is 35 ml.
 (D) The volume of oxygen used for combustion is 75 ml
7. 10 ml of a gaseous mixture containing C_2H_x and C_3H_8 exactly requires 40 ml O_2 for complete combustion and produces 25 ml CO_2 and 30 ml H_2O vapour. The correct information (s) is/are -
- (A) Total volume contraction = 5 ml
 (B) Volume contraction due to combustion of $C_2H_x = 0$
 (C) $x = 4$
 (D) Volume of C_2H_x in the initial mixture = 5 ml
8. 100 ml mixture of CO and CO_2 mixed with 30 mL of O_2 and sparked in eudiometer tube. The residual gas after treatment with aq. KOH has a volume of 10 mL which remains unchanged when treated with alkaline pyrogallol. If all the volumes are under the same conditions, point out **correct** options(s):
- (A) The volume of CO that reacts, is 60 mL
 (B) The volume of CO that remains unreacted, is 10 mL
 (C) The volume of O_2 that remains unreacted, is 10 mL
 (D) The volume of CO_2 that gets absorbed by aq.KOH, is 90 mL.

Paragraph for Q.9 to Q.11

For the given series of reaction



9. If 20 ml of NH_3 is mixed with 100 ml of O_2 . Volume contraction at the completion of above reactions is
- (A) 20 ml (B) 85 ml (C) 35 ml (D) 100 ml
10. To obtain maximum mass of NO_2 from a given mass of a mixture of NH_3 and O_2 , the ratio of mass of NH_3 to O_2 should be
- (A) $\frac{4}{7}$ (B) $\frac{17}{56}$ (C) $\frac{17}{40}$ (D) None of these
11. Total volume of O_2 used if 20 ml NH_3 is mixed with 100 ml O_2
- (A) 40 (B) 60 (C) 35 (D) None of these

Table type question :

Column-I (Gas taken)	Column-II (O ₂ needed for complete combustion)	Column-III (Contraction in volume)
(1) 20 ml C ₂ H ₄	(i) 60 ml	(I) 50 ml
(2) 25 ml C ₃ H ₄	(ii) 100 ml	(II) 40 ml
(3) 30 ml C ₂ H ₆	(iii) 70 ml	(III) 75 ml
(4) 35 ml CH ₄	(iv) 105 ml	(IV) 70 ml

All volumes are measured at 25°C and 1 atm.

12. Which of the following is correct match -

- (A) 1 – i – II (B) 1 – iii – IV (C) 2 – iv – II (D) 2 – ii – III

13. Which of the following is correct match -

- (A) 3 – iii – III (B) 3 – iv – III (C) 4 – iii – III (D) 4 – iv – IV

14. Which of the following is incorrect (**One or more than one correct**)

- (A) 2 – ii – I (B) 4 – iii – IV (C) 3 – iv – IV (D) 1 – iii – II

Match the column :

15. Gaseous alkane (C_nH_{2n+2}) exploded with oxygen. Ratio of the mol of O₂ for complete combustion to the mole of CO₂ formed is given in column-I & in column II formula is given.

Column-I	Column-II
(A) 7 : 4	(P) C ₃ H ₈
(B) 2 : 1	(Q) C ₄ H ₁₀
(C) 5 : 3	(R) C ₂ H ₆
(D) 13 : 8	(S) CH ₄

ANSWER-KEY**EXERCISE # S-I**

- | | | |
|---|---|---------------|
| 1. Ans. 60 ml | 2. Ans. 140 ml | 3. Ans. 30 ml |
| 4. Ans. 70 ml | 5. Ans. (10) | |
| 6. Ans. NO = 44 ml ; N ₂ O = 16 ml | 7. Ans. 10 ml | |
| 8. Ans. C ₂ H ₂ = 6 ml, CO = 14 ml | 9. Ans. 5 ml, 2 ml, 3 ml | |
| 10. Ans. CH ₄ = 4.5 ml, CO ₂ = 1.5 ml | 11. Ans. (N ₂ O ₅) | |
| 12. Ans. (alkene) | 13. Ans. (C ₂ H ₆) | |
| 14. Ans. (C ₂ H ₆) | 15. Ans. (3) | |

EXERCISE # S-II

- | | | |
|--|------------------------------------|-------------|
| 1. Ans. (5) | 2. Ans. (896 ml) | 3. Ans. (8) |
| 4. C ₂ H ₂ = 16 ml, CO = 24 ml | 5. C ₂ H ₆ O | |

EXERCISE # O-I

- | | | |
|--------------|--------------|--------------|
| 1. Ans. (D) | 2. Ans. (2) | 3. Ans. (B) |
| 4. Ans. (B) | 5. Ans. (C) | 6. Ans. (A) |
| 7. Ans. (D) | 8. Ans. (B) | 9. Ans. (A) |
| 10. Ans. (B) | 11. Ans. (C) | 12. Ans. (D) |
| 13. Ans. (B) | 14. Ans. (C) | 15. Ans. (B) |

EXERCISE # O-II

- | | | |
|--|-----------------|---------------|
| 1. Ans. (C) | 2. Ans. (A) | 3. Ans. (C) |
| 4. Ans. (A) | 5. Ans. (A,C) | 6. Ans. (B,C) |
| 7. Ans. (B,C,D) | 8. Ans. (A,B,D) | 9. Ans. (C) |
| 10. Ans. (B) | 11. Ans. (C) | 12. Ans. (A) |
| 13. Ans. (B) | 14. Ans. (C,D) | |
| 15. Ans. A - R ; B - S ; C - P ; D - Q | | |

THERMODYNAMICS-01

Session 2019-20

ALLERK
Session 2019-20

THERMODYNAMICS-01

1. INTRODUCTION

Thermodynamics is concerned with energy and its transformation in various forms in different physical and chemical processes.

Thermodynamics $\equiv$ Thermo + dynamics

Dynamics $\equiv$ Study of change due to a driving force

Thermo $\equiv$ Thermal which is related to temperature or energy.

The laws of thermodynamics deal with energy changes of macroscopic systems involving a large number of molecules rather than microscopic systems containing a few molecules. Thermodynamics is not concerned about how and at what rate these energy transformations are carried out, but is based on initial and final states of a system undergoing the change. Laws of thermodynamics apply only when a system is in equilibrium or moves from one equilibrium state to another equilibrium state. Macroscopic properties like pressure and temperature do not change with time for a system in equilibrium state.

2. SOME BASIC DEFINITIONS (Thermodynamic terms)

2.1 SYSTEM :

The macroscopic part of the universe under study is called a system. Rest of the universe outside the system is called surroundings. The actual or imaginary surface that separates the system from the surroundings is called the boundary.

However, the entire universe other than the system is not affected by the changes taking place in the system. Therefore, for all practical purposes, the surroundings are that portion of the remaining universe which can interact with the system. Usually, the region of space in the neighbourhood of the system constitutes its surroundings.

2.1.1 Types of boundary :

- (i) Real or imaginary
- (ii) Rigid (fixed) or movable (flexible)
- (iii) Permeable (allow mass transfer) or impermeable (does not allow mass transfer).
- (iv) Diathermal (allow heat transfer) or Adiabatic (does not allow heat transfer).

2.1.2 Types of systems :

- (i) **Isolated system** : A system is said to be isolated if it cannot exchange matter and energy with the surroundings. Example : coffee in a thermos flask.
- (ii) **Closed system** : A system is said to be closed if it can exchange energy but not matter. Example : Coffee in a closed stainless steel flask.
- (iii) **Open system** : A system is said to be open if it can exchange matter and energy. Example : A thermos flask or a steel flask if not closed.

Note : A perfectly isolated system is only a theoretical system.

- 2.2 **State of system** : A system is called in a particular state where all the macroscopic properties of the system have defined value.

2.3 Properties of system : The state of a system is defined by a particular set of its measurable quantities called properties. They can be categorised into extensive and intensive properties.

Intensive property is one whose value is independent of the size (or mass) of the system. An extensive property is one whose value depends on the size (or mass) of the system.

- * Extensive properties are additive but intensive properties are non additive.
- * Ratio of two extensive property gives as intensive property.

Extensive Properties

Volume
Number of moles
Mass
Free Energy (G)
Entropy (S)
Enthalpy (H)
Internal energy (E & U)
Heat capacity

Intensive Properties

Molar volume
Density
Refractive index
Surface tension
Viscosity
Free energy per mole
Specific heat
Pressure
Temperature
Boiling point, freezing point etc

2.4 State function or state variable : Variables like P, V, T are State Functions or State Variables because their values depend only on the present state of a system and not on how the state was reached

Condition for a function to be a state function :

- (i) If ϕ is state function, $\int_A^B d\phi = \phi_B - \phi_A$
- (ii) If ϕ is a state function, $\oint d\phi = 0$

2.5 Path function : Function which depend on the path, i.e. how the process is carried out e.g. work & heat.

2.6 Thermodynamic process : A thermodynamic process involves change of a system from one state to another state.

EXERCISE # I

1. Which of the following are extensive and which are intensive properties ?

Temperature, boiling point, melting point, pressure, density, viscosity, surface tension, refractive index, molar volume, free energy/mole, specific heat, Specific volume, Mass, Volume, number of moles, Heat capacity, internal energy, enthalpy, entropy, ΔG , concentration, dipole moment, pH, gas constant, vapour pressure, specific gravity, E.M.F. of the dry cell, molarity, molality.

Sol. Intensive property : Temperature, (boiling point, melting point), pressure, density, viscosity, surface tension, refractive index, molar volume, free energy/mole, specific heat, Specific volume, concentration, dipole moment, pH, gas constant, vapour pressure, specific gravity, E.M.F. of the dry cell, molarity, molality.

Extensive property : Mass, Volume, number of moles, Heat capacity, internal energy, enthalpy, entropy

2. Which of the following are state function & path function ?

Pressure, Volume, Enthalpy, Work, Heat, Gibbs energy, temperature, Internal energy, Entropy

Sol. State function : Pressure, volume, enthalpy, Gibbs energy, temperature, Internal energy, Entropy

Path function : Work, Heat

3. A state function is that :

- (A) which is used in thermochemistry
- (B) which obeys all laws of thermodynamics
- (C) quantity whose value depends only upon the state of the system
- (D) quantity which is used in measuring thermal change

Ans. (C)

4. Which amongst the following is an extensive property of the system -

- (A) Temperature
- (B) Volume
- (C) Viscosity
- (D) Refractive index

Ans. (B)

5. Which of the following is not a state function of thermodynamic system -

- (A) Internal energy(E)
- (B) Free energy(G)
- (C) Enthalpy(H)
- (D) Work(W)

Ans. (D)

6. What is true for a cyclic process

- (A) $W = 0$
- (B) $\Delta E = 0$
- (C) $\Delta H = 0$
- (D) B & C both

Ans. (D)

7. The internal energy change when a system goes from state A to B is 40 kJ/mole. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy ?

[AIEEE-2003]

- (A) < 40 kJ
- (B) Zero
- (C) 40 kJ
- (D) > 40 kJ

Ans. (B)

Type of process :

(i) **Isothermal process :**

A process in which temperature of the system remains constant is called isothermal process.



(ii) **Isobaric process :**

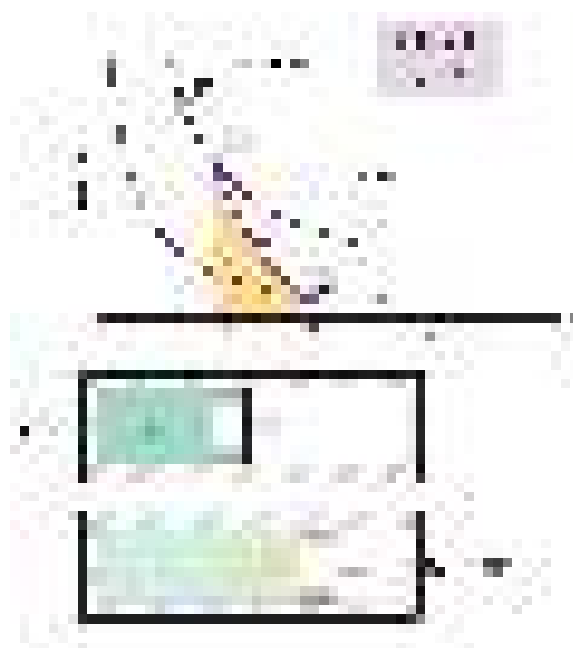
A process in which pressure of the system remains constant is called isobaric process. Temperature and volume of the system may change.



Ex. All the reactions or processes taking place in open vessel like boiling of water in open vessel, burning of charcoal, melting of wax take place at constant pressure (1 atm.)

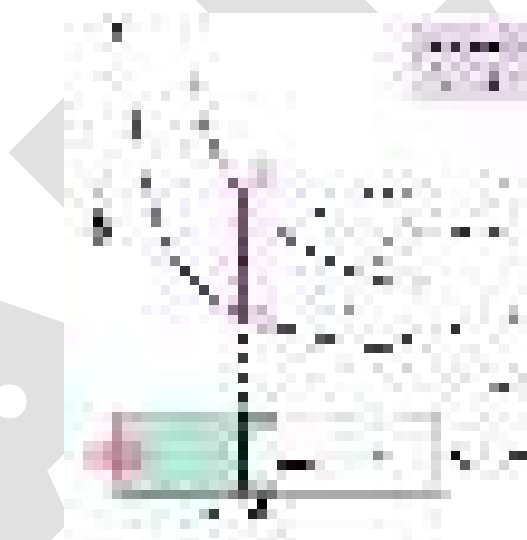
(iii) Adiabatic process :

A process in which no heat exchange takes place is called adiabatic process. Adiabatic process occurs in systems with insulated walls.



(iv) Isochoric process :

The process for which volume of the system remains constant is called isochoric process i.e., Heating of gas in closed or rigid vessel.



(v) **Cyclic process :** It is combination of two or more process in which the final state of system, becomes identical to the initial. The net change in all thermodynamic properties of system must be zero.

$$\Delta T = \Delta P = \Delta V = \Delta H = \Delta S = \Delta G \dots\dots\dots = 0$$

However Q_{net} or W_{net} may or may not be zero.

(vi) **Reversible or Irreversible process :** If the initial state may be received just by reversing the direction of process at any state, process is called reversible. In irreversible process the initial state can never be achieved just by reversing the direction of process.

In the reversible process, the driving force is only infinitesimally greater than the opposing force. In the irreversible process, they differ largely.

A perfectly reversible process is a theoretical process because it takes infinite time but reversible processes are important because it results maximum efficiency in any machine. The actual process occurring in the machine is quasi static which may tend to reversible or irreversible process.

During irreversible process, one of the equation of state like $PV = nRT$, is valid. Such equation is valid only at initial & final state. However during reversible process such equation are valid throughout the process.

A process is reversible when the system throughout remain in thermodynamic equilibrium with the surrounding. A system is said in thermodynamic equilibrium when it simultaneously satisfy the following equilibria.

- (a) **Thermal equilibrium:** Same temperature throughout (No heat transfer within the system)
- (b) **Mechanical equilibrium :** Same pressure or force throughout the system (No work should be performed by one part on the other)
- (c) **Material equilibrium :** No change in the composition of system with time. No mass transfer within the system.

If a system is in thermodynamic equilibrium there is no net energy or mass transfer within the system.

3. WORK

Thermodynamically, work may be defined as the form of energy which appears only when, there is some change in the boundary of the system. Such work is called mechanical or PV work.

Work may also be non-mechanical like electric work.

Presently, we will discuss only mechanical work.

- (i) It is not thermodynamic property of system.
- (ii) It depends on the quantity of system.
- (iii) Sign convention, work on system = (+)ve.

In physics, work is calculate from the force applied by system, but in chemistry due to external force.

$W = \int_{V_1}^{V_2} P \cdot dV = - \int_{V_1}^{V_2} P_{\text{ext}} dV$
Physics Chemistry

4. HEAT

Heat is defined as the energy that flows into or out of a system because of a difference in temperature between the system and its surrounding.

According to IUPAC convention

heat lost by system is expressed with –ive sign

heat given to system is expressed with +ive sign

- * $q_v = nC_{v,m} dT$
- * $q_p = nC_{p,m} dT$
- * $C_{p,m} - C_{v,m} = R$ (for ideal gas)
- * C_v & C_p depends on temperature even for an ideal gas. ($C = a + bT + cT^2 \dots\dots\dots$)

General values of C_v & C_p for an ideal gas can be taken as follows.

Atomicity		n _{tr}	n _{Rot}	n _{Vib}	C _v		C _p		γ	
						Incl. Vib	Excl. Vib	Incl. Vib	Excl. Vib	Incl. Vib
Mono		3	0	0	$\frac{3}{2}R$	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{2}R$	$\frac{5}{3}$	$\frac{5}{3}$
Di		3	2	1	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{2}R$	$\frac{9}{2}R$	$\frac{7}{5}$	$\frac{9}{7}$
Tri	Linear	3	2	4	$\frac{5}{2}R$	$\frac{13}{2}R$	$\frac{7}{2}R$	$\frac{15}{2}R$	$\frac{7}{5}$	$\frac{15}{13}$
	Non Linear	3	3	3	3R	6R	4R	7R	$\frac{4}{3}$	$\frac{7}{6}$

5. INTERNAL ENERGY (E & U)

Every system having some quantity of matter is associated with a definite amount of energy, called internal energy.

$$U = U_{\text{Kinetics}} + U_{\text{Potential}} + U_{\text{Electronic}} + U_{\text{nuclear}} + \dots$$

U is a state function & is an extensive property.

$$\Delta U = U_{\text{final}} - U_{\text{initial}}$$

For a given closed system

$$U = f(T, V)$$

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

6. ENTHALPY

Chemical reactions are generally carried out at constant pressure (atmospheric pressure) so it has been found useful to define a new state function **Enthalpy** (H) as.

$$H = U + PV$$

$$\Delta H = \Delta U + \Delta(PV)$$

at constant pressure

$$\Delta H = \Delta U + P \Delta V$$

combining with first law.

$$\Delta H = q_p$$

Hence transfer of heat at constant volume brings about a change in the internal energy of the system whereas that at constant pressure brings about a change in the enthalpy of the system.

6.1 RELATIONSHIP BETWEEN ΔH & ΔU :

The difference between ΔH & ΔU becomes significant only when gases are involved (insignificant in solids and liquids)

$$\Delta H = \Delta U + \Delta(PV)$$

If substance is not undergoing chemical reaction or phase change.

$$\Delta H = \Delta U + nR\Delta T$$

In case of chemical reaction

$$\Delta H = \Delta U + (\Delta n_g)RT$$

7. ZEROth LAW OF THERMODYNAMICS

Two system in thermal equilibrium with a third system are also in thermal equilibrium with each other. It introduces temperature as a state function.

8. FIRST LAW OF THERMODYNAMICS

"Total energy of universe remains constant." It is law of conservation of energy.

Let us consider a system whose internal energy is U_1 . If the system is supplied with heat q , the internal energy of the system increases to $U_1 + q$. If work (w) is done on the system, the internal energy in the final state of the system, U_2 is given by

$$U_2 = U_1 + q + w$$

$$\text{or } U_2 - U_1 = q + w$$

$$\Delta U = q + w$$

According to IUPAC, heat added to the system and work done on the system are assigned positive values as both these modes increase the internal energy of the system.

EXERCISE # II

8. Represent the following observations in terms of proper IUPAC symbol?

- Heat absorbed by a system is 20 Joule.
- Work done by a system is 40 Joule.
- Work done on a system is 5 Joule.
- Heat given out by system is 50 Joule.

Ans. It is standard practice to represent both types of heat and work (in/out or on/by) by single symbols q and w -

(a) $q = +20$ Joule.

(b) $w = -40$ Joule.

(c) $w = +5$ Joule.

(d) $q = -50$ Joule.

9. For certain processes the heat and work exchanged between system and surrounding is given in standard format. Describe the physical interpretation of each observation.

(a) $q = +10$ kJ

(b) $w = -20$ kJ

Ans. (a) $q = +10$ kJ :

Since numerical value of q is positive, this shows heat is absorbed by the system from surrounding resulting in gain of energy by system.

(b) $w = -20$ kJ :

Since numerical value of work is negative, this shows work is done by the system on surrounding resulting in loss of energy of system.

10. Predict sign of work done in following reactions at constant pressure.

Initial state		Final state
(i) $H_2O(g)$	$\longrightarrow$	$H_2O(l)$
(ii) $H_2O(s)$	$\longrightarrow$	$H_2O(g)$
(iii) $H_2O(l)$	$\longrightarrow$	$H_2O(s)$
(iv) $CaCO_3(s)$	$\longrightarrow$	$CaO(s) + CO_2(g)$

Ans. (i) +, (ii) -, (iii) -, (iv) -

11. Explain why variation of enthalpy for a process involving an ideal gas is given by $dH = nC_p dT$, irrespective of process ?

Sol. Because

$$H_{\text{ideal gas}} = f(T), \text{ independent of pressure or volume}$$

- 12.** The heat capacity of a molecule depends upon complexity of the molecule. Explain ?

Sol. The heat capacity can be defined as energy needed to raise the temperature of a body by 1°C . The molecule which have large number of degree's of freedom requires larger quantity of heat to raise the temperature by 1°C . This is because energy supplied is distributed in each degree of freedom equally.

13. If work done by the system is 300 joule when 100 cal heat is supplied to it. The change in internal energy during the process is :-

- (A) -200 Joule
(B) 400 Joule
(C) 720 Joule
(D) 120 Joule

Ans. (D)

14. One mole of a gas absorbs 200J of heat at constant volume. Its temperature rises from 298 K to 308 K. The change in internal energy is :-

- (A) 200 J (B) -200 J (C) $200 \times \frac{308}{298}$ J (D) $200 \times \frac{298}{308}$ J

Ans. (A)

- 15.** A system has internal energy equal to U_1 , 450 J of heat is taken out of it and 600 J of work is done on it. The final energy of the system will be -

- (A) $(U_1 + 150 \text{ J})$ (B) $(U_1 + 1050 \text{ J})$ (C) $(U_1 - 150 \text{ J})$ (D) $(U_1 - 1050 \text{ J})$

Ans. (A)

- 16.** The work done by a system is 8J when 40J heat is supplied to it. The change in internal energy of the system during the process :

- (A) 32 J (B) 40 J (C) 48 J (D) -32 J

Ans. (A)

17. Two moles of an ideal gas expand spontaneously into vacuum. The work done is :—

- (A) Zero (B) $2J$ (C) $4J$ (D) $8J$

Ans. (A)

18. The work done during the expansion of a gas from a volume of 4 dm^3 to 6 dm^3 against a constant external pressure of 3 atm is -

- (A) -608 J (B) $+304\text{ J}$ (C) -304 J (D) -6 J

Ans. (A)

- 19.** Assuming that water vapour is an ideal gas, the internal energy change (ΔU) when 1 mol of water is vapourised at 1 bar pressure and 100°C , (Given : Molar enthalpy of vapourisation of water at 1 bar and 373 K = 41 kJ mol^{-1} and $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$ will be) :- **[AIEEE-2007]**

- (A) 4.100 kJ mol⁻¹ (B) 3.7904 kJ mol⁻¹ (C) 37.904 kJ mol⁻¹ (D) 41.00 kJ mol⁻¹

Ans.(3)

20. A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0 mL to 375 mL at a constant temperature of 37.0°C. As it does so, it absorbs 208 J of heat. The values of q and w for the process will be :-

[JEE-MAIN-2013]

(R = 8.314 J/mol K) (ln 7.5 = 2.01)

(A) $q = + 208 \text{ J}$, $w = - 208 \text{ J}$

(B) $q = - 208 \text{ J}$, $w = - 208 \text{ J}$

(C) $q = - 208 \text{ J}$, $w = + 208 \text{ J}$

(D) $q = + 208 \text{ J}$, $w = + 208 \text{ J}$

Ans. (A)

Sol. $q = + 208 \text{ Joule}$

for isothermal process

$$\Delta U = 0$$

 $\therefore$ from 1st law of thermodynamics

$$\Delta U = q + w$$

$$0 = q + w$$

$$\Rightarrow w = -q = -208 \text{ Joule}$$

9. CALCULATION OF q , w , ΔU & ΔH IN VARIOUS PROCESS :**9.1 ISOTHERMAL EXPANSION OR COMPRESSION OF IDEAL GAS :**For isothermal process : $\Delta T = 0$

$$\therefore \Delta U = n.C_{p,m}.\Delta T = 0$$

$$\Delta H = n.C_{p,m}.\Delta T = 0$$

and $q = \Delta U - w = -w$

Now, w depends on path (reversible or irreversible).**9.1.1 Reversible process :**

$$w_{\text{rev.}} = - \int_{V_1}^{V_2} P_{\text{ext}} \cdot dV = - \int_{V_1}^{V_2} (P \pm dP) \cdot dV = - \int_{V_1}^{V_2} P \cdot dV = - \int_{V_1}^{V_2} \frac{nRT}{V} \cdot dV$$

$$\therefore w_{\text{rev.}} = -nRT \cdot \ln \frac{V_2}{V_1} = -nRT \cdot \ln \frac{P_1}{P_2}$$

9.1.2 Irreversible process, against a constant external pressure :

$$w_{\text{irr.}} = -P_{\text{ext}} \int_{V_1}^{V_2} dV = -P_{\text{ext}}(V_2 - V_1)$$

9.1.3 Free expansion (or expansion in vacuum) :

$$P_{\text{ext}} = 0 \text{ but } dV = \text{finite, Hence, } w = 0$$

9.1.4 Comparison of magnitude of work :In expansion, magnitude of work = $-w_{\text{exp}}$ In compression, magnitude of work = w_{comp}

$$\text{Now, } w_{\text{rev}} - w_{\text{irr}} = (-P \cdot dV) - (-P_{\text{ext}} \cdot dV) = (P_{\text{ext}} - P) \cdot dV \\ = -ve, \text{ always}$$

In expansion, $P_{\text{ext}} < P$ and $dV = +ve$ In compression, $P_{\text{ext}} > P$ and $dV = -ve$

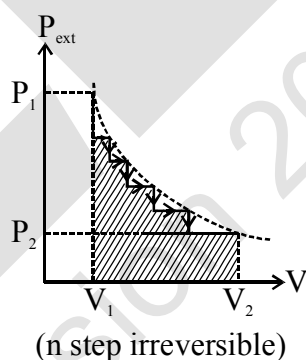
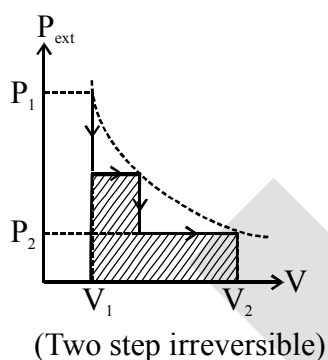
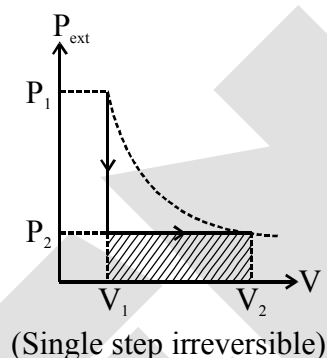
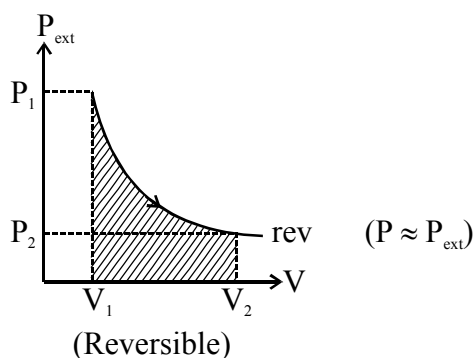
Hence, $w_{\text{rev}} < w_{\text{irr}}$ (Always, with proper sign).

Now, as magnitude of work in expansion is $-w_{\text{exp}}$, hence, $(-w_{\text{rev, exp}}) > (-w_{\text{irr, exp}})$, i.e., magnitude of work is greater when expansion is reversible.

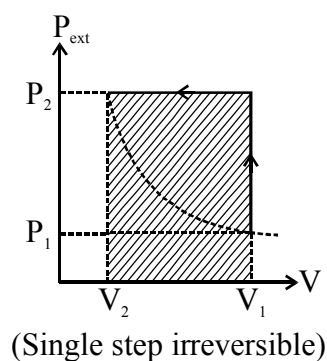
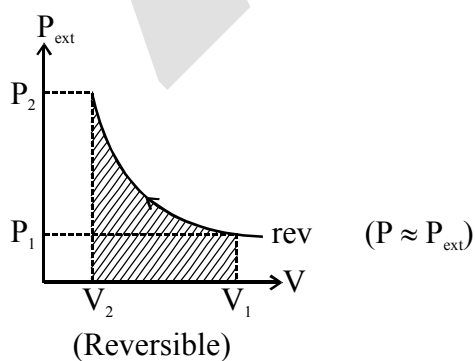
And, as magnitude of work in compression is w_{comp} , hence, $(w_{\text{rev, comp}}) < (w_{\text{irr, comp}})$, i.e., magnitude of work is greater when compression is irreversible.

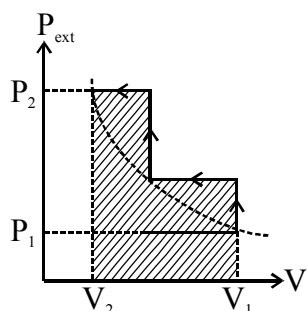
9.1.5 Graphical comparison :

(a) Expansion :

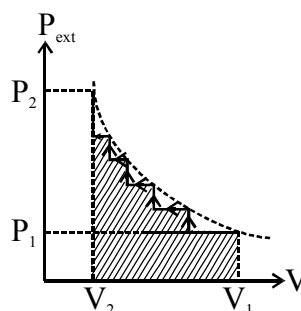

$$(-w_{\text{exp}}) : \text{Rev} > n \text{ step} > \dots > 2 \text{ step} > \text{single step}$$

(b) Compression :





(Two step irreversible)



(n step irreversible)

$(w_{\text{comp}}) : \text{Rev} < n \text{ step} < \dots < 2 \text{ step} < \text{single step}$

Ex.1 Two moles of an ideal gas undergoes isothermal expansion from 4 L to 20 L at 27°C. Calculate q , w , ΔU and ΔH , if the process is performed.

- reversibly
- irreversibly, against a constant external pressure of 1 atm.
- as free expansion

Sol. For isothermal process, $\Delta T = 0$. Hence,

$$\Delta U = n \cdot C_{v,m} \cdot \Delta T = 0$$

$$\Delta H = n \cdot C_{p,m} \cdot \Delta T = 0$$

$$(i) \quad w = -nRT \cdot \ln \frac{V_2}{V_1} = -2 \times 8.314 \times 300 \times \ln \frac{20}{4} = -8028.52 \text{ J}$$

$$\text{and } q = -w = 8028.52 \text{ J}$$

$$(ii) \quad w = -P_{\text{ext}} \cdot (V_2 - V_1) = -1 \text{ atm} (20 \text{ L} - 4 \text{ L}) = -16 \text{ L-atm}$$

$$= -16 \times 101.3 \text{ J} = -1620.8 \text{ J}$$

$$q = -w = 1620.8 \text{ J}$$

$$(iii) \quad w = 0 \text{ and } q = 0$$

9.2 ADIABATIC EXPANSION OR COMPRESSION OF IDEAL GAS :

$$q = 0$$

$$\therefore \Delta U = w = n \cdot C_{v,m} \cdot (T_2 - T_1)$$

$$\text{and } \Delta H = n \cdot C_{p,m} \cdot (T_2 - T_1)$$

The change in temperature may be calculated as

$$n \cdot \int_{T_1}^{T_2} C_{v,m} \cdot dT = - \int_{V_1}^{V_2} P_{\text{ext}} \cdot dV$$

- (a) If $C_{v,m}$ is temperature independent and the process is reversible, then

$$T \cdot V^{\gamma-1} = \text{constant}$$

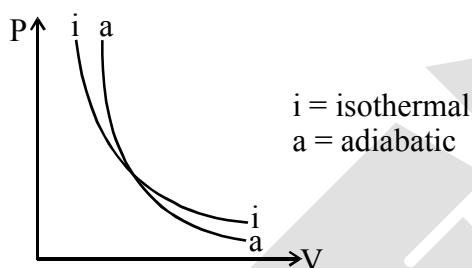
$$\text{or, } P \cdot V^{\gamma} = \text{constant}$$

$$\text{or, } T^{\gamma} P^{1-\gamma} = \text{constant}$$

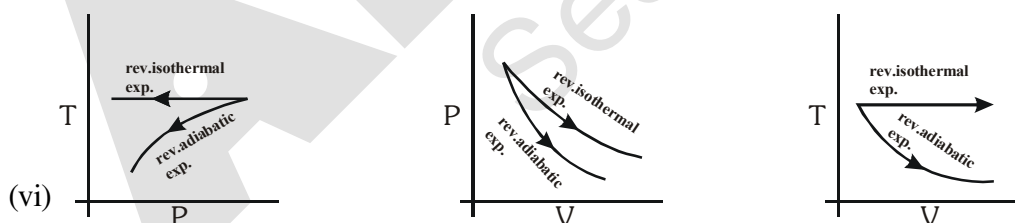
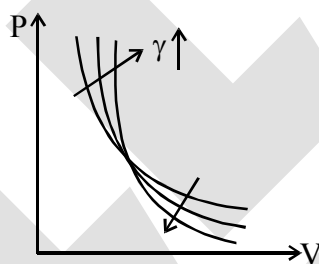
- (b) If $C_{v,m}$ is temperature independent and the process is irreversible, then

$$n.C_{v,m}.(T_2 - T_1) = -P_{ext}.(V_2 - V_1)$$

- (i) The temperature of ideal gas decreases in adiabatic expansion (except free expansion, which is isothermal too) and the temperature of ideal gas increases in adiabatic compression.
- (ii) The final temperature of gas is always higher in irreversible process (expansion or compression) relative to reversible process, for the same change in volume.
- (iii) Just like isothermal process, the magnitude of work is maximum when expansion is reversible and compression is irreversible (single step).
- (iv) Work in reversible isothermal and adiabatic processes may be compared graphically.



- (v) Work in reversible adiabatic process may be compared graphically.



Comparison of reversible adiabatic expansion and reversible isothermal expansion on TP, PV and TV diagram

Ex.2. Two moles of an ideal monoatomic gas undergoes adiabatic expansion from 5L, 127°C to 40L. Calculate q , ΔU , w and ΔH , if the process is performed.

- (i) reversibly
 (ii) irreversibly, against a constant external pressure of 0.1 atm.
 (iii) as free expansion

Sol. : For adiabatic process, $q = 0$

$$(i) \quad T \cdot V^{\gamma-1} = \text{constant} \Rightarrow T_1 \cdot V_1^{\gamma-1} = T_2 \cdot V_2^{\gamma-1}$$

$$\therefore T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 400 \times \left(\frac{5}{40} \right)^{\frac{5}{3}-1} = 100\text{K}$$

$$\text{Now, } w = \Delta U = nC_v \Delta T = -7482.6 \text{ J}$$

$$\Delta H = \gamma \Delta U = -12471 \text{ J}$$

$$(ii) \quad \Delta U = w = -P_{\text{ext}} \cdot (V_2 - V_1) = -0.1 \times (40 - 5) \times 101.3 = -354.55 \text{ J}$$

$$\text{and } \Delta H = \frac{5}{3} \cdot \Delta U = -590.92 \text{ J}$$

$$(iii) \quad \Delta U = w = 0 \\ \text{and } \Delta H = 0$$

Ex.3. Five moles of an ideal monoatomic gas undergoes adiabatic expansion from 12 atm to 1 atm, against a constant external pressure of 1 atm. If the initial temperature of gas is 27°C, calculate the final temperature. Also calculate q , ΔU , w and ΔH .

Sol. : $q = 0$

$$\Delta U = w$$

$$\text{or, } n \cdot C_{v,m} \cdot (T_2 - T_1) = -P_{\text{ext}} \cdot (V_2 - V_1)$$

$$\text{or, } n \cdot \frac{3}{2} R (T_2 - T_1) = -P_2 \left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right)$$

$$\text{or, } \frac{3}{2} (T_2 - 300) = - \left(T_2 - 300 \times \frac{1}{12} \right) \Rightarrow T_2 = 190\text{K}$$

$$\text{or, } \Delta U = w = 5 \times \frac{3}{2} R \times (190 - 300) = -6859.05 \text{ J}$$

$$\text{and } \Delta H = \gamma \cdot \Delta U = -11431.75 \text{ J}$$

9.3 ISOBARIC PROCESS FOR AN IDEAL GAS :

$$q = \Delta H = n \cdot C_{v,m} \cdot \Delta T$$

$$\therefore \Delta U = n \cdot C_{v,m} \cdot \Delta T$$

$$w = -P \cdot (V_2 - V_1) = -nR \cdot \Delta T$$

Ex.4. One mole of an ideal diatomic undergoes isobaric expansion from 27°C to 87°C. Calculate q, ΔU, w and ΔH.

Sol. : $q = \Delta H = n.C_{v,m}.\Delta T = 1 \times \frac{7}{2}R \times 60 = 1745.94 \text{ J}$

$$\Delta U = \frac{\Delta H}{\gamma} = 1247.1 \text{ J}$$

and $w = -nR.\Delta T = -1 \times 8.314 \times 60 = -498.84\text{J}$

9.4 ISOCHORIC PROCESS FOR AN IDEAL GAS :

$$\mathbf{w} = 0$$

$$q = \Delta U = n.C_{vm}.\Delta T$$

$$\therefore \Delta U = n.C_{v,m}.\Delta T$$

Ex.5. Ten moles of an ideal gas ($\gamma = 1.2$) is heated from 27°C to 47°C at constant volume. Calculate q , ΔU , w and ΔH .

Sol. : $w = 0$

$$q = \Delta U = n.C_{v,m}.\Delta T = 2 \times \frac{R}{1.2-1} \times 20 = 200 R$$

$$\Delta H = \gamma . \Delta U = 240R$$

9.5 POLYTROPIC PROCESS FOR AN IDEAL GAS :

$$\Delta U = n.C_{vm}.\Delta T$$

$$\therefore \Delta H = n.C_{p,m}.\Delta T$$

For ideal gas is reversible polytropic process and $C_{v,m}$ temperature independent,

$$P.V^x = \text{constant}$$

or, $T.V^{x-1} = \text{constant}$

where x = polytropic index

$$W_{\text{rev}} = \frac{P_2 V_2 - P_1 V_1}{1 - x} = -\frac{nR(T_2 - T_1)}{1 - x} \quad (x \neq 1)$$

$$= -nRT \cdot \ln \frac{V_2}{V_1} \quad (x = 1)$$

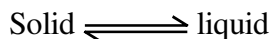
and $q = n.C_m.\Delta T$

$$\text{The molar heat capacity, } C_m = C_{v,m} + \frac{P \cdot dV}{n \cdot dT} = C_{v,m} + \frac{R}{1-x}$$

9.6 CYCLIC PROCESS :



9.7 CHANGE IN PHYSICAL STATE :



Changes in physical state occurs at constant pressure and temperature conditions.

$$\Delta U = q + w$$

$$q = \Delta H = m.L$$

$$\text{and } w = -P(V_{\text{final state}} - V_{\text{initial state}})$$

Ex.6. 90 gm water is completely converted into steam at 100°C and 1 atm. Calculate q , ΔU , w and ΔH . Latent heat of vaporisation of water at 100°C is 540 cal/gm.

Sol. : $\Delta H = m.L = 90 \times 540 = 48600 \text{ cal}$

$$w = -P(V_{\text{vap}} - V_{\text{water}}) = -P.V_{\text{vap}} = -nRT = -\frac{90}{18} \times 2 \times 373 = -3730 \text{ cal}$$

$$\Delta U = q + w = 48600 + (-3730) = 44870 \text{ cal}$$

EXERCISE # III

21. Find the work done when one mole of the gas is expanded reversibly and isothermally from 5 atm to 1 atm at 27°C. ($\ln 5 = 1.6$)

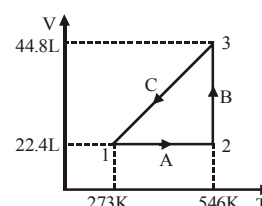
Ans. $w = -3.99 \text{ kJ}$

22. Five moles of an ideal gas at 300 K, expanded isothermally from an initial pressure of 4 atm to a final pressure of 1 atm against a constant external pressure of 1 atm. Calculate q , w , ΔU & ΔH . Calculate the corresponding value of all if the above process is carried out reversibly. ($\ln 2 = 0.7$)

Ans. $w_{\text{irr}} = -225 \text{ R}$, $w_{\text{rev}} = -210 \text{ R}$, $\Delta U = \Delta H = 0$, $q = -w$

23. One mole of an ideal monoatomic gas is carried through the reversible cycle of the given figure consisting of step A, B and C and involving state 1, 2 and 3. Fill in the blank space in the table given below assuming reversible steps.

Table-1			
State	P	V	T
1			
2			
3			



Step	Name of process	q	w	ΔU	ΔH
A					
B					
C					
overall					

Table-1

State	P	V	T
1	1 atm	22.4	273
2	2 atm	22.4	546
3	1 atm	44.8	546

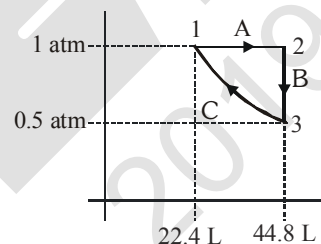
Ans.

Step	Name of process	q	w	ΔE	ΔH
A	Isochoric	$3/2 R (273)$	0	$3/2 R (273)$	$5/2 R(273)$
B	Isothermal	$546 R \ln 2$	$-546 R \ln 2$	0	0
C	Isobaric	$-5/2 R (273)$	$R(273)$	$-3/2 R (273)$	$-5/2 R(273)$

24. One mole of an ideal monoatomic gas is put through reversible path as shown in figure. Fill in the blank in the tables given below.

Table-1

Table-1			
State	P	V	T
1			
2			
3			



Step	Name of process	q	w	ΔU	ΔH
A					
B					
C					
	cyclic				

Table-1

State	P	V	T
1	1 atm	22.4	273
2	1	44.8	546
3	0.5	44.8	273

Ans.

Step	Name of process	q	w	ΔE	ΔH
A	Isobaric	$5/2 R (273)$	$-R(273)$	$3/2 R (273)$	$5/2 R(273)$
B	Isochoric	$- 3/2 R (273)$	0	$- 3/2 R (273)$	$- 5/2 R (273)$
C	Isothermal	$- 273 R \ln 2$	$273 R \ln 2$	0	0
	Cyclic	$R(273) - 273 R \ln 2$	$-R(273)+ 273R \ln 2$	0	0

25. $\frac{1}{22.4}$ mol of an ideal monoatomic gas undergoes a reversible process for which $PV^2 = C$. The gas is expanded from initial volume of 1 L to final volume of 2 L starting from initial temperature of 273 K. Find the heat exchanged q during the process. Express your answer in litre atm.

$$R = \frac{22.4}{273} \text{ L atm mole}^{-1} \text{ degree}^{-1}.$$

Ans. -0.25

26. An ideal gas is allowed to expand both adiabatic reversibly and adiabatic irreversibly. If T_i is the initial temperature and T_f is the final temperature, which of the following statements is correct :-

[AIEEE-2006]

- (1) $T_f > T_i$ for reversible process but $T_f = T_i$ for irreversible process
- (2) $(T_f)_{\text{rev}} = (T_f)_{\text{irrev}}$
- (3) $T_f = T_i$ for both reversible and irreversible processes
- (4) $(T_f)_{\text{irrev}} > (T_f)_{\text{rev}}$

Ans. (4)

27. Which of the following statements/relationships is not correct in thermodynamic changes ?

- (1) $q = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas) [JEE-MAINS(online)-2014]
- (2) For a system at constant volume, heat involved merely changes to internal energy.
- (3) $w = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)
- (4) $\Delta U = 0$ (isothermal reversible expansion of a gas)

Ans. (1)

10. SUMMARY

Process	Expression for w	Expression for q	ΔU	ΔH	Work on PV-graph
Reversible isothermal	$w = -nRT \ln \frac{V_2}{V_1}$ $= -nRT \ln \frac{P_1}{P_2}$	$q = nRT \ln \left(\frac{V_2}{V_1} \right)$ $q = nRT \ln \left(\frac{P_1}{P_2} \right)$	0 process	0	
Irreversible isothermal	$w = -P_{\text{ext}} (V_2 - V_1)$ $= -P_{\text{ext}} \left(\frac{nRT}{P_2} - \frac{nRT}{P_1} \right)$	$q = P_{\text{ext}} (V_2 - V_1)$	0	0	
Isobaric process	$w = -P_{\text{ext}} (V_2 - V_1)$ $= -nR\Delta T$	$q = \Delta H = nC_p \Delta T$	$\Delta U = nC_v \Delta T$	$\Delta H = nC_p \Delta T$	
Isochoric process	$w = 0$	$q = \Delta U = nC_v \Delta T$	$\Delta U = nC_v \Delta T$	$\Delta H = nC_p \Delta T$	
Reversible adiabatic process	$w = nC_v (T_2 - T_1)$ $= \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$	$q = 0$ $PV^\gamma = \text{constant}$ $TV^{\gamma-1} = \text{constant}$ $TP^{1-\gamma/\gamma} = \text{constant}$	$\Delta U = nC_v \Delta T$	$\Delta H = nC_p \Delta T$	
Irreversible adiabatic process	$w = nC_v (T_2 - T_1)$ $= \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$	$q = 0$ $nC_v (T_2 - T_1) =$ $-P_{\text{ext}} \left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right)$	$\Delta U = nC_v \Delta T$	$\Delta H = nC_p \Delta T$	
Polytropic	$w = \frac{P_2 V_2 - P_1 V_1}{n - 1}$ $w = \frac{R(T_2 - T_1)}{(n - 1)}$	$q = \int_{T_1}^{T_2} C_v dT$ $w = \frac{R(T_2 - T_1)}{(n - 1)}$	$\Delta U = nC_v \Delta T$	$\Delta H = nC_p \Delta T$	
Cyclic Process	Area enclosed in PV-diagram For clockwise -ive For anticlockwise +ive	$q = -w$	0	0	

11. LIMITATION OF FIRST LAW :

It can predict the change in energy as a result of change in state or vice-versa, but can not predict the natural direction of change (whether a change can happen on it's own or not).

MISCELLANEOUS PREVIOUS YEARS QUESTION

1. A sample of argon gas at 1 atm pressure and 27°C expands reversibly and adiabatically from 1.25 dm³ to 2.50 dm³. Calculate the enthalpy change in this process. $C_{v,m}$ for argon is 12.48 JK⁻¹ mol⁻¹. [JEE 2000]

Ans. $\Delta H \approx -115 \text{ J}$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

$$T_2 = 300 (1/2)^{8.3/12.48} \left[\gamma^{-1} \frac{R}{C_{v,m}} \right]$$

$$T_2 = 189.2 \text{ Kelvin.}$$

$$n = \frac{P_1 V_1}{RT_1} = \frac{(1)(1.25)}{(0.0821)(300)} = 0.507$$

$$\begin{aligned} \Delta H &= n C_{p,m} (T_2 - T_1) \\ &= 0.507 (20.78) (189.2 - 300) \\ &= -116.732 \text{ Joule.} \end{aligned}$$

2. Which of the following statement is false ?

[JEE 2001]

- (A) Work is a state function
(B) Temperature is a state function
(C) Change of state is completely defined when initial and final states are specified.
(D) Work appears at the boundary of the system

Ans. (A)

3. One mole of non-ideal gas undergoes a change of state (2.0 atm, 3.0 L, 95 K) to (4.0 atm, 5.0 L, 245 K) with a change in internal energy (ΔU) = 30.0 L-atm. The change in enthalpy (ΔH) of the process in L-atm. [JEE 2002]

- (A) 40.0 (B) 42.3
(C) 44.0 (D) Not defined, because pressure is not constant

Ans. (C)

$$\Delta H = \Delta U + P_2 Y_2 - P_1 Y_1$$

$$\begin{aligned} \Delta H &= 30 + 20 - 6 \\ &= 44 \end{aligned}$$

4. One mole of a liquid (1 bar, 100 ml) is taken in an adiabatic container and the pressure increases steeply to 100 bar. Then at a constant pressure of 100 bar, volume decrease by 1 ml. Find ΔU and ΔH .

Ans. $\Delta U = 10 \text{ J}$, $\Delta H = 990 \text{ J}$

[JEE 2004]

$$0 + w = \Delta U$$

$$\Delta U = -P_{\text{ext}} (Y_2 - Y_1)$$

$$= -100 \left(\frac{-1}{1000} \right)$$

$$= 0.1 \text{ bar litre.}$$

$$\Delta H = \Delta U + P_2 Y_2 - P_1 Y_1$$

$$\Delta H = 0.1 + \left[\frac{100(99) - (1)(100)}{1000} \right]$$

$$\Delta H = 0.1 + 9.8$$

$$\Delta H = 9.9 \text{ bar. litre}$$

$$\Delta H = 990 \text{ Joules}$$

5. Two mole of an ideal gas is expanded isothermally and reversibly from 1 litre to 10 litre at 300 K. The enthalpy change (in KJ) for the process is :- **[JEE 2004]**
 (A) 11.4 kJ (B) - 11.4 kJ (C) 0 kJ (D) 4.8 kJ

Ans. (C)

In isothermal process $\Delta T = 0$

$$\Delta H = nC_{v,m} \Delta T$$

$$\Delta H = 0$$

6. One mole of monoatomic ideal gas expands adiabatically at initial temp. T against a constant external pressure of 1 atm from one litre to two litre. Find out the final temperature. ($R = 0.0821 \text{ litre. atm K}^{-1} \text{ mol}^{-1}$) **[JEE 2005]**

(A) T (B) $\frac{T}{(2)^{\frac{5}{3}-1}}$ (C) $T - \frac{2}{3 \times 0.0821}$ (D) $T + \frac{2}{3 \times 0.0821}$

Ans. (C)

$$-P_{\text{ext}} (V_2 - V_1) = nC_{v,m} (T_2 - T_1)$$

$$-1(2 - 1) = 1 \left(\frac{3}{2} R \right) (T_2 - T)$$

$$T_2 = T - \frac{2}{3R}$$

7. The molar heat capacity of a monoatomic gas for which the ratio of pressure and volume is one.
 (A) $\frac{4}{2} R$ (B) $\frac{3}{2} R$ (C) $\frac{5}{2} R$ (D) zero **[JEE 2006]**

Ans. (A)

$$\frac{P}{V} = C$$

$$PV^{-1} = C$$

$$C_m = C_{v,m} + \frac{R}{1 - (-1)}$$

$$= \frac{3}{2} R + \frac{R}{2} = \frac{4R}{2}$$

8. Among the following, the state function(s) is (are) **[JEE 2009]**
 (A) Internal energy (B) Irreversible expansion work
 (C) Reversible expansion work (D) Molar enthalpy

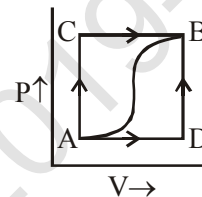
Ans. (A,D)

EXERCISE # S-I

BASIC

- Find the work done in each case :
 - When one mole of ideal gas in 10 litre container at 1 atm is allowed to enter an very large evacuated bulb of capacity 100 litre.
 - When 1 mole of gas expands from 1 litre to 5 litre against constant one atmospheric pressure.
 - Find the work done in atm-litre when 18 mL of water is getting vapourised at 373 K in open vessel.
(Assume the ideal behaviour of water vapour).
 - Find total degree of freedom and break up as translational, rotational or vibrational degree of freedom in following cases.
 - CO_2
 - SO_2
 - He
 - NH_3
- ❑ Problems related to First law of thermodynamics :
- The gas is cooled such that it loses 65 J of heat. The gas contracts as it cools and work done on the system is equal to 20 J. What are q , w and ΔU ?
 - When a system is taken from state A to state B along the path ACB, 80J of heat flows into the system and the system does 30J of work.
 - How much heat flows into the system along path ADB if the work done by the system is 10J ?
 - When the system is returned from state B to A along the curved path, the work done on the system is 20J. Does the system absorb or liberate heat, and how much ?
 - If $U_D - U_A = 40\text{J}$, find the heat absorbed in the process AD and DB if work done by system in ADB is 10 Joule.
 - An ideal gas undergoes expansion from A(10 atm, 1 litre) to B(1 atm, 10 litre), first against 5 atm and then against 1 atm, isothermally. Calculate the amount of heat absorbed (in litre. atm)
 - 2 moles of an ideal gas is compressed from (1 bar, 2L) to 2 bar isothermally. Calculate magnitude of minimum possible work involved in the change (in Joules). (Given : 1 bar L = 100 J) ($\ln 2 = 0.7$)
 - 2 mole of an ideal gas undergoes isothermal compression along three different paths if it's initial state is (2 bar, 4 litre) in all three process.
 - Reversible compression from till 20 bar.
 - A single stage compression against a constant external pressure of 20 bar
 - A two stage compression consisting initially of compression against a constant external pressure of 10 bar until $P_{\text{gas}} = P_{\text{ext}}$, followed by compression against a constant pressure of 20 bar until $P_{\text{gas}} = P_{\text{ext}}$.

Calculate the work (in bar. L) for each of these processes and for which of the irreversible processes is the magnitude of the work greater ? [Given : $R = 0.08 \text{ bar. L/mole.K}$]
 - A monoatomic ideal gas undergoing irreversible adiabatic compression from 4L to 1L against 1 bar pressure. Calculate the enthalpy change of gas. [Given : 1 bar-L = 100 J]
 - Two mole of ideal diatomic gas ($C_{V,m} = 5/2 R$) at 300K and 5 atm expanded irreversibly & adiabatically to a final pressure of 2 atm against a constant pressure of 1 atm. Calculate q , w , ΔH & ΔU .



11. Calculate the work done by system in an irreversible (single step) adiabatic expansion of 1 mole of a polyatomic gas ($\gamma = 4/3$) from 400 K and pressure 10 atm to 1 atm.
12. 1 mole of CO_2 gas at 300 K is expanded under reversible adiabatic condition such that its volume becomes 27 times.

(a) What is the final temperature ?

(b) What is work done ?

Given $\gamma = 1.33$ and $C_V = 25.08 \text{ J mol}^{-1} \text{ K}^{-1}$ for CO_2 .

13. Molar heat capacity of an ideal gas at constant volume is given by $C_V = (16.5 + 10^{-2}T) \frac{\text{J}}{\text{K} - \text{mol}}$. If 2 moles of this ideal gas are heated at constant volume from 300K to 400K. Then the change in internal energy (in kJ) will be approximately.
14. Three moles of an ideal gas at 200 K and 2.0 atm pressure undergoes reversible adiabatic compression until the temperature becomes 250 K. For the gas C_V is $27.5 \text{ JK}^{-1} \text{ mol}^{-1}$ in this temperature range. Calculate q , w , ΔU , ΔH and final V and final P .

Given $\left(\frac{5}{4}\right)^{1/0.3} = 2.1$, $\left(\frac{5}{4}\right)^{35.8/8.3} = 2.61$

[$R = 8.3 \text{ Joule/mol.K}$ or $R = 0.0821 \text{ atm litre/mol.K}$]

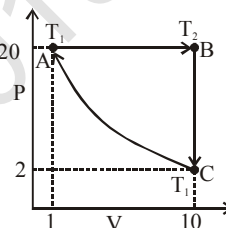
❑ Problem related to cyclic process

15. One mole of a perfect monoatomic gas is put through a cycle consisting of the following three reversible steps :

(CA) Isothermal compression from 2 atm and 10 litres to 20 atm and 1 litre.

(AB) Isobaric expansion to return the gas to the original volume of 10 litres with

T going from T_1 to T_2 . (BC) Cooling at constant volume to bring the gas to the original pressure and temperature. The steps are shown schematically in the figure shown.



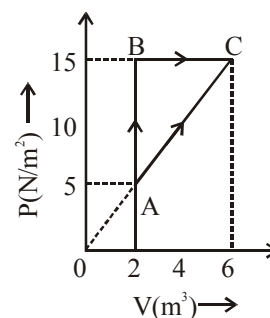
(a) Calculate T_1 and T_2

(b) Calculate ΔU , q and w (in calories) for the cycle.

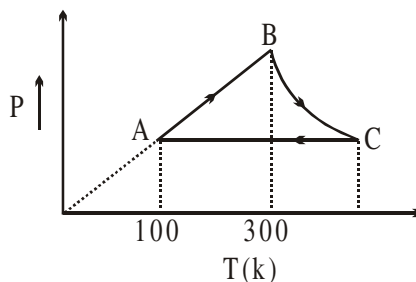
[Given : $\ln 10 = 2.3$, $R = 0.821 \text{ atm.litre/mol.K}$ or $R = 2 \text{ cal/mol.K}$]

16. The given figure shows a change of state A to state C by two paths ABC and AC for an ideal gas. Calculate the :

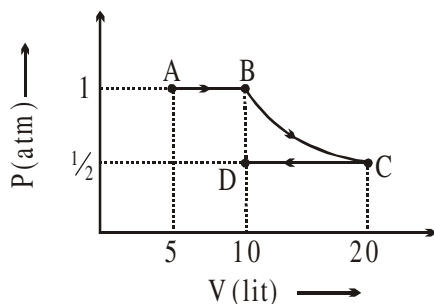
- (a) Path along which magnitude of work done is least ABC or AC
- (b) Internal energy at C if the internal energy of gas at A is 10 J and amount of heat supplied to change its state to C through the path AC is 200J.
- (c) Amount of heat supplied to the gas to go from A to B, if internal energy change of gas from A to B is 10 J.



17. Calculate the net work done in the following cycle for one mol of an ideal gas (in calorie), where in process BC, $PT = \text{constant}$. ($R = 2 \text{ cal/mol-K}$).



18.



Calculate magnitude of total work done (in atm.lit) for the above process ABCD involving a monoatomic ideal gas. [Given : $\ln 2 = 0.7$]

❑ **Problem based on polytropic process**

19. For 1 mole of ideal monoatomic gas if $\frac{P}{V^2} = \text{constant}$ and initial temperature is 100 K. If gas is expanded from 1 L to 2 L then find (a) heat capacity (b) total heat absorbed (c) work (d) change in internal energy.

(Assume $R = 0.0821 \frac{\text{L-atm}}{\text{mole-K}}$, $1 \text{ L-atm} = 24 \text{ cal}$)

20. One mole of an ideal monoatomic gas undergoes expansion along a straight line on P-V curve from initial state A(3L, 8 atm) to final state B(7.5 L, 2 atm). Calculate q for the above process in L atm.

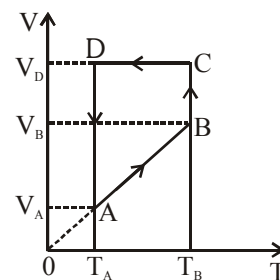
❑ **Problems based on change in physical state :**

21. Water expands when it freezes. Determine amount of work done, in joules, when a system consisting of 1.0 L of liquid water freezes under a constant pressure of 1.0 atm and forms 1.1 L of ice.
22. What is ΔU when 2.0 mole of liquid water vaporises at 100°C ? The heat of vaporisation, ΔH_{vapour} of water at 100°C is $40.66 \text{ kJ mol}^{-1}$.
23. When 1 mole of ice melt at 0°C and at constant pressure of 1 atm, 1440 calories of heat are absorbed by the system. The molar volumes of ice and water are 0.0196 and 0.0180 litre respectively. Calculate ΔH and ΔU for the reaction.

EXERCISE # S-II

1. A monoatomic ideal gas of two moles is taken through a reversible cyclic process starting from A as shown in figure. The volume ratios

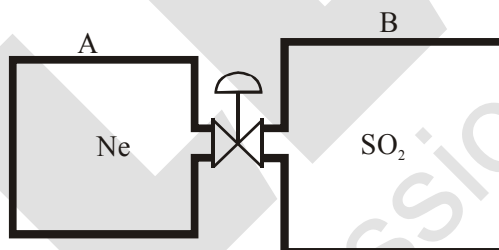
are $\frac{V_B}{V_A} = 2$ and $\frac{V_D}{V_A} = 4$. If the temperature T_A at A is 27°C , calculate:



- The temperature of the gas at point B.
 - Heat absorbed or released by the gas in each process.
 - The total work done by the gas during complete cycle. ($\ln 2 = 0.7$)
2. One mole of an ideal gas (not necessarily monoatomic) is subjected to the following sequence of steps.
- It is heated at constant volume from 298 K to 373 K
 - It is expanded freely into a vacuum to double volume at 373 K.
 - It is cooled reversibly at constant pressure to 298K.
- Calculate q , w , ΔU and ΔH for the overall process.
3. Derive a mathematical expression for the work done when n -mole a gas that has the equation of state $PV = nRT - \frac{n^2a}{V}$ expands reversibly from V_i to V_f at constant temperature T .
4. 3 dm^3 of an ideal monoatomic gas at 600 K and 32 bar expands until pressure of the gas is 1bar Calculate q , w , ΔU and ΔH for the process if the expansion is :
- (Use : $R = 0.08 \text{ bar litre/mol-K}$, $= 2 \text{ cal / mol-K}$)
- Isothermal and reversible
 - Adiabatic and reversible
 - Isothermal and adiabatic
 - Against 1 bar and adiabatic
 - Against 1 bar and isothermal.
5. Pressure over 1000 mL of a liquid is gradually increased from 1 bar to 1001 bar under adiabatic conditions. If the final volume of the liquid is 990 mL, calculate ΔU and ΔH of the process, assuming linear variation of volume with pressure.
6. At 500 kilobar pressure, density of diamond and graphite are 3 g/cc and 2 g/cc respectively, at certain temperature 'T'. Find the value $|\Delta H - \Delta U|$ (kJ/mole) for the conversion of 1 mole of graphite to 1 mole of diamond at temperature 'T' :
7. 2 mole of ideal monoatomic gas was subjected to reversible adiabatic compression from initial state of $P = 1 \text{ atm}$ and $T = 300 \text{ K}$ till the pressure is $4\sqrt{2} \text{ atm}$ and temperature is TK . The gas is then subjected to reversible isothermal expansion at TK till the internal pressure is one atm. The gas is now isobarically cooled to attain initial state. Find W_{Net} (in calorie) for whole process.

[Use $\ln 2 = 0.7$, $R = 2 \text{ cal/K-mol}$]

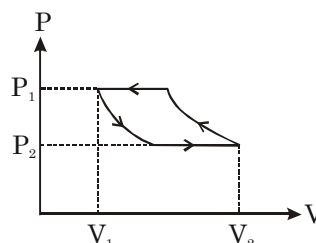
8. One mole of an ideal gas at 300 K is heated at constant volume (V_1) until its temperature is doubled, then it is subjected to isothermal reversible expansion till it reaches the original pressure. Finally the gas is cooled at constant pressure till system reached to the half of original volume ($V_1/2$). Determine total work done ($|w|$) in cal. [Use : $\ln 2 = 0.70$; $R = 2 \text{ Cal/K.mol.}$]
9. How many times volume of diatomic gas should be expanded reversibly & adiabatically in order to reduce its r.m.s velocity to half.
10. 0.5 mole each of two ideal gases A ($C_v = \frac{3}{2}R$) and B ($C_v = \frac{5}{2}R$) are taken in a container and expanded reversibly and adiabatically from $V = 1 \text{ L}$ to $V = 4 \text{ L}$ starting from initial temperature $T = 300 \text{ K}$. Find $|\Delta H|$ for the process in (cal/mol).
11. 0.50 mol of an ideal gas initially at a temperature of 300 K and at a pressure of 2 atm is expanded isothermally in three steps. In each step, the pressure is dropped suddenly and held constant until equilibrium is re-established. The pressure at each of the three stages of expansion are 1.6, 1.2 and 1 atm. Calculate the work done ($|W|$) (in atm-litre) in this process.
[Use $R = 0.08 \text{ atm litre/ mole. K}$]
12. Two rigid adiabatic vessels A and B which initially, contains two gases at different temperatures are connected by pipe line with valve of negligible volume. The vessel 'A' contains 2 moles Ne gas ($C_{p,m} = \frac{5}{2}R$) at 300 K, vessel 'B' contains 3 moles of SO_2 gas ($C_{p,m} = 4R$) at 400 K. The volume of A & B vessel is 4 and 6 litre respectively. The final total pressure (in atm) when valve is opened and 12 Kcal heat supplied through it to vessels.
[Use : $R = 2 \text{ cal/mol. K}$ and $R = 0.08 \text{ L. atm / mol K}$ as per desire]



13. One mole of an ideal monoatomic gas initially at 1200 K and 64 atm is expanded to a final state at 300 K and 1.00 atm. To achieve change, a reversible path is constructed that involve an adiabatic expansion in the beginning followed by an isothermal expansion to the final state. Determine the magnitude of net work done by the gas, in cal. ($R = 2 \text{ cal/K-mol. } \ln 2 = 0.7$)
14. An ideal gas is carried through a thermodynamic cycle.

$$P_1 = 10 \text{ bar} \quad V_1 = \frac{1}{\ln 10} \text{ litre}$$

$$P_2 = 1 \text{ bar} \quad V_2 = \frac{4}{\ln 10} \text{ litre}$$

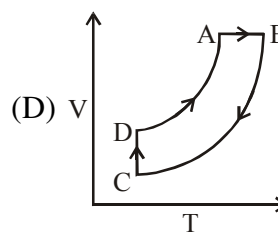
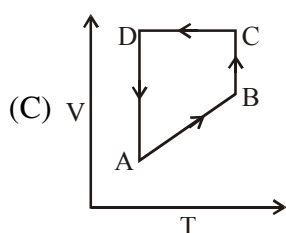
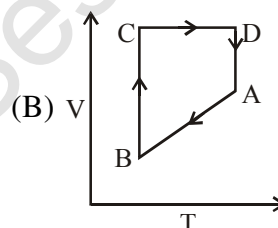
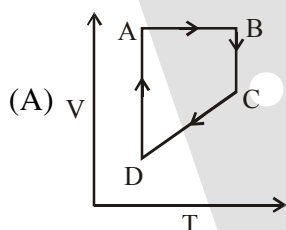
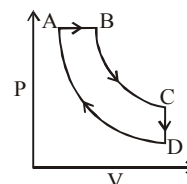
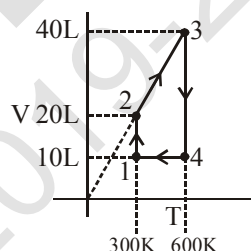


Consisting of two isobaric and two isothermal processes. Calculate the net work in the entire cycle in litre bar.

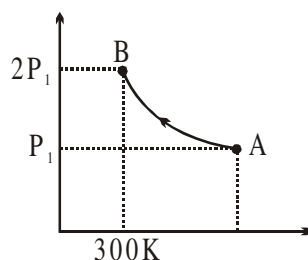
10. A gas is expanded from volume V_1 to V_2 through three different process :
 (a) Reversible adiabatic (b) Reversible isothermal (c) Irreversible adiabatic

The correct statements is -

- (A) $(T_f)_{\text{Reversible Isothermal}} > (T_f)_{\text{Reversible adiabatic}} > (T_f)_{\text{Irreversible adiabatic}}$
 (B) $(T_f)_{\text{Reversible Isothermal}} > (T_f)_{\text{Irreversible adiabatic}} > (T_f)_{\text{Reversible adiabatic}}$
 (C) $W_{\text{Reversible Isothermal}} > W_{\text{Irreversible adiabatic}} > W_{\text{reversible adiabatic}}$
 (D) $(P_f)_{\text{Reversible Isothermal}} > (P_f)_{\text{Reversible adiabatic}} > (P_f)_{\text{Irreversible adiabatic}}$
11. What is the magnitude of work performed by one mole of an ideal gas when its volume increases eight times in irreversible adiabatic expansion if the initial temperature of the gas is 300 K ? C_v for the gas is $1.5 R$. ($R = 2 \text{ Cal / mol / K}$)
 (A) 900 Cal (B) 450 Cal (C) 675 Cal (D) 331.57 Cal
12. For a process which follows the equation $PV^3 = C$, the work done when one mole of ideal gas was taken from 1 atm to $2\sqrt{2}$ atm starting from initial temperature of 300 K is
 (A) 300 R (B) 150 R (C) 600 R (D) 900 R
13. What is the net work done (w) when 1 mole of monoatomic ideal gas undergoes in a process described by 1, 2, 3, 4 in given V - T graph
 Use : $R = 2 \text{ cal/mole K}$
 $\ln 2 = 0.7$
 (A) -600 cal (B) - 660 cal
 (C) + 660 cal (D) + 600 cal
14. A cyclic process ABCD is shown in PV diagram for an ideal gas.
 which of the following diagram represents the same process ?

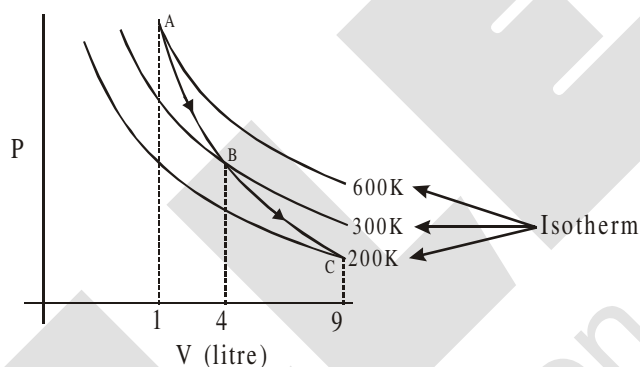


- 15.** 2 moles of an ideal monoatomic gas is taken from state A to state B through a process AB in which $PT = \text{constant}$. The process can be represented on a P-T graph as follows:

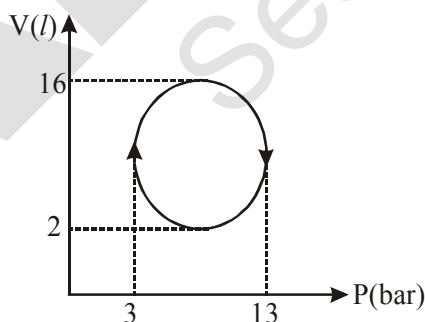


Select the incorrect option(s):

- (A) Heat evolved by the gas during process AB = 2100 R
(B) Heat absorbed by the gas during process AB = 2100 R
(C) $\Delta U = -900 R$
(D) $W = 1200R$
- 16.** The given figure shows a polytropic process ABC for one mole of an ideal gas. Calculate the polytropic index (x) for the process-

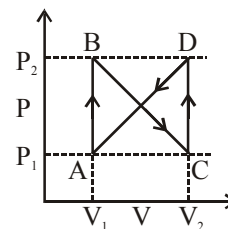


- (A) $1/2$ (B) $-1/2$ (C) $3/2$ (D) None
17. Work in the following cyclic process is



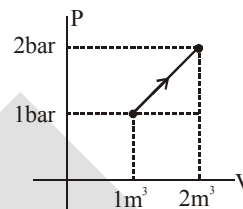
- (A) -11 (B) -11000 (C) 11 (D) 11000

18. An ideal gas is taken around the cycle ABCDA as shown in figure. The net work done during the cycle is equal to :-



- (A) Zero
(B) Positive
(C) Negative
(D) We cannot predict

19. What is ΔU for the process described by figure. Heat supplied during the process $q = 200$ kJ.



- (A) +50 kJ
(B) -50 kJ
(C) -150 kJ
(D) +150 kJ

20. A diatomic ideal gas initially at 273 K is given 100 cal heat due to which system did 209 J work, Molar heat capacity (C_m) of gas for the process is [1 cal = 4.18 Joule]

- (A) $\frac{3}{2}R$ (B) $\frac{5}{2}R$ (C) $\frac{5}{4}R$ (D) $5R$

21. For an ideal monoatomic gas during any process $T = kV$, find out the molar heat capacity of the gas during the process. (Assume vibrational degree of freedom to be active)

- (A) $\frac{5}{2}R$ (B) $3R$ (C) $\frac{7}{2}R$ (D) $4R$

22. An amount Q of heat is added to a monoatomic ideal gas in a process in which the gas performs a work $Q/2$ on its surrounding. The molar heat capacity of gas (in cal/K-mol) for the process is.

- (A) $3R$ (B) $5R$ (C) $4R$ (D) $2R$

EXERCISE # O-II

❑ ONE OR MORE THAN ONE MAY BE CORRECT :

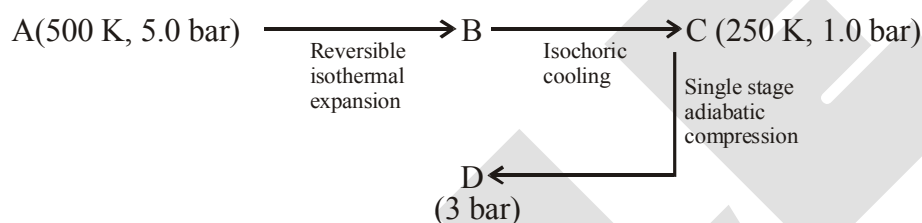
1. An insulated container of gas has two chambers separated by an insulating partition. One of the chambers has volume V_1 and contains ideal gas at pressure P_1 and temperature T_1 . The other chamber has volume V_2 and contains same ideal gas at pressure P_2 and temperature T_2 . If the partition is removed without doing any work on the gas, the final equilibrium temperature of the gas in the container will be :-

(A) $\frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_2 + P_2 V_2 T_1}$ (B) $\frac{P_1 V_1 T_1 + P_2 V_2 T_2}{P_1 V_1 + P_2 V_2}$ (C) $\frac{P_1 V_1 T_2 + P_2 V_2 T_1}{P_1 V_1 + P_2 V_2}$ (D) $\frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_1 + P_2 V_2 T_2}$

2. The heat capacity of liquid water is 75.6 J/mol.K, while the enthalpy of fusion of ice is 6.0 kJ/mol. What is the smallest number of ice cubes at 0°C, each containing 9.0 g of water, needed to cool 500 g of liquid water from 20°C to 0°C ?

(A) 1 (B) 7 (C) 14 (D) 252

3. Two moles of an ideal gas ($C_{v,m} = \frac{3}{2}R$) is subjected to following change of state



The correct statement is/are

(A) The pressure at B is 2.0 bar (B) The temperature at D is 450 K
(C) $\Delta H_{CD} = 1000 R$ (D) $\Delta U_{BC} = 375 R$

4. A real gas is subjected to an adiabatic process from (2 bar, 40 lit., 300 K) to (4 bar, 30 lit., 300 K) against a constant pressure 4 bar the enthalpy change for the process is

(A) Zero (B) 6000 J (C) 8000 J (D) 80 J

5. Which of the following statement is/are correct ?

- (A) Enthalpy can be written as $H = f(V, T)$ for a substance (no physical or chemical change)
(B) Absolute value of enthalpy can not be determined
(C) The heat absorbed during the isothermal expansion of an ideal gas against vacuum is zero
(D) During compression of an ideal gas at constant pressure the temperature of gas decreases.

6. Choose the correct statement(s) :-

- (A) During adiabatic expansion of an ideal gas, magnitude of work obtained is equal to ΔH of gas.
(B) For same change in temperature of ideal gas through adiabatic process, magnitude of W will be same in reversible as well as irreversible process.
(C) During an adiabatic reversible expansion of an ideal gas, temperature of the system increases.
(D) For same change in volume of ideal gas, ΔU is less in adiabatic expansion than in isobaric expansion if expansion starts with same initial state.

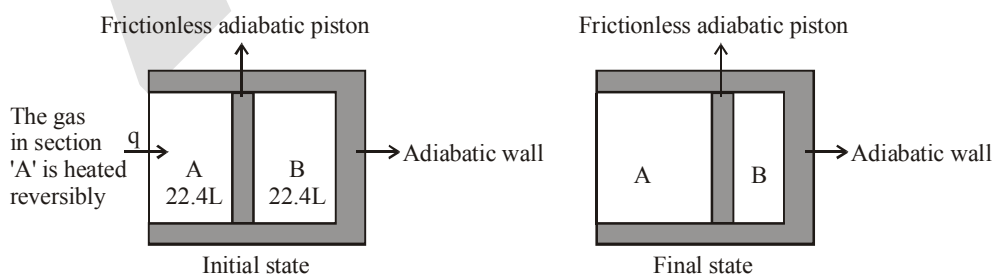
7. Choose the correct statement(s) among the following :-
- (A) Internal energy of H_2O remains constant during conversion of liquid into its vapour at constant temperature.
- (B) During fusion of ice into water, enthalpy change and internal energy change are almost same at constant temperature.
- (C) Molar heat capacity of gases are temperature dependent
- (D) During boiling of H_2O at 1 atm, 100°C , average kinetic energy of H_2O molecules increases.
8. Suppose that the volume of a certain ideal gas is to be doubled by one of the following processes :
- (A) isothermal expansion (B) adiabatic expansion
- (C) free expansion in insulated condition (D) expansion at constant pressure.
- If E_1, E_2, E_3 and E_4 respectively are the changes in average kinetic energy of the molecules for the above four processes, then-
- (A) $E_2 = E_3$ (B) $E_1 = E_3$ (C) $E_1 > E_4$ (D) $E_4 > E_3$

❑ **Assertion / reason :**

9. **Statement-1 :** There is no change in enthalpy of an ideal gas during compression at constant temperature.
- Statement-2 :** Enthalpy of an ideal gas is a function of temperature and pressure.
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.
10. **Statement-1 :** Due to adiabatic expansion, the temperature of an ideal gas always decreases.
- Statement-2 :** For an adiabatic process, $\Delta U = w$.
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

Paragraph for Q.11 to Q.13

A cylindrical container of volume 44.8 litres is containing equal no. of moles (in integer no.) of an ideal monoatomic gas in two sections A and B separated by an adiabatic frictionless piston as shown in figure. The initial temperature and pressure of gas in both section is 27.3K and 1 atm. Now gas in section 'A' is slowly heated till the volume of section B becomes $(1/8)^{\text{th}}$ of initial volume.



Given : $R = 2 \text{ cal/mol-K}$, $C_{v,m}$ of monoatomic gas $= \frac{3}{2} R$,

At 1 atm & 0°C ideal gas occupy 22.4 litre.

11. What will be the final pressure in container B.
 (A) 2 atm (B) 8 atm (C) 16 atm (D) 32 atm
12. Final temperature in container A will be
 (A) 1638 K (B) 6988 K (C) 3274 K (D) 51 K
13. Change in enthalpy for section A in Kcal
 (A) 48.3 (B) 80.53 (C) 4.83 (D) 8.05

Paragraph for Q.14 to 16

As a result of the isobaric heating by $\Delta T = 72\text{K}$, one mole of a certain ideal gas obtains an amount of heat $Q = 1.60\text{ kJ}$.

14. The work performed by the gas is -
 (A) 8.60 kJ (B) 0.60 kJ (C) 16.60 kJ (D) 4.60 kJ
15. The increment of its internal energy (in kJ) is -
 (A) 1.0 (B) 1.6 (C) 2.2 (D) 2.0
16. The value of γ for the gas is -
 (A) 0.6 (B) 1.4 (C) 1.6 (D) 1.5

MATCH THE COLUMN :

17. Match Column-I with Column-II

Column-I (Ideal Gas)

- (A) Reversible isothermal process
 (B) Reversible adiabatic process
 (C) Irreversible adiabatic process
 (D) Irreversible isothermal process

Column-II (Related equations)

- (P) $W = 2.303 n RT \log(P_2/P_1)$
 (Q) $W = nC_{V,m} (T_2 - T_1)$
 (R) $W = -2.303nRT \log(V_2/V_1)$
 (S) $W = -\int_{V_i}^{V_f} P_{\text{ext.}} dV$

18. Match the column

Column-I

- (A) Isothermal vaporisation of water at 100°C & 1 atm
 (B) Isothermal reversible expansion of an ideal gas
 (C) Adiabatic free expansion of ideal gas
 (D) Isochoric heating of an ideal gas

Column-II

- (P) $\Delta T = 0$
 (Q) $\Delta U = 0$
 (R) $\Delta H = 0$
 (S) $q = 0$
 (T) $w = 0$

MATCH THE LIST :

19. Match the following, and select the correct code :-

Column-I

- (P) Isothermal process (reversible)
 (Q) Adiabatic process (reversible)
 (R) Isochoric process
 (S) Isothermal process (irreversible)

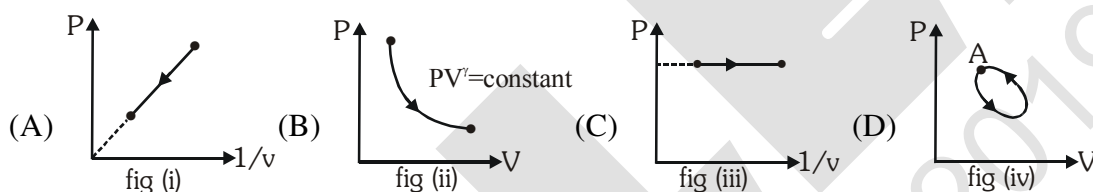
Column-II

- (1) $q = 2.303 nRT \log \frac{P_1}{P_2}$
 (2) $PV^\gamma = \text{const.}$
 (3) $q = P_{\text{ext.}} (V_2 - V_1)$
 (4) Area under P-V curve is zero

Code:

	P	Q	R	S
(A)	2	1	4	3
(B)	1	2	4	3
(C)	2	1	3	4
(D)	1	2	3	4

20. The figures given below depict different processes for a given amount of an ideal gas.



Column-I

- (P) In Fig (i)
 (Q) In Fig (ii)
 (R) In Fig (iii)
 (S) In Fig (iv)

Column-II

- (1) Net heat is absorbed by the system
 (2) Net work is done on the system
 (3) Net heat is rejected by the system
 (4) Net work is done by the system
 (5) Net internal energy remains constant

Code:

	P	Q	R	S
(A)	1, 4, 5	4	2, 3	2, 3, 5
(B)	1, 4, 5	4	2, 3	2, 3
(C)	1, 2, 4	2	1, 4	2, 3
(D)	1, 2	2	1, 4	3, 5

EXERCISE # J-MAIN

1. A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0 mL to 375 mL at a constant temperature of 37.0°C. As it does so, it absorbs 208 J of heat. The values of q and w for the process will be :- **[Jee-Main-2013]**

(R = 8.314 J/mol K) (ln 7.5 = 2.01)

- (1) $q = + 208 \text{ J}$, $w = - 208 \text{ J}$
 - (2) $q = - 208 \text{ J}$, $w = - 208 \text{ J}$
 - (3) $q = - 208 \text{ J}$, $w = + 208 \text{ J}$
 - (4) $q = + 208 \text{ J}$, $w = + 208 \text{ J}$
2. Which of the following statements/relationships is not correct in thermodynamic changes ?

(1) $q = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas) [Jee-Main_(Online)-2014]

(2) For a system at constant volume, heat involved merely changes to internal energy.

(3) $w = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

(4) $\Delta U = 0$ (isothermal reversible expansion of a gas)

3. A gas undergoes change from state A to state B. In this process, the heat absorbed and work done by the gas is 5 J and 8 J, respectively. Now gas is brought back to A by another process during which 3 J of heat is evolved. In this reverse process of B to A:- **[Jee-Main (Online)-2017]**

(1) 10 J of the work will be done by the surrounding on gas.

(2) 6 J of the work will be done by the surrounding on gas.

(3) 10 J of the work will be done by the gas.

(4) 6 J of the work will be done by the gas.

4. The enthalpy change on freezing of 1 mol of water at 5°C to ice at -5°C is: [Jee-Main (Online)-2017]

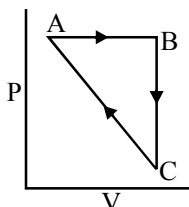
(Given $\Delta_{\text{fus}}H = 6 \text{ kJ mol}^{-1}$ at 0°C , $C_p(\text{H}_2\text{O}, l) = 75.3 \text{ J mol}^{-1} \text{ K}^{-1}$, $C_p(\text{H}_2\text{O}, s) = 36.8 \text{ J mol}^{-1} \text{ K}^{-1}$)

(1) $-6.56 \text{ kJ mol}^{-1}$ (2) $-5.81 \text{ kJ mol}^{-1}$ (3) $-6.00 \text{ kJ mol}^{-1}$ (4) $-5.44 \text{ kJ mol}^{-1}$

- 5.** ΔU is equal to **[Jee-Main (Online)-2017]**

(1) Isochoric work (2) Isobaric work (3) Adiabatic work (4) Isothermal work

6. An ideal gas undergoes a cyclic process as shown in figure. [Jee-Main (Online)-2018]



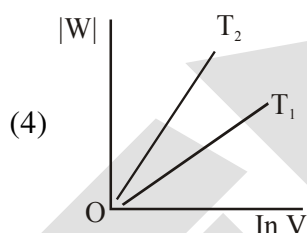
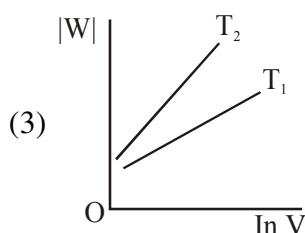
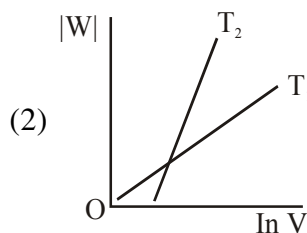
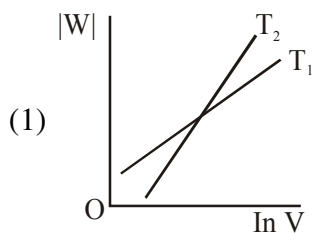
$$\Delta U_{BC} = -5 \text{ kJ mol}^{-1}, q_{AB} = 2 \text{ kJ mol}^{-1}$$

$$W_{AB} = -5 \text{ kJ mol}^{-1}, W_{CA} = 3 \text{ kJ mol}^{-1}$$

Heat absorbed by the system during process CA is :-

(1) 18 kJ mol^{-1} (2) $+5 \text{ kJ mol}^{-1}$ (3) -5 kJ mol^{-1} (4) -18 kJ mol^{-1}

7. Consider the reversible isothermal expansion of an ideal gas in a closed system at two different temperatures T_1 and T_2 ($T_1 < T_2$). The correct graphical depiction of the dependence of work done (w) on the final volume (V) is: [Jee-Main_(Online)-2019]



8. An ideal gas undergoes isothermal compression from 5 m^3 to 1 m^3 against a constant external pressure of 4 Nm^{-2} . Heat released in this process is used to increase the temperature of 1 mole of Al. If molar heat capacity of Al is $24 \text{ J mol}^{-1} \text{ K}^{-1}$, the temperature of Al increases by [JEE-MAINS(online)-2019]

(1) $\frac{3}{2} \text{ K}$

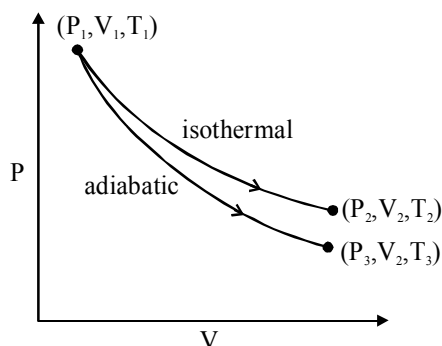
(2) $\frac{2}{3} \text{ K}$

(3) 1 K

(4) 2 K

EXERCISE # J-ADVANCED

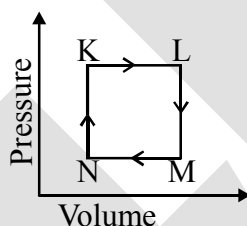
1. The reversible expansion of an ideal gas under adiabatic and isothermal conditions is shown in the figure. Which of the following statement(s) is (are) correct ? [JEE 2012]



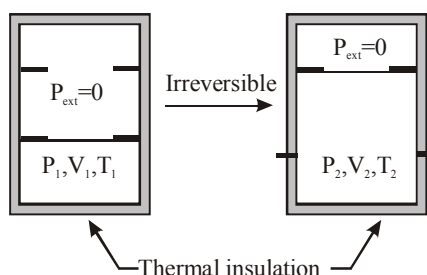
- (A) $T_1 = T_2$ (B) $T_3 > T_1$ (C) $w_{\text{isothermal}} > w_{\text{adiabatic}}$ (D) $\Delta U_{\text{isothermal}} > \Delta U_{\text{adiabatic}}$

Paragraph for Q.02 and 03

A fixed mass 'm' of a gas is subjected to transformation of states from K to L to M to N and back to K as shown in the figure. [JEE 2013]

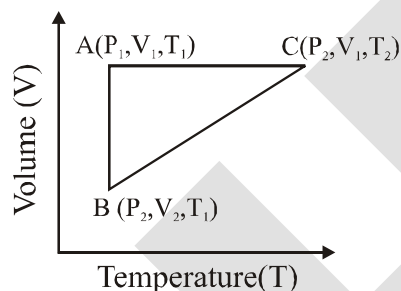


2. The pair of isochoric processes among the transformation of states is
 (A) K to L and L to M (B) L to M and N to K
 (C) L to M and M to N (D) M to N and N to K
3. The succeeding operations that enable this transformation of states are
 (A) Heating, cooling, heating, cooling (B) cooling, heating, cooling, heating
 (C) Heating, cooling, cooling, heating (D) Cooling, heating, heating, cooling
4. An ideal gas in thermally insulated vessel at internal pressure = P_1 , volume = V_1 and absolute temperature = T_1 expands irreversibly against zero external pressure, as shown in the diagram. The final internal pressure, volume and absolute temperature of the gas are P_2 , V_2 and T_2 , respectively. For this expansion, [JEE 2014]



- (A) $q = 0$ (B) $T_2 = T_1$ (C) $P_2 V_2 = P_1 V_1$ (D) $P_2 V_2^\gamma = P_1 V_1^\gamma$

5. An ideal gas is expanded from (p_1, V_1, T_1) to (p_2, V_2, T_2) under different conditions. The correct statement(s) among the following is(are) [JEE 2017]
- (A) The work done on the gas is maximum when it is compressed irreversibly from (p_2, V_2) to (p_1, V_1) against constant pressure p_1
- (B) The work done by the gas is less when it is expanded reversibly from V_1 to V_2 under adiabatic conditions as compared to that when expanded reversibly from V_1 to V_2 under isothermal conditions.
- (C) The change in internal energy of the gas (i) zero, if it is expanded reversibly with $T_1 = T_2$, and (ii) positive, if it is expanded reversibly under adiabatic conditions with $T_1 \neq T_2$
- (D) If the expansion is carried out freely, it is simultaneously both isothermal as well as adiabatic
6. A reversible cyclic process for an ideal gas is shown below. Here, P , V and T are pressure, volume and temperature, respectively. The thermodynamic parameters q , w , H and U are heat, work, enthalpy and internal energy, respectively. [JEE 2018]



The correct option(s) is (are)

- (A) $q_{AC} = \Delta U_{BC}$ and $w_{AB} = P_2 (V_2 - V_1)$ (B) $w_{BC} = P_2 (V_2 - V_1)$ and $q_{BC} = \Delta H_{AC}$
- (C) $\Delta H_{CA} < \Delta U_{CA}$ and $q_{AC} = \Delta U_{BC}$ (D) $q_{BC} = \Delta H_{AC}$ and $\Delta H_{CA} > \Delta U_{CA}$

ANSWER-KEY

EXERCISE # S-I

1. Ans. (a) $W = -P\Delta V$ (b) $W = -P\Delta V = -1(5 - 1) = -4$ litre-atm.
2. Ans. -3.1 kJ
3. Ans. (i) 9, 3, 2, 4 (ii) 9, 3, 3, 3 (iii) 3, 3, 0, 0 (iv) 12, 3, 3, 6
4. Ans. $q = -65$ J; $w = 20$ J; $\Delta U = -45$ J
5. Ans. (a) $+60$ J (b) -70 J (c), $+50$ J, $+10$ J
6. Ans. (13)
7. Ans. (140)
8. Ans. (i) 18.42 bar.L; (ii) 72 bar.L; (iii) 40 bar.L
9. Ans. (500 J)
10. Ans. $\Delta U = w = -1050$ R ; $\Delta H = -1470$ R , $q = 0$
11. Ans. 2244.78 Joule
12. Ans. $T_2 = 100$ K; $w = -5.016$ kJ
13. Ans. (7)
14. Ans. $q = 0$, $w = \Delta U = 4.125$ kJ ; $\Delta H = 5.372$ kJ ; $V_f = 11.8$ dm³ ; $P = 5.22$ atm
15. Ans. (a) $T_1 = 243.60$ K ; $T_2 = 2436$ K, (b) $\Delta U = 0$; $q = -w = 3264.24$ cal
16. Ans. (a) AC, (b) 170 J, (c) 10 J
17. Ans. $[-800$ cal]
18. Ans. (7)
19. Ans. (a) $\frac{11}{3}$ cal /K-mol. (b) 2566.67 cal (c) -466.67 cal (d) 2100 cal
20. Ans. (9)
21. Ans. -10.13 J
22. Ans. $\Delta E = 75.12$ kJ
23. Ans. $\Delta H \cong \Delta U = 1440$ calories

EXERCISE # S-II

1. Ans. (a) 600 K, (b) 3 kcal, 1.680 kcal, -1.8 kcal, -1.680 kcal (c) -1.2 kcal
2. Ans. ΔU & $\Delta H = 0$; $w = 623.55$ J ; $q = -623.55$ J
3. Ans. $w = -nRT \ln \frac{V_f}{V_i} - n^2 a \left(\frac{1}{V_f} - \frac{1}{V_i} \right)$
4. Ans. (i) $w = -8400$ cal, $q = 8400$ cal, $\Delta U = \Delta H = 0$
 (ii) $w = -2700$ cal, $q = 0$ cal, $\Delta U = -2700$, $\Delta H = -4500$ cal
 (iii) $w = 0$ cal, $q = 0$ cal, $\Delta U = 0$, $\Delta H = 0$
 (iv) $w = -1395$ cal, $q = 0$, $\Delta U = -1395$ cal, $\Delta H = -2325$ cal
 (v) $w = -2325$ cal, $q = 2325$ cal, $\Delta U = 0$, $\Delta H = 0$
5. Ans. $\Delta U = 501$ J ; $\Delta H = 99.5$ kJ
6. Ans. 100 KJ/mole
7. Ans. -1200 cal
8. Ans. 60 cal
9. Ans. (32)
10. Ans. (-900 calorioes)
11. Ans. 7.4 atm litre
12. Ans. 35 atm
13. Ans. 3120
14. Ans. (6)

EXERCISE # O-I

- | | | | |
|-------------|-------------|-------------|-------------|
| 1. Ans.(C) | 2. Ans.(B) | 3. Ans.(C) | 4. Ans.(D) |
| 5. Ans.(C) | 6. Ans.(C) | 7. Ans.(B) | 8. Ans.(C) |
| 9. Ans.(D) | 10. Ans.(B) | 11. Ans.(D) | 12. Ans.(B) |
| 13. Ans.(C) | 14. Ans.(C) | 15. Ans.(B) | 16. Ans.(C) |
| 17. Ans.(C) | 18. Ans.(A) | 19. Ans.(A) | 20. Ans.(D) |
| 21. Ans.(A) | 22. Ans.(A) | | |

EXERCISE # O-II

- | | | | |
|---|--------------|----------------|--------------|
| 1. Ans.(A) | 2. Ans.(C) | 3. Ans.(A,B,C) | 4. Ans.(C) |
| 5. Ans.(A,B,C,D) | 6. Ans.(B,D) | 7. Ans.(B,C) | 8. Ans.(B,D) |
| 9. Ans.(C) | 10. Ans.(D) | 11. Ans.(D) | 12. Ans.(A) |
| 13. Ans.(B) | 14. Ans.(B) | 15. Ans.(A) | 16. Ans.(C) |
| 17. Ans.(A)→(P,R,S) ; (B)→(Q,S) ; (C)→(Q,S) ; (D)→(S) | | | |
| 18. Ans.(A)→(P) ; (B)→(P,Q,R) ; (C)→(P,Q,R,S,T) ; (D)→(T) | | | |
| 19. Ans.(B) | 20. Ans.(A) | | |

EXERCISE # J-MAIN

- | | | | |
|------------|------------|------------|------------|
| 1. Ans.(1) | 2. Ans.(1) | 3. Ans.(2) | 4. Ans.(1) |
| 5. Ans.(3) | 6. Ans.(2) | 7. Ans.(2) | 8. Ans.(2) |

EXERCISE # J-ADVANCED

- | | | | |
|----------------|--------------|------------|----------------|
| 1. Ans.(A,D) | 2. Ans.(B) | 3. Ans.(C) | 4. Ans.(A,B,C) |
| 5. Ans.(A,B,D) | 6. Ans.(B,C) | | |

ALLERK
Session 2019-20

THERMODYNAMICS-02

1. LIMITATION OF FIRST LAW OF THERMODYNAMICS

The first law of thermodynamics can evaluate the value of ΔU and ΔH (i.e. energy difference between two states), but can not predict the feasibility of a process. Many processes can take place on their own and many others require an external force to carry out the process. The first law is clueless about why some processes happens on their own and why some others do not.

❖ Importance of Second law :

What is the driving force for a process which takes system from one state to other on it's own? Why such processes (*called spontaneous process*) take place at all? Answers to these questions are given by second law.

2. SPONTANEOUS PROCESS

A spontaneous process is the one which takes place on its own & can drive some other process as well. However, in some cases, initiation may be required.

Ex :

- (i) Mixing of two gases.
- (ii) Cooling of a hot object by transferring heat to surrounding.
- (iii) Water flowing down the hill.
- (iv) Rusting of iron.
- (v) Burning of coal, coke, hydrocarbons.
- (vi) Volcanic eruptions.

The synonyms of spontaneous processes are **natural processes** and irreversible processes.

It is a common observation that natural processes occur with finite rate. All spontaneous processes take place due to a finite driving force (which can be temperature difference, height difference, potential energy difference, etc.). Also the change brought about by a spontaneous process can not be restored without making some effort (which brings about permanent change in surrounding). Hence it is quite natural to call these processes as irreversible processes.

Ex :

When water flows from a dam, a part of the potential energy is converted into electrical energy and a part is converted into kinetic energy. The kinetic energy part is lost as heat energy. This loss of extra work is the permanent change in universe, because this heat energy lost cannot be completely converted back into P.E. of water.

3. Disorder :

The term 'disorder' is related to randomness. Any macroscopic property of a system is an average of large number of microscopic forces which are random in nature. Greater the randomness associated with microscopic forces, greater is the uncertainty about exact microscopic description of a macroscopic state and greater will be the disorder associated with the macroscopic state.

As a rule :

- (i) Greater the number of molecules in which a given energy is distributed, greater is the disorder.
- (ii) Greater the spread of given number of molecules with given energy in space, greater will be the disorder.

4. SECOND LAW OF THERMODYNAMICS

The entropy of the universe increases in course of every spontaneous process. The entropy of system is a state function. The change in entropy of a system is given by -

$$dS = \frac{dq_{rev}}{T} \quad \text{where ; } dq_{rev} = \text{heat exchanged by system in a very small step (reversibly).}$$

4.1 Entropy :

Entropy is a measure of disorder. The entropy is a state function and a basic quantity directly defined by second law. In every spontaneous process, entropy of universe increases.

There are three types of entropy changes :

- (i) ΔS_{System} : Entropy change of system
 - (ii) $\Delta S_{\text{Surrounding}}$: Entropy change of surrounding
 - (iii) ΔS_{Total} : Entropy change of universe
- $$\Delta S_{\text{system}} + \Delta S_{\text{surrounding}} = \Delta S_{\text{universe}} = \Delta S_{\text{total}} = \Delta S_{\text{isolated}}$$
- Normally, ΔS means entropy change of system.

4.2 Second law in terms of entropy of universe :

For any spontaneous process (irreversible process) -

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surrounding}} > 0$$

During a reversible process -

$$\Delta S_{\text{universe}} = 0$$

Thus during irreversible process, entropy of universe increases while during reversible process, entropy of universe remains constant.

4.3 Physical significance of quantity $\frac{dq_{rev}}{T}$:

dq_{rev} = heat absorbed by the system, T = absolute temperature at which heat is absorbed. Suppose, similar quantity of heat is absorbed at two different temperatures T_1 and T_2 ($T_2 \gg T_1$). When heat is supplied at high temperature, the % age increase in kinetic energy is less compared to heat supplied at low temperature. Hence, disorder will increase by greater value, when heat is supplied at lower temperature.

4.4 Determining ' $\Delta S_{\text{surr.}}$ ' for irreversible processes :

The change in entropy is given by : $\frac{\text{heat absorbed by an entity}}{\text{Exact temperature at which heat is absorbed}}$

The irreversible processes are fast and uncontrolled. The exact temperature of system during exchange of heat cannot be known. Hence, reversible path is needed to evaluate the entropy of system. Along reversible path, the heat exchange and exact temperature both are known.

Surrounding is generally a big entity and heat absorbed by surrounding is comparatively very small as compared to its size. Hence, internal equilibrium of surrounding is not disturbed during exchange of heat.

$\therefore$ Actual heat exchange of system is reversible heat exchange of surroundings

$\therefore (q_{\text{rev}})_{\text{surr}} = -q_{\text{irr}}$ [q_{irr} = Actual heat gained by system in a irreversible process.]

Hence, exact temperature and heat exchange both are known as far as surrounding is concerned even for an irreversible process.

$\therefore$ **For irreverisble process :** $\Delta S_{\text{surr}} = \frac{-q_{\text{irr}}}{T}$

	ΔS_{system}	$\Delta S_{\text{surrounding}}$
(i)	For reversible process : $\Delta S_{\text{system}} = \int_{T_1}^{T_2} \frac{dq_{\text{rev}}}{T}$	For reversible process : $\Delta S_{\text{surr}} = \int_{T_1}^{T_2} \frac{dq_{\text{rev}}}{T} = -\Delta S_{\text{system}}$
(ii)	For irreversible process : $\Delta S_{\text{system}} = \int_{T_1}^{T_2} \frac{dq_{\text{rev}}}{T}$ [Where ; dq_{rev} = heat gained by system for imaginary reversible path between same states]	For irreversible process : $\Delta S_{\text{surr}} = -\frac{q_{\text{irr}}}{T}$
(iii)	Always evaluated along reversible path (for both reversible and irreversible process)	For reversible process, ΔS_{surr} is evaluated along reversible path, while for irreversible process, irreversible heat is used to calculate ΔS_{surr} .
(iv)	Since no actual process is reversible, ΔS_{system} is always evaluated by hypothetical path.	For hypothetical process, ΔS_{surr} is evaluated by hypothetical heat, while for real irreversible process, it is evaluated by heat exchanged in irreversible process.

5. SECOND LAW (IN TERMS OF ENGINE)

- (i) No cyclic engine is possible whose only effect is complete conversion of heat into work without making any change in surrounding on its own.
- (ii) Heat can not flow from cold body to hot body on its own without external intervention.
- (iii) Complete conversion of heat (taken from a single source) into work is not possible in a cyclic process while complete conversion of work into heat is possible in a cyclic process. Thus, heat and work are not equivalent from the view point of II law.

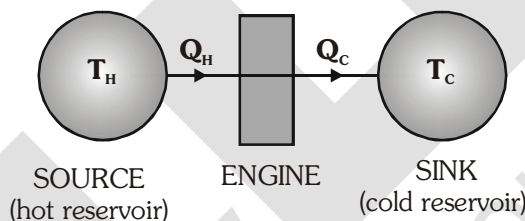
All above statements can be easily justified by realising that heat is a disordered form of energy while work is an ordered form of energy.

Complete conversion of [Heat $\longrightarrow$ Work] on its own without any external assistance means [Disorder $\longrightarrow$ Order] on its own - which is impossible.

5.1 Heat engine :

A heat engine is a device which can work continuously and cyclically without any external help absorbing heat from a source and rejecting a part of heat into sink. The net heat absorbed by engine (heat absorbed from source – heat given out to sink) is converted into work.

$\Rightarrow$ Heat engine must work in cycles, otherwise it can not work continuously.



Q_H = Heat gained by engine per cycle

Q_C = Heat rejected by engine per cycle

$$\text{Efficiency of engine : } \eta = \frac{Q_H - Q_C}{Q_H} = \frac{W_{by}}{Q_H}$$

W_{by} = Work done by engine per cycle.

During operation of heat engine, heat is transferred from source (high temperature reservoir) to sink (low temperature reservoir) thus degrading the quality of energy. Note that net energy of universe remains conserved, but its quality is degraded (i.e. more disordered energy) when heat engine runs spontaneously.

5.2 CARNOT ENGINE

A carnot engine (a hypothetical engine) runs in reversible manner. Thus during expansion (work producing process) maximum work is done. However during compression (work requiring process) minimum work need to be done. Thus no engine can be more efficient than carnot engine.

The carnot engine works in following steps. (Assume ideal gas as working material).

- (i) Reversible isothermal expansion $(P_1, V_1, T_2 \rightarrow P_2, V_2, T_2)$
- (ii) Reversible adiabatic expansion $(P_2, V_2, T_2 \rightarrow P_3, V_3, T_1)$
- (iii) Reversible isothermal compression $(P_3, V_3, T_1 \rightarrow P_4, V_4, T_1)$
- (iv) Reversible adiabatic compression $(P_3, V_3, T_1 \rightarrow P_1, V_4, T_2)$

Now: $Q_{NET} = -w_{NET}$

$$w_{NET} = w_{12} + w_{23} + w_{34} + w_{41}$$

$$w_{12} = -nRT_H \ln\left(\frac{V_2}{V_1}\right)$$

$$w_{23} = nC_V(T_C - T_H)$$

$$w_{34} = -nRT_C \ln\left(\frac{V_4}{V_3}\right)$$

$$w_{41} = -nC_V(T_H - T_C)$$

Also since process 2 – 3 and 4 – 1 are reversible adiabatic process.

$$T_H V_2^{\gamma-1} = T_C V_3^{\gamma-1}$$

$$T_C V_1^{\gamma-1} = T_H V_4^{\gamma-1}$$

from above equations

$$\left(\frac{V_2}{V_1}\right) = \left(\frac{V_3}{V_4}\right)$$

Substituting the value of $\left(\frac{V_4}{V_3}\right)$ in eq.(v) and adding w in individual step.

$$w_{NET} = -nRT_H \ln\left(\frac{V_2}{V_1}\right) + nC_V(T_C - T_H) - nRT_C \ln\left(\frac{V_4}{V_3}\right) - nC_V(T_C - T_H)$$

$$w_{NET} = -nRT_H \ln\left(\frac{V_2}{V_1}\right) - nRT_C \ln\left(\frac{V_1}{V_2}\right)$$

$$w_{NET} = -nR [T_C - T_H] \ln\left(\frac{V_2}{V_1}\right)$$

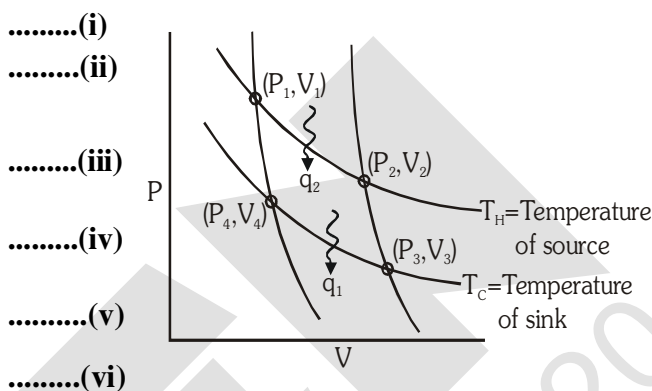
$$\Rightarrow \text{Efficiency } \eta = -\frac{w_{NET}}{q_2}$$

$$\eta = \frac{nR \ln\left(\frac{V_2}{V_1}\right) [T_H - T_C]}{nRT_H \ln\left(\frac{V_2}{V_1}\right)}$$

$$\therefore \eta = \frac{T_H - T_C}{T_H}$$

Thus efficiency of carnot engine depends only upon temperature of source and sink.

Note : If working substance is changed (eg. ideal gas by real gas) efficiency of carnot engine remains unaffected.



.....(vii) for process 2 – 3

.....(viii) for process 1 – 4

.....(ix)

EXERCISE # 1

1. A heat engine absorbs 760 kJ heat from a source at 380K. It rejects (i) 650 kJ, (ii) 560 kJ, (iii) 504 kJ of heat to sink at 280K. State which of these represent a reversible, an irreversible or an impossible cycle.
2. The maximum efficiency of a heat engine operating between 100°C and 25°C is :-
(A) 20.11% (B) 22.2% (C) 25.17% (D) None
3. A carnot engine operating between 227°C and 27°C absorbs 2 Kcal of heat from the 227°C reservoir reversibly per cycle. The amount of work done in one cycle is :-
(A) 0.4 Kcal (B) 0.8 Kcal (C) 4 Kcal (D) 8 Kcal
4. An ideal gas heat engine operates in carnot's cycle between 227°C and 127°C. It absorbs 6×10^4 cal of heat at high temperature. Amount of heat converted to work is :
(A) 2.4×10^4 cal (B) 4.8×10^4 cal (C) 1.2×10^4 cal (D) 6.0×10^4 cal

6. ENTROPY CHANGE OF AN IDEAL GAS

From definition of entropy of system -

$$dS_{\text{system}} = \frac{dq_{\text{rev}}}{T} \quad \text{.....(i)}$$

from first law, $dq_{\text{rev}} = dU + PdV$

For a reversible change involving an ideal gas -

$$dq_{\text{rev}} = nC_{V,m}dT + \frac{nRT}{V}dV \quad \text{.....(ii)}$$

$$\Rightarrow \int_1^2 \frac{dq_{\text{rev}}}{T} = \int_1^2 \frac{nC_{V,m}dT}{T} + \int_{V_1}^{V_2} nR \frac{dV}{V}$$

$$\Delta S_{\text{system}} = nC_{V,m} \ln \frac{T_2}{T_1} + nR \ln \left(\frac{V_2}{V_1} \right) = nC_{p,m} \ln \frac{T_2}{T_1} - nR \ln \frac{P_2}{P_1}$$

Note : This formula is applicable to all ideal gas process.

6.1 Ideal gas processes :

(i) For isothermal change :

$$\Delta S = \int_1^2 \frac{dq_{\text{rev}}}{T} = \frac{1}{T} \int_1^2 dq_{\text{rev}} \quad (\because T = \text{constant})$$

$$\Rightarrow \Delta S = \frac{q_{\text{rev}}}{T} = nR \ln \frac{V_2}{V_1} \quad \left[\because q_{\text{rev}} = nRT \ln \left(\frac{V_2}{V_1} \right) \right]$$

(ii) For isochoric change :

$$\Delta S = \int_1^2 \frac{dq_v}{T} = \int_1^2 \frac{nC_{V,m}dT}{T} = nC_{V,m} \int_1^2 \frac{dT}{T} \quad [\because dq_v = nC_{V,m}dT]$$

$$\Rightarrow \Delta S = nC_{V,m} \ln \frac{T_2}{T_1}$$

(iii) For isobaric change :

$$\Delta S = \int_1^2 \frac{dq_p}{T} = \int_1^2 \frac{nC_{p,m}dT}{T} = nC_{p,m} \int_1^2 \frac{dT}{T} \quad [\because dq_p = nC_{p,m}dT]$$

$$\Rightarrow \Delta S = nC_{p,m} \ln \left(\frac{T_2}{T_1} \right)$$

(iv) **Reversible adiabatic process :**

$$q_{\text{rev}} = 0$$

$$\Rightarrow \Delta S = \underbrace{nC_v \ln \frac{T_2}{T_1}}_{\text{entropy change due to change in temperature}} + \underbrace{nR \ln \frac{V_2}{V_1}}_{\text{entropy change due to change in volume}} = 0$$

(a) **During reversible adiabatic expansion :**

$$T_2 < T_1 \text{ and } V_2 > V_1$$

$\Rightarrow$ Sign of two terms are opposite and magnitudes are same because for reversible adiabatic process -

$$nC_{v,m} \ln \frac{T_2}{T_1} = -nR \ln \frac{V_2}{V_1}$$

$\Rightarrow$ +ive and -ive term cancel out each other and hence, $\Delta S = 0$

Thus, decrease in entropy during expansion due to decrease in temperature is exactly cancelled out by increase in entropy due to increase in volume of system in reversible adiabatic expansion.

(b) **During reversible adiabatic compression :**

$$T_2 > T_1 \text{ and } V_2 < V_1$$

$$\Rightarrow \Delta S_{\text{Total}} = \underbrace{nC_{v,m} \ln \frac{T_2}{T_1}}_{\text{+ive}} + \underbrace{nR \ln \frac{V_2}{V_1}}_{\text{? ive}} = 0$$

Thus, increase in entropy due to increase in temperature is compensated by decrease in entropy due to decrease in volume of system in reversible adiabatic compression.

Hence, $\Delta S = 0$ for reversible adiabatic compression.

(v) **Irreversible adiabatic process :**

Since process is irreversible, $\Delta S_{\text{Total}} > 0$.

$$\Rightarrow \Delta S_{\text{system}} + \Delta S_{\text{surr}} > 0$$

$$\therefore \Delta S_{\text{surr}} = 0$$

$$\therefore \Delta S_{\text{system}} > 0$$

The physical interpretation is -

(a) During irreversible adiabatic compression, large quantity of heat is produced while decrease in volume is less.

Thus, increase in entropy due to rise in temperature exceeds the decrease in entropy due to decrease in volume. Hence net entropy change is positive.

(b) During irreversible adiabatic expansion, due to sudden decrease in external pressure, the system remains less challenged during expansion. Hence system do less work-resulting in lesser decrease in temperature.

The increase in volume of system is relatively larger (because system was less challenged)

Thus increase in entropy due to increase in volume exceeds decrease in entropy due to decrease in temperature. The net change in entropy is positive in both processes $\Delta S_{\text{surr.}} = 0$ ($\because$ no heat is exchanged with surrounding) and $\Delta S_{\text{total}} > 0$.

(vi) Free expansion of an ideal gas against vacuum :

Free expansion of an ideal gas is an example of irreversible adiabatic as well as irreversible isothermal process.

During free expansion, the $P_{\text{ext.}} = 0$. Hence, the ideal gas is not challenged at all during expansion. The kinetic energy of ideal gas remains constant. Hence, no temperature difference is created during expansion between system and surrounding.

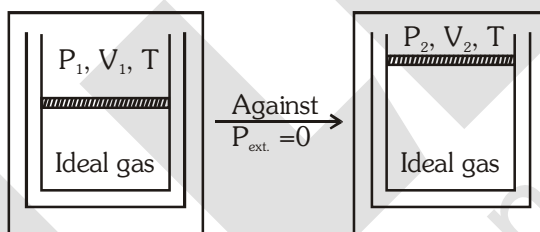
$\because \Delta T = 0$, the heat absorbed during the process $= 0$

Since, $V_f > V_i$ and $T = \text{constant}$, $\Delta S_{\text{system}} > 0$

The free expansion is an example where $q_{\text{actual}} = 0$, still $\Delta S_{\text{system}} > 0$.

This is reasonable, because : $\Delta S_{\text{system}} = \int \frac{dq_{\text{rev.}}}{T}$

This, example clearly shows that even if q_{actual} between two states is zero, this does not mean that $\int_1^2 \frac{dq_{\text{rev.}}}{T}$ between two states is zero. This outlines the importance of reversible path and $q_{\text{rev.}}$ in calculation of entropy of system.



$$q_{\text{irr}} = 0 ; w_{\text{irr}} = 0$$

$$\Delta S_{\text{system}} = \int_1^2 \frac{dq_{\text{rev.}}}{T} = nR \ln \frac{V_2}{V_1}$$

$$\Delta S_{\text{surr.}} = \frac{-q_{\text{irr}}}{T} = 0$$

$$\Delta S_{\text{total.}} = nR \ln \frac{V_2}{V_1} > 0$$

ENTROPY CALCULATION

Process	$\Delta S_{\text{Sys.}}$	$\Delta S_{\text{Surr.}}$
Isothermal reversible	$\Delta S_{\text{Sys.}} = nR \ln \frac{V_2}{V_1}$	$\Delta S_{\text{Surr.}} = -\Delta S_{\text{Sys.}}$
Isothermal irreversible	$\Delta S_{\text{Sys.}} = nR \ln \frac{V_2}{V_1}$	$\Delta S_{\text{Surr.}} = \frac{-q_{\text{sys}}}{T} = \frac{W_{\text{Sys}}}{T} = \frac{-P_{\text{ext}}(V_2 - V_1)}{T}$
Adiabatic reversible	$\Delta S_{\text{Sys.}} = 0$	$\Delta S_{\text{Surr.}} = 0$
Adiabatic irreversible	$\Delta S_{\text{Sys.}} = nC_{P,m} \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$	$\Delta S_{\text{Surr.}} = 0$
Isochoric reversible	$\Delta S_{\text{Sys.}} = nC_{V,m} \ln \frac{T_2}{T_1}$	$\Delta S_{\text{Surr.}} = -\Delta S_{\text{Sys.}}$
Isochoric irreversible	$\Delta S_{\text{Sys.}} = nC_{V,m} \ln \frac{T_2}{T_1}$	$\Delta S_{\text{Surr.}} = \frac{-q_{\text{sys}}}{T_{\text{surr}}} = \frac{-nC_{V,m} \Delta T}{T_{\text{final}}}$

Ex.1 Three moles of an ideal gas is expanded isothermally from 2L to 10L, at 27°C. Calculate ΔS_{sys} and ΔS_{surr} and $\Delta S_{\text{univ.}}$ if the process is preformed.

(i) Reversibly

(ii) Irreversibly, against a constant external pressure of 1 atm

(iii) As free expansion

Sol: (i) $\Delta S_{\text{univ.}} = 0$, $\Delta S_{\text{surr.}} = -\Delta S_{\text{sys.}}$

$$\text{and } \Delta S_{\text{sys}} = nR \ln \frac{V_2}{V_1} = 3R \cdot \ln \frac{10}{2} = 40.14 \text{ J/K}$$

(ii) $\Delta S_{\text{sys}} = 40.14 \text{ J/K}$

$$\Delta S_{\text{surr.}} = -\frac{P_{\text{ext.}}(V_2 - V_1)}{T} = -\frac{1 \times (10 - 2) \times 101.3}{300} = -2.7 \text{ J/K}$$

$$\text{and } \Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 37.44 \text{ J/K}$$

(iii) $\Delta S_{\text{sys}} = 40.14 \text{ J/K}$

$$\Delta S_{\text{surr.}} = 0 \quad \text{as } P_{\text{ext}} = 0$$

$$\text{and } \Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 40.14 \text{ J/K}$$

Ex.2. Two moles of an ideal monoatomic gas is expanded adiabatically from 10L, 27°C to 80L. Calculate ΔS_{sys} , ΔS_{surr} and ΔS_{univ} , if the process is performed.

(i) Reversibly

(ii) Irreversibly, against a constant external pressure of 0.1 atm.

(iii) As free expansion

Sol.: $\Delta S_{surr} = 0$ for all

(i) $\Delta S_{sys} = \Delta S_{surr} = \Delta S_{univ} = 0$

(ii) $n \cdot C_{v,m} \cdot (T_2 - T_1) = -P_{ext}(V_2 - V_1)$

or $2 \times \frac{3}{2} \times 0.0821 \times (T_2 - 300) = -0.1 \times (80 - 10) \Rightarrow T_2 = 271.58 \text{ K}$

Now, $\Delta S_{sys} = n \cdot C_{v,m} \cdot \ln \frac{T_2}{T_1} + nR \cdot \ln \frac{V_2}{V_1}$
 $= 2 \times \frac{3}{2} R \times \ln \frac{271.58}{300} + 2 \times R \times \ln \frac{80}{10} = 32.90 \text{ J/K}$

and $\Delta S_{univ} = \Delta S_{sys}$

(iii) $T_1 = T_2 \Rightarrow \Delta S_{sys} = nR \cdot \ln \frac{V_2}{V_1} = 2 \times R \times \ln \frac{80}{10} = 34.58 \text{ J/K}$

6.2 Entropy change of solids and liquids upon heating :

$$\Delta S = \int \frac{dq_{rev}}{T} = \int \frac{mS \cdot dT}{T} = mS \cdot \ln \frac{T_2}{T_1} \quad [S = \text{specific heat of solid/ liquid}]$$

For molar change in entropy, $m = M(\text{g/mole})$; where $M = \text{molar mass}$.

Note, for solids and liquids ($MS \cong C_p \cong C_v$)

Ex.3 Two blocks of equal masses and heat capacity but at different temperatures are taken in an adiabatic vessel. Show that total change in entropy when both attain thermal equilibrium is always positive.

Sol. Let the temperature of hot body be T_H , temperature of cold body be T_C & the final temperature be T_F .

Now, net heat exchanged by two bodies must be zero.

$$\Rightarrow m \cdot S (T_F - T_H) + mS (T_F - T_C) = 0$$

$$\Rightarrow T_F = \frac{T_H + T_C}{2}$$

Since both bodies are of finite mass, entropy of both bodies are evaluated by reversible paths.

$$\Delta S_{Hot} = \int_{T_H}^{T_F} mS \cdot \frac{dT}{T} = mS \cdot \ln \frac{T_F}{T_H}$$

$$\Delta S_{Cold} = \int_{T_C}^{T_F} mS \cdot \frac{dT}{T} = mS \cdot \ln \frac{T_F}{T_C}$$

$$\Rightarrow \Delta S_{\text{total}} = (mS) \ln \left(\frac{T_F^2}{T_H T_C} \right)$$

$$\Rightarrow T_H^2 + T_C^2 + 2T_H T_C - 4T_H T_C = (T_H - T_C)^2 > 0$$

$$\Rightarrow (T_H + T_C)^2 > 4T_H T_C$$

$$\Rightarrow \Delta S_{\text{total}} > 0$$

6.3 Reversible phase transitions :

Reversible phase transitions are always isothermal & isobaric.

$$\Delta S_{\text{system, P,T}} = \frac{q_{\text{rev.}}}{T} = \frac{\Delta H_{P,T}}{T_{\text{Transition}}}$$

$$\Delta S_{\text{surr}} = \frac{-\Delta H_{P,T}}{T_{\text{Transition}}}$$

$$\Delta S_{\text{total}} = 0$$

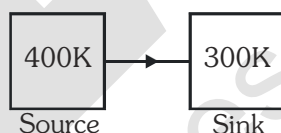
Thus, $\Delta S_{\text{vaporisation}}$ at boiling point = $\frac{\Delta H_{\text{vaporisation}}}{T_{\text{vaporisation}}}$

$$\Delta S_{\text{fusion}}$$
 at melting point = $\frac{\Delta H_{\text{fusion}}}{T_{\text{fusion}}}$

$$\Delta S_{\text{sublimation}}$$
 at sublimation point = $\frac{\Delta H_{\text{sublimation}}}{T_{\text{sublimation}}}$

$$\Delta S_{\text{allotropic modification}} = \frac{\Delta H_{\text{allotropic modification}}}{T_{\text{transition}}}$$

Ex.4 100 kJ heat is transferred from a larger heat reservoir at 400 K to another large heat reservoir at 300 K. Suppose there is no change in temperature due to exchange of heat :



Find ΔS_{source} , ΔS_{sink} and ΔS_{total} . Comment on spontaneity of process.

Ans. (i) $\Delta S_{\text{source}} = \frac{\text{heat absorbed by source}}{\text{Temperature of source}}$

$$\Delta S_{\text{source}} = \frac{-100\text{kJ}}{400} = -250\text{J/K}$$

(ii) $\Delta S_{\text{sink}} = \frac{\text{heat absorbed by sink}}{\text{Temperature of sink}}$

$$= \frac{+100000\text{J}}{300} = 333.33\text{J/K}$$

(iii) $\Delta S_{\text{Total}} = \Delta S_{\text{source}} + \Delta S_{\text{sink}} = +83.33\text{J/K}$

Ex.5 Trouton's rule states that the ratio of the molar heat of vaporisation of a liquid to its normal

boiling point is approximately the same for all liquids, $\frac{\Delta H_{\text{vap}}}{T_{\text{bp}}} = 88 \text{ JK}^{-1} \text{ mol}^{-1} = \Delta S_{\text{vap}}$.

Which of the following liquids do not follow Trouton's rule : NH_3 , C_6H_6 , CCl_4 , CHCl_3 , Hg . Explain.

Ans. NH_3 , due to H-bonding : $\Delta S_{\text{vap}} > 88 \text{ J/K-mol}$

EXERCISE # 2

5. Identify the correct statement regarding a spontaneous process :- [AIEEE-2007]
 (A) For a spontaneous process in an isolated system, the change in entropy is positive
 (B) Endothermic processes are never spontaneous
 (C) Exothermic processes are always spontaneous
 (D) Lowering of energy in the reaction process is the only criterion for spontaneity
6. Find the change in entropy (in cal/k) of 1 mole of O_2 gas ($C_V = 5/2 R$), when it is
 (a) heated from 300 K to 400 K isobarically
 (b) heated from 300 K to 400 K isochorically (Given : $\ln 3 = 1.1$, $\ln 2 = 0.7$)
7. One mole of NaCl(s) on melting at its melting point absorbed 32.76 kJ of heat and its entropy is increased by 30 JK^{-1} . What is the melting point of sodium chloride ?
8. The entropy change when two moles of ideal monoatomic gas is heated from 200°C to 300°C reversibly and isochorically :-
 (A) $\frac{3}{2}R \ln\left(\frac{300}{200}\right)$ (B) $\frac{5}{2}R \ln\left(\frac{573}{273}\right)$ (C) $3R \ln\left(\frac{573}{473}\right)$ (D) $\frac{3}{2}R \ln\left(\frac{573}{473}\right)$
9. Predict the sign entropy change for the following processes:
 (A) $\text{O}_2(\text{g}) \rightarrow 2\text{O}(\text{g})$ (B) $2\text{O}_3(\text{g}) \rightarrow 3\text{O}_2(\text{g})$
 (C) $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ (D) $\text{NaCl(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
 (E) $\text{C}_2\text{H}_5\text{OH(l)} \rightarrow \text{C}_2\text{H}_5\text{OH(g)}$ (F) $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl(s)}$
10. 5 mole of an ideal gas expands reversibly from a volume of 8 dm^3 to 80 dm^3 at a constant temperature of 27°C . The change in entropy is :-
 (A) 41.57 JK^{-1} (B) -95.73 JK^{-1} (C) 95.73 JK^{-1} (D) -41.57 JK^{-1}
11. ΔS for the reaction ; $\text{MgCO}_3(\text{s}) \longrightarrow \text{MgO(s)} + \text{CO}_2(\text{g})$ will be :
 (A) 0 (B) -ve (C) +ve (D) ∞
12. Change in entropy is negative for :-
 (A) Bromine (ℓ) $\longrightarrow$ Bromine (g)
 (B) $\text{C(s)} + \text{H}_2\text{O(g)} \longrightarrow \text{CO(g)} + \text{H}_2(\text{g})$
 (C) $\text{N}_2(\text{g}, 10 \text{ atm}, 298 \text{ K}) \longrightarrow \text{N}_2(\text{g}, 1 \text{ atm}, 298 \text{ K})$
 (D) $\text{Fe(at } 400 \text{ K)} \longrightarrow \text{Fe(at } 300 \text{ K)}$
13. For which reaction from the following, ΔS will be maximum ?
 (A) $\text{Ca(s)} + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{CaO(s)}$ (B) $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO(s)} + \text{CO}_2(\text{g})$
 (C) $\text{C(s)} + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$ (D) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO(g)}$
14. When two gases are mixed the entropy :-
 (A) Remains constant (B) Decreases (C) Increases (D) Becomes zero

15. For the process, $\text{CO}_2(\text{s}) \longrightarrow \text{CO}_2(\text{g})$:

- (A) Both ΔH and ΔS are +ve (B) ΔH is -ve and ΔS is +ve
(C) ΔH is +ve and ΔS is -ve (D) Both ΔH and ΔS are -ve

7. THIRD LAW OF THERMODYNAMICS

“At absolute zero, the entropy of a perfectly crystalline substance is zero”, which means that at absolute zero, every crystalline solid is in a state of perfect order and its entropy should be zero. By virtue of the third law, the absolute value of entropy (unlike absolute value of enthalpy) for any pure substance can be calculated at room temperature.

$$S_T - S_{0K} = \int_0^T \frac{q_{\text{rev}}}{T}$$

Since $S_{0K} = 0$

$$S_T = \int_0^T \frac{q_{\text{rev}}}{T}$$

The variation of entropy of a substance with temperature is given by graph :

At 1, entropy of substance is zero.

- from 1-2 ; The temperature of solid increases upon heating -

$$\Rightarrow S_2 - S_1 = \int \frac{C_{P,\text{solid}}}{T} dT \quad \text{.....(i)}$$

- from 2-3 ; The temperature remains constant during phase transition

$$\Rightarrow S_3 - S_2 = \Delta S_{\text{fusion}} = \frac{\Delta H_{\text{fusion}}}{T_f} \quad \text{.....(ii)}$$

- from 3-4 ; The temperature of liquid increases

$$\Rightarrow S_4 - S_3 = \int_{T_3}^{T_4} \frac{C_{P,\text{liquid}}}{T} dT \quad \text{.....(iii)}$$

- from 4-5 ; Reversible vaporisation takes place at constant temperature -

$$\Rightarrow S_5 - S_4 = \Delta S_{\text{vap.}} = \frac{\Delta H_{\text{vap.}}}{T_b} \quad \text{.....(iv)}$$

- from 5-6 ; The gas is heated -

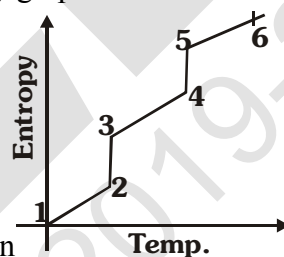
$$\Rightarrow S_6 - S_5 = \int_{T_5}^{T_6} \frac{C_{P,\text{gas}}}{T} dT \quad \text{.....(v)}$$

$$\Delta S_{1-6} = (S_2 - S_1) + (S_3 - S_2) + (S_4 - S_3) + (S_5 - S_4) + (S_6 - S_5) = S_6 - S_1 \quad [\because S_1 = 0]$$

$\therefore S_6$ = absolute entropy at 6.

$$= \int_{T_1}^{T_2} \frac{C_{P,\text{Solid}} dT}{T} + \frac{\Delta H_{\text{fusion}}}{T_f} + \int_{T_3}^{T_4} \frac{C_{P,\text{liquid}} dT}{T} + \frac{\Delta H_{\text{vap}}}{T_b} + \int_{T_5}^{T_6} \frac{C_{P,\text{gas}} dT}{T}$$

- Comment :** Absolute entropy of a substance can be calculated using third law by reversible path.



Change in entropy of a pure substance with temperature at constant pressure.

7.1 Standard absolute molar entropy :

It is the entropy in one mole of substance at standard temperature at 1 bar.

It is calculated assuming that substance is heated isobarically and reversibly at 1 bar from 0 K to standard temperature.

7.2 Comparing absolute entropies of substance :

(i) The absolute entropies of substance follow the order,

$$S(g) \gg S(l) > S(s)$$

(ii) Entropy increases when solid and liquid is dissolved into solvent,

e.g.	CH ₃ OH(l)	S° = 127 J/K-mole
	CH ₃ OH(aq)	S° = 132.3 J/K-mole
	NaCl(s)	S° = 72.4 J/K-mole
	NaCl(aq.)	S° = 115.4 J/K-mole

(iii) The entropy of dissolved gas is less than pure gas.

e.g.	HCl(g)	S° = 186.7 J/K-mole
	HCl(aq)	S° = 55.2 J/K-mole

(iv) Entropy rises with increasing mass, (if other factors are similar).

e.g.	F ₂ (g)	S° = 203 J/K-mole
	Cl ₂ (g)	S° = 223 J/K-mole
	Br ₂ (g)	S° = 245 J/K-mole

(v) Entropy is lower in covalently bonded solids, with strong directional bonds, than in solids with partial metallic character.

e.g.	C(diamond)	S° = 2.44 J/K-mole
	C(graphite)	S° = 5.69 J/K-mole
	Sn(gray)	S° = 44.8 J/K-mole
	Sn(white)	S° = 51.5 J/K-mole

(vi) Entropy increases with increasing softness and with weakness of bonds between atoms.

e.g.	C(diamond)	Be(s)	SiO ₂ (s)	Pb(s)	Hg(l)
	2.44	9.54	41.8	64.9	77.4
	diamond	hard metal	quartz	soft metal	liquid

(vii) Entropy increases with chemical complexity

Sub.	NaCl	MgCl ₂	AlCl ₃
	72.4	89.5	167

For CuSO₄.nH₂O

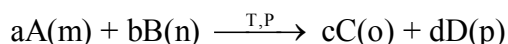
n = 0	n = 1	n = 3	n = 5
S° = 113	150	225	305

Note : Above data is given in SI unit.

7.3 Entropy change ($\Delta_r S$) of a chemical reaction :

The change in entropy when reactants turn into products according to a balanced chemical reaction with specified physical state of each reactant and product is called entropy of reaction.

• **Consider a reaction :**



where m, n, o, p are physical states at temperature T and pressure P.

$$\Delta_r S = \sum S(\text{products}) - \sum S(\text{reactants})$$

$$\Delta_r S = cS_C + dS_D - aS_A - bS_B$$

S_A, S_B, S_C and S_D are absolute molar entropies at temperature T and pressure P (from third law)

So, $\Delta_r S$ is evaluated from table of data of third law entropies of various substances.

7.4 $\Delta S_{\text{Surrounding}}$ for chemical reaction :

$\Delta_r H = q_p$ = heat absorbed by the system during chemical reaction.

$\Rightarrow -\Delta_r H = -q_p = q_{\text{surr}}$ = heat absorbed by the surrounding.

Now,

$$\Delta S_{\text{Surr}} = \frac{q_{\text{surr}}}{T} = -\frac{\Delta_r H}{T}$$

If $\Delta_r H = -\text{ive}$ (exothermic) heat is lost by the system and gained by the surrounding. Hence, ΔS_{surr} increases.

If $\Delta_r H = +\text{ive}$ (endothermic process), entropy of surrounding decreases.

$$\therefore \Delta S_{\text{surr}} = -\frac{\Delta_r H}{T} = -\text{ive}$$

- We can not comment on entropy change of system from knowledge of $\Delta_r H$. The entropy change of system ($\Delta_r S$) is estimated by third law.

7.5 Driving force of a spontaneous chemical process :

The driving force of a spontaneous chemical process is increase in entropy of universe.

In simple terms, negative value of $\Delta_r H$ and positive value of $\Delta_r S$ favours the chemical process.

If $\Delta_r H$ is positive, the enthalpy factor don't support the forward process. Under these conditions, $\Delta_r S$ should be sufficiently positive to drive the reaction in forward direction. If $\Delta_r H$ is +ive and $\Delta_r S$ is -ive, under a given situation, the process become non-spontaneous because both factors tends to decrease the entropy of the universe.

7.6 Prediction the sign of $\Delta_r S$ by inspecting a balanced chemical reaction.

(i) If more no. of gaseous moles are present on product side, $\Delta_r S$ will be +ive (since gas is more disordered than solid or liquid).

(ii) If

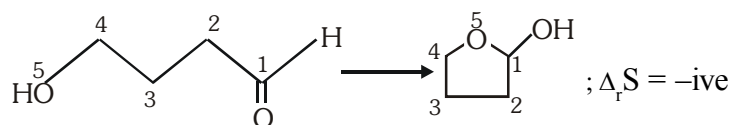
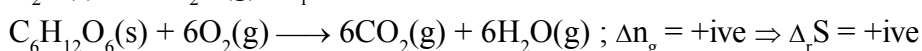
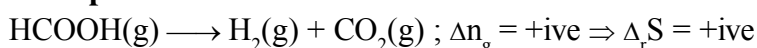
solid	$\longrightarrow$	liquid
solid	$\longrightarrow$	gas
liquid	$\longrightarrow$	gas

then, $\Delta_r S = +\text{ive}$.

(iii) If a molecule is undergoing cyclisation, it's rotational modes of motion are replaced by vibrational modes of motion.

Since quantum energy separation of rotational motion is closely spaced compared to vibrational energy state, rotational energy is more disordered. Hence, $\Delta_r S = -\text{ive}$.

Example :



7.7 Table for spontaneity of chemical process and sign of $\Delta_r H$, $\Delta_r S$ and dependence on temperature.

Sign of ΔH	Sign of ΔS	Comment	Example	ΔH_{298}°	ΔS_{298}°
–	+	Spontaneous at all temperature	$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$ $C(s) + O_2(g) \rightarrow CO_2(g)$	–185 –394	14.1 3
–	–	Spontaneous at low temperature	$H_2(g) + 1/2 O_2(g) \rightarrow H_2O(l)$ $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	–44 –198	–119 –187
+	+	Spontaneous at high temperature	$NH_4Cl(s) \rightarrow NH_3(g) + HCl(g)$ $N_2(g) + O_2(g) \rightarrow 2NO(g)$	176 180	284 25
+	–	Non spontaneous at all temperature	$3O_2 \rightarrow 2O_3$ $2H_2O(l) + O_2(g) \rightarrow 2H_2O_2(l)$	286 196	–137 –126

7.8 State of equilibrium :

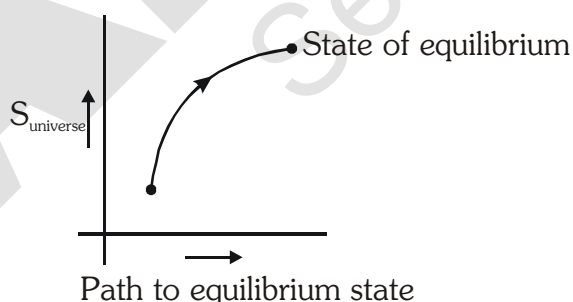
A system at equilibrium is a stable system. Hence, its state variables become constant (do not vary with time) at equilibrium.

If a system is in non-equilibrium state, it moves spontaneously in the direction of equilibrium (because the equilibrium state is stable). During the spontaneous movement towards equilibrium, the entropy of universe increases.

As the system gets closer to equilibrium state, the driving force to attain equilibrium becomes small and so does the increase in disorder of universe. At equilibrium, the driving force to move ahead finishes and entropy of the universe can not increase further.

In nutshell, when a system attains equilibrium, its capacity to increase the disorder of universe is lost.

- At equilibrium, the entropy of universe is maximum.



8. GIBB'S FUNCTION

Gibb's function 'G' is defined as $G = H - TS$.

Since 'G' is a combination of state variables (a compound state variable), it is a state function. The dimension of Gibb's function is the dimension of energy. 'G' is an extensive quantity.

Gibb's function provide a very useful criteria for spontaneity of a process taking place at constant temperature and pressure (isobaric, isothermal process).

The entropy of universe is also a very useful criteria of spontaneity, but it is based on properties of system as well as surrounding.

The Gibb's function provides criteria for spontaneity based only on the properties of system.

$$\Delta G = \Delta H - \Delta(TS)$$

$$\Delta G = \Delta H - T\Delta S$$

(Isothermal process)

Ex.6 The thermodynamic stability of a substance is dependent upon the value of Gibb's function. Explain the stability of solid at low temperature and stability of gas at very high temperature.

Sol. Any stable state tries to minimise it's enthalpy while maximising its entropy.

For solids - $G_{\text{solid}} = H_{\text{solid}} - TS_{\text{solid}}$ (at certain T & P)

For liquids - $G_{\text{liquid}} = H_{\text{liquid}} - TS_{\text{liquid}}$ (at certain T & P)

For gas - $G_{\text{gas}} = H_{\text{gas}} - TS_{\text{gas}}$ (at certain T & P)

At any given T and P, $H_{\text{solid}} < H_{\text{liquid}} < H_{\text{gas}}$

and $S_{\text{solid}} < S_{\text{liquid}} < S_{\text{gas}}$

At very low temperature, TS term contributes very little towards Gibb's function. Hence -

$$G_{\text{solid}} < G_{\text{liquid}} < G_{\text{gas}} \quad (\text{at very low temperature}).$$

This explains why solids are most stable at low temperature.

At very high temperature, TS term is very important. Since S_{gas} is large, for gas TS term becomes very large at large temperature. Hence -

$\Rightarrow G_{\text{solid}} > G_{\text{liquid}} > G_{\text{gas}}$ (at very high temperature). Thus, gaseous state is most stable at high temperature.

Hence, the low enthalpies of solid are responsible for stability of solid at low temperature, while high entropy of gas is responsible for stability of gas at high temperature.

8.1 $\Delta G_{T,P}$ and $\Delta S_{\text{universe}}$

Decrease in Gibb's function at constant temperature and pressure is related to ΔS_{total} (total entropy change of system and surrounding).

$$-\Delta G = T(\Delta S_{\text{total}})$$

At constant P, T

if $\Delta G < 0 \Rightarrow$ spontaneous process

if $\Delta G > 0 \Rightarrow$ non-spontaneous process

if $\Delta G = 0 \Rightarrow$ reversible process

At constant P, T, Gibbs energy of a system spontaneously decreases.

EXERCISE # 3

16. Animals operate under conditions of constant pressure and most of the processes that maintain life are electrical (in a broad sense). How much energy is available for sustaining this type of muscular and nervous activity from the combustion of 1 mol of glucose molecules under standard conditions at 37°C (blood temperature) ? The entropy change is + 210 JK⁻¹ for the reaction as stated.

$$\Delta H_{\text{combustion [glucose]}} = -2808 \text{ kJ / mol.}$$

17. Which of the following pairs of a chemical reaction is certain to result in a spontaneous reaction?
 (A) endothermic and decreasing disorder (B) exothermic and increasing disorder
 (C) endothermic and increasing disorder (D) exothermic and decreasing disorder
18. Standard entropy of X₂, Y₂ and XY₃ are 60, 40 and 50 JK⁻¹mol⁻¹, respectively. for the reaction,
 $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, $\Delta H = -30 \text{ kJ}$ to be at equilibrium, the temperature will be :-
 (A) 1250 K (B) 500 K (C) 750 K (D) 1000 K
19. For a particular reversible reaction at temperature T, ΔH and ΔS were found to be both +ve. If T_e is the temperature at equilibrium, then reaction would be spontaneous when :-
[AIEEE-2010]
 (A) T = T_e (B) T_e > T (C) T > T_e (D) T_e is 5 times T
20. For a reaction at 25°C, enthalpy change (ΔH) and entropy change (ΔS) are $-11.7 \times 10^3 \text{ J mol}^{-1}$ and $-105 \text{ J mol}^{-1} \text{ K}^{-1}$ respectively. The reaction is :
 (A) Spontaneous (B) Non spontaneous (C) At equilibrium (D) Can't say anything
21. Identify the correct statement for change of Gibbs energy for a system at constant temperature and pressure.
 (A) If $\Delta G_{\text{system}} > 0$, the process is spontaneous.
 (B) If $\Delta G_{\text{system}} = 0$, the system has attained equilibrium.
 (C) If $\Delta G_{\text{system}} = 0$, the system is still moving in a particular direction.
 (D) If $\Delta G_{\text{system}} < 0$, the process is non spontaneous.
22. In conversion of lime-stone to lime, $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
 the values of ΔH° and ΔS° are +179.1 kJ mol⁻¹ and 160.2 J/K respectively at 298 K and 1 bar. Assuming that ΔH° and ΔS° do not change with temperature, temperature above which conversion of limestone to lime will be spontaneous is :-
[AIEEE-2007]
 (A) 1008 K (B) 1200 K (C) 845 K (D) 1118 K

8.2 CALCULATE ΔG FOR VARIOUS PROCESSES

- (i) **Ideal gas subjected to isothermal compression or expansion :**

For isothermal process :

$$\Delta G = \Delta H - T\Delta S$$

For ideal gas, $\Delta H = nC_{p,m}\Delta T = 0$

(As $\Delta T = 0$, for isothermal change)

$$\& \quad \Delta S = nR \ln \frac{V_2}{V_1}$$

(for isothermal change)

$$\therefore \quad \Delta G = -nRT \ln \frac{V_2}{V_1}$$

Hence, during isothermal expansion Gibb's function decreases while during isothermal compression Gibb's function increases.

(ii) **Isothermal and isobaric phase transitions :**

At transition temperature, the stability (or Gibb's functions) of two phases are equal. Hence, there is no driving force for conversion of one phase into other. This means phase transitions are examples of reversible processes at transition temperature.

Now, for reversible, isothermal and isobaric change, $\Delta G = 0$

$\Rightarrow \Delta G = 0$, for phase transitions at transition temperature.

Thus, $\Delta G_{\text{vap}} = 0$ at boiling point :

$$\Rightarrow \Delta H_{\text{vap}} = T_b \Delta S_{\text{vap}}$$

$$\Rightarrow T_b = \frac{\Delta H_{\text{vap}}}{\Delta S_{\text{vap}}}$$

Also, $\Delta G_{\text{fusion}} = 0$ at melting point :

$$\Rightarrow \Delta H_{\text{fusion}} = T_f \Delta S_{\text{fusion}}$$

$$T_f = \frac{\Delta H_{\text{fusion}}}{\Delta S_{\text{fusion}}}$$

Ex.7 The enthalpy of vaporization of a liquid is 30 kJ mol^{-1} and entropy of vaporization is $75 \text{ J mol}^{-1} \text{ K}^{-1}$. The boiling point of the liquid at 1 atm is :

- (A) 250 K (B) 400 K (C) 450 K (D) 600 K

Ans. [B]

Sol. $T = \frac{\Delta H_{\text{vap}}}{\Delta S_{\text{vap}}} = \frac{30 \times 10^3}{75} = 400 \text{ K}$

Ex.8 At what minimum temperature would a given reaction become spontaneous if $\Delta H = +119 \text{ kJ}$ and $\Delta S = +263 \text{ J/K}$?

- (A) 452 K (B) 2210 K (C) 382 K (D) 2.21 K

Ans. [A]

Sol. $T = \frac{\Delta H}{\Delta S} = \frac{119 \times 10^3}{263} = 452.47 \text{ K}$

iii. $\Delta_r G$ for a chemical reaction.



The change in Gibb's function for a chemical reaction is change in ΔG when reactants react according to a given balanced chemical reaction with specified physical state of each reactant and product at given temperature and pressure.

$$\Delta_r G = cG_C + dG_D - aG_A - bG_B \quad \text{.....(i)}$$

Where G_A , G_B , G_C and G_D are molar Gibb's function of A, B, C and D respectively at given T and P.

also, $G_A = H_A - TS_A$

$$\left. \begin{aligned} G_B &= H_B - TS_B \\ G_C &= H_C - TS_C \\ G_D &= H_D - TS_D \end{aligned} \right\} \dots\dots(ii)$$

Substituting (ii) in (i)

$$\Delta_r G = (cH_C + dH_D - aH_A - bH_B) - T(cS_C + dS_D - aS_A - bS_B)$$

$$\Rightarrow \boxed{\Delta_r G = \Delta_r H - T\Delta_r S}$$

$(\Delta H_r)_{T,P}$	$(\Delta S_r)_{T,P}$	(ΔG_r)	Remarks
- ve	+ ve	Always -ve	Reaction is spontaneous
+ ve	- ve	Always +ve	Reaction non-spontaneous
+ ve	+ ve	At low temperature, $\Delta G = +ve$	Non-spontaneous
		At high temperature, $\Delta G = -ve$	Spontaneous
- ve	- ve	At low temperature, - ve	Spontaneous
- ve	- ve	At high temperature, + ve	Non spontaneous

(iv) **Gibb's functions of pure substances :**

Consider a pure substance having a Gibb's function as follows :

$$G = H - TS = U + PV - TS$$

Let, the pressure and temperature are varied by infinitesimally small value :

$$dG = dU + PdV + VdP - TdS - SdT$$

As, $dU = dq - PdV$ and $TdS = dq$ [Assume only PV- work and reversible process.]

$$dG = VdP - SdT$$

(a) If pressure is varied keeping temperature constant -

$$dG = VdP \Rightarrow \left(\frac{dG}{dP} \right)_T = V$$

The physical significance of above relation is when pressure over a substance is increased, it's Gibb's function increases & the increase is proportional to molar volume. Since molar volume of gas, liquid and solid are in order, as follows :

$$V_{\text{Gas}} > V_{\text{Liquid}} > V_{\text{Solid}}$$

Hence, Gibb's function of gases increases most rapidly on increasing pressure.

That's why, gas phase becomes unstable at high pressure.

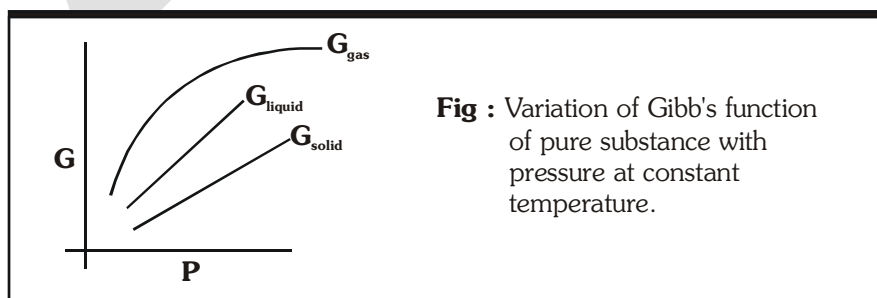


Fig : Variation of Gibb's function of pure substance with pressure at constant temperature.

(b) If temperature is changed keeping pressure constant -

$$dG = -SdT \Rightarrow \left(\frac{dG}{dT} \right)_p = -S$$

Since entropy of pure substances are always positive, the slope of G vs T curve is negative. Also since,

$$S_{\text{gas}} > S_{\text{liquid}} > S_{\text{solid}}$$

$\Rightarrow$ the slope is more negative for gas.

Thus, on increasing temperature, the Gibb's function of pure substances decreases, but this decrement is maximum for gaseous phase. Thus, gas phase acquire stability on increasing temperature.

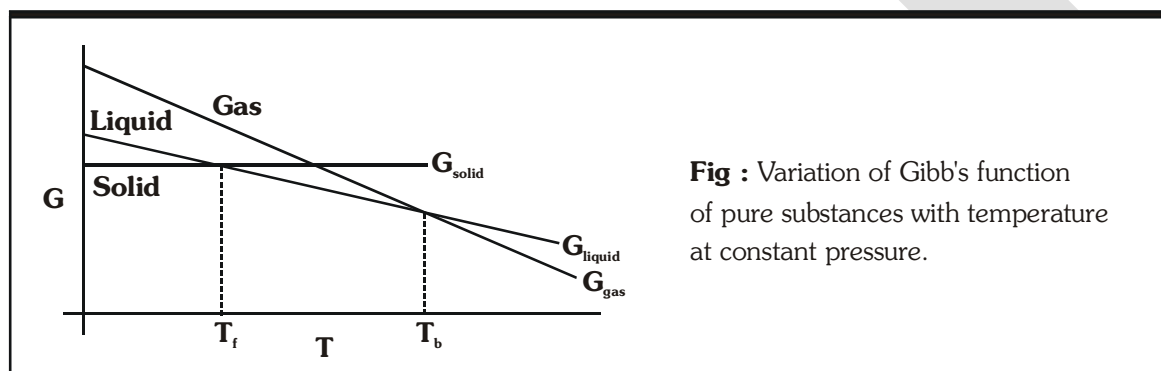
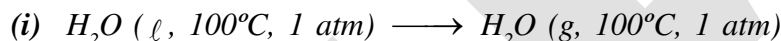
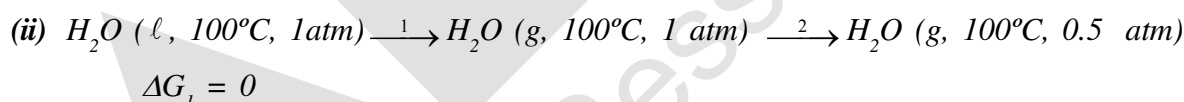


Fig : Variation of Gibb's function of pure substances with temperature at constant pressure.

Ex. 9. Calculate ΔG_m for the following process:



Sol. : (i) $\Delta G = 0$ (Equilibrium condition)



$$\Delta G_2 = nRT \cdot \ln \frac{P_2}{P_1} = 1 \times R \times 373 \times \ln \frac{0.5}{1} = -2149.5 \text{ J}$$

$$\therefore \Delta G = \Delta G_1 + \Delta G_2 = -2149.5 \text{ J}$$

8.3 APPLICATION OF GIBB'S ENERGY (AS MEASURE OF NON-PV WORK)

For a pure substance

$$dG = d(H - TS) = dH - TdS - SdT$$

also $dH = dU + PdV + VdP$

also $dU = dq - PdV - w_{\text{non-PV}}$

($w_{\text{non-PV}}$ = non-PV work done by system)

$\Rightarrow dG = dq - PdV - w_{\text{non-PV}} + PdV + VdP - TdS - SdT$

for a reversible change at constant T and P.

$dq = TdS$; and SdT and VdP term vanishes.

$$\Rightarrow dG = TdS - w_{\text{non-PV}} - TdS$$

$$(dG)_{T,P} = -w_{\text{non-PV}}$$

$$\Rightarrow \boxed{-(dG)_{T,P} = w_{\text{non-PV}}}$$

Thus decrease in Gibbs function at constant temperature and pressure is equal to maximum non-PV work obtainable from system.

9. Chemical equilibrium :

The state of chemical equilibrium is characterised by constant value of temperature, pressure and composition of a closed system. The state of equilibrium represent most stable state acquired by chemical system under given conditions.

The characteristics of state of equilibrium are :

- (i) The equilibrium is attained in closed system.
- (ii) At equilibrium, the chemical system is at minimum value of Gibbs function possible under given conditions.
- (iii) A state of equilibrium is attained spontaneously.
- (iv) At equilibrium, the rate of change of Gibbs function with progress of reaction under given conditions becomes zero $\Rightarrow \Delta_r G = 0$.
- (v) At state of equilibrium, the total Gibbs function of products becomes equal to total Gibbs function of reactants.

9.1 EQUILIBRIUM CONSTANT

Gas phase reactions :

Consider a reaction : $aA(g) + bB(g) \longrightarrow cC(g) + dD(g)$

Now molar Gibbs functions of A, B, C and D are given by :

$$G_A = G_A^\circ + RT \ln \left(\frac{P_A}{P_0} \right)$$

$$G_B = G_B^\circ + RT \ln \left(\frac{P_B}{P_0} \right)$$

$$G_C = G_C^\circ + RT \ln \left(\frac{P_C}{P_0} \right)$$

$$G_D = G_D^\circ + RT \ln \left(\frac{P_D}{P_0} \right)$$

Now, $\Delta_r G = cG_C + dG_D - aG_A - bG_B$

Substituting value of G_A , G_B , G_C and G_D in above equation :

$$\Delta_r G = (cG_C^\circ + dG_D^\circ - bG_B^\circ - aG_A^\circ) + RT \ln \left[\frac{\left(\frac{P_C}{P_0} \right)^c \left(\frac{P_D}{P_0} \right)^d}{\left(\frac{P_A}{P_0} \right)^a \left(\frac{P_B}{P_0} \right)^b} \right]$$

$$\Delta_r G = \Delta_r G^\circ + RT \ln Q$$

Where, $Q = \left(\frac{P_C^c \cdot P_D^d}{P_A^a \cdot P_B^b} \right) \left(\frac{1}{P_0} \right)^{(c+d-a-b)}$

At equilibrium, $\Delta_r G = 0$ and $Q = K_p = \left(\frac{P_C^c \cdot P_D^d}{P_A^a \cdot P_B^b} \right) \left(\frac{1}{P_0} \right)^{\Delta n_g}$

$$\Delta_r G^\circ = -RT \ln K_p$$

Note : Q contains partial pressures at that moment but K_p contain partial pressures at equilibrium.

10. RELATIONSHIP BETWEEN, ΔG° AND EQUILIBRIUM CONSTANT

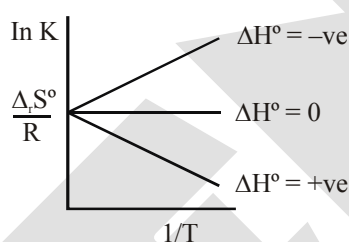
$$\Delta G^\circ = -RT \ln K_{eq}$$

$$\Delta H^\circ - T\Delta S^\circ = -RT \ln K_{eq}$$

$$\ln K_{eq} = \frac{-\Delta_r H^\circ}{RT} + \frac{\Delta_r S^\circ}{R}$$

$$\ln K_1 = \frac{\Delta_r S^\circ}{R} - \frac{\Delta_r H^\circ}{RT_1}$$

$$\ln K_2 = \frac{\Delta_r S^\circ}{R} - \frac{\Delta_r H^\circ}{RT_2}$$



[where K_1 & K_2 are values of K_p at temperature T_1 K & T_2 K respectively]

$$\ln \left(\frac{K_2}{K_1} \right) = \frac{\Delta H^\circ}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \quad (\text{Van't Hoff's equation})$$

EXERCISE # 4

23. In a irreversible process taking place at constant T and P and in which only pressure-volume work is being done the change in Gibbs free energy (dG) and change in entropy (dS) satisfy the criteria:-
 (A) $(dS)_{V,E} = 0$, $(dG)_{T,P} = 0$ (B) $(dS)_{V,E} = 0$, $(dG)_{T,P} > 0$ [AIEEE-2003]
 (C) $(dS)_{V,E} < 0$, $(dG)_{T,P} < 0$ (D) $(dS)_{V,E} > 0$, $(dG)_{T,P} < 0$
24. The correct relationship between free energy change in a reaction and the corresponding equilibrium constant K_c is :- [AIEEE-2003]
 (A) $\Delta G^\circ = RT \ln K_c$ (B) $-\Delta G^\circ = RT \ln K_c$ (C) $\Delta G = RT \ln K_c$ (D) $-\Delta G = RT \ln K_c$
25. For the reaction at 298 K
 $A(g) + B(g) \rightleftharpoons C(g) + D(g)$
 $\Delta H^\circ = -29.8 \text{ kcal}$; $\Delta S^\circ = -0.1 \text{ kcal/K}$
 Calculate ΔG° and K.
26. The correct relationship between free energy change in a reaction and the corresponding equilibrium constant K is :-
 (A) $-\Delta G^\circ = RT \ln K$ (B) $\Delta G = RT \ln K$
 (C) $-\Delta G = RT \ln K$ (D) $\Delta G^\circ = RT \ln K$

ANSWER KEY

EXERCISE-1

1. Ans. (i) irreversible, (ii) reversible, (iii) impossible
2. Ans.(A) 3. Ans.(B) 4. Ans.(C)

EXERCISE-2

5. Ans.(A)
6. Ans. (a) 2.1 cal/K , (b) 1.5 cal/K 7. Ans 819°C
8. Ans.(C)
9. Ans.
(A) (+ive) (B) (+ive) (C) ≈ 0 (D) (+ive)
(E) (+ive) (F) (-ive)
10. Ans.(C) 11. Ans.(C) 12. Ans.(D) 13. Ans.(B)
14. Ans.(C) 15. Ans.(A)

EXERCISE-3

16. Ans. -2873.1 kJ 17. Ans. (B) 18. Ans.(C) 19. Ans.(C)
20. Ans.(B) 21. Ans.(B) 22. Ans.(D)

EXERCISE-4

23. Ans.(D) 24. Ans.(B) 25. Ans. $\Delta G^\circ = 0$; $K = 1$
26. Ans.(A)

MISCELLANEOUS SOLVED QUESTION

1. When 1-pentyne(A) is treated with 4N alcoholic KOH at 175°C, it is converted slowly into an equilibrium mixture of 1.3% 1-pentyne (A), 95.2% 2-pentyne(B) & 3.5% of 1, 2-pentadiene (C). The equilibrium was maintained at 175°C. Calculate ΔG° for the following equilibria.



From the calculated value of ΔG_1° & ΔG_2° indicate the order of stability of A, B & C. Write a reasonable reaction mechanism sharing all intermediate leading to A, B & C. [JEE 2001]

2. Show that the reaction $\text{CO(g)} + \frac{1}{2} \text{O}_2\text{(g)} \longrightarrow \text{CO}_2\text{(g)}$ at 300 K is spontaneous and exothermic, when the standard entropy is $-0.094 \text{ kJ mol}^{-1} \text{ K}^{-1}$. The standard Gibbs free energies of formation for CO_2 and CO are -394.4 and $-137.2 \text{ kJ mol}^{-1}$, respectively. [JEE 2001]

3. $\text{N}_2\text{O}_4\text{(g)} \rightleftharpoons 2\text{NO}_2\text{(g)}$

This reaction is carried out at 298 K and 20 bar. 5 mol each of N_2O_4 and NO_2 are taken initially.

Given : $(\Delta G_f^\circ)_{\text{N}_2\text{O}_4} = 100 \text{ kJ mol}^{-1}$; $(\Delta G_f^\circ)_{\text{NO}_2} = 50 \text{ kJ mol}^{-1}$

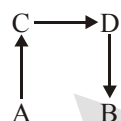
(i) Find ΔG for reaction at 298 K under given condition.

(ii) Find the direction in which the reaction proceeds to achieve equilibrium. [JEE 2004]

4. The enthalpy of vapourization of a liquid is 30 kJ mol^{-1} and entropy of vapourization is $75 \text{ J mol}^{-1} \text{ K}^{-1}$. The boiling point of the liquid at 1 atm is :- [JEE 2004]

(A) 250 K (B) 400 K (C) 450 K (D) 600 K

5. A process $A \longrightarrow B$ is difficult to occur directly instead it takes place in three successive steps.



$$\Delta S (A \longrightarrow C) = 50 \text{ e.u.}$$

$$\Delta S (C \longrightarrow D) = 30 \text{ e.u.}$$

$$\Delta S (B \longrightarrow D) = 20 \text{ e.u.}$$

[JEE 2006]

Where e.u. is entropy unit.

Then the entropy change for the process $\Delta S (A \longrightarrow B)$ is :-

(A) + 100 e.u. (B) - 60 e.u.

(C) - 100 e.u. (D) + 60 e.u.

6. For the process $\text{H}_2\text{O (l)} (1 \text{ bar}, 373 \text{ K}) \longrightarrow \text{H}_2\text{O (g)} (1 \text{ bar}, 373 \text{ K})$, the correct set of thermodynamic parameters is :- [JEE 2007]

(A) $\Delta G = 0$, $\Delta S = + \text{ve}$

(B) $\Delta G = 0$, $\Delta S = - \text{ve}$

(C) $\Delta G = + \text{ve}$, $\Delta S = 0$

(D) $\Delta G = - \text{ve}$, $\Delta S = + \text{ve}$

7. **Statement-1** : There is a natural asymmetry between converting work to heat and converting heat to work.

Statement-2 : No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work. [JEE 2008]

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

8. **Statement-1** : For every chemical reaction at equilibrium, standard Gibbs energy of reaction is zero.

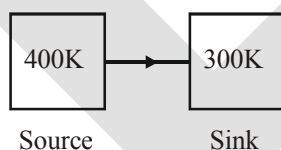
Statement-2 : At constant temperature and pressure, chemical reactions are spontaneous in the direction of decreasing Gibbs energy. [JEE 2008]

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

EXERCISE (S-I)

□ Carnot Cycle

1. The efficiency of a carnot cycle is $1/6$. On decreasing the temperature of the sink by 65°C , the efficiency increases to $1/3$. Calculate the temperature of source and sink.
2. A carnot cycle has an efficiency of 40%. Its low temperature reservoir is at 9°C . What is the temperature of source ?
3. A diatomic ideal gas is expanded isothermally to 32 times of its initial volume then it is cooled to restore to initial entropy at constant volume. Calculate ratio of initial temperature to final temperature
4. One mole of an ideal monoatomic gas was taken through reversible isochoric heating from 100 K to 1000 K. Calculate ΔS_{system} , ΔS_{surr} and ΔS_{total}
 - (i) when the process is carried out reversibly
 - (ii) when the process is carried out irreversibly (one step)
5. 100 kJ heat is transferred from a large heat reservoir at 400 K to another large heat reservoir at 300 K. Suppose there is no change in temperature due to exchange of heat and combinedly source and sink form isolate system.

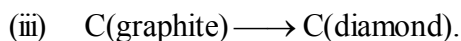
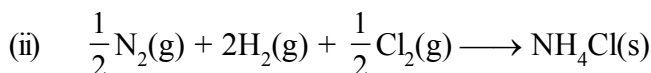
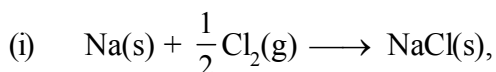


Find (a) ΔS_{source} , (b) ΔS_{sink} and (c) ΔS_{total} . Also comment on spontaneity of process.

6. A system of 100 kg mass undergoes a process in which its specific entropy increases from $0.3 \text{ kJkg}^{-1}\text{K}^{-1}$ to $0.4 \text{ kJkg}^{-1}\text{K}^{-1}$. At the same time, the entropy of the surrounding decreases from $80 \text{ kJ}\text{K}^{-1}$ to $75 \text{ kJ}\text{K}^{-1}$. Find the $(\Delta S)_{\text{universe}}$ in kJK^{-1} .
7. Calculate $(\Delta S)_{\text{universe}}$ (in cal/K) when 1kg water at 300K is brought into contact with a heat reservoir at 600K. Specific heat capacity of water is $1 \text{ cal}/\text{gm}^\circ\text{C}$. Assume no change in physical state of water.
($\ln 2 = 0.7$)
8. 1 mole of ideal monoatomic gas is heated by supplying 5 kJ heat from a reservoir maintained at 400 K from 300 K to 400 K. In the process volume of gas increased from 1 L to 10 L. Find ΔS_{total} (in $\text{J}/\text{K} \cdot \text{mol}$) in the process

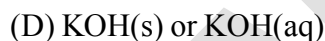
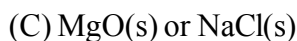
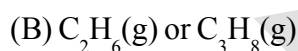
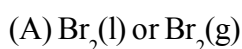
Use : $\ln \left(\frac{4}{3} \right) = 0.3$, $\ln 10 = 2.3$, $R = 8.3 \text{ J/K-mol}$ and $\ln x = 2.3 \log x$

9. Calculate ΔS_r° at 298K of



The values of S° of Na, Cl_2 , NaCl, NH_4Cl , N_2 , H_2 diamond & graphite are 51, 223, 72, 95, 192, 131, 2.43 & 5.69 $\text{JK}^{-1} \text{mol}^{-1}$ respectively.

10. Identify the substance in each of the following pairs of samples that has the higher entropy?

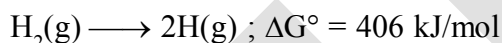


□ Gibbs's Function

11. Calculate the ΔG° change at 300 K for the reaction;

$\text{Br}_2(\text{l}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{BrCl(g)}.$ For the reaction $\Delta H^\circ = 29.3 \text{ kJ}$ & the standard entropies of $\text{Br}_2(\text{l})$, $\text{Cl}_2(\text{g})$ & BrCl(g) at the 300 K are 150, 220, 240 $\text{J mol}^{-1}\text{K}^{-1}$ respectively.

12. The standard entropies of $\text{H}_2(\text{g})$ and H(g) are 130 and 115 $\text{J mol}^{-1}\text{K}^{-1}$ respectively at 300K. Using the data given below calculate the bond energy of H_2 (in kJ/mol) :-



13. 5 mole $\text{H}_2\text{O}(\text{l})$ at 373K and 1 atm is converted into $\text{H}_2\text{O(g)}$ at 373K and 5 atm. Calculate ΔG for this process. [Given : $R = 2 \text{ Cal/K-mol}$, $\ln 5 = 1.6$]

14. Calculate ΔG (in bar-L) when a definite mass of a monoatomic ideal gas at 1 bar & 27°C is expanded adiabatically against vacuum from 10 L to 20 L ($\ln 2 = 0.7$)

15. Find $(\Delta S)_{\text{universe}}$ (in Joule/mole/K) at 1 bar for a chemical reaction at 300 K if $\Delta H_{300 \text{ K}}^\circ = 75 \text{ kJ/mol}$
 $\Delta S_{300 \text{ K}}^\circ = 300 \text{ J/K}$

16. A liquid freezes into a solid ($\Delta H = -1000 \text{ J/mole}$) at 200 K and 1 atm, (it's normal melting point).

(i) What is the value of ΔG at 200 K ?

(ii) What is the ΔS value at 200 K ?

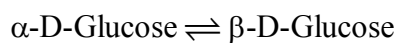
(iii) Will the freezing be spontaneous at 150 K and 1 atm ? Calculate ΔS_{total} assuming ΔS and ΔH do not depend on temperature.

(iv) What is the value of ΔG at 250 K and 1 atm ? Assuming ΔS and ΔH do not depend on temperature.

(Assume ΔS and ΔH do not depend on temperature)

❑ Thermodynamic and equilibrium constant

17. α -D Glucose undergoes mutarotation to β -D-Glucose in aqueous solution. If at 300K there is 60% conversion. Calculate ΔG° of the reaction. ($\ln 2 = 0.7$, $\ln 3 = 1.1$)



- 18.** The equilibrium constant of the reaction $2\text{C}_3\text{H}_6(\text{g}) \rightleftharpoons \text{C}_2\text{H}_4(\text{g}) + \text{C}_4\text{H}_8(\text{g})$ is found to fit the expression

$$\ln K = -1.04 - \frac{1088K}{T}$$

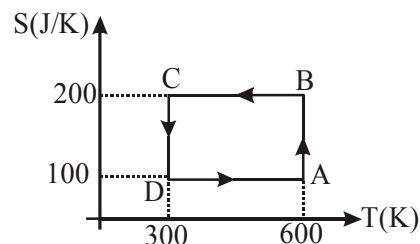
Calculate the standard reaction enthalpy and entropy at 400 K.

- 19.** For the reaction $\text{SO}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$; $\Delta H^\circ_{300} = -95 \text{ kJ/mole}$, $\Delta S^\circ_{300} = -95.0 \text{ J/K mole}$. Find the value of $\ln k_p$ for this reaction at 300 K.

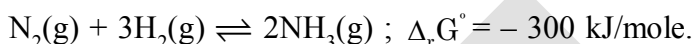
EXERCISE-(S-II)

1. From the given T-S diagram of a reversible carnot engine, find

- (i) work delivered by engine in one cycle
 (ii) heat taken from the source in each cycle.
 (iii) ΔS_{sink} in each cycle.



2. At 300 K, $\Delta H^{\circ}_{\text{combustion}}(\text{sucrose}) = -5000 \text{ kJ/mol}$ & $\Delta G^{\circ}_{\text{combustion}}(\text{sucrose}) = -6000 \text{ kJ/mol}$. Estimate additional non-PV work that is obtained by raising temperature to 309 K. Assume $\Delta_r C_p = 0$ for this temperature change.
3. The K_p for reaction $A + B \rightleftharpoons C + D$ is 2.0 at 27°C and 8.0 at 127°C . Determine the standard free energy change of this reaction at each temperature and ΔH° for the reaction over this range of temperature ($\ln 2 = 0.7$) ?
4. Pressure of 10 moles of an ideal gas is changed from 2 atm to 1 atm against constant external pressure without change in temperature. If surrounding temperature (300 K) and pressure (1 atm) always remains constant then calculate total entropy change ($\Delta S_{\text{system}} + \Delta S_{\text{surrounding}}$) for given process.
5. What is $\Delta_r G$ for synthesis of ammonia at 300 K at following sets of partial pressure :



[Take $R = \frac{25}{3} \text{ J/K mole}$, $\ln 2 = 0.7$, $\ln 3 = 1.1$, $\ln 5 = 1.6$]

Gas	N_2	H_2	NH_3
Pressure (atm)	1	3	0.02

6. $2\text{A}(\text{s}) \rightleftharpoons \text{B}(\text{g}) + 2\text{C}(\text{g}) + 3\text{D}(\text{g})$

Total pressure developed in a closed container by decomposition of A at equilibrium is 12 bar at 727°C . Calculate $|\Delta G^{\circ}|$ (in kcal) of the reaction at 727°C ($R = 2 \text{ cal/mole-K}$, $\ln 2 = 0.7$, $\ln 3 = 1.1$)

7. Find $\Delta G(\text{J/mol})$ for the reaction at 300 kPa & 27°C when all gases are in stoichiometric ratio of moles.
 $\text{N}_2\text{O}_4(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$ [$\ln 2 = 0.7$]

Given: $\Delta G_f^{\circ}(\text{N}_2\text{O}_4) = 100 \text{ kJ/mol}$

$\Delta G_f^{\circ}(\text{NO}_2) = 50 \text{ kJ/mol}$

$R = 8 \text{ J/mol-K}$

8. Oxygen is heated from 300 to 600 K at a constant pressure of 1 bar. What is the increases in molar entropy ? The molar heat capacity (in $\text{JK}^{-1} \text{mol}^{-1}$) for the O_2 is (Given : $\ln 2 = 0.7$)

$$C_{P,m} = 10 + 10^{-2}T$$

9. Calculate the molar entropy of a substance at 600 K and 1 bar using the following data.

(i) Heat capacity of solid from 0K to normal melting point 200 K

$$C_{p,m}(s) = 0.035 T \text{ JK}^{-1}\text{mol}^{-1}.$$

(ii) Enthalpy of fusion = 7.5 kJ mol^{-1} ,

(iii) Enthalpy of vaporisation = 30 kJ mol^{-1} ,

(iv) Heat capacity of liquid from 200K to normal boiling point 300K

$$C_{p,m}(\ell) = 60 + 0.016T \text{ JK}^{-1}\text{mol}^{-1}.$$

(v) Heat capacity of gas from 300K to 600K at 1 atm

$$C_{p,m}(g) = 50.0 \text{ JK}^{-1}\text{mol}^{-1}.$$

10. For a reaction : $2A(s) + B(g) \longrightarrow 3C(l)$, standard entropy change of reaction is 2 kJ/mol-K and standard enthalpy of combustion of $A(s)$, $B(g)$ and $C(l)$ are -100 , -60 -285 kJ/mol respectively, then find the maximum useful work that can be obtained at 27°C and 1 bar pressure from the reaction (in kJ/mol).

EXERCISE (O-I)

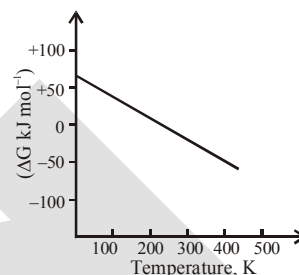
- A reversible heat engine A (based on Carnot cycle) absorbs heat from a reservoir at 1000 K and rejects heat to a reservoir at T_2 . A second reversible engine B (based on Carnot cycle) absorbs the same amount of heat as rejected by the engine A, from the reservoir at T_2 and rejects energy to a reservoir at 360 K.
If the efficiencies of engines A and B are the same then the temperature T_2 is :-
(A) 680 K (B) 640 K (C) 600 K (D) 670 K
- Which of the following is incorrect for a closed system in which an irreversible process is occurring?
(A) $Q_{\text{sys}} + Q_{\text{surr}} = 0$ (B) $W_{\text{sys}} + W_{\text{surr}} = 0$
(C) $\Delta U_{\text{sys}} + \Delta U_{\text{surr}} = 0$ (D) $\Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$
- For conversion $\text{C}(\text{graphite}) \rightarrow \text{C}(\text{Diamond})$, the ΔS is :-
(A) Zero (B) Positive
(C) Negative (D) Can not be predicted
- Considering entropy (S) as a thermodynamic parameter, the criteria for the spontaneity of any process is :-
(A) $\Delta S_{\text{system}} - \Delta S_{\text{surroundings}} > 0$ (B) $\Delta S_{\text{system}} > 0$ only
(C) $\Delta S_{\text{surroundings}} > 0$ only (D) $\Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$
- If $\Delta H_{\text{vaporisation}}$ of substance $X(\ell)$ (molar mass = 30 g/mol) is 300 J/g at its boiling point 300 K, then molar entropy change for reversible condensation process is :-
(A) 30 J/mol.K (B) -300 J/mol.K (C) -30 J/mol.K (D) -10 J/mol.K
- For 1 mole of an ideal monoatomic gas on moving from one state to other, the temperature is doubled but pressure becomes $\sqrt{2}$ times. Then entropy change in the process will be
($R = 2 \text{ Cal/mol.K}$)
(A) $R \ln 2$ (B) $2R \ln 2$ (C) $3R \ln 2$ (D) $\frac{R}{2} \ln 2$
- If S° for H_2 , Cl_2 and HCl are 0.13, 0.22 and 0.19 $\text{kJ K}^{-1} \text{mol}^{-1}$ respectively. The total change in standard entropy for the reaction, $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ is :
(A) 30 $\text{JK}^{-1} \text{mol}^{-1}$ (B) 40 $\text{JK}^{-1} \text{mol}^{-1}$ (C) 60 $\text{JK}^{-1} \text{mol}^{-1}$ (D) 20 $\text{JK}^{-1} \text{mol}^{-1}$
- For a perfectly crystalline solid $C_{p,m} = aT^3$, where a is constant. If $C_{p,m}$ is 0.42 J/K-mol at 10 K, molar entropy at 10 K is
(A) 0.42 J/K-mol (B) 0.14 J/K-mol (C) 4.2 J/K-mol (D) Zero
- Identify the correct statement regarding entropy.
(A) At absolute zero, the entropy of perfectly crystalline substance is taken to be +ve
(B) At absolute zero, entropy of perfectly crystalline substance is taken to be zero.
(C) At 0 °C, the entropy of a perfectly crystalline substance is taken to be zero.
(D) At absolute zero, the entropy of all crystalline substances is taken to be zero.

10. In which of the following reactions do you expect to have a decrease in entropy?

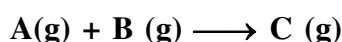
- (A) $\text{Fe(s)} \longrightarrow \text{Fe(l)}$
 (B) $2 \text{Fe(s)} + 3/2 \text{O}_2\text{(g)} \longrightarrow \text{Fe}_2\text{O}_3\text{(s)}$
 (C) $\text{HF(l)} \longrightarrow \text{HF(g)}$
 (D) $2 \text{H}_2\text{O}_2\text{(l)} \longrightarrow 2 \text{H}_2\text{O(l)} + \text{O}_2\text{(g)}$

11. What can be concluded about the values of ΔH and ΔS from this graph?

- (A) $\Delta H > 0, \Delta S > 0$ (B) $\Delta H > 0, \Delta S < 0$
 (C) $\Delta H < 0, \Delta S > 0$ (D) $\Delta H < 0, \Delta S < 0$



12. For the reaction at 300 K



$$\Delta U = -3.0 \text{ kcal} ; \Delta S = -10.0 \text{ cal/K}$$

value of ΔG is ?

- (A) -600 cal (B) -6600 cal (C) -6000 cal (D) none

13. For the reaction that taking place at certain temperature $\text{NH}_4\text{HS(s)} \rightleftharpoons \text{NH}_3\text{(g)} + \text{H}_2\text{S(g)}$, if equilibrium pressure is X bar, then $\Delta_r G^\circ$ would be :-

- (A) $-2 RT \ln X$ (B) $-RT (\ln X - \ln 2)$
 (C) $-2 RT (\ln X - \ln 2)$ (D) $-0.5 RT (\ln X - \ln 2)$

14. What is the free energy change (ΔG) when 1.0 mole of water at 100°C and 1 atm pressure is converted into steam at 100°C and 1 atm pressure ?

- (A) 80 cal (B) 540 cal (C) 620 cal (D) zero

15. What is the free energy change (ΔG) when 1.0 mole of water at 100°C and 1 atm pressure is converted into steam at 100°C and 2 atm pressure ?

- (A) Zero (B) 540 cal (C) 517.13 cal (D) 510 cal

16. If $\Delta G^\circ > 0$ for a gaseous reaction then :

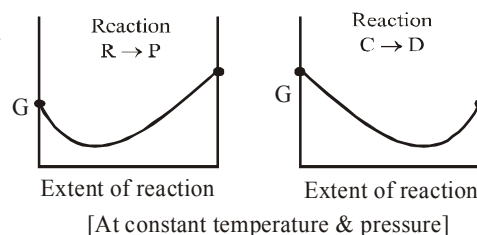
- (A) $K_p > 1$
 (B) $K_p < 1$
 (C) The products predominate in the equilibrium mixture
 (D) Mole of product must be less than mole of reactant

17. Heat liberated for an ideal gas undergoing reversible isothermal process is 1200 cal at 300 K. What will be Gibb's free energy change for the process ?

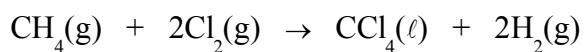
- (A) 1200 cal (B) -1200 cal (C) 400 cal (D) -400 cal

18. With the help of given curves select the correct statement.

- (A) C will partially convert into D.
 (B) R will not convert into P
 (C) R will be completely converted into P.
 (D) C will be completely converted in D.



19. Evaluate ΔS° for the reaction below at 25°C .



Given :

	$\text{CH}_4(\text{g})$	$\text{Cl}_2(\text{g})$	$\text{CCl}_4(\ell)$	$\text{H}_2(\text{g})$
$\Delta H_f^\circ (\text{kJ/mol})$	-74.81	0	-135.4	0
$\Delta G_f^\circ (\text{kJ/mol})$	-50.75	0	-65.27	0
(A) -360 J/K	(B) -66.9 J/K	(C) -155 J/K	(D) -487 J/K	

20. All of the following have $\Delta G_f^\circ = 0$, except -

(A) $\text{O}_2(\text{g})$ (B) $\text{Br}_2(\text{g})$ (C) $\text{H}_2(\text{g})$ (D) $\text{Ca}(\text{s})$

EXERCISE (O-II)

Single correct :

1. When two equal sized pieces of the same metal at different temperatures T_h (hot piece) and T_c (cold piece) are brought into thermal contact and isolated from its surrounding. The total change in entropy of system is given by ? Suppose heat capacity of each piece is C .

(A) $C \ln \frac{T_c + T_h}{2T_c}$ (B) $C \ln \frac{T_2}{T_1}$ (C) $C \ln \frac{(T_c + T_h)^2}{2T_h \cdot T_c}$ (D) $C \ln \frac{(T_c + T_h)^2}{4T_h \cdot T_c}$

2. The change in entropy of 2 moles of an ideal gas upon isothermal expansion at 243.6 K from 20 litre until the pressure becomes 1 atm, is :-

(A) 1.385 cal/K (B) - 1.2 cal/K (C) 1.2 cal/K (D) 2.77 cal/K

3. For the hypothetical reaction, $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$, $\Delta_r G$ and $\Delta_r S$ are 20 kJ/mole and $-20 \text{ JK}^{-1} \text{ mol}^{-1}$ respectively at 200 K. If $\Delta_r C_p$ is $20 \text{ JK}^{-1} \text{ mol}^{-1}$ then $\Delta_r H$ at 400 K is :-

(A) 20 kJ/mole (B) 7.98 kJ/mole (C) 28 kJ/mole (D) 16 kJ/mole

4. A mole of steam is condensed at 100°C , the water is cooled to 0°C and frozen to ice. What is the difference in entropies of the ice and steam ? The heats of vaporization and fusion are 540 cal gm^{-1} and 80 cal gm^{-1} respectively. Use the average heat capacity of liquid water as $1 \text{ cal gm}^{-1} \text{ degree}^{-1}$.

(A) $18 \left(\frac{80}{373} + \frac{540}{373} + \ln \frac{373}{273} \right)$ (B) $18 \left(\frac{80}{273} + \frac{540}{373} + \ln \frac{373}{273} \right)$

(C) $18 \left(\frac{80}{273} + \frac{540}{373} + \ln \frac{273}{373} \right)$ (D) $18 \left(\frac{80}{273} + \frac{540}{373} + 100 \right)$

5. The value of ΔG_f° of gaseous mercury is 38 kJ/mole. At what total external pressure mercury start boiling at 27°C . ($R \ln 10 = 19 \text{ J/K-mol}$)

(A) $10^{-6.67}$ (B) $10^{-1.67}$ (C) $10^{-13.33}$ (D) $10^{-3.33}$

6. If molar internal energy for a gas in a closed rigid vessel given by

$$U = a + bT + cT^2$$

find the entropy change (in J/K) at constant volume when 1 mol of gas are heated from 200 K to 400K. [Given : $a = 20 \text{ J/mol}$; $b = 10 \text{ J/K-mol}$; $c = 2 \times 10^{-2} \text{ J/K}^2\text{-mol}$] ($\ln 2 = 0.7$)

(A) 15 (B) 30 (C) 60 (D) 9.78

7. Statement - 1 : When process $\text{H}_2\text{O}(s) \rightleftharpoons \text{H}_2\text{O}(l)$, reaches equilibrium in a closed system at constant temperature and pressure, Gibb's function of $\text{H}_2\text{O}(s)$ & $\text{H}_2\text{O}(l)$ become same.

Statement - 2 : For reversible phase change at constant temperature & pressure, change in Gibb's free energy will be zero.

(A) Statement-1 is true, statement- 2 is true and statement- 2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

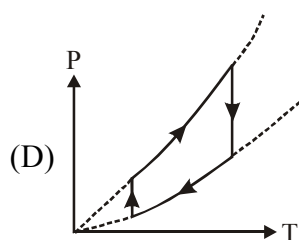
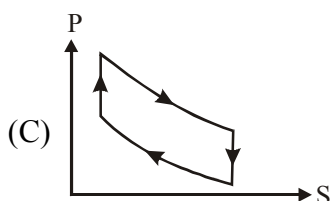
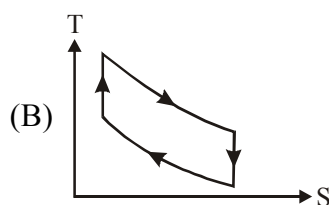
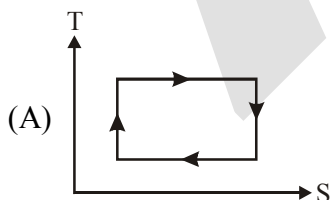
(C) Statement-1 is true, statement-2 is false

(D) Statement-1 is false, statement-2 is true.

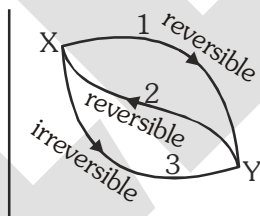
8. Statement-1 : Absolute entropy of an ion in aqueous solution at 298 K may be negative.
Statement -2 : Absolute entropy at 298 K of any substance can never be negative.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false
(D) Statement-1 is false, statement-2 is true.
9. **Statement-1** : Net heat absorbed in a cyclic process must be always equal to net work done by the system in the cyclic process.
Statement-2 : Internal energy of system is a function of state.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
10. **Statement-1** : Entropy change in reversible adiabatic expansion of an ideal gas is zero.
Statement-2 : The increase in entropy due to volume increase just compensate the decrease in entropy due to fall in temperature.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

MORE THAN ONE MAY BE CORRECT :

11. Select the correct statement(s).
(A) In a reversible process, ΔG is always zero in a closed system.
(B) In a reversible process, ΔS_{univ} is always zero in a closed system.
(C) In a reversible process, ΔS_{sys} is always zero in a closed system.
(D) In a reversible process, ΔS_{sys} is always zero in an isolated system.
12. Which of the following statement (s) is/ are correct ?
(A) The quantities E, H and G have the same dimension
(B) Gibb's free energy of 10 gm ice at 0°C and 1.0 atm is less than the Gibb's free energy of 10 gm water at 0°C and 1 atm.
(C) $\Delta S_{\text{sys}} = 0$ for every adiabatic process in a closed system.
(D) For every reversible process in a closed system, $\Delta S_{\text{sys}} = \frac{\Delta H_{\text{sys}}}{T}$
13. Which of the following represents the carnot cycle-



14. The normal boiling point of a liquid 'A' is 350 K. ΔH_{vap} at normal boiling point is 35 kJ/mole. Pick out the correct statement(s). (Assume ΔH_{vap} to be independent of pressure).
- (A) $\Delta S_{\text{vaporisation}} > 100 \text{ J/K mole}$ at 350 K and 0.5 atm
(B) $\Delta S_{\text{vaporisation}} < 100 \text{ J/K mole}$ at 350 K and 0.5 atm
(C) $\Delta S_{\text{vaporisation}} < 100 \text{ J/K mole}$ at 350 K and 2 atm
(D) $\Delta S_{\text{vaporisation}} = 100 \text{ J/K mole}$ at 350 K and 2 atm
15. In isothermal ideal gas compression :
- (A) w is + ve (B) ΔH is zero (C) ΔS_{gas} is + ve (D) ΔG is + ve
16. Which of the following statement (s) is/are false :
- (A) $\Delta_r S$ for $\frac{1}{2}\text{N}_2(\text{g}) \longrightarrow \text{N}(\text{g})$ is positive
(B) ΔG_{system} is always zero for a reversible process in a closed system
(C) ΔG° for an ideal gas is a function of temperature and pressure
(D) Entropy of a closed system is always maximized at equilibrium
17. Which of the following processes are spontaneous ?
- (A) Burning of fossil fuel
(B) Decomposition of water into H_2 and O_2 gas at room temperature
(C) Spreading of perfume in a room
(D) Diffusion of gas from high pressure to low pressure
18. Suppose a system make a transition from state X to state Y.



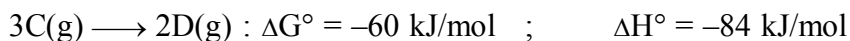
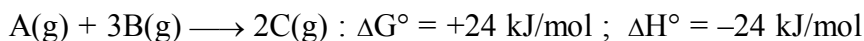
Given : $\Delta S_{\text{xy}} = 10 \text{ J/K}$

- (A) The state Y is more disordered than state X.
- (B) ΔS_{XY} for path 1 and 3 is same.
- (C) $\Delta S_{YX} = -10 \text{ J/K}$
- (D) The transition $X \rightarrow Y$ must be spontaneous.
- 19.** In which of the following processes involving an ideal gas, entropy of system remains constant?
- (A) Reversible isothermal expansion (B) Irreversible adiabatic expansion
- (C) Reversible adiabatic expansion (D) Free expansion
- 20.** In which of the following process involving ideal gas, entropy of surrounding remains constant?
- (A) Reversible isobaric heating (B) Reversible adiabatic expansion
- (C) Irreversible adiabatic compression (D) Free expansion

Paragraph for Question 21 to 22

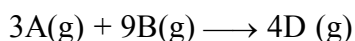
The reactions whose ΔG° are positive can not take place under standard state conditions. However another reaction whose ΔG° is negative can be coupled with the former type of reaction to give overall spontaneous process.

Consider the given reactions whose ΔG° at 300 K are provided to answer following questions.



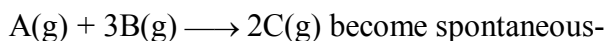
All data at 300 K temperature.

21. What is ΔS° at 300 K of reaction (in J/Kmole) :



- (A) -640 (B) -480 (C) +640 (D) -240

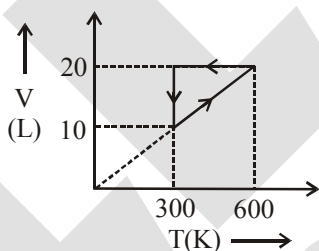
22. Assuming ΔH° and ΔS° do not vary with temperature. At what temperature reaction



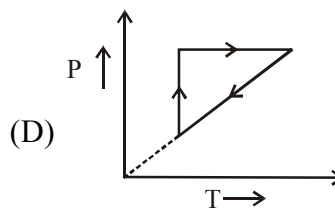
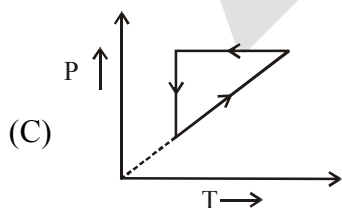
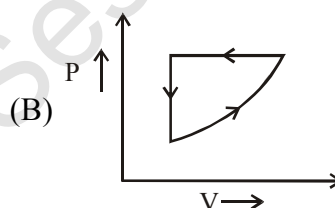
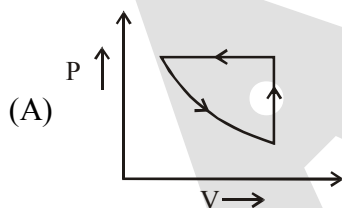
- (A) 6250 K (B) 625 K (C) 150 K (D) 1000 K

Paragraph for Q.23 to Q.24

One mol of ideal monoatomic gas undergo the state change as shown in the following graph ($\ln 2 = 0.7$)



23. Correct graph for the process in paragraph is -



24. Efficiency of the cycle will be -

- (A) 50% (B) 25% (C) 13.63% (D) 12%

Table type question :

Column-I	Column-II	Column-III
(A) Combustion of butane gas at S.T.P.	(P) Entropy change of universe constant	(I) Low temperature conditions favourable
(B) Dissolution of sodium chloride in water at STP	(Q) Entropy change of system positive	(II) More spontaneous at high temperature
(C) Adding carbon dioxide gas to water to form carbonated water at high pressure and 298K	(R) Enthalpy change of system negative	(III) High pressure condition favourable
(D) Vaporisation of liquid water at 1 atm and 373K	(S) Entropy change of universe positive	(IV) More spontaneous at low pressure

25. Which of following option is incorrect -

- (A) A – S – I (B) B – Q – IV (C) C – R – I (D) D – P – IV

26. In column-I (A) if the reaction was carried at 400 K which option would be correct -

- (A) A – P – II (B) A – Q – III (C) A – S – IV (D) A – R – III

27. In column-I (D), if vaporisation is carried out at 15 atm which option is correct -

- (A) D – P – IV (B) D – S – II (C) D – R – I (D) D – Q – II

MATCH THE LIST :

28. Match the Column :

Column-I	Column-II
(P) $\text{H}_2\text{O} (\ell, 1 \text{ atm}, 363 \text{ K}) \rightarrow \text{H}_2\text{O} (\text{g}, 1 \text{ atm}, 363 \text{ K})$	(1) $\Delta_r S > 0$
(Q) $\text{H}_2\text{O} (\text{s}, 1 \text{ atm}, 373 \text{ K}) \rightarrow \text{H}_2\text{O} (\text{g}, 1 \text{ atm}, 373 \text{ K})$	(2) $\Delta_r G > 0$
(R) $\text{H}_2\text{O} (\ell, 1 \text{ atm}, 273 \text{ K}) \rightarrow \text{H}_2\text{O} (\text{s}, 1 \text{ atm}, 273 \text{ K})$	(3) $\Delta_r H < 0$
(S) $\text{H}_2\text{O} (\text{s}, 1 \text{ atm}, 353 \text{ K}) \rightarrow \text{H}_2\text{O} (\ell, 1 \text{ atm}, 353 \text{ K})$	(4) $\Delta_r U > 0$

Code:

	P	Q	R	S
(A)	2	4	1	3
(B)	4	2	3	1
(C)	2	1	3	4
(D)	4	3	1	2

MATCH THE COLUMN :

29. Column-I

- (A) Reversible adiabatic compression
(B) Reversible vaporisation
(C) Adiabatic free expansion of ideal gas in vacuum
(D) Dissociation of $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

Column-II

- (P) $\Delta S_{\text{system}} > 0$
(Q) $\Delta S_{\text{system}} < 0$
(R) $\Delta S_{\text{surrounding}} < 0$
(S) $\Delta S_{\text{surrounding}} = 0$

30. Column-I

(Related to process)

- (A) Fusion at melting point
(B) Vapourisation at boiling point
(C) Condensation at triple point
(D) Melting at normal boiling point

Column-II

(Related to system)

- (P) $\Delta G = 0$
(Q) $\Delta G < 0$
(R) $\Delta S > 0$
(S) $\Delta H \approx \Delta U$

EXERCISE (J-MAIN)

- The entropy change involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 dm³ to a volume of 100 dm³ at 27°C is :- [AIEEE-2011]
 - (1) 32.3 J mol⁻¹ K⁻¹
 - (2) 42.3 J mol⁻¹ K⁻¹
 - (3) 38.3 J mol⁻¹ K⁻¹
 - (4) 35.8 J mol⁻¹ K⁻¹
- The incorrect expression among the following is :- [AIEEE-2012]
 - (1) $K = e^{-\Delta G^\circ/RT}$
 - (2) $\frac{\Delta G_{\text{system}}}{\Delta S_{\text{total}}} = -T$
 - (3) In isothermal process, $W_{\text{reversible}} = -nRT \ln \frac{V_f}{V_i}$
 - (4) $\ln K = \frac{\Delta H^\circ - T\Delta S^\circ}{RT}$
- The entropy (S°) of the following substances are : [JEE-MAINS-(online) 2014]

CH₄ (g) 186.2 J K⁻¹ mol⁻¹

O₂ (g) 205.0 J K⁻¹ mol⁻¹

CO₂ (g) 213.6 J K⁻¹ mol⁻¹

H₂O (l) 69.9 J K⁻¹ mol⁻¹

The entropy change (ΔS°) for the reaction

CH₄(g) + 2O₂(g) → CO₂(g) + 2H₂O (ℓ) is :-

 - (1) -312.5 JK⁻¹ mol⁻¹
 - (2) -37.6 JK⁻¹ mol⁻¹
 - (3) -108.1 JK⁻¹ mol⁻¹
 - (4) -242.8 JK⁻¹ mol⁻¹
- The molar heat capacity (C_p) of CD₂O is 10 cal at 1000 K. The change in entropy associated with cooling of 32 g of CD₂O vapour from 1000 K to 100 K at constant pressure will be [JEE-MAINS-(online) 2014]

(D = deuterium, at. mass = 2u)

 - (1) -23.03 cal deg⁻¹
 - (2) 2.303 cal deg⁻¹
 - (3) 23.03 cal deg⁻¹
 - (4) -2.303 cal deg⁻¹
- Δ_fG° at 500 K for substance 'S' in liquid state and gaseous state are + 100.7 kcal mol⁻¹ and + 103 kcal mol⁻¹, respectively. Vapour pressure of liquid 'S' at 500 K is approximately equal to : [JEE-MAINS-(online) 2018]

(R = 2 cal K⁻¹ mol⁻¹) -

 - (1) 0.1 atm
 - (2) 10 atm
 - (3) 100 atm
 - (4) 1 atm

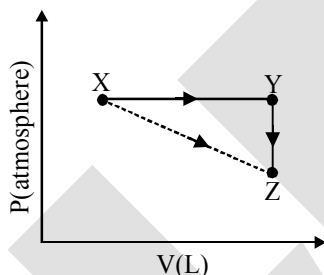
6. For which of the following processes, ΔS is negative? [JEE-MAINS-(online) 2018]
- (1) $C(\text{diamond}) \rightarrow C(\text{graphite})$ (2) $N_2(g, 273\text{ K}) \rightarrow N_2(g, 300\text{ K})$
(3) $H_2(g) \rightarrow 2H(g)$ (4) $N_2(g, 1\text{ atm}) \rightarrow N_2(g, 5\text{ atm})$
7. At 320 K, a gas A_2 is 20% dissociated to $A(g)$. The standard free energy change at 320 K and 1 atm in $J\text{ mol}^{-1}$ is approximately : [JEE-MAINS-(online) 2018]
- ($R = 8.314\text{ JK}^{-1}\text{ mol}^{-1}$; $\ln 2 = 0.693$; $\ln 3 = 1.098$)
- (1) 4281 (2) 4763 (3) 2068 (4) 1844

EXERCISE (J-ADVANCE)

1. Match the transformations in **Column-I** with appropriate option in **Column-II** [JEE 2011]

Column-I	Column-II
(A) $\text{CO}_2(\text{s}) \rightarrow \text{CO}_2(\text{g})$	(p) phase transition
(B) $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$	(q) allotropic change
(C) $2\text{H}\cdot \rightarrow \text{H}_2(\text{g})$	(r) ΔH is positive
(D) $\text{P}_{(\text{white, solid})} \rightarrow \text{P}_{(\text{red, solid})}$	(s) ΔS is positive
	(t) ΔS is negative

2. For an ideal gas, consider only P-V work in going from an initial state X to the final state Z. The final state Z can be reached by either of the two paths shown in the figure. Which of the following choice(s) is (are) correct ? [take ΔS as change in entropy and w as work done] [JEE 2012]



- (A) $\Delta S_{x \rightarrow z} = \Delta S_{x \rightarrow y} + \Delta S_{y \rightarrow z}$ (B) $W_{x \rightarrow z} = W_{x \rightarrow y} + W_{y \rightarrow z}$
 (C) $W_{x \rightarrow y \rightarrow z} = W_{x \rightarrow y}$ (D) $\Delta S_{x \rightarrow y \rightarrow z} = \Delta S_{x \rightarrow y}$

3. For the process [JEE 2014]



at $T = 100^\circ\text{C}$ and 1 atmosphere pressure, the correct choice is

- (A) $\Delta S_{\text{system}} > 0$ and $\Delta S_{\text{surroundings}} > 0$ (B) $\Delta S_{\text{system}} > 0$ and $\Delta S_{\text{surroundings}} < 0$
 (C) $\Delta S_{\text{system}} < 0$ and $\Delta S_{\text{surroundings}} > 0$ (D) $\Delta S_{\text{system}} < 0$ and $\Delta S_{\text{surroundings}} < 0$

4. Match the thermodynamic processes given under Column-I with the expressions given under Column-II. [JEE 2015]

Column - I	Column - II
(A) Freezing of water at 273 K and 1 atm	(P) $q = 0$
(B) Expansion of 1 mol of an ideal gas into a vacuum under isolated conditions	(Q) $w = 0$
(C) Mixing of equal volumes of two ideal gases at constant temperature and pressure in an isolated container	(R) $\Delta S_{\text{sys}} < 0$
(D) Reversible heating of $\text{H}_2(\text{g})$ at 1 atm from 300 K to 600 K followed by reversible cooling to 300 K at 1 atm	(S) $\Delta U = 0$
	(T) $\Delta G = 0$

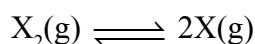
5. One mole of an ideal gas at 300 K in thermal contact with surroundings expands isothermally from 1.0 L to 2.0 L against a constant pressure of 3.0 atm. In this process, the change in entropy of surroundings (ΔS_{surr}) in J K^{-1} is - [JEE 2016]

(1 L atm = 101.3 J)

- (A) 5.763 (B) 1.013 (C) -1.013 (D) -5.763

Paragraph for Q.6 & Q.7

Thermal decomposition of gaseous X_2 to gaseous X at 298 K takes place according to the following equation : [JEE 2016]



The standard reaction Gibbs energy, $\Delta_r G^\circ$, of this reaction is positive. At the start of the reaction, there is one mole of X_2 and no X. As the reaction proceeds, the number of moles of X formed is given by β . Thus, $\beta_{\text{equilibrium}}$ is the number of moles of X formed at equilibrium. The reaction is carried out at a constant total pressure of 2 bar. Consider the gases to behave ideally.

(Given : $R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$)

6. The equilibrium constant K_p for this reaction at 298 K, in terms of $\beta_{\text{equilibrium}}$, is
- (A) $\frac{8\beta_{\text{equilibrium}}^2}{2-\beta_{\text{equilibrium}}}$ (B) $\frac{8\beta_{\text{equilibrium}}^2}{4-\beta_{\text{equilibrium}}^2}$ (C) $\frac{4\beta_{\text{equilibrium}}^2}{2-\beta_{\text{equilibrium}}}$ (D) $\frac{4\beta_{\text{equilibrium}}^2}{4-\beta_{\text{equilibrium}}^2}$
7. The **INCORRECT** statement among the following for this reaction is
- (A) Decrease in the total pressure will result in formation of more moles of gaseous X
- (B) At the start of the reaction, dissociation of gaseous X_2 takes place spontaneously
- (C) $\beta_{\text{equilibrium}} = 0.7$
- (D) $K_c < 1$
8. The standard state Gibbs free energies of formation of C(graphite) and C(diamond) at $T = 298 \text{ K}$ are [JEE 2017]

$$\Delta_f G^\circ [\text{C(graphite)}] = 0 \text{ kJ mol}^{-1}$$

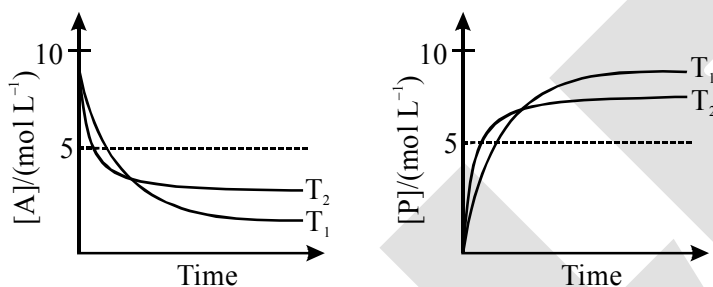
$$\Delta_f G^\circ [\text{C(diamond)}] = 2.9 \text{ kJ mol}^{-1}$$

The standard state means that the pressure should be 1 bar, and substance should be pure at a given temperature. The conversion of graphite [C(graphite)] to diamond [C(diamond)] reduces its volume by $2 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}$. If C(graphite) is converted to C(diamond) isothermally at $T = 298 \text{ K}$, the pressure at which C(graphite) is in equilibrium with C(diamond), is

[Useful information : $1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$; $1 \text{ Pa} = 1 \text{ kg m}^{-1} \text{ s}^{-2}$; $1 \text{ bar} = 10^5 \text{ Pa}$]

- (A) 14501 bar (B) 29001 bar (C) 58001 bar (D) 1405 bar

9. For a reaction taking place in a container in equilibrium with its surroundings, the effect of temperature on its equilibrium constant K in terms of change in entropy is described by
- With increase in temperature, the value of K for exothermic reaction decreases because the entropy change of the system is positive
 - With increase in temperature, the value of K for endothermic reaction increases because unfavourable change in entropy of the surroundings decreases
 - With increase in temperature, the value of K for exothermic reaction decreases because favourable change in entropy of the surroundings decreases
 - With increase in temperature, the value of K for endothermic reaction increases because the entropy change of the system negative
10. For a reaction, $A \rightleftharpoons P$, the plots of $[A]$ and $[P]$ with time at temperatures T_1 and T_2 are given below. [JEE 2018]



If $T_2 > T_1$, the correct statement(s) is (are)

(Assume ΔH° and ΔS° are independent of temperature and ratio of $\ln K$ at T_1 to $\ln K$ at T_2 is greater than T_2/T_1 . Here H , S , G and K are enthalpy, entropy, Gibbs energy and equilibrium constant, respectively.)

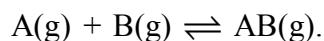
- $\Delta H^\circ < 0$, $\Delta S^\circ < 0$
 - $\Delta G^\circ < 0$, $\Delta H^\circ > 0$
 - $\Delta G^\circ < 0$, $\Delta S^\circ < 0$
 - $\Delta G^\circ < 0$, $\Delta S^\circ > 0$
11. The surface of copper gets tarnished by the formation of copper oxide. N_2 gas was passed to prevent the oxide formation during heating of copper at 1250 K. However, the N_2 gas contains 1 mole % of water vapour as impurity. The water vapour oxidises copper as per the reaction given below
- $$2\text{Cu(s)} + \text{H}_2\text{O(g)} \rightarrow \text{Cu}_2\text{O(s)} + \text{H}_2\text{(g)}$$
- p_{H_2} is the minimum partial pressure of H_2 (in bar) needed to prevent the oxidation at 1250 K. The value of $\ln(p_{\text{H}_2})$ is ____.

(Given : total pressure = 1 bar, R (universal gas constant) = $8 \text{ JK}^{-1}\text{mol}^{-1}$, $\ln(10) = 2.3$. Cu(s) and $\text{Cu}_2\text{O(s)}$ are mutually immiscible.

At 1250 K : $2\text{Cu(s)} + 1/2\text{O}_2\text{(g)} \rightarrow \text{Cu}_2\text{O(s)}$; $\Delta G^\circ = -78,000 \text{ J mol}^{-1}$

$\text{H}_2\text{(g)} + 1/2\text{O}_2\text{(g)} \rightarrow \text{H}_2\text{O(g)}$; $\Delta G^\circ = -1,78,000 \text{ J mol}^{-1}$; G is the Gibbs energy) [JEE 2018]

12. Consider the following reversible reaction, [JEE 2018]



The activation energy of the backward reaction exceeds that of the forward reaction by $2RT$ (in J mol^{-1}). If the pre-exponential factor of the forward reaction is 4 times that of the reverse reaction, the absolute value of ΔG° (in J mol^{-1}) for the reaction at 300 K is ____.

(Given ; $\ln(2) = 0.7$, $RT = 2500 \text{ J mol}^{-1}$ at 300 K and G is the Gibbs energy)

ANSWER KEY

EXERCISE (S-I)

1. Ans. 117°C, 52°C
2. Ans. 197°C
3. Ans. (4)
4. Ans. (i) $\Delta S_{\text{syst}} = 28.72 \text{ J/K}$; $\Delta S_{\text{surr}} = -28.72 \text{ J/K}$; $\Delta S_{\text{total}} = 0$
(ii) $\Delta S_{\text{syst}} = 28.72 \text{ J/K}$; $\Delta S_{\text{surr}} = -11.22 \text{ J/K}$, $\Delta S_{\text{total}} = 17.50 \text{ J/K}$
5. Ans. (a) -250 J/K (b) 333.33 J/K (c) $+83.33 \text{ J/K}$. Process is spontaneous.
6. Ans.(5)
7. Ans (200)
8. Ans. 10.325
9. Ans. (i) -90.5 (ii) -374.5 (iii) -3.26 (all in $\text{J mol}^{-1} \text{ K}^{-1}$)
10. Ans. [A] - II ; [B] - II ; [C] - II ; [D] - II
11. Ans. -3.7 kJ
11. Ans 436 kJ
13. Ans. 5.968 kcal
14. Ans.(-7)
15. Ans. (50)
16. Ans.
(i) $\Delta G = 0$
(ii) $\Delta S = -5 \text{ J/mole.K}$
(iii) $\Delta S_{\text{total}} = +1.66 \text{ J/ mole. K}$; process is spontaneous
(iv) $\Delta S_{\text{total}} = -\text{J/mole. K} = \Delta G = +250 \text{ J/mole. K}$]
17. Ans. -997.68 J/mol
18. Ans. $\Delta H^\circ = 9.04 \text{ kJ/mol}$; $\Delta S^\circ = -8.64 \text{ J/mol}^{-1} \text{ K}^{-1}$
19. Ans. 26.7

EXERCISE-(S-II)

- | | | |
|--|------------|---|
| 1. Ans. (i) 30 kJ | (ii) 60 kJ | (iii) 100 J/K |
| 2. Ans. 30 kJ/mole | | |
| 3. Ans. -1745.94 J/mol ; -6983.76 J/mol and 13.97 kJ/mol | | |
| 4. Ans. (16 J/K) | | 5. Ans. (-57.75 kJ/ mol) |
| 6. Ans. (17.8) | | 7. Ans.(3360) |
| 8. Ans 10 J/K | | 9. Ans. $205.08 \text{ JK}^{-1}\text{mol}^{-1}$ |
| 10. Ans.(5) | | |

EXERCISE (O-I)

- | | | | |
|-------------|--------------|--------------|-------------|
| 1. Ans.(C) | 2. Ans.(D) | 3. Ans.(C) | 4. Ans.(D) |
| 5. Ans.(C) | 6. Ans.(B) | 7. Ans.(A) | 8. Ans.(B) |
| 9. Ans.(B) | 10. Ans.(B) | 11. Ans.(A) | 12. Ans.(A) |
| 13. Ans.(C) | 14. Ans.(D) | 15. Ans.(C) | 16. Ans.(B) |
| 17. Ans.(A) | 18. Ans. (A) | 19. Ans. (C) | 20. Ans.(B) |

EXERCISE (O-II)

- | | | | |
|--|-----------------|-----------------|-----------------|
| 1. Ans.(D) | 2. Ans.(D) | 3. Ans.(A) | 4. Ans.(B) |
| 5. Ans.(A) | 6. Ans.(A) | 7. Ans.(A) | 8. Ans.(C) |
| 9. Ans.(A) | 10. Ans.(A) | 11. Ans.(B,D) | 12. Ans.(A) |
| 13. Ans. (A,C,D) | 14. Ans.(A,C) | 15. Ans.(A,B,D) | 16. Ans.(B,C,D) |
| 17. Ans.[A,C,D] | 18. Ans.[A,B,C] | 19. Ans.[C] | 20. Ans.[B,C,D] |
| 21. Ans.(A) | 22. Ans.(C) | 23. Ans.(D) | 24. Ans.(D) |
| 25. Ans.(B) | 26. Ans.(C) | 27. Ans.(D) | 28. Ans.(C) |
| 29. Ans. (A) - S ; (B) - P,R ; (C) - P, S (D) - P, R | | | |
| 30. Ans. (A) - P,S,R ; (B) - P,R ; (C) - P (D) - Q, R, S | | | |

EXERCISE (J-MAIN)

- | | | | |
|------------|------------|------------|------------|
| 1. Ans.(3) | 2. Ans.(4) | 3. Ans.(4) | 4. Ans.(1) |
| 5. Ans.(1) | 6. Ans.(4) | 7. Ans.(2) | |

EXERCISE (J-ADVANCE)

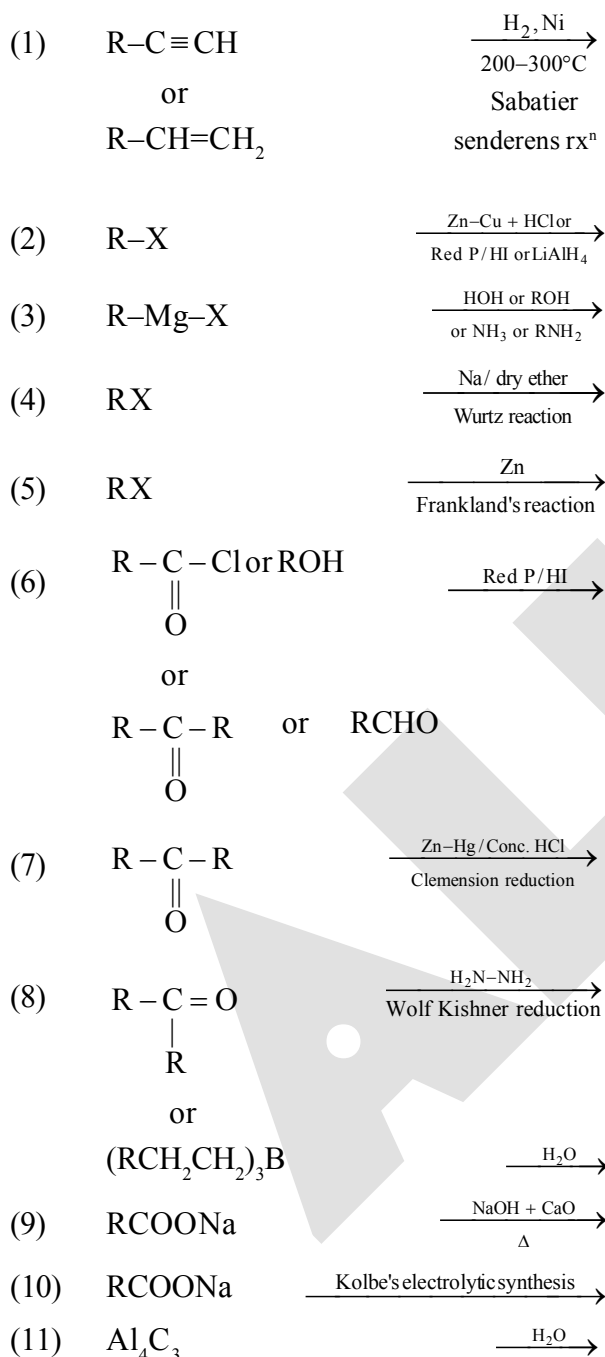
- | | | | |
|---|---------------|-----------------|----------------|
| 1. Ans.(A)→(p, r, s) ; (B)→(r, s) ; (C)→(t) ; (D)→(p, q, t) | | | |
| 2. Ans.(A) , (C) | 3. Ans. (B) | | |
| 4. Ans. (A) → R, T ; (B) → P, Q, S ; (C) → P, Q, S ; (D) → P, Q, S, T | | | |
| 5. Ans. (C) | 6. Ans.(B) | 7. Ans.(C) | 8. Ans.(A) |
| 9. Ans.(BC) | 10. Ans.(A,C) | 11. Ans.(−14.6) | 12. Ans.(8500) |

HYDROCARBON

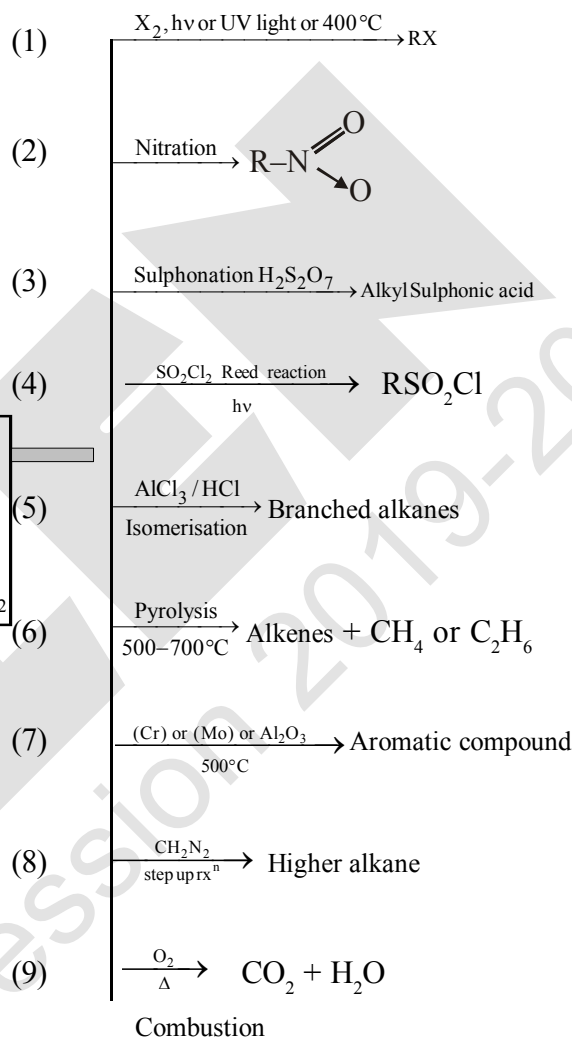
REACTION CHART FOR ALKANES

GMP

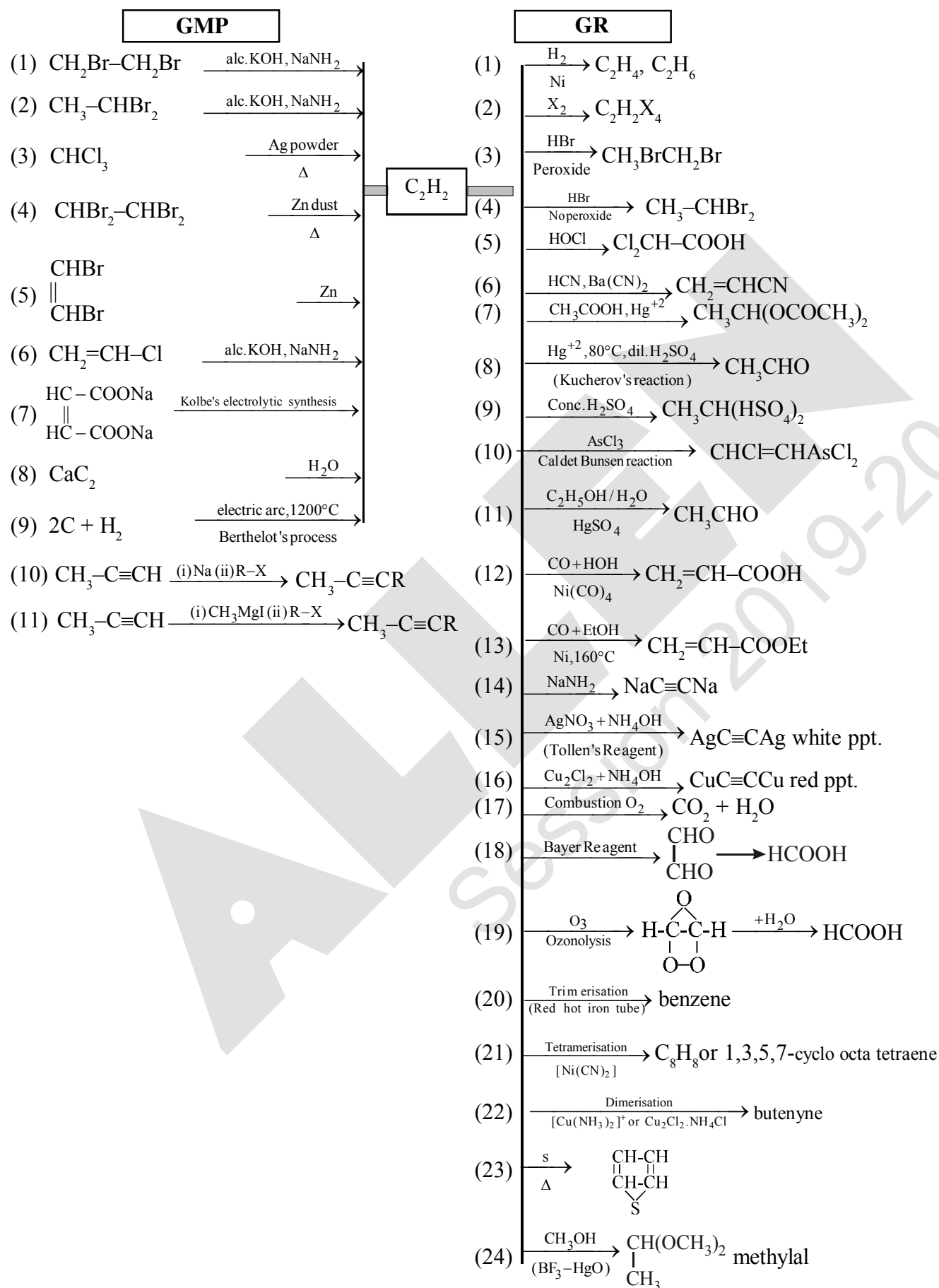
GR



R-H
or
R-R
or
 C_nH_{2n+2}



REACTION CHART FOR ALKYNES



EXERCISE # I

1. During the preparation of ethane by Kolbe's electrolytic method using inert electrodes the pH of the electrolyte –
 (A) Increases progressively as the reaction proceeds
 (B) Decreases progressively as the reaction proceeds
 (C) Remains constant throughout the reaction
 (D) May decrease if the concentration of the electrolyte is not very high

2. $\text{BrCH}_2\text{---CH}_2\text{---CH}_2\text{Br}$ reacts with Na in the presence of ether at 100°C to produce –

- (A) $\text{BrCH}_2\text{---CH=CH}_2$ (B) $\text{CH}_2\text{=C=CH}_2$ (C) $\begin{array}{c} \text{CH}_2\text{---CH}_2 \\ \diagup \quad \diagdown \\ \text{CH}_2 \end{array}$ (D) All of these

3. How many products will be formed excluding stereo when cis-1,3,5-trimethyl cyclohexene reacts with NBS?

- (A) 3 (B) 4 (C) 5 (D) 6

4. $\text{Cyclohexyl-Br} \xrightarrow{\text{Mg/dry ether}} (\text{X}) \xrightarrow{\text{CH}_3\text{---C(=O)---NH}_2} (\text{Y})$

The structures of (X) and (Y) respectively are

- (A) Cyclohexyl-MgBr ; Cyclohexyl-OH (B) Cyclohexyl-Br ; Cyclohexyl-Mg-OH
 (C) Cyclohexyl-MgBr ; Cyclohexyl (D) $\text{BrMg-Cyclohexyl-MgBr}$; HO-Cyclohexyl-OH

5. When n-butane is heated in the presence of AlCl_3/HCl it will be converted into –
 (A) Ethane (B) Propane (C) Butene (D) Isobutane
6. How many dibromo derivatives are formed when bromine is added to 3-Methyl Cyclohexene in 1,2-dichloroethane.
 (A) 2 (B) 3 (C) 4 (D) 6

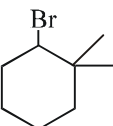
7. $\begin{array}{c} \text{H} \\ | \\ \text{Ph---C---CH}_3 \\ | \\ \text{H}_3\text{C---C---Ph} \\ | \\ \text{OCOR} \end{array} \xrightarrow{\Delta} \text{P(Alkene)} \xrightarrow[\text{CCl}_4]{\text{Br}_2} \xrightarrow[\text{Acetone}]{\text{NaI}} \text{Q(Alkene)}$

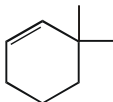
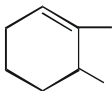
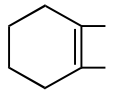
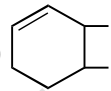
Alkene (P) & (Q) respectively are

- (A) Both $\begin{array}{c} \text{Ph} \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H}_3\text{C} \quad \text{Ph} \end{array}$ (B) $\begin{array}{c} \text{Ph} \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H}_3\text{C} \quad \text{Ph} \end{array}$, $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{Ph} \quad \text{Ph} \end{array}$
 (C) Both $\begin{array}{c} \text{H} \\ | \\ \text{Ph---C---CH}_3 \\ | \\ \text{Ph---C} \\ | \\ \text{CH}_2 \end{array}$ (D) Both $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{Ph} \quad \text{Ph} \end{array}$

8. Anti-Markownikoff's addition of HBr is not observed in –

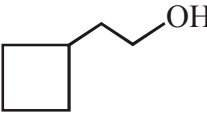
- (A) Propene (B) But-2-ene (C) But-1-ene (D) Pent-2-ene

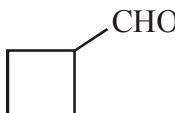
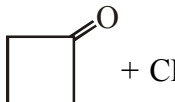
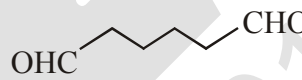
9.  $\xrightarrow[\Delta]{\text{alcoholic KOH}}$ Product ; Product is:

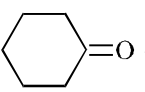
- (A)  (B)  (C)  (D) 

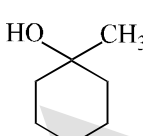
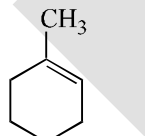
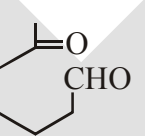
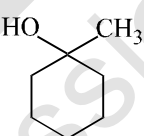
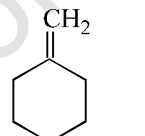
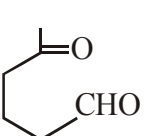
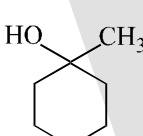
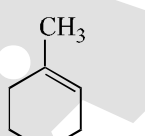
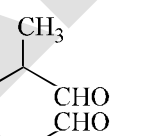
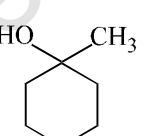
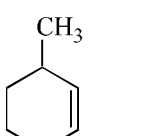
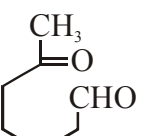
10. Which is expected to react most readily with bromine –

- (A) $\text{CH}_3\text{CH}_2\text{CH}_3$ (B) $\text{CH}_2=\text{CH}_2$ (C) $\text{CH}\equiv\text{CH}$ (D) $\text{CH}_3-\text{CH}=\text{CH}_2$

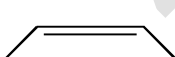
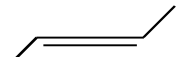
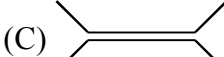
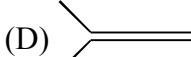
11.  $\xrightarrow[\Delta]{\text{H}^+}$ A $\xrightarrow[\text{(ii) Zn/H}_2\text{O}]{\text{(i) O}_3}$ Product(s). Product is/are?

- (A)  + HCHO (B) 
(C)  + CH_3CHO (D) 

12.  + $\text{CH}_3\text{MgBr} \xrightarrow{\text{H}_3\text{O}^+}$ A $\xrightarrow[\Delta]{\text{Con. H}_2\text{SO}_4}$ B $\xrightarrow[\text{(ii) Zn, H}_2\text{O}]{\text{(i) O}_3}$ C ; A, B and C are –

- (A)    (B)   
(C)    (D)   

13. Which has least heat of hydrogenation –

- (A)  (B)  (C)  (D) 

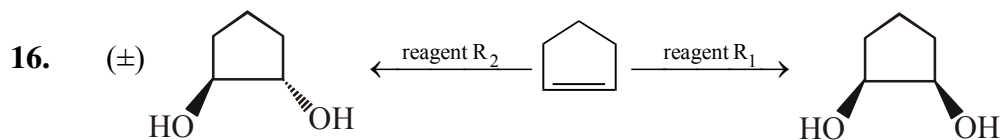
14. For the ionic reaction of hydrochloric acid with the following alkenes, predict the correct sequence of reactivity as measured by reaction rates:

- (I) $\text{ClCH}=\text{CH}_2$ (II) $(\text{CH}_3)_2\text{C}=\text{CH}_2$ (III) $\text{OHC}\cdot\text{CH}=\text{CH}_2$ (IV) $(\text{NC})_2\text{C}=\text{C}(\text{CN})_2$
(A) $\text{IV} > \text{I} > \text{III} > \text{II}$ (B) $\text{I} > \text{IV} > \text{II} > \text{III}$ (C) $\text{III} > \text{II} > \text{IV} > \text{I}$ (D) $\text{II} > \text{I} > \text{III} > \text{IV}$

15. $\text{CH}_3\text{-CH=CH-CH}_3 \xrightarrow{\text{X}}$ Product is Y (non-resolvable) then X can be –

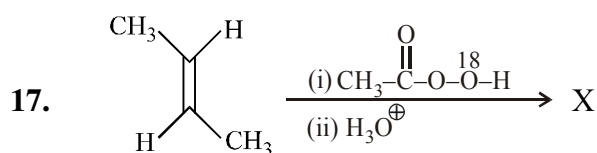
cis

- (A) Br_2 water (B) HCO_3H
(C) Cold alkaline KMnO_4 (D) all of the above

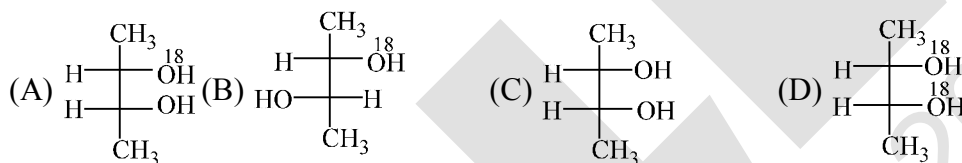


R_1 and R_2 respectively are –

- (A) Cold alkaline KMnO_4 , $\text{OsO}_4/\text{H}_2\text{O}_2$ (B) Cold alkaline KMnO_4 , HCO_3H
(C) Cold alkaline KMnO_4 , $\text{CH}_3\text{-O-O-CH}_3$ (D) $\text{C}_6\text{H}_5\text{CO}_3\text{H}$, HCO_3H

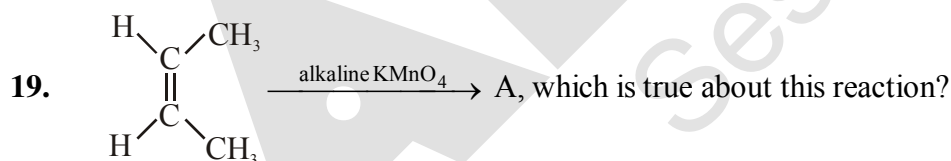


The probable structure of 'X' is



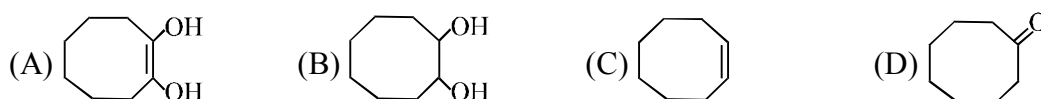
18. Which alkene on heating with alkaline KMnO_4 solution gives acetone and a gas, which turns lime water milky –

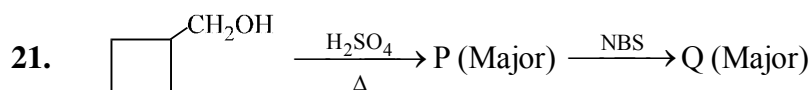
- (A) 2-Methyl-2-butene (B) Isobutylene
(C) 1-Butene (D) 2-Butene



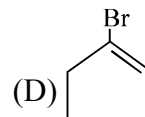
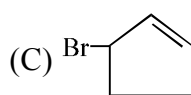
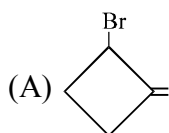
- (A) A is meso 2, 3-butanediol formed by *syn* addition
(B) A is meso 2, 3-butanediol formed by *anti* addition
(C) A is a racemic mixture of d and l, 2, 3-butanediol formed by anti addition
(D) A is a racemic mixture of d and l 2,3-butanediol formed by *syn* addition

20. The reaction of cyclooctyne with HgSO_4 in the presence of a H_2SO_4 gives

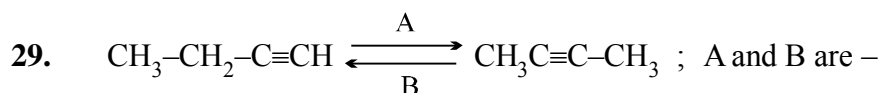




The structure of Q is



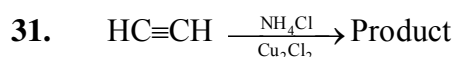
22. Mixture of one mole each of ethene and propyne on reaction with Na will form H_2 gas at S.T.P. –
 (A) 22.4 L (B) 11.2 L (C) 33.6 L (D) 44.8 L
23. Acetylene may be prepared using Kolbe's electrolytic method employing –
 (A) Pot. acetate (B) Pot. succinate (C) Pot. fumarate (D) None of these
24. $B \xleftarrow{\text{Lindlar}} R-C\equiv C-R \xrightarrow{Na/NH_3} A$
 A and B are geometrical isomers –
 (A) A is trans, B is cis (B) A and B both are cis
 (C) A and B both are trans (D) A is cis, B is trans
25. A mixture of CH_4 , C_2H_4 and C_2H_2 gaseous are passed through a Wolf bottle containing ammonical cuprous chloride. The gas coming out is
 (A) Methane (B) Acetylene
 (C) Mixture of methane and ethylene (D) original mixture
26. $B \xleftarrow[H_2O_2, OH^-]{BH_3, THF} CH_3-C\equiv CH \xrightarrow{HgSO_4/H_2SO_4} A$
 A and B are –
 (A) CH_3CH_2CHO , $CH_3-C(=O)-CH_3$ (B) $CH_3-C(=O)-CH_3$, CH_3CH_2CHO
 (C) CH_3CH_2CHO (both) (D) $CH_3-C(=O)-CH_3$ (both)
27. Which of the following reagents cannot be used to locate the position of triple bond in $CH_3-C\equiv C-CH_3$
 (A) Br_2/CCl_4 (B) O_3/H_2O (C) Cu_2Cl_2/NH_4OH (D) $KMnO_4/H^+$
28. In the presence of strong bases, triple bonds will migrate within carbon skeletons by the
 (A) removal of protons (B) addition of protons
 (C) removal and readdition of protons (D) addition and removal of protons.



- (A) alcoholic KOH and NaNH_2 (B) NaNH_2 and alcoholic KOH
(C) NaNH_2 and Lindlar catalyst (D) Lindlar and NaNH_2 catalyst

30. If a mixture of iso-octane (70%) & n-heptane (30%) is present in sample. The octane number of this sample is :

- (A) 40 (B) 70 (C) 30 (D) 85



Product is –

- (A) $\text{Cu--C}\equiv\text{C--Cu}$ (B) $\text{H}_2\text{C=CH--C}\equiv\text{CH}$ (C) $\text{HC}\equiv\text{C--Cu}$ (D) $\text{Cu--C}\equiv\text{C--NH}_4$

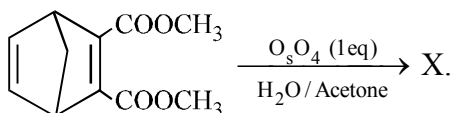
32. Which of the following process is not good for the preparation of open chain alkane having odd number of carbons :

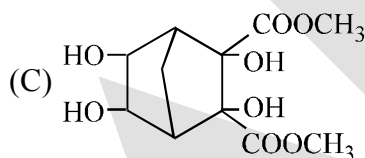
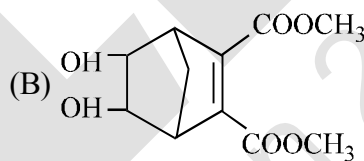
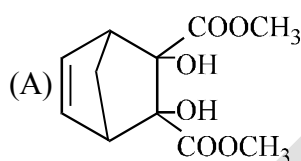
- (A) Wurtz process (B) Kolbe electrolysis
(C) Corey house synthesis (D) Both (A) & (B)

EXERCISE # II

- How many moles of O_2 required for complete combustion of one mole of propane –
(A) 7 (B) 5 (C) 16 (D) 10
- How much volume of air will be needed for complete combustion of 10 lit. of ethane –
(Assuming that approx 20% O_2 is present in air)
(A) 135 lit. (B) 35 lit. (C) 175 lit. (D) 205 lit.

- $CH_3CH=CH_2 \xrightarrow[H_2O_2/OH^-]{B_2D_6}$ product X ; X is –
(A) $CH_3-\underset{\substack{| \\ OH}}{CH}-CH_2D$ (B) $CH_3-\underset{\substack{| \\ D}}{CH}-CH_2OH$
(C) $CH_3-\underset{\substack{| \\ OD}}{CH}-CH_3$ (D) none is correct

-  Identify 'X'.



(D) Reaction will not occur

- Aqueous solution of potassium propanoate is electrolysed. Possible organic products are:
(A) n-Butane (B) $C_2H_5COOC_2H_5$
(C) CH_3-CH_3 (D) $CH_2=CH_2$

- $Ph-\overset{\overset{O}{||}}{C}-CH_3 \xrightarrow{A} Ph-CH_2-CH_3$

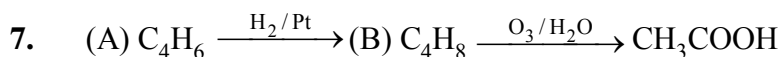
A could be:

(A) NH_2NH_2 , glycol/ OH^- , Δ

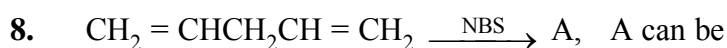
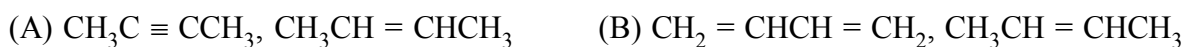
(B) $Na(Hg)/conc. HCl$

(C) Red P/HI

(D) CH_2-CH_2 ; Raney Ni, H_2 , Δ
 $\begin{array}{cc} | & | \\ SH & SH \end{array}$



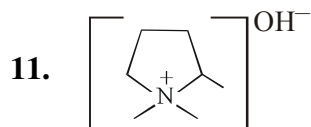
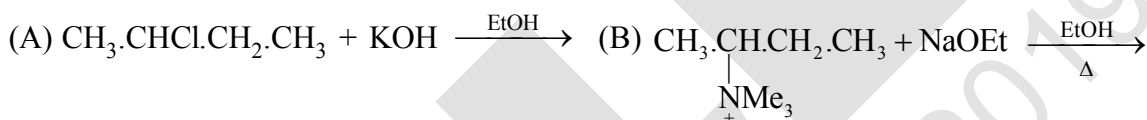
Hence A and B are



9. An alkene on ozonolysis yields only ethanal. The ozonolysis of isomer of this alkene yields :



10. Which of the following elimination reactions will occur to give but-1-ene as the major product?



The above compound undergoes elimination on heating can yield which of the following products?



12. Which of the following will give same product with HBr in presence or absence of peroxide.

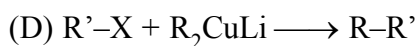
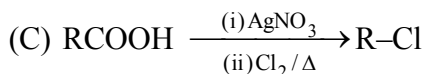
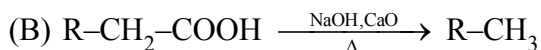
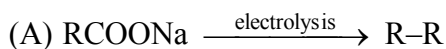


13. The ionic addition of HCl to which of the following compounds will produces a compound having Cl on carbon next to terminal.



6. Match the column

Column I



Column II

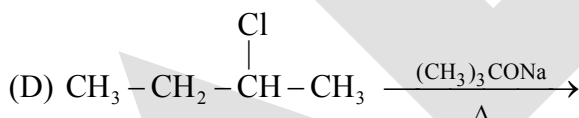
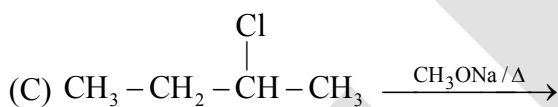
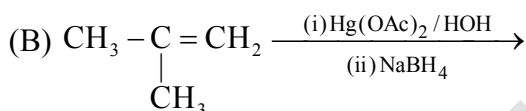
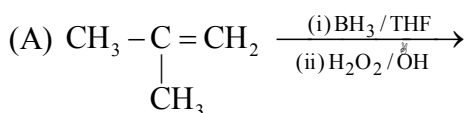
(P) Correy-House reaction

(Q) Kolbe electrolysis

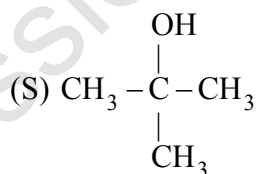
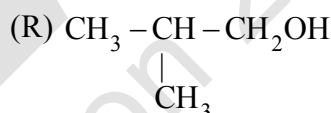
(R) Oakwood degradation / Soda lime process

(S) Hunsdiecker reaction

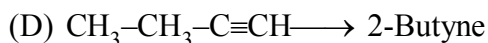
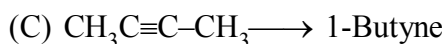
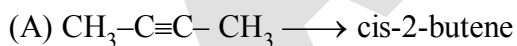
7. Column I



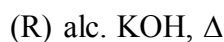
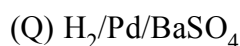
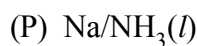
Column II



8. Column I

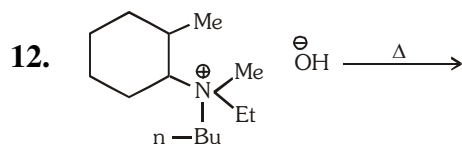


Column II



EXERCISE # (IV) (J-MAINS)

1. Which of these will not react with acetylene - [AIEEE-2002]
 (1) NaOH (2) ammonical AgNO_3 (3) Na (4) HCl
 2. What is the product formed when acetylene reacts with hypochlorous acid - [AIEEE-2002]
 (1) CH_3COCl (2) ClCH_2CHO (3) Cl_2CHCHO (4) ClCH_2COOH
 3. 1-Butene may be converted to butane by reaction with - [AIEEE-2003]
 (1) Pd/H_2 (2) $\text{Zn} - \text{HCl}$ (3) $\text{Sn} - \text{HCl}$ (4) $\text{Zn} - \text{Hg/HCl}$
 4. On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be - [AIEEE-2003]
 (1) neopentane (2) propane (3) pentane (4) isopentane
 5. Which one of the following is reduced with Zn-Hg/HCl to give the corresponding hydrocarbon [AIEEE-2004]
 (1) Butan-2-one (2) Acetic acid
 (3) Acetamide (4) Ethyl acetate
 6. Which one of the following has the minimum boiling point : [AIEEE-2004]
 (1) isobutane (2) 1-butyne (3) 1-butene (4) n-butane
 7. 2-Methylbutane on reacting with bromine in the presence of sunlight gives mainly [AIEEE-2005]
 (1) 2-bromo-2-methylbutane (2) 1-bromo-2-methylbutane
 (3) 1-bromo-3-methylbutane (4) 2-bromo-3-methylbutane
 8. Alkyl halides react with dialkyl copper reagent to give [AIEEE-2005]
 (1) alkyl copper halides (2) alkenes
 (3) alkenyl halides (4) alkanes
- Reaction of one molecule of HBr with one molecule of 1,3-butadiene at 40°C gives predominantly
- (1) 1-bromo-2-butene under thermodynamically controlled conditions [AIEEE-2005]
 - (2) 3-bromobutene under kinetically controlled conditions
 - (3) 1-bromo-2-butene under kinetically controlled conditions
 - (4) 3-bromobutene under thermodynamically controlled conditions
10. Acid catalyzed hydration of alkenes except ethene leads to the formation of [AIEEE-2005]
 (1) secondary or tertiary alcohol
 (2) primary alcohol
 (3) mixture of secondary and tertiary alcohols
 (4) mixture of primary and secondary alcohols
 11. Elimination of bromine from 2-bromobutane results in the formation of [AIEEE-2005]
 (1) predominantly 2-butene (2) equimolar mixture of 1 and 2-butene
 (3) predominantly 2-butyne (4) predominantly 1-butene



The alkene formed as a major product in the above elimination reaction is- [AIEEE-2006]



13. Reaction of trans-2-phenyl-1-bromocyclopentane on reaction with alcoholic KOH produces-

- (1) 4-phenyl cyclopentene (2) 2-phenyl cyclopentene [AIEEE-2006]
(3) 1-phenyl cyclopentene (4) 3-phenyl cyclopentene

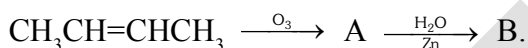
14. Phenyl magnesium bromide reacts with methanol to give- [AIEEE-2006]

- (1) A mixture of anisole and $\text{Mg}(\text{OH})\text{Br}$ (2) A mixture of benzene and $\text{Mg}(\text{OMe})\text{Br}$
(3) A mixture of toluene and $\text{Mg}(\text{OH})\text{Br}$ (4) A mixture of phenol and $\text{Mg}(\text{Me})\text{Br}$

15. Which of the following reactions will yield, 2, 2-dibromopropane [AIEEE-2007]

- (1) $\text{CH}_3-\text{C}\equiv\text{CH} + 2\text{HBr} \longrightarrow$ (2) $\text{CH}_3\text{CH}=\text{CHBr} + \text{HBr} \longrightarrow$
(3) $\text{CH}\equiv\text{CH} + 2\text{HBr} \longrightarrow$ (4) $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{HBr} \longrightarrow$

16. In the following sequence of reactions, the alkene affords the compound 'B' :- [AIEEE-2008]



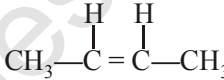
The compound B is

- (1) $\text{CH}_3\text{CH}_2\text{CHO}$ (2) CH_3COCH_3 (3) $\text{CH}_3\text{CH}_2\text{COCH}_3$ (4) CH_3CHO

17. The hydrocarbon which can react with sodium in liquid ammonia is [AIEEE-2008]

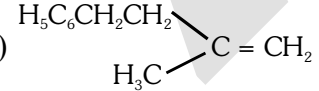
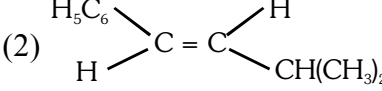
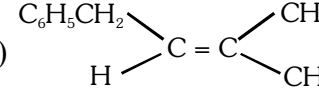
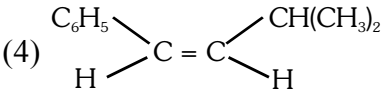
- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CCH}_2\text{CH}_2\text{CH}_3$ (2) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$
(3) $\text{CH}_3\text{CH}=\text{CHCH}_3$ (4) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CCH}_2\text{CH}_3$

18. The treatment of CH_3MgX with $\text{CH}_3\text{C}\equiv\text{C}-\text{H}$ produces [AIEEE-2008]

- (1) $\text{CH}_3-\text{CH}=\text{CH}_2$ (2) $\text{CH}_3\text{C}\equiv\text{C}-\text{CH}_3$ (3)  (4) CH_4

19. The main product of the following reaction is [AIEEE-2010]



- (1)  (2) 
(3)  (4) 

20. One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44 u. The alkene is :- [AIEEE-2010]

- (1) Ethene (2) Propene (3) 1-Butene (4) 2-Butene

21. Ozonolysis of an organic compound gives formaldehyde as one of the products. This confirms the presence of :- [AIEEE-2011]

- (1) an isopropyl group (2) an acetylenic triple bond
(3) two ethylenic double bonds (4) a vinyl group

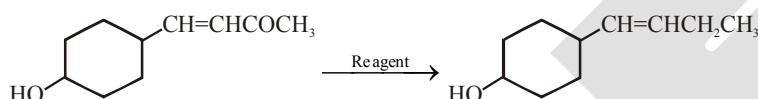
22. Ozonolysis of an organic compound 'A' produces acetone and propionaldehyde in equimolar mixture. Identify 'A' from the following compounds :- [AIEEE-2011]

- (1) 2-Methyl-1-pentene (2) 1-Pentene
(3) 2-Pentene (4) 2-Methyl-2-pentene

23. 2-Hexyne gives trans-2-Hexene on treatment with :- [AIEEE-2012]

- (1) LiAlH_4 (2) Pt/H_2 (3) Li/NH_3 (4) Pd/BaSO_4

24. In the given transformation, which of the following is the most appropriate reagent ? [AIEEE-2012]



- (1) NaBH_4 (2) $\text{NH}_2\text{NH}_2 \cdot \text{OH}^-$ (3) $\text{Zn} - \text{Hg} / \text{HCl}$ (4) $\text{Na}, \text{Liq. NH}_3$

25. The major organic compound formed by the reaction of 1, 1, 1-trichloroethane with silver powder is :- (JEE MAIN OFFLINE 2013)

- (1) 2-Butyne (2) 2-Butene (3) Acetylene (4) Ethene

26. The number and type of bonds in C_2^{2-} ion in CaC_2 are: (JEE MAIN ONLINE 2014)

- (1) Two σ bonds and one π - bond (2) Two σ bonds and two π - bonds
(3) One σ bond and two π - bonds (4) One σ bond and one π - bond

27. In the hydroboration - oxidation reaction of propene with diborane, H_2O_2 and NaOH , the organic compound formed is : (JEE MAIN ONLINE 2014)

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (2) $(\text{CH}_3)_3\text{COH}$ (3) $\text{CH}_3\text{CHOHCH}_3$ (4) $\text{CH}_3\text{CH}_2\text{OH}$

28. Which one of the following class of compounds is obtained by polymerization of acetylene ? (JEE MAIN ONLINE 2014)

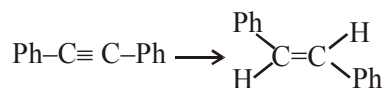
- (1) Poly-ene (2) Poly-yne (3) Poly-amide (4) Poly-ester

29. The gas liberated by the electrolysis of Dipotassium succinate solution is : (JEE MAIN ONLINE 2014)

- (1) Ethyne (2) Ethene (3) Propene (4) Ethane

30. The reagent needed for converting

(JEE MAIN ONLINE 2014)



is :

(1) $\text{H}_2/\text{Lindlar Cat.}$

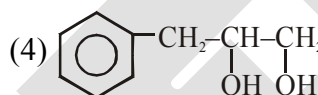
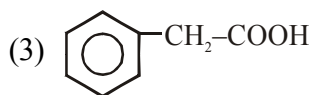
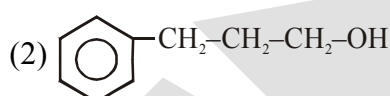
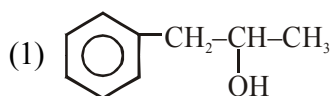
(2) Cat. Hydrogenation

(3) LiAlH_4

(4) Li/NH_3

31. - $\text{CH}_2-\text{CH}=\text{CH}_2$ on mercuration- demercuration produces the major product :-

(Jee Main online 2014)



32. The major product obtained in the photo catalysed bromination of 2-methylbutane is :-

(Jee Main online 2014)

(1) 2-bromo-2-methylbutane

(2) 2-bromo-3-methylbutane

(3) 1-bromo-2-methylbutane

(4) 1-bromo-3-methylbutane

33. In the presence of peroxide, HCl and HI do not give anti-Markownikoff's addition to alkenes because

(Jee Main online 2014)

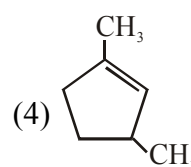
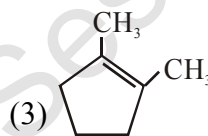
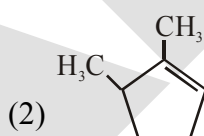
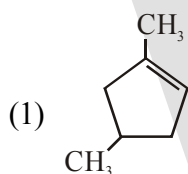
(1) All the steps are exothermic in HCl and HI

(2) One of the steps is endothermic in HCl and HI

(3) HCl is oxidizing and the HI is reducing

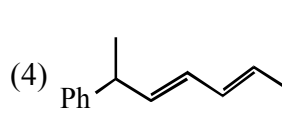
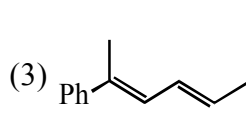
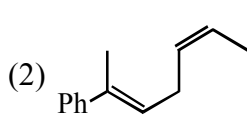
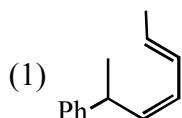
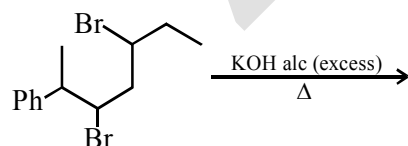
(4) Both HCl and HI are strong acids

34. Which compound would give 5-keto-2-methyl hexanal upon ozonolysis? (Jee Main offline 2015)



35. The major product of the following reaction is

(Jee Main (Jan) 2019)



$$\text{CH}_3 - \text{CH}_2 - \underset{\gamma}{\text{CH}} = \underset{\alpha}{\text{CH}_2}$$

- ClC1=CC(=O)C=CC=C1>>i HBr<>[i](ii) alc. KOH<>O=C1C=CC=CC=C1

- (1)  (2)  (3)  (4) 

CC1=C2CCCCC2C1 + HCl →

- (1)  (2)  (3)  (4) 

$$\text{CH}_3\text{CH}_2\underset{\text{Br}}{\text{CH}}-\underset{\text{Br}}{\text{CH}_2} \xrightarrow[\text{in liq NH}_3]{\begin{array}{l} \text{(i) KOH alc.} \\ \text{(ii) NaNH}_2 \end{array}}$$

- $$\text{CH}_3\text{C}\equiv\text{CH} \xrightarrow[\text{(ii) DI}]{\text{(i) DCl (1 equiv.)}}$$

- code06\B0Al-B0\Kata\JEE(Advanced)\Enthusiast\Chem\Sheet\Hydrocarbon\Eng.p65

42. The major product of the following addition reaction is :

(Jee Main (April) 2019)



- (1) $\text{CH}_3 - \underset{\text{Cl}}{\text{CH}} - \underset{\text{OH}}{\text{CH}_2}$ (2) $\text{H}_3\text{C} - \underset{\text{OH}}{\text{CH}} - \underset{\text{Cl}}{\text{CH}_2}$ (3) $\text{H}_3\text{C} - \triangle_{\text{O}}$ (4) $\text{H}_3\text{C} - \overset{\text{O}}{\parallel} - \text{CH}_3$

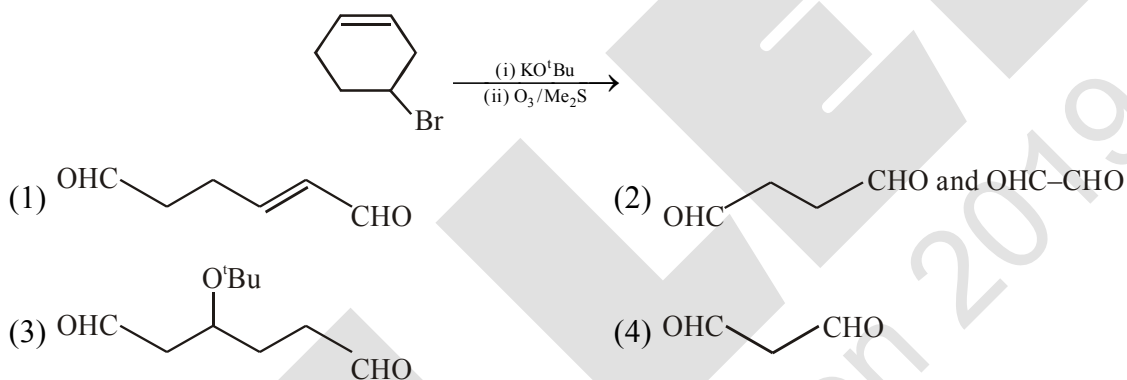
43. But-2-ene on reaction with alkaline KMnO_4 at elevated temperature followed by acidification will give :

(Jee Main (April) 2019)

- (1) one molecule of CH_3CHO and one molecule of CH_3COOH
 (2) $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$
 (3) 2 molecules of CH_3COOH
 (4) 2 molecules of CH_3CHO

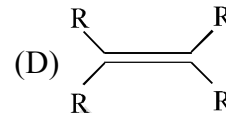
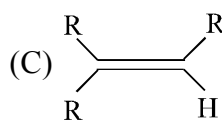
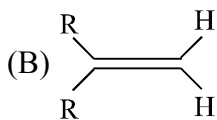
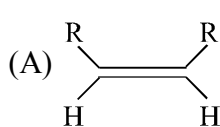
44. The major product(s) obtained in the following reaction is/are :

(Jee Main (April) 2019)

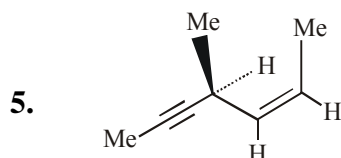


EXERCISE # IV (A) (JEE ADVANCE)

1. Propyne and propene can be distinguished by – [IIT '2000]
 (A) conc. H_2SO_4 (B) Br_2 in CCl_4 (C) dil. KMnO_4 (D) AgNO_3 in ammonia
2. Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition [IIT '2000]

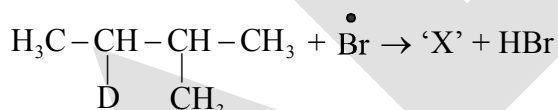


3. In the presence of peroxide, hydrogen chloride and hydrogen iodide do not give anti-Markovnikov addition to alkene because – [IIT S'2001]
 (A) both are highly ionic (B) one is oxidising and the other is reducing
 (C) one of the step is endothermic in both the cases (D) All the steps are exothermic in both cases
4. The reaction of propene with HOCl proceeds via the addition of – [IIT '2001]
 (A) H^+ in first step (B) Cl^+ in first step
 (C) OH^- in first step (D) Cl^+ and OH^- in single step

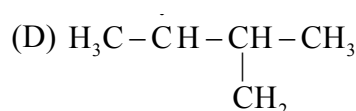
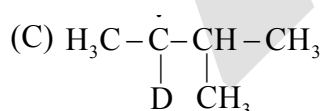
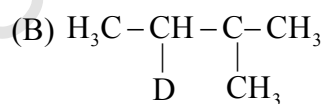
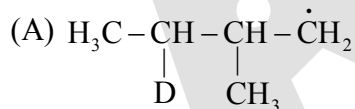


Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives –

- [IIT '2001]
 (A) An optically active compound (B) An optically inactive compound
 (C) A racemic mixture (D) A diastereomeric mixture
6. Consider the following reactions – [IIT '2002]



Identify the structure of the major product 'X'

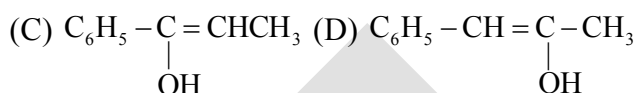
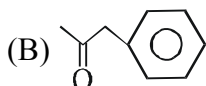
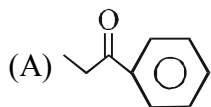


7. The nodal plane in the π -bond of ethene is located in – [IIT '2002]
 (A) The molecular plane
 (B) A plane parallel to the molecular plane
 (C) A plane perpendicular to the molecular plane which contains the carbon-carbon σ -bond at right angle
 (D) A plane perpendicular to the molecular plane which contains the carbon-carbon σ -bond

8. Identify a reagent from the following list which can easily distinguish between 1-butyne and 2-butyne- [IIT '2002]

(A) Bromine, CCl_4 (B) H_2 , Lindlar catalyst
(C) Dilute H_2SO_4 , HgSO_4 (D) Ammonical Cu_2Cl_2 solution

9. $\text{C}_6\text{H}_5-\text{C}\equiv\text{C}-\text{CH}_3 \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4} \text{A}$ [IIT '2003]



10. [IIT '2003]

Number of compounds in X will be:

- (A) 2 (B) 3 (C) 4 (D) 5 [IIT '2004]

11. 2-hexyne can be converted into trans-2-hexene by the action of: [IIT '2004]

(A) H_2 -Pd-BaSO₄ (B) Li in li NH₃ (C) H_2 -PtO₂ (D) NaBH₄

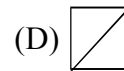
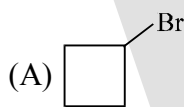
12. Cyclohexene is best prepared from cyclohexanol by which of the following: [IIT '2004]

(A) conc. H_3PO_4 (B) conc. HCl/ZnCl_2 (C) conc. HCl (D) conc. HBr

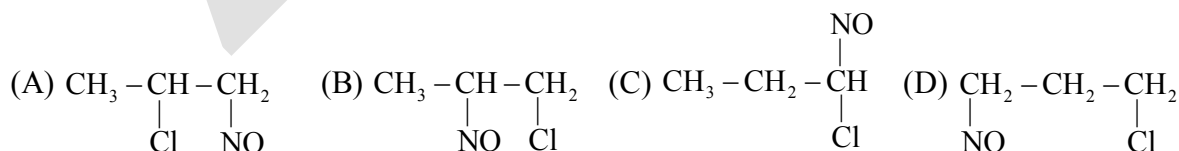
13. When Phenyl Magnesium Bromide reacts with tert. butanol, which of the following is formed?

(A) Tert. butyl methyl ether (B) Benzene [IIT '2005]
(C) Tert. butyl benzene (D) Phenol

14. 1-Bromo-3-chlorocyclobutane when treated with two equivalents of Na, in the presence of ether which of the following will be formed? [IIT '2005]



15. $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{NOCl} \rightarrow \text{P}$ [IIT 2006]
Identify the adduct.



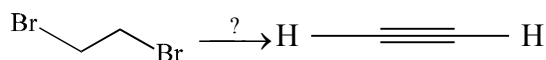
16. The number of stereoisomers obtained by bromination of trans-2-butene is [IIT 2007]

(A) 1 (B) 2 (C) 3 (D) 4

17. The number of structural isomers for C_6H_{14} is [IIT 2007]

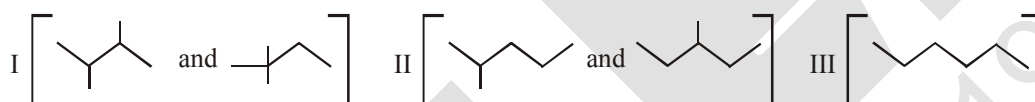
(A) 3 (B) 4 (C) 5 (D) 6

- 18.** The reagent(s) for the following conversion, **[IIT 2007]**



is / are

- (A) alcoholic KOH (B) alcoholic KOH followed by NaNH_2
(C) aqueous KOH followed by NaNH_2 (D) $\text{Zn} / \text{CH}_3\text{OH}$
19. The synthesis of 3-octyne is achieved by adding a bromoalkane into a mixture of sodium amide and an alkyne. The bromoalkane and alkyne respectively are [IIT-2010]
- (A) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$
(B) $\text{BrCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{C} \equiv \text{CH}$
(C) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{C} \equiv \text{CH}$
(D) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$
20. Isomers of hexane, based on their branching, can be divided into three distinct classes as shown in the figure. [IIT-2014]

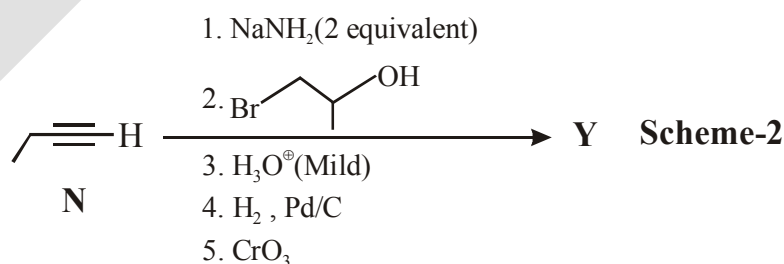
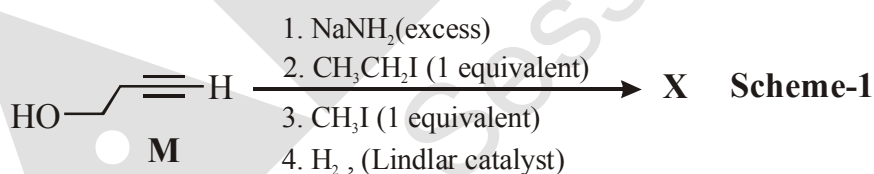


The correct order of their boiling point is

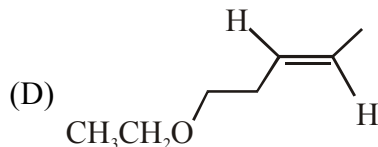
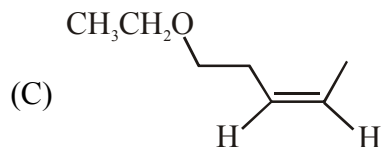
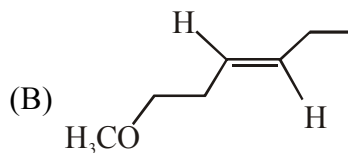
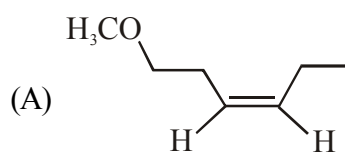
- (A) $I > II > III$ (B) $III > II > I$ (C) $II > III > I$ (D) $III > I > II$

Paragraph For Question 21 and 22

Schemes 1 and 2 describe sequential transformation of alkynes M and N. Consider only the major products formed in each step for both the schemes. [IIT-2014]



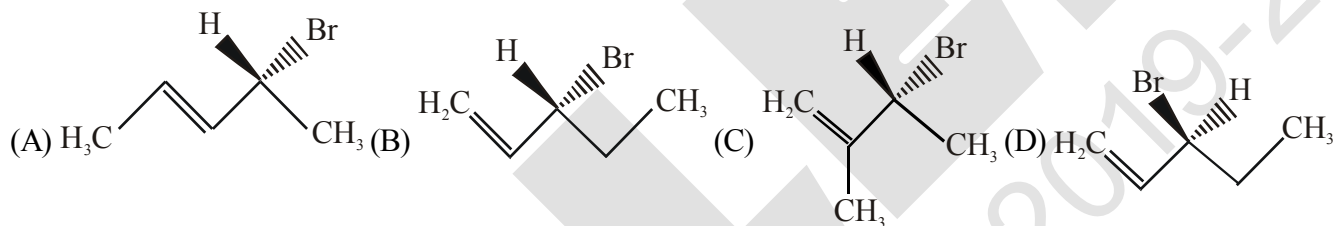
21. The product X is -



22. The correct statement with respect to product Y is -

- (A) It gives a positive Tollens test and is a functional isomer of X
 (B) It gives a positive Tollens test and is a geometrical isomer of X
 (C) It gives a positive Iodoform test and is a functional isomer of X
 (D) It gives a positive Iodoform test and is a geometrical isomer of X

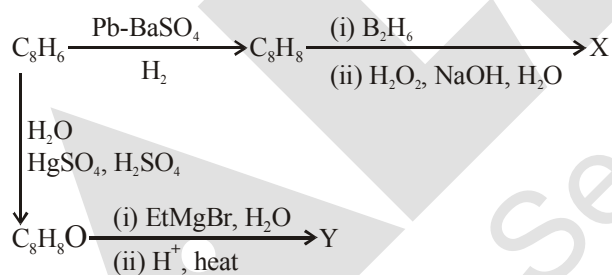
23. Compound(s) that on hydrogenation produce(s) optically inactive compound(s) is (are) [IIT-2015]



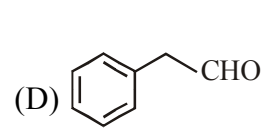
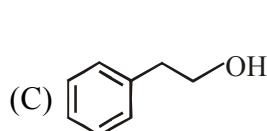
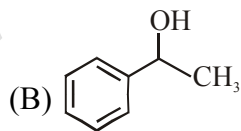
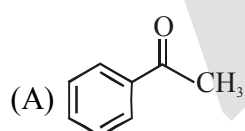
Paragraph For Questions 24 and 25

In the following reaction

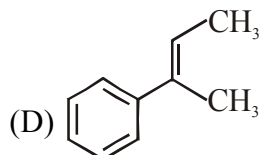
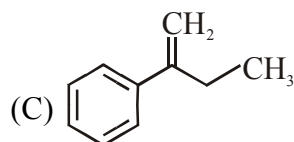
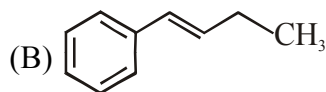
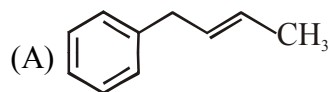
[IIT-2015]



24. Compound X is :

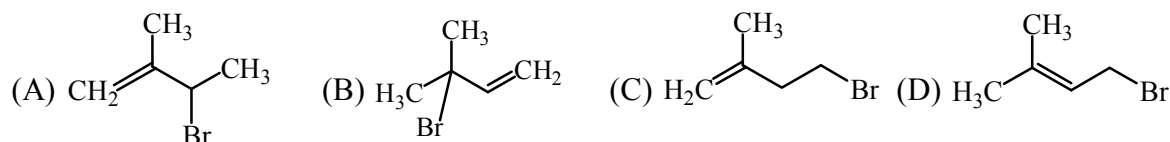
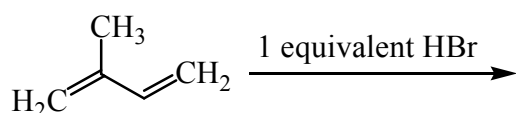


25. The major compound Y is :



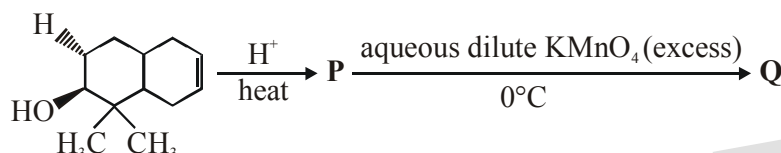
26. In the following reaction, the major product is -

[IIT-2015]



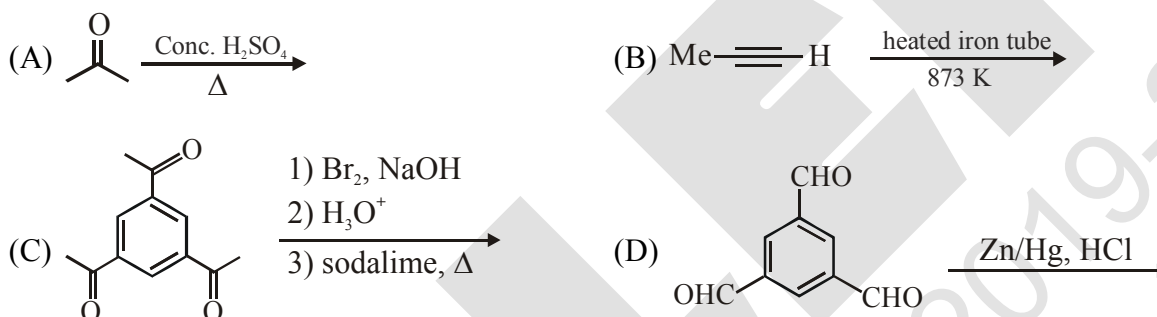
27. The number of hydroxyl group(s) in Q is

[IIT-2015]



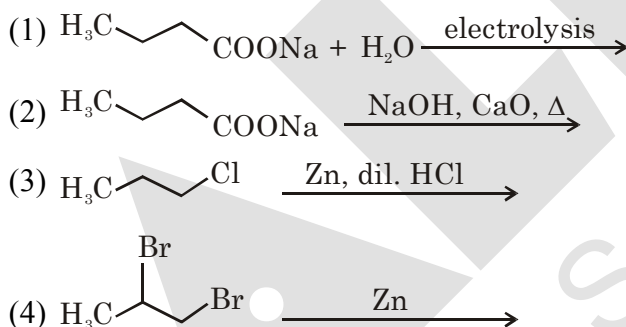
28. The reaction(s) leading to the formation of 1,3,5-trimethylbenzene is (are)

[IIT-2018]

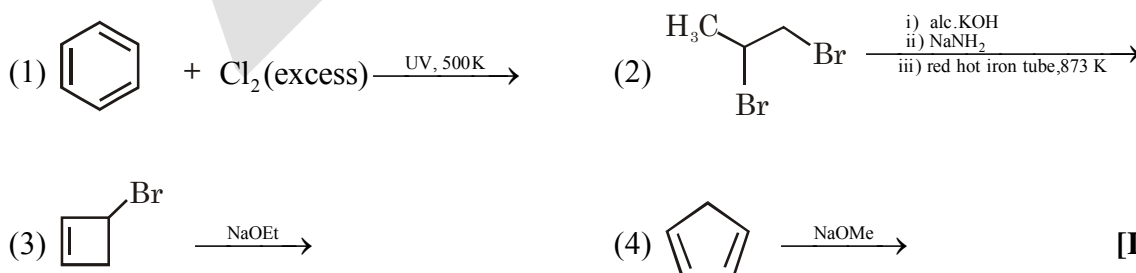


29. Which of the following reactions produce(s) propane as a major product?

[IIT-2019]



30. Choose the correct option(s) that give(s) an aromatic compound as the major product.



[IIT-2019]

ANSWER KEY

EXERCISE # I

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (A) | 2. (C) | 3. (B) | 4. (C) | 5. (D) |
| 6. (C) | 7. (C) | 8. (B) | 9. (A) | 10. (D) |
| 11. (B) | 12. (A) | 13. (C) | 14. (D) | 15. (C) |
| 16. (B) | 17. (A) | 18. (B) | 19. (A) | 20. (D) |
| 21. (C) | 22. (B) | 23. (C) | 24. (A) | 25. (C) |
| 26. (B) | 27. (A) | 28. (C) | 29. (A) | 30. (B) |
| 31. (B) | 32. (D) | | | |

EXERCISE # II

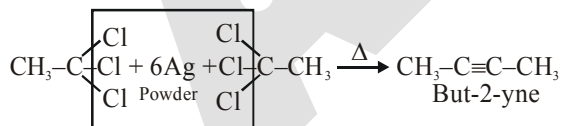
- | | | | | |
|--------------|-----------|-------------|------------|--------------|
| 1. (B) | 2. (C) | 3. (B) | 4. (B) | 5. (A,B,C,D) |
| 6. (A,B,C,D) | 7. (A,B) | 8. (A,B) | 9. (A,B,C) | 10. (B,C) |
| 11. (B,C,D) | 12. (A,C) | 13. (A,B,D) | | |

EXERCISE # III

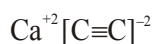
- | | | |
|----------------------------------|------------------------------------|----------|
| 1. (A,B,C,D) | 2. (D) | 3. (A,C) |
| 4. (A)→P ; (B)→R ; (C)→S ; (D)→Q | 5. (A)→S ; (B)→R ; (C)→P ; (D)→R,S | |
| 6. (A)→Q ; (B)→R ; (C)→S ; (D)→P | 7. (A)→R ; (B)→S ; (C)→Q ; (D)→P | |
| 8. (A)→Q ; (B)→P ; (C)→S ; (D)→R | | |

EXERCISE # (IV) (J-MAINS)

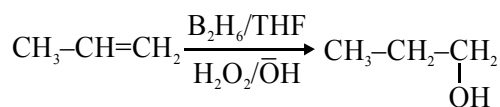
- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (1) | 2. (3) | 3. (1) | 4. (1) | 5. (1) |
| 6. (1) | 7. (1) | 8. (4) | 9. (1) | 10. (1) |
| 11. (1) | 12. (2) | 13. (4) | 14. (2) | 15. (1) |
| 16. (4) | 17. (2) | 18. (4) | 19. (2) | 20. (4) |
| 21. (4) | 22. (4) | 23. (3) | 24. (2) | |
| 25. (1) | | | | |



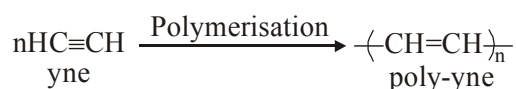
26. (3)



27. (1)



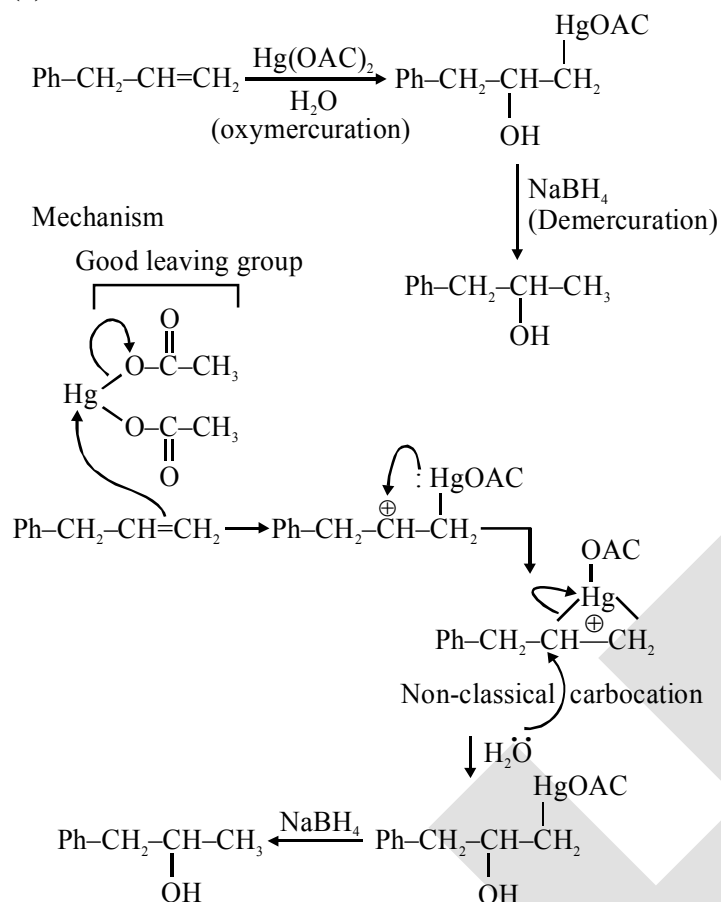
28. (2)



29. (2)

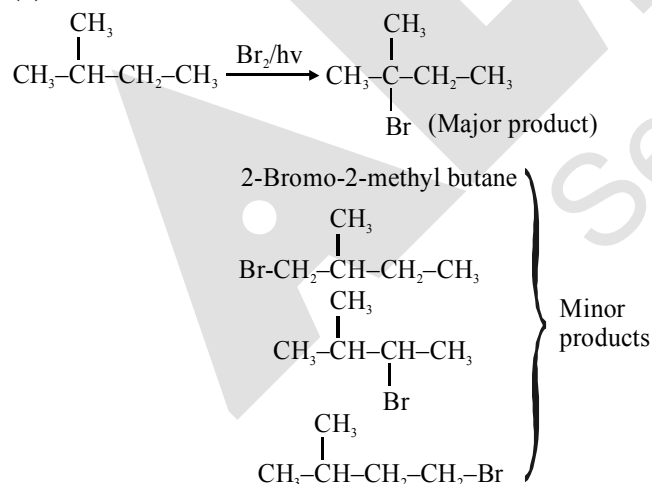
30. (4)

31. (1)



Rearrangement of carbocation formed is not possible due to formation of cyclic non-classical carbocation.

32. (1)



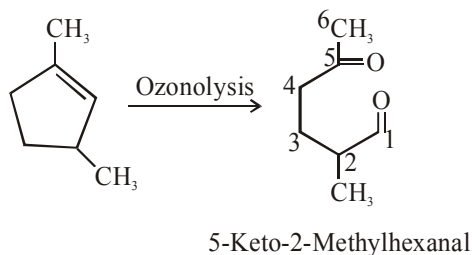
selectivity ratio for bromination is

$1^\circ : 2^\circ : 3^\circ :: 1 : 82 : 1600$

Hence 3° product will be major product.

33. (2)

34. (4)



35. Ans. (3)

36. Ans. (2)

37. Ans. (1)

38. Ans. (1)

39. Ans. (1)

40. Ans. (1)

41. Ans. (4)

42. Ans. (2)

43. Ans. (3)

44. Ans. (2)

EXERCISE # IV (A) (JEE ADVANCE)

1. (D)

2. (A)

3. (C)

4. (B)

5. (B)

6. (B)

7. (A)

8. (D)

9. (A)

10. (B)

11. (B)

12. (A)

13. (B)

14. (D)

15. (A)

16. (A)

17. (C)

18. (B)

19. (D)

20. (B)

21. (A)

22. (C)

23. (B,D)

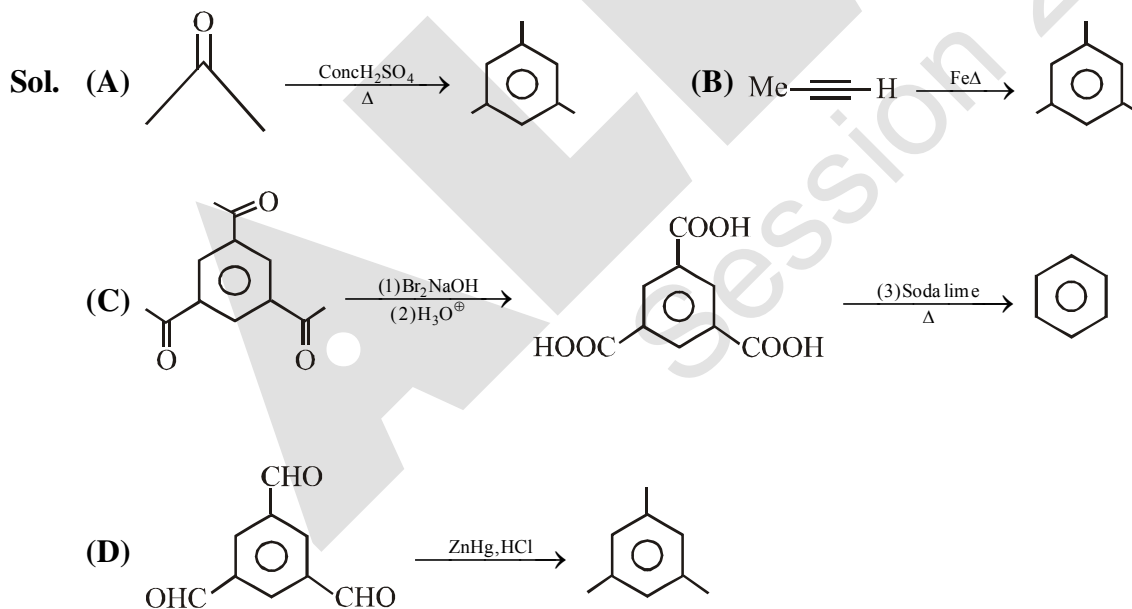
24. (C)

25. (D)

26. (D)

27. (4)

28. (A,B,D)



29. Ans. (2,3)

30. Ans. (2,4)

Important Notes

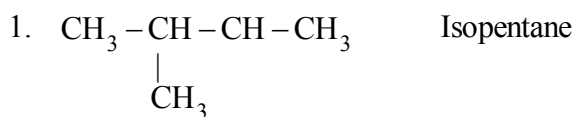
NURTURE COURSE

**NOMENCLATURE OF ORGANIC COMPOUND
AND COMMON NAMES**

NOMENCLATURE OF ORGANIC COMPOUND AND COMMON NAMES

-ESSENTIAL COMMON NAMES-

ALKANE



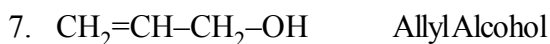
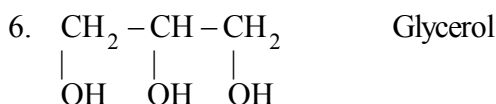
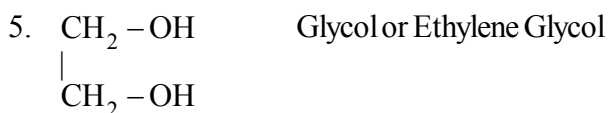
ALKENE



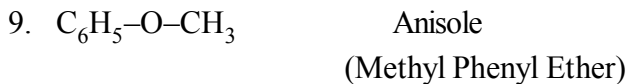
ALKYL HALIDE



ALCOHOL



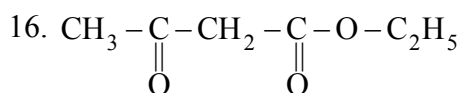
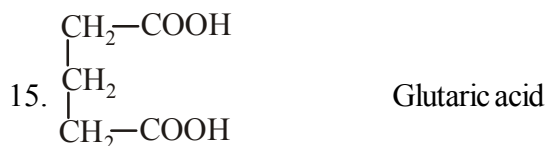
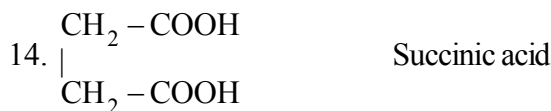
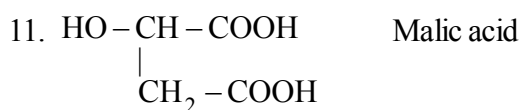
ETHER



KETONE

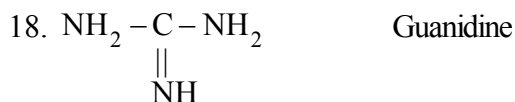


CARBOXYLIC ACID

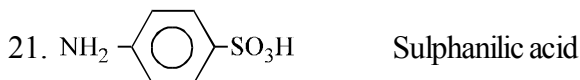
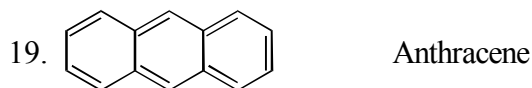


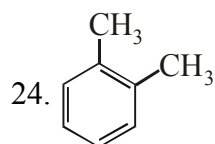
Aceto Acetic Ester (AAE)
or Ethyl Aceto Acetate

N-DERIVATIVES

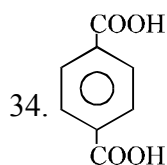


AROMATIC COMPOUNDS

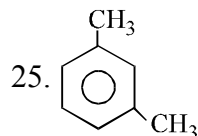




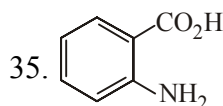
o-xylene



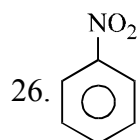
Terephthalic acid



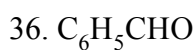
m-xylene



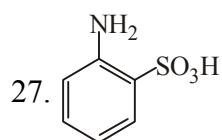
Anthranilic acid (o-aminobenzoic acid)



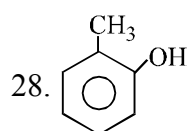
Nitrobenzene (oil of mirbane)



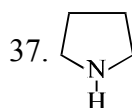
Benzaldehyde



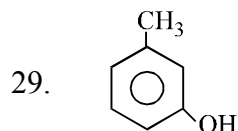
Ortho-aniline

HETEROCYCLIC COMPOUNDS

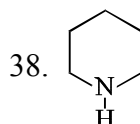
o-Cresol



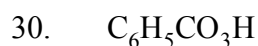
Pyrrolidine



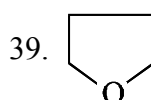
m-Cresol



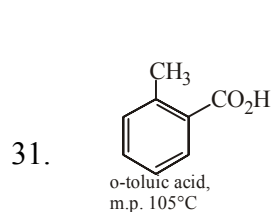
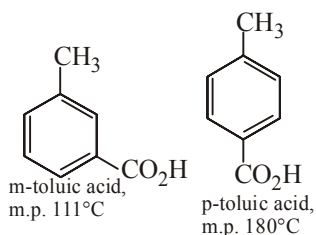
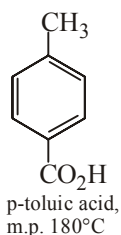
Piperidine



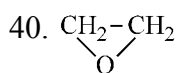
Perbenzoic acid



Tetrahydrofuran (THF)

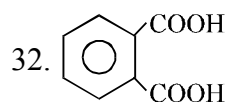
o-toluic acid,
m.p. 105°Cm-toluic acid,
m.p. 111°Cp-toluic acid,
m.p. 180°C

Toluic acids

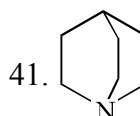


Oxirane or Ethylene Oxide or

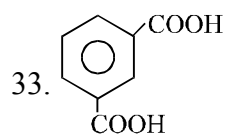
Oxo Cyclo Propane



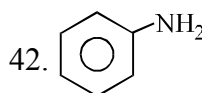
Phthalic acid



Quinuclidine



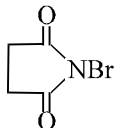
Isophthalic acid



Aniline

SOME REAGENTS

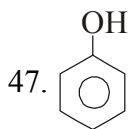
43. Grignard's reagent RMgX

44. NBS N-Bromosuccinimide

POLAR PROTIC SOLVENTS

45. $\text{H}-\text{O}-\text{H}$ Water

46. R-O-H Alcohol



47.  Phenol

48. $\text{CH}_3-\text{C}-\text{OH}$ Acetic acid
 $\parallel$
 O

49. HF Hydrogen Fluoride

50. NH_3 Ammonia

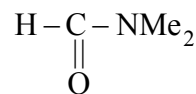
POLAR APROTIC SOLVENTS

51. DMS Dimethyl sulphide $\text{CH}_3\text{-S-CH}_3$

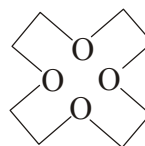
52. DMSO Dimethyl sulphoxide $\text{Me}_2\text{S}=\text{O}$

53. HMPT	Hexamethylphosphoramide
or	
HMPTA	$\text{O}=\text{P}-(\text{NMe}_2)_3$

54. DMF Dimethyl formamide



55. Crown ethers Cyclic polyethers



(12 – C – 4)

-DESIRABLE COMMON NAMES-**ALKANES**

1. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH} - \text{CH}_3 \\ | \quad | \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$ Triptane
2. $\begin{array}{c} -\text{CH}_2 - \text{CH}_2 - \text{CH} - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$ Isopentyl Group

ALKENES

3. $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{CH}_2$ α -Butylene
4. $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ β -Butylene
5. $\begin{array}{c} \text{CH}_3 - \text{C} = \text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$ Iso Butylene

ALKYNES

6. $\text{HC} \equiv \text{CH}$ Purified Acetylene or Norcelyne
7. $\text{CH}_3 - \text{C} \equiv \text{CH}$ Allylene

ETHER

8. $\text{CH}_3\text{CH}(\text{OCH}_3)_2$ Methylal

ALDEHYDE

9. $\begin{array}{c} \text{CHO} \\ | \\ \text{COOH} \end{array}$ Glyoxalic acid
10. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CHO} \\ | \\ \text{CH}_3 \\ \text{or} \\ (\text{CH}_3)_3\text{C} - \text{CHO} \end{array}$ Pivaldehyde
11. $(\text{CH}_3)_2\text{CHCHO}$ Isobutyraldehyde
12. $\begin{array}{c} \text{CH}_3 - \text{C} - \text{C} - \text{CH}_3 \\ || \quad || \\ \text{O} \quad \text{O} \end{array}$ Dimethyl Glyoxal

13. $\begin{array}{c} \text{CH}_3 - \text{C} - \text{C} - \text{H} \\ || \quad || \\ \text{O} \quad \text{O} \end{array}$ Methyl Glyoxal or Pyruvialdehyde

KETONE

14. $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{C} = \text{CH} - \text{C} - \text{CH} = \text{C} \diagdown \\ \diagdown \quad || \quad \diagup \\ \text{CH}_3 \quad \text{O} \quad \text{CH}_3 \end{array}$ Phorone
15. $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{C} = \text{CH} - \text{C} - \text{CH}_3 \\ \diagdown \quad || \\ \text{CH}_3 \quad \text{O} \end{array}$ Mesityl Oxide
16. $\text{H}_2\text{C} = \text{C} = \text{O}$ Ketene

CARBOXYLIC ACID

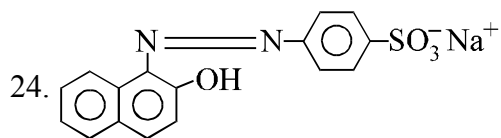
17. $\text{CH}_3 - \text{CO} - \text{COOH}$ Pyruvic Acid
18. $\begin{array}{c} \text{C}_6\text{H}_5 - \text{CH} - \text{COOH} \\ | \\ \text{OH} \end{array}$ Mandelic Acid
19. NH_2COOH Carbamic Acid (Amino formic Acid)

20. $\begin{array}{c} \text{COOH} \\ | \\ \text{COOH} \end{array}$ Oxalic acid

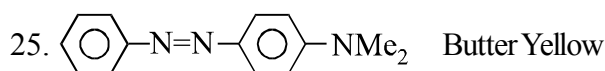
ACID DERIVATIVES

21. $\begin{array}{c} \text{Cl} - \text{C} - \text{C} - \text{Cl} \\ || \quad || \\ \text{O} \quad \text{O} \end{array}$ Oxalyl Chloride
22. $\text{NH}_2\text{COONH}_4$ Ammonium Carbamate
23. $\begin{array}{c} \text{NH}_2 - \text{C} - \text{C} - \text{NH}_2 \\ || \quad || \\ \text{O} \quad \text{O} \end{array}$ Oxanamide

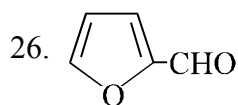
AROMATIC COMPOUNDS



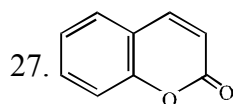
Orange II



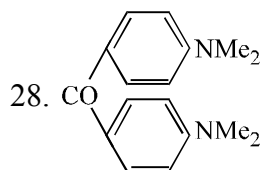
Butter Yellow



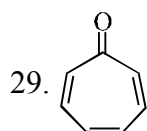
Furfural



Coumarine

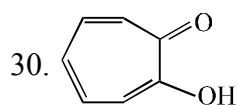


Michler's Ketone



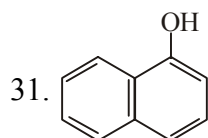
Tropone

(Cycloheptatrienone)

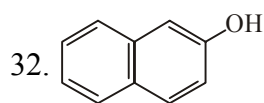


Tropolone

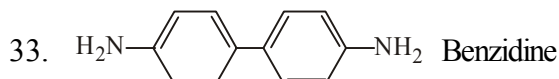
(Cycloheptatrienolone)



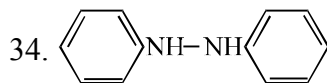
α -naphthol



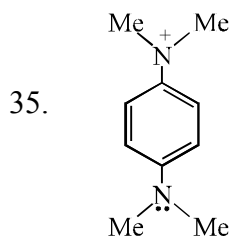
β -naphthol



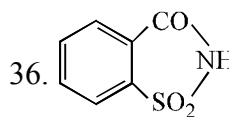
Benzidine



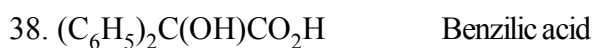
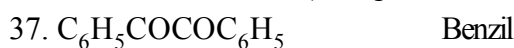
Hydrazobenzene



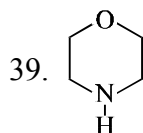
Wurster salts



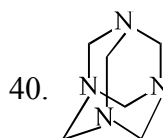
Saccharin (o-sulphobenzoic imide)



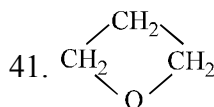
HETEROCYCLIC COMPOUNDS



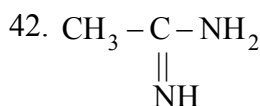
Morpholine



Hexa-methylenetetramine or Urotropine

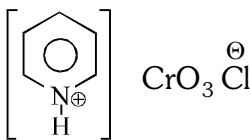
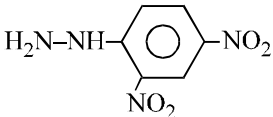


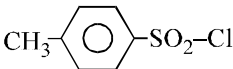
Oxirane



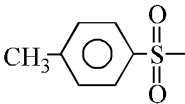
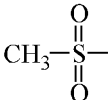
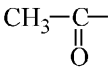
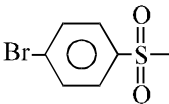
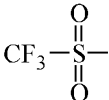
Amidine

SOME REAGENTS

43. LAH	Lithium aluminium hydride : LiAlH_4
44. SBH	Sodium borohydride NaBH_4
45. PCC	Pyridinium chlorochromate 
46. Raney Nickel	Ni-Al alloy
47. Wilkinson's	Tris(Triphenylphosphine) catalyst chlororhodium (I) $(\text{PPh}_3)_3\text{RH}^+\text{Cl}^-$
48. Bayer's reagent	1% dil. alkaline aq. sol. of KMnO_4
49. Braddy's reagent 2,4 DNP	
50. Liemieux reagent	$\text{NaIO}_4 + \text{dil. alk. KMnO}_4$

51. TEL	Tetra ethyl lead
52. Gillman's reagent	$\text{R}_2\text{CuLi}/[\text{R}_2\text{Cu}]^-\text{Li}^+$
53. Tollen's reagent	alk. sol. of AgNO_3
54. Fehling's reagent	alk. sol. of CuSO_4
55. Hinsberg's reagent	

SOME GROUPS

56. Ts	Tosyl	
57. Ms	Mesyl	
58. Ac	Acyl	
59. Bs	Brosyl	
60. Tf	Triflate	

EXERCISE # I

Q.1 How many 1° carbon atom will be present in a simplest open chain hydrocarbon having two 3° and one 2° carbon atom ?

- (A) 3 (B) 4 (C) 5 (D) 6

Q.2 Alicyclic compounds are :

- (A) Aromatic compounds (B) Aliphatic cyclic compounds
(C) Heterocyclic compounds (D) None of the above

Q.3 How many 1° , 2° , 3° C atoms does 1, 3, 5-Trimethyl cyclohexane have?

- (A) 3, 6, 0 (B) 3, 4, 2 (C) 0, 3, 6 (D) 3, 3, 3

Q.4 The compound which has one isopropyl group is:

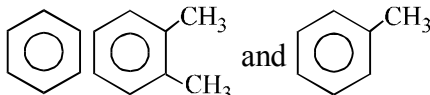
- (A) 2,2,3,3-Tetramethyl pentane (B) 2,2-Dimethyl pentane
(C) 2,2,3-Trimethyl pentane (D) 2-Methyl pentane

Q.5 Which of the following is the first member of ester homologous series?

- (A) Ethyl ethanoate (B) Methyl ethanoate
(C) Methyl methanoate (D) Ethyl methanoate

Q.6 A group closely related compounds which can be expressed by a general formula & in which two consecutive members differ by 14 in their molecular masses is called

- (A) a heterogeneous series (B) a homologous series
(C) a homogeneous series (D) a electrochemical series

Q.7  and

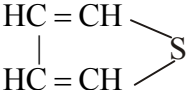
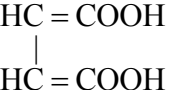
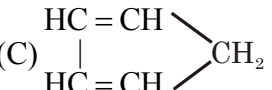
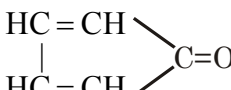
Number of secondary carbon atoms present in the above compounds are respectively:

- (A) 6,4,5 (B) 4,5,6 (C) 5,4,6 (D) 6,2,1

Q.8 The molecular formula of the first member of the family of alkenynes and its name is given by the set

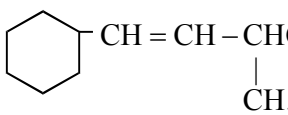
- (A) C_3H_6 , Alkene (B) C_5H_6 , Pent-1-en-3-yne
(C) C_6H_8 , Hex-1-en-5-yne (D) C_4H_4 , Butenyne

Q.9 Which of the following is a heterocyclic compound :

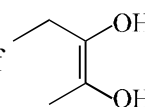
- (A)  (B)  (C)  (D) 

Q.10 The correct IUPAC name of the compound $\text{CH}_3 - \text{CH}_2 - \overset{\text{CH}_3}{\underset{\text{C}_2\text{H}_5}{\text{C}}} = \text{C} - \text{CH} - \overset{\text{CH}_3}{\text{C}} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$:

- (A) 5-Ethyl-3, 6-dimethyl non-3-ene (B) 5-Ethyl-4, 7-dimethyl non-3-ene
(C) 4-Methyl-5, 7-diethyl oct-2-ene (D) 2,4-Ethyl-5-methyl oct-2-ene

Q.11 The IUPAC name of  is:

- (A) 1-Cyclohexyl-3-methyl pent-1-ene (B) 3-Methyl-5-cyclohexyl pent-1-ene
(C) 1-Cyclohexyl-3-ethyl but-1-ene (D) 1-Cyclohexyl-3,4-dimethyl but-1-ene

Q.12 IUPAC name of  is:

- (A) But-2-ene-2,3-diol (B) Pent-2-ene-2,3-diol
(C) 2-Methylbut-2-ene-2,3-diol (D) Pent-3-ene-3,4-diol

Q.13 IUPAC name of $\text{CH}_2=\text{CH}-\text{CN}$ is:

- (A) Ethenenitrile (B) Vinyl cyanide (C) Cyano ethene (D) Prop-2-enenitrile

Q.14 The IUPAC name of $\text{CH}_3\text{CH}_2-\underset{\text{CH}_3}{\text{N}}-\text{CH}_2\text{CH}_3$ is:

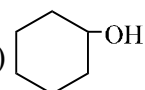
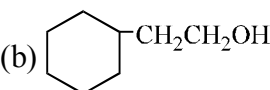
- (A) N-Methyl-N-ethyl ethanamine (B) Diethyl methanamine
(C) N-Ethyl-N-methyl ethanamine (D) Methyl diethyl ethanamine

Q.15 The IUPAC name of acetyl acetone is :

- (A) Pentane-2,5- dione (B) Pentane -2,4-dione
(C) Hexane-2,4-dione (D) Butane-2,4-dione

Q.16 When vinyl & allyl are joined each other, we get

- (A) Conjugated alkadiene (B) cumulative alkadiene
(C) Isolated alkadiene (D) Allenes

Q.17 (a)  and (b) 

True statement for the above compounds is :

- (A) (a) is phenol while (b) is alcohol (B) Both (a) and (b) are primary alcohol
(C) (a) is primary and (b) is secondary alcohol (D) (a) is secondary and (b) is primary alcohol

Q.18 The IUPAC name of the following structure $(\text{CH}_3)\text{C}.\text{C}.\text{C}.\text{C}(\text{CH}_3)\text{CH}(\text{CH}_3)$ is:

- (A) 3-Methylhex-4-yn-2-ene (B) 3-Methylhex-2-en-4-yne
(C) 4-Methylhex-4-en-4-yne (D) All are correct

Q.19 The IUPAC name of the following structure is $[\text{CH}_3\text{CH}(\text{CH}_3)]_2 \text{C}(\text{CH}_2\text{CH}_3)\text{C}(\text{CH}_3) \text{C}(\text{CH}_2\text{CH}_3)_2$

- (A) 3,5-Diethyl-4,6-dimethyl-5-[1-methylethyl]hept-3-ene
(B) 3,5-Diethyl-5-isopropyl-4,6-dimethylhept-2-ene
(C) 3,5-Diethyl-5-propyl-4,6-dimethylhept-3-ene
(D) None of these

Q.20 The correct IUPAC name of $\text{CH}_3 - \text{CH}_2 - \underset{\text{CH}_2}{\overset{\text{||}}{\text{C}}} - \text{COOH}$ is:

- (A) 2-Methyl butanoic acid (B) 2-Ethylprop-2-enoic acid
(C) 2-Carboxybutene (D) None of the above

Q.21 The correct IUPAC name of 2-ethylpent-3-yne is:

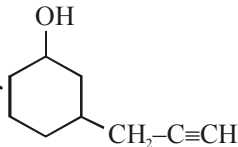
- (A) 3-Methyl hex-4-yne (B) 4-Ethyl pent-2-yne
(C) 4-methyl hex-2 yne (D) None of these

Q.22 All the following IUPAC names are correct except:

- (A) 1-Chloro-1-ethoxy propane (B) 1-Amino-1-ethoxypropane
(C) 1-Ethoxy-2-propanol (D) 1-Ethoxy-1-propanamine

Q.23 The IUPAC name of the compound $\text{CH}_3\text{CH} = \text{CHCH} = \text{CHC} \equiv \text{CCH}_3$ is:

- (A) Octa-4,6-diene-2-yne (B) Octa-2,4-diene-6-yne
(C) Oct-2-yne-4,6-diene (D) Oct-6-yne-2,4-diene

Q.24 The correct IUPAC name of 

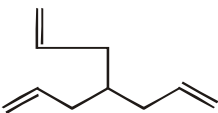
- (A) 3-Cyclohexanol Propyne (B) 3-[3-Hydroxy Cyclohexyl] Propyne
(C) 3-Propynyl Cyclohexanol (D) 3-(2-propynyl) Cyclohexanol

Q.25 The IUPAC name of β -ethoxy- α -hydroxy propionic acid (trivial name) is:

- (A) 1,2-Dihydroxy-1-oxo-3-ethoxy propane (B) 1-Carboxy-2-ethoxy ethanol
(C) 3-Ethoxy-2-hydroxy propanoic acid (D) All above

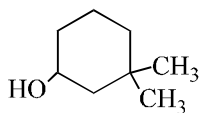
Q.26 As per IUPAC rules, which one of the following groups, will be regarded as the principal functional group ?

- (A) $-\text{C} \equiv \text{C}-$ (B) $-\text{OH}$ (C) $-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-$ (D) $-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{H}$

Q.27 The IUPAC name of the compound  is :

- (A) 4-Prop-1-enyl hepta-1,6-diene (B) 4-Propylidene hepta-1,6-diene
(C) 4-Propenyl hepta-1,6-diene (D) 4-[Prop-2-enyl] hepta-1,6-diene

Q.28 The IUPAC name of the given compound is:



- (A) 1,1-Dimethyl-3-hydroxy cyclohexane (B) 3,3-Dimethyl-1-hydroxy cyclohexane
(C) 3,3-Dimethylcyclohexanol (D) 1,1-Dimethylcyclohexan-3-ol

Q.29 The IUPAC name of $(C_2H_5)_2NCH_2CH(Cl)COOH$ is:

- (A) 2-Chloro-4-N-ethylpentanoic acid (B) 2-Chloro-3-(N,N-diethyl amino)-propanoic acid
(C) 2-Chloro-2-oxo diethylamine (D) 2-Chloro-2-carboxy-N-ethyl ethane

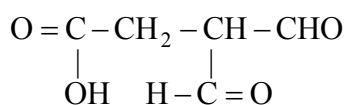
Q.30 The IUPAC name of the compound is $CH_3 - \underset{\substack{| \\ CH_3}}{CH} - \overset{\substack{| \\ Ph}}{CH} - NH_2$

- (A) 1-Amino-1-phenyl-2-methyl propane (B) 2-Methyl-1-phenyl propan-1-amine
(C) 2-Methyl-1-amino-1-phenyl propane (D) 1-Isopropyl-1-phenyl methyl amine

Q.31 Which of the following compound is wrongly named ?

- (A) $CH_3CH_2CH_2\underset{\substack{| \\ Cl}}{CH}COOH$; 2-Chloro pentanoic acid
(B) $CH_3C \equiv C\underset{\substack{| \\ CH_3}}{CH}COOH$; 2-Methyl hex-3-enoic acid
(C) $CH_3CH_2CH=CHCOCH_3$; Hex-3-en-2-one
(D) $CH_3 - \underset{\substack{| \\ CH_3}}{CH}CH_2CH_2CHO$; 4-Methyl pentanal

Q.32 The correct IUPAC name of the following compound is:



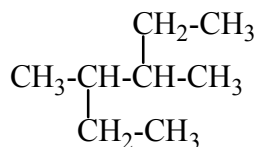
- (A) 3,3-Diformylpropanoic acid (B) 3-Formyl-4-oxo-butanoic acid
(C) 3,3-Dioxo propanoic acid (D) 3,3-Dicarbaldehyde propanoic acid

Q.33 The correct IUPAC name of compound $\text{CH}_3 - \text{CH}_2 - \text{C} - \text{CH} - \text{CHO}$ is :



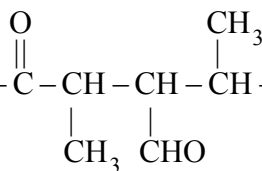
- (A) 2-Cyano-3-oxopentanal (B) 2-Formyl-3-oxopentanenitrile
(C) 2-Cyanopentane-1,3-dione (D) 1,3-Dioxo-2-cyanopentane

Q.34 IUPAC name of compound



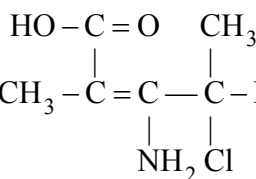
- (A) 2, 3-diethyl butane (B) 2-ethyl-3-methyl pentane
(C) 3-methyl-2-ethyl pentane (D) 3,4-dimethyl hexane

Q.35 The IUPAC name of compound $\text{CH}_3 - \text{C} - \text{CH} - \text{CH} - \text{CH} - \text{CH}_3$ is:



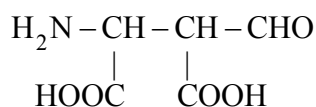
- (A) 3,5-Dimethyl-4-Formyl pentanone (B) 1-Isopropyl-2-methyl-4-oxo butanal
(C) 2-Isopropyl-3-methyl-4-oxo pentanal (D) None of the above

Q.36 The IUPAC name of compound $\text{CH}_3 - \text{C} = \text{C} - \text{C} - \text{H}$ is :



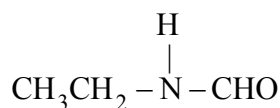
- (A) 2-Amino-3-chloro-2-methylpent-2-enoic acid (B) 3-Amino-4-chloro-2-methylpent-2-enoic acid
(C) 4-Amino-3-chloro-2-methylpent-2-enoic acid (D) All of the above

Q.37 The IUPAC name of the structure is:



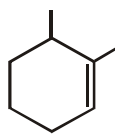
- (A) 3-Amino-2-formyl butane-1, 4-dioic acid (B) 3-Amino-2, 3-dicarboxy propanal
(C) 2-Amino-3-formyl butane-1, 4-dioic acid (D) 1-Amino-2-formyl succinic acid

Q.38 One among the following is the correct IUPAC name of the compound



- (A) N-Formyl aminoethane (B) N-Ethyl formyl amine
(C) N-Ethyl methanamide (D) Ethylamino methanal

Q.39 The IUPAC name of the structure is :



- (A) 1,2-Dimethyl-Cyclohexane (B) 1,6-Dimethyl-Cyclohexene
(C) 1,2-Dimethyl-Cyclohex-2-ene (D) 2,3-Dimethyl-Cyclohexane

Q.40 The IUPAC name of $\text{C}_6\text{H}_5\text{CH}=\text{CH}-\text{COOH}$ is :

- (A) Cinnamic acid (B) 1-Phenyl-2-carboxy ethane
(C) 3-Phenyl prop-2-enoic acid (D) Dihydroxy-3-phenyl propionic acid

Q.41 The IUPAC name of $\text{BrCH}_2-\text{CH}(\text{CONH}_2)-\text{CO}-\text{CH}_2-\text{CH}_2\text{CH}_3$ is:

- (A) 2-Bromomethyl-3-oxohexanamide (B) 1-Bromo-2-amino-3-oxohexane
(C) 1-Bromo-2-amino-n-propyl ketone (D) 3-Bromo-2-propyl propanamide

Q.42 IUPAC name will be $\text{CH}_2(\text{CN})-\text{CH}(\text{CN})-\text{CH}_2(\text{CN})$

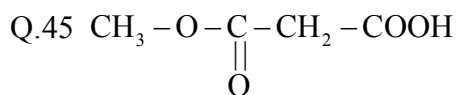
- (A) 1,2,3-Tricyano propane (B) Propane-1,2,3- trinitrile
(C) 1,2,3-Cyano propane (D) Propane-1,2,3-tricarbonitrile

Q.43 The IUPAC name of compound

- (A) 3-Carbonyl methoxy -5- Ethanoyl oxy cyclohexanoic acid
(B) 3-Ethanoyl oxy -5- Methoxy carbonyl cyclohexane carboxylic acid
(C) 5-Ethanoyl oxy -5- Methoxy carbonyl cyclohexanoic acid
(D) 3-Methoxy carbonyl -5- Ethanoyl oxy cyclohexane carboxylic acid

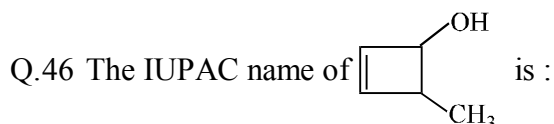
Q.44 The IUPAC name of $\text{CH}_3-\text{C}(=\text{O})-\text{O}-\text{CH}_2-\text{C}(=\text{O})-\text{OH}$ is:

- (A) 1-Acetoxy acetic acid (B) 2-Acetoxy ethanoic acid
(C) 2-Ethanoyloxyacetic acid (D) 2-Ethanoyloxyethanoic acid

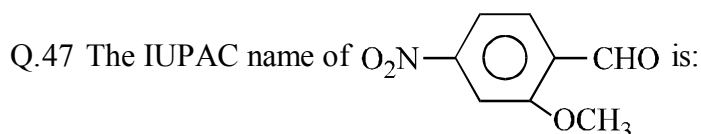


The correct IUPAC systematic name of the above compound is:

- (A) 2-Acetoxy ethanoic acid (B) 2-Methoxy carbonyl ethanoic acid
(C) 3-Methoxy formyl ethanoic acid (D) 2-Methoxy formyl acetic acid

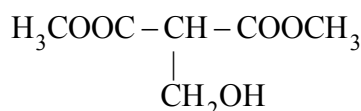


- (A) 3-Methyl cyclobut-1-ene-2-ol (B) 4-Methyl cyclobut-2-ene-1-ol
(C) 4-Methyl cyclobut-1-ene-3-ol (D) 2-Methyl cyclobut-3-ene-1-ol

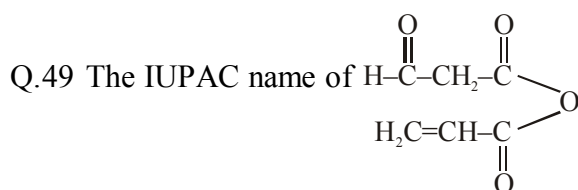


- (A) 2-Methoxy-4-nitro benzaldehyde (B) 4-Nitro anisaldehyde
(C) 3-Methoxy-4-formyl nitro benzene (D) 2-Formyl-4-nitro anisole

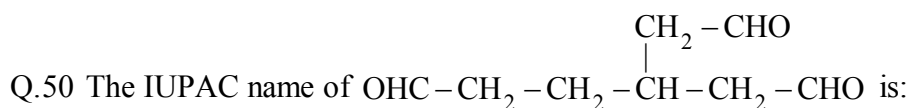
Q.48 The IUPAC name of compound



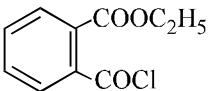
- (A) 2-(Hydroxy methyl) methyl propanedioate (B) Methyl-2-(hydroxy methyl) propanedioate
(C) 2-(Hydroxy methyl) dimethyl propanedioate (D) None of these



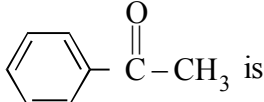
- (A) 2-Formyl ethanoic propanoic Anhydride (B) 2-Oxo-propanoic prop-2-enoic Anhydride
(C) Prop-2-enoic-2-formyl propanoic Anhydride (D) 2-Formyl ethanoic prop-2-enoic Anhydride



- (A) 4,4-Di(formylmethyl) butanal (B) 2-(Formylmethyl) butane-1, 4-dicarbaldehyde
(C) Hexane-3-acetal-1, 6-dial (D) 3-(Formylmethyl) hexane-1, 6-dial

Q.51 The IUPAC name of  is :

- (A) 2-Chlorocarbonyl ethylbenzoate (B) 2-Carboxyethyl benzoyl chloride
(C) Ethyl-2-(chlorocarbonyl)benzoate (D) Ethyl-1-(chlorocarbonyl)benzoate

Q.52 The IUPAC name of  is:

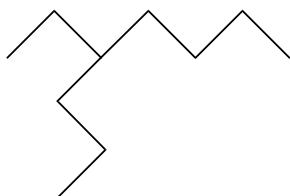
- (A) Phenyl ethanone (B) Methyl phenyl ketone
(C) Acetophenone (D) Phenyl methyl ketone

Q.53 Structural formula of isopropyl methanoate is :

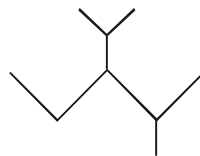
- (A) $\text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{O} - \underset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_3$ (B) $\text{H} - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{O} - \text{CH}_2 - \underset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_3$
(C) $\text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{O} - \text{CH}_2 - \underset{\text{CH}_3}{\underset{|}{\text{CH}_2}}$ (D) $\text{H} - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{O} - \underset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_3$

Give the IUPAC names for each of the following :

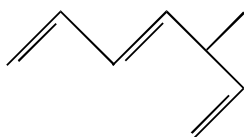
Q.1



Q.2



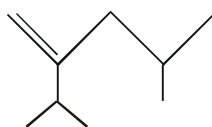
Q.3



Q.4



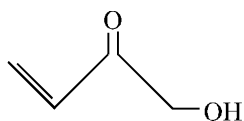
Q.5



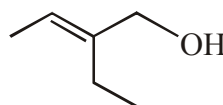
Q.6



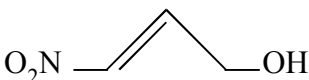
Q.7



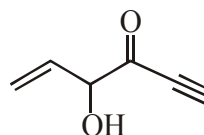
Q.8



Q.9



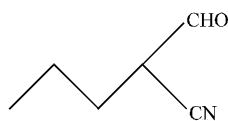
Q.10



Q.11



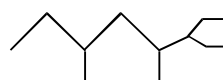
Q.12



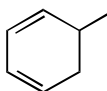
Q.13



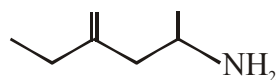
Q.14



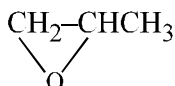
Q.15



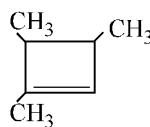
Q.16

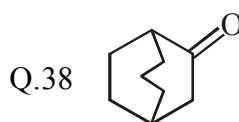
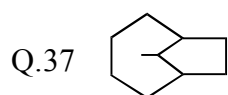
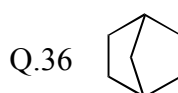
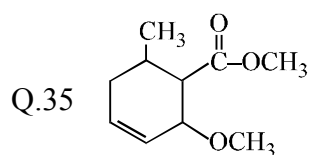
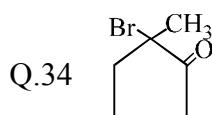
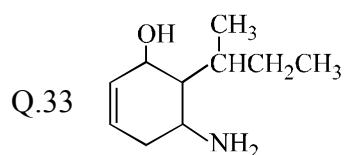
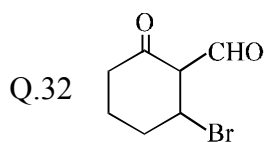
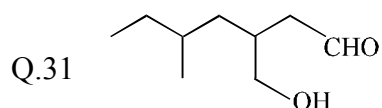
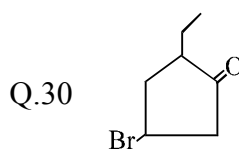
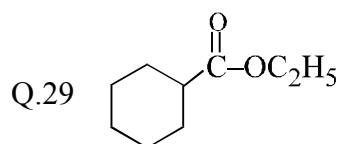
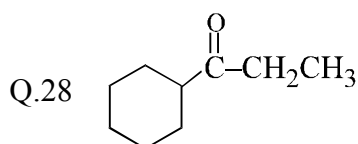
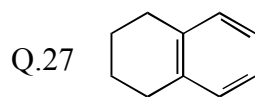
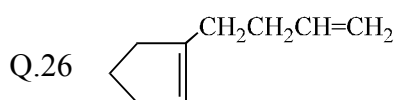
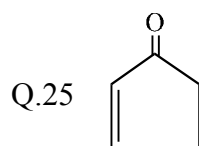
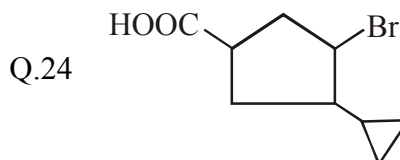
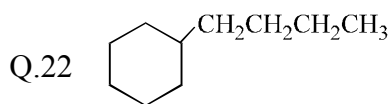
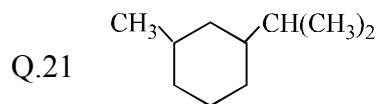
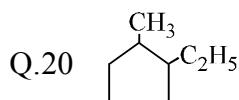
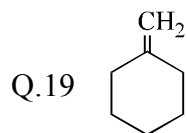


Q.17



Q.18

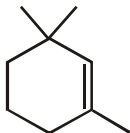




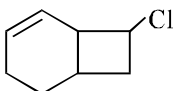
Q.39



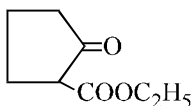
Q.41



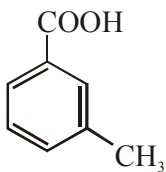
Q.43



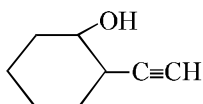
Q.45



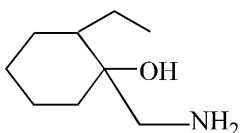
Q.47



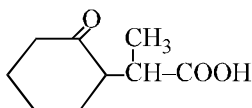
Q.49



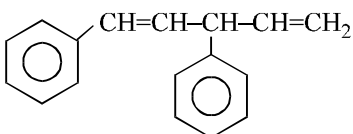
Q.51



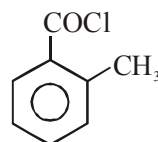
Q.53



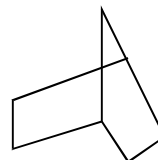
Q.55



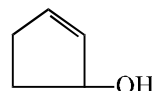
Q.40



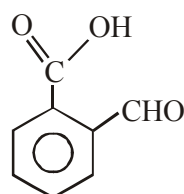
Q.42



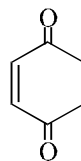
Q.44



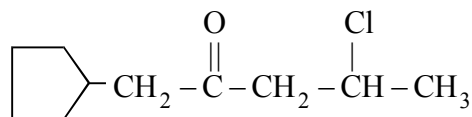
Q.46



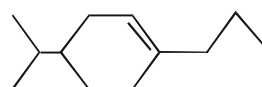
Q.48



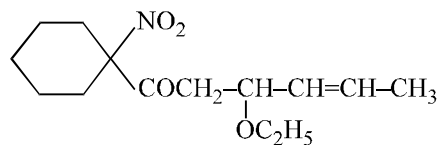
Q.50



Q.52



Q.54



EXERCISE # III

- Q.1 Which of the following pairs have absence of carbocyclic ring in both compounds?
 (A) Pyridine, Benzene (B) Benzene, Cyclohexane
 (C) Cyclohexane, Furane (D) Furane, Pyridine
- Q.2 The commercial name of trichloroethene is:
 (A) Westron (B) Perclene (C) Westrosol (D) Orlone
- Q.3 A substance containing an equal number of primary, secondary and tertiary carbon atoms is:
 (A) Mesityl Oxide (B) Mesitylene (C) Maleic acid (D) Malonic acid
- Q.4 The IUPAC name of the compound Glycerine $\text{CH}_2 - \text{CH} - \text{CH}_2$ is:
 $\begin{array}{ccc} | & | & | \\ \text{OH} & \text{OH} & \text{OH} \end{array}$
 (A) 1,2,3-Tri hydroxy propane (B) 3-Hydroxy pentane-1,5-diol
 (C) 1,2,3-Hydroxy propane (D) Propane-1,2,3-triol
- Q.5 Which of the following is crotonic acid:
 (A) $\text{CH}_2 = \text{CH} - \text{COOH}$ (B) $\text{C}_6\text{H}_5 - \text{CH} = \text{CH} - \text{COOH}$
 (C) $\text{CH}_3 - \text{CH} = \text{CH} - \text{COOH}$ (D) $\begin{array}{c} \text{CH} - \text{COOH} \\ || \\ \text{CH} - \text{COOH} \end{array}$
- Q.6 The group of heterocyclic compounds is:
 (A) Phenol, Furane (B) Furane, Thiophene
 (C) Thiophene, Phenol (D) Furane, Aniline
- Q.7 **Column - I** **Column - II**
(Common Name) **(Structural formula)**
- (A) Isooctane (P) $\begin{array}{cc} \text{CH}_2 & - & \text{CH}_2 \\ | & & | \\ \text{Cl} & & \text{Cl} \end{array}$
- (B) Neopentane (Q) $\begin{array}{ccccccc} & & \text{CH}_3 & & & & \\ & & | & & & & \\ \text{CH}_3 & - & \text{C} & - & \text{CH} & - & \text{CH} & - & \text{CH}_3 \\ & & | & & | & & \\ & & \text{CH}_3 & & \text{CH}_3 & & \end{array}$
- (C) Ethylidene chloride (Geminal dihalide) (R) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_3$
- (D) Ethylene Dichloride (Vicinal dihalide) (S) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
- (T) $\text{CH}_3 - \text{CH} \begin{array}{l} \nearrow \text{Cl} \\ \searrow \text{Cl} \end{array}$

Q.8 Column - I
(Common Name)

- (A) Acetone
- (B) Acetaldehyde
- (C) Crotonaldehyde
- (D) Acrolein

Column - II
(Structural formula)

- (P) $\text{CH}_2 = \text{CH} - \overset{\text{O}}{\parallel} \text{C} - \text{H}$
- (Q) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$
- (R) $\text{CH}_3 - \text{CH} = \text{CH} - \overset{\text{O}}{\parallel} \text{C} - \text{H}$
- (S) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{H}$

Q.9 Column - I
(Common Name)

- (A) $\begin{array}{c} \text{CHO} \\ | \\ \text{CHO} \end{array}$
- (B) $\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2 - \text{OH} \end{array}$
- (C) $\text{H}_2\text{N} - \text{CH}_2 - \text{COOH}$
- (D) $\begin{array}{c} \text{H} \\ | \\ \text{CH}_3 - \text{C} - \text{COOH} \\ | \\ \text{OH} \end{array}$

Column - II
(Structural formula)

- (P) Lactic acid (In milk)
- (Q) Glyoxal
- (R) Glyceraldehyde
- (S) Glycine
- (T) Glycerol

Q.10 Column - I
(Common Name)

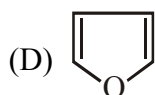
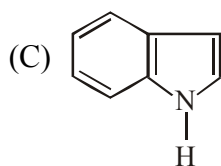
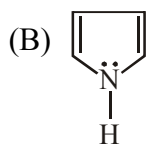
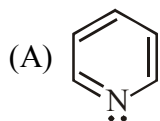
- (A) Fumaric acid
- (B) Adipic acid
- (C) Maleic acid
- (D) Tartaric acid

Column - II
(Structural formula)

- (P) $\begin{array}{c} \text{HO} - \text{CH} - \text{COOH} \\ | \\ \text{HO} - \text{CH} - \text{COOH} \end{array}$
- (Q) $\begin{array}{c} \text{HC} - \text{COOH} \\ \parallel \\ \text{HOOC} - \text{CH} \end{array}$
- (R) $\begin{array}{c} \text{H} - \text{C} - \text{COOH} \\ \parallel \\ \text{H} - \text{C} - \text{COOH} \end{array}$
- (S) $\text{COOH}(\text{CH}_2)_4\text{COOH}$

Q.11 Column - I

(Common Name)



Column - II

(Structural formula)

(P) Pyrrole

(Q) Furan

(R) Thiophene

(S) Indol

(T) Pyridine

Q.12 Column - I

(Common Name)

(A) p-Cresol

(B) p-Xylene

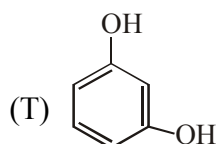
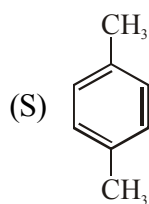
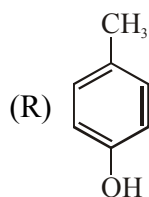
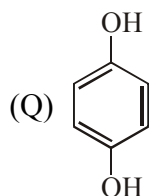
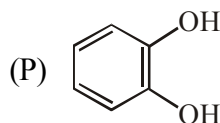
(C) Resorcinol

(D) Quinol

(E) Catechol

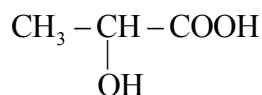
Column - II

(Structural formula)

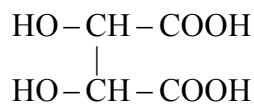


Q.13 Which of the following is not correctly matched:

(A) Lactic acid



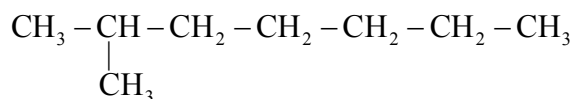
(B) Tartaric acid



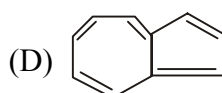
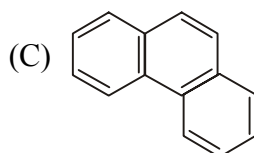
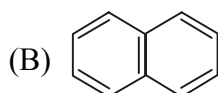
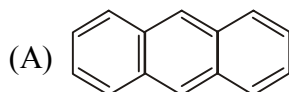
(C) Pivaldehyde



(D) Iso-octane



Q.14 **Column - I**



Column - II

(P) Phenanthrene

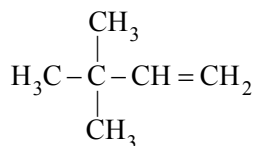
(Q) Anthracene

(R) Azulene

(S) Naphthalene

EXERCISE # IV

- Q.1 The IUPAC name of the compound having the formula is : [JEE 1984]



- (A) 3,3,3-trimethyl-1-propene
(B) 1,1,1-trimethyl-2-propene
(C) 3,3-dimethyl-1-butene
(D) 2,2-dimethyl-3-butene

- Q.2 Write the IUPAC name of $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}\cdot\text{COOH}$ [JEE 1986]

- Q.3 The IUPAC name of the compound $\text{CH}_2=\text{CH}-\text{CH}(\text{CH}_3)_2$ is :

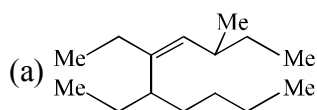
- (A) 1,1-dimethyl-2-propene (B) 3-methyl-1-butene
(C) 2-vinyl propane (D) None of the above

[JEE 1987]

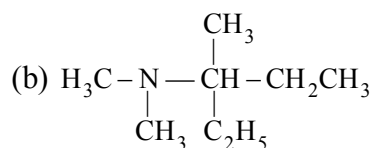
- Q.4 The number of sigma and pi-bonds in 1-butene 3-yne are: [JEE 1989]

- (A) 5 sigma and 5 pi (B) 7 sigma and 3 pi
(C) 8 sigma and 2 pi (D) 6 sigma and 4 pi

- Q.5 Write I.U.P.A.C name of following :



Me = methyl group



[JEE 1990]

- Q.6 Write IUPAC name of succinic acid. [JEE 1994]

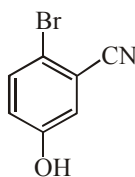
- Q.7 The IUPAC name of $\text{C}_6\text{H}_5\text{COCl}$ is

- (A) Benzoyl chloride
(B) Benzene chloro ketone
(C) Benzene carbonyl chloride
(D) Chloro phenyl ketone

[JEE 2006]

Q.8 The IUPAC name of the following compound is

[JEE 2009]



(A) 4-Bromo-3-cyanophenol

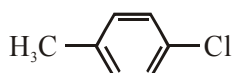
(B) 2-Bromo-5-hydroxybenzonitrile

(C) 2-Cyano-4-hydroxybromobenzene

(D) 6-Bromo-3-hydroxybenzonitrile

Q.9 The IUPAC name(s) of the following compound is(are) :

[JEE 2017]



(A) 4-methylchlorobenzene

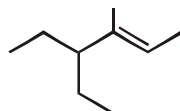
(B) 4-chlorotoluene

(C) 1-chloro-4-methylbenzene

(D) 1-methyl-4-chlorobenzene

10. The IUPAC name of the following compound is :

[JEE Mains On_line 2018]



(A) 4-methyl-3-ethylhex-4-ene

(B) 4,4-diethyl-3-methylbut-2-ene

(C) 3-ethyl-4-methylhex-4-ene

(D) 4-ethyl-3-methylhex-2-ene

ANSWER KEY**EXERCISE # I**

Q.1	B	Q.2	B	Q.3	D	Q.4	D	Q.5	C	Q.6	B	Q.7	A
Q.8	D	Q.9	A	Q.10	A	Q.11	A	Q.12	B	Q.13	D	Q.14	C
Q.15	B	Q.16	C	Q.17	D	Q.18	B	Q.19	A	Q.20	B	Q.21	C
Q.22	B	Q.23	B	Q.24	D	Q.25	C	Q.26	D	Q.27	D	Q.28	C
Q.29	B	Q.30	B	Q.31	B	Q.32	B	Q.33	B	Q.34	D	Q.35	C
Q.36	B	Q.37	C	Q.38	C	Q.39	B	Q.40	C	Q.41	A	Q.42	D
Q.43	B	Q.44	D	Q.45	B	Q.46	B	Q.47	A	Q.48	B	Q.49	D
Q.50	D	Q.51	C	Q.52	A	Q.53	D						

EXERCISE # II

Q.1	4-Ethyl octane	Q.17	1,2-epoxy propane
Q.2	3-Ethyl-2,4-dimethyl pentane	Q.18	1,3,4-trimethyl cyclobutene
Q.3	5-Methyl hepta-1,3,6-triene	Q.19	Methylene cyclohexane
Q.4	Hepta-1,5-dien-3-yne	Q.20	1-ethyl-2-methylcyclopentane
Q.5	2-Isopropyl-4-methyl pent-1-ene	Q.21	1-methyl-3-(methyl ethyl) cyclohexane or 1-isopropyl-3-methylcyclohexane
or	4-Methyl-2-(methyl ethyl) pent-1-ene	Q.22	Butyl cyclohexane
Q.6	3-Methoxypropene	Q.23	Isopropylidenecyclopentane or 1-methyl ethylidene cyclopentane
Q.7	1-Hydroxybut-3-en-2-one	Q.24	3-Bromo-4-cyclopropyl cyclopentane carboxylic acid
Q.8	2-Ethylbut-2-en-1-ol	Q.25	Cyclopent-2-en-1-one
Q.9	3-nitroprop-2-en-1-ol	Q.26	1-(3-butenyl) cyclopentene
Q.10	4-hydroxyhex-5-en-1-yn-3-one	Q.27	1,2-diethenyl cyclohexene
Q.11	4,6-Bis-[1,1-Dimethyl ethyl] Nonane	Q.28	1-cyclohexyl-1-propanone
Q.12	2-Formyl pentane nitrile	Q.29	Ethyl cyclohexanecarboxylate
Q.13	2,2,6,7-tetramethyloctane	Q.30	4-Bromo-2-ethyl cyclopentanone
Q.14	3-Ethyl-4,6-dimethyloctane	Q.31	3-(hydroxymethyl)-5-methylheptanal
Q.15	5-Methyl cyclohexa-1,3-diene		
Q.16	4-Ethyl Pent-4-en-2-amine		

- | | |
|---|---|
| Q.32 2-Bromo-6-oxocyclohexanecarbaldehyde | Q.43 8-chloro bicyclo(4,2,0) oct-2-ene |
| Q.33 5-amino-6-(1-methyl propyl)
cyclo hex-2-enol | Q.44 2-cyclopenten-1-ol |
| Q.34 2-bromo-2-methyl cyclopentanone | Q.45 Ethyl-2-oxo cyclo pentane carboxylate |
| Q.35 Methyl-2-methoxy-6-methyl-3- cyclohexene
carboxylate | Q.46 2-Formyl Benzoic acid |
| Q.36 Bicylo(2,2,1)heptane | Q.47 3-Mthyl Benzoic acid |
| Q.37 9-methyl bicyclo(4,2,1) nonane | Q.48 Cyclohex-2-en-1,4-dione |
| Q.38 Bicyclo [3,2,2] Non-6-one | Q.49 2-ethynyl cyclohexanol |
| Q.39 spiro(4,5) decane | Q.50 4-chloro-1-cyclopentyl pentane-2-one |
| Q.40 2-Methyl Benzoyl Chloride | Q.51 1-Amino methyl-2-ethyl cyclohexanol |
| Q.41 1,3,3-Trimethyl cyclohexene | Q.52 4-isopropyl -1-propyl cyclohexene
or 4-(methyl ethyl)-1-propyl cyclohexene |
| Q.42 Bicyclo(2,2,1) heptane | Q.53 2-(2-oxo-cyclohexyl) propanoic acid |
| | Q.54 3-ethoxy-1(1-nitrocyclohexyl)-hex-4-en-1-one |
| | Q.55 1,3-diphenyl-1,4-pentadiene |

EXERCISE # III

- Q.1 D Q.2 C Q.3 B Q.4 D Q.5 C Q.6 B
- Q.7 (A)-Q ; (B)-S ; (C)-T ; (D)-P Q.8 (A)-Q ; (B)-S ; (C)-R ; (D)-P
- Q.9 (A)-Q ; (B)-R ; (C)-S ; (D)-P Q.10 (A)-Q ; (B)-S ; (C)-R ; (D)-P
- Q.11 (A)-T ; (B)-P ; (C)-S ; (D)-Q Q.12 (A)-R ; (B)-S ; (C)-T ; (D)-Q ; (E)-P
- Q.13 D Q.14 (A)-Q ; (B)-S ; (C)-P ; (D)-R

EXERCISE # IV

- Q.1 C
- Q.2 $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{CH} - \text{COOH}$
 5 4 3 2 1
 2-pentene-1-oic acid and or 2-pentenoic acid
- Q.3 B Q.4 B
- Q.5 (a) 5,6-diethyl-3-methyl-dec-4-ene
 (b) N,N, 3-trimethyl-3-pentanamine
- Q.6 Butane-1,4-dioic acid Q.7 C Q.8 B Q.9 B,C
10. D

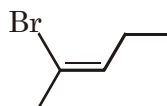
Important Notes

[illegible]

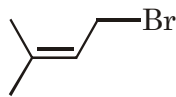
STEREO ISOMERISM

EXERCISE # I (MAINS ORIENTED)

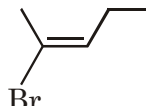
- 1.** Which one of the following statements concerning compounds **V–Z** is true :



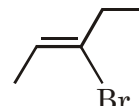
(V)



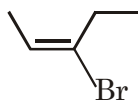
(W)



(X)

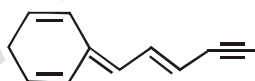
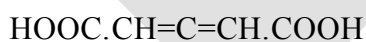


(Y)

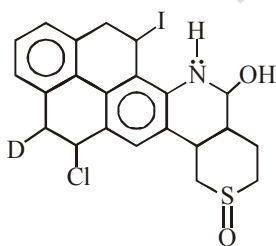


(Z)

- (A) **V** and **X** are conformational isomers
(B) **Y** and **Z** are constitutional isomers
(C) **X** and **Y** are constitutional isomers
(D) **V** and **Y** are stereoisomers
2. Which of the following compound has no isomer?
(A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$
(B) CH_3CHO
(C) $\text{CH}_2=\text{CH}-\text{Cl}$
(D) $\text{ClCH}_2\text{CH}_2\text{Cl}$
3. Compound CH_2Cl_2 contain :
(A) Plane of symmetry
(B) Centre of symmetry
(C) Axis of symmetry
(D) Both (A) & (C)
4. Number of POS present in CH_4 :
(A) 3
(B) 4
(C) 5
(D) 6
5. How many stereoisomers of the following molecule are possible ?
 $\text{HOOC}.\text{CH}=\text{C}=\text{CH}.\text{COOH}$
(A) Two optical isomers
(B) Two geometrical isomers
(C) Two optical and two geometrical isomers
(D) None
6. The number of cis-trans isomer possible for the following compound.

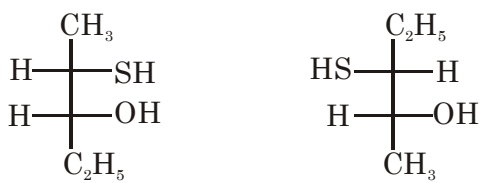


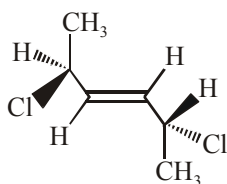
- (A) 2 (B) 4 (C) 6 (D) 8



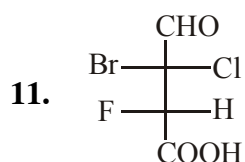
7.  has 'x' chiral centre then find the value of x :

- (A) 7 (B) 8 (C) 6 (D) 5

8. 
- (A) Diastereomers (B) Enantiomers (C) Identical (D) Constitutional isomers
9. The number of optically active compounds in the isomers of C_4H_9Br is :
- (A) 1 (B) 2 (C) 3 (D) 4
10. Compound have :

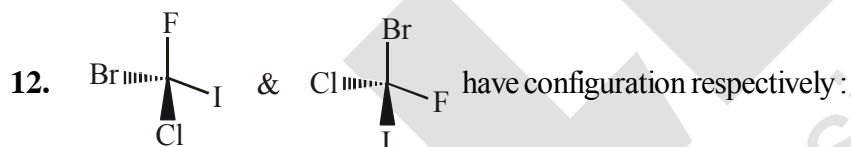


- (A) Plane of symmetry (B) Centre of symmetry
(C) Axis of symmetry (D) None



Configuration of compound is :

- (A) 2S, 3S (B) 2R, 3S (C) 2R, 3R (D) 2S, 3R

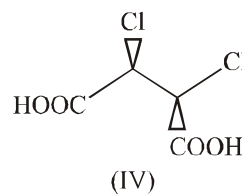
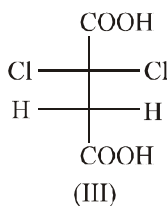
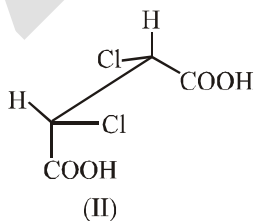
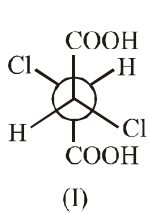


- (A) R, S (B) S, S (C) S, R (D) R, R

13. Minimum molecular weight of a hydrocarbon containing minimum number of C-atom to show optical isomerism :

- (A) 100 (B) 80 (C) 68 (D) 70

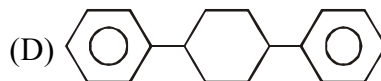
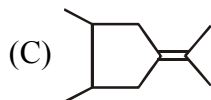
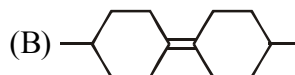
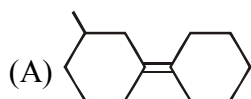
14. For the given configuration :



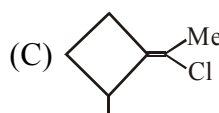
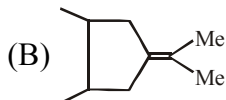
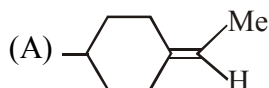
Which of the compound/configuration are optically active :

- (A) I (B) II (C) III (D) IV

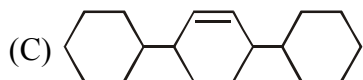
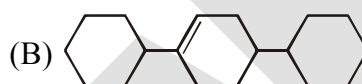
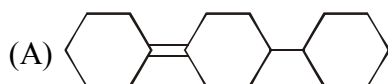
15. Compounds which can show both optical as well as geometrical isomerism :



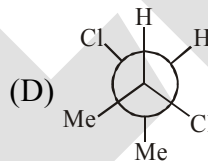
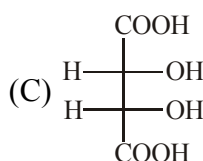
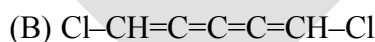
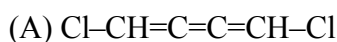
16. Which of the following will not show optical isomerism :



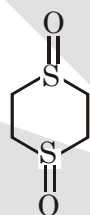
17. Optical & geometrical isomerism both can be shown by :-



18. Which of the following will not show optical isomerism :



19. The correct statement for the given compound is

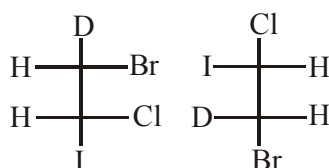


- (A) It can show geometrical isomerism (B) It can show optical isomerism
(C) It contains a chiral centre (D) None of these

20. Meso-tartaric acid and d-tartaric acid are :-

- (A) Positional isomers (B) Enantiomers (C) Diastereomers (D) Racemic mixture

21. The two compounds given below are :



- (A) Enantiomers (B) Diastereomers (C) Optically inactive (D) Identical